CHAPTER 11
FLIGHT CONTROLS

CHAPTER OVERVIEW

SECTION 11-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

11-1-1 FLIGHT CONTROL SYSTEM DESCRIPTION AND DATA.

11-1-1.1 System Description.

The flight control system is hydraulically power-boosted mechanical control system. The flight control system consists of the collective, cyclic, and tail rotor (directional) control subsystems. These subsystems use a series of push-pull rods, bellcranks, cables, pulleys, and servos that transmit control movements from cockpit to the main and tail rotors. The pilot and copilot have dual controls. Cyclic sticks control forward, rearward, and lateral helicopter movements; collective sticks control vertical helicopter movements; and tail rotor pedals control helicopter headings. Hydraulic power is supplied by the first stage, second stage, and backup hydraulic systems. Electrical power is supplied by the ac and dc electrical system. Assistance for the pilot or copilot in pitch, roll, yaw control is provided by the stability augmentation system (SAS), flight path stabilization (FPS), and electromechanical trim. For a complete description of these systems, refer to TM 11-70-23.

11-1-1.2 Principles Of Operation.

The flight controls are either manually operated, by the pilot or copilot moving the cyclic stick, collective stick, and tail rotor pedals, or automatically by the SAS. Movement of the cyclic or collective stick is transmitted by mechanical linkage to hydraulic servos for power assist, and then to the mixing unit. The mixing unit mechanically combines inputs to the main rotor and provides proportional control movements to the tail rotor. This takes place through the collective to yaw coupling and through the hydraulic primary servos. The primary servos move the main rotor swashplate, which changes blade pitch. The tail rotor pedals are connected by bellcranks, idlers, and control rods to the hydraulic yaw boost servo for power assist, and then through the mixing unit. Control cables transmit this movement to the rear control quadrant, then to a control rod to the hydraulic tail rotor servo. This moves the pitch change beam, which changes the tail rotor blade angles.

11-1-1.3 Subsystem Description.

11-1-1.3.1 Flight Control Self-Retaining Bolts.

Self-retaining (impedance) bolts are used as the primary connections in the flight controls system to prevent components from disconnecting accidently. These bolts are identified by a split collar on the bolt shank at the threaded end. The collar provides the self-retaining feature of the bolt. The split collar is compressed into a groove during installation and expands on the outside of the hole when the bolt is completely installed.

11-1-1.3.2 Collective Control Subsystem.

This subsystem gives vertical helicopter control. The collective sticks are connected through a series of control rods, bellcranks, the collective boost servo and the mixing unit to the primary servos. These all raise or lower the main rotor swashplate, independent of the cyclic position of the swashplate. This causes the pitch angle of all blades to change equally. The collective boost servo is powered by the second stage hydraulic system.

11-1-1.3.3 Collective Stick Assembly.

The pilot's collective stick assembly consists of a grip assembly, friction lock boot assembly, tube assembly, socket assembly, drag strut assembly, and associated wiring. The copilot's collective stick assembly consists of a grip assembly, telescoping tube assembly, socket assembly, and associated wiring. Both stick assemblies use the same grip assembly. The grip assembly has a LDG LT (landing light control) switch, SVO OFF (servo shutoff) toggle switch, SRCH LT (searchlight) control switch, searchlight control thumb switch, ENG RPM (engine speed trim) switch, and hook emergency release switch.

11-1-1.3.4 Cyclic Control Subsystem.

This subsystem provides forward, rearward, and lateral control of the helicopter. The cyclic sticks are mechanically-coupled, lever-type controls for both pilot and copilot. The cyclic sticks are connected through a torque shaft, a series of control rods, bellcranks, pitch trim assembly, roll assembly SAS actuator and a mixing unit to the primary servos. These control movement of the main rotor blades. The servos are powered by the first stage and second stage hydraulic systems.

11-1-1.3.5 Cyclic Stick Assembly.

The cyclic stick assembly consists of a grip assembly, tube assembly, socket assembly, and associated wiring. The grip assembly has a stick trim thumb switch, RADIO - ICS rocker switch, trim release pushbutton, PNL LTS (panel lights) kill switch, GA or GO ARND (go around enable) pushbutton, CARGO REL (cargo hook release) pushbutton. The HOIST thumb switch (hoist control) and HOIST REL pushbutton (emergency release switch) applies to cyclic grip assembly 70400-87261-101. The cyclic stick also houses a manual slew-up switch.

11-1-1.3.6 Tail Rotor Control Subsystem.

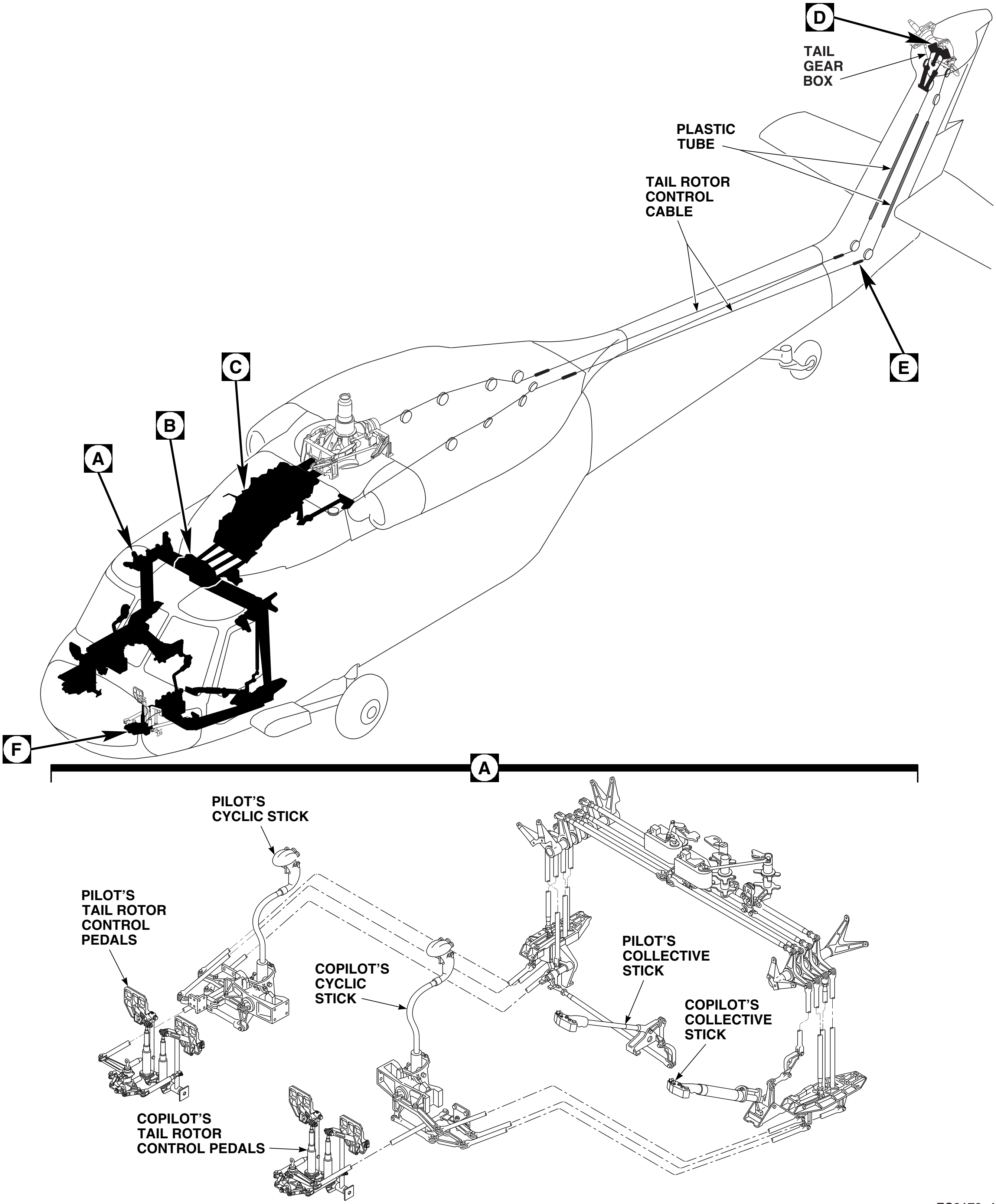
The tail rotor (directional) control subsystem determines helicopter heading, or yaw by controlling pitch of the tail rotor blades. The control pedals are connected through a series of control rods, bellcranks, yaw boost servo, the mixing unit, cables, and quadrants to the tail rotor servo. This moves the pitch change beam to change tail rotor blade angles. The tail rotor controls are powered by the first stage or backup hydraulic systems.

11-1-1.3.7 Directional Control Pedals.

The pedals are mechanically coupled and permit the pilot and copilot to control helicopter headings. The pedals contain independent toe-operated wheel brake controls. Each set of pedals can be adjusted to the pilot’s leg length.

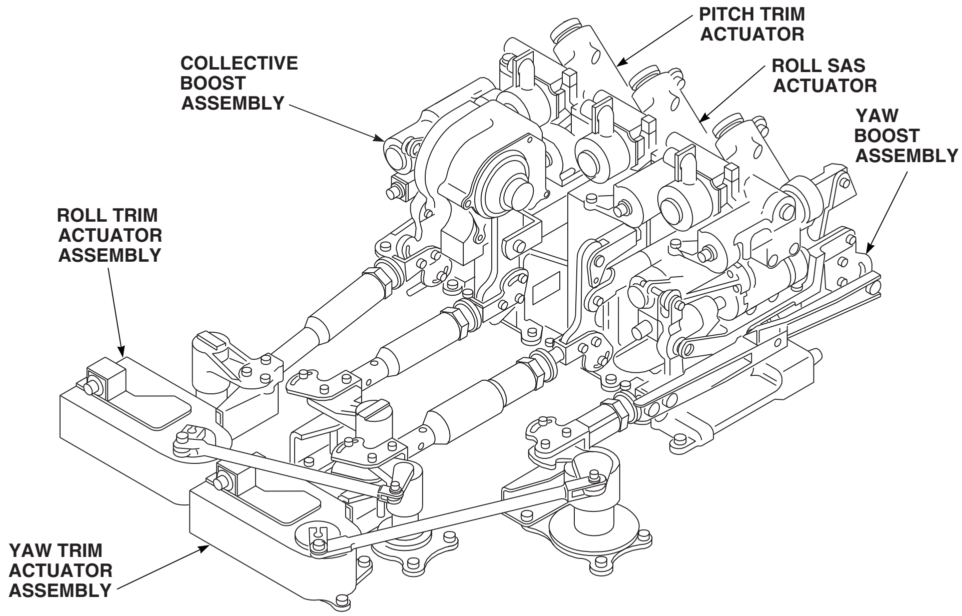
11-1-1.3.8 Tail Rotor Quadrant.

A tail rotor quadrant, mounted on the tail gear box, transmits tail rotor cable movements into the tail rotor servo. Two spring cylinders are connected to the quadrant. If one cable is broken, the spring cylinders allow the quadrant to operate normally. The TAIL ROTOR QUADRANT caution capsule on the caution/advisory panel will go on if a cable breaks. The remaining cable will unlatch when the helicopter is shut down.

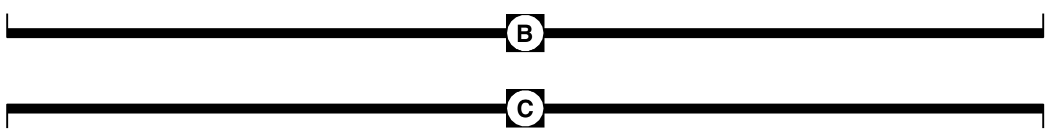


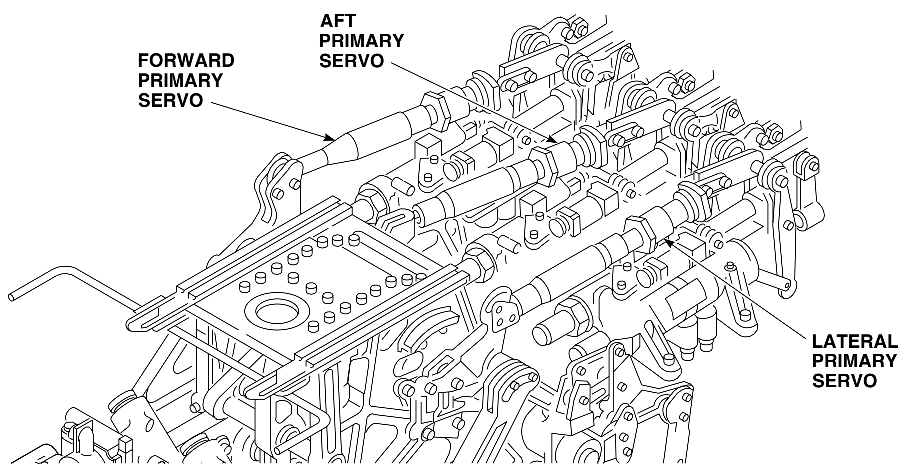
TYPICAL FLIGHT CONTROL ARRANGEMENT

FIGURE 11-1-1-1. FLIGHT CONTROL SYSTEM LOCATION DIAGRAM (SHEET 1 OF 3)



PILOT ASSIST ASSEMBLY





MIXER ASSEMBLY

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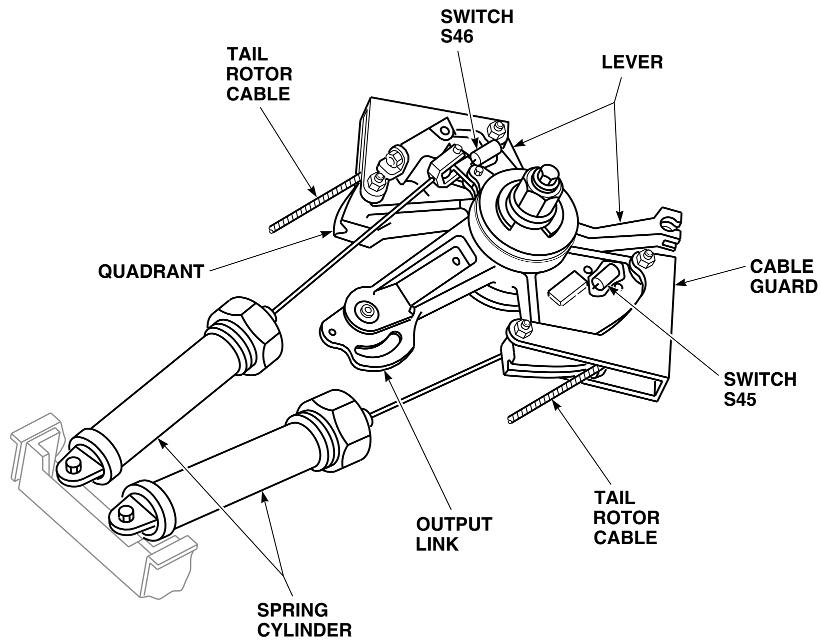
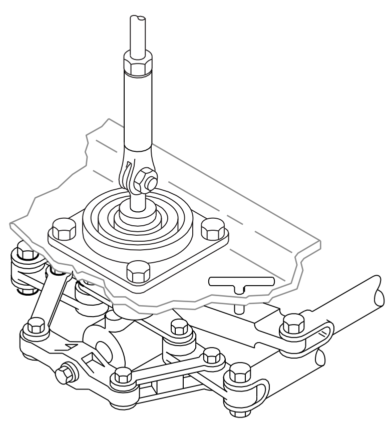
FIGURE 11-1-1-1. FLIGHT CONTROL SYSTEM LOCATION DIAGRAM (SHEET 2 OF 3)



TAIL ROTOR QUADRANT

D



PILOT'S/COPILOT'S
PEDAL ADJUSTOR

F

E

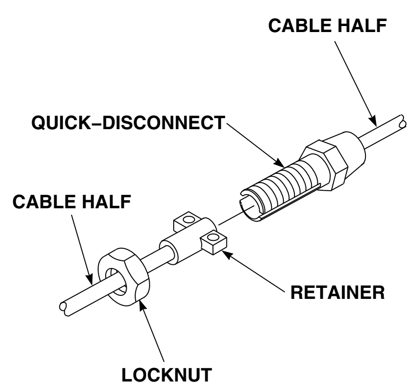


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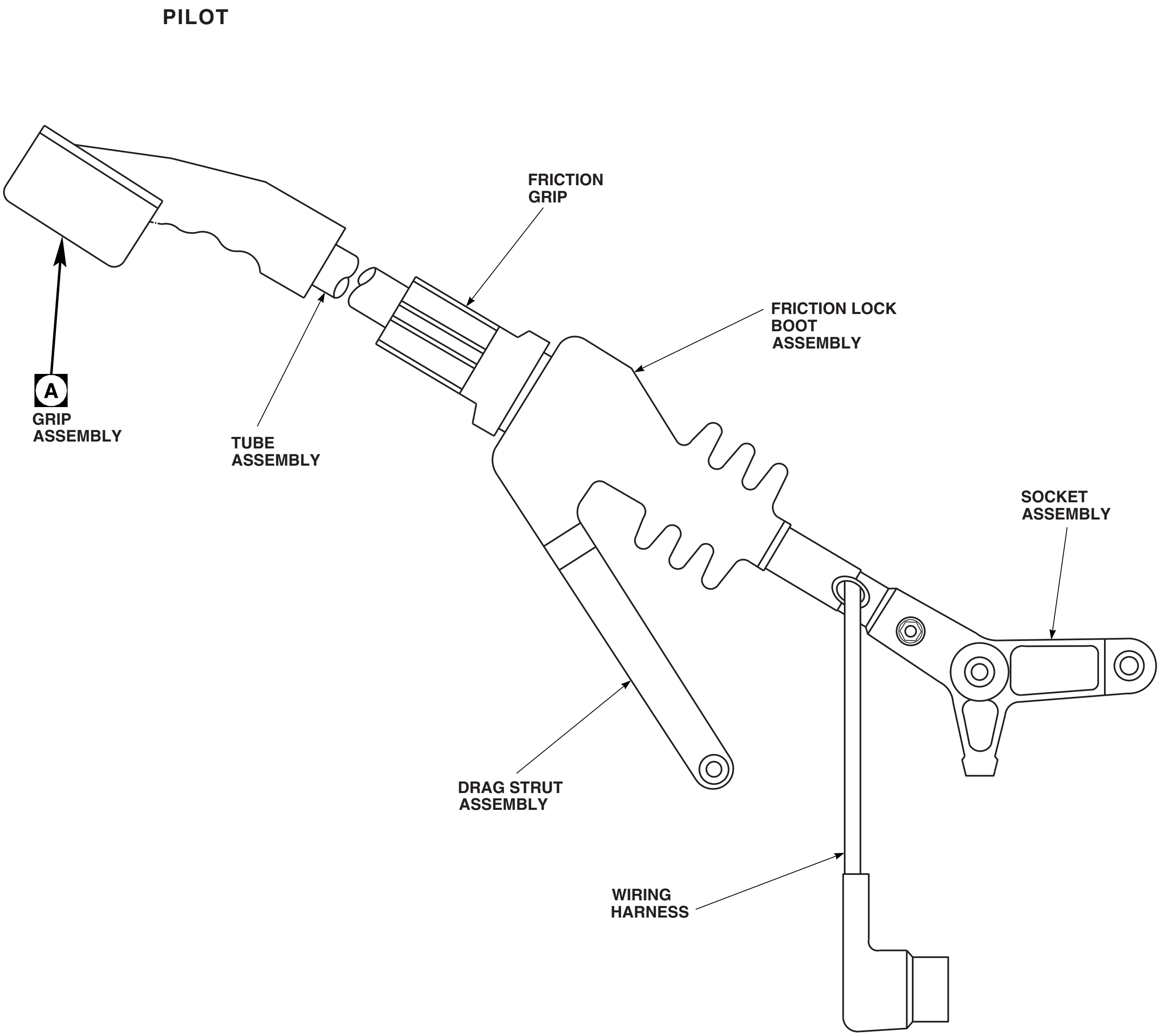
FIGURE 11-1-1-1. FLIGHT CONTROL SYSTEM LOCATION DIAGRAM (SHEET 3 OF 3)



NOTES
1. APPLICABLE FOR GRIP ASSEMBLIES 70400–01405–103 AND –104 INSTALLED ON COLLECTIVE STICK ASSEMBLIES 70400–01401–042, –044, –046, –048, 70400–01402–044 AND –046.
2. APPLICABLE FOR GRIP ASSEMBLIES 70400–01405–105 AND –106 INSTALLED ON COLLECTIVE STICK ASSEMBLIES 70400–01401–052, –054, 70400–01402–048 AND –050.

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FIGURE 11-1-1-2. COLLECTIVE STICK ASSEMBLY (SHEET 1 OF 2)

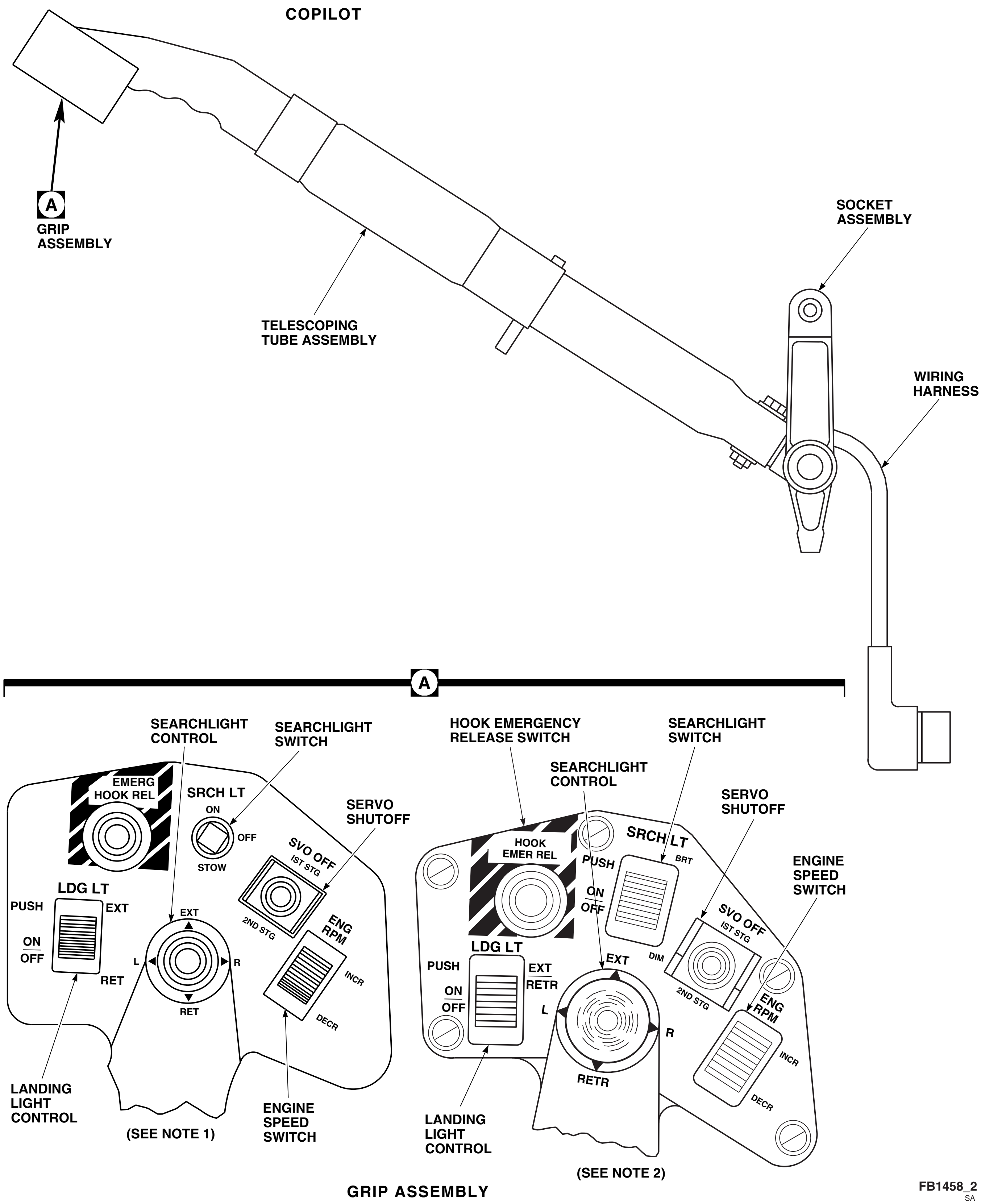
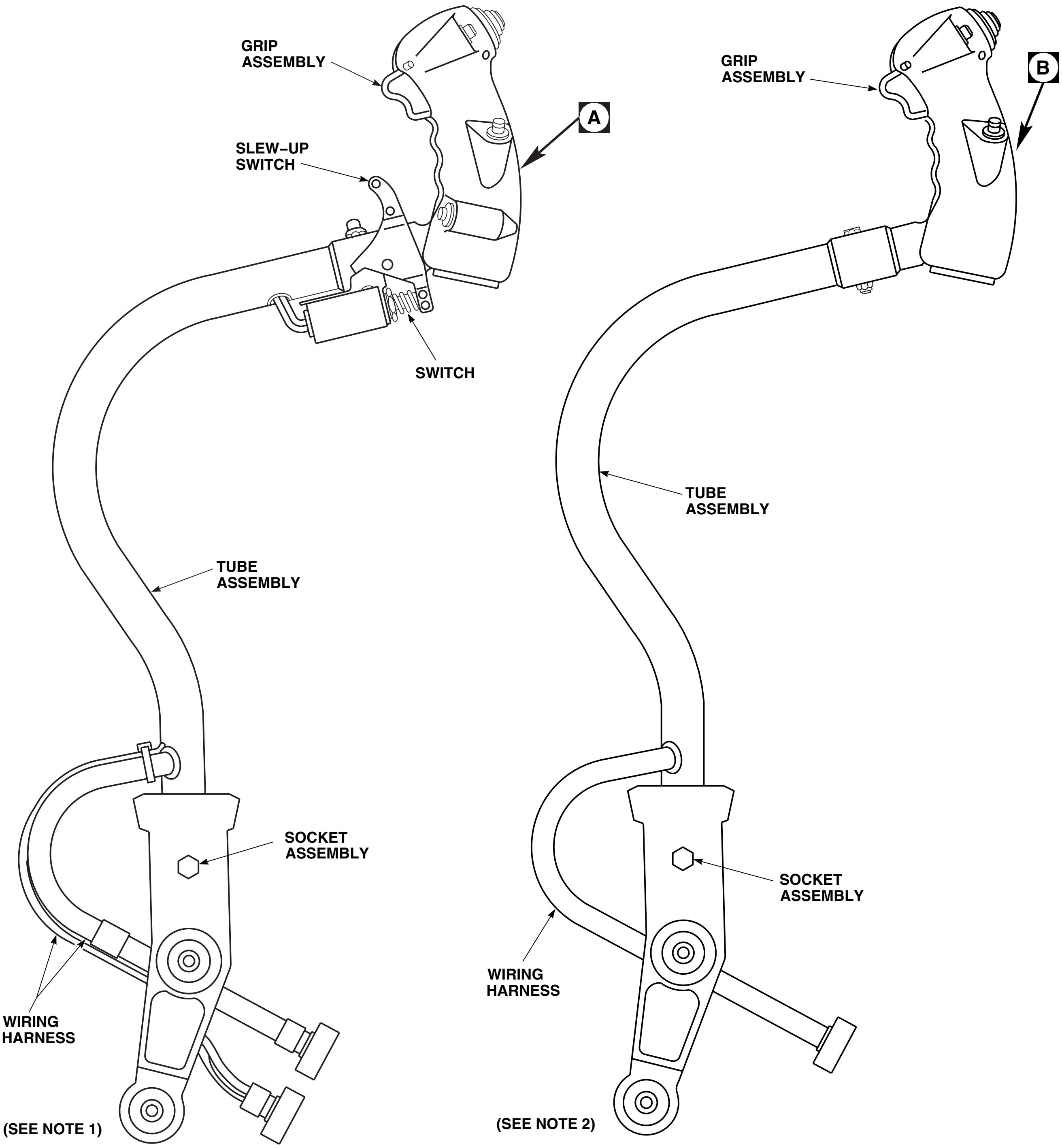


FIGURE 11-1-1-2. COLLECTIVE STICK ASSEMBLY (SHEET 2 OF 2)



NOTES

1. 70923 – SUBQ
2. 70583 – 701263
3. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–102.
4. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–101.

PILOT'S AND COPILOT'S CYCLIC STICK GRIP ASSEMBLY

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FIGURE 11-1-1-3. CYCLIC STICK ASSEMBLY (SHEET 1 OF 3)

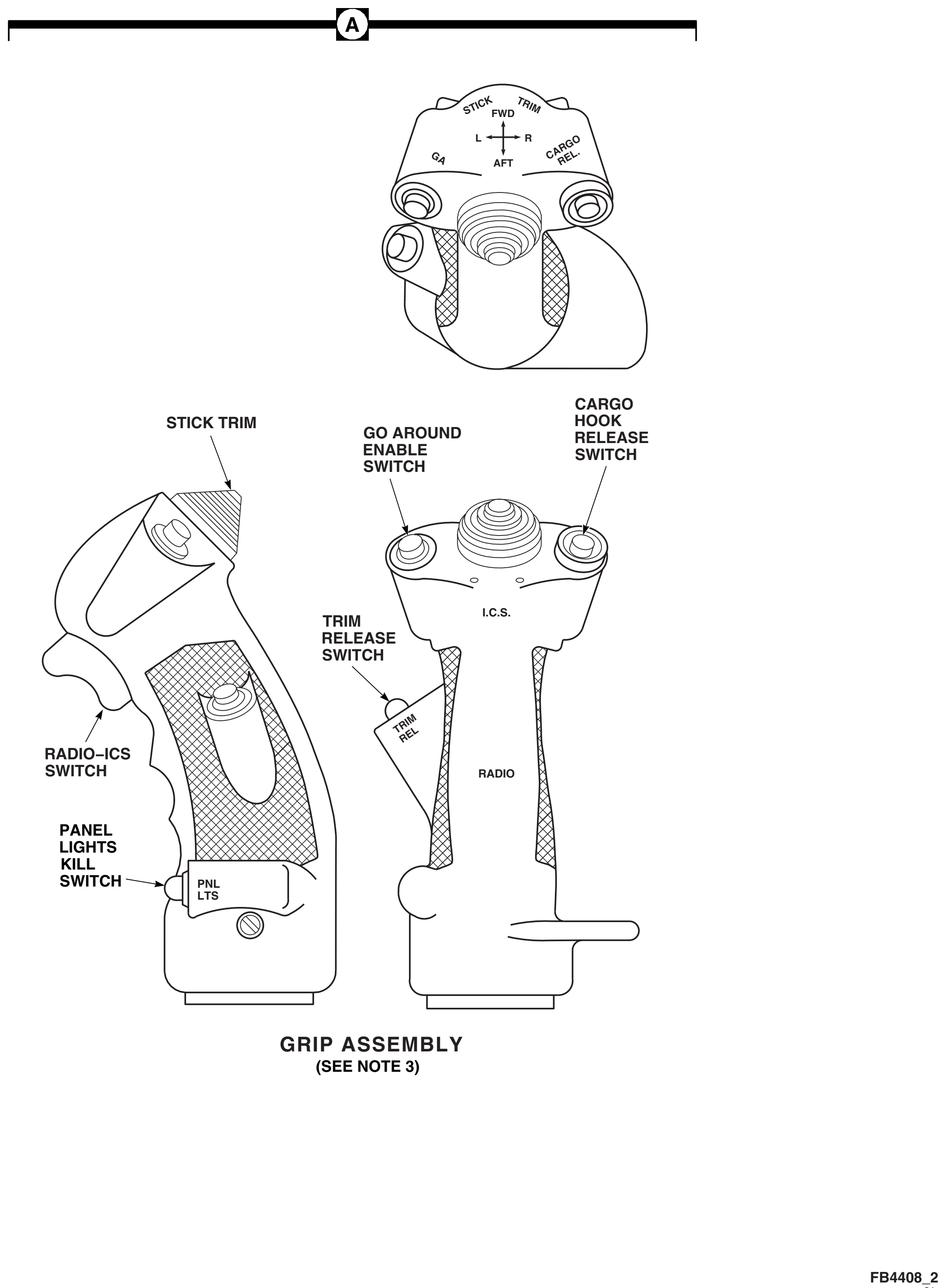


FIGURE 11-1-1-3. CYCLIC STICK ASSEMBLY (SHEET 2 OF 3)

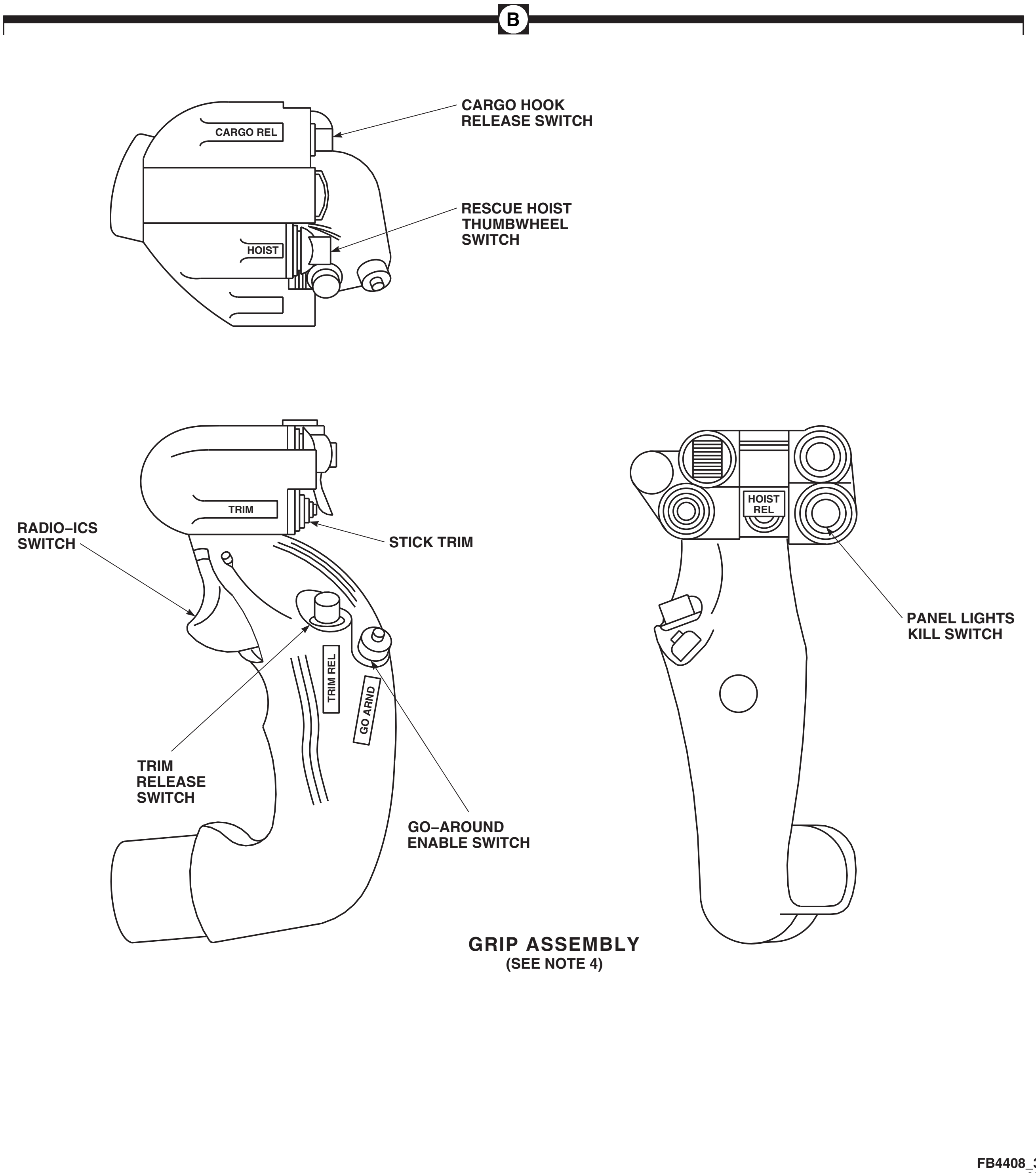
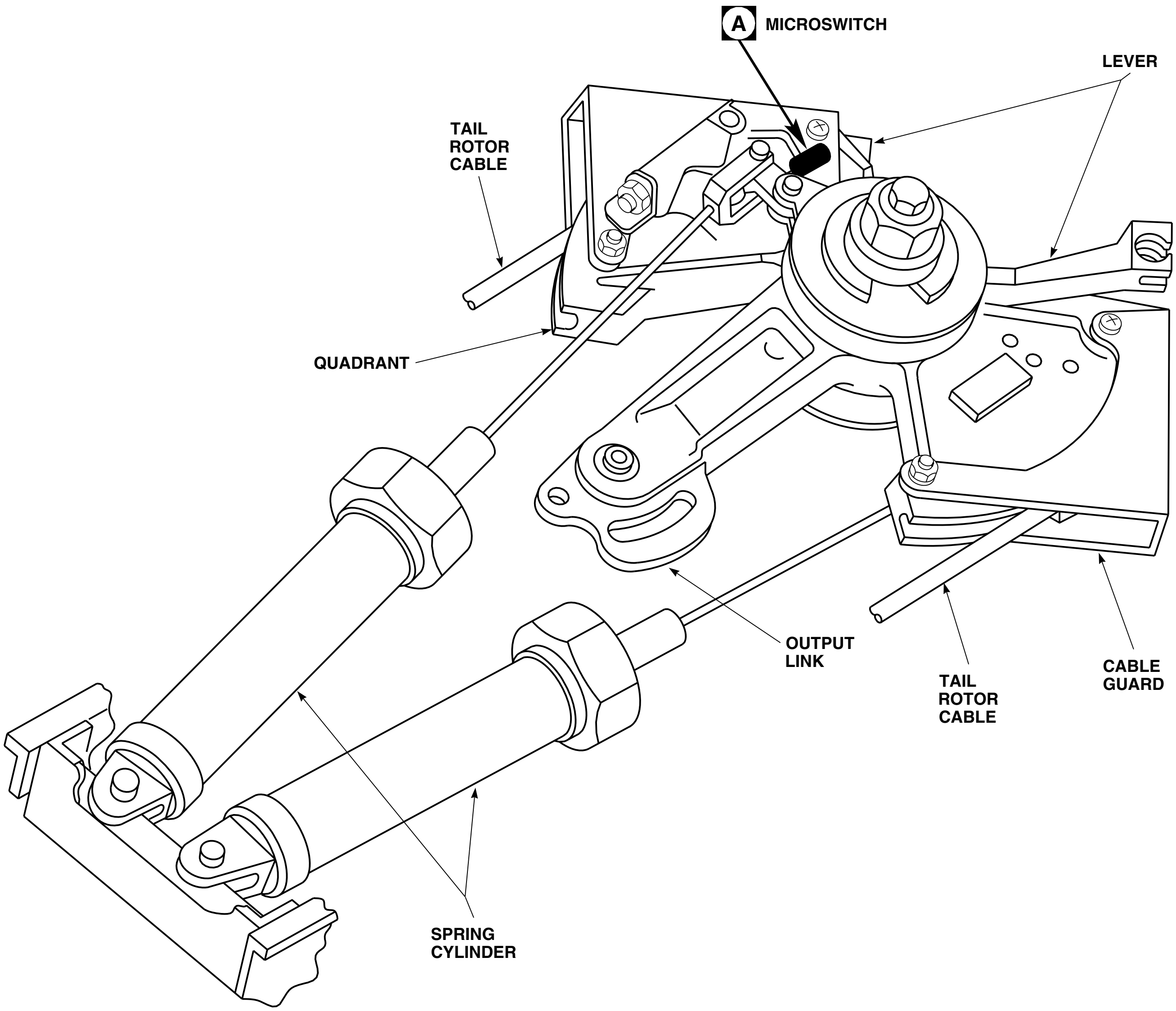
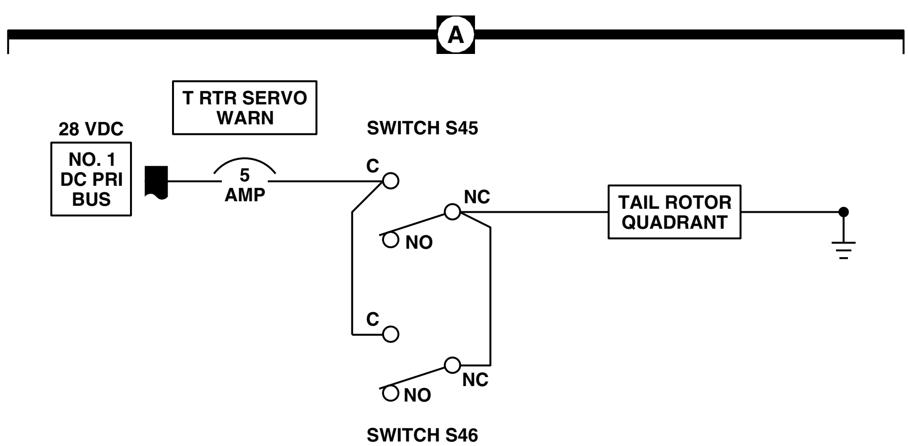


FIGURE 11-1-1-3. CYCLIC STICK ASSEMBLY (SHEET 3 OF 3)





TAIL ROTOR QUADRANT
WARNING SCHEMATIC

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FIGURE 11-1-1-4. TAIL ROTOR QUADRANT

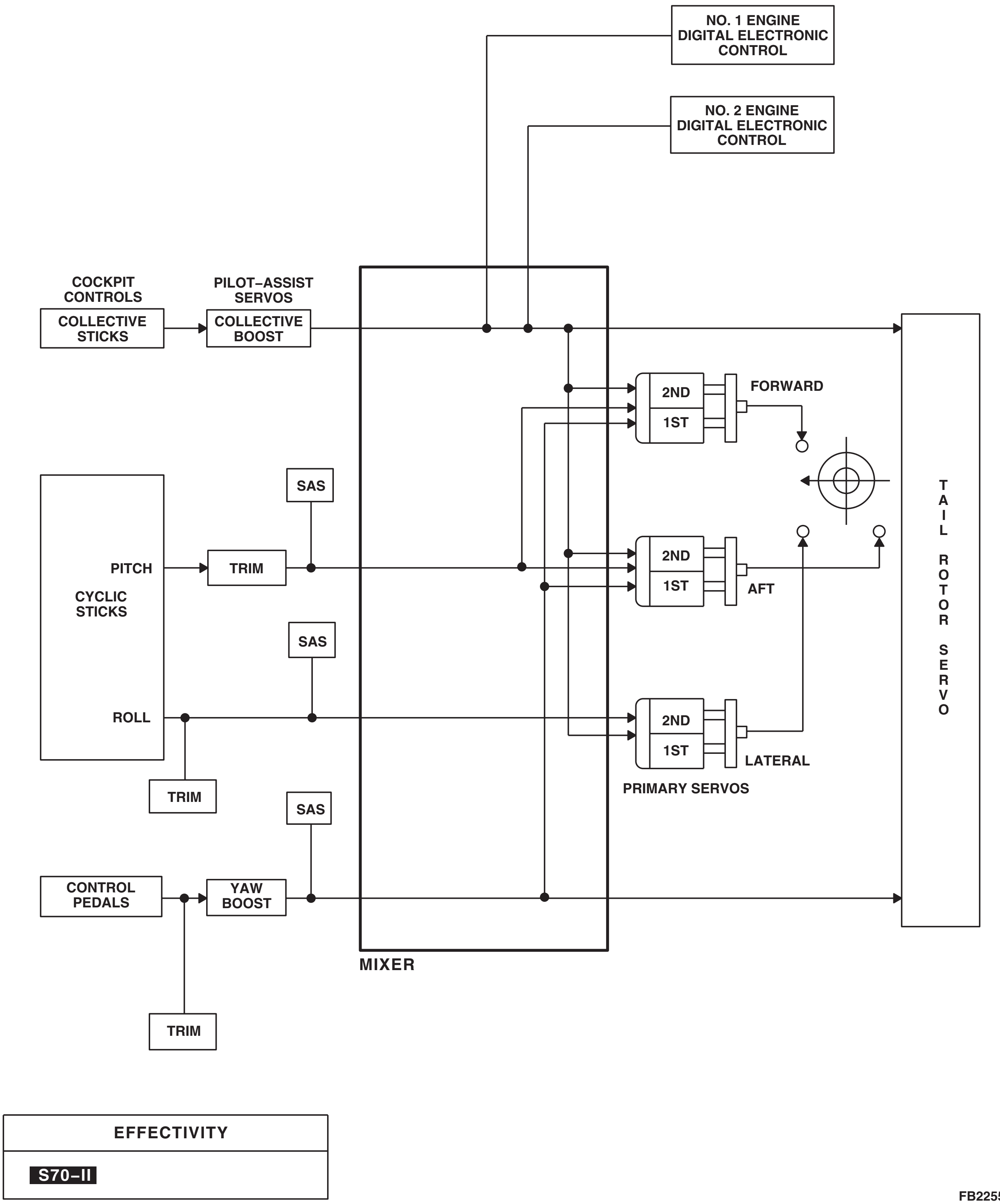


FIGURE 11-1-1-5. FLIGHT CONTROL SYSTEM BLOCK DIAGRAM

SECTION 11-2

TROUBLESHOOTING

SECTION OVERVIEW

11-2-1 CYCLIC STICK ASSEMBLY.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 27/AN

Tools

- Electrical Repairer Toolkit
- Torque Wrench, S-70 Torque Kit

Special Instructions

The cyclic stick assembly operational/troubleshooting procedure is subdivided as follows:

(1) Grip Assembly Electrical

(2) Slew-Up Switch Electrical 701263–SUBQ

(3) Grip Assembly Mechanical Inspection

(4) Slew-Up Switch Mechanical Inspection 701263–SUBQ

11-2-1.1 Grip Assembly Electrical.

a. Check continuity between the following pins while operating switches as listed below:

CHART NO. 11-2-1-1. Cyclic Stick Grip (70400-87261-101)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER READING (Ohms)	CORRECTIVE ACTION (Continuity) CONNECTOR PIN - SWITCH TERMINAL
TRIM REL	*Push	A - D	Infinity	**Replace switch
TRIM REL	Release	A - D	0	A - 1 D - 2
TRIM REL	*Push	B - C	0	B - 4 C - 3
TRIM REL	Release	B - C	Infinity	**Replace switch
CARGO REL	*Push	F - H	0	F - CARGO REL (F-20) H - CARGO REL (H-20)
CARGO REL	Release	F - H	Infinity	**Replace switch
ICS/RADIO	*First detent	k - m	0	k - ICS/RADIO (k - 22) m - ICS/RADIO (m-22)

CHART NO. 11-2-1-1. Cyclic Stick Grip (70400-87261-101) (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER READING (Ohms)	CORRECTIVE ACTION (Continuity) CONNECTOR PIN - SWITCH TERMINAL
ICS/RADIO	*Release	k - m	Infinity	**Replace switch
ICS/RADIO	*Second detent	k - s	0	k - ICS/RADIO (k-22) s - ICS/RADIO (s-22)
ICS/RADIO	*Release	k - s	Infinity	**Replace switch
HOIST	Center (off)	q - t	Infinity	**Replace switch
HOIST	Center (off)	q - p	Infinity	**Replace switch
HOIST	*Up	q - p	0	q - E p - A
HOIST	*Down	q - t	0	q - E t - C
HOIST REL	*Push	h - P	0	h - 3 P - 4
HOIST REL	*Push	S - P	Infinity	**Replace switch
HOIST REL	Release	S - P	0	S - 2 P - 1
HOIST REL	Release	h - P	Infinity	**Replace switch
TRIM	Center (off)	j - i	Infinity	**Replace switch
TRIM	Center (off)	j - U	Infinity	**Replace switch
TRIM	Center (off)	j - V	Infinity	**Replace switch
TRIM	Center (off)	j - W	Infinity	**Replace switch
TRIM	*Left	j - i	0	j - E i - B
TRIM	*Right	j - U	0	j - E U - D

CHART NO. 11-2-1-1. Cyclic Stick Grip (70400-87261-101) (Cont)

SWITCH	SWITCH POSITION	CHECK RE-SISTANCE BETWEEN CONNECTOR PINS:	METER READING (Ohms)	CORRECTIVE AC-TION (Continuity) CONNECTOR PIN - SWITCH TERMINAL
TRIM	*Forward	j - V	0	j - E V - A
TRIM	*Aft	j - W	0	j - E W - C
GO ARND	*Push	Y - Z	Infinity	**Replace switch
GO ARND	Release	Y - Z	0	Y - 1 Z - 2
GO ARND	*Push	L - M	0	M - 3 L - 4
GO ARND	Release	L - M	Infinity	**Replace switch
PANEL LTS	Push (on)	c - d	0	c - PANEL LTS (c-20) d - PANEL LTS (d-20)
PANEL LTS	Push (on)	J - K	Infinity	**Replace switch
PANEL LTS	Push (off)	J - K	0	J - PANEL LTS (J-20) K - PANEL LTS (K-20)
PANEL LTS	Push (off)	c - d	Infinity	**Replace switch

*Indicates momentary contact. Switch must be held at this position.
**PARA 11-4-18, PARA 11-4-19, or PARA 11-4-20

CHART NO. 11-2-1-2. Cyclic Stick Grip (70400-87261-102)

SWITCH	SWITCH POSITION	CHECK RE-SISTANCE BETWEEN CONNECTOR PINS:	METER READING (Ohms)	CORRECTIVE AC-TION (Continuity) CONNECTOR PIN - SWITCH TERMINAL
TRIM REL	*Push	A - D	Infinity	**Replace switch

CHART NO. 11-2-1-2. Cyclic Stick Grip (70400-87261-102) (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER READING (Ohms)	CORRECTIVE ACTION (Continuity) CONNECTOR PIN - SWITCH TERMINAL
TRIM REL	Release	A - D	0	A - 1 D - 2
CARGO REL	*Push	E - F	0	E - CARGO REL. (E-20) F - CARGO REL. (F-20)
CARGO REL	Release	E - F	Infinity	**Replace switch
RADIO - ICS	*RADIO (Push)	G - H	0	G - RADIO - ICS (G-20) H - RADIO - ICS (H-20)
RADIO - ICS	*RADIO	J - K	Infinity	**Replace switch
RADIO - ICS	*ICS (Push)	J - K	0	J - RADIO - ICS (J-20) K - RADIO - ICS (K-20)
RADIO - ICS	*ICS	G - H	Infinity	**Replace switch
STICK TRIM	Center (off)	T - U	Infinity	**Replace switch
STICK TRIM	Center (off)	T - V	Infinity	**Replace switch
STICK TRIM	Center (off)	T - W	Infinity	**Replace switch
STICK TRIM	Center (off)	T - X	Infinity	**Replace switch
STICK TRIM	*L	T - U	0	T - 16 U - 15
STICK TRIM	*R	T - X	0	T - 6 X - 5
STICK TRIM	*FWD	T - V	0	T - 2 V - 1
STICK TRIM	*AFT	T - W	0	T - 12 W - 11
GA	*Push	Y - Z	Infinity	**Replace switch
GA	Release	Y - Z	0	Y - 1 Z - 2

CHART NO. 11-2-1-2. Cyclic Stick Grip (70400-87261-102) (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER READING (Ohms)	CORRECTIVE ACTION (Continuity) CONNECTOR PIN - SWITCH TERMINAL
PNL LTS	Push (on)	N - P	0	N - 3 P - 4
PNL LTS	Push (on)	a - b	Infinity	**Replace switch
PNL LTS	Push (off)	a - b	0	a - 1 b - 2
PNL LTS	Push (off)	N - P	Infinity	**Replace switch

*Indicates momentary contact. Switch must be held at this position.
**PARA 11-4-18, PARA 11-4-19, or PARA 11-4-20

RESULT: Meter shall indicate corresponding resistance.

- If result is not as specified, do corresponding corrective action.
  (a) If a meter indication of infinity is specified, and continuity is found, replace switch (PARA 11-4-18, PARA 11-4-19, or PARA 11-4-20).
  (b) If a meter indication of 0 is specified, and continuity is not found, repair/replace wiring (Appendix F).

11-2-1.2 Slew-Up Switch Electrical 701263–SUBQ .

a. Check continuity between the following pins while operating switches as listed below:

SLEW- UP SWITCH POSITION	CHECK RESISTANCE BETWEEN PINS: (P7R or P8R)	METER INDICATION (Ohms)	CORRECTIVE ACTION
*Push	A - G	0	A to 1 NO 1 NO to 1 Wiper 1 Wiper to 2 Wiper 2 Wiper to G
Release	A - G	Infinity	**Replace switch

SLEW- UP SWITCH POSITION	CHECK RESISTANCE BETWEEN PINS: (P7R or P8R)	METER INDICATION (Ohms)	CORRECTIVE ACTION
*Push	G - H	0	G to 2 NO 2 NO to 2 Wiper
Release	G - H	Infinity	**Replace switch
*Push	C - E	0	C to 4 NO 3 Wiper to 4 Wiper
Release	C - E	Infinity	**Replace switch
*Push	D - E	0	D to 3 NO 3 NO to 3 Wiper 3 Wiper to 4 Wiper 4 Wiper to E
Release	D - E	Infinity	**Replace switch
*Push	E - C	0	C to 4 NO 4 NO to 4 Wiper
Release	E - C	Infinity	**Replace switch

*Indicates momentary contact. Switch must be held at this position.
**TM 11-70-23

RESULT: Meter shall indicate corresponding resistance.

- If result is not as specified, do corresponding corrective action.

  (a) If a meter indication of infinity is specified, and continuity is found, replace switch (TM 11-70-23).

  (b) If a meter indication of 0 is specified, and continuity is not found, repair/replace wiring (Appendix F).

11-2-1.3 Grip Assembly Mechanical Inspection.

a. Visually check grip for cracks.

RESULT: Grip shall not be cracked.

- If result is not as specified, replace grip assembly (PARA 11-4-17).

b. Visually check tube assembly for cracks.

RESULT: Tube assembly shall not be cracked.

- If result is not as specified, replace tube assembly (PARA 11-4-24).

c. Visually check grip for loose or cracked controls.

RESULT: Grip controls shall not be loose or cracked.

- If result is not as specified, replace grip assembly (PARA 11-4-17).

d. If tag attached to assembly indicates rough or sloppy stick movement, do the following:

(1) Inspect socket assembly bearings for ratcheting or play.

• If ratcheting or play is found, replace malfunctioning bearing(s) (PARA 11-4-23) and/or socket assembly (PARA 11-4-22).

(2) Check each bearing for binding by turning with torque wrench. Bearing shall turn with 2 inch-ounce torque or less.

• If bearing does not turn, replace malfunctioning bearing(s) (PARA 11-4-23) and/or socket assembly (PARA 11-4-22).

11-2-1.4 Slew-Up Switch Mechanical Inspection 701263–SUBQ .

a. Inspect spring for proper tension by determining if trigger returns to static stop (full open position) after pressing trigger.

RESULT: Trigger shall return to full open position.

• If result is not as specified, replace spring (TM 11-70-23).

b. Inspect trigger guide and switch button for proper alignment.

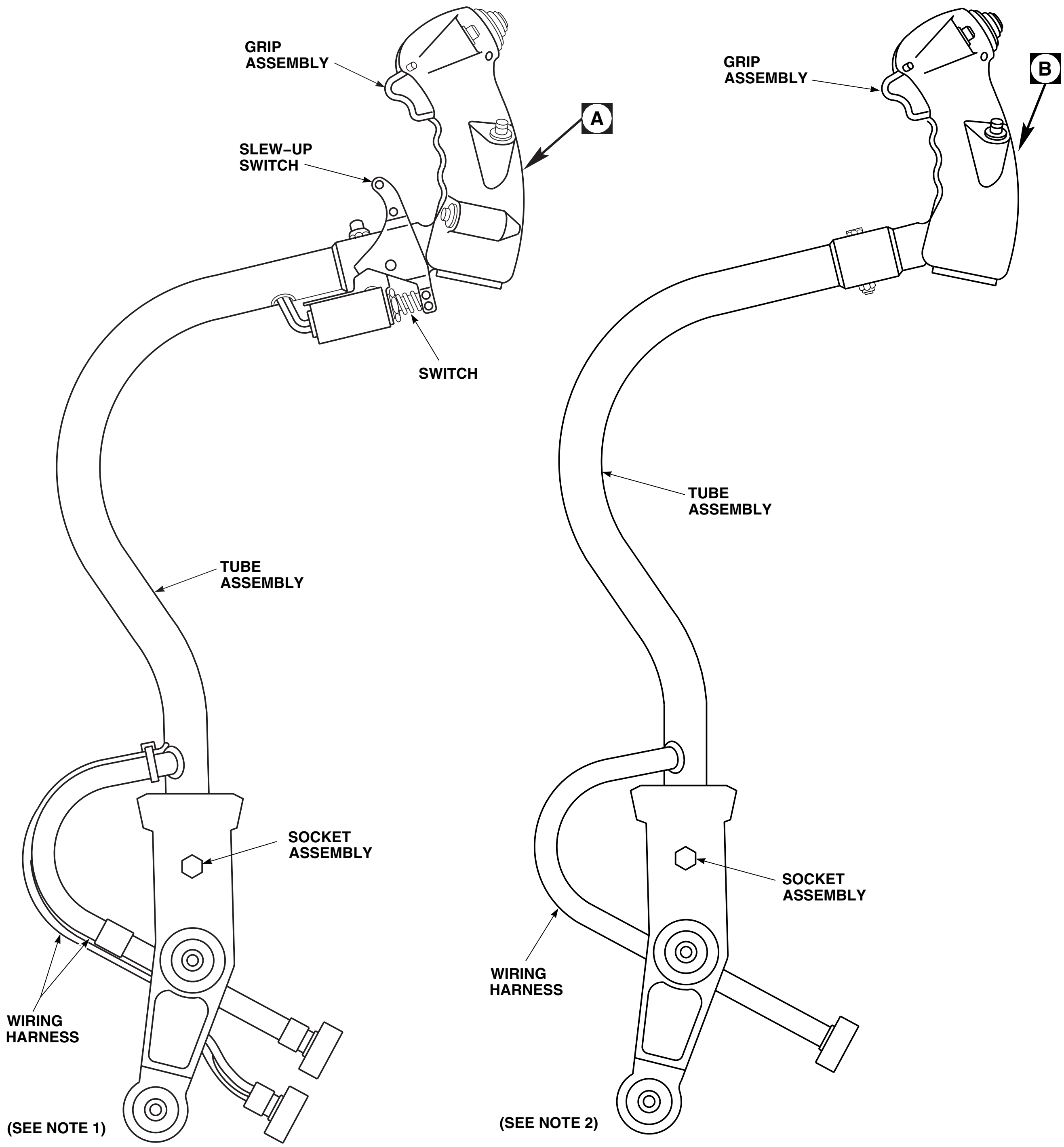
RESULT: Trigger guide shall remain on switch button and cause no binding.

• If result is not as specified, replace trigger assembly (TM 11-70-23).

c. Inspect trigger assembly for looseness or play.

RESULT: No looseness or play shall be found.

• If result is not as specified, replace trigger assembly (TM 11-70-23).



NOTES

1. 701263 – SUBQ
2. 70583 – 701262
3. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–102.
4. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–101.

PILOT'S AND COPILOT'S CYCLIC STICK GRIP ASSEMBLY

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FIGURE 11-2-1-1. CYCLIC STICK ASSEMBLY (SHEET 1 OF 3)

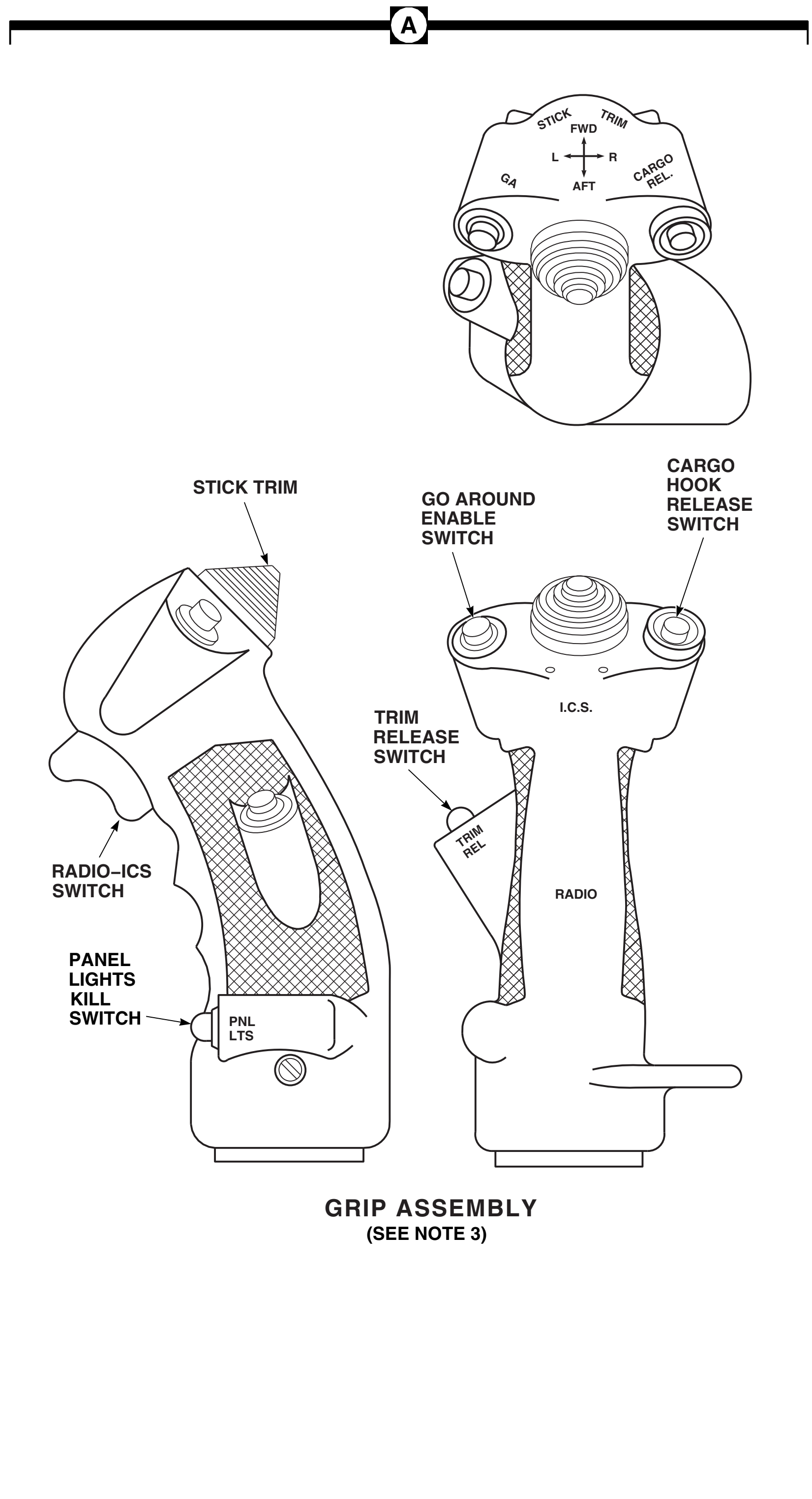


FIGURE 11-2-1-1. CYCLIC STICK ASSEMBLY (SHEET 2 OF 3)

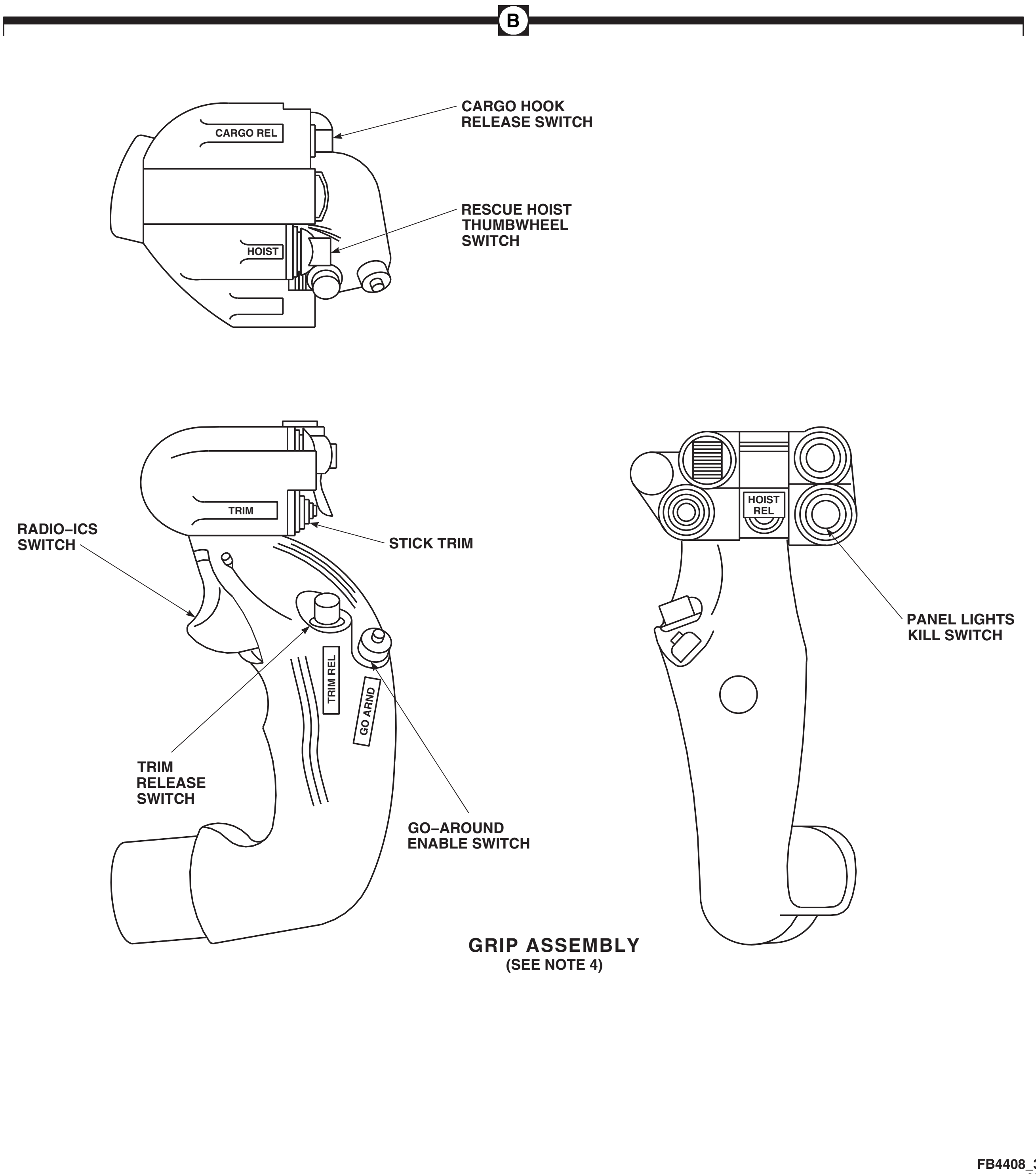
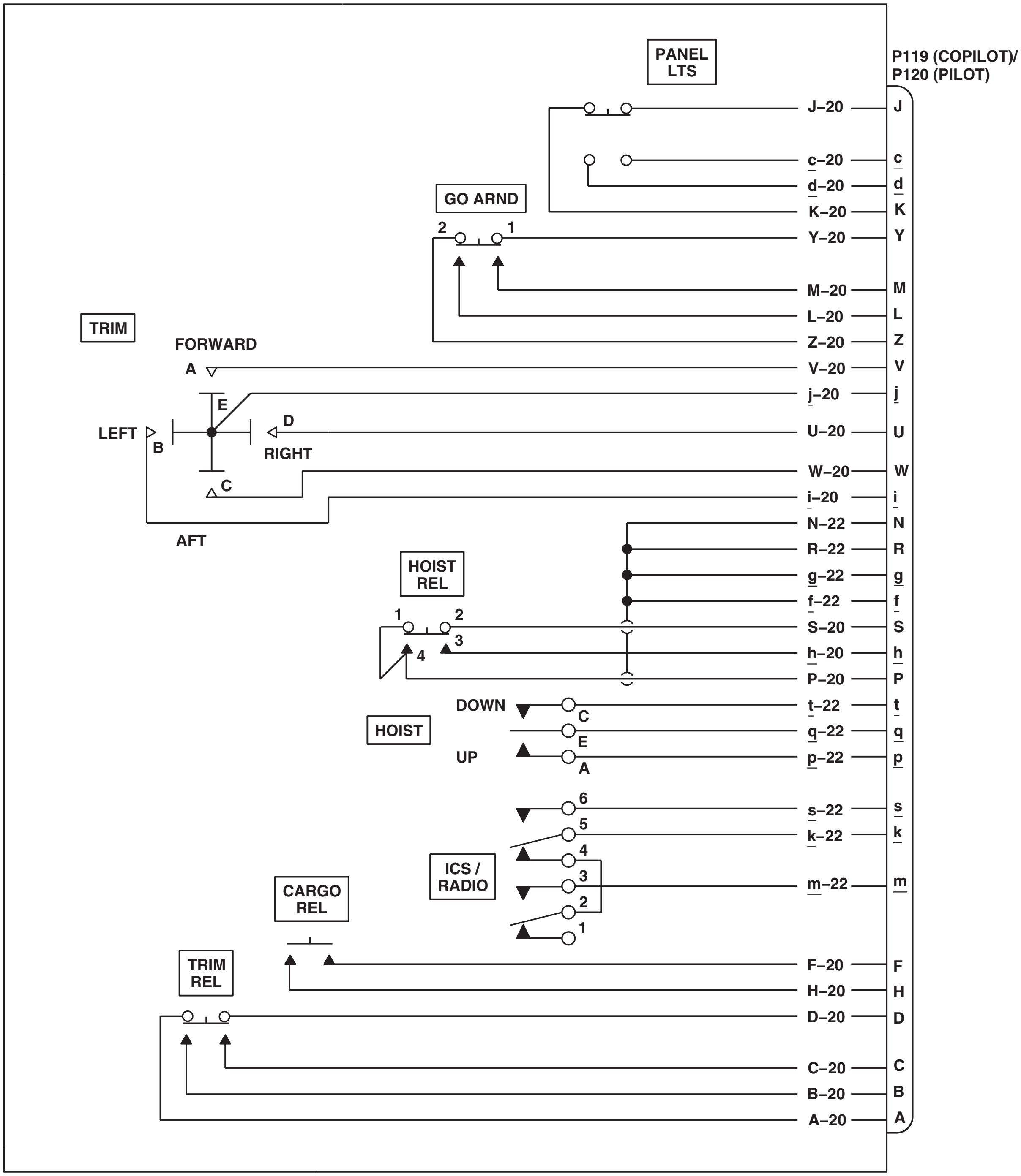


FIGURE 11-2-1-1. CYCLIC STICK GRIP ASSEMBLY (SHEET 3 OF 3)



NOTES
1. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–101.
2. WIRES B–20, C–20, L–20, M–20 ARE BOMBED.
3. CYCLIC GRIP ASSEMBLY PART NO. 70400–87261–102.

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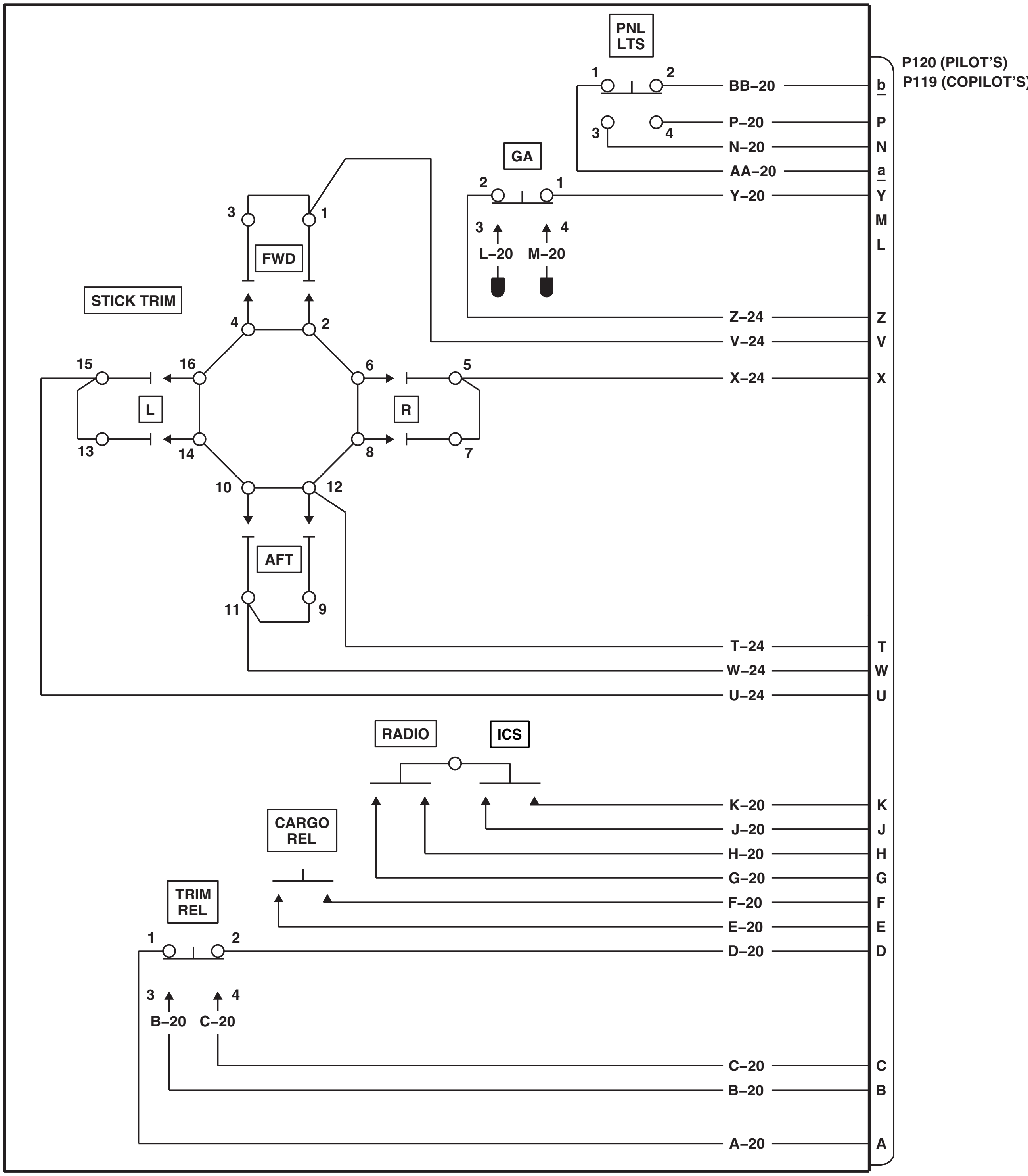
FIGURE 11-2-1-2. CYCLIC STICK GRIP ASSEMBLY SCHEMATIC DIAGRAM (SHEET 1 OF 2)



EFFECTIVITY
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FIGURE 11-2-1-2. CYCLIC STICK GRIP ASSEMBLY SCHEMATIC DIAGRAM (SHEET 2 OF 2)

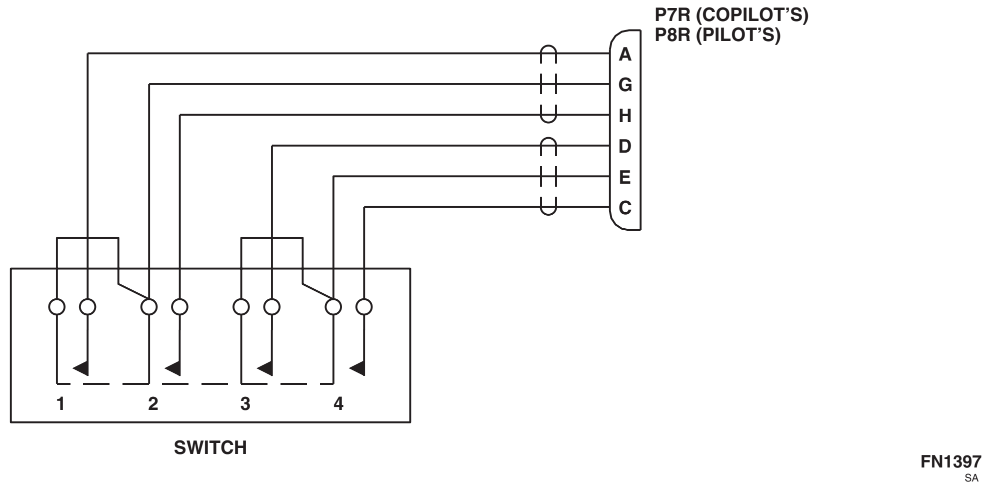


FIGURE 11-2-1-3. CYCLIC STICK SLEW-UP SWITCH SCHEMATIC DIAGRAM

11-2-2 COLLECTIVE STICK ASSEMBLY.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Electrical Repairer Toolkit
- Torque Wrench, S-70 Torque Kit

References

- PARA 1-6-16

Equipment Conditions

- External Electrical Power Available

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (PARA 9-2-3).

Special Instructions

The collective stick assembly operational/ troubleshooting procedure is subdivided as follows:

(1) Electrical

(2) Mechanical Inspection, Pilot

(3) Mechanical Inspection, Copilot

11-2-2.1 Electrical.

a. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

b. Turn on electrical power (PARA 1-6-16).

c. Adjust both upper console INST LT PILOT FLT and CPLT FLT INST LTS to BRT.

RESULT: Collective sticks shall be evenly lit.

- If result is not as specified, go to TABLE NO. 11-2-2-4.

d. Turn off electrical power (PARA 1-6-16).

e. Do TABLE NO. 11-2-2-1, TABLE NO. 11-2-2-2, or TABLE NO. 11-2-2-3:

TABLE NO. 11-2-2-1. Continuity Check (Collective Stick Assemblies: 70400-01401-052, -054 And 70400-01402-048, -050).

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
HOOK EMER REL	Push (on)	F - H	0	H - 1 F - 2
HOOK EMER REL	Push (off)	F - H	Infinity	Replace switch (PARA 11-4-34)

TABLE NO. 11-2-2-1. Continuity Check (Collective Stick Assemblies: 70400-01401-052, -054 And 70400-01402-048, -050). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
ENG RPM	Center (off)	h - g	0	h - 5 g - 6
ENG RPM	Center (off)	h - i	Infinity	Replace switch (PARA 11-4-35)
ENG RPM	Center (off)	k - j	0	k - 2 j - 1
ENG RPM	Center (off)	k - m	Infinity	Replace switch (PARA 11-4-35)
ENG RPM	*INCR	k - m	0	m - 3
ENG RPM	*INCR	g - h	0	Replace switch (PARA 11-4-35)
ENG RPM	*DECR	h - i	0	i - 4
ENG RPM	*DECR	j - k	0	Replace switch (PARA 11-4-35)
LDG LT	*EXT	U - W	0	U - 1 W - 3
LDG LT	*EXT	U - S	Infinity	Replace switch (PARA 11-4-34)
LDG LT	Center (off)	U - W	Infinity	Replace switch (PARA 11-4-34)
LDG LT	Center (off)	U - S	Infinity	Replace switch (PARA 11-4-34)
LDG LT	*RETR	U - S	0	S - 2
LDG LT	*RETR	U - W	Infinity	Replace switch (PARA 11-4-34)
LDG LT	*Pressed	X - Y	0	X - 5 Y - 4

TABLE NO. 11-2-2-1. Continuity Check (Collective Stick Assemblies: 70400-01401-052, -054 And 70400-01402-048, -050). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
LDG LT	Not Pressed	X - Y	Infinity	Replace switch (PARA 11-4-34)
SRCH LT	Center (off)	L - M	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	Center (off)	M - N	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	*DIM	M - N	0	M - 1 N - 3
SRCH LT	*DIM	L - M	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	*BRT	L - M	0	L - 2 M - 1
SRCH LT	*BRT	M - N	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	*Pressed	K - P	0	K - 5 P - 4
SRCH LT	Not Pressed	K - P	Infinity	Replace switch (PARA 11-4-33)
Searchlight control	Center (off)	C - A	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - B	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - D	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - e	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	*EXT	C - e	0	e - P C - E

TABLE NO. 11-2-2-1. Continuity Check (Collective Stick Assemblies: 70400-01401-052, -054 And 70400-01402-048, -050). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (Continuity)
Searchlight control	*RETR	C - B	0	B - B
Searchlight control	*L	C - A	0	A - A
Searchlight control	*R	C - D	0	D - C
SVO OFF	1ST STG	f - J	0	J - 6 f - 5 or replace switch (PARA 11-4-33)
SVO OFF	1ST STG	c - n	0	n - 3 c - 2 or replace switch (PARA 11-4-33)
SVO OFF	2ND STG	f - p	0	p - 4 or replace switch (PARA 11-4-33)
SVO OFF	2ND STG	c - d	0	d - 1 or replace switch (PARA 11-4-33)

*Indicates momentary contact. Switch must be held at this position.

TABLE NO. 11-2-2-2. Continuity Check (Collective Stick Assembly: 70400-01402-046).

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
EMERG HOOK REL	Push (on)	F - H	0	H - 1 F - 2
EMERG HOOK REL	Push (off)	F - H	Infinity	Replace switch (PARA 11-4-34)

TABLE NO. 11-2-2-2. Continuity Check (Collective Stick Assembly: 70400-01402-046). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
ENG RPM	Center (off)	h - g	0	h - 5 g - 6
ENG RPM	Center (off)	h - i	Infinity	Replace switch (PARA 11-4-35)
ENG RPM	Center (off)	k - j	0	k - 2 j - 1
ENG RPM	Center (off)	k - m	Infinity	Replace switch (PARA 11-4-35)
ENG RPM	*INCR	k - m	0	m - 3
ENG RPM	*INCR	g - h	0	Replace switch (PARA 11-4-35)
ENG RPM	*DECR	h - i	0	i - 4
ENG RPM	*DECR	j - k	0	Replace switch (PARA 11-4-35)
LDG LT	*EXT	U - W	0	U - 1 W - 3
LDG LT	*EXT	U - S	Infinity	Replace switch (PARA 11-4-34)
LDG LT	Center (off)	U - W	Infinity	Replace switch (PARA 11-4-34)
LDG LT	Center (off)	U - S	Infinity	Replace switch (PARA 11-4-34)
LDG LT	*RET	U - S	0	S - 2
LDG LT	*RET	U - W	Infinity	Replace switch (PARA 11-4-34)
LDG LT	*Pressed	X - Y	0	X - 5 Y - 4

TABLE NO. 11-2-2-2. Continuity Check (Collective Stick Assembly: 70400-01402-046). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
LDG LT	Not Pressed	X - Y	Infinity	Replace switch (PARA 11-4-34)
Searchlight control	Center (off)	C - A	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - B	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - D	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - e	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	*EXT	C - e	0	e - P C - E
Searchlight control	*RET	C - B	0	B - B
Searchlight control	*L	C - A	0	A - A
Searchlight control	*R	C - D	0	D - C
SRCH LT	OFF or STOW	R - P	0	P - 1 R - 2
SRCH LT	OFF or STOW	R - K	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	OFF or STOW	L - M	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	OFF or STOW	M - N	0	N - 4 M - 5
SRCH LT	ON	R - K	0	K - 3
SRCH LT	ON	L - M	0	L - 6 M - 5

TABLE NO. 11-2-2-2. Continuity Check (Collective Stick Assembly: 70400-01402-046). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
SRCH LT	ON	M - N	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	ON	P - R	Infinity	Replace switch (PARA 11-4-33)
SVO OFF	1ST STG	f - J	0	J - 6 f - 5 or replace switch (PARA 11-4-33)
SVO OFF	1ST STG	c - n	0	n - 3 c - 2 or replace switch (PARA 11-4-33)
SVO OFF	2ND STG	f - p	0	p - 4
SVO OFF	2ND STG	c - d	0	d - 1 or replace switch (PARA 11-4-33)

*Indicates momentary contact. Switch must be held at this position.

TABLE NO. 11-2-2-3. Continuity Check (Collective Stick Assemblies: 70400-01401-042, -044, -046, -048, And 70400-01402-044).

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
EMERG HOOK REL	Push (on)	F - G	0	F - 1 G - 2
EMERG HOOK REL	Push (off)	F - G	Infinity	Replace switch (PARA 11-4-34)
ENG RPM	Center (off)	h - g	0	h - 5 g - 6

TABLE NO. 11-2-2-3. Continuity Check (Collective Stick Assemblies: 70400-01401-042, -044, -046, -048, And 70400-01402-044). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
ENG RPM	Center (off)	h - i	Infinity	Replace switch (PARA 11-4-35)
ENG RPM	Center (off)	k - j	0	k - 2 j - 1
ENG RPM	Center (off)	k - m	Infinity	Replace switch (PARA 11-4-35)
ENG RPM	*INCR	k - m	0	m - 3
ENG RPM	*INCR	g - h	0	Replace switch (PARA 11-4-35)
ENG RPM	*DECR	h - i	0	i - 4
ENG RPM	*DECR	j - k	0	Replace switch (PARA 11-4-35)
LDG LT	*EXT	U - W	0	U - 1 W - 3
LDG LT	*EXT	U - S	Infinity	Replace switch (PARA 11-4-34)
LDG LT	Center (off)	U - W	Infinity	Replace switch (PARA 11-4-34)
LDG LT	Center (off)	U - S	Infinity	Replace switch (PARA 11-4-34)
LDG LT	*RET	U - S	0	S - 2
LDG LT	*RET	U - W	Infinity	Replace switch (PARA 11-4-34)
LDG LT	*Pressed	X - Y	0	X - 5 Y - 4
LDG LT	Not Pressed	X - Y	Infinity	Replace switch (PARA 11-4-34)

TABLE NO. 11-2-2-3. Continuity Check (Collective Stick Assemblies: 70400-01401-042, -044, -046, -048, And 70400-01402-044). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
Searchlight control	Center (off)	C - A	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - B	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - D	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	Center (off)	C - E	Infinity	Replace switch (PARA 11-4-36)
Searchlight control	*EXT	C - E	0	C - E E - P
Searchlight control	*RET	C - B	0	B - B
Searchlight control	*L	C - A	0	A - A
Searchlight control	*R	C - D	0	D - C
SRCH LT	OFF or STOW	R - P	0	P - 1 R - 2
SRCH LT	OFF or STOW	R - K	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	OFF or STOW	L - M	Infinity	Replace switch (PARA 11-4-33)
SRCH LT	OFF or STOW	M - N	0	N - 4 M - 5
SRCH LT	ON	R - K	0	K - 3
SRCH LT	ON	L - M	0	L - 6 M - 5
SRCH LT	ON	M - N	Infinity	Replace switch (PARA 11-4-33)

TABLE NO. 11-2-2-3. Continuity Check (Collective Stick Assemblies: 70400-01401-042, -044, -046, -048, And 70400-01402-044). (Cont)

SWITCH	SWITCH POSITION	CHECK RESISTANCE BETWEEN CONNECTOR PINS:	METER INDICATION (Ohms)	CORRECTIVE ACTION (CONTINUITY)
SVO OFF	Center (off)	f - a	Infinity	Replace switch (PARA 11-4-33)
SVO OFF	Center (off)	f - e	Infinity	Replace switch (PARA 11-4-33)
SVO OFF	Center (off)	c - b	Infinity	Replace switch (PARA 11-4-33)
SVO OFF	Center (off)	c - d	Infinity	Replace switch (PARA 11-4-33)
SVO OFF	1ST STG	f - a	0	f - 5 a - 6
SVO OFF	1ST STG	c - b	0	c - 2 b - 3
SVO OFF	2ND STG	f - e	0	e - 4
SVO OFF	2ND STG	c - d	0	d - 1

*Indicates momentary contact. Switch must be held at this position.

RESULT: Meter shall indicate corresponding resistance.

- If result is not as specified, do corresponding corrective action.

  (a) If meter indication of infinity is specified and continuity is present, replace switch (PARA 11-4-33, PARA 11-4-34, PARA 11-4-35, or PARA 11-4-36), as required.

  (b) If meter indication of 0 is specified and continuity is not present, repair/ replace wiring (Appendix F), as required.

11-2-2.2 Mechanical Inspection, Pilot.

a. Visually check grip for cracks.

RESULT: Grip shall not be cracked.

- If result is not as specified, replace grip assembly (PARA 11-4-31).

b. Visually check friction lock boot for tears or cracks.

RESULT: Boot shall not be torn or cracked.

- If result is not as specified, replace boot (PARA 11-4-26).

c. Check stick friction operation.

RESULT: Friction grip shall work and stick shall move smoothly.

- If result is not as specified, go to TABLE NO. 11-2-2-5.

11-2-2.3 Mechanical Inspection, Copilot.

a. Visually check grip for cracks.

RESULT: Grip shall not be cracked.

- If result is not as specified, replace grip assembly (PARA 11-4-31).

b. Check assembly for rough or sloppy stick movement.

RESULT: Assembly movement shall not be rough or sloppy.

- If result is not as specified, do the following:

  (a) Inspect socket assembly bearings for ratcheting or play.

- If ratcheting or play is found, replace malfunctioning bearing(s) (PARA 11-4-41) and/or socket assembly (PARA 11-4-40).

  (a) Check each bearing for binding by turning with torque wrench. Bearing shall turn with 8 inch-ounce torque or less.

- If bearing binds, replace malfunctioning bearing(s) (PARA 11-4-41) and/or socket assembly (PARA 11-4-40).

TABLE NO. 11-2-2-4. Lighted Panel Is Not Lit Correctly.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is lighted panel lit?

Step 1. If panel is not lit, go to **2.**
Step 2. If panel is lit, go to **4.**

2. Check for 115 vac between:
J103-T and J103-V (Copilot's)
J104-T and J104-V (Pilot's)

Step 1. If voltage is as specified, go to **3.**
Step 2. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (PARA 9-2-10). Go to **5.**

3. Check wiring between:
P103-T and panel lights (Copilot's)
P103-V and panel lights (Copilot's)
P104-T and panel lights (Pilot's)
P104-V and panel lights (Pilot's)

Step 1. If wiring is good, replace lighted panel assembly (PARA 11-4-32). Go to **5.**

TABLE NO. 11-2-2-4. Lighted Panel Is Not Lit Correctly. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If wiring is not good, repair/replace wiring (Appendix F), as required. Go to **5.**

4. Check for burnt-out lamps.

Step 1. If lamps are good, replace lighted panel assembly (PARA 11-4-32). Go to **5.**
Step 2. If lamps are not good, replace lamps (PARA 11-4-32), as required. Go to **5.**

5. Procedure completed.

TABLE NO. 11-2-2-5. Stick Has Friction.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Does friction grip cause friction malfunction?

Step 1. If grip causes malfunction, go to **2.**
Step 2. If grip does not cause malfunction, go to **3.**

2. Is over-tightening of grip possible?

Step 1. If over-tightening is possible, replace friction lockring and stop assembly (PARA 11-4-39). Go to **6.**
Step 2. If over-tightening is not possible, replace friction lock (PARA 11-4-39). Go to **6.**

3. Inspect socket assembly bearings for ratcheting or play.

Step 1. If bearings have ratcheting or play, replace malfunctioning bearings (PARA 11-4-41) and/or socket assembly (PARA 11-4-40). Go to **6.**
Step 2. If bearings do not have ratcheting or play, go to **4.**

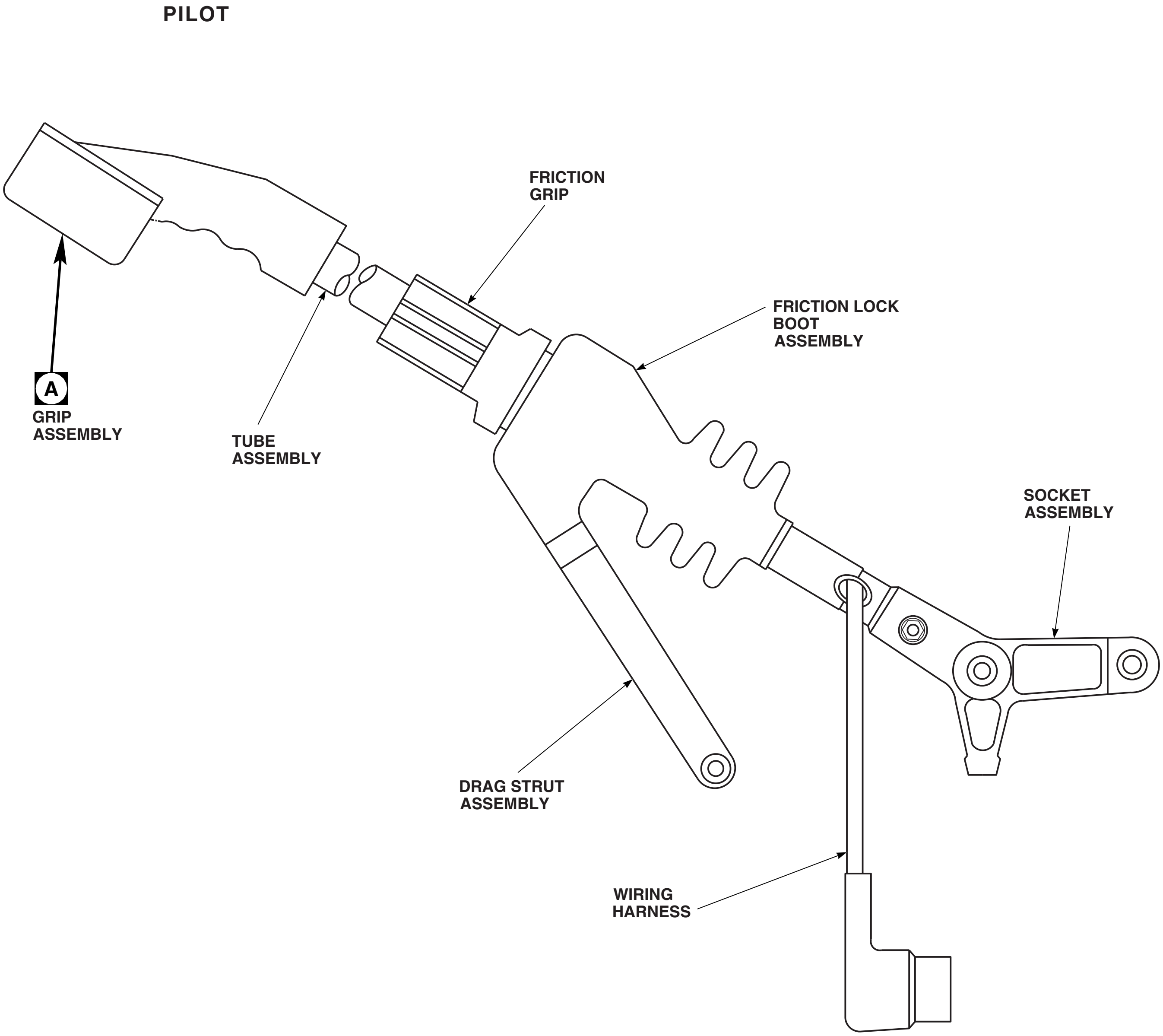
4. Check each bearing for binding by turning with torque wrench. Go to 5.

5. Do bearings turn with less than 8 inch-ounces torque?

Step 1. If bearings turn with less than 8 inch-ounces torque, replace friction lock (PARA 11-4-39). Go to **6.**

TABLE NO. 11-2-2-5. Stick Has Friction. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
Step 2. If bearings do not turn with less than 8 inch-ounces torque, replace malfunctioning bearings (PARA 11-4-41) and/or socket assembly (PARA 11-4-40). Go to **6.**
6. Procedure completed.



NOTES
1. APPLICABLE FOR GRIP ASSEMBLIES 70400–01405–103 AND –104 INSTALLED ON COLLECTIVE STICK ASSEMBLIES 70400–01401–042, –044, –046, –048, 70400–01402–044 AND –046.
2. APPLICABLE FOR GRIP ASSEMBLIES 70400–01405–105 AND –106 INSTALLED ON COLLECTIVE STICK ASSEMBLIES 70400–01401–052, –054, 70400–01402–048 AND –050.

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FIGURE 11-2-2-1. COLLECTIVE STICK ASSEMBLY (SHEET 1 OF 2)

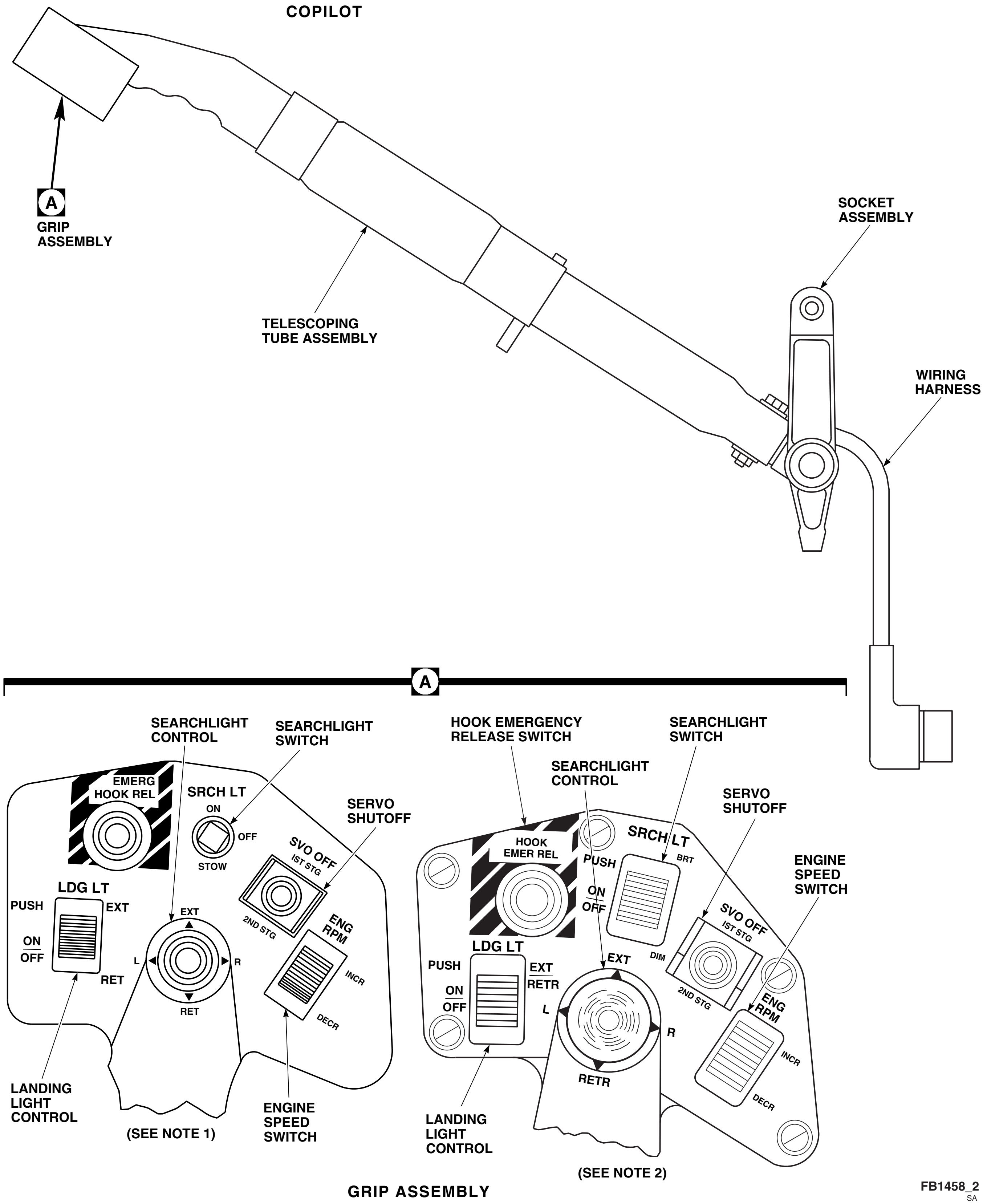
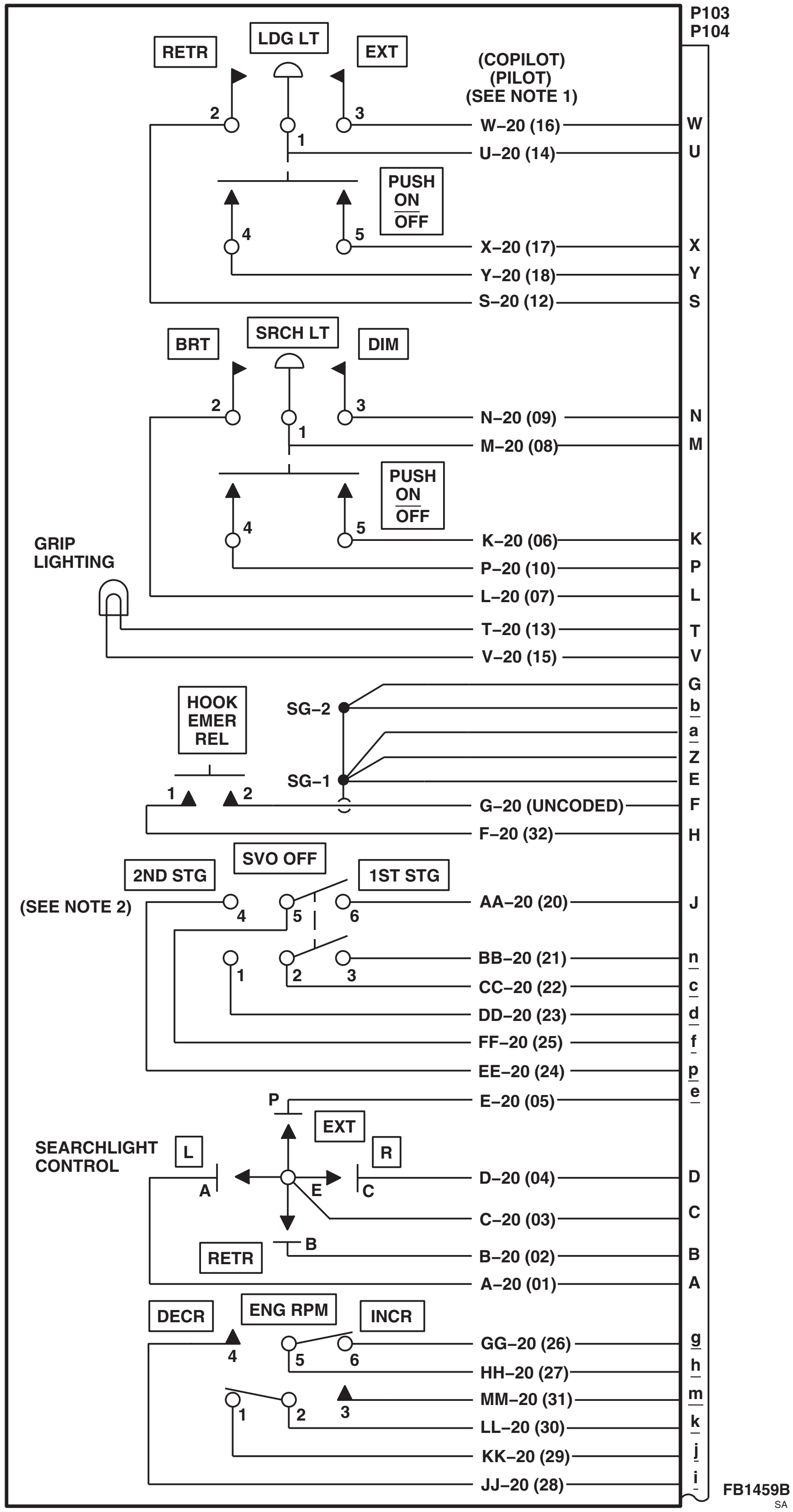


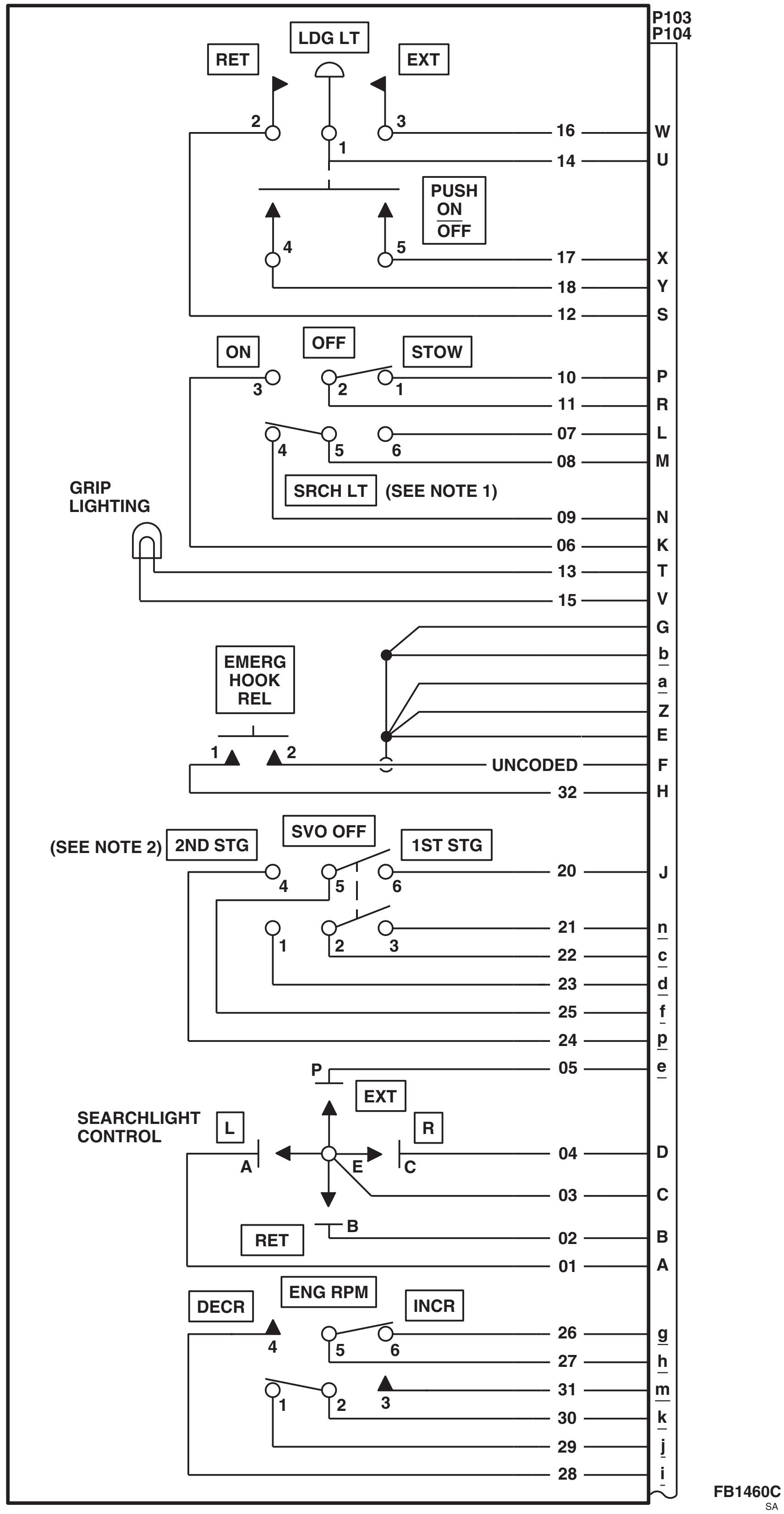
FIGURE 11-2-2-1. COLLECTIVE STICK ASSEMBLY (SHEET 2 OF 2)



EFFECTIVITY
COLLECTIVE STICK ASSEMBLIES 70400–01401–052, –054 AND 70400–01402–048, –050.

NOTES
1. PILOT STICK ASSEMBLY WIRE NUMBERS SHOWN WITHOUT PARENTHESES. CO–PILOT STICK ASSEMBLY WIRE NUMBERS SHOWN IN PARENTHESES.   2. SVO OFF SWITCH SHOWN IN OFF POSITION. WITH SWITCH IN 2ND STG, CONNECTIONS BETWEEN 5 AND 4, AND 2 AND 1 ARE MADE. WITH SWITCH IN 1ST STG, CONNECTIONS BETWEEN 5 AND 6, AND 2 AND 3 ARE MADE.

FIGURE 11-2-2-2. COLLECTIVE STICK ASSEMBLY SCHEMATIC DIAGRAM (SHEET 1 OF 3)



EFFECTIVITY
COLLECTIVE STICK ASSEMBLY 70400–01402–046.

NOTES

1. SRCH LT SWITCH SHOWN IN OFF POSITION. WITH SWITCH IN ON, CONNECTIONS BETWEEN TERMINALS 2 AND 3, AND 5 AND 6 ARE MADE. WITH SWITCH IN STOW, CONNECTIONS BETWEEN 2 AND 1, AND 4 AND 5 ARE MADE.
2. SVO OFF SWITCH SHOWN IN OFF POSITION. WITH SWITCH IN 2ND STG, CONNECTIONS BETWEEN 5 AND 4, AND 2 AND 1 ARE MADE. WITH SWITCH IN 1ST STG, CONNECTIONS BETWEEN 5 AND 6, AND 2 AND 3 ARE MADE.

FIGURE 11-2-2-2. COLLECTIVE STICK ASSEMBLY SCHEMATIC DIAGRAM (SHEET 2 OF 3)

EFFECTIVITY
COLLECTIVE STICK ASSEMBLIES 70400–01401–042, –044, –046, –048, AND 70400–01402–044.

NOTES
1. PILOT STICK ASSEMBLY WIRE NUMBERS SHOWN WITHOUT PARENTHESES. CO–PILOT STICK ASSEMBLY WIRE NUMBERS SHOWN IN PARENTHESES. 2. SRCH LT SWITCH SHOWN IN OFF POSITION. WITH SWITCH IN ON, CONNECTIONS BETWEEN TERMINALS 2 AND 3, AND 5 AND 6 ARE MADE. WITH SWITCH IN STOW, CONNECTIONS BETWEEN 2 AND 1, AND 4 AND 5 ARE MADE. 3. SVO OFF SWITCH SHOWN IN OFF POSITION. WITH SWITCH IN 2ND STG, CONNECTIONS BETWEEN 5 AND 4, AND 2 AND 1 ARE MADE. WITH SWITCH IN 1ST STG, CONNECTIONS BETWEEN 5 AND 6, AND 2 AND 3 ARE MADE.

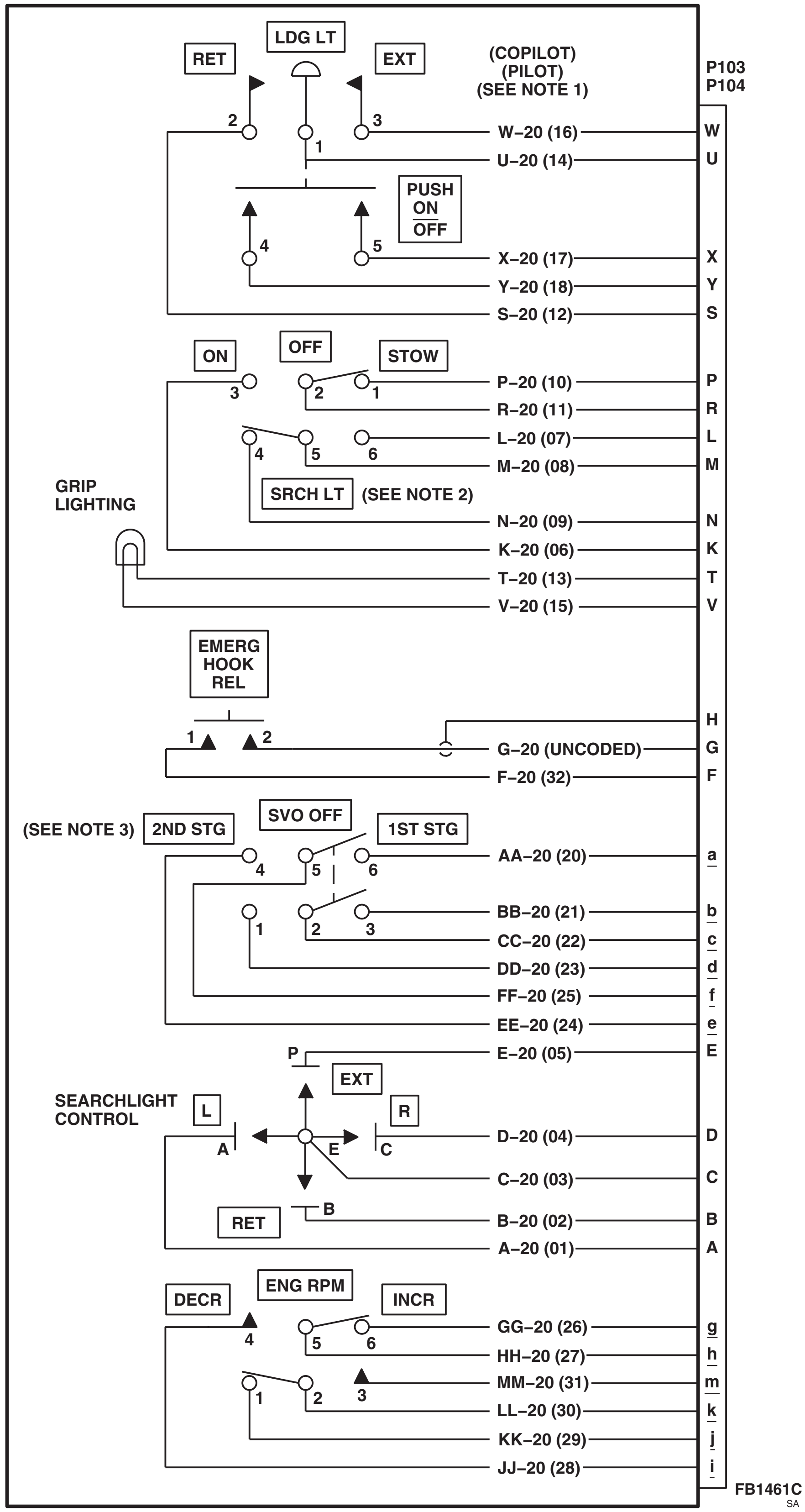


FIGURE 11-2-2-2. COLLECTIVE STICK ASSEMBLY SCHEMATIC DIAGRAM (SHEET 3 OF 3)

11-2-3 MECHANICAL FLIGHT CONTROLS.

INITIAL SETUP

Test Equipment

- Multimeter, Fluke 45

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer Toolkit

Materials/Parts

- Insulated Jumper Wire

References

- Appropriate MTF
- PARA 1-6-16
- PARA 11-4-87
- TM 11-70-23

Equipment Conditions

- All Flight Control Components Installed
- Backup Hydraulic Pump Operational
- External Electrical Power Available or APU Operational

CAUTION

Flight Control System will be damaged if cockpit controls are moved while control rods are disconnected. Loose control rods will strike other components. Make sure disconnected control rods are secured and flight controls are not moved when performing maintenance in this area.

CAUTION

Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to the helicopter. Do not remove primary servo input control rods with hydraulic power on.

CAUTION

To prevent overheating Backup Pump Hydraulic System, use this chart for limits when running backup pump.

AIR TEMPERATURE F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
32° (90°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
101° - 125° (39° - 52°)	16	48

NOTE

If any one of the following capsules on the caution/advisory panel go on during the operational/troubleshooting procedure, go to PARA 7-2-1:

#1 PRI SERVO PRESS
#2 PRI SERVO PRESS
#1 TAIL RTR SERVO

NOTE

If any one of the following capsules on the caution/advisory panel go on during the operational/troubleshooting procedure, go to TM 11-70-23:

SAS OFF
BOOST SERVO OFF

Special Instructions

The mechanical flight controls operational/troubleshooting procedure is subdivided as follows:

(1) Binding

(2) In-Flight Attitude Oscillations

(3) Flight Control Noise Checkout

11-2-3.1 Binding.

NOTE

If any unusual noises are heard and no binding is felt while moving pilot or copilot pedals, collective sticks, or cyclic sticks, perform Flight Control Noise Checkout procedure.

a. Visually check complete Flight Control System to make sure all flight control rods are connected.

b. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

c. Turn on electrical power (PARA 1-6-16).

d. Make sure BOOST, TRIM, and FPS buttons on stabilator control/auto flight (AFCS) control panel are disengaged (capsules not on).

e. Place upper console BACKUP HYD PUMP switch to ON.

f. Move cyclic stick forward, aft, right, left, and back to center position.

RESULT: Cyclic stick shall move freely in all directions with no restrictions.

- If cyclic stick is difficult to move forward, but moves aft freely (or the reverse), visually check for objects interfering with flight controls.
- If cyclic stick is difficult to move forward and aft, go to TABLE NO. 11-2-3-1.
- If cyclic stick is difficult to move left and right, go to TABLE NO. 11-2-3-2.

g. Push on left and right pedals through their full range.

RESULT: Pedals shall move freely with no restrictions.

- If TAIL ROTOR QUADRANT capsule goes on, go to TABLE NO. 11-2-3-3.
- If there is too much free play in pedals, go to TABLE NO. 11-2-3-4.
- If pedals are difficult to move, go to TABLE NO. 11-2-3-5.

h. Completely unlock friction lock on collective stick by turning friction grip until friction lock moves forward.

NOTE

It is normal for pedals to move when collective stick is moved.

i. Move collective stick up and down through full range.

RESULT: Collective stick shall move freely with no restrictions.

- If collective stick requires much greater force to raise than to lower, go to TABLE NO. 11-2-3-6.

- If collective stick is difficult to move up and down, go to TABLE NO. 11-2-3-7.

j. On pilot's collective stick, set SVO OFF/1ST STG/2ND STG switch to 1ST STG.

k. Move pilot collective stick from full down to full up position in about 1 second.

RESULT:

(1) #1 PRI SERVO PRESS capsule on caution/advisory panel shall go on.

(2) #2 PRI SERVO PRESS capsule on caution/advisory panel shall stay off.

(3) Cyclic stick shall have no longitudinal (forward and aft) control feedback.

(4) Cyclic stick shall have no lateral (left and right) control feedback.

- If result (1) is not as specified, troubleshoot hydraulic systems (PARA 7-2-1).
- If result (2) is not as specified, go to TABLE NO. 11-2-3-8.
- If cyclic stick has longitudinal feedback in the aft direction when collective stick is moved to the up position, replace aft primary servo (PARA 11-4-87).
- If cyclic stick has longitudinal feedback in the forward direction when collective stick is moved to the up position, replace forward primary servo (PARA 11-4-87).
- If cyclic stick has longitudinal feedback in the aft direction when collective stick is moved to the down position, replace forward primary servo (PARA 11-4-87).
- If cyclic stick has longitudinal feedback in the forward direction when collective stick is moved to the down position, replace aft primary servo (PARA 11-4-87).
- If the cyclic stick has lateral feedback when the collective stick is moved up and down, go to TABLE NO. 11-2-3-9.

l. On pilot's collective stick, set SVO OFF/1ST STG/2ND STG switch to 2ND STG.

m. Move pilot collective stick from full down to full up position in about 1 second.

RESULT:

(1) #2 PRI SERVO PRESS capsule on caution/advisory panel shall go on.

(2) #1 PRI SERVO PRESS capsule on caution/advisory panel shall stay off.

(3) Cyclic stick shall have no longitudinal (forward and aft) control feedback.

(4) Cyclic stick shall have no lateral (left and right) control feedback.

- If result (1) is not as specified, troubleshoot hydraulic systems (PARA 7-2-1).
- If result (2) is not as specified, go to TABLE NO. 11-2-3-10.
- If cyclic stick has longitudinal feedback in the aft direction when collective stick is moved to the up position, replace aft primary servo (PARA 11-4-87).
- If cyclic stick has longitudinal feedback in the forward direction when collective stick is moved to the up position, replace forward primary servo (PARA 11-4-87).
- If cyclic stick has longitudinal feedback in the aft direction when collective stick is moved to the down position, replace forward primary servo (PARA 11-4-87).
- If cyclic stick has longitudinal feedback in the forward direction when collective stick is moved to the down position, replace aft primary servo (PARA 11-4-87).

- If the cyclic stick has lateral feedback when the collective stick is moved up and down, go to TABLE NO. 11-2-3-9.

n. Install an insulated jumper between J611-A and J611-C.

RESULT: TAIL ROTOR QUADRANT capsule on caution/advisory panel shall go on.

- If result is not as specified, go to TABLE NO. 11-2-3-11.

o. Remove insulated jumper from J611-A and J611-B, then install insulated jumper between J611-B and J611-C.

RESULT: TAIL ROTOR QUADRANT capsule on caution/advisory panel shall go on.

- If result is not as specified, repair/replace wiring between terminal 1 (T RTR SERVO WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and J611-B (Appendix F).

p. Remove jumper wire and reconnect electrical connector.

q. Place upper console BACKUP HYD PUMP switch to OFF.

r. Turn off electrical power (PARA 1-6-16).

11-2-3.2 In-Flight Attitude Oscillations.

NOTE

During maintenance test flight, it is necessary to troubleshoot AFCS to isolate problem area of flight controls (TM 11-70-23).

a. Perform maintenance test flight to determine type of attitude oscillation and troubleshoot AFCS (TM 11-70-23).

NOTE

- If type of attitude oscillation cannot be determined, troubleshoot AFCS (TM 11-70-23).
- If type of attitude oscillation has most effect on pitch, roll, collective, or yaw, check that system first.

b. Fly helicopter straight-and-level in forward flight.

c. Disengage AFCS TRIM.

RESULT: Oscillation shall persist.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

d. Engage AFCS TRIM.

e. Disengage SAS 1.

RESULT: Oscillation shall persist.

- If result is not as specified, troubleshoot analog SAS units (TM 11-70-23).

f. Disengage SAS 2.

RESULT: Oscillation shall persist.

- If result is not as specified, troubleshoot digital AFCS (TM 11-70-23).

g. If oscillation has a noticeable pitch component, disengage stabilator AUTO CONTROL.

RESULT: Oscillation shall persist.

- If result is not as specified, troubleshoot stabilator units (TM 11-70-23).

h. Check primary servo valve crank assembly for play (PARA 11-4-87).

RESULT: Primary servo valve crank assembly shall not have play.

- If result is not as specified, replace primary servo (PARA 11-4-87).

i. Check all Flight Control System components, connections and hardware between primary servos and swashplate for security, damage, and corrosion.

RESULT: Flight Control System components between primary servos and swashplate shall not have damage, corrosion, or loose hardware beyond limits.

- If result is not as specified, repair/replace Flight Control System components, as required.

j. Check all Flight Control System components, connections and hardware between primary servos, and pilot's and copilot's controls for security, damage, and corrosion.

RESULT: Flight Control System components between primary servos and pilot's and copilot's controls shall not have damage, corrosion, or loose hardware beyond limits.

- If result is not as specified, repair/replace system components and hardware, as required.

k. If no deficiencies were found in Flight Control System being inspected, inspect all other flight control systems.

l. Do a rigging check (PARA 11-4-87).

m. Do a maintenance test flight (Appropriate MTF).

11-2-3.3 Flight Control Noise Checkout.

NOTE

- In flight control systems, some slight noise can be expected as the system ages. Some of these noises are normal and do not indicate a problem. Other noises indicate that a problem exists which requires investigation and correction. The following procedures will help localize and correct known sources of unusual noises.
- During the following checkout procedure, two personnel will be required: one to move flight controls and another to listen for location of unusual noises.

a. Turn on electrical power (PARA 1-6-16).

b. Place upper console BACKUP HYD PUMP switch to ON.

c. On stabilator control/automatic flight control panel, press BOOST switch (capsule on).

d. While slowly moving pilot's cyclic stick fully left then fully right, carefully listen for any unusual noises in the lateral cyclic control channel.

RESULT: No unusual noises shall be heard while moving pilot's cyclic stick from left to right.

- If noise is heard while moving pilot's cyclic stick from left to right, go to TABLE NO. 11-2-3-12.

e. While slowly moving pilot's cyclic stick fully forward then fully aft, carefully listen for any unusual noises in the pitch cyclic control channel.

RESULT: No unusual noises shall be heard while moving pilot's cyclic stick from forward to aft.

- If noise is heard while moving pilot's cyclic stick from forward to aft, go to TABLE NO. 11-2-3-13.

NOTE

If noises in the collective control channel are suspected to be coming from forward of the boost actuators, disengage stabilator control/automatic flight control panel BOOST switch (capsule off) and repeat step f.

f. While slowly moving pilot's collective stick fully up then fully down, carefully listen for any unusual noises in the collective control channel.

RESULT: No unusual noises shall be heard while moving pilot's collective stick up then down.

- If noise is heard while moving pilot's collective stick up then down, go to TABLE NO. 11-2-3-14.

NOTE

- In order to reproduce unusual noises in the yaw control channel, momentary movement of the collective stick from the down position to the up position may be required.

- If noises in the yaw control channel are suspected to be coming from forward of the boost actuators, disengage stabilator control/automatic flight control panel BOOST switch (capsule off) and repeat step g.

g. While slowly pushing on pilot's left and right pedals, carefully listen for any unusual noises in the yaw control channel.

RESULT: No unusual noises shall be heard while pushing pilot's left and right pedals.

- If noise is heard while pushing pilot's left and right pedals, go to TABLE NO. 11-2-3-15.

h. On stabilator control/automatic flight control panel ensure SAS 1 and SAS 2 are disengaged (capsule off).

i. On stabilator control/automatic flight control panel ensure BOOST is disengaged (capsule off).

j. Place upper console BACKUP HYD PUMP switch to OFF.

k. Turn off electrical power (PARA 1-6-16).

TABLE NO. 11-2-3-1. Cyclic Stick Is Difficult To Move Forward And Aft.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Visually inspect mixer assembly for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (PARA 11-4-77). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **2.**

2. Visually inspect pitch trim assembly input and output linkage for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace pitch trim assembly (PARA 11-4-68). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **3.**

3. Visually inspect forward primary servo input and output linkage for corrosion, damage, or an interfering object.

TABLE NO. 11-2-3-1. Cyclic Stick Is Difficult To Move Forward And Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace forward primary servo (PARA 11-4-87). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **4.**

4. Visually inspect rear primary servo input and output linkage for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace rear primary servo (PARA 11-4-87). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **5.**

5. Visually inspect lateral primary servo for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace lateral primary servo (PARA 11-4-87). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **6.**

6. Remove cabin soundproofing and visually inspect cyclic stick balance spring for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace cyclic stick balance spring (PARA 11-4-60). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **7.**

7. Check adjustment of cyclic stick balance spring.

Step 1. If cyclic stick balance spring is good, go to **8.**
Step 2. If cyclic stick balance spring is not good, adjust cyclic stick balance spring (PARA 11-4-61). Go to **10.**

8. Visually inspect pitch torque shaft and levers for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace pitch torque shaft and levers (PARA 11-4-56). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **9.**

9. Visually inspect all flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.

TABLE NO. 11-2-3-1. Cyclic Stick Is Difficult To Move Forward And Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks, as required. Go to **10.**

Step 2. If corrosion, damage, or an interfering object is not present, rig and adjust Flight Control System (PARA 11-4-2). Go to **10.**

10. Procedure completed.

TABLE NO. 11-2-3-2. Cyclic Stick Is Difficult To Move Left And Right.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Visually inspect mixer assembly for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (PARA 11-4-77). Go to **8.**

Step 2. If corrosion, damage, or an interfering object is not present, go to **2.**

2. Visually inspect roll trim assembly output linkage (under boot) for evidence of scoring.

Step 1. If scoring is visible, replace roll trim assembly (PARA 11-4-69). Go to **8.**

Step 2. If scoring is not visible, go to **3.**

3. Visually inspect roll trim assembly for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace roll trim assembly (PARA 11-4-69). Go to **8.**

Step 2. If corrosion, damage, or an interfering object is not present, go to **4.**

4. Visually inspect lateral primary servo for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace lateral primary servo (PARA 11-4-87). Go to **8.**

Step 2. If corrosion, damage, or an interfering object is not present, go to **5.**

TABLE NO. 11-2-3-2. Cyclic Stick Is Difficult To Move Left And Right. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

5. Visually inspect roll torque shaft and levers for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace roll torque shaft and levers (PARA 11-4-57). Go to **8.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **6.**

6. With BACKUP PUMP switch ON, slowly move cyclic stick forward, aft, left, and right, while visually inspecting flight control area for binding against wire bundles.

Step 1. If binding is present, reposition wire bundles, as required. Go to **8.**
Step 2. If binding is not present, go to **7.**

7. Visually inspect all flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks, as required. Go to **8.**
Step 2. If corrosion, damage, or an interfering object is not present, rig and adjust Flight Control System (PARA 11-4-2). Go to **8.**

8. Procedure completed.

TABLE NO. 11-2-3-3. TAIL ROTOR QUADRANT Capsule Is On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Visually inspect tail rotor cables for broken or disconnected cables.

Step 1. If broken or disconnected cables are present, repair/replace broken or disconnected cables (PARA 11-4-105 or PARA 11-4-106). Go to **9.**
Step 2. If broken or disconnected cables are not present, go to **2.**

2. Disconnect J611. Go to 3.

3. Does TAIL ROTOR QUADRANT capsule go off?

TABLE NO. 11-2-3-3. TAIL ROTOR QUADRANT Capsule Is On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If capsule is off, go to **4.**
Step 2. If capsule is not off, go to **5.**

4. Check adjustment of tail rotor quadrant warning switches.

Step 1. If warning switches are properly adjusted, replace tail pylon cable (harness) (PARA 11-4-121). Go to **9.**
Step 2. If warning switches are not properly adjusted, adjust rotor quadrant warning switches (PARA 11-4-110). Go to **9.**

5. Check helicopter configuration.

Step 1. EMEP Go to **6.**
Step 2. W/O EMEP Go to **8.**

EMEP **6. Remove and check pin filter adapter P117 (PARA 9-4-98).**

Step 1. If pin filter adapter is good, go to **7.**
Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to **9.**

EMEP **7. Install pin filter adapter (PARA 9-4-98). Go to 8.**

8. Check for 28 VDC between P117-9 and ground with J611 disconnected.

Step 1. If voltage is as specified, repair/replace short circuit between J611-A, J611-B, and J611-C (Appendix F). Go to **9.**
Step 2. If voltage is not as specified, replace caution/advisory panel (PARA 8-4-19). Go to **9.**

9. Procedure completed.

TABLE NO. 11-2-3-4. Excessive Free Play In Pedals.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Inspect pedal assembly mounting hardware for incorrect torque or wear.

TABLE NO. 11-2-3-4. Excessive Free Play In Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If pedal assembly hardware is incorrectly torqued, replace hardware, as required and torque to specifications. Go to **6.**
Step 2. If pedal assembly hardware is correctly torqued, go to **2.**

2. Inspect pedal pivot shaft and directional control rods for looseness or wear.

Step 1. If pedal pivot shaft and directional control rods are loose or worn, repair/replace components or replace hardware, as required. Go to **6.**
Step 2. If pedal pivot shaft and directional control rods are not loose or worn, go to **3.**

3. Inspect pedal adjuster for looseness or wear.

Step 1. If pedal adjuster is loose or worn, repair/replace components or replace hardware (PARA 11-4-11), as required. Go to **6.**
Step 2. If pedal adjuster is not loose or worn, go to **4.**

4. Inspect both cable disconnect fittings in tail cone. Go to 5.

5. Make sure retainers are bottomed out in quick disconnect fittings. Go to 6.

6. Procedure completed.

TABLE NO. 11-2-3-5. Pedals Are Difficult To Move.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Visually inspect pedal floor cover for binding (around pedal shaft on cockpit floor).

Step 1. If binding is present, replace pedal floor covers (PARA 11-4-10). Go to **10.**
Step 2. If binding is not present, go to **2.**

2. Visually inspect mixer assembly for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (PARA 11-4-77). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **3.**

TABLE NO. 11-2-3-5. Pedals Are Difficult To Move. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

3. Visually inspect Tail Rotor System for incorrectly routed, kinked, or broken cables, damaged cable pulleys, or an object interfering with cable movement.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace cables (PARA 11-4-105 or PARA 11-4-106), or pulleys (PARA 11-4-107). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **4.**

4. Visually inspect plastic tubes and cables on tail pylon for binding.

Step 1. If binding is present, repair/replace tail rotor cables or tubes (PARA 11-4-106). Go to **10.**
Step 2. If binding is not present, go to **5.**

5. Visually inspect yaw boost assembly input and output linkage for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace yaw boost assembly (PARA 11-4-70). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **6.**

6. Check cables for proper tension.

Step 1. If proper tension is present, go to **7.**
Step 2. If proper tension is not present, adjust cable tension (PARA 11-4-104). Go to **10.**

7. Visually inspect tail rotor servo for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace tail rotor servo (PARA 11-4-113). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **8.**

8. Visually inspect directional (yaw) trim servo for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace directional (yaw) trim servo (PARA 11-4-52). Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **9.**

TABLE NO. 11-2-3-5. Pedals Are Difficult To Move. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Visually inspect flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks, as required. Go to **10.**
Step 2. If corrosion, damage, or an interfering object is not present, rig and adjust Flight Control System (PARA 11-4-2). Go to **10.**

10. Procedure completed.

TABLE NO. 11-2-3-6. Collective Stick Requires Greater Force To Raise Than To Lower.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Remove soundproofing and visually inspect collective balance spring for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective balance spring (PARA 11-4-60). Go to **5.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **2.**

2. Check adjustment of collective balance spring.

Step 1. If collective balance spring is properly adjusted, go to **3.**
Step 2. If collective balance spring is not properly adjusted, adjust collective stick balance spring (PARA 11-4-62). Go to **5.**

3. Visually inspect boot at base of collective stick.

Step 1. If boot at base of collective stick is good, go to **4.**
Step 2. If boot at base of collective stick is not good, replace collective stick boot (PARA 11-4-26). Go to **5.**

4. Visually inspect all flight control rods, bearings, and bellcranks for corrosion, damage, or an interfering object.

TABLE NO. 11-2-3-6. Collective Stick Requires Greater Force To Raise Than To Lower. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace control rods, bearings, and bellcranks, as required. Go to **5.**

Step 2. If corrosion, damage, or an interfering object is not present, rig and adjust Flight Control System (PARA 11-4-2). Go to **5.**

5. Procedure completed.

TABLE NO. 11-2-3-7. Collective Stick Is Difficult To Move Up And Down.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Visually inspect boot at base of collective stick.

Step 1. If boot at base of collective stick is good, go to **2.**

Step 2. If boot at base of collective stick is not good, replace collective stick boot (PARA 11-4-26). Go to **11.**

2. Remove soundproofing and visually inspect collective stick balance spring for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective stick balance spring (PARA 11-4-60). Go to **11.**

Step 2. If corrosion, damage, or an interfering object is not present, go to **3.**

3. Check adjustment of collective balance spring.

Step 1. If collective balance spring is properly adjusted, go to **4.**

Step 2. If collective balance spring is not properly adjusted, adjust collective stick balance spring (PARA 11-4-62). Go to **11.**

4. Visually inspect mixer assembly for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace mixer assembly (PARA 11-4-77). Go to **11.**

Step 2. If corrosion, damage, or an interfering object is not present, go to **5.**

5. Visually inspect collective boost assembly for corrosion, damage, or an interfering object.

TABLE NO. 11-2-3-7. Collective Stick Is Difficult To Move Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace collective boost assembly (PARA 11-4-67). Go to **11.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **6.**

6. Visually inspect collective trim servo for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object. Go to **11.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **7.**

7. Visually inspect forward primary servo input and output linkage for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace forward primary servo (PARA 11-4-87). Go to **11.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **8.**

8. Visually inspect rear primary servo input and output linkage for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace rear primary servo (PARA 11-4-87). Go to **11.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **9.**

9. Visually inspect lateral primary servo for corrosion, damage, or an interfering object.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace lateral primary servo (PARA 11-4-87). Go to **11.**
Step 2. If corrosion, damage, or an interfering object is not present, go to **10.**

10. Visually inspect Tail Rotor System for incorrectly routed, kinked, or broken cables, damaged cable pulleys, or an object interfering with cable movement.

Step 1. If corrosion, damage, or an interfering object is present, remove interfering object or replace cables (PARA 11-4-105 or PARA 11-4-106) or pulleys (PARA 11-4-107). Go to **11.**
Step 2. If corrosion, damage, or an interfering object is not present, rig and adjust Flight Control System (PARA 11-4-2). Go to **11.**

TABLE NO. 11-2-3-7. Collective Stick Is Difficult To Move Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

11. Procedure completed.

TABLE NO. 11-2-3-8. #2 PRI SERVO PRESS Capsule On Caution/Advisory Panel Goes On When Collective Stick Is Moved Full Down To Full Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect connector P427 from lateral primary servo 2nd stage pressure switch, and install an insulated jumper wire between P427-4 and P427-9. Go to 2.

2. With BACKUP HYD PUMP switch set to ON and pilot's collective stick SVO OFF/1ST STG/2ND STG switch set to 2ND STG, move collective stick from full down to full up position. #2 PRI SERVO PRESS capsule on caution/advisory panel shall remain off.

Step 1. If #2 PRI SERVO PRESS capsule remains off, replace lateral primary servo (PARA 11-4-87). Go to **8.**
Step 2. If #2 PRI SERVO PRESS capsule goes on, go to **3.**

3. Remove insulated jumper wire from P427, and reconnect P427 to lateral primary servo 2nd stage pressure switch. Go to 4.

4. Disconnect connector P428 from aft primary servo 2nd stage pressure switch, and install an insulated jumper wire between P428-4 and P428-9. Go to 5.

5. With BACKUP HYD PUMP switch set to ON and pilot's collective stick SVO OFF/1ST STG/2ND STG switch set to 2ND STG, move collective stick from full down to full up position. #2 PRI SERVO PRESS capsule on caution/advisory panel shall remain off.

Step 1. If #2 PRI SERVO PRESS capsule remains off, replace aft primary servo (PARA 11-4-87). Go to **8.**
Step 2. If #2 PRI SERVO PRESS capsule goes on, go to **6.**

6. Remove insulated jumper wire from P428, and reconnect P428 to aft primary servo 2nd stage pressure switch. Go to 7.

7. Replace forward primary servo (PARA 11-4-87). Go to 8.

TABLE NO. 11-2-3-8. #2 PRI SERVO PRESS Capsule On Caution/Advisory Panel Goes On When Collective Stick Is Moved Full Down To Full Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

8. Procedure completed.

TABLE NO. 11-2-3-9. Cyclic Stick Has Lateral Movement When Collective Stick Is Moved From Full Down To Full Up Position.

TEST OR INSPECTION
CORRECTIVE ACTION

CAUTION

Do not move any cockpit controls while servo is disconnected and pressurized. Unrestricted travel of primary servos can cause damage to swashplate components.

1. With BACKUP HYD PUMP switch set to OFF, disconnect forward, aft, and lateral primary servo outputs. Go to 2.

2. Disconnect lateral primary servo input. Go to 3.

3. Set BACKUP HYD PUMP switch to ON and pressurize forward, aft, and lateral servos. Go to 4.

4. Reconnect lateral primary servo input. Go to 5.

5. Reconnect forward, aft, and lateral primary servo outputs. Go to 6.

6. Carefully grasp lateral primary servo input link, and displace and hold input link in either extended or retracted position. Full stroke travel shall be possible in one second.

Step 1. If full stroke travel is possible in one second, go to **7.**
Step 2. If full stroke travel is not possible in one second, replace lateral primary servo (PARA 11-4-87). Go to **8.**

7. With BACKUP HYD PUMP switch ON, move collective stick from full down to full up position while visually inspecting mixer assembly for corrosion, damage, or an interfering object.

TABLE NO. 11-2-3-9. Cyclic Stick Has Lateral Movement When Collective Stick Is Moved From Full Down To Full Up Position. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If corrosion damage, or an interfering object is found, remove object or replace mixer assembly (PARA 11-4-77). Go to **8.**

Step 2. If corrosion damage, or an interfering object is not found, rig and adjust Flight Control System (PARA 11-4-2). Go to **8.**

8. Procedure completed.

TABLE NO. 11-2-3-10. #1 PRI SERVO PRESS Capsule On Caution/Advisory Panel Goes On When Collective Stick Is Moved Full Down To Full Up.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Disconnect connector P424 from lateral primary servo 1st stage pressure switch, and install an insulated jumper wire between P424-4 and P424-9. Go to 2.

2. With BACKUP HYD PUMP switch set to ON and pilot's collective stick SVO OFF/1ST STG/2ND STG switch set to 1ST STG, move collective stick from full down to full up position. #1 PRI SERVO PRESS capsule on caution/advisory panel shall remain off.

Step 1. If #1 PRI SERVO PRESS capsule remains off, replace lateral primary servo (PARA 11-4-87). Go to **8.**

Step 2. If #1 PRI SERVO PRESS capsule goes on, go to **3.**

3. Remove insulated jumper wire from P424, and reconnect P424 to lateral primary servo 1st stage pressure switch. Go to 4.

4. Disconnect connector P425 from aft primary servo 1st stage pressure switch, and install an insulated jumper wire between P425-4 and P425-9. Go to 5.

5. With BACKUP HYD PUMP switch set to ON and pilot's collective stick SVO OFF/1ST STG/2ND STG switch set to 1ST STG, move collective stick from full down to full up position. #1 PRI SERVO PRESS capsule on caution/advisory panel shall remain off.

Step 1. If #1 PRI SERVO PRESS capsule remains off, replace aft primary servo (PARA 11-4-87). Go to **8.**

TABLE NO. 11-2-3-10. #1 PRI SERVO PRESS Capsule On Caution/Advisory Panel Goes On When Collective Stick Is Moved Full Down To Full Up. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If #1 PRI SERVO PRESS capsule goes on, go to **6.**

6. **Remove insulated jumper wire from P425, and reconnect P425 to aft primary servo 1st stage pressure switch. Go to 7.**

7. **Replace forward primary servo (PARA 11-4-87). Go to 8.**

8. **Procedure completed.**

TABLE NO. 11-2-3-11. TAIL ROTOR QUADRANT Capsule Does Not Go On.

TEST OR INSPECTION
CORRECTIVE ACTION

1. **Check helicopter configuration.**

Step 1. EMEP Go to **2.**
Step 2. W/O EMEP Go to **4.**

EMEP

2. **Remove and check pin filter adapter P117 (PARA 9-4-98).**

Step 1. If pin filter adapter is good, go to **3.**
Step 2. If pin filter adapter is not good, replace pin filter adapter (PARA 9-4-98). Go to **8.**

EMEP

3. **Install pin filter adapter (PARA 9-4-98). Go to 4.**

4. **With insulated jumper wire connected between J611-A or J611-B and J611-C, check for 28 VDC between P117-9 and ground.**

Step 1. If voltage is as specified, troubleshoot caution/advisory panel (PARA 8-2-4). Go to **9.**
Step 2. If voltage is not as specified, go to **5**

5. **Remove insulated jumper. Go to 6.**

6. **Check for 28 VDC between J611-A and ground.**

TABLE NO. 11-2-3-11. TAIL ROTOR QUADRANT Capsule Does Not Go On. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If voltage is as specified, repair/replace wiring between J611-C and P117-9. Go to **8.**
Step 2. If voltage is not as specified, go to **7.**

7. Check continuity between terminal 1 (T RTR SERVO WARN circuit breaker, NO. 1 DC PRI BUS - copilot's circuit breaker panel) and J611-A.

Step 1. If continuity is present, replace circuit breaker (PARA 9-4-1). Go to **8.**
Step 2. If continuity is not present, repair/replace wiring (Appendix F). Go to **8.**

8. Procedure completed.

TABLE NO. 11-2-3-12. Noise Is Heard While Moving Pilot's Cyclic Stick From Left To Right.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is noise coming from forward of vertical torque shaft?

Step 1. If noise is coming from forward of vertical torque shaft, go to **7.**
Step 2. If noise is not coming from forward of vertical torque shaft, go to **2.**

2. Is noise coming from control shaft in upper broom closet?

Step 1. If noise is coming from control shaft in upper broom closet, go to **10.**
Step 2. If noise is not coming from control shaft in upper broom closet, go to **3.**

3. Is noise coming from vertical torque shaft?

Step 1. If noise is coming from vertical torque shaft, go to **14.**
Step 2. If noise is not coming from vertical torque shaft, go to **4.**

4. Is noise coming from aft of vertical torque shaft?

Step 1. If noise is coming from aft of vertical torque shaft, go to **20.**
Step 2. If noise is not coming from aft of vertical torque shaft, go to **5.**

5. Is noise coming from mixer assembly?

TABLE NO. 11-2-3-12. Noise Is Heard While Moving Pilot's Cyclic Stick From Left To Right. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If noise is coming from mixer assembly, go to **23.**
Step 2. If noise is not coming from mixer assembly, go to **6.**

6. Is noise coming from aft of mixer assembly but forward of stationary swashplate?

Step 1. If noise is coming from aft of mixer assembly but forward of stationary swashplate, go to **26.**
Step 2. If noise is not coming from aft of mixer assembly but forward of stationary swashplate, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

7. Visually inspect area forward of vertical torque shafts for missing or loose mounting hardware.

Step 1. If missing or loose mounting hardware is present, replace hardware, as required. Go to **29.**
Step 2. If missing or loose mounting hardware is not present, go to **8.**

8. Inspect the following components forward of the vertical torque tubes for excessive radial play in bearings:
Pilot's and Copilot's Cyclic Sticks (PARA 11-3-4)
Cockpit Lateral Bellcrank (PARA 11-3-22)
Cabin Lateral Bellcrank (PARA 11-3-22)

Step 1. If excessive radial play is present in cyclic stick, cockpit lateral bellcrank, or cabin lateral bellcrank bearings, repair/replace cyclic stick bearings or lateral bellcranks (PARA 11-4-23, PARA 11-4-47, or PARA 11-4-48), as required. Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **9.**

9. Visually inspect pilot's cyclic lateral control rods, copilot's cyclic lateral control rods, and cabin lateral control rod forward of vertical torque shafts for loose swaged inserts (Figure 11-2-3-2).

Step 1. If swaged inserts on control rods are loose, replace pilot's directional and cyclic control rods, copilot's directional and cyclic control rods, or cabin control rods (PARA 11-4-42, PARA 11-4-43, or PARA 11-4-46), as required. Go to **29.**
Step 2. If swaged inserts on control rods are not loose, perform forward, aft and lateral flight control inspection, and bearing friction wear inspection (PARA 11-3-24 or PARA 11-3-2). Go to **29.**

TABLE NO. 11-2-3-12. Noise Is Heard While Moving Pilot's Cyclic Stick From Left To Right. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

10. Visually inspect lateral control rods leading to upper broom closet for loose or missing mounting hardware.

Step 1. If missing or loose mounting hardware is present, replace hardware, as required (PARA 11-4-92). Go to **29.**
Step 2. If missing or loose mounting hardware is not present, go to **11.**

11. Inspect midsection bellcranks and shafts for excessive radial play in bearings (PARA 11-4-50).

Step 1. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (PARA 11-4-50). Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **12.**

12. Inspect pilot's and copilot's lateral control rods leading to upper broom closet for loose swaged inserts (Figure 11-2-3-2).

Step 1. If lateral control rod swaged inserts are loose, repair/replace lateral control rods leading to upper broom closet (PARA 11-4-46). Go to **29.**
Step 2. If lateral control rod swaged inserts are not loose, go to **13.**

13. Inspect midsection bellcranks split spacers for proper gap (PARA 11-4-50).

Step 1. If split spacer gaps are within tolerance, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**
Step 2. If split spacer gaps are not within tolerance, repair/replace split spacers (PARA 11-4-50), as required. Go to **29.**

14. Visually inspect lateral vertical torque shaft for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware (PARA 11-4-46), as required. Go to **29.**
Step 2. If missing or loose hardware is not present, go to **15.**

15. Inspect lateral vertical torque shafts for excessive radial play in bearings.

Step 1. If excessive radial play in bearings is present, replace vertical torque shafts (PARA 11-4-46), as required. Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **16.**

TABLE NO. 11-2-3-12. Noise Is Heard While Moving Pilot's Cyclic Stick From Left To Right. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

16. Inspect swaged inserts on lateral vertical torque shafts for loose swaged inserts (Figure 11-2-3-2).

Step 1. If swaged inserts are loose, replace swaged insert vertical torque shafts (PARA 11-4-46), as required. Go to **29.**
Step 2. If swaged inserts are not loose, go to **17.**

17. Inspect riveted-type lateral vertical torque shafts for loose rod ends (Figure 11-2-3-2).

Step 1. If riveted-type vertical torque shafts rod ends are loose, replace riveted-type vertical torque shafts (PARA 11-4-46), as required. Go to **29.**
Step 2. If riveted-type vertical torque shafts rod ends are not loose, go to **18.**

18. Inspect lateral torque shaft and levers for loose or missing tapered pins (PARA 11-4-57).

Step 1. If loose or missing tapered pins are present, replace tapered pins (PARA 11-4-59), as required. Go to **29.**
Step 2. If loose or missing tapered pins are not present, go to **19.**

19. Inspect airframe around roll trim actuator and lateral vertical torque tubes for loose or missing rivets.

Step 1. If rivets are working loose in airframe near torque shaft or roll trim actuator, replace rivets or repair airframe structure (PARA 2-6-2), as required. Go to **29.**
Step 2. If rivets are not working loose in airframe near torque shaft or roll trim actuator, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

20. Visually inspect area aft of lateral vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware (PARA 11-4-46), as required. Go to **29.**
Step 2. If missing or loose hardware is not present, go to **21.**

21. Inspect roll trim actuator assembly for excessive radial play in bearings.

Step 1. If excessive radial play in bearings is present, replace roll trim assembly (PARA 11-4-69). Go to **29.**

TABLE NO. 11-2-3-12. Noise Is Heard While Moving Pilot's Cyclic Stick From Left To Right. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If excessive radial play in bearings is not present, go to **22.**

22. Inspect riveted-type pushrods for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type pushrod ends are loose, replace riveted-type pushrods (PARA 11-4-65 or PARA 11-4-84), as required. Go to **29.**
Step 2. If riveted-type pushrod ends are not loose, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

23. Visually inspect mixer assembly for loose or missing hardware (PARA 11-3-8).

Step 1. If loose or missing hardware is present, replace hardware (PARA 11-4-77), as required. Go to **29.**
Step 2. If loose or missing hardware is not present, go to **24.**

24. Inspect mixer assembly bearing for excessive radial play (PARA 11-4-77).

Step 1. If excessive radial play in bearings is present, repair mixer (PARA 11-4-77). Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **25.**

25. Inspect mixer assembly area for loose or missing taper pins.

Step 1. If loose or missing taper pins are present, repair/replace taper pins (PARA 11-4-59), as required. Go to **29.**
Step 2. If loose or missing taper pins are not present, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

26. Visually inspect area aft of mixer assembly but forward of stationary swash plate for loose or missing hardware.

Step 1. If loose or missing hardware is present, replace hardware, as required. Go to **29.**
Step 2. If loose or missing hardware is not present, go to **27.**

27. Inspect lateral swashplate link uniball bearings for excessive radial play and freedom of movement (PARA 11-3-12).

Step 1. If excessive radial play or sticky bearings are present, replace lateral swash-plate link (PARA 11-4-95). Go to **29.**
Step 2. If excessive radial play or sticky bearings are not present, go to **28.**

TABLE NO. 11-2-3-12. Noise Is Heard While Moving Pilot's Cyclic Stick From Left To Right. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

28. Inspect the following components for worn journal bearings:
Lateral Primary Servo (PARA 11-4-87)
Mixer to Lateral Primary Servo Control Rod (PARA 11-4-82)
Lateral Control Rod (PARA 11-4-92)
Lateral Bellcrank (PARA 11-3-22)
Left and Right Tie Rods (PARA 11-3-16)

Step 1. If worn journal bearings are present, replace lateral primary servo, mixer to lateral primary servo control rod, lateral control rod, lateral bellcrank, or left and right tie rod (PARA 11-4-87, PARA 11-4-82, PARA 11-4-92, PARA 11-4-97, or PARA 11-4-99), as required. Go to **29.**
Step 2. If worn journal bearings are not present, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

29. Procedure completed.

TABLE NO. 11-2-3-13. Noise Is Heard While Moving Pilot Cyclic Stick From Forward To Aft.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is noise coming from forward of vertical torque shaft?

Step 1. If noise is coming from forward of vertical torque shaft, go to **7.**
Step 2. If noise is not coming from forward of vertical torque shaft, go to **2.**

2. Is noise coming from control shaft in upper broom closet?

Step 1. If noise is coming from control shaft in upper broom closet, go to **10.**
Step 2. If noise is not coming from control shaft in upper broom closet, go to **3.**

3. Is noise coming from vertical torque shaft?

Step 1. If noise is coming from vertical torque shaft, go to **14.**
Step 2. If noise is not coming from vertical torque shaft, go to **4.**

4. Is noise coming from aft of vertical torque shaft?

Step 1. If noise is coming from aft of vertical torque shaft, go to **20.**

TABLE NO. 11-2-3-13. Noise Is Heard While Moving Pilot Cyclic Stick From Forward To Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If noise is not coming from aft of vertical torque shaft, go to **5.**

5. Is noise coming from mixer assembly?

Step 1. If noise is coming from mixer assembly, go to **23.**
Step 2. If noise is not coming from mixer assembly, go to **6.**

6. Is noise coming from aft of mixer assembly but forward of stationary swashplate?

Step 1. If noise is coming from aft of mixer assembly but forward of stationary swashplate, go to **26.**
Step 2. If noise is not coming from aft of mixer assembly but forward of stationary swashplate perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

7. Visually inspect area forward of pitch vertical torque shafts for missing or loose mounting hardware.

Step 1. If missing or loose mounting hardware is present, replace hardware, as required. Go to **29.**
Step 2. If missing or loose mounting hardware is not present, go to **8.**

8. Inspect the following components forward of the pitch vertical torque tubes for excessive radial play in bearings:
Pilot and Copilot Cyclic Sticks (PARA 11-3-4)
Cockpit Forward and Aft Bellcranks (PARA 11-3-21 and PARA 11-3-20)
Cabin Longitudinal Bellcrank (PARA 11-4-48)

Step 1. If excessive radial play is present in cyclic stick, cockpit forward and aft bellcrank, or cabin longitudinal bellcrank bearings, repair/replace cyclic stick bearings, cockpit forward and aft bellcrank, or cabin longitudinal bellcrank (PARA 11-4-23, PARA 11-4-47, or PARA 11-4-48), as required. Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **9.**

9. Visually inspect pilot cyclic pitch control rods, copilot cyclic pitch control rods, and cabin pitch control rod forward of vertical torque shafts for loose swaged inserts (Figure 11-2-3-3).

TABLE NO. 11-2-3-13. Noise Is Heard While Moving Pilot Cyclic Stick From Forward To Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If swaged inserts on control rods are loose, replace pilot's directional and cyclic control rods, copilot's directional and cyclic control rods, or cabin control rods (PARA 11-4-42, PARA 11-4-43, or PARA 11-4-46), as required. Go to **29.**

Step 2. If swaged inserts on control rods are not loose, perform forward, aft and lateral flight control inspection (PARA 11-3-24). Go to **29.**

10. Visually inspect pitch control rods leading to upper broom closet for loose or missing mounting hardware.

Step 1. If loose or missing mounting hardware is present, replace hardware, as required. Go to **29.**

Step 2. If loose or missing mounting hardware is not present, go to **11.**

11. Inspect midsection bellcranks and shafts for excessive radial play in bearings (PARA 11-4-50).

Step 1. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (PARA 11-4-50). Go to **29.**

Step 2. If excessive radial play in bearings is not present, go to **12.**

12. Inspect pilot and copilot pitch control rods leading to upper broom closet for loose swaged inserts (Figure 11-2-3-3).

Step 1. If pitch control rod swaged inserts are loose, repair/replace pitch control rods leading to upper broom closet (PARA 11-4-46), as required. Go to **29.**

Step 2. If pitch control rod swaged inserts are not loose, go to **13.**

13. Inspect midsection bellcranks split spacers for proper gap (PARA 11-4-50).

Step 1. If split spacer gaps are within tolerance, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

Step 2. If split spacer gaps are not within tolerance, repair/replace split spacers (PARA 11-4-50), as required. Go to **29.**

14. Visually inspect pitch vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware, as required. Go to **29.**

Step 2. If missing or loose hardware is not present, go to **15.**

15. Inspect pitch vertical torque shafts for excessive radial play in bearings.

TABLE NO. 11-2-3-13. Noise Is Heard While Moving Pilot Cyclic Stick From Forward To Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If excessive radial play in bearings is present, replace pitch vertical torque shafts (PARA 11-4-46), as required. Go to **29.**

Step 2. If excessive radial play in bearings is not present, go to **16.**

16. Inspect swaged inserts on pitch vertical torque shafts for loose swaged inserts (Figure 11-2-3-3).

Step 1. If swaged inserts on pitch vertical torque shafts are loose, replace swaged insert pitch vertical torque shafts (PARA 11-4-46), as required. Go to **29.**

Step 2. If swaged insert on pitch vertical torque shafts are not loose, go to **17.**

17. Inspect riveted-type pitch vertical torque shafts for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type pitch vertical torque shafts rod ends are loose, replace riveted-type pitch vertical torque shafts (PARA 11-4-46), as required. Go to **29.**

Step 2. If riveted-type pitch vertical torque shafts rod ends are not loose, go to **18.**

18. Inspect pitch torque shaft and levers for loose or missing tapered pins (PARA 11-4-56).

Step 1. If loose or missing tapered pins are present, replace tapered pins (PARA 11-4-59), as required. Go to **29.**

Step 2. If loose or missing tapered pins are not present, go to **19.**

19. Inspect airframe around pitch trim actuator and pitch vertical torque tubes for loose or missing rivets.

Step 1. If rivets are working loose in airframe near torque shaft or pitch trim actuator, replace rivets or repair airframe structure (PARA 2-6-2), as required. Go to **29.**

Step 2. If rivets are not working loose in airframe near torque shaft or pitch trim actuator, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

20. Visually inspect area aft of pitch vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware, as required. Go to **29.**

Step 2. If missing or loose hardware is not present, go to **21.**

TABLE NO. 11-2-3-13. Noise Is Heard While Moving Pilot Cyclic Stick From Forward To Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

21. Inspect pitch trim actuator assembly for excessive radial play in bearings.

Step 1. If excessive radial play in bearings is present, replace pitch trim assembly (PARA 11-4-68). Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **22.**

22. Inspect riveted-type pushrods for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type pushrod ends are loose, replace riveted-type pushrods (PARA 11-4-64 or PARA 11-4-74), as required. Go to **29.**
Step 2. If riveted-type pushrod ends are not loose, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

23. Visually inspect mixer assembly for loose or missing hardware (PARA 11-3-8).

Step 1. If loose or missing hardware is present, replace hardware, as required (PARA 11-4-77). Go to **29.**
Step 2. If loose or missing hardware is not present, go to **24.**

24. Inspect mixer assembly bearing for excessive radial play (PARA 11-4-77).

Step 1. If excessive radial play in bearings is present, repair mixer assembly (PARA 11-4-77). Go to **29.**
Step 2. If excessive radial play in bearings is not present, go to **25.**

25. Inspect mixer assembly area for loose or missing taper pins.

Step 1. If loose or missing taper pins are present, repair/replace taper pins, as required (PARA 11-4-59). Go to **29.**
Step 2. If loose or missing taper pins are not present, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

26. Visually inspect area aft of mixer assembly but forward of stationary swash plate for loose or missing hardware.

Step 1. If loose or missing hardware is present, replace hardware, as required. Go to **29.**
Step 2. If loose or missing hardware is not present, go to **27.**

27. Inspect forward and aft swashplate link uniball bearings for excessive radial play and freedom of movement (PARA 11-3-12).

TABLE NO. 11-2-3-13. Noise Is Heard While Moving Pilot Cyclic Stick From Forward To Aft. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If excessive radial play or sticky bearings are present, replace forward and aft swashplate link (PARA 11-4-95), as required. Go to **29.**
Step 2. If excessive radial play or sticky bearings are not present, go to **28.**

28. Inspect the following components for worn journal bearings:
Forward and Aft Primary Servos (PARA 11-4-87)
Mixer to Aft Primary Servo Control Rod (PARA 11-4-81)
Aft Control Rod (PARA 11-4-91)
Long Control Rod (PARA 11-4-94)
Aft Bellcrank (PARA 11-3-20)

Step 1. If worn journal bearings are present, replace forward primary servo, aft primary servo, mixer to aft primary servo control rod, aft control rod, long control rod, aft bellcrank (PARA 11-4-87, PARA 11-4-81, PARA 11-4-91, or PARA 11-4-98), as required. Go to **29.**
Step 2. If worn journal bearings are not present, perform forward, aft, and lateral flight control channel inspection (PARA 11-3-24). Go to **29.**

29. Procedure completed.

TABLE NO. 11-2-3-14. Noise Is Heard While Moving Pilot's Collective Stick Up And Down.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is noise coming from forward of vertical torque shaft?

Step 1. If noise is coming from forward of vertical torque shaft, go to **7.**
Step 2. If noise is not coming from forward of vertical torque shaft, go to **2.**

2. Is noise coming from collective control shaft in upper broom closet?

Step 1. If noise is coming from control shaft in upper broom closet, go to **10.**
Step 2. If noise is not coming from control shaft in upper broom closet, go to **3.**

3. Is noise coming from vertical torque shaft?

Step 1. If noise is coming from vertical torque shaft, go to **16.**

TABLE NO. 11-2-3-14. Noise Is Heard While Moving Pilot's Collective Stick Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If noise is not coming from vertical torque shaft, go to **4.**

4. Is noise coming from aft of vertical torque shaft?

Step 1. If noise is coming from aft of vertical torque shaft, go to **22.**
Step 2. If noise is not coming from aft of vertical torque shaft, go to **5.**

5. Is noise coming from mixer assembly?

Step 1. If noise is coming from mixer assembly, go to **25.**
Step 2. If noise is not coming from mixer assembly, go to **6.**

6. Is noise coming from aft of mixer assembly but forward of stationary swashplate?

Step 1. If noise is coming from aft of mixer assembly but forward of stationary swashplate, go to **28.**
Step 2. If noise is not coming from aft of mixer assembly but forward of stationary swashplate, perform bearing friction wear inspection (PARA 11-3-2). Go to **31.**

7. Visually inspect area forward of collective vertical torque shafts for missing or loose mounting hardware.

Step 1. If missing or loose mounting hardware is present, replace hardware, as required. Go to **31.**
Step 2. If missing or loose mounting hardware is not present, go to **8.**

8. Inspect the following components forward of the collective vertical torque tubes for excessive radial play in bearings:
Pilot and Copilot Collective Sticks (PARA 11-3-5)
Cabin Collective Bellcrank (PARA 11-4-48)

Step 1. If excessive radial play in bearings is present, repair/replace collective stick bearings, or cabin collective bellcrank (PARA or PARA 11-4-48), as required. Go to **31.**
Step 2. If excessive radial play in bearings is not present, go to **9.**

9. Visually inspect pilot collective control rods, copilot collective control rods, and cabin collective control rod forward of vertical torque shafts for loose swaged inserts (Figure 11-2-3-3).

TABLE NO. 11-2-3-14. Noise Is Heard While Moving Pilot's Collective Stick Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If swaged inserts on control rods are loose, replace pilot's collective control rods, copilot's collective control rods, or cabin collective control rods (PARA 11-4-44, PARA 11-4-45, or PARA 11-4-46), as required. Go to **31.**

Step 2. If swaged inserts on control rods are not loose, perform bearing friction wear inspection (PARA 11-3-2). Go to **31.**

10. Visually inspect collective control rods leading to upper broom closet for loose or missing mounting hardware.

Step 1. If loose or missing mounting hardware is present, replace hardware, as required. Go to **31.**

Step 2. If loose or missing mounting hardware is not present, go to **11.**

NOTE

The popping sound generated between the shaft and forward attachment fitting is considered benign and acceptable.

11. Check for a small amount of motion between the shaft and forward attachment fitting by cycling high collective left pedal to low collective right pedal.

Step 1. If a large amount of motion is present, replace collective torque shafts (PARA 11-4-55). Go to **31.**

Step 2. If a large amount of motion is not present, go to **12.**

12. Inspect midsection bellcranks and shafts for excessive radial play in bearings (PARA 11-4-50).

Step 1. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (PARA 11-4-50). Go to **31.**

Step 2. If excessive radial play in bearings is not present, go to **13.**

13. Inspect pilot and copilot collective control rods leading to upper broom closet for loose swaged inserts (Figure 11-2-3-3).

Step 1. If collective control rod swaged inserts are loose, repair/replace collective control rods leading to upper broom closet (PARA 11-4-46), as required. Go to **31.**

Step 2. If collective control rod swaged inserts are not loose, go to **14.**

14. Inspect midsection bellcranks split spacers for proper gap (PARA 11-4-50).

Step 1. If split spacer gaps are within tolerance, go to **15.**

TABLE NO. 11-2-3-14. Noise Is Heard While Moving Pilot's Collective Stick Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If split spacer gaps are not within tolerance, repair/replace split spacers (PARA 11-4-50), as required. Go to **31.**

15. Perform bearing friction wear inspection (PARA 11-3-2). Go to 31.

16. Visually inspect collective vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware (PARA 11-4-55), as required. Go to **31.**
Step 2. If missing or loose hardware is not present, go to **17.**

17. Inspect collective vertical torque shafts for excessive radial play in bearings.

Step 1. If excessive radial play in bearings is present, replace collective vertical torque shafts (PARA 11-4-55), as required. Go to **31.**
Step 2. If excessive radial play in bearings is not present, go to **18.**

18. Inspect swaged inserts on collective vertical torque shafts for loose swaged inserts (Figure 11-2-3-3).

Step 1. If swaged inserts on collective vertical torque shafts are loose, replace collective vertical torque shafts (PARA 11-4-55), as required. Go to **31.**
Step 2. If swaged inserts on collective vertical torque shafts are not loose, go to **19.**

19. Inspect riveted-type collective vertical torque shafts for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type collective vertical torque shafts rod ends are loose, replace riveted-type collective vertical torque shafts (PARA 11-4-55), as required. Go to **31.**
Step 2. If riveted-type collective vertical torque shafts rod ends are not loose, go to **20.**

20. Inspect collective torque shaft and levers for loose or missing tapered pins (PARA 11-4-55).

Step 1. If loose or missing tapered pins are present, replace tapered pins (PARA 11-4-59), as required. Go to **31.**
Step 2. If loose or missing tapered pins are not present, go to **21.**

21. Inspect airframe around collective vertical torque tubes for loose or missing rivets.

TABLE NO. 11-2-3-14. Noise Is Heard While Moving Pilot's Collective Stick Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If rivets are working loose in airframe near torque shaft, replace rivets or repair airframe structure (PARA 2-6-2), as required. Go to **31.**

Step 2. If rivets are not working loose in airframe near torque shaft, perform bearing friction wear inspection (PARA 11-3-2). Go to **31.**

22. Visually inspect area aft of collective vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware, as required. Go to **31.**

Step 2. If missing or loose hardware is not present, go to **23.**

23. Inspect collective boost assembly for excessive radial play in bearings.

Step 1. If excessive radial play in bearings is present, replace collective boost assembly (PARA 11-4-67). Go to **31.**

Step 2. If excessive radial play in bearings is not present, go to **24.**

24. Inspect riveted-type pushrods for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type pushrod ends are loose, replace riveted-type pushrods (PARA 11-4-64 or PARA 11-4-74), as required. Go to **31.**

Step 2. If riveted-type pushrod ends are not loose, perform bearing friction wear inspection (PARA 11-3-2). Go to **31.**

25. Visually inspect mixer assembly for loose or missing hardware (PARA 11-3-8).

Step 1. If loose or missing hardware is present, replace hardware (PARA 11-4-77), as required. Go to **31.**

Step 2. If loose or missing hardware is not present, go to **26.**

26. Inspect mixer assembly bearing for excessive radial play (PARA 11-4-77).

Step 1. If excessive radial play in bearings is present, repair mixer assembly (PARA 11-4-77). Go to **31.**

Step 2. If excessive radial play in bearings is not present, go to **27.**

27. Inspect mixer assembly area for loose or missing taper pins.

Step 1. If loose or missing taper pins are present, repair/replace taper pins (PARA 11-4-59), as required. Go to **31.**

TABLE NO. 11-2-3-14. Noise Is Heard While Moving Pilot's Collective Stick Up And Down. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 2. If loose or missing taper pins are not present, perform bearing friction wear inspection (PARA 11-3-2). Go to **31.**

28. Visually inspect area aft of mixer assembly but forward of stationary swash plate for loose or missing hardware.

Step 1. If loose or missing hardware is present, replace hardware, as required. Go to **31.**
Step 2. If loose or missing hardware is not present, go to **29.**

29. Inspect forward and aft swashplate link uniball bearings for excessive radial play and freedom of movement (PARA 11-3-12).

Step 1. If excessive radial play or sticky bearings are present, replace forward and aft swashplate link (PARA 11-4-95), as required. Go to **31.**
Step 2. If excessive radial play or sticky bearings are not present, go to **30.**

30. Inspect the following components for worn journal bearings:
Forward Primary Servo (PARA 11-4-87)
Mixer to Forward Primary Servo Control Rod (PARA 11-4-80)
Forward Control Rod (PARA 11-4-90)
Forward Bellcrank (PARA 11-3-21)

Step 1. If worn journal bearings are present, replace forward primary servo, mixer to forward primary servo control rod, forward control rod, forward bellcrank (PARA 11-4-87, PARA 11-4-80, PARA 11-4-90, or PARA 11-4-96), as required. Go to **31.**
Step 2. If worn journal bearings are not present, perform bearing friction wear inspection (PARA 11-3-2). Go to **31.**

31. Procedure completed.

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals.

TEST OR INSPECTION
CORRECTIVE ACTION

1. Is noise coming from vicinity of pedals?

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If noise is in the vicinity of pedals, go to **8.**
Step 2. If noise is not coming from vicinity of pedals, go to **2.**

2. Is noise between pedals and vertical torque shafts?

Step 1. If noise is coming from between pedals and torque shafts, go to **13.**
Step 2. If noise is not coming from between pedals and torque shafts, go to **3.**

3. Is noise coming from control shafts in upper broom closet?

Step 1. If noise is coming from control shafts in upper broom closet, go to **16.**
Step 2. If noise is not coming from control shafts in upper broom closet, go to **4.**

4. Is noise coming from vertical torque shaft?

Step 1. If noise is coming from vertical torque shaft, go to **23.**
Step 2. If noise is not coming from vertical torque shaft, go to **5.**

5. Is noise coming from aft of vertical torque shafts?

Step 1. If noise is aft of vertical torque shafts, go to **29.**
Step 2. If noise is not coming from aft of vertical torque shafts, go to **6.**

6. Is noise coming from mixer assembly?

Step 1. If noise is coming from mixer assembly, go to **32.**
Step 2. If noise is not coming from mixer assembly, go to **7.**

7. Is noise coming from mixer assembly and forward quadrant?

Step 1. If noise is coming from mixer assembly and forward quadrant, go to **35.**
Step 2. If noise is not coming from mixer assembly and forward quadrant, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

8. Visually inspect pilot and copilot pedal adjusters for missing or loose mounting hardware.

Step 1. If missing or loose mounting hardware is present, replace hardware (PARA 11-4-13 or PARA 11-4-14), as required. Go to **38.**
Step 2. If missing or loose mounting hardware is not present, go to **9.**

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

9. Inspect pilot and copilot pedal adjuster bearings for excessive radial play (PARA 11-4-11).

Step 1. If excessive radial play is present in pilot or copilot pedal adjuster bearings, repair/replace pedal adjuster bearings (PARA 11-4-11), as required. Go to **38.**

Step 2. If excessive radial play is not present in pilot or copilot pedal adjuster bearings, go to **10.**

10. Inspect pilot and copilot pedal and adjuster control rods for loose swaged inserts (Figure 11-2-3-3).

Step 1. If swaged insert on pilot or copilot pedal is loose and adjuster control rods are loose, replace control rods (PARA 11-4-13 or PARA 11-4-14), as required. Go to **38.**

Step 2. If swaged insert on pilot or copilot control rods is not loose, go to **11.**

11. Inspect pilot and copilot pedal adjuster taper pins for security.

Step 1. If loose tapered pins are present, repair/replace tapered pins (PARA 11-4-59), as required. Go to **38.**

Step 2. If loose tapered pins are not present, go to **12.**

12. Inspect pilot and copilot pedal adjusters for proper shimming.

Step 1. If pilot and copilot pedal adjusters are properly shimmed, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

Step 2. If pilot or copilot pedal adjusters are not properly shimmed, repair/replace shims (PARA 11-4-11), as required. Go to **38.**

13. Visually inspect area between pilot and copilot pedals and upper broom closet for missing or loose hardware.

Step 1. If missing or loose hardware is present, replace hardware, as required. Go to **38.**

Step 2. If missing or loose hardware is not present, go to **14.**

14. Inspect the following components in pilot and copilot yaw channel for excessive radial play in bearings:

Yaw Lever (PARA 11-3-11)
Yaw Bellcrank (PARA 11-4-48)

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If pilot yaw lever, pilot yaw bellcrank, copilot yaw lever, or copilot yaw bellcrank bearings have excessive radial play, repair/replace component (PARA 11-4-47 or PARA 11-4-48), as required. Go to **38.**

Step 2. If pilot yaw lever, pilot yaw bellcrank, copilot yaw lever, or copilot yaw bellcrank bearings do not have excessive radial play, go to **15.**

15. Inspect pilot and copilot yaw directional control rods in cockpit for loose swaged inserts (Figure 11-2-3-3).

Step 1. If control rod swaged inserts are loose, repair/replace control rods (PARA 11-4-42 or PARA 11-4-43), as required. Go to **38.**

Step 2. If control rod swaged inserts are not loose, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

16. Visually inspect upper broom closet for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware, as required. Go to **38.**

Step 2. If missing or loose hardware is not present, go to **17.**

NOTE

The popping sound generated between the bellcrank shaft and forward attachment fitting is considered benign and acceptable.

17. Check for a small amount of motion between the bellcrank shaft and forward attachment fitting by cycling the pedals.

Step 1. If a large amount of motion is present, replace yaw torque shaft (PARA 11-4-58). Go to **38.**

Step 2. If a large amount of motion is not present, go to **18.**

18. With upper console BACKUP HYD PUMP switch OFF, check midsection bellcranks and shafts for excessive radial play in bearings (PARA 11-4-50).

Step 1. If excessive radial play in bearings is present, repair/replace midsection bellcranks and shaft (PARA 11-4-50). Go to **38.**

Step 2. If excessive radial play in bearings is not present, go to **19.**

19. Inspect pilot and copilot yaw control rods leading to upper broom closet for loose swaged inserts (Figure 11-2-3-3).

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

Step 1. If yaw control rods swaged inserts are loose, repair/replace control rods leading to upper broom closet (PARA 11-4-46), as required. Go to **38.**
Step 2. If yaw control rod swaged inserts are not loose, go to **20.**

20. Inspect midsection bellcranks split spacers for proper gap (PARA 11-4-50).

Step 1. If split spacer gaps are within tolerance, go to **21.**
Step 2. If split spacer gaps are not within tolerance, repair/replace split spacers (PARA 11-4-50), as required. Go to **38.**

21. Inspect airframe around bellcrank attachment fittings for loose or missing rivets.

Step 1. If rivets are working loose in airframe near bellcrank attachment fittings, repair airframe structure (PARA 2-6-2), as required. Go to **38.**
Step 2. If rivets are not working loose in airframe near bellcrank attachment fittings, go to **22.**

22. Perform bearing friction wear inspection (PARA 11-3-2). Go to 38.

23. Visually inspect pilot and copilot yaw vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware (PARA 11-4-58), as required. Go to **38.**
Step 2. If missing or loose hardware is not present, go to **24.**

24. With upper console BACKUP HYD PUMP switch OFF, check yaw torque shaft and levers for excessive radial play in bearings (PARA 11-4-58).

Step 1. If excessive radial play in bearings is present, replace yaw torque shaft and levers (PARA 11-4-58). Go to **38.**
Step 2. If excessive radial play in bearings is not present, go to **25.**

25. Inspect swaged inserts on yaw vertical torque shafts for loose swaged inserts (Figure 11-2-3-3).

Step 1. If swaged inserts on yaw vertical torque shafts are loose, replace yaw vertical torque shafts (PARA 11-4-58), as required. Go to **38.**
Step 2. If swaged inserts on yaw vertical torque shafts are not loose, go to **26.**

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

26. Inspect riveted-type yaw vertical torque shafts for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type yaw vertical torque shafts rod ends are loose, replace riveted-type yaw vertical torque shafts (PARA 11-4-58), as required. Go to **38.**
Step 2. If riveted-type yaw vertical torque shafts rod ends are not loose, go to **27.**

27. Inspect yaw torque shaft and levers for loose or missing tapered pins (PARA 11-4-58).

Step 1. If loose or missing tapered pins are present, replace tapered pins (PARA 11-4-59), as required. Go to **38.**
Step 2. If loose or missing tapered pins are not present, go to **28.**

28. Inspect airframe around yaw trim actuator, and yaw vertical torque tubes for loose or missing rivets.

Step 1. If rivets are working loose in airframe near torque tubes or yaw trim actuator, replace rivets or repair airframe structure (PARA 2-6-2), as required. Go to **38.**
Step 2. If rivets are not working loose in airframe near torque tubes or yaw trim actuator, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

29. Visually inspect area aft of yaw vertical torque shafts for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware (PARA 11-4-58), as required. Go to **38.**
Step 2. If missing or loose hardware is not present, go to **30.**

30. Inspect yaw boost assembly for excessive radial play in bearings.

Step 1. If excessive radial play in bearings is present, replace yaw boost assembly (PARA 11-4-70). Go to **38.**
Step 2. If excessive radial play in bearings is not present, go to **31.**

31. Inspect riveted-type pushrods for loose rod ends (Figure 11-2-3-3).

Step 1. If riveted-type pushrod ends are loose, replace riveted-type pushrods (PARA 11-4-64 or PARA 11-4-74), as required. Go to **38.**
Step 2. If riveted-type pushrod ends are not loose, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION
CORRECTIVE ACTION

32. Visually inspect mixer assembly for loose or missing hardware.

Step 1. If missing or loose hardware is present, replace hardware (PARA 11-4-77), as required. Go to **38.**
Step 2. If missing or loose hardware is not present, go to **33.**

33. Inspect mixer assembly bearing for excessive radial play.

Step 1. If excessive radial play in bearings is present, repair mixer assembly (PARA 11-4-77). Go to **38.**
Step 2. If excessive radial play in bearings is not present, go to **34.**

34. Inspect mixer assembly area for loose or missing taper pins.

Step 1. If loose or missing taper pins are present, repair/replace taper pins (PARA 11-4-59), as required. Go to **38.**
Step 2. If loose or missing taper pins are not present, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

35. Visually inspect area between mixer and forward quadrant for loose or missing hardware.

Step 1. If loose or missing hardware is present, replace hardware, as required. Go to **38.**
Step 2. If loose or missing hardware is not present, go to **36.**

36. Inspect forward quadrant and support bearings for excessive radial play.

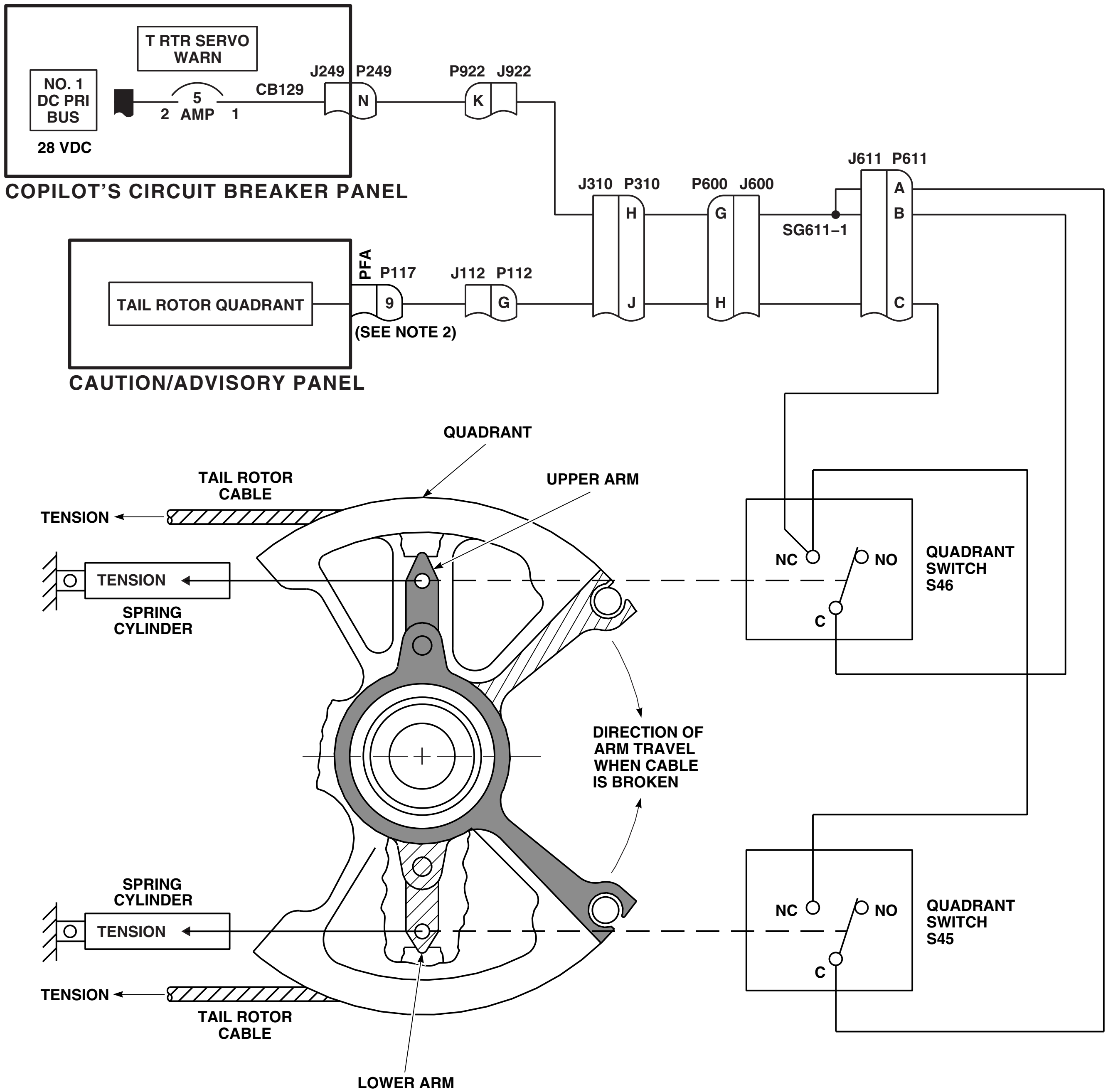
Step 1. If excessive radial play in bearings is present, replace forward quadrant and supports (PARA 11-4-86), as required. Go to **38.**
Step 2. If excessive radial play in bearings is not present, go to **37.**

37. Inspect the following components for loose rod ends:
Mixer to Yaw Lever Pushrod (PARA 11-4-83)
Yaw Lever to Forward Quadrant Control Rod (PARA 11-4-84)

Step 1. If loose rod ends are present, replace control rod (PARA 11-4-83 or PARA 11-4-84), as required. Go to **38.**
Step 2. If loose rod ends are not present, perform bearing friction wear inspection (PARA 11-3-2). Go to **38.**

TABLE NO. 11-2-3-15. Noise Is Heard While Pushing Pilot's Left And Right Pedals. (Cont)

TEST OR INSPECTION CORRECTIVE ACTION
38. Procedure completed.



NOTES

1. WHEN A CABLE IS BROKEN, SPRING CYLINDER PULLS ARM (UPPER OR LOWER) AND CAUSES QUADRANT SWITCH CONTACTS TO CLOSE.
2. **W/O EMEP** PIN FILTER ADAPTER (PFA) IS NOT INSTALLED.

LEGEND

———	ELECTRICAL
— — —	MECHANICAL

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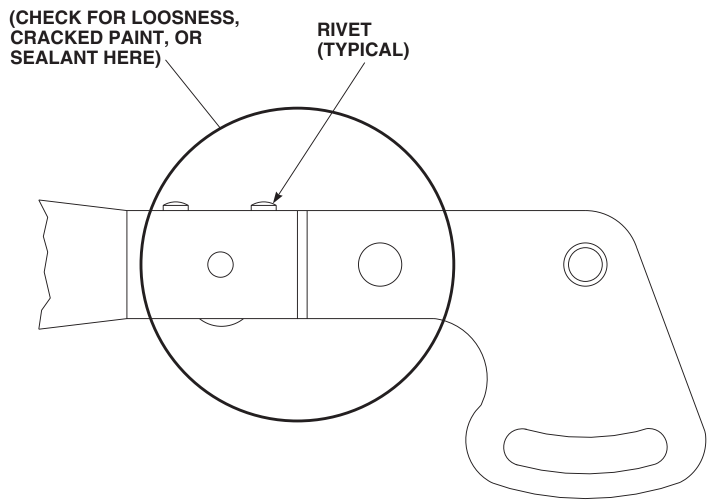
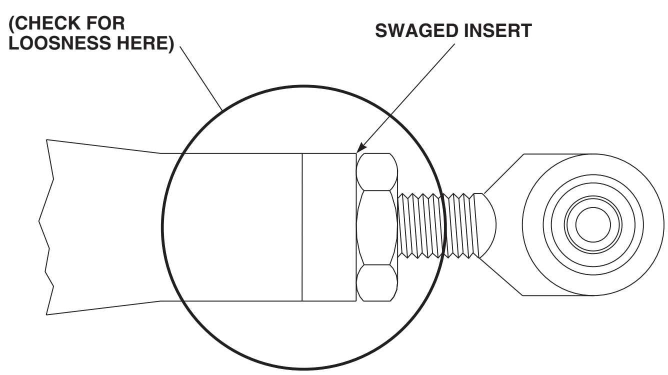
FIGURE 11-2-3-1. TAIL ROTOR QUADRANT WARNING SYSTEM SCHEMATIC DIAGRAM



RIVETED–TYPE PUSHROD

(SEE NOTE 1)



SWAGED INSERT–TYPE PUSHROD

(SEE NOTE 2)

NOTES

1. USUALLY FOUND IN THE YAW SYSTEM, BETWEEN THE YAW BOOST SERVO AND THE YAW SYSTEM STOP IN THE CABIN OVERHEAD.
2. USUALLY FOUND IN THE YAW SYSTEM, MOST COMMONLY THE FOUR RODS CONNECTING THE PEDALS TO THE PEDAL ADJUSTERS.

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SA

FIGURE 11-2-3-2. CONTROL ROD INSPECTION DIAGRAM

11-2-4 ROLL SAS ASSEMBLY (BENCH).

INITIAL SETUP

Test Equipment

- Dial Indicator, CM 6400
- Nulling Fixture Assembly, SAS Actuator, 70700-20675-041 (Figure C-37, Appendix C)

Tools

- Hydraulic Repairer Toolkit
- Magnet, GGG-F-0036

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix C
- Appendix D

Equipment Conditions

- Hydraulic (3000 psi) Bench Power Available
- 28 vdc and 115 vac, Single-Phase, 60 Hz Bench Power Available

CAUTION

To prevent leakage between the nulling fixture and servo, make sure manifold and servo self-sealing couplings are clean.

NOTE

If a servovalve or actuator has to be replaced, repeat operational/troubleshooting procedure.

11-2-4.1 Operational/Troubleshooting Procedure.

a. Position roll SAS assembly on nulling fixture base. Carefully slide roll assembly toward manifold and engage self-sealing couplings.

b. Install three bolts securing roll SAS assembly to base. Tighten bolts.

c. Place all test box pilot-assist/nulling switches to OFF and place TRIM COIL SELECT switch to BOTH.

d. Connect test box cables to pilot-assist module and roll SAS assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)

CABLE CONNECTOR	CONNECT TO
J3	Roll assembly (Servo valve) (P/N 70700-20682-042 cable assembly)

e. Connect test box power cables to electrical power source.

f. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.

g. Turn on hydraulic power source and set pressure for 3000 psi at a maximum of 6 gpm.

h. Bleed air from inside assembly in accordance with steps i. through t.

CAUTION

To prevent damage to test equipment, do not apply electrical power until hydraulic pressure is applied.

i. Place test box POWER DC and AC switches to ON.

RESULT: POWER DC and AC ON indicators shall light and SAS PRESS ON light shall go on.

- If result is not as specified, make sure test box AC and DC circuit breakers are pushed in.

j. Place test box SAS switch to ON and then to OFF.

RESULT: SAS PRESS ON and OFF lights shall cycle from on to off.

- If result is not as specified, replace test box.

k. Repeat step j. several times and leave switch at ON (SAS PRESS ON light on).

l. Turn TRIM switch on.

m. Vary test box CURRENT CONTROL until VALVE CURRENT meter indicates 0.

n. Vary test box CURRENT CONTROL from 0 to 4 mA, back to 0, then to - 4 mA, and back to 0 several times, to stroke SAS piston. Return CURRENT CONTROL to 0.

CAUTION

When using the P/N 70700-20682-042 cable assembly, make certain that the connector marked SAS ACTUATOR is correctly connected. If the incorrect connector is used, servo valve may become damaged.

o. Disconnect cable assembly, 70700-20682-041, from test box and trim servovalve. Connect cable assembly 70700-20682-042 to test set and SAS pitch trim assembly SAS ACTUATOR.

p. Cycle SAS switch several times ON and OFF. After cycling, leave switch in ON position.

q. Vary test box CURRENT CONTROL from 0 to 4 mA, back to 0, then to - 4 mA, and back to 0 several times, to stroke SAS piston.

CAUTION

If bleed screw is not backed out properly, packing on bleed screw will be damaged.

r. Decrease hydraulic pressure to 500 psi. Back out bleed screw until bleed port is visible and a good flow is established.

s. Allow fluid to flow from bleed port until fluid is free of air. Decrease pressure to 0 psi. Close bleed port. **TORQUE TO 19 - 21 INCH-POUNDS** and lockwire.

t. Increase hydraulic pressure to 3000 psi and watch roll SAS assembly output piston for any movement.

RESULT: Piston shall not move.

- If result is not as specified, there is still air trapped in servo. Repeat steps i. through o.

u. Install dial indicator against SAS actuator link. Zero indicator.

v. Adjust indicator to read zero.

w. Cycle test box SAS switch several times from ON to OFF. Watch actuator piston for any jump or movement.

RESULT: Actuator piston jump shall be below 0.005 inches to be nulled.

- If result is not as specified, do the following:

  (a) Remove cover from inside of jamnut using magnet.

  (b) Loosen jamnut 1/4 turn.

  (c) Insert screwdriver through jamnut and turn eccentric screw as needed to reduce piston jump below 0.005 inch. Cycle SAS switch to check for jump.

  (d) If piston jump is out of limits, replace SAS module (PARA 11-5-6), and repeat operational/troubleshooting procedure.

  (e) If piston jump is below 0.005 inch, **TORQUE JAMNUT 47 - 53 INCH-POUNDS**. Install cover on inside of jamnut and lockwire.

x. Repeat piston jump check after torquing jamnut. Readjust eccentric screw if necessary.

NOTE

This verifies if SAS output rod moves properly when a specific current is applied to servovalve.

y. On test box, vary CURRENT CONTROL and do VALVE CURRENT check as shown in Figure 11-2-4-2.

RESULT: Valve movement shall correspond to chart on Figure 11-2-4-2.

- If SAS output rod movement does not correspond to milliamp input from test box and if nulling check of servo is acceptable, replace SAS module (SAS actuator) (PARA 11-5-6) and/or electrohydraulic servo valve (PARA 11-5-5).

z. Place test box POWER DC and AC switches to OFF.

aa. Turn off hydraulic pressure and electrical power.

ab. Remove dial indicator.

ac. Disconnect hydraulic lines from nulling fixture.

ad. Remove test box cable from power source.

ae. Remove roll SAS assembly from nulling fixture by removing three bolts.

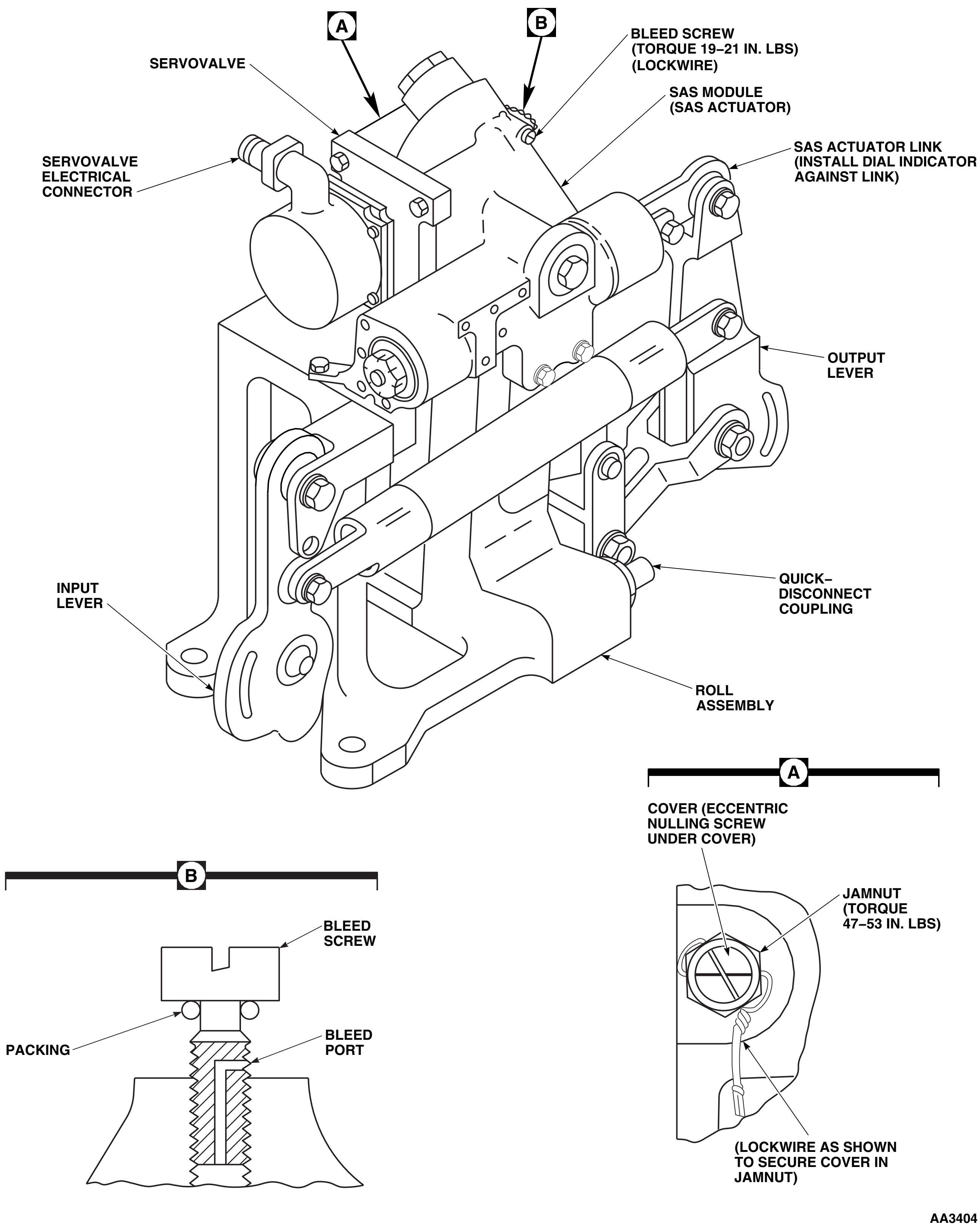
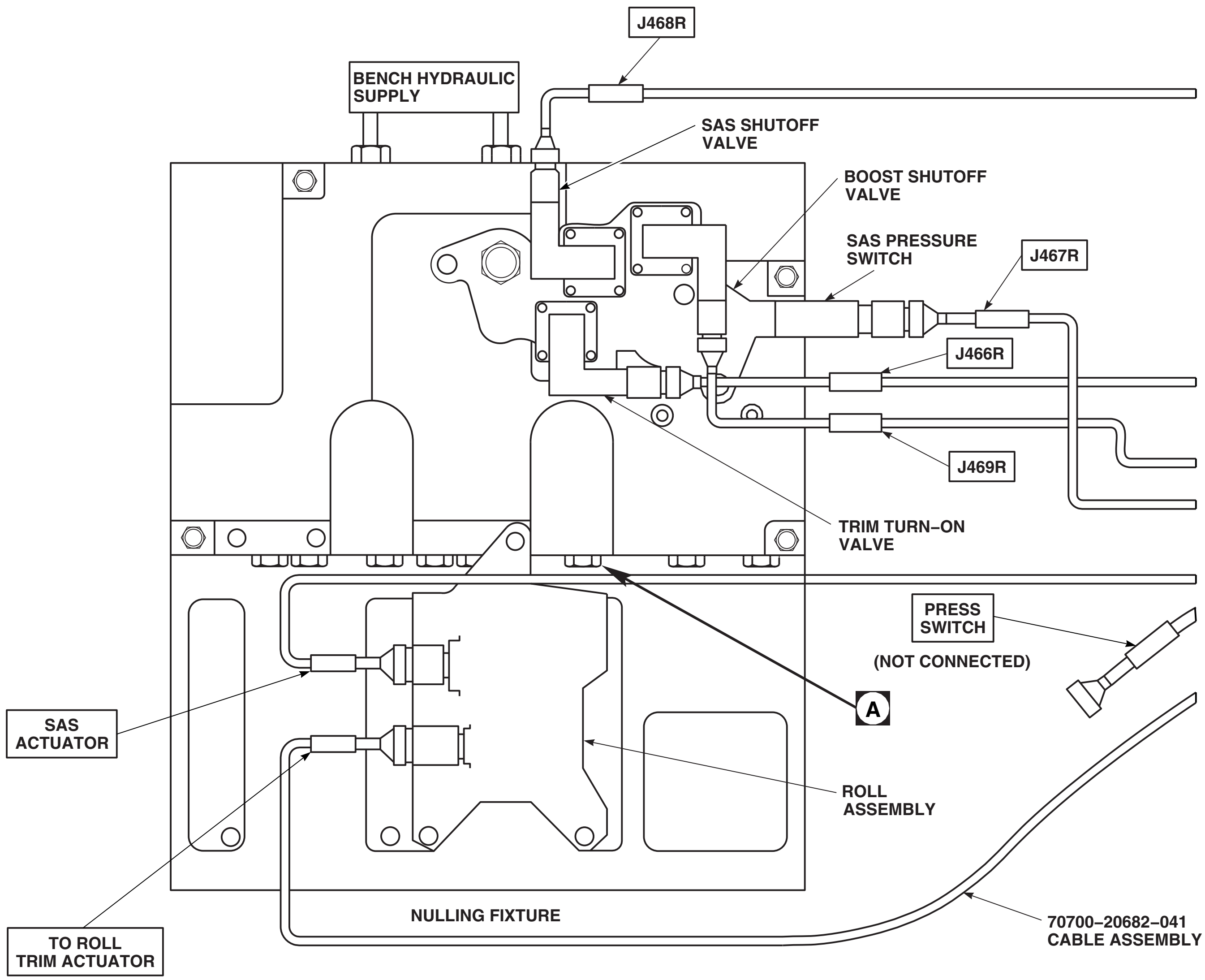


FIGURE 11-2-4-1. ROLL SAS ASSEMBLY



VALVE CURRENT CHART

TRIM COIL SELECT SWITCH IN BOTH POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–3.70	0.090 TO 0.110 (EXTENDED FROM NULL)
–6.60	0.155 TO 0.200 (EXTENDED FROM NULL)
+3.70	0.090 TO 0.110 (RETRACTED FROM NULL)
+6.60	0.155 TO 0.200 (RETRACTED FROM NULL)

TRIM COIL SELECT SWITCH IN 1 OR 2 POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–1.85	0.045 TO 0.055 (EXTENDED FROM NULL)
–3.30	0.0775 TO 0.100 (EXTENDED FROM NULL)
+1.85	0.045 TO 0.055 (RETRACTED FROM NULL)
+3.30	0.0775 TO 0.100 (RETRACTED FROM NULL)

NOTE

CONNECT CONNECTOR AS SPECIFIED IN OPERATIONAL CHECK.

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SA

FIGURE 11-2-4-2. ROLL SAS ASSEMBLY TEST SETUP DIAGRAM (SHEET 1 OF 2)

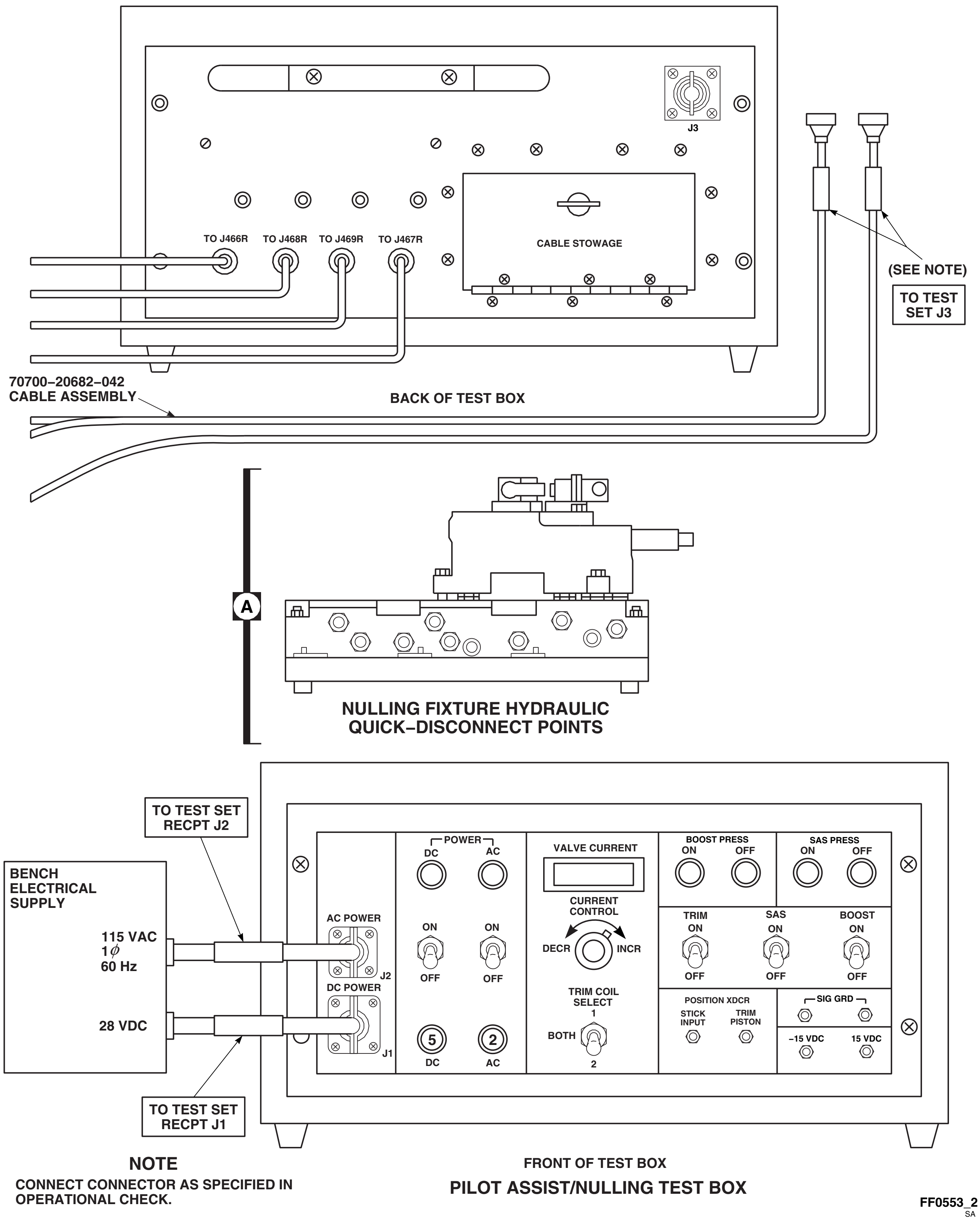


FIGURE 11-2-4-2. ROLL SAS ASSEMBLY TEST SETUP DIAGRAM (SHEET 2 OF 2)

11-2-5 YAW BOOST ASSEMBLY (BENCH).

INITIAL SETUP

Test Equipment

- Dial Indicator, CM 6400
- Nulling Fixture Assembly, SAS Assembly, 70700-20675-041 (Figure C-37, Appendix C)

Tools

- Hydraulic Repairer Toolkit
- Magnet, GGG-F-0036

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix C
- Appendix D

Equipment Conditions

- Hydraulic (3000 psi) Bench Power Available
- 28 vdc and 115 vac, Single-Phase, 60 Hz Bench Power Available

CAUTION

To prevent leakage between the nulling fixture and servo, make sure manifold and servo self-sealing couplings are clean.

NOTE

If a SAS module or actuator has to be replaced, repeat operational/troubleshooting procedure.

11-2-5.1 Operational/Troubleshooting Procedure.

a. Position yaw boost assembly on nulling fixture base. Carefully slide yaw boost assembly toward manifold and engage self-sealing couplings.

b. Install four bolts securing yaw boost assembly to base. Tighten bolts.

c. Place all test box pilot-assist/nulling switches to OFF and place TRIM COIL SELECT switch to BOTH.

CAUTION

When using the 70700-20682-042 cable, make sure that the connectors marked PRESS SWITCH and SAS ACTUATOR are correctly connected. If an incorrect connection is made, servo valve may become damaged.

d. Connect test box cables to pilot-assist module and yaw boost assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)

CABLE CONNECTOR	CONNECT TO
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)
J3	Yaw boost assembly (SAS actuator and pressure switch)

e. Connect test box power cables to electrical power source.

f. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.

g. Turn on hydraulic power source and set pressure for 3000 psi at a maximum of 6 gpm.

h. Bleed air from inside assembly in accordance with steps i. through t.

CAUTION

To prevent damage to test equipment, do not apply electrical power until hydraulic pressure is applied.

i. Place test box POWER DC and AC switches to ON.

RESULT: POWER DC and AC ON indicators shall light and SAS PRESS ON light shall go on.

- If result is not as specified, make sure test box AC and DC circuit breakers are pushed in.

j. Place test box BOOST switch to ON and then to OFF several times (BOOST PRESS ON and OFF lights cycle), then leave ON. Move servo input lever back and forth several times to cycle output piston.

k. Place test box SAS switch to ON and then to OFF.

RESULT: SAS PRESS ON and OFF lights shall cycle from on to off.

- If result is not as specified, replace test box.

l. Repeat step k. several times and leave switch at ON (SAS PRESS ON light on).

m. Vary test box CURRENT CONTROL until VALVE CURRENT meter indicates 0.

n. Vary test box CURRENT CONTROL from 0 to 5.0, back to 0, then to - 5.0, and back to 0 several times to stroke the SAS piston. Return CURRENT CONTROL to 0.

CAUTION

If bleed screw is not backed out properly, packing on bleed screw will be damaged.

o. Decrease hydraulic pressure to 500 psi. Back out bleed screw until bleed port is visible and a good flow is established.

p. Allow fluid to flow from bleed port until fluid is free of air. Decrease pressure to 0 psi. Close bleed port. **TORQUE TO 19 - 21 INCH-POUNDS** and lockwire.

q. Increase hydraulic pressure to 3000 psi and watch SAS assembly output piston for any movement.

RESULT: Piston shall not move.

- If result is not as specified, there is still air trapped in servo. Repeat steps h. through o.

r. Install dial indicator against SAS actuator link.

s. Adjust indicator to read zero.

t. Cycle test box SAS switch several times from ON to OFF. Watch actuator piston for any jump or movement.

RESULT: Actuator piston jump shall be less than 0.005 inches to be nulled.

- If result is not as specified, do the following:

  (a) Remove cover from inside of jamnut using magnet.

  (b) Loosen jamnut 1/4 turn.

  (c) Insert screwdriver through jamnut and turn eccentric screw as needed to reduce piston jump below 0.005 inch. Cycle SAS switch and check for jump.

  (d) If piston jump is out of limits, replace servovalve (PARA 11-5-5) and repeat operational/troubleshooting procedure.

  (e) If piston jump is below 0.005 inch, **TORQUE JAMNUT 47 - 53 INCH-POUNDS**. Install cover on inside of jamnut and lockwire.

u. Repeat piston jump check after torquing jamnut. Readjust eccentric screw if necessary.

NOTE

This verifies if SAS output rod moves properly when a specific current is applied to servo valve.

v. On test box, vary CURRENT CONTROL and do VALVE CURRENT check as shown in Figure 11-2-5-2.

RESULT: Valve movement shall correspond to chart on Figure 11-2-5-2.

- If SAS output rod movement does not correspond to milliamp input from test box and if nulling check of servo is acceptable, replace SAS module (PARA 11-5-6) and/or electrohydraulic servo valve (PARA 11-5-5).

w. Place test box POWER DC and AC switches to OFF.

x. Turn off hydraulic pressure and electrical power.

y. Remove dial indicator.

z. Disconnect hydraulic lines from nulling fixture.

aa. Remove test box cable from power source.

ab. Remove yaw boost assembly from nulling fixture by removing four bolts.

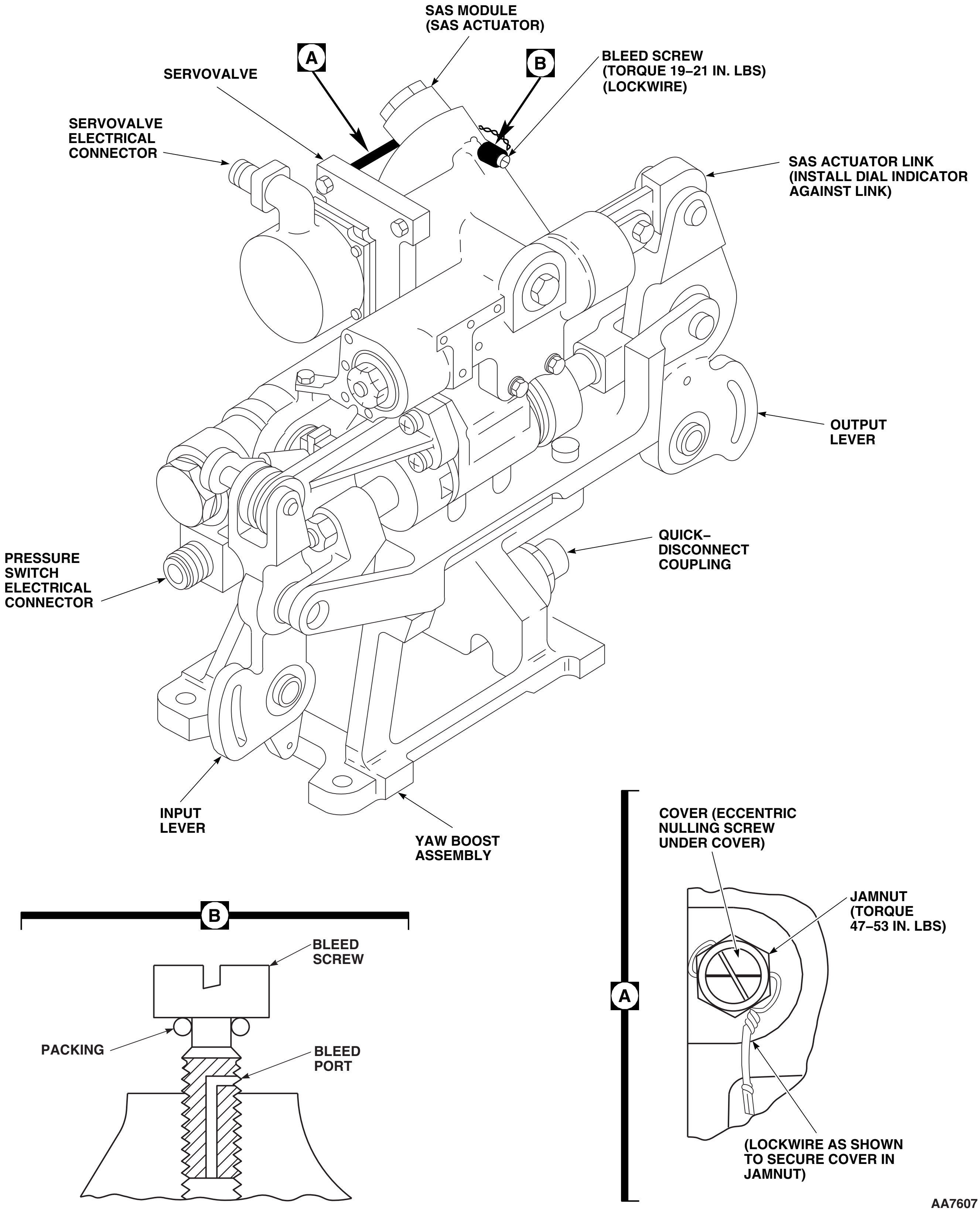
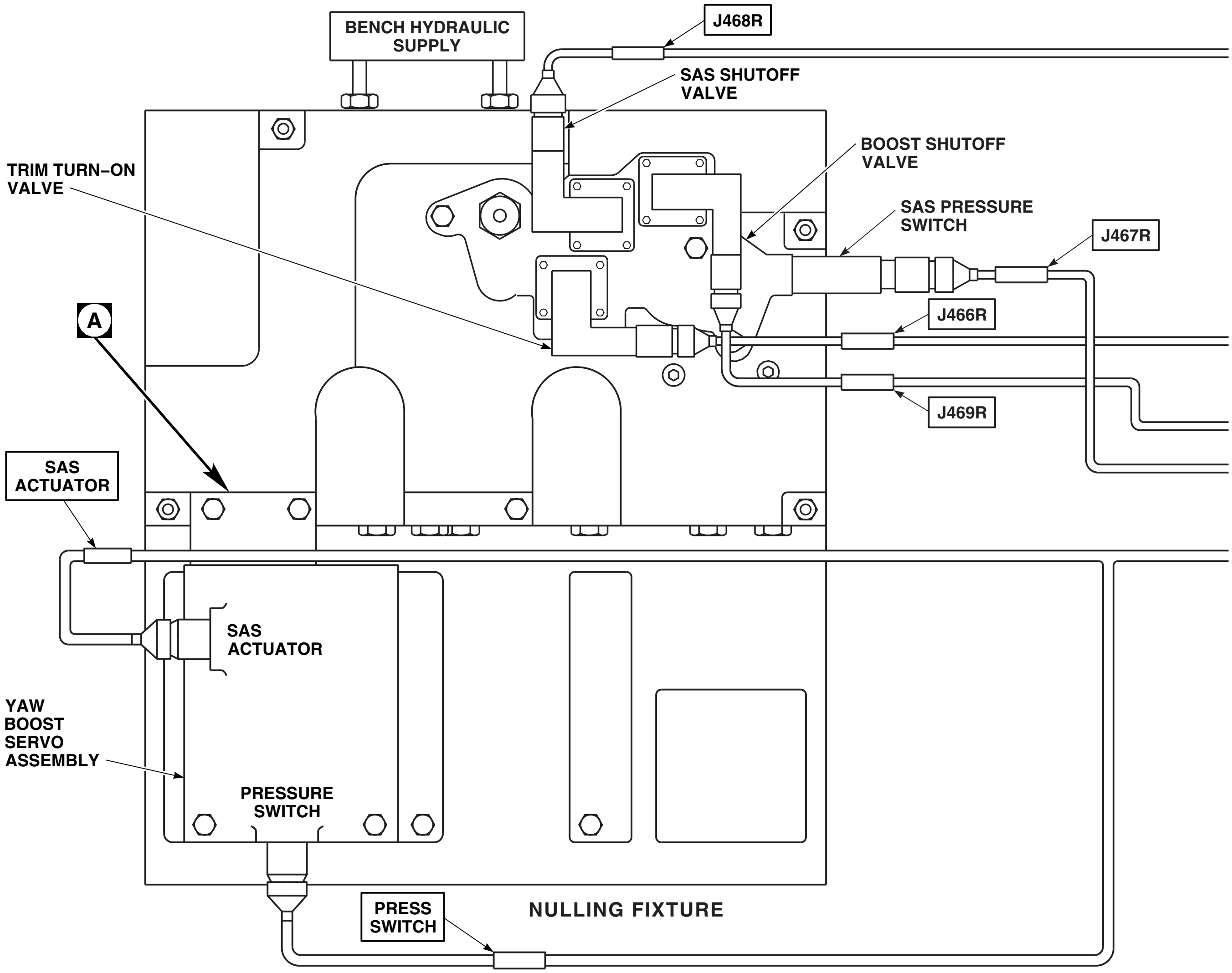


FIGURE 11-2-5-1. YAW BOOST ASSEMBLY



VALVE CURRENT CHART

TRIM COIL SELECT SWITCH IN BOTH POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–3.70	0.090 TO 0.110 (EXTENDED FROM NULL)
–6.60	0.155 TO 0.200 (EXTENDED FROM NULL)
+3.70	0.090 TO 0.110 (RETRACTED FROM NULL)
+6.60	0.155 TO 0.200 (RETRACTED FROM NULL)

TRIM COIL SELECT SWITCH IN 1 OR 2 POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–1.85	0.045 TO 0.055 (EXTENDED FROM NULL)
–3.30	0.0775 TO 0.100 (EXTENDED FROM NULL)
+1.85	0.045 TO 0.055 (RETRACTED FROM NULL)
+3.30	0.0775 TO 0.100 (RETRACTED FROM NULL)

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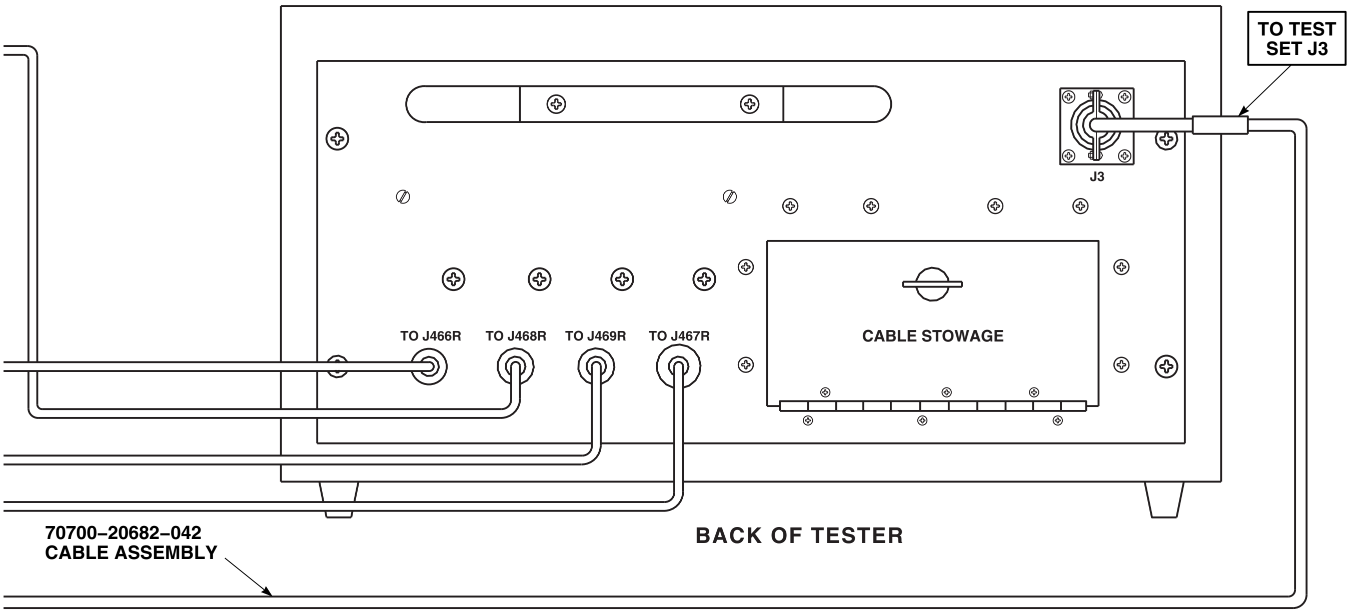
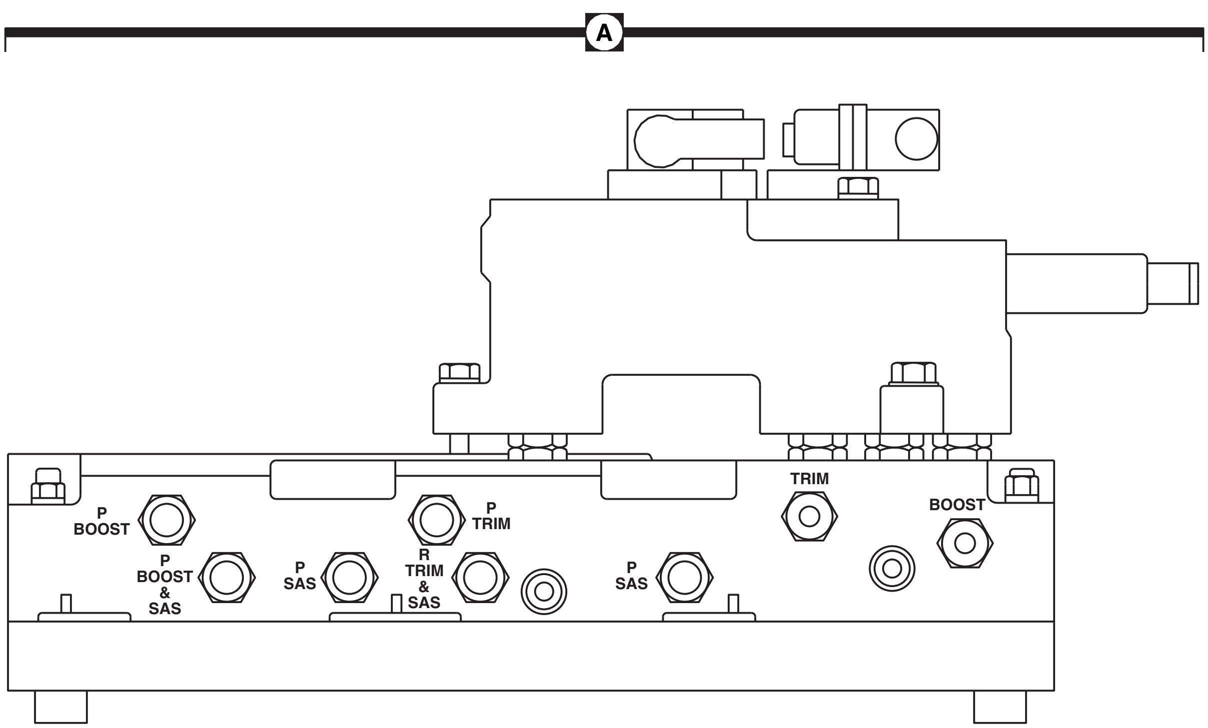
FIGURE 11-2-5-2. YAW BOOST ASSEMBLY TEST SETUP (SHEET 1 OF 3)





NULLING FIXTURE HYDRAULIC QUICK–DISCONNECT POINTS

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FIGURE 11-2-5-2. YAW BOOST ASSEMBLY TEST SETUP (SHEET 2 OF 3)

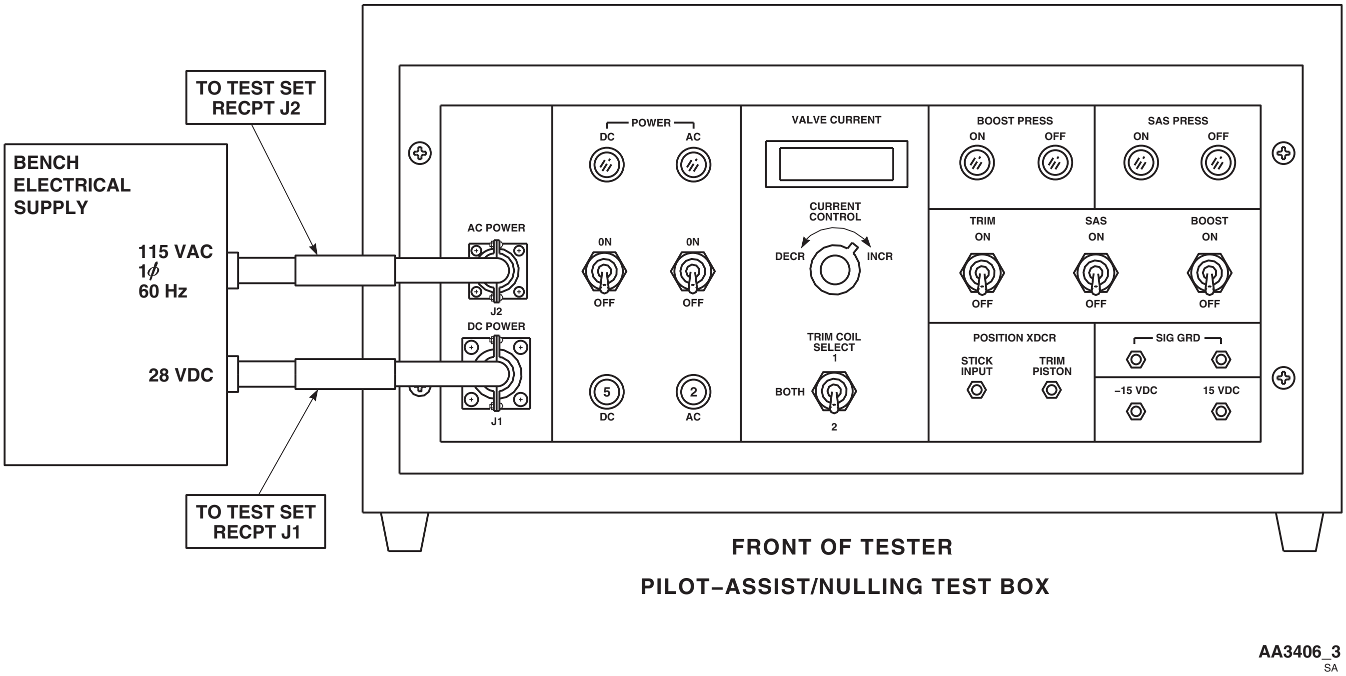


FIGURE 11-2-5-2. YAW BOOST ASSEMBLY TEST SETUP (SHEET 3 OF 3)

11-2-6 PITCH TRIM ASSEMBLY (BENCH).

INITIAL SETUP

Test Equipment

- Dial Indicator, CM 6400
- Nulling Fixture Assembly, SAS Actuator, 70700-20675-041 (Figure C-37, Appendix C)

Tools

- Hydraulic Repairer Toolkit
- Magnet, GGG-F-0036
- Torque Wrench, S-70 Torque Kit

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix C
- Appendix D

Equipment Conditions

- Hydraulic (3000 psi) Bench Power Available
- 28 vdc and 115 vac, Single-Phase, 60 Hz Bench Power Available

CAUTION

To prevent leakage between the nulling fixture and servo, make sure manifold and servo self-sealing couplings are clean.

NOTE

If a servovalve and actuator has to be replaced, repeat operational/troubleshooting procedure.

11-2-6.1 Operational/Troubleshooting Procedure.

a. Position pitch trim assembly on nulling fixture base. Carefully slide pitch trim assembly toward manifold and engage self-sealing couplings.

b. Install three bolts securing pitch trim assembly to base. Tighten bolts.

c. Place all test box pilot-assist/nulling switches to OFF and place TRIM COIL SELECT switch to BOTH.

d. Connect test box cables to pilot-assist module and pitch trim assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)

CABLE CONNECTOR	CONNECT TO
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)
J3	Pitch trim actuator (Servo valve) (70700-20682-041 cable assembly)

e. Connect test box power cables to electrical power source.

f. Connect test box power cables to electrical power source.

g. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.

h. Turn on hydraulic power source and set pressure for 3000 psi at a maximum of 6 gpm.

i. Bleed air from inside assembly in accordance with steps i. through t.

CAUTION

To prevent damage to test equipment, do not apply electrical power until hydraulic pressure is applied.

j. Place test box POWER DC and AC switches to ON.

RESULT: POWER DC and AC ON indicators shall light and SAS PRESS ON light shall go on.

- If result is not as specified, make sure test box AC and DC circuit breakers are pushed in.

k. Place test box SAS switch to ON and then to OFF.

RESULT: SAS PRESS ON and OFF lights shall cycle from on to off.

- If result is not as specified, replace test box.

l. Repeat step j. several times and leave switch at ON (SAS PRESS ON light on).

m. Turn TRIM switch ON.

n. Vary CURRENT CONTROL until VALVE CURRENT meter indicates 0.

o. Vary CURRENT CONTROL from 0 to 5.0, back to 0, then to -5.0, and back to 0 several times to stroke trim piston. Return CURRENT CONTROL to 0.

CAUTION

When using the 70700-20682-042 cable assembly, make sure that the connector marked SAS ACTUATOR is correctly connected. If the incorrect connector is used, servovalve may become damaged.

p. Disconnect cable assembly, 70700-20682-041, from test box and trim servovalve. Connect cable assembly, 70700-20682-042 to test set and SAS servovalve.

q. Cycle SAS switch several times ON and OFF. After cycling switch leave in ON position.

r. Vary CURRENT CONTROL from 0 to 5.0, back to 0, then to -5.0, and back to 0 several times, to stroke SAS position.

CAUTION

If bleed screw is not backed out properly, packing on bleed screw will be damaged.

s. Decrease hydraulic pressure to 500 psi. Back out bleed screw until bleed port is visible and a good flow is established.

t. Allow fluid to flow from bleed port until fluid is free of air. Decrease pressure to 0 psi. Close bleed port. **TORQUE TO 19 - 21 INCH-POUNDS** and lockwire.

u. Increase hydraulic pressure to 3000 psi and watch SAS assembly output piston for any movement.

RESULT: Piston shall not move.

- If piston moves there is still air trapped in servo. Repeat steps h. through q.

v. Install dial indicator against SAS actuator link.

w. Adjust indicator to read zero.

x. Cycle SAS switch several times from ON to OFF. Watch actuator piston for any jump or movement.

RESULT: Actuator piston jump shall be less than 0.005 inches to be nulled.

- If result is not as specified, do the following:

  (a) Remove cover from inside of jamnut using magnet.

  (b) Loosen jamnut 1/4 turn.

  (c) Insert screwdriver through jamnut and turn eccentric screw as needed to reduce piston jump below 0.005 inch. Cycle SAS switch and check for jump.

  (d) If piston jump is out of limits, replace servovalve (PARA 11-5-5) and repeat operational/troubleshooting procedure.

  (e) If piston jump is below 0.005 inch, **TORQUE JAMNUT 47 - 53 INCH-POUNDS**. Install cover on inside of jamnut and lockwire.

y. Disconnect hydraulic lines from nulling fixture.

z. Repeat piston jump check after torquing jamnut. Readjust eccentric screw if necessary.

NOTE

This verifies if SAS output rod moves properly when a specific current is applied to servo valve.

aa. On test box, vary CURRENT CONTROL and do VALVE CURRENT check as shown in Figure 11-2-6-2.

RESULT: Valve movement shall correspond to chart on Figure 11-2-6-2.

- If SAS output rod movement does not correspond to milliamp input from test box and if nulling check of servo is acceptable, replace SAS module (PARA 11-5-6) and/or electrohydraulic servo valve (PARA 11-5-5).

ab. Place test box POWER DC and AC switches to OFF.

ac. Turn off hydraulic pressure and electrical power.

ad. Remove dial indicator.

ae. Disconnect hydraulic lines from nulling fixture.

af. Remove test box cable from power source.

ag. Remove pitch trim assembly from nulling fixture by removing three bolts.

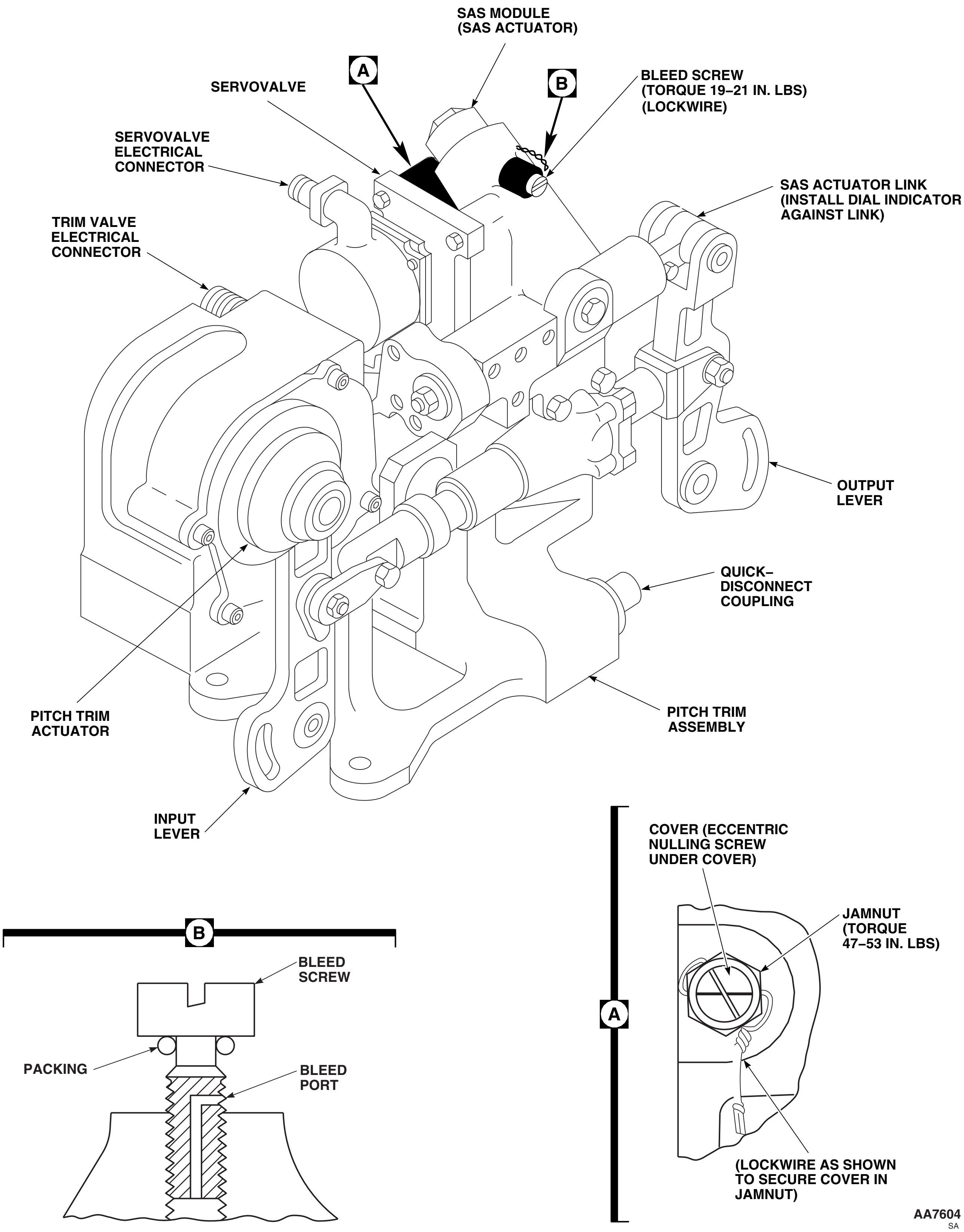
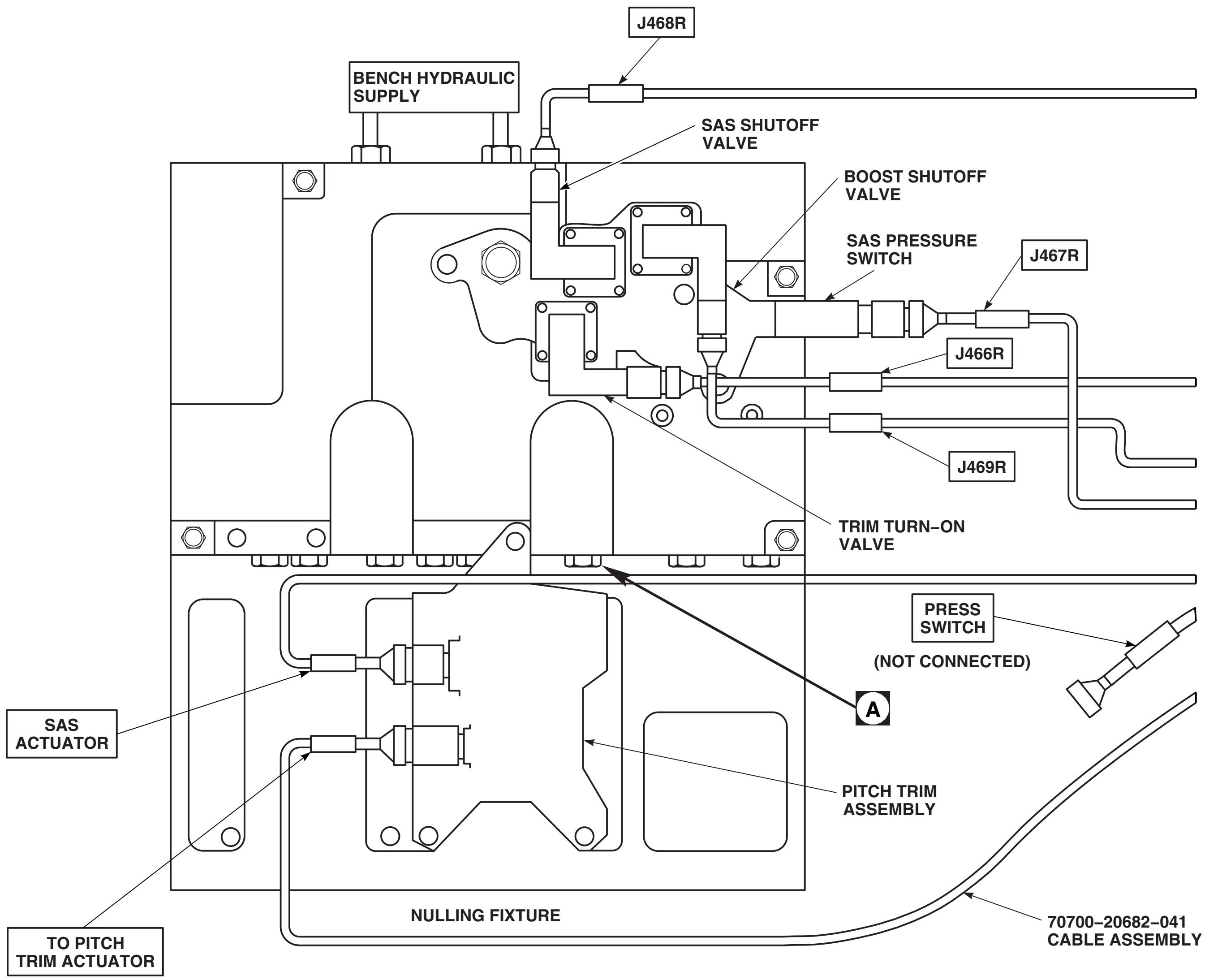


FIGURE 11-2-6-1. PITCH TRIM ASSEMBLY



VALVE CURRENT CHART

TRIM COIL SELECT SWITCH IN BOTH POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–3.70	0.090 TO 0.110 (EXTENDED FROM NULL)
–6.60	0.155 TO 0.200 (EXTENDED FROM NULL)
+3.70	0.090 TO 0.110 (RETRACTED FROM NULL)
+6.60	0.155 TO 0.200 (RETRACTED FROM NULL)

TRIM COIL SELECT SWITCH IN 1 OR 2 POSITION

SIGNAL MILLI–AMPS (MA)	DISPLACEMENT MEASURED FROM NULL (INCHES)
0	0 (NULLED)
–1.85	0.045 TO 0.055 (EXTENDED FROM NULL)
–3.30	0.0775 TO 0.100 (EXTENDED FROM NULL)
+1.85	0.045 TO 0.055 (RETRACTED FROM NULL)
+3.30	0.0775 TO 0.100 (RETRACTED FROM NULL)

NOTE

CONNECT CONNECTOR AS SPECIFIED IN OPERATIONAL CHECK.

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FIGURE 11-2-6-2. PITCH TRIM ASSEMBLY TEST SETUP DIAGRAM (SHEET 1 OF 2)

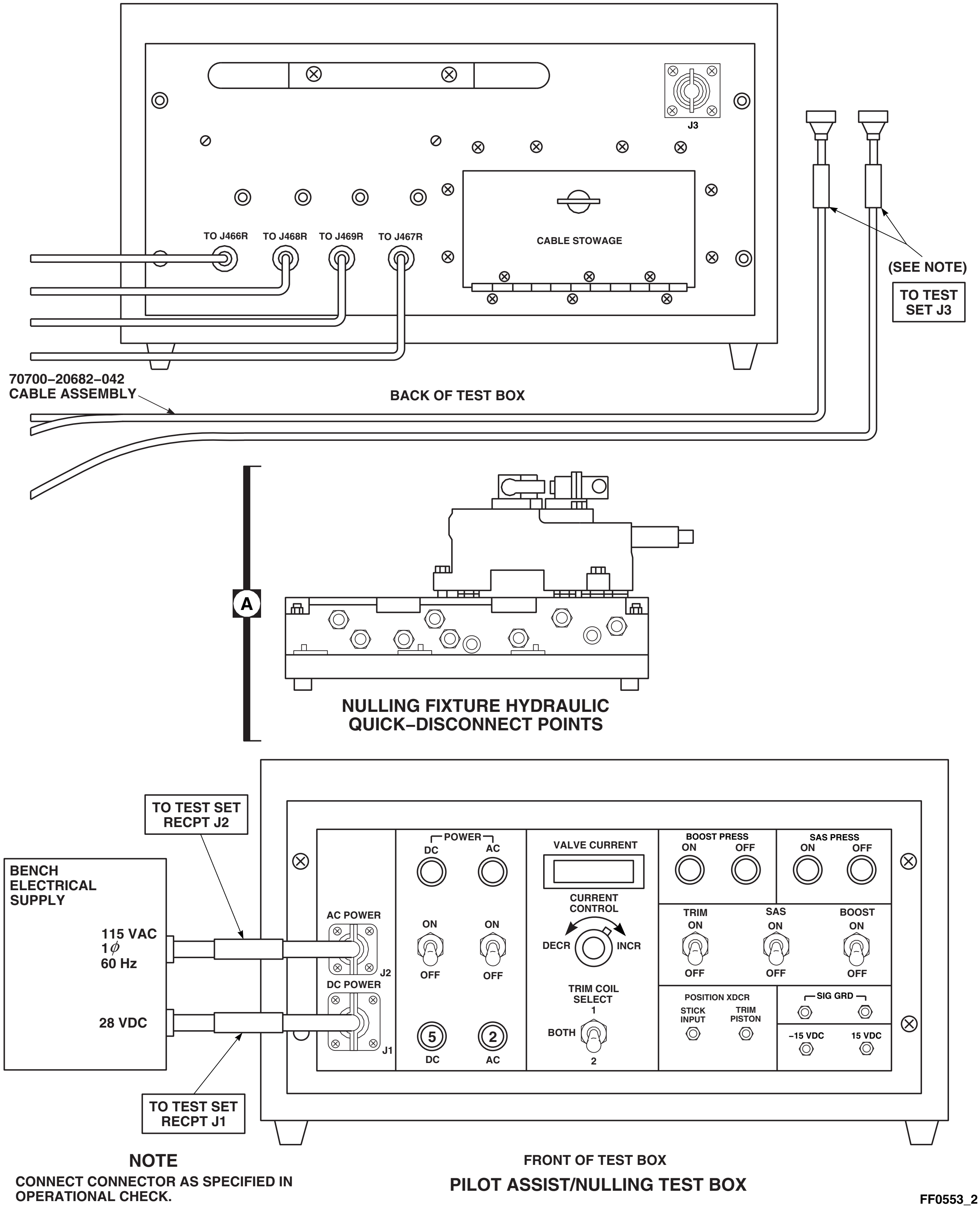


FIGURE 11-2-6-2. PITCH TRIM ASSEMBLY TEST SETUP DIAGRAM (SHEET 2 OF 2)

11-2-7 PILOT-ASSIST MODULE (BENCH).

INITIAL SETUP

Test Equipment

- Nulling Fixture Assembly, SAS Actuator, 70700-20675-041 (Figure C-37, Appendix C)

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, GGG-W-00686

Equipment Conditions

- Hydraulic Power Source (3000 psi) Available
- 28 vdc Electrical Power Available

CAUTION

To prevent leakage between the nulling fixture and pilot-assist module, make sure manifold and servo self-sealing couplings are clean.

NOTE

If a three-way valve or pressure reducer has to be replaced, repeat operational/ troubleshooting procedure.

11-2-7.1 Operational/Troubleshooting Procedure.

a. Remove existing pilot-assist module from nulling fixture by removing three bolts and electrical connectors. Lift module from nulling fixture.

b. Install pilot-assist module to be tested on nulling fixture, making sure quick-disconnects are lined up. Secure module with three bolts. **TORQUE BOLTS TO 180 - 200 INCH-POUNDS.**

c. Install a test block on each servo position on nulling fixture.

d. Install a pressure gauge in TRIM and BOOST pressure ports of test fixture.

e. Place all test box pilot-assist/nulling switches on to OFF and place TRIM COIL SELECT switch to BOTH.

f. Connect test box cables to pilot-assist module assembly on nulling fixture as follows:

CABLE CONNECTOR	CONNECT TO
J466R	Pilot-assist module (Trim turn-on valve)
J467R	Pilot-assist module (SAS pressure switch)
J468R	Pilot-assist module (SAS shutoff valve)
J469R	Pilot-assist module (Boost shutoff valve)

g. Connect test box power cables to electrical power source.

h. Connect hydraulic lines from hydraulic power source to PRESSURE and RETURN ports on nulling fixture.

i. Turn on hydraulic power source and set pressure to 3000 psi at 6 gpm.

CAUTION

To prevent damage to test equipment, do not apply electrical power until hydraulic pressure is applied.

j. Place test box POWER DC and AC switches to ON.

RESULT:

(1) Pressure gauge connected to TRIM pressure port shall indicate 0 psi, and pressure gauge connected to BOOST pressure port shall indicate 0 psi.

(2) POWER DC and AC on indicators shall light.

(3) SAS PRESS OFF light shall be on.

- If result (1) is not as specified, replace boost three-way valve (PARA 11-5-8).
- If results (2) and (3) are not as specified, make sure that test box AC and DC circuit breakers are pushed in.
- If result (3) is not as specified, check SAS pressure switch and replace if necessary (PARA 7-4-35).

k. Place test box TRIM switch to ON.

RESULT: Pressure gauge connected to TRIM pressure port shall indicate 950 to 1050 psi.

- If gauge remains at 0 psi, replace trim three-way valve (PARA 11-5-8).
- If gauge indicates more than 1050 or less than 950 psi, replace pressure reducer (PARA 11-5-9).

l. Place test box TRIM switch to OFF.

RESULT:

(1) TRIM pressure gauge shall indicate 0 psi.

(2) Button on pressure reducer shall not pop.

- If result (1) is not as specified, replace trim three-way valve (PARA 11-5-8).
- If result (2) is not as specified and will not remain reset, replace pressure reducer (PARA 11-5-9).

m. Place test box SAS switch to ON.

RESULT: SAS PRESS ON light shall be on.

- If SAS PRESS OFF light stays lit, check SAS pressure switch and replace if necessary (PARA 7-4-35).
- If trouble remains, replace SAS three-way valve (PARA 11-5-8).

n. Place test box SAS switch to OFF.

RESULT: SAS PRESS ON light shall go off.

- If SAS PRESS OFF light is not on, check SAS pressure switch and replace if necessary (PARA 7-4-35).
- If trouble remains, replace SAS three-way valve (PARA 11-5-8).

o. Place test box BOOST switch to ON.

RESULT: Pressure gauge connected to BOOST pressure port shall indicate 3000 psi.

- If result is not as specified, replace boost three-way valve (PARA 11-5-8).

p. Place test box BOOST switch to OFF.

RESULT: BOOST pressure gauge shall indicate 0 psi.

- If result is not as specified, replace boost three-way valve (PARA 11-5-8).

q. Place test box POWER DC and AC switches to OFF.

r. Turn off hydraulic pressure and electrical power.

s. Disconnect hydraulic lines from nulling fixture.

t. Remove test box cable from power source.

u. Disconnect electrical cables from pilot-assist module.

v. Remove pilot-assist module from nulling fixture by removing three bolts and lifting module.

w. Install previously attached pilot-assist module onto nulling fixture. Install three bolts. Tighten bolts.

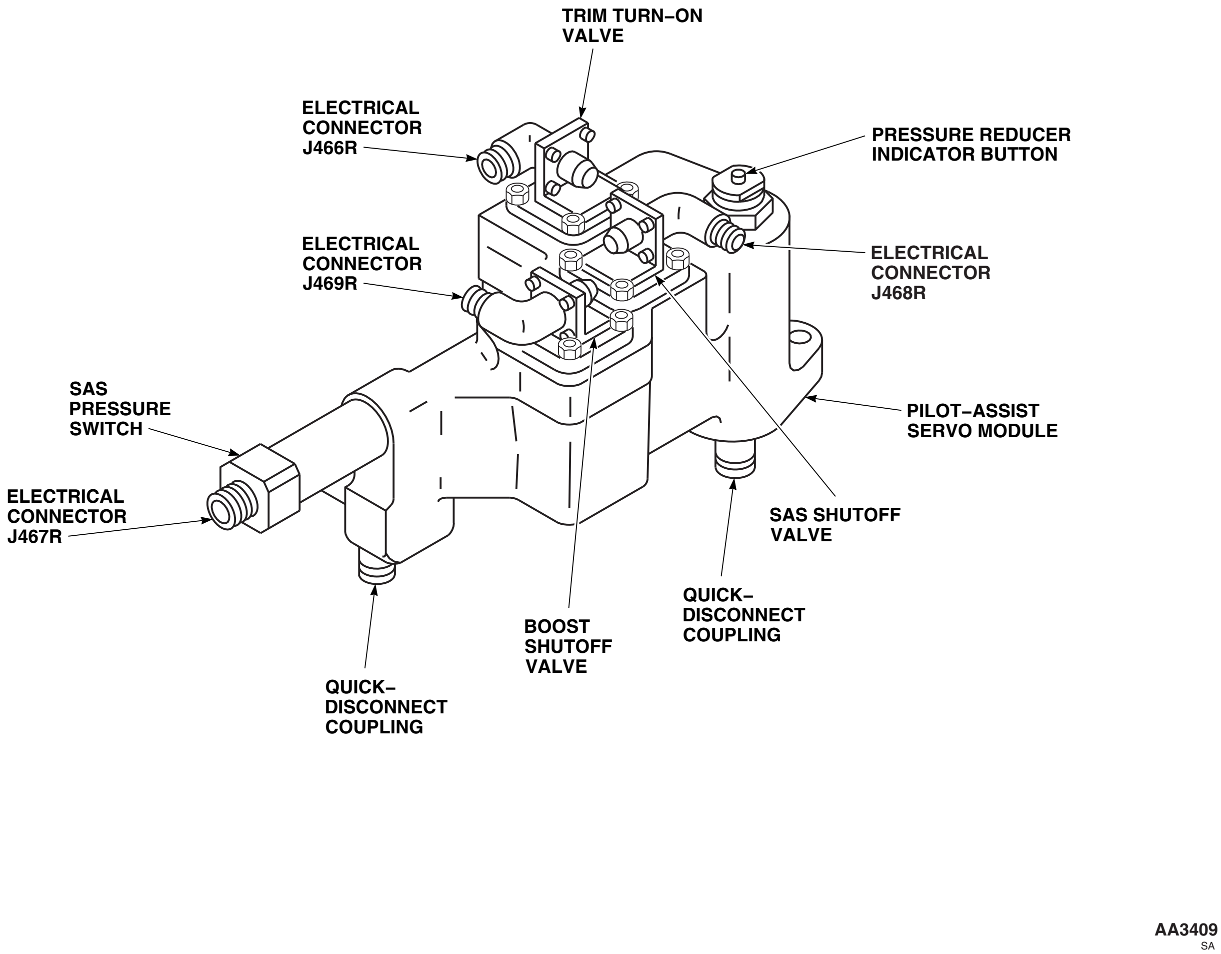


FIGURE 11-2-7-1. PILOT-ASSIST MODULE

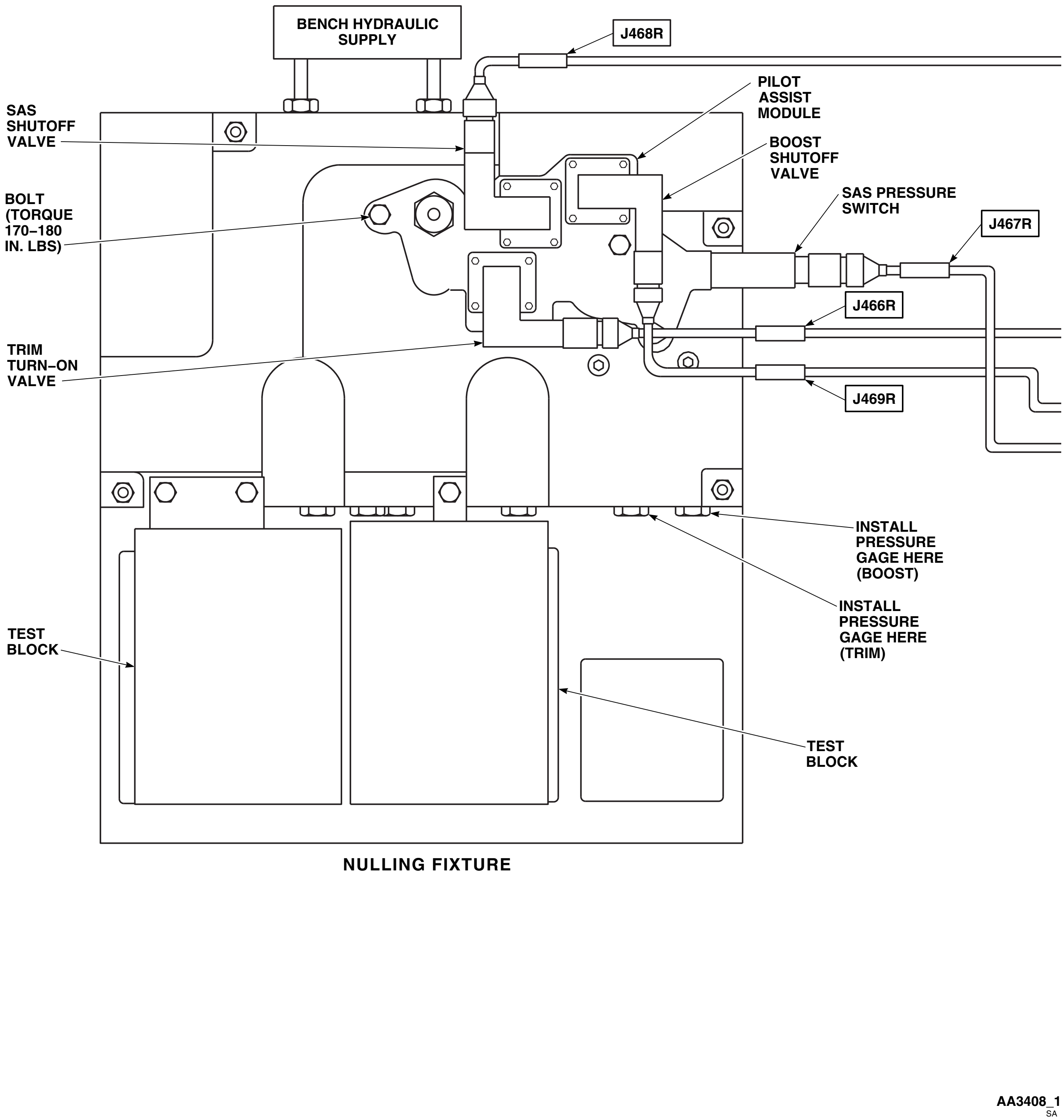


FIGURE 11-2-7-2. PILOT-ASSIST MODULE TEST SETUP (SHEET 1 OF 2)

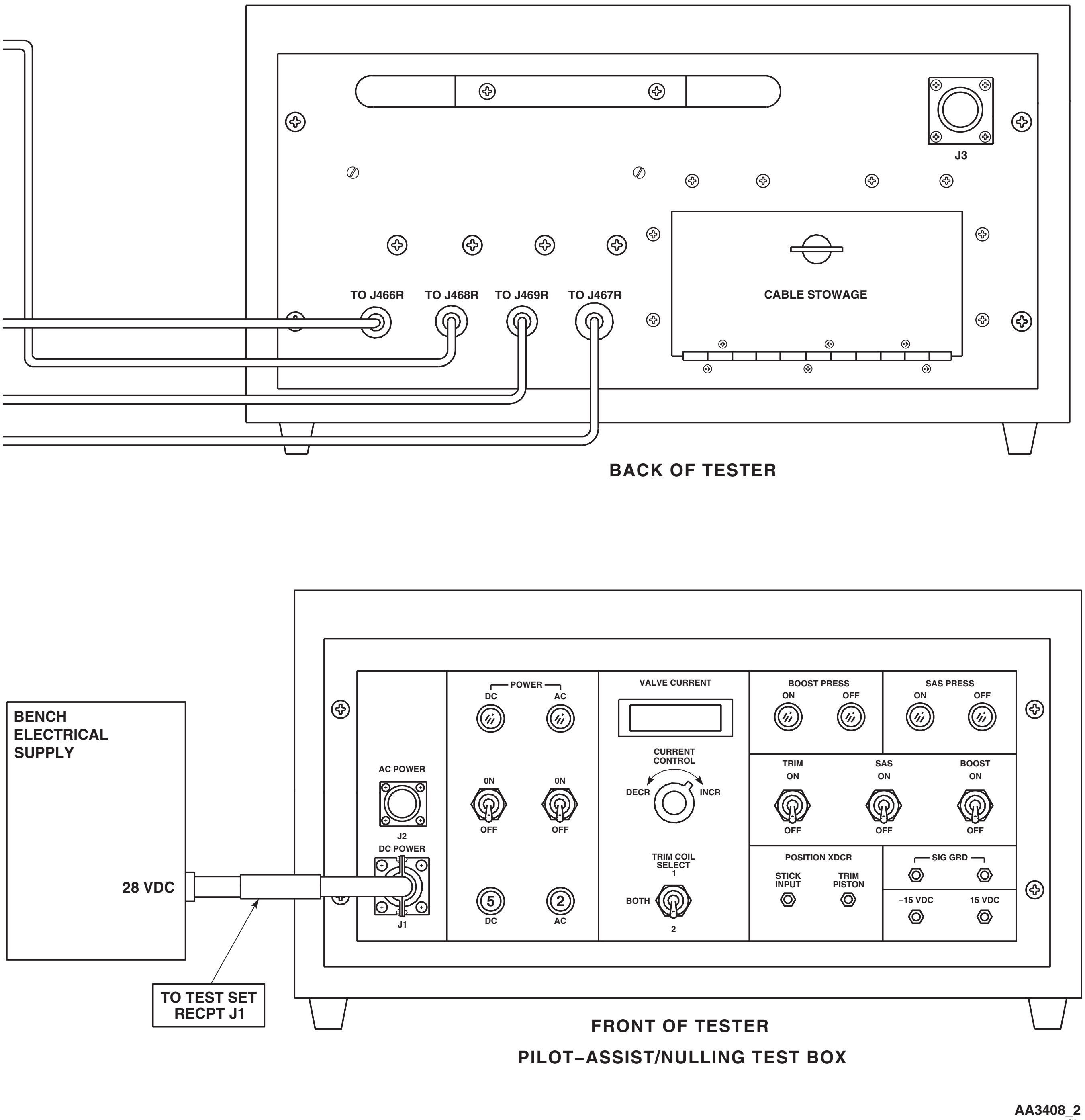


FIGURE 11-2-7-2. PILOT-ASSIST MODULE TEST SETUP (SHEET 2 OF 2)

SECTION 11-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

11-3-1 SELF-RETAINING BOLT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107

11-3-1.1 Inspect Self-Retaining Bolts.

WARNING

FSCAP

Outside diameter and bolthead shoulder of self-retaining bolts are critical surfaces which must not be damaged.

NOTE

Two types of self-retaining bolts are used in the flight control system. Impedance-type self-retaining bolts may have a retaining ring or spring-loaded retaining balls. Impedance-type bolts do not have a plunger in head. Locking-type self-retaining bolts have retaining balls which are locked in place until a plunger in bolthead is depressed.

a. For impedance-type bolts, make sure retaining ring or retaining balls are intact. When installing bolt, retaining ring or retaining balls must provide resistance when going into bolthole.

b. For locking-type bolts, make sure retaining balls cannot be pushed in without first depressing button in center of bolthead.

c. Self-retaining bolts may be reused provided bolt shows no sign of corrosion, fretting, nicks, or damaged threads, and retaining mechanism is not damaged.

11-3-1.2 Special Self-Retaining Bolt FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in critical area is limited to a one-time repair and is to be done only when bolt replacement is not possible.
- All rework of critical area which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

c. Check critical area for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED.** If damage is found, repair bolt as follows (Figure 11-3-1-1, Detail A):

(1) Check log card (SA Form 7343-21) to see if bolt critical area has been previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREA ALLOWED.** If bolt critical area has been previously reworked into base metal, and damage penetrates into base metal, replace bolt.

NOTE

Minimum bolthead to shank radius is 0.015-inch.

(2) Damage in critical area may be blended up to 0.003-inch deep provided that critical characteristic minimum shank diameter and bolthead to shank radius are not exceeded. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE BOLT** (Table 11-3-1-1). Record any critical area rework into base metal on component log card (SA Form 7343-21).

(3) Using fluorescent penetrant inspection kit, inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bolt.

(4) Apply solid film lubricant, Item 201, Appendix D, to all reworked areas of bolt.

d. Clean bolt using cleaning cloth, Item 102, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Inspect bolt as follows:

(1) Inspect bolt for evidence of corrosion in the area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt.

(2) Visually inspect bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack(s) is visible, replace bolt.

CAUTION

If resistance or binding of the retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free the mechanism. The inner core of the bolt is corroded and must be rejected.

e. Release retaining balls by holding retaining pin in. If resistance or binding of the retaining ball mechanism is observed when pin is actuated, discard bolt. If no resistance or binding of the retaining ball mechanism is observed when pin is actuated, apply corrosion preventive compound, Item 114, Appendix D, to bolt as follows:

(1) Work corrosion preventive compound through one retaining ball hole until

excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.

(2) While working retaining pin, spray corrosion preventive compound, Item 114, Appendix D, into hole in bolthead and hole in end of bolt.

(3) Wipe excessive corrosion preventive compound from bolt using cleaning cloth, Item 102, Appendix D, prior to installation.

(4) Apply corrosion preventive compound, Item 109, Appendix D, to the exterior retaining ball area and threads.

Table 11-3-1-1. Special Self-Retaining Bolt Critical Dimensions

Part Number	Critical Minimum Diameter
70400-08159-101 through 70400-08159-106	0.745-inch
70400-08159-107 and 70400-08159-108	0.870-inch

f. Install main rotor pylon transmission cover (PARA 2-4-107).

g. Close main rotor pylon sliding cover.

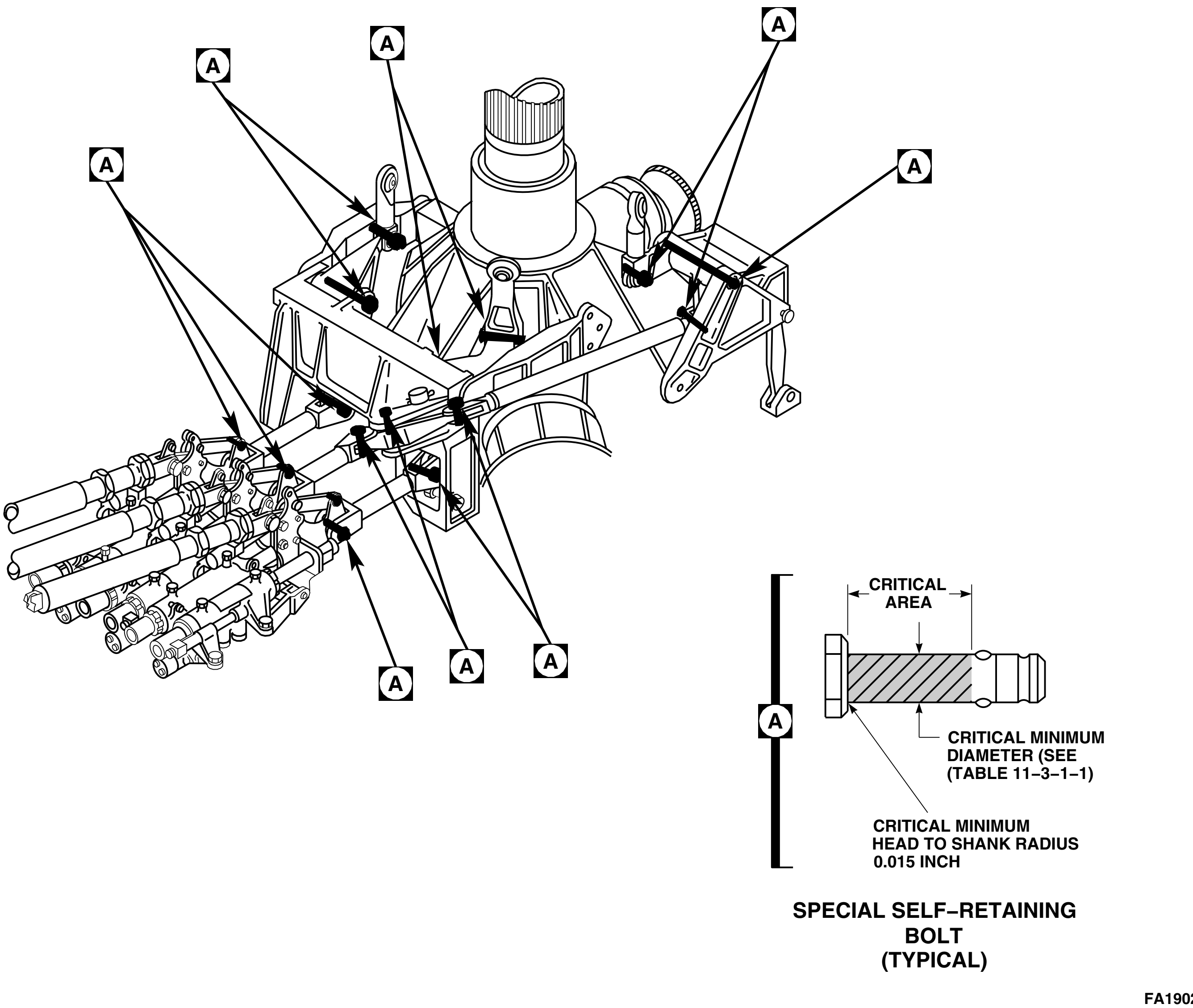


FIGURE 11-3-1-1. INSPECT SPECIAL SELF-RETAINING BOLTS

11-3-2 BEARING FRICTION WEAR INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Spring Scale

References

- PARA 1-6-16
- PARA 4-4-27
- PARA 11-4-63
- PARA 11-4-64
- PARA 11-4-65
- PARA 11-4-66
- PARA 11-4-118

11-3-2.1 Inspect Bearing Friction Wear.

a. Disconnect both engine load-demand push/pull cables from mixer (PARA 4-4-27).

b. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

c. Apply external electrical power (PARA 1-6-16).

d. On upper console, place BACKUP HYD PUMP switch to ON.

e. On STABILATOR CONTROL/AUTO FLT CONT panel, engage BOOST switch.

f. Place spring scale on collective grip. Measure and record force necessary to just start collective stick moving.

g. Place collective stick in mid-to-high position.

h. Place cyclic stick in midposition.

i. With spring scale on cyclic grip, measure and record force necessary to just start cyclic moving left, right, front, and back.

j. Place spring scale on left, then right pedal. Measure and record force necessary to just start pedal moving.

k. Compare recorded forces with Table 11-3-2-1.

Table 11-3-2-1. Maximum Control Forces

Channel	Position	Maximum Force (Ounces)
Lateral	Push right	15.0
	Push left	15.0
Longitudinal	Push forward	15.0
	Push back	15.0
Directional	Left and right	64.0
Collective	Up and down	48.0

l. If any channel measures more than Table 11-3-2-1 force values, perform the following:

(1) On upper console, place BACKUP HYD PUMP switch to OFF.

(2) Leave both trim actuator rods connected. On STABILATOR CONTROL/AUTO FLT CONT panel, push TRIM button OFF.

(3) Disconnect collective input control rod from collective torque shaft lever (PARA 11-4-63).

(4) Disconnect pitch/trim input control rod from pitch torque shaft lever (PARA 11-4-64).

(5) Disconnect lateral input control rod from lateral torque shaft lever (PARA 11-4-65).

(6) Disconnect yaw boost servo input control rod from yaw torque shaft lever (PARA 11-4-66).

(7) Place spring scale, on collective stick grip, cyclic stick grip, and yaw pedals.

(8) Measure and record force necessary to just move each control, as done with hydraulic power on.

(9) Compare recorded forces with Table 11-3-2-2.

Table 11-3-2-2. Maximum Control Forces

Channel	Position	Maximum Forces In Ounces
Lateral	Push right	7.0
	Push left	7.0
Longitudinal	Push forward	7.0
	Push back	7.0
Directional	Left and right	48.0
Collective	Up and down	40.0

m. If maximum forces of Table 11-3-2-2 are exceeded, check each bearing in system from torque shafts to cockpit controls for cracks, displacement, seal damage, or binding. If bearing is cracked, loose, displaced, or binding, replace part having a damaged bearing.

n. If more force than limits shown in Table 11-3-2-2 is needed, but not more than limits shown in Table 11-3-2-1, check each bearing in system from pilot-assist servos to main rotor spindles and tail rotor servo quadrant. If bearing is binding, cracked, loose, or displaced, replace part having a damaged bearing.

o. If disconnected, connect the following control rods:

(1) Collective input control rod to collective torque shaft lever (PARA 11-4-63).

(2) Pitch/trim input control rod to pitch torque shaft lever (PARA 11-4-64).

(3) Lateral input control rod to lateral torque shaft lever (PARA 11-4-65).

(4) Yaw boost servo input control rod to yaw torque shaft lever (PARA 11-4-66).

p. Connect engine load-demand push/pull cables to mixer (PARA 4-4-27).

q. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

CAUTION

Damage to control rods, servos, etc., will result if excessive force is used to move controls. Use no force greater than light hand pressure to move controls with hydraulic power off.

r. Check system free play at each control (cyclic, collective, and yaw pedals) with hydraulic pressure off.

s. If there is more than 1/8-inch total free play, check each bearing in system for play. Replace bearing(s) that have most play and then recheck system for 1/8-inch free play.

11-3-3 CONTROL ROD CONNECTION INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

11-3-3.1 Inspect Control Rod Connection.

a. Open main rotor pylon sliding cover.

NOTE

Refer to Table 11-3-3-1 for proper washer stackup.

b. Check for free rotation of long bushing using finger force, and for a clearance of 0.002 - 0.125-inch between washer and control rod clevis (Figure 11-3-3-1, Detail A). Proceed as follows:

(1) **IF BUSHING CANNOT BE ROTATED AND CLEARANCE IS LESS THAN 0.002-INCH, REMOVE BOLTS, WASHERS, AND NUTS, AND REPLACE BUSHING WITH A LONGER BUSHING.**

(2) **IF BUSHING CAN BE ROTATED AND CLEARANCE IS MORE THAN 0.125-INCH, REMOVE BOLTS, WASHERS, AND NUTS, AND REPLACE BUSHING WITH A SHORTER BUSHING.**

c. Check for no rotation of short bushings and for a clearance of 0.002 - 0.125-inch between washers and control rod clevis on each side. Proceed as follows:

(1) **IF BUSHINGS CAN BE ROTATED AND CLEARANCE IS LESS THAN 0.002-INCH ON EACH SIDE, REMOVE BOLTS, WASHERS, AND NUTS, AND REPLACE BUSHINGS WITH LONGER BUSHINGS.**

(2) **IF BUSHINGS CAN BE ROTATED AND CLEARANCE IS 0.002 - 0.125-INCH, RETORQUE NUTS; THEN RECHECK FOR NO ROTATION OF BUSHINGS AND FOR A CLEARANCE OF 0.002 - 0.125-INCH.**

(3) **IF BUSHINGS CANNOT BE ROTATED AND CLEARANCE IS MORE THAN 0.125-INCH, REMOVE BOLTS, WASHERS, AND NUTS, AND REPLACE BUSHINGS WITH SHORTER BUSHINGS.**

d. The following bushings may be used to achieve proper stack-up and spacing:

Part Number	Length (Inch)
70400-00513-101	0.330 - 0.325
70400-00513-102	0.250 - 0.245
70400-00513-104	0.406 - 0.401
70400-00513-105	0.206 - 0.201
70400-00513-106	0.378 - 0.371
70400-00513-107	0.469 - 0.464

e. Install bolts, washers, and nuts. Torque nuts to value shown in maintenance task, and install cotter pin.

f. Where a rubbing or wearing condition exists between short bushings and control rod clevis slot, **A MAXIMUM OF 0.015-INCH MATERIAL REMOVAL IS ALLOWED.**

g. Where control rod clevis holes are elongated by long bushings contact, **A MAXIMUM ELONGATION OF 0.015-INCH IS ALLOWED.**

h. Close and latch main rotor pylon sliding cover.

Table 11-3-3-1. Washer Stackups (Tri-Pivot Bolt)

Combination	Washer Thickness (Inch)	Quantity
A	0.016	2
B	0.016	3
C	0.063	1
D	0.016 0.063	1 1
E	0.016 0.063	2 1

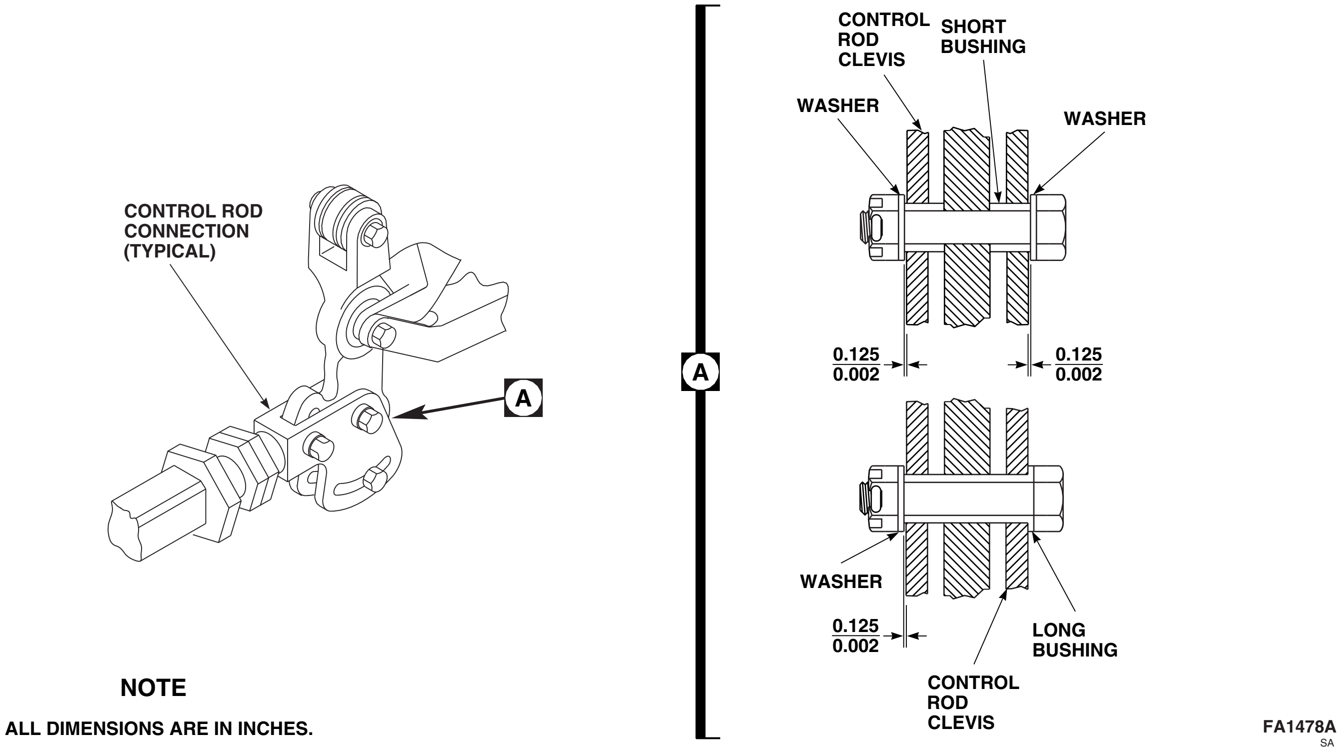


FIGURE 11-3-3-1. INSPECT CONTROL ROD CONNECTION

11-3-4 CYCLIC STICK INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator

References

- PARA 11-4-15
- PARA 11-4-23

11-3-4.1 Inspect Cyclic Stick Bearings.

a. Remove cyclic stick boot (PARA 11-4-15).

b. Inspect bearings using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARINGS IS 0.005-INCH.** If beyond limits, replace bearing(s) (PARA 11-4-23).

c. Install cyclic stick boot (PARA 11-4-15).

11-3-4.2 Inspect Cyclic Stick Boot.

Inspect boot for rips, tears, deterioration, and obvious damage. **NONE ALLOWED.** If damage is found, replace boot (PARA 11-4-15).

11-3-5 COLLECTIVE STICK INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Electrical Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Spiral Wrap, Item 314, Appendix D
- Tape, Item 317B, Appendix D

References

- Appendix D
- PARA 11-2-2
- PARA 11-4-26
- PARA 11-4-37
- PARA 11-4-41

11-3-5.1 Inspect Copilot's Collective Stick Grip Coil Cord.

a. Remove bolt, washers, special washers, and nut that connects socket assembly to stick at end of collective stick index pin.

b. Carefully pull stick upper tube assembly about 3 inches away from bottom half of stick. Do not attempt to extend stick upper tube assembly beyond point where plug is snug against grommet.

c. Inspect coiled cable exposed between upper tube assembly and bottom half of stick for worn, chaffed, torn, or damaged coiled cable.

d. If coiled cable sleeving shows scuff marks but no individual wires are in view, reassemble two halves of stick.

e. If coiled cable sleeving is worn so individual wires are in view but insulation on wire does not have scuff marks, repair coiled cable as follows:

(1) Add spiral wrap, Item 314, Appendix D, to damaged area. Make sure tubing covers at least ¾-inch but not more than 1 ¼-inches beyond damaged area at each end.

(2) Secure tubing with tape, Item 317B, Appendix D.

f. If coiled cable sleeving is worn and any individual wire insulation is scuffed or bare wire is exposed, replace coil cord (PARA 11-4-37).

g. Carefully reassemble stick upper tube assembly to bottom half of stick.

h. Extend plug and cable from stick until cable is extended about 3 inches. Do not apply enough force to separate wire from connection.

i. Mate stick to socket assembly.

NOTE

Some copilot's collective stick and socket assemblies may have been assembled with self-locking nuts. When reassembling stick and socket, use bolt, AN175-22, nut, MS17825-5, and cotter pin.

j. Place special washers on each side of tube and install bolt, washers and nut. **TORQUE NUT TO 20 - 35 INCH-POUNDS.** Install cotter pin.

k. Perform operational check of collective stick assembly (PARA 11-2-2).

11-3-5.2 Inspect Collective Stick Bearings.

a. Remove collective stick boot and cover (PARA 11-4-26).

b. Inspect bearings, MKSP4A-A1173, MKSP4A-400, MKSP4A-5A2922, and MKP5A-A1173, in pilot's and copilot's collective stick socket, 70400-01412-041, -042, using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARINGS, MKSP4A-A1173, MKSP4A-400, AND MKSP4A-5A2922 IS 0.0025-INCH. MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARING, MKP5A-A1173, IS 0.005-INCH.** If beyond limits, replace bearing(s) (PARA 11-4-41).

c. Install collective stick boot and cover (PARA 11-4-26).

11-3-5.3 Inspect Collective Stick Boot.

Inspect boot for rips, tears, deterioration, and obvious damage. **NONE ALLOWED.** If damage is found, replace boot (PARA 11-4-26).

11-3-6 YAW BOOST SERVO INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage

References

- PARA 11-4-70

11-3-6.1 Inspect Yaw Boost Servo.

a. Open main rotor pylon sliding cover.

b. Visually inspect both input and output power piston shafts for cracks or evidence of separation of shaft from clevis end. **NONE ALLOWED.** If cracks or evidence of separation is found, replace yaw boost servo (PARA 11-4-70).

c. Visually inspect cylinder for evidence of cracks. **NONE ALLOWED.** If cracks are found, replace yaw boost servo (PARA 11-4-70).

d. Check all servo pin-rivet connections as follows (Figure 11-3-6-1, Detail A):

(1) Check pin-rivets securing redundant link to input arm and output lever for rotation in link assembly. **NO ROTATION OR AXIAL MOTION OF PIN-RIVET ASSEMBLIES ALLOWED.** If rotation or axial motion is present, replace yaw boost servo (PARA 11-4-70).

(2) Check pin-rivet in upper connection of input arm for rotation. **PIN-RIVET IN UPPER CONNECTION OF INPUT ARM MAY OR MAY NOT ROTATE IN ARM ASSEMBLY.**

(3) Check collars for rotation or axial movement. **COLLARS SHALL NOT ROTATE OR MOVE AXIALLY WITH RESPECT TO PIN-RIVETS ON ANY SERVO CONNECTIONS.** If rotation or axial motion is present, replace yaw boost servo (PARA 11-4-70).

(4) Using feeler gage, measure axial play of upper bushings and pin-rivet connections. **COMBINED READINGS BETWEEN PINNED PARTS MUST NOT EXCEED 0.040-INCH ON ANY CONNECTION.** If limit is exceeded, replace yaw boost servo (PARA 11-4-70).

e. Close main rotor pylon sliding cover.

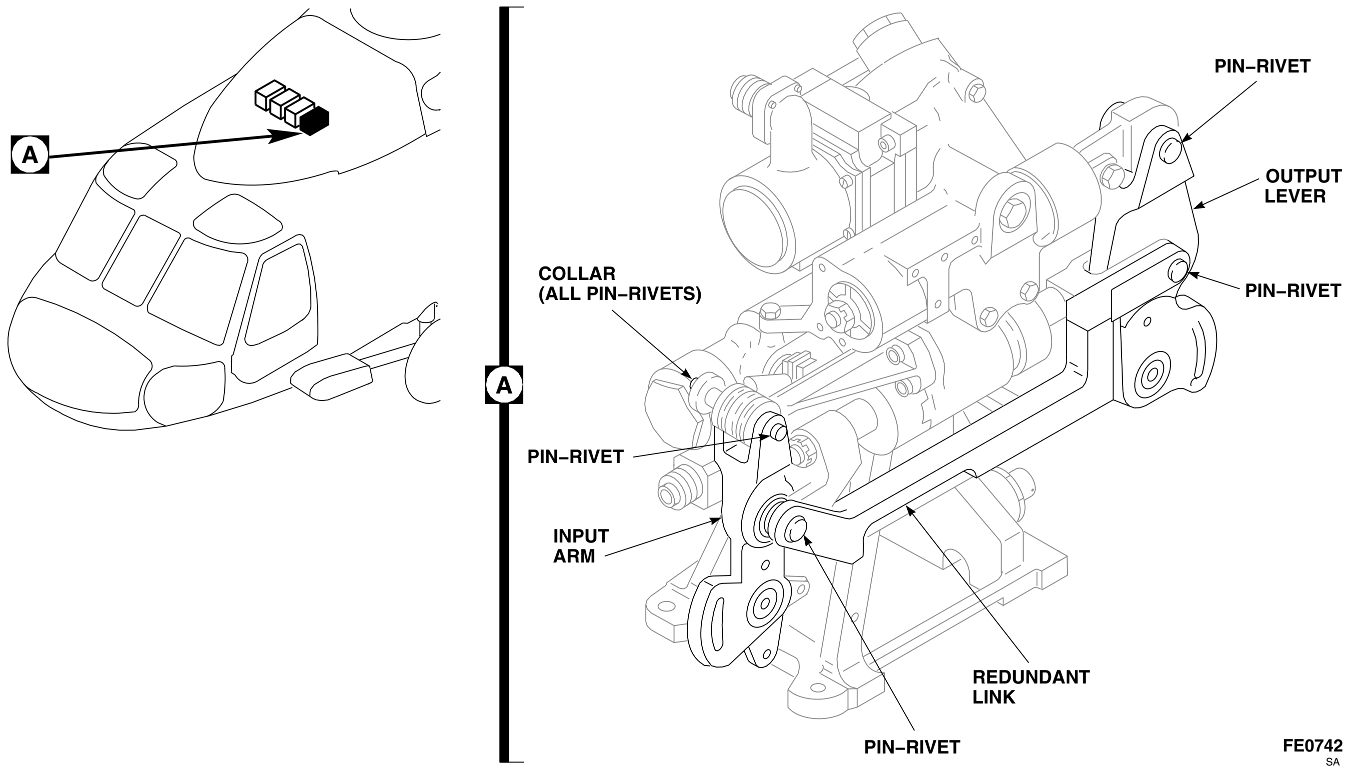


FIGURE 11-3-6-1. INSPECT YAW BOOST SERVO

11-3-7 COLLECTIVE BOOST ASSEMBLY INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage

References

- PARA 11-4-67

11-3-7.1 Inspect Collective Boost Assembly.

a. Open main rotor pylon sliding cover.

b. Visually inspect both input and output power piston shafts for cracks or evidence of separation of shaft from clevis end. **NONE ALLOWED.** If cracks or evidence of separation is found, replace collective boost assembly (PARA 11-4-67).

c. Visually inspect cylinder for evidence of cracks. **NONE ALLOWED.** If cracks are found, replace collective boost assembly (PARA 11-4-67).

d. Check all boost assembly pin-rivet connections (Figure 11-3-7-1, Detail A) as follows:

(1) Pin-rivets attaching redundant link to input arm and output lever shall not rotate in link assembly when finger pressure is applied. **NO AXIAL MOTION OF PIN-RIVET ASSEMBLIES ALLOWED.** If not within limits, replace collective boost assembly (PARA 11-4-67).

(2) Check pin-rivet in upper connection of input arm as follows:

NOTE

Pin-rivet in upper connection of input arm may or may not rotate in arm assembly.

(a) Using feeler gage, measure axial play of upper bushing and pin-rivet connection at upper connection of input arm. **NO AXIAL MOTION OF PIN-RIVET ASSEMBLY ALLOWED. NO REPAIR ALLOWED.**

(b) If combined feeler gage readings between pinned parts exceeds 0.040-inch on connection point, replace collective boost assembly (PARA 11-4-67).

(3) Pin-rivets in output link connections with output lever and grounding support may or may not rotate in output lever and grounding support.

(4) Bushings and collars shall not rotate or move axially with respect to pin-rivets on any of the boost assembly connections. If not within limits, replace collective boost assembly (PARA 11-4-67).

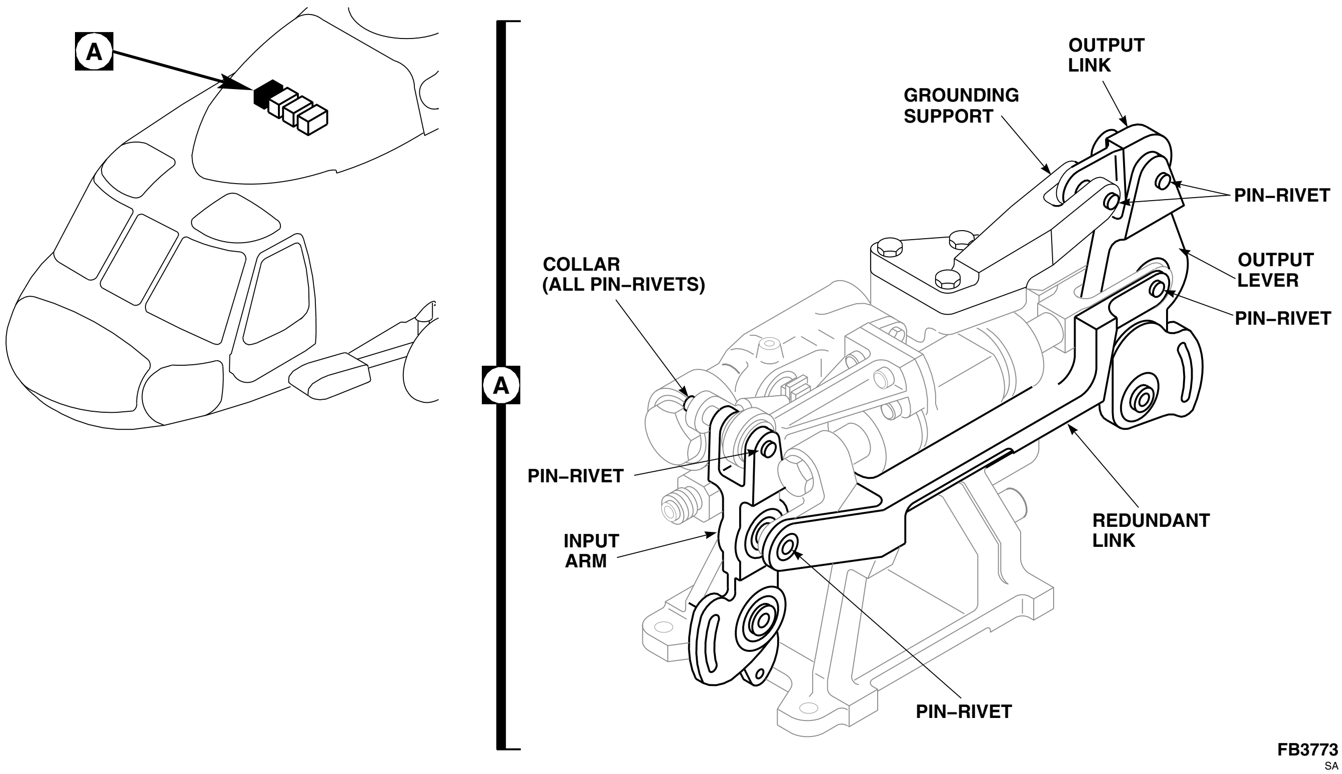


FIGURE 11-3-7-1. INSPECT COLLECTIVE BOOST ASSEMBLY

e. Close and latch main rotor pylon sliding cover.

11-3-8 MIXER INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, Dial Indicator, 0 - 30 in. lbs
- Torque Wrench, Dial Indicator, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-69
- PARA 2-6-5

11-3-8.1 Mixer Mounting Bolt Torque Check.

a. Open main rotor pylon sliding cover.

b. Using nonmetallic scraper, remove sealing compound from mixer mounting bolts.

c. If main rotor blades are installed, turn main rotor head to make torque check of mixer easier.

WARNING

FSCAP

Verification of minimum torque of bolts is a critical characteristic.

d. Using dial indicating torque wrench, apply torque to bolt in tightening direction and note torque value just prior to bolt rotation. **DO NOT EXCEED 128 INCH-POUNDS.** Proceed as follows:

(1) If bolt does not rotate, torque check is complete. Proceed to step h.

(2) If bolt requires less than 5 inch-pounds of torque to start rotation, perform the following:

(a) In cabin, remove soundproofing panels, as required, from ceiling to gain access to nutplates in mixer supports (PARA 2-4-69).

(b) Check to see if bolt is fully threaded into nutplate and that two threads are showing past nut.

(c) If bolt is installed properly, replace nutplate.

(d) If less than two threads are showing past nut, retorque bolt per step d. (3) through step h.

(3) **IF BOLT REQUIRED LESS THAN 128 INCH-POUNDS, BUT MORE THAN 5**

INCH-POUNDS TO START ROTATION, RETORQUE BOLT TO 129 - 141 INCH-POUNDS. Apply torque stripe (PARA 2-6-5).

e. Make sure bolt is fully threaded into nutplate and that two threads are showing past nut.

NOTE

There must be at least one washer but not more than three washers under bolthead.

f. If more than four threads are showing past nut, add washers under bolthead as required or remove washers if bolt does not fully engage nut.

g. Seal areas around boltheads with sealing compound, Item 292, Appendix D.

h. If required, install soundproofing panels (PARA 2-4-69).

i. Close main rotor pylon sliding cover.

11-3-9 TORQUE SHAFT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

11-3-9.1 Torque Shaft Torque Check.

NOTE

This check applies to collective, pitch, lateral, and yaw torque shafts.

a. Open main rotor pylon sliding cover.

NOTE

Check torque in tightening direction using crisscross pattern.

b. Check torque on bolts and nuts attaching torque shaft to lower support structure. **NUT TORQUE SHALL BE 95 - 105 INCH-POUNDS. IF CHECKING TORQUE OF BOLT, TORQUE SHALL BE 105 - 115 INCH-POUNDS.** If torques are not as specified, torque bolts or nuts to proper value.

c. Close and latch main rotor pylon sliding cover.

11-3-10 YAW TORQUE SHAFT LOWER SUPPORT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 2-6-5
- PARA 11-4-58

11-3-10.1 Inspect Yaw Torque Shaft Lower Support.

a. Open main rotor pylon sliding cover.

b. Inspect yaw torque shaft lower support structure straps for loose rivets. If loose rivets are found, perform the following:

(1) Remove yaw torque shaft (PARA 11-4-58).

(2) Remove rivets and strap.

(3) Install new strap with rivets.

(4) Touch up paint repaired area (PARA 2-6-5).

(5) Install yaw torque shaft (PARA 11-4-58).

c. Close and latch main rotor pylon sliding cover.

11-3-11 YAW LEVER AND SUPPORT (STA 301.0) INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Micrometer

References

- PARA 11-4-85

11-3-11.1 Inspect Yaw Lever and Support (STA 301.0).

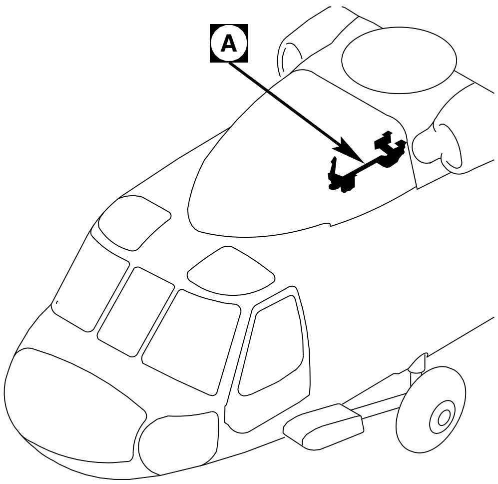
a. Open main rotor pylon sliding cover.

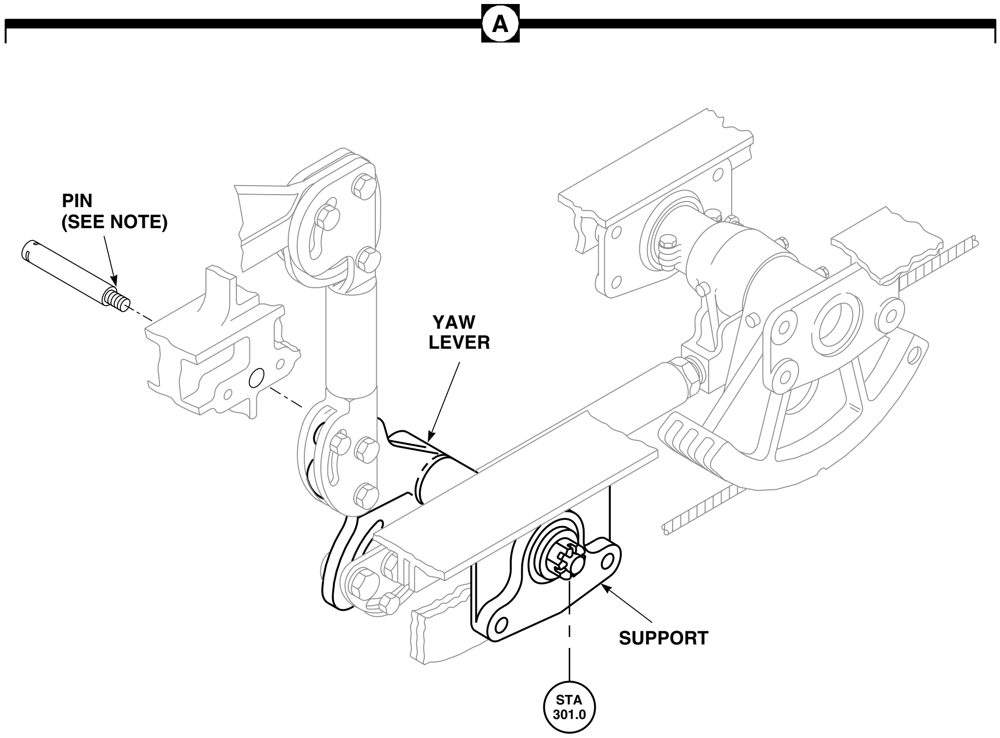
b. Measure installed radial slot of pin by manually moving yaw lever vertically (up and down) (Figure 11-3-11-1, Detail A). If any movement is found, inspect pin as follows:

(1) Using micrometer, measure diameter of pin where it contacts bearing. Set micrometer near shoulder of pin to obtain basic diameter. **MAXIMUM ALLOWABLE WEAR IS 0.005-INCH.**

(2) If maximum allowable wear is exceeded, replace pin (PARA 11-4-85).

c. Close and latch main rotor pylon sliding cover.





NOTE

PIN SHOWN REMOVED FOR CLARITY.

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FIGURE 11-3-11-1. INSPECT YAW LEVER AND SUPPORT (STA 301.0)

11-3-12 SWASHPLATE LINK INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ¼-inch
- Dial Indicator
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-24
- PARA 11-4-95

11-3-12.1 Swashplate Link FSCAP Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

WARNING

FSCAP

Do not attempt to repair damage that penetrates into base metal in a critical characteristic area. Damage to paint or protective finishes not penetrating into base metal may be touched up. Parts with damage into base metal that are within the critical characteristic area are to be replaced.

NOTE

If damage is found in shaded areas on swashplate link, swashplate link must be removed and critical characteristic areas inspected (Figure 11-3-12-1, Detail A and Detail B).

b. Check critical characteristic areas of swashplate link for minor surface corrosion. Surface corrosion may be carefully blended out with abrasive cloth, Item 10, Appendix D. Keep blend depth to absolute minimum required to remove corrosion. **REWORK SHALL NOT PENETRATE INTO BASE METAL.** If rework penetrates into base metal, replace swashplate link (PARA 11-4-95).

c. Visually inspect critical characteristic areas of link for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OR**

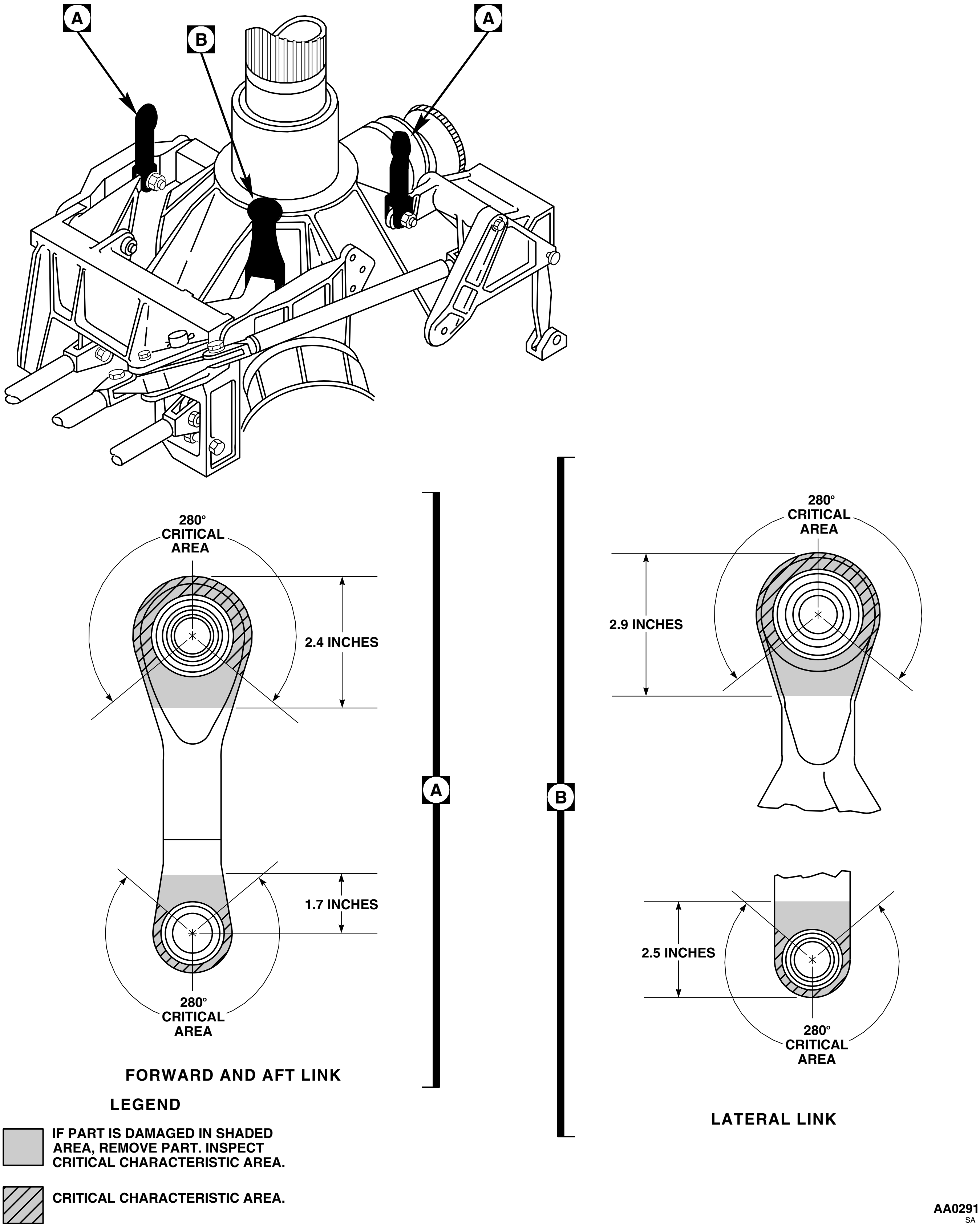


FIGURE 11-3-12-1. INSPECT SWASHPLATE LINKS

PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT (PARA 2-6-5). **REWORK SHALL NOT PENETRATE INTO BASE METAL.** If rework penetrates into base metal, replace swashplate link (PARA 11-4-95).

d. Check for surface damage on noncritical areas of link. Blend any damage found with abrasive cloth, Item 10, Appendix D. Keep blend depth to absolute minimum required to remove damage. **DAMAGE IS NOT TO EXCEED 0.020-INCH AFTER BLENDING.** If damage exceeds limit, replace swashplate link (PARA 11-4-95).

e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace swashplate link (PARA 11-4-95).

f. Touch up finish all reworked areas (PARA 2-6-5).

g. Install main rotor pylon transmission cover (PARA 2-4-107).

11-3-12.2 Swashplate Link Bearing Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- If bearing is cracked do not remove expandable pin.
- Spherical bearing with one crack may remain in service for an additional 10 flight hours if all other inspection criteria is acceptable.

b. Inspect spherical bearings on all three swashplate links for cracks. **ANY LINK SHALL NOT HAVE MORE THAN ONE CRACK.** If limit is exceeded, replace swashplate link (PARA 11-4-95).

WARNING

FSCAP

- Verification of proper hardware installation and torque of nut is a critical characteristic.
- Verification of gap under washer, and installation of expandable pin, split lockring, and half male spacer are critical characteristics.

c. Using feeler gage, measure gap between expandable pin and washer. **GAP MUST NOT EXCEED 0.002-INCH.** Evaluate as follows:

(1) If gap exceeds 0.002-inch and bearing is not cracked, remove expandable pin and reinstall (PARA 11-4-95).

(2) If gap exceeds 0.002-inch and bearing is cracked, replace swashplate link (PARA 11-4-95).

(3) If gap is less than 0.002-inch and bearing is not cracked, bolt is locked in place.

(4) If gap is less than 0.002-inch and bearing is cracked, perform the following:

(a) Inspect spherical bearing for radial play using dial indicator. **NONE ALLOWED.** If radial movement is found, replace swashplate link (PARA 11-4-95).

(b) Measure combined total radial play from servos to swashplate links (PARA 11-3-24).

(c) Inspect spherical bearing for axial play using dial indicator. **AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace swashplate link (PARA 11-4-95).

(d) Inspect spherical bearing for extrusion of Teflon liner. **NO EXTRU-**

SION ALLOWED. If extrusion is present, replace swashplate link (PARA 11-4-95).

d. Perform expandable pin torque check as follows:

(1) Remove cotter pin. Insert ¼-inch Allen wrench in center of swashplate expandable pin.

WARNING

FSCAP

Verification of proper hardware installation and torque of nut is a critical characteristic.

(2) **HOLD ALLEN WRENCH AND PROOF TORQUE NUT TO 300 INCH-POUNDS.** Do not allow expandable pin to turn. If nut moves under 300 inch-pounds, replace link (PARA 11-4-95).

(3) Remove Allen wrench. Install cotter pin.

e. If all of the above conditions are met, swashplate link with one crack may remain in service for an additional 10 flight hours before replacement is required.

f. Install main rotor pylon transmission cover (PARA 2-4-107).

11-3-13 MAIN ROTOR CONTROL SELF-RETAINING BOLT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107

11-3-13.1 Main Rotor Control Self-Retaining Bolt FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in critical area is limited to a one-time repair and is to be done only when bolt replacement is not possible.
- All rework of critical area which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

c. Check critical area for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED.** If damage is found, repair bolt as follows (Figure 11-3-13-1, Detail A):

(1) Check log card (SA Form 7343-21) to see if bolt critical area has been previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREA ALLOWED.** If bolt critical area has been previously reworked into base metal, and damage penetrates into base metal, replace bolt.

NOTE

Minimum bolthead to shank radius is 0.015-inch.

(2) Damage in critical area may be blended up to 0.003-inch deep provided that criti-

cal characteristic minimum shank diameter and bolthead to shank radius are not exceeded. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE BOLT** (Table 11-3-13-1). Record any critical area rework into base metal on component log card (SA Form 7343-21).

(3) Using fluorescent penetrant inspection kit, inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bolt.

(4) Apply solid film lubricant, Item 201, Appendix D, to all reworked areas of bolt.

Table 11-3-13-1. Main Rotor Control Self-Retaining Bolt Critical Dimensions

Part Number	Critical Minimum Diameter
70400-08159-101 through 70400-08159-106	0.745-inch
70400-08159-107 and 70400-08159-108	0.870-inch

d. Install main rotor pylon transmission cover (PARA 2-4-107).

e. Close main rotor pylon sliding cover.

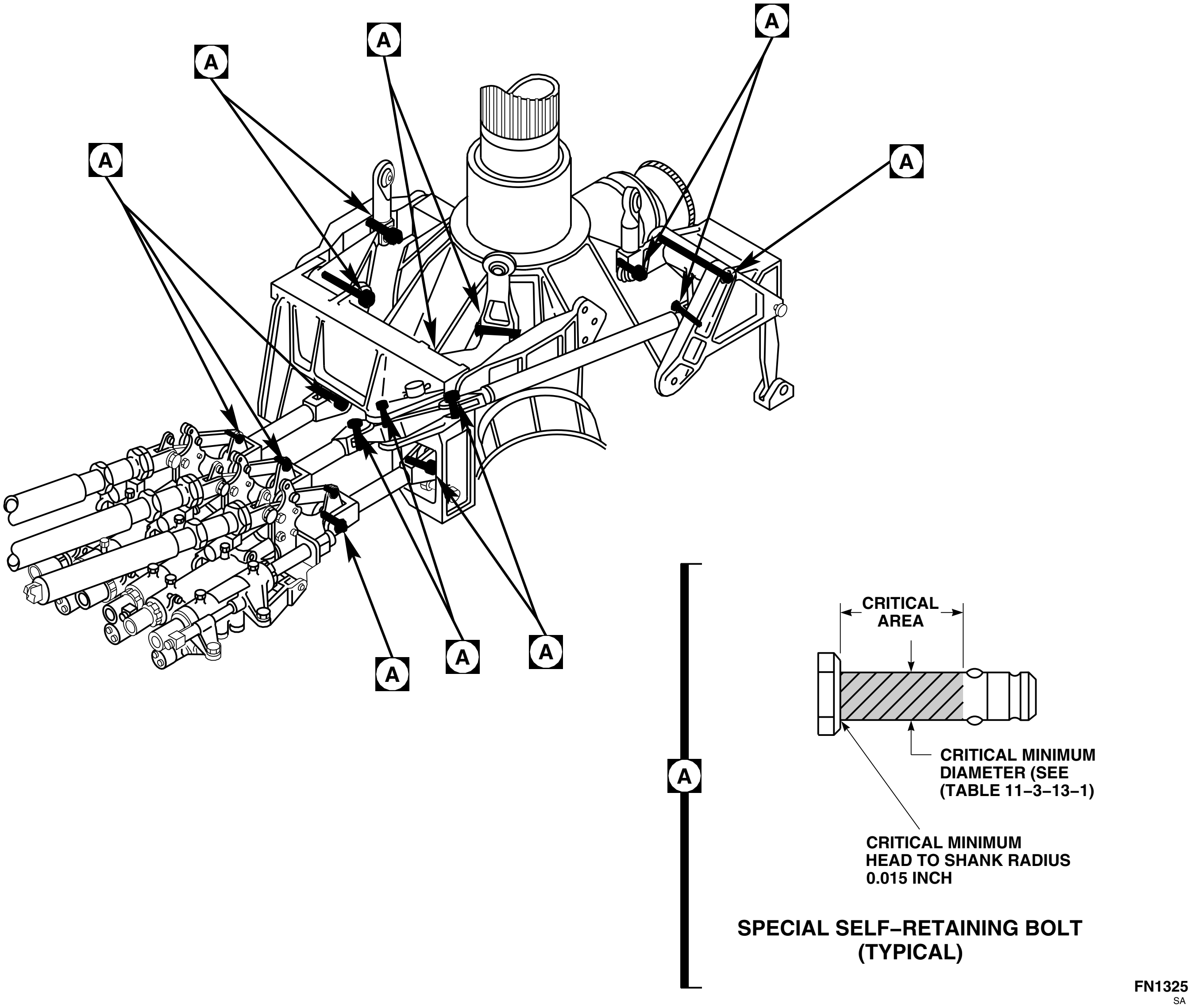


FIGURE 11-3-13-1. INSPECT MAIN ROTOR CONTROL SELF-RETAINING BOLTS

11-3-14 WALKING BEAM INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-24
- PARA 11-4-93

11-3-14.1 Walking Beam FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in critical areas are limited to a one-time repair.
- All rework of critical areas which penetrate into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.
- If damage is found in shaded areas on walking beam, walking beam must be removed (PARA 11-4-93), and critical characteristic areas inspected (Figure 11-3-14-1, Detail A).

c. Check critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-14-1, View A). **NONE ALLOWED.** If damage is found, repair walking beam as follows:

(1) Check log card (SA Form 7343-21) to see if walking beam critical areas have been previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREAS ALLOWED.** If walking beam critical areas have been previously

reworked into base metal, and damage penetrates into base metal, replace walking beam (PARA 11-4-93).

(2) **DAMAGE IN ZONE I CRITICAL AREAS MAY BE BLENDED UP TO 0.003-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace part (PARA 11-4-93). Record any zone I rework into base metal on component log card (SA Form 7343-21).

(3) **DAMAGE IN ZONE II CRITICAL AREAS MAY BE BLENDED UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace part (PARA 11-4-93). Record any zone II rework into base metal on component log card (SA Form 7343-21).

d. Check noncritical zone III areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG, WEB, OR FLANGE IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace part (PARA 11-4-93).

e. Using fluorescent penetrant inspection kit, inspect all reworked areas (ASTM E1417). **NONE ALLOWED.** If crack is found, replace part (PARA 11-4-93).

f. Apply corrosion compound, Item 108, Appendix D, to reworked areas of walking beam with acid brush, Item 84, Appendix D. Allow to dry for 30 seconds then rinse thoroughly with water. Allow walking beam to dry completely.

g. Touch up finish all reworked areas (PARA 2-6-5).

h. Install main rotor pylon transmission cover (PARA 2-4-107).

i. Close main rotor pylon sliding cover.

11-3-14.2 Walking Beam Bearing Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Inspect spherical bearing connections for maximum radial and axial wear limits.

d. Using dial indicator, measure radial play. **MAXIMUM RADIAL PLAY SHALL NOT EXCEED 0.020-INCH PER CONNECTION.** If radial play exceeds 0.020-inch, replace walking beam (PARA 11-4-93).

e. Measure combined total radial play from servos to swashplate links (PARA 11-3-24).

f. Using dial indicator, measure axial play. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds 0.020-inch, replace walking beam (PARA 11-4-93).

g. Measure axial play between walking beam and forward bellcrank support. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds 0.020-inch, replace walking beam (PARA 11-4-93).

h. Install main rotor pylon transmission cover (PARA 2-4-107).

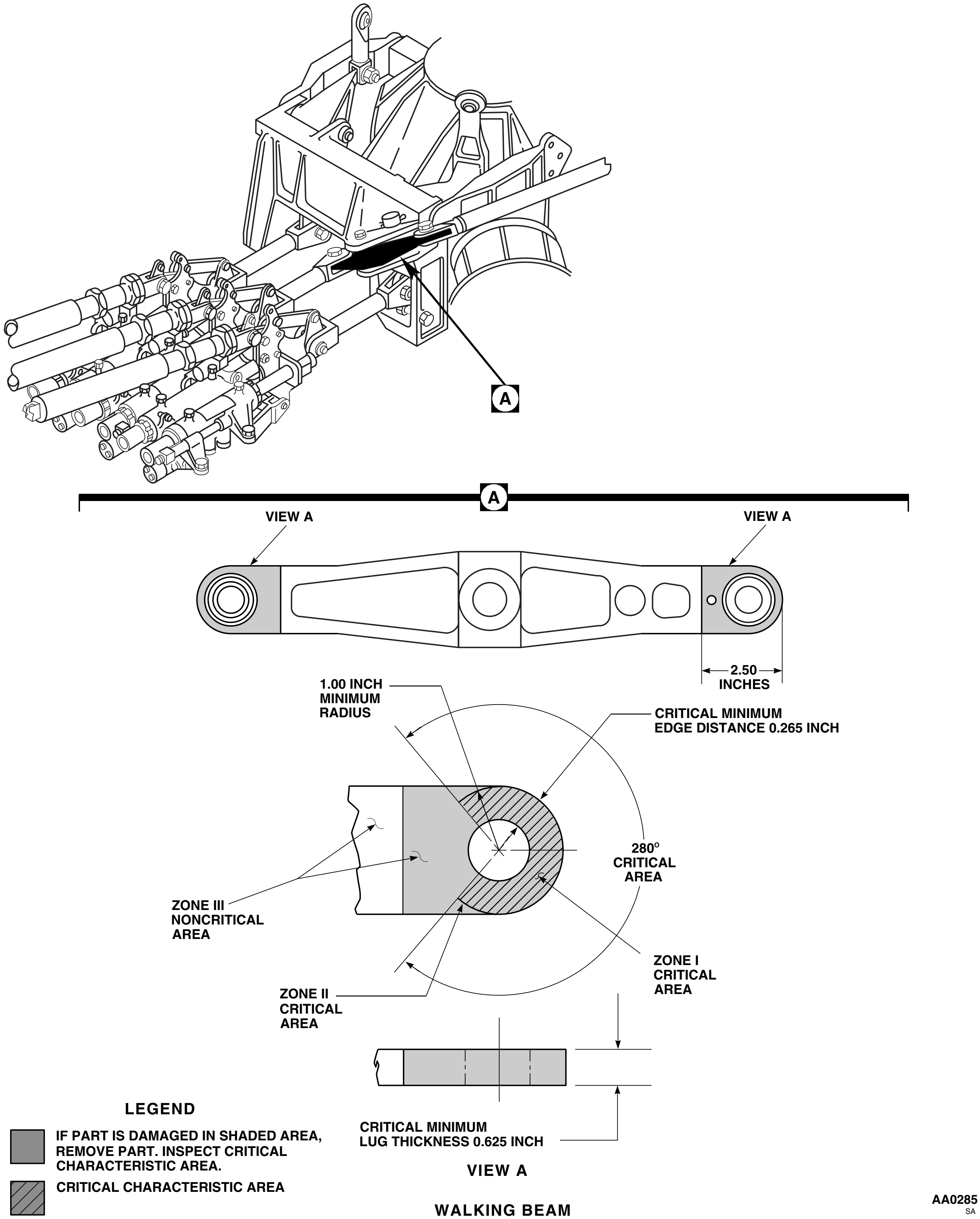


FIGURE 11-3-14-1. INSPECT WALKING BEAM

i. Close main rotor pylon sliding cover.

11-3-15 FORWARD BELLCRANK SUPPORT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-4-100

11-3-15.1 Forward Bellcrank Support FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

WARNING

FSCAP

Do not attempt to repair damage that penetrates into base metal in zone I critical area. Damage to paint or protective finishes not penetrating into base metal may be touched up. Parts with damage into base metal that are within zone I critical area are to be replaced.

NOTE

If damage is found in shaded areas on forward bellcrank support, forward bellcrank support must be removed (PARA 11-4-100), and critical characteristic areas inspected (Figure 11-3-15-1, Sheet 2, Detail A or Figure 11-3-15-2, Sheet 2, Detail A).

c. Inspect lugs, A, B, C, D, E, and F as follows (Figure 11-3-15-1, Sheet 2, Detail B and Detail C, and Sheet 3, Detail D, Detail E, and Detail F, and Figure 11-3-15-2, Sheet 2, Detail B and Detail C, and Sheet 3, Detail D, Detail E, and Detail F):

(1) Check zone I critical area of lugs A, B, C, D, E, and F for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If re-

work penetrates into base metal, replace support (PARA 11-4-100).

(2) Check zone I critical area of lugs A, B, C, D, E, and F for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OR PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT** (PARA 2-6-5). If damage penetrates into base metal, replace support (PARA 11-4-100).

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if lugs A, B, C, D, E, or F zone II critical areas were previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If lugs A, B, C, D, E, or F zone II areas have been previously reworked into base metal, no further repair is allowed and part must be replaced (PARA 11-4-100).

(4) Check zone II critical areas of lugs A, B, C, D, E, and F for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-100). Record zone II repairs on component log card (SA Form 7343-21).

(5) Check noncritical zone II area of lugs A, B, C, D, E, and F for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-100).

d. Inspect flange A as follows (Figure 11-3-15-1, Sheet 4, Detail H and Figure 11-3-15-2, Sheet 4, Detail H):

(1) Check zone I critical area of flange A for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If rework penetrates into base metal, replace support (PARA 11-4-100).

(2) Check zone I critical area of flange A for nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED.** Damage in paint or protective finishes not penetrating into base metal may be repaired with touchup paint (PARA 2-6-5). If damage penetrates into base metal, replace support (PARA 11-4-100).

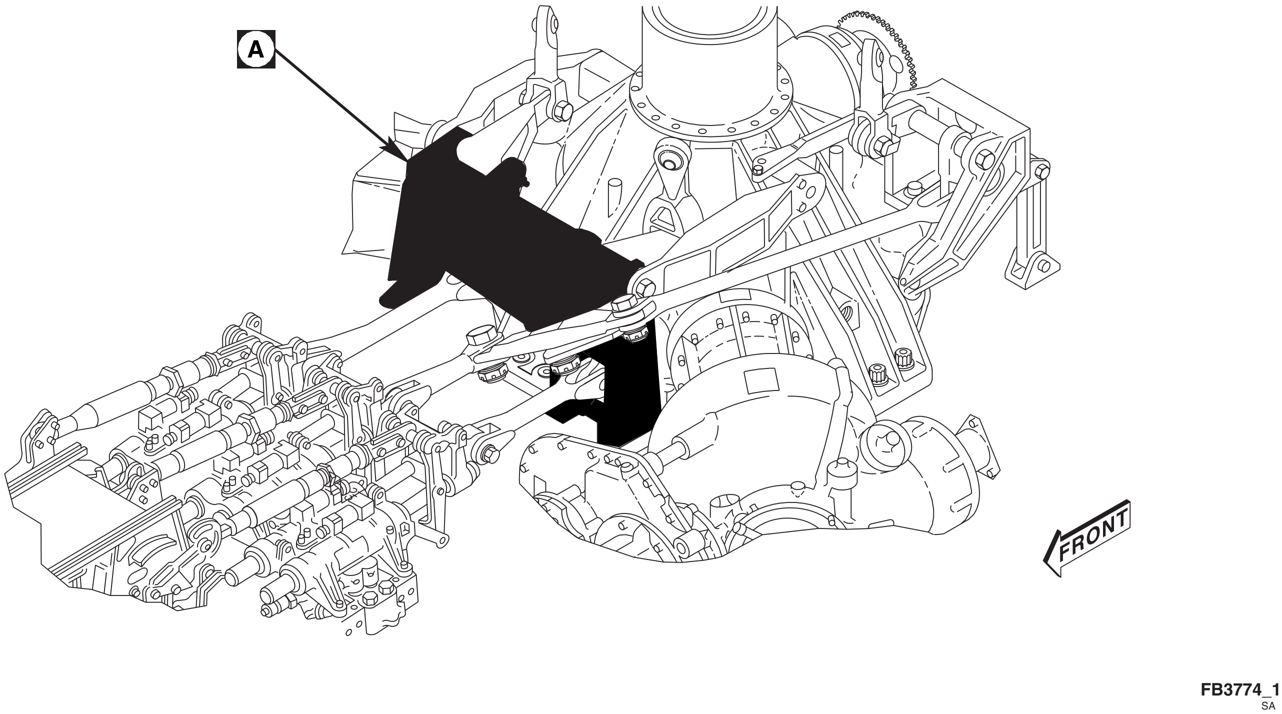


FIGURE 11-3-15-1. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 1 OF 4)

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if flange A zone II critical area was previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If flange A zone II area has been previously reworked into base metal no further repair is allowed and support must be replaced (PARA 11-4-100).

(4) Check zone II critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. Damage may be blended out up to 0.005-inch deep, 0.100-inch wide, and 0.380-inch long, provided that minimum radius and flange thickness are maintained, and that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of flange in same area. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100). Record zone II repairs on component log card (SA Form 7343-21).

e. Inspect flanges B and C as follows (Figure 11-3-15-1, Sheet 4, Detail J and Figure 11-3-15-2, Sheet 4, Detail J):

(1) Check zone A on flanges B and C for surface damage. Damage may be blended out up to 0.010-inch deep, 0.100-inch wide, and 0.380-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of flange in same area. Blend damage using abrasive cloth, Item

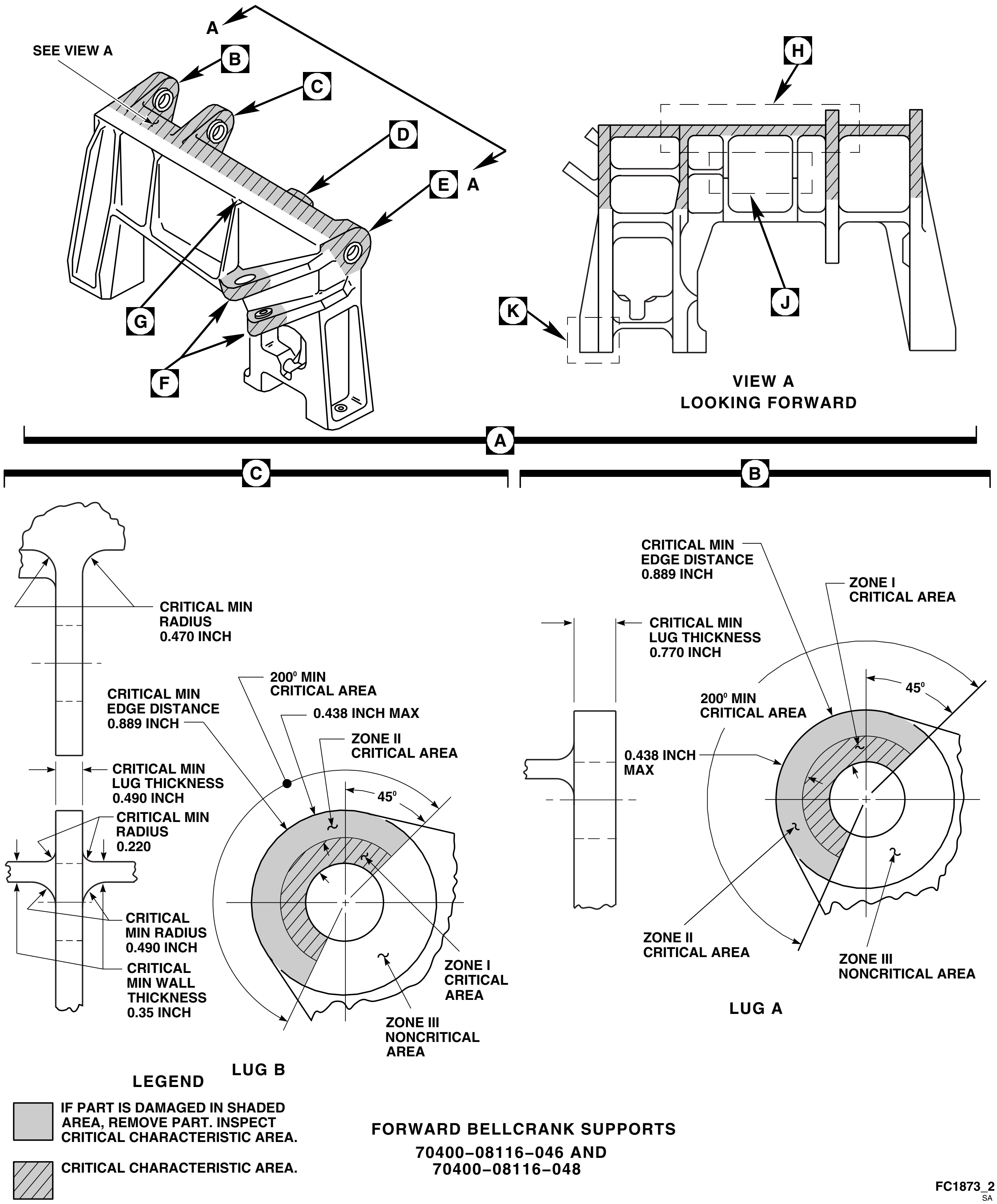


FIGURE 11-3-15-1. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 2 OF 4)

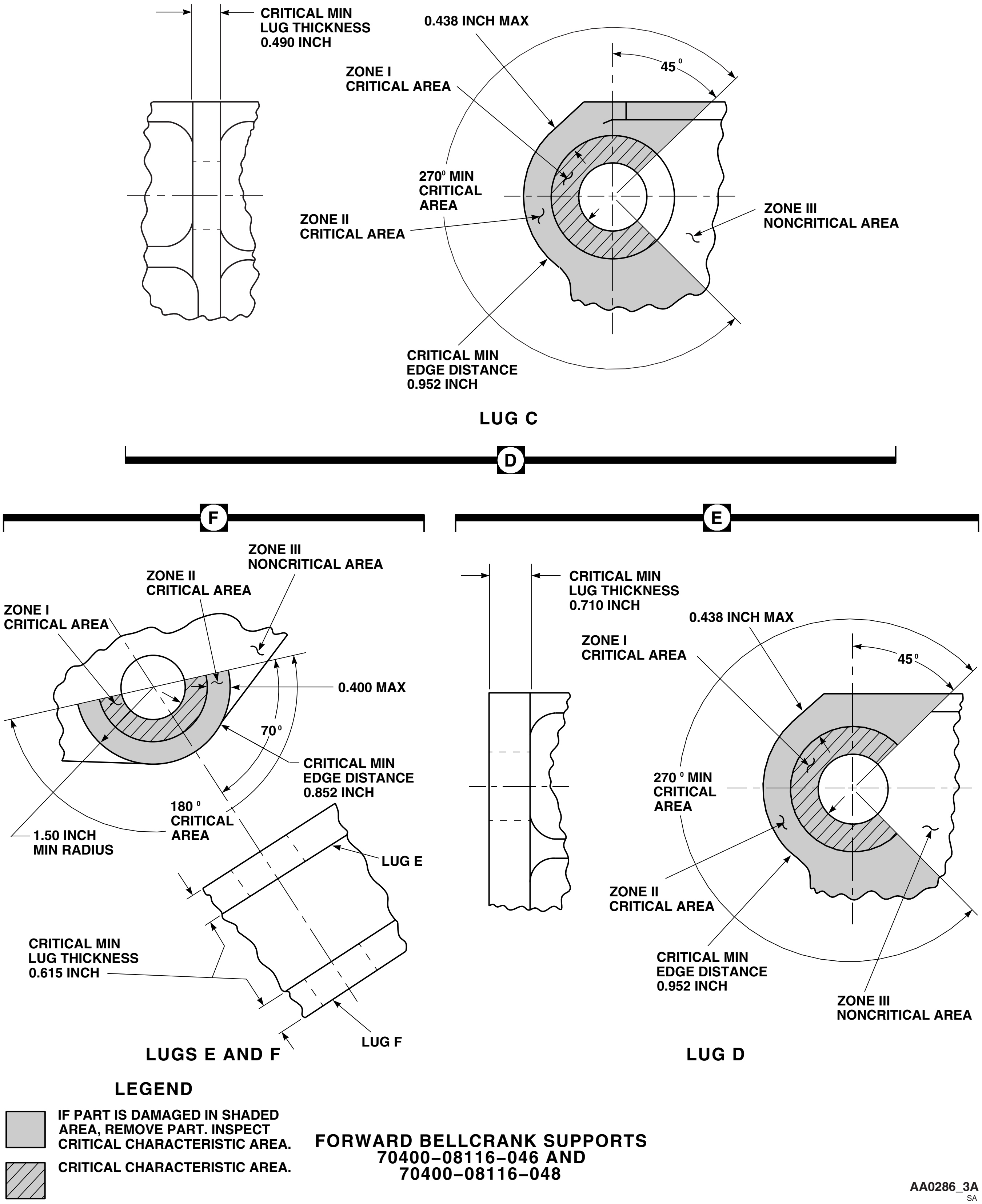


FIGURE 11-3-15-1. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 3 OF 4)

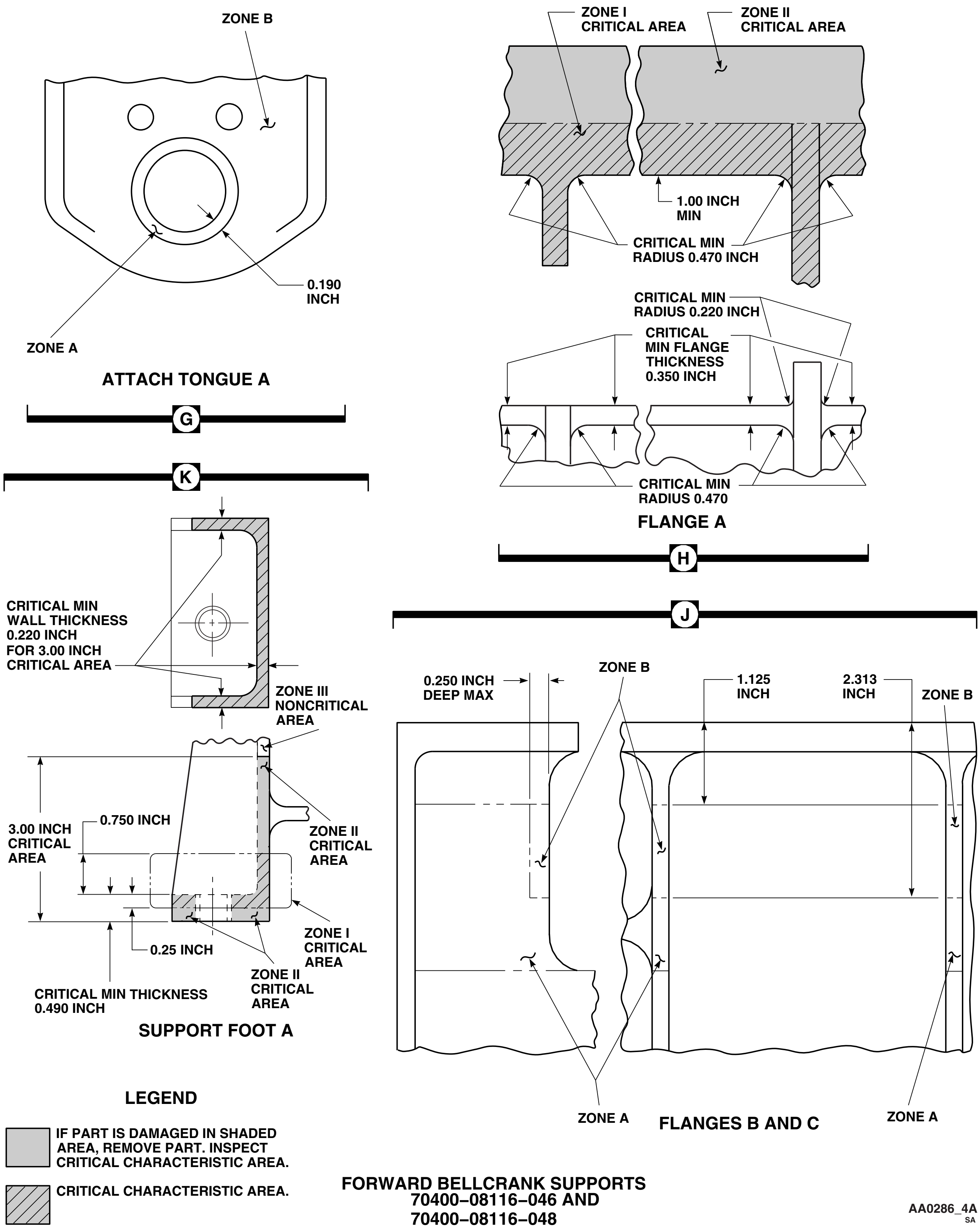


FIGURE 11-3-15-1. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 4 OF 4)

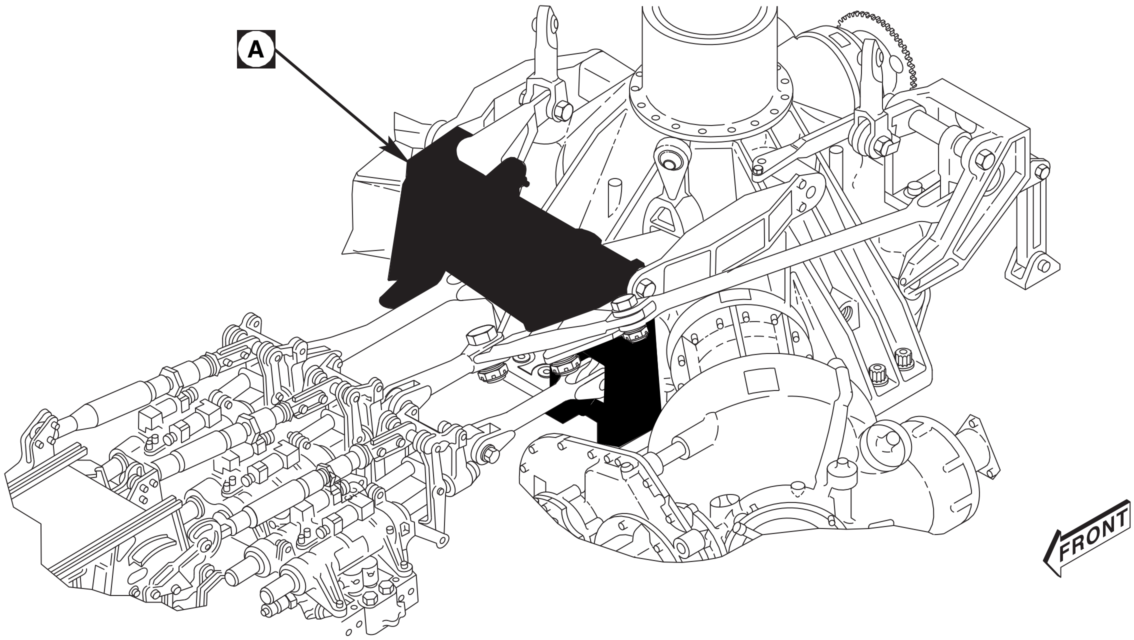


FIGURE 11-3-15-2. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 1 OF 4)

8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100).

(2) Check zone B of flanges B and C for surface corrosion, nicks, dents, scratches, and tool or scuff marks. Damage may be blended out up to 0.040-inch deep, 0.250-inch wide, and 0.750-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of flange in same area. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100).

f. Inspect tongue A as follows (Figure 11-3-15-1, Sheet 4, Detail G and Figure 11-3-15-2, Sheet 3, Detail G):

(1) Check zone A of tongue A for surface corrosion, nicks, dents, scratches, and tool or scuff marks. Damage may be blended out up to 0.010-inch deep, 0.100-inch wide, and 0.380-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of tongue in same area. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100).

(2) Check zone B on tongue A for surface damage. Damage may be blended out up to 0.020-inch deep, 0.150-inch wide, and 0.750-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of tongue in same area. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100).

g. On forward bellcrank support, 70400-08116-046 and 70400-08116-048, inspect support foot as follows (Figure 11-3-15-1, Sheet 4, Detail K):

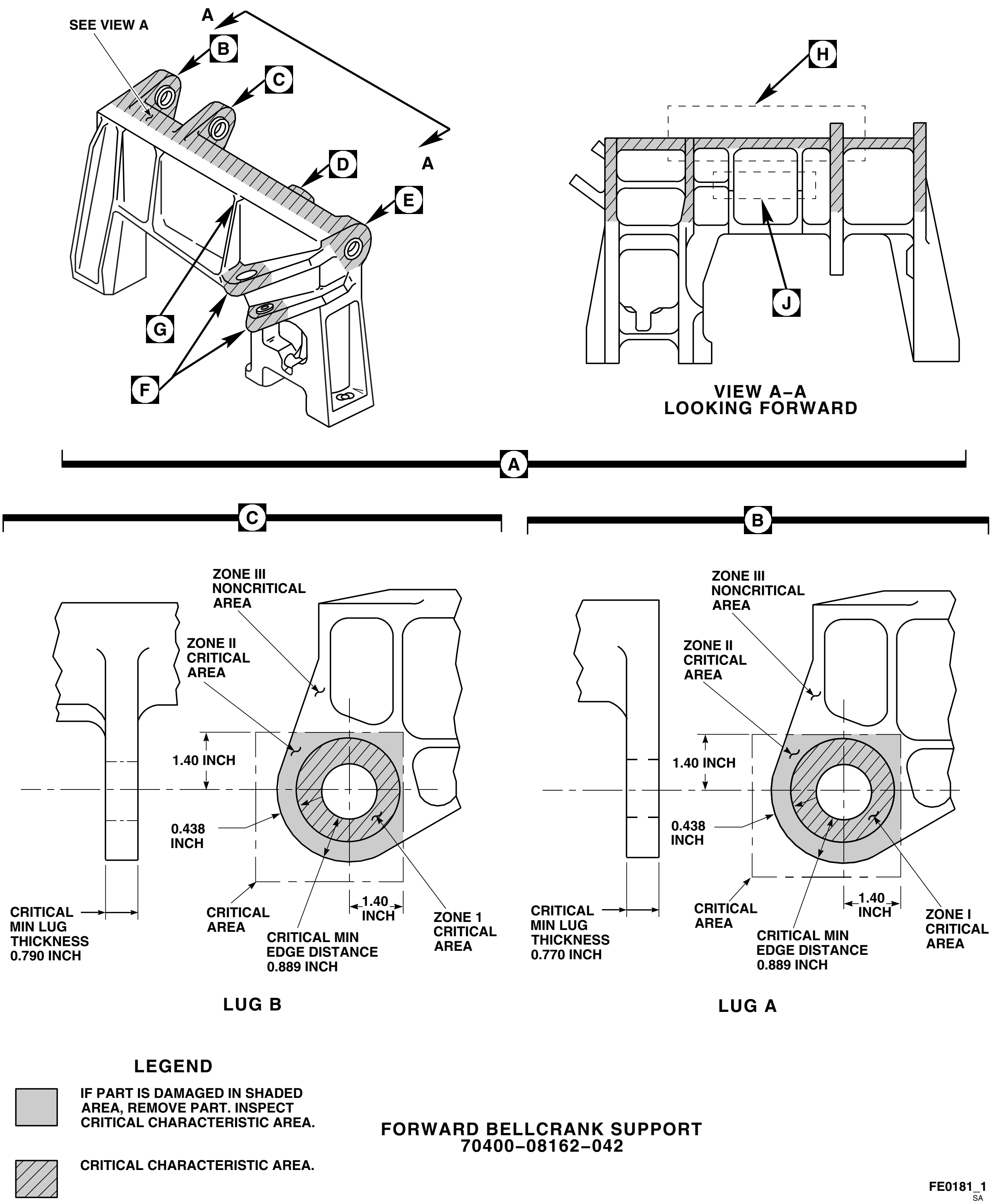


FIGURE 11-3-15-2. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 2 OF 4)

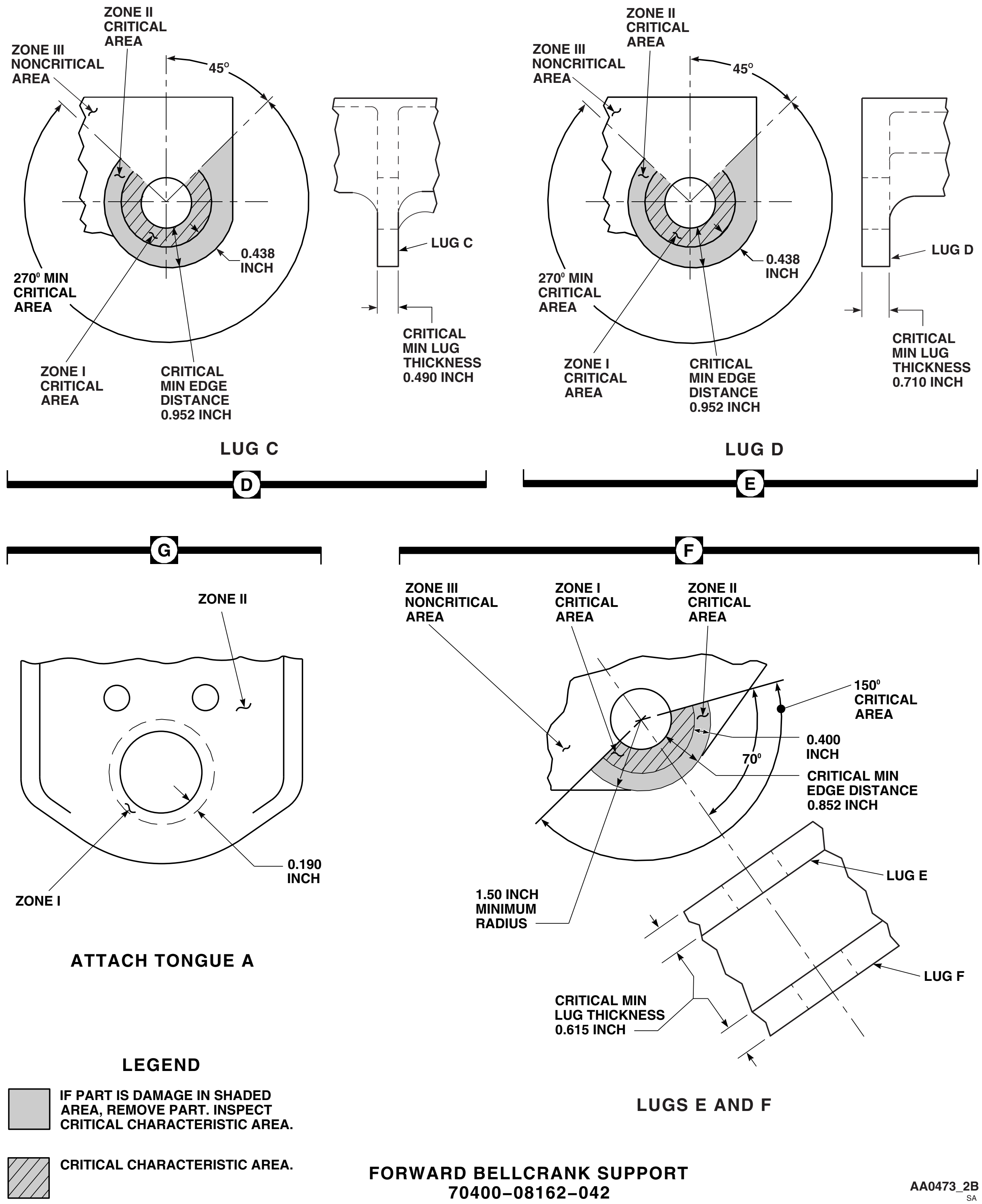


FIGURE 11-3-15-2. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 3 OF 4)

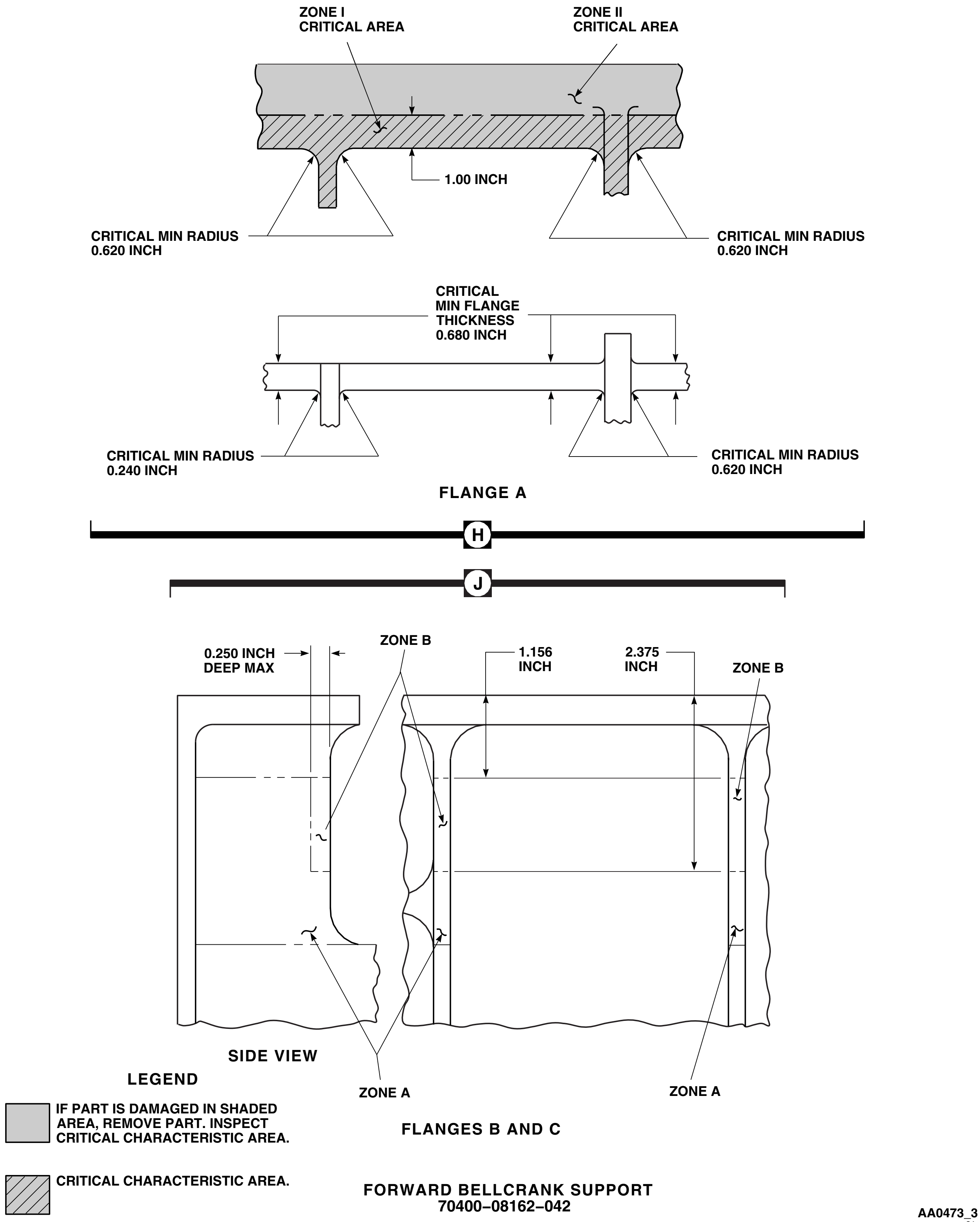


FIGURE 11-3-15-2. INSPECT FORWARD BELLCRANK SUPPORT (SHEET 4 OF 4)

(1) Check zone I critical area for minor surface corrosion. Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. **IF REWORK PENETRATES INTO BASE METAL, REPLACE SUPPORT.**

(2) Check zone I critical area for nicks, dents, scratches, and tool or scuff marks. Damage in paint or protective finishes not penetrating into base metal may be repaired with touchup paint (PARA 2-6-5). **IF DAMAGE PENETRATES INTO BASE METAL, REPLACE SUPPORT** (PARA 11-4-100).

(3) Check areas outside of zone I but inside the critical characteristic area for surface corrosion, nicks, dents, scratches, and tool or scuff marks. Damage may be blended out up to 0.005-inch deep, 0.100-inch wide, and 0.380-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of support foot in same area. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100).

(4) Check area outside of critical characteristic area for surface damage. Damage may be blended out up to 0.010-inch deep, 0.100-inch wide, and 0.380-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of support foot in same area. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (PARA 11-4-100).

h. Check for surface damage on noncritical zone III areas of forward bellcrank support that were not previously inspected. Damage may be blended out up to 0.010-inch deep, 0.100-inch wide, and 0.380-inch long, provided that no two damaged areas are within 0.250-inch of each other or less than 0.500-inch on opposite side of lug, flange, web, or wall section. Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT** (Figure 11-3-15-1, Sheet 2, Detail B and Detail C, Sheet 3, Detail D, and Detail E, and Sheet 4, Detail G and Detail K, and Figure 11-3-15-2, Sheet 2, Detail B and Detail C, and Sheet 3, Detail D, Detail E, and Detail G) (PARA 11-4-100).

i. Using fluorescent penetrant inspection kit, inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support (PARA 11-4-100).

j. Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas.

k. Touch up finish on all reworked areas (PARA 2-6-5).

l. Install main rotor pylon transmission cover (PARA 2-4-107).

m. Close main rotor pylon sliding cover.

11-3-16 RIGHT AND LEFT TIE ROD INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-4-99

11-3-16.1 Right and Left Tie Rod FSCAP Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

WARNING

FSCAP

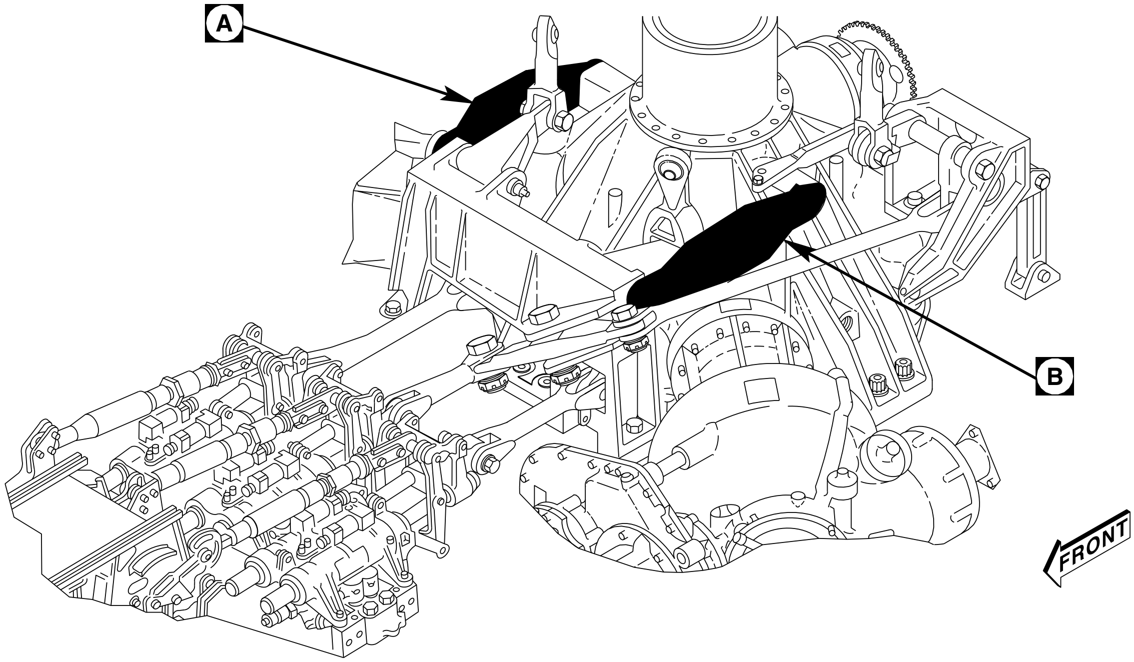
Do not attempt to repair damage that penetrates into base metal in zone I critical area. Damage to paint or protective finishes not penetrating into base metal may be touched up. Parts with damage into base metal that are within the zone I critical area are to be replaced.

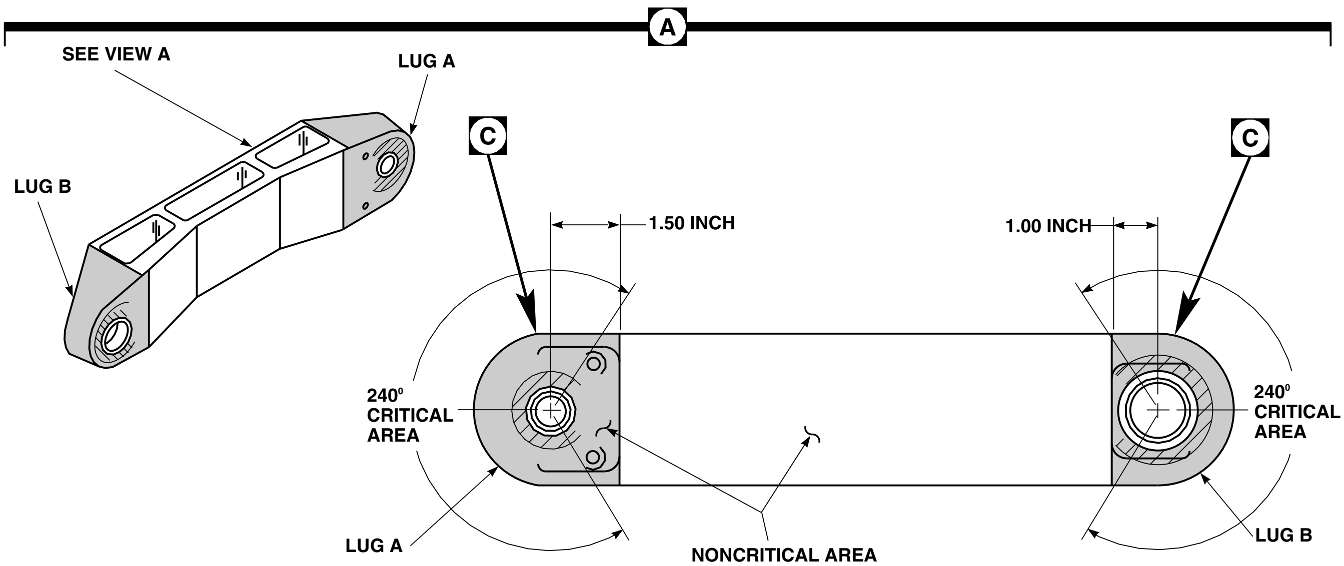
NOTE

If damage is found in shaded areas on tie rod, tie rod must be removed and critical characteristic areas inspected (Figure 11-3-16-1, Sheet 2, Detail A and Detail B and Sheet 3, Detail C and Detail D).

b. Check zone I critical area for minor surface corrosion (Figure 11-3-16-1, Sheet 3, Detail C and Detail D). **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If rework penetrates into base metal, replace tie rod (PARA 11-4-99).

c. Check zone I critical area for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OR PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT** (PARA 2-6-5). If damage penetrates into base metal, replace tie rod (PARA 11-4-99).





LEGEND

- (shaded) IF PART IS DAMAGED IN SHADED AREA, REMOVE PART. INSPECT CRITICAL CHARACTERISTIC AREA.
- (hatched) CRITICAL CHARACTERISTICS AREA.

RIGHT TIE RODS
70400–08114–047
70400–08114–049
AND 70400–08114–051

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FIGURE 11-3-16-1. INSPECT RIGHT AND LEFT TIE RODS (SHEET 1 OF 3)

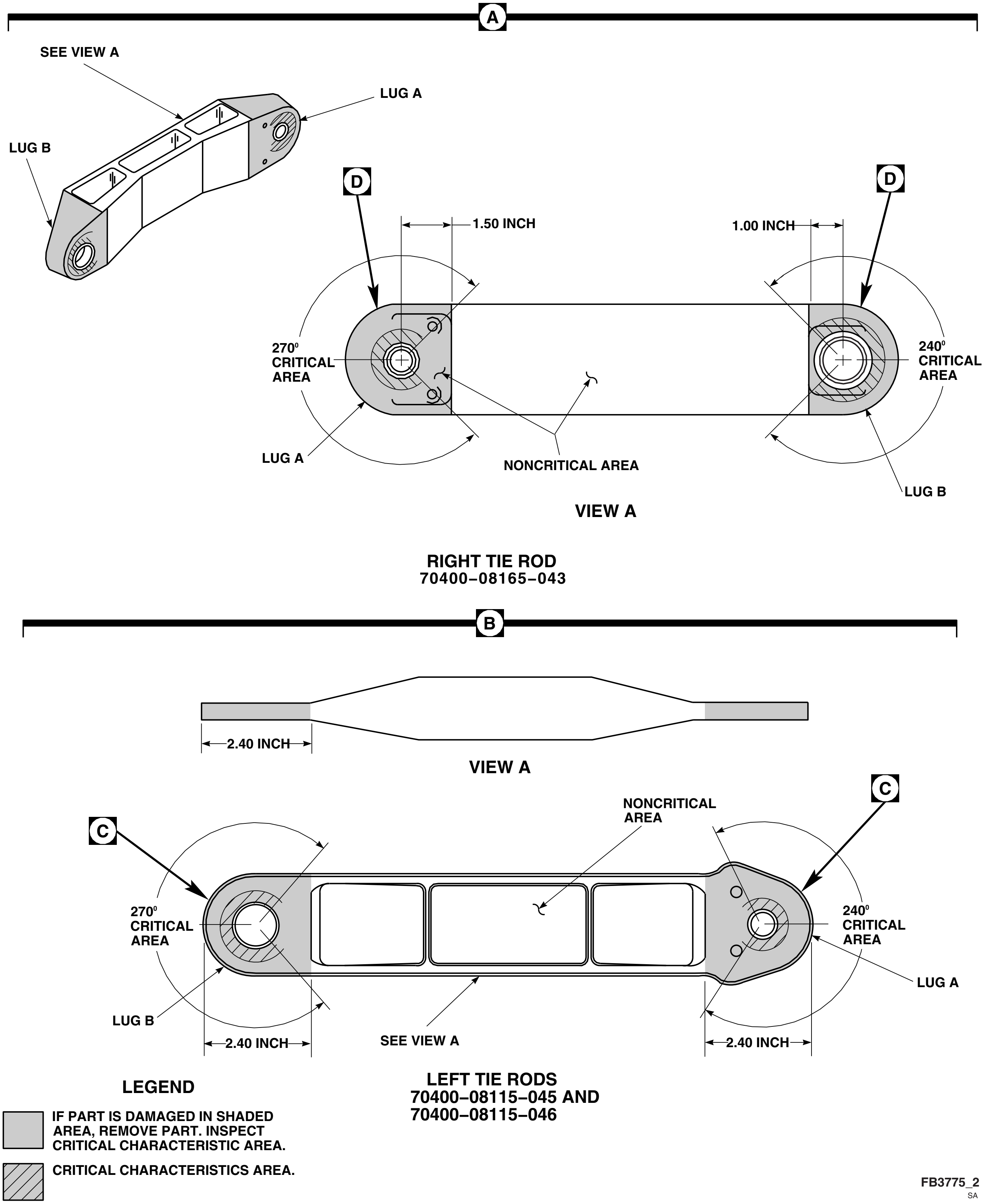
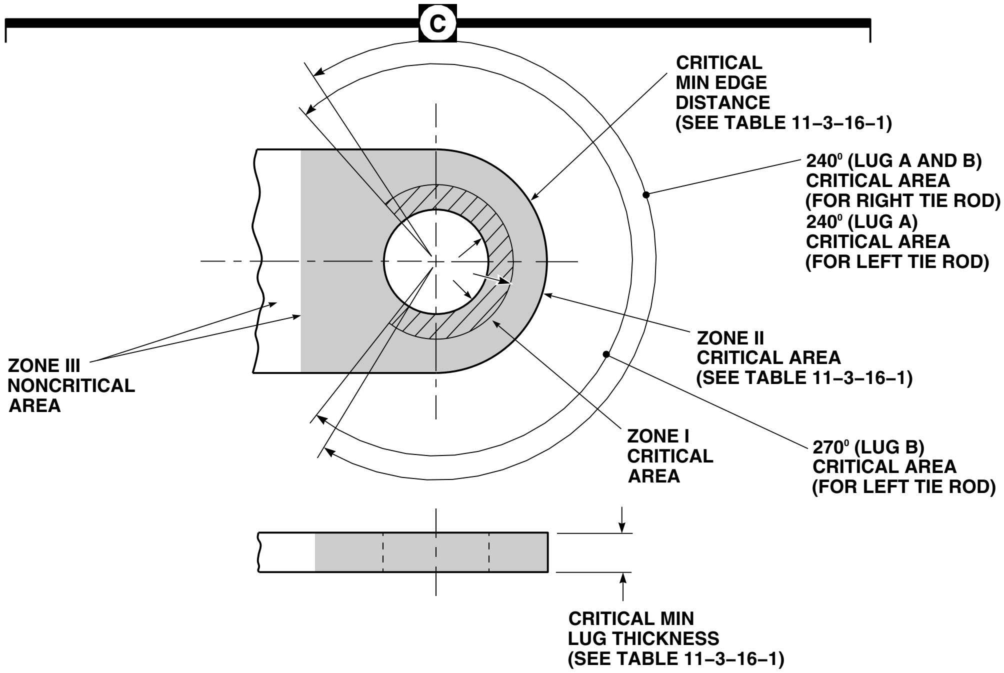
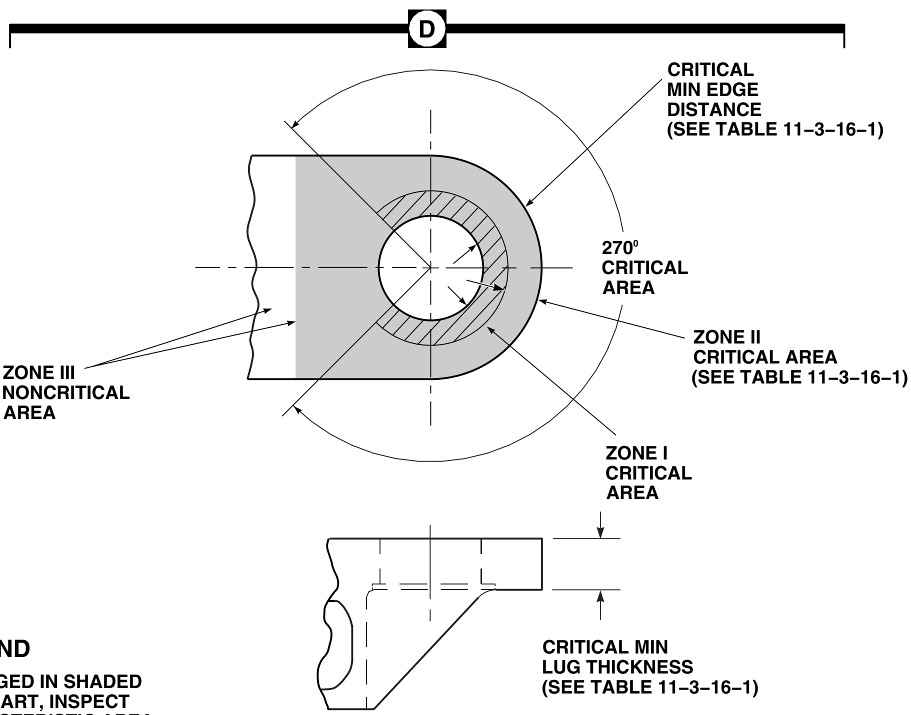


FIGURE 11-3-16-1. INSPECT RIGHT AND LEFT TIE RODS (SHEET 2 OF 3)



RIGHT TIE RODS 70400–08114–047,
70400–08114–049, 70400–08114–051, AND
LEFT TIE RODS 70400–08115–045,
70400–08115–046 LUG AREA (TYPICAL)



LEGEND

IF PART IS DAMAGED IN SHADED AREA, REMOVE PART, INSPECT CRITICAL CHARACTERISTIC AREA.

CRITICAL CHARACTERISTIC AREA.

RIGHT TIE ROD
70400–08165–043
LUG AREA (TYPICAL)

FB3775_3
SA

FIGURE 11-3-16-1. INSPECT RIGHT AND LEFT TIE RODS (SHEET 3 OF 3)

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

d. Check log card (SA Form 7343-21) to see if tie rod zone II critical area was previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If tie rod zone II area has been previously reworked into metal, no further repair is allowed and tie rod must be replaced (PARA 11-4-99).

e. Check zone II critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace tie rod (Table 11-3-16-1) (PARA 11-4-99). Record zone II repairs on component log card (SA Form 7343-21).

f. Check noncritical zone III area for surface damage (Figure 11-3-16-1, Sheet 1, Detail A, Sheet 2, Detail A and Detail B, and Sheet 3, Detail C and Detail D). **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF PART IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace tie rod (Table 11-3-16-1) (PARA 11-4-99).

g. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace tie rod (PARA 11-4-99).

h. Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas.

i. Touch up finish all reworked areas (PARA 2-6-5).

j. Install main rotor pylon transmission cover (PARA 2-4-107).

Table 11-3-16-1. Tie Rod Critical Dimensions

Component	Lug	Critical Minimum Lug Thickness	Critical Minimum Edge Distance	Zone II (Maximum)
Right Tie Rod 70400-08114-047, 70400-08114-049, 70400-08114-051	A	0.490-inch	0.922-inch	0.410-inch
	B	0.780-inch	0.601-inch	0.250-inch

Table 11-3-16-1. Tie Rod Critical Dimensions (Cont)

		Critical Minimum	Critical Minimum	Zone II
Right Tie Rod 70400-08165-043	A	0.620-inch	1.024-inch	0.512-inch
	B	0.820-inch	0.820-inch	0.410-inch
Left Tie Rod 70400-08115-045, 70400-08115-046	A	0.490-inch	0.714-inch	0.320-inch
	B	0.490-inch	0.707-inch	0.320-inch

11-3-17 MAIN ROTOR CONTROL ROD INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-4-90
- PARA 11-4-91
- PARA 11-4-92
- PARA 11-4-94

11-3-17.1 Main Rotor Control Rod FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in critical areas are limited to a one-time repair.
- All rework of critical areas which penetrate into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.
- If damage is found in shaded areas on main rotor control rod, main rotor control rod must be removed (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, or PARA 11-4-94), and critical characteristic areas inspected (Figure 11-3-17-1, Detail A).

c. Check critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED.** If damage is found, repair control rod as follows:

(1) Check log card (SA Form 7343-21) to see if control rod critical areas have been

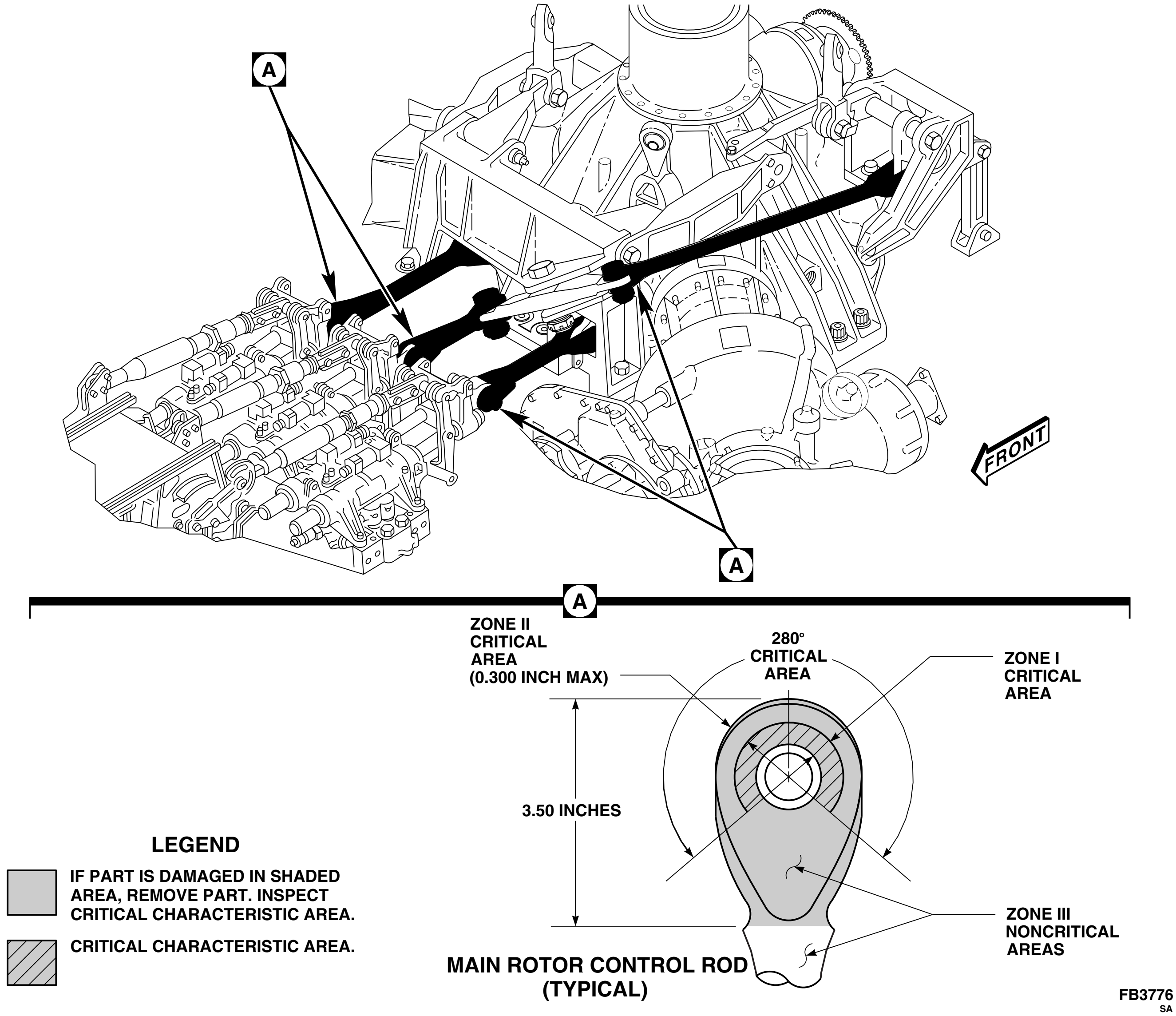


FIGURE 11-3-17-1. INSPECT MAIN ROTOR CONTROL RODS

previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREAS ALLOWED.** If control rod critical areas have been previously reworked into base metal and damage penetrates into base metal, replace control rod (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, or PARA 11-4-94).

(2) **DAMAGE IN ZONE I CRITICAL AREA MAY BE BLENDED OUT UP TO 0.003-INCH DEEP.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace control rod (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, or PARA 11-4-94). Record any zone I rework into base metal on component log card (SA Form 7343-21).

(3) **DAMAGE IN ZONE II CRITICAL AREA MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG OR TUBE WALL IN SAME AREA.**

Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace control rod (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, or PARA 11-4-94). Record any zone II rework into base metal on component log card (SA Form 7343-21).

d. Check noncritical zone III areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.020-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG OR TUBE WALL IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace control rod (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, or PARA 11-4-94).

e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace control rod (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, or PARA 11-4-94).

f. Apply corrosion compound, Item 108, Appendix D, to reworked areas of control rod with acid brush, Item 84, Appendix D. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow control rods to dry completely.

g. Touch up finish all reworked areas (PARA 2-6-5).

h. Install main rotor pylon transmission cover (PARA 2-4-107).

i. Close main rotor pylon sliding cover.

11-3-18 AFT BELLCRANK SUPPORT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-4-101

11-3-18.1 Aft Bellcrank Support FSCAP Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

WARNING

FSCAP

Do not attempt to repair damage that penetrates into base metal in zone I critical area. Damage to paint or protective finishes not penetrating into base metal may be touched up. Parts with damage into base metal that are within the zone I critical area are to be replaced.

NOTE

If damage is found in shaded areas on aft bellcrank support, aft bellcrank support must be removed and critical characteristic areas inspected (Figure 11-3-18-1, Sheet 1, Detail A or 11-3-18-2, Sheet 1, Detail A).

b. Inspect lugs A, B, C, and D, as follows (Figure 11-3-18-1, Sheet 1, Detail B and Sheet 2, Detail C and Detail D, and Figure 11-3-18-2, Sheet 1, Detail B and Detail C and Sheet 2, Detail D):

(1) Check zone I critical area of lugs A, B, C, and D for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If rework penetrates into base metal, replace support (PARA 11-4-101).

(2) Check zone I critical area of lugs A, B, C, and D for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OF PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL**

MAY BE REPAIRED WITH TOUCHUP PAINT (PARA 2-6-5). If damage penetrates into base metal, replace support (PARA 11-4-101).

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if lugs A, B, C, or D zone II critical areas were previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If lugs A, B, C, or D zone II areas have been previously reworked into base metal no further repair is allowed and part must be replaced (PARA 11-4-101).

(4) Check zone II critical areas of lugs A, B, C, and D for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101). Record zone II repairs on component log card (SA Form 7343-21).

(5) Check noncritical zone III areas of lugs A, B, C, and D for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101).

c. Inspect support base A as follows (Figure 11-3-18-1, Sheet 3, Detail F and Figure 11-3-18-2, Sheet 2, Detail E):

(1) Check zone I critical area of support base A for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If rework penetrates into base metal, replace support (PARA 11-4-101).

(2) Check zone I critical area of support base A for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OF PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT** (PARA 2-6-5). If damage penetrates into base metal, replace support (PARA 11-4-101).

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if support base A zone II critical area was

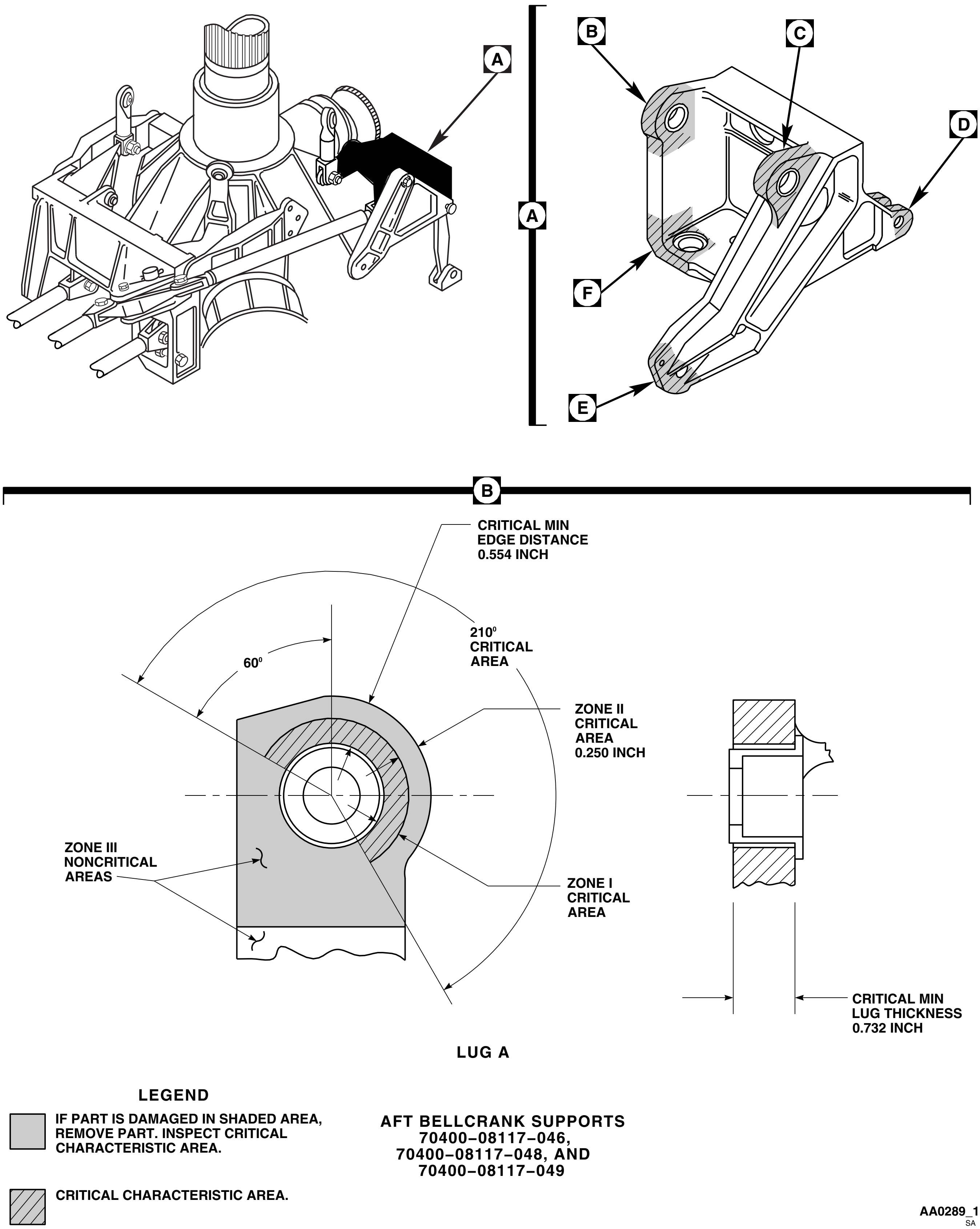


FIGURE 11-3-18-1. INSPECT AFT BELLCRANK SUPPORT (SHEET 1 OF 3)

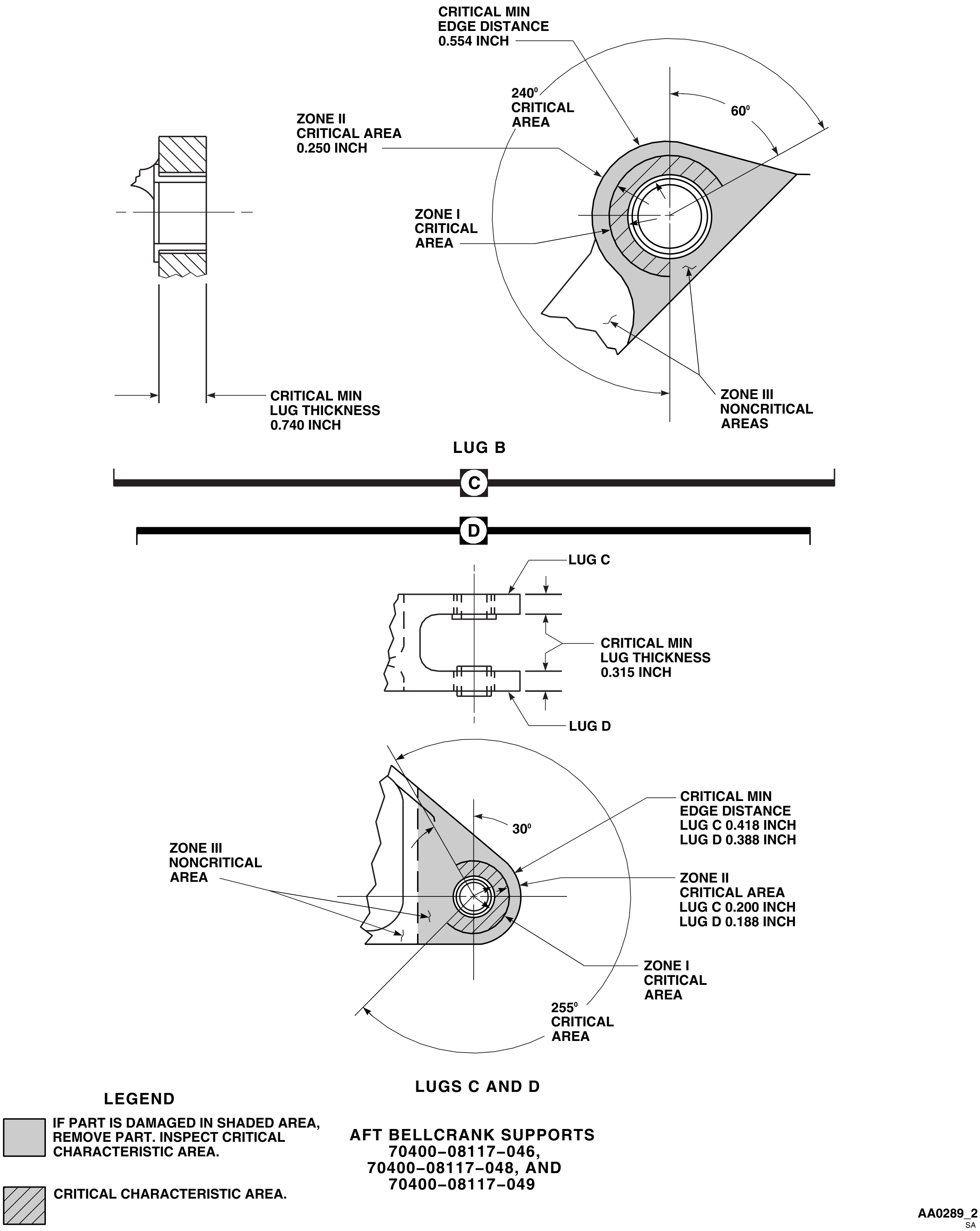


FIGURE 11-3-18-1. INSPECT AFT BELLCRANK SUPPORT (SHEET 2 OF 3)

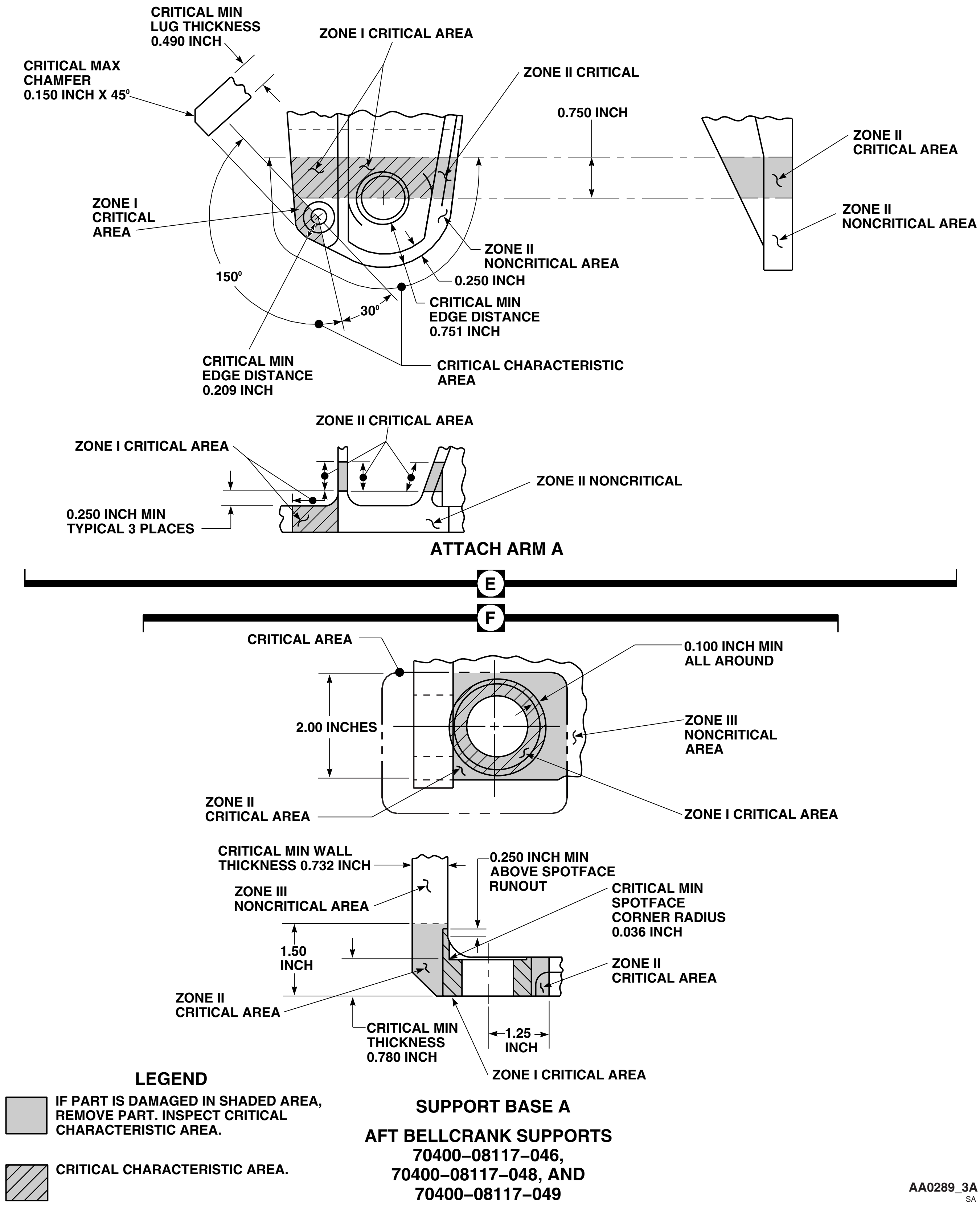


FIGURE 11-3-18-1. INSPECT AFT BELLCRANK SUPPORT (SHEET 3 OF 3)

previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If support base A zone II area has been previously reworked into base metal no further repair is allowed and part must be replaced (PARA 11-4-101).

(4) Check zone II critical areas of support base A for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM SPOT-FACE CORNER RADIUS, WALL THICKNESS AND BASE THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF WALL SECTION IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101). Record zone II repairs on component log card (SA Form 7343-21).

(5) Check noncritical zone III area of support base A for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF PART IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101).

d. On aft bellcrank supports, 70400-08117-046, 70400-08117-048, and 70400-08117-049, inspect attach arm A as follows (Figure 11-3-18-1, Sheet 3, Detail E):

(1) Check zone I critical area of attach arm A for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101).

(2) Check zone I critical area of attach arm A for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OF PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT** (PARA 2-6-5). If damage penetrates into base metal, replace support (PARA 11-4-101).

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an approved overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if attach arm A zone II critical area was previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If attach arm A zone II area has been previously reworked into base metal no further repair is allowed and support must be replaced (PARA 11-4-101).

(4) Check zone II critical areas of attach arm A for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE, LUG THICKNESS, AND MAXIMUM CHAMFER ARE MAINTAINED, AND**

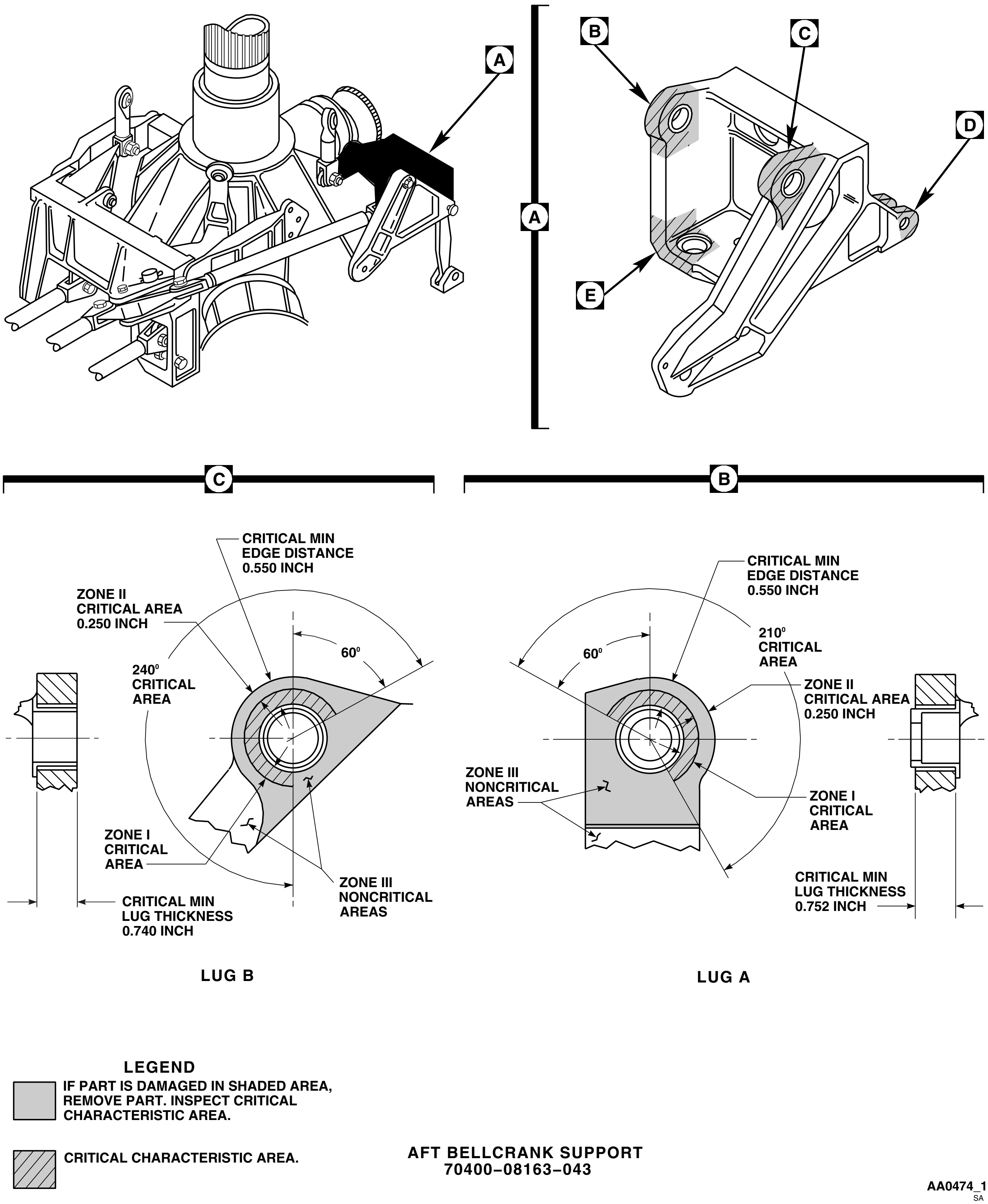


FIGURE 11-3-18-2. INSPECT AFT BELLCRANK SUPPORT (SHEET 1 OF 2)

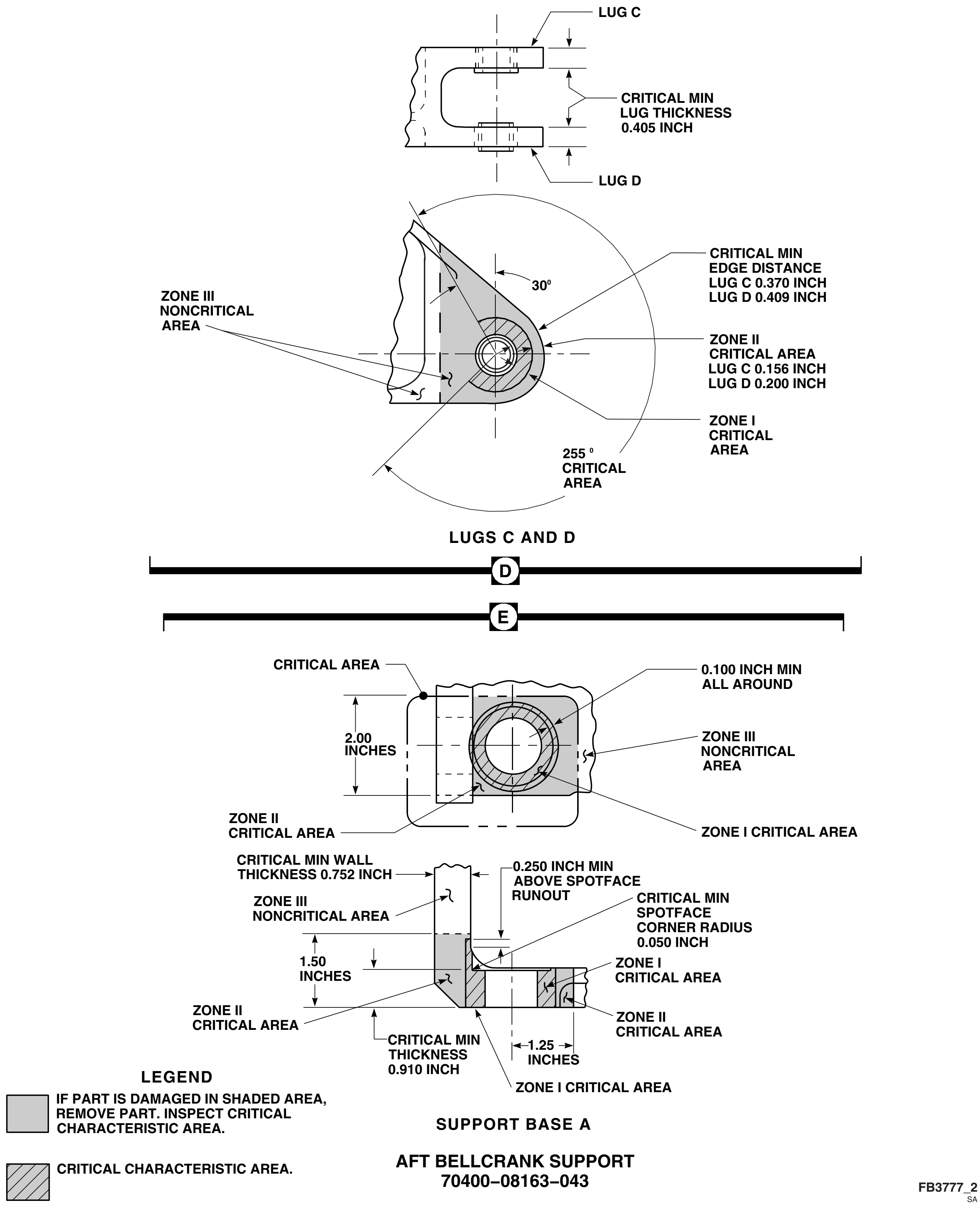


FIGURE 11-3-18-2. INSPECT AFT BELLCRANK SUPPORT (SHEET 2 OF 2)

THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF ATTACH ARM IN SAME AREA. Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101). Record zone II repairs on component log card (SA Form 7343-21).

(5) Check noncritical zone III area of attach arm A for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF ATTACH ARM IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (PARA 11-4-101).

e. Check for surface damage on noncritical zone III areas of aft bellcrank which were not previously inspected. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF ATTACH ARM IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support (Figure 11-3-18-1, Sheet 1, Detail B, Sheet 2, Detail C and Detail D, and Sheet 3, Detail E and Detail F, and Figure 11-3-18-2, Sheet 1, Detail B and Detail C and Sheet 2, Detail D and Detail E) (PARA 11-4-101).

f. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support (PARA 11-4-101).

g. Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas.

h. Touch up finish on all reworked areas (PARA 2-6-5).

i. Install main rotor pylon transmission cover (PARA 2-4-107).

11-3-19 AFT TIE ROD AND SUPPORT FITTING INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-4-103

11-3-19.1 Aft Tie Rod and Support Fitting FSCAP Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

WARNING

FSCAP

Do not attempt to repair damage that penetrates into base metal in zone I critical area. Damage to paint or protective finishes not penetrating into base metal may be touched up. Parts with damage into base metal that are within the zone I critical area are to be replaced.

NOTE

If damage is found in shaded areas on aft tie rod or support fitting, aft tie rod or support fitting must be removed (PARA 11-4-103), and critical characteristic areas inspected (Figure 11-3-19-1, Sheet 2, Detail A and Sheet 3, Detail B).

b. Inspect aft tie rod as follows (Figure 11-3-19-1, Sheet 2, Detail A):

(1) Check zone I critical areas of aft tie rod for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If rework penetrates into base metal, replace aft tie rod (PARA 11-4-103).

(2) Check zone I critical areas for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OR PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT** (PARA 2-6-5). If damage penetrates into

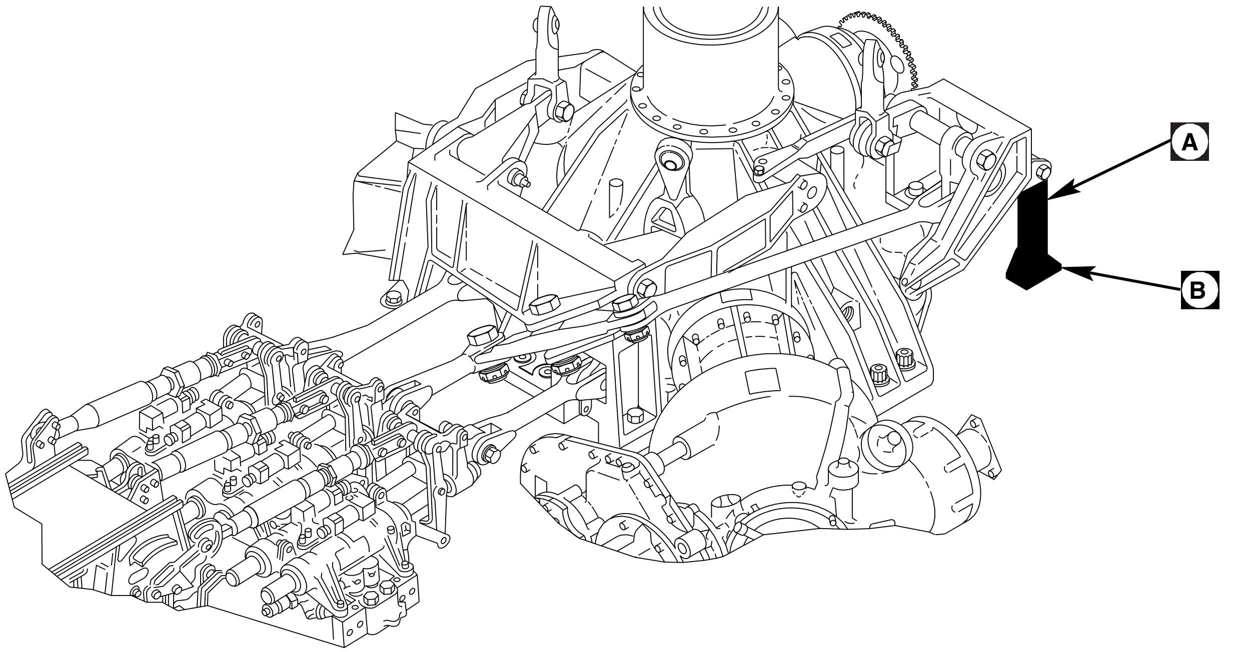


FIGURE 11-3-19-1. INSPECT AFT TIE ROD AND SUPPORT FITTING (SHEET 1 OF 3)

base metal, replace aft tie rod (PARA 11-4-103).

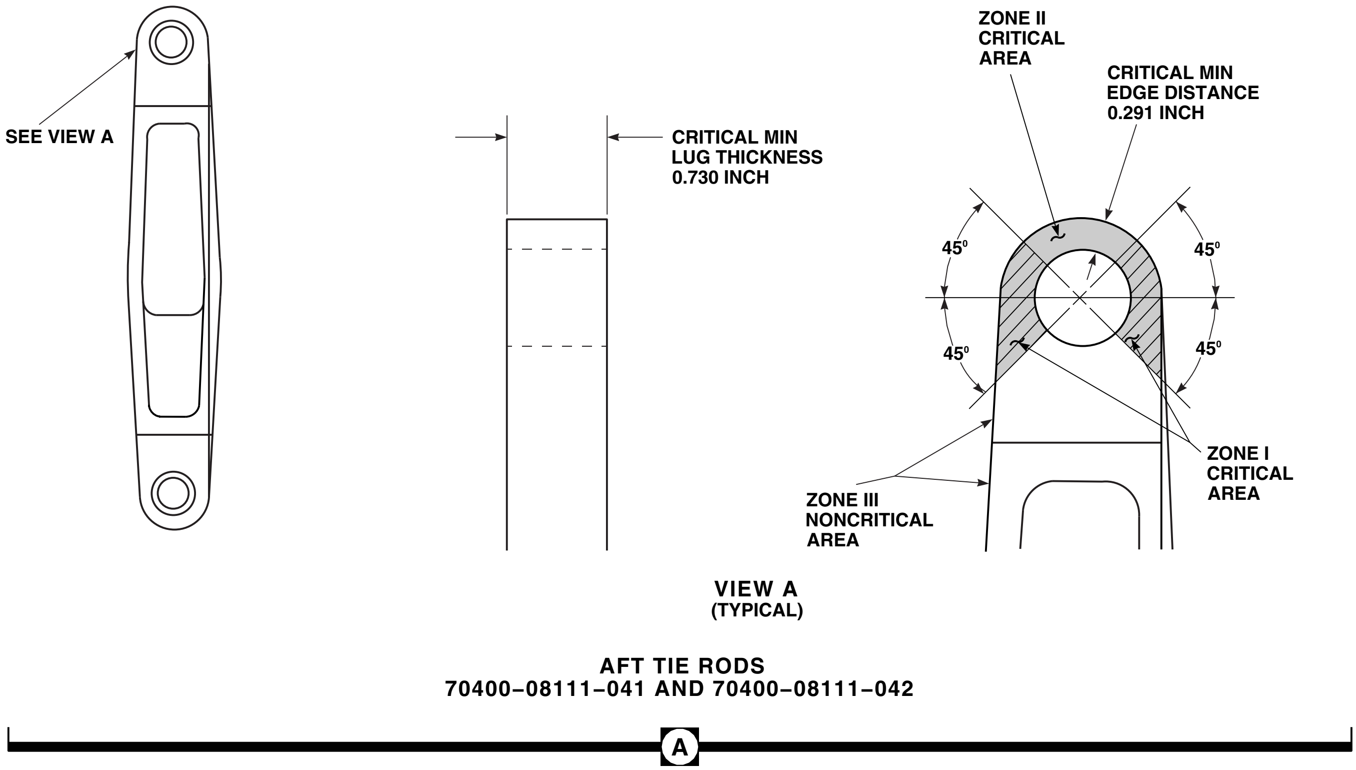
NOTE

- Rework of zone II critical areas is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if aft tie rod zone II critical area was previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If aft tie rod zone II area has been previously reworked into base metal no further repair is allowed and part must be replaced (PARA 11-4-103).

(4) Check zone II critical areas of aft tie rod for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF TIE ROD IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace aft tie rod (PARA 11-4-103). Record zone II repairs on component log card (SA Form 7343-21).

(5) Check noncritical zone III area of aft tie rod for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO**



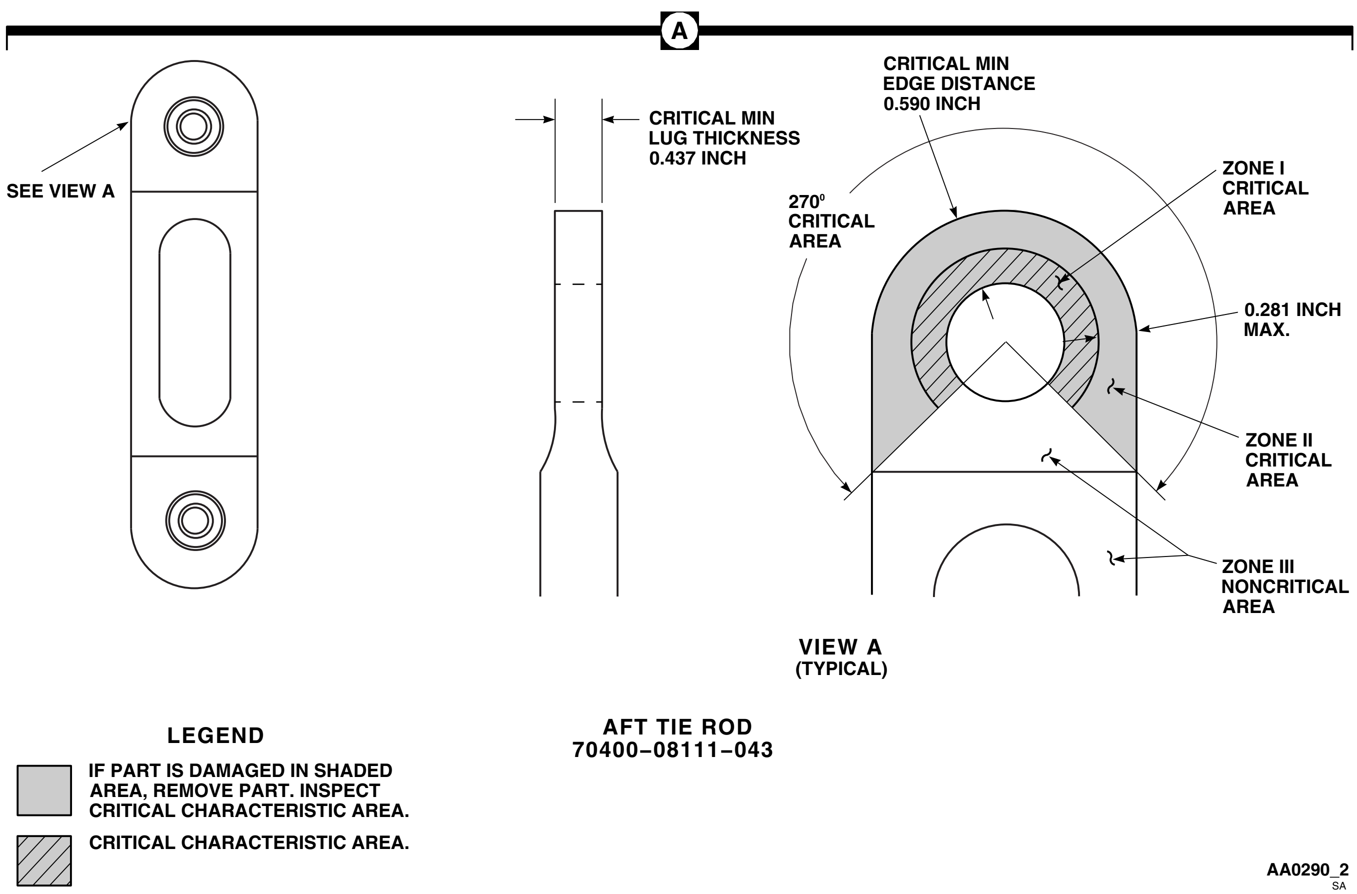
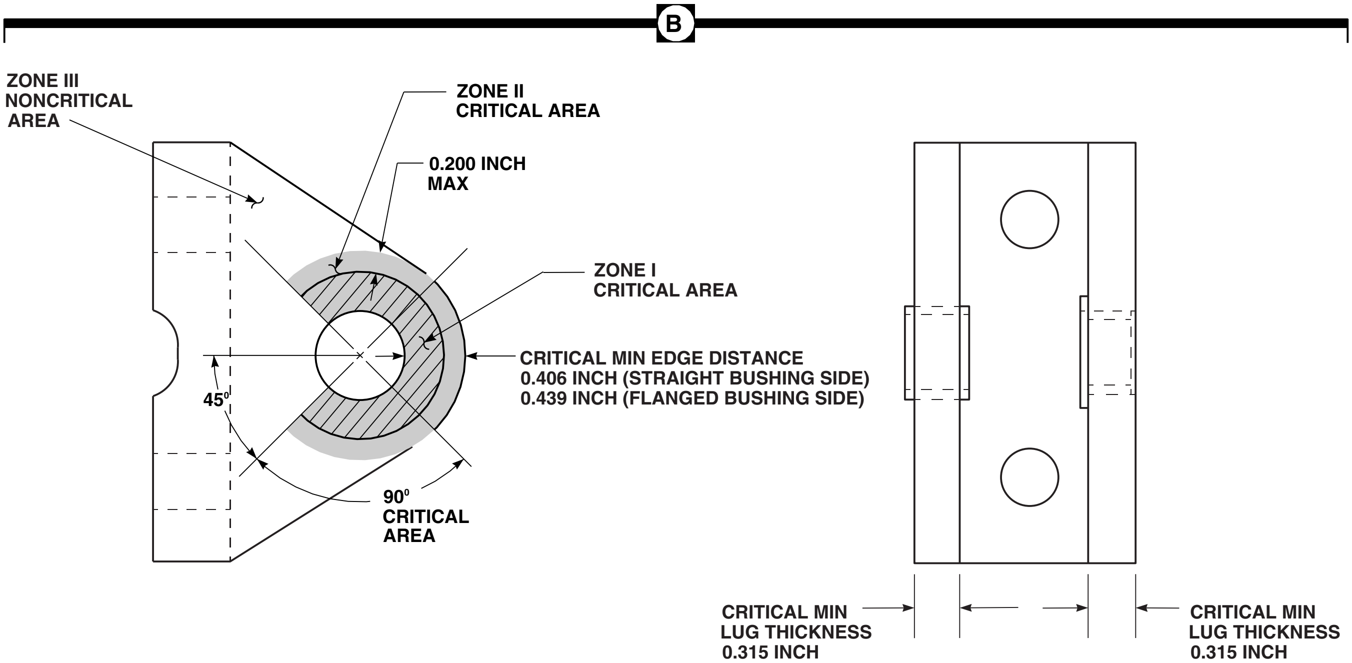
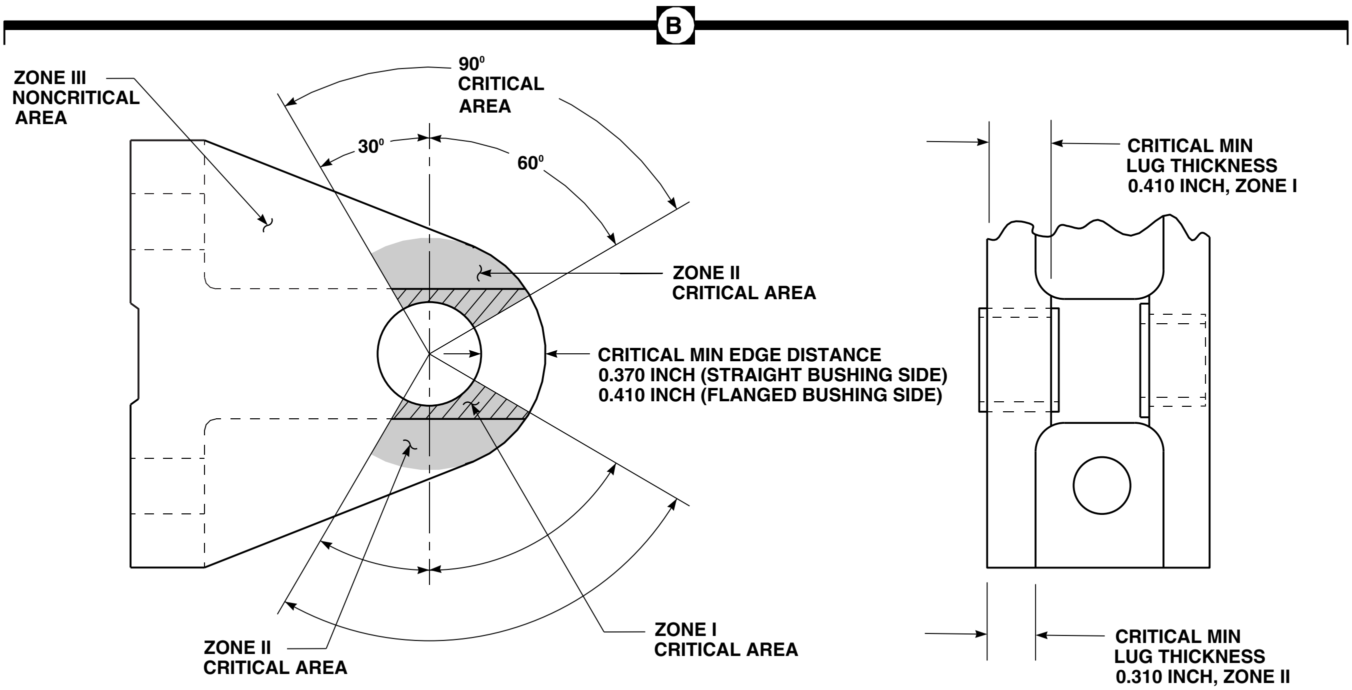


FIGURE 11-3-19-1. INSPECT AFT TIE ROD AND SUPPORT FITTING (SHEET 2 OF 3)



AFT SUPPORT FITTINGS
70400–08112–044 AND
70400–08112–045



LEGEND

IF PART IS DAMAGED IN SHADED AREA, REMOVE PART. INSPECT CRITICAL CHARACTERISTIC AREA.

CRITICAL CHARACTERISTIC AREA.

AFT SUPPORT FITTING
70400–08112–046

FB3778
SA

FIGURE 11-3-19-1. INSPECT AFT TIE ROD AND SUPPORT FITTING (SHEET 3 OF 3)

TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF PART IN SAME AREA. Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace aft tie rod (PARA 11-4-103).

(6) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace aft tie rod (PARA 11-4-103).

(7) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas.

(8) Touch up finish all reworked areas (PARA 2-6-5).

c. Inspect aft tie rod, 70400-08111-043, for bearing wear as follows:

(1) Measure radial play at each end of tie rod using dial indicator. **MAXIMUM RADIAL PLAY SHALL NOT EXCEED 0.005-INCH.** If radial play exceeds 0.005-inch, replace aft tie rod (PARA 11-4-103).

(2) Measure axial play at each end of tie rod using dial indicator. **MAXIMUM AXIAL PLAY SHALL NOT EXCEED 0.008-INCH.** If axial play exceeds 0.008-inch, replace aft tie rod (PARA 11-4-103).

d. Inspect aft tie rod support fitting as follows (Figure 11-3-19-1, Sheet 3, Detail B):

WARNING

FSCAP

Do not attempt to repair damage that penetrates into base metal in zone I critical area. Damage to paint or protective finishes not penetrating into base metal may be touched up. Parts with damage into base metal that are within the zone I critical area are to be replaced.

(1) Check zone I critical areas on each lug of support fitting for minor surface corrosion. **NONE ALLOWED.** Surface corrosion may be carefully blended out using abrasive cloth, Item 8, Appendix D. If rework penetrates into base metal, replace support fitting (PARA 11-4-103).

(2) Check zone I critical areas of tie rod support fitting for nicks, dents, scratches, and tool or scuff marks. **DAMAGE IN PAINT OR PROTECTIVE FINISHES NOT PENETRATING INTO BASE METAL MAY BE REPAIRED WITH TOUCHUP PAINT** (PARA 2-6-5). If damage penetrates into base metal, replace support fitting (PARA 11-4-103).

NOTE

- Rework of zone II critical area is limited to a one-time repair. All rework of zone II which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone II critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.

(3) Check log card (SA Form 7343-21) to see if support fitting zone II critical area was previously reworked into base metal. **ONE-TIME REWORK OF ZONE II CRITICAL AREA ALLOWED.** If support fitting zone II area has been previously reworked into base metal, no further repair is allowed and part must be replaced (PARA 11-4-103).

(4) Check zone II critical areas of each lug on support fitting for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND**

LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA. Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support fitting (PARA 11-4-103). Record zone II repairs on component log card (SA Form 7343-21).

(5) Check noncritical zone III area for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF PART IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support fitting (PARA 11-4-103).

(6) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support fitting (PARA 11-4-103).

(7) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas.

(8) Touch up finish all reworked areas (PARA 2-6-5).

e. Install main rotor pylon transmission cover (PARA 2-4-107).

11-3-20 AFT BELLCRANK INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-24
- PARA 11-4-98

11-3-20.1 Aft Bellcrank FSCAP Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in critical areas are limited to a one-time repair.
- All rework of critical areas which penetrate into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.
- If damage is found in shaded areas on aft bellcrank, aft bellcrank must be removed (PARA 11-4-98), and critical characteristic areas inspected (Figure 11-3-20-1, Detail A).

b. Check critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-20-1, View A). **NONE ALLOWED.** If damage is found, repair aft bellcrank as follows:

(1) Check log card (SA Form 7343-21) to see if aft bellcrank critical areas have been previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREAS ALLOWED.** If aft bellcrank critical areas have been previously reworked into base metal, and damage penetrates into base metal, replace aft bellcrank (PARA 11-4-98).

(2) **DAMAGE IN ZONE I CRITICAL AREA MAY BE BLENDED UP TO 0.003-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace aft bellcrank (PARA 11-4-98). Record any zone I rework into base metal on component log card (SA Form 7343-21).

(3) **DAMAGE IN ZONE II CRITICAL AREA MAY BE BLENDED UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace aft bellcrank (PARA 11-4-98). Record any zone II rework into base metal on component log card (SA Form 7343-21).

c. Check noncritical zone III area for surface corrosion, nicks, dents, scratches, and tool or scuff marks. This includes the inside diameter of the bellcrank body.**NONE ALLOWED. DAMAGE IN ZONE II CRITICAL AREA MAY BE BLENDED UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace aft bellcrank (PARA 11-4-98).

d. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace aft bellcrank (PARA 11-4-98).

e. Apply corrosion compound, Item 108, Appendix D, to reworked areas of aft bellcrank with acid brush, Item 84, Appendix D. Allow to dry for 30 seconds then rinse thoroughly with water. Allow aft bellcrank to dry completely.

f. Touch up finish all reworked areas (PARA 2-6-5).

g. Install main rotor pylon transmission cover (PARA 2-4-107).

11-3-20.2 Aft Bellcrank Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Using dial indicator, measure radial play. **MAXIMUM RADIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If radial play exceeds limit, replace aft bellcrank (PARA 11-4-98).

c. Measure combined total radial play from servos to swashplate links (PARA 11-3-24).

d. Using dial indicator, measure axial play. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace aft bellcrank (PARA 11-4-98).

e. Using dial indicator, measure axial play between aft bellcrank and aft bellcrank support. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.040-INCH.** If axial play exceeds limit, replace aft bellcrank (PARA 11-4-98).

f. Install main rotor pylon transmission cover (PARA 2-4-107).

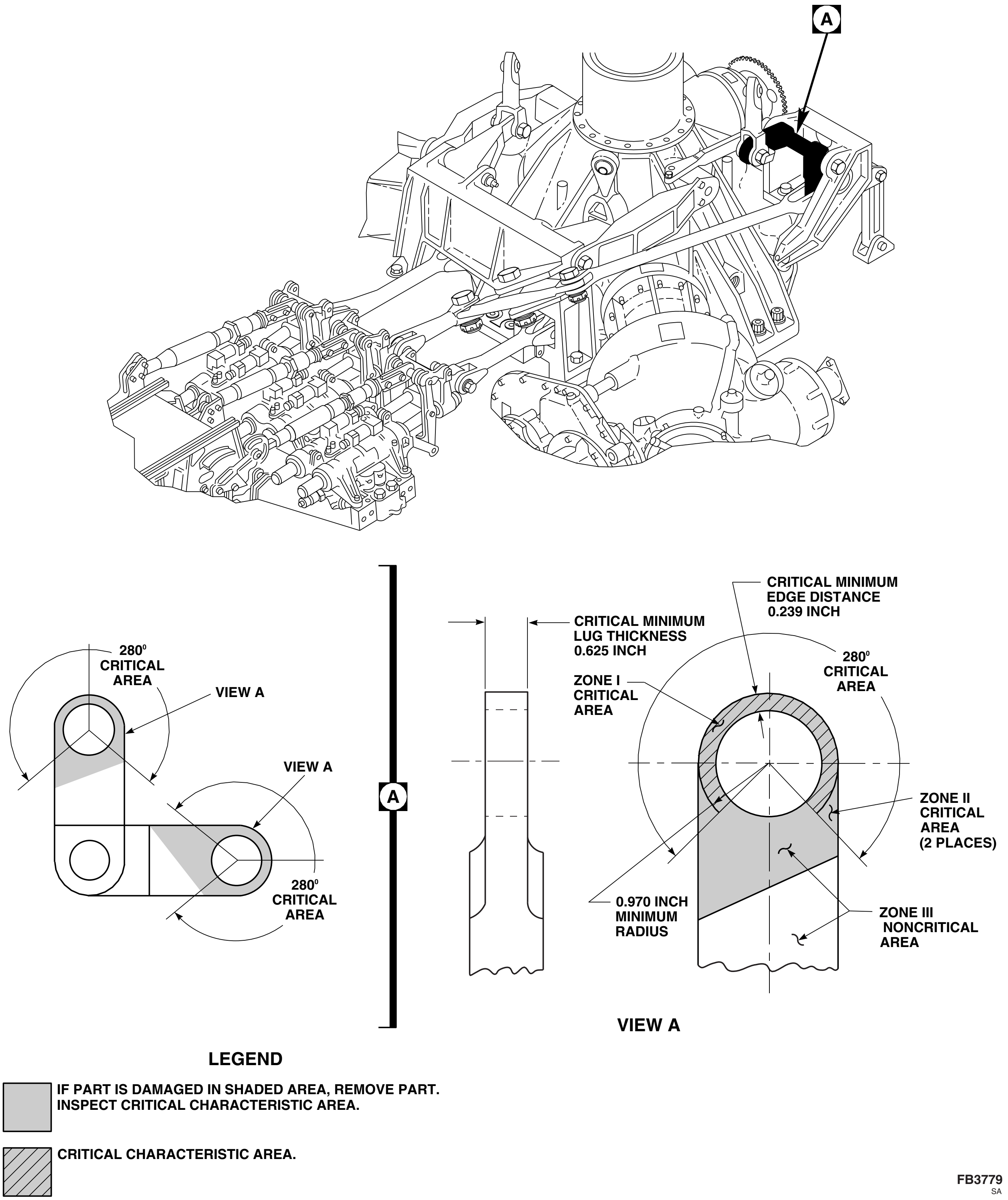


FIGURE 11-3-20-1. INSPECT AFT BELLCRANK

11-3-21 FORWARD BELLCRANK INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-24
- PARA 11-4-96

11-3-21.1 Forward Bellcrank FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in a critical area is limited to a one-time repair.
- All rework of critical areas which penetrate into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.
- If damage is found in shaded areas on forward bellcrank, forward bellcrank must be removed (PARA 11-4-96), and critical characteristic areas inspected (Figure 11-3-21-1, Detail A).

c. Check zone I critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-21-1, View A and View B). **NONE ALLOWED.** If damage is found, repair forward bellcrank as follows:

(1) Check log card (SA Form 7343-21) to see if forward bellcrank critical areas have been previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREAS ALLOWED.** If bellcrank critical

areas have been previously reworked into base metal, and damage penetrates into base metal, replace forward bellcrank (PARA 11-4-96).

(2) **DAMAGE IN ZONE I CRITICAL AREA MAY BE BLENDED UP TO 0.003-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace forward bellcrank (PARA 11-4-96). Record any zone I rework into base metal on component log card (SA Form 7343-21).

d. Check zone II noncritical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.010-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER OR LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace forward bellcrank (PARA 11-4-96).

e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace forward bellcrank (PARA 11-4-96).

f. Apply corrosion compound, Item 108, Appendix D, to reworked areas of forward bellcrank with acid brush, Item 84, Appendix D. Allow to dry for 30 seconds then rinse thoroughly with water. Allow to dry completely.

g. Touch up finish all reworked areas (PARA 2-6-5).

h. Install main rotor pylon transmission cover (PARA 2-4-107).

i. Close main rotor pylon sliding cover.

11-3-21.2 Forward Bellcrank Bearing Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Measure radial play, using dial indicator. **MAXIMUM RADIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If radial play exceeds limit, replace forward bellcrank (PARA 11-4-96).

d. Measure combined total radial play from servos to swashplate links (PARA 11-3-24).

e. Measure axial play, using dial indicator. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace forward bellcrank (PARA 11-4-96).

f. Measure axial play, between forward bellcrank and forward bellcrank support, using dial indicator. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace forward bellcrank (PARA 11-4-96).

g. Install main rotor pylon transmission cover (PARA 2-4-107).

h. Close main rotor pylon sliding cover.

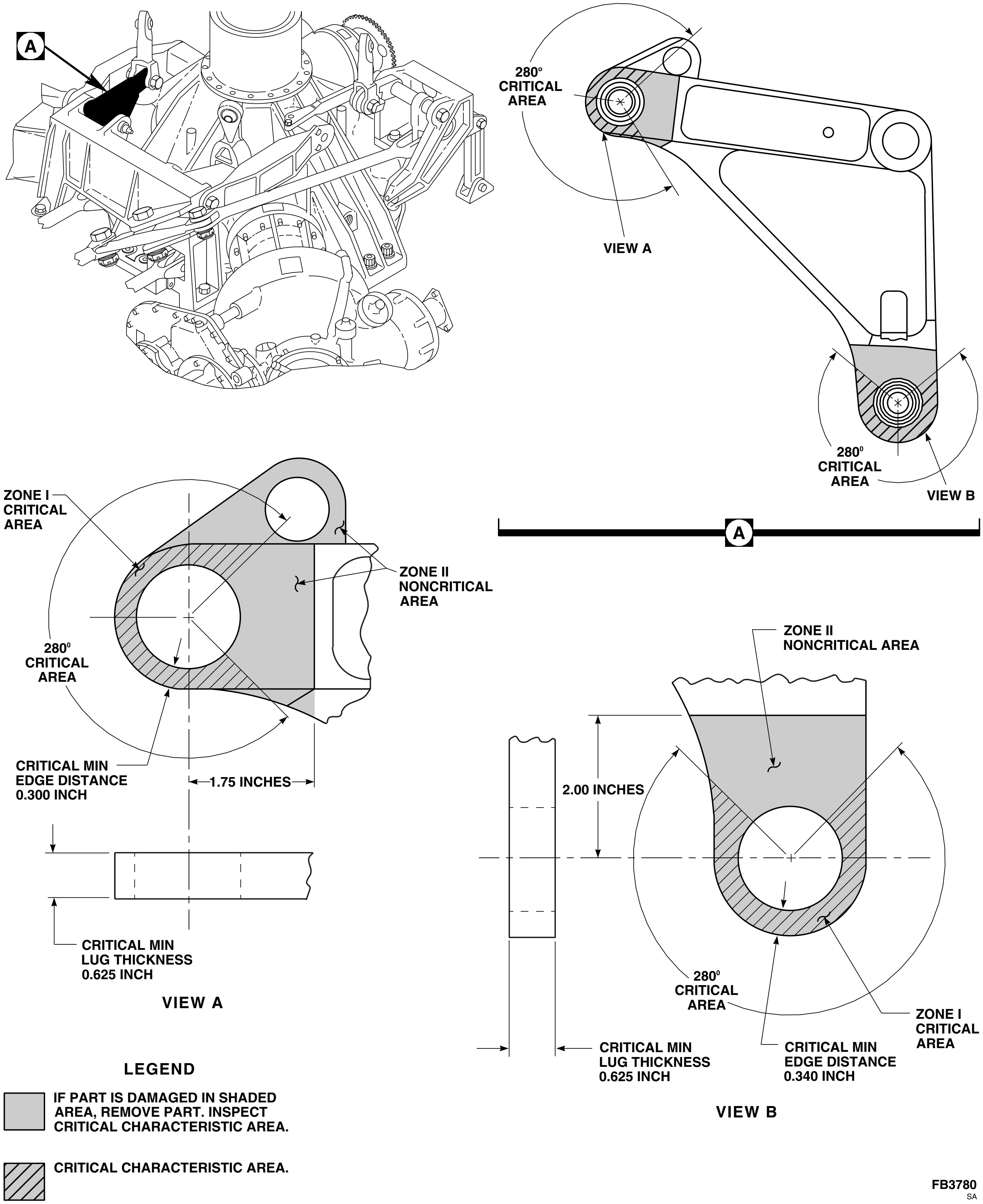


FIGURE 11-3-21-1. INSPECT FORWARD BELLCRANK

11-3-22 LATERAL BELLCRANK INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108 Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-24
- PARA 11-4-97

11-3-22.1 Lateral Bellcrank FSCAP Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in zone I critical area is limited to a one-time repair.
- All rework of zone I critical areas which penetrate into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in zone I critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than next 500 hour periodic inspection.
- If damage is found in shaded areas on lateral bellcrank, lateral bellcrank must be removed (PARA 11-4-97), and critical characteristic areas inspected (Figure 11-3-22-1, Sheet 1, Detail A and Sheet 2, Detail A).

c. Check zone I critical area for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-22-1, Sheet 1, View A and Sheet 2, View A). **NONE ALLOWED.** If damage is found, repair bellcrank as follows:

(1) Check log card (SA Form 7343-21) to see if zone I critical area has been previously reworked into base metal. **ONE-TIME REWORK OF ZONE I CRITICAL**

AREA ALLOWED. If zone I critical area has been previously reworked into base metal, replace lateral bellcrank (PARA 11-4-97).

(2) **DAMAGE IN ZONE I CRITICAL AREA MAY BE BLENDED UP TO 0.003-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT MINIMUM EDGE DISTANCE AND LUG THICKNESS ARE MAINTAINED, AND THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER AND LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace lateral bellcrank (PARA 11-4-97). Record any zone I rework into base metal on component log card (SA Form 7343-21).

d. On lateral bellcrank, 70400-08150-043 and 70400-08150-045, check zone II noncritical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-22-1, Sheet 1, View A and Sheet 2, View A). **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER AND LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace lateral bellcrank (PARA 11-4-97).

e. On lateral bellcrank, 70400-08166-041, check zone II critical area for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-22-1, Sheet 1, View A and Sheet 2, View A). **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER AND LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG IN SAME AREA.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace lateral bellcrank (PARA 11-4-97).

f. On lateral bellcrank, 70400-08166-041, check zone III noncritical area for surface damage. **NONE ALLOWED. DAMAGE MAY BE BLENDED OUT UP TO 0.020-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG, PROVIDED THAT NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER AND LESS THAN 0.500-INCH ON OPPOSITE SIDE OF LUG, FLANGE, WEB OR WALL SECTION.** Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace bellcrank (PARA 11-4-97).

g. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace lateral bellcrank (PARA 11-4-97).

h. Apply corrosion compound, Item 108, Appendix D, to reworked areas of bellcrank with acid brush, Item 84, Appendix D. Allow to dry for 30 seconds, then rinse thoroughly with water. Allow to dry completely.

i. Touch up finish all reworked areas (PARA 2-6-5).

j. Install main rotor pylon transmission cover (PARA 2-4-107).

k. Close main rotor pylon sliding cover.

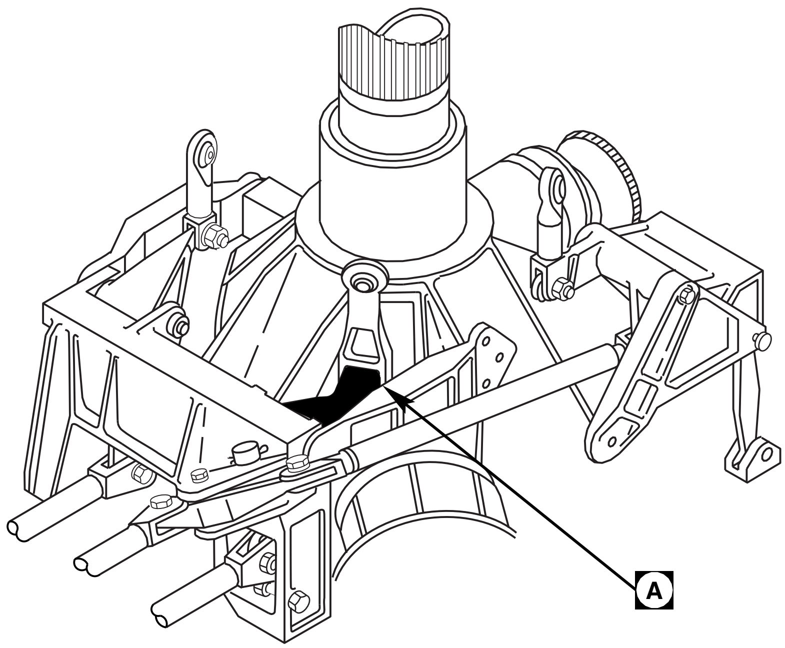
11-3-22.2 Lateral Bellcrank Bearing Inspection.

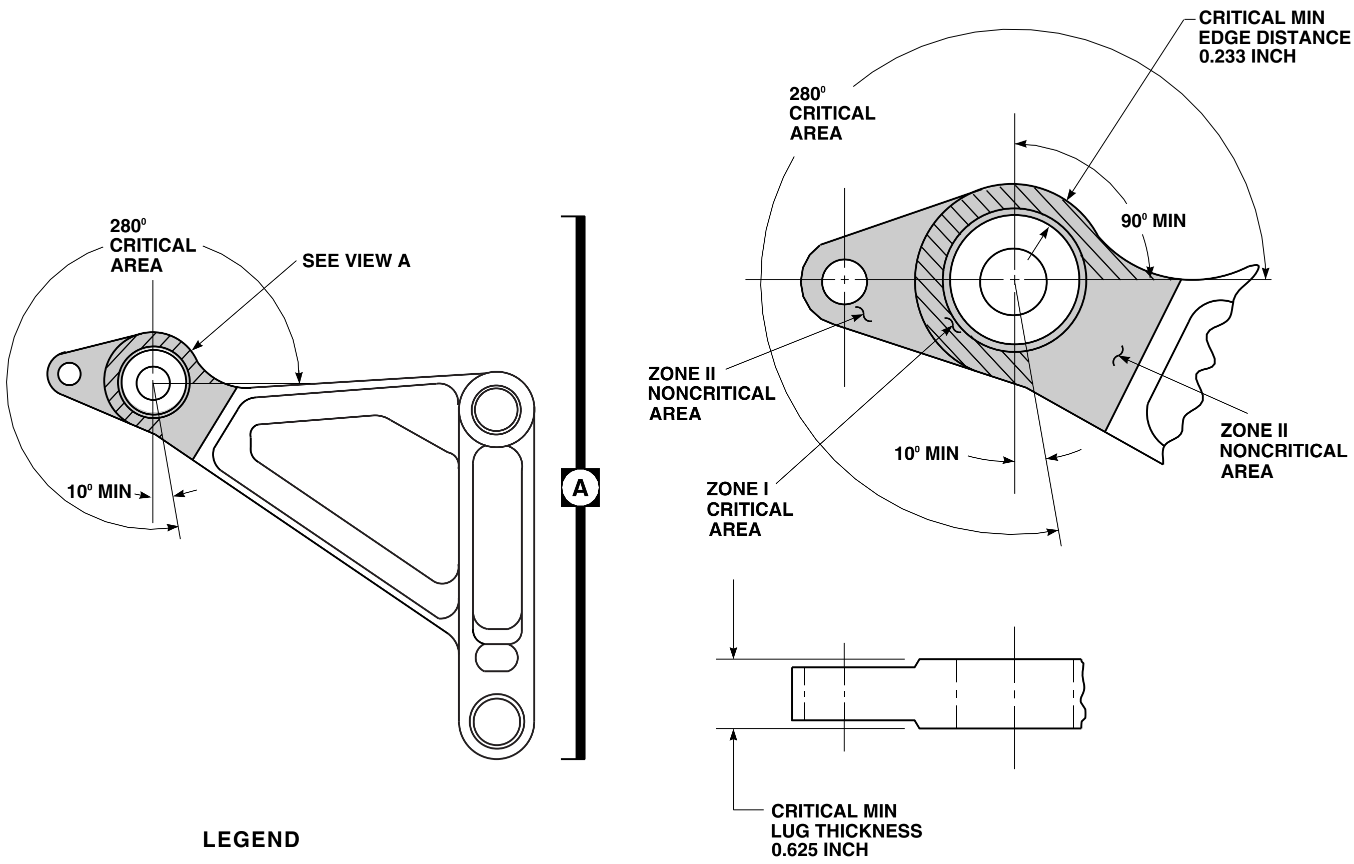
a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Using dial indicator, measure radial play. **MAXIMUM RADIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If radial play exceeds limit, replace lateral bellcrank (PARA 11-4-97).

d. Measure combined total radial play from servos to swashplate links (PARA 11-3-24).





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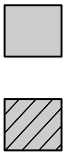


IF PART IS DAMAGED IN SHADED AREA, REMOVE PART. INSPECT CRITICAL CHARACTERISTIC AREA.

CRITICAL CHARACTERISTIC AREA.

LATERAL BELLCRANKS
70400–08150–043 AND
70400–08150–045

AA0295_1
SA

FIGURE 11-3-22-1. INSPECT LATERAL BELLCRANK (SHEET 1 OF 2)

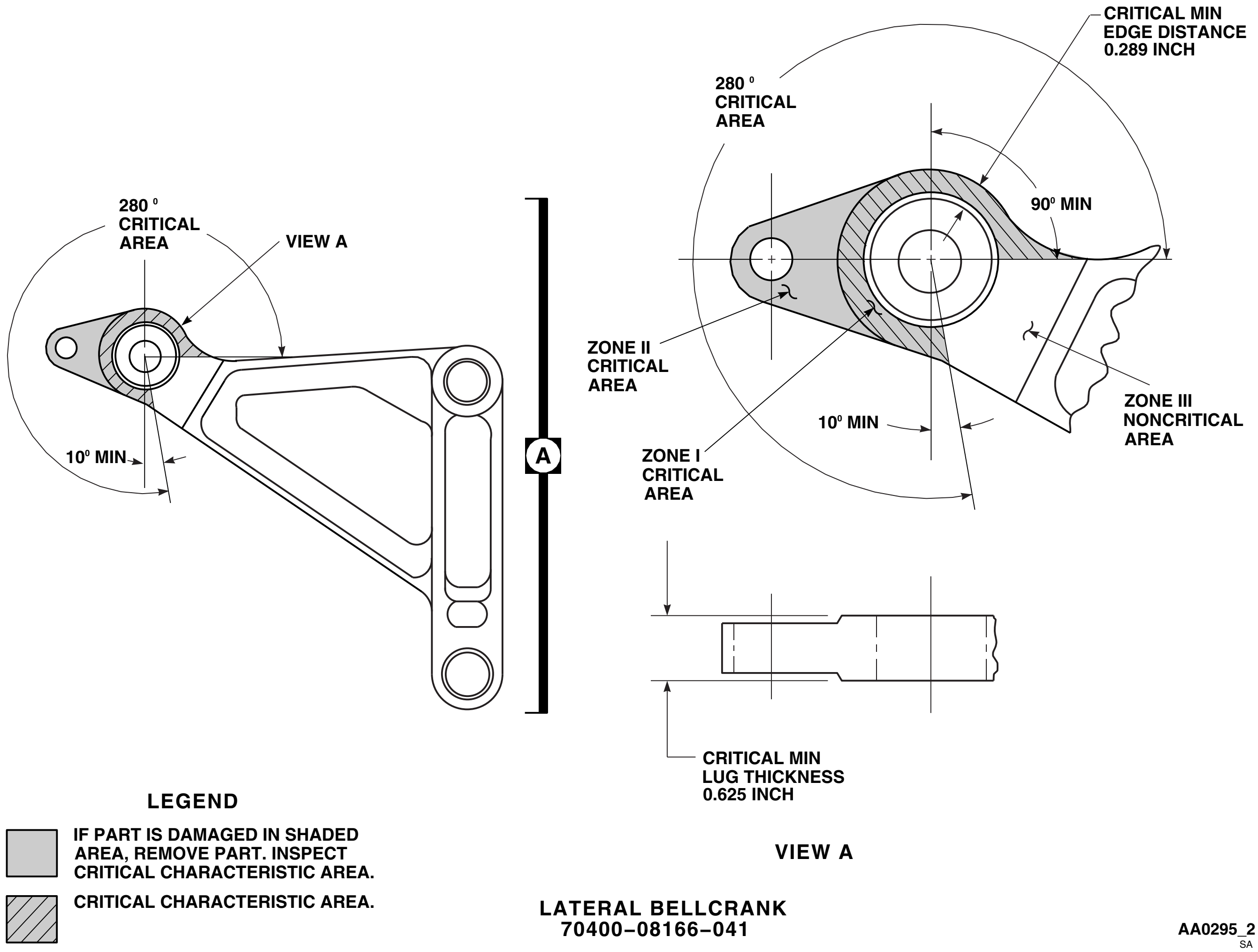


FIGURE 11-3-22-1. INSPECT LATERAL BELLCRANK (SHEET 2 OF 2)

e. Using dial indicator, measure axial play. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace lateral bellcrank (PARA 11-4-97).

f. Using dial indicator, measure axial play between lateral bellcrank and forward bellcrank support. **TOTAL AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace lateral bellcrank (PARA 11-4-97).

g. Install main rotor pylon transmission cover (PARA 2-4-107).

h. Close main rotor pylon sliding cover.

11-3-23 AFT LONGITUDINAL BELLCRANK SUPPORT ARM INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-4-102

11-3-23.1 Aft Longitudinal Bellcrank Support Arm FSCAP Inspection.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

NOTE

- Rework into base metal in critical areas are limited to a one-time repair.
- All rework of critical areas which penetrates into base metal must be recorded on component log card (SA Form 7343-21).
- Any part which is reworked into base metal in a critical area should be replaced and sent to an authorized overhaul facility as soon as possible, but no later than the next 500 hour periodic inspection.
- If damage is found in shaded areas on aft longitudinal bellcrank support arm, support arm must be removed (PARA 11-4-102), and critical characteristic areas inspected (Figure 11-3-23-1, Detail B and Detail C).

b. Check critical areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks (Figure 11-3-23-1, Detail B and Detail C). **NONE ALLOWED.** If damage is found, repair support arm as follows:

(1) Check log card (SA Form 7343-21) to see if support arm critical areas have been previously reworked into base metal. **ONE-TIME REWORK OF CRITICAL AREAS ALLOWED.** If support arm critical areas have been previously reworked into base metal and damage penetrates into base metal, replace support arm (PARA 11-4-102).

(2) **DAMAGE TO LARGE HOLE IN LUG A MAY BE BLENDED OUT UP TO 0.001-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG PROVIDED NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER** (Figure 11-3-23-1, Detail B). Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support arm (PARA 11-4-102). Record any rework into base metal on component log card (SA Form 7343-21).

(3) **DAMAGE TO HOLE IN LUG B MAY BE BLENDED OUT UP TO 0.005-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG PROVIDED NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER** (Figure 11-3-23-1, Detail C). Blend damage using abrasive cloth, Item 8, Appendix D. If damage or rework is beyond limits, replace support arm (PARA 11-4-102). Record any rework into base metal on component log card (SA Form 7343-21).

(4) **DAMAGE TO LUGS A AND B MAY BE BLENDED OUT UP TO 0.010-INCH DEEP, 0.100-INCH WIDE, AND 0.380-INCH LONG PROVIDED NO TWO DAMAGED AREAS ARE WITHIN 0.250-INCH OF EACH OTHER** (Figure 11-3-23-1, Detail B and Detail C). Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT ARM** (PARA 11-4-102). Record any rework into base metal on component log card (SA Form 7343-21).

c. Check noncritical surface areas for surface corrosion, nicks, dents, scratches, and tool or scuff marks. Damage may be blended out up to 0.020-inch deep, 0.150-inch wide, and 0.750-inch long. **MINIMUM DIMENSIONS AFTER CLEANUP SHALL BE 0.046-INCH AND 0.960-INCH** (Figure 11-3-23-1, Section A-A). Blend damage using abrasive cloth, Item 8, Appendix D. **IF DAMAGE OR REWORK IS BEYOND LIMITS, REPLACE SUPPORT ARM** (PARA 11-4-102).

d. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace support arm (PARA 11-4-102).

e. Apply corrosion compound, Item 108, Appendix D, to reworked area of support arm with acid brush, Item 84, Appendix D. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow support arm to dry completely.

f. Touch up finish all reworked areas (PARA 2-6-5).

g. Install main rotor pylon transmission cover (PARA 2-4-107).

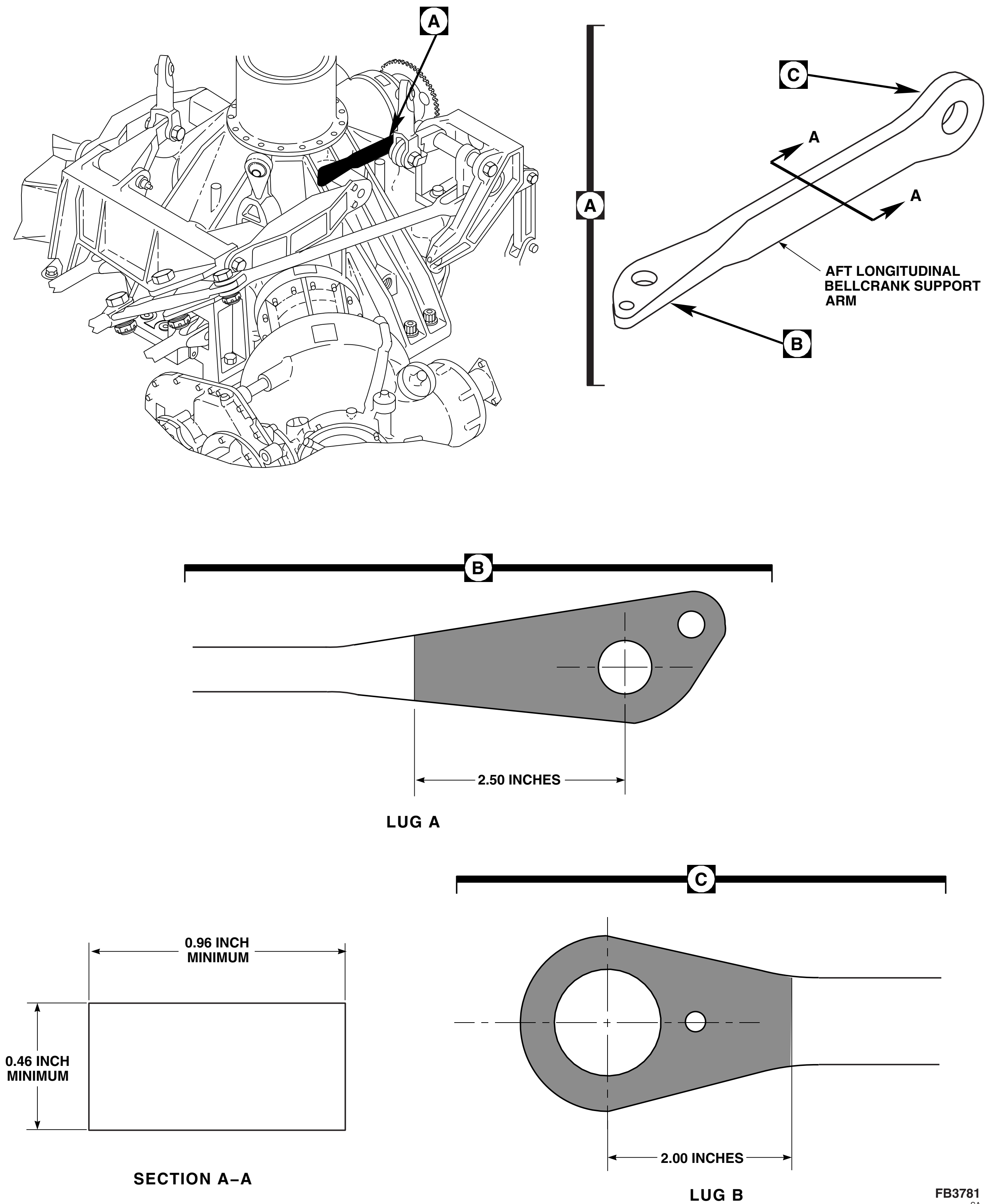


FIGURE 11-3-23-1. INSPECT AFT LONGITUDINAL BELLCRANK SUPPORT ARM

11-3-24 FORWARD, AFT, AND LATERAL FLIGHT CONTROL CHANNEL INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator

References

- PARA 2-4-107

11-3-24.1 Radial Play Inspection.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Inspect spherical bearing connections in the forward, aft, and lateral flight control channels for radial play using dial indicator (Figure 11-3-24-1, Detail A).

d. **MAXIMUM RADIAL PLAY MUST NOT EXCEED 0.020-INCH PER CONNECTION.**

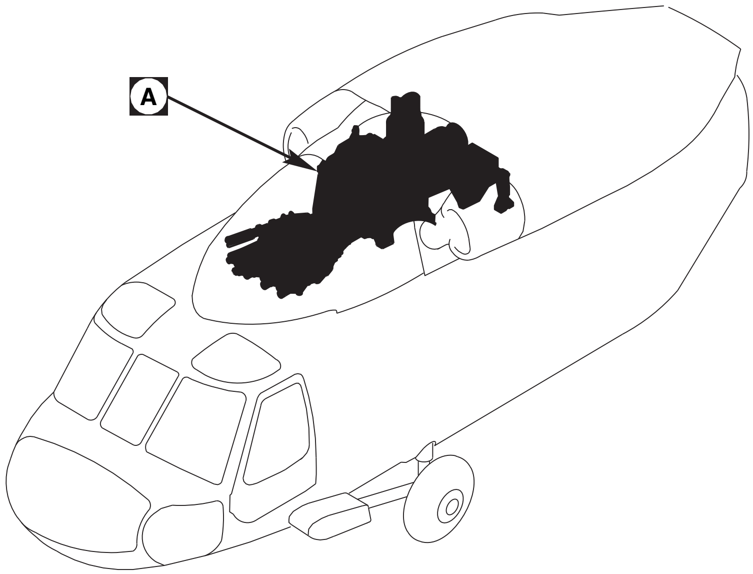
e. **THE COMBINED TOTAL RADIAL PLAY, OF ALL INDIVIDUAL SPHERICAL BEARING CONNECTIONS ON FORWARD LINK, FORWARD BELLCRANK, AND FORWARD PRIMARY SERVO SHALL NOT EXCEED 0.040-INCH.** If combined total radial play exceeds 0.040-inch, replace bearing(s) with greatest play to bring combined total radial play within limits.

f. **THE COMBINED TOTAL RADIAL PLAY, OF ALL INDIVIDUAL SPHERICAL BEARING CONNECTIONS ON AFT LINK, AFT BELLCRANK, WALKING BEAM, AND AFT PRIMARY SERVO SHALL NOT EXCEED 0.040-INCH.** If combined total radial play exceeds 0.040-inch, replace bearing(s) with greatest play to bring combined total radial play within limits.

g. **THE COMBINED TOTAL RADIAL PLAY, OF ALL INDIVIDUAL SPHERICAL BEARING CONNECTIONS ON LATERAL LINK, LATERAL BELLCRANK, AND LATERAL PRIMARY SERVO SHALL NOT EXCEED 0.040-INCH.** If combined total radial play exceeds 0.040-inch, replace bearing(s) with greatest play to bring combined total radial play within limits.

h. Install main rotor pylon transmission cover (PARA 2-4-107).

i. Close and latch main rotor pylon sliding cover.



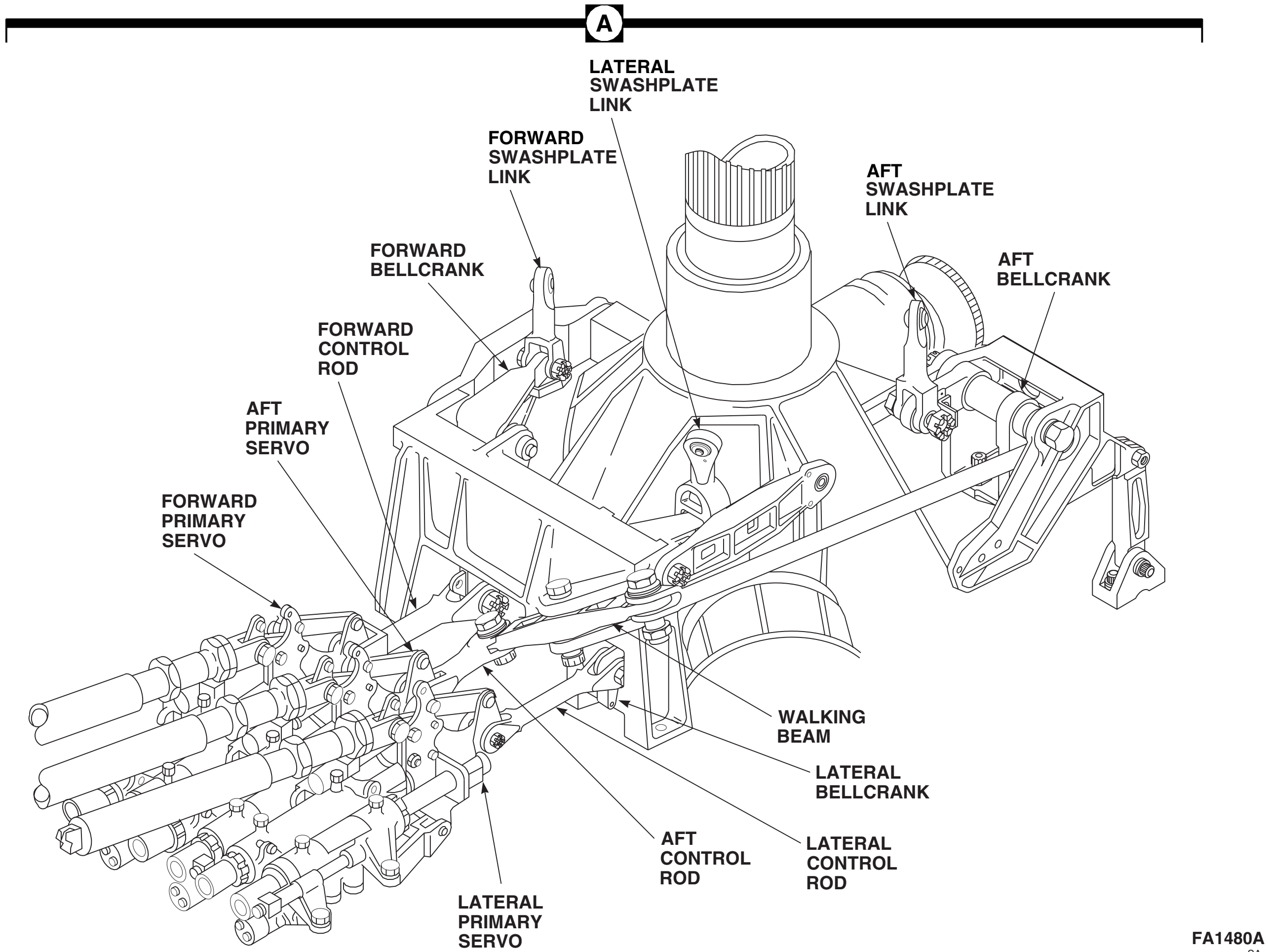


FIGURE 11-3-24-1 INSPECT RADIAL PLAY

11-3-25 FLIGHT CONTROL CABLE INSPECTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Adhesive, Item 49, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-69
- PARA 2-4-137
- PARA 2-6-44
- PARA 11-4-104
- PARA 11-4-105
- PARA 11-4-106
- PARA 11-4-107
- PARA 11-4-108

11-3-25.1 Cable Inspection.

a. Gain access to flight control cables as follows:

(1) For cabin flight control cables, perform the following:

(a) Remove soundproofing panels from cabin ceiling, STA 295.0 to rear of cabin (PARA 2-4-69).

(b) Remove cargo net from fuel panel enclosure.

(2) For tail cone flight control cables, perform the following:

(a) Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0 (Figure 11-3-25-1, Sheet 2, Detail A). Remove access panel.

(b) Remove soundproofing panels from rear bulkhead covering left or right stowage compartments (PARA 2-4-69).

(c) Remove cargo net from fuel panel enclosure.

(3) For tail rotor pylon flight control cables, perform the following:

(a) Open tail rotor pylon leading edge cover/antenna.

(b) Remove tail gear box front fairing (PARA 2-4-137).

(c) Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0. Remove access panel.

b. Inspect cable plastic coatings for wear. **WEAR MUST NOT EXPOSE BARE METAL.** If plastic coating is worn through exposing bare cable, replace cable (PARA 11-4-104) for cabin cables, (PARA 11-4-105) for tail cone cables, and (PARA 11-4-106) for tail pylon cables.

c. Inspect plastic cable grommets at STA 311.0, STA 327.0, STA 360.0, STA 379.0, STA 386.0, STA 389.0, STA 398.0, STA 433.0, and STA 443.0 for condition and security (Figure 11-3-25-1, Sheet 2, Detail B). If worn or loose, replace grommet(s) as follows:

(1) Remove old grommet from airframe. Remove any remaining adhesive with nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(2) Lightly scuff grommet-to-airframe mounting surface with abrasive cloth, Item 1A, Appendix D.

(3) Thoroughly clean airframe mounting surface of any contaminants using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(4) Apply a thin, even film of adhesive, Item 49, Appendix D, to mating surfaces of grommet and airframe. Allow adhesive to cure for approximately 10 minutes or until tacky.

(5) Install grommet to airframe with cut at top using firm, even pressure, making sure all mating surfaces are in direct contact. Do not move or adjust grommet after pressing onto airframe.

(6) Allow adhesive to cure for 24 hours at room temperature before installing cables.

d. Inspect to make sure locking clips are installed on turnbuckles. Inspect threaded end of each terminal for alignment of keyway and witness mark on swaged fitting. **NO MISALIGNMENT ALLOWED.** If keyway is not visible, use turnbuckle locking clip to provide a reference point.

e. Inspect turnbuckle for safety. **NO MORE THAN THREE THREADS OF TERMINAL TO SHOW BEYOND END OF TURNBUCKLE.** If more than three threads are showing, check cable for proper installation.

f. Inspect cable guard pins on pulleys for wear (Figure 11-3-25-1, Sheet 3, Detail C). **NO BENDS OR NOTCHES ALLOWED.** If guard pins are bent or notched, replace guard pins.

g. Perform the following:

(1) For cabin flight control cables, perform the following:

(a) Install soundproofing panels (PARA 2-4-69).

(b) Install cargo net.

(2) For tail cone flight control cables, perform the following:

(a) Install tail cone flight controls access panel on tail cone and secure with screws and washers (Figure 11-3-25-1, Sheet 2, Detail A).

(b) Remove soundproofing panels (PARA 2-4-69).

(c) Install cargo net.

(3) For tail rotor pylon flight control cables, perform the following:

(a) Close tail rotor pylon leading edge cover/antenna.

(b) Install tail gear box front fairing (PARA 2-4-137).

(c) Install tail cone flight controls access panel on tail cone and secure with screws and washers.

11-3-25.2 Tail Rotor Pylon Cable Conduit Inspection.

a. Open tail rotor pylon leading edge cover/antenna.

b. Inspect cable conduit for damage or wear (Figure 11-3-25-1, Sheet 1). **WEAR SHALL NOT EXCEED ½ OF TUBE THICKNESS.** If wear exceeds limit, replace conduit (PARA 11-4-108).

c. Close tail rotor pylon leading edge cover/antenna.

11-3-25.3 Cable Pulley Bracket Inspection.

a. Gain access to flight control cable pulleys as follows:

(1) Remove soundproofing panels as required (PARA 2-4-69).

(2) Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0 (Figure 11-3-25-1, Sheet 2, Detail A). Remove access panel.

(3) Open tail rotor pylon leading edge cover/antenna.

(4) Remove tail gear box front fairing (PARA 2-4-137).

b. Visually inspect pulleys for damage (Figure 11-3-25-1, Sheet 2, Detail B). **NO CRACKS OR CHIPS ALLOWED. NO DAMAGE/WEAR THAT EXPOSES FABRIC ALLOWED.** If pulley is cracked, chipped, or has damage/wear that exposes fabric, replace pulley (PARA 11-4-107).

c. Visually inspect cable pulley brackets for cracks and sheet metal bracket flanges for deformation (Figure 11-3-25-1, Sheet 3, Detail D). **NONE ALLOWED.** If cracks or deformation is found, replace cable pulley bracket (PARA 2-6-44).

d. For suspect cracks, strip paint from part. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect any suspect cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace cable pulley (PARA 2-6-44).

e. Perform the following:

(1) Install soundproofing panels (PARA 2-4-69).

(2) Install tail cone flight controls access panel on tail cone and secure with screws and washers (Figure 11-3-25-1, Sheet 2, Detail A).

(3) Close tail rotor pylon leading edge cover/antenna.

(4) Install tail gear box front fairing (PARA 2-4-137).

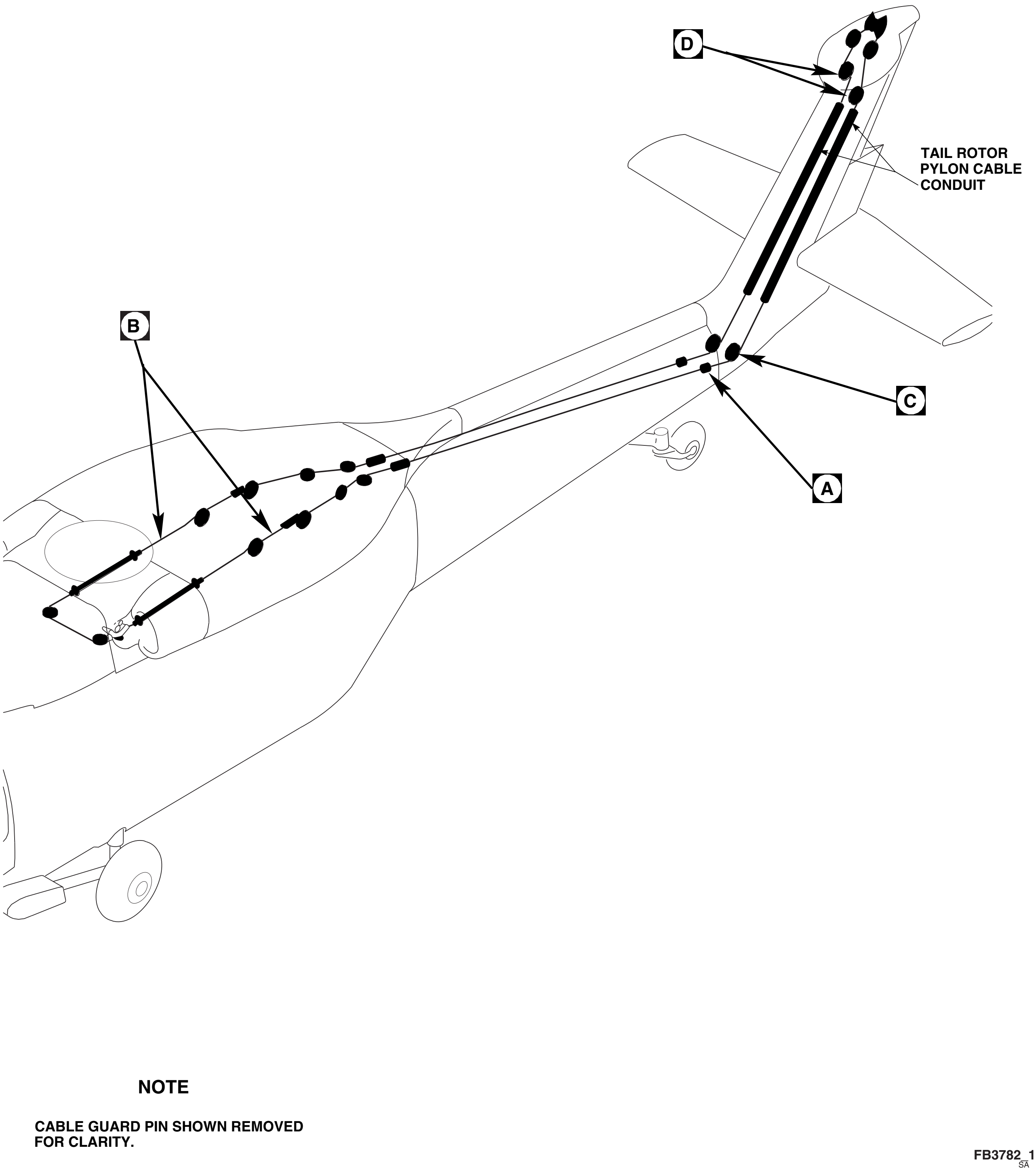
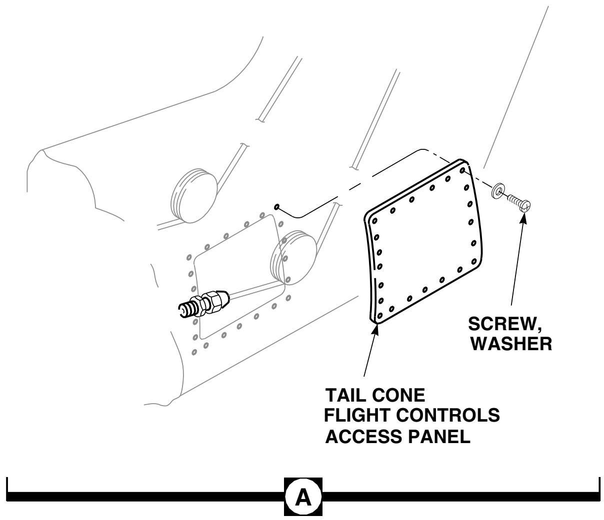


FIGURE 11-3-25-1. INSPECT FLIGHT CONTROL CABLES (SHEET 1 OF 3)



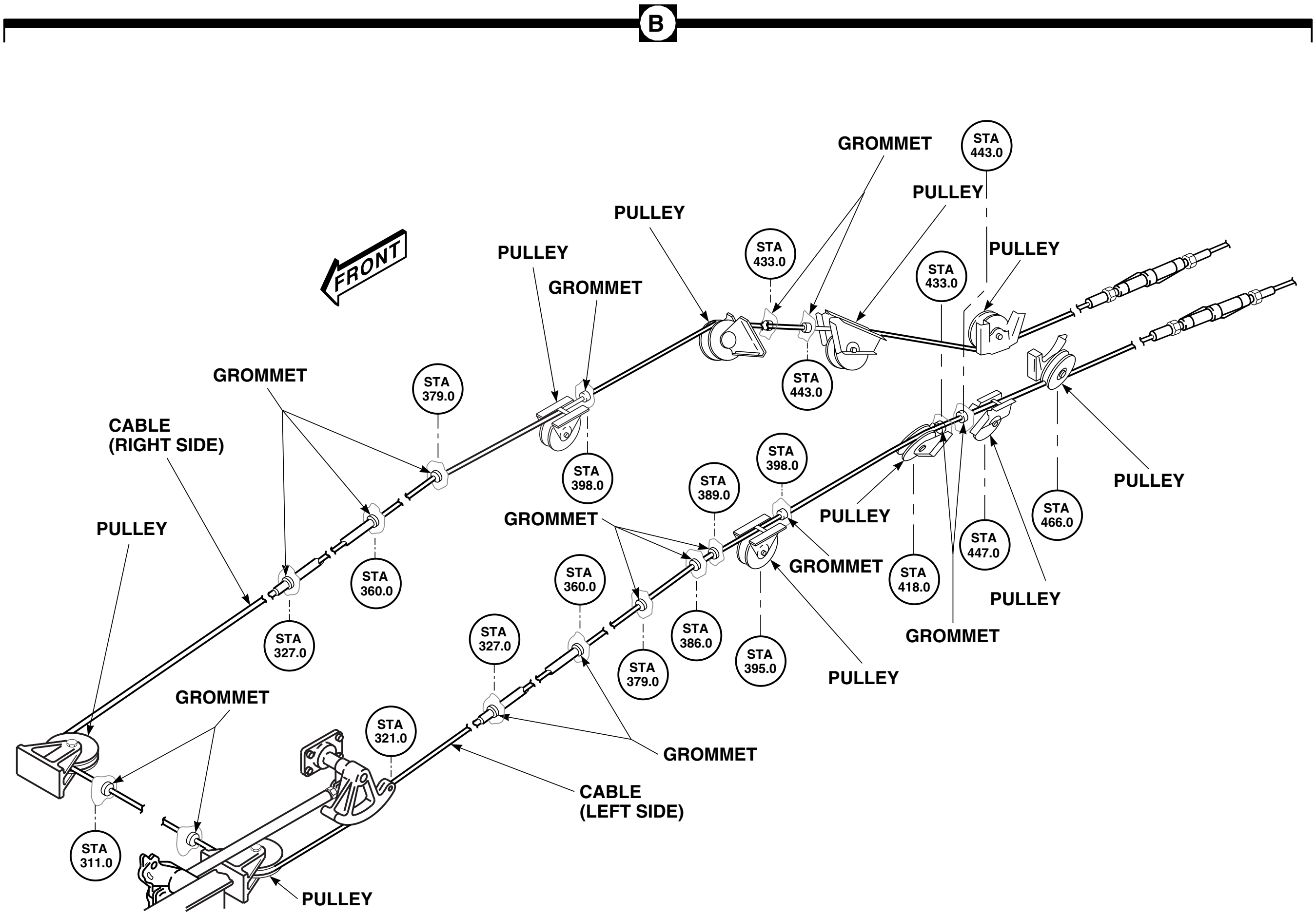


FIGURE 11-3-25-1. INSPECT FLIGHT CONTROL CABLES (SHEET 2 OF 3)

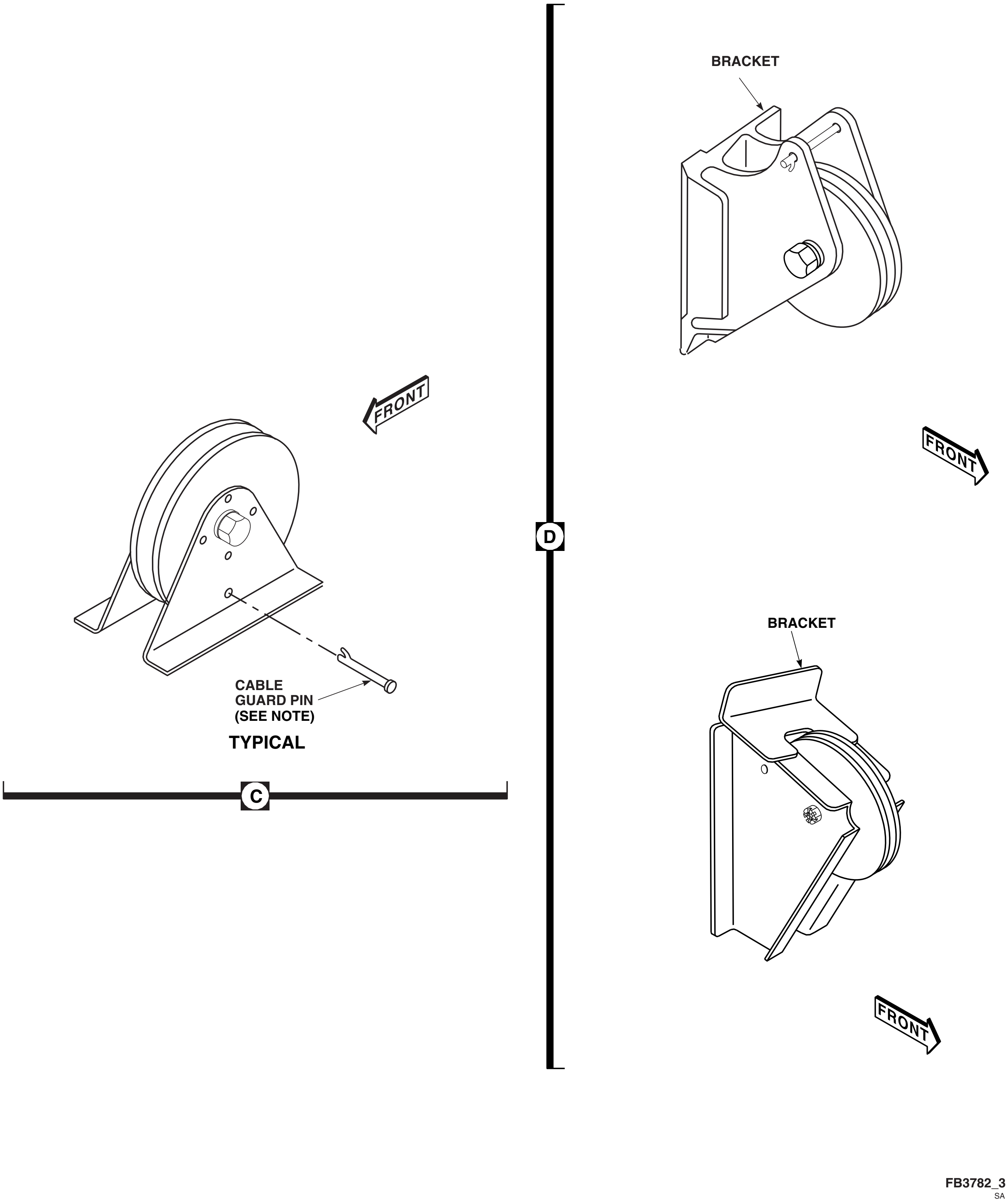


FIGURE 11-3-25-1. INSPECT FLIGHT CONTROL CABLES (SHEET 3 OF 3)

11-3-26 TAIL ROTOR QUADRANT INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Cord
- Inspection Mirror
- Spring Scale
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-137
- PARA 11-2-3
- PARA 11-4-2
- PARA 11-4-6
- PARA 11-4-118

11-3-26.1 Tail Rotor Quadrant Function Check.

a. Remove tail gear box front fairing (PARA 2-4-137).

b. Remove screws and washers securing left and right tail cone flight controls access panels to tail cone at STA 640.0. Remove access panels.

c. Open main rotor pylon sliding cover.

d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16) or run APU (Appropriate Flight Manual).

e. On STABILATOR CONTROL/AUTO FLT CONT panel, engage BOOST switch.

f. Install rigging pins in pitch, roll, yaw, and collective upper torque shafts (PARA 11-4-2).

g. Tie a length of cord to left control cable with knots in front and behind cable disconnect.

WARNING

Injury to personnel and damage to equipment will result if tail rotor cable quick-disconnect ends are not restrained prior to cable separation. Make sure quick-disconnect ends are restrained before separating cable. Tail rotor quadrant spring cylinders maintain 51 to 223 pounds tension on cables.

h. Back off jamnut that secures quick-disconnect and retainer. Separate quick-disconnect and retainer.

i. Visually check right tail rotor servo quadrant toggle and spring cylinder. Toggle will be disengaged and spring cylinder relaxed.

j. On caution/advisory panel, check that TAIL ROTOR QUADRANT light is on.

k. Disconnect right tail rotor quadrant spring cylinder from support (PARA 11-4-118).

WARNING

FSCAP

Verification of toggle installation is a critical characteristic.

l. Install toggle in quadrant notch.

m. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire jamnut to retainer. Remove cable restraint cord.

n. Remove rigging pins from pitch, roll, yaw, and collective upper torque shafts.

o. Have assistant hold high collective, left pedal, and mid-cyclic. Connect right tail rotor quadrant spring cylinder to support (PARA 11-4-118). On caution/advisory panel, check that TAIL ROTOR QUADRANT light is off.

p. Using inspection mirror, inspect left cable at each pulley installation to make sure cable rests in pulley grooves.

q. Install rigging pins in pitch, roll, yaw, and collective upper torque shafts (PARA 11-4-2).

r. Tie a length of cord to right control cable with knots in front and behind cable disconnect.

WARNING

Injury to personnel and damage to equipment will result if tail rotor cable quick-disconnect ends are not restrained prior to cable separation. Make sure quick-disconnect ends are restrained before separating cable. Tail rotor quadrant spring cylinders maintain 51 to 223 pounds tension on cables.

s. Back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer.

t. Visually check left tail rotor servo quadrant toggle and spring cylinder. Toggle will be disengaged and spring cylinder relaxed.

u. On caution/advisory panel, check that TAIL ROTOR QUADRANT light is on.

v. Disconnect left tail rotor quadrant spring cylinder from support (PARA 11-4-118).

WARNING

FSCAP

Verification of toggle installation is a critical characteristic.

w. Install toggle in quadrant notch.

x. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire jamnut to retainer. Remove cable restraint cord.

y. Remove rigging pins from pitch, roll, yaw, and collective upper torque shafts.

z. Have assistant hold high collective, right pedal, and mid-cyclic. Connect left tail rotor quadrant spring cylinder to support (PARA 11-4-118). On caution/advisory panel, check that TAIL ROTOR QUADRANT light is off.

aa. Using inspection mirror, inspect right cable at each pulley installation to make sure cable rests in pulley grooves.

ab. On STABILATOR CONTROL/AUTO FLT CONT panel, engage SAS 1 switch. Disengage SAS 2, TRIM, and FPS switches.

ac. Attach spring scale to pilot's pedals. **FORCE REQUIRED TO MOVE PEDALS LEFT AND RIGHT THROUGH FULL RANGE OF TRAVEL SHALL NOT BE MORE THAN 4 POUNDS.** If pedals need more than 4 pounds to be moved, check each pulley installation for correct cable routing and cable guard installation. Repair cable system, as required.

ad. On STABILATOR CONTROL/AUTO FLT CONT panel, disengage SAS 1 switch.

ae. Perform tail rotor system rig check (PARA 11-4-6).

af. Perform operational check of mechanical flight controls (PARA 11-2-3).

ag. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16) or shutdown APU (Appropriate Flight Manual).

ah. Make sure area is clean and free of foreign material.

ai. Close and latch main rotor pylon sliding cover.

aj. Install left and right tail cone flight control access panels on tail cone and secure with screws and washers.

ak. Install tail gear box front fairing (PARA 2-4-137).

al. Perform maintenance test flight (Appropriate MTF).

11-3-27 TAIL ROTOR QUADRANT SPRING CYLINDER SUPPORT TERMINAL ASSEMBLY INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-137
- PARA 2-6-5
- PARA 11-4-119

11-3-27.1 Inspect Tail Rotor Quadrant Spring Cylinder Support Terminal Assembly.

a. Remove tail gear box front fairing (PARA 2-4-137).

b. If required, remove tail rotor quadrant spring cylinder support (PARA 11-4-119).

c. Inspect spherical bearings in terminal assembly as follows (Figure 11-3-27-1, Detail A):

(1) Inspect spherical bearings for axial play using dial indicator. **AXIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If axial play exceeds limit, replace tail rotor quadrant spring cylinder support (PARA 11-4-119).

(2) Inspect spherical bearings for radial play using dial indicator. **RADIAL PLAY SHALL NOT EXCEED 0.020-INCH.** If radial play exceeds limit, replace tail rotor quadrant spring cylinder support (PARA 11-4-119).

d. Inspect terminal assembly for wear/damage as follows:

(1) **MAXIMUM WEAR/DAMAGE ON THE TERMINAL ASSEMBLY IS 0.020-INCH DEEP, 0.025-INCH LONG, OR 0.025-INCH WIDE.** If wear/damage exceeds limits, replace tail rotor quadrant spring cylinder support (PARA 11-4-119).

(2) Damage may be blended out up to 0.020-inch deep, 0.025-inch deep, and 0.025-inch long using abrasive cloth, Item 8, Appendix D. Rework shall be blended to a smooth transition with adjacent areas.

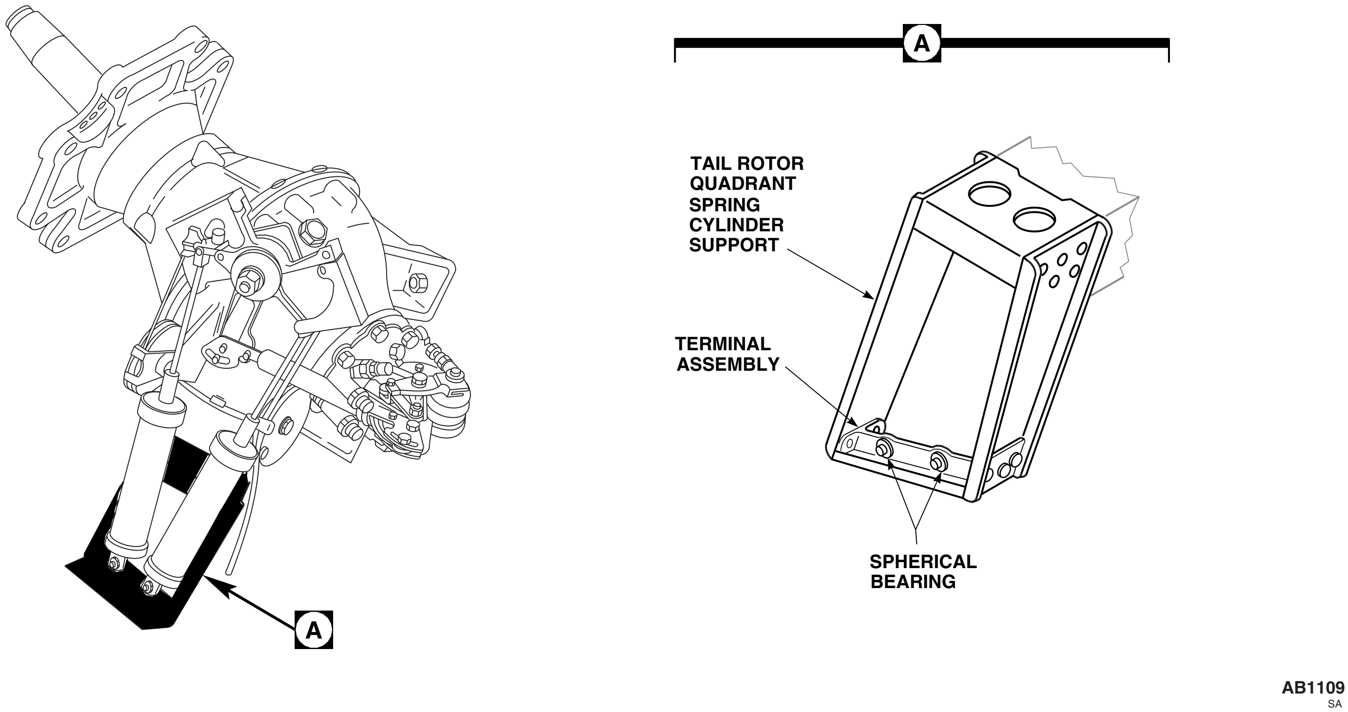


FIGURE 11-3-27-1. INSPECT TAIL ROTOR QUADRANT SPRING CYLINDER SUPPORT TERMINAL ASSEMBLY

(3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace terminal assembly (PARA 11-4-119).

(4) Touch up all reworked areas with corrosion resistant coating, Item 118, Appendix D.

(5) Touch up finish all reworked areas (PARA 2-6-5).

e. If removed, install tail rotor quadrant spring cylinder support (PARA 11-4-119).

f. If necessary, install tail gear box front fairing (PARA 2-4-137).

11-3-28 SAS ACTUATOR INSPECTION.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Flashlight
- Inspection Mirror

References

- PARA 11-4-68
- PARA 11-4-69
- PARA 11-4-70

11-3-28.1 Inspect SAS Actuator Nut.

a. Open main rotor pylon sliding cover.

b. Using flashlight and inspection mirror, visually inspect SAS actuator nut for cracks (Figure 11-3-28-1, Detail A). **NONE ALLOWED.** If cracks are found, replace pitch/trim assembly (PARA 11-4-68), roll trim actuator (PARA 11-4-69), or yaw boost servo (PARA 11-4-70), as required.

c. Close and latch main rotor pylon sliding cover.

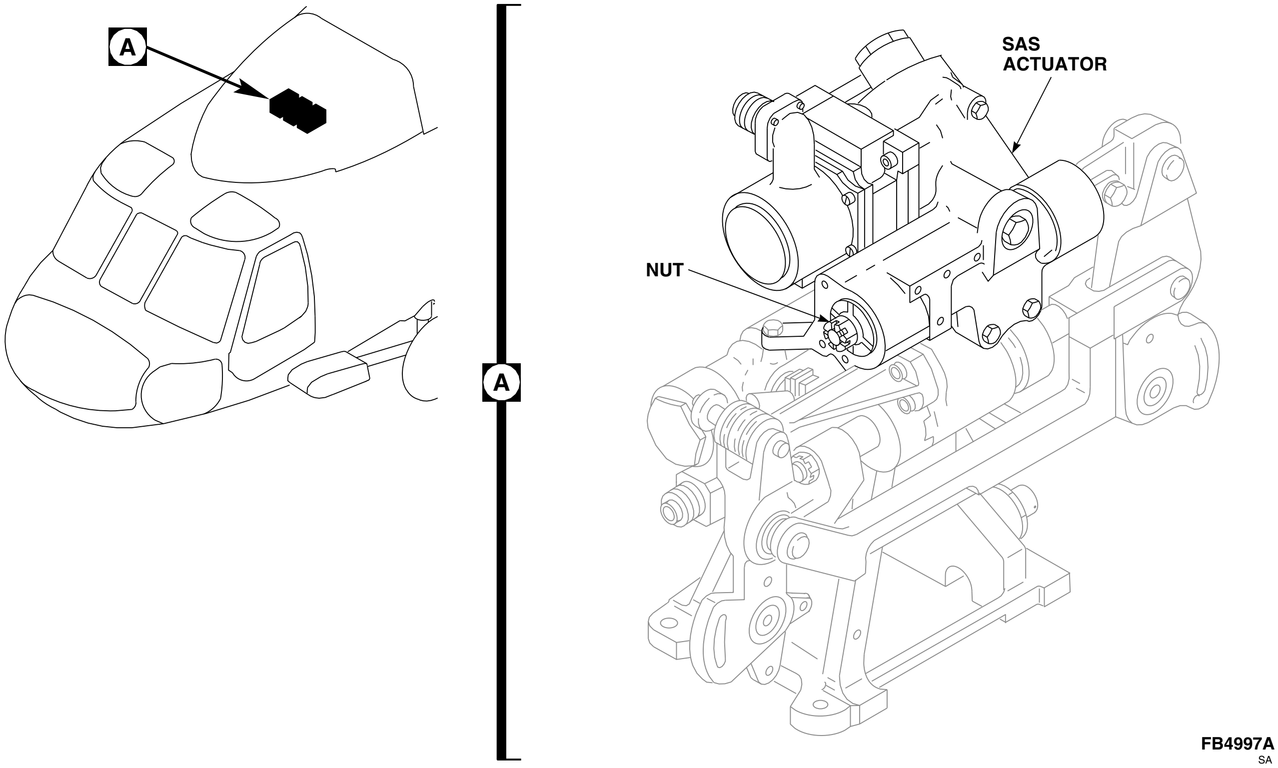


FIGURE 11-3-28-1. INSPECT SAS ACTUATOR NUT

SECTION 11-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
11-4-1	Control Rods
11-4-2	Flight Control System General Rigging Instructions
11-4-3	Main Rotor System Complete Rig
11-4-4	Main Rotor System Rig Check
11-4-5	Tail Rotor System Complete Rig
11-4-6	Tail Rotor System Rig Check
11-4-7	Yaw Control Pedal Boots
11-4-8	Yaw Control Pedal
11-4-9	Yaw Control Pedal Trim Switch
11-4-10	Yaw Control Pedal Support Assembly
11-4-11	Yaw Control Pedal Adjuster
11-4-12	Yaw Control Pedal Adjuster Cable and Handle
11-4-13	Pilot’s Yaw Pedals and Adjuster Control Rods
11-4-14	Copilot’s Yaw Pedals and Adjuster Control Rods
11-4-15	Cyclic Stick Boot
11-4-16	Cyclic Stick
11-4-17	Cyclic Stick Grip
11-4-18	Cyclic Stick Switches
11-4-19	Cyclic STICK TRIM Switch
11-4-20	Cyclic Stick ICS/RADIO Switch

PARAGRAPH	TITLE
11-4-71	Collective SAS Assembly/Yaw Boost Servo Pressure Switch and Thermal Relief Valve
11-4-72	SAS Actuator Bleed Screw and Packing
11-4-73	Collective Servo to Mixer Control Rod
11-4-74	Pitch/Trim Assembly to Mixer Control Link
11-4-75	Roll Trim Assembly to Mixer Control Rod
11-4-76	Yaw Boost Servo to Mixer Control Rod
11-4-77	Mixer
11-4-78	Mixer Repair
11-4-79	No. 3 Collective Stick Position Sensor
11-4-80	Mixer to Forward Primary Servo Control Rod
11-4-81	Mixer to Aft Primary Servo Control Rod
11-4-82	Mixer to Lateral Primary Servo Control Rod
11-4-83	Mixer to Yaw Lever Pushrod
11-4-84	Yaw Lever to Forward Quadrant Control Rod
11-4-85	Yaw Lever and Support (STA 301.0)
11-4-86	Forward Quadrant and Supports
11-4-87	Primary Servo
11-4-88	Primary Servo Pressure Switch
11-4-89	Primary Servo Bypass Valve Cap Packing
11-4-90	Forward Control Rod
11-4-91	Aft Control Rod
11-4-92	Lateral Control Rod
11-4-93	Walking Beam
11-4-94	Long Control Rod
11-4-95	Swashplate Links

PARAGRAPH	TITLE
11-4-96	Forward Bellcrank
11-4-97	Lateral Bellcrank
11-4-98	Aft Bellcrank
11-4-99	Right and Left Tie Rods
11-4-100	Forward Bellcrank Support
11-4-101	Aft Bellcrank Support
11-4-102	Aft Longitudinal Bellcrank Support Arm
11-4-103	Aft Tie Rod and Support Fitting
11-4-104	Cabin Flight Control Cables
11-4-105	Tail Cone Flight Control Cables
11-4-106	Tail Rotor Pylon Flight Control Cables
11-4-107	Flight Control Cable Pulleys
11-4-108	Flight Control Cable Conduit
11-4-109	Flight Control Cable Lockout Blocks
11-4-110	Tail Rotor Quadrant
11-4-111	Tail Rotor Quadrant Repair
11-4-112	Tail Rotor Servo to Tail Rotor Quadrant Pushrod
11-4-113	Tail Rotor Servo and Pitch Control Shaft
11-4-114	Tail Rotor Servo Pressure Switch
11-4-115	Tail Rotor Servo Thermal Relief Valve
11-4-116	Tail Rotor Servo Hydraulic Connectors
11-4-117	Tail Rotor Quadrant Support
11-4-118	Tail Rotor Quadrant Spring Cylinders
11-4-119	Tail Rotor Quadrant Spring Cylinder Support
11-4-120	Servo to Swashplate Control Rod Nylon Washer

PARAGRAPH	TITLE
11-4-121	Tail Pylon Cable

11-4-1 CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Crowfoot Attachment

Materials/Parts

- Corrosion Preventive Compound, Item 108B, Appendix D
- Corrosion Preventive Compound, Item 110, Appendix D

References

- Appendix D
- PARA 11-4-63 through PARA 11-4-66
- PARA 11-4-73
- PARA 11-4-75
- PARA 11-4-76
- PARA 11-4-80 through PARA 11-4-82
- PARA 11-4-84
- PARA 11-4-112

11-4-1.1 Control Rod Installation.

NOTE

- A maximum of three washers, only in thickness combinations indicated in table, can be installed under bolthead so correct nut torque, self-retaining features, and cotter pin installation can be obtained.
- If more than three washers are required to obtain correct nut torque, self-retaining features, and cotter pin installation, substitution of a shorter or longer bolt of the same part number, but with a different dash number is allowed.

a. Refer to Table 11-4-1-1 for proper washer stackup on bolt.

NOTE

Washers are installed under bolthead.

b. Install self-retaining bolt and washers.

c. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup using Table 11-4-1-1, or use next longer or shorter bolt.

d. Install nut on bolt. Torque nut to value shown in task, and install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. Proof torque bolthead to value shown in task. If bolt rotates, adjust using washers in Table 11-4-1-1, or use next longer or shorter bolt.

Table 11-4-1-1. Washer Stackups

Combination	Washer Thickness (Inch)	Quantity
A	0.016	3
B	0.016	2
C	0.063	1
D	0.016 0.063	1 1
E	0.016 0.063	2 1

11-4-1.2 Control Rod Corrosion Protection.

a. Application of corrosion protection coating is required for the following:

(1) Torque shaft to boost servo control rods (PARA 11-4-63, PARA 11-4-64, PARA 11-4-65, and PARA 11-4-66).

(2) Boost servo to mixer control rods (PARA 11-4-73, PARA 11-4-75, and PARA 11-4-76).

(3) Mixer to primary servo control rods (PARA 11-4-80, PARA 11-4-81, and PARA 11-4-82).

(4) Yaw lever to forward quadrant control rod (PARA 11-4-84).

(5) Tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

NOTE

- Corrosion preventive compound, Item 108B, Appendix D, is an amber soft gel.
- Corrosion preventive compound, Item 110, Appendix D, is a firm black coating.

b. Determine which corrosion protection coating is currently applied to the control rod.

NOTE

- Application of corrosion protection coating shall be performed after installation, adjustment, jamnut torquing, and lockwiring of control rod is complete.
- Do not mix corrosion protection coatings. Corrosion preventive compound, Item 108B, Appendix D, will not adhere to corrosion preventive compound, Item 110, Appendix D. If coatings are mixed, this will result in unacceptable corrosion protection.

c. Touch up corrosion protection coating, using corrosion preventive compound, Item 108B, Appendix D, or corrosion preventive compound, Item 110, Appendix D. Apply corrosion protection coating to the following areas:

(1) All threaded portions.

(2) Jamnuts.

(3) Lockwashers.

(4) Slotted keyways.

(5) Inspection holes.

11-4-2 FLIGHT CONTROL SYSTEM GENERAL RIGGING INSTRUCTIONS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Protractor
- Rigging Kit, 70700-20389-042
- Universal Propeller Protractor

References

- PARA 1-6-3
- PARA 11-4-63
- PARA 11-4-64
- PARA 11-4-65
- PARA 11-4-66
- PARA 11-4-87

11-4-2.1 Introduction.

11-4-2.1.1 General.

a. Rigging of the flight control system consists of coordinating the pilot's/copilot's stick and pedal movements in the cockpit, with the correct blade angles at the main and tail rotors.

b. Rigging shall be done by qualified mechanics who are familiar with the helicopter and flight control system.

c. Rigging of the main rotor must be done with blades installed.

d. A complete main rotor rig is required:

(1) After major flight control overhaul.

(2) After major repair due to ballistic or crash damage.

(3) At the direction of maintenance supervisor.

CAUTION

Damage to flight control components will result if hydraulic power is shut down during rigging with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

e. A main rotor system complete rig and/or rig check is required whenever a main transmission, main rotor head, bellcrank support, or any flight control component is replaced, except as noted:

(1) Fixed length pushrods, bellcranks, and idlers are interchangeable without a rigging check.

(2) Adjustable length rods may be changed without a rigging check, provided the replacement is trammeled to exact length (set to dimension measured pivot bolthole to pivot bolthole). Whenever possible, controls should be pinned immediately at the front and rear of the component being replaced.

(3) Any work done on longitudinal and collective requires a recheck of yaw system due to coupling in mixer.

f. When replacing the primary servos, refer to PARA 11-4-87 for rigging requirements.

g. Procedures in PARA 11-4-2.1.2 and PARA 11-4-2.1.3 are examples of how to use a protractor (universal propeller or digital) to read or check main rotor blade angles.

11-4-2.1.2 Zero Protractor.

NOTE

If using universal propeller protractor, proceed to step a. If using digital protractor, proceed to step b.

a. Zero universal propeller protractor as follows:

NOTE

For accurate readings, keep bubble centered during all adjustments.

(1) Hinge corner spirit level out to right angles with frame (Figure 11-4-2-1, Sheet 1, Detail B and Detail C).

(2) Line up vernier scale zero with degree scale zero and set disc-to-ring lock. This locks ring to disc (Figure 11-4-2-2, Sheet 1, Detail C).

(3) Loosen ring-to-frame lock so ring can be turned in frame by ring adjuster.

(4) Set bottom edge of protractor on sleeve boss (Figure 11-4-2-1, Sheet 1, Detail A and Detail B).

NOTE

Protractor must be held on sleeve boss (Figure 11-4-2-1, Sheet 1, Detail A and Detail B) in the same position it will be in when checking blade angles on blade cuff (Figure 11-4-2-1, Sheet 2, Detail E and Detail F).

(5) Hold protractor firmly on sleeve boss and hold vertical (Figure 11-4-2-1, Sheet 1, Detail A and Detail B). Check corner spirit level bubble (Figure 11-4-2-1, Sheet 1, Detail C).

(6) Turn ring adjuster until center spirit level bubble centers in spirit level. Tighten ring-to-frame lock (Figure 11-4-2-1, Sheet 1, Detail C).

(7) Pull disc-to-ring lock and turn it one-quarter turn to keep it out.

b. Zero digital protractor as follows:

NOTE

- It is essential to keep helicopter perfectly still while zeroing and taking blade readings with digital protractor. A limit of three persons is recommended on helicopter - one to position controls in cockpit, one to take readings, and one to assist in rotating blades to the 90°, 180°, 270°, and 0° positions.
- Helicopter must be jacked (just enough to relieve load on landing gear struts) so that helicopter will not rock if personnel enter or exit helicopter while blade angle readings are being taken.

(1) Jack helicopter slightly (just enough to relieve load on landing gear struts) (PARA 1-6-3).

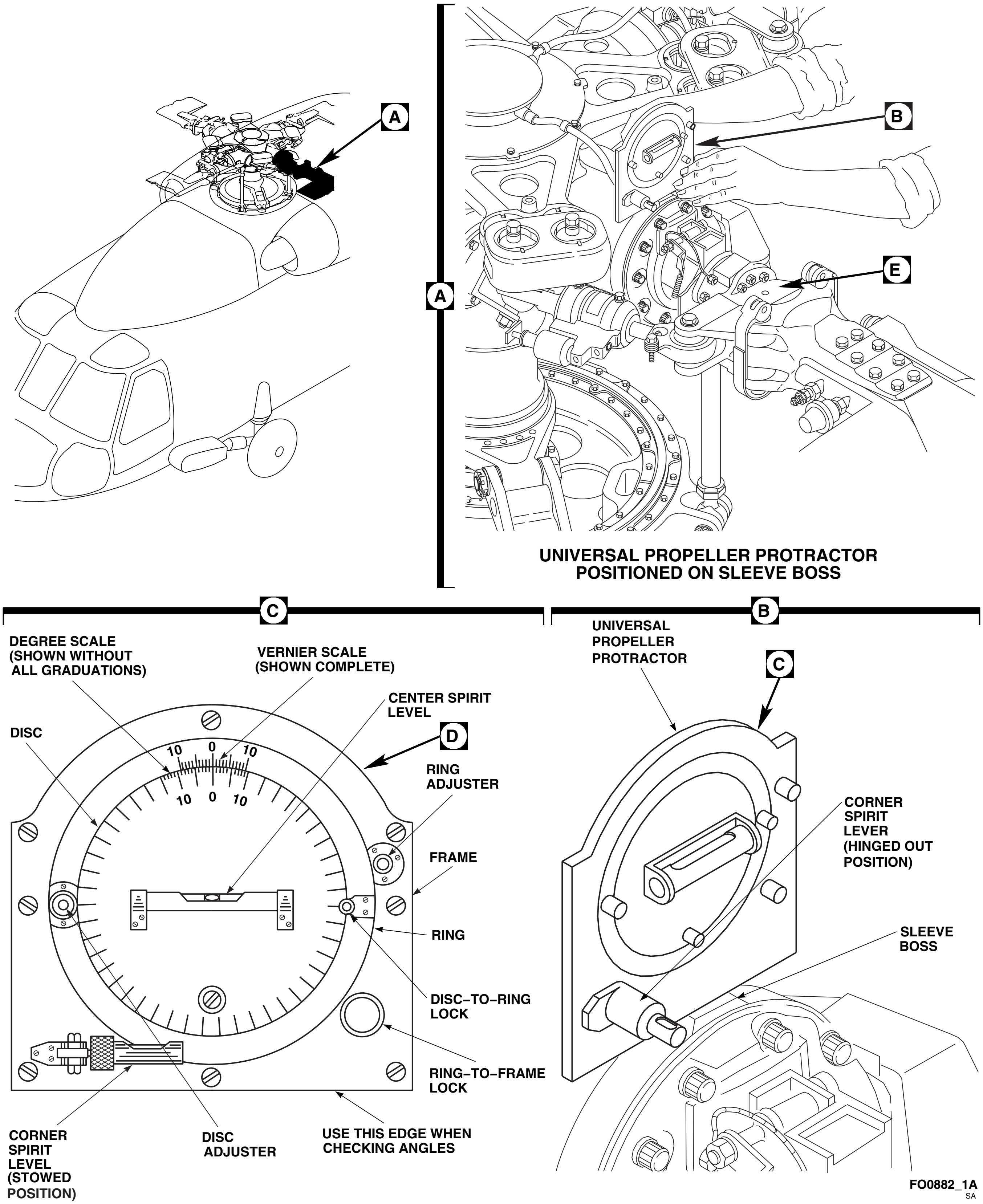


FIGURE 11-4-2-1. MEASURE BLADE ANGLE USING UNIVERSAL PROPELLER PROTRACTOR (SHEET 1 OF 2)

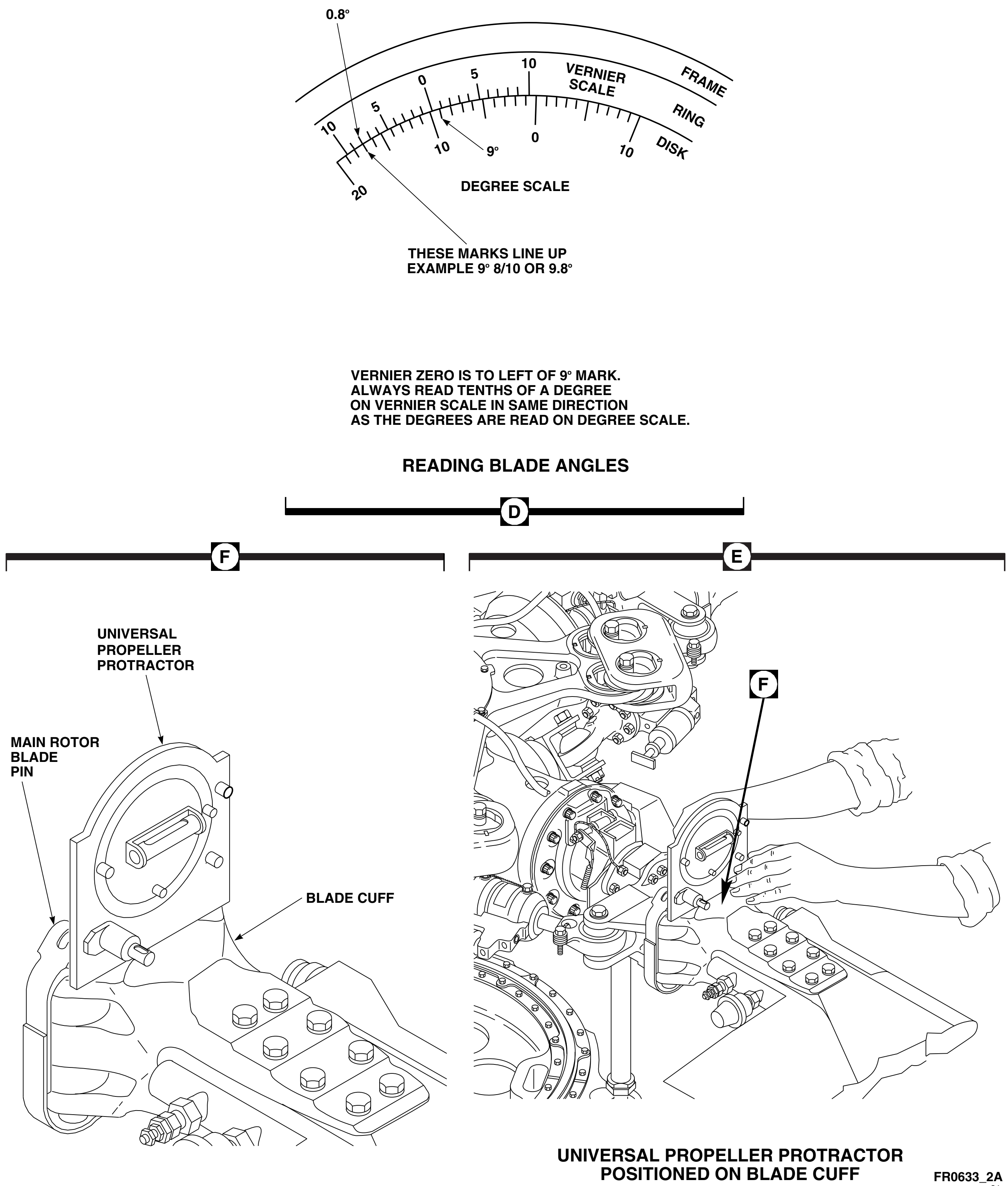


FIGURE 11-4-2-1. MEASURE BLADE ANGLE USING UNIVERSAL PROPELLER PROTRACTOR (SHEET 2 OF 2)

(2) Push ON/OFF switch one time to turn on digital protractor (Figure 11-4-2-2, Sheet 1, Detail C).

NOTE

Low battery voltage will affect protractor accuracy.

(3) Check digital display for LO BAT message. If LO BAT warning is displayed, replace protractor battery.

(4) Press reading and protractor zero button (ALT REF) to set ALT REF. Press second time to zero protractor.

(5) Set bottom edge of protractor on sleeve boss (Figure 11-4-2-2, Sheet 1, Detail A and Detail B).

NOTE

Protractor must be held on sleeve boss (Figure 11-4-2-2, Sheet 1, Detail A and Detail B) in the same position it will be in when checking blade angles on blade cuff (Figure 11-4-2-2, Sheet 2, Detail E and Detail F).

(6) Hold protractor firmly on sleeve boss and hold vertical (Figure 11-4-2-2, Sheet 1, Detail A and Detail B).

11-4-2.1.3 Measure Blade Angle.

NOTE

- Refer to Figure 11-4-2-3 for sample main rotor rigging worksheet and calculations.
- Difference between blade cuff and hub attaching surface (where blade angle is measured) is 9°. Therefore, it is possible for blade to be negative and still read a positive angle at attaching surface. All calculations are made using angle measured at blade cuff attaching surface.
- Protractor must be held on sleeve boss (Figure 11-4-2-1, Sheet 1, Detail A and Detail B or Figure 11-4-2-2, Sheet 1, Detail A and Detail B) in the same position it will be in when checking blade angles on blade cuff (Figure 11-4-2-1, Sheet 2, Detail E and Detail F or Figure 11-4-2-2, Sheet 2, Detail E and Detail F).
- External hydraulic power should be applied whenever available. Backup pump should be used only as an alternate source to power hydraulic systems.
- Make sure edge of protractor is not setting on washers under main rotor blade pins.

a. Place protractor on blade cuff next to main rotor blade pins (Figure 11-4-2-1, Sheet 2, Detail E and Detail F or Figure 11-4-2-2, Sheet 2, Detail E and Detail F).

b. If using universal propeller protractor, turn disc adjuster so center spirit level is level again (bubble centered) (Figure 11-4-2-1, Sheet 1, Detail C).

c. Read blade angle on protractor (Figure 11-4-2-1, Sheet 2, Detail D or Figure 11-4-2-2, Sheet 2, Detail D).

NOTE

Rotor head and swashplate are not level in any rigging position; blade angles must be average readings.

d. Check collective angles as follows:

(1) Set collective stick to full down (low pitch) position.

(2) Install rigging pin in pedals (PARA 11-4-2.2).

(3) Install rigging pin in yaw boost servo (PARA 11-4-2.2).

(4) Install rigging pins in cyclic stick fore-and-aft and lateral (roll) positions (PARA 11-4-2.2).

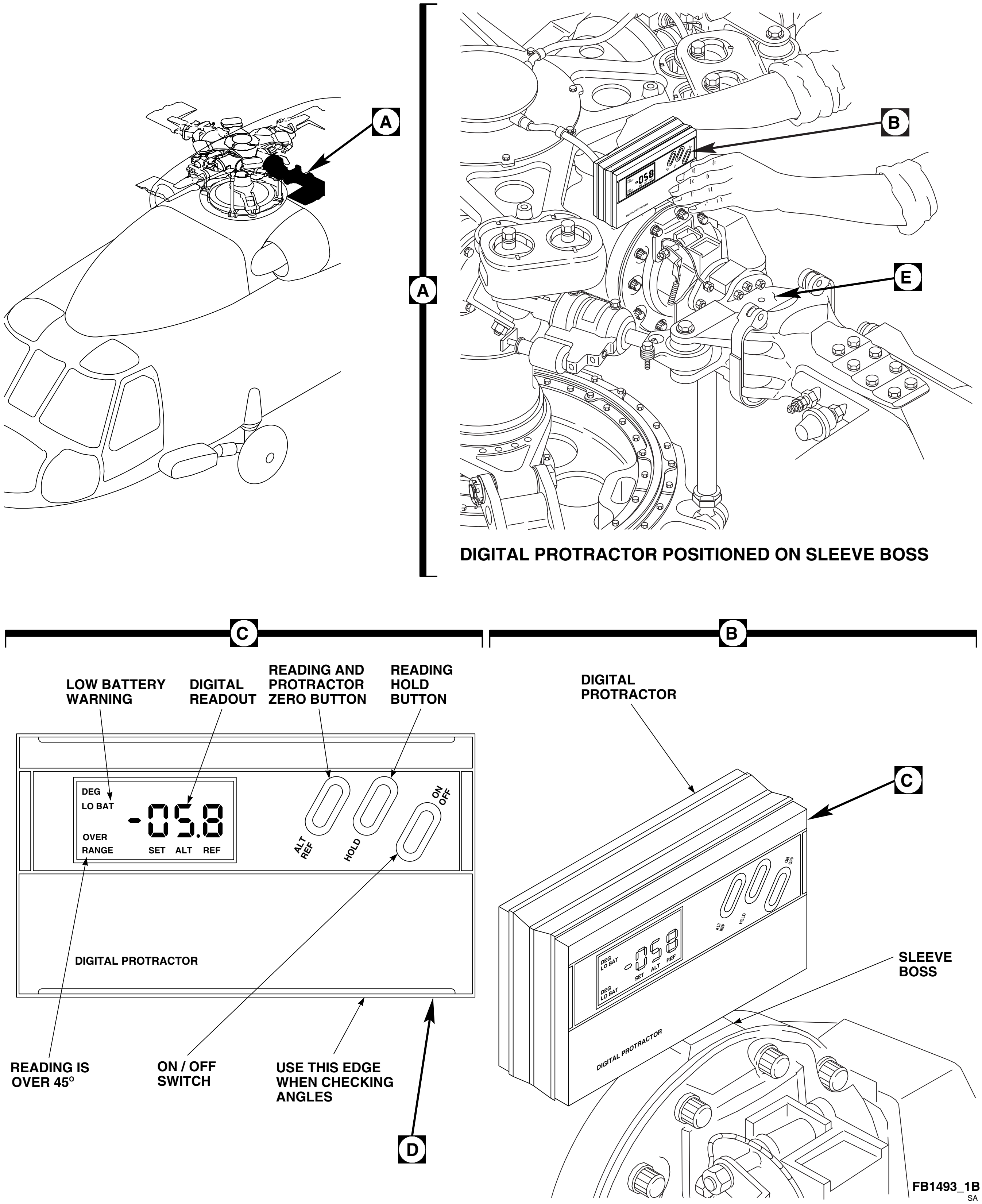


FIGURE 11-4-2-2. MEASURE BLADE ANGLE USING DIGITAL PROTRACTOR (SHEET 1 OF 2)

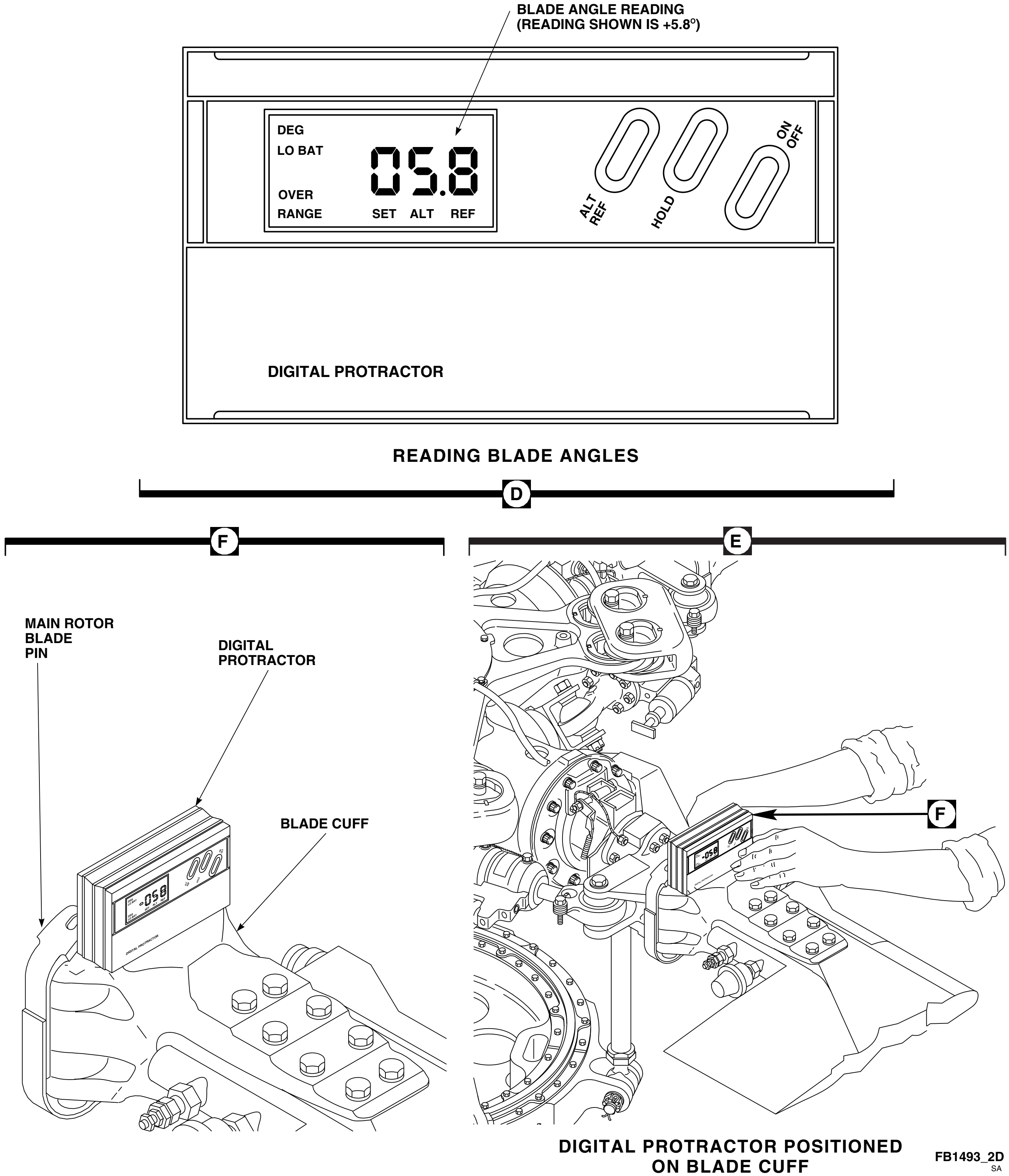


FIGURE 11-4-2-2. MEASURE BLADE ANGLE USING DIGITAL PROTRACTOR (SHEET 2 OF 2)

NOTE

Protractor must be zeroed out on top of main rotor sleeve boss in line with each blade location for each reading.

(5) Move any blade to 90° position (turning pitch control rod over forward longitudinal input, and black marks on swashplate lined up). Check and record blade angle. Turn rotor head and repeat readings on same blade at all four control positions (Figure 11-4-2-4). Add four readings and divide by 4. This is low collective angle.

(6) For high collective angle, move collective stick to full up (high pitch) position and repeat step d. (2) through step d. (5).

e. For longitudinal (pitch) ranges, average two readings taken on same blade in 90° and 270° positions.

f. For lateral (roll) ranges, average two readings taken on same blade in 180° and 0° locations.

g. When taking the following readings, make sure copilot's collective stick is extended. Have assistant hold light pressure on controls to eliminate lost motion as follows:

(1) Low Collective Reading:

(a) Cyclic stick aft and left.

(b) Pedals left.

(c) Collective stick full down.

(2) High Collective Reading:

(a) Cyclic stick aft and left.

(b) Pedals left.

(c) Collective stick full up.

(3) High Collective, Forward Cyclic Reading:

(a) Cyclic stick forward and left.

(b) Pedals left.

(c) Collective stick full up.

(4) High Collective, Aft Cyclic Reading:

(a) Cyclic stick aft and left.

(b) Pedals left.

(c) Collective stick full up.

(5) Right Cyclic Reading:

(a) Cyclic stick aft and right.

(b) Pedals left.

(c) Collective stick full down.

(6) Left Cyclic Reading:

(a) Cyclic stick aft and left.

(b) Pedals left.

(c) Collective stick full down.

11-4-2.2 Flight Control Rigging Pin Installation.

a. Rigging kit, 70700-20389-042, consists of the following pins/blocks which are used in checking helicopter rigging (Table 11-4-2-1).

CAUTION

Each self-retaining bolt used for attaching control rods and links in the flight control system has a ring that expands above the diameter of the bolt. This means the bolt has to be tapped in or out when it is removed or installed. Do not install a self-retaining bolt in a rod end just to keep the bolt with the rod. It is possible to damage the bearing and rod end by popping out the bearing, using the force needed to install the bolt. At normal installation, the bearing is restrained inside a clevis.

AIRCRAFT TAIL NO. *79 – 23278*

WORKSHEET

#1 (RED BLADE POSITION)

DATE *30 Jun 95*

CONTROL POSITIONS	90° RT	180° FWD	270° LFT	0° AFT	TOTAL°	AVG	REQUIRED TOLERANCE
LOW COLL ANGLE (NOTE 1) LOW COLL NEUTRAL CYCLIC NEUTRAL YAW BLADES	#1 *+10.1* #2 *+10.2* #3 *+10.2* #4 *+10.1*	*–2.3*	*–11.4*	*+1.5*	*–2.1*	*–0.525*	–0.6° ± 0.125° (–0.475° TO –0.725°) BLADES #2, #3, #4 SAME AS #1 BLADE
HIGH COLL ANGLE (NOTE 2) HIGH COLL NEUTRAL CYCLIC NEUTRAL YAW	*+21.2*	*+11.0*	*+9.9*	*+20.0*	*+62.1*	*+15.525*	+15° TO +16°
FWD CYCLIC ANGLE HIGH COLL NEUTRAL LAT FWD CYCLIC NEUTRAL YAW	*–2.0*		*+32.0*		*+34.0*		DIFFERENCE BETWEEN 270° & 90° POSITIONS +33.0° MIN.
AFT CYCLIC ANGLE HIGH COLL NEUTRAL LAT AFT CYCLIC NEUTRAL YAW	*+26.2*		*+4.8*		*+21.4*		DIFFERENCE BETWEEN 90° & 270° POSITIONS NOT LESS THAN +19°
LEFT CYCLIC ANGLE LOW COLL NEUTRAL LONG LEFT CYCLIC NEUTRAL YAW		*–9.0*		*+7.4*	*+16.4*		SUM OF 180° & 0° POSITION TO BE 15° TO 17° AVG RANGE +7.5° TO 8.5°
RIGHT CYCLIC ANGLE LOW COLL NEUTRAL LONG RIGHT CYCLIC NEUTRAL YAW		*+7.1*		*–8.1*	*+15.2*		SUM OF 180° & 0° POSITION TO BE 14° TO 16° AVG RANGE +7.0° TO 8.0°

(FWD) 180°

(LEFT) 270° — 90° (RIGHT)

0° (AFT)

NOTES

1. ON NEW MAIN ROTOR HEADS AT 90° RIGHT POSITION, ADJUST NO. 1 PITCH CONTROL ROD TO OBTAIN AVERAGE ANGLE. REMAINING THREE PITCH CONTROL RODS MUST BE WITHIN 0.3 DEGREES OF NO. 1.
2. CHECK SWASHPLATE LARGE SPHERICAL BEARING (UNI–BALL) LIMIT (0.250 IN. MAX).

PRIMARY SERVO CONTROL ROD BIAS:	
FWD:	*1/2 Turn Shorter*
AFT:	*1/2 Turn Longer*
LAT:	*No change*

SAMPLE RIG SHEET ANGLE CALCULATIONS

LOW COLLECTIVE BLADE ANGLE –

THE LOW COLLECTIVE BLADE ANGLE IS THE AVERAGE OF THE READINGS TAKEN AT THE 90°, 180°, 270°, 0° POSITIONS WITH THE COLLECTIVE STICK IN THE FULL DOWN POSITION:

(+10.1 –2.3 –11.4 +1.5) / 4 = –2.1 / 4 = –0.525 DEGREES

HIGH COLLECTIVE BLADE ANGLE –

THE HIGH COLLECTIVE BLADE ANGLE IS THE AVERAGE OF THE READINGS TAKEN AT THE 90°, 180°, 270°, 0° POSITIONS WITH THE COLLECTIVE STICK IN THE FULL UP POSITION:

(+21.1 +11.0 +9.9 +20.0) / 4 = 62.1 / 4 = +15.525 DEGREES

FORWARD CYCLIC BLADE ANGLE –

THE FORWARD CYCLIC BLADE ANGLE IS THE DIFFERENCE IN THE READINGS TAKEN AT THE 270° AND 90° POSITIONS WITH THE CYCLIC STICK IN THE FULL FORWARD POSITION:

(+32.0 –(–2.0)) = +34.0 DEGREES

AFT CYCLIC BLADE ANGLE –

THE AFT CYCLIC BLADE ANGLE IS THE DIFFERENCE IN THE READINGS TAKEN AT THE 90° AND 270° POSITIONS WITH THE CYCLIC STICK IN THE FULL AFT POSITION:

(+26.2 –4.8) = +21.4 DEGREES

LEFT CYCLIC BLADE ANGLE –

THE LEFT LATERAL BLADE ANGLE IS THE SUM OF THE READINGS TAKEN AT THE 180° AND 0° POSITIONS WITH THE CYCLIC STICK IN THE FULL LEFT POSITION (NOTE THAT NEGATIVE SIGNS ARE IGNORED):

(+9.0 +7.4) = +16.4 DEGREES

RIGHT CYCLIC BLADE ANGLE –

THE RIGHT LATERAL BLADE ANGLE IS THE SUM OF THE READINGS TAKEN AT THE 180° AND 0° POSITIONS WITH THE CYCLIC STICK IN THE FULL RIGHT POSITION (NOTE THAT NEGATIVE SIGNS ARE IGNORED):

(+7.1 +8.1) = 15.2 DEGREES

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FIGURE 11-4-2-3. SAMPLE MAIN ROTOR RIGGING WORKSHEET WITH CALCULATIONS

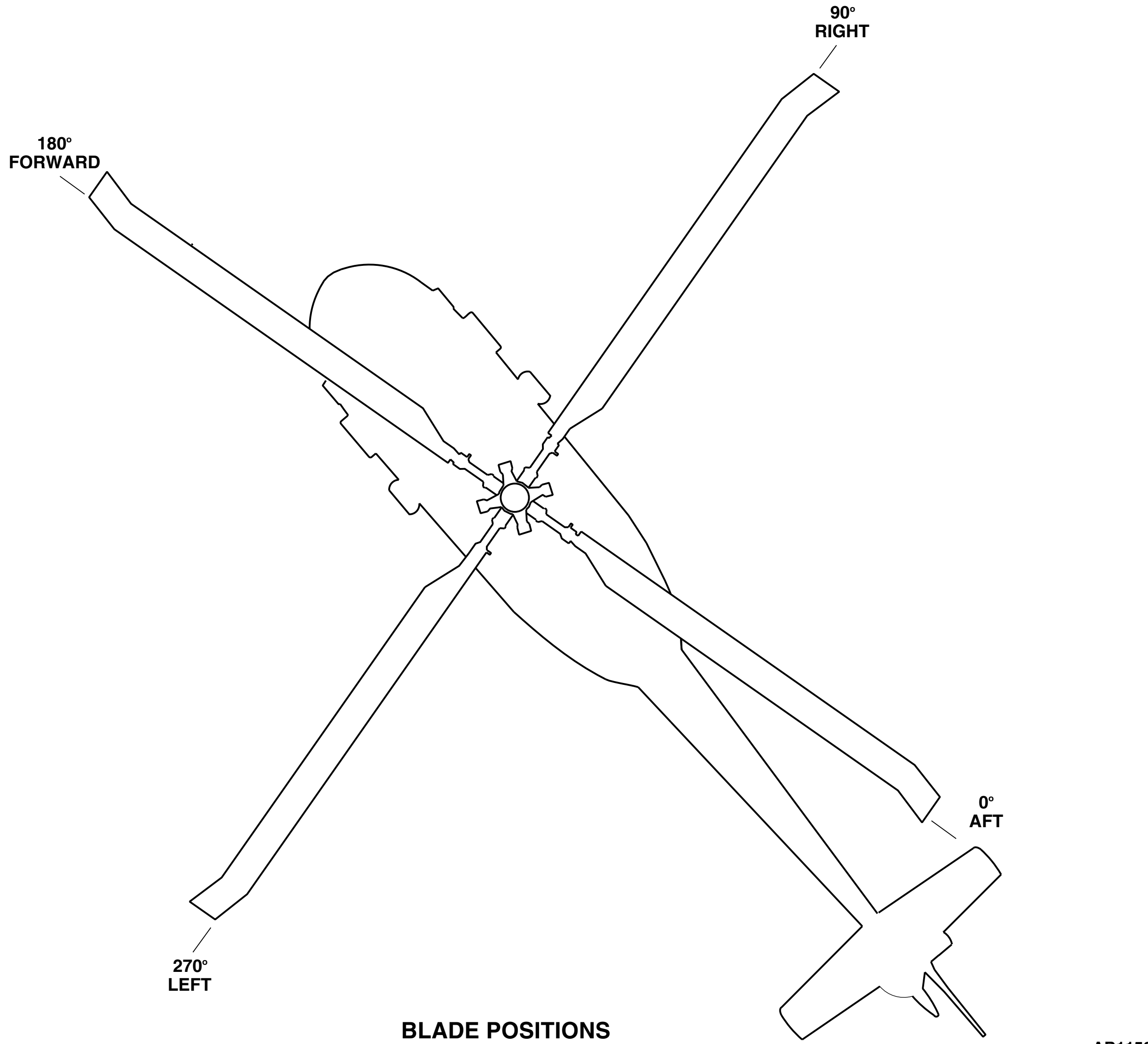


FIGURE 11-4-2-4. BLADE POSITIONS

NOTE

Make sure self-retaining element on bolt is completely out of bolthole before installing nut and cotter pin.

b. When replacing control rods, it may be necessary to move controls to allow clearance when removing hardware.

CAUTION

Damage to flight control components will result if hydraulic power is shut down during rigging with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

c. If hydraulic power is not available, disconnect input control rods between torque shafts and servos on upper deck (PARA 11-4-63, PARA 11-4-64, PARA 11-4-65, and PARA 11-4-66).

d. Refer to Figure 11-4-2-5, Sheet 1, Sheet 2, Sheet 3, and Sheet 4 for location of rigging pins.

Table 11-4-2-1. Rigging Kit, 70700-20389-042

Rigging Pin Part Number	Location	Diameter (Inch)	Length (Inch)	Quantity
70700-20380-065	Pitch Cyclic Sticks	¼	4	2
70700-20380-066	Roll Cyclic Sticks	¼	4	2
70700-20380-042	Yaw Pedal Adjuster	5⁄16	6	2
70700-20380-043	Torque Shafts	5⁄16	4	4
70700-20380-044	Pitch/Roll Assemblies	5⁄16	1.5	2
70700-20380-046	Mixer (Right Side)	5⁄16	36	1
70700-20380-068	Mixer (Left Side) Yaw	5⁄16	12	1
70700-20342-041	Swashplate Bellcrank	⅞	7.5	2
70700-20342-042	Swashplate Bellcrank	½	7	1
70700-20380-071	Tail Rotor Servo	¼	6	1
70700-20378-103	Damper Blocks	-	-	4
70700-20380-062	Yaw Boost Servo	Pin	-	1
70700-20380-062	Collective Boost Assembly	Pin	-	1

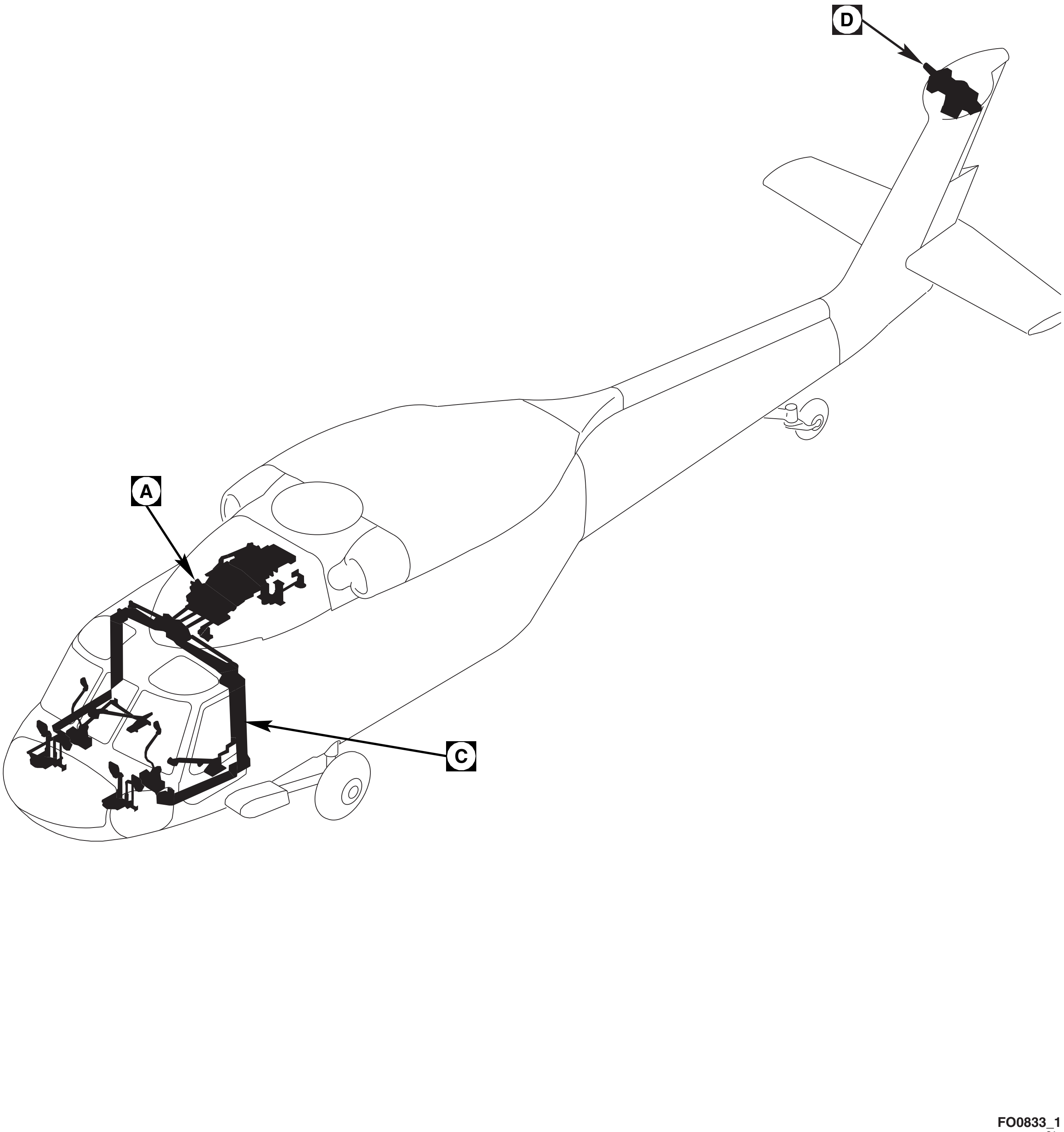
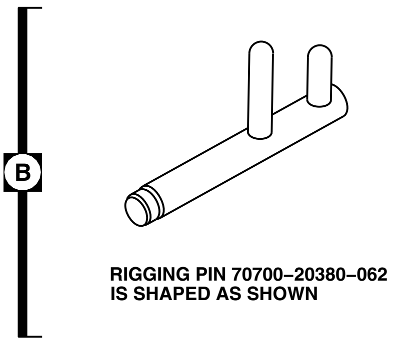


FIGURE 11-4-2-5. FLIGHT CONTROL RIGGING PIN INSTALLATION (SHEET 1 OF 4)



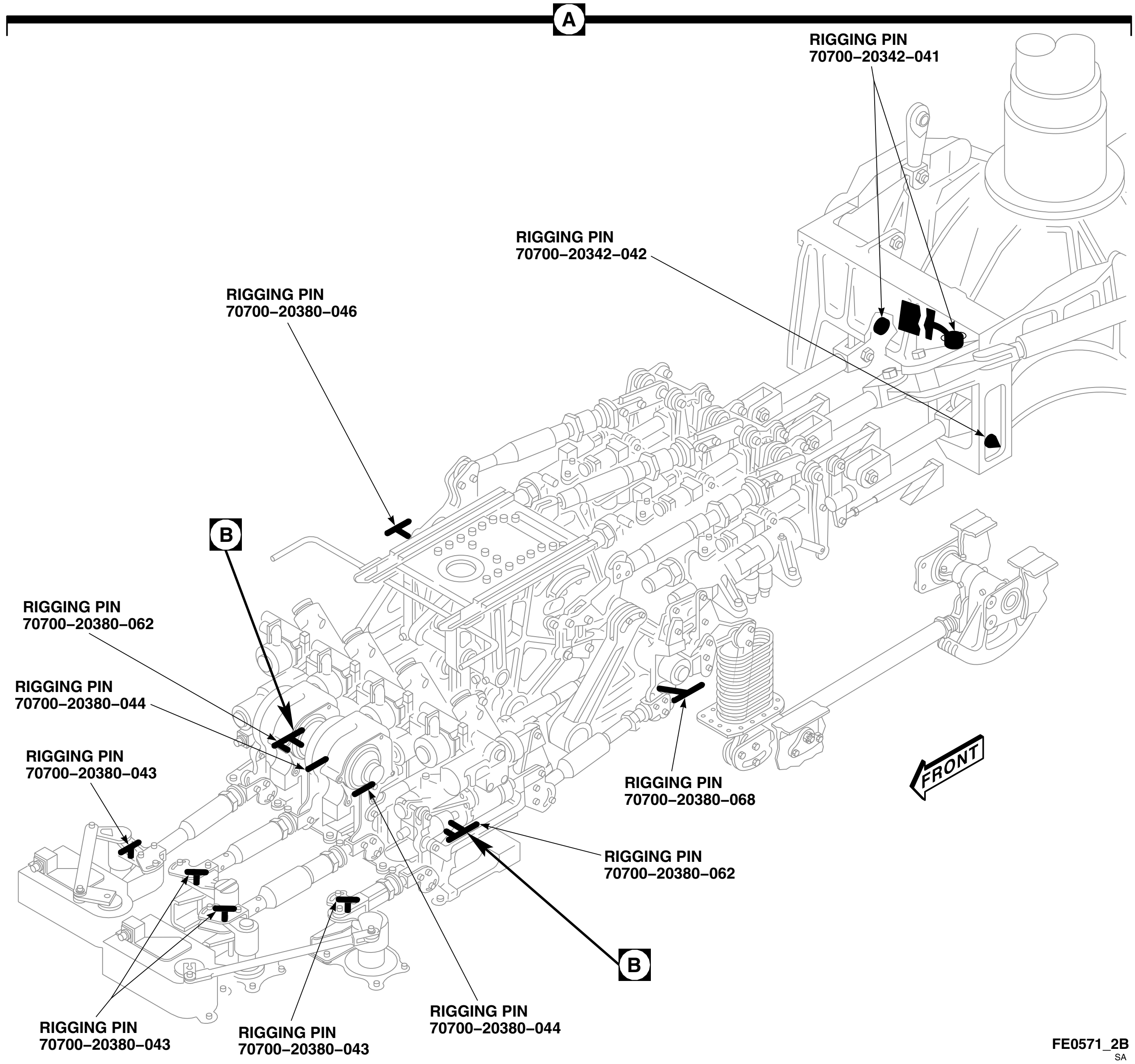


FIGURE 11-4-2-5. FLIGHT CONTROL RIGGING PIN INSTALLATION (SHEET 2 OF 4)

C

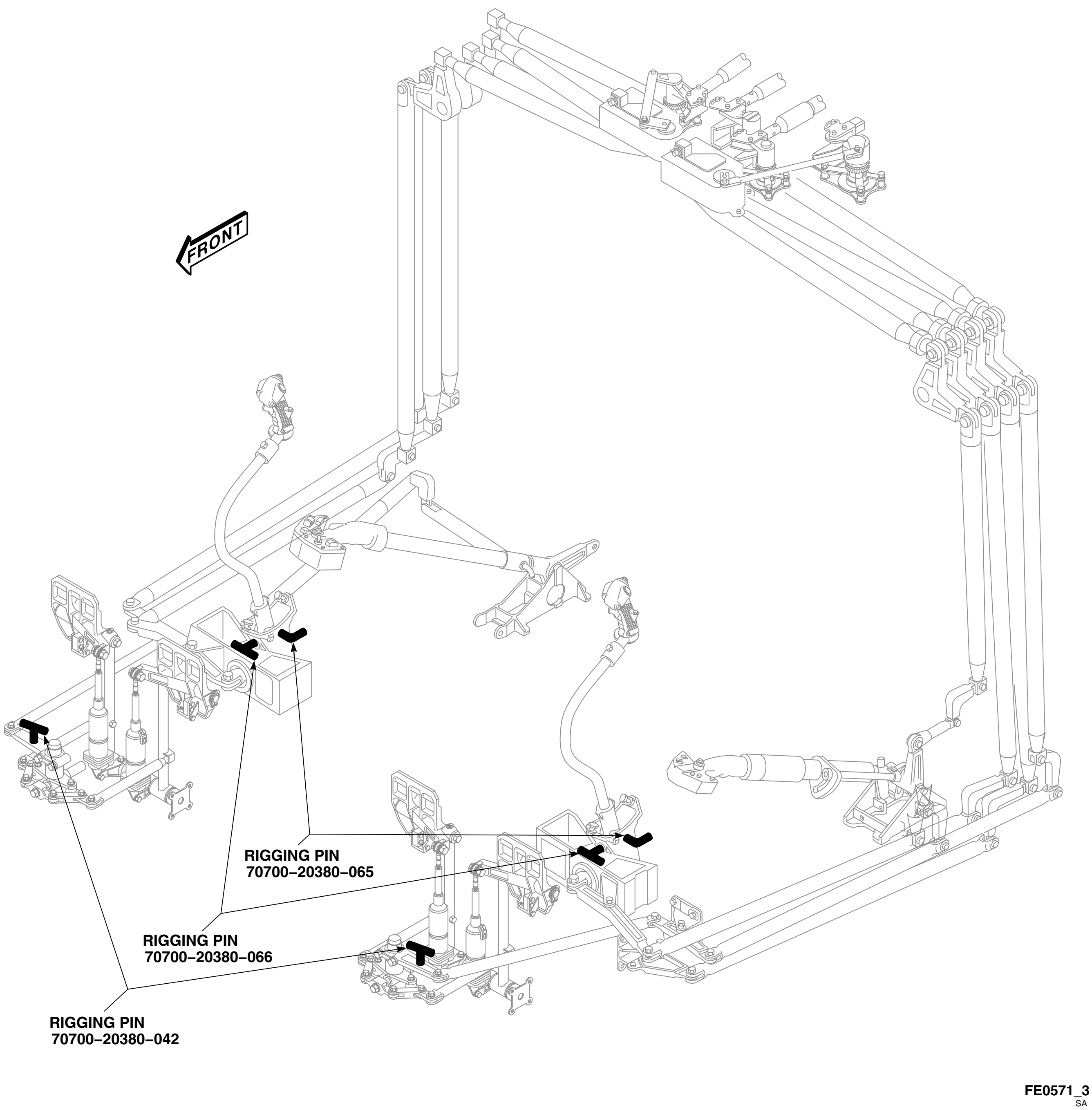


FIGURE 11-4-2-5. FLIGHT CONTROL RIGGING PIN INSTALLATION (SHEET 3 OF 4)

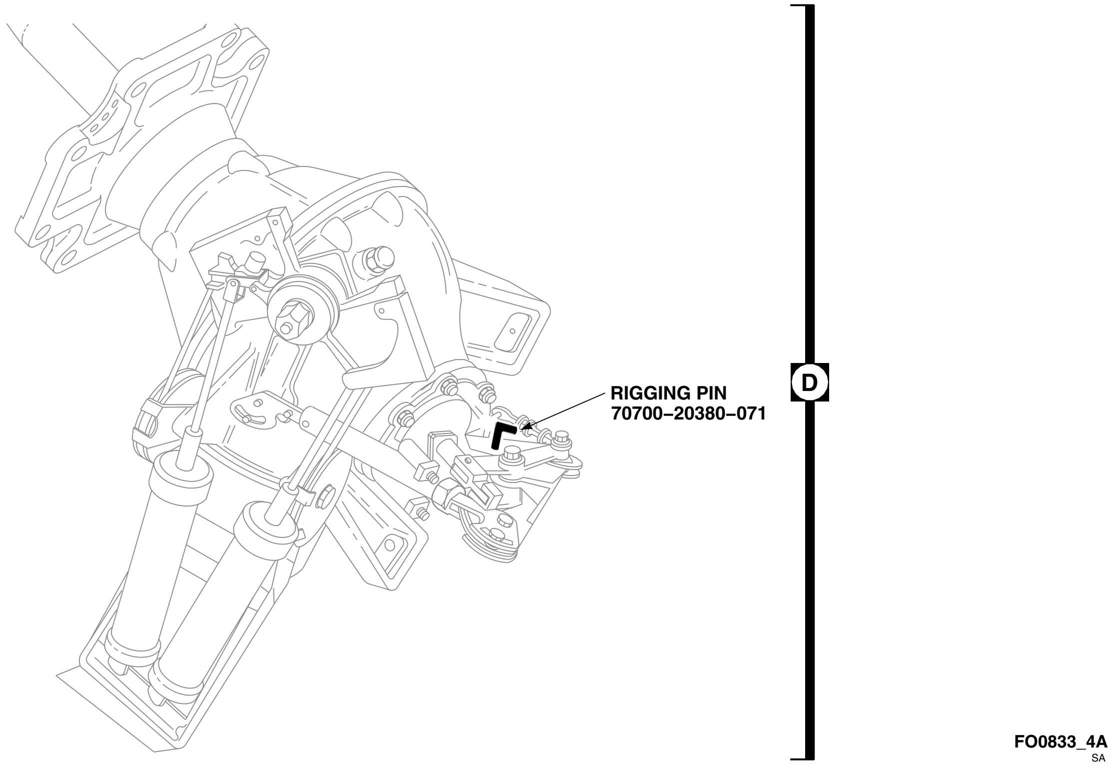


FIGURE 11-4-2-5. FLIGHT CONTROL RIGGING PIN INSTALLATION (SHEET 4 OF 4)

11-4-3 MAIN ROTOR SYSTEM COMPLETE RIG.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Protractor
- Feeler Gage, 0.020-inch
- Feeler Gage, 0.030-inch
- Rigging Kit, 70700-20389-042
- Torque Wrench, 150 - 750 in. lbs
- Universal Propeller Protractor

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 194, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate Flight Manual
- PARA 1-3-8
- PARA 1-6-3
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-5
- PARA 1-7-20
- PARA 2-4-43
- PARA 2-4-69
- PARA 2-4-101
- PARA 2-4-105
- PARA 4-4-27
- PARA 11-2-3
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-4
- PARA 11-4-5
- PARA 11-4-7
- PARA 11-4-13
- PARA 11-4-14
- PARA 11-4-15
- PARA 11-4-26
- PARA 11-4-46
- PARA 11-4-61

- PARA 11-4-62
- PARA 11-4-63
- PARA 11-4-64
- PARA 11-4-65
- PARA 11-4-66
- PARA 11-4-80
- PARA 11-4-81
- PARA 11-4-82
- PARA 11-4-83
- PARA 12-4-26
- TM 1-70-VIB
- TM 11-70-23

11-4-3.1 Main Rotor System Complete Rig.

a. A complete main rotor rig is required:

(1) After major flight control overhaul.

(2) After major repair due to ballistic or crash damage.

(3) At the direction of maintenance supervisor.

b. Remove main rotor pylon sliding cover (PARA 2-4-101).

c. Remove main rotor pylon work platform (PARA 2-4-105).

d. Disconnect load-demand rotary inputs from engine hydromechanical unit (PARA 4-4-27).

e. Disconnect engine load-demand spindle (LDS) cables from mixer levers (PARA 4-4-27).

f. Remove ceiling soundproofing panels at STA 247.0 through STA 295.0, and left and right control enclosure covers at STA 247.0 through STA 265.0 (PARA 2-4-69).

g. Remove pilot's and copilot's cyclic stick boots (PARA 11-4-15).

h. Remove pilot's and copilot's collective stick covers and boots (PARA 11-4-26).

i. Remove pedal boots (PARA 11-4-7) and access panels (PARA 2-4-43) around pilot's and copilot's pedal assemblies.

j. Disconnect the following control rods from torque shafts:

(1) Collective (PARA 11-4-63).

(2) Pitch/Trim (PARA 11-4-64).

(3) Lateral (PARA 11-4-65).

(4) Yaw Boost (PARA 11-4-66).

k. Extend copilot's collective stick to its flight ready position and check low collective stopbolt. **LOW COLLECTIVE STOPBOLT BOLTHEAD SHALL BE 0.630 TO 0.670-INCH ABOVE SUPPORT** (Figure 11-4-3-1, Sheet 1, Detail B and Detail C). If copilot's low collective stopbolt is beyond limit, perform the following:

(1) Back off jamnut on low collective stopbolt (Figure 11-4-3-1, Sheet 1, Detail B).

(2) Adjust stopbolt to obtain 0.630 - 0.670-inch gap between bolthead and support (Figure 11-4-3-1, Sheet 1, Detail C).

(3) **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(4) Using lockwire, Item 197, Appendix D, lockwire jamnut.

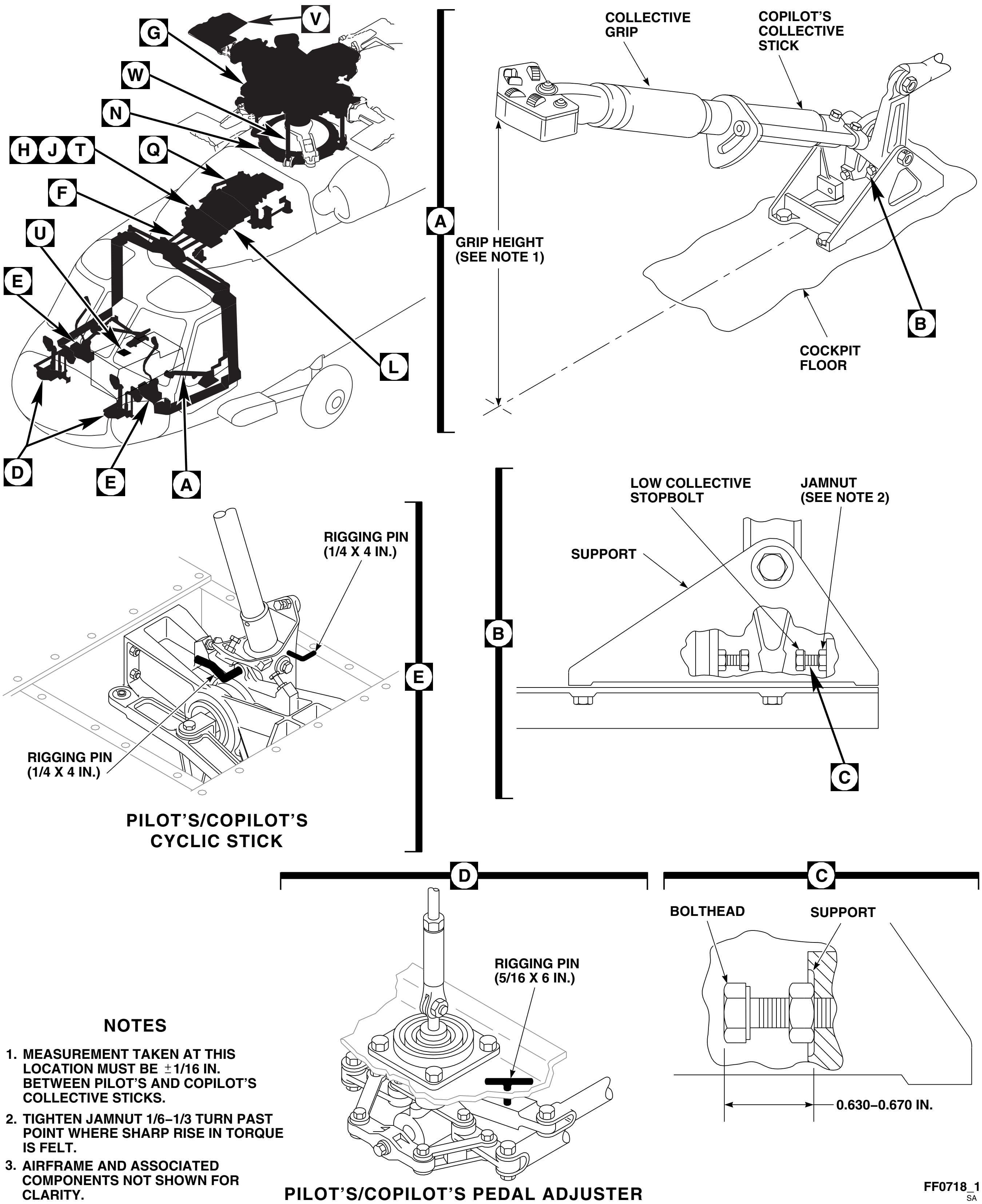


FIGURE 11-4-3-1. RIG MAIN ROTOR SYSTEM (SHEET 1 OF 6)

NOTE

Make sure that pilot's low collective stopbolt does not interfere with copilot's stick stop. Copilot's collective stick should be resting on 0.020-inch feeler gage when checking pilot's collective stick grip height.

l. Insert 0.020-inch feeler gage between copilot's collective stick and low collective stopbolt. With collective sticks in full down position, check that grip height of pilot's stick from cockpit floor is same as grip height of copilot's stick ± 1/16-inch. If pilot's stick height is beyond limit, perform the following:

(1) Disconnect pilot's and copilot's collective input control rod at torque shaft lever, located in cabin ceiling behind pilot's seat (PARA 11-4-46).

(2) Adjust pilot's low collective stopbolt until grip height of pilot's stick is same as grip height of copilot's stick ± 1/16-inch (Figure 11-4-3-1, Sheet 1, Detail A and Detail B).

(3) **TIGHTEN PILOT'S STOPBOLT JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(4) Using lockwire, Item 197, Appendix D, lockwire stopbolt.

(5) Install collective torque shaft rigging pin, 70700-20380-043, in collective torque shaft (Figure 11-4-3-1, Sheet 2, Detail F).

(6) Adjust and connect pilot's and copilot's overhead control rod to collective torque shaft lever (PARA 11-4-46).

(7) Check to make sure that lockwire, Item 193, Appendix D, will not pass through inspection hole in rod end.

(8) If lockwire will pass through inspection hole, shorten rod until wire will not pass through, and adjust collective control rods forward of overhead control rod until all rods remain in safety after adjustment.

m. Install rigging pins from rigging kit in the following components (Figure 11-4-3-1, Sheet 1, Detail D and Detail E and Sheet 2, Detail F and Detail G):

(1) Longitudinal (Pitch) System:

(a) Pilot's cyclic, 70700-20380-065.

(b) Copilot's cyclic, 70700-20380-065.

(c) Torque shaft at top of cabin, 70700-20380-043.

(2) Lateral (Roll) System:

(a) Pilot's cyclic, 70700-20380-066.

(b) Copilot's cyclic, 70700-20380-066.

(c) Torque shaft at top of cabin, 70700-20380-043.

(3) Collective system: torque shaft at top of cabin, 70700-20380-043 (if not already installed).

(4) Directional (Yaw) System:

(a) Arm between pilot's pedals, 70700-20380-042.

(b) Arm between copilot's pedals, 70700-20380-042.

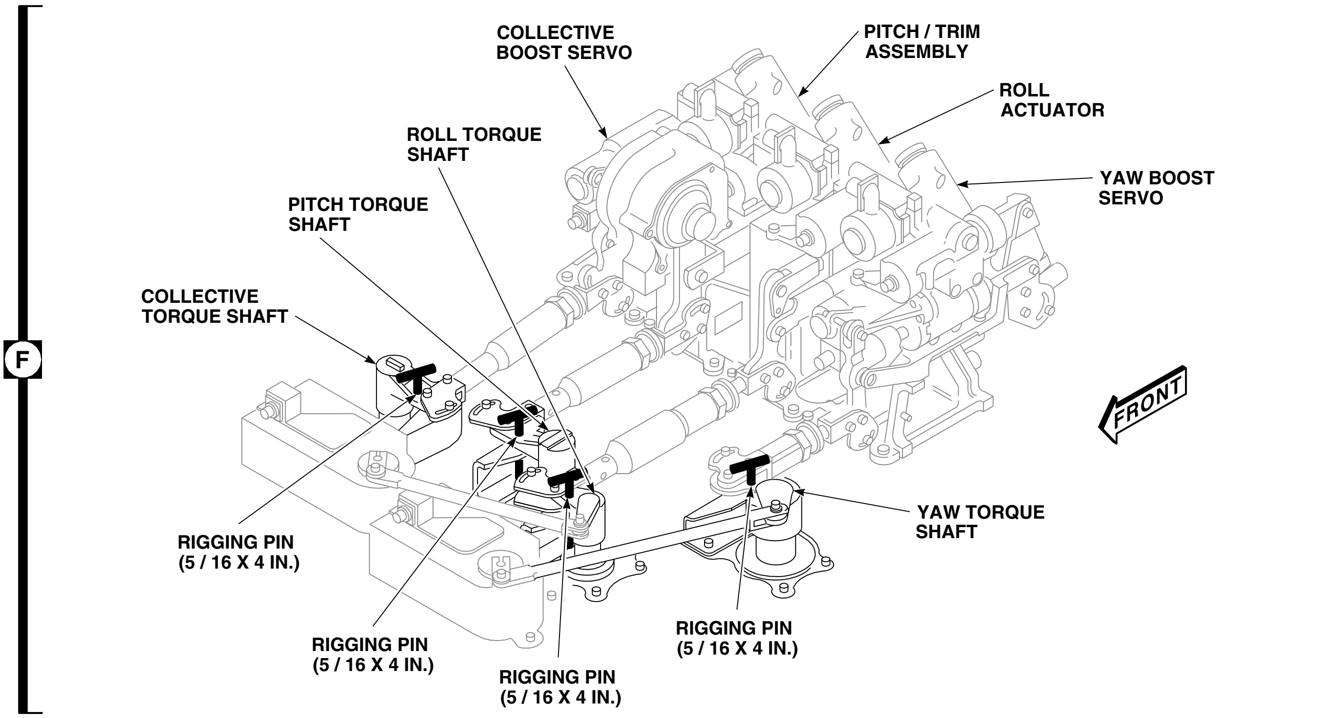
(c) Torque shaft at top of cabin, 70700-20380-043.

n. Adjust cyclic stick front stopbolt as follows:

NOTE

Cyclic stick grip is allowed to touch glare shield padding or instrument knobs with forward stop on cyclic stick properly set.

(1) Push cyclic stick to front.



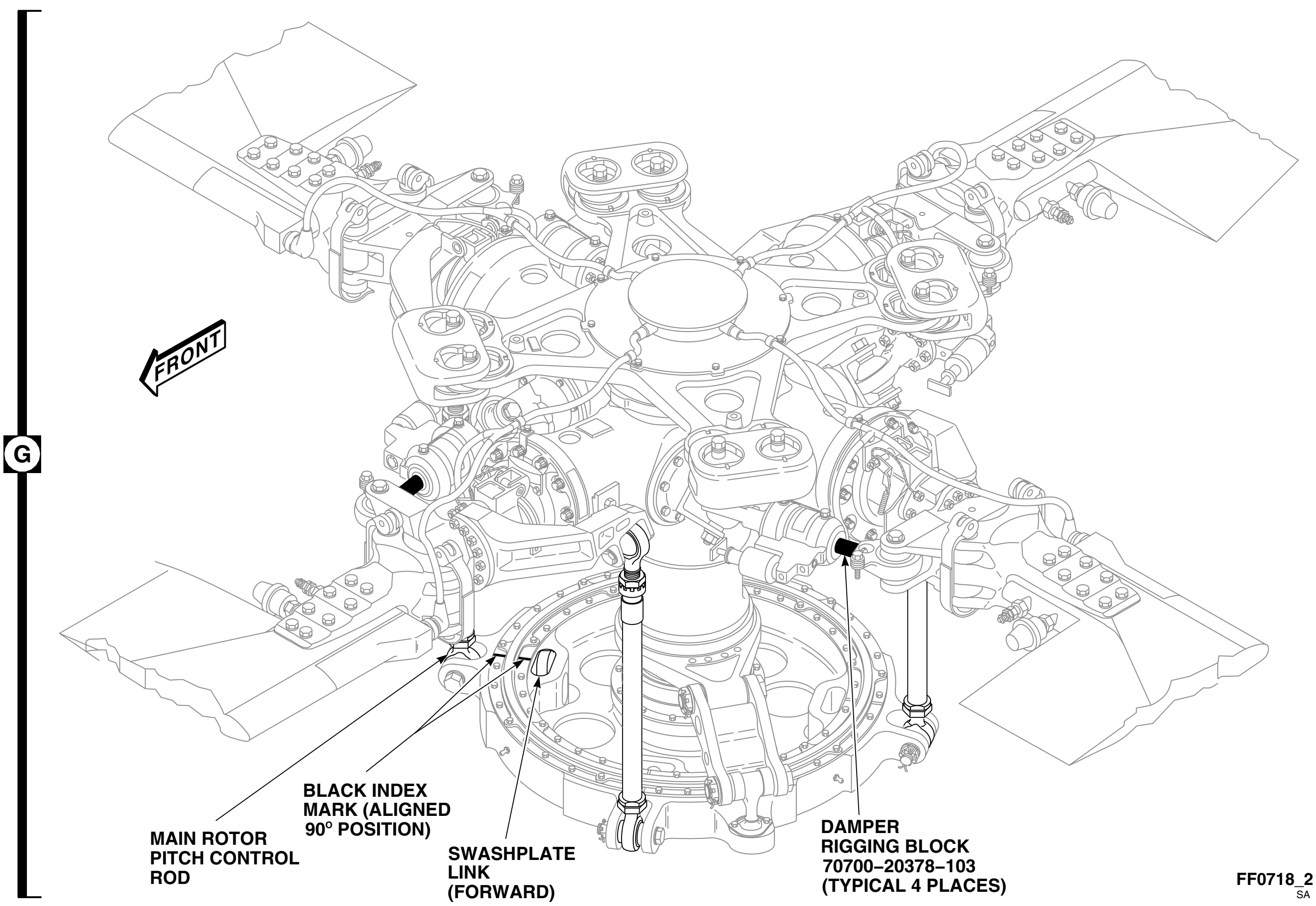


FIGURE 11-4-3-1. RIG MAIN ROTOR SYSTEM (SHEET 2 OF 6)

(2) Adjust front stopbolt to obtain stick grip dimension of 0.730 - 0.790-inch from instrument panel.

(3) **TIGHTEN FRONT STOPBOLT JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(4) Using lockwire, Item 194, Appendix D, lockwire stopbolt.

o. With pedal rigging pins installed in both pedal adjusters, view pedals from side. If pedals are aligned across helicopter, no further adjustment is required. If pedals are not in alignment across helicopter, adjust control rods as follows:

NOTE

Nominal length of control rod from pedal adjuster to pedal support is 10 13⁄16 - 10 7⁄8-inches from bolthole-center to bolthole-center.

(1) Disconnect yaw pedal control rods from pedal supports (PARA 11-4-13 for pilot's rods and PARA 11-4-14 for copilot's rods).

(2) Adjust rod ends 1⁄2 turn in one control rod and 1⁄2 turn out in other until pedals are even.

(3) Check to make sure lockwire, Item 193, Appendix D, will not pass through inspection hole in rod end.

(4) **TIGHTEN JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(5) Connect yaw pedal control rods to pedal supports (PARA 11-4-13 for pilot's rods and PARA 11-4-14 for copilot's rods).

p. If torque shaft rigging pin will not slide in and out easily, adjust collective, pitch, roll, or yaw channel as follows:

(1) Adjust cabin overhead control rods until rigging pins will slide freely in and out of rigging holes in torque shafts.

(2) Check that lockwire, Item 193, Appendix D, will not pass through inspection hole in rod end.

(3) If lockwire will pass through inspection hole, shorten rod until wire will not pass through, and adjust control rods forward of overhead rods until rigging pin slides freely in and out of rigging pin hole in torque shaft.

(4) Check that lockwire, Item 193, Appendix D, will not pass through inspection hole in all control rods that were adjusted.

q. Set cyclic sticks for maximum aft longitudinal (pitch) range of travel as follows:

(1) Remove pilot's and copilot's longitudinal (pitch) rigging pins from cyclic sticks (Figure 11-4-3-1, Sheet 1, Detail E).

(2) Remove pitch torque shaft rigging pin (Figure 11-4-3-1, Sheet 2, Detail F).

(3) Back off pilot's and copilot's aft cyclic stick stops.

(4) Move cyclic stick aft until binding is felt.

(5) Adjust longitudinal (pitch) control rods between pitch torque shaft and cyclic stick until binding occurs at pitch link/pitch lever connection (Figure 11-4-3-1, Sheet 3, Detail H).

(6) While holding aft cyclic, adjust pilot's cyclic rear stopbolt until it touches stick; then turn out approximately 1⁄2 turn more until binding is removed.

(7) **TIGHTEN PILOT'S STOPBOLT JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(8) Using lockwire, Item 197, Appendix D, lockwire stopbolt.

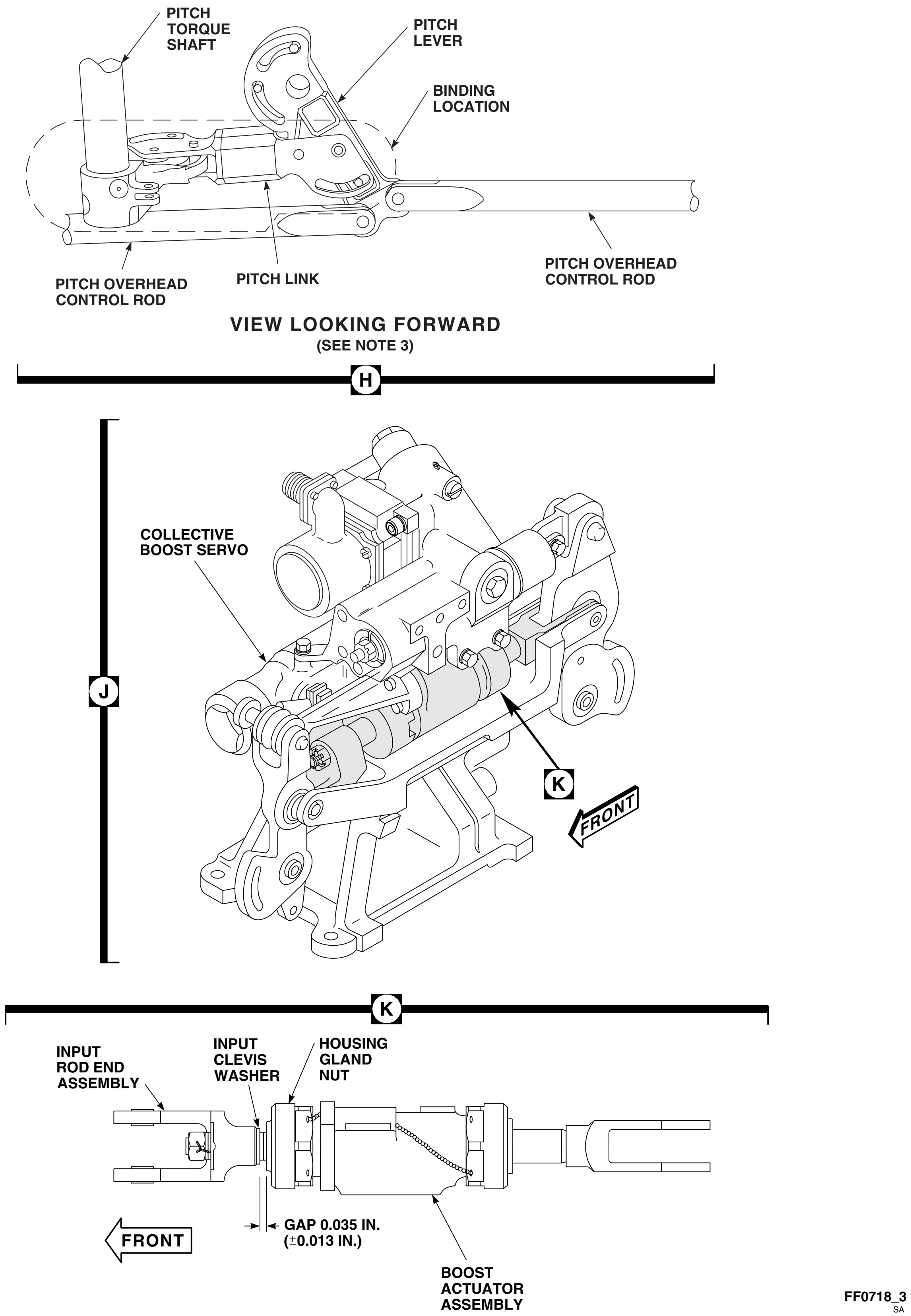


FIGURE 11-4-3-1. RIG MAIN ROTOR SYSTEM (SHEET 3 OF 6)

(9) Adjust copilot's cyclic stick rear stop so it touches stick at same time as pilot's.

(10) **TIGHTEN COPILOT'S STOPBOLT JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(11) Using lockwire, Item 197, Appendix D, lockwire stopbolt.

(12) Install pilot's and copilot's longitudinal (pitch) rigging pins, 70700-20380-065, in cyclic sticks (Figure 11-4-3-1, Sheet 1, Detail E).

(13) Install pitch torque shaft rigging pin, 70700-20380-043, in pitch torque shaft. If rigging pin will not go into torque shaft, repeat step p. and step q. for pitch channel (Figure 11-4-3-1, Sheet 2, Detail F).

r. Check to make sure that lockwire, Item 193, Appendix D, will not pass through inspection holes on all control rods that were adjusted. If lockwire will pass through inspection hole, readjust control rods until all control rods are within limits.

s. Connect the following control rods to torque shafts:

(1) Pitch/Trim (PARA 11-4-64).

(2) Lateral (PARA 11-4-65).

(3) Yaw Boost (PARA 11-4-66).

CAUTION

Damage to flight controls will result if hydraulic power is applied with mixer to primary servo control rods disconnected. Make sure control rods are connected before applying hydraulic power to helicopter.

t. Disconnect the following controls rods from primary servos:

(1) Forward (PARA 11-4-80).

(2) Aft (PARA 11-4-81).

(3) Lateral (PARA 11-4-82).

u. Disconnect mixer to yaw lever pushrod from mixer lever (PARA 11-4-83).

NOTE

There may be two rigging pin holes in pitch/trim assembly. Use aft rigging pin hole for flight control rigging.

v. Bottom collective boost servo in full low collective position (push control rod to rear until piston bottoms internally in the servo housing). **THE EXTERNAL GAP BETWEEN THE INPUT CLEVIS WASHER AND THE HOUSING GLAND NUT SHOULD BE 0.035 ± 0.013-INCH** (Figure 11-4-3-1, Sheet 3, Detail J and Detail K).

w. Adjust and connect collective input control rod to torque shaft as follows:

(1) Loosen jamnut and slide locking device away from sleeve.

(2) Loosen sleeve jamnut.

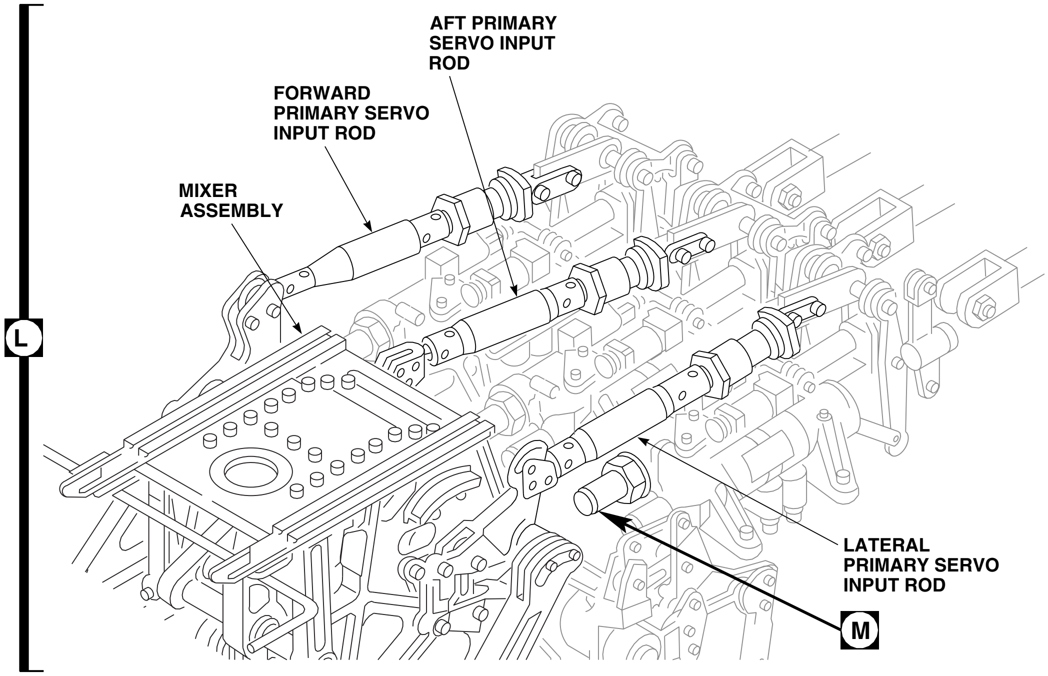
(3) Thread sleeve into tube assembly to make control rod shorter or out to make longer.

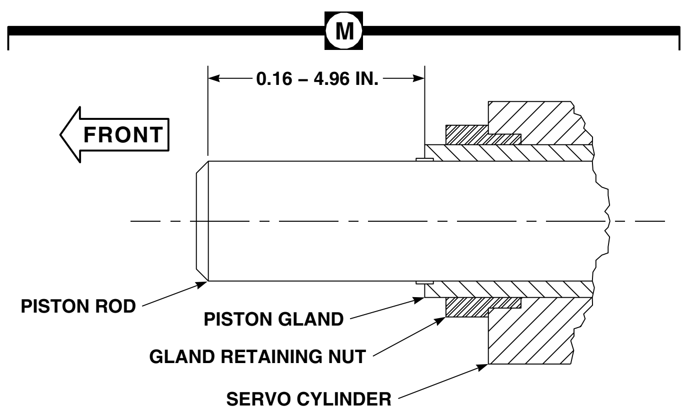
(4) **TIGHTEN SLEEVE JAMNUT 1/6 TO 1/3 TURN PAST WHERE SHARP RISE IN TORQUE IS FELT.**

(5) Line up key of locking device on sleeve side of control rod with keyway in sleeve. Line up and mesh rod end side of locking device with sleeve side of locking device and tighten jamnut.

(6) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.**

(7) Check that lockwire, Item 193, Appendix D, will not pass through inspection holes on all control rods that were adjusted. If





TYPICAL FOR ALL PRIMARY SERVOS

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FIGURE 11-4-3-1. RIG MAIN ROTOR SYSTEM (SHEET 4 OF 6)

lockwire will pass through inspection hole, repeat step w. (1) through step w. (6) until control rod is within limits.

(8) Check that witness hole in rod end keyway cannot be seen. If necessary, repeat step w. (1) through step w. (7) until condition is met.

(9) Using lockwire, Item 197, Appendix D, lockwire jamnuts to locking device tab on torque shaft output control rod.

NOTE

The following adjustment is required to make sure the collective controls bottom on the collective stick stop and not at the collective boost servo.

x. Check to see if pilot's collective stick is resting on low collective stopbolt (Figure 11-4-3-1, Sheet 1, Detail B):

(1) If pilot's collective stick is touching low collective stopbolt, no adjustment is required.

(2) If pilot's collective stick is not touching low collective stopbolt, adjust stop until it just touches stick, then lengthen ¼ turn.

(3) Check for 0.010 - 0.030-inch gap between copilot's collective stick and low collective stopbolt. If gap is beyond limits, perform the following:

(a) Insert 0.030-inch feeler gage between copilot's collective stick and low collective stopbolt.

(b) Adjust copilot's low collective stopbolt to obtain 0.030-inch gap between copilot's collective stick and low stopbolt.

(4) **TIGHTEN COPILOT'S/PILOT'S STOPBOLT JAMNUT ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(5) Using lockwire, Item 197, Appendix D, lockwire stopbolt.

y. Install the following rigging pins (PARA 11-4-2):

(1) Pitch/trim assembly, 70700-20380-044.

(2) Roll actuator, 70700-20380-044.

(3) Yaw boost servo, 70700-20380-062.

z. If necessary, adjust pitch, roll, and yaw input control rods (PARA 11-4-63).

aa. Connect engine LDS cables to mixer levers (PARA 4-4-27).

ab. Connect load-demand rotary inputs to engine hydromechanical unit (PARA 4-4-27).

ac. Install long mixer rigging pin, 70700-20380-046, in right side of mixer (PARA 11-4-2). If rigging pin cannot be installed, adjust collective, roll, and yaw boost servo to mixer control rods until rigging pin can be installed as follows:

(1) Loosen jamnuts.

(2) Turn adjuster to lengthen or shorten control rod.

(3) Insert lockwire, Item 193, Appendix D, into inspection holes in tube and rod end. Adjuster threads should stop wire from passing through inspection holes.

(4) **TIGHTEN ASSEMBLY JAMNUT ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(5) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.**

(6) Using lockwire, Item 197, Appendix D, lockwire adjuster to tube jamnut and rod end jamnut to tab on locking device.

ad. Remove all rigging pins (PARA 11-4-2).

ae. Connect the following control rods to primary servos:

NOTE

Do not adjust primary servo input control rods when connecting them to servos. Move controls as required to align control rod to servo mounting holes.

(1) Forward (PARA 11-4-80).

(2) Aft (PARA 11-4-81).

(3) Lateral (PARA 11-4-82).

af. Connect mixer to yaw lever pushrod to mixer lever (PARA 11-4-83).

CAUTION

- Damage to flight control components will result if normal cockpit control forces are applied against a pin point aft of boost servos when not restrained by an intermediate pin point. Make sure all rigging pin requirements are carefully followed when hydraulic pressure is applied to helicopter.
- Damage to flight control components will result if hydraulic power is shut down during rigging with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

NOTE

Backup pump should be used only as an alternate source to power hydraulic systems. External hydraulic power should be applied whenever available.

ag. Turn on all hydraulic power (PARA 1-6-15).

ah. Install the following rigging pins:

(1) Pilot's and copilot's longitudinal pitch pins, 70700-20380-065 (Figure 11-4-3-1, Sheet 1, Detail E).

(2) Pilot's and copilot's cyclic stick, 70700-20380-066 (Figure 11-4-3-1, Sheet 1, Detail E).

(3) Pilot's and copilot's pedal adjuster, 70700-20380-065 (Figure 11-4-3-1, Sheet 1, Detail D).

(4) Pitch torque shaft, 70700-20380-043 (Figure 11-4-3-1, Sheet 2, Detail F).

(5) Roll torque shaft, 70700-20380-043.

(6) Yaw boost servo, 70700-20380-062.

ai. Place collective stick in full down position and lock in place.

aj. Set swashplate to its initial position as follows:

CAUTION

Damage to flight controls will result if controls in cockpit are moved while attempting to install three large rigging pins. Make sure all personnel are out of cockpit before attempting to install any of the three large rigging pins.

NOTE

Correct alignment of rigging pin holes in forward bellcrank support with holes in forward bellcrank, lateral bellcrank, and walking beam is required to properly set initial position of swashplate relative to cockpit controls. This is only a preliminary adjustment and any future changes in length to primary servo input control rods will result in the inability to install one or more of the three large rigging pins.

(1) Adjust primary servo input control rods so that rigging pin holes in forward bellcrank support can be visually seen to line up with rigging pin holes in forward bellcrank, lateral bellcrank, and walking beam.

NOTE

Do not leave any of the three large rigging pins installed in forward bellcrank support. The following checks are for slip-fit only and pin should be removed after attempting to install it.

(2) Attempt to install the following rigging pins:

(a) Forward bellcrank support to forward bellcrank rigging pin, 70700-20342-041.

(b) Forward bellcrank support to walking beam rigging pin, 70700-20342-041.

(c) Forward bellcrank support to lateral bellcrank rigging pin, 70700-20342-042.

(3) If pins slide in easily, setting of swashplate initial position is complete.

CAUTION

Damage to flight controls will result if control rod is adjusted while large rigging pin is partially installed. Make sure rigging pin is completely removed before attempting to adjust primary servo input control rod.

(4) If pin(s) will not slide in easily, remove rigging pin and adjust primary servo input control rod(s) until all three large rigging pins will slide easily into rigging pin hole. Once all three large rigging pins will slide easily into rigging pin holes, setting of swashplate initial position is complete.

ak. Remove rigging pins from pitch torque shaft, roll torque shaft, and yaw torque shaft. Release friction from collective stick.

al. Install damper rigging blocks. Turn main rotor head until black index marks on swashplate line up and No. 1 (red) blade (master blade) is over 90° position, with pitch control rod over front longitudinal (pitch) input (Figure 11-4-3-1, Sheet 2, Detail G).

am. Turn on all electrical power (PARA 1-6-16).

an. Perform BOOST ON, SAS OFF procedure (PARA 11-4-4).

ao. Install the following rigging pins:

(1) Pilot's and copilot's longitudinal pin, 70700-20380-065.

(2) Pilot's and copilot's cyclic stick, 70700-20380-066 (Figure 11-4-3-1, Sheet 1, Detail E).

(3) Pilot's and copilot's pedal adjuster, 70700-20380-065 (Figure 11-4-3-1, Sheet 1, Detail D).

(4) Pitch torque shaft, 70700-20380-043 (Figure 11-4-3-1, Sheet 2, Detail F).

(5) Roll torque shaft, 70700-20380-043.

(6) Yaw boost servo, 70700-20380-062.

NOTE

- It is essential to keep helicopter perfectly still while zeroing and taking blade readings with digital protractor. A limit of three persons is recommended on helicopter-one to position controls in cockpit, one to take readings, and one to assist in rotating blades to the 90°, 180°, 270°, and 0° positions.
- Helicopter must be jacked (just enough to relieve load on landing gear struts) so that helicopter will not rock if personnel enter or exit helicopter while blade angle readings are being taken.

ap. Jack helicopter slightly (just enough to relieve load on landing gear struts) (PARA 1-6-3).

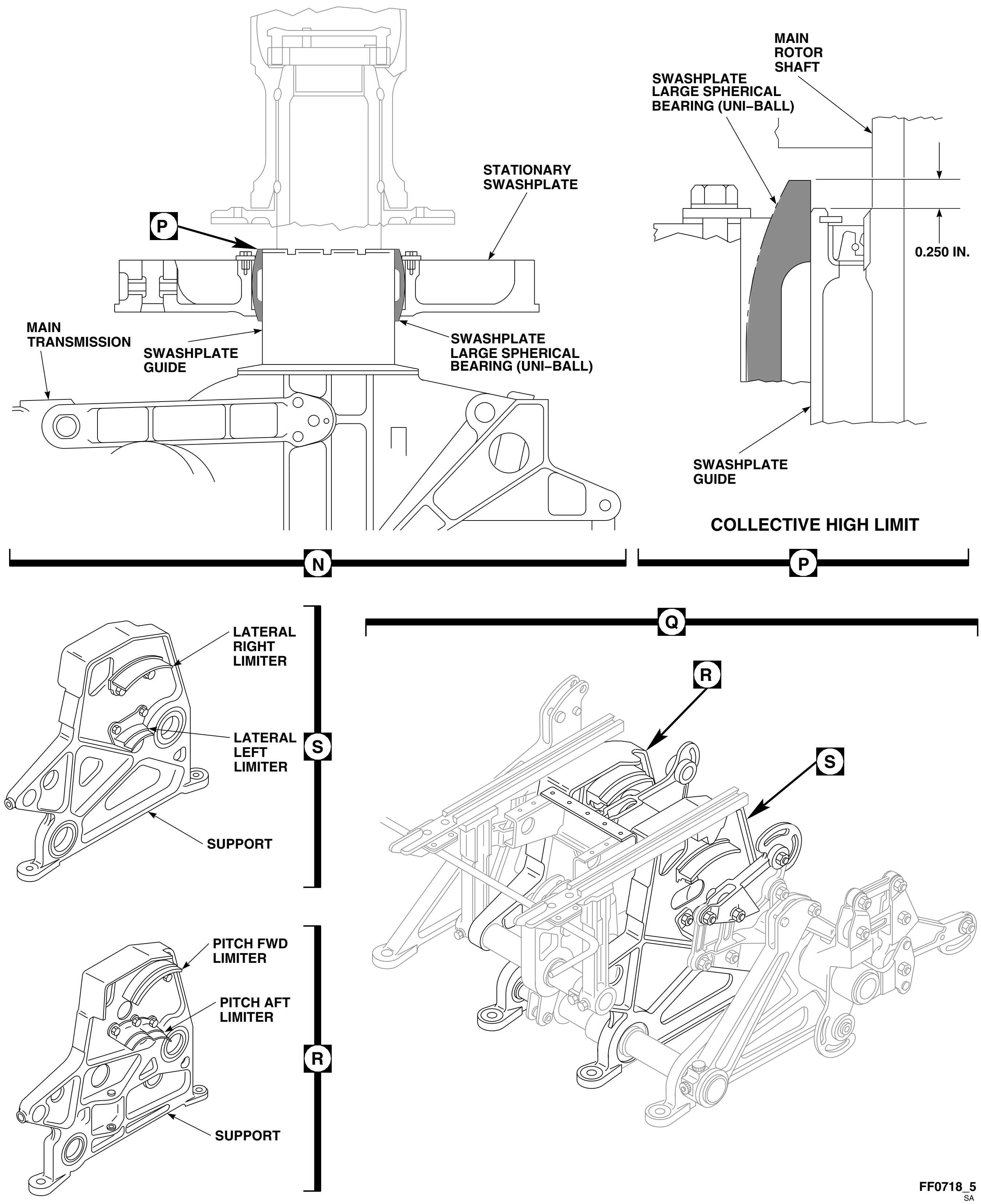

FIGURE 11-4-3-1. RIG MAIN ROTOR SYSTEM (SHEET 5 OF 6)

NOTE

Use main rotor rigging worksheet to record readings (Figure 11-4-3-2). Refer to Figure 11-4-3-3 for sample rigging worksheet with calculations.

aq. Turn main rotor until stationary and rotating swashplate alignment marks line up at 90° position for No. 1 (red) blade.

NOTE

After cycling controls three times, do not recycle controls until all readings have been taken for a particular cyclic/ collective control position.

ar. Cycle collective at least three times before taking collective pitch readings.

as. Unless otherwise instructed, apply 2 - 5 pounds hand/foot force, in following directions, while taking blade angle readings:

(1) Cyclic stick - left and rear.

(2) Yaw pedals - left.

(3) Collective - full down.

at. Disconnect blade deice electrical connectors from main rotor blades (PARA 12-4-26).

au. Install protective caps and plugs on electrical connectors.

av. With collective stick in full down position, perform the following:

NOTE

One turn of rotating pitch control rod is equal to about 0.5° blade pitch.

(1) Adjust pitch control rod of No. 1 (red) blade until average blade angle (as blade is positioned at 90°, 180°, 270°, and 0° locations) is -0.6° ± 0.125° (Figure 11-4-3-1, Sheet 6, Detail V).

(2) Once this angle is satisfied for red blade, turn each of remaining three blades to 90° location and adjust each pitch control rod to be same as red blade angle ± 0.3°.

(3) Check to make sure that exposed threads on each pitch control rod are not more than ¾-inch (Figure 11-4-3-1, Sheet 6, Detail W).

aw. Back off copilot's collective stick high pitch stop so that they will not interfere with adjustment of pilot's stick stop.

CAUTION

Damage to swashplate large spherical bearing (uni-ball) will result if uni-ball is allowed to extend past swashplate guide by more than 0.250-inch (Figure 11-4-3-1, Sheet 5, Detail N and Detail P). Make sure that swashplate large spherical bearing (uni-ball) does not extend past swashplate guide by more than 0.250-inch when applying full up collective (Figure 11-4-3-1, Sheet 5, Detail N and Detail P).

NOTE

A preliminary setting of the high collective stopbolt may be obtained by raising the collective stick until the swashplate ball extends by 0.125-inch over the top of the transmission guide. Care must be taken not to exceed the 0.250-inch maximum extension.

ax. Raise and hold collective stick in full up position and perform the following:

(1) While maintaining upward pressure on stick, adjust pilot's collective stick stop until four reading average (taken as blade is positioned at 90°, 180°, 270°, and 0° locations) is 15.0° to 16.0° (Figure 11-4-3-1, Sheet 6, Detail V).

(2) **AFTER ADJUSTMENT, TIGHTEN STOP JAMNUT.**

(3) Using lockwire, Item 197, Appendix D, lockwire jamnut.

ay. After adjusting pilot's stick stop, adjust copilot's collective stick stop to agree with pilot's. Using lockwire, Item 197, Appendix D, lockwire jamnut.

az. Check that swashplate large spherical bearing (uni-ball) does not extend past guide by more than 0.250-inch (Figure 11-4-3-1, Sheet 5, Detail N and Detail P).

ba. Remove longitudinal (pitch) cyclic stick rigging pin and pitch trim pin.

bb. Adjust cyclic stick forward (pitch) angle as follows:

(1) Raise and hold collective stick to full up position and move cyclic stick to full forward position.

(2) Check that pitch forward limiter roller on mixer cannot be freely turned by hand (Figure 11-4-3-1, Sheet 5, Detail Q and Detail R) and that cyclic stick is not touching front stopbolt on yoke. Measure blade angle at 90° and 270° locations, using universal propeller or digital protractor (Figure 11-4-3-1, Sheet 6, Detail V). If range difference between both readings is 33.0° or greater and pitch forward limiter roller cannot be freely turned, no further adjustment is required.

(3) If forward limiter roller can be turned by hand, check for interference with linkage forward of mixer. If no interference with linkage forward of mixer can be found, check primary servo for bottoming. **FORWARD END OF PRIMARY SERVO PISTON ROD MUST EXTEND 0.16 - 4.96-INCHES BEYOND SERVO PISTON GLAND** (Figure 11-4-3-1, Sheet 4, Detail L and Detail M). If forward limiter is contacted and roller cannot be turned by hand, but blade angle is less than 33.0°, perform the following:

NOTE

When biasing servo input rods, make sure that forward and aft primary servo control rods are adjusted in equal and opposite directions. If one rod is lengthened by ½ turn, the other must be shortened by ½ turn. A ½ turn bias of these rods is normal and expected.

(a) Shorten forward primary servo control rod by turning barrel to right (while facing rear) and lengthen aft primary servo control rod by turning barrel to left, the same amount (while facing rear).

(b) Perform adjustment until range difference between both readings is greater than 33.0° and pitch forward limiter roller cannot be freely turned.

NOTE

Over biasing the primary servo rods, and obtaining angles much greater than 33°, may result in the inability to obtain the high collective aft blade angle requirement.

(c) Enter amount of adjustment (bias) that was made to forward and aft primary servo control rods at bottom of rigging worksheet (Figure 11-4-3-2).

(d) Check to make sure main rotor primary servo is not bottomed. **FORWARD END OF PRIMARY SERVO PISTON ROD MUST EXTEND 0.16 - 4.96-INCHES BEYOND SERVO PISTON GLAND** (Figure 11-4-3-1, Sheet 4, Detail L and Detail M).

bc. While holding high collective, pull cyclic stick to full aft position. **VERIFY THE CYCLIC STICK IS CONTACTING THE AFT STICK STOP.** If the stick stop is not contacted, check the control system for interference.

bd. Position No. 1 (red) blade in 90° and 270° position and read blade angles in each position. **DIFFERENCE BETWEEN 90° AND 270° POSITIONS MUST BE 19.0° OR GREATER.**

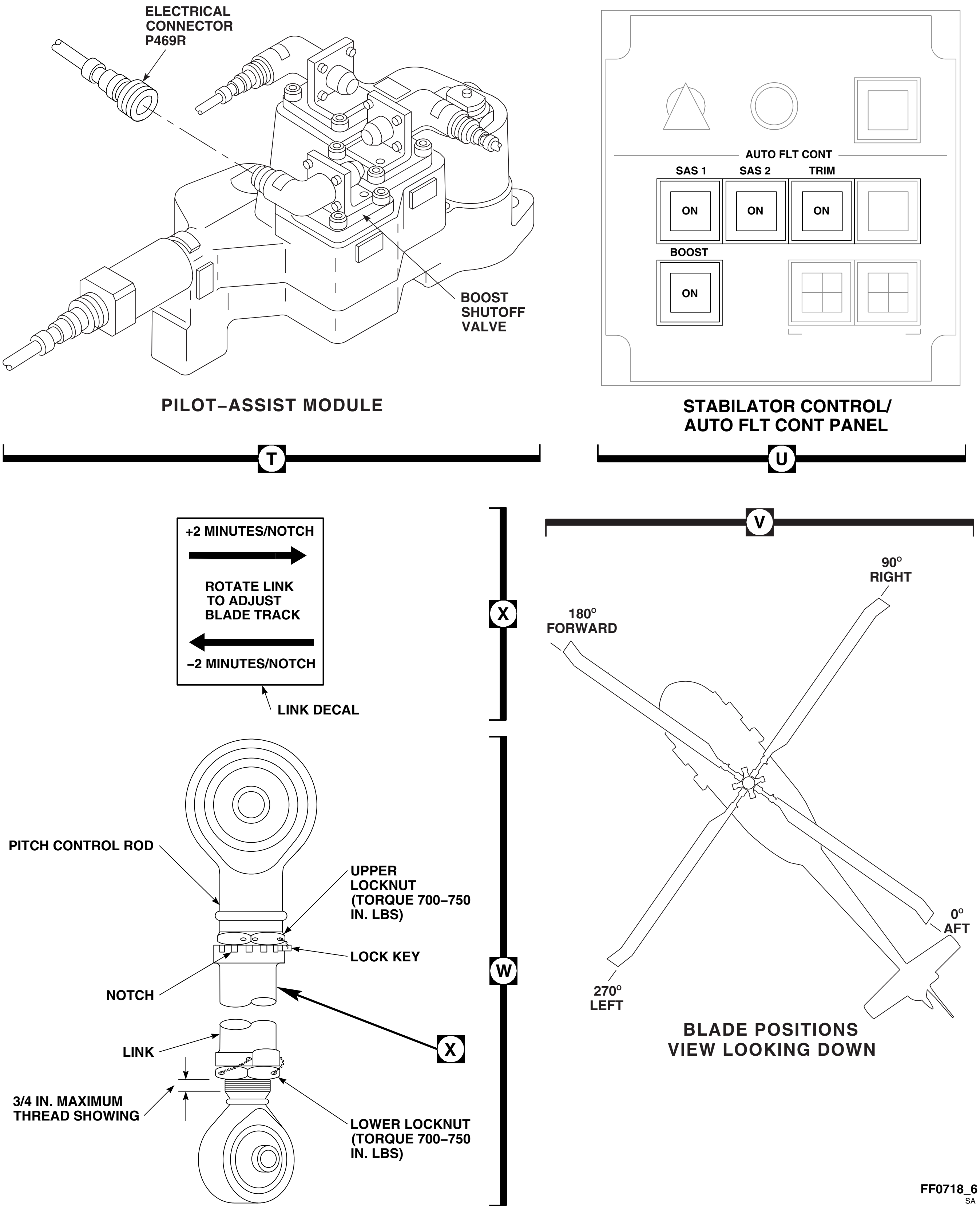

FIGURE 11-4-3-1. RIG MAIN ROTOR SYSTEM (SHEET 6 OF 6)

WORKSHEET
#1 (RED BLADE POSITION)

AIRCRAFT TAIL NO. ______________ BLADES ON–OFF ______________ DATE ______________

CONTROL POSITION (YAW PINNED)		90° RT	180° FWD	270° LFT	0° AFT	TOTAL°	AVG	REQUIRED TOLERANCE
LOW COLL ANGLE LOW COLL NEUTRAL CYCLIC NEUTRAL YAW	BLADES #1							–0.60° ± 0.125° (BLADES ON) –0.10 ± 0.125° (BLADES OFF)
	#2 #3 #4							BLADES #2, #3, #4 SAME AS #1 BLADE
HIGH COLL ANGLE HIGH COLL NEUTRAL CYCLIC NEUTRAL YAW								15.5 ± 0.500 (BLADES ON) 16.0 ± 0.500 (BLADES OFF)
FWD CYCLIC ANGLE HIGH COLL NEUTRAL LAT FWD CYCLIC NEUTRAL YAW								DIFFERENCE BETWEEN 270° & 90° POSITIONS +33.0° MIN.
AFT CYCLIC ANGLE HIGH COLL NEUTRAL LAT AFT CYCLIC NEUTRAL YAW								DIFFERENCE BETWEEN 90° & 270° POSITIONS NOT LESS THAN +19°
LEFT CYCLIC ANGLE LOW COLL NEUTRAL LONG LEFT CYCLIC NEUTRAL YAW								DIFFERENCE OF 180° & 0° POSITION TO BE 15° TO 17°
RIGHT CYCLIC ANGLE LOW COLL NEUTRAL LONG RIGHT CYCLIC NEUTRAL YAW								DIFFERENCE OF 180° TO 0° POSITION TO BE 14° TO 16°
(FWD) 180° (LEFT) 270° — 90° (RIGHT) 0° (AFT)		**PRIMARY SERVO CONTROL ROD BIAS:** FWD ADDITIONAL ADJUSTMENT: AFT ADDITIONAL ADJUSTMENT: LAT ADDITIONAL ADJUSTMENT:						

FA1632B
SA

FIGURE 11-4-3-2. MAIN ROTOR RIGGING WORKSHEET

If difference between 90° and 270° positions is less than 19.0°, adjust as follows:

(1) Lengthen pitch/trim assembly input control rod until 19.0° is obtained.

(2) Using lockwire, Item 197, Appendix D, lockwire jamnuts to locking device tab.

(3) Recheck reading with cyclic stick both forward and aft. If necessary, repeat adjustments until conditions are met.

(4) If correct forward and aft cyclic reading cannot be obtained, check for interference in control system between torque shafts and primary servos. Repeat adjustments as required until correct readings are obtained.

(5) Check primary servos for bottoming.

NOTE

Adjustment of pitch/trim assembly input control rod may result in the inability to install long mixer rigging pin in right side of mixer.

(6) If pitch/trim assembly input control rod was adjusted, check long mixer rigging pin installation as follows:

(a) Lower collective stick to full down position and hold in place.

(b) Install pilot's cyclic stick fore-and-aft (pitch) rigging pin, 70700-20380-065 (Figure 11-4-3-1, Sheet 1, Detail E).

(c) Install pitch torque shaft rigging pin, 70700-20380-043 (Figure 11-4-3-1, Sheet 2, Detail F).

(d) Attempt to install long mixer rigging pin in right side of mixer. If unable to install rigging pin, record this in helicopter logbook for future reference.

(e) Remove rigging pins previously installed in step bd. (6) (b) and step bd. (6) (c) in reverse order.

be. Move collective stick to full down position, and hold in place.

bf. Install pilot's cyclic stick fore-and-aft (pitch) rigging pin, 70700-20380-065, and pitch trim rigging pin, 70700-20380-044.

bg. Remove pilot's cyclic stick lateral (roll) rigging pin and roll trim pin.

bh. Move cyclic stick full left until roller touches lateral (roll) left limiter in mixer (Figure 11-4-3-1, Sheet 5, Detail Q and Detail S). If roller does not contact lateral left limiter in mixer, check for interference with linkage forward of mixer, and clear before proceeding. With the roller contacting limiter, adjust pilot's and copilot's left stick stops, at base of cyclic stick, to 0.010 - 0.030-inch clearance between bolt and yoke.

NOTE

Left roll cyclic range is found by adding 180° reading to 0° reading, ignoring that one reading is a negative number.

bi. With cyclic stick in full left position, position No. 1 (red) blade to 180° and 0° locations (Figure 11-4-3-1, Sheet 6, Detail V). Using universal propeller or digital protractor, measure blade angle at each location. **SUM OF BOTH READINGS SHOULD BE 15° - 17°.** If sum of both readings is less than 15°, adjust lateral primary servo input control rod as follows:

(1) Shorten lateral primary servo control rod by turning barrel to right (while facing rear).

(2) Continue to shorten control rod in small increments until total angle between 0° and 180° positions reads 15° to 17°.

(3) Enter amount of adjustment (bias) that was made to lateral primary servo control rod at bottom of rigging worksheet (Figure 11-4-3-2).

bj. Move cyclic stick full right until roller touches lateral (roll) right limiter in mixer (Figure 11-

4-3-1, Sheet 5, Detail Q and Detail S). If roller does not contact lateral left limiter in mixer, check for interference with linkage forward of mixer, and clear before proceeding. With the roller contacting the limiter, adjust pilot's and copilot's right stick stops, at base of each cyclic stick, to 0.010 - 0.030-inch clearance between bolt and yoke. **TIGHTEN JAMNUTS 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 197, Appendix D, lockwire jamnuts.

NOTE

Add 0° reading to 180° reading, ignoring that this reading is negative.

bk. With cyclic stick in full right position, move No. 1 (red) blade to 180° and 0° locations, and measure blade angle at both locations, using universal propeller or digital protractor (Figure 11-4-3-1, Sheet 6, Detail V). **SUM OF BOTH READINGS SHOULD BE 14° - 16°.** If sum of both readings is less than 14°, adjust lateral primary servo input control rod as follows:

(1) Lengthen lateral primary servo control rod by turning barrel to left (while facing aft).

(2) Continue to lengthen control rod in small increments until total angle between 0° and 180° positions reads 14° - 16°.

(3) Recheck low collective left cyclic range to make sure it is still within limits. Adjust lateral primary servo input control rod until both lateral range requirements are within limits.

(4) Enter amount of adjustment (bias) that was made to lateral primary servo control rod at bottom of rigging worksheet (Figure 11-4-3-2).

(5) Using lockwire, Item 197, Appendix D, lockwire jamnuts to locking device tab on lateral primary servo input pushrod.

bl. Place a copy of rigging worksheet with aircraft historical records.

bm. Remove pilot's fore-and-aft (pitch) cyclic rigging pin (Figure 11-4-3-1, Sheet 1, Detail E).

bn. Move collective stick to full down position.

bo. Maintain downward pressure on collective stick and push cyclic stick fully forward. Check to make sure that roller on mixer output crank is contacting forward limiter and that roller cannot be rotated using finger pressure. Check forward and aft primary servo for bottoming.

bp. Maintain full down collective and pull cyclic stick fully aft. Check to make sure that roller on mixer output crank is contacting aft limiter and that roller cannot be rotated using finger pressure. Check forward and aft primary servo for bottoming.

bq. Remove yaw boost servo rigging pin and pedal adjuster pin.

br. Remove damper rigging blocks.

bs. Connect blade deice electrical connectors to main rotor blades (PARA 12-4-26).

bt. Center cyclic stick and yaw pedals.

CAUTION

Damage to equipment may result if forward longitudinal (pitch) bellcrank comes into contact with main transmission housing. A minimum clearance of 1⁄16-inch is required.

bu. Place collective in full up position and cyclic stick against rear stop.

bv. Slowly move pedals in short increments to full left pedal, making sure that 1⁄16-inch minimum clearance between bellcrank and housing is maintained throughout entire pedal movement.

bw. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

bx. Remove protective caps and plugs from electrical connectors.

by. Inspect and treat electrical connectors (PARA 1-7-20).

bz. Connect electrical connector, P469R, to boost shutoff valve on top of pilot-assist module (Figure 11-4-3-1, Sheet 6, Detail T).

ca. Check collective and cyclic balance springs as follows:

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) On STABILATOR CONTROL/AUTO FLT CONT panel, engage BOOST switch (Figure 11-4-3-1, Sheet 6, Detail U).

(3) Pull collective stick up to high collective. **STICK SHOULD STAY UP AND NOT MOVE.**

(4) Push collective stick down to low collective. **UPWARD MOVEMENT MUST NOT EXCEED ½-INCH.**

(5) On STABILATOR CONTROL/AUTO FLT CONT panel, disengage TRIM switch (Figure 11-4-3-1, Sheet 6, Detail U).

(6) Move cyclic stick to neutral position.

(7) Push stick forward about 6-inches. Let go of stick. **STICK SHOULD NOT MOVE FORWARD.** Move it forward again until it is almost fully forward. Let go of stick. **STICK SHOULD NOT MOVE FORWARD.**

(8) If collective or cyclic sticks move when they are let go, adjust balance springs (PARA 11-4-61 and PARA 11-4-62).

cb. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

cc. Service hydraulic systems (PARA 1-3-8).

cd. If main rotor pitch control rods have been adjusted for collective angles, adjust rods for pretrack as follows:

(1) Divide number stenciled on bottom of blade by 2. This number is the number of notches to turn rod.

(2) Loosen upper and lower locknuts and slide lock key upward enough to allow link to be free to rotate (Figure 11-4-3-1, Sheet 6, Detail W).

(3) Turn rod in +2 MINUTES/NOTCH (counterclockwise) direction if number on blade is positive, or -2 MINUTES/NOTCH (clockwise) direction if number on blade is negative (Figure 11-4-3-1, Sheet 6, Detail X).

(4) Check for no more than ¾-inch of threads showing on lower end of pitch control rod (Figure 11-4-3-1, Sheet 6, Detail W).

(5) Tighten upper and lower locknuts as follows:

NOTE

To prevent improper torque of upper locknut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

(a) **SECURE LOCK KEY IN NOTCH BY TIGHTENING UPPER LOCKNUT.**

(b) **TORQUE UPPER LOCKNUT TO 700 - 750 INCH-POUNDS.**

(c) **TORQUE LOWER LOCKNUT TO 700 - 750 INCH-POUNDS.**

(6) Using lockwire, Item 197, Appendix D, lockwire lower locknut to link, and then back to lower locknut (double-safety).

(7) Using lockwire, Item 197, Appendix D, lockwire from upper locknut to lock key, and then back to opposite side of upper locknut (double-safety).

(8) On helicopters with pitch control rods equipped with elastomeric rod ends,

check helicopter logbook to make sure pitch control rod upper and lower locknuts retorque is shown due 9 to 11 flight hours after installation (PARA 1-7-5).

ce. Rig LDS system (PARA 4-4-27).

cf. Perform the following quality control checks on main rotor flight control system:

(1) No passage of lockwire, Item 193, Appendix D, through control rod inspection holes.

(2) Main rotor pitch control rods above rotating swashplate for no more than ¾-inch thread showing.

(3) Swashplate large spherical bearing (uniball) extends no more than 0.250-inch above swashplate guide with collective in full up position (Figure 11-4-3-1, Sheet 5, Detail N and Detail P).

(4) Primary servo input control rods for lockwire.

(5) Control rods for lockwire.

(6) Cotter pins installed in nuts, where required.

(7) All flight control system attaching hardware installed and safetied.

(8) All control system tri-pivot hardware is properly installed and safetied.

(9) Cyclic and collective stick stopbolts, and high pitch collective mixer stopbolt for lockwire.

(10) All rigging pins and damper rigging blocks removed.

(11) Cycle controls through full range. Other than previously adjusted stops, no interference with any component is allowable except cyclic stick grip may touch glare shield padding or instrument knobs with forward stop on cyclic stick properly set.

cg. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

ch. Perform tail rotor system complete rig (PARA 11-4-5).

ci. Perform flight control friction check (PARA 11-2-3) and digital AFCS operational check (TM 11-70-23).

cj. Make sure all areas are clean and free of foreign objects before installing panels, boots, and covers.

ck. Install access panels (PARA 2-4-43) and pedal boots (PARA 11-4-7) around pilot’s and copilot’s pedal assemblies.

cl. Install pilot’s and copilot’s collective stick covers and boots (PARA 11-4-26).

cm. Install pilot’s and copilot’s cyclic stick boots (PARA 11-4-15).

cn. Install left and right control enclosure covers, STA 247.0 through STA 265.0, and ceiling soundproofing panels, STA 247.0 through STA 295.0 (PARA 2-4-69).

co. Install main rotor pylon work platform (PARA 2-4-105).

cp. Install main rotor pylon sliding cover (PARA 2-4-101).

cq. Perform track and balance of main rotor blades (TM 1-70-VIB).

cr. Perform maintenance test flight of autorotational RPM and high collective stop (Appropriate Flight Manual).

AIRCRAFT TAIL NO. *79 – 23278*

WORKSHEET
#1 (RED BLADE POSITION)

DATE *30 Jun 95*

CONTROL POSITIONS	90° RT	180° FWD	270° LFT	0° AFT	TOTAL°	AVG	REQUIRED TOLERANCE
LOW COLL ANGLE (NOTE 1) LOW COLL NEUTRAL CYCLIC NEUTRAL YAW BLADES – #1	#1 *+10.1*	*–2.3*	*–11.4*	*+1.5*	*–2.1*	*–0.525*	–0.6° ± 0.125° (–0.475° TO –0.725°)
BLADES – #2, #3, #4	#2 *+10.2* #3 *+10.2* #4 *+10.1*						BLADES #2, #3, #4 SAME AS #1 BLADE
HIGH COLL ANGLE (NOTE 2) HIGH COLL NEUTRAL CYCLIC NEUTRAL YAW	*+21.2*	*+11.0*	*+9.9*	*+20.0*	*+62.1*	*+15.525*	+15° TO +16°
FWD CYCLIC ANGLE HIGH COLL NEUTRAL LAT FWD CYCLIC NEUTRAL YAW	*–2.0*		*+32.0*		*+34.0*		DIFFERENCE BETWEEN 270° & 90° POSITIONS +33.0° MIN.
AFT CYCLIC ANGLE HIGH COLL NEUTRAL LAT AFT CYCLIC NEUTRAL YAW	*+26.2*		*+4.8*		*+21.4*		DIFFERENCE BETWEEN 90° & 270° POSITIONS NOT LESS THAN +19°
LEFT CYCLIC ANGLE LOW COLL NEUTRAL LONG LEFT CYCLIC NEUTRAL YAW		*–9.0*		*+7.4*	*+16.4*		SUM OF 180° & 0° POSITION TO BE 15° TO 17° AVG RANGE +7.5° TO 8.5°
RIGHT CYCLIC ANGLE LOW COLL NEUTRAL LONG RIGHT CYCLIC NEUTRAL YAW		*+7.1*		*–8.1*	*+15.2*		SUM OF 180° & 0° POSITION TO BE 14° TO 16° AVG RANGE +7.0° TO 8.0°

(FWD) 180°
(LEFT) 270° — 90° (RIGHT)
0° (AFT)

NOTES

1. ON NEW MAIN ROTOR HEADS AT 90° RIGHT POSITION, ADJUST NO. 1 PITCH CONTROL ROD TO OBTAIN AVERAGE ANGLE. REMAINING THREE PITCH CONTROL RODS MUST BE WITHIN 0.3 DEGREES OF NO. 1.
2. CHECK SWASHPLATE LARGE SPHERICAL BEARING (UNI–BALL) LIMIT (0.250 IN. MAX).

PRIMARY SERVO CONTROL ROD BIAS:	
FWD:	*1/2 Turn Shorter*
AFT:	*1/2 Turn Longer*
LAT:	*No change*

SAMPLE RIG SHEET ANGLE CALCULATIONS

LOW COLLECTIVE BLADE ANGLE –

THE LOW COLLECTIVE BLADE ANGLE IS THE AVERAGE OF THE READINGS TAKEN AT THE 90°, 180°, 270°, 0° POSITIONS WITH THE COLLECTIVE STICK IN THE FULL DOWN POSITION:

(+10.1 –2.3 –11.4 +1.5) / 4 = –2.1 / 4 = –0.525 DEGREES

HIGH COLLECTIVE BLADE ANGLE –

THE HIGH COLLECTIVE BLADE ANGLE IS THE AVERAGE OF THE READINGS TAKEN AT THE 90°, 180°, 270°, 0° POSITIONS WITH THE COLLECTIVE STICK IN THE FULL UP POSITION:

(+21.1 +11.0 +9.9 +20.0) / 4 = 62.1 / 4 = +15.525 DEGREES

FORWARD CYCLIC BLADE ANGLE –

THE FORWARD CYCLIC BLADE ANGLE IS THE DIFFERENCE IN THE READINGS TAKEN AT THE 270° AND 90° POSITIONS WITH THE CYCLIC STICK IN THE FULL FORWARD POSITION:

(+32.0 –(–2.0)) = +34.0 DEGREES

AFT CYCLIC BLADE ANGLE –

THE AFT CYCLIC BLADE ANGLE IS THE DIFFERENCE IN THE READINGS TAKEN AT THE 90° AND 270° POSITIONS WITH THE CYCLIC STICK IN THE FULL AFT POSITION:

(+26.2 –4.8) = +21.4 DEGREES

LEFT CYCLIC BLADE ANGLE –

THE LEFT LATERAL BLADE ANGLE IS THE SUM OF THE READINGS TAKEN AT THE 180° AND 0° POSITIONS WITH THE CYCLIC STICK IN THE FULL LEFT POSITION (NOTE THAT NEGATIVE SIGNS ARE IGNORED):

(+9.0 +7.4) = +16.4 DEGREES

RIGHT CYCLIC BLADE ANGLE –

THE RIGHT LATERAL BLADE ANGLE IS THE SUM OF THE READINGS TAKEN AT THE 180° AND 0° POSITIONS WITH THE CYCLIC STICK IN THE FULL RIGHT POSITION (NOTE THAT NEGATIVE SIGNS ARE IGNORED):

(+7.1 +8.1) = 15.2 DEGREES

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FIGURE 11-4-3-3. SAMPLE RIGGING WORKSHEET WITH CALCULATIONS

11-4-4 MAIN ROTOR SYSTEM RIG CHECK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Protractor
- Rigging Kit, 70700-20389-042
- Universal Propeller Protractor

Materials/Parts

- Protective Caps and Plugs

References

- Appropriate MTF
- PARA 1-6-3
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 11-4-2
- PARA 11-4-3
- TM 1-70-VIB

11-4-4.1 Perform Main Rotor System Rig Check.

NOTE

- A main rotor system rig check is required whenever a main transmission, rotor head, primary servo or bellcrank support has been replaced.
- Fixed-length pushrods, bellcranks, and idlers can be replaced with like items without a rigging check, if replacement part is adjusted to the exact length (pivot hole-to-pivot hole).

a. Turn on all electrical power (PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down during rigging with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

b. Connect 2800 - 3000 psi external hydraulic pressure to No. 2 hydraulic system (PARA 1-6-15).

c. Perform BOOST ON, SAS OFF procedure as follows:

NOTE

If helicopter electrical power is not available for use, an alternate method to achieve BOOST ON, SAS OFF is to apply 28 VDC to pin 1, and ground pin 2 on pilot-assist module side of electrical connector, P468R.

(1) On lower console, STABILATOR CONTROL/AUTO FLT CONT panel, disengage BOOST, SAS 1, and SAS 2 switches (Figure 11-4-4-1, Sheet 1, Detail A).

(2) On caution/advisory panel, verify that BOOST SERVO OFF and SAS OFF capsules are on.

(3) Turn off all electrical power (PARA 1-6-16).

(4) Open main rotor pylon sliding cover.

(5) Disconnect electrical connector, P469R, from boost shutoff valve on pilot-assist module(Figure 11-4-4-1 , Sheet 1, Detail B).

(6) Install protective caps and plugs on electrical connectors.

(7) Turn on all electrical power (PARA 1-6-16).

NOTE

When electrical connector, P469R, is disconnected, STABILATOR CONTROL/AUTO FLT CONT panel BOOST ON light does not illuminate when boost is on. Electrical connector, P469R, is disconnected to prevent inadvertent boost disengagement during rig. If BOOST SERVO OFF capsule is on, no hydraulic boost servo pressure exists.

(8) On caution/advisory panel, verify that SAS OFF capsule is on and BOOST SERVO OFF capsule is off.

d. Check that yaw/pitch coupling link is installed.

e. Install damper rigging blocks (PARA 11-4-3).

f. Install the following rigging pins:

(1) Pilot's longitudinal (pitch) cyclic pin, 70700-20380-065.

(2) Pilot's lateral (roll) cyclic pin, 70700-20380-066.

(3) Yaw boost servo rigging pin, 70700-20380-062.

NOTE

- It is essential to keep helicopter perfectly still while zeroing and taking blade readings with digital protractor. A limit of three persons is recommended on helicopter - one to take readings, one to assist in rotating blades, and one in cockpit to position controls.
- Helicopter must be jacked (just enough to relieve load on landing gear struts) so that helicopter will not rock if personnel enter or exit helicopter while blade angle readings are being taken.

g. Jack entire helicopter slightly (just enough to relieve load on landing gear struts) (PARA 1-6-3).

h. Unless otherwise directed, apply slight (2 - 5 pounds) hand/foot force, in following directions, while taking blade readings to remove lost motion in controls:

(1) Cyclic stick - left and rear.

(2) Yaw pedals - left.

(3) Collective stick - full down.

i. Use main rotor rigging worksheet to record blade angle readings (Figure 11-4-4-2).

j. With collective full down, use digital or universal propeller protractor to measure blade angles (PARA 11-4-2). **AVERAGE NO. 1**

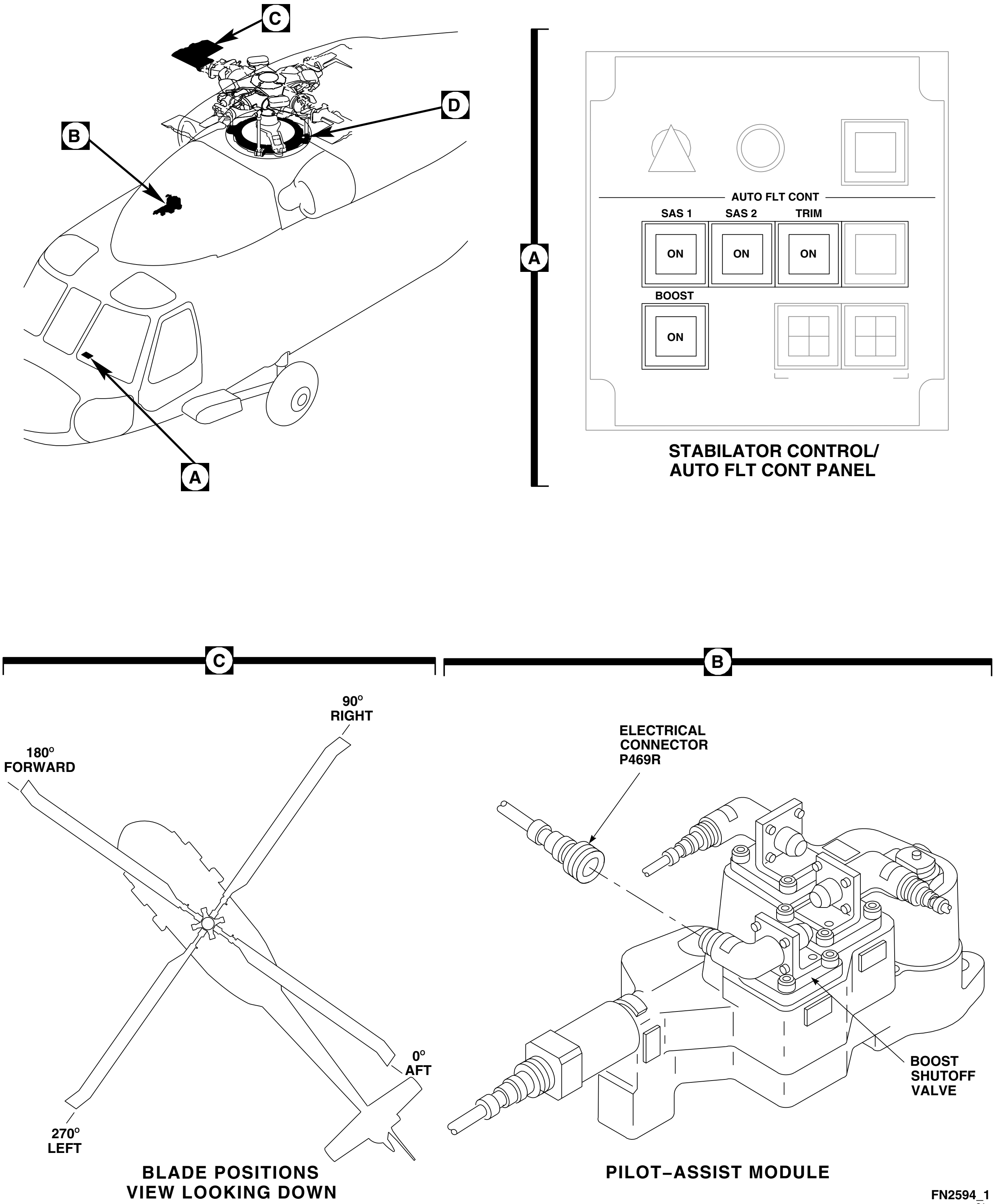

FIGURE 11-4-4-1. CHECK MAIN ROTOR SYSTEM RIG (SHEET 1 OF 2)

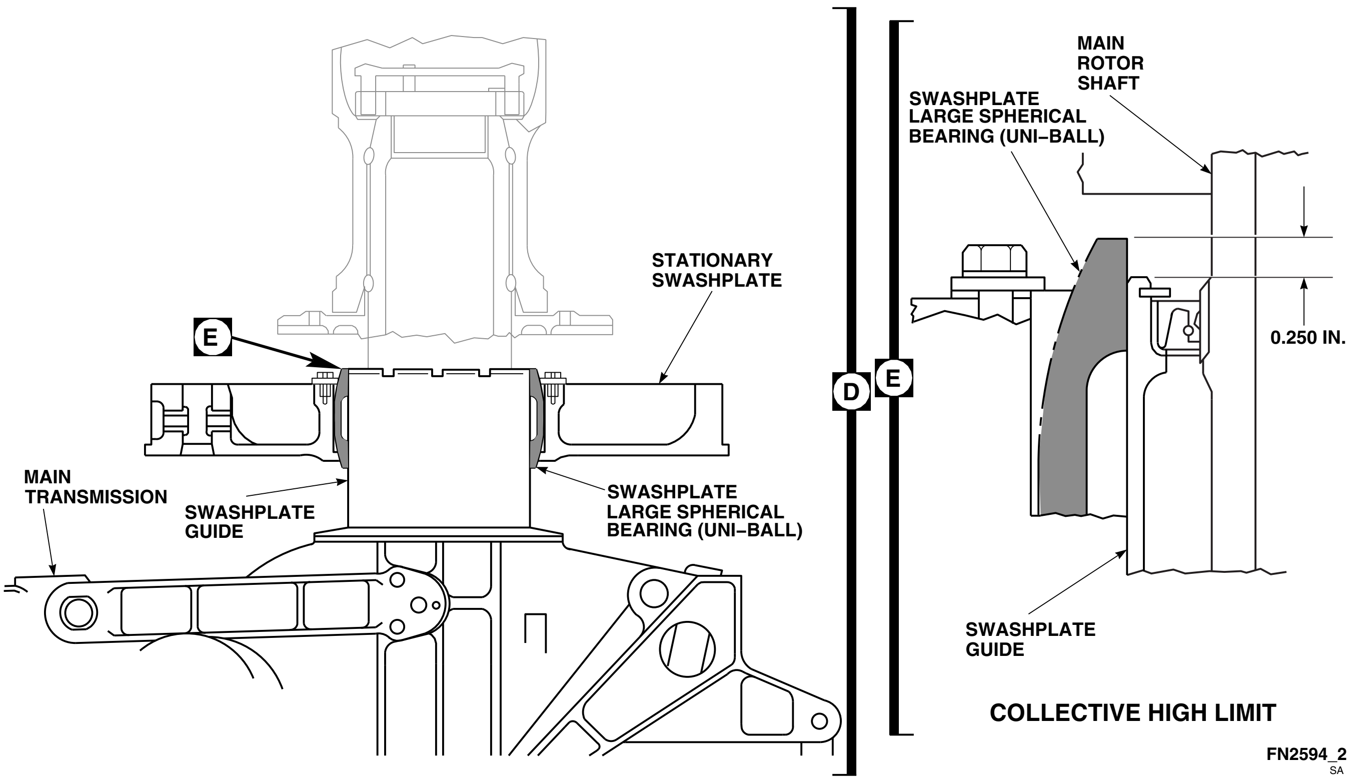

FIGURE 11-4-4-1. CHECK MAIN ROTOR SYSTEM RIG (SHEET 2 OF 2)

(RED) BLADE ANGLES AT 90°, 180°, 270°, 360° (0°) POSITIONS. (Figure 11-4-4-1, Sheet 1, Detail C). If helicopter has been rigged, but not had final adjustments for autorotation performed, with blades on, average reading should be -0.6 ± 0.125°. If helicopter is in operational status, collective angle average of -0.8° to +0.5° is acceptable.

CAUTION

Damage to swashplate large spherical bearing (uni-ball) will result if uni-ball is allowed to extend past swashplate guide by more than 0.250-inch. Make sure that swashplate large spherical bearing (uni-ball) does not extend past swashplate guide by more than 0.250-inch when applying full up collective (Figure 11-4-4-1, Sheet 2, Detail D and Detail E).

k. With collective in full up position, measure blade angles (PARA 11-4-2). **AVERAGE NO. 1 (RED) BLADE ANGLE AT 90°, 180°, 270°, AND 0° POSITIONS MUST BE 15.0° TO 16.0°** (Figure 11-4-4-1, Sheet 1, Detail C). If blade angle average is not within limits, refer to PARA 11-4-3. If helicopter has been adjusted for autorotation, differences between low and high collective averages of 15.5° - 17.5° are acceptable.

l. Remove pilot’s longitudinal (pitch) cyclic pin.

m. With collective in full up position, move cyclic stick fully forward. Check forward limiter for contact. **DIFFERENCE BETWEEN NO. 1 (RED) BLADE ANGLES AT 90° AND 270° POSITION MUST NOT BE LESS THAN 33.0°.** If blade angle difference is not within limits, refer to PARA 11-4-3.

n. With collective in full up position, move cyclic stick fully rearward. Check for aft stick stop for contact. **DIFFERENCE BETWEEN NO. 1 (RED) BLADE ANGLES AT 90° AND 270° LOCATION MUST NOT BE LESS THAN 19.0°.** If blade angle difference is not within limits, refer to PARA 11-4-3.

WORKSHEET

AIRCRAFT TAIL NO. ____________ #1 (RED BLADE POSITION) DATE ____________

CONTROL POSITIONS	90° RT	180° FWD	270° LFT	0° AFT	TOTAL°	AVG	REQUIRED TOLERANCE
LOW COLL ANGLE LOW COLL NEUTRAL CYCLIC NEUTRAL YAW BLADES	#1						–0.6°± 0.125° (–0.475° TO –0.725°)
	#2 #3 #4						BLADES #2, #3, #4 SAME AS #1 BLADE
HIGH COLL ANGLE HIGH COLL NEUTRAL CYCLIC NEUTRAL YAW							+15° TO +16° (SEE NOTE)
FWD CYCLIC ANGLE HIGH COLL NEUTRAL LAT FWD CYCLIC NEUTRAL YAW							DIFFERENCE BETWEEN 270° & 90° POSITIONS +33.0° MIN.
AFT CYCLIC ANGLE HIGH COLL NEUTRAL LAT AFT CYCLIC NEUTRAL YAW							DIFFERENCE BETWEEN 90° & 270° POSITIONS NOT LESS THAN +19°
LEFT CYCLIC ANGLE LOW COLL NEUTRAL LONG LEFT CYCLIC NEUTRAL YAW							SUM OF 180° & 0° POSITION TO BE 15° TO 17° AVG RANGE +7.5° TO 8.5°
RIGHT CYCLIC ANGLE LOW COLL NEUTRAL LONG RIGHT CYCLIC NEUTRAL YAW							SUM OF 180° TO 0° POSITION TO BE 14° TO 16° AVG RANGE +7.0° TO 8.0°
(FWD) 180° (LEFT) 270° — 90° (RIGHT) 0° (AFT)	PRIMARY SERVO CONTROL ROD BIAS: FWD: AFT: LAT:						

NOTE

WHEN PERFORMING RIG CHECK, SEE TEXT FOR TOLERANCE WHEN PITCH CONTROL RODS HAVE BEEN ADJUSTED FOR AUTOROTATIONAL RPM AND / OR BLADE TRACK.

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FIGURE 11-4-4-2. MAIN ROTOR RIGGING WORKSHEET

o. Move collective stick to full down position.

p. Maintain downward pressure on collective stick and push cyclic stick fully forward. Check to make sure that roller on mixer output crank is contacting forward limiter and that roller cannot be rotated using finger pressure. Check forward and aft primary servo for bottoming.

q. Maintain full down collective and pull cyclic stick fully aft. Check to make sure that roller on mixer output crank is contacting aft limiter and that roller cannot be rotated using finger pressure. Check forward and aft primary servo for bottoming.

r. Install pilot's longitudinal (pitch) cyclic pin, 70700-20380-065.

s. Place collective in full down position, and hold in place.

t. Remove pilot's lateral (roll) cyclic pin and install pilot's longitudinal (pitch) cyclic pin.

u. Move cyclic stick full left. Check for left limiter contact. **SUM OF NO. 1 (RED) BLADE ANGLES AT 180° AND 360° (0°)**

POSITIONS MUST BE 15° - 17°. If blade angle sum is not within limits, refer to PARA 11-4-3.

v. Move cyclic stick full right. Check for right limiter contact. **SUM OF NO. 1 (RED) BLADE ANGLES AT 180° AND 0° POSITIONS MUST BE 14° - 16°.** If blade angle sum is not within limits, refer to PARA 11-4-3.

w. Remove pilot's longitudinal (pitch) cyclic pin, yaw boost servo rigging pin, and damper rigging blocks.

x. Make sure all rigging pins and warning envelopes are removed.

y. Turn off all electrical and hydraulic power (PARA 1-6-15 and PARA 1-6-16).

z. Lower helicopter (PARA 1-6-3).

aa. Remove protective caps and plugs from electrical connectors.

ab. Inspect and treat electrical connectors (PARA 1-7-20).

ac. Connect electrical connector, P469R, to boost shutoff valve on top of pilot-assist module.

ad. Make sure area is clean and free of foreign material.

ae. Perform main rotor blade track and balance (TM 1-70-VIB).

af. Perform autorotation test flight (Appropriate MTF).

11-4-5 TAIL ROTOR SYSTEM COMPLETE RIG.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Tensiometer
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D
- Tiedown Straps, Item 338, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-10
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-4-137
- PARA 11-2-3
- PARA 11-4-1
- PARA 11-4-106
- PARA 11-4-112
- PARA 11-4-118

11-4-5.1 Perform Tail Rotor System Complete Rig.

CAUTION

- Place sign in cockpit to notify personnel that flight controls should not be moved during rigging, as damage to components could result.

- To prevent damage to antiflap cam and antiflap stop attaching bolts, antiflap cam must be held in flight position when cycling flight controls with main rotor blades not installed.

a. If main rotor blades are not installed, pull up antiflap cam levers (flight position) and tie them to horn with tiedown straps, Item 338, Appendix D.

NOTE

- Main rotor must be in proper rig before rigging tail rotor.

- A tail rotor system complete rig is required whenever major changes in tail rotor flight controls are made or when adjustments must be done to system that are not covered in tail rotor system rig check.

b. Open main rotor pylon sliding cover.

c. Remove tail gear box front fairing (PARA 2-4-137).

d. Remove soundproofing panels from cabin ceiling at STA 311.0 (PARA 2-4-69).

e. Remove passenger's seats from rear cabin (PARA 2-4-45).

f. Remove soundproofing panel from storage compartment (PARA 2-4-69).

g. Turn on all electrical power (PARA 1-6-16).

NOTE

Backup pump should be used only as an alternate source to power hydraulic systems. External hydraulic power should be applied whenever available.

h. Pressurize backup hydraulic system (PARA 1-6-15).

i. If using external hydraulic power, pull BACKUP HYD CONTR circuit breaker in DC ESNTL BUS on overhead upper console (Figure 11-4-5-1, Sheet 1, Detail A).

j. On STABILATOR CONTROL/AUTO FLT CONT panel, disengage SAS 1, SAS 2, and TRIM switches. Engage BOOST switch (Figure 11-4-5-1, Sheet 2, Detail B).

k. Disconnect spring cylinders from support at tail rotor quadrant (PARA 11-4-118).

l. Disconnect tail rotor cables (PARA 11-4-106).

m. Disconnect electrical connectors, P609 and P610, on tail rotor servo pressure switches (Figure 11-4-5-1, Sheet 4, Detail J).

n. Remove pushrod from tail rotor servo and install rigging pin (1/4 in. x 6 in.) in tail rotor servo.

o. Inspect and treat electrical connectors (PARA 1-7-20).

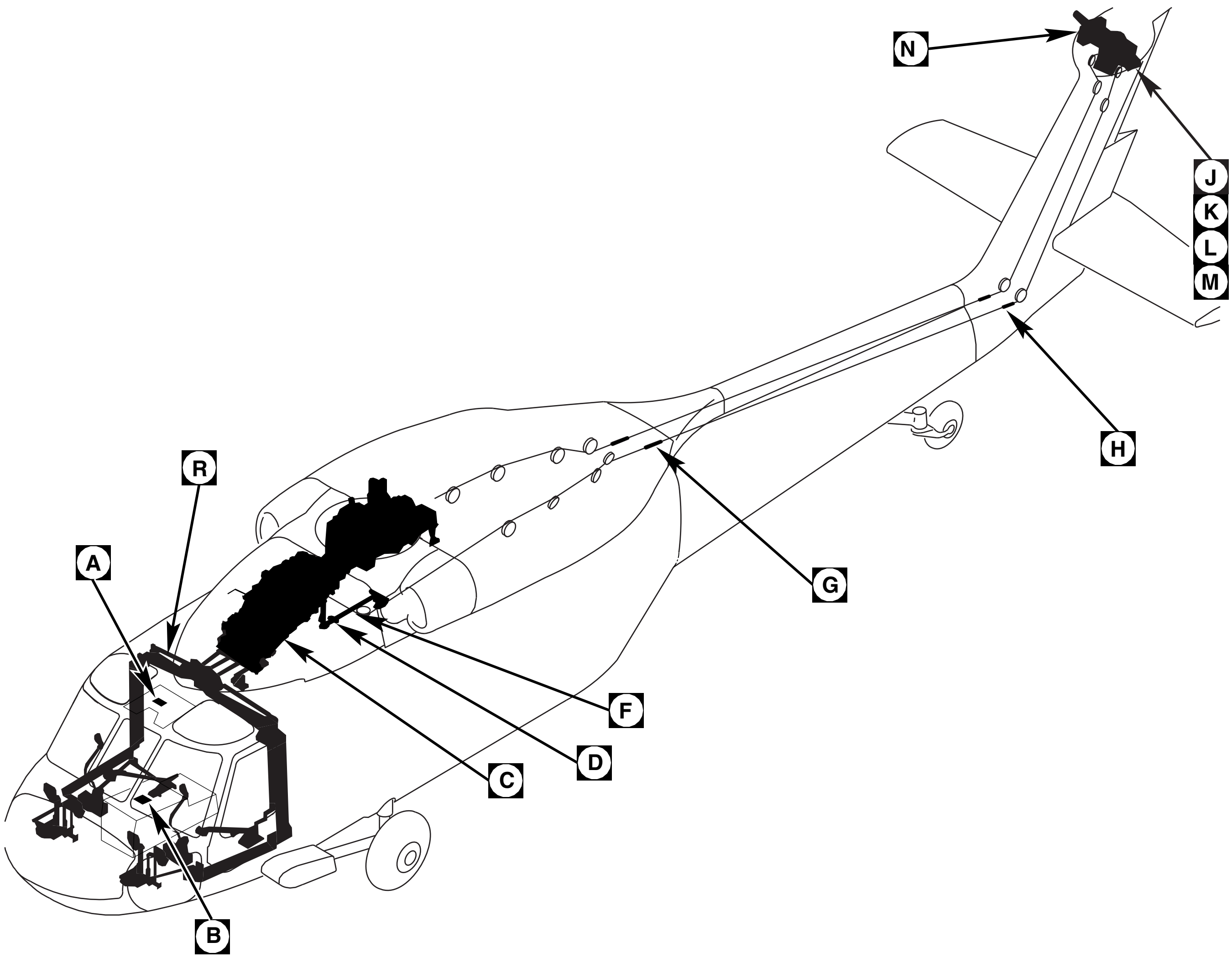
p. Connect electrical connectors, P609 and P610, on tail rotor servo pressure switches.

q. Install rigging pin (F-pin), 70700-20380-062, on yaw boost servo (Figure 11-4-5-1, Sheet 7, Detail S).

r. Install rigging pin (5/16 in. x 4 in.) in yaw torque shaft. Adjust yaw input control rod if necessary.

s. Move pilot's collective stick to midposition. Install rigging pin (F-pin), 70700-20380-062, on collective boost servo.

t. Install two rigging pins (5/16 in. x 6 in.) in both pilot's and copilot's yaw pedal adjusters. If rig-



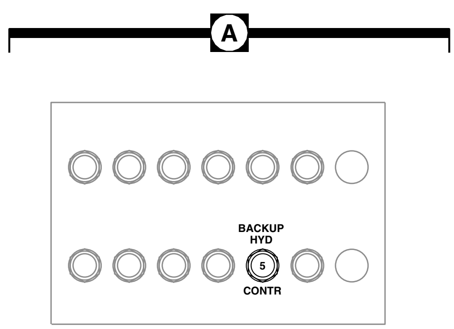


FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 1 OF 7)

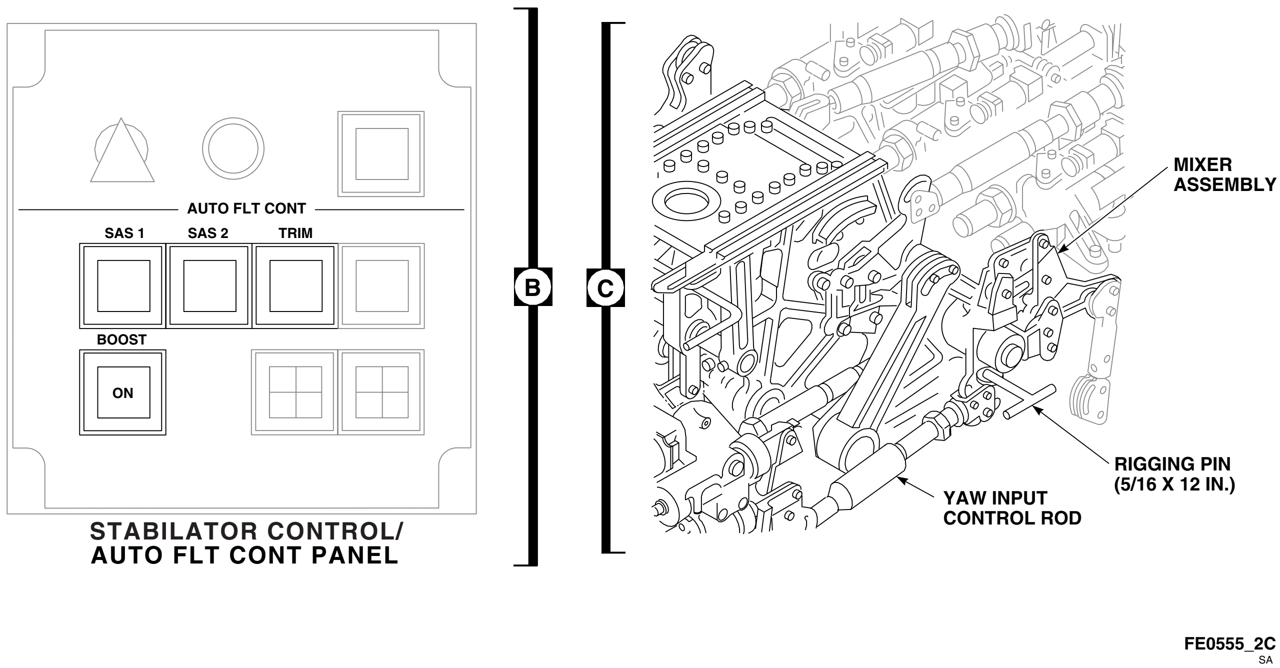


FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 2 OF 7)

ging pins will not go in easily, adjust torque shaft input rods in cabin as required (Figure 11-4-5-1, Sheet 7, Detail T).

u. Install rigging pin (5⁄16 in. x 12 in.) in left side of mixer assembly (Figure 11-4-5-1, Sheet 2, Detail C).

v. At this time, make sure all control rods are adjusted and connected from cockpit to mixer.

w. Check that output rod from mixer to torque shaft is properly connected (Figure 11-4-5-1, Sheet 3, Detail F).

NOTE

Control rod is adjusted to correct length when notch in forward quadrant outside diameter is lined up with hole in airframe which holds cable guard. It may be necessary to screw right pedal stop all the way in to perform rod adjustment.

x. Adjust and connect control rod from torque shaft to quadrant. **TIGHTEN JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

y. Adjust tail rotor quadrant to tail rotor servo pushrod to a length between 12.56 - 12.68-inches (PARA 11-4-112).

z. Connect cable disconnects (Figure 11-4-5-1, Sheet 4, Detail H). Bottom forward cable retainer jamnut in aft cable slot. **TORQUE LOCKNUT TO 40 - 60 INCH-POUNDS,** and lockwire, Item 197, Appendix D. Do not use these connections as an adjustment for cable tension.

aa. Connect tail rotor pushrod between tail rotor servo and quadrant (PARA 11-4-112).

ab. Connect spring cylinders to support at tail rotor quadrant (PARA 11-4-118).

ac. In forward end of tail cone, adjust cable tension using Table 11-4-5-1 and tensiometer.

ad. Make sure rigging pins in mixer and tail rotor servo remain free at all times. Install new lock-

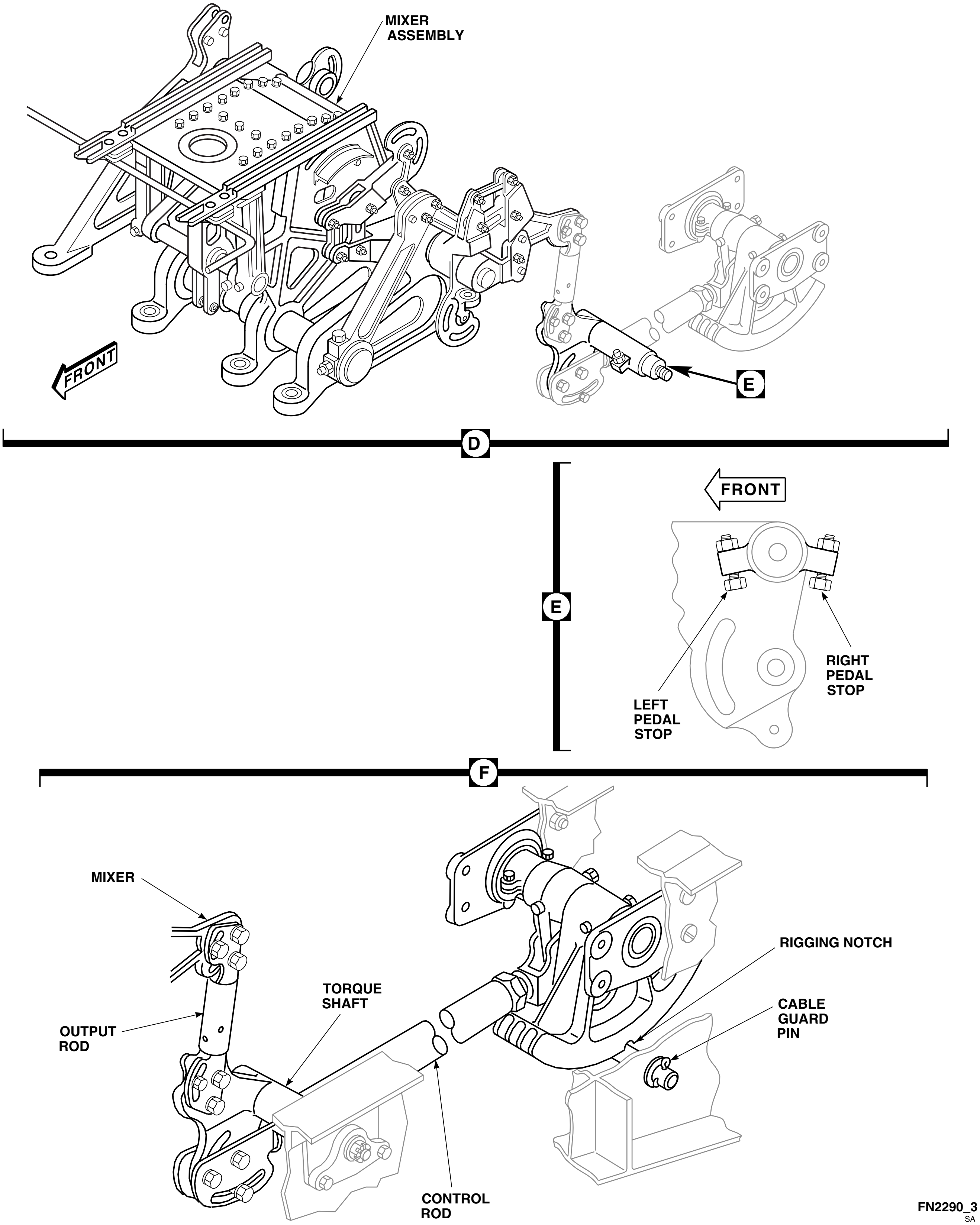


FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 3 OF 7)

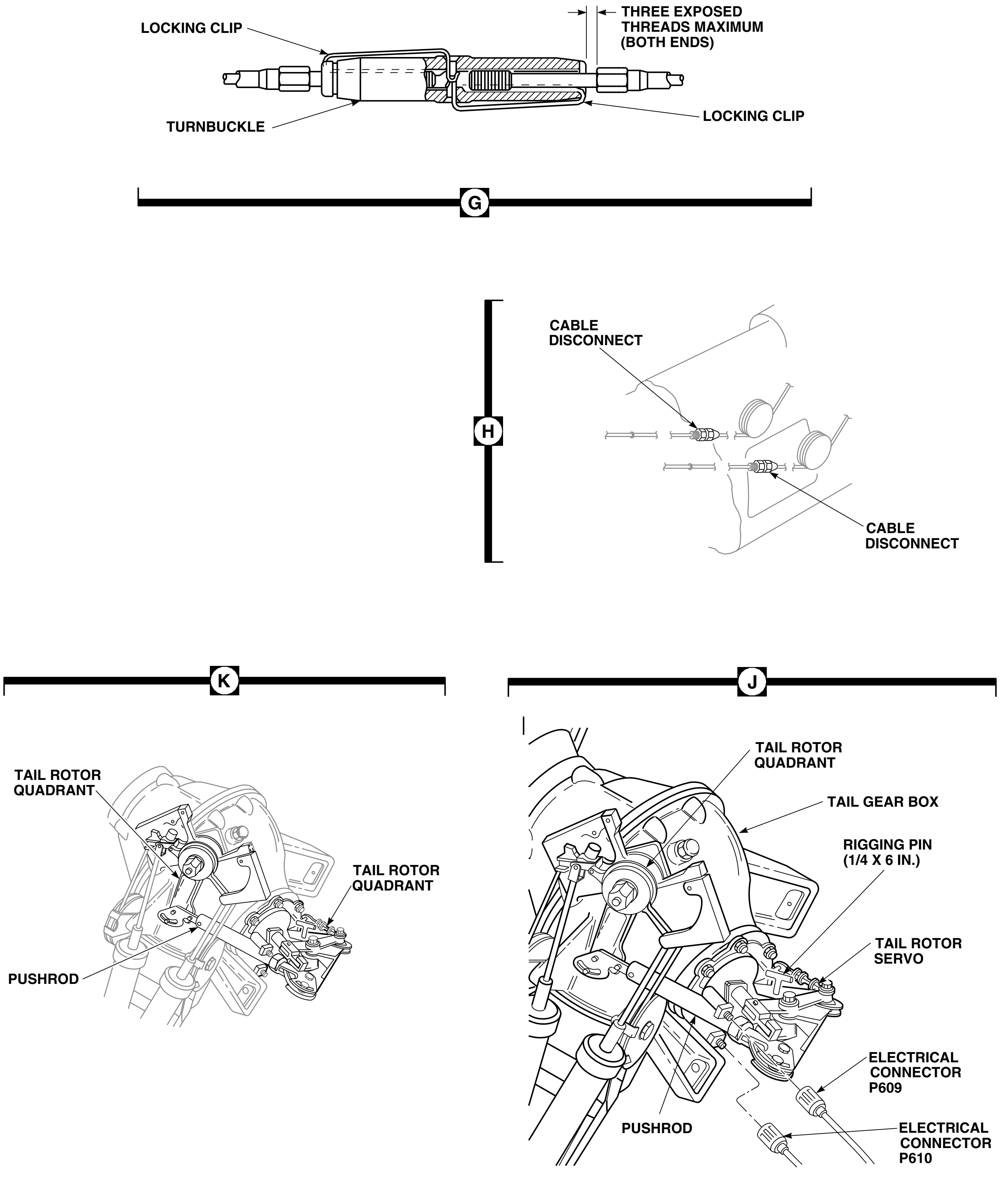


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FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 4 OF 7)

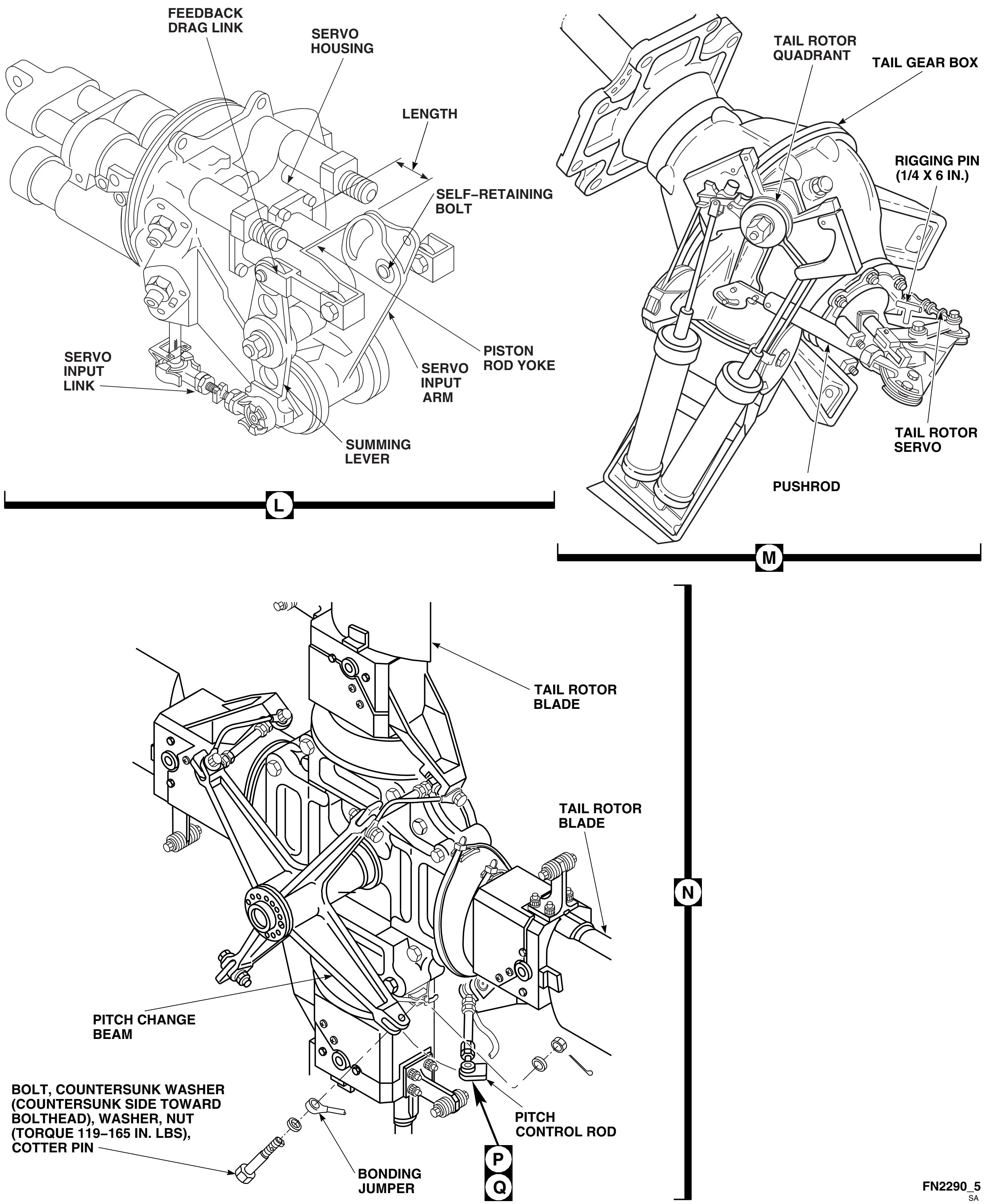


FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 5 OF 7)

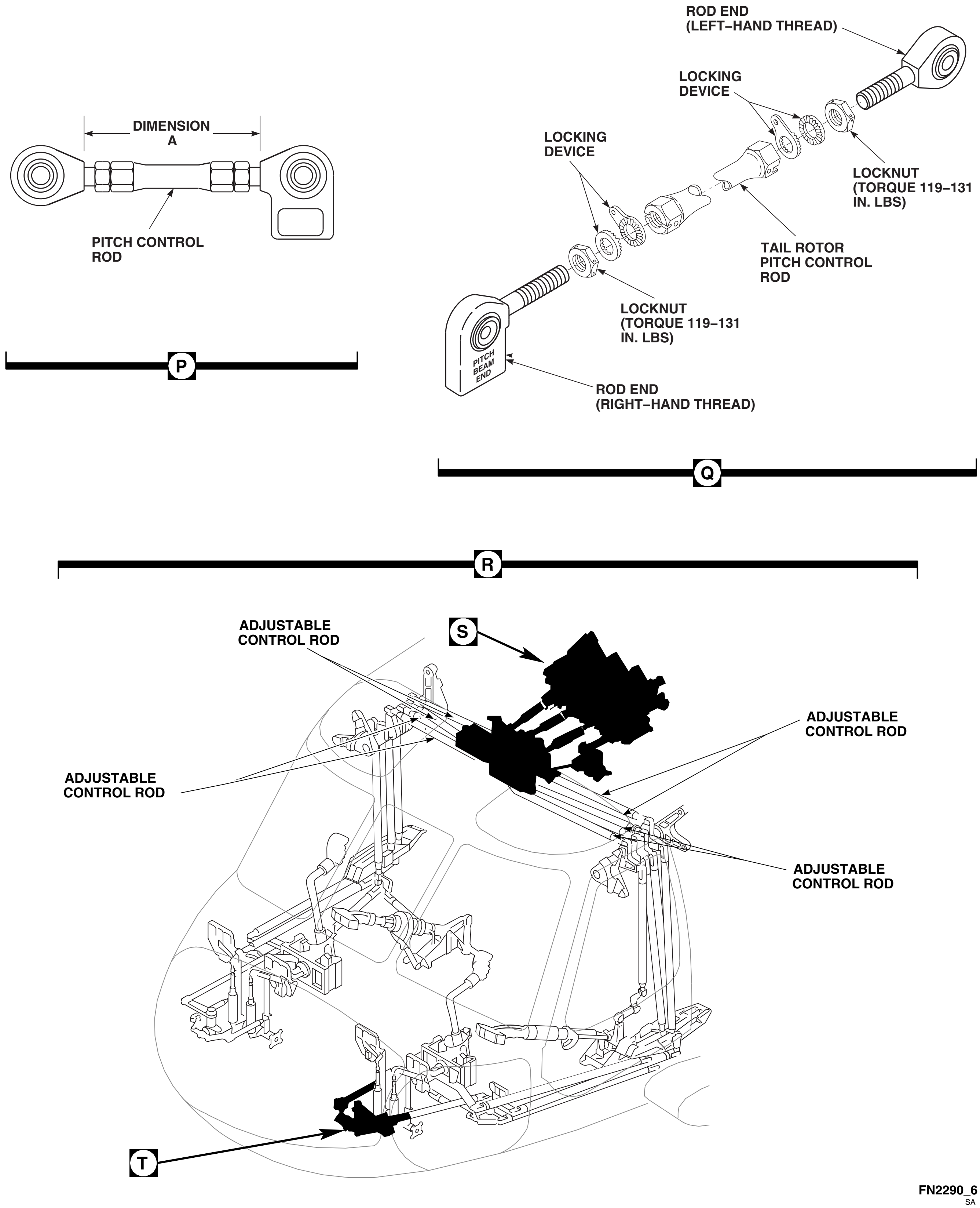


FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 6 OF 7)

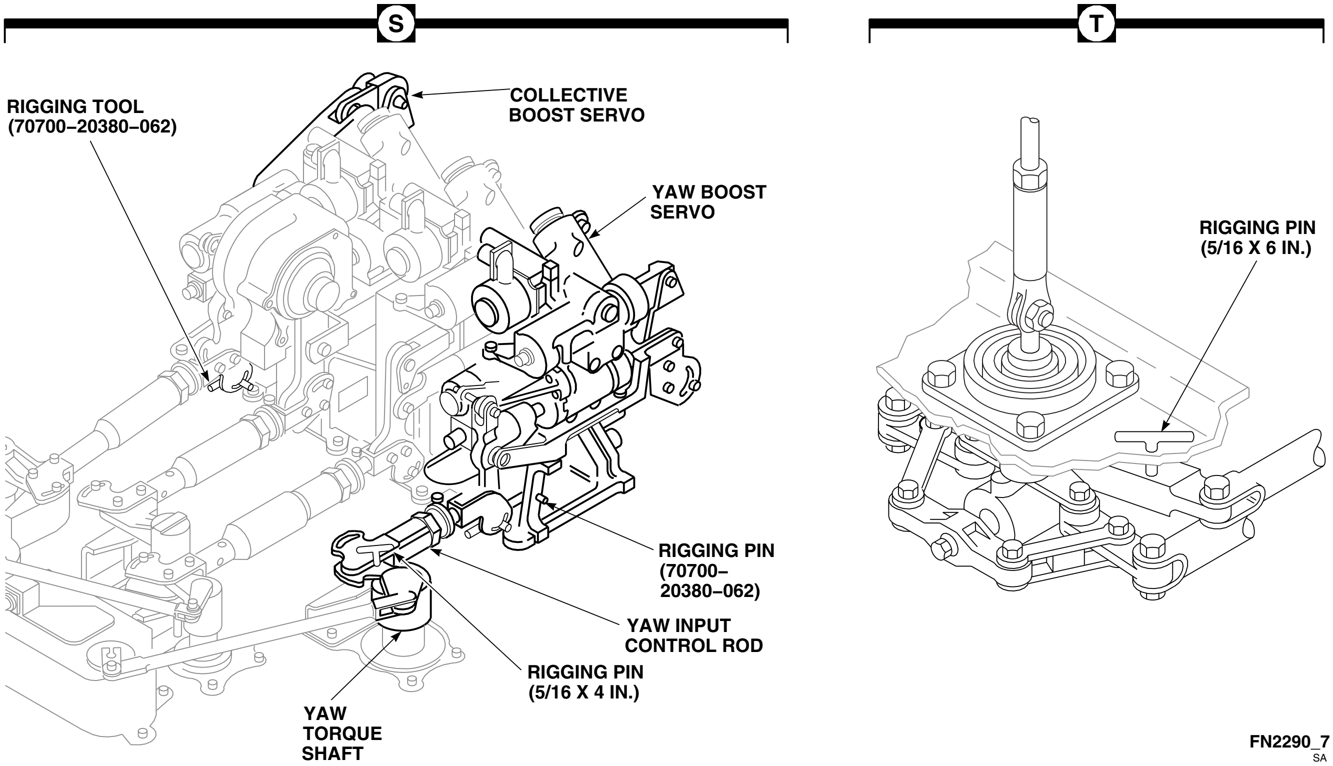


FIGURE 11-4-5-1. TAIL ROTOR SYSTEM COMPLETE RIG (SHEET 7 OF 7)

ing clips in turnbuckles (Figure 11-4-5-1, Sheet 4, Detail G).

ae. After cable tension has been set, check that rigging pin in tail rotor servo can be removed and reinstalled easily without pulling or pushing on servo input linkage (Figure 11-4-5-1, Sheet 4, Detail J). If not, readjust cable tension. Safety turnbuckles with locking clips (Figure 11-4-5-1, Sheet 4, Detail G).

af. With each blade, in turn, in horizontal position pointing forward, remove bolt, countersunk washer, bonding jumper, washer, and nut securing pitch control rod to pitch change beam (Figure 11-4-5-1, Sheet 5, Detail N).

ag. Adjust pitch control rod as required until attaching bolts drop freely into holes on pitch change beam. Do not put any pressure on blades to line up attaching boltholes.

ah. Install pitch control rods to pitch change beam with bolts, countersunk washers, bonding jumpers, washers, and nuts. **HANDTIGHTEN LOCKNUTS.**

ai. Position each blade, in turn, in horizontal position pointing forward.

aj. Measure each pitch control rod to the nearest ± 0.050-inch. Record dimension A and corresponding blade color in helicopter logbook (Figure 11-4-5-1, Sheet 6, Detail P).

ak. Back off locknuts on both ends of pitch control rods approximately 0.050-inch (Figure 11-4-5-1, Sheet 6, Detail Q).

al. Set tail rotor bias by holding locknut in place and lengthening each control rod by rotating pitch control rod barrel 4 complete turns.

Table 11-4-5-1. Tail Rotor Cable Temperature Correction Chart

Temperature Range Degrees (°F)	Temperature Range Degrees (°C)	Cable Tension (lbs)
+5 to +24	-15.0 to -5	61 to 85
+25 to +44	-4.9 to +7	85 to 109
+45 to +64	+7.2 to +18	109 to 133
+65 to +84	+18.3 to +29	133 to 168
+85 to +104	+29.4 to +40	168 to 191
+105 to +124	+41.6 to +51	191 to 215
+125 to +144	+52.7 to +63	215 to 237
+145 to +160	+63.8 to +71	237 to 250

NOTES

1. Check cable tension only after helicopter has been at an even temperature for 1 hour. Do not check tension with helicopter parked in hot sun. Tolerance on specified tension values is ± 2% difference between two cables.

2. If temperatures below +5°F (-15°C) are anticipated, set tension to 75 pounds. Reset tension to chart before helicopter is operated at temperatures above 40°F (4°C).

3. Flight control cables are 5⁄32-inch diameter, and with a plastic coating, total diameter is 7/32 inch. Measure cable tension with proper tensiometer setting.

   (a) When using adjustable risers, the No. 3 riser adapts to 7/32-inch total-diameter cable.

   (b) When tensiometer does not have adjustable risers, use the 5⁄32-inch-diameter tension scale.

4. Record temperature and cable tension on helicopter logbook for use when performing future tension check.

5. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 - 11 flight hours (PARA 1-7-10).

NOTE

Metal-to-metal contact between rod end and pitch horn may be present while in the static position.

am. Make sure that rod ends are aligned within pitch beam ears and pitch horn. Check to see that rod ends are parallel to each other.

an. **TORQUE PITCH CONTROL ROD NUT TO 119 - 165 INCH-POUNDS.** Install cotter pins in castellated nut, with head horizontal to nut slot and bend pins over hex flats (Figure 11-4-5-1, Sheet 5, Detail N).

WARNING

To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

ao. Tighten jamnut against barrel. **TORQUE LOCKNUTS TO 119 - 131 INCH-POUNDS.** Make sure locking devices are engaged (Figure 11-4-5-1, Sheet 6, Detail Q). Lockwire, Item 197, Appendix D, locknut to tang and then back to opposite side of locknut.

ap. Position each blade, in turn, in horizontal position pointing forward.

aq. Remeasure each pitch control rod. Record dimension A and corresponding blade color in helicopter logbook (Figure 11-4-5-1, Sheet 6, Detail P).

NOTE

Proper tail rotor bias is obtained using 4 complete turns of the rotating pitch control rods. Do not adjust the number of turns to achieve specified dimensions.

ar. Subtract first measurement from second for each blade. **VALUE SHALL BE 0.280 - 0.380-INCH.**

as. If value is not within this range, disconnect pitch control rod from that blade and repeat step af. through step ar. for that blade only.

at. Remove all rigging pins in this order:

(1) 1 tail rotor servo rigging pin (Figure 11-4-5-1, Sheet 5, Detail M).

(2) 1 mixer rigging pin (Figure 11-4-5-1, Sheet 2, Detail C).

(3) 1 collective boost servo rigging pin (Figure 11-4-5-1, Sheet 7, Detail S).

(4) 1 yaw boost servo rigging pin.

(5) 1 yaw torque shaft rigging pin.

(6) 2 yaw pedal adjuster rigging pins (Figure 11-4-5-1, Sheet 7, Detail T).

au. Raise collective stick to full up position and lock in place.

av. Thread in left and right pedal stopbolts at front torque shaft lever installation just below mixer (Figure 11-4-5-1, Sheet 3, Detail E).

aw. Disconnect spring cylinders from support at tail rotor quadrant (PARA 11-4-118).

ax. Push left pedal until tail rotor servo fully extends. While left pedal is fully pressed, note length of exposed piston rod between servo housing and piston rod yoke. **LENGTH OF EXPOSED PISTON MUST BE GREATER THAN 3.50-INCHES** (Figure 11-4-5-1, Sheet 5, Detail L).

ay. With left pedal fully pressed, adjust left pedal stopbolt at front torque shaft lever installation just below mixer until stop is contacted. Turn stop out in ½-turn increments while cycling pedals. Recheck extended dimension between adjustments until noted dimension is decreased by 0.010 - 0.060-inch (Figure 11-4-5-1, Sheet 3, Detail E).

az. **EXPOSED PISTON ROD BETWEEN SERVO HOUSING AND PISTON ROD YOKE WILL READ 3.44-INCHES OR**

GREATER AT FULL LEFT PEDAL AFTER ABOVE ADJUSTMENTS ARE PERFORMED (Figure 11-4-5-1, Sheet 5, Detail L). Lockwire, Item 197, Appendix D, bolt.

ba. Unlock and move collective stick to full down position and lock in place.

bb. Push right pedal until tail rotor servo bottoms (less than 0.030-inch gap between servo housing and piston rod yoke). Note this dimension.

bc. With right pedal fully pressed, adjust right pedal stopbolt at front torque shaft lever installation until stop is contacted. Turn stop out in ½-turn increments while cycling pedals. Recheck dimension between adjustments until noted dimension is increased by 0.010 - 0.060-inch and tail rotor servo no longer bottoms.

bd. **GAP BETWEEN SERVO HOUSING AND PISTON ROD YOKE WILL BE LESS THAN 0.090-INCH AT FULL RIGHT PEDAL AFTER ABOVE ADJUSTMENTS ARE PERFORMED.** Lockwire, Item 197, Appendix D, bolt.

be. Connect spring cylinders to support at tail rotor quadrant (PARA 11-4-118).

bf. Perform the following quality control checks:

(1) Yaw pedals stopbolts lockwired.

(2) No passage of lockwire, Item 193, Appendix D, through control rod inspection holes.

(3) Turnbuckles safetied with locking clips.

(4) Cables, pulleys, and fair-leads for condition and proper routing.

(5) Pitch control rod measurement calculation between a value of 0.280 - 0.380-inch.

(6) Make sure all rigging pins are removed.

(7) All flight control system attaching hardware installed and safetied.

(8) All control system tri-pivot hardware is properly installed and safetied.

(9) Perform operational check on mechanical flight controls (PARA 11-2-3).

bg. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

bh. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

bi. Make sure area is clean and free of foreign material.

bj. Install soundproofing panels in cabin ceiling and storage compartment (PARA 2-4-69).

bk. Install passenger's seats in rear cabin (PARA 2-4-45).

bl. Install tail gear box front fairing (PARA 2-4-137).

bm. Close and latch main rotor pylon sliding cover.

bn. Perform maintenance test flight (Appropriate MTF).

bo. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 - 11 flight hours (PARA 1-7-10).

11-4-6 TAIL ROTOR SYSTEM RIG CHECK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Tensiometer

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D
- Tiedown Straps, Item 338, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-10
- PARA 2-4-69
- PARA 2-4-137
- PARA 11-2-3
- PARA 11-4-3
- PARA 11-4-112

11-4-6.1 Perform Tail Rotor System Rig Check.

WARNING

To prevent damage to equipment or injury to personnel, a tail rotor system rig check is required after cable tensioning or removal/replacement of parts listed in step a.

NOTE

Main rotor must be in proper rig before rigging tail rotor.

a. Perform tail rotor system rig check if cable tensioning is required, or removal/replacement of the following parts:

(1) Forward quadrant.

(2) Torque lever to forward quadrant push-rod.

(3) Control cables.

(4) Cable pulleys.

(5) Tail gear box.

(6) Certain tail rotor components as specified in Chapter 5.

(7) Tail rotor servo.

b. If tail rotor system rig check is required, make sure main rotor system complete rig is performed (PARA 11-4-3).

CAUTION

- To prevent damage to flight control components, place sign in cockpit to notify personnel that flight controls should not be moved during rigging, as damage to components could result.
- To prevent damage to bolts securing antiflap cam and antiflap stop, antiflap cam must be held in flight position when cycling flight controls with main rotor blades not installed.

c. If main rotor blades are not installed, pull up antiflap cam levers (flight position) and tie them to horn with tiedown straps, Item 338, Appendix D.

d. Open main rotor pylon sliding cover.

e. Remove tail gear box front fairing (PARA 2-4-137).

f. Remove soundproofing panels from ceiling at STA 311.0 (PARA 2-4-69).

g. Connect a source of electrical power (PARA 1-6-16).

NOTE

Backup pump should be used only as an alternate source to power hydraulic systems. External hydraulic power should be applied whenever available.

h. If not using external hydraulic power, pressurize backup hydraulic system (PARA 1-6-15).

i. If using external hydraulic power, pull BACKUP HYD CONTR circuit breaker (DC ESNTL BUS - copilot's side upper console circuit breaker panel) (Figure 11-4-6-1, Sheet 1, Detail A).

j. On STABILATOR CONTROL/AUTO FLT CONT panel, disengage SAS 1, SAS 2, and TRIM switches. Engage BOOST switch (Figure 11-4-6-1, Sheet 2, Detail B).

k. Install rigging pin (F-pin) 70700-20380-062, in yaw boost servo (Figure 11-4-6-1, Sheet 5, Detail J).

l. Move pilot's collective stick to midposition. Install rigging pin (F-pin) in collective boost servo.

m. Install rigging pin (5/16-in. x 6-in.) in either pilot's or copilot's yaw pedal adjustor (Figure 11-4-6-1, Sheet 5, Detail K).

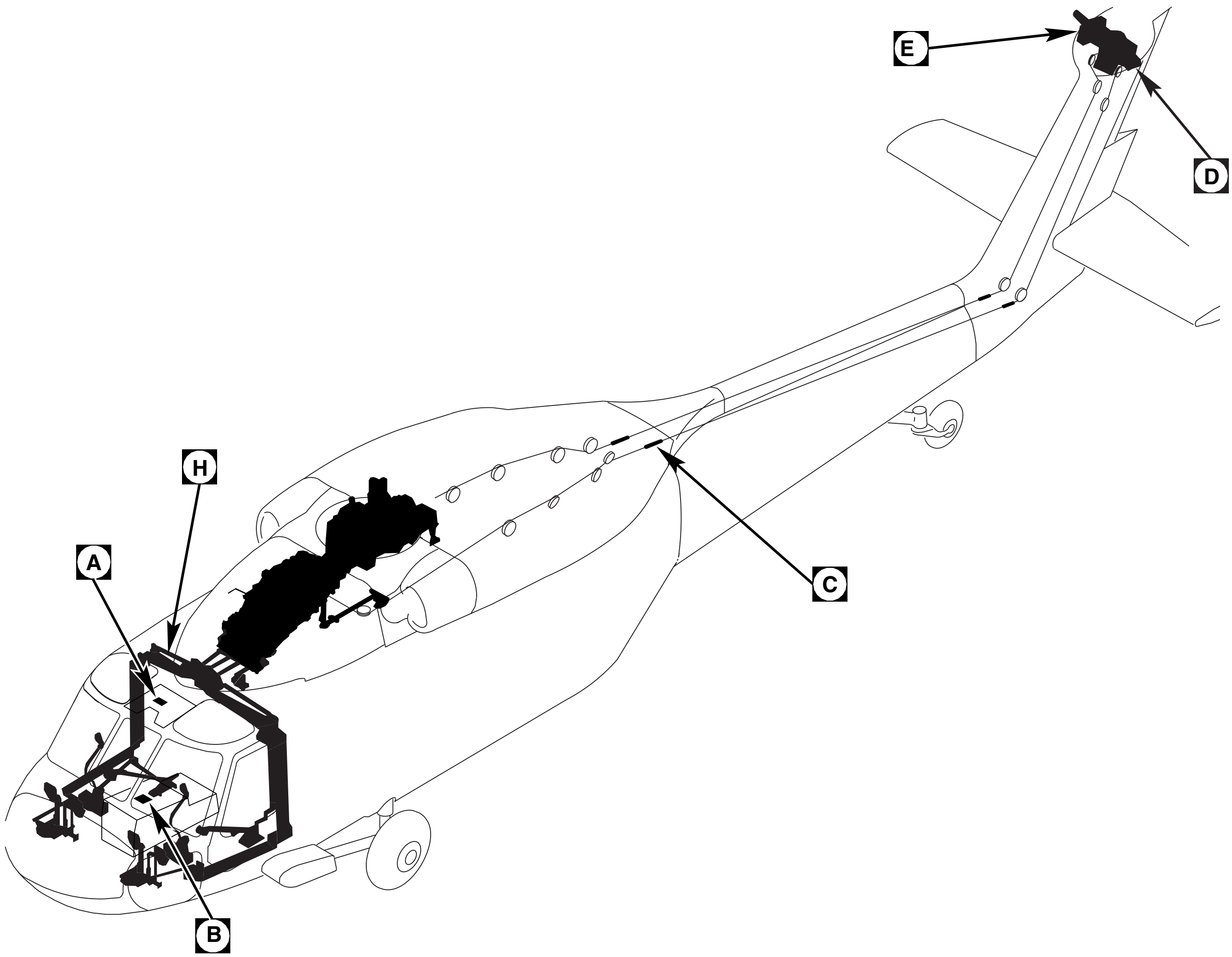
NOTE

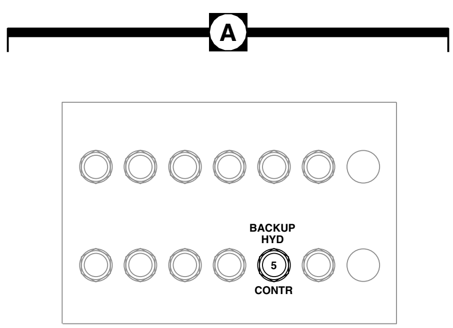
Make sure spring capsules are connected.

n. Attempt to install rigging pin (1/4-in. x 6-in.) in tail rotor servo (Figure 11-4-6-1, Sheet 3, Detail D). If pin will not slide in easily, perform the following:

(1) Check tail rotor servo to tail rotor quadrant pushrod for correct length. **PUSHROD SHALL BE 12.56 - 12.68-INCHES BETWEEN BOLTHOLES.** If beyond limits, adjust pushrod (PARA 11-4-112).

(2) In forward end of tail cone (Figure 11-4-6-1, Sheet 2, Detail C), adjust turnbuckles so that cable tension is in limits and tail rotor servo rig pin can be removed and reinstalled easily without pushing or pulling on servo input linkage. Refer to Table 11-4-6-1 for cable tension limits.





COPILOT'S SIDE UPPER CONSOLE
CIRCUIT BREAKER PANEL

FN2291_1A
SA

FIGURE 11-4-6-1. TAIL ROTOR SYSTEM RIG CHECK (SHEET 1 OF 5)

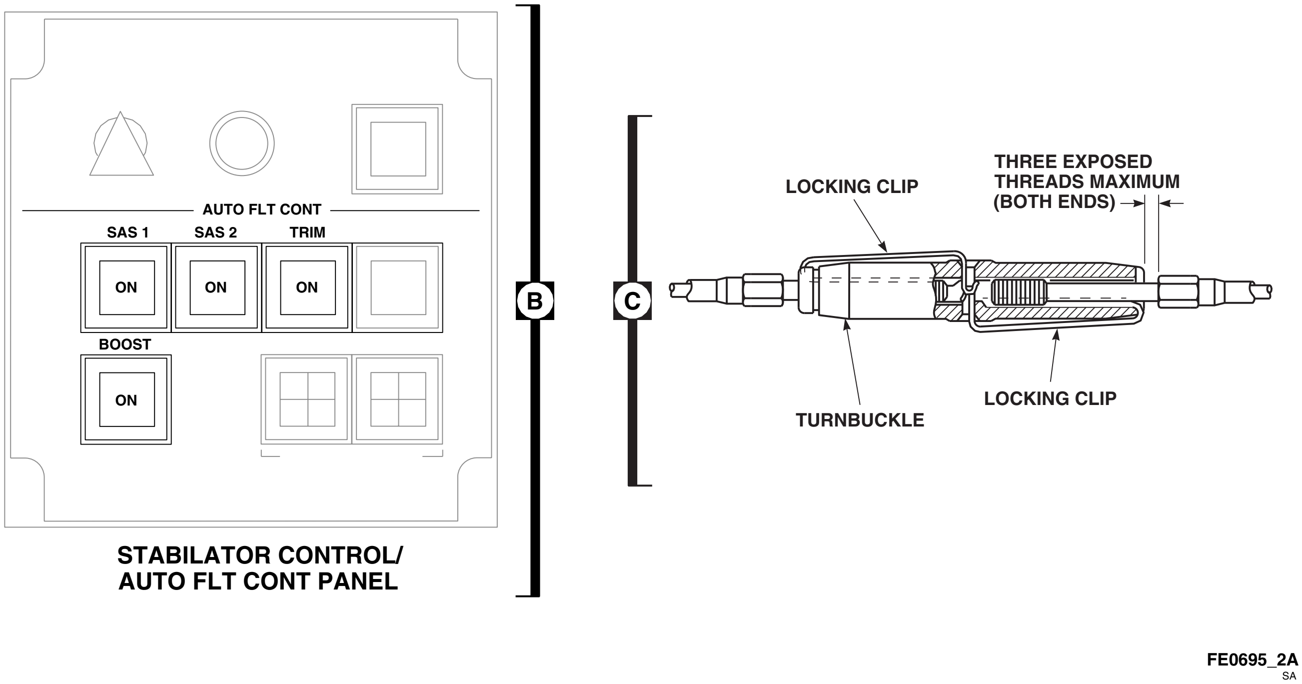


FIGURE 11-4-6-1. TAIL ROTOR SYSTEM RIG CHECK (SHEET 2 OF 5)

Table 11-4-6-1. Tail Rotor Cable Temperature Correction Chart

Temperature Range Degrees (°F)	Temperature Range Degrees (°C)	Cable Tension (lbs)
+5 to +24	-15.0 to -5	61 to 85
+25 to +44	-4.9 to +7	85 to 109
+45 to +64	+7.2 to +18	109 to 133
+65 to +84	+18.3 to +29	133 to 168
+85 to +104	+29.4 to +40	168 to 191
+105 to +124	+41.6 to +51	191 to 215
+125 to +144	+52.7 to +63	215 to 237
+145 to +160	+63.8 to +71	237 to 250

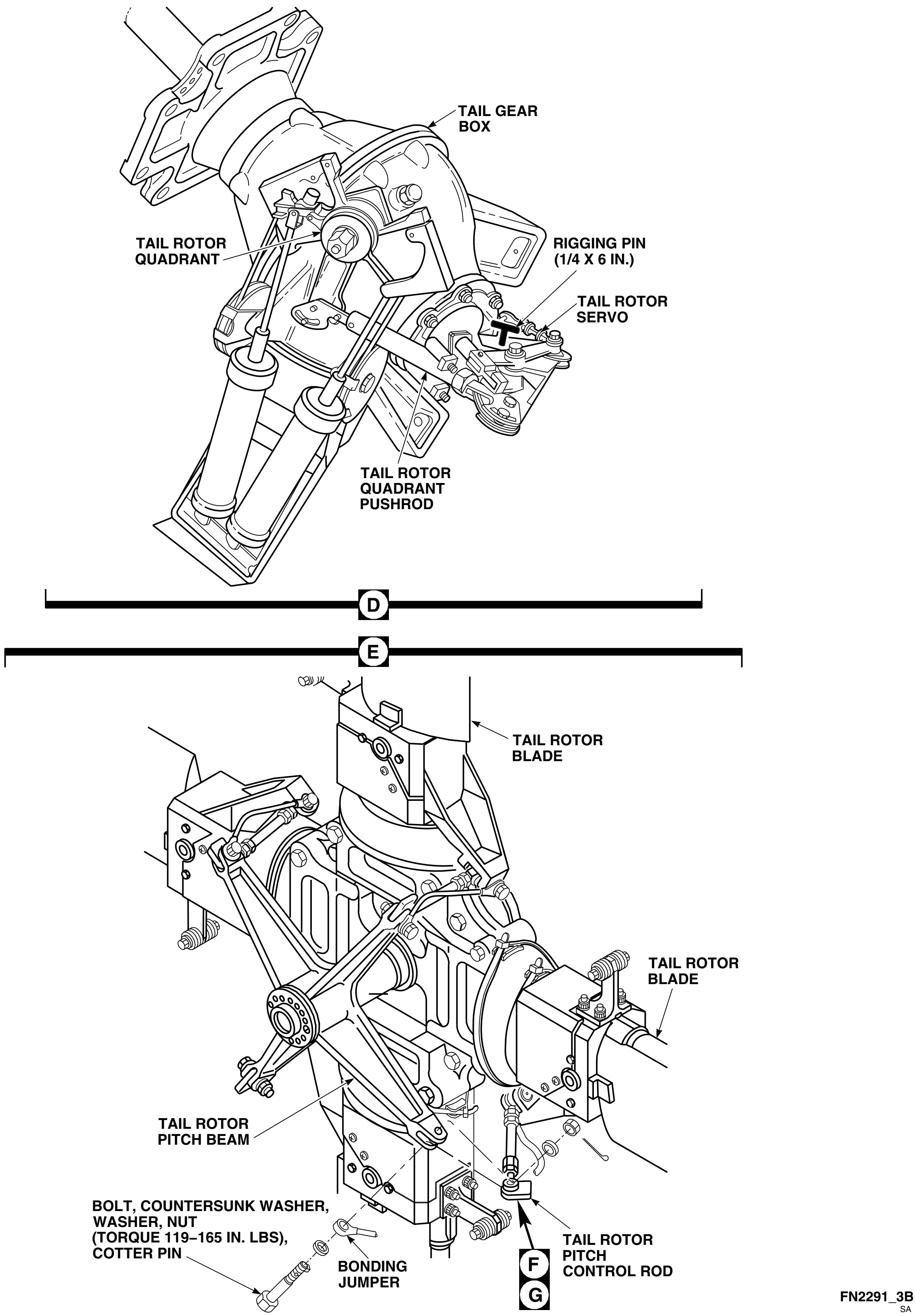


FIGURE 11-4-6-1. TAIL ROTOR SYSTEM RIG CHECK (SHEET 3 OF 5)

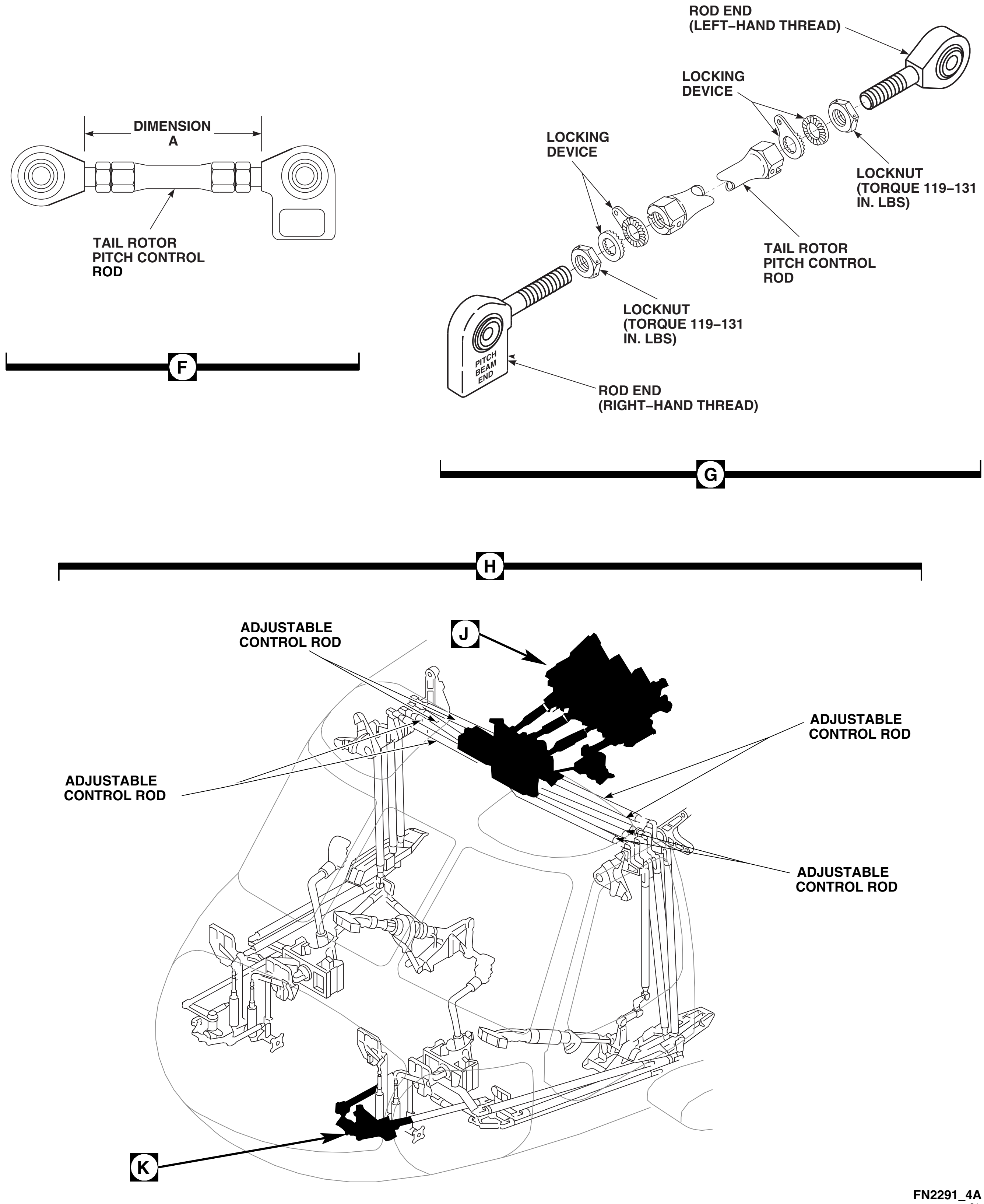
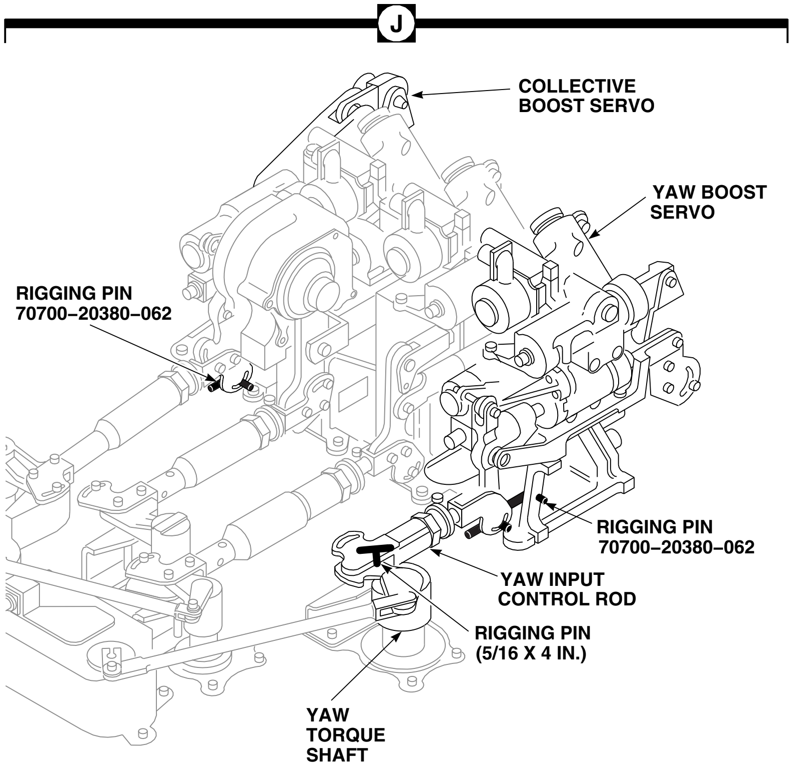


FIGURE 11-4-6-1. TAIL ROTOR SYSTEM RIG CHECK (SHEET 4 OF 5)



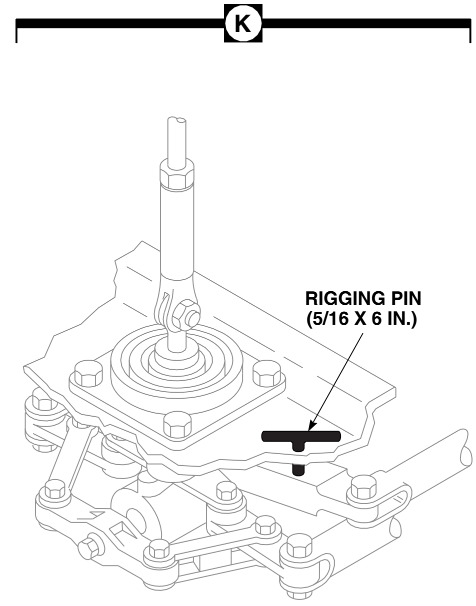


FIGURE 11-4-6-1. TAIL ROTOR SYSTEM RIG CHECK (SHEET 5 OF 5)

NOTES

1. Check cable tension only after helicopter has been at an even temperature for 1 hour. Do not check tension with helicopter parked in hot sun. Tolerance on specified tension values is ±2% difference between two cables.

2. If temperatures below +5°F (-15°C) are anticipated, set tension to 75 pounds. Reset tension to chart before helicopter is operated at temperatures above 40°F (4°C).

3. Flight control cables are 5⁄32-inch diameter, and with a plastic coating, total diameter is 7⁄32-inch. Measure cable tension with proper tensiometer setting.

   (a) When using adjustable risers, No. 3 riser adapts to 7⁄32-inch total-diameter cable.

   (b) When tensiometer does not have adjustable risers, use 5⁄32-inch-diameter tension scale.

4. Record temperature and cable tension on helicopter logbook for use when performing future tension check.

5. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 - 11 flight hours (PARA 1-7-10).

o. With each blade, in turn, in horizontal position pointing forward, remove bolt, countersunk washer, bonding jumper, washer, and nut securing pitch control rod to pitch change beam (Figure 11-4-6-1, Sheet 3, Detail E).

p. Adjust pitch control rod as required until attaching bolts drop freely into holes on pitch change beam. Do not put any pressure on blades to line up attaching boltholes.

q. Install pitch control rods to pitch change beam with bolts, countersunk washers, bonding jumpers, and nuts. **TIGHTEN LOCKNUTS HANDTIGHT.**

r. Position each blade, in turn, in horizontal position pointing forward.

s. Measure each pitch control rod to nearest ± 0.050-inch. Record dimension A and corresponding blade color in helicopter logbook (Figure 11-4-6-1, Sheet 4, Detail F).

t. Back off locknuts on both ends of pitch control rods approximately 0.050-inch (Figure 11-4-6-1, Sheet 4, Detail G).

u. While holding locknut in place, lengthen each control rod by rotating pitch control rod barrel 4 complete turns.

v. Make sure rod ends are in safety.

w. **TORQUE PITCH CONTROL ROD NUT TO 119 - 165 INCH-POUNDS.** Install cotter pins in castellated nut, with head horizontal to nut slot and bend pins over hex flats (Figure 11-4-6-1, Sheet 3, Detail E).

WARNING

To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

x. Tighten jamnut against barrel. **TORQUE LOCKNUTS TO 119 - 131 INCH-POUNDS.** Make sure locking devices are engaged (Figure 11-4-6-1, Sheet 4, Detail G). Using lockwire, Item 197, Appendix D, lockwire locknut to tang and then back to opposite side of locknut.

y. Position each blade, in turn, in horizontal position pointing forward.

z. Remeasure each pitch control rod. Record Dimension A and corresponding blade color helicopter logbook.

NOTE

Proper tail rotor bias is obtained using 4 complete turns of the rotating pitch control rods. Do not adjust the number of turns to achieve specified dimensions.

aa. Subtract first measurement from second for each blade. **VALUE SHALL BE 0.280 - 0.380-INCH.**

ab. If value is not within range, disconnect pitch control rod from that blade and repeat step o. through step aa. for that blade only.

ac. Remove all rigging pins in the following order:

(1) 1 tail rotor servo rigging pin (Figure 11-4-6-1, Sheet 3, Detail D).

(2) 1 collective boost servo rigging pin (Figure 11-4-6-1, Sheet 5, Detail J).

(3) 1 yaw boost servo rigging pin.

(4) 1 yaw pedal adjustor rigging pin (Figure 11-4-6-1, Sheet 5, Detail K).

ad. Perform operational check of tail rotor flight control system (PARA 11-2-3).

ae. Perform the following quality control checks:

(1) Yaw pedals stopbolts lockwired.

(2) No passage of lockwire, Item 193, Appendix D, through control rod inspection holes.

(3) Turnbuckles safetied with locking clips.

(4) Cables, pulleys, and fair-leads for condition and proper routing.

(5) Pitch control rod measurement calculation between a value of 0.280 - 0.380-inch.

(6) Make sure all rigging pins are removed.

(7) All flight control system attaching hardware is installed and safetied.

(8) All control system tri-pivot hardware is properly installed and safetied.

(9) Perform operational check on tail rotor flight control system (PARA 11-2-3).

af. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ag. Make sure area is clean and free of foreign material.

ah. Install soundproofing panels in ceiling at STA 311.0 (PARA 2-4-69).

ai. Install tail gear box front fairing (PARA 2-4-137).

aj. Close main rotor pylon sliding cover.

ak. Perform maintenance test flight (Appropriate MTF).

11-4-7 YAW CONTROL PEDAL BOOTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

11-4-7.1 Replace Yaw Control Pedal Boots.

NOTE

Replacement of all yaw control pedal boots is the same.

11-4-7.1.1 Remove.

a. Unzip boot.

b. Remove screws and washers securing boot to floor panels around pedal assembly (Figure 11-4-7-1, Detail A).

c. Remove boot up and over pedal assembly and master cylinder.

11-4-7.1.2 Install.

a. Make sure area is clean and free of foreign material.

NOTE

Install boots so zippers face away from the center of individual pilot positions.

b. Install boot over pedal assembly and master cylinder. Fasten to floor panels with screws and washers (Figure 11-4-7-1, Detail A).

c. Zip up boot.

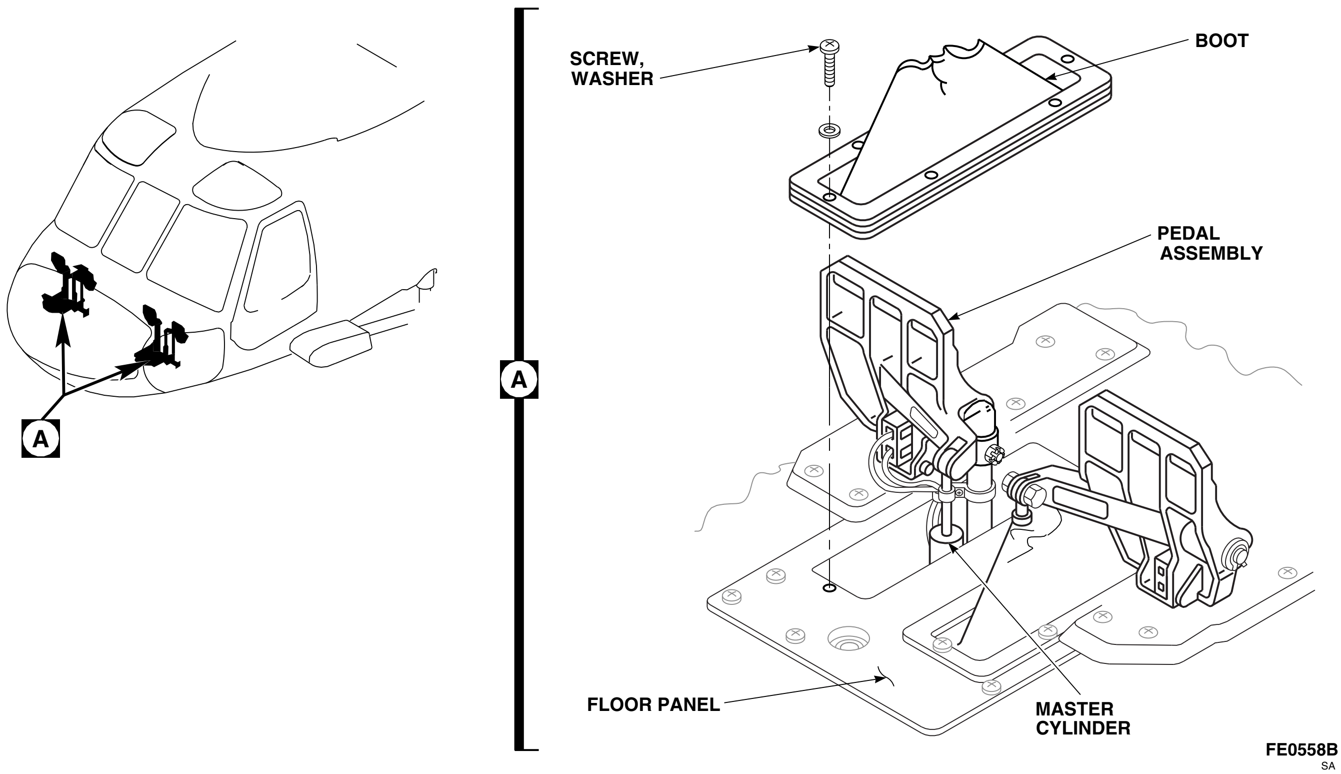


FIGURE 11-4-7-1. REPLACE YAW CONTROL PEDAL BOOTS

11-4-8 YAW CONTROL PEDAL.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Crowfoot Attachment
- Feeler Gage
- Retaining Ring Pliers
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 0 - 300 ft lbs

References

- PARA 1-6-16
- PARA 2-6-5
- PARA 11-4-1

11-4-8.1 Replace Yaw Control Pedal.

NOTE

Replacement of all yaw control pedals is the same.

11-4-8.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws, washers, and nuts securing pedal trim switch to pedal. Carefully lower switch and wiring and let them hang from clamp (Figure 11-4-8-1, Detail A or Figure 11-4-8-2, Detail A).

c. Remove self-retaining bolt, washer, and nut securing master brake cylinder to pedal clevis.

d. On helicopters with pedal support, 70400-01604-044, remove pedal as follows:

(1) Remove nut and washers securing pedal to pedal support. Pull pedal from pedal support (Figure 11-4-8-1, Detail A).

(2) Remove retaining ring using retaining ring pliers. Slide shim, washer, and pedal from shaft (Figure 11-4-8-1, Detail B).

e. On helicopters with pedal support, 70400-01604-049, remove pedal as follows (Figure 11-4-8-2, Detail A):

(1) Remove bolts, washers, and nuts. Lift pedal and pedal support from lower pedal support.

(2) Remove retaining ring using retaining ring pliers. Slide wave washer, washers, and pedal from pedal support. If required, remove washer and retaining ring from pedal support.

11-4-8.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. On helicopters with pedal support, 70400-01604-044, install pedal as follows:

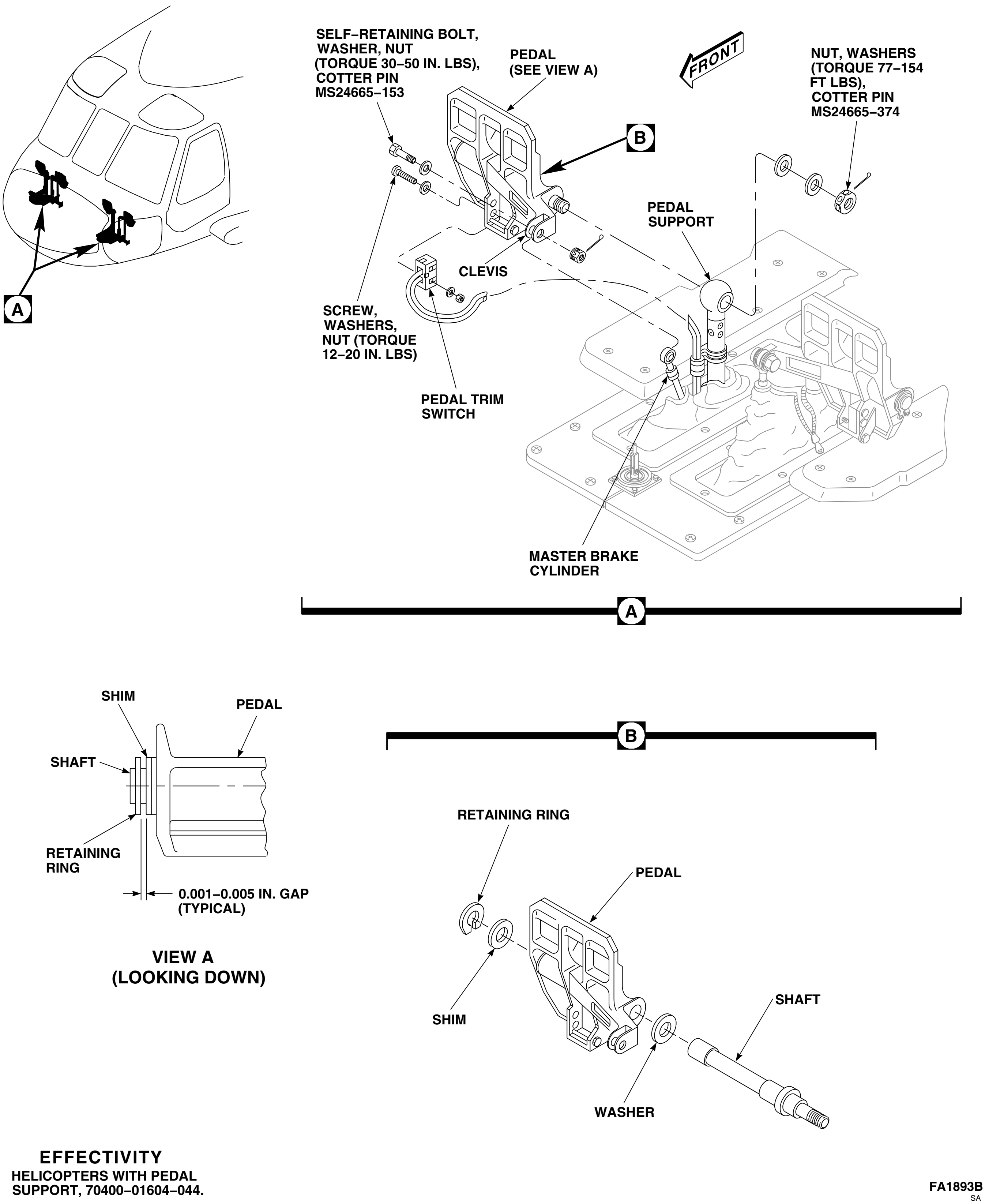


FIGURE 11-4-8-1. REPLACE YAW CONTROL PEDAL

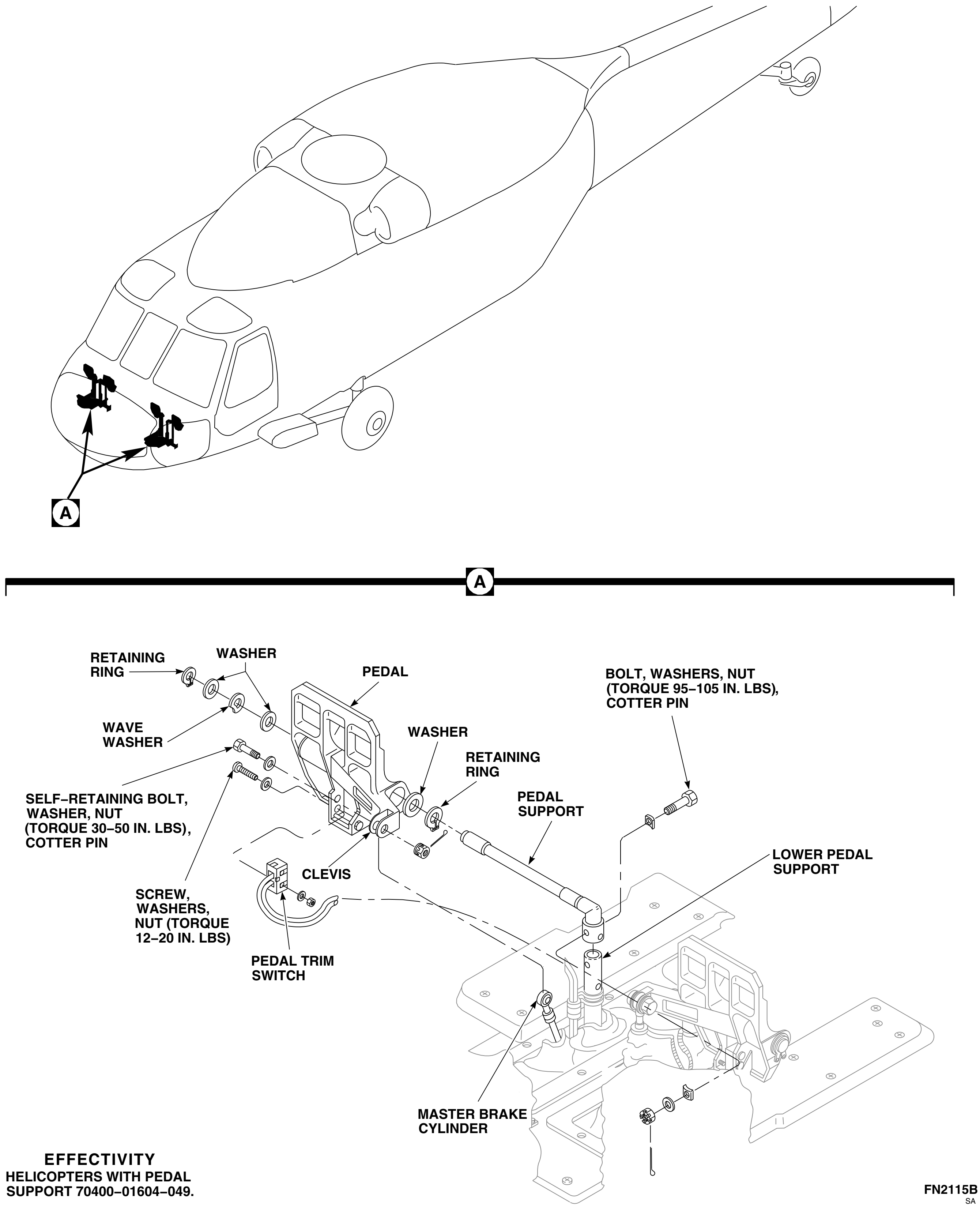


FIGURE 11-4-8-2. REPLACE YAW CONTROL PEDAL

(1) Install washer on shaft. Slide shaft into pedal (Figure 11-4-8-1, Detail B).

(2) Install shim and retaining ring on shaft.

(3) Install pedal in pedal support (Figure 11-4-8-1, Detail A).

(4) Install washers and nut. **TORQUE NUT TO 77 - 154 FOOT-POUNDS.**

(5) Using feeler gage, inspect gap between shim and retaining ring. **GAP TO BE BETWEEN 0.001 - 0.005-INCH** (Figure 11-4-8-1, View A).

(6) If gap is not between 0.001 - 0.005-inch, perform the following:

(a) Remove nut and washers securing pedal to pedal support (Figure 11-4-8-1, Detail A).

(b) Remove retaining ring using retaining ring pliers. Slide shim, washers, and pedal from shaft (Figure 11-4-8-1, Detail B).

(c) If gap was greater than 0.005-inch, replace shim. Repeat step c. (1) through c. (5).

(d) If gap was less than 0.001-inch, peel shim as required. Repeat step c. (1) through c. (5).

(7) If gap is between 0.001 - 0.005-inch, install cotter pin (Figure 11-4-8-2).

c. On helicopters with pedal support, 70400-01604-049, install pedal as follows (Figure 11-4-8-2, Detail A):

(1) If removed, install retaining ring and washer on support.

(2) Slide pedal, washers, and wave washer on pedal support. Install retaining ring using retaining ring pliers.

(3) Assemble pedal support and lower pedal support. Line up boltholes. Install bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.** Install cotter pin.

d. Install master brake cylinder in pedal clevis (Figure 11-4-8-1, Detail A or Figure 11-4-8-2, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

e. Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

f. Install nut on self-retaining bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to proof torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

g. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

h. Install pedal trim switch on pedal and secure with screws, washers, and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**

i. Make sure area is clean and free of foreign material.

11-4-9 YAW CONTROL PEDAL TRIM SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- TM 11-70-23

11-4-9.1 Replace Yaw Control Pedal Trim Switch.

11-4-9.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws, washers, and nuts, and remove switch. Slide enclosure back on wires (Figure 11-4-9-1).

c. Tag wires and disconnect them from switch.

d. Remove enclosure from wires.

11-4-9.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install replacement switch against support in pedal and loosely install screws, washers, and nuts. Slide enclosure on wires (Figure 11-4-9-1).

c. Connect electrical wiring to replacement switch. Remove tags from wires. Place switch enclosure over switch. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**

d. Perform operational check of pedal trim switches (TM 11-70-23).

e. Make sure area is clean and free of foreign material.

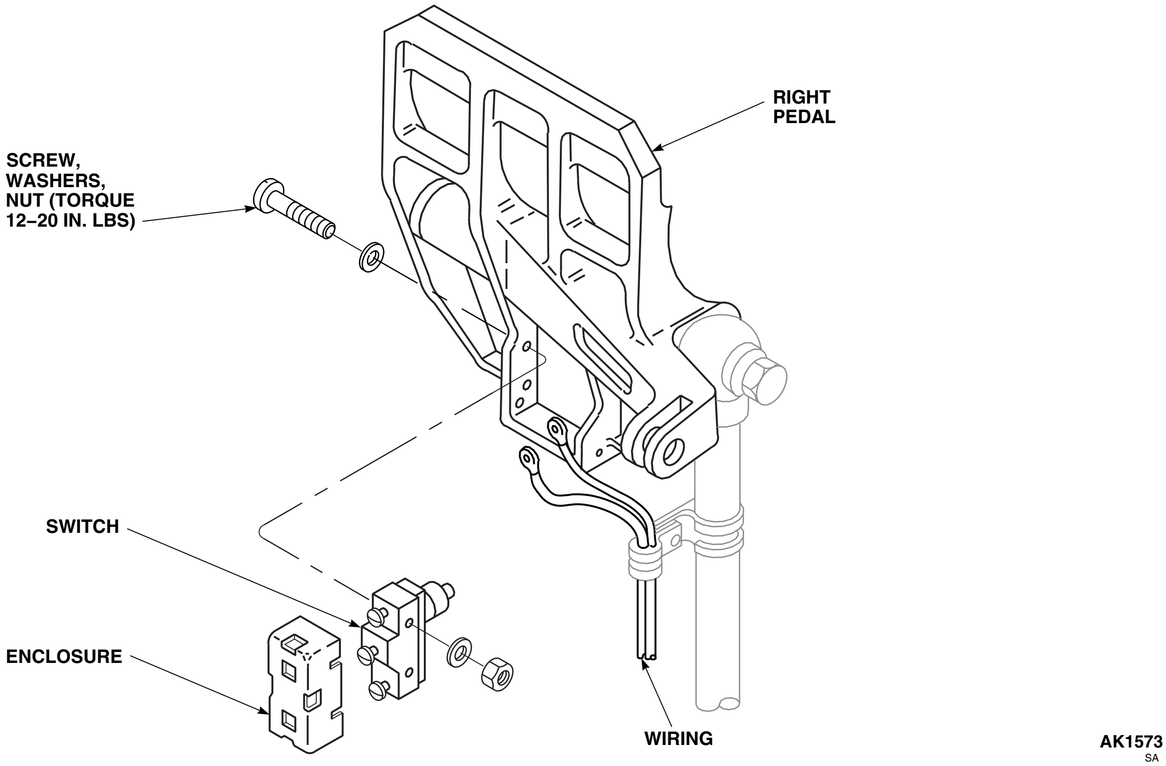


FIGURE 11-4-9-1. REPLACE YAW CONTROL PEDAL TRIM SWITCH

11-4-10 YAW CONTROL PEDAL SUPPORT ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Tags

References

- PARA 1-6-16
- PARA 2-4-43
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-7
- TM 11-70-23

11-4-10.1 Replace Yaw Control Pedal Support Assembly.

NOTE

Replacement of all yaw control pedal support assemblies is the same.

11-4-10.1.1 Remove.

NOTE

Pedal pivot shaft may become discolored (reddish-brown) while in service. Surface discoloration to this component is not a cause for replacement.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove left and right boots (PARA 11-4-7).

c. Remove floor panel around pedal assembly (PARA 2-4-43).

d. Tag switch for left or right pedal. Remove screws, washers, and nuts securing switch to pedal. Remove switch (Figure 11-4-10-1, Detail A).

e. Remove screws, washers, and nuts securing clamps to pedal supports. Remove clamps and wiring from pedal supports (Figure 11-4-10-1, Detail B).

NOTE

It is not necessary to disconnect hydraulic lines from master cylinders.

f. Remove self-retaining bolts, washers, and nuts securing master cylinders to pedal assemblies and master cylinder support arms. Move cylinders aside. Do not let cylinders hang by hoses (Figure 11-4-10-1, Detail A).

g. Remove self-retaining bolts, washers, and nuts securing directional control rods to pedal supports. Move control rod out of way.

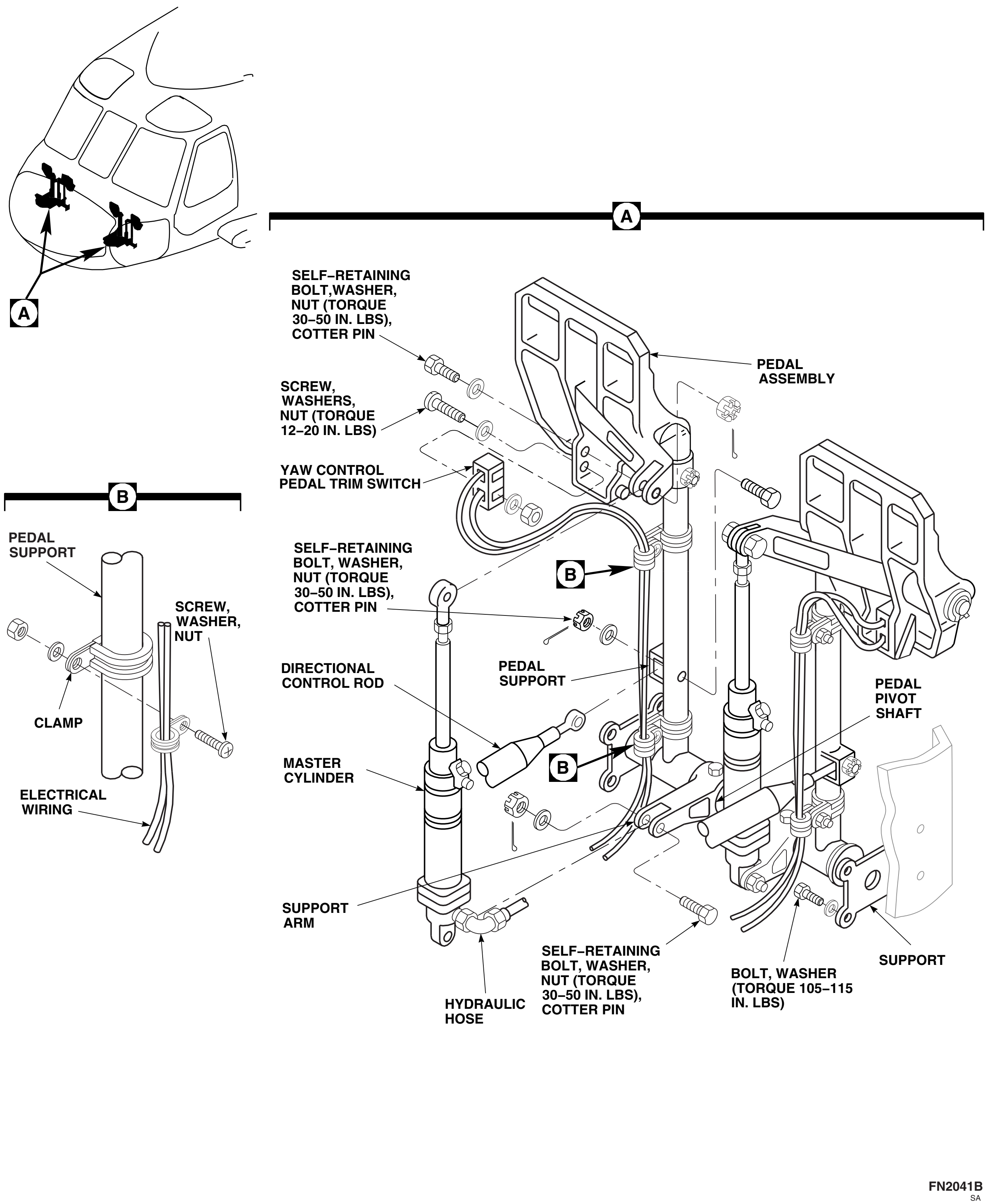


FIGURE 11-4-10-1. REPLACE YAW CONTROL PEDAL SUPPORT ASSEMBLY

h. Remove bolts and washers securing support to floor beams.

i. Slide pedal support assembly out from between floor beams.

11-4-10.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install pedal support and support between floor beams. Install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-10-1, Detail A).

c. Position directional control rods on pedal supports. Secure rods as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

d. Position master cylinders between support arms and pedal assemblies. Secure master cylinders as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. Remove tags and install pedal switch on pedal with screw, washers, and nut. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**

f. Install wiring to pedal supports with clamps, screws, washers, and nuts (Figure 11-4-10-1, Detail B).

g. Perform operational check of pedal trim switches (TM 11-70-23).

h. Make sure area is clean and free of foreign material.

i. Install floor panel around pedal assembly (PARA 2-4-43).

j. Install left and right boots (PARA 11-4-7).

11-4-11 YAW CONTROL PEDAL ADJUSTER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-16
- PARA 2-4-43
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-7

11-4-11.1 Replace Yaw Control Pedal Adjuster.

NOTE

Replacement of both yaw control pedal adjusters is the same.

11-4-11.1.1 Remove.

a. Remove both left or right yaw control pedal boots (PARA 11-4-7).

b. Remove cockpit floor access panels around pedal assembly (PARA 2-4-43).

NOTE

Pedal adjuster should be in retracted position prior to removal.

c. Remove screws and cover over directional control rod.

d. Remove self-retaining bolt, washer, and nut securing directional control rod to pedal adjuster arm (Figure 11-4-11-1, Sheet 2, Detail B).

e. Remove self-retaining bolt, washer, and nut securing pedal directional control rods to pedal adjuster links.

f. Remove bolt and nut securing clevis of release cable to pedal adjuster assembly (Figure 11-4-11-1, Sheet 1, Detail A).

g. Remove bolts and washers securing upper and lower support bearings and shims, if installed, to support flange (Figure 11-4-11-1, Sheet 2, Detail B).

h. Remove adjuster and support bearings from between airframe supports. Remove support bearing from adjuster.

11-4-11.1.2 Inspect.

a. Check support bearings for freedom of operation. Replace if necessary.

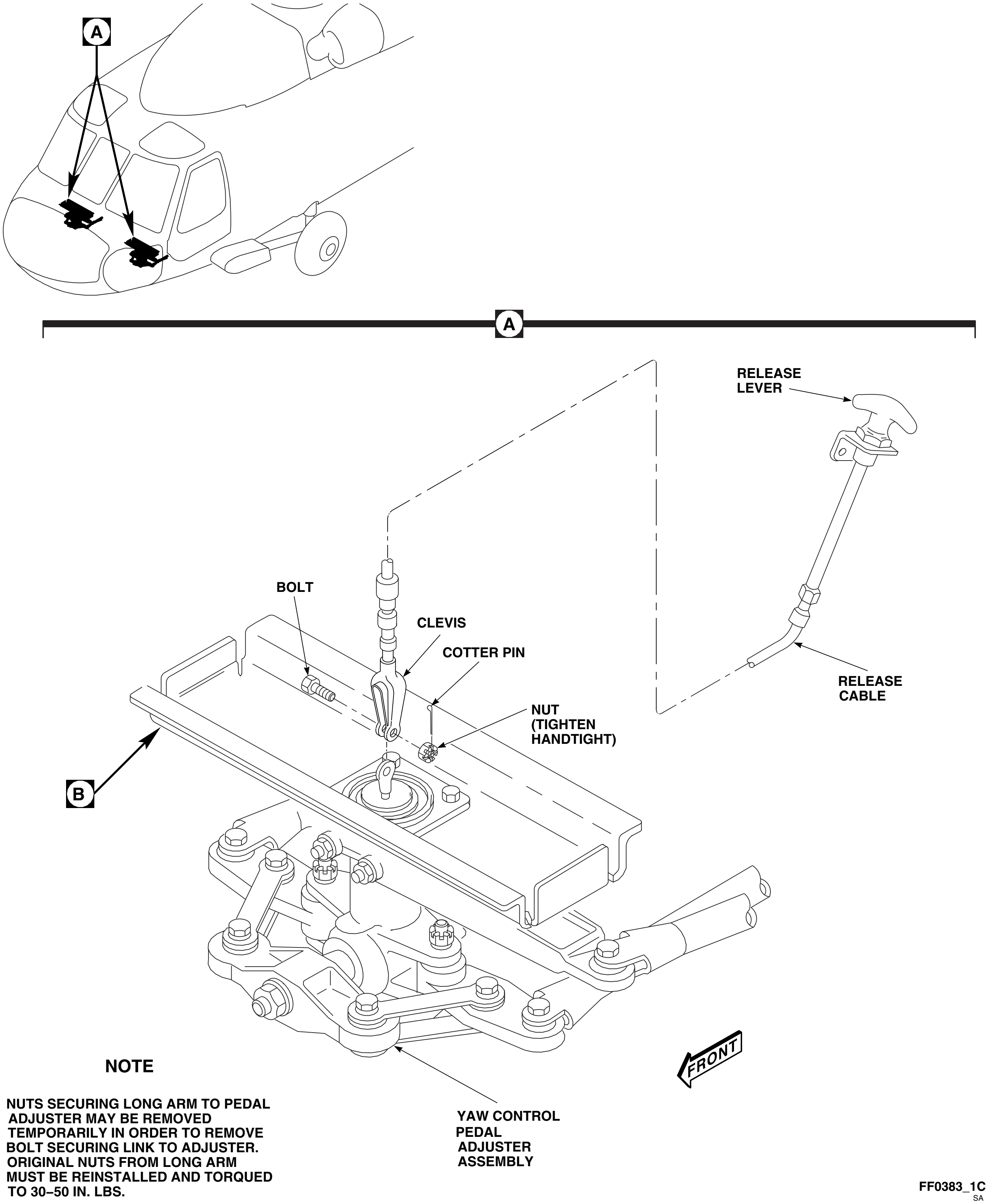


FIGURE 11-4-11-1. REPLACE YAW CONTROL PEDAL ADJUSTER (SHEET 1 OF 3)

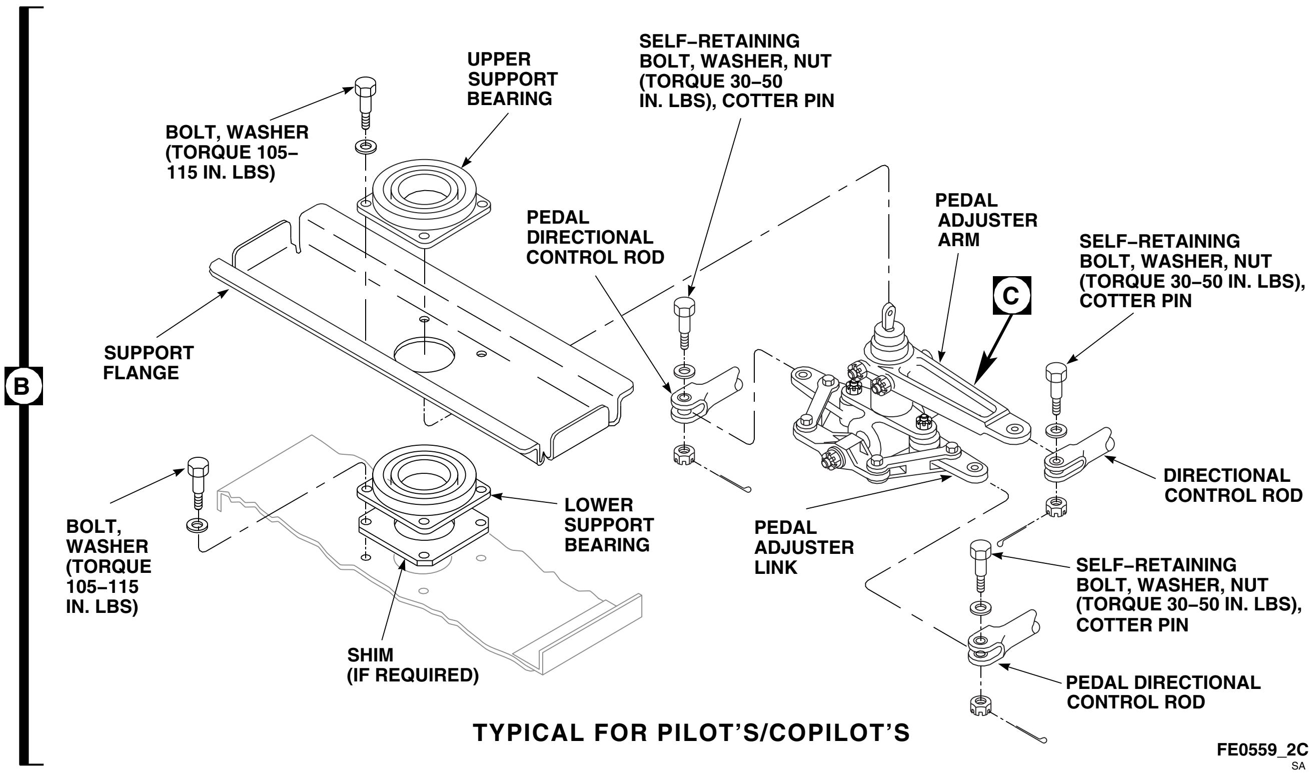


FIGURE 11-4-11-1. REPLACE YAW CONTROL PEDAL ADJUSTER (SHEET 2 OF 3)

b. Inspect upper and lower support bearings using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005 INCH.**

c. Inspect bearings in small arm and links using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005 INCH.**

d. Inspect bearing in long arm using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0025 INCH.**

11-4-11.1.3 Disassemble.

a. Remove self-retaining bolts, washers, and nuts from rigid links and small arm (Figure 11-4-11-1, Sheet 3, Detail C).

b. Remove nut and washer securing small arm from shaft of pedal adjuster. Remove small arm from pedal adjuster being careful not to lose key from shaft.

c. Remove self-retaining bolt, washer, and nut securing rigid links to links. Remove rigid links.

NOTE

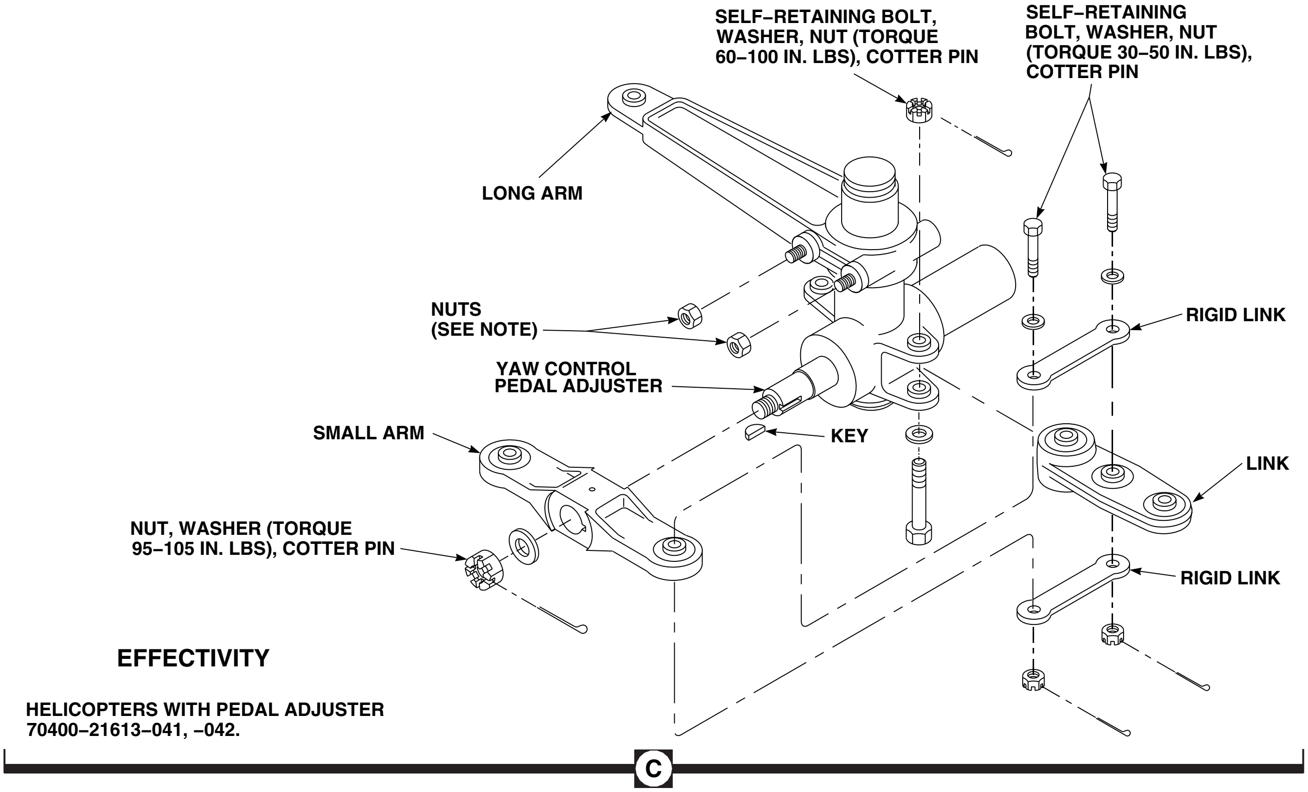
Hardware securing long arm to pedal adjuster may be removed temporarily in order to remove bolt securing link to adjuster. Original hardware from long arm must be reinstalled.

d. If required, remove nuts securing long arm to pedal adjuster.

e. Remove self-retaining bolts, washers, and nuts securing links to pedal adjuster. Remove links from pedal adjuster.

11-4-11.1.4 Assemble.

a. If required, install nuts securing long arm to pedal adjuster. **TORQUE NUTS TO 30 - 50**



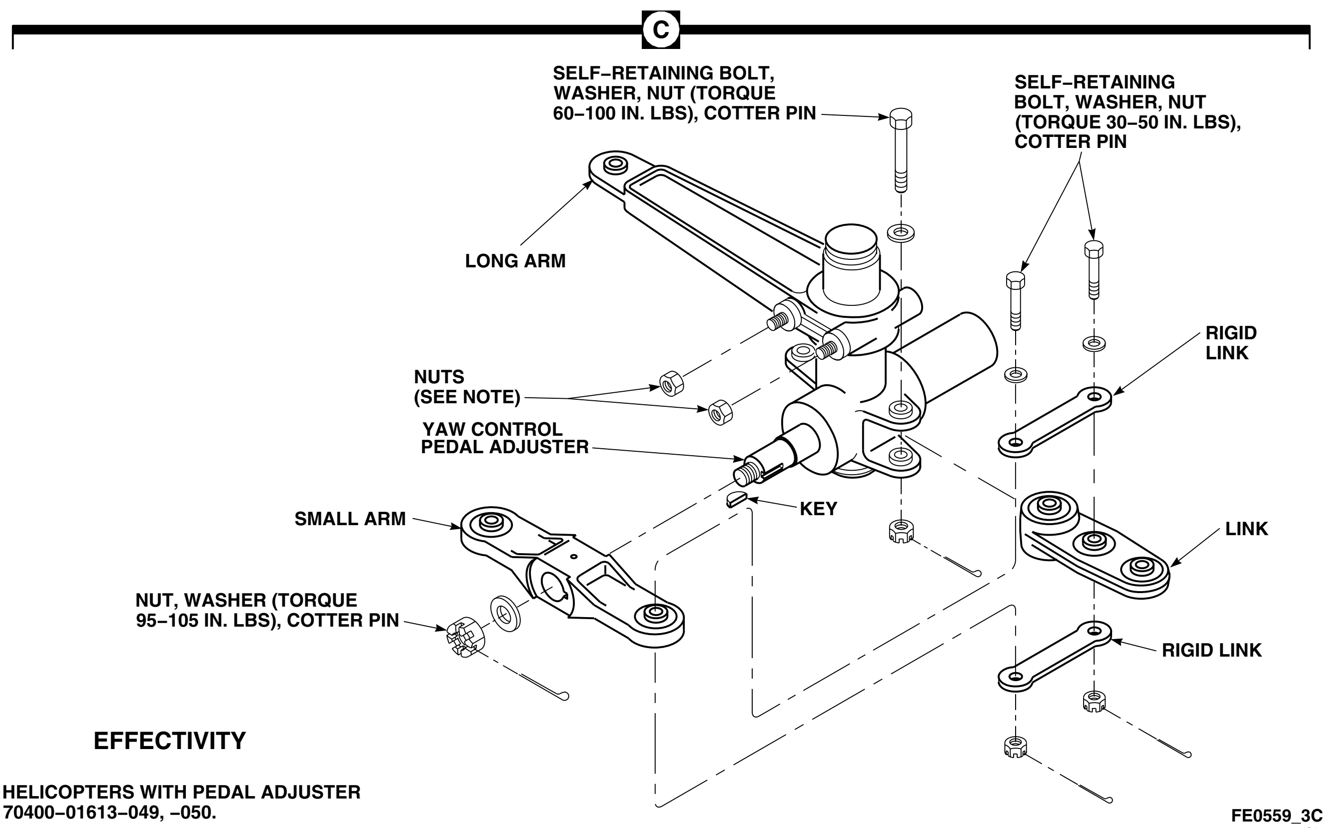


FIGURE 11-4-11-1. REPLACE YAW CONTROL PEDAL ADJUSTER (SHEET 3 OF 3)

INCH-POUNDS (Figure 11-4-11-1, Sheet 3, Detail C).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation and proof torque of self-retaining bolts are critical characteristics.

b. Connect links to pedal adjuster. Secure links as follows:

NOTE

Self-retaining bolt, securing links to pedal adjustor, 70400-21613-041, -042, is installed with bolthead down. Self-retaining bolt securing links to pedal adjustor, 70400-01613-049, -050, is installed with bolthead up.

(1) Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

(2) Install nuts on bolts. **TORQUE NUTS TO 60 - 100 INCH-POUNDS.** Install cotter pins.

(3) **PROOF TORQUE BOLTHEADS TO 30 - 32 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation and proof torque of self-retaining bolts are critical characteristics.

c. Connect rigid links to links. Secure rigid links as follows:

(1) Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

(2) Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

(3) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

d. Position key in slot of shaft and slide small arm on shaft. Install washer and nut. **TORQUE NUT TO 95 - 105 INCH-POUNDS. IF BOLTHOLE DOES NOT LINE UP WITH NUT, TIGHTEN NUT ONE CASTELLATION.** Install cotter pin.

e. Connect rigid links to small arm. Secure rigid links as follows:

(1) Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

(2) Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

(3) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

11-4-11.1.5 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Pedal adjuster should be in retracted position prior to installation.

b. Install lower support bearing with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-11-1, Sheet 2, Detail B).

c. Place adjuster in lower bearing.

d. Install upper support bearing with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

e. Check adjuster endplay (up and down). **MINIMUM ENDPLAY ALLOWED IS 0.005 INCH. MAXIMUM ENDPLAY ALLOWED IS 0.040 INCH.** If endplay is less than 0.005 inch, install shim under upper bearing support. If endplay is greater than 0.040 inch, install shim under lower bearing support. Peel shim(s) as necessary.

f. Connect directional control rod to pedal adjuster arm. Secure directional control rod as follows:

  (1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

  (2) Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

  (3) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

g. Connect pedal directional control rods to pedal adjuster links. Secure pedal directional control rods as follows:

  (1) Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

  (2) Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

  (3) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

h. Position clevis of release cable to pedal adjuster assembly and secure with bolt and nut. **TIGHTEN NUT HANDTIGHT.** Install cotter pin (Figure 11-4-11-1, Sheet 1, Detail A).

11-4-11.1.6 Checkout.

a. Push pedals fully to the front.

b. Lock release lever. Pedals will stay.

c. Pull release lever out to unlock. Pedals will move back toward you.

d. If pedals do not operate properly, perform the following:

  (1) Unlock pedals.

  (2) Remove bolt and nut securing clevis to adjuster (Figure 11-4-11-1, Sheet 1, Detail A).

  (3) Release jamnut on clevis end of cable.

  (4) Adjust clevis to shorten or lengthen release cable. Shorten clevis to tighten cable; lengthen clevis to loosen cable.

  (5) Install bolt to hold clevis to adjuster. **INSTALL NUT HANDTIGHT.** Install cotter pin.

e. Repeat step a. through step c.

f. Make sure area is clean and free of foreign material.

g. Install cockpit floor access panels around pedal assembly (PARA 2-4-43).

h. Install left and right boots (PARA 11-4-7).

i. Install cover over directional control rod and adjuster arm with screws.

11-4-12 YAW CONTROL PEDAL ADJUSTER CABLE AND HANDLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 113, Appendix D
- Sealing Primer, Item 305, Appendix D
- Spiral Wrap, Item 314, Appendix D
- Tiedown Strap, Item 328, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 11-4-11

11-4-12.1 Replace Yaw Control Pedal Adjuster Cable and Handle.

NOTE

Replacement of both yaw control pedal adjuster cables and handles is the same.

11-4-12.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove bolt and nut securing clevis of release cable to pedal adjuster assembly (Figure 11-4-12-1, Detail A).

c. Remove screws and washers securing clamps to brackets. Remove clamps.

d. Remove screws and washers securing release handle and spacer to instrument panel.

e. To remove handle, loosen setscrew in handle and remove handle from release cable (Figure 11-4-12-1, Detail B).

11-4-12.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. To install handle, perform the following:

(1) Place a light coat of corrosion preventive compound, Item 113, Appendix D, on threads of cable.

(2) Install handle on release cable (Figure 11-4-12-1, Detail B).

(3) Place a light coat of sealing primer, Item 305, Appendix D, on threads of setscrew. Tighten setscrew in handle until seated against cable.

c. Install spiral wrap, Item 314, Appendix D, on cable where it contacts support on end of instrument panel. Install tiedown straps, Item 328, Appendix D, on each end of tubing.

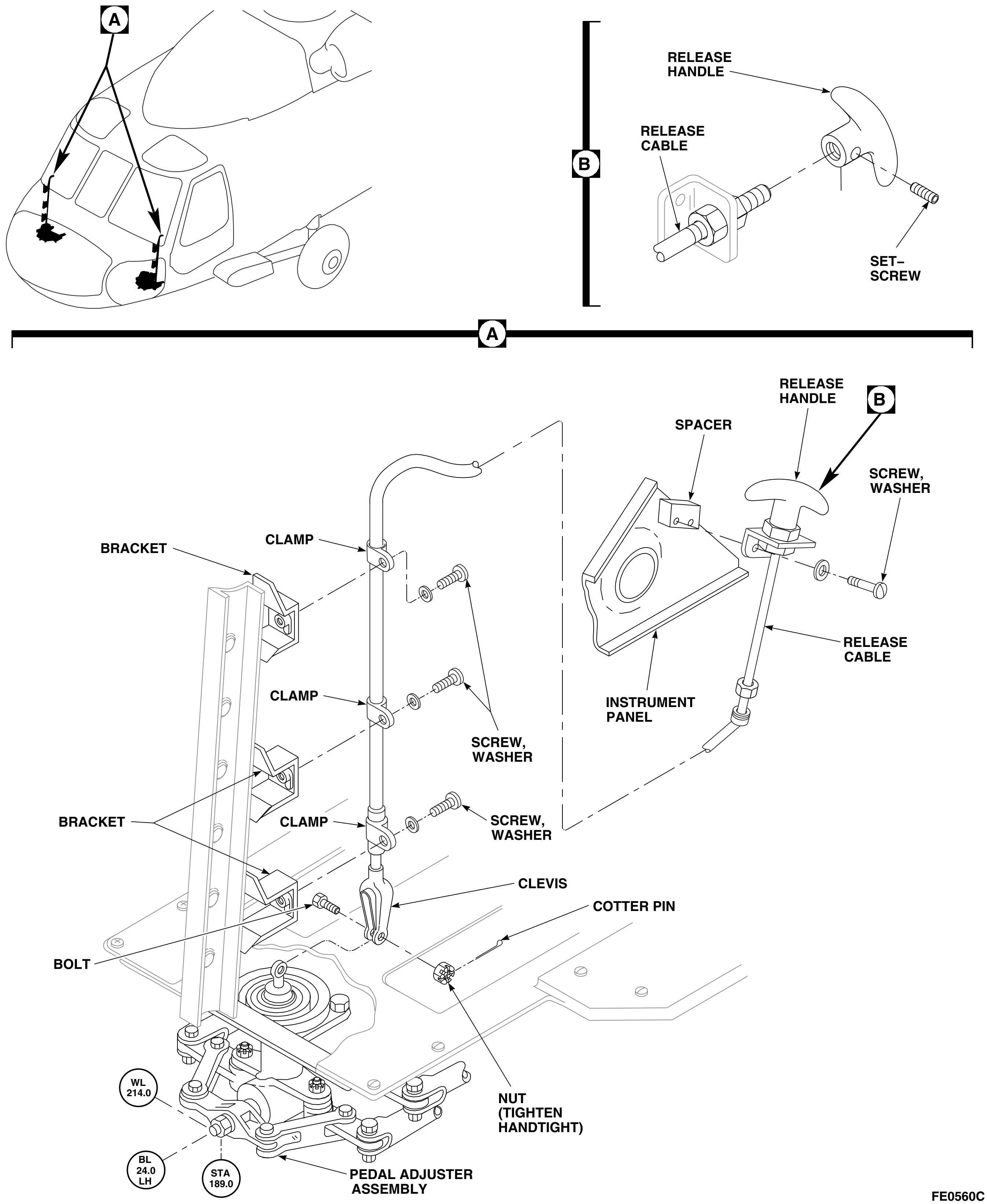


FIGURE 11-4-12-1. REPLACE YAW CONTROL PEDAL ADJUSTER CABLE AND HANDLE

d. Place clamps on cable (Figure 11-4-12-1, Detail A).

e. Install release handle and spacer to side of instrument panel using screws and washers.

f. Install clamps and cable on airframe brackets using screws and washers.

g. Position clevis of release cable on pedal adjuster assembly and secure with bolt and nut. **TIGHTEN NUT HANDTIGHT.** Install cotter pin.

h. Check cable operation and clevis adjustment (PARA 11-4-11).

11-4-13 PILOT'S YAW PEDALS AND ADJUSTER CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D
- PARA 2-4-38
- PARA 2-4-43
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-4
- PARA 11-4-7
- PARA 11-4-8
- PARA 11-4-15

11-4-13.1 Replace Pilot's Yaw Pedals and Adjuster Control Rods.

NOTE

- Pilot's yaw pedals and adjuster control rods consists of one pilot's directional control rod and two yaw pedal control rods.
- Replacement of both yaw pedal control rods is the same.

11-4-13.1.1 Remove.

a. Remove yaw control pedal boot (PARA 11-4-7).

b. Remove pedal assembly from pedal support (PARA 11-4-8).

c. Remove cyclic stick boot (PARA 11-4-15).

d. Remove cockpit floor access panels (PARA 2-4-43).

e. If removing pilot's directional control rod, remove right lower window (PARA 2-4-38).

f. Remove bolts, washers, and nuts securing yaw pedal control rod(s) to pedal adjuster and pedal support(s) (Figure 11-4-13-1, Detail A). Remove control rods.

g. Remove bolts, washers, and nuts securing pilot's directional control rod to pedal adjuster

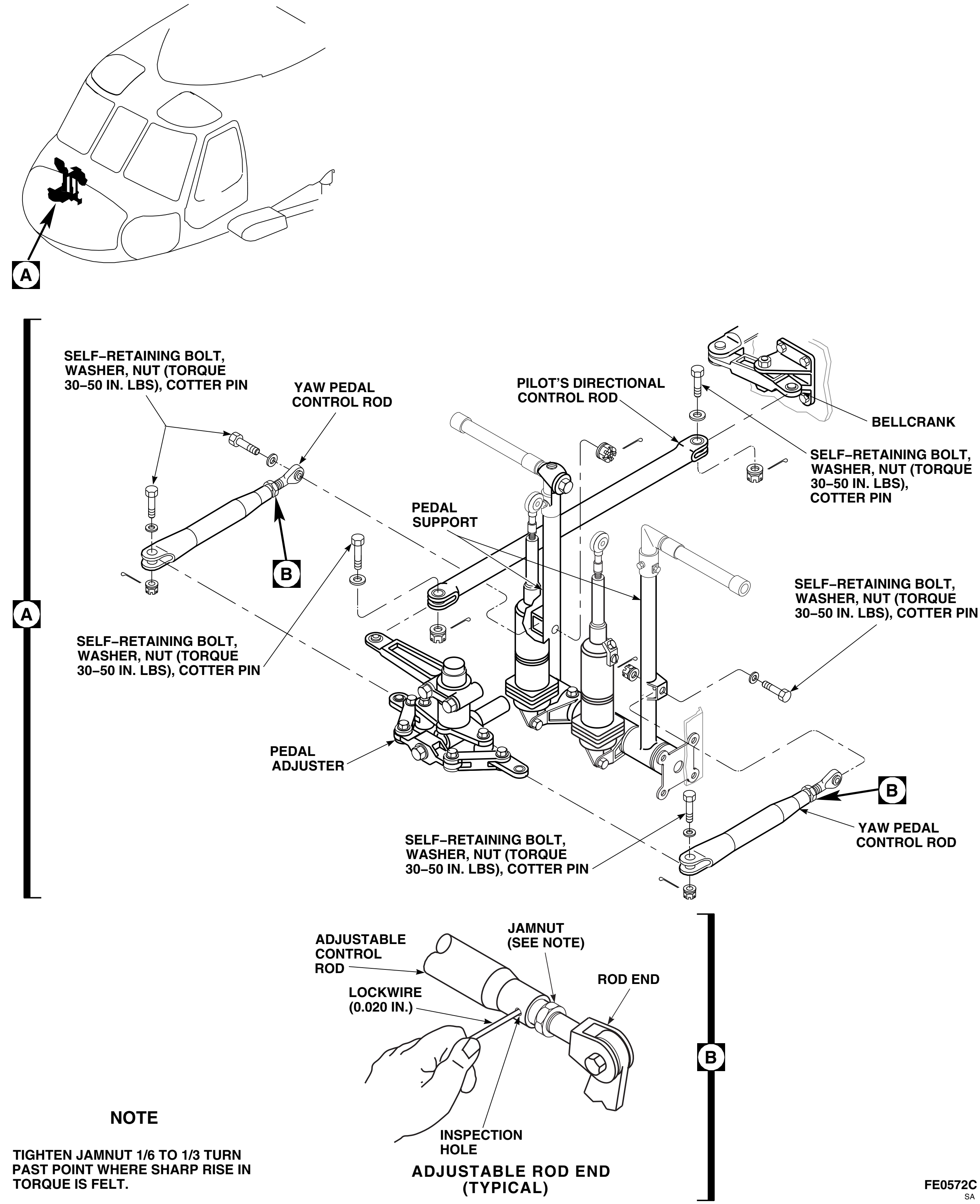


FIGURE 11-4-13-1. REPLACE PILOT'S YAW PEDALS AND ADJUSTER CONTROL RODS and bellcrank. Remove control rod through frame of lower window.

11-4-13.1.2 Install.

a. Install pilot's directional control rod as follows (Figure 11-4-13-1, Detail A):

(1) Install replacement control rod between bellcrank and pedal adjuster through frame of lower window.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

b. Install yaw pedal control rod(s) as follows (Figure 11-4-13-1, Detail A):

(1) Check yaw pedal control rod length as follows:

(a) Measure adjustable control rod dimensions from bolthole-center to bolthole-center. **INITIAL LENGTH BETWEEN BOLTHOLE CENTERS IS TO BE 10.81 - 10.87-INCHES.** Loosen jamnut and adjust as required.

(b) Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-13-1, Detail B).

(c) Secure rod end in place with jamnut. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(2) Install replacement yaw pedal control rod between pedal adjuster and pedal support (Figure 11-4-13-1, Detail A). **VERIFY PEDALS ARE EVEN ACROSS HELICOPTER.** If required, adjust length as follows:

(a) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(b) Loosen jamnut and adjust length of replacement control rod until pedals are even across helicopter.

(c) Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-13-1, Detail B).

(d) Secure rod end in place with jamnut. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(3) Install nuts on self-retaining bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins (Figure 11-4-13-1, Detail A).

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

c. If rod ends were adjusted, check control rigging between cyclic sticks, pedal adjusters, and torque shafts (PARA 11-4-4).

d. Make sure area is clean and free of foreign material.

e. Install cockpit floor access panels (PARA 2-4-43).

f. If removed, install right lower window (PARA 2-4-38).

g. Install yaw control pedal boot (PARA 11-4-7).

h. Install pedal assembly on pedal support (PARA 11-4-8).

i. Install cyclic stick boot (PARA 11-4-15).

11-4-14 COPILOT'S YAW PEDALS AND ADJUSTER CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D
- PARA 2-4-38
- PARA 2-4-43
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-4
- PARA 11-4-7
- PARA 11-4-8
- PARA 11-4-15

11-4-14.1 Replace Copilot's Yaw Pedals and Adjuster Control Rods.

NOTE

- Copilot's yaw pedals and adjuster control rods consists of one copilot's directional control rod and two yaw pedal control rods.
- Replacement of both yaw pedal control rods is the same.

11-4-14.1.1 Remove.

a. Remove yaw control pedal boot (PARA 11-4-7).

b. Remove pedal assembly from pedal support (PARA 11-4-8).

c. Remove cyclic stick boot (PARA 11-4-15).

d. Remove cockpit floor access panels (PARA 2-4-43).

e. If removing copilot's directional control rod, remove left lower window (PARA 2-4-38).

f. Remove bolts, washers, and nuts securing yaw pedal control rod(s) to pedal adjuster and pedal support(s) (Figure 11-4-14-1, Detail A). Remove control rods.

g. Remove bolts, washers, and nuts securing copilot's directional control rod to pedal

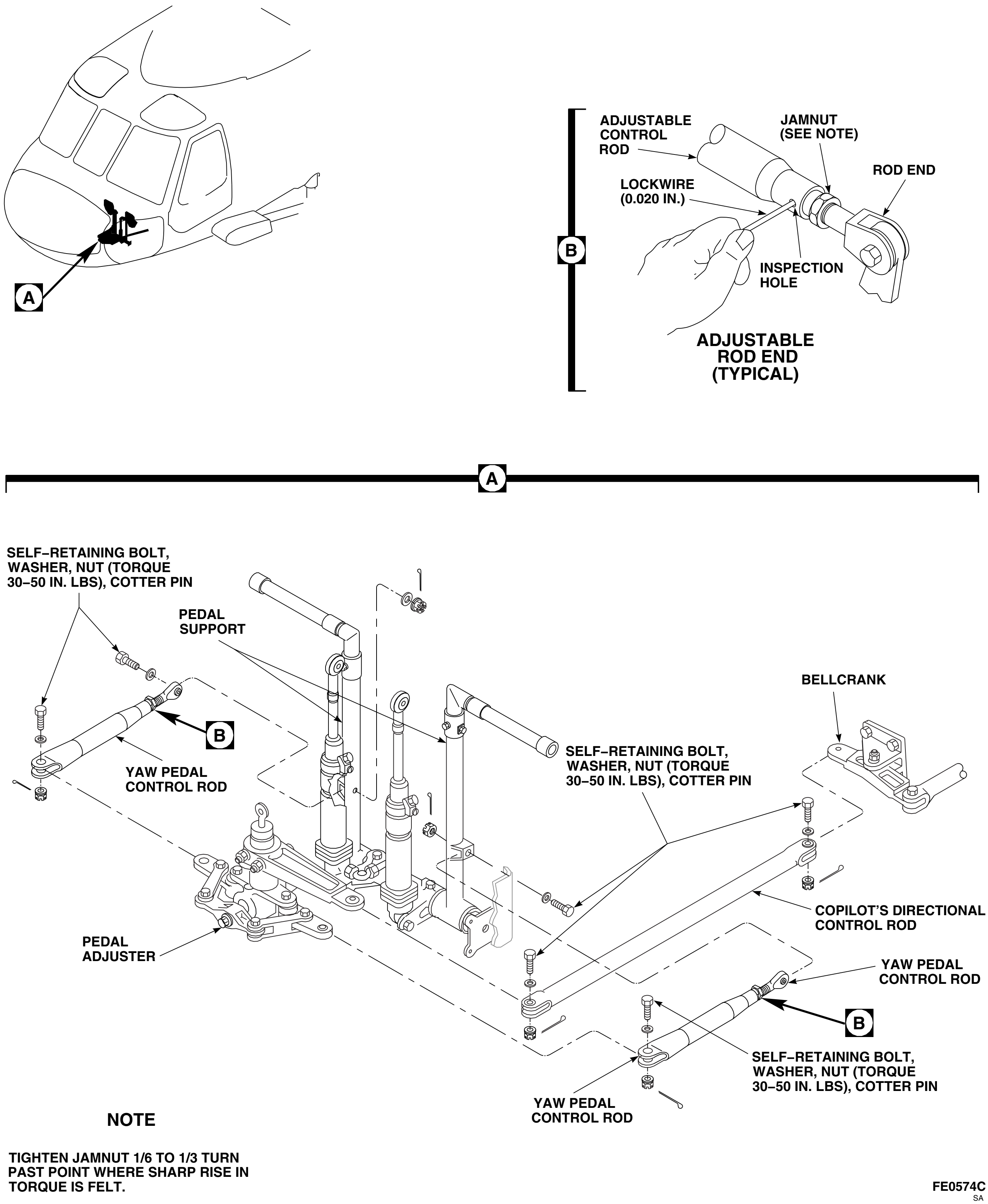


FIGURE 11-4-14-1. REPLACE COPILOT'S YAW PEDALS AND ADJUSTER CONTROL RODS adjuster and bellcrank. Remove control rod through frame of lower window.

11-4-14.1.2 Install.

a. Install copilot's directional control rod as follows (Figure 11-4-14-1, Detail A):

(1) Position replacement control rod between bellcrank and pedal adjuster through frame of lower window.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

b. Install yaw pedal control rods as follows (Figure 11-4-14-1, Detail A):

(1) Check yaw pedal control rod length as follows:

(a) Measure adjustable control rod dimensions from bolthole-center to bolthole-center. **INITIAL LENGTH BETWEEN BOLTHOLE CENTERS MUST BE 10.65 - 10.71-INCHES.** Loosen jamnut and adjust as required.

(b) Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-14-1, Detail B).

(c) Secure rod end in place with jamnut. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(2) Install replacement yaw pedal control rod between pedal adjuster and pedal support (Figure 11-4-14-1, Detail A). **VERIFY PEDALS ARE EVEN ACROSS HELICOPTER.** If required, adjust length as follows:

(a) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(b) Loosen jamnut and adjust length of replacement control rod until pedals are even across helicopter.

(c) Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-14-1, Detail B).

(d) Secure rod end in place with jamnut. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(3) Install nuts on self-retaining bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins (Figure 11-4-14-1, Detail A).

(4) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

c. If rod ends were adjusted, check control rigging between cyclic sticks, pedal adjusters, and torque shafts (PARA 11-4-4).

d. Make sure area is clean and free of foreign material.

e. Install cockpit floor access panels (PARA 2-4-43).

f. If removed, install left lower window (PARA 2-4-38).

g. Install yaw control pedal boot (PARA 11-4-7).

h. Install pedal assembly on pedal support (PARA 11-4-8).

i. Install cyclic stick boot (PARA 11-4-15).

11-4-15 CYCLIC STICK BOOT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16

11-4-15.1 Replace Cyclic Stick Boot.

NOTE

Replacement of both cyclic stick boots is the same.

11-4-15.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws and washers securing boot retainer and boot to floor panel (Figure 11-4-15-1, Detail A).

c. Lift up and remove boot retainer over cyclic stick.

d. Unzip and remove boot.

11-4-15.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Make sure area is clean and free of foreign material.

c. Install boot over cyclic stick and around base of stick (Figure 11-4-15-1, Detail A).

d. Install boot retainer over boot.

e. Install screws and washers securing boot retainer and boot to floor panel.

f. Zip boot around cyclic stick.

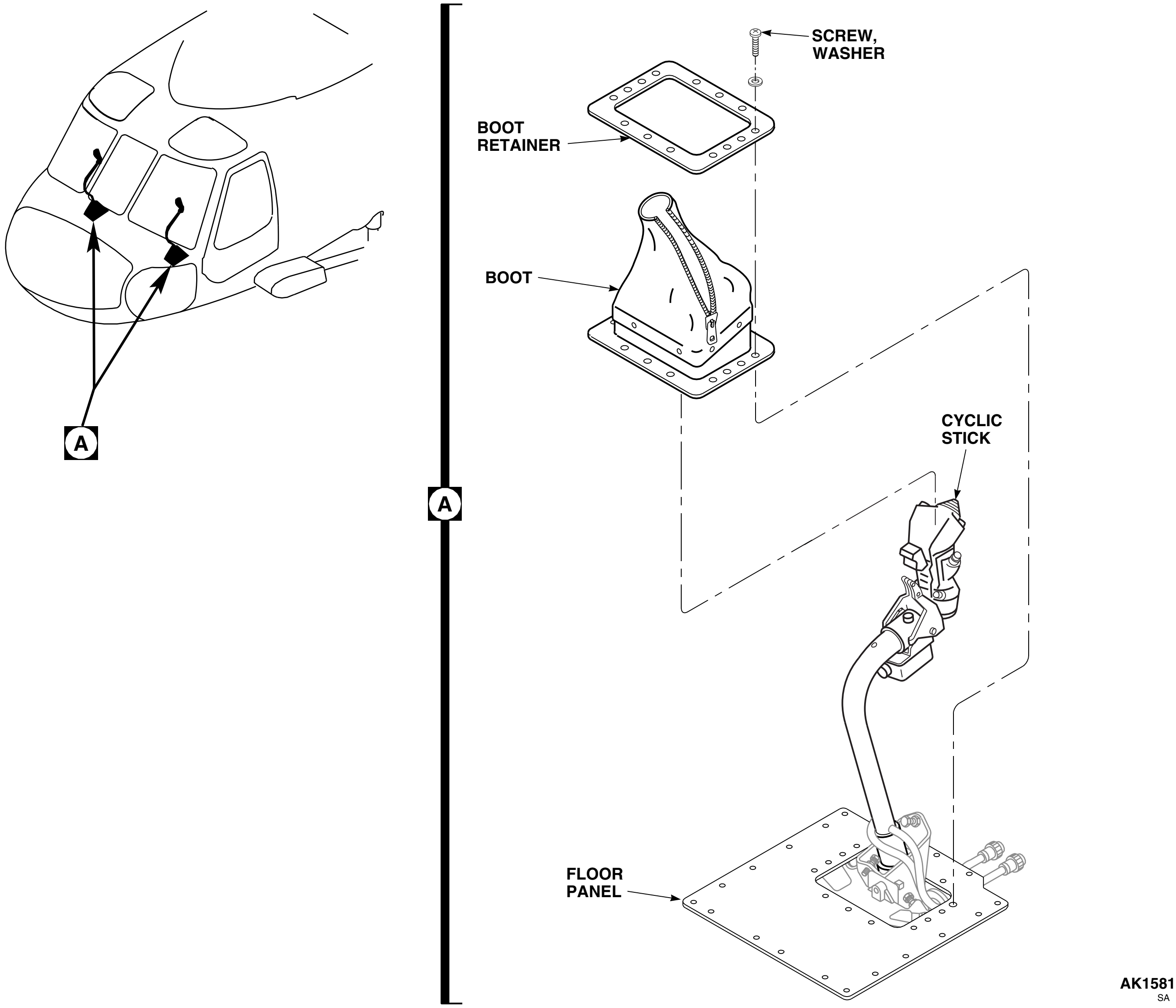


FIGURE 11-4-15-1. REPLACE CYCLIC STICK BOOT

11-4-16 CYCLIC STICK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 49, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-2-1
- PARA 11-4-1
- PARA 11-4-15

11-4-16.1 Replace Cyclic Stick.

NOTE

Replacement of both cyclic sticks is the same, except as noted.

11-4-16.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove soundproofing panels covering flight control rods in cabin side walls or ceiling, as necessary (PARA 2-4-69).

NOTE

The below step will allow cyclic stick to be moved without using hydraulic power.

c. Remove self-retaining bolts, washers, and nuts securing fore-and-aft control rod to bellcrank in midsection inside cabin (Figure 11-4-16-1, Detail C).

d. Remove cyclic stick boot (PARA 11-4-15).

e. Remove screws and washers securing floor panel, remove floor panel (Figure 11-4-16-1, Detail A).

f. On helicopters with cyclic stick, 70400-01222-041, disconnect electrical connectors as follows:

(1) On pilot's cyclic stick, disconnect P120 from J120.

(2) On copilot's cyclic stick, disconnect P119 from J119.

(3) Install protective caps and plugs on electrical connectors.

g. On helicopters with cyclic stick, 70400-01222-045, disconnect electrical connectors as follows (Figure 11-4-16-1, Detail A):

(1) On pilot's cyclic stick, disconnect P8R and P120 from J8R and J120.

(2) On copilot's cyclic stick, disconnect P7R and P119 from J7R and J119.

(3) Install protective caps and plugs on electrical connectors.

h. On helicopters with cyclic stick, 70400-01222-049, or 70400-01222-050, disconnect electrical connectors as follows (Figure 11-4-16-1, Detail A):

(1) On pilot's cyclic stick, disconnect P8R and P120 from J8R and J120, and P908R from dummy receptacle.

(2) On copilot's cyclic stick, disconnect P7R and P119 from J7R and J119, and P909R from dummy receptacle.

(3) Install protective caps and plugs on electrical connectors.

i. Remove screw, washer, and nut securing electrical cable clamp to beam.

j. Working through access hole in yoke, remove self-retaining bolts, washers, and nuts securing pitch pushrod to cyclic stick (Figure 11-4-16-1, Detail B).

k. Remove self-retaining bolt, washer, and nut securing socket to yoke.

l. Remove cyclic stick. Guide wiring through beam as stick is removed.

11-4-16.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Guide wiring through beam and place cyclic stick into yoke, with curve in stick facing front of helicopter (Figure 11-4-16-1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

c. Install cyclic stick socket in yoke. Secure cyclic stick socket as follows:

(1) Install self-retaining bolt (bolthead facing left) and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

d. Line up pitch pushrod with cyclic stick socket. Secure pitch pushrod as follows:

NOTE

Bolthead may face either direction when installing bolt in pitch pushrod.

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. On helicopters with cyclic stick, 70400-01222-041, connect electrical connectors as follows (Figure 11-4-16-1, Detail A):

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

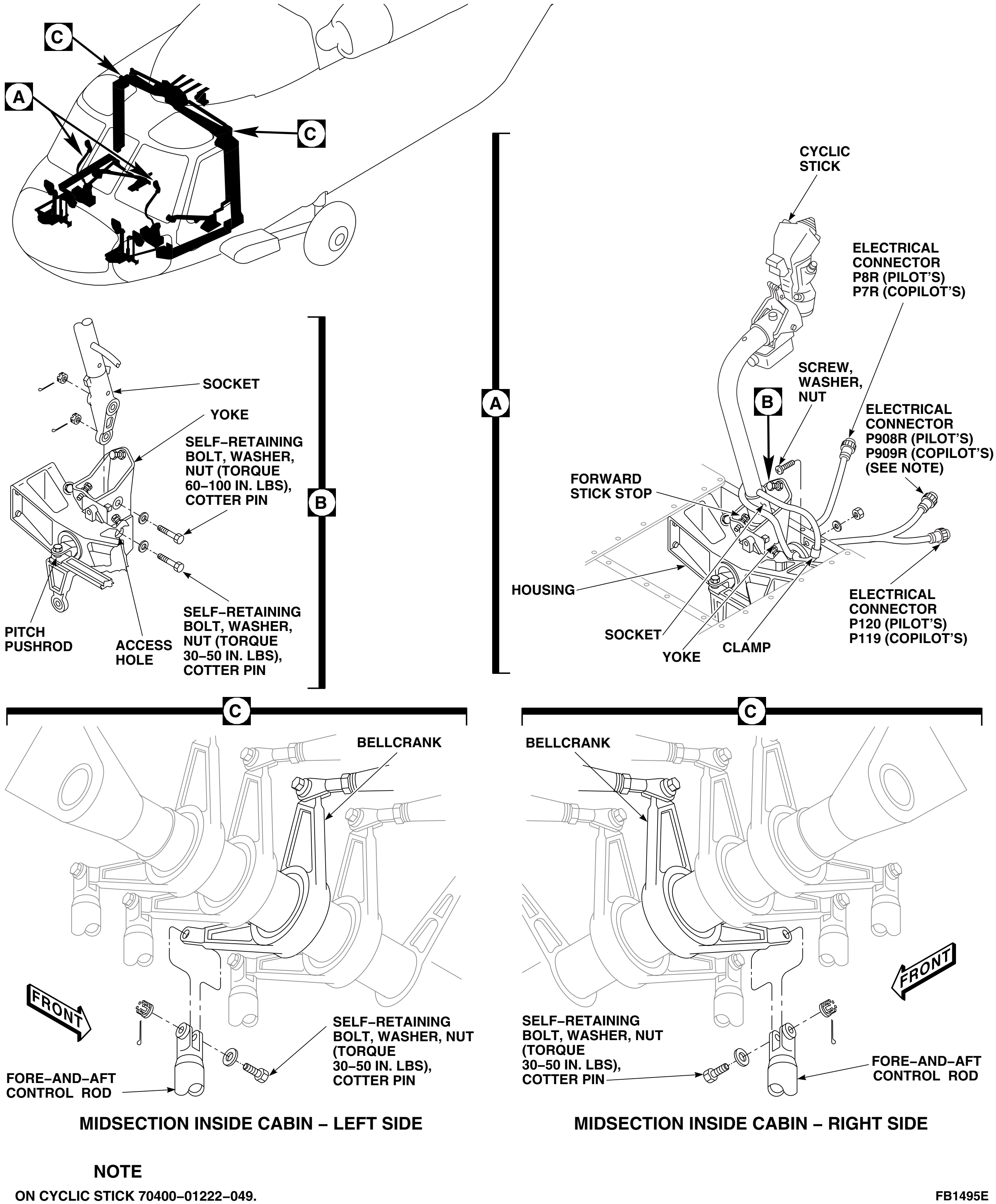


FIGURE 11-4-16-1. REPLACE CYCLIC STICK

(3) On pilot’s cyclic stick, connect P120 to J120.

(4) On copilot’s cyclic stick, connect P119 to J119.

f. On helicopters with cyclic stick, 70400-01222-045, connect electrical connectors as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) On pilot’s cyclic stick, connect P8R and P120 to J8R and J120.

(4) On copilot’s cyclic stick, connect P7R and P119 to J7R and J119.

g. On helicopters with cyclic stick, 70400-01222-049, or 70400-01222-050, connect electrical connectors as follows (Figure 11-4-16-1, Detail A):

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) On pilot’s cyclic stick, connect P8R and P120 to J8R and J120, and P908R to dummy receptacle.

(4) On copilot’s cyclic stick, connect P7R and P119 to J7R and J119, and P909R to dummy receptacle.

h. Make sure area is clean and free of foreign material.

i. Line up fore-and-aft control rod with bellcrank (Figure 11-4-16-1, Detail C). Secure control rod as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

CAUTION

To prevent interference of electrical cable with cyclic lateral stop, location of clamp should be 7.00 - 7.25-inches from where the wire harness exits the cyclic stick at ″clamp here″ decal on wire bundle.

j. Bond wire bundle to ″clamp here″ tag using adhesive, Item 49, Appendix D, and verify clamp is located 7.00 - 7.25-inches from attachment point of wire bundle to cyclic stick.

CAUTION

To prevent possible interference between wire harness and cyclic stick stops, make sure of proper wire harness clamping.

k. Install electrical cable clamp on beam with screws, washer, and nut, making sure wire bundle is secure enough to prevent wire harness clamping (Figure 11-4-16-1, Detail A).

l. While at approximately mid-collective and mid-pedal, cycle the cyclic stick through the full range of motion, full left to full right and full forward to full aft. Throughout the full control range of motion, the pilot and copilot wire harness must remain clear of all the cyclic stops.

m. Cycle stick three times and make sure wiring bundle does not jam, tangle, or shear against stopbolts, or any structure/mechanism in area of wiring bundle.

n. Install cyclic stick boot (PARA 11-4-15).

o. Install soundproofing panels over flight control rods in cabin side walls or ceiling, as necessary (PARA 2-4-69).

p. Perform operational check of cyclic stick (PARA 11-2-1).

11-4-17 CYCLIC STICK GRIP.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cap, Electrical, Item 87A, Appendix D
- Fibrous Cord, Item 139, Appendix D
- Protective Caps and Plugs
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 11-2-1
- TM 11-70-23

11-4-17.1 Replace Cyclic Stick Grip.

NOTE

Replacement of both cyclic stick grips is the same, except as noted.

11-4-17.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. On helicopters with cyclic stick, 70400-01222-041, disconnect electrical connectors as follows (Figure 11-4-17-1):

(1) On pilot's cyclic stick, disconnect connector, P120, from connector, J120.

(2) On copilot's cyclic stick, disconnect connector, P119, from connector, J119.

(3) Install protective caps and plugs on electrical connectors.

c. On helicopters with cyclic stick, 70400-01222-045, disconnect electrical connectors as follows:

(1) On pilot's cyclic stick, disconnect connectors, P8R and P120, from connectors, J8R and J120.

(2) On copilot's cyclic stick, disconnect connectors, P7R and P119, from connectors, J7R and J119.

(3) Install protective caps and plugs on electrical connectors.

d. On helicopters with cyclic stick, 70400-01222-049, or 70400-01222-050, disconnect electrical connectors as follows:

(1) On pilot's cyclic stick, disconnect electrical connectors, P8R and P120, from connectors, J8R and J120, and P908R from dummy receptacle.

(2) On copilot's cyclic stick, disconnect electrical connectors, P7R and P119, from connectors, J7R and J119, and P909R from dummy receptacle.

(3) Install protective caps and plugs on electrical connectors.

e. Loosen backshell on electrical connector and slide back along wiring.

f. Tag and remove pins from electrical connectors, P119 (copilot's) or P120 (pilot's), as required and remove backshell.

g. On helicopters with cyclic stick, 70400-01222-049, tag and remove pins from electrical connectors, P908R (pilot's) or P909R (copilot's) as required and remove connector backshell. On helicopters with cyclic stick, 70400-01222-049, or 70400-01222-050, tag and remove pins from electrical connectors, P908R (pilot's) or P909R (copilot's) as required and remove connector backshell.

h. Remove pins from wires.

i. If installed, remove stabilator slew-up switch (TM 11-70-23).

j. Remove grip from cyclic stick by removing screw, washer, and capnut (Figure 11-4-17-1, Detail A).

k. If grip is to be replaced, tie fibrous cord, Item 139, Appendix D, approximately 72-inches in length, securely around lower end of wire bundle.

l. Remove grip from cyclic stick making sure wire bundle is supported.

NOTE

Do not let cord fall into cyclic stick.

m. If grip is to be replaced, remove cord from wire bundle and leave cord in cyclic stick.

11-4-17.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If same grip is being installed in replacement cyclic stick, perform the following:

(1) Tie fibrous cord, Item 139, Appendix D, approximately 72-inches in length, to end of grip wire bundle.

(2) Feed cord into cyclic stick and route cord through grommet in stick.

c. If replacement grip is being installed, tie cord found in cyclic stick securely around lower end of wire bundle.

d. Pull cord and route wiring through grommet in stick until grip is seated (Figure 11-4-17-1, Detail A).

e. If installed, position trigger assembly on stick before securing grip to stick.

f. Position and secure grip to stick using screw, washer, and capnut. **TORQUE CAPNUT TO 43 - 47 INCH-POUNDS.**

g. If required, install stabilator slew-up switch (TM 11-70-23).

h. Slide backshell over cyclic stick wiring.

i. On helicopters with cyclic stick, 70400-01222-041 or 70400-01222-045, connect wiring to electrical connector, P119 (copilot's) or P120 (pilot's) as follows:

Wire Number	To	Electrical Connector Pin
A-20		A
B-20		See Note
C-20		See Note
D-20		D
E-20		E

Wire Number	To	Electrical Connector Pin
F-20		F
G-20		G
H-20		H
J-20		J
K-20		K
L-20		See Note
M-20		See Note
N-20		N
P-20		P
T-24		T
U-24		U
V-24		V
W-24		W
X-24		X
Y-20		Y
Z-20		Z
AA-20		a
BB-20		b

NOTE: The indicated wires should not be brought out to stick assembly connector. These wires are to be encapped using electrical cap, Item 87A, Appendix D, and stowed in flex tubing.

j. On helicopters with cyclic stick, 70400-01222-049 or 70400-01222-050, connect wiring to electrical connectors, P119 and P909R (copilot's), or P120 and P908R (pilot's), as follows:

Wire Number	To	Electrical Connector Pin
A-20		(P119 or P120)-A
B-20		See Note
C-20		See Note
D-20		(P119 or P120)-D
E-20		(P119 or P120)-E
F-20		(P119 or P120)-F
G-20		(P119 or P120)-G
H-20		(P119 or P120)-H
J-20		(P119 or P120)-J
K-20		(P119 or P120)-K
L-20		(P908R or P909R)-A
M-20		(P908R or P909R)-B
N-20		(P119 or P120)-N
P-20		(P119 or P120)-P
T-24		(P119 or P120)-T
U-24		(P119 or P120)-U
V-24		(P119 or P120)-V
W-24		(P119 or P120)-W
X-24		(P119 or P120)-X
Y-20		(P119 or P120)-Y
Z-20		(P119 or P120)-Z

Wire Number	To	Electrical Connector Pin
AA-20		(P119 or P120)-a
BB-20		(P119 or P120)-b

NOTE:
The indicated wires should not be brought out to stick assembly connector. These wires are to be encapped using electrical cap, Item 87A, Appendix D, and stowed in flex tubing.

k. Slide connector backshell over wire bundle. **TIGHTEN BACKSHELL HANDTIGHT.**

l. Secure electrical wiring with fibrous cord, Item 139, Appendix D.

m. Inspect and treat electrical connectors (PARA 1-7-20).

n. On helicopters with cyclic stick, 70400-01222-041, connect electrical connectors as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) On pilot's cyclic stick, connect connector, P120, to connector, J120.

(4) On copilot's cyclic stick, connect connector, P119, to connector, J119.

o. On helicopters with cyclic stick, 70400-01222-045, connect electrical connectors as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) On pilot's cyclic stick, connect connectors, P8R and P120, to connectors, J8R and J120.

(4) On copilot's cyclic stick, connect connectors, P7R and P119, to connectors, J7R and J119.

p. On helicopters with cyclic stick, 70400-01222-049, or 70400-01222-050, connect electrical connectors as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) On pilot's cyclic stick, connect electrical connectors, P8R and P120 to J8R, and J120 and P908R to dummy receptacle.

(4) On copilot's cyclic stick, connect electrical connectors, P7R and P119 to J7R, and J119 and P909 to dummy receptacle.

q. Perform operational check of cyclic stick grip (PARA 11-2-1).

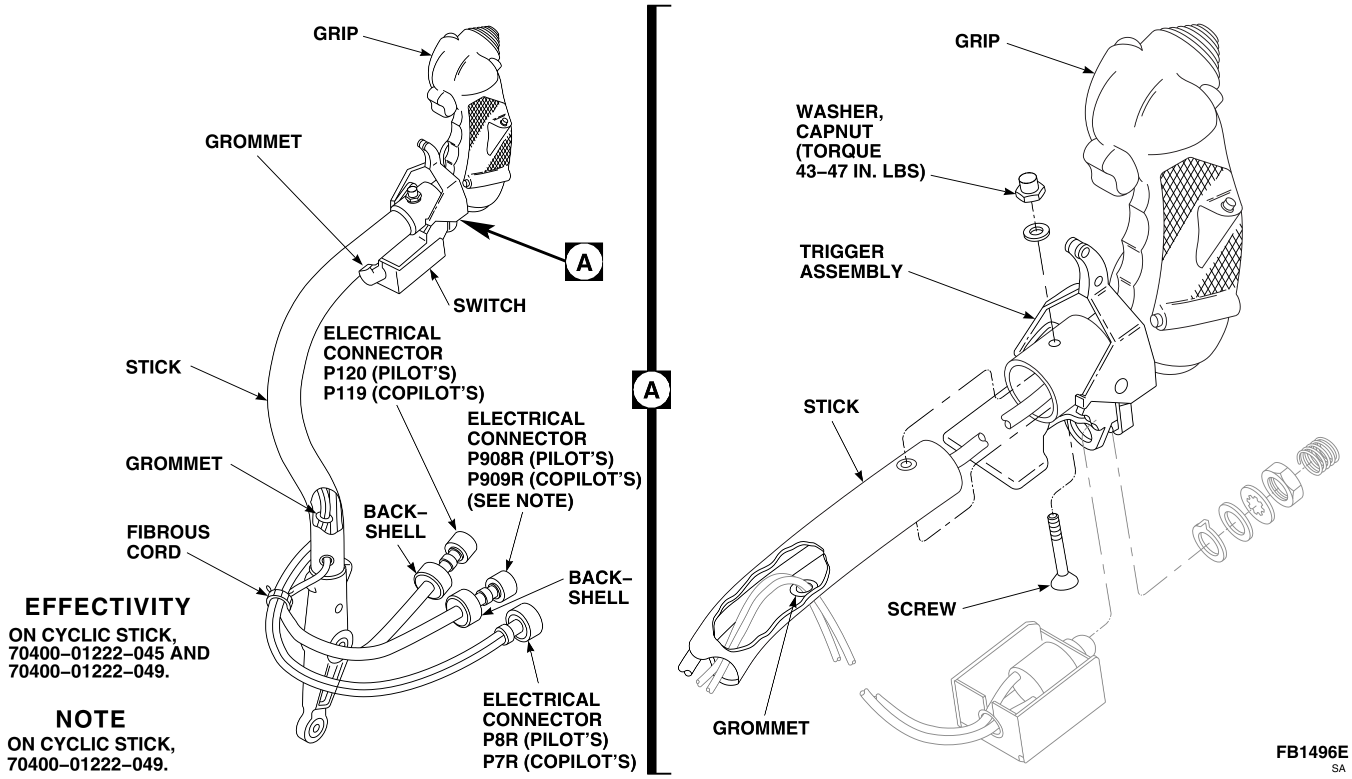


FIGURE 11-4-17-1. REPLACE CYCLIC STICK GRIP

11-4-18 CYCLIC STICK SWITCHES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Heat Gun
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 31, Appendix D
- Sealing Compound, Item 289, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-1

11-4-18.1 Replace Cyclic Stick PNL LTS, CARGO REL, TRIM REL, and GA Switches.

NOTE

Replacement of cyclic stick PNL LTS, CARGO REL, TRIM REL, and GA switches on both cyclic sticks is the same.

11-4-18.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Using heat gun, heat area of switch for about 30 seconds.

c. Carefully pull switch from cyclic stick grip to expose solder terminals (Figure 11-4-18-1, Detail A).

d. Remove sealing compound from soldered terminals.

e. Tag switch wires to identify switch connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

f. Using soldering iron, unsolder wires from switch.

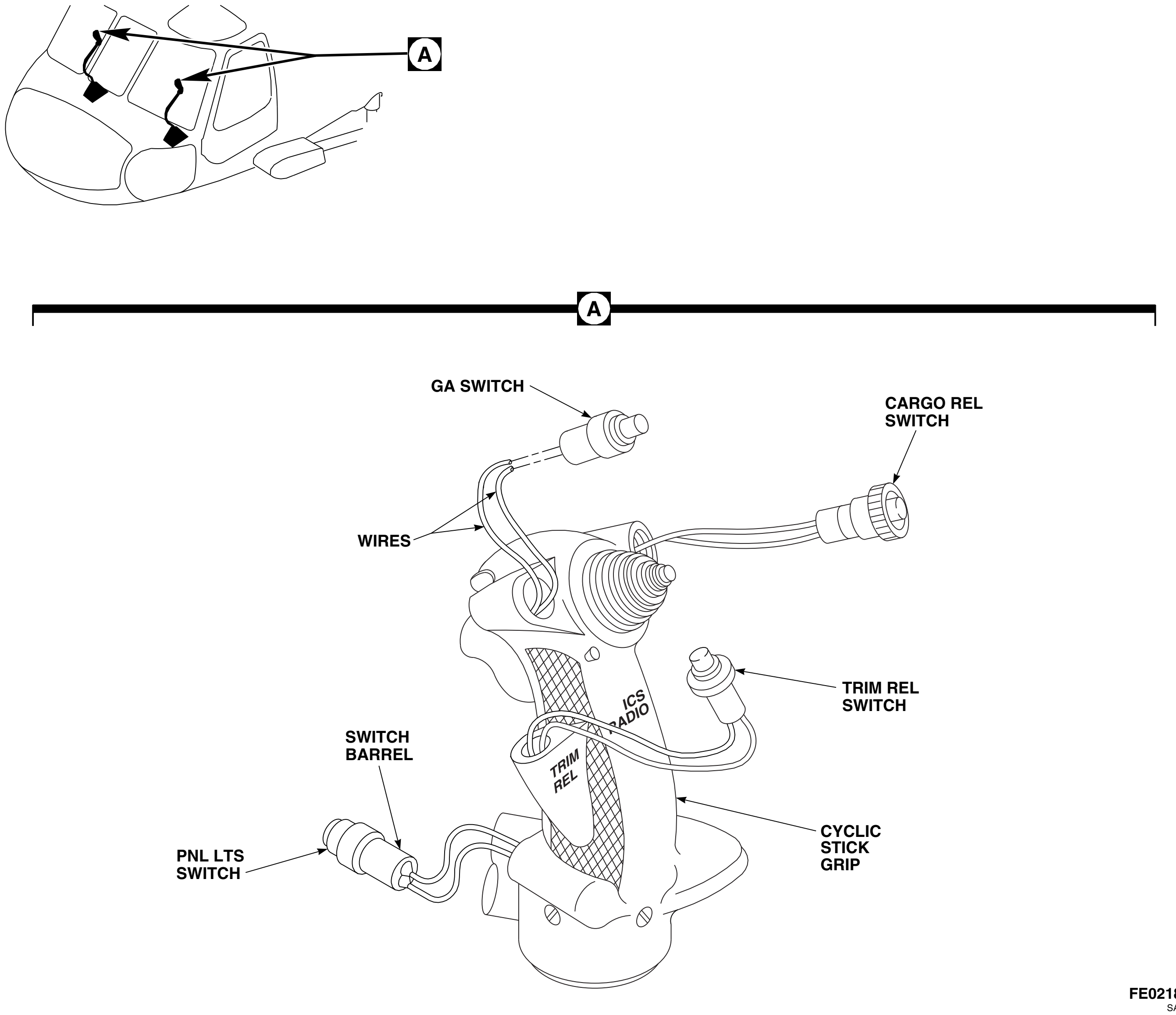


FIGURE 11-4-18-1. REPLACE CYCLIC STICK PNL LTS, CARGO REL, TRIM REL, AND GA SWITCHES

11-4-18.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering iron and solder, Item 313, Appendix D, solder wires to switch (Figure 11-4-18-1, Detail A). Remove tags.

c. Coat all exposed solder terminals with sealing compound, Item 289, Appendix D.

d. Coat barrel of switch with adhesive, Item 31, Appendix D.

e. Carefully press switch into cyclic stick grip, routing wires to prevent interference.

f. Perform operational check of cyclic stick grip assembly (PARA 11-2-1).

11-4-19 CYCLIC STICK TRIM SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Sealing Compound, Item 289, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-1

11-4-19.1 Replace Cyclic STICK TRIM Switch.

NOTE

Replacement of cyclic STICK TRIM switch on both cyclic sticks is the same.

11-4-19.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove sealing compound from pin hole.

c. Drive out pin securing STICK TRIM switch in cyclic stick grip (Figure 11-4-19-1, Detail A).

d. Carefully pull STICK TRIM switch from cyclic stick grip to expose solder terminals.

e. Remove sealing compound from soldered terminals.

f. Tag switch wires to identify switch connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

g. Using soldering iron, unsolder wires from switch.

11-4-19.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

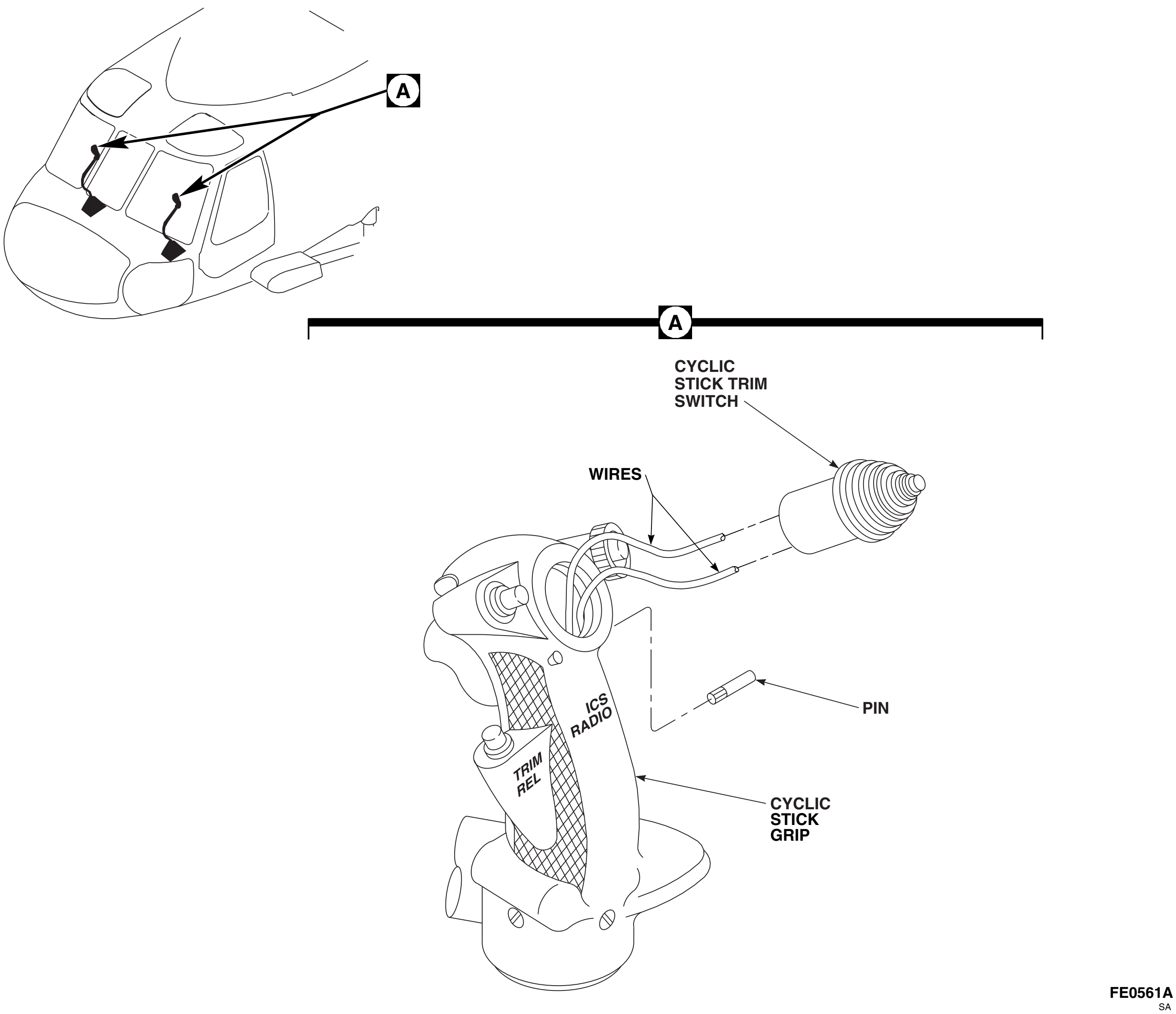


FIGURE 11-4-19-1. REPLACE CYCLIC STICK TRIM SWITCH

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering iron and solder, Item 313, Appendix D, solder wires to switch (Figure 11-4-19-1, Detail A). Remove tags.

c. Coat all exposed solder terminals with sealing compound, Item 289, Appendix D.

d. Carefully press STICK TRIM switch into grip assembly, routing wires to prevent interference.

e. Install pin securing STICK TRIM switch to cyclic stick grip.

f. Fill pin hole with sealing compound, Item 289, Appendix D.

g. Perform operational check of cyclic stick grip assembly (PARA 11-2-1).

11-4-20 CYCLIC STICK ICS/RADIO SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Sealing Compound, Item 289, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-1

11-4-20.1 Replace Cyclic Stick ICS/RADIO Switch.

NOTE

Replacement of cyclic stick ICS/RADIO switch on both cyclic sticks is the same.

11-4-20.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove sealing compound from screw and pin hole.

c. Remove screw and drive out pin (Figure 11-4-20-1, Detail A).

d. Carefully pull ICS/RADIO switch from cyclic stick grip to expose solder terminals.

e. Remove sealing compound from soldered terminals.

f. Tag switch wires to identify switch connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

g. Using soldering iron, unsolder wires from switch.

11-4-20.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

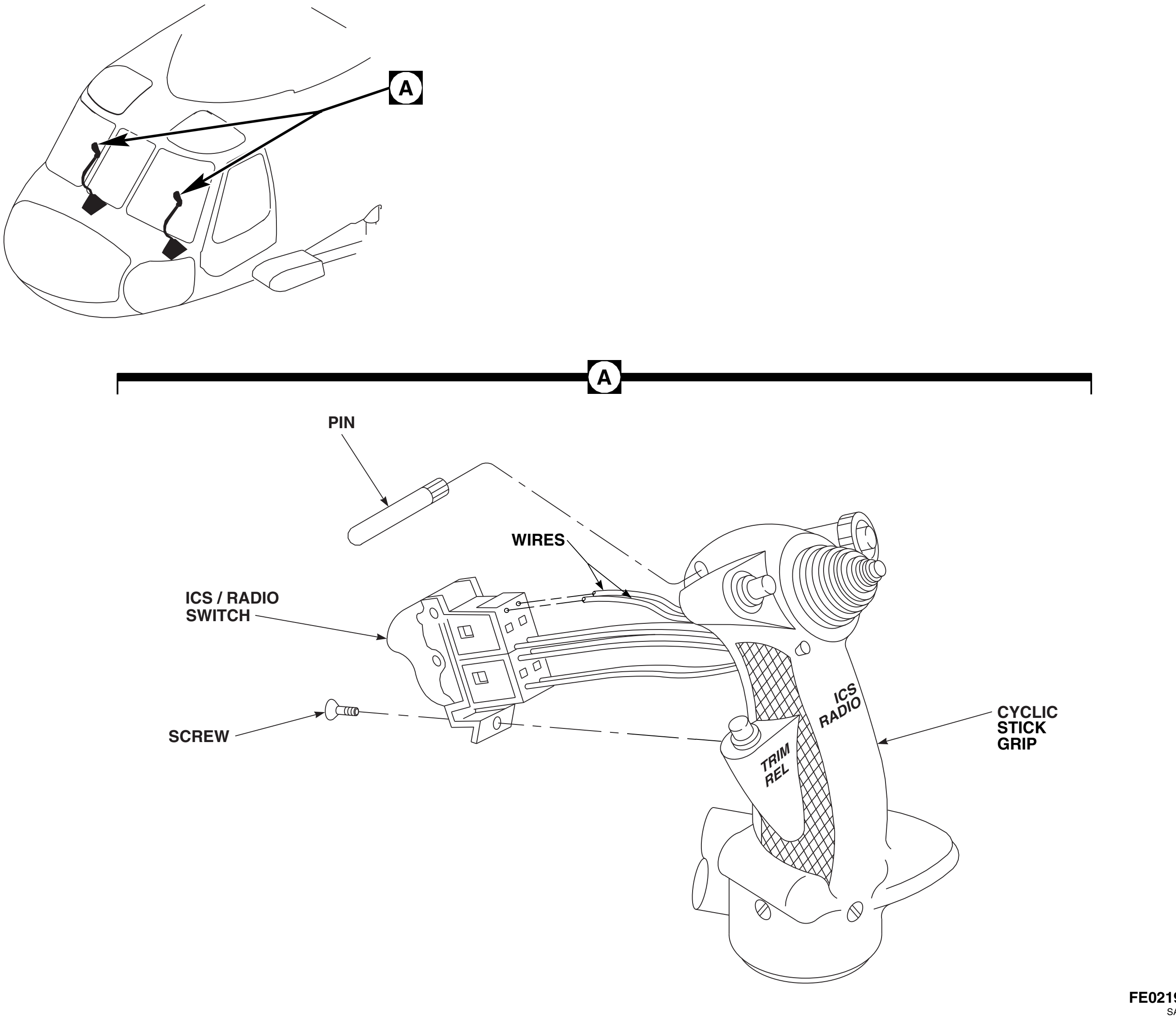


FIGURE 11-4-20-1. REPLACE CYCLIC STICK ICS/RADIO SWITCH

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

b. Using soldering iron and solder, Item 313, Appendix D, solder wires to switch (Figure 11-4-20-1, Detail A). Remove tags.

c. Coat all exposed solder terminals with sealing compound, Item 289, Appendix D.

d. Carefully press ICS/RADIO switch into cyclic stick grip, routing wires to prevent interference.

e. Install screw and pin securing ICS/RADIO switch to cyclic stick grip.

f. Fill screw and pin hole with sealing compound, Item 289, Appendix D.

g. Perform operational check of cyclic stick grip assembly (PARA 11-2-1).

11-4-21 CYCLIC STICK WIRING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Fibrous Cord, Item 139, Appendix D
- Sealing Compound, Item 289, Appendix D
- Solder, Item 313, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-1
- PARA 11-4-17
- PARA 11-4-18
- PARA 11-4-19
- PARA 11-4-20
- TM 11-70-23

11-4-21.1 Replace Cyclic Stick Wiring.

NOTE

Replacement of wiring is the same for both cyclic sticks.

11-4-21.1.1 Remove.

a. Remove cyclic stick grip (PARA 11-4-17).

b. Remove fibrous cord securing electrical cables.

NOTE

Do not unsolder wires when removing switch.

c. Remove affected switch (PARA 11-4-18, PARA 11-4-19, or PARA 11-4-20).

d. Once switch is removed, perform the following:

(1) Identify damaged/worn wire.

(2) Remove sealing compound from soldered terminals.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

(3) Using soldering iron, unsolder wire from switch.

e. Pull damaged/worn wire through harness.

11-4-21.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Pull replacement wire through harness.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Using soldering iron and solder, Item 313, Appendix D, solder replacement wire to switch.

d. Coat exposed solder terminal with sealing compound, Item 289, Appendix D.

e. Install switch (PARA 11-4-18, PARA 11-4-19, or PARA 11-4-20).

f. Secure electrical cables together with fibrous cord, Item 139, Appendix D.

g. Install cyclic stick grip (PARA 11-4-17).

h. Perform operational check on switch (PARA 11-2-1 or TM 11-70-23).

11-4-22 CYCLIC STICK SOCKET.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Reamer Set
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 11-4-16
- PARA 11-4-23

11-4-22.1 Replace Cyclic Stick Socket.

NOTE

Replacement of both cyclic stick sockets is the same.

11-4-22.1.1 Remove.

a. Remove cyclic stick (PARA 11-4-16).

b. Remove bolt, washers, and nut securing socket to stick (Figure 11-4-22-1). Remove socket.

11-4-22.1.2 Inspect.

a. Inspect socket for nicks, gouges, cracks, and corrosion. Nicks and small gouges up to 0.020 inch deep by 1/16 inch wide by 1/2 inch long should be blended out with abrasive cloth, Item 10, Appendix D. **NO CRACKS OR CORROSION ALLOWED.** If damage is more than allowed, replace socket.

b. Apply corrosion resistant coating, Item 118, Appendix D, to reworked area. Touch up paint (PARA 2-6-5).

c. Check bearing rotational drag (PARA 11-4-23).

d. Inspect bearings for ratcheting, binding, and radial play. Check bearing radial play with dial indicator. **MAXIMUM RADIAL PLAY IS 0.005 INCH.** Replace worn or damaged bearings (PARA 11-4-23).

11-4-22.1.3 Install.

a. Place socket on stick and line up boltholes (Figure 11-4-22-1).

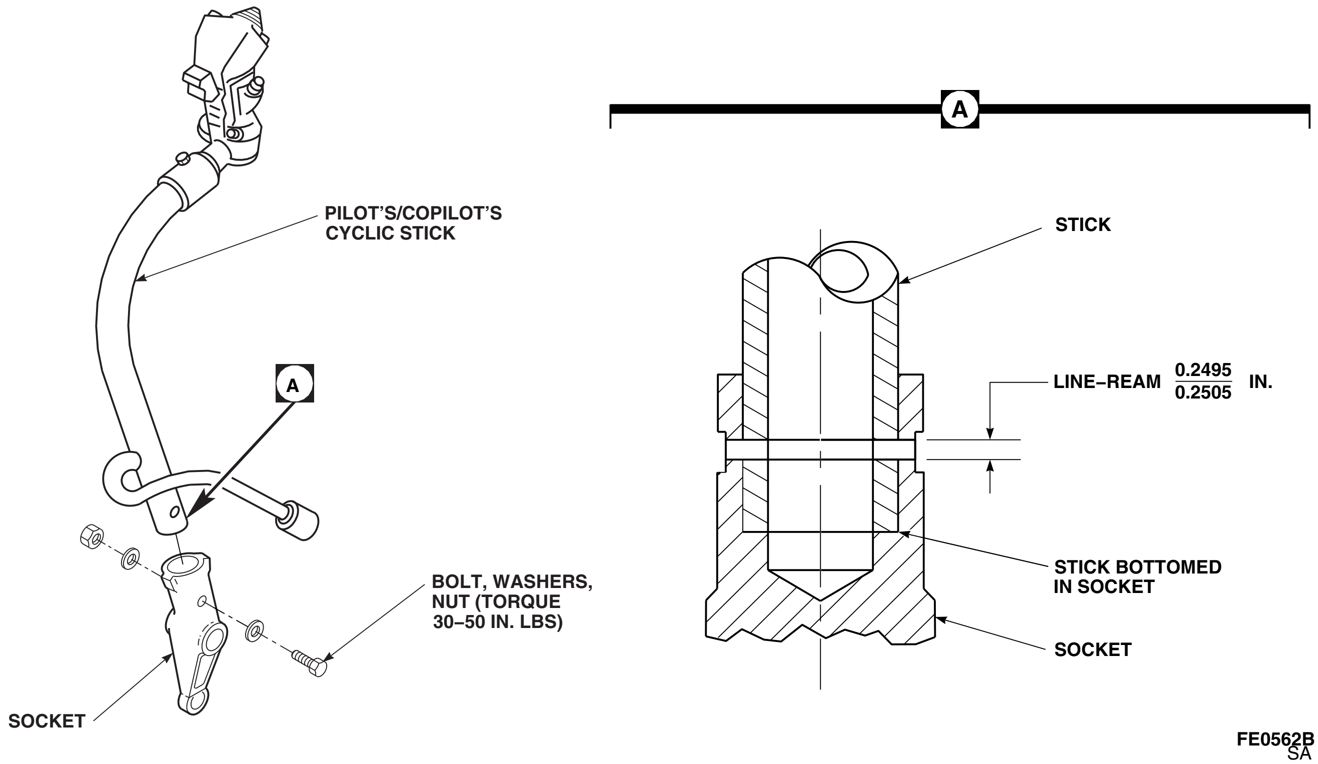


FIGURE 11-4-22-1. REPLACE CYCLIC STICK SOCKET

b. Put bolt through socket and stick. If bolt fits, do not ream hole. If bolt does not fit, perform the following:

(1) Using reamer set, line-ream holes to 0.2495 - 0.2505 inch diameter (Figure 11-4-24-1, Detail A).

(2) Remove socket from stick and clean out metal chips.

c. Using epoxy primer coating kit, Item 135, Appendix D, coat inside of socket hole with epoxy primer.

d. Install socket on stick using bolt, washers, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS** (Figure 11-4-22-1).

e. Install cyclic stick (PARA 11-4-16).

11-4-23 CYCLIC STICK BEARINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Bearing Remover, Step-Type
- Bolt, 1/4-inch diameter, 2-inch shank
- Nut, 1/4-inch diameter
- Torque Wrench, 0 - 30 in. oz

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Cleaning Cloth, Item 102, Appendix D
- Cotton Swab, Item 119, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Primer, Item 305, Appendix D

References

- Appendix D
- PARA 11-4-22

11-4-23.1 Replace Cyclic Stick Bearings.

NOTE

Replacement of bearings is the same for both cyclic sticks

11-4-23.1.1. Remove.

a. Remove cyclic stick socket (PARA 11-4-22).

b. Remove upper bearings and spacer from socket using arbor press and step-type bearing remover (Figure 11-4-23-1).

c. Remove lower bearing from socket using arbor press and step-type bearing remover.

11-4-23.1.2. Install.

NOTE

If new bearing is being installed, do not sand outside diameter.

a. If required, remove residual sealing compound by lightly sanding socket inside diameter(s) and outside diameter of bearing(s) with abrasive cloth, Item 8, Appendix D.

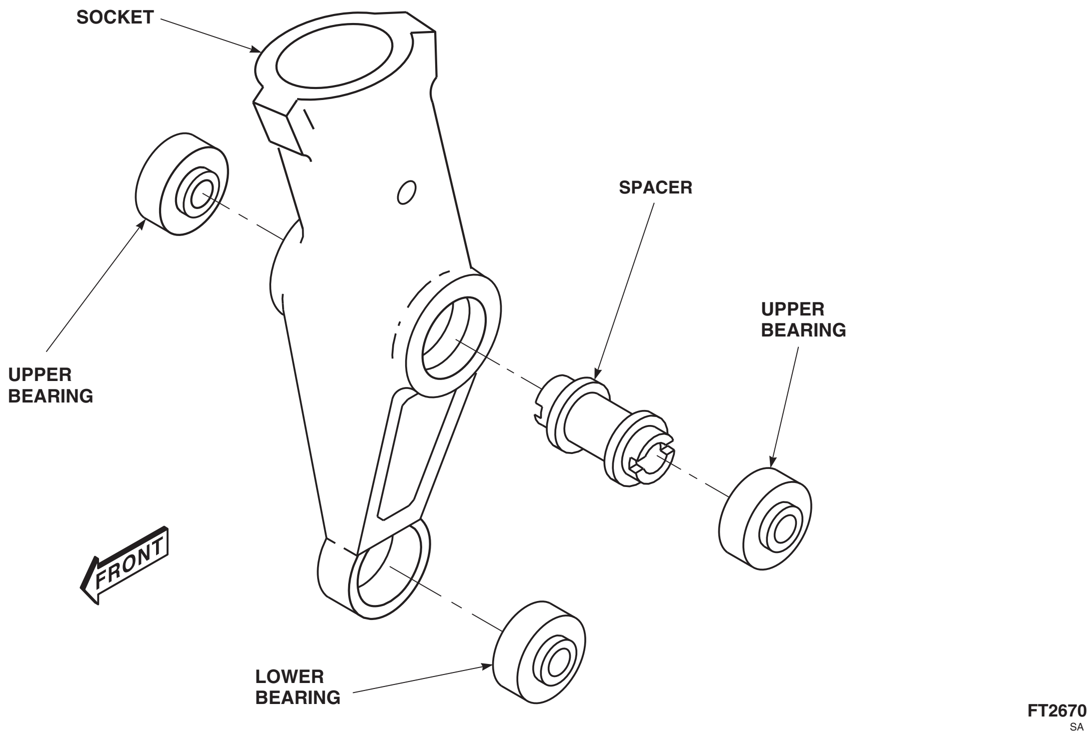


FIGURE 11-4-23-1. REPLACE CYCLIC STICK BEARINGS

CAUTION

Damage to bearing will result if sealing primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

NOTE

- Make sure all sanding residue is removed.
- After cleaning, do not handle bearing by outside diameter with bare hands.

b. Clean socket inside diameter(s) and outside diameter of bearing(s) using cotton swab, Item 119, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

CAUTION

Damage to bearing will result if sealing primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

c. Using clean cotton swab, Item 119, Appendix D, apply a light coat of sealing primer, Item

305, Appendix D, to inside diameter(s) of socket.

NOTE

- Protect primed surfaces from contamination during drying.
- Do not touch primed surfaces with bare hands.

d. Allow primer to dry 15 minutes minimum. Primer must be visibly dry. If any break in primer coat occurs, repeat step b. and step c.

e. Install upper bearings and spacer as follows (Figure 11-4-23-1):

CAUTION

Damage to bearing will result if primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

NOTE

Bearing must be installed within 4 minutes of application of sealing compound to socket inside diameters. If this time is exceeded, step b. through step d. must be repeated.

(1) Using clean cotton swab, Item 119, Appendix D, apply an even layer of sealing compound, Item 273, Appendix D to outside diameter of bearings and inside diameters of socket.

(2) Using arbor press, install bearing in one side of socket.

(3) Install spacer in socket and using arbor press, install bearing in socket.

(4) Allow socket assembly to cure for 6 hours minimum.

f. Install lower bearing as follows:

CAUTION

Damage to bearing will result if primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

NOTE

Bearing must be installed within 4 minutes of application of sealing compound to socket inside diameters. If this time is exceeded, step b. through step d. must be repeated.

(1) Using clean cotton swab, Item 119, Appendix D, apply an even layer of sealing compound, Item 273, Appendix D to outside diameter of bearing and inside diameter of socket.

(2) Using arbor press, install bearing in socket. Bearing must be installed flush to 0.010-inch below left surface of socket.

(3) Allow socket assembly to cure for 6 hours minimum.

CAUTION

Damage to bearing will result if sealing primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

g. Remove sealing compound squeezeout using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

h. Perform rotational drag check of bearings as follows:

(1) Install 1/4-inch diameter bolt through bearings and secure with nut.

(2) Using torque wrench and socket, turn bolthead and measure rotational drag of bearings.

(3) **IF ROTATIONAL DRAG IS GREATER THAN 2 INCH-OUNCES, REPLACE BEARINGS.**

(4) Remove bolt and nut from socket.

i. Install cyclic stick socket (PARA 11-4-22).

11-4-24 CYCLIC STICK TUBE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Reamer Set
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 11-4-16
- PARA 11-4-17
- PARA 11-4-21
- PARA 11-4-22

11-4-24.1 Replace Cyclic Stick Tube.

NOTE

Replacement of both cyclic stick tubes is the same.

11-4-24.1.1 Remove.

a. Remove cyclic stick socket (PARA 11-4-22).

b. Remove cyclic stick grip (PARA 11-4-17).

c. Remove wiring from cyclic stick tube (PARA 11-4-21).

11-4-24.1.2 Install.

a. Place socket on stick and line up boltholes (Figure 11-4-24-1).

b. Put bolt through socket and stick. If bolt fits, do not ream hole. If bolt does not fit, perform the following:

(1) Using reamer set, line-ream holes to 0.2495 - 0.2505-inch diameter (Figure 11-4-24-1, Detail A).

(2) Remove socket from stick and clean out metal chips.

c. Install wiring in cyclic stick tube (PARA 11-4-21).

d. Install cyclic stick grip (PARA 11-4-17).

e. Using epoxy primer coating kit, Item 135, Appendix D, coat inside of socket hole with epoxy primer.

f. Install socket on stick using bolt, washers, and nut. **TORQUE NUT 30 - 50 INCH-POUNDS** (Figure 11-4-24-1).

g. Install cyclic stick (PARA 11-4-16).

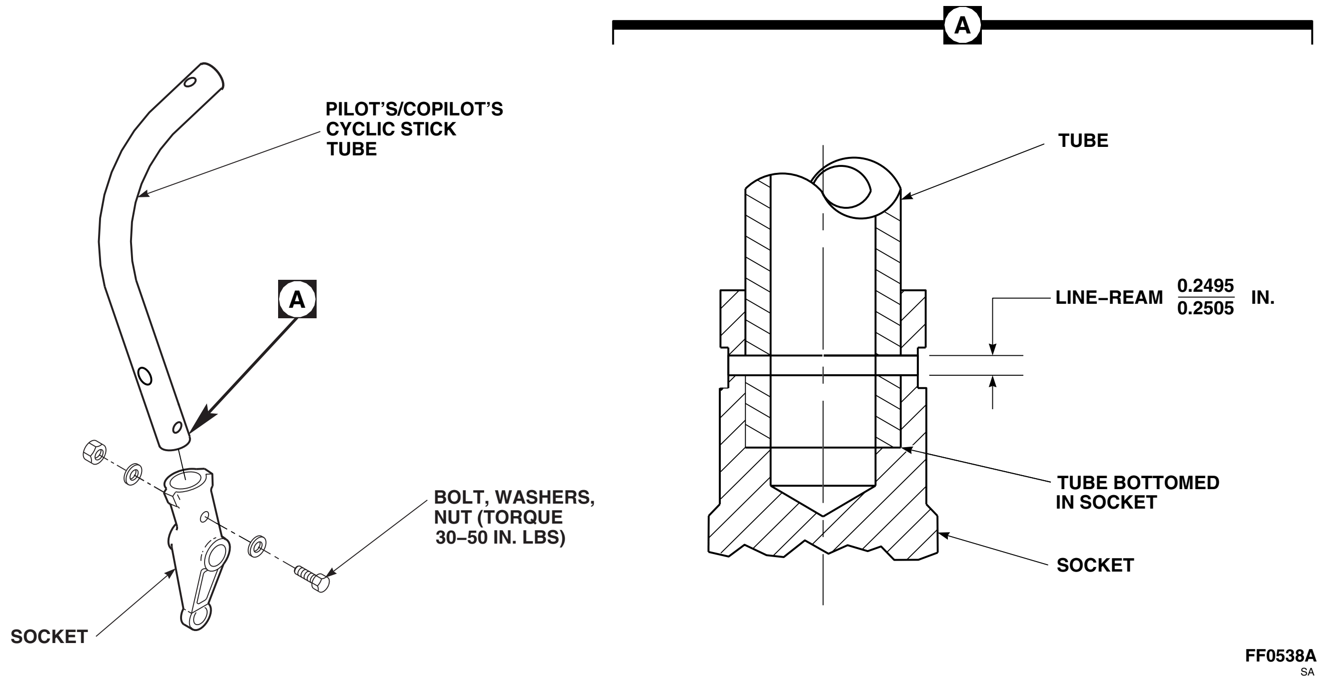


FIGURE 11-4-24-1. REPLACE CYCLIC STICK TUBE

11-4-25 CYCLIC STICK YOKE AND HOUSING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 194, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-16

11-4-25.1 Replace Cyclic Stick Yoke and Housing.

NOTE

Replacement of both cyclic stick yokes and housings is the same.

11-4-25.1.1 Remove.

a. Remove cyclic stick (PARA 11-4-16).

b. On left side, remove bolts, washers, and nuts securing fore-and-aft control rod to lever (Figure 11-4-25-1, Sheet 1). Remove control rod.

c. Remove self-retaining bolts, washers, and nuts securing lever to pitch pushrod.

d. Remove self-retaining bolts, washers, and nuts securing lever to yoke and housing.

e. On right side, remove self-retaining bolts, washers, and nuts securing bellcrank to link, pitch pushrod, and yoke housing (Figure 11-4-25-1, Sheet 1, Detail A).

f. Remove self-retaining bolts, washers, and nuts securing lateral control rod to crank (Figure 11-4-25-1, Sheet 1).

g. Remove bolts, washers, and nuts securing yoke and housing to airframe (Figure 11-4-25-1, Sheet 1, Detail B). Remove yoke and housing.

11-4-25.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

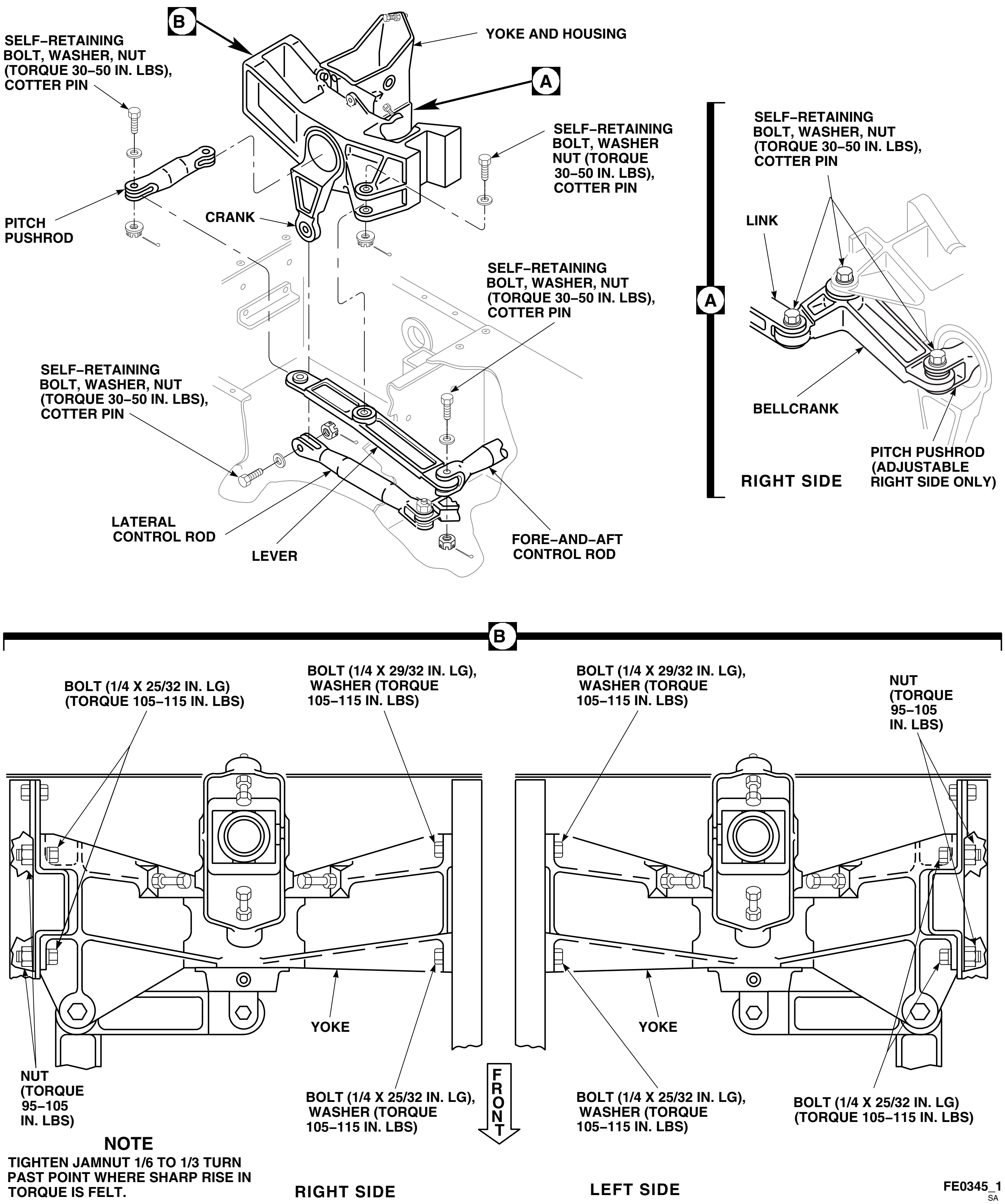


FIGURE 11-4-25-1. REPLACE CYCLIC STICK YOKE AND HOUSING (SHEET 1 OF 2)

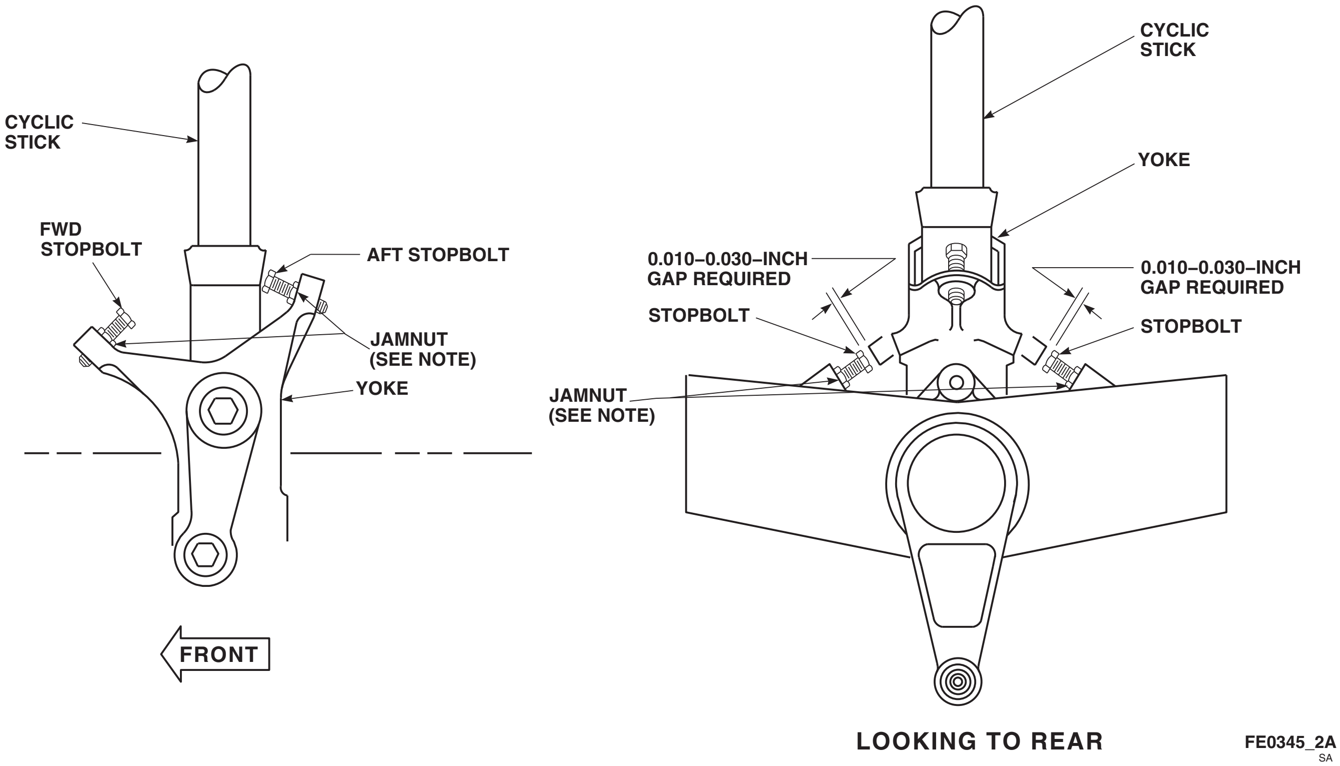


FIGURE 11-4-25-1. REPLACE CYCLIC STICK YOKE AND HOUSING (SHEET 2 OF 2)

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Place yoke and housing between walls of airframe. Install bolts and washers and bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS AND TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-25-1, Sheet 1, Detail B).

NOTE

Cyclic stick grip is allowed to touch glare shield padding or instrument knobs with forward stop on cyclic stick properly set.

c. Adjust cyclic stick front stopbolt as follows:

(1) Push cyclic stick to front.

(2) **ADJUST FRONT STOPBOLT TO ATTAIN STICK GRIP DIMENSION OF 0.730 - 0.790 INCH FROM INSTRUMENT PANEL. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(3) Lockwire, Item 194, Appendix D, stopbolt.

d. Install lateral control rod to crank (Figure 11-4-25-1, Sheet 1). Secure control rod as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. On left side, install lever in yoke and housing support. Secure lever as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Install pitch pushrod in yoke.

g. Connect lever to pitch pushrod. Secure pushrod as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

h. Install fore-and-aft control rod on lever. Secure rod as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

i. On right side, install bellcrank in support (Figure 11-4-25-1, Sheet 1, Detail A). Secure bellcrank as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. Install adjustable pitch pushrod in yoke with adjustable end inside. Connect bellcrank to pushrod as follows (Figure 11-4-25-1, Sheet 1):

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

k. Connect link to bellcrank as follows (Figure 11-4-25-1, Sheet 1, Detail A):

NOTE

Bolt may be installed with bolthead either up or down.

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

l. Install cyclic stick (PARA 11-4-16). Do not install boot, boot retainer, and floor panels at this time.

m. Adjust cyclic stick stopbolts (PARA 11-4-25.1.3).

11-4-25.1.3 Adjust.

a. Turn on all electrical power (PARA 1-6-16).

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
100° - 125° (39° - 52°)	16	48

b. Turn on backup hydraulic system (PARA 1-6-15).

NOTE

Pilot stops are adjusted first; then copilots are adjusted. Copilot's cyclic stick aft stop should be adjusted so it touches stick at same time as pilot's.

c. Adjust aft stopbolt as follows:

(1) Loosen jamnut and turn aft stopbolt into yoke as far as possible (Figure 11-4-25-1, Sheet 2).

(2) Raise collective stick to full up position and hold.

(3) Pull cyclic stick full aft until binding is felt. Adjust aft stopbolt out until it contacts cyclic stick; then turn out approximately 1/2 turn more until binding is removed. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(4) Lockwire, Item 194, Appendix D, stopbolt.

d. Adjust lateral (roll) stops as follows:

(1) Push collective stick to full low position and hold.

(2) Push cyclic stick full right and adjust right stopbolt to have 0.010 - 0.030 inch gap between stopbolt and cyclic stick. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 194, Appendix D, stopbolt.

(3) With collective stick in full low position, push cyclic stick full left and adjust left stopbolt to have 0.010 - 0.030 inch gap between stopbolt and cyclic stick. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 194, Appendix D, stopbolt.

e. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

f. Make sure area is clean and free of foreign material.

g. Install boot, boot retainer, and floor panels using screws and washers.

11-4-26 COLLECTIVE STICK BOOT AND COVER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 61, Appendix D
- Fastener Tape, Hook, Item 137A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-16

11-4-26.1 Replace Collective Stick Boot and Cover.

NOTE

Replacement of both collective stick boots and covers is the same.

11-4-26.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. To remove pilot's boot and cover, perform the following:

(1) Remove top panel from around collective stick, and lower panel halves (Figure 11-4-26-1, Detail B).

(2) Remove screws and washers and remove cover from side of center console (Figure 11-4-26-1, Detail A).

(3) Remove top and bottom panel halves (Figure 11-4-26-1, Detail B).

c. To remove copilot's boot and cover, perform the following:

(1) Remove top panel from collective stick and center panel (Figure 11-4-26-1, Detail D).

(2) Remove center panel from bottom panel.

(3) Remove screws and washers and remove cover from floor and rear bulkhead (Figure 11-4-26-1, Detail C).

(4) Remove screws and washers and remove retainer and bottom panel from cover (Figure 11-4-26-1, Detail D).

(5) Remove side panel from top of cover assembly and bulkhead by pulling Velcro straps.

(6) Pull Velcro strap on seam of side cover and lift side cover forward away from bulkhead and electrical cable.

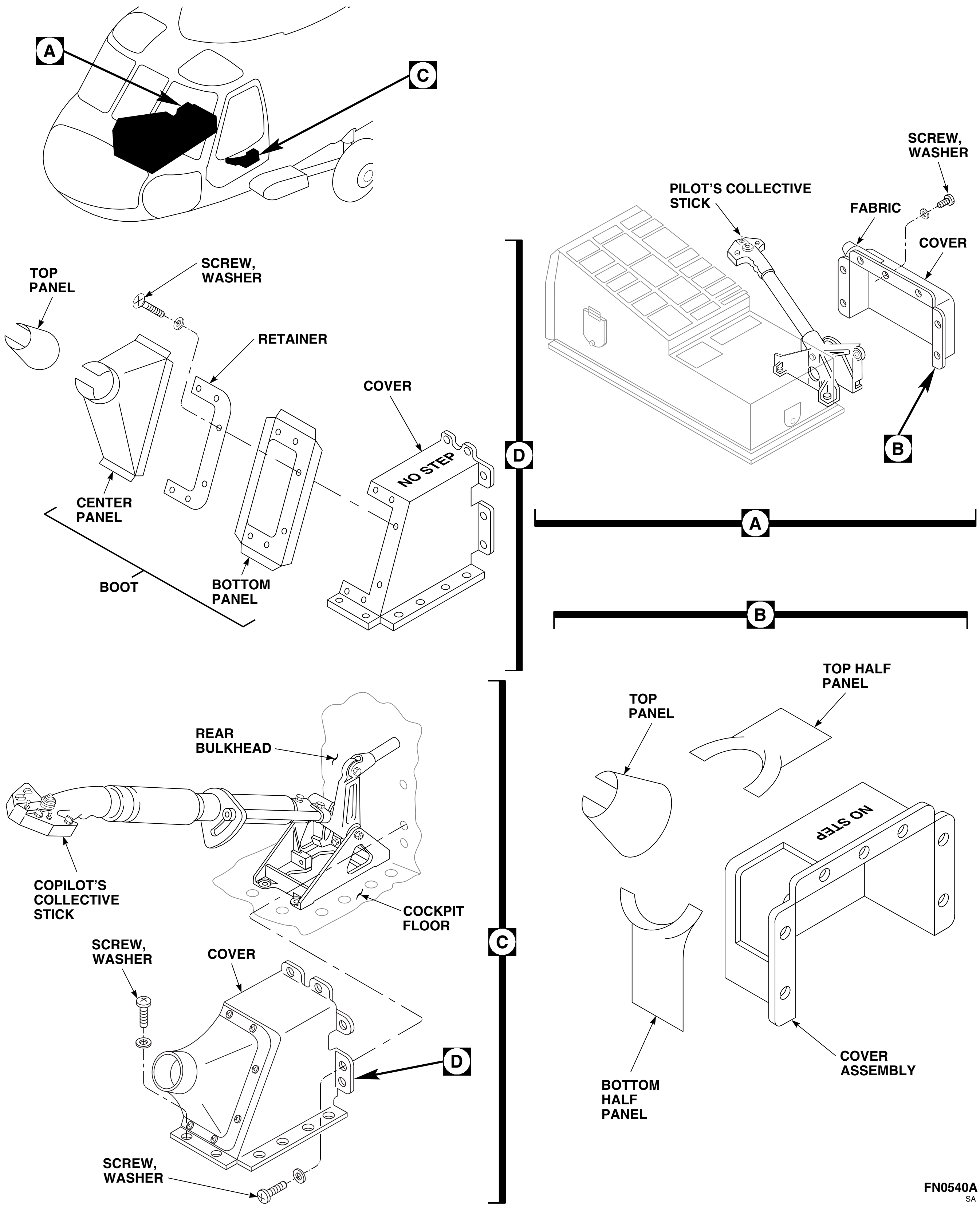


FIGURE 11-4-26-1. REPLACE COLLECTIVE STICK BOOT AND COVER

11-4-26.1.2 Repair.

a. Using nonmetallic scraper, remove damaged hook fastener tape from structure.

b. Using clean machinery towel, Item 211, Appendix D, dampened in methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, remove remaining adhesive from structure.

c. Within 15 minutes after cleaning, apply coat of adhesive, Item 61, Appendix D, to hook fastener tape, Item 137A, Appendix D, and structure. Allow adhesive to dry for 10 minutes or until tacky.

d. Press hook fastener tape, Item 137A, Appendix D, onto adhesive on structure. Make sure of good bond between hook fastener tape and structure. Allow adhesive to dry at room temperature for 12 - 16 hours.

11-4-26.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install pilot's boot and cover as follows:

(1) Make sure area is clean and free of foreign material.

(2) Install replacement top and bottom panel halves on cover. Secure cover to center console with screws and washers. Join panels (Figure 11-4-26-1, Detail A and Detail B).

(3) Install top panel around collective stick and panels (Figure 11-4-26-1, Detail B).

c. Install copilot's boot and cover as follows:

(1) Make sure area is clean and free of foreign material.

(2) Put replacement bottom panel on cover and install retainer on cover with screws and washers (Figure 11-4-26-1, Detail D).

(3) Secure cover to floor and rear bulkhead with screws and washers (Figure 11-4-26-1, Detail C).

(4) Install center panel on bottom panel and close at seam (Figure 11-4-26-1, Detail D).

(5) Install top panel around collective stick and around center panel.

(6) Place hole in side cover over electrical cable and fasten Velcro strap.

(7) Fasten top of side cover to cover assembly by fastening Velcro strap.

(8) Fasten end of side cover to airframe by fastening Velcro strap.

11-4-27 PILOT'S COLLECTIVE STICK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 11-2-2
- PARA 11-4-1
- PARA 11-4-26
- PARA 11-4-29
- PARA 11-4-31
- PARA 11-4-40

11-4-27.1 Replace Pilot's Collective Stick.

11-4-27.1.1 Remove.

a. Remove collective stick boot and cover (PARA 11-4-26).

b. Disconnect electrical connector, P104, from base of pilot's collective stick (Figure 11-4-27-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

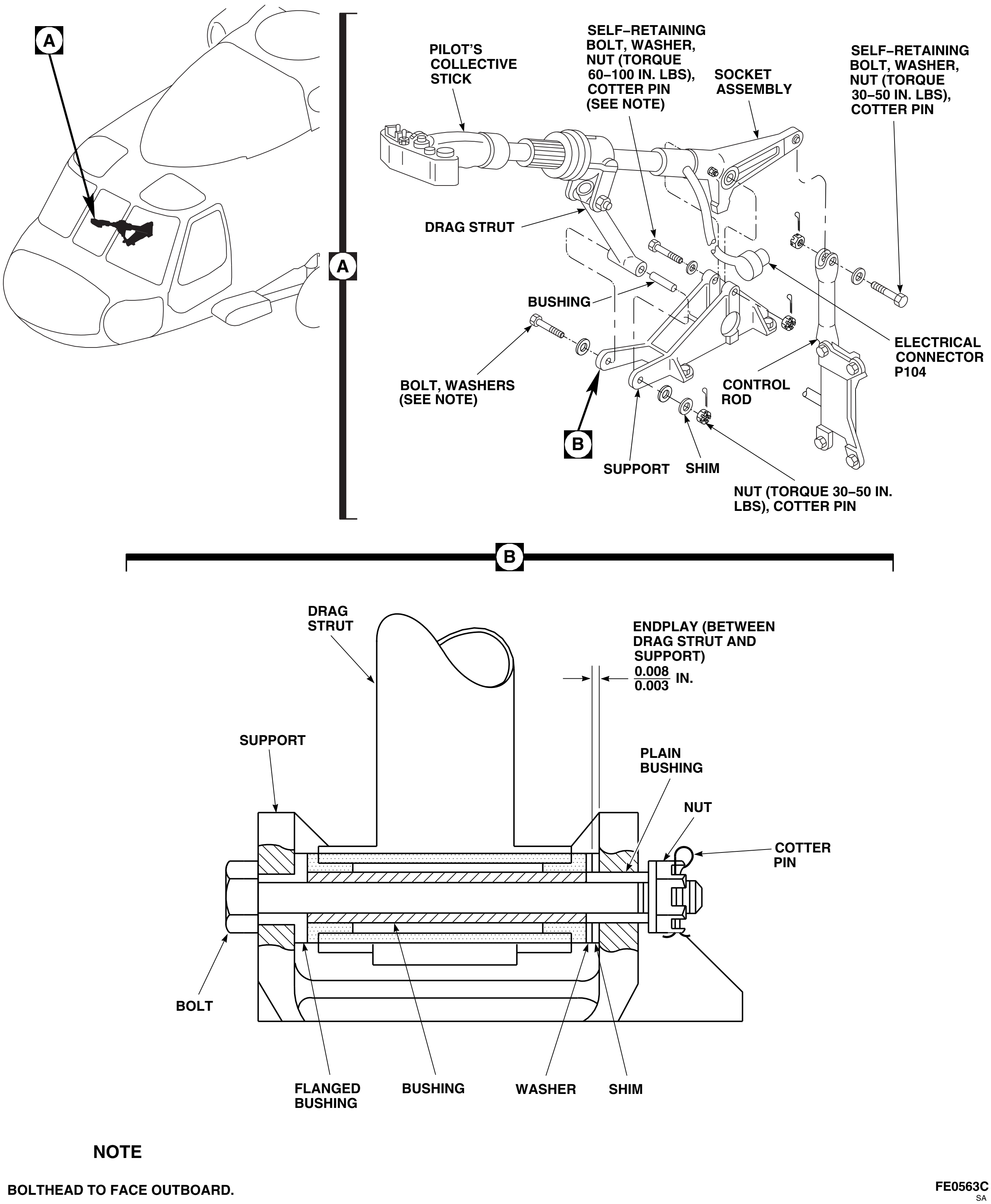


FIGURE 11-4-27-1. REPLACE PILOT'S COLLECTIVE STICK

d. Remove self-retaining bolt, washers, and nut securing control rod to socket assembly. Move control rod out of way.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in the seat. Extension springs are under load at all times. With seat at lowest position, vertical preload could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

e. Tilt pilot's seat back.

f. Remove bolt, bushing, washer, shim, and nut securing drag strut to support. Remove strut from support (Figure 11-4-27-1, Detail A and Detail B).

g. Remove self-retaining bolt, washers, and nut securing socket assembly to support.

h. Slide collective stick and socket assembly out of support.

11-4-27.1.2 Inspect/Repair.

a. Remove collective stick grip assembly (PARA 11-4-31).

b. Remove collective stick socket (PARA 11-4-40).

NOTE

No repairs are allowed to tube assembly.

c. Check outside diameter of tube assembly for cleanness, nicks, scratches, corrosion or other damage.

NOTE

No repairs are allowed to tube assembly.

d. Check that tubes slide easily together or apart.

e. Replace tube if necessary.

f. Install collective stick grip assembly (PARA 11-4-31).

g. Install collective stick socket on lower tube (PARA 11-4-40).

11-4-27.1.3 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all electrical power (PARA 1-6-16).

b. Position socket assembly in support (Figure 11-4-27-1, Detail A). Secure socket assembly to support as follows:

WARNING

To prevent damage to collective stick cover, make sure bolt is installed with bolthead facing outboard.

(1) Install self-retaining bolt (with bolthead facing outboard) and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

c. Install bushing in drag strut on pilot's collective stick.

d. Install shim, then washer, over plain bushing in support.

e. Secure drag strut to pilot's collective stick support with bolt (with bolthead facing outboard), washer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS. CHECK FOR 0.003 - 0.008-INCH ENDPLAY** (Figure 11-4-27-1, Detail B). Install cotter pin.

f. Position control rod on socket assembly (Figure 11-4-27-1, Detail A). Secure rod to socket assembly as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in the seat. Extension springs are under load at all times. With seat at lowest position, vertical preload could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat or with hands on guide tubes above linear bearings, will be seriously injured.

g. Move pilot's seat upright.

h. Remove protective caps and plugs from electrical connectors.

i. Inspect and treat electrical connectors (PARA 1-7-20).

j. Connect electrical connector, P104, to base of collective stick.

k. Check friction grip for proper adjustment (PARA 11-4-29).

l. Install collective stick boot and cover (PARA 11-4-26).

m. Perform operational check of all collective stick assemblies (PARA 11-2-2).

11-4-28 COPILOT'S COLLECTIVE STICK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Electrical Repairer's Toolkit
- Puller
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Dry Ice, Item 125, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 11-2-2
- PARA 11-4-1
- PARA 11-4-26
- PARA 11-4-31
- PARA 11-4-40

11-4-28.1 Replace Copilot's Collective Stick.

11-4-28.1.1 Remove.

a. Remove collective stick cover and boot (PARA 11-4-26).

b. Disconnect electrical connector, P103, at base of copilot's collective stick (Figure 11-4-28-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Remove clamp and spacer from airframe.

e. Remove clamp from electrical cable.

f. Remove self-retaining bolt, washer, and nut securing collective control rod to collective stick socket.

g. Remove bolt, washer, and nut securing collective stick to support.

h. Slide collective stick out of support.

11-4-28.1.2 Inspect/Repair.

a. Remove collective stick grip assembly (PARA 11-4-31).

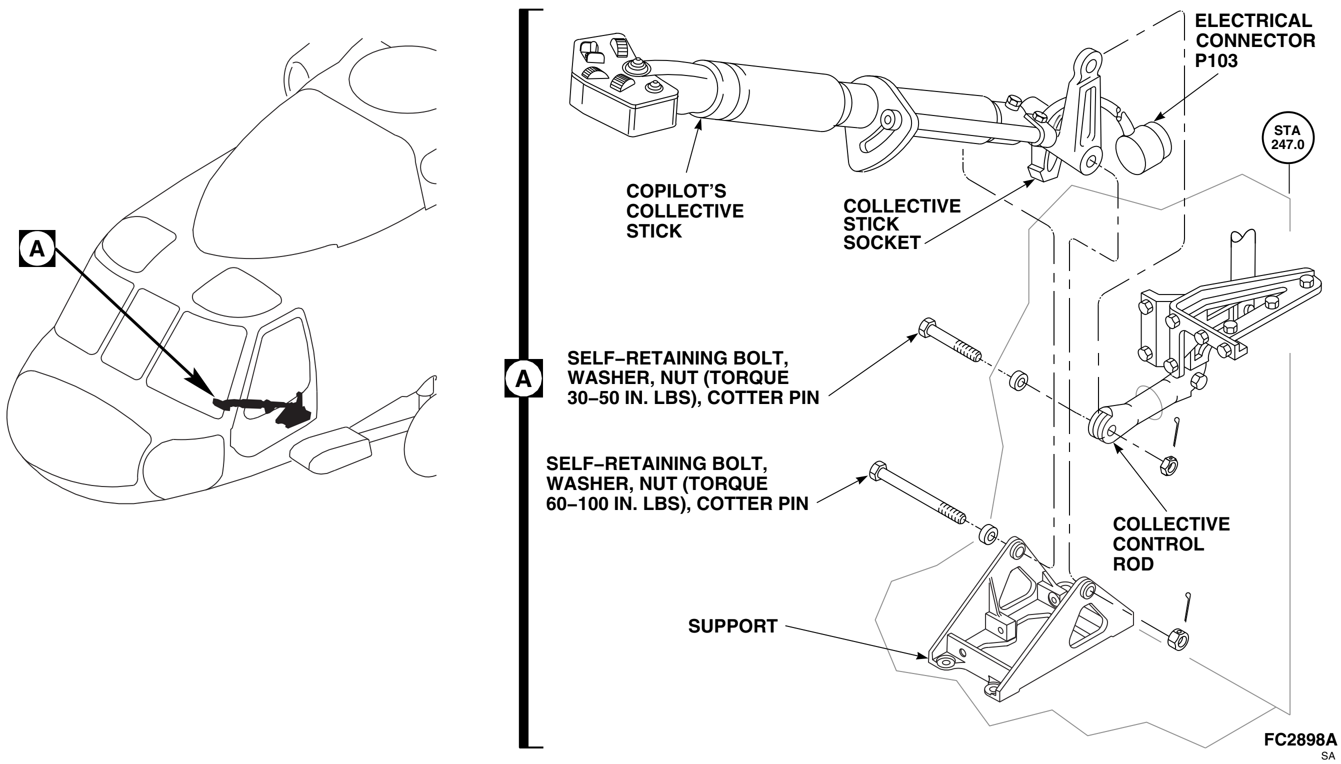


FIGURE 11-4-28-1. REPLACE COPILOT'S COLLECTIVE STICK

b. Remove collective stick socket (PARA 11-4-40).

NOTE

No repairs are allowed to tube assembly.

c. Check outside diameter of tube assembly for cleanness, nicks, scratches, corrosion, or other damage.

NOTE

No repairs are allowed to tube assembly.

d. Check that tubes slide easily together or apart.

e. If necessary, replace tube assembly (PARA 11-4-28.2).

f. Install collective stick grip assembly (PARA 11-4-31).

g. Install collective stick socket (PARA 11-4-40).

11-4-28.1.3 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Position collective stick in support (Figure 11-4-28-1, Detail A). Secure collective stick socket as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

c. Place collective control rod on collective stick. Secure control rod as follows:

(1) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connector (PARA 1-7-20).

f. Connect electrical connector, P103, at base of stick.

g. Place clamp on electrical cable.

h. Install clamp on airframe using screw, spacer, washer, and nut.

i. Install collective stick cover and boot (PARA 11-4-26).

j. Perform operational check of collective stick assembly (PARA 11-2-2).

11-4-28.2 Replace Copilot's Collective Stick Tube Assembly.

a. Position upper tube to unlock.

b. Remove bolt, washers, and nut securing tube to lower tube (Figure 11-4-28-2, Sheet 1, Detail A).

c. Using arbor press and puller, press out and remove tube from upper tube.

d. Remove upper tube from lower tube.

e. Inspect tube assembly (Table 11-4-28-1).

f. Place lower tube assembly onto upper tube assembly in collapsed position.

g. Pack tube in dry ice, Item 125, Appendix D.

h. Remove tube from dry ice. Install tube through locking slot in upper tube assembly and press into lower tube assembly, using arbor press.

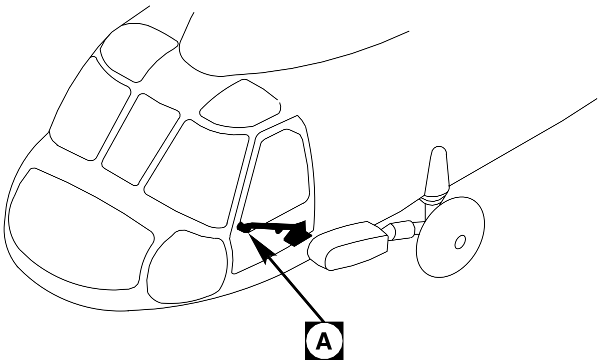
i. Secure lower tube assembly to tube with bolt, washers, and nut. **TORQUE NUT TO 45 INCH-POUNDS.**

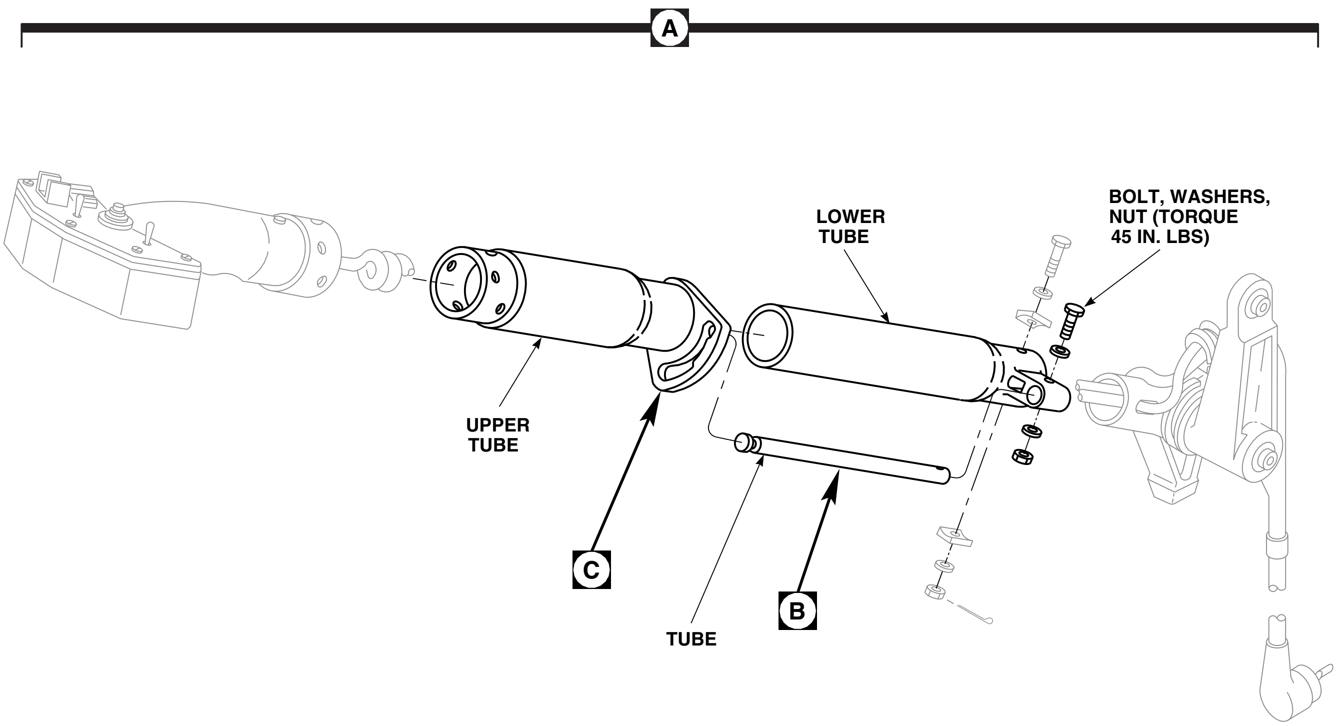
Table 11-4-28-1. INSPECTION CRITERIA

Nomenclature and Part No.	Inspection	New Parts Min./Max. Limits	Max. Serviceable Limits	Max. Reparable Limits	Corrective Action
Tube	Corrosion	-/-	None allowed	Not reparable	Replace tube
	Damage or Wear	-/-	Dimension A is 0.584 in. minimum Dimension B is 0.325 in. minimum Dimension C is 0.100 in. minimum (Figure 11-4-28-2, Sheet 2, Detail B)	Not reparable	Replace tube
	Straightness	-/-	Not more than 0.010 in.	Not reparable	Replace tube
	Nicks and scratches	-/-	None allowed	Not reparable	Replace tube
	Missing or damaged protective coating	-/-	-	-	Apply chemical conversion coating (PARA 2-6-5)
Upper tube	Damage or Wear	-/-	Cam surface A, B, C, or D is 0.040 in. minimum (Figure 11-4-28-2, Sheet 2, Detail C)	Not reparable	Replace tube
	Nicks and scratches	-/-	-	Not more than 0.010 in. deep or not more than 1% of total surface area	Blend out nicks and scratches using abrasive cloth, Item 8, Appendix D

Table 11-4-28-1. INSPECTION CRITERIA (Cont)

Nomenclature and Part No.	Inspection	New Parts Min./Max. Limits	Max. Serviceable Limits	Max. Reparable Limits	Corrective Action
	Dents	-/-	Not deeper than 0.005 in. or less than 0.060 radius. No dents allowed in area of bonded liner	Not reparable	Replace upper tube
	Corrosion	-/-	-	Not more than 0.005 in. deep or not more than 2% of surface area	Repair damaged area with cleaning compound (PARA 2-6-5)
Lower tube	Nicks and scratches	-/-	-	Not more than 0.005 in. deep	Blend out nicks and scratches using abrasive cloth, Item 8, Appendix D
	Dents	-/-	None allowed	Not reparable	Replace lower tube
	Corrosion	-/-	-	Not more than 0.005 in. deep or not more than 2% of surface area	Repair damaged area with cleaning compound (PARA 2-6-5)





NOTE

MAXIMUM DEPTH OF DAMAGE OR WEAR FOR SURFACES A, B, C, AND D IS 0.040 INCH.

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FIGURE 11-4-28-2. REPLACE COPILOT'S COLLECTIVE STICK TUBE ASSEMBLY (SHEET 1 OF 2)

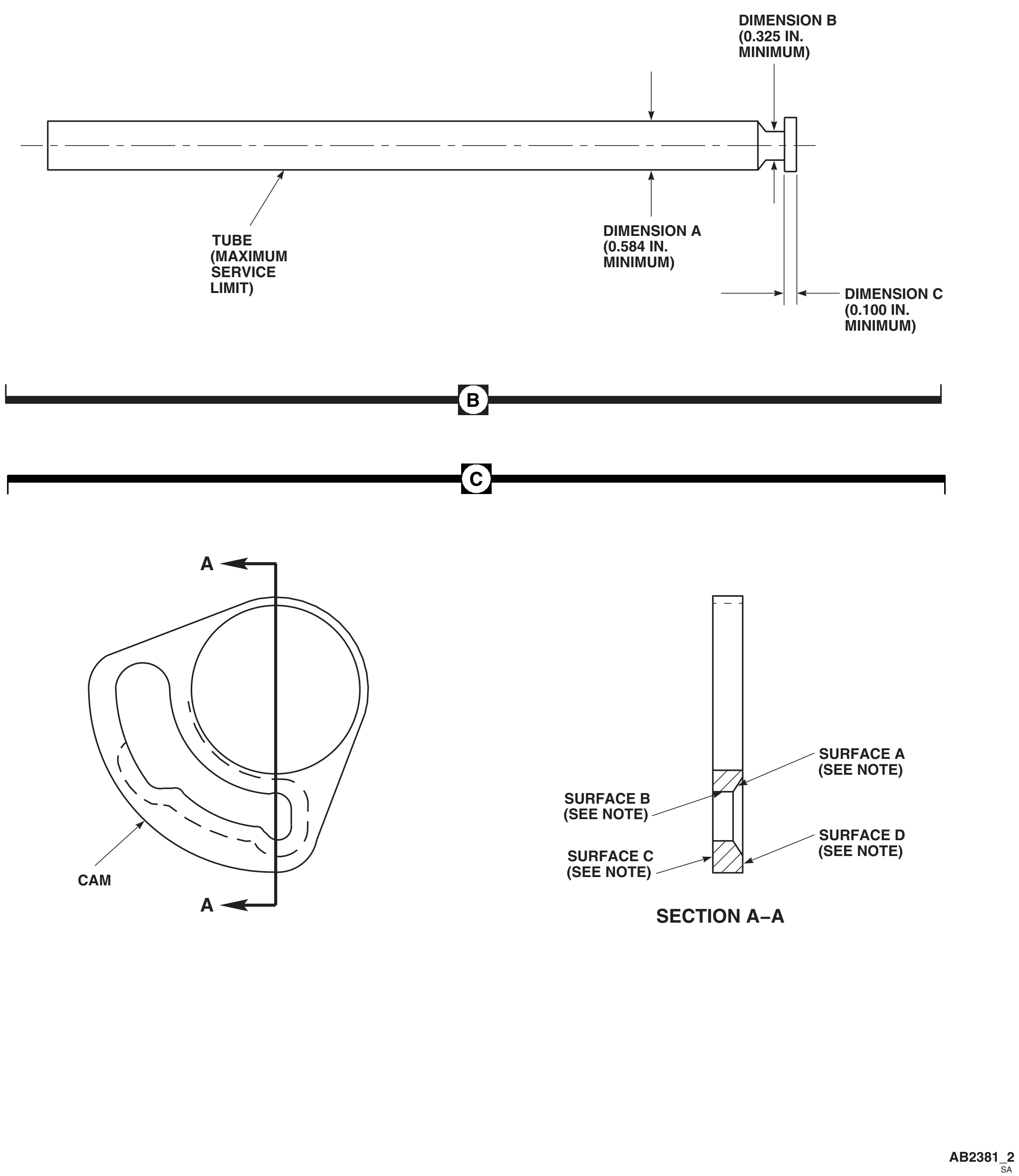


FIGURE 11-4-28-2. REPLACE COPILOT'S COLLECTIVE STICK TUBE ASSEMBLY
(SHEET 2 OF 2)

11-4-29 PILOT'S COLLECTIVE STICK COLLET BLOCKS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Drift
- Pencil
- Spring Scale, 0 - 50 pounds
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Damping Fluid, Item 121, Appendix D
- Lockwire, Item 194, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-16

11-4-29.1 Replace Pilot's Collective Stick Collet Blocks.

11-4-29.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove bolt, washer, and adjustable stop from friction grip (Figure 11-4-29-1, Detail A).

c. Unscrew friction grip from sleeve.

d. Remove special bolts, washers, and spacer(s) from stop and guard, sleeve, and collet block halves.

e. Using thin nonmetallic drift, carefully tap collet blocks and seal from sleeve.

11-4-29.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

WARNING

Injury to personnel and damage to equipment will result if phenolic spacer strips are not removed from collet block halves prior to installation. Make sure spacer strips are removed and discarded prior to installing collet block halves.

b. Make sure collet block halves are clean and free of any dirt, nicks, scratches, corrosion, or other damage that could stop them from sliding smoothly on stick.

c. Check collective stick for dirt, nicks, scratches, corrosion, or other damage that could stop friction lock and sleeve from sliding smoothly on stick (Figure 11-4-29-1, Detail A).

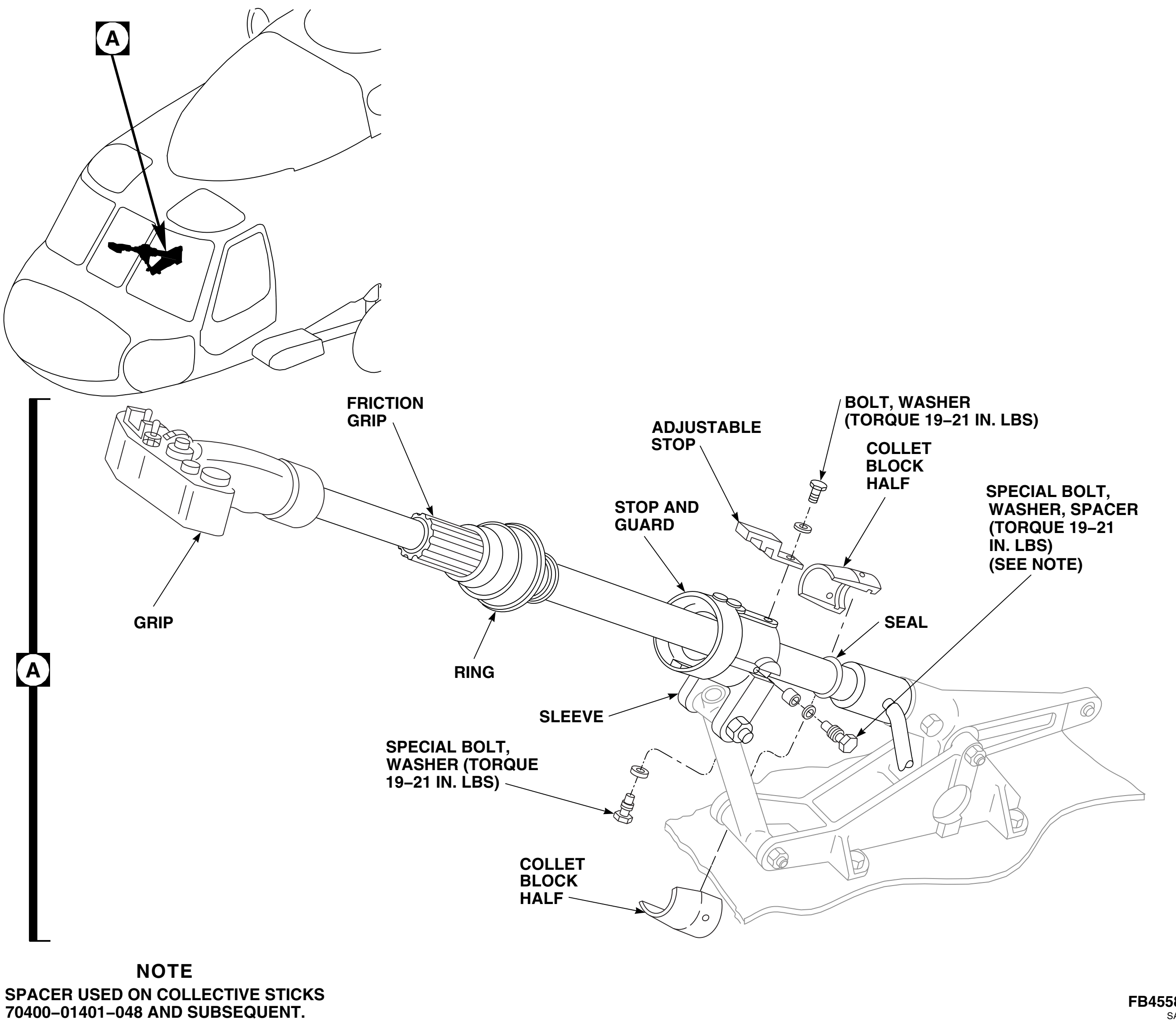


FIGURE 11-4-29-1. REPLACE PILOT'S COLLECTIVE STICK COLLET BLOCKS

d. Make sure stick, friction grip, and sleeve are clean and dry.

e. Place replacement collet block halves on collective stick, and on seal, being careful not to damage seal as it is seated in groove inside collet.

CAUTION

Damage to collective stick assembly will result if damping fluid is allowed to penetrate into inside diameter of collet block halves. Make sure no fluid gets between collective stick and inside diameter of collet block halves.

f. Apply a light coat of damping fluid, Item 121, Appendix D, to outside diameter of collet block halves at tapered ends of collets.

g. Slide collet block halves into sleeve (Figure 11-4-29-2, Detail A and Figure 11-4-29-3, Detail A).

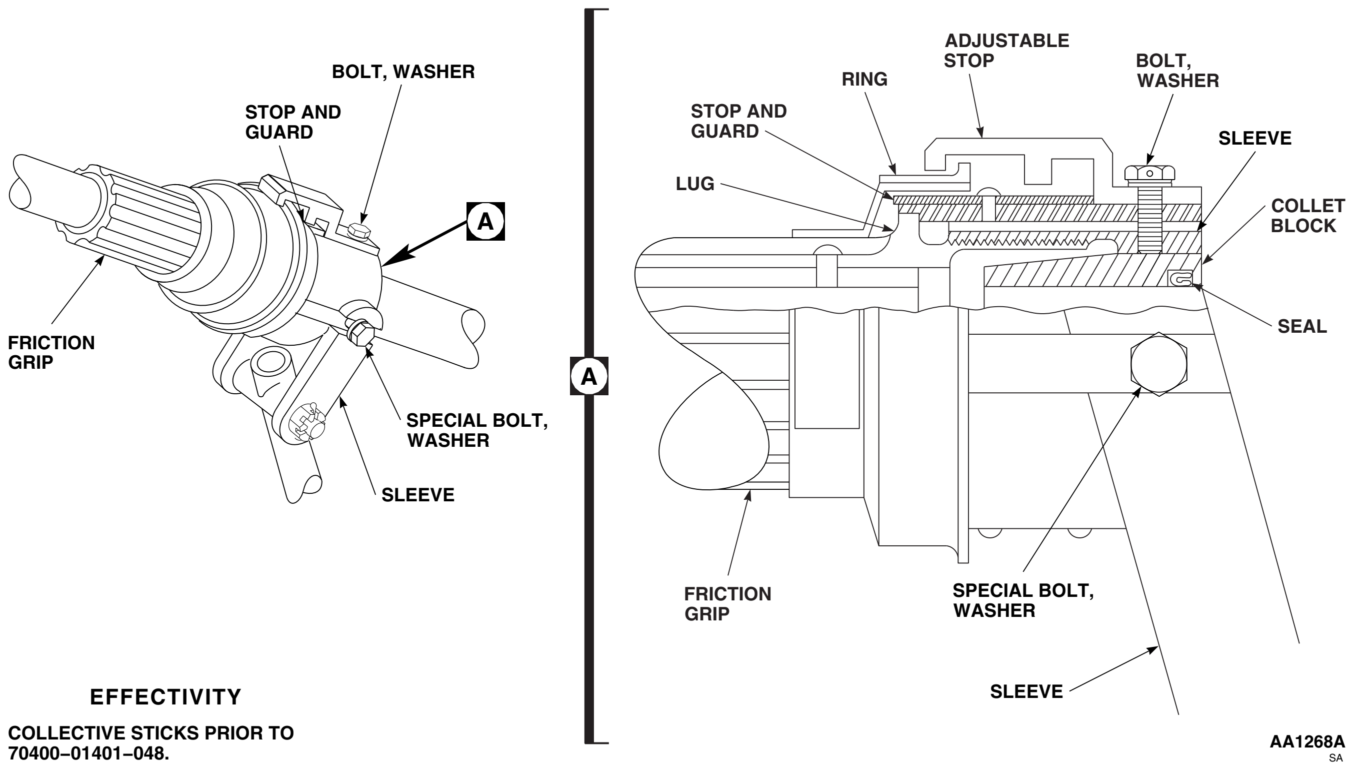


FIGURE 11-4-29-2. ADJUST FRICTION LOCK

WARNING

Injury to personnel and damage to equipment will result if hardware securing collet block halves and adjustable stop is intermixed. Friction lock assembly may jam, resulting in stiff or sticking collective stick movement. Make sure correct hardware is used when installing collet block halves and adjustable stop.

h. Line up boltholes in collet block halves, stop and guard, and sleeve, and install special bolts, washers, and spacer(s). Make sure collet block halves are secure (Figure 11-4-29-1, Detail A, Figure 11-4-29-2, Detail A, and Figure 11-4-29-3, Detail A).

i. Thread friction grip into sleeve.

j. Install adjustable stop on sleeve with bolt and washer.

k. Turn on all electrical power (PARA 1-6-16).

WARNING

Injury to personnel and damage to equipment will result if friction lock is adjusted incorrectly, resulting in a collective stick which cannot be moved with friction lock full on. Make sure friction lock is adjusted so that collective stick retains full range of motion when friction lock is full on.

l. On collective sticks prior to 70400-01401-048, adjust friction locks as follows (Figure 11-4-29-2):

(1) On upper console, place BACKUP HYD PUMP switch ON.

NOTE

- As friction grip threads into sleeve it squeezes collet on collective stick.
- Amount of squeeze is set when lug on friction grip contacts stop and guard.
- If stop and guard is moved toward grip, lug will contact stop in less turns, squeeze is less.
- If stop and guard is moved away from grip, lug will contact stop in more turns, squeeze is more.
- Adjustable stop (or bonded stop) is set, so when ring touches stop, friction on stick is fully released. It prevents grip from coming out of sleeve completely and allows pilot to apply friction in only a few turns.
- If friction does not fully release when ring touches stop, check to make sure correct hardware was used to install collet block halves and adjustable stop.
- High or fading friction forces may be eliminated by lightly scuffing ID of collet block halves using abrasive cloth, Item 8, Appendix D, or equivalent.

(2) With lock full on, use spring scale attached to collective grip, at stick grip just behind lighted panel and move pilot's collective stick up and down.

(3) **STICK MUST MOVE WITH 20 - 40 POUNDS PULL.**

(4) If pull is above or below limit, back off friction grip, loosen bolt and special bolts, slide stop and guard in slots in sleeve, forward to loosen, back to tighten. Check force required to move stick.

(5) If more adjustment is needed to loosen friction, turn stop and guard in same direction as loosening friction grip, and install in next set of slots in sleeve.

(6) To tighten friction, turn stop and guard in same direction as tightening friction grip, and install in sleeve.

(7) Adjust stop and guard so friction grip may be backed off without having lug on friction grip strike stop and guard, after backing off one full turn.

(8) Set adjustable stop so friction grip may be backed off from locked position 2 ½ - 4 full turns, before ring touches stop.

(9) Tighten stop and guard attaching bolt. **TORQUE BOLT TO 19 - 21 INCH-POUNDS** (Figure 11-4-29-1, Detail A).

(10) Using lockwire, Item 197, Appendix D, lockwire stop and guard attaching bolt.

(11) Tighten special bolts. **TORQUE TO 19 - 21 INCH-POUNDS.**

(12) Move collective stick full up and full down and check for binds and smoothness of operation.

(13) Place BACKUP HYD PUMP switch OFF. Turn off all electrical power (PARA 1-6-16).

m. On collective sticks, 70400-01401-048 and subsequent, adjust friction lock as follows (Figure 11-4-29-3):

WARNING

Injury to personnel and damage to equipment will result if friction lock is adjusted incorrectly, resulting in a collective stick which cannot be moved with friction lock full on. Make sure friction lock is adjusted so that collective stick retains full range of motion when friction lock is full on.

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
100° - 125° (39° - 52°)	16	48

(1) On upper console, place BACKUP HYD PUMP switch ON.

NOTE

- As friction grip threads into sleeve it squeezes collet block halves on collective stick.
- Amount of squeeze is set when stop segment on friction grip contacts stop and guard.
- If friction grip is turned toward stop and guard, friction is increased.
- If friction grip is turned away from stop and guard, friction is decreased.
- Adjustable stop is set so when ring touches stop, friction on stick is fully released. It prevents friction grip from coming fully out of sleeve and allows pilot to apply friction in only a few turns.

(2) With lock full on, use spring scale attached to collective grip, move pilot's collective stick up and down. **STICK MUST MOVE WITH 20 - 40 POUNDS PULL.**

(3) On upper console, place BACKUP HYD PUMP switch OFF.

(4) If pull is above or below limits, perform the following:

(a) Make a pencil mark on outside of ring on friction grip to show position of tang on inside of friction grip.

(b) On upper console, place BACKUP HYD PUMP switch ON.

(c) Using spring scale attached to friction grip, turn friction grip until desired friction is obtained (20 - 40 pounds pull).

(d) On upper console, place BACKUP HYD PUMP switch OFF.

(e) Note distance between marked position of tang and nearest edge of first slot in sleeve that was passed. This distance will determine hole in stop segment that will be used when installing stop segment.

(f) Unscrew friction grip from sleeve and install stop of stop and guard in slot of sleeve noted in previous step.

(g) Install stop segment on friction grip with edge of stop segment as close as possible to stop of stop and guard, using self-locking screw (Figure 11-4-29-3, Detail B).

NOTE

Do not reuse self-locking screw if friction must be readjusted.

(h) Slide stop and guard to front of collective stick, to make positive contact between stop and stop segment. Check that stop and guard has not slid far enough to prevent friction grip to be backed off at least one turn. If one turn cannot be obtained, use nonmetallic drift and tap stop and guard back until grip can be turned one full turn.

(i) On upper console, place BACKUP HYD PUMP switch ON.

(j) Check to make sure friction grip is in positive stop.

(k) Using spring scale attached to collective grip, check for 20 - 40 pounds pull to move collective stick.

(l) If desired friction was not obtained, unscrew friction grip far enough to get to stop segment. Remove self-locking screw from stop segment and adjust stop segment as needed. Install new self-locking screw in stop segment.

NOTE

Do not reuse self-locking screw.

(m) Reinstall friction grip in sleeve and recheck force to move collective stick.

(n) If correct friction cannot be obtained after adjusting stop segment, remove friction grip from sleeve and reposition stop of stop and guard in next slot of sleeve past pencil mark. Repeat adjustment procedure until correct friction is obtained or replace collet block halves.

(o) On upper console, place BACKUP HYD PUMP switch OFF. Turn off all electrical power (PARA 1-6-16).

(p) Set adjustable stop so friction grip can be backed off from locked position 2 ½ - 4 full turns before ring touches stop.

WARNING

Injury to personnel and damage to equipment will result if hardware securing collet block halves and adjustable stop is intermixed. Friction lock assembly may jam, resulting in stiff or sticking collective stick movement. Make sure correct hardware is used when installing collet block halves and adjustable stop.

n. Install bolt and washer in adjustable stop (Figure 11-4-29-1, Detail A). **TORQUE BOLT TO 19 - 21 INCH-POUNDS.**

o. Install special bolt and washer in stop and guard. Install special bolts, washers, and spacers in sleeve and collet block halves. **TORQUE SPECIAL BOLTS TO 19 - 21 INCH-POUNDS.** Using lockwire, Item 194, Appendix D, lockwire bolts in pairs.

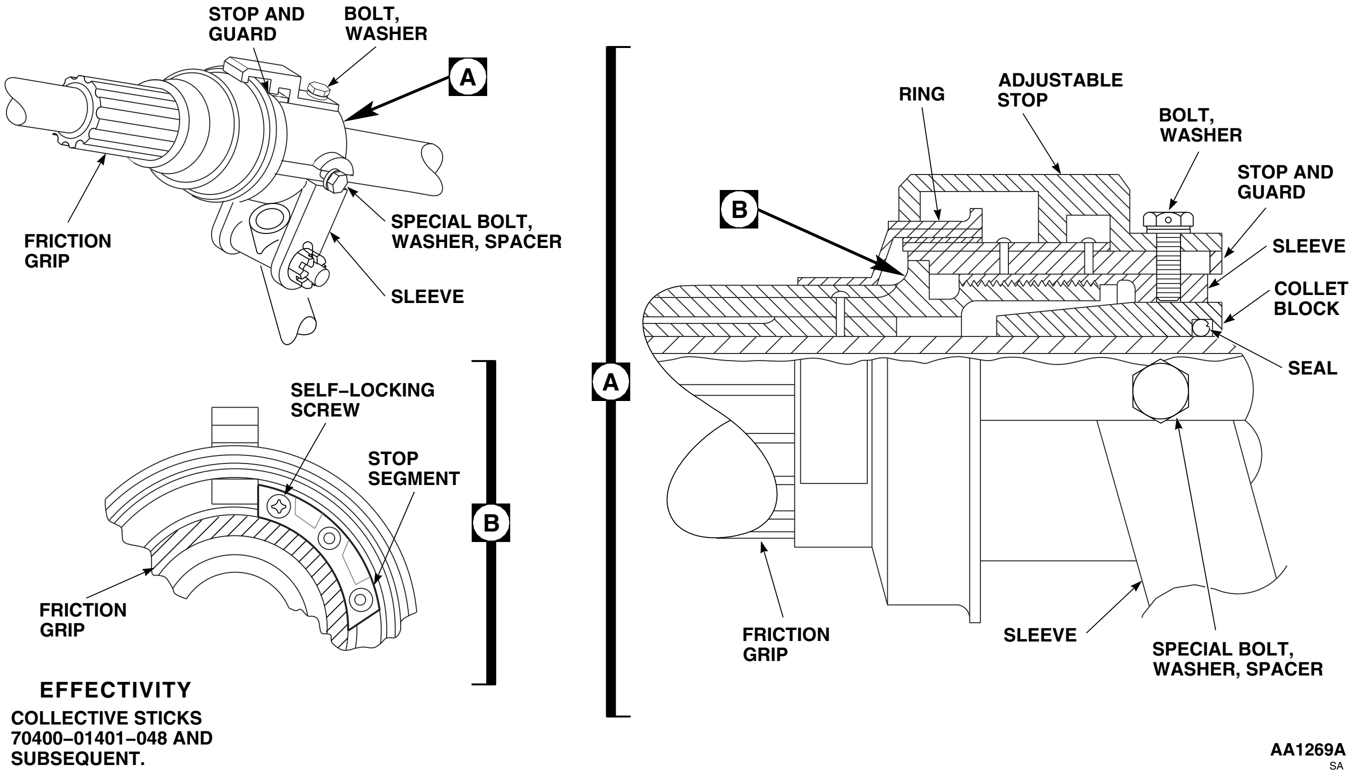


FIGURE 11-4-29-3. ADJUST FRICTION LOCK

11-4-30 PILOT'S AND COPILOT'S COLLECTIVE STICK SUPPORT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage, 0.020-inch
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Cheesecloth, Item 90, Appendix D
- Lockwire, Item 197, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 11-4-4
- PARA 11-4-27
- PARA 11-4-28
- PARA 11-4-44
- PARA 11-4-45

11-4-30.1 Replace Pilot's and Copilot's Collective Stick Support.

NOTE

Replacement of both collective stick supports is the same.

11-4-30.1.1. Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove pilot's or copilot's collective stick (PARA 11-4-27 or PARA 11-4-28).

c. Remove bolts and washers securing collective stick support to cabin floor. Remove support (Figure 11-4-30-1 or Figure 11-4-30-2).

NOTE

Shim washers are bonded to floor under stick support. Make sure shims are kept in same position on installation.

d. Check stick support for loose shim material. Peel from support and replace shim stock on stack it came from.

e. Loosely install bolts in stick support boltholes to protect shims.

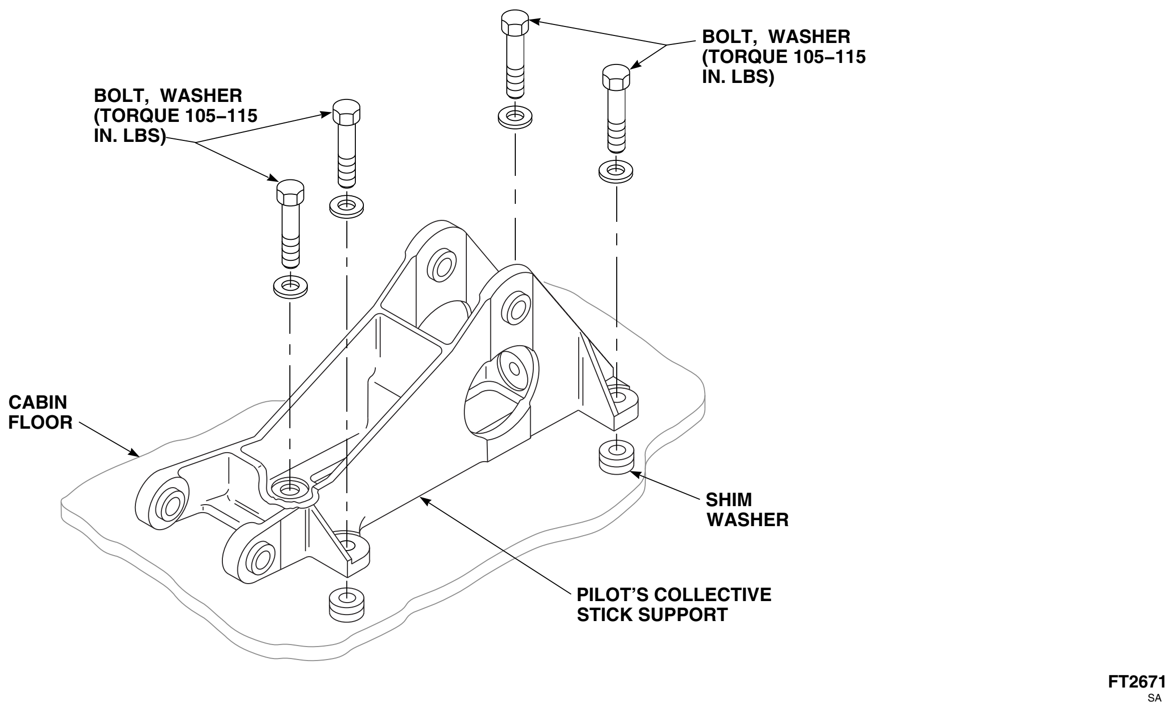


FIGURE 11-4-30-1. REPLACE PILOT'S COLLECTIVE STICK SUPPORT

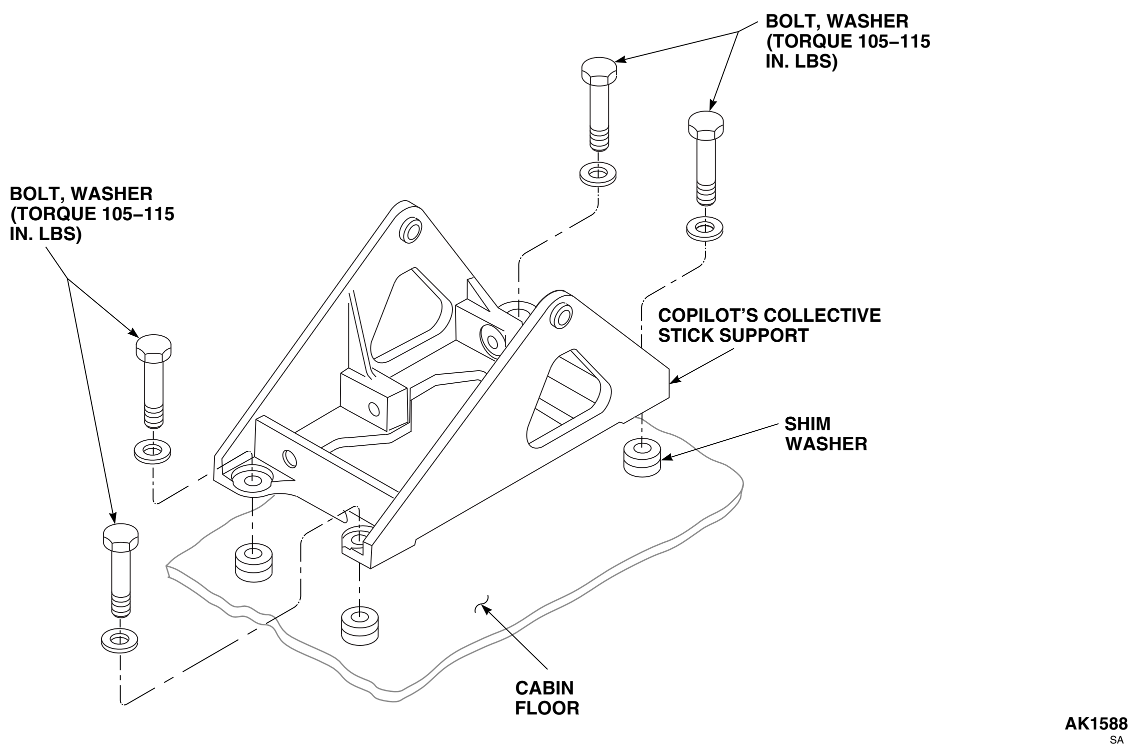
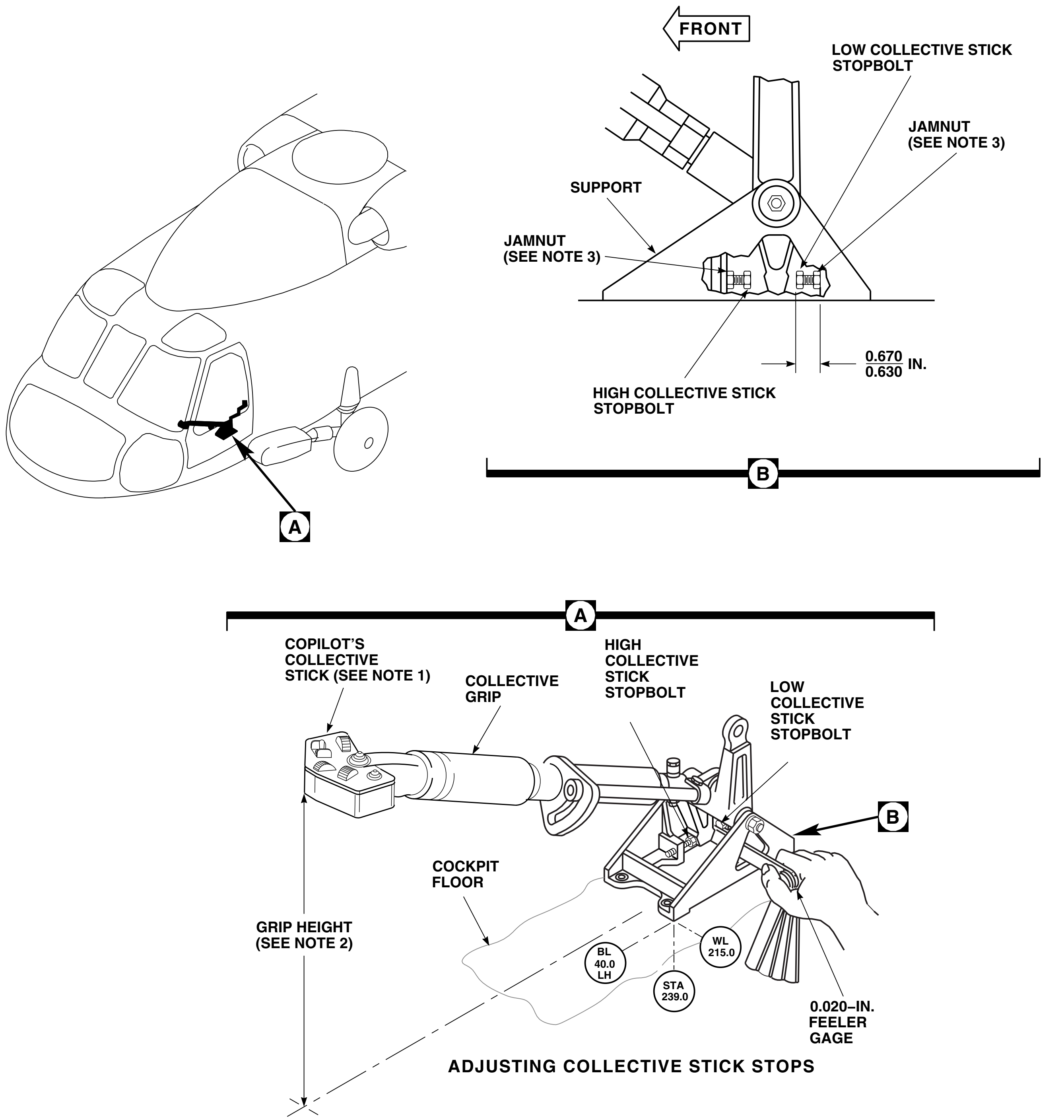


FIGURE 11-4-30-2. REPLACE COPILOT'S COLLECTIVE STICK SUPPORT



NOTES

1. PILOT'S COLLECTIVE STICK ADJUSTED THE SAME.
2. MEASURMENT TAKEN AT THIS LOCATION MUST BE WITHIN ± 0.063 INCH BETWEEN PILOT'S AND COPILOT'S COLLECTIVE STICKS.
3. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 11-4-30-3. RESET PILOT'S AND COPILOT'S COLLECTIVE STICK STOP

11-4-30.1.2. Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Check mounting pad boltholes for damaged or missing shim washers. Replace washers as follows:

(1) Remove shim and adhesive debris with nonmetallic scraper and scouring pad, Item 270, Appendix D.

(2) Wipe washer and floor bonding contact surfaces clean with methyl ethyl ketone, Item 217, Appendix D, and clean cheesecloth, Item 90, Appendix D. Wipe until no stain shows on cloth after last wipe. Do not touch treated surfaces.

(3) Apply adhesive, Item 32, Appendix D. to shim washer and floor contact areas. Do not get adhesive in boltholes.

(4) Using bolt as a guide to center washer over bolthole, press shim washer onto floor (Figure 11-4-30-1 or Figure 11-4-30-2).

(5) Loosely install pivot bolt through collective stick support pivot bushings.

c. Install pilot's or copilot's collective stick support to cabin floor with washers and bolts. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

d. Turn stick pivot bolt with fingers. **NO BINDING ALLOWED.** A stuck or binding bolt shows that stick support is twisted due to one or more of the mount pads being higher than the rest. Remove bolts and washers securing support to floor and peel shims as required until pivot bolt turns free with mounting bolts torqued properly.

e. Install pilot's or copilot's collective stick (PARA 11-4-27 or PARA 11-4-28).

NOTE

- Reset procedures for pilot's and copilot's collective stick stops are the same. Pilot's and copilot's designations are reversed where applicable.
- The following procedure adjusts the pilot's collective stick to match the copilot's collective stick positions. Do not attempt to readjust copilot's stick stops or control rods while following this procedure. If copilot's stick stops or control rods are adjusted, aircraft rig must be rechecked.

f. Reset pilot's collective stick stop as follows:

(1) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Back off jamnuts on pilot's collective stick stopbolts (Figure 11-4-30-3, Detail B).

(3) Disconnect forward end of the copilot's collective control rod (PARA 11-4-45).

(4) Insert 0.020 inch feeler gage between the copilot's collective stick and the low collective stick stop (Figure 11-4-30-3, Detail A). With the copilot's collective stick up against the 0.020 inch feeler gage and stop, check the height of pilot's collective stick grip to the copilot's collective stick grip height. **THE PILOT'S STICK GRIP HEIGHT IS TO BE WITHIN ±0.063 INCHES OF THE COPILOT'S STICK GRIP HEIGHT. IF THE PILOT'S GRIP HEIGHT IS WITHIN THE ±0.063 INCHES, TIGHTEN PILOT'S LOW COLLECTIVE STICK STOPBOLT JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 197, Appendix D. Remove feeler gage.

(5) If pilot's stick grip height is not within limits, disconnect pilot's collective control rod (PARA 11-4-44) and **ADJUST THE ROD END IN OR OUT TO OBTAIN THE ±0.063 INCHES.** Reconnect pilot's collective control rod (PARA 11-4-44). With the copilot's collec-

tive stick up against the 0.020 inch feeler gage and low collective stick stop, **TIGHTEN PILOT'S LOW COLLECTIVE STICK STOPBOLT JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 197, Appendix D. Remove feeler gage.

(6) Move copilot's and pilot's collective sticks to full high collective. Adjust the pilot's high collective stick stop to match the copilot's high collective stick stop. **TIGHTEN PILOT'S HIGH COLLECTIVE STICK STOPBOLT JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Lockwire, Item 197, Appendix D.

(7) Reconnect forward end of the copilot's collective control rod (PARA 11-4-45).

g. Perform main rotor system rig check (PARA 11-4-4).

11-4-31 COLLECTIVE STICK GRIP ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Electrical Repairer's Toolkit
- Heat Gun

Materials/Parts

- Fibrous Cord, Item 139, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 169, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 11-2-2

11-4-31.1 Replace Collective Stick Grip Assembly.

NOTE

Replacement of both collective stick grips is the same, except as noted.

11-4-31.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Disconnect electrical connector, P103 (copilot's), or P104 (pilot's) (Figure 11-4-31-1, Detail A).

c. Install protective caps and plugs on electrical connectors.

d. Slide boot away from electrical connector.

e. Loosen backshell on electrical connector and slide back along wiring.

f. Remove pins from electrical connector and remove pins from wires. Tag wires.

g. Remove screw, lockwasher, and locknut securing grip to collective stick.

h. If grip is to be replaced, tie suitable length fibrous cord, Item 139, Appendix D, securely around lower end of wire bundle.

i. Remove grip from collective stick, making sure wire bundle is supported.

NOTE

Do not let cord fall into collective stick.

j. If grip is to be replaced, remove cord from wire bundle and leave cord in collective stick.

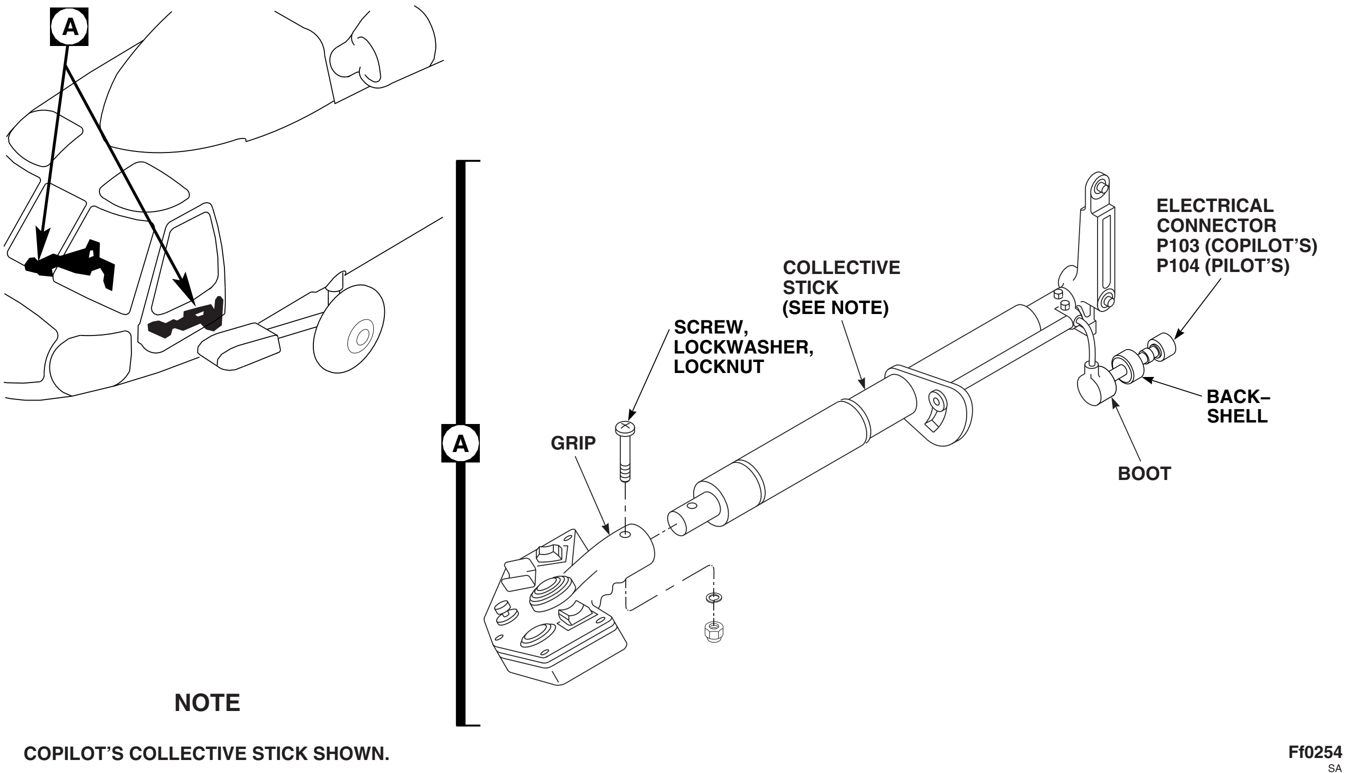


FIGURE 11-4-31-1. REPLACE COLLECTIVE STICK GRIP ASSEMBLY

11-4-31.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If same grip is being installed in replacement collective stick, perform the following:

(1) Tie suitable length fibrous cord, Item 139, Appendix D, securely around lower end of wire bundle.

(2) Feed cord into collective stick tube and route cord through grommet hole of stick.

c. If replacement grip is being installed, tie cord found in collective stick securely around lower end of wire bundle.

d. Pull cord and route wiring through grommet hole of stick until grip is seated.

e. Position and secure grip to stick using screw, lockwasher, and locknut (Figure 11-4-31-1, Detail A).

f. Slide 0.625-inch inside diameter heat-shrinkable insulation sleeving, Item 169, Appendix D, over wire bundle and heat-shrink using heat gun.

g. Fill cavity around nut with sealing compound, Item 292, Appendix D, flush with grip surface.

h. Slide backshell over wire bundle.

i. Connect wiring to electrical connector, remove tags.

j. Remove protective caps and plugs from electrical connectors.

k. Inspect and treat electrical connector (PARA 1-7-20).

l. Connect electrical connector, P103 (copilot's), or P104 (pilot's).

m. Perform operational check of collective stick assembly (PARA 11-2-2).

n. Slide boot onto back of electrical connector.

11-4-32 COLLECTIVE STICK GRIP LIGHTED PANEL AND LAMPS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit

References

- PARA 1-6-16
- PARA 9-2-4

11-4-32.1 Replace Collective Stick Grip Lighted Panel and Lamps.

NOTE

Replacement of both collective stick lighted panels and lamps is the same.

11-4-32.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws securing lighted panel to stick grip and remove panel (Figure 11-4-32-1, Detail A).

c. Peel off and discard moisture seal from lamp being replaced.

d. Turn lamp assembly 1/4 turn counterclockwise and remove from lighted panel.

11-4-32.1.2. Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Place lamp in lighted panel socket (Figure 11-4-32-1, Detail A).

c. Line up slot in lamp with lighted panel guides.

d. Turn lamp 1/4 turn clockwise to lock in place.

e. Seal lamp by applying moisture seal.

f. Secure lighted panel to stick grip using screws.

g. Make sure area is clean and free of foreign objects.

h. Perform operational check of instrument panel lights (PARA 9-2-4).

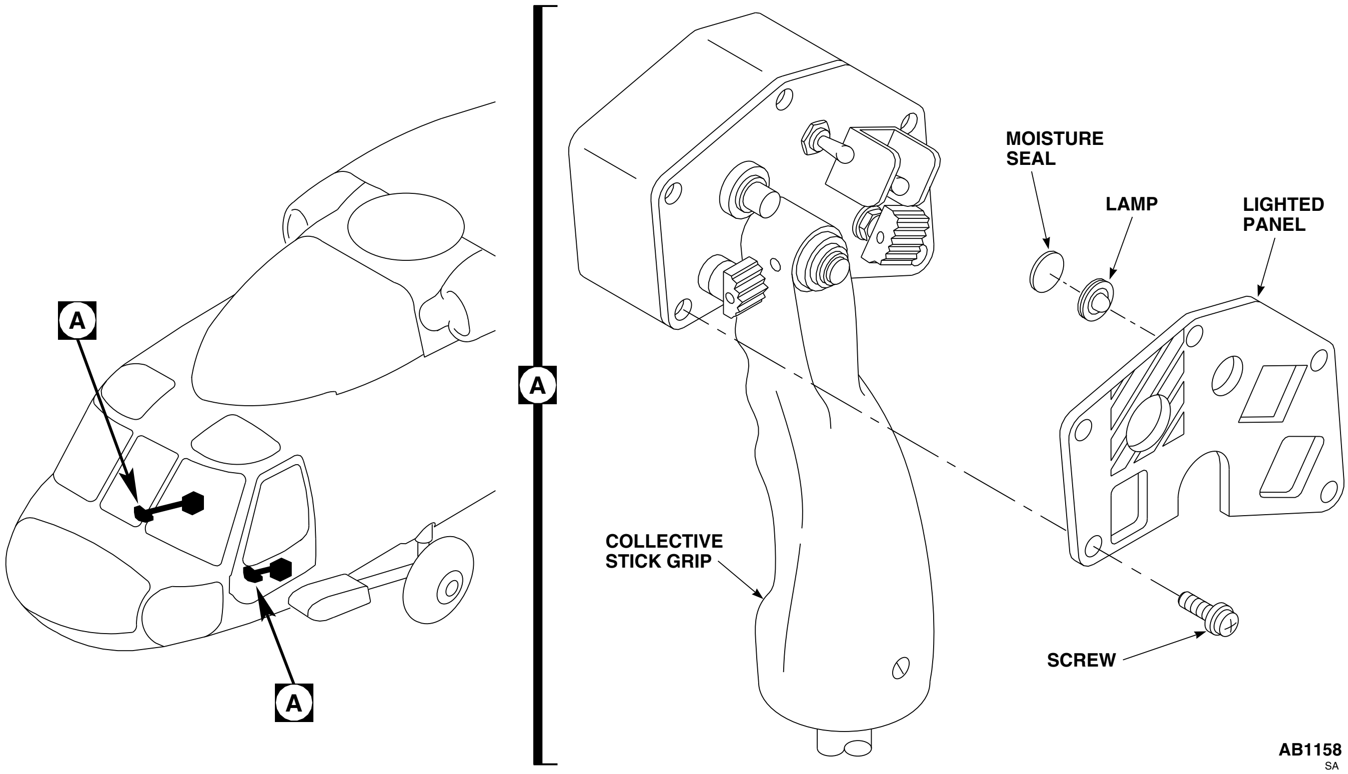


FIGURE 11-4-32-1. REPLACE COLLECTIVE STICK GRIP LIGHTED PANEL AND LAMPS

11-4-33 COLLECTIVE STICK SRCH LT AND SVO OFF SWITCHES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-2

11-4-33.1 Replace Collective Stick SRCH LT and SVO OFF Switches.

NOTE

Replacement of collective stick SRCH LT and SVO OFF switches on both collective sticks is the same, except as noted.

11-4-33.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws securing lighted panel and cover to stick grip and remove lighted panel and cover (Figure 11-4-33-1, Detail A).

c. Tag switch wires to identify switch connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Unsolder wires from switch, using soldering iron.

e. Remove SVO OFF switch by removing nut, guard, and keywasher.

f. W/O SLS Remove SRCH LT switch by removing nut and key washer.

g. SLS If required, remove pin securing thumb grip to SRCH LT switch. Remove thumb grip (Figure 11-4-33-1, Detail B).

h. SLS Remove SRCH LT switch by pushing out from inside of grip assembly.

11-4-33.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install SVO OFF switch using keywasher, guard, and nut (Figure 11-4-33-1, Detail A).

c. W/O SLS Install SRCH LT switch using key washer and nut.

d. SLS Install SRCH LT switch by pushing in place from front of grip assembly. Use adhesive, Item 55, Appendix D, to hold switch in place if required (Figure 11-4-33-1, Detail B).

e. SLS If required, install thumb grip on SRCH LT switch and secure with pin.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

f. Solder, Item 313, Appendix D, wires to switch using soldering iron. Remove tags.

g. Coat all exposed solder terminals with adhesive, Item 55, Appendix D.

h. Secure lighted panel and cover to grip using screws.

i. Perform operational check of collective stick assembly (PARA 11-2-2).

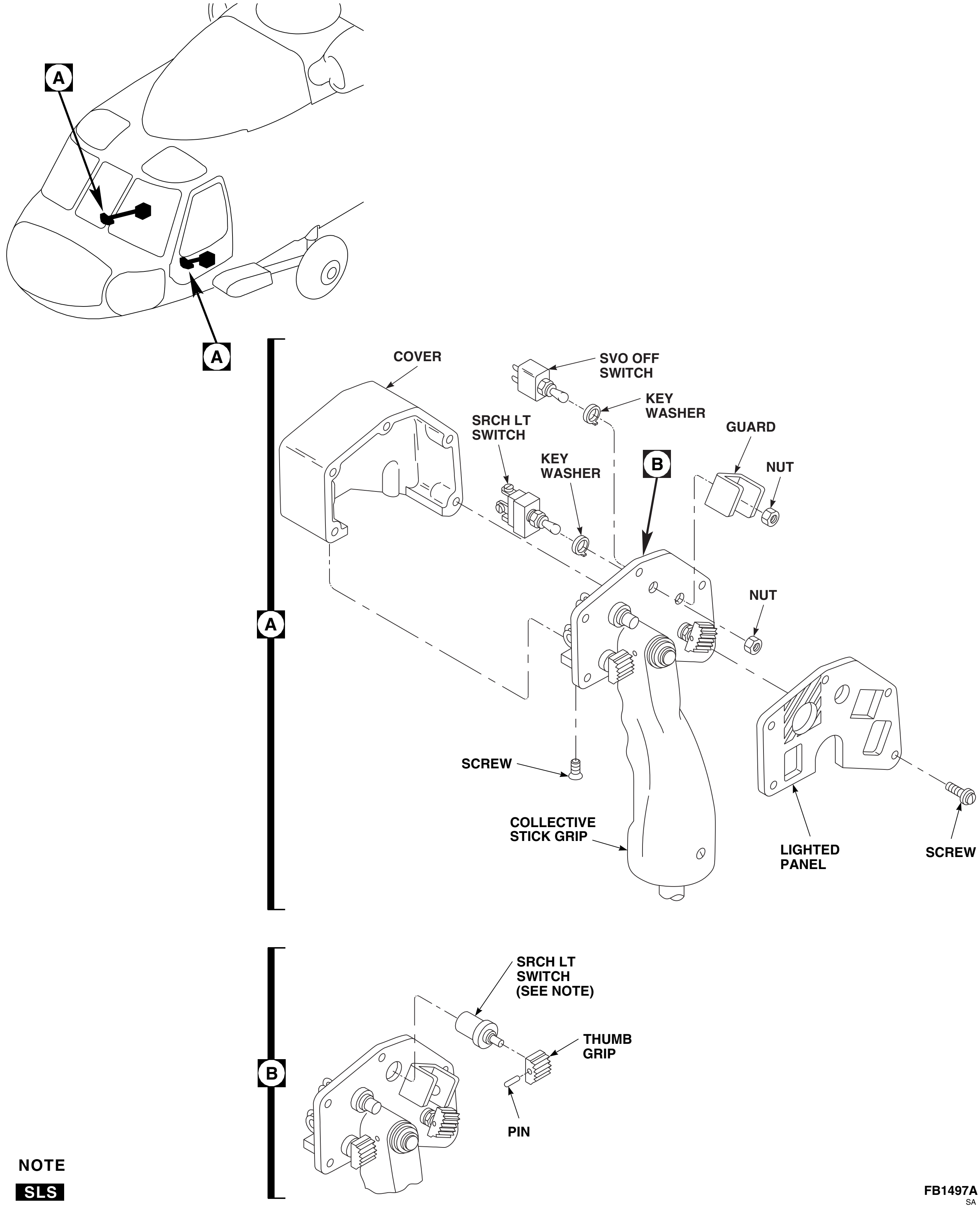


FIGURE 11-4-33-1. REPLACE COLLECTIVE STICK SRCH LT AND SVO OFF SWITCHES

11-4-34 COLLECTIVE STICK LDG LT AND EMERG HOOK REL SWITCHES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-2

11-4-34.1 Replace Collective Stick LDG LT and EMERG HOOK REL Switches.

NOTE

Replacement of collective stick LDG LT and EMERG HOOK REL switches on both collective sticks is the same.

11-4-34.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws securing lighted panel and cover to collective stick grip and remove lighted panel and cover (Figure 11-4-34-1, Detail A).

c. Tag switch wires to identify switch connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering iron, unsolder wires from switch.

e. Remove pin from thumb grip on switch and remove thumb grip.

f. Remove switch by pushing out from inside of grip assembly.

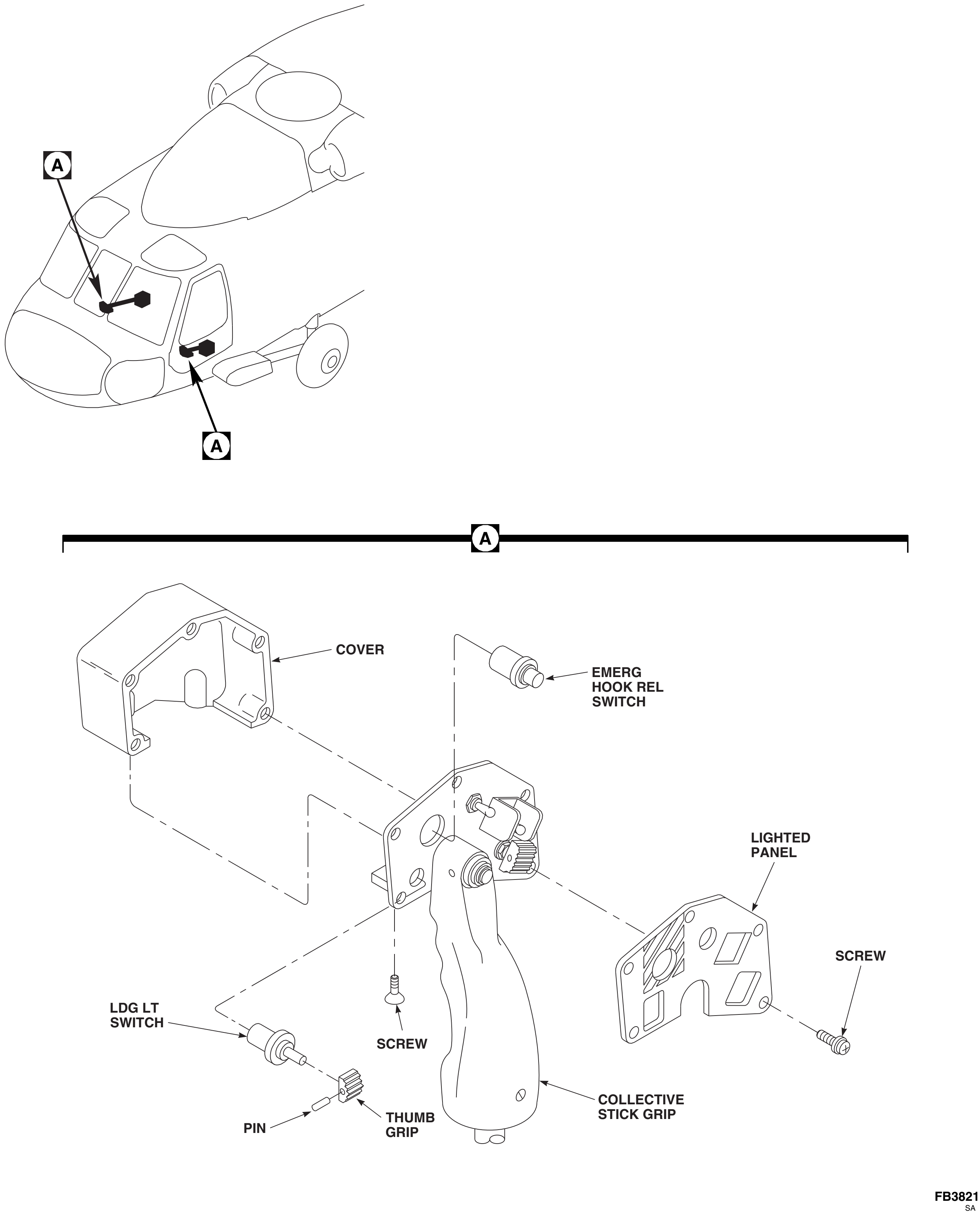


FIGURE 11-4-34-1. REPLACE COLLECTIVE STICK LDG LT AND EMERG HOOK REL SWITCHES

11-4-34.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Install switch by pushing switch in place from front of grip assembly (Figure 11-4-34-1, Detail A).

c. Use adhesive, Item 55, Appendix D, to hold switch in place if required.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering iron and solder, Item 313, Appendix D, solder wires to switch and remove tags.

e. Coat all exposed solder terminals with adhesive, Item 55, Appendix D.

f. Install thumb grip on switch and secure with pin.

g. Secure lighted panel and cover to grip using screws.

h. Perform operational check of collective stick assembly (PARA 11-2-2).

11-4-35 COLLECTIVE STICK ENG RPM SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-2

11-4-35.1 Replace Collective Stick ENG RPM Switch.

NOTE

Replacement of collective stick ENG RPM switches on both collective sticks is the same.

11-4-35.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws securing lighted panel and cover to grip and remove lighted panel and cover (Figure 11-4-35-1, Detail A).

c. Remove pin from switch thumb grip and remove thumb grip.

d. Tag switch wires to identify switch connections.

WARNING

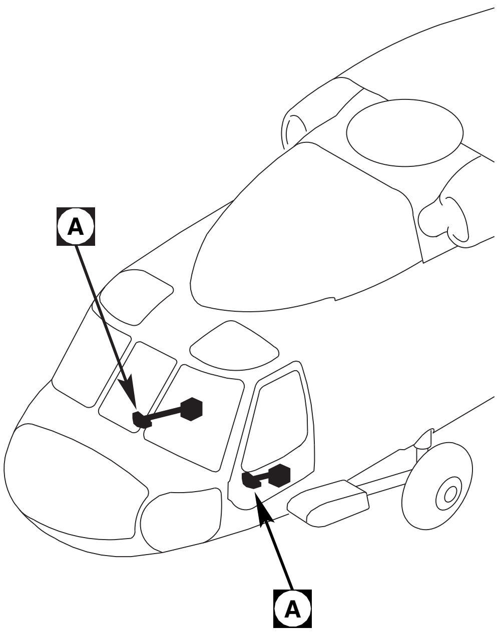
Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

e. Unsolder wires from switch, using soldering iron.

f. Remove switch by pushing out from inside of grip assembly.

11-4-35.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).



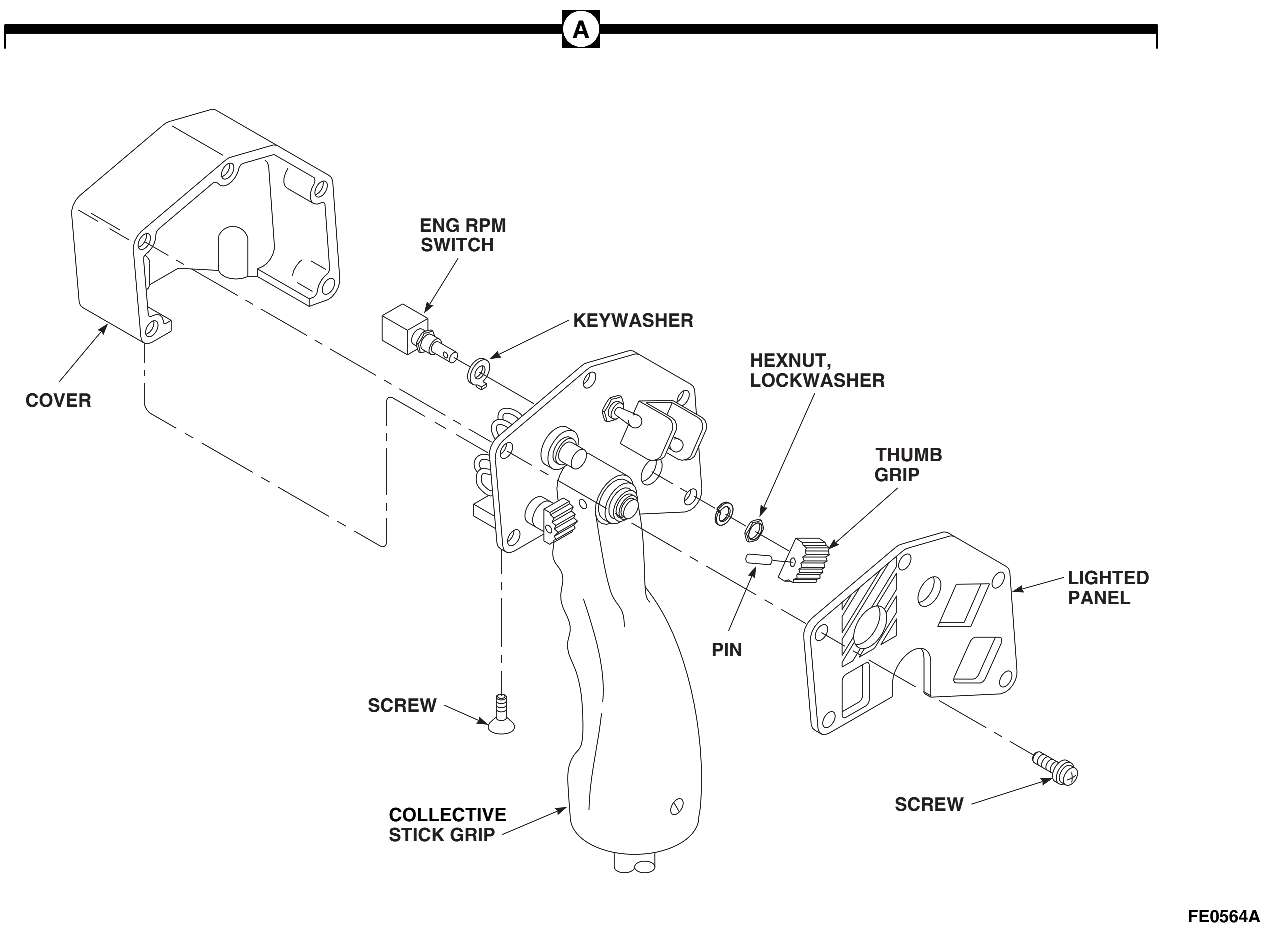


FIGURE 11-4-35-1. REPLACE COLLECTIVE STICK ENG RPM SWITCH

b. Replace switch by pushing switch in place from front of grip assembly. Use adhesive, Item 55, Appendix D, to hold switch in place if required (Figure 11-4-35-1, Detail A).

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Solder, Item 313, Appendix D, wires to switch using soldering iron. Remove tags.

d. Coat all exposed solder terminals with adhesive, Item 55, Appendix D.

e. Secure lighted panel and cover to grip using screws.

f. Install thumb grip on switch and secure with pin.

g. Perform operational check of collective stick assembly (PARA 11-2-2).

11-4-36 COLLECTIVE STICK SRCH LT SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-2

11-4-36.1 Replace Collective Stick SRCH LT Switch.

NOTE

Replacement of collective stick SRCH LT switches on both collective sticks is the same.

11-4-36.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove screws securing lighted panel and cover to grip and remove lighted panel and cover (Figure 11-4-36-1, Detail A).

c. Remove pin holding SRCH LT switch in place. Slide switch from grip by pushing out from inside of grip.

d. Tag switch wires to identify switch connections.

e. Remove adhesive from solder terminals.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

f. Using soldering iron, unsolder wires from switch.

g. Remove switch by pushing out from inside of grip assembly.

11-4-36.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

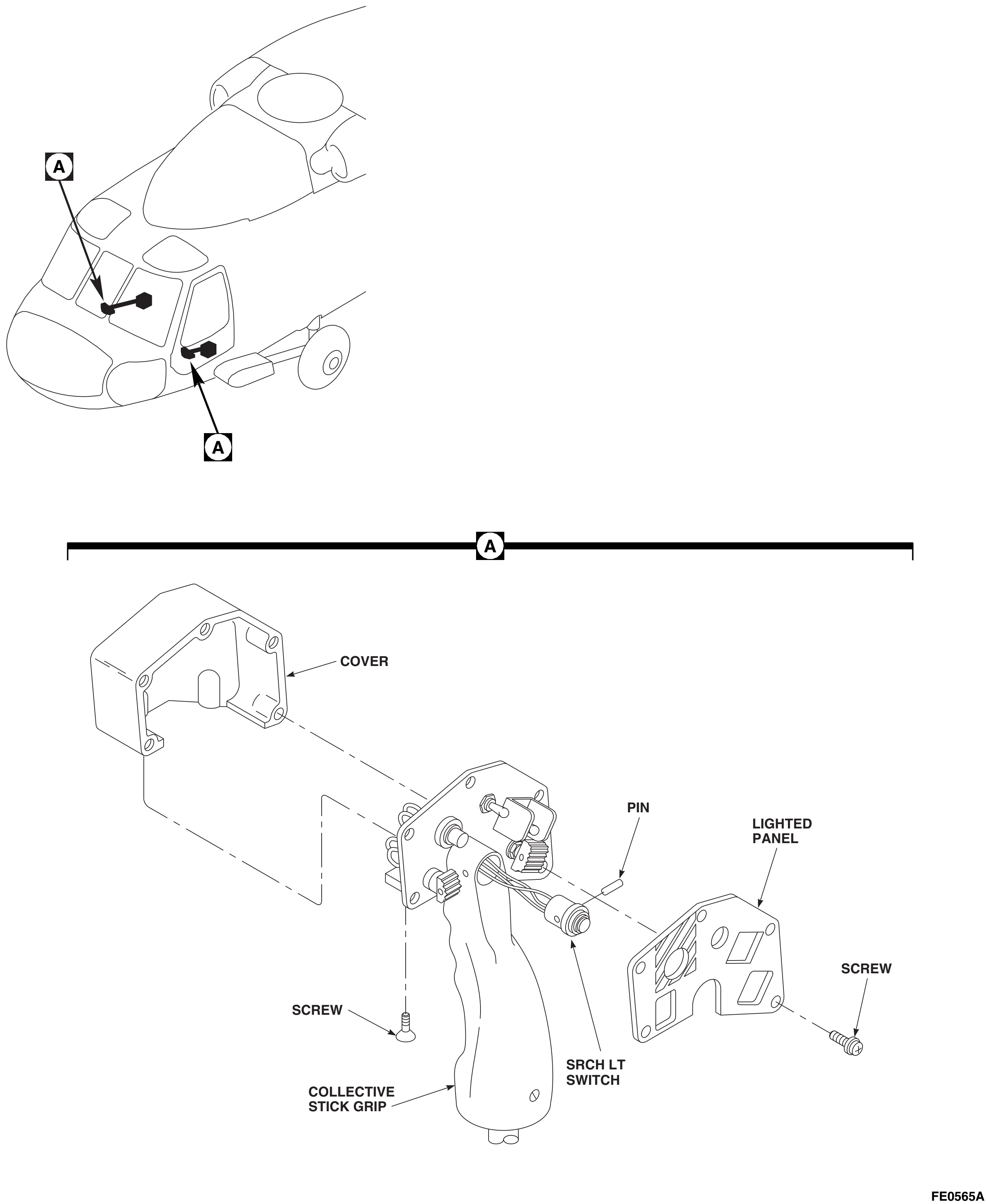


FIGURE 11-4-36-1. COLLECTIVE STICK SRCH LT SWITCH

b. Install switch by pushing switch in place from front of grip assembly. Use adhesive, Item 55, Appendix D, to hold switch in place if required (Figure 11-4-36-1, Detail A).

c. Secure switch to grip with pin.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering iron, solder, Item 313, Appendix D, wires to switch. Remove tags.

e. Coat all exposed solder terminals with adhesive, Item 55, Appendix D.

f. Install lighted panel and cover to grip using screws.

g. Perform operational check of collective stick assembly (PARA 11-2-2).

11-4-37 COPILOT'S COLLECTIVE STICK GRIP COIL CORD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 11-2-2
- PARA 11-4-31

11-4-37.1 Replace Copilot's Collective Stick Grip Coil Cord.

11-4-37.1.1 Remove.

a. Remove copilot's collective stick grip assembly (PARA 11-4-31).

b. Remove screws securing cover and lighted panel to grip and remove cover and lighted panel (Figure 11-4-37-1, Detail A).

c. Tag wires to identify switch and light connections.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Using soldering iron, unsolder all wires from switches and lights.

e. Remove coil cord from grip.

11-4-37.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Position replacement coil cord in grip (Figure 11-4-37-1, Detail A).

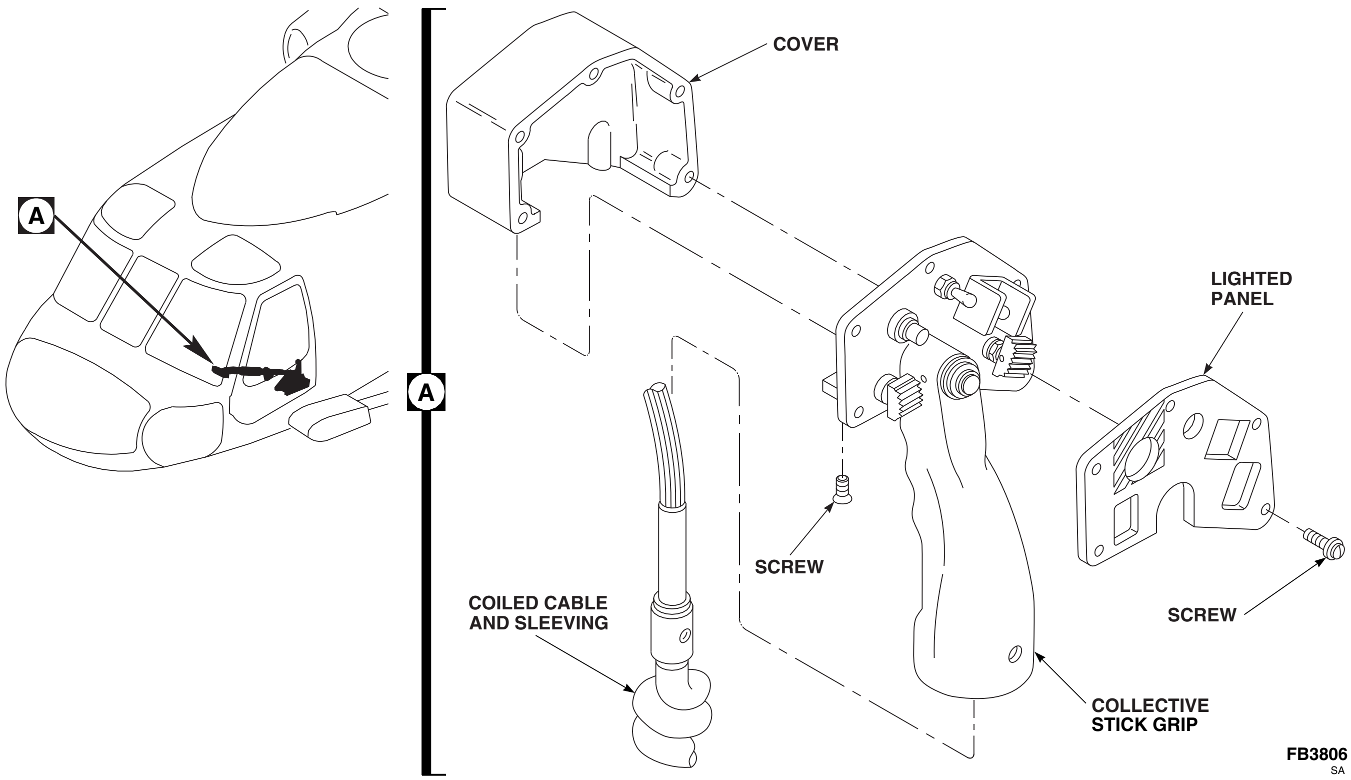


FIGURE 11-4-37-1. REPLACE COPILOT'S COLLECTIVE STICK GRIP COIL CORD

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Using solder, Item 313, Appendix D, solder wires to switches and lights using soldering iron. Remove tags.

d. Coat all exposed solder terminals with adhesive, Item 55, Appendix D.

e. Secure cover and lighted panel to grip using screws.

f. Install copilot's collective stick grip assembly (PARA 11-4-31).

g. Perform operational check of collective stick assembly (PARA 11-2-2).

11-4-38 PILOT'S COLLECTIVE STICK GRIP WIRE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Electrical Repairer's Toolkit
- Protection, Eye and Skin
- Soldering Iron

Materials/Parts

- Adhesive, Item 55, Appendix D
- Solder, Item 313, Appendix D

References

- Appendix D
- PARA 11-2-2
- PARA 11-4-31

11-4-38.1 Replace Pilot's Collective Stick Grip Wire.

a. Remove pilot's collective stick grip assembly (PARA 11-4-31).

b. Remove screws securing grip assembly cover and remove cover.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

c. Using soldering iron, unsolder faulty wire from switch in grip.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

d. Solder, Item 313, Appendix D, replacement wire to bad wire, using soldering iron.

e. Pull replacement wire through harness with bad wire.

WARNING

Hot metal droppings occur when soldering, causing both a fire hazard and injury to personnel. Use protective clothing, including eye protection while soldering. Make sure hot metal droppings do not come in contact with skin or clothing.

f. Using solder, Item 313, Appendix D, solder replacement wire in place to switch, using soldering iron.

g. Coat exposed solder terminals with adhesive, Item 55, Appendix D.

h. Secure cover to grip assembly using screws.

i. Install pilot's collective stick grip assembly (PARA 11-4-31).

j. Install new wire in plug, and remove bad wire.

k. Perform operational check of collective stick assembly (PARA 11-2-2).

11-4-39 PILOT'S COLLECTIVE STICK FRICTION LOCK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Drift
- Pencil
- Protection, Eye
- Spring Scale, 0 - 50 lbs
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Adhesive, Item 32, Appendix D
- Damping Fluid, Item 121, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 149, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Tiedown Strap, Item 330, Appendix D
- Tiedown Strap, Item 332, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 11-4-31

11-4-39.1 Replace Pilot's Collective Stick Friction Lock.

11-4-39.1.1 Remove.

a. Remove pilot's collective stick grip assembly (PARA 11-4-31).

b. If installed, remove rear tiedown strap from boot. Push boot back to expose upper end of drag strut.

c. Remove bolt, washers, and nut. Remove drag strut and shim from sleeve clevis (Figure 11-4-39-1).

d. Remove spacer from drag strut.

e. Slide friction grip and sleeve from stick.

f. If installed, remove front tiedown strap and remove boot from sleeve.

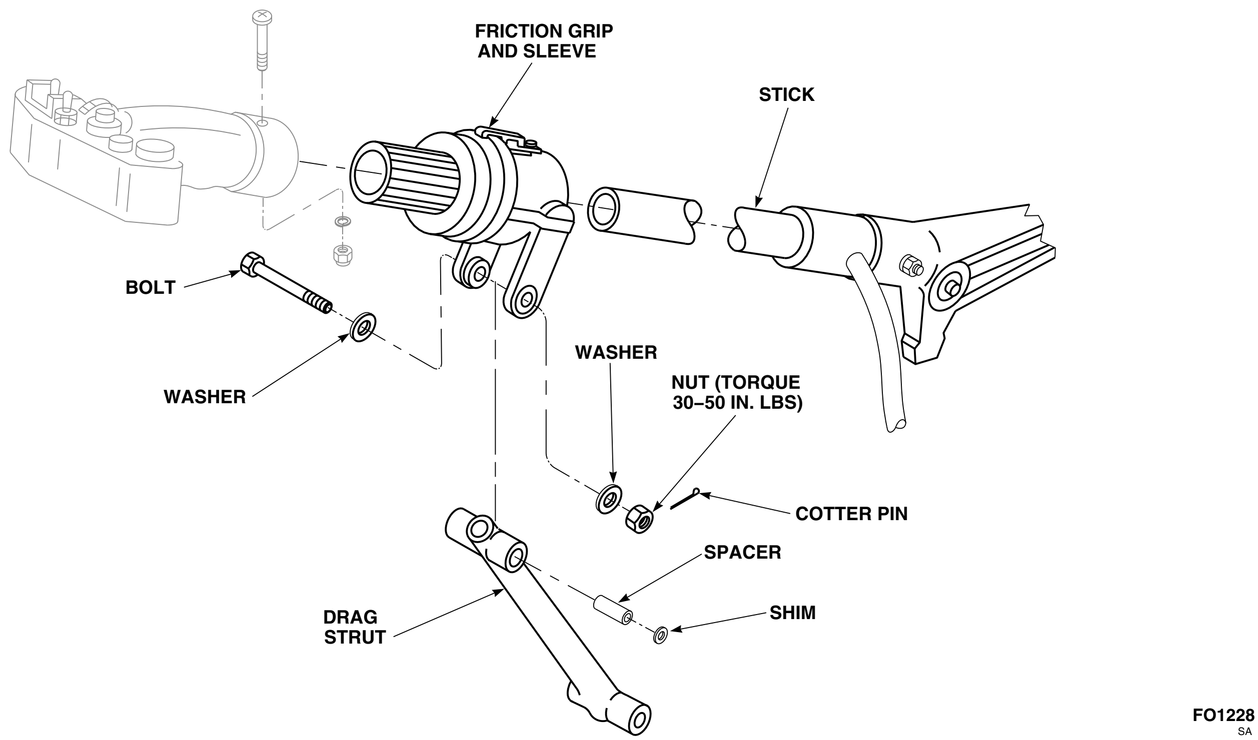


FIGURE 11-4-39-1. REPLACE PILOT'S COLLECTIVE STICK FRICTION LOCK

11-4-39.1.2 Repair.

NOTE

Pilot's friction grip is repaired by replacing parts.

a. On collective sticks prior to 70400-01401-048, repair friction grip as follows (Figure 11-4-39-2):

(1) Remove bolt and washer. Remove adjustable stop.

(2) Unscrew friction grip from sleeve.

(3) Remove special bolts, washers, and spacers. Remove stop and guard and collet halves from sleeve.

(4) Remove retaining ring, retainer and scraper ring from front end of friction grip.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(5) Clean all parts with dry-cleaning solvent, Item 124, Appendix D. Blow dry with dry compressed air.

(6) Inspect threads of friction grip and sleeve for nicks, dents, or any damage that would cause them to seize if screwed together.

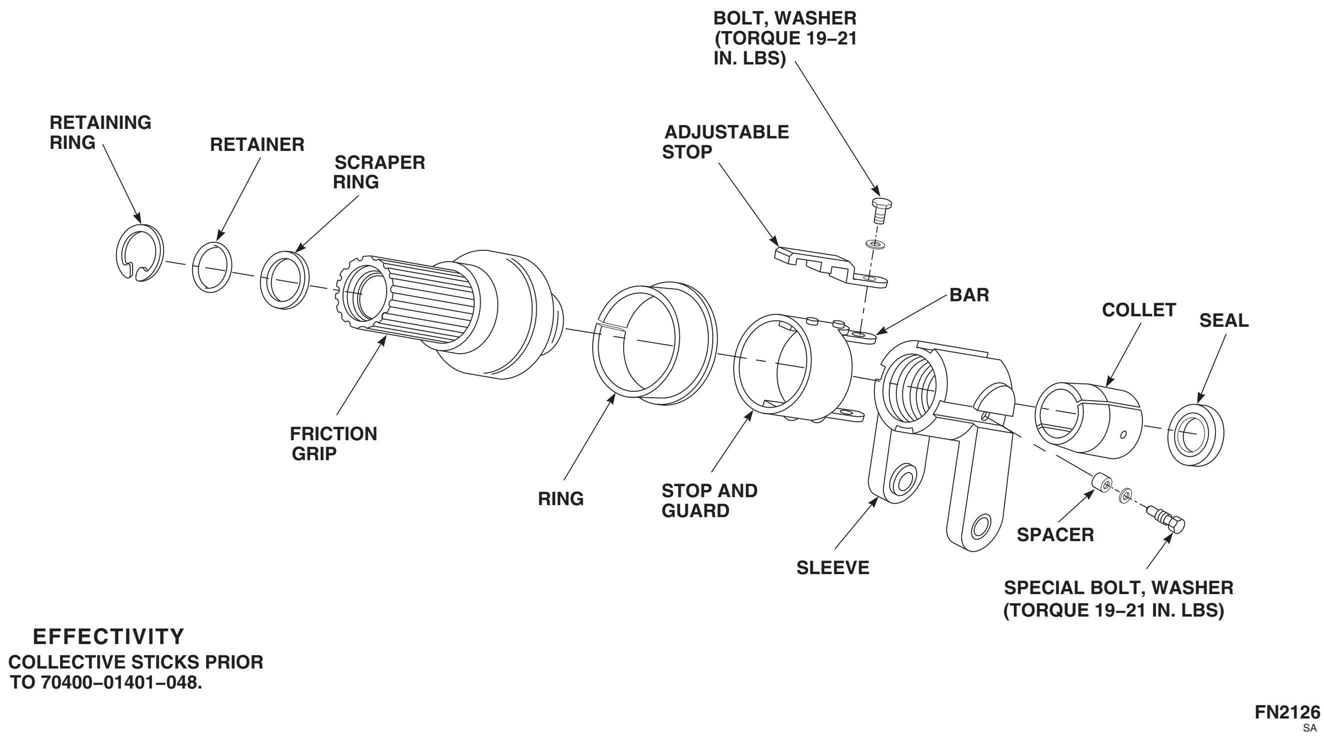


FIGURE 11-4-39-2. REPAIR PILOT'S COLLECTIVE STICK FRICTION LOCK

(7) Thread friction grip in and out of sleeve to make sure it turns easily and smoothly for entire length of thread. Replace both parts if there is any doubt.

(8) Make sure collets are clean and free from any damage that might stop them from sliding smoothly on collective stick.

(9) Check that ring on friction grip is bonded to grip. If loose, or if lip is damaged, remove ring from friction grip.

(10) Clean all bonding material from friction lock with methyl ethyl ketone, Item 217, Appendix D.

(11) Install replacement scraper ring in front of friction grip. Install retainer and retaining ring.

CAUTION

Damage to collective stick assembly will result if damping fluid is allowed to penetrate into ID of collet halves. Make sure no fluid gets between collective stick and inner diameter of collet halves.

(12) Apply a light coat of damping fluid, Item 121, Appendix D, to outside diameter of collet halves at tapered ends of collets.

WARNING

Injury to personnel and damage to equipment will result if new collets are installed without removing phenolic spacer strips. Make sure spacer strips are removed from collets prior to installation.

(13) Install seal in collet halves and install collet in sleeve.

(14) Install stop and guard on sleeve, and thread friction grip in sleeve.

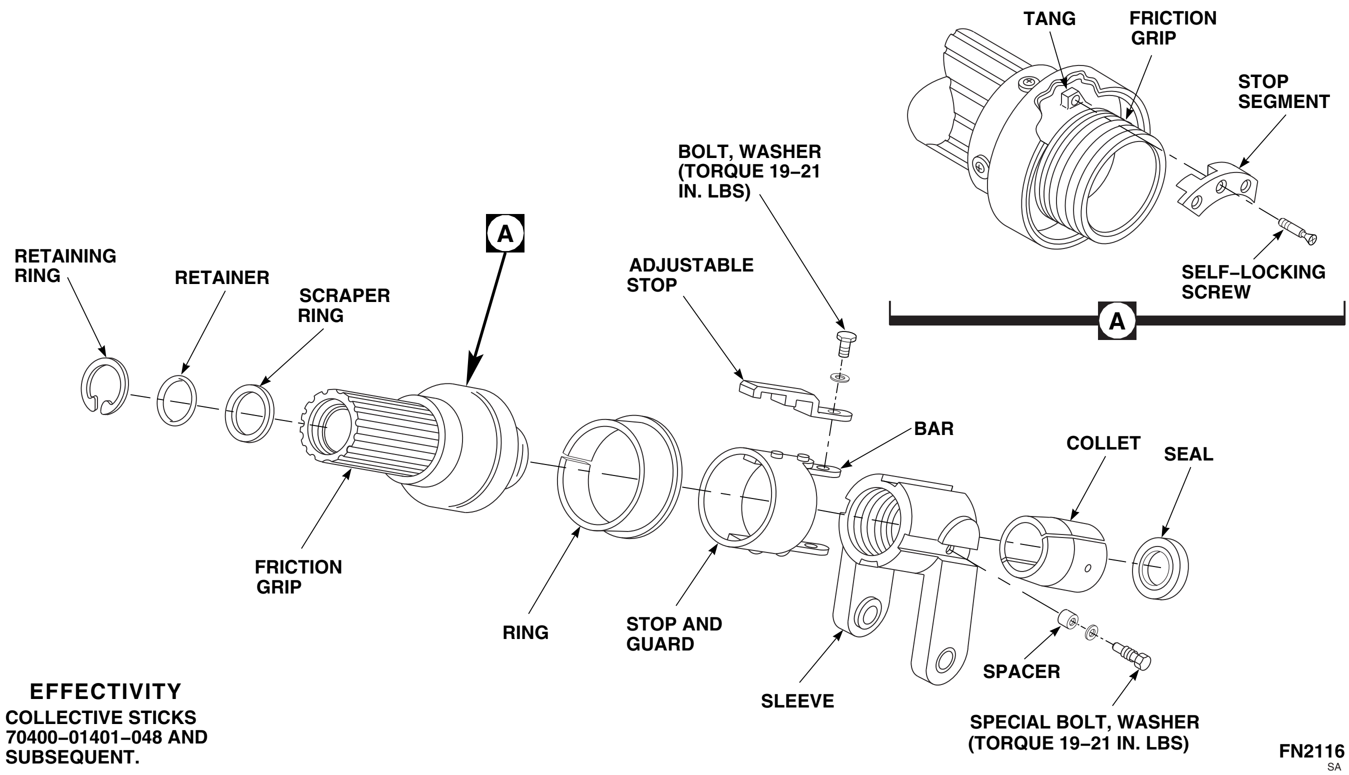


FIGURE 11-4-39-3. REPAIR PILOT'S COLLECTIVE STICK FRICTION LOCK

WARNING

Injury to personnel and damage to equipment will result if hardware securing collet halves and adjustable stop is intermixed. Friction lock assembly may jam, resulting in stiff or sticking collective stick movement. Make sure correct hardware is used when installing collet halves and adjustable stop.

NOTE

Ring is bonded to friction grip after adjusting friction.

(15) Temporarily install adjustable stop on bar of stop and guard with bolt and washer. Install special bolts and washers and special bolts, spacers, and washers through stop, stop and guard, sleeve, and collet halves.

NOTE

Pilot's friction grip is repaired by replacing parts.

b. On collective sticks, 70400-01401-048 and subsequent, repair friction grip as follows (Figure 11-4-39-3):

(1) Remove bolt, washer, and adjustable stop from friction grip.

(2) Unscrew friction grip from sleeve.

(3) Remove retaining ring, retainer, and scraper ring from friction grip.

(4) Remove stop segment from friction grip (Figure 11-4-39-3, Detail A).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

(5) Clean all parts in dry-cleaning solvent, Item 124, Appendix D. Blow dry with dry compressed air.

(6) Remove special bolts, washers, and spacer from sleeve (Figure 11-4-39-3).

(7) Remove stop and guard from sleeve.

(8) Remove collet halves and seal from sleeve.

(9) Inspect threads of friction grip and sleeve for nicks, dents, or any damage that would cause them to seize when screwed together.

(10) Check that ring on friction grip is bonded to grip. If loose, or if lip is damaged, remove ring from friction grip.

(11) Thread friction grip in and out of sleeve to make sure it turns easily and smoothly for entire length of threads. Replace both parts if there is any doubt.

(12) Make sure collet halves are clean and free from any damage that might stop them from sliding smoothly on collective stick.

NOTE

Make sure that phenolic spacers are installed.

(13) Install collet halves and seal in sleeve.

WARNING

Injury to personnel and damage to equipment will result if hardware securing collet halves and adjustable stop is intermixed. Friction lock assembly may jam, resulting in stiff or sticking collective stick movement. Make sure correct hardware is used when installing collet halves and adjustable stop.

(14) Temporarily install stop and guard and adjustable stop on sleeve with bolt, special bolt, and washers.

(15) Temporarily install special bolts, washers, and spacers in sleeve and collet halves.

(16) Temporarily install stop segment on end of friction grip using old self-locking screw (Figure 11-4-39-3, Detail A).

NOTE

When adjusting friction, a new self-locking screw must be used for stop segment (Figure 11-4-39-3).

(17) Install scraper ring, retainer, and retaining ring in front of friction grip.

NOTE

When coating threads of friction grip and sleeve, make sure no grease will get between collective stick and collet halves when installing friction grip.

(18) Lightly coat threads of friction grip and sleeve with grease, Item 149, Appendix D, and thread friction grip in sleeve.

11-4-39.1.3 Install.

a. Check inside replacement friction grip and sleeve. Make sure bushing in grip and collets in sleeve are clean and free of any dirt, nicks, scratches, corrosion or other damage that could stop them from sliding smoothly on stick.

b. Check collective stick for dirt, nicks, scratches, corrosion or other damage that could stop friction lock and sleeve from sliding smoothly on stick.

c. Make sure stick, friction grip, and sleeve are clean and dry.

d. Slide sleeve on collective stick and measure force needed using spring scale. If force is more than 1 ½ pounds, check stick and collets for dirt or damage. Wipe clean if necessary and try sliding force again. Use a different sleeve or collective stick if necessary. Remove sleeve.

e. Slide friction grip on collective stick and measure force needed using spring scale. If force is more then 1 ½ pounds, check stick and grip for dirt or damage. Wipe clean if necessary and try sliding force again. Use a different friction grip or collective stick if necessary. Remove sleeve.

f. If boot was removed, position new boot on sleeve and install tiedown strap, Item 330, Appendix D.

g. Pull boot up to expose sleeve clevis.

h. Install spacer in drag strut. Place drag strut end between ears of sleeve clevis. Install shims between spacer and clevis ears to allow 0.003 - 0.008-inch endplay (Figure 11-4-39-1).

i. Install bolt, washers, nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.**

j. Check endplay again and add shims if necessary. Install cotter pin.

k. If a boot was removed, slide end of boot over clevis and tube.

l. Slide sleeve and friction lock on stick.

WARNING

Injury to personnel and damage to equipment will result if friction lock is adjusted incorrectly, resulting in a collective stick which cannot be moved with friction lock full on. Make sure friction lock is adjusted so that collective stick retains full range of motion when friction lock is full on.

m. On collective sticks prior to 70400-01401-048, adjust friction locks as follows (Figure 11-4-39-4):

NOTE

- As friction grip threads into sleeve it squeezes collet on collective stick.
- Amount of squeeze is set when lug on friction grip contacts stop and guard.
- If stop and guard is moved toward grip, lug will contact stop in less turns; squeeze is less.
- If stop and guard is moved away from grip, lug will contact stop in more turns; squeeze is more.
- Adjustable stop (or bonded stop) is set, so when ring touches stop, friction on stick is fully released. It prevents grip from coming out of sleeve completely and allows pilot to apply friction in only a few turns.
- If friction does not fully release when ring touches stop, check to make sure correct hardware was used to install collet halves and adjustable stop.
- High or fading friction forces may be eliminated by lightly scuffing ID of collet blocks using abrasive cloth, Item 8, Appendix D, or equivalent.

(1) With lock full on, use spring scale, at stick grip just behind lighted panel and move pilot's collective stick up and down. **STICK MUST MOVE WITH 20 - 40 POUNDS PULL.**

(2) If pull is above or below limit, back off friction grip, loosen bolt and special bolts,

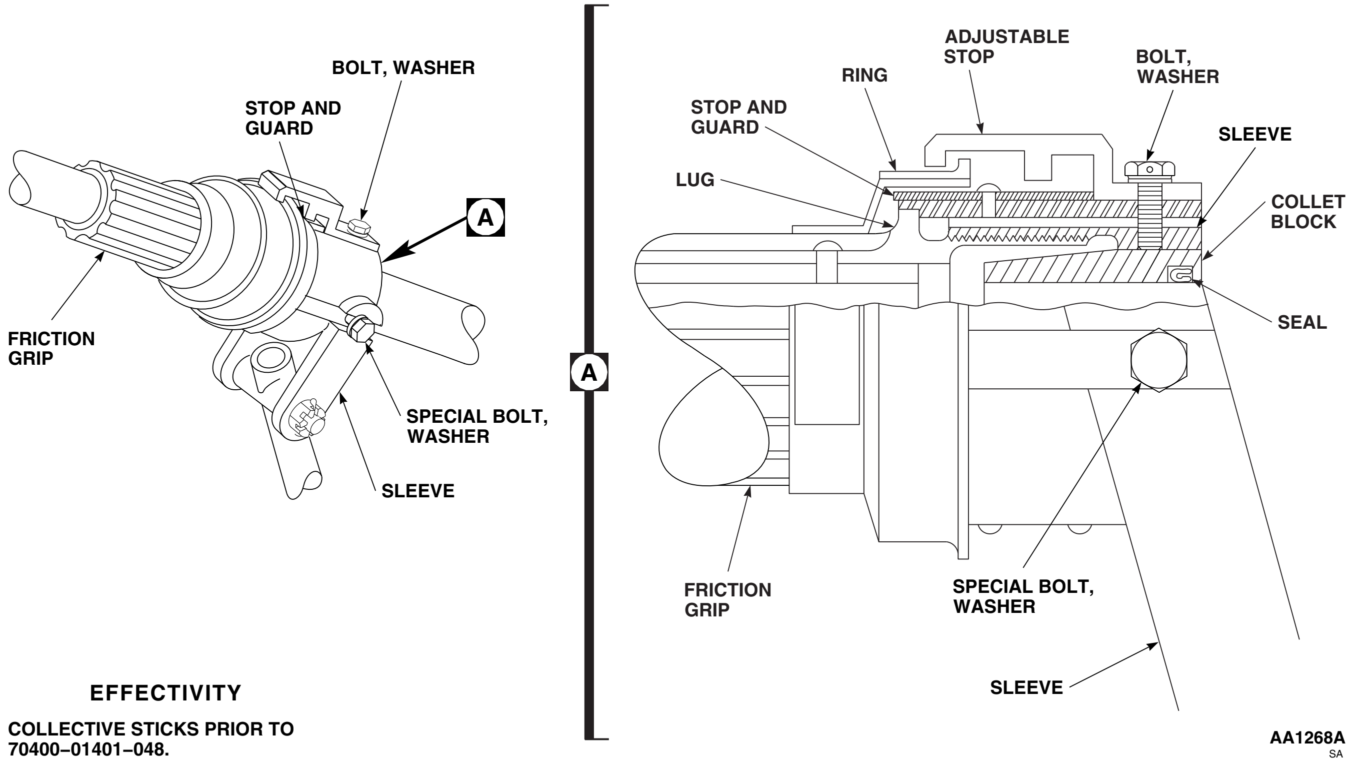


FIGURE 11-4-39-4. ADJUST PILOT'S COLLECTIVE STICK FRICTION LOCK

slide stop and guard in slots in sleeve, forward to loosen, back to tighten. Check force required to move stick.

(3) If more adjustment is needed to loosen friction, turn stop and guard in same direction as loosening friction grip, and install in next set of slots in sleeve.

(4) To tighten friction, turn stop and guard in same direction as tightening friction grip, and install in sleeve.

(5) Adjust stop and guard so friction grip may be backed off without having lug on friction grip strike stop and guard, after backing off one full turn.

(6) Set adjustable stop so friction grip may be backed off from locked position 2 ½ - 4 full turns, before ring touches stop.

(7) Tighten stop and guard attaching bolt (Figure 11-4-39-2). **TORQUE BOLT TO 19 - 21 INCH-POUNDS.**

(8) Using lockwire, Item 197, Appendix D, lockwire stop and guard attaching bolt.

(9) Tighten special bolts. **TORQUE TO 19 - 21 INCH-POUNDS.**

(10) Move collective stick full up and full down and check for binds and smoothness of operation.

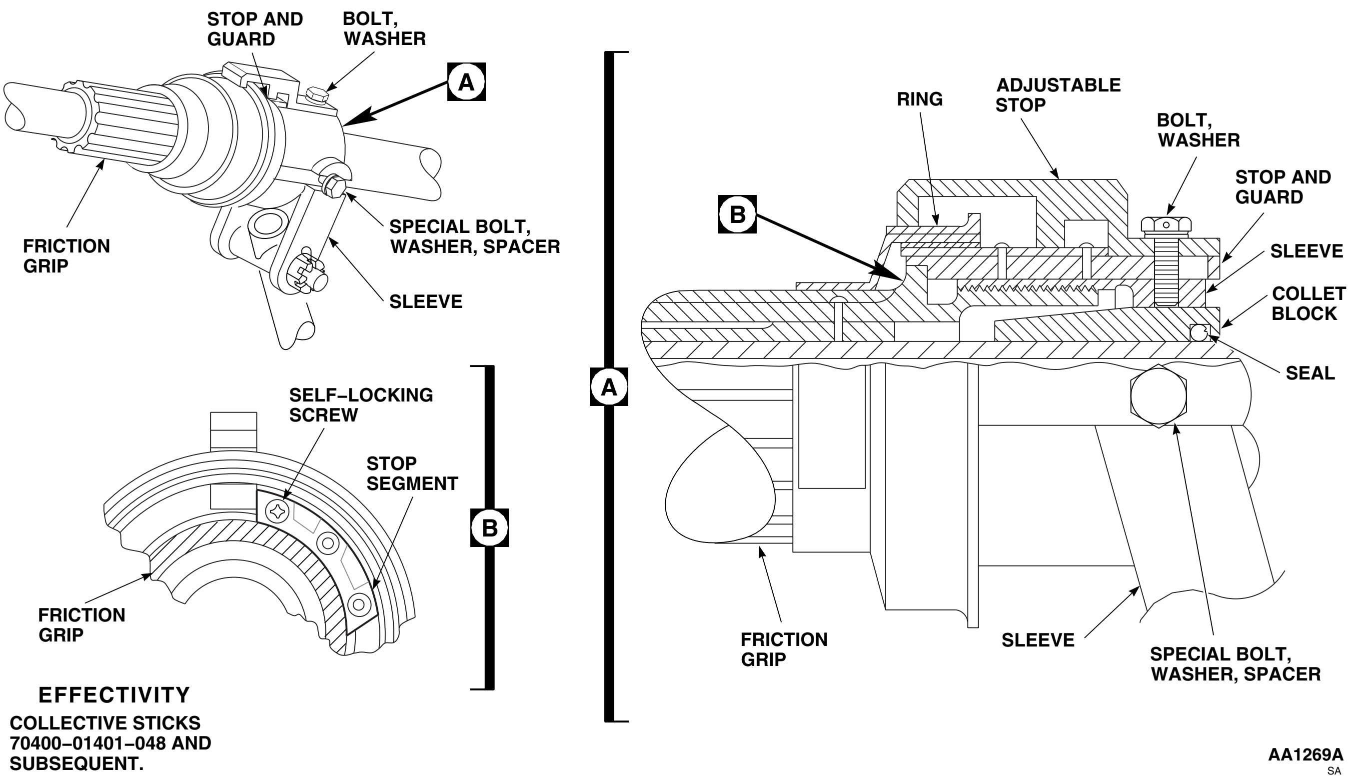


FIGURE 11-4-39-5. ADJUST PILOT'S COLLECTIVE STICK FRICTION LOCK

(11) Place BACKUP HYD PUMP switch OFF. Remove external electrical power (PARA 1-6-16).

WARNING

Injury to personnel and damage to equipment will result if friction lock is adjusted incorrectly, resulting in a collective stick which cannot be moved with friction lock full on. Make sure friction lock is adjusted so that collective stick retains full range of motion when friction lock is full on.

n. On collective sticks, 70400-01401-048 and subsequent, adjust friction lock as follows (Figure 11-4-39-5):

NOTE

- As friction grip threads into sleeve it squeezes collet halves on collective stick.
- Amount of squeeze is set when stop segment on friction grip contacts stop and guard.
- If friction grip is turned toward stop and guard, friction is increased.
- If friction grip is turned away from stop and guard, friction is decreased.
- Adjustable stop is set so when ring touches stop, friction on stick is fully released. This prevents friction grip from coming fully out of sleeve and allows pilot to apply friction in only a few turns.

(1) On upper console, place BACKUP HYD PUMP switch ON.

(2) With lock full on, use spring scale attached to the front end of collective stick grip, move pilot's collective stick up and down. **STICK MUST MOVE WITH 20 - 40 POUNDS PULL.**

(3) On upper console, place BACKUP HYD PUMP switch OFF.

(4) If pull is above or below limits, perform the following:

(a) Make a pencil mark on outside of ring on friction grip to show position of tang on inside of friction grip.

(b) On upper console, place BACKUP HYD PUMP switch ON.

(c) While pulling spring scale attached to collective grip, turn friction grip until desired friction is obtained (20 - 40 pounds pull).

(d) On upper console, place BACKUP HYD PUMP switch OFF.

(e) Note distance between marked position of tang and nearest edge of first slot in sleeve that was passed. This distance will determine hole in stop segment that will be used when installing stop segment.

(f) Unscrew friction grip from sleeve and install stop of stop and guard in slot of sleeve noted in previous step.

(g) Using self-locking screw, install stop segment on friction grip with edge of stop segment as close as possible to stop of stop and guard (Figure 11-4-39-5, Detail B).

NOTE

Do not reuse self-locking screw if friction must be readjusted.

(h) Slide stop and guard to front of collective stick, to make positive contact between stop and stop segment. Check that stop and guard has not slid far enough to prevent friction grip from being backed off at least one turn. If one turn cannot be obtained, use nonmetallic drift and tap stop and guard back until grip can be turned one full turn.

(i) On upper console, place BACKUP HYD PUMP switch ON.

(j) Check to make sure friction grip is in positive stop.

(k) Using spring scale attached to collective grip, check for 20 - 40 pounds pull to move collective stick.

(l) If desired friction was not obtained, unscrew friction grip far enough to get to stop segment. Remove self-locking screw from stop segment and adjust stop segment as needed. Install new self-locking screw in stop segment.

NOTE

Do not reuse self-locking screw.

(m) Reinstall friction grip in sleeve and recheck force to move collective stick.

(n) If correct friction cannot be obtained after adjusting stop segment, remove friction grip from sleeve and reposition stop of stop and guard in next slot of sleeve past pencil mark. Repeat adjustment procedure until correct friction is obtained or replace collet halves.

(o) On upper console, place BACKUP HYD PUMP switch OFF.

(p) Set adjustable stop so friction grip can be backed off from locked position 2 ½ - 4 full turns before ring touches stop.

WARNING

Injury to personnel and damage to equipment will result if hardware securing collet halves and adjustable stop is intermixed. Friction lock assembly may jam, resulting in stiff or sticking collective stick movement. Make sure correct hardware is used when installing collet halves and adjustable stop.

o. Install bolt and washer in adjustable stop (Figure 11-4-39-3). **TORQUE BOLT TO 19 - 21 INCH-POUNDS.**

p. Install special bolt and washer in stop and guard. Install special bolts, washers, and spacers in sleeve and collet halves. **TORQUE BOLTS TO 19 - 21 INCH-POUNDS.** Using lockwire, Item 196, Appendix D, lockwire bolts in pairs.

q. Bond ring in this position using adhesive, Item 32, Appendix D, as follows:

(1) Clean area where ring is to be bonded using clean machinery towel, Item 211, Appendix D, dampened in methyl ethyl ketone, Item 217, Appendix D.

(2) Mix adhesive, Item 32, Appendix D, in ratio of 100 parts by weight of part A to 22 parts by weight of part B.

(3) Apply a light coat of adhesive to area of friction grip where ring is installed. Install ring on friction grip.

(4) Allow adhesive to cure at 120° ± 10°F (49° ± 5°C) for at least 2 hours.

r. If previously installed, attach boot to stick using tiedown strap, Item 332, Appendix D.

s. Install pilot's collective stick grip assembly (PARA 11-4-31).

11-4-40 COLLECTIVE STICK SOCKET.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Reamer Set
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Paint Thinner, Item 232, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 11-4-27
- PARA 11-4-28
- PARA 11-4-41

11-4-40.1 Replace Collective Stick Socket.

NOTE

Replacement of both collective stick sockets is the same, except as noted.

11-4-40.1.1 Remove.

a. Remove pilot's or copilot's collective stick (PARA 11-4-27 or PARA 11-4-28).

b. On pilot's collective stick, remove bolt, washers, and nut securing socket to stick. Remove socket from stick (Figure 11-4-40-1).

c. On copilot's collective stick, remove bolt, washers, special washers, and nut securing socket to stick. Remove socket from stick.

d. If necessary, use paint thinner, Item 232, Appendix D, to dissolve old primer.

e. Check socket assembly as follows:

(1) Inspect socket for nicks, gouges, cracks, and corrosion. Nicks and small gouges up to 0.020-inch deep by 1/16 inch wide x 1/2 inch long should be blended out with abrasive cloth, Item 10, Appendix D. **NO CRACKS OR CORROSION AL-**

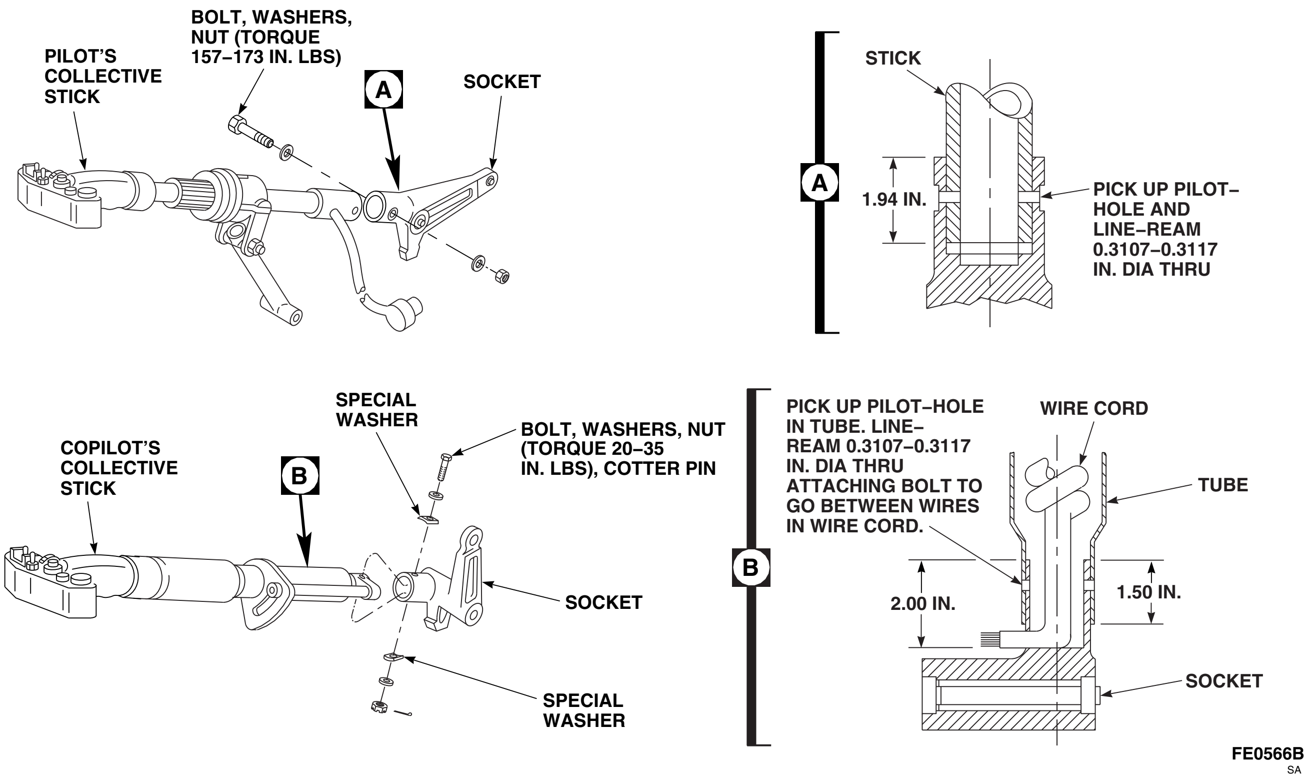


FIGURE 11-4-40-1. REPLACE COLLECTIVE STICK SOCKET

LOWED. If damage is more than allowed, replace socket.

(2) Apply corrosion resistant coating, Item 118, Appendix D, to reworked area. Touch up paint (PARA 2-6-5).

(3) Check bearing rotational drag (PARA 11-4-41).

(4) Inspect bearing for ratcheting, binding and side play. Check bearing radial play with dial indicator. **MAXIMUM RADIAL PLAY IS 0.005 INCH.** Replace worn or damaged bearings (PARA 11-4-41).

11-4-40.1.2 Install.

a. Place socket on stick and line up boltholes (Figure 11-4-40-1).

b. Put bolt through socket and stick. If bolt does not fit, perform the following:

(1) Place stick in socket to depth shown (Figure 11-4-40-1, Detail A and Detail B) and line up pilot-hole and boltholes, and line-ream hole to 0.3107 to 0.3117-inch diameter, using reamer set.

(2) Remove socket and clean away metal chips.

(3) Coat inside of socket hole with primer, Item 257, Appendix D.

c. On pilot's stick, install socket on stick. Install bolt (bolthead facing outboard), washers, and nut (Figure 11-4-40-1). **TORQUE NUT TO 157 - 173 INCH-POUNDS.**

NOTE

- Some copilot's collective stick and socket assemblies may have been assembled with self-locking nuts. When reassembling stick and socket, use bolt, AN175-22, nut, MS17825-5, and cotter pin.
- Make sure bolt goes between wire cord (Figure 11-4-40-1, Detail B).

d. On copilot's collective stick, install socket on stick. Place special washers on each side of tube and install bolt, washers and nut. **TORQUE NUT TO 20 - 35 INCH-POUNDS.** Install cotter pin (Figure 11-4-40-1).

e. Install pilot's or copilot's collective stick (PARA 11-4-27 or PARA 11-4-28).

11-4-41 COLLECTIVE STICK BEARINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Bearing Remover, Step-Type
- Bolt, 5/16-inch diameter, 1-1/2-inch shank
- Nut, 5/16-inch diameter
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Cleaning Cloth, Item 102, Appendix D
- Cotton Swab, Item 119, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Primer, Item 305, Appendix D

References

- Appendix D
- PARA 11-4-40

11-4-41.1 Replace Collective Stick Bearings.

11-4-41.1.1. Remove.

a. Remove socket from stick (PARA 11-4-40).

b. Remove lower bearings and spacer (copilot's) or front bearings and spacer (pilot's) from socket using arbor press and step-type bearing remover (Figure 11-4-41-1).

c. Remove upper bearing (copilot's) or rear bearing (pilot's) from socket using arbor press and step-type bearing remover.

11-4-41.1.2. Install.

NOTE

If new bearing is being installed, do not sand outside diameter.

a. If required, remove residual sealing compound by lightly sanding socket inside diameter(s) and outside diameter of bearing(s) with abrasive cloth, Item 8, Appendix D.

CAUTION

Damage to bearing will result if sealing primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

NOTE

- Make sure all sanding residue is removed.
- After cleaning, do not handle bearing by outside diameter with bare hands.

b. Clean socket inside diameter(s) and outside diameter of bearing(s) using cotton swab, Item 119, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

CAUTION

Damage to bearing will result if sealing primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

c. Using clean cotton swab, Item 119, Appendix D, apply a light coat of sealing primer, Item 305, Appendix D, to inside diameter(s) of socket.

NOTE

- Protect primed surfaces from contamination during drying.
- Do not touch primed surfaces with bare hands.

d. Allow primer to dry 15 minutes minimum. Primer must be visibly dry. If any break in primer coat occurs, repeat step b. and step c.

e. Install lower bearings and spacer (copilot's) or front bearings and spacer (pilot's) as follows (Figure 11-4-41-1):

CAUTION

Damage to bearing will result if primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

NOTE

Bearing must be installed within 4 minutes of application of sealing compound to socket inside diameters. If this time is exceeded, step b. through step d. must be repeated.

(1) Using clean cotton swab, Item 119, Appendix D, apply an even layer of sealing compound, Item 273, Appendix D to outside diameter of bearings and inside diameters of socket.

(2) Using arbor press, install bearing in one side of socket.

(3) Install spacer in socket and using arbor press, install bearing in socket.

(4) Allow socket assembly to cure for 6 hours minimum.

f. Install upper bearing (copilot's) or rear bearing (pilot's) as follows:

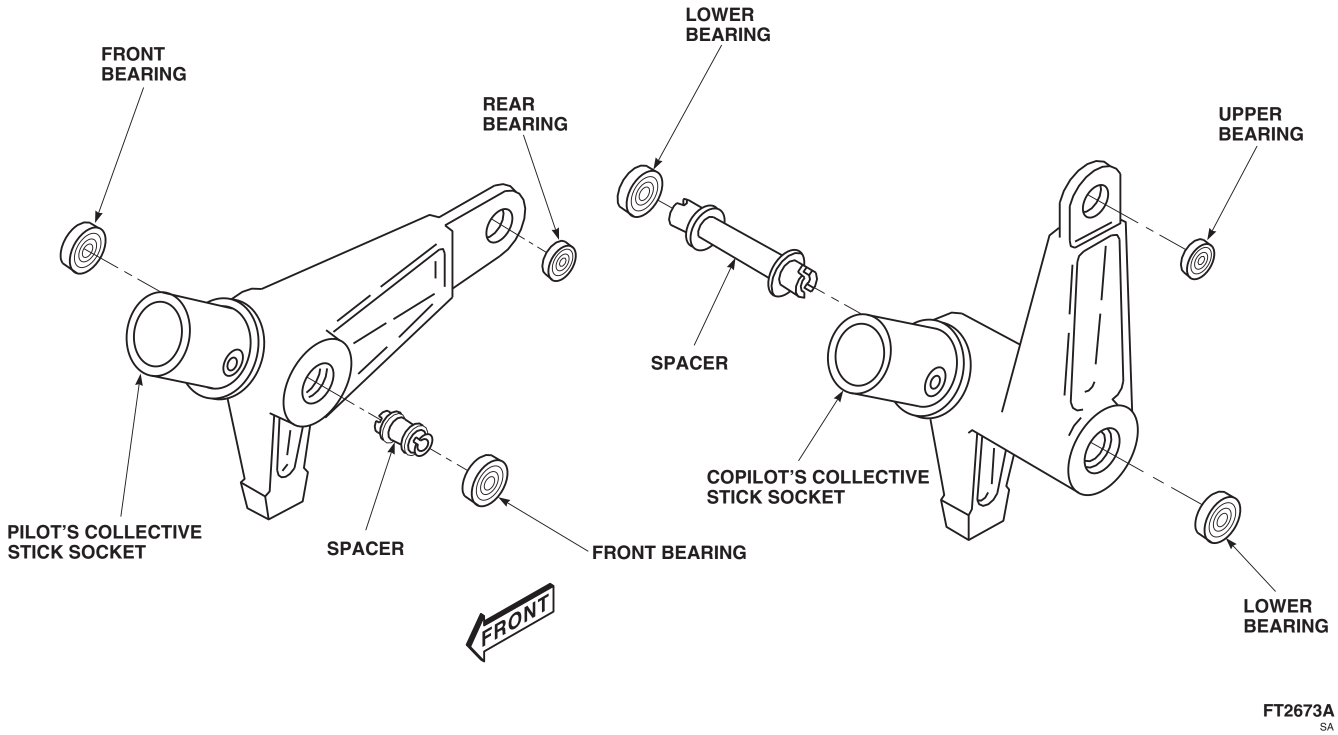


FIGURE 11-4-41-1. REPLACE COLLECTIVE STICK BEARINGS

CAUTION

Damage to bearing will result if primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

NOTE

Bearing must be installed within 4 minutes of application of sealing compound to socket inside diameters. If this time is exceeded, step b. through step d. must be repeated.

(1) Using clean cotton swab, Item 119, Appendix D, apply an even layer of sealing compound, Item 273, Appendix D to outside diameter of bearing and inside diameter of socket.

(2) Using arbor press, install bearing in socket. Bearing must be installed 0.010 to 0.020-inch above left surface of socket.

(3) Allow socket assembly to cure for 6 hours minimum.

CAUTION

Damage to bearing will result if sealing primer, solvent, or sealing compound is allowed to seep into bearings. Do not allow primer, solvent, or sealing compound to come in contact with bearings.

g. Remove sealing compound squeezeout using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

h. Perform rotational drag check of bearings as follows:

(1) Install 5/16-inch diameter bolt through bearing(s) and secure with nut.

(2) Using torque wrench and socket, turn bolthead and measure rotational drag of bearings.

(3) **IF DRAG IS GREATER THAN 8 INCH-OUNCES, REPLACE BEARINGS.**

(4) Remove bolt and nut from socket.

11-4-42 PILOT'S DIRECTIONAL AND CYCLIC CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-43
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-15

11-4-42.1 Replace Pilot's Directional and Cyclic Control Rods.

11-4-42.1.1 Remove.

a. Remove soundproofing panels right side of cabin to gain access to right lower support (PARA 2-4-69).

b. Remove cyclic stick boot (PARA 11-4-15).

c. Remove cockpit floor access panels (PARA 2-4-43).

d. Remove access panel at STA 214.0, WL 208.0.

e. Remove self-retaining bolts, washers, and nuts securing lateral, pitch, or yaw control rods to bellcranks at cyclic stick and bellcranks in lower support (Figure 11-4-42-1, Sheet 1, Detail B and Sheet 2, Detail C).

f. Remove control rods through cabin.

g. Remove self-retaining bolts, washers, and nuts securing push-pull rod to cyclic stick yoke bellcrank and lateral bellcrank on frame. Remove push-pull rod (Figure 11-4-42-1, Sheet 1, Detail B).

11-4-42.1.2 Install.

a. Adjust replacement yaw control as follows:

(1) **SET REPLACEMENT YAW CONTROL ROD TO 45.44 INCHES BETWEEN HOLE CENTERS.** Loosen jamnut and adjust as required.

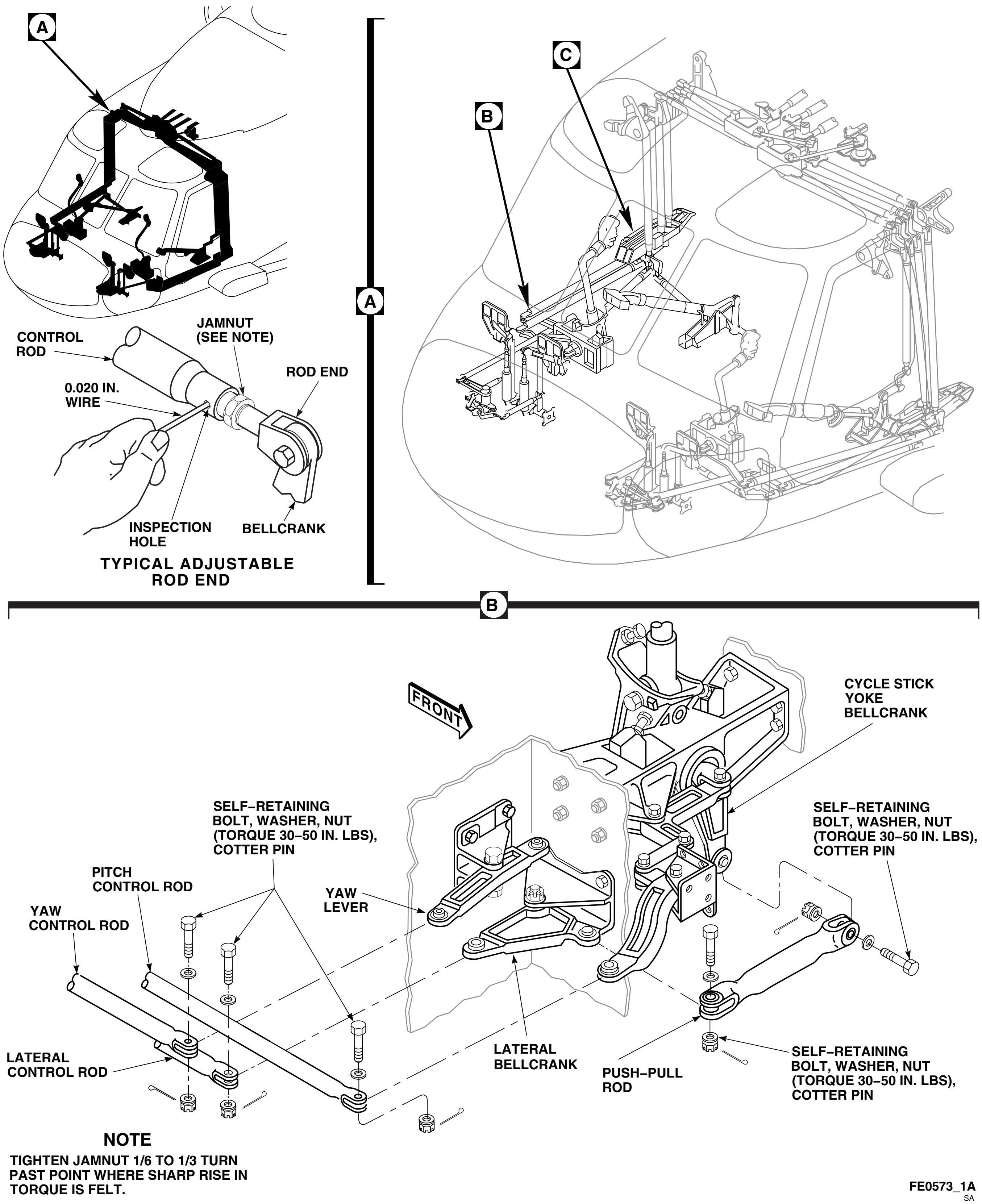


FIGURE 11-4-42-1. REPLACE PILOT'S DIRECTIONAL AND CYCLIC CONTROL RODS (SHEET 1 OF 2)

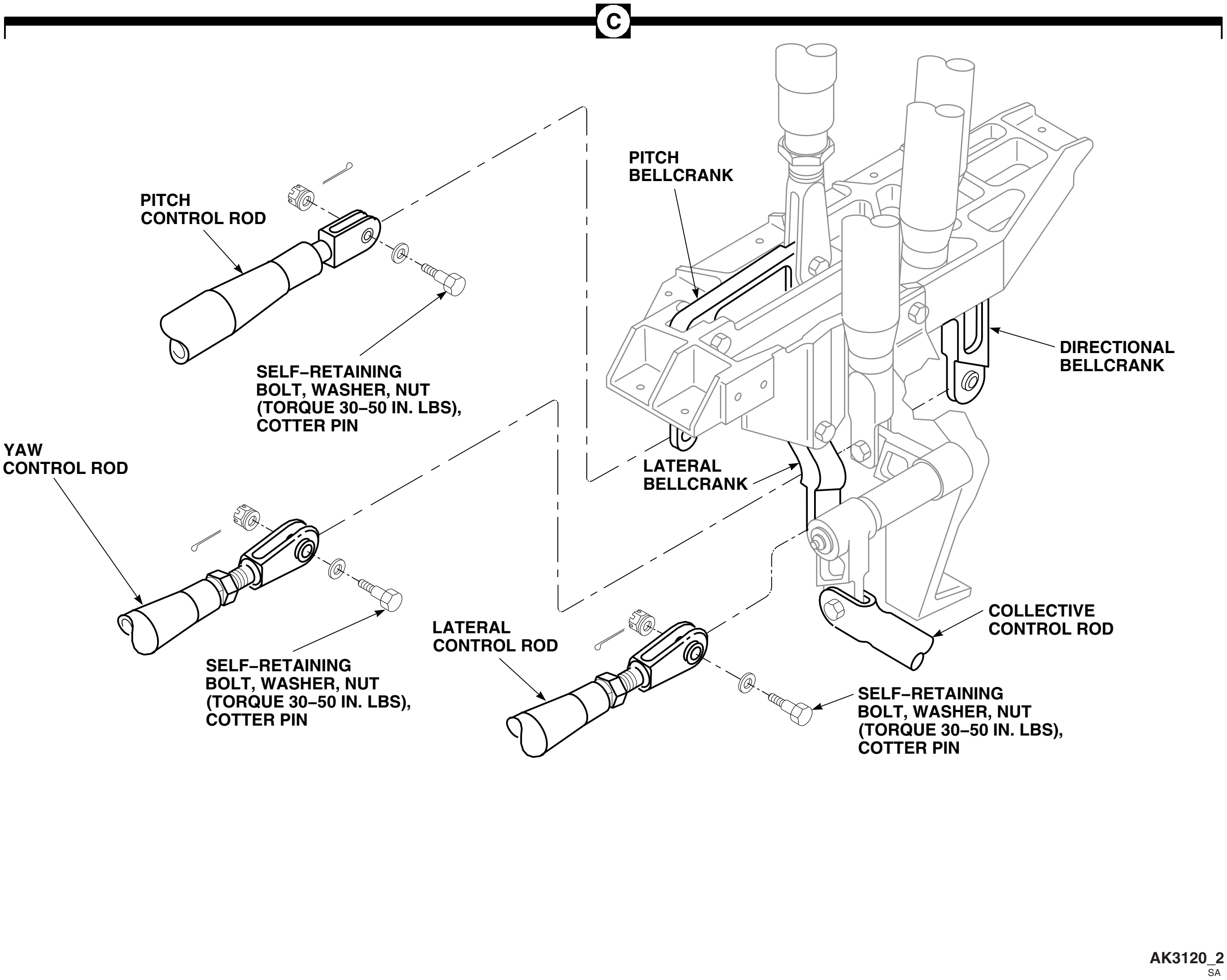


FIGURE 11-4-42-1. REPLACE PILOT'S DIRECTIONAL AND CYCLIC CONTROL RODS (SHEET 2 OF 2)

(2) Insert lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-42-1).

(3) **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

WARNING

Injury to personnel and damage to equipment will result if yaw control rod is improperly installed. Directional control movement will be limited. Make sure control rod is installed with plain bushing side facing up.

b. Install yaw control rod with plain bushing side-up (Figure 11-4-42-1, Sheet 1, Detail B and Sheet 2, Detail C).

c. Install replacement lateral or pitch control rod between bellcranks in lower support and bellcranks at cyclic stick. Install from cabin.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

d. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

e. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

f. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. Install replacement push-pull rod between cyclic stick yoke bellcrank and bellcrank on frame (Figure 11-4-42-1, Sheet 1, Detail B).

h. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

i. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

j. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

k. After replacing pilot's cyclic and directional control rods, perform the following:

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Install rigging pins in yaw pedal adjusters, and in pilot's cyclic stick, so controls are centered (PARA 11-4-2).

(3) Remove lateral rigging pin from cyclic sticks, and push pilot's stick 6.5 inches to right, measured at STICK TRIM switch.

(4) Remove rigging pins from adjusters and depress right pedal 2 3/4 inches measured at pivot on foot plate.

(5) Check lateral bellcrank clearance on right support. **CLEARANCE BETWEEN LATERAL BELLCRANK ARM AND YAW PUSHROD MUST BE 0.090 - 0.120 INCH.** Adjust lateral pushrod, as necessary, to get this clearance.

(6) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

l. Make sure area is clean and free of foreign material.

m. Install cockpit floor access panels (PARA 2-4-43).

n. Install access panel at STA 214.0, WL 208.0.

o. Install cyclic stick boot (PARA 11-4-15).

p. Install soundproofing on right side of cabin (PARA 2-4-69).

11-4-43 COPILOT'S DIRECTIONAL AND CYCLIC CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-43
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-15

11-4-43.1 Replace Copilot's Directional and Cyclic Control Rods.

11-4-43.1.1 Remove.

a. Remove soundproofing panel on left side to get to lower left bellcrank support (PARA 2-4-69).

b. Remove cyclic stick boot (PARA 11-4-15).

c. Remove cockpit floor access panels (PARA 2-4-43).

d. Remove access panel at STA 214.0, WL 208.0.

e. Remove self-retaining bolts, washers, and nuts securing lateral, yaw, and pitch control rods to bellcranks in lower left support, and bellcranks at cyclic stick (Figure 11-4-43-1, Sheet 1, Detail B and Sheet 2, Detail C).

f. Remove control rods through cabin.

g. Remove self-retaining bolts, washers, and nuts securing push-pull rod to cyclic stick yoke bellcrank and bellcrank on frame. Remove push-pull rod (Figure 11-4-43-1, Sheet 2, Detail C).

11-4-43.1.2 Install.

a. Adjust replacement yaw control as follows:

(1) **SET REPLACEMENT YAW CONTROL ROD TO 45.44 INCHES BETWEEN HOLE CENTERS.** Loosen jamnut and adjust as required.

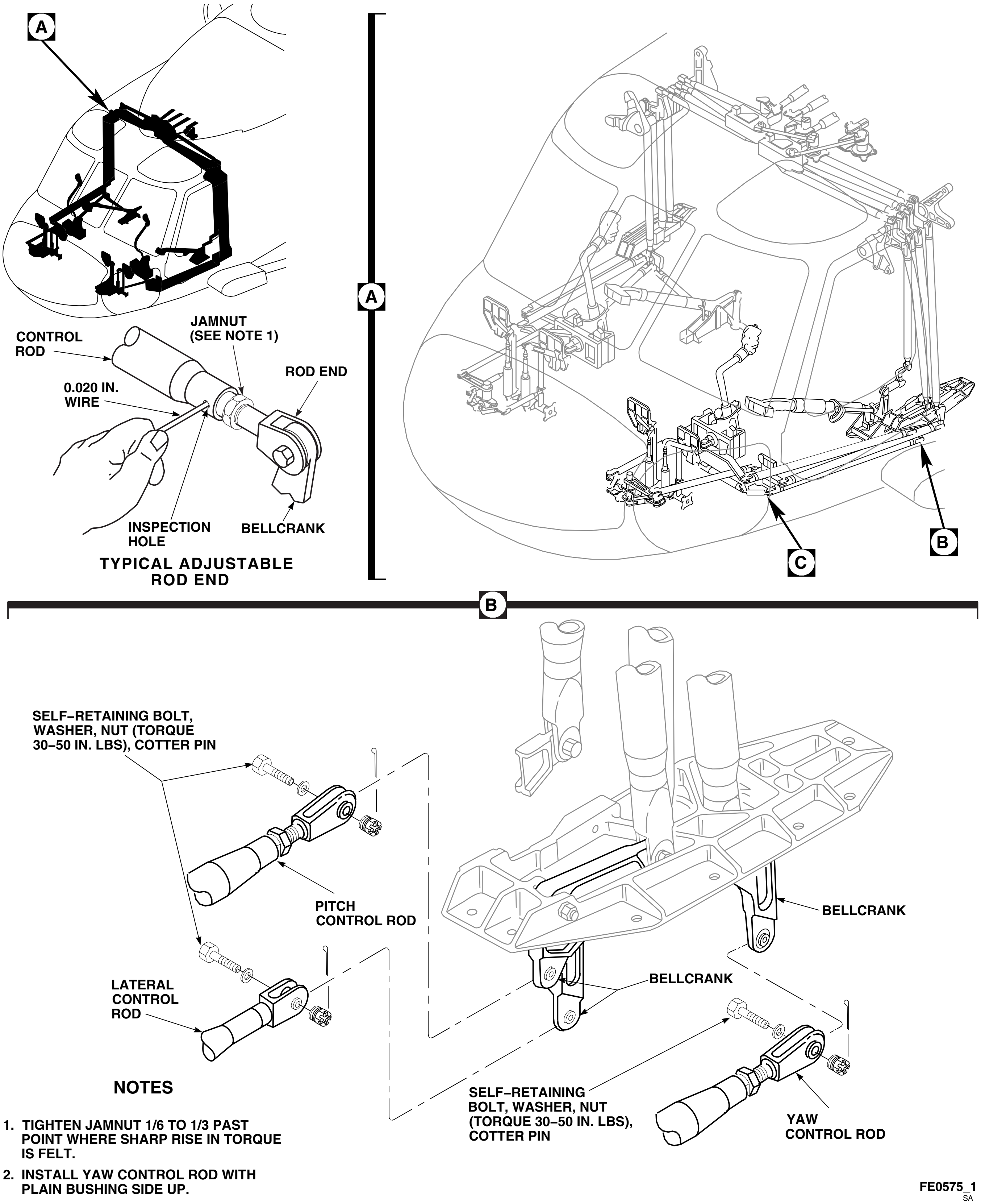


FIGURE 11-4-43-1. REPLACE COPILOT'S DIRECTIONAL AND CYCLIC CONTROL RODS (SHEET 1 OF 2)

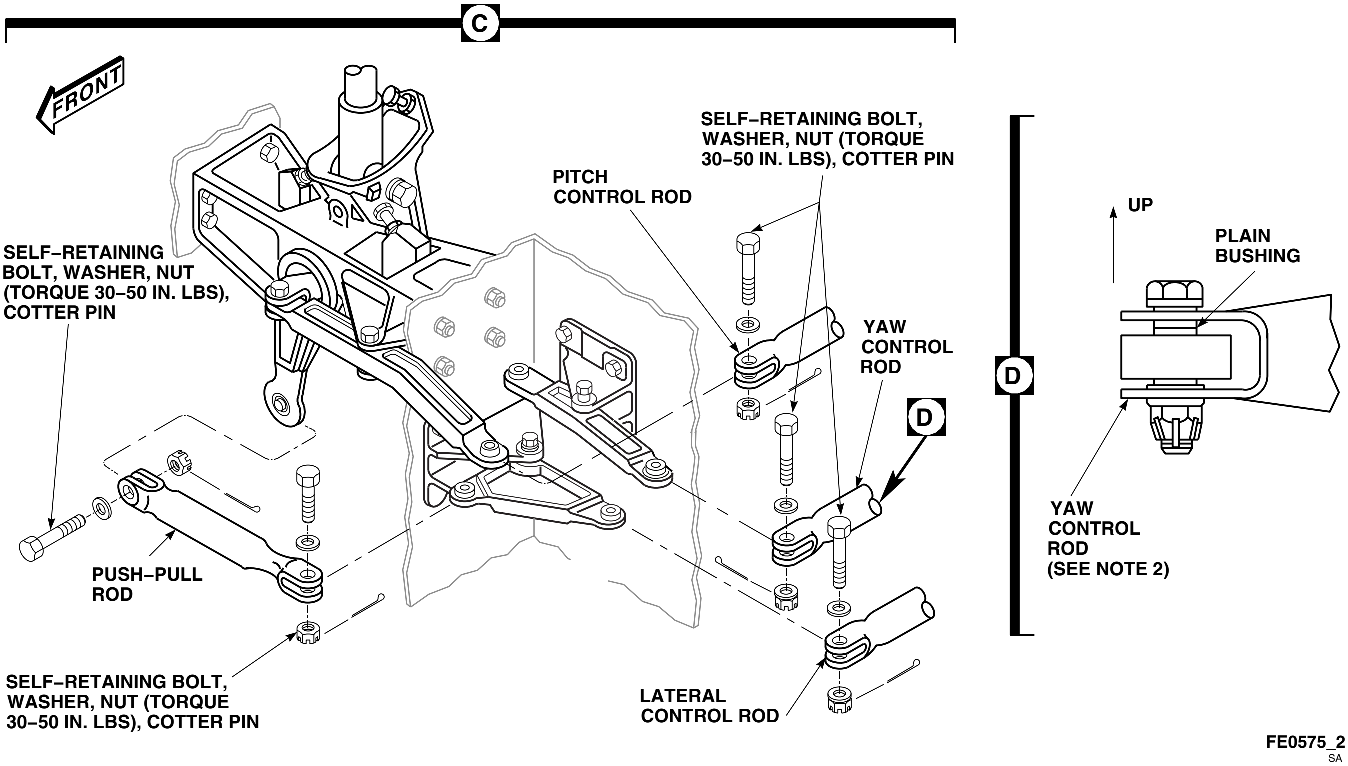


FIGURE 11-4-43-1. REPLACE COPILOT'S DIRECTIONAL AND CYCLIC CONTROL RODS (SHEET 2 OF 2)

(2) Insert lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-43-1, Sheet 1).

(3) **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

WARNING

Injury to personnel and damage to equipment will result if yaw control rod is improperly installed. Directional control movement will be limited. Make sure control rod is installed with plain bushing side facing up.

b. Install yaw control rod with plain bushing side facing up (Figure 11-4-43-1, Sheet 1, Detail B and Sheet 2, Detail C and Detail D).

c. Install replacement lateral or pitch control rod between bellcranks in lower support and bellcranks at cyclic stick. Install from cabin (Figure 11-4-43-1, Sheet 1, Detail B and Sheet 2, Detail C).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

d. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

e. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

f. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. Install replacement push-pull rod between cyclic stick yoke bellcrank and bellcrank on frame (Figure 11-4-43-1, Sheet 2, Detail C).

h. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

i. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

j. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

k. After replacing copilot's cyclic and directional control rods, perform the following:

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Install rigging pins in yaw pedal adjusters and in cyclic sticks so controls are centered (PARA 11-4-2).

(3) Remove lateral rigging pin from cyclic sticks and push copilot's stick 3.5 inches to left, measured at STICK TRIM switch.

(4) Remove rigging pins from pedal adjusters and depress right pedal 2 3⁄4 inches measured at pivot on foot plate.

(5) Check lateral bellcrank clearance on left support. **MAKE SURE THAT LATERAL BELLCRANK ARM CLEARS YAW PUSHROD BY 0.09 TO 0.12 INCH.** Adjust lateral control rod, as necessary, to get this clearance.

(6) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

l. Make sure area is clean and free of foreign material.

m. Install cockpit floor access panels (PARA 2-4-43).

n. Install access panel at STA 214.0, WL 208.0.

o. Install cyclic stick boot (PARA 11-4-15).

p. Install soundproofing panel (PARA 2-4-69).

11-4-44 PILOT'S COLLECTIVE CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1

11-4-44.1 Replace Pilot's Collective Control Rods.

11-4-44.1.1 Remove.

a. Remove soundproofing panels on right side to gain access to lower right support bellcranks (PARA 2-4-69).

b. Remove self-retaining bolts, washers, and nuts securing pilot's short control rod to collective stick and bellcrank (Figure 11-4-44-1, Detail B).

c. Remove self-retaining bolts, washers, and nuts securing pilot's long control rod to bellcrank under collective stick and bellcrank in right support.

11-4-44.1.2 Install.

a. Install replacement long control rod between bellcrank in right support and bellcrank under collective stick (Figure 11-4-44-1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

c. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

d. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. Install replacement short control rod between collective stick and bellcrank.

f. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

g. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

h. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

i. Make sure area is clean and free of foreign material.

j. Install soundproofing panels on right side (PARA 2-4-69).

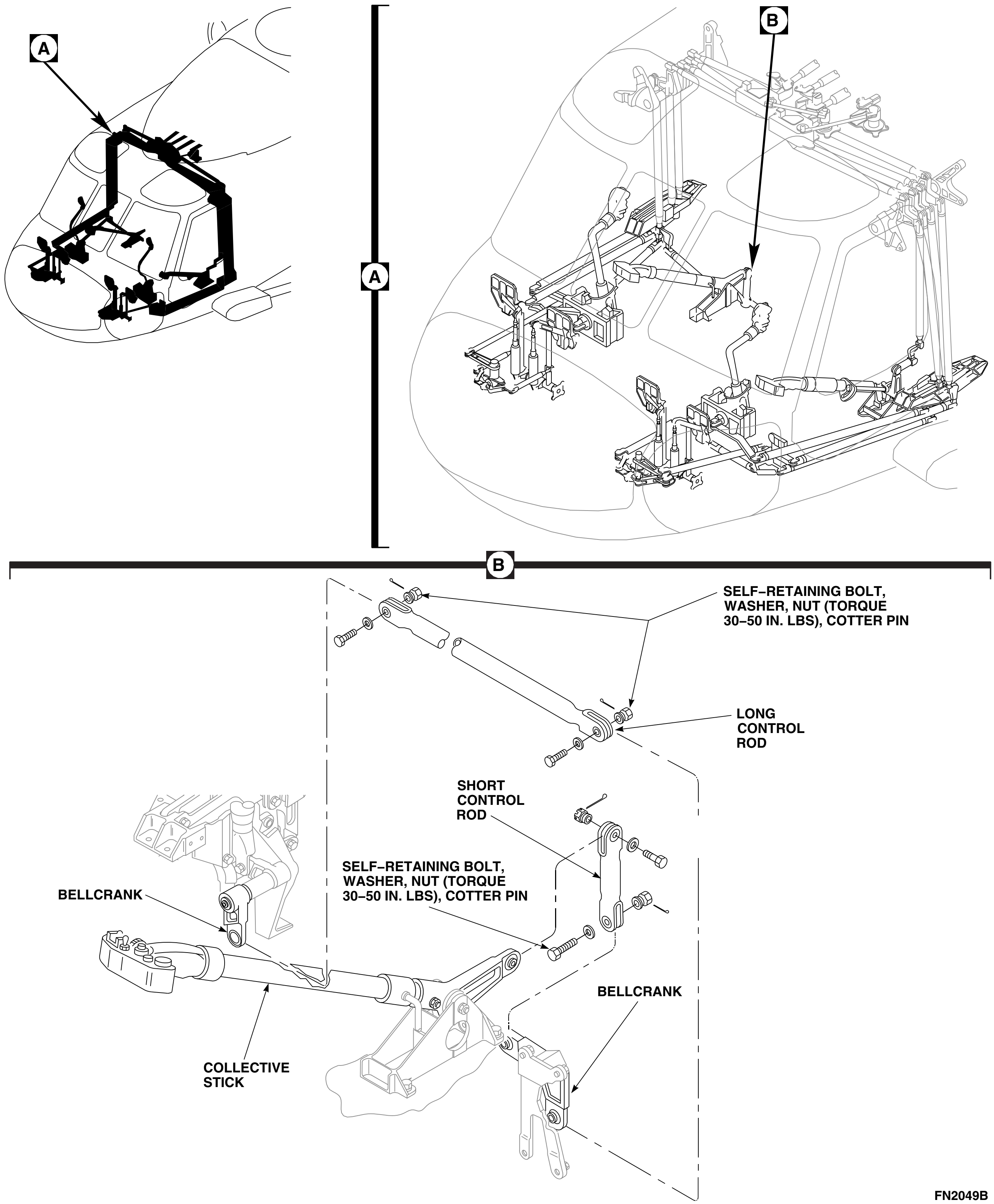


FIGURE 11-4-44-1. REPLACE PILOT'S COLLECTIVE CONTROL RODS

11-4-45 COPILOT'S COLLECTIVE CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1

11-4-45.1 Replace Copilot's Collective Control Rod.

11-4-45.1.1 Remove.

a. Remove soundproofing panels on left side to gain access to collective bellcrank (PARA 2-4-69).

b. Remove self-retaining bolts, washers, and nuts securing collective control rod to collective stick socket and bellcrank on bulkhead (Figure 11-4-45-1, Detail B).

11-4-45.1.2 Install.

a. Install collective control rod between collective stick and bellcrank on bulkhead (Figure 11-4-45-1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

c. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

d. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. Make sure area is clean and free of foreign material.

f. Install soundproofing panels (PARA 2-4-69).

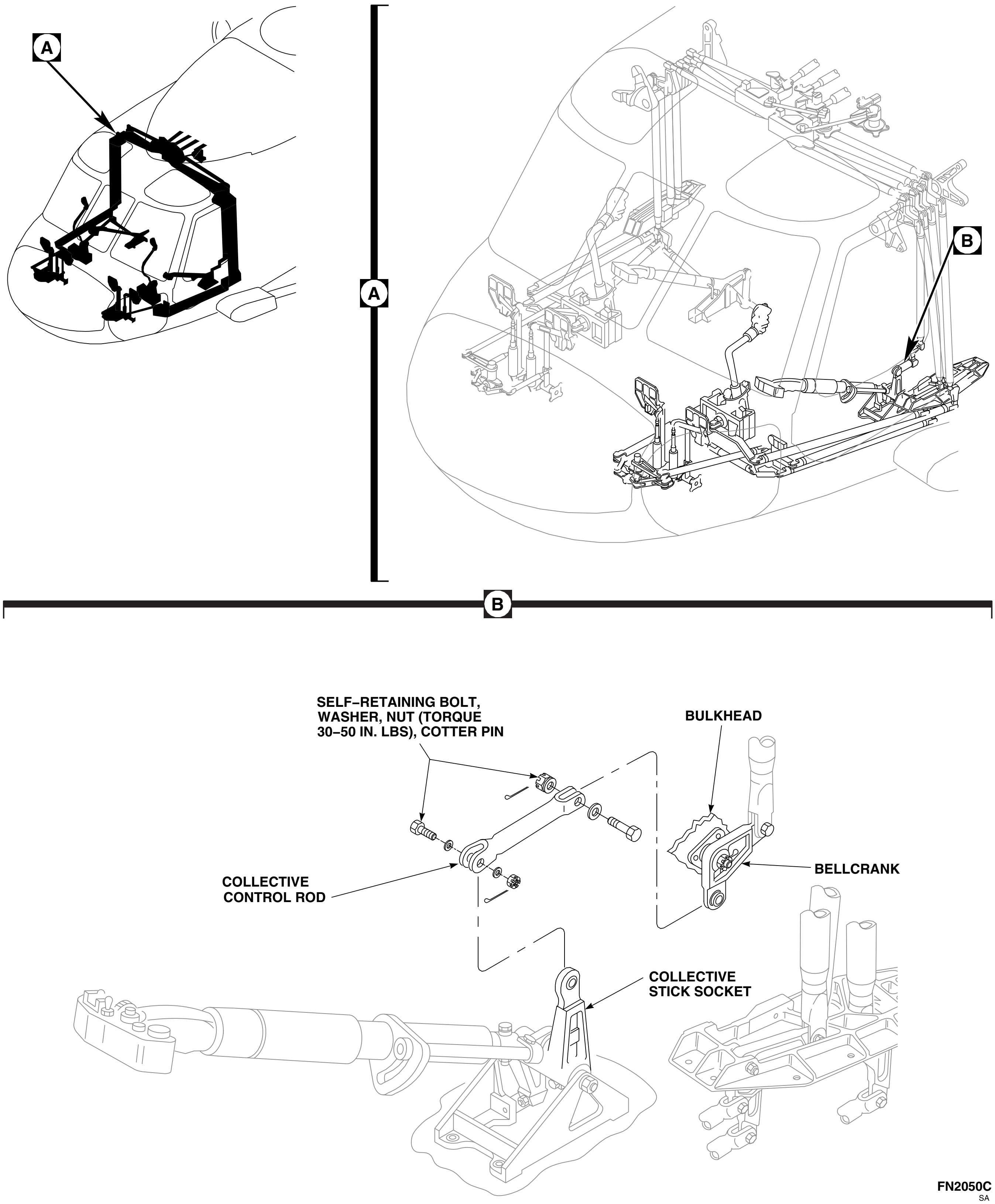


FIGURE 11-4-45-1. REPLACE COPILOT'S COLLECTIVE CONTROL ROD

11-4-46 CABIN CONTROL RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-3
- PARA 11-4-61
- PARA 11-4-62

11-4-46.1 Replace Cabin Control Rods.

NOTE

Replacement of all control rods is the same, except as noted.

11-4-46.1.1 Remove.

a. Remove soundproofing panels from cabin sides and overhead between STA 247.0 and STA 265.0 (PARA 2-4-69).

b. Open main rotor pylon sliding cover.

c. Remove right side control rods as follows:

(1) Remove self-retaining bolts, washers, and nuts securing control rods to bellcranks at lower support (Figure 11-4-46-1, Sheet 1, Detail B).

(2) Remove self-retaining bolts, washers, and nuts securing control rods to upper midsection bellcranks (Figure 11-4-46-1, Sheet 2, Detail C). Remove control rod.

(3) Measure removed adjustable rod from bolthole-center to bolthole-center. Record measurement.

d. Remove overhead control rods as follows:

(1) Release tension from cyclic and collective balance springs (PARA 11-4-61 and PARA 11-4-62).

(2) Remove self-retaining bolts, washers, and nuts securing control rods to pitch lever, lateral lever, or torque shaft lower lever as required (Figure 11-4-46-1, Sheet 2, Detail D, and Sheet 3, Detail E).

(3) Remove self-retaining bolts, washers, and nuts securing control rods to upper midsection bellcrank (Figure 11-4-46-1, Sheet 2, Detail C and Sheet 3, Detail F).

(4) Measure removed adjustable rod from bolthole-center to bolthole-center. Record measurement.

e. Remove left side control rods as follows:

(1) Remove self-retaining bolts, washers, and nuts securing control rods to upper midsection bellcranks (Figure 11-4-46-1, Sheet 3, Detail F).

(2) Remove self-retaining bolts, washers, and nuts securing control rod to bellcranks at lower support (Figure 11-4-46-1, Sheet 4, Detail G).

(3) Measure removed adjustable control rod bolthole-center to bolthole-center. Record measurement.

11-4-46.1.2 Inspect/Repair.

a. Visually inspect control rods for nicks, scratches, and corrosion. If necessary, remove damage using abrasive cloth, Item 10, Appendix D, or abrasive mat, Item 14, Appendix D. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM EXTERIOR DAMAGE IS 0.010-INCH.** If maximum depth is exceeded, replace control rod.

b. Using fluorescent penetrant inspection kit, inspect damaged area (ASTM E1417). Treat repaired surface using corrosion resistant coating, Item 118, Appendix D.

c. Touch up repaired areas with epoxy primer coating kit, Item 135, Appendix D.

d. Touch up primer areas with paint (PARA 2-6-5).

11-4-46.1.3 Install.

WARNING

FSCAP

- Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

- Injury to personnel and damage to equipment will result if lateral control rod is installed incorrectly. To prevent binding of lateral clevis on lower bellcrank during operation, install lateral control tube with adjustable clevis end on top.

a. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Move cyclic or directional controls in cockpit to line up rigging pin holes in control system.

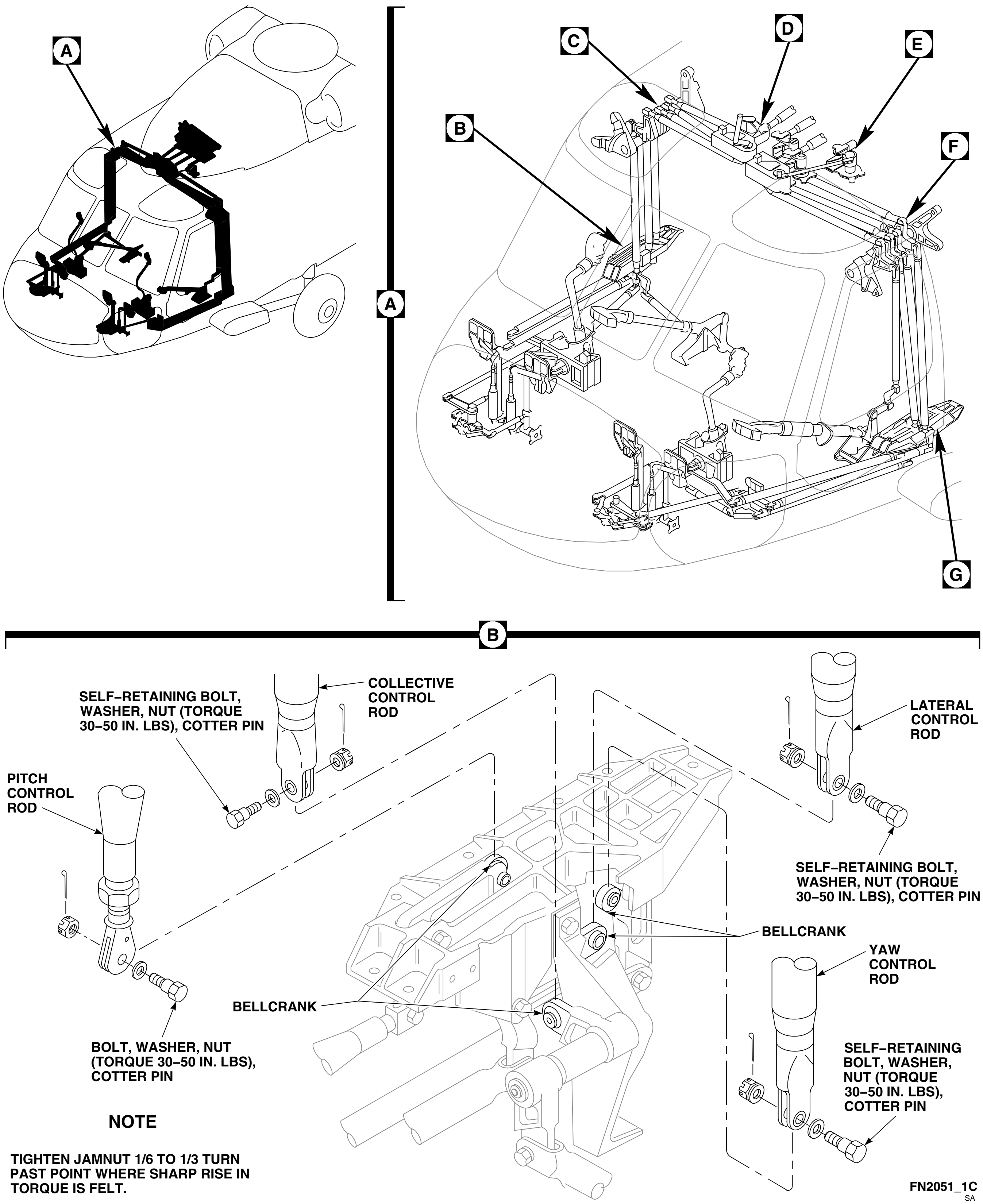


FIGURE 11-4-46-1. REPLACE CABIN CONTROL RODS (SHEET 1 OF 4)

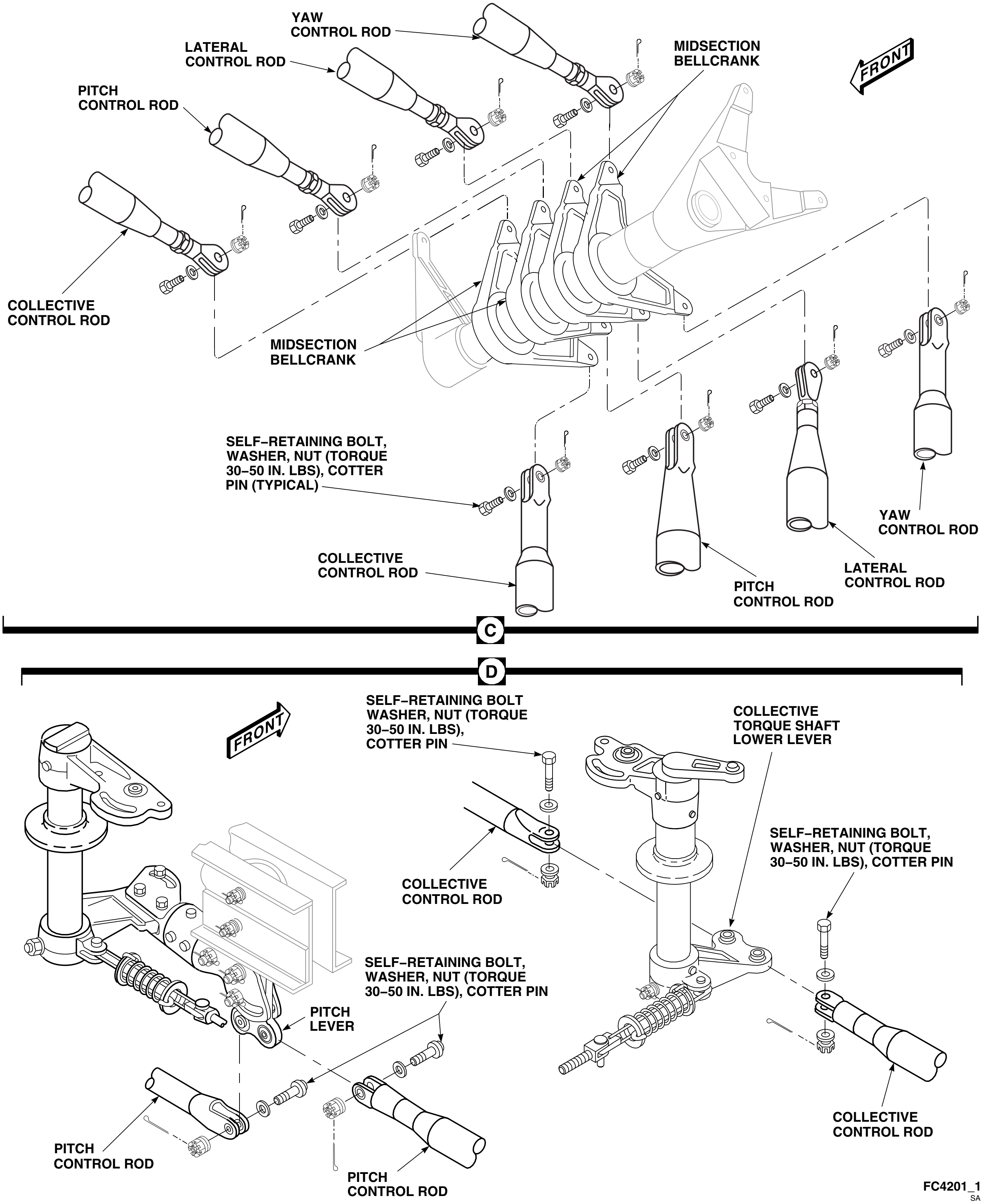


FIGURE 11-4-46-1. REPLACE CABIN CONTROL RODS (SHEET 2 OF 4)

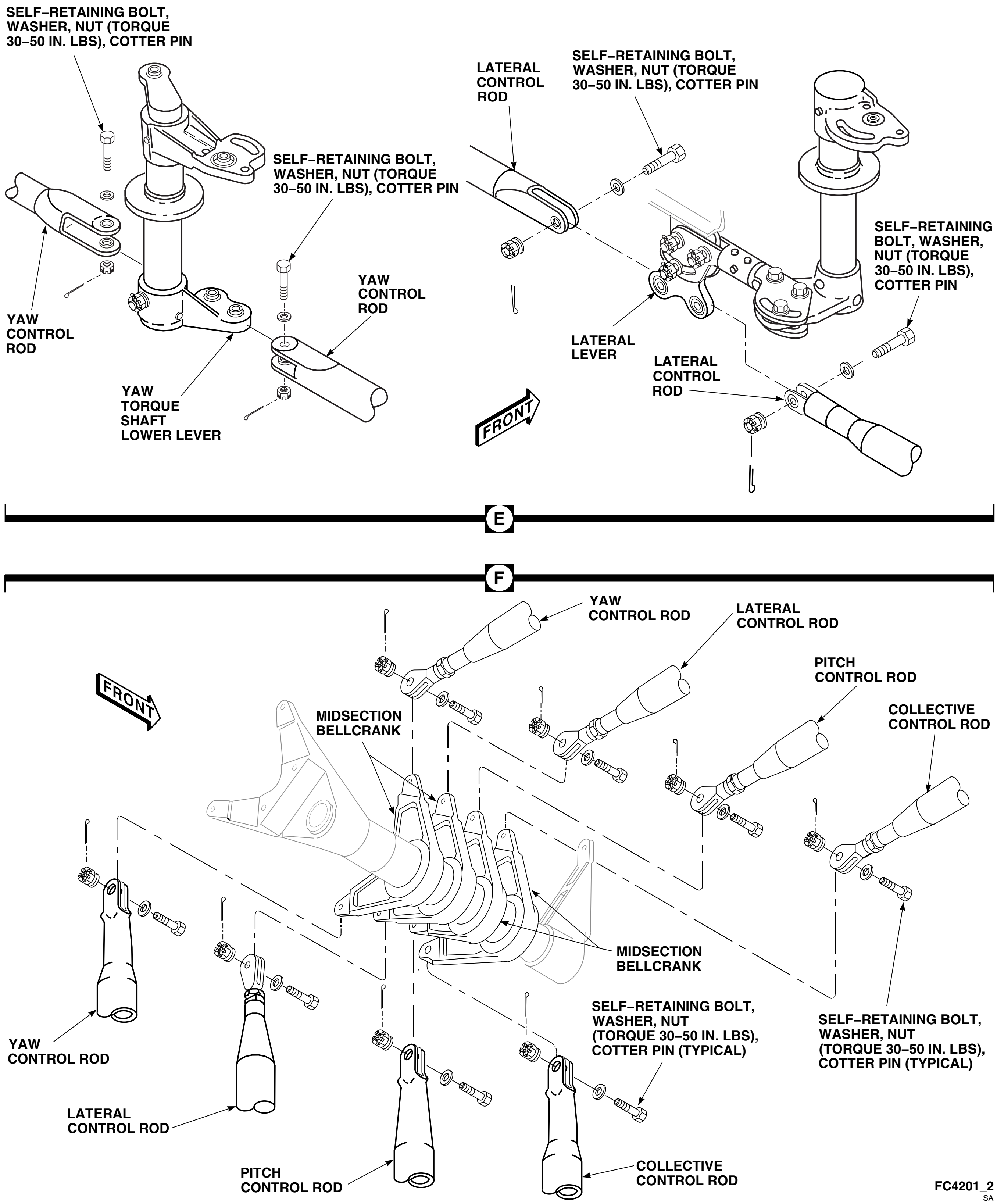
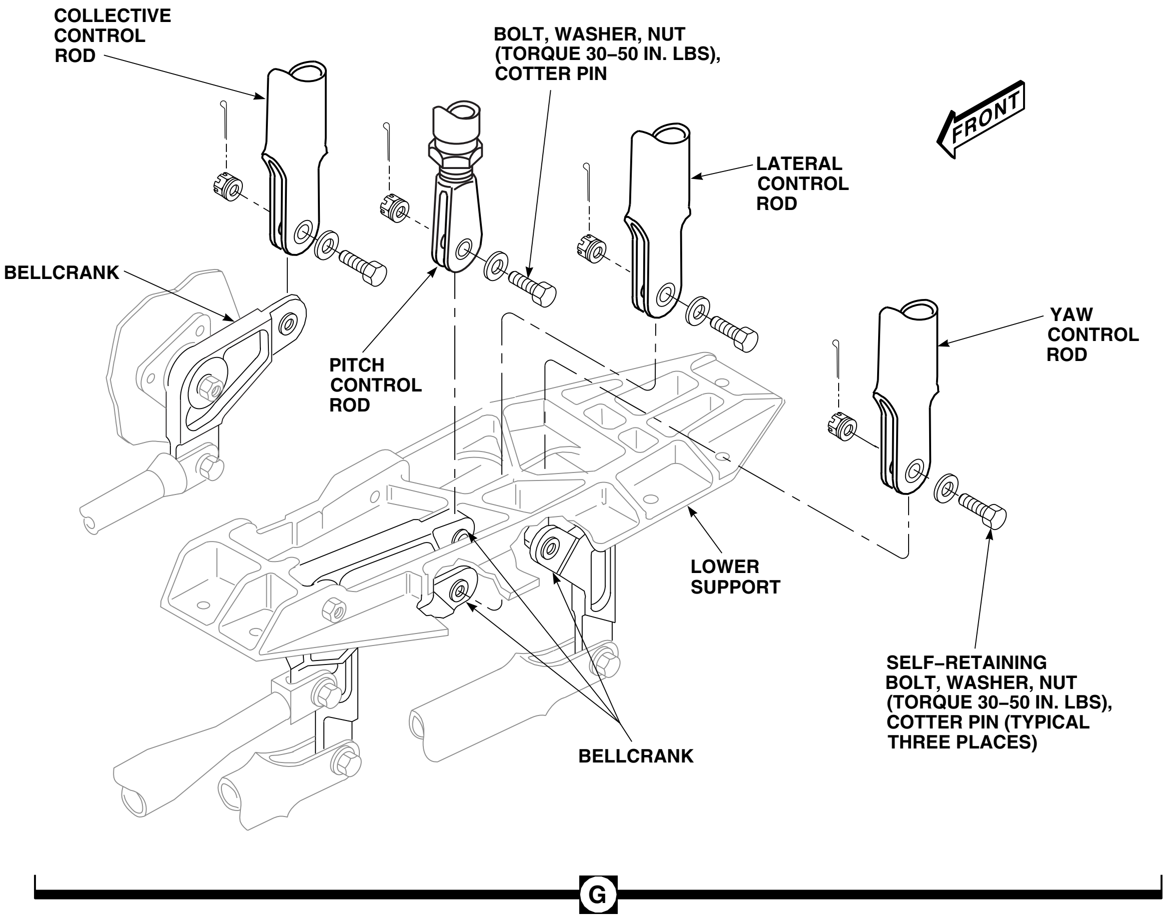
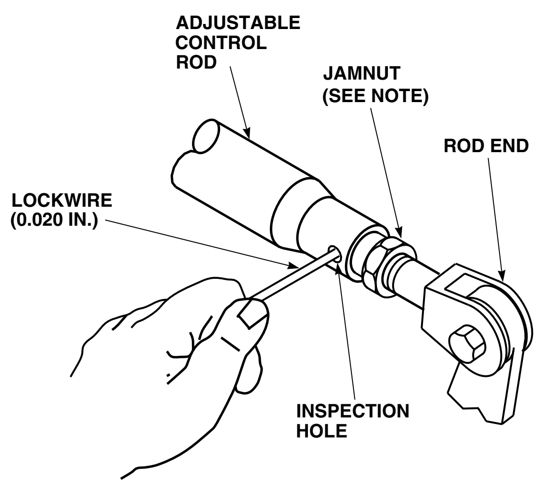


FIGURE 11-4-46-1. REPLACE CABIN CONTROL RODS (SHEET 3 OF 4)





ADJUSTABLE ROD END
(TYPICAL)

FN2051_4B
SA

FIGURE 11-4-46-1. REPLACE CABIN CONTROL RODS (SHEET 4 OF 4)

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

c. Insert rigging pin at rigging pin hole nearest each end of control rod to be installed (PARA 11-4-2).

d. Measure adjustable control rod dimensions from bolthole-center to bolthole-center. If necessary, loosen jamnut and adjust threaded rod end to dimensions of removed rod.

NOTE

If removed control rod is not available for measurement, replacement rod may be installed between bellcranks, and threaded rod end adjusted in place on airframe.

e. If necessary, adjust control rod end in place as follows:

(1) Loosen jamnut and adjust length as required.

(2) Insert 0.020-inch lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-46-1, Sheet 4).

(3) Secure rod end in place with jamnut. **TIGHTEN JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Position control rod in place.

f. Install self-retaining bolt and washers (under bolthead) (Figure 11-4-46-1, Sheet 2, Detail C and Detail D and Sheet 3, Detail E and Detail F) (PARA 11-4-1).

g. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

h. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

i. Remove all rigging pins.

NOTE

If a rigging pin binds during removal, you may have to adjust one of the adjustable control rods. Repeat the above procedures again.

j. Move collective control in cockpit to line up rigging pin hole in control system.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

k. Insert rigging pin at rigging pin hole in collective control system (PARA 11-4-2).

l. Measure adjustable control rod dimensions from bolthole-center to bolthole-center. If necessary, loosen jamnut and adjust threaded rod end to dimensions of removed rod.

NOTE

If removed control rod is not available for measurement, replacement rod may be installed between bellcranks, and threaded rod end adjusted in place on airframe.

m. This adjustment requires that low pitch stop gage dimension at collective stick be held (PARA 11-4-3). Perform the following:

(1) Place 0.010 - 0.030-inch feeler gage in low pitch stop of applicable collective stick. Have assistant hold down collective stick to hold gage dimension, and adjust threaded rod end in place.

(2) Insert lockwire, Item 193, Appendix D, into inspection hole in adjustable control rod. Adjusted threaded rod end must prevent wire from passing through inspection hole (Figure 11-4-46-1, Sheet 4).

(3) Secure rod end in place with jamnut. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Install control rod in place.

n. Install self-retaining bolt and washers (under bolthead) (Figure 11-4-46-1, Sheet 2, Detail C and Sheet 3, Detail F) (PARA 11-4-1).

o. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

p. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

q. Remove rigging pin. Have assistant relax collective stick and remove feeler gage from low pitch stop.

NOTE

If rigging pin binds during removal, you may have to adjust one of the adjustable control rods. Repeat above procedures.

r. On lower end of vertical pitch control rod install bolt, washer, and nut (Figure 11-4-46-1, Sheet 1, Detail B and Sheet 4, Detail G). **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin. Using lockwire, Item 197, Appendix D, lockwire bolthead to clevis.

s. Adjust cyclic and collective balance springs (PARA 11-4-61 and PARA 11-4-62).

t. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

u. Make sure area is clean and free of foreign material.

v. Install soundproofing panels (PARA 2-4-69).

w. Close main rotor pylon sliding cover.

11-4-47 COCKPIT BELLCRANKS AND LEVERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-43
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-15

11-4-47.1 Replace Cockpit Bellcranks and Levers.

11-4-47.1.1 Remove.

NOTE

The length of each arm and measurement from pivot hole to control rod attaching point is not the same on the bellcranks or on the levers. Make sure each one is positioned as shown and as instructed in the steps following, so flight controls rigging will not be changed.

a. Remove control rods, as necessary, from bellcrank or lever to be replaced.

b. At pilot's cyclic stick, perform the following (Figure 11-4-47-1, Sheet 1, Detail A):

(1) Remove self-retaining bolts, washers, and nuts securing yaw lever and lateral bellcrank to airframe supports. Remove yaw lever and lateral bellcrank.

(2) Remove self-retaining bolts, washers, and nuts securing link to forward and aft bellcranks. Remove link.

(3) Remove self-retaining bolts, washers, and nuts securing outboard forward and aft bellcrank to support. Remove outboard forward and aft bellcrank.

(4) Remove self-retaining bolts, washers, and nuts securing forward and aft bellcrank to support and cyclic stick push-pull rod. Remove forward and aft bellcrank.

c. At copilot's cyclic stick, perform the following (Figure 11-4-47-1, Detail B):

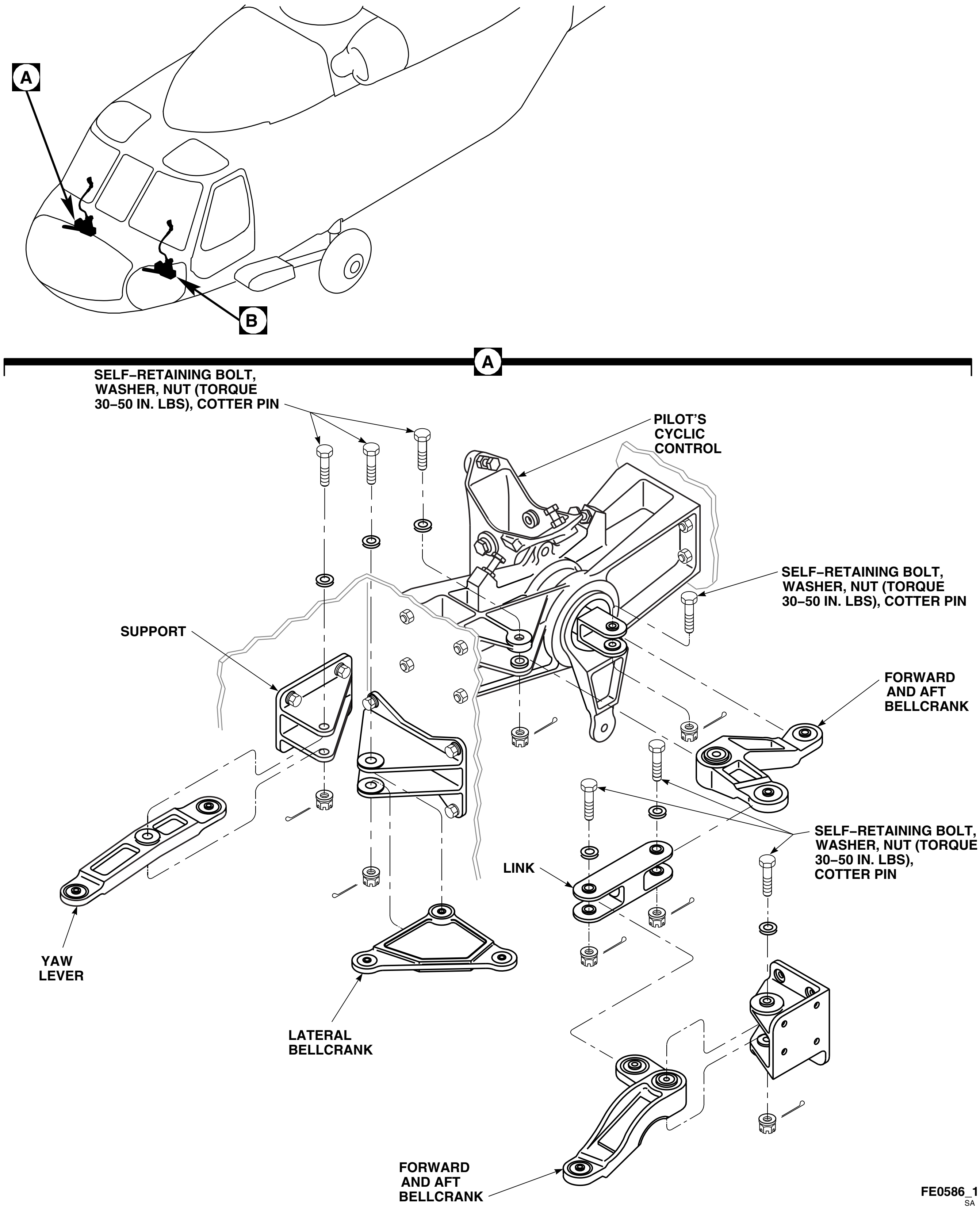


FIGURE 11-4-47-1. REPLACE COCKPIT BELLCRANKS AND LEVERS (SHEET 1 OF 2)

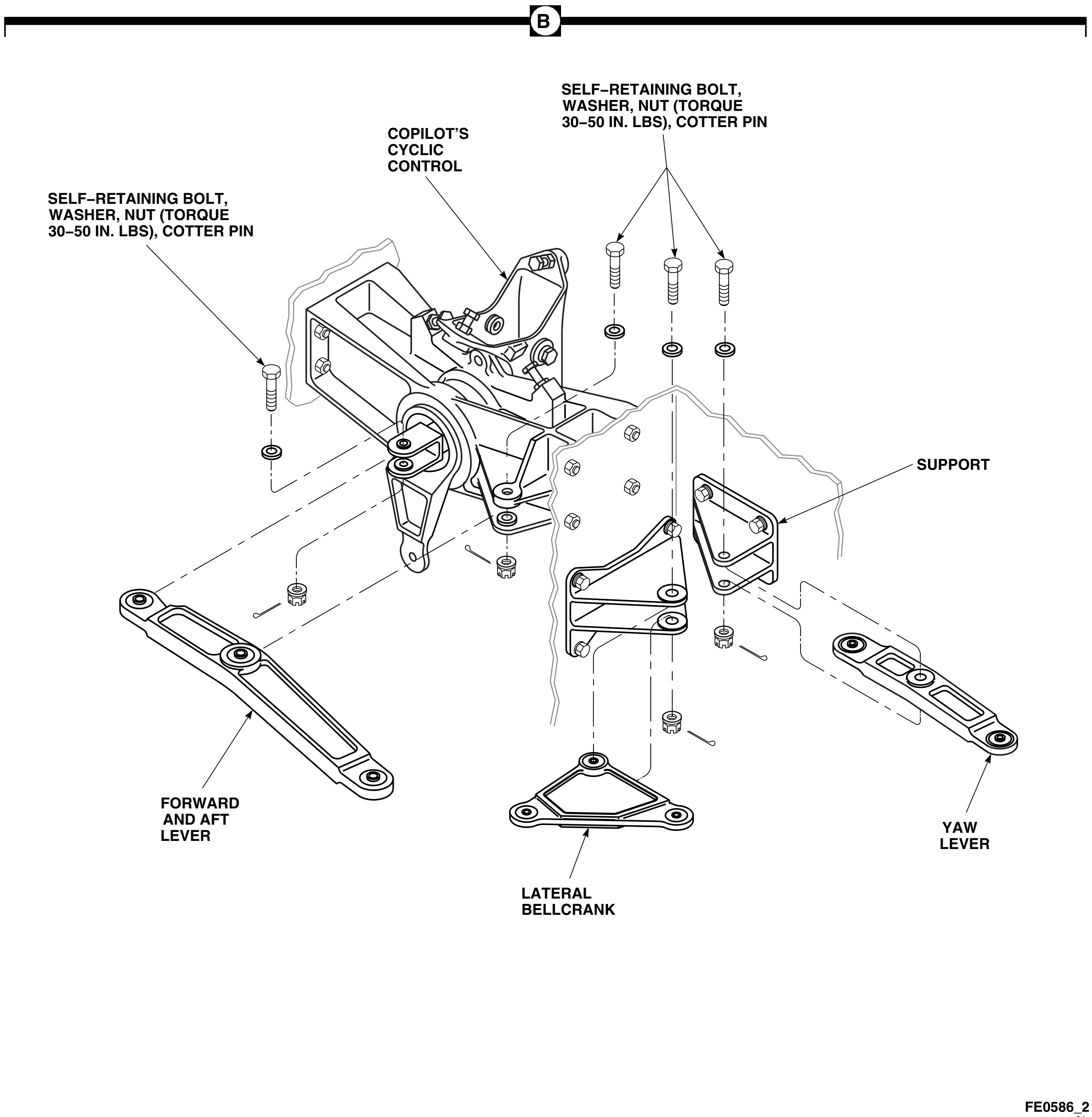


FIGURE 11-4-47-1. REPLACE COCKPIT BELLCRANKS AND LEVERS (SHEET 2 OF 2)

(1) Remove self-retaining bolts, washers, and nuts securing lateral bellcrank and yaw lever to airframe supports. Remove lateral bellcrank and yaw lever.

(2) Remove self-retaining bolts, washers, and nuts securing forward and aft lever to support and cyclic stick push-pull rod. Remove forward and aft lever.

11-4-47.1.2 Inspect.

a. Inspect bearings, 38950-00903-102 and 38950-00902-105, in lateral bellcrank, 70400-01210-041, and bearings, 38950-00903-104 and 38950-00902-105 in bellcrank, 70400-01210-042, using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARING, 38950-00902-105, IS 0.005-INCH. MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARINGS, 38950-00903-102 AND 38950-00903-104, IS 0.0025-INCH.**

b. Inspect bearings, 38950-00903-102, 38950-00903-104, and 38950-00902-105, in yaw lever using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00902-105, IS 0.005-INCH. MAXIMUM ALLOWABLE PLAY IN BEARINGS, 38950-00903-102, AND 38950-00903-104, IS 0.0025-INCH.**

c. Inspect bearings in pilot's forward and aft bellcranks, 70400-01212-042 and 70400-01212-043, using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

d. Inspect bearings in copilot's forward and aft lever using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN ALL BEARINGS IS 0.005-INCH.**

11-4-47.1.3 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. At pilot's cyclic stick, install forward and aft bellcrank as follows (Figure 11-4-47-1, Sheet 1, Detail A):

(1) Install forward and aft bellcrank between cyclic stick push-pull rod and support.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

b. At pilot's cyclic stick, install outboard forward and aft bellcrank as follows:

(1) Install outboard forward and aft bellcrank in support with long arm outboard and curved down.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

c. At pilot's cyclic stick, install link as follows:

(1) Install link between forward and aft bellcranks.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

d. At pilot's cyclic stick, install lateral bellcrank as follows:

(1) Install lateral bellcrank in lower support on airframe with long arm outboard.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. At pilot's cyclic stick, install yaw lever as follows:

(1) Install yaw lever in upper support on airframe with long arm outboard.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. At copilot's cyclic stick, install forward and aft lever as follows (Figure 11-4-47-1, Sheet 2, Detail B):

(1) Install forward and aft lever between cyclic stick push-pull rod and support with long arm outboard and curved down.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. At copilot's cyclic stick, install lateral bellcrank as follows:

(1) Install lateral bellcrank in lower airframe support with long arm outboard.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

h. At copilot's cyclic stick, install yaw lever as follows:

(1) Install yaw lever in upper airframe support with long arm outboard.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

i. Install control rods as required. Do not install floor panels or boots now.

j. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

NOTE

If rigging pins cannot be installed, check that replaced bellcrank or lever is installed correctly. There should be no need to adjust any control rod if bellcrank or lever is installed correctly.

k. Install rigging pins in pedal adjustor, cyclic stick, and torque shafts (PARA 11-4-2).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

l. Remove rigging pins.

m. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

n. Make sure area is clean and free of foreign material.

o. Install floor panels and boots (PARA 2-4-43 and PARA 11-4-15).

11-4-48 CABIN BELLCRANKS AND BELLCRANK SUPPORT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-42
- PARA 11-4-43
- PARA 11-4-44
- PARA 11-4-45
- PARA 11-4-46

11-4-48.1 Replace Cabin Bellcranks and Bellcrank Support.

NOTE

Replacement of pilot's and copilot's cabin bellcrank and bellcrank support is the same, except as noted.

11-4-48.1.1. Remove.

a. Remove soundproofing panels covering flight controls left or right side of cabin (PARA 2-4-69).

b. Remove control rods, as required, from bellcranks (PARA 11-4-42, PARA 11-4-43, PARA 11-4-44, PARA 11-4-45, or PARA 11-4-46).

c. At left support, remove bolts, washers, and nuts securing lateral, longitudinal, and yaw bellcranks to support (Figure 11-4-48-1, Sheet 1, Detail A).

d. Remove bolt, washer, and nut securing copilot's collective bellcrank to bulkhead support.

NOTE

- To aid in assembly, loosely bolt forward support shims to same side of support they were installed on.
- Make sure to note location of rear support shims.

e. Remove bolts, washers, and nuts securing front support and rear support, to right support and structure. Remove shims (Figure 11-4-48-1, Sheet 2, Detail B).

f. Remove bolt, washer, and nut securing right collective bellcrank to front and rear supports.

g. At right support, remove bolts, washers, and nuts securing lateral, longitudinal, and yaw bellcranks to support (Figure 11-4-48-1, Sheet 1, Detail A).

h. Remove bolt, washer, and nut securing pilot's collective bellcrank to support on bulkhead behind pilot's collective stick.

i. Remove bolts, washers, and nuts securing pilot's bellcrank support to bulkhead.

11-4-48.1.2. Inspect.

a. Inspect bearings, 38950-00902-102 or 38950-00903-102, and 38950-00903-104, in yaw and lateral bellcranks using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARING, 38950-00902-102, IS 0.005 INCH. MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARINGS, 38950-00903-102 AND 38950-00903-104, IS 0.0025 INCH.**

b. Inspect bearings, 38950-00903-102 and 38950-00902-105, in longitudinal bellcrank, 70400-02503-041 or 70400-02503-042, using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00903-102, IS 0.0025 INCH. MAXIMUM ALLOWABLE RADIAL PLAY FOR BEARING, 38950-00902-105, IS 0.005 INCH.**

c. Inspect bearings, 38950-00903-104 and 38950-00902-105, in longitudinal bellcrank, 70400-02503-043, -044, using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00903-104, IS 0.0025 INCH. MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00902-105, IS 0.005 INCH.**

d. Inspect bearings, 38950-00903-102 and 38950-00907-102, in right collective bellcrank, 70400-02504-042, and bearings, 38950-00903-104 and 38950-00907-102, in right collective bellcrank, 70400-02504-043, using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARINGS, 38950-00903-102 AND 38950-00903-104, IS 0.0025 INCH. MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00907-102, IS 0.005 INCH.**

e. Inspect bearings, 38950-00903-104 and 38950-00902-105, in copilot's collective bellcrank using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00903-104, IS 0.0025 INCH. MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, 38950-00902-105, IS 0.005 INCH.**

11-4-48.1.3. Install.

NOTE

Length of each arm from pivot hole to control rod attaching point is not the same on each bellcrank. Make sure each one is positioned as shown and as instructed in steps following, so flight controls rigging will not be changed.

a. Install pilot's bellcrank support on bulkhead with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-48-1, Sheet 1, Detail A).

b. Install pilot's collective bellcrank as follows:

(1) Position pilot's collective bellcrank in support, on bulkhead behind pilot's collective stick, with long arm down.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

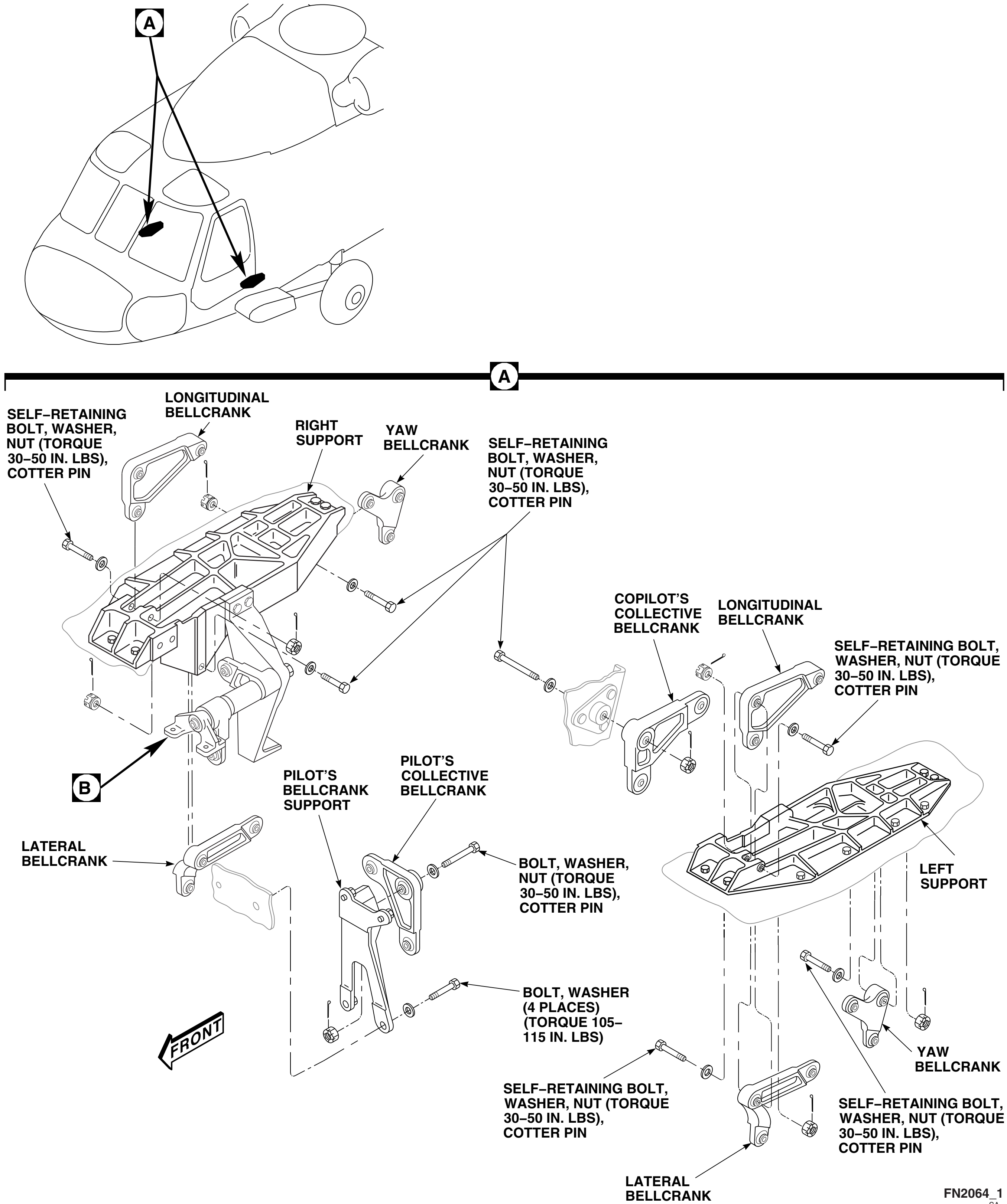


FIGURE 11-4-48-1. REPLACE CABIN BELLCRANKS AND BELLCRANK SUPPORT (SHEET 1 OF 2)

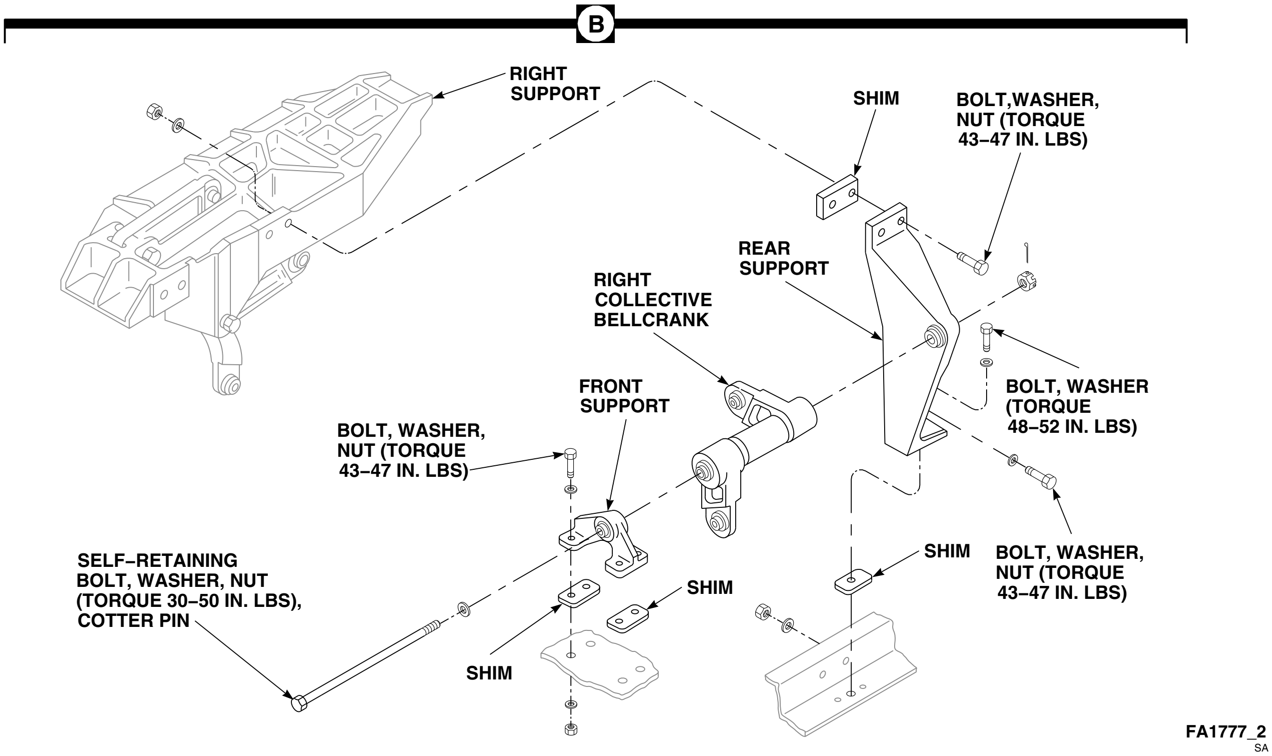


FIGURE 11-4-48-1. REPLACE CABIN BELLCRANKS AND BELLCRANK SUPPORT (SHEET 2 OF 2)

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

c. Install right yaw bellcrank as follows:

(1) Position yaw bellcrank in rear end of right support with long arm down and short arm toward front.

(2) Install self-retaining bolt (bolthead inboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

d. Install right longitudinal bellcrank as follows:

(1) Position longitudinal bellcrank in right support with long arm toward rear.

(2) Install self-retaining bolt (bolthead outboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. Install right lateral control bellcrank as follows:

(1) Position lateral control bellcrank in right support with long arm toward rear.

(2) Install self-retaining bolt (bolthead inboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Install right collective bellcrank as follows (Figure 11-4-48-1, Sheet 2, Detail B):

(1) Position right collective bellcrank between front and rear supports with arm that is oblique closest to front of helicopter.

NOTE

Bellcrank attaching bolthole is not in center between ends of front support. End farthest from bellcrank attaching bolthole faces outboard.

(2) With bolthead at front support, install long self-retaining bolt, washer (under bolthead) and nut. Tighten nut only enough to stop endplay. Do not install cotter pin now.

(3) Install assembly against right support and on structure. Install upper and lower side bolts, washers, and nuts in rear support. Do not torque now.

(4) Check that front support holes line up with holes in structure. If not, front support will have to be turned around.

(5) Install shim under rear support and install bolt and washer. Do not torque now.

(6) Install shims under front support in same place they were removed from. Install bolts, washers, and nuts. Do not torque now.

(7) **TORQUE REAR SUPPORT LOWER BOLT TO 48 - 52 INCH-POUNDS.**

(8) Check that upper and lower side bolts can be turned freely. If they cannot be turned freely, adjust thickness of shim under support so side bolts can turn freely when lower bolt is fully torqued.

(9) **TORQUE REAR SUPPORT NUTS TO 43 - 47 INCH-POUNDS.**

(10) **TORQUE FRONT SUPPORT NUTS TO 43 - 47 INCH-POUNDS.**

(11) Check that self-retaining bolt can move in and out without binding and that bellcrank swings freely. If not, adjust shims under front support to line up bellcrank attaching boltholes in rear support and front support.

(12) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(13) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(14) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. Install copilot's collective bellcrank as follows:

(1) Position copilot's collective bellcrank in support on bulkhead behind copilot, with long arm toward rear.

(2) Install self-retaining bolt (bolthead inboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

h. Install left yaw bellcrank as follows:

(1) Position yaw bellcrank in rear end of left support, with long arm down and short arm toward front.

(2) Install self-retaining bolt (bolthead inboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

i. Install left longitudinal bellcrank as follows:

(1) Position longitudinal bellcrank in left support, with long arm toward rear.

(2) Install self-retaining bolt (bolthead outboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. Install left lateral bellcrank as follows:

(1) Position lateral bellcrank in left support, with long arm toward rear.

(2) Install self-retaining bolt (bolthead inboard), and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

k. Install control rods (PARA 11-4-42, PARA 11-4-43, PARA 11-4-44, PARA 11-4-45, or PARA 11-4-46).

l. Open main rotor pylon sliding cover.

m. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

n. Place collective stick in low pitch.

NOTE

If rigging pins cannot be installed, check that bellcrank has been installed correctly. There should be no need to adjust any control rod if bellcrank has been installed correctly.

o. Install rigging pins in cyclic sticks, pedal adjustors, and overhead torque shafts (PARA 11-4-2).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

p. Remove rigging pins.

q. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

r. Make sure area is clean and free of foreign material.

s. Install floor panels and soundproofing panels as necessary (PARA 2-4-69).

t. Close and latch main rotor pylon sliding cover.

11-4-49 LOWER CABIN BELLCRANK SUPPORTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Micrometer
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Acetone, Item 25, Appendix D
- Adhesive, Item 32, Appendix D
- Cloth, Cleaning, Item 102A, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 11-4-48

11-4-49.1 Replace Lower Cabin Bellcrank Supports.

NOTE

Replacement of left and right lower supports is the same, except as noted.

11-4-49.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove longitudinal, lateral, and yaw bellcranks from lower support being removed (PARA 11-4-48).

c. If removing lower right support, remove front support, rear support, and right collective bellcrank (PARA 11-4-48).

d. If removing lower left support, remove bolts, washers, and nuts securing support assembly to lower support and airframe (Figure 11-4-49-1, Detail A).

e. Remove bolts, washers, and nuts securing lower support to airframe.

f. Inspect shims under lower support for delaminations, disbonding, and corrosion. **NONE ALLOWED.** If delamination, disbonding, or corrosion is found, perform the following:

NOTE

Make sure to note location and thickness of each shim.

(1) Remove shim(s) using nonmetallic scraper.

(2) Measure shim thickness using micrometer.

(3) Clean lower support mounting area using cleaning cloth, Item 102A, Appendix D, dampened with acetone, Item 25, Appendix D. Wipe area dry using clean lint-free cleaning cloth, Item 102A, Appendix D.

11-4-49.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. If replacing shims, perform the following:

(1) Peel shim laminations from replacement shim as necessary to match measurement recorded during removal.

NOTE

- Make sure lower support mounting area is clean before bonding shim(s).
- If more than one shim is used, all shims must be bonded. Maximum of two shims allowed at any bonding point.

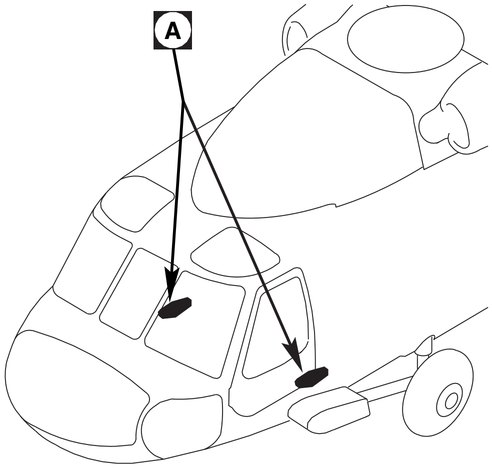
(2) Bond shims, 70400-02283-101, 70400-02283-102, and 70400-02283-103, to airframe at previously recorded locations using adhesive, Item 32, Appendix D (Figure 11-4-49-1, Detail A).

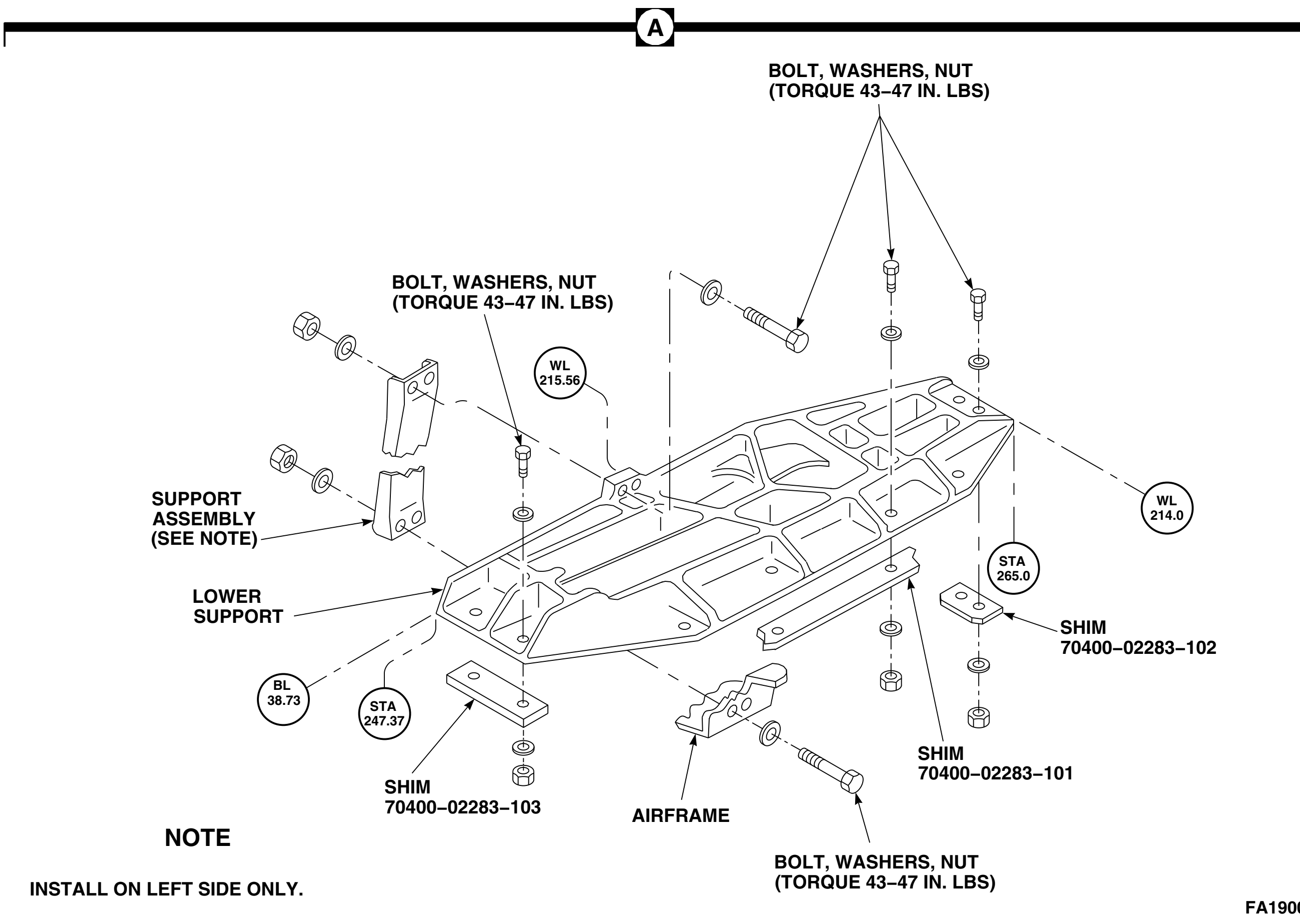
c. Install bolts, washers, and nuts securing lower support to airframe. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

d. If installing left support, install bolts, washers, and nuts securing support assembly to lower support and airframe. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

e. If installing lower right support, install front support, rear support, and right collective bellcrank (PARA 11-4-48).

f. Install longitudinal, lateral, and yaw bellcranks (PARA 11-4-48).





NOTE

INSTALL ON LEFT SIDE ONLY.

FIGURE 11-4-49-1. REPLACE LOWER CABIN BELLCRANK SUPPORTS

11-4-50 MIDSECTION BELLCRANKS AND SHAFT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Feeler Gage
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Frame Shim, 70211-02104-138, Locally-Made, Figure H-146, Appendix H
- Frame Shim, 70211-02104-139, Locally-Made, Figure H-146, Appendix H
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Paint Remover, Item 229, Appendix D
- Tags
- Tiedown Strap, Item 337, Appendix D
- Washers, NAS1149D0416J
- Washers, NAS1149D0463J

References

- Appendix D
- Appendix H
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-4
- PARA 11-5-16

11-4-50.1 Replace Midsection Bellcranks and Shaft.

NOTE

Replacement of left and right midsection bellcranks and shaft is the same, except as noted.

11-4-50.1.1 Remove.

a. Remove soundproofing panels from cabin overhead and sidewalls, STA 247.0 to STA 265.0 (PARA 2-4-69).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

c. Install rig pins in pilot's and copilot's cyclic stick (PARA 11-4-2). If pins do not slide in holes, perform main rotor system rig check (PARA 11-4-4).

d. Install rig pins in pilot's and copilot's yaw pedals (PARA 11-4-2). If pins do not slide in holes, perform main rotor system rig check (PARA 11-4-4).

e. Install rig pins in collective, pitch, roll, and yaw torque tubes (PARA 11-4-2). If pins do not slide in holes, perform main rotor system rig check (PARA 11-4-4).

f. Make sure there is no gap between pilot's collective stick and low stopbolt (PARA 11-4-2).

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Remove bolts, washers, and nuts securing control rods to both ends of bellcranks. Place side control rods to one side. Tie upper control rods up to overhead supports with tiedown straps, Item 337, Appendix D (Figure 11-4-50-1, Sheet 1, Detail A).

i. Remove bolts, washers, and nuts securing spacer assembly. Remove spacer assembly (Figure 11-4-50-1, Sheet 3, Detail C and Detail D).

j. Remove bolt, washer, nut, and radius washer securing shaft to rear support.

k. Remove bolts, washers, and nuts securing both front and rear supports to airframe (Figure 11-4-50-1, Sheet 1, Detail A and Sheet 2, Detail B).

l. Slide supports toward center of shaft. Remove shaft, bellcranks, and supports.

m. Remove components as follows:

(1) On right side installation, slide shaft supports, spacers, and bellcranks from shaft. Tag each piece when removing.

(2) On left side installation, slide shaft supports, spacers, and bellcranks from shaft. Tag each piece when removing.

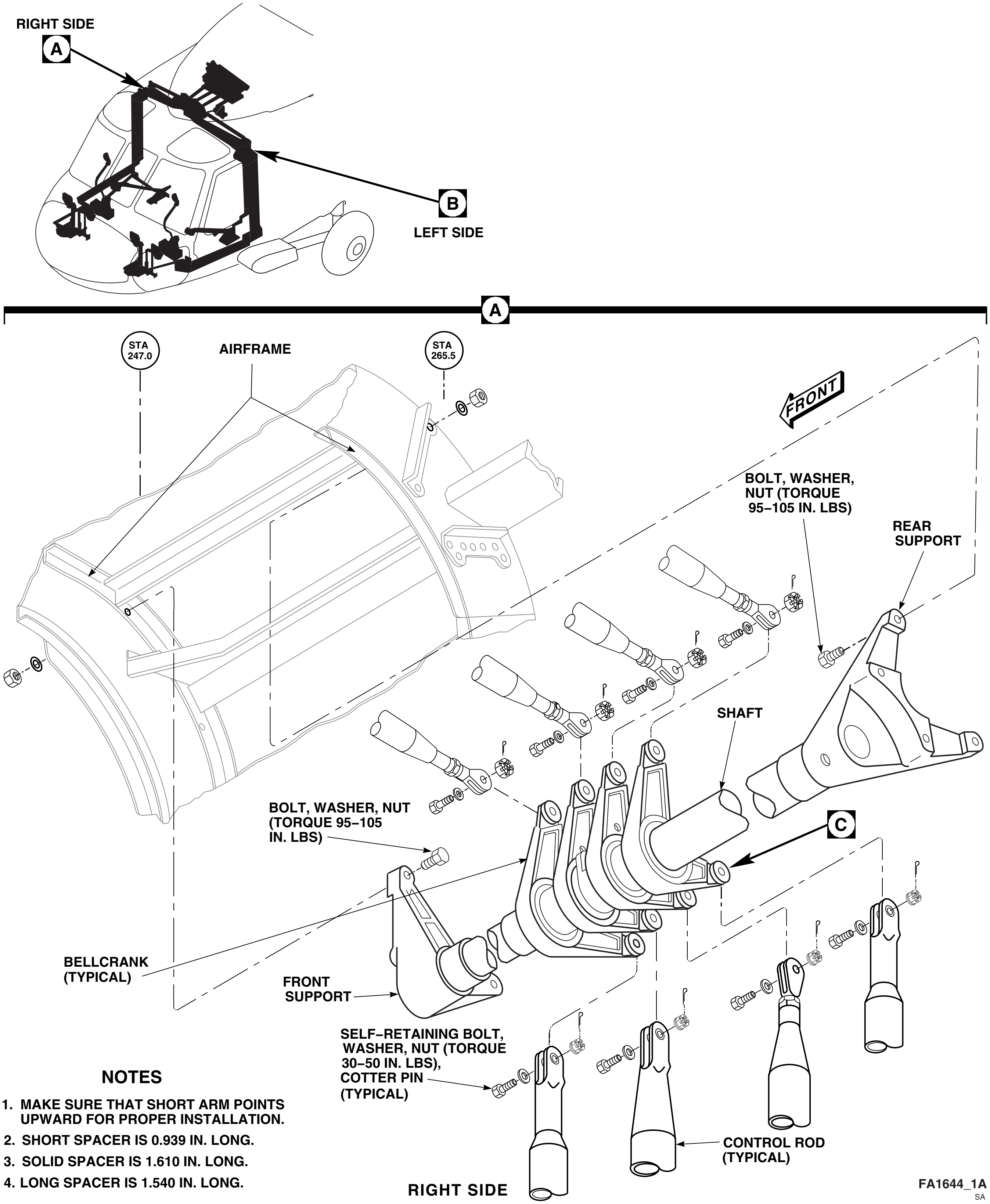
11-4-50.1.2 Inspect.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

a. Inspect large bearings, MKP25B-A1173 (superseded by, 38950-00908-103), in directional, roll, pitch, and collective bellcranks using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play is beyond limits, replace bearing (PARA 11-5-16).

b. Inspect small bearing, SA4-14A1-504 (superseded by, SA4-14A1-512), in collective bellcrank using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play is beyond limits, replace bearing (PARA 11-5-16).



NOTES

1. MAKE SURE THAT SHORT ARM POINTS UPWARD FOR PROPER INSTALLATION.
2. SHORT SPACER IS 0.939 IN. LONG.
3. SOLID SPACER IS 1.610 IN. LONG.
4. LONG SPACER IS 1.540 IN. LONG.

FIGURE 11-4-50-1. REPLACE MIDSECTION BELLCRANKS AND SHAFT (SHEET 1 OF 3)

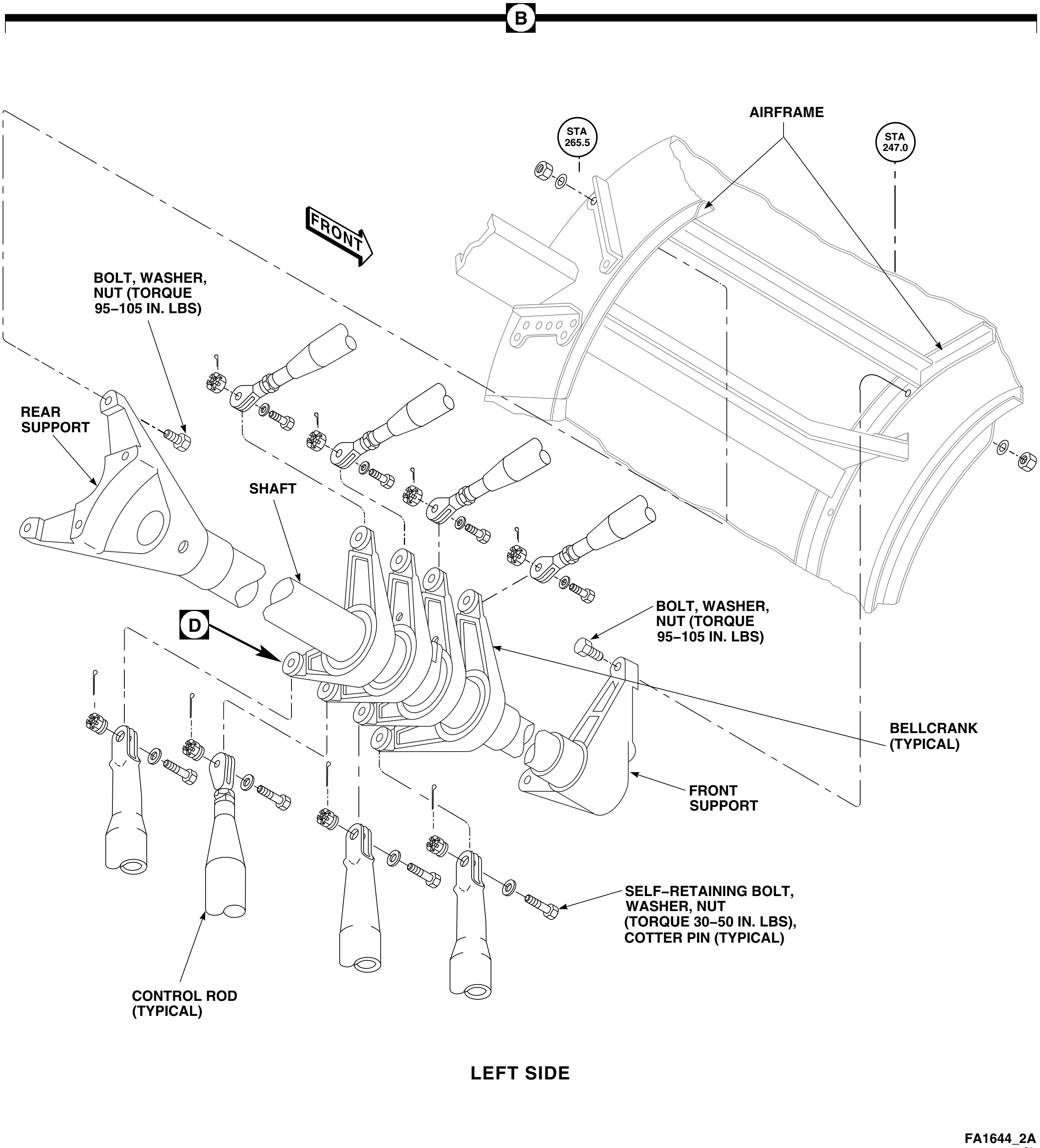
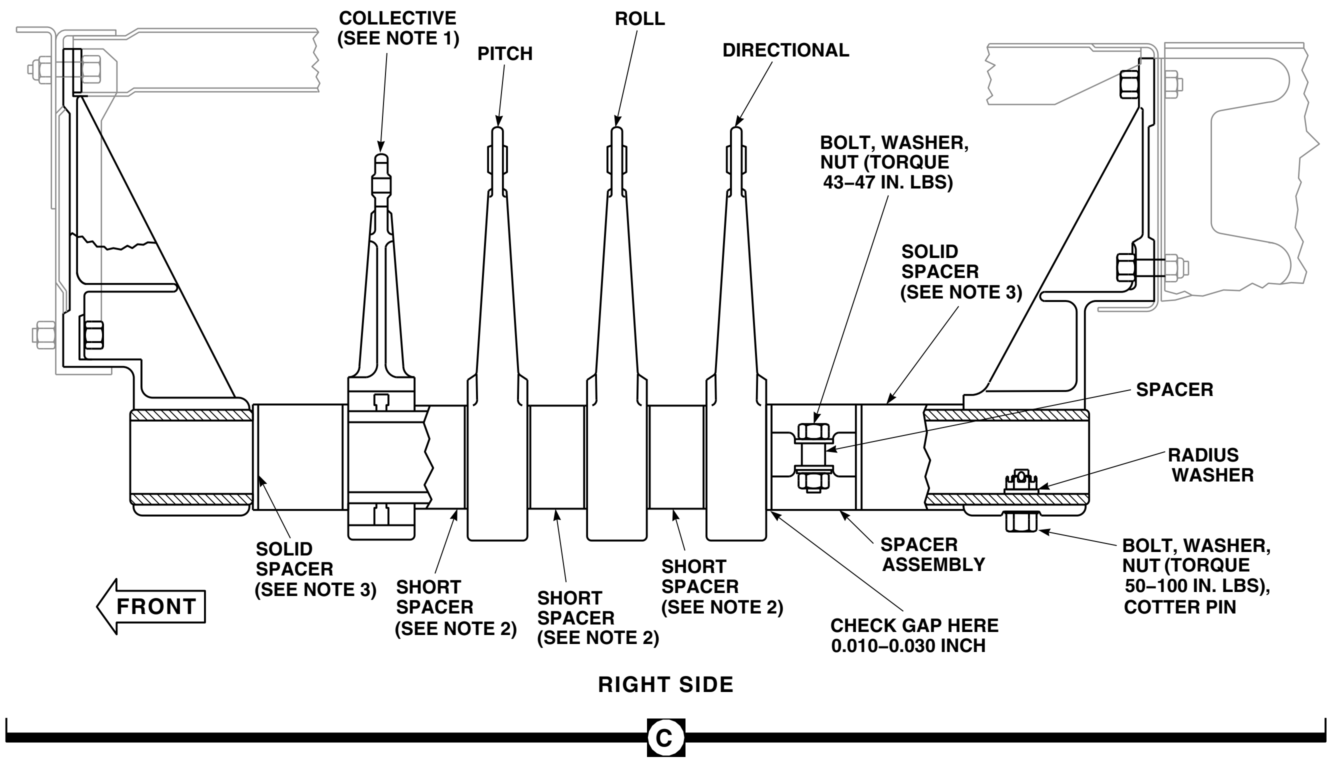


FIGURE 11-4-50-1. REPLACE MIDSECTION BELLCRANKS AND SHAFT (SHEET 2 OF 3)



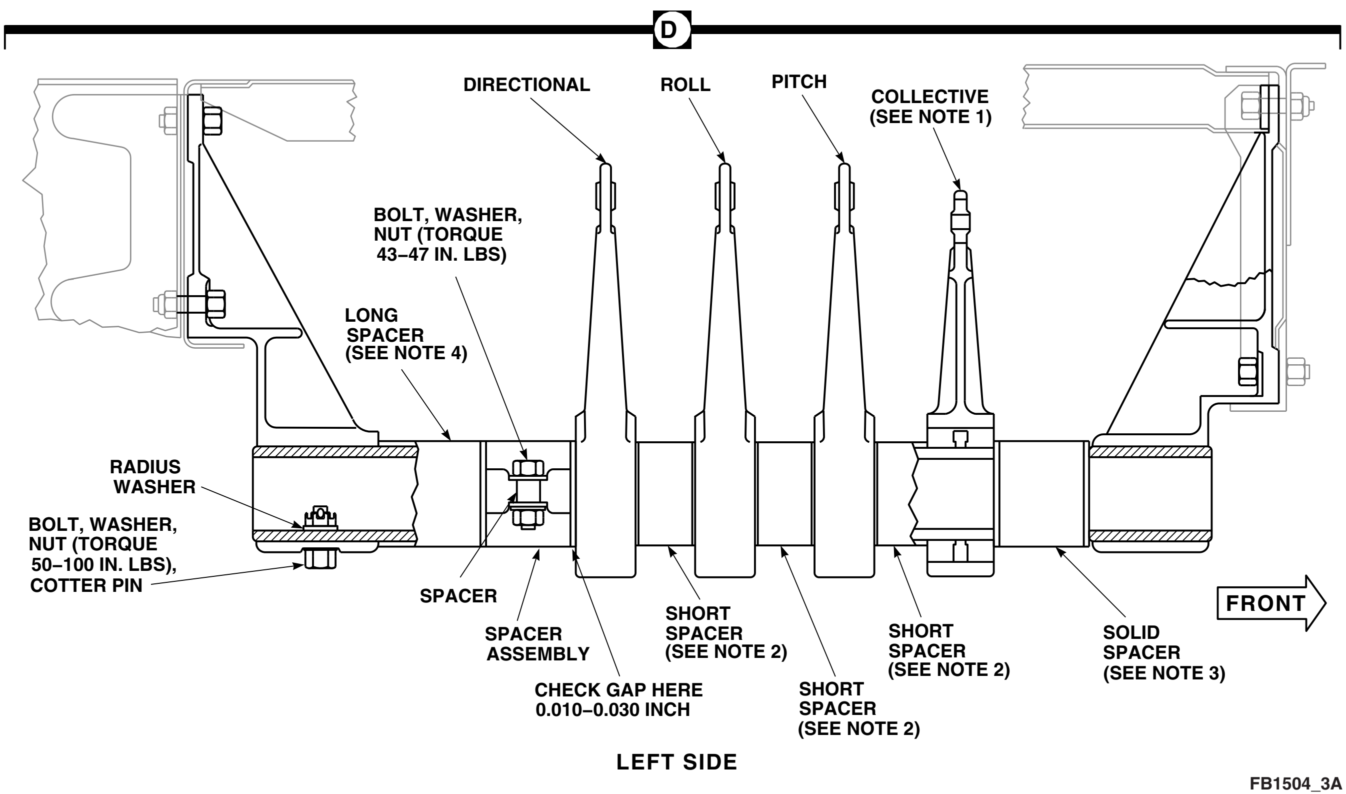


FIGURE 11-4-50-1. REPLACE MIDSECTION BELLCRANKS AND SHAFT (SHEET 3 OF 3)

c. Check shaft for gouges. **GOUGES MUST NOT EXCEED 0.010-INCH.** If damage is beyond limits, replace shaft.

11-4-50.1.3 Install.

NOTE

- Some bellcranks will be identified by a painted letter for proper installation. D for directional, R for roll, L for longitudinal, CR for collective right side, CL for collective left side. Others will be identified with a white dot, on upper arm.
- For proper installation, make sure that long arms with letter, or white dot, on pitch, roll, and directional bellcranks point upward.
- For proper installation, make sure that short arm with letter, or white dot, on collective bellcrank points upward.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. If installing a new rear support and no shims are bonded to frame at STA 265.5, perform the following:

(1) Install rear support on frame at STA 265.5 with bolts, washers, and nuts. Tighten nuts snug against support.

(2) Inspect for gaps between mounting pads of support and airframe.

c. If gap is found, perform the following:

(1) Locally make frame shims, 70211-02104-138, and 70211-02104-139 (Figure H-146, Appendix H). Shim as needed.

(2) Strip paint from front side of frame using paint remover, Item 229, Appendix D, and acid brush, Item 84, Appendix D. Wait 15 minutes then wipe dry with machinery towel, Item 211, Appendix D.

(3) Using epoxy primer coating kit, Item 135, Appendix D, and acid brush, Item 84, Appendix D, apply epoxy primer.

(4) Bond shim to frame with adhesive, Item 32, Appendix D, where bellcrank support mounts to frame. Make sure holes in shim align with holes in frame.

(5) Secure shim in place using 1-inch masking tape, Item 212, Appendix D.

(6) Allow adhesive, Item 32, Appendix D, to cure at room temperature for 24 hours.

(7) Touch up paint as required (PARA 2-6-5).

d. Install correct stackup on shaft as tagged.

NOTE

Any combination of washers, NAS1149D0416J and NAS1149D0463J, may be used provided a minimum of one washer, NAS1149D0416J, is used under bolthead and nut. A total of three washers may be used under bolthead and nut provided a minimum of two full threads are visible after torquing nut.

e. Slide supports toward center of shaft. Install assembly on airframe with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS** (Figure 11-4-50-1, Sheet 1, Detail A and Sheet 2, Detail B).

f. Install bolt and washer in rear support and shaft. Install radius washer and nut on bolt. **TORQUE NUT TO 50 - 100 INCH-POUNDS.** Install cotter pin (Figure 11-4-50-1, Sheet 3, Detail C and Detail D).

g. Install spacer assembly on left side bellcrank installation as follows (Figure 11-4-50-1, Sheet 3, Detail D):

(1) Push bellcranks and spacers to front of helicopter.

NOTE

Make sure bolthead is located on same side as spacer, 70400-02510-101.

(2) Install rear spacer assembly on shaft with bolts, washers, nuts, and spacer. **SET**

0.010 - 0.030 INCH GAP BETWEEN SPACER AND DIRECTIONAL BELL-CRANK WITH FEELER GAGE. TORQUE NUTS TO 43 - 47 INCH-POUNDS.

(3) With bellcranks hanging free, push each bellcrank gently one way or other. Bell-crank should return to first position with no sign of binding.

(4) If bellcrank hangs up or has a jerky motion, check gap at spacer assembly. If gap is correct, replace bellcrank.

h. Install spacer assembly on right side bellcrank installation as follows (Figure 11-4-50-1, Sheet 3, Detail C):

(1) Push bellcranks and spacer to front of helicopter.

NOTE

Make sure bolthead is located on same side as spacer, 70400-02510-101.

(2) Install rear spacer assembly on shaft with bolts, washers, nuts, and spacer. **SET 0.010 - 0.030 INCH GAP BETWEEN SPACER AND DIRECTIONAL BELL-CRANK WITH FEELER GAGE. TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

(3) With bellcranks hanging free, push each bellcrank gently one way or other. Bell-crank should return to first position with no sign of binding.

(4) If bellcrank hangs up or has a jerky motion, check gap at spacer assembly. If gap is correct, replace bellcrank.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

i. Install control rods on bellcranks as follows (Figure 11-4-50-1, Sheet 1, Detail A and Sheet 2, Detail B):

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Connect control rods to bellcranks.

(3) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(4) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(5) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

NOTE

Control rods should bolt to bellcranks without adjustment. If control rods do not line up with bellcranks, check bell-cranks for proper installation.

(6) Check pilot's collective stick for no gap between stick and low stopbolt. If there is a gap, recheck collective bellcrank installation.

j. Remove rig pins from collective, pitch, roll, and yaw torque shafts.

k. Remove rig pins from pilot's and copilot's yaw pedals.

l. Remove rig pins from pilot's and copilot's cyclic sticks.

m. Cycle flight controls, to make sure they are free for full range of travel.

n. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

o. Make sure area is clean and free of foreign material.

p. Install soundproofing panels in cabin overhead (PARA 2-4-69).

11-4-51 UPPER CABIN LINKS AND LEVERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Tiedown Strap, Item 338, Appendix D

References

- Appendix D
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-46

11-4-51.1 Replace Pitch Link.

11-4-51.1.1 Remove.

a. Remove soundproofing panels from overhead cabin (PARA 2-4-69).

b. Support control rods from overhead supports with tiedown strap, Item 338, Appendix D.

c. Disconnect cabin control rods from pitch lever (PARA 11-4-46).

d. Remove bolts, washers, bushings, and nuts securing pitch link to pitch torque shaft lower lever and pitch lever (Figure 11-4-51-1, Sheet 1, Detail A). Remove pitch lever.

11-4-51.1.2 Install.

a. Position pitch link between pitch lever and pitch torque shaft lower lever (Figure 11-4-51-1, Sheet 1, Detail A). Secure pitch link as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Self-retaining bolt securing pitch link to pitch lever is installed with bolthead facing front.

(1) Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

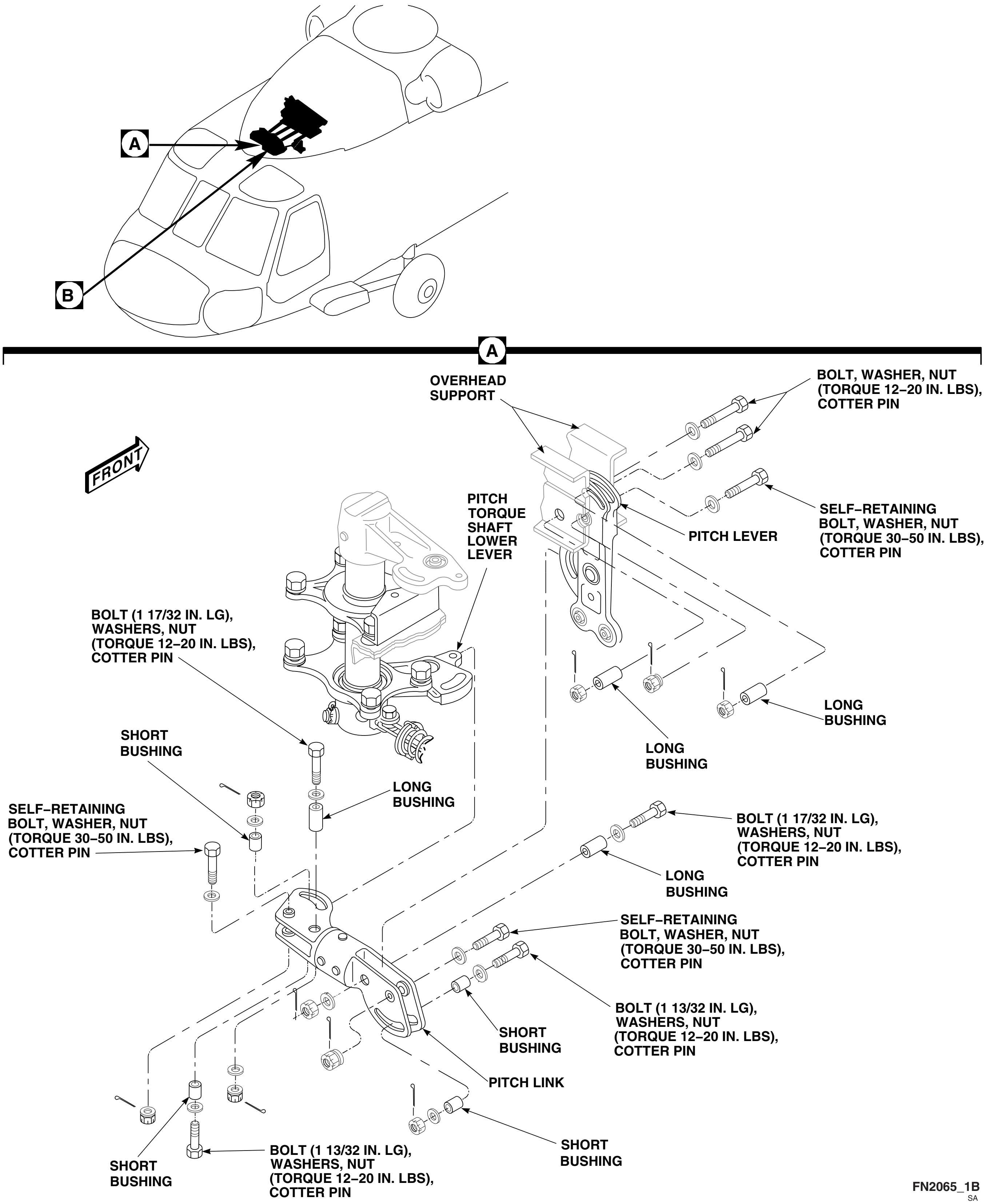


FIGURE 11-4-51-1. REPLACE UPPER CABIN LINKS AND LEVERS (SHEET 1 OF 2)

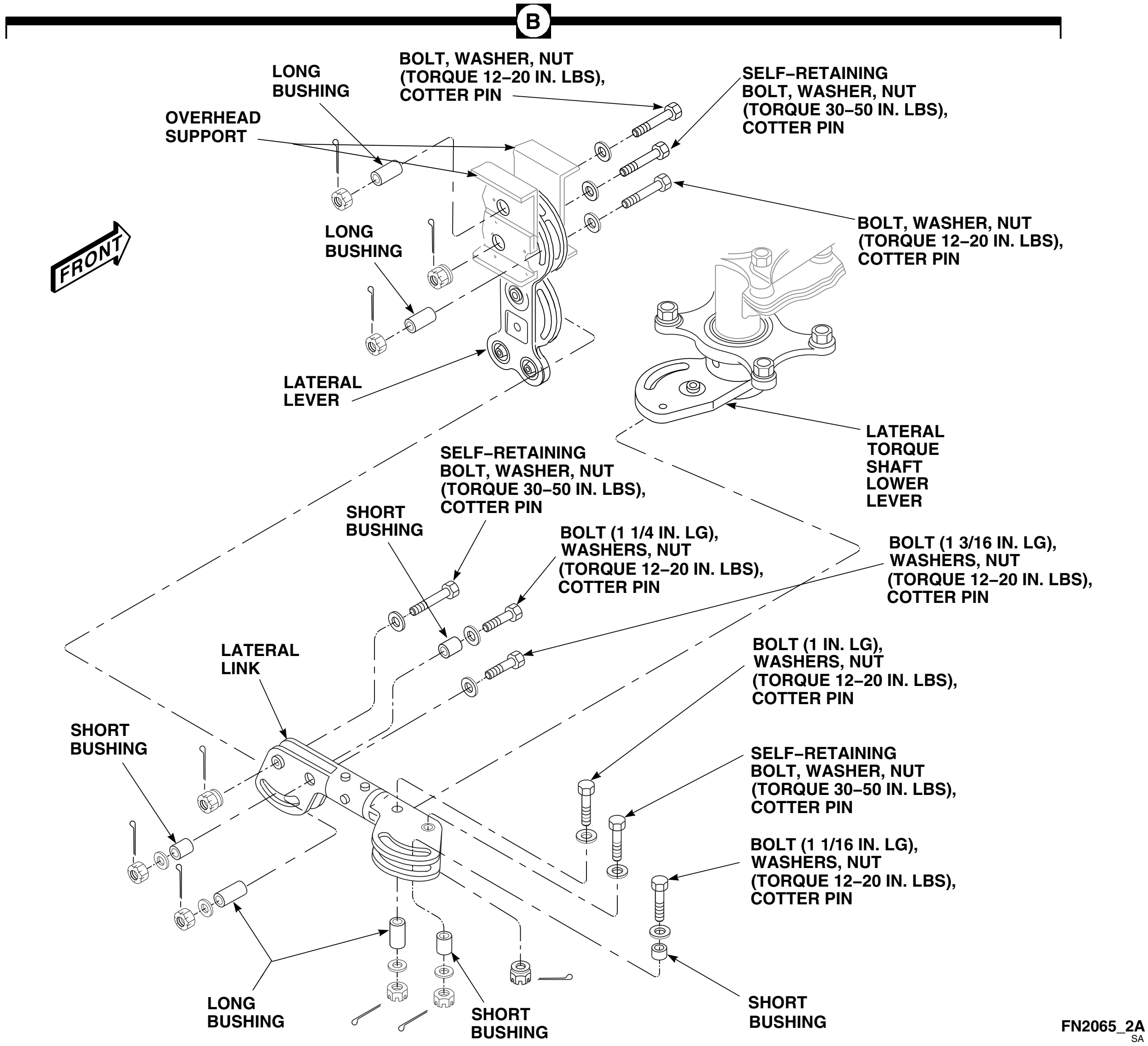


FIGURE 11-4-51-1. REPLACE UPPER CABIN LINKS AND LEVERS (SHEET 2 OF 2)

(2) Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

(3) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

(4) At pitch torque shaft lower lever, install long bushing through link and slot in lever, and install a 1 17⁄32-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

(5) At pitch torque shaft lower lever, install a short bushing in each slot of link. Install a 1 13⁄32-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

NOTE

Bolts securing pitch link to pitch lever are installed with boltheads facing front.

(6) At pitch lever, install a short bushing in each slot in link. Install a 1 13⁄32-inch long bolt, washers, and nuts. **TORQUE NUT**

TO 12 - 20 INCH-POUNDS. Install cotter pin.

(7) At pitch lever, install long bushing in pitch link and slot in pitch lever. Install 1 17/32-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

b. Connect cabin control rods to pitch lever (PARA 11-4-46).

c. Make sure area is clean and free of foreign material.

d. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

e. Install soundproofing panels (PARA 2-4-69).

11-4-51.2 Replace Pitch Lever.

11-4-51.2.1 Remove.

a. Remove soundproofing panels from overhead cabin (PARA 2-4-69).

b. Support control rods from overhead supports with tiedown strap, Item 338, Appendix D.

c. Disconnect cabin control rods from pitch lever (PARA 11-4-46).

d. Remove bolts, washers, bushings, and nuts securing pitch link to pitch lever (Figure 11-4-51-1, Sheet 1, Detail A).

e. Remove bolts, washers, bushings, and nuts securing pitch lever to overhead supports. Remove pitch lever and bushings.

11-4-51.2.2 Install.

a. Install pitch lever between overhead supports (Figure 11-4-51-1, Sheet 1, Detail A). Secure pitch lever as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Bolts securing pitch link to overhead supports are installed with boltheads facing front.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(4) Install long bushings through rear side of overhead support and through slots in pitch lever. Install bolts, washers and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pin.

b. Connect pitch link to pitch lever as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Bolts securing pitch link to pitch lever are installed with boltheads facing front.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1). Secure as follows:

(a) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(b) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(2) Install a short bushing in each slot in link. Install a 1 13⁄32-inch long bolt, washers, and nuts. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

(3) Install long bushing in pitch link and slot in pitch lever. Install 1 17⁄32-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

c. Connect cabin control rods to pitch lever (PARA 11-4-46).

d. Make sure area is clean and free of foreign material.

e. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

f. Install soundproofing panels (PARA 2-4-69).

11-4-51.3 Replace Lateral Link.

11-4-51.3.1 Remove.

a. Remove soundproofing panels from overhead cabin (PARA 2-4-69).

b. Support control rods from overhead supports with tiedown strap, Item 338, Appendix D.

c. Disconnect cabin control rods from lateral lever (PARA 11-4-46).

d. Remove bolts, washers, bushings, and nuts securing lateral link to lateral torque shaft lower lever and lateral lever (Figure 11-4-51-1, Sheet 2, Detail B). Remove lateral link.

11-4-51.3.2 Install.

NOTE

Make sure large end of lateral link is installed with large slot over lateral lever.

a. Position lateral link between lateral lever and lateral torque shaft lower lever (Figure 11-4-51-1, Sheet 2, Detail B). Secure lateral link as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Self-retaining bolt securing lateral link to lateral lever is installed with bolthead facing front.

(1) Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

(2) Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

(3) **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

(4) At lateral torque shaft lower lever, install long bushing through link and slot in lever and install 1-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

(5) At lateral torque shaft lower lever, install a short bushing in each slot of link. Install a 1 1⁄16-inch bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

NOTE

Bolts securing lateral link to lateral lever are installed with boltheads facing front.

(6) At lateral lever, install a short bushing in each slot. Install 1 1⁄4-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

(7) At lateral lever, install long bushing in lateral link and slot in pitch lever. Install a

1 3/16-inch long bolt, washers, and nut. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pin.

b. Connect cabin control rods to lateral lever (PARA 11-4-46).

c. Make sure area is clean and free of foreign material.

d. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

e. Install soundproofing panels (PARA 2-4-69).

11-4-51.4 Replace Lateral Lever.

11-4-51.4.1 Remove.

a. Remove soundproofing panels from overhead cabin (PARA 2-4-69).

b. Support control rods from overhead supports with tiedown strap, Item 338, Appendix D.

c. Disconnect cabin control rods from lateral lever (PARA 11-4-46).

d. Remove bolts, washers, bushings, and nuts securing lateral link to lateral lever (Figure 11-4-51-1, Sheet 2, Detail B).

e. Remove bolts, washers, bushings, and nuts securing lateral lever to overhead supports. Remove lateral lever and bushings.

11-4-51.4.2 Install.

a. Install replacement lateral lever between overhead supports (Figure 11-4-51-1, Sheet 2, Detail B). Secure lateral lever as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Bolts securing lateral link to overhead supports are installed with boltheads facing front.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(4) Install long bushings through rear side of overhead support and through slots in lateral lever. Install bolts, washers and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pin.

b. Connect lateral link to lateral lever as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Bolts securing lateral link to lateral lever are installed with boltheads facing front.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1). Secure as follows:

(a) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(b) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(2) Install a short bushing in each slot. Install 1 1/4-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

(3) Install long bushing in lateral link and slot in pitch lever. Install a 1 3/16-inch long bolt, washers, and nut. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pin.

c. Connect cabin control rods to lateral lever (PARA 11-4-46).

d. Make sure area is clean and free of foreign material.

e. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

f. Install soundproofing panels (PARA 2-4-69).

11-4-52 YAW TRIM SERVO.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 231, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Washers, NAS1149D0416J

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-101
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-4-1
- PARA 11-4-54
- TM 11-70-23

11-4-52.1 Replace Yaw Trim Servo.

CAUTION

To prevent contamination, do not loosen or remove plug while replacing yaw trim servo.

11-4-52.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove main rotor pylon sliding cover (PARA 2-4-101).

c. Disconnect electrical connector, P476R, from yaw trim servo (Figure 11-4-52-1, Detail A or Figure 11-4-52-2, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove self-retaining bolt, washer, and nut securing roll trim servo control rod to crank arm on roll trim servo and move control rod away from servo.

f. Remove yaw trim servo control rod (PARA 11-4-54).

g. Remove bolts and washers securing servo to deck. Remove servo.

h. Clean off any remaining adhesive on mating surfaces of deck skin using paint remover, Item 231, Appendix D, and nonmetallic scraper.

i. Clean off paint remover residue using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry using clean machinery towel.

11-4-52.1.2 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all electrical power (PARA 1-6-16).

b. **W/O EMEP** Install yaw trim servo to deck with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-52-1, Detail A).

c. **EMEP** Install yaw trim servo to deck as follows:

(1) Inspect chemical conversion coated (mating) surfaces for breaks or scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

(2) Position yaw trim servo on deck.

(3) Measure for gaps between servo mounting feet and deck. **NO GAPS ALLOWED.**

(4) If gaps exist, measure washers, NAS1149D0416J, required to fill gaps between servo mounting feet and deck mounting surface. **MAXIMUM THICKNESS 0.064-INCH.**

(5) Install yaw trim servo to deck with bolts and washers. Install washers, NAS1149D0416J, if required, to fill gap. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(6) Perform EME resistance check as follows:

NOTE

- Resistance measurement must be taken at least three times.
- Resistance measurement must be taken at each of the four servo mounting pads.

(a) Using digital low-resistance ohmmeter, check resistance between face of servo and deck area as close to servo as possible. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(b) If resistance is greater than 2.5 milliohms, remove servo and apply chemical conversion coating (PARA 2-6-5).

d. Apply a bead of sealing compound, Item 292, Appendix D, around bottom edge of yaw trim servo and deck.

e. Install yaw trim servo control rod (PARA 11-4-54).

f. Connect roll trim servo control rod to roll trim servo as follows:

(1) Position control rod on crank arm of roll trim servo.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

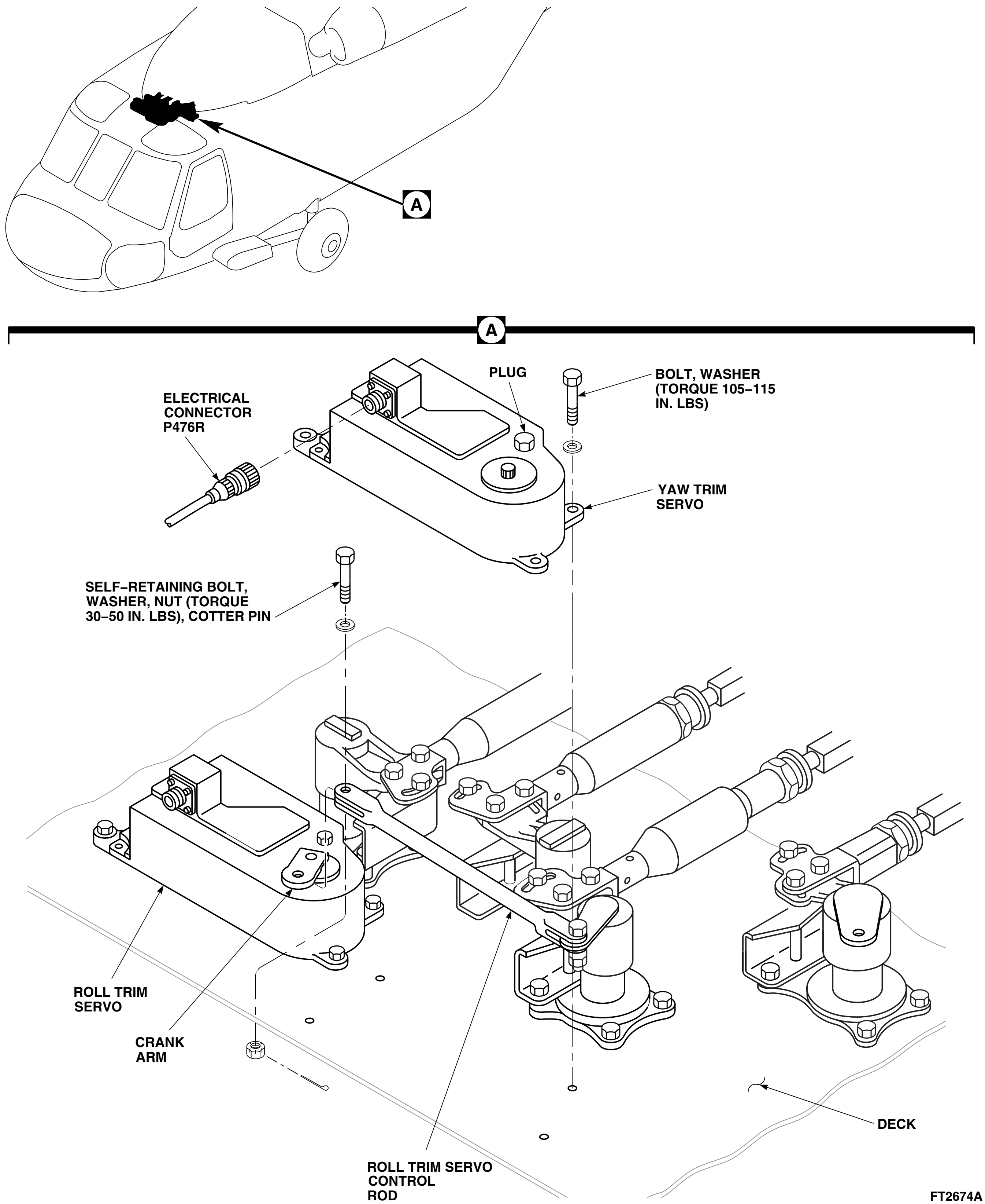
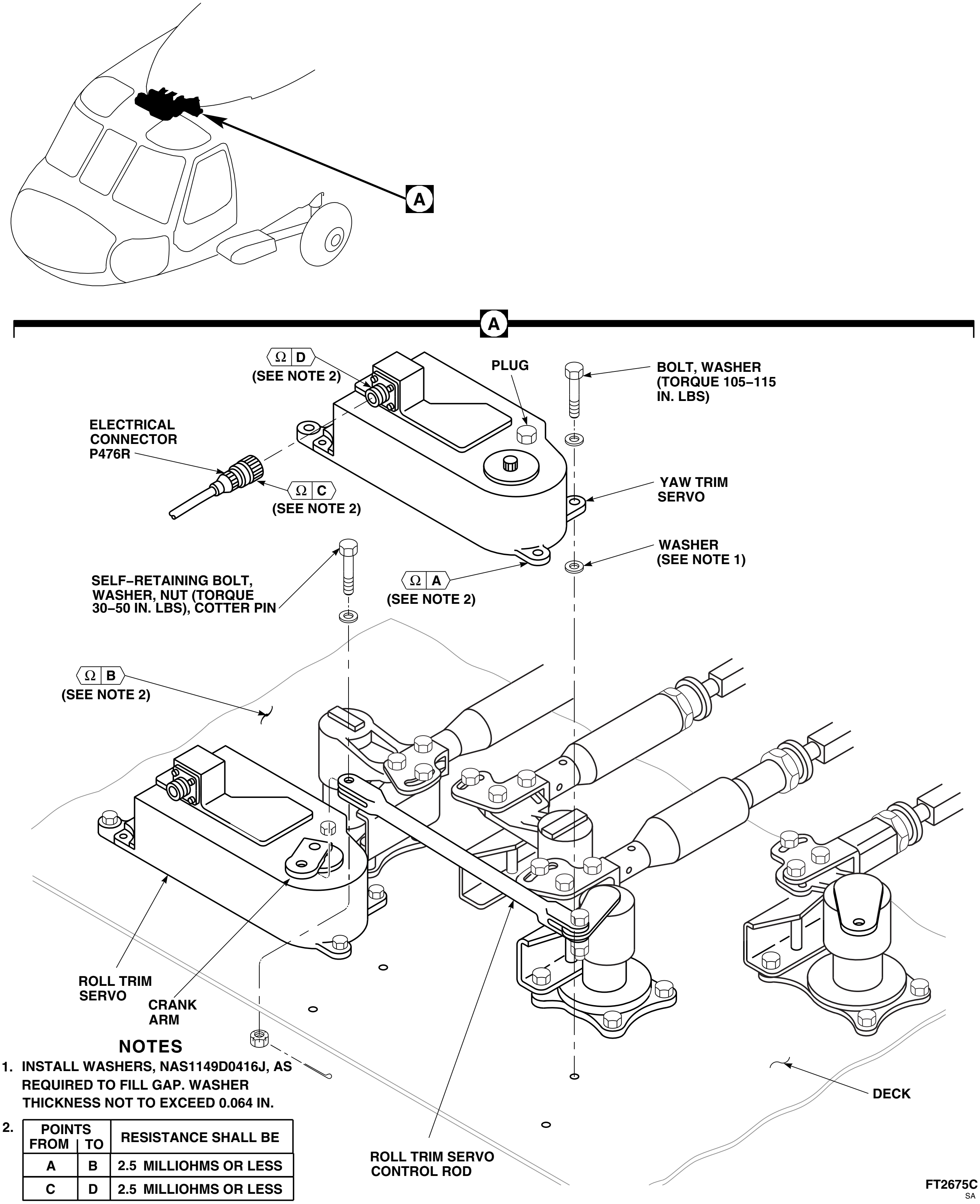


FIGURE 11-4-52-1. REPLACE YAW TRIM SERVO W/O EMEP



NOTES

1. INSTALL WASHERS, NAS1149D0416J, AS REQUIRED TO FILL GAP. WASHER THICKNESS NOT TO EXCEED 0.064 IN.
2.

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS
C	D	2.5 MILLIOHMS OR LESS

FIGURE 11-4-52-2. REPLACE YAW TRIM SERVO EMEP

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. Remove protective caps and plugs from electrical connectors.

h. Inspect and treat electrical connectors (PARA 1-7-20).

i. Connect electrical connector, P476R, to yaw trim servo (Figure 11-4-52-1, Detail A or Figure 11-4-52-2, Detail A).

j. EMEP Perform EME resistance check of pin filter connector as follows:

NOTE

Resistance measurement must be taken at least three times.

(1) Using digital low-resistance ohmmeter, check resistance between pin filter connector and yaw trim servo. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(2) If resistance is greater than 2.5 milliohms, disconnect pin filter connector and clean both pin filter connector and yaw trim servo receptacle threads as required, using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

k. Install main rotor pylon sliding cover (PARA 2-4-101).

l. Perform operational check of DAFCS (TM 11-70-23).

11-4-53 ROLL TRIM SERVO.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer's Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 231, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tape
- Tie Wrap
- Washers, NAS1149D0416J

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-101
- PARA 2-6-5
- PARA 11-4-1
- TM 11-70-23

11-4-53.1 Replace Roll Trim Servo.

CAUTION

To prevent contamination, do not loosen or remove plug while replacing roll trim servo.

11-4-53.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove main rotor pylon sliding cover (PARA 2-4-101).

NOTE

EMEP Do not remove pin filter adapter from electrical connector, P465R (Figure 11-4-53-2, Detail B).

c. Disconnect electrical connector, P465R, from roll trim servo (Figure 11-4-53-1, Detail A or Figure 11-4-53-2, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove self-retaining bolt, washer, and nut securing roll trim servo control rod to crank arm on roll trim servo and move control rod away from servo.

f. Remove bolts and washers securing servo to deck. Remove servo.

g. Clean off any remaining adhesive on mating surfaces of deck skin using paint remover, Item 231, Appendix D, and nonmetallic scraper.

h. Clean off paint remover residue using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry using clean machinery towel.

11-4-53.1.2 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all electrical power (PARA 1-6-16).

b. **W/O EMEP** Install roll trim servo to deck with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-53-1, Detail A).

c. **EMEP** Install roll trim servo to deck as follows:

(1) Inspect chemical conversion coated (mating) surfaces for breaks or scratches. **NONE ALLOWED.** If coated surfaces have breaks or scratches, apply chemical conversion coating (PARA 2-6-5).

(2) Place roll trim servo on deck mounting surface.

(3) Measure for gaps between servo mounting feet and deck mounting surface. **NO GAPS ALLOWED.**

(4) If gaps exist, measure washers, NAS1149D0416J, required to fill gaps between servo mounting feet and deck mounting surface (Figure 11-4-53-2, Detail A). **MAXIMUM THICKNESS 0.064-INCH.**

(5) Install roll trim servo to deck with bolts and washers. Install washers, NAS1149D0416J, if required, to fill gap. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(6) Perform EME resistance check as follows:

NOTE

Resistance measurement must be taken at least three times.

(a) Check resistance from face of servo and deck area as close to servo as possible using digital low-resistance ohmmeter. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(b) If resistance is greater than 2.5 milliohms, remove servo and apply chemical conversion coating (PARA 2-6-5).

d. Apply a bead of sealing compound, Item 292, Appendix D, around bottom edge of roll trim servo and cabin skin.

e. Connect roll trim servo control rod to roll trim servo as follows (Figure 11-4-53-1, Detail A or Figure 11-4-53-2, Detail A):

(1) Position control rod on crank arm of roll trim servo.

(2) Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

(3) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

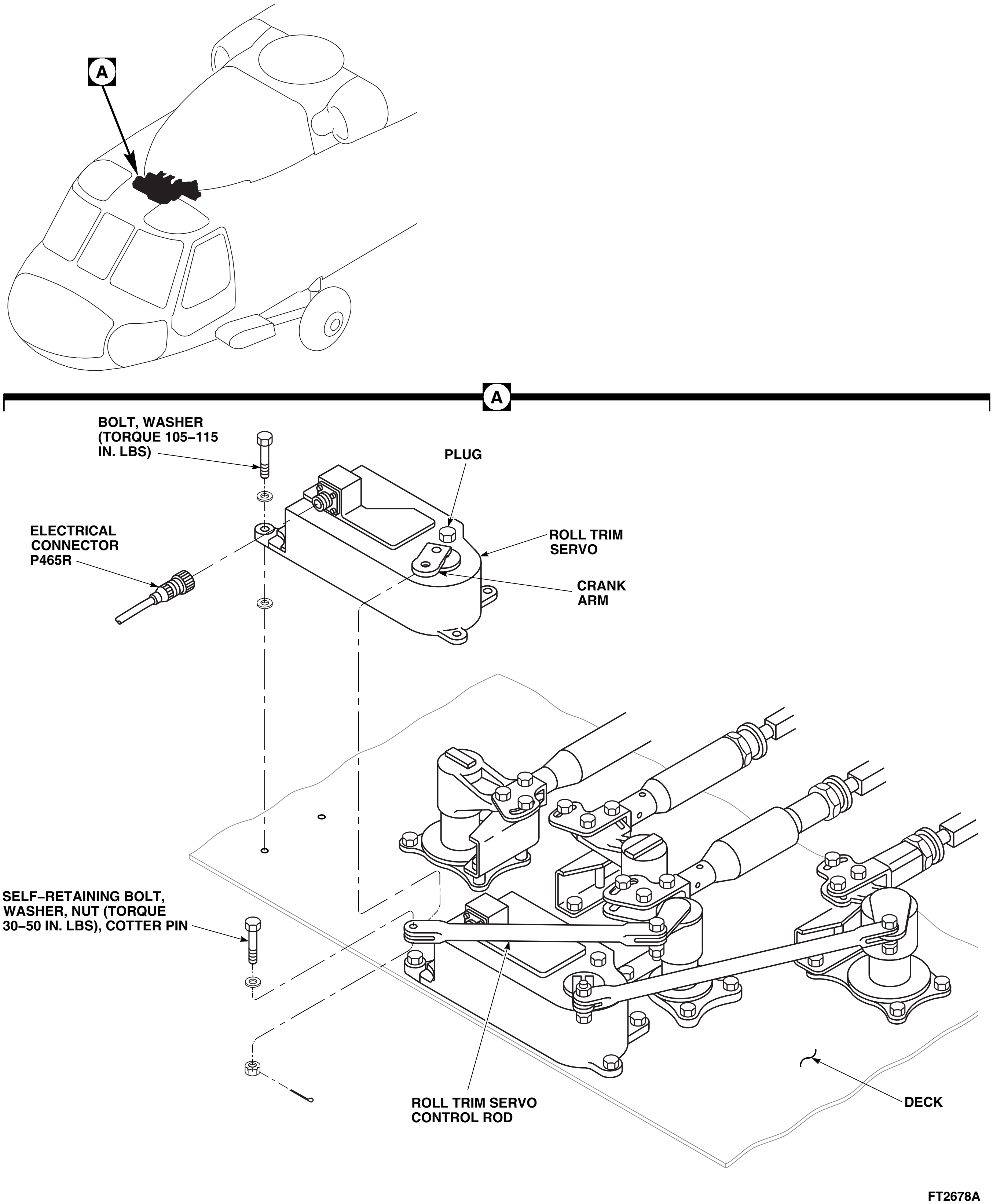
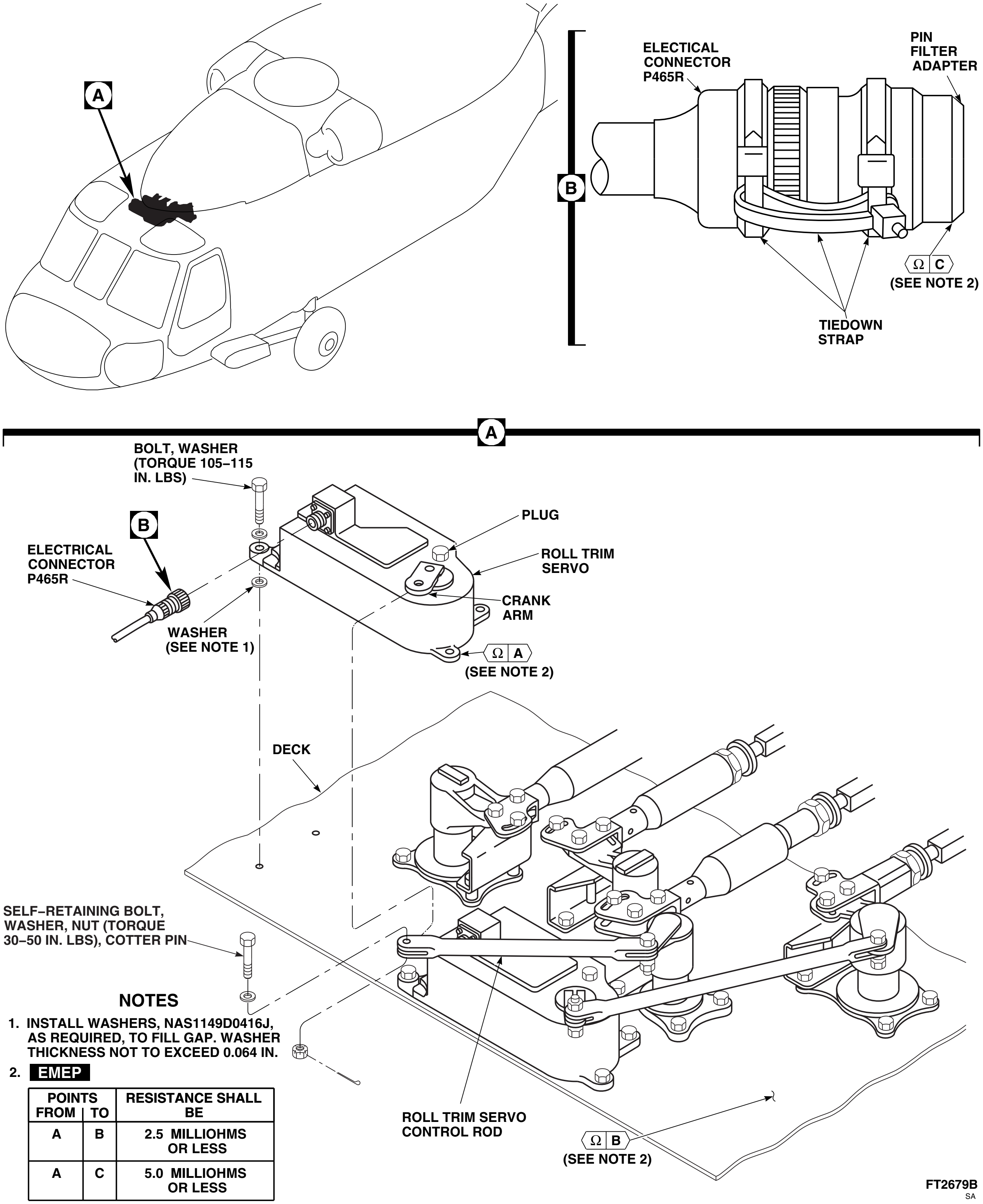


FIGURE 11-4-53-1. REPLACE ROLL TRIM SERVO W/O EMEP



NOTES

1. INSTALL WASHERS, NAS1149D0416J, AS REQUIRED, TO FILL GAP. WASHER THICKNESS NOT TO EXCEED 0.064 IN.
2. EMEP

POINTS FROM	TO	RESISTANCE SHALL BE
A	B	2.5 MILLIOHMS OR LESS
A	C	5.0 MILLIOHMS OR LESS

FIGURE 11-4-53-2. REPLACE ROLL TRIM SERVO EMEP

(4) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connector, P465R, to roll trim servo.

i. EMEP Perform EME resistance check of pin filter adapter as follows (Figure 11-4-53-2, Detail A and Detail B):

NOTE

Resistance measurement must be taken at least three times.

(1) Measure resistance between shell of pin filter adapter and mounting surface of roll trim servo connector. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**

(2) If resistance is greater than 5.0 milliohms, measure resistance between shell of pin filter adapter and connector on roll trim servo. **RESISTANCE SHALL BE 2.5 MILLIOHMS OR LESS.**

(3) If resistance is still beyond limits, clean surface between pin filter adapter and roll trim servo connector.

j. Install main rotor pylon sliding cover (PARA 2-4-101).

k. Perform operational check of AFCS system (TM 11-70-23).

11-4-54 TRIM SERVO CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-101
- PARA 2-6-5
- PARA 11-4-1

11-4-54.1 Replace Trim Servo Control Rod.

NOTE

Replacement of either trim servo control rod is the same.

11-4-54.1.1. Remove.

a. Remove main rotor pylon sliding cover (PARA 2-4-101).

b. Remove self-retaining bolt, washer, and nut securing control rod to torque shaft (Figure 11-4-54-1, Detail B).

c. Remove socket head screw, and slide crank arm from shaft on trim servo.

d. Remove self-retaining bolt, washer, and nut securing control rod to crank arm. Remove control rod.

11-4-54.1.2. Install.

a. Position crank arm on control rod (Figure 11-4-54-1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

c. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

d. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. Slide crank arm on shaft. Secure with socket head screw using socket wrench attachment. **TORQUE SOCKET HEAD SCREW TO 22 - 25 INCH-POUNDS.**

f. Position control rod on torque shaft.

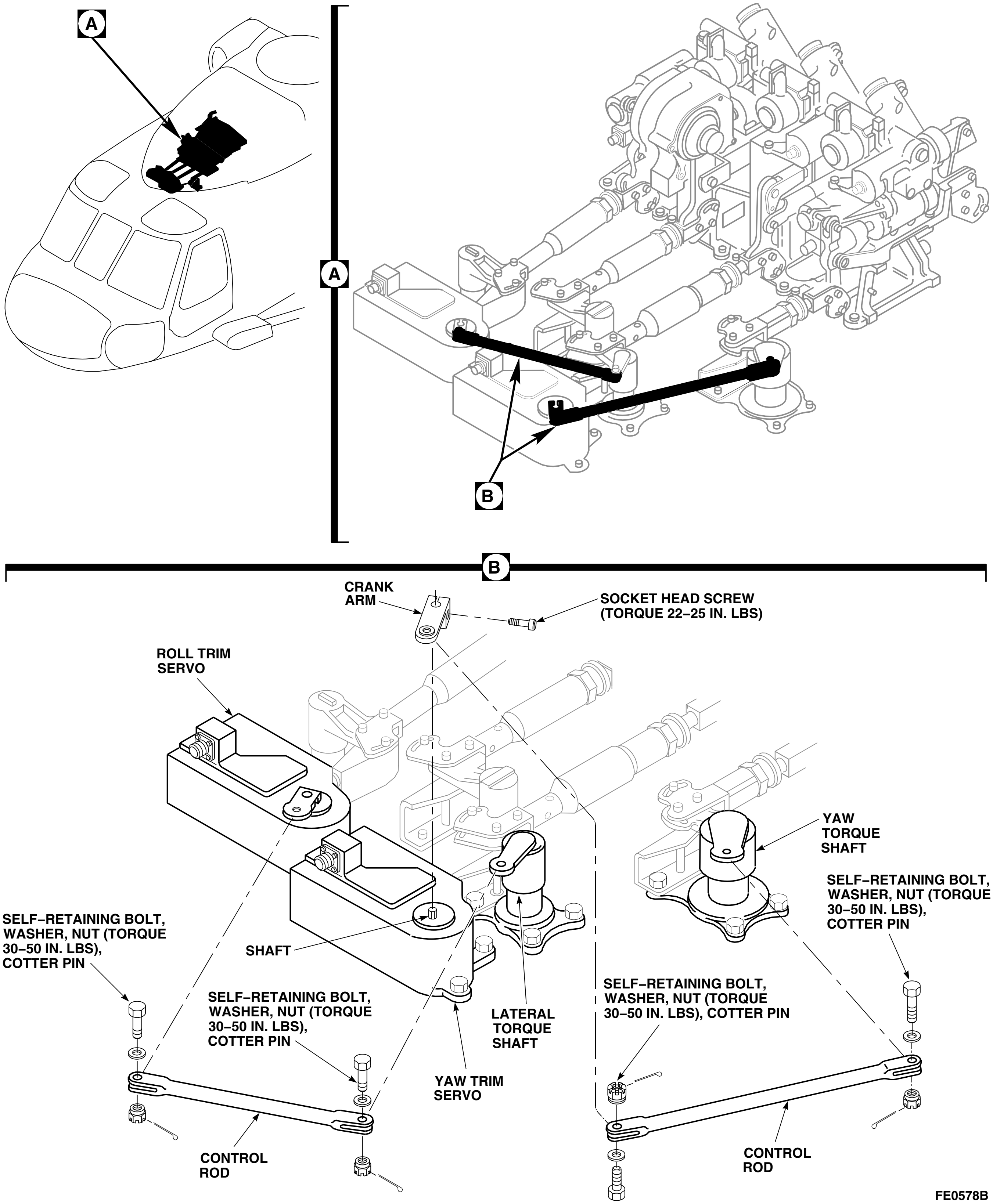


FIGURE 11-4-54-1. REPLACE TRIM SERVO CONTROL ROD

g. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

h. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

i. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. Install main rotor pylon sliding cover (PARA 2-4-101).

11-4-55 COLLECTIVE TORQUE SHAFT AND LEVERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Brass Drift
- Dial Indicator
- Feeler Gage
- Feeler Gage, 0.020-inch
- Hammer
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. oz
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 37, Appendix D
- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-46
- PARA 11-4-61
- PARA 11-4-62
- PARA 11-4-63

11-4-55.1 Replace Collective Torque Shaft and Levers.

11-4-55.1.1 Remove.

a. Remove collective input control rod (PARA 11-4-63).

b. Disconnect overhead control rods from lower lever on collective torque shaft (PARA 11-4-46).

c. Before removing balance spring, measure distance between end (shoulder) of adjusting rod and spring bracket. Record measurement (Figure 11-4-55-1, Sheet 2, Detail E).

d. At spring bracket, loosen balance spring by backing off jamnut. Remove pin and washer. Disconnect balance spring from lower lever (Figure 11-4-55-1, Sheet 1, Detail B).

e. Remove nuts and washers from tapered pins.

f. Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

g. Remove tapered pins using hammer and brass drift. Remover lower lever from torque shaft.

h. Remove torque shaft and upper lever from supports (Figure 11-4-55-1, Sheet 1, Detail B). Remove spacer and boot assembly (Figure 11-4-55-1, Sheet 2, Detail C).

i. At spring bracket bolted to lower support, measure distance between rod end (shoulder) of adjusting rod and spring bracket. Record measurement (Figure 11-4-55-1, Sheet 2, Detail E).

j. Loosen jamnut to remove tension from balance spring connected to pitch torque shaft lever.

k. Remove bolts, washers, and nuts securing lower support and shims to airframe. Remove lower support and spacer tube from airframe (Figure 11-4-55-1, Sheet 2, Detail D).

l. Remove sealing compound from parts as follows:

(1) Remove sealing compound from lower support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

m. Remove bolts and washers securing upper support, shims, spacers, and rigging bracket to airframe. Remove spacers.

n. Remove sealing compound from parts as follows:

(1) Remove sealing compound from upper support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

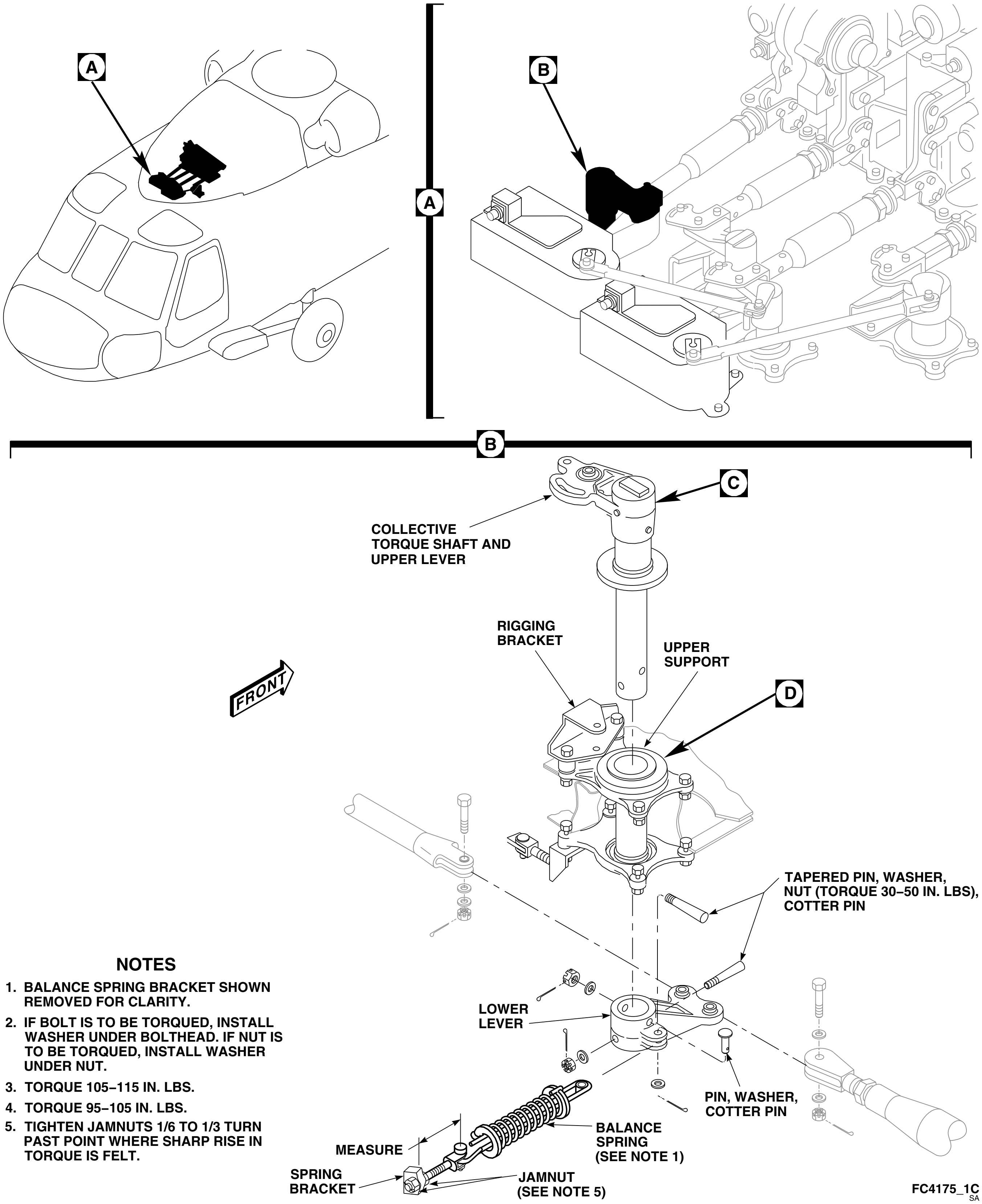
11-4-55.1.2 Inspect.

a. Inspect tapered pin shank. Tapered surface must be smooth and free of damage, nicks, dents, scratches, or scuff marks. **NO DAMAGE ALLOWED.** If damaged, replace pin.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

b. Using dial indicator, inspect bearing in lower lever. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0025-INCH.**



NOTES

1. BALANCE SPRING BRACKET SHOWN REMOVED FOR CLARITY.
2. IF BOLT IS TO BE TORQUED, INSTALL WASHER UNDER BOLTHEAD. IF NUT IS TO BE TORQUED, INSTALL WASHER UNDER NUT.
3. TORQUE 105–115 IN. LBS.
4. TORQUE 95–105 IN. LBS.
5. TIGHTEN JAMNUTS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

FIGURE 11-4-55-1. REPLACE COLLECTIVE TORQUE SHAFT AND LEVERS (SHEET 1 OF 2)

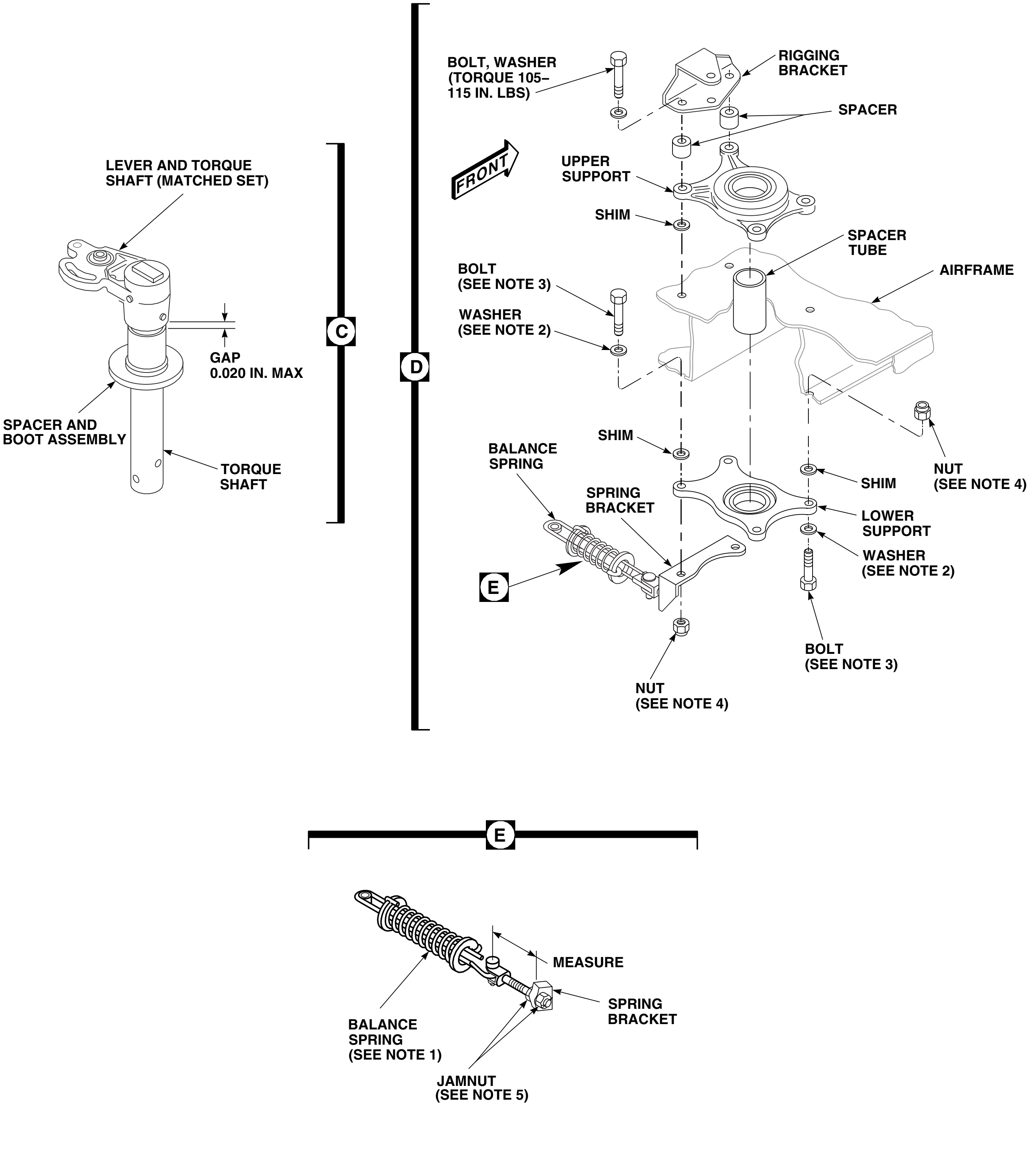


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FIGURE 11-4-55-1. REPLACE COLLECTIVE TORQUE SHAFT AND LEVERS (SHEET 2 OF 2)

c. Using dial indicator, inspect bearing in upper lever and torque shaft. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

d. Inspect bearing in upper and lower supports as follows:

(1) Using dial indicator, inspect radial play. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

(2) Check bearings for smooth operation. **IF BEARING FEELS ROUGH OR CANNOT BE ROTATED, REPLACE BEARING.**

e. Inspect rigging bracket for deformation. **NONE ALLOWED.** If bracket shows evidence of deformation, replace bracket.

11-4-55.1.3 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Bond replacement boot to spacer as follows:

(1) If old bond is on spacer, clean with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and wipe dry with clean cloth.

(2) Apply a coat of adhesive, Item 37, Appendix D, on one end of spacer, about as wide as boot collar. Apply a coat of adhesive on inside of boot collar.

(3) Allow adhesive to dry for at least 10 minutes.

(4) Apply a second coat of adhesive to mating surfaces and allow at least 10 minutes for it to dry.

(5) Slide boot on end of spacer and keep lower face of boot and end face of spacer flush. Press collar on boot against spacer all around to make sure of complete contact.

(6) Allow to dry for at least 8 hours.

c. Install upper support, lower support, and torque shaft as follows:

(1) Install two bolts and washers in right mounting holes of upper support (Figure 11-4-55-1, Sheet 2, Detail D).

(2) Place spacers and rigging bracket on left side of upper support. Face rigging hole in bracket toward center of support.

(3) Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

(4) Install spacer tube through opening in airframe.

(5) Install replacement lower support (flat side up), against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.
- When securing right side of lower support, install boltheads facing down. When securing left side of lower support, install boltheads facing up.

(6) Install bolts, washers, and nuts securing lower support to airframe using a maxi-

mum of three washers in any combination. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(7) Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

(8) On replacement torque shaft, tap out tapered pins using hammer and brass drift. Remove lower lever from torque shaft.

(9) Install spacer and boot assembly on torque shaft (Figure 11-4-55-1, Sheet 2, Detail C), and slide shaft in upper and lower supports (Figure 11-4-55-1, Sheet 1, Detail B).

(10) Install lower lever on torque shaft, line up tapered pin holes and install one tapered pin. Tap pin in to make sure it is fully seated. Do not install nut and washer now.

d. On upper deck, pull up on torque shaft and use feeler gage to measure gap between upper lever and spacer (Figure 11-4-55-1, Sheet 2, Detail C). Record measurement.

e. Remove tapered pin and remove lower lever from torque shaft. Remove torque shaft from supports (Figure 11-4-55-1, Sheet 1, Detail B).

f. Remove bolts, washers, and nuts securing upper support, lower support, and spacer tube to airframe (Figure 11-4-55-1, Sheet 2, Detail D).

g. Shim torque shaft and levers as follows:

NOTE

- If measurement taken in step d. is 0.020-inch or less, shimming is not required.
- Shims may be installed at the upper support location, lower support location, or both locations.
- Total thickness of shims at the upper support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims at the lower support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims installed cannot exceed 0.162-inch.

(1) If measurement taken was 0.021-inch or more, add shims to obtain a maximum gap of 0.020-inch.

(2) Apply a light coat of adhesive, Item 32, Appendix D, on shims and around mounting holes on airframe. Place shims on airframe.

(3) Clean extra adhesive from holes in shims and lower support.

(4) Allow sealing compound to dry for 24 hours or use heat gun at temperature of 110° - 130°F (43° - 54°C) for two hours.

h. Install two bolts and washers in right mounting holes of upper support.

i. Place spacers and rigging bracket on left side of upper support. Face rigging hole in bracket toward center of support.

j. Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

k. Install spacer tube through opening in airframe.

l. Install replacement lower support (flat side up), against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut, is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

m. Install bolts (bolthead down), washers, and nuts on right side of lower support, using a maximum of three washers in any combination. Do not torque at this time.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut, is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

n. Position spring bracket on lower support left mounting holes and install bolts (bolthead up), washers, and nuts. Do not torque at this time.

o. Install spacer and boot assembly on torque shaft (Figure 11-4-55-1, Sheet 2, Detail C), and slide shaft in upper and lower supports (Figure 11-4-55-1, Sheet 1, Detail B).

NOTE

Record hole and tapered pin positions in shaft and lever before cleaning and priming.

p. Install lower lever on torque shaft and line up tapered pin holes.

q. Wipe holes and surrounding surfaces using dry machinery towel, Item 211, Appendix D.

NOTE

Do not allow technical trichloroethylene to dry on parts.

r. Clean holes, shaft, and lever using clean machinery towel, Item 211, Appendix D, wet with technical trichloroethylene, Item 343, Appendix D. Wipe dry.

s. Using epoxy primer coating kit, Item 135, Appendix D, and acid brush, Item 84, Appendix D, apply one coat of epoxy primer to tapered pin holes in shaft and lever. Allow to dry.

t. Using acid brush, Item 84, Appendix D, apply a thin coat (0.002 - 0.003-inch) of sealing compound, Item 292 or Item 301A, Appendix D, covering tapered pins.

u. Install tapered pins, wet with sealing compound, into proper holes in shaft and lever. **SMALL DIAMETER END OF TAPERED PIN SHANK SHALL NOT EXTEND MORE THAN 1/16-INCH ABOVE LEVER SURFACE.** If end of tapered pin extends more than 1/16-inch, replace pin to obtain correct dimension.

v. Install washers and nuts making sure that sealing compound is visible between pin surfaces, washers, and both sides of lever. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins. If sealing compound, Item 301A, Appendix D, was used, make sure to wipe off excess sealing compound with clean machinery towel, Item 211, Appendix D.

w. **TORQUE LOWER SUPPORT NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-55-1, Sheet 2, Detail D).

x. Rotate torque shaft 360°. **TORQUE SHAFT MUST TURN FREELY AND FRICTION MUST NOT EXCEED 20 INCH-OUNCES.** If shaft does not turn freely or friction exceeds 20 inch-ounces, torque shaft is preloaded (less than zero gap). Disassemble and repeat step g. though step x.

y. Install rigging pin through torque shaft upper lever and rigging bracket.

z. Insert 0.020-inch feeler gage at copilot's low collective stick stop and lower stick until contact. Place pilot's collective stick in low position contacting the low stick stop and lock in place.

aa. Connect overhead control rods to lower lever on collective torque shaft (PARA 11-4-46).

ab. Connect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ac. Bottom collective boost in extend direction by pushing input arm aft until servo stops.

ad. Install collective input control rod (PARA 11-4-63).

ae. Remove collective torque shaft rigging pin.

af. Move collective sticks to full up position. **VERIFY HIGH COLLECTIVE STICK STOPS ARE CONTACTED.** Remove 0.020-inch feeler gage from copilot's low collective stop.

ag. Move collective sticks to full down position. **VERIFY LOW COLLECTIVE STICK STOPS ARE CONTACTED.**

ah. Disconnect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

NOTE

If measurement is not known, balance spring will need to be adjusted.

ai. Adjust balance spring connected to pitch torque shaft lever by adjusting balance spring rod to measurement recorded when jamnut was loosened (Figure 11-4-55-1, Sheet 2, Detail E). **TIGHTEN JAMNUTS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

aj. Install balance spring as follows (Figure 11-4-55-1, Sheet 1, Detail A):

(1) Position balance spring between lever and spring bracket.

(2) Install pin, washer, and cotter pin in lever end and jamnut on adjustable end.

NOTE

If measurement is not known, balance spring will need to be adjusted.

(3) Adjust balance spring rod to measurement recorded when removed.

(4) **TIGHTEN JAMNUTS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

ak. If necessary, adjust balance spring(s) (PARA 11-4-61 or PARA 11-4-62).

al. Make sure area is clean and free of foreign material.

am. Install soundproofing panels in cabin overhead (PARA 2-4-69).

an. Close and latch main pylon rotor pylon sliding cover.

NOTE

Make sure proper autorotation RPM and high stick stop setting are checked during the maintenance test flight.

ao. Perform maintenance test flight (Appropriate MTF).

11-4-56 PITCH TORQUE SHAFT AND LEVERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Brass Drift
- Dial Indicator
- Feeler Gage
- Hammer
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. oz
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 37, Appendix D
- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-51
- PARA 11-4-61
- PARA 11-4-64

11-4-56.1 Replace Pitch Torque Shaft and Levers.

11-4-56.1.1 Remove.

a. Remove pitch/trim input control rod (PARA 11-4-64).

b. Remove pitch link (PARA 11-4-51).

c. Before removing balance spring, measure distance between end (shoulder) of adjusting rod and spring bracket. Record measurement (Figure 11-4-56-1, Sheet 2, Detail D).

d. At spring bracket, loosen balance spring by backing off jamnut. Remove pin and washer. Disconnect balance spring from lower lever (Figure 11-4-56-1, Sheet 1, Detail B).

e. Remove nuts and washers from tapered pins.

f. Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

g. Remove tapered pins using hammer and brass drift. Remove lower lever from torque shaft (Figure 11-4-56-1, Sheet 2, Detail C).

h. Remove torque shaft and upper lever from supports (Figure 11-4-56-1, Sheet 2, Detail C). Remove spacer and boot assembly (Figure 11-4-56-1, Sheet 2, Detail E).

i. Remove bolts, washers, and nuts securing lower support and shims to airframe. Remove lower support and spacer tube from airframe (Figure 11-4-56-1, Sheet 2, Detail F).

j. Remove sealing compound from parts as follows:

(1) Remove sealing compound from lower support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

k. Remove bolts and washers securing upper support, shims, spacers, and rigging bracket to airframe. Remove spacers.

l. Remove sealing compound from parts as follows:

(1) Remove sealing compound from upper support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

11-4-56.1.2 Inspect.

a. Inspect tapered pin shank. Tapered surface must be smooth and free of damage, nicks, dents, scratches, or scuff marks. **NONE ALLOWED.** If damaged, replace pin.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

b. Using dial indicator, inspect bearings in pitch lever as follows:

(1) **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, MDW4K-A1173 OR MDW4K-400, IS 0.005-INCH.**

(2) **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, SA4-14A1-504, IS 0.005-INCH.**

(3) **MAXIMUM ALLOWABLE RADIAL PLAY IN BEARING, MKSP4A-A1173 OR MKSP4A-400, IS 0.0025-INCH.**

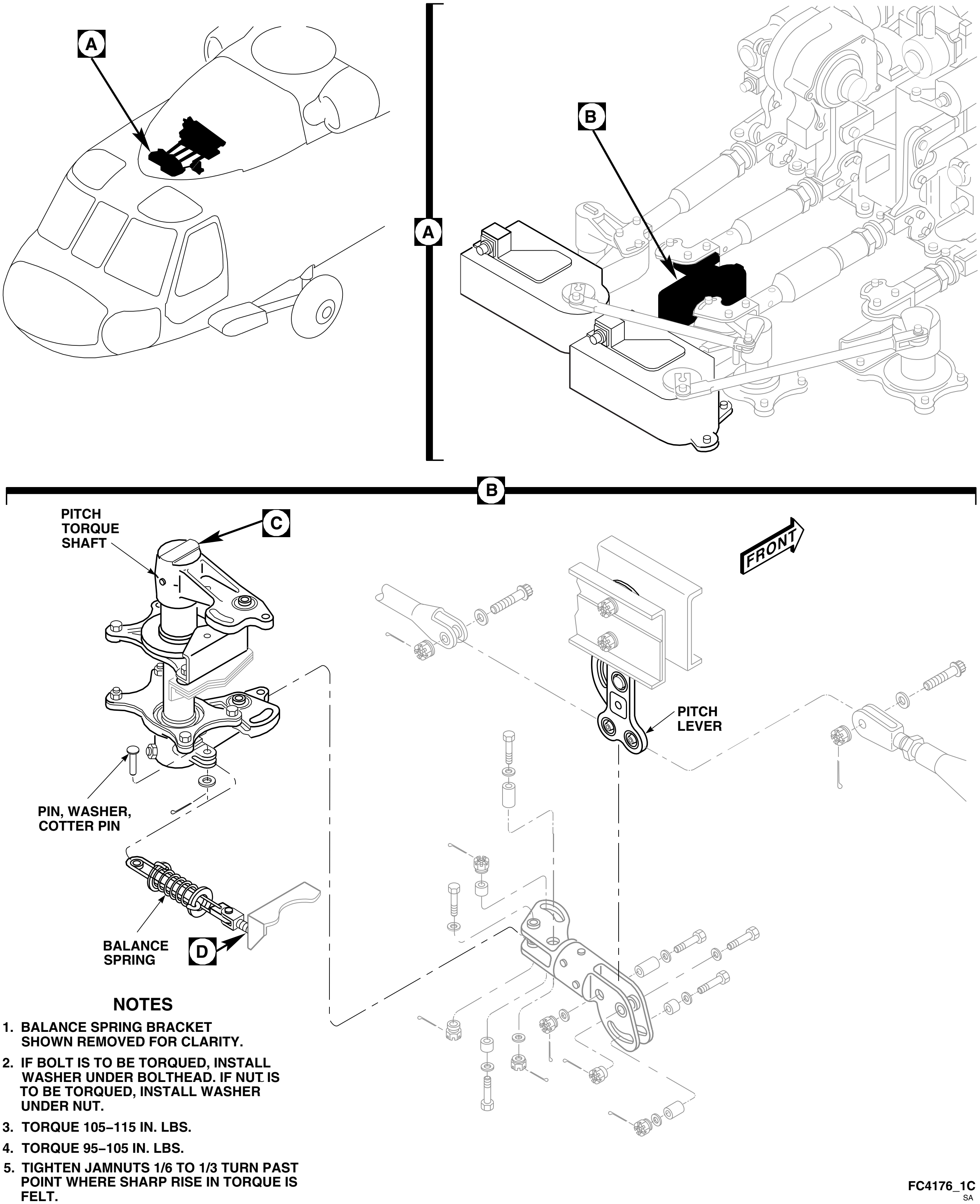


FIGURE 11-4-56-1. REPLACE PITCH TORQUE SHAFT AND LEVERS (SHEET 1 OF 2)

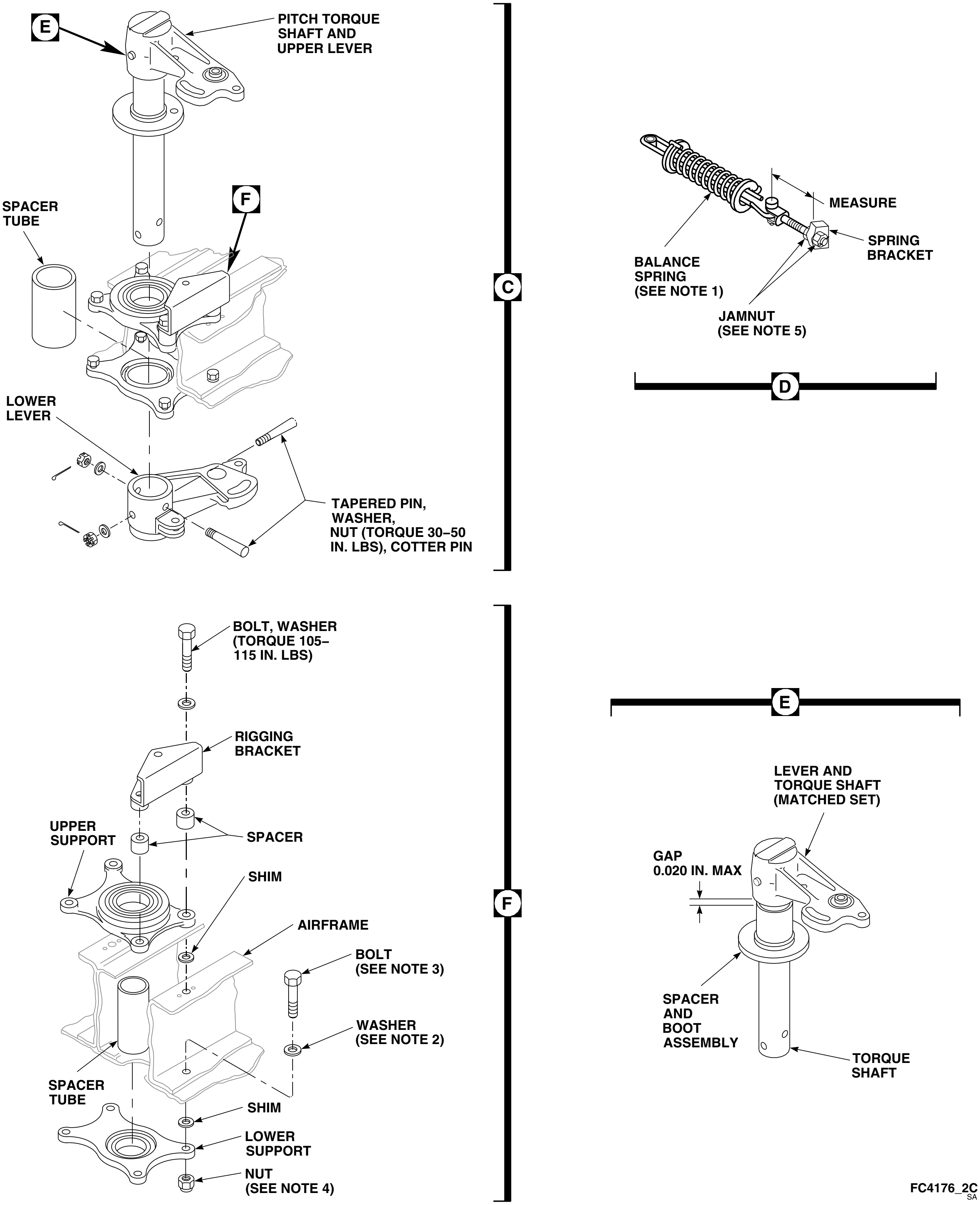


FIGURE 11-4-56-1. REPLACE PITCH TORQUE SHAFT AND LEVERS (SHEET 2 OF 2)

c. Using dial indicator, inspect bearing in upper and lower levers. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

d. Inspect bearing in upper and lower supports as follows:

(1) Using dial indicator, inspect radial play. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

(2) Check bearings for smooth operation. **IF BEARING FEELS ROUGH OR CANNOT BE ROTATED, REPLACE BEARING.**

e. Inspect rigging bracket for deformation. **NONE ALLOWED.** If bracket shows evidence of deformation, replace bracket.

11-4-56.1.3 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Bond replacement boot to spacer as follows:

(1) If old bond is on spacer, clean with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and wipe dry with clean cloth.

(2) Apply a coat of adhesive, Item 37, Appendix D, on one end of spacer, about as wide as boot collar. Apply a coat of adhesive on inside of boot collar.

(3) Allow adhesive to dry for at least 10 minutes.

(4) Apply a second coat of adhesive to mating surfaces and allow at least 10 minutes for it to dry.

(5) Slide boot on end of spacer and keep lower face of boot and end face of spacer flush. Press collar on boot against spacer all around to make sure of complete contact.

(6) Allow to dry for at least 8 hours.

c. Install upper support, lower support, and torque shaft as follows:

(1) Install two bolts and washers in left mounting holes of upper support (Figure 11-4-56-1, Sheet 2, Detail F).

(2) Place spacers and rigging bracket on right side of upper support. Face rigging hole in bracket toward center of support.

(3) Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

(4) Install spacer tube through opening in airframe.

(5) Install replacement lower support (flat side up) against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

(6) Install bolts (bolthead up), washers, and nuts securing lower support to airframe using a maximum of three washers in any combination. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(7) Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

(8) On replacement torque shaft, tap out tapered pins using hammer and brass drift. Remove lower lever from torque shaft.

(9) Install spacer and boot assembly on torque shaft (Figure 11-4-56-1, Sheet 2, Detail E), and slide shaft in upper and lower supports (Figure 11-4-56-1, Sheet 2, Detail C).

(10) Install lower lever on torque shaft, line up tapered pin holes and install one tapered pin (Figure 11-4-56-1, Sheet 2, Detail C). Tap pin in to make sure it is fully seated. Do not install nut and washer now.

d. On upper deck, pull up on torque shaft and use feeler gage to measure gap between upper lever and spacer (Figure 11-4-56-1, Sheet 2, Detail E). Record measurement.

e. Remove tapered pin and remove lower lever from torque shaft. Remove torque shaft from supports (Figure 11-4-56-1, Sheet 2, Detail C).

f. Remove bolts, washers, and nuts securing upper support, lower support, and spacer tube to airframe (Figure 11-4-56-1, Sheet 2, Detail F).

g. Shim torque shaft and levers as follows:

NOTE

- If measurement taken in step d. is 0.020-inch or less, shimming is not required.
- Shims may be installed at the upper support location, lower support location, or both locations.
- Total thickness of shims at the upper support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims at the lower support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims installed cannot exceed 0.162-inch.

(1) If measurement taken was 0.021-inch or more, add shims to obtain a maximum gap of 0.020-inch.

(2) Apply a light coat of adhesive, Item 32, Appendix D, on shims and around mounting holes on airframe. Place shims on airframe.

(3) Clean extra adhesive from holes in shims and lower support.

(4) Allow sealing compound to dry for 24 hours or use heat gun at temperature of 110° - 130°F (43° - 54°C) for two hours.

h. Install two bolts and washers in left mounting holes of upper support.

i. Place spacers and rigging bracket on right side of upper support. Face rigging hole in bracket toward center of support.

j. Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

k. Install spacer tube through opening in airframe.

l. Install replacement lower support (flat side up) against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

m. Install bolts (bolthead up), washers, and nuts on lower support, using a maximum of three washers in any combination. Do not torque at this time.

n. Install spacer and boot assembly on torque shaft (Figure 11-4-56-1, Sheet 2, Detail E), and slide shaft in upper and lower supports (Figure 11-4-56-1, Sheet 2, Detail C).

NOTE

Record hole and tapered pin positions in shaft and lever before cleaning and priming.

o. Install lower lever on torque shaft and line up tapered pin holes.

p. Wipe holes and surrounding surfaces using dry machinery towel, Item 211, Appendix D.

NOTE

Do not allow technical trichloroethylene to dry on parts.

q. Clean holes, shaft, and lever using clean machinery towel, Item 211, Appendix D, wet with technical trichloroethylene, Item 343, Appendix D. Wipe dry.

r. Using epoxy primer coating kit, Item 135, Appendix D, and acid brush, Item 84, Appendix D, apply one coat of epoxy primer to tapered pin holes in shaft and lever. Allow to dry.

s. Using acid brush, Item 84, Appendix D, apply a thin coat (0.002 - 0.003-inch) of sealing compound, Item 292 or Item 301A, Appendix D, covering tapered pins.

t. Install tapered pins, wet with sealing compound, into proper holes in shaft and lever. **SMALL DIAMETER END OF TAPERED PIN SHANK SHALL NOT EXTEND MORE THAN 1/16-INCH ABOVE LEVER SURFACE.** If end of tapered pin extends more than 1/16-inch, replace pin to obtain correct dimension.

u. Install washers and nuts making sure that sealing compound is visible between pin surfaces, washers, and both sides of lever. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins. If sealing compound, Item 301A, Appendix D, was used, make sure to wipe off excess sealing compound with clean machinery towel, Item 211, Appendix D.

v. **TORQUE LOWER SUPPORT NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-56-1, Sheet 2, Detail F).

w. Rotate torque shaft 360°. **TORQUE SHAFT MUST TURN FREELY AND FRICTION MUST NOT EXCEED 20 INCH-OUNCES.** If shaft does not turn freely or friction exceeds 20 inch-ounces, torque shaft is preloaded (less than zero gap). Disassemble and repeat step g. through step w.

x. Install pilot’s and copilot’s longitudinal stick rigging pins.

y. Install pitch torque shaft rigging pin.

z. Install pitch link (PARA 11-4-51).

aa. Install pitch/trim input control rod (PARA 11-4-64).

ab. Connect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ac. Install pilot’s and copilot’s lateral stick rigging pins.

ad. Install pilot’s and copilot’s pedal adjustor rigging pins and yaw boost actuator rigging pin.

ae. Move collective stick to full down position, and move cyclic stick to full forward position.

VERIFY FORWARD LIMITER IN MIXER IS CONTACTED BY ROLLER.

af. While holding collective stick in full down position, move cyclic stick to full aft position. **VERIFY AFT LIMITER IN MIXER IS CONTACTED BY ROLLER.**

ag. Move collective sticks to full up position.

ah. Move cyclic stick to full forward position. **VERIFY FORWARD LIMITER IN MIXER IS CONTACTED BY ROLLER.**

ai. Move cyclic stick to full aft position. **VERIFY AFT STICK STOPS ARE CONTACTED.**

aj. Remove all rigging pins.

ak. Disconnect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

al. Install balance spring as follows (Figure 11-4-56-1, Sheet 1, Detail B):

(1) Position balance spring in lever.

(2) Install pin, washer, and cotter pin.

NOTE

If measurement is not known, balance spring will need to be adjusted.

(3) Adjust balance spring rod to measurement recorded when removed.

(4) **TIGHTEN JAMNUTS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

am. If necessary, adjust balance spring (PARA 11-4-61).

an. Make sure area is clean and free of foreign material.

ao. Install soundproofing panels in cabin overhead (PARA 2-4-69).

ap. Close and latch main rotor pylon sliding cover.

aq. Perform maintenance test flight (Appropriate MTF).

11-4-57 LATERAL TORQUE SHAFT AND LEVERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Brass Drift
- Dial Indicator
- Feeler Gage
- Hammer
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. oz
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 37, Appendix D
- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-51
- PARA 11-4-54
- PARA 11-4-65

11-4-57.1 Replace Lateral Torque Shaft and Levers.

11-4-57.1.1 Remove.

a. Remove trim servo control rod (PARA 11-4-54).

b. Remove lateral input control rod (PARA 11-4-65).

c. Remove lateral link (PARA 11-4-51).

d. Remove nuts and washers from tapered pins (Figure 11-4-57-1, Sheet 1, Detail B).

e. Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

f. Remove tapered pins using hammer and brass drift. Remove lower lever from torque shaft.

g. Remove torque shaft and upper lever from supports (Figure 11-4-57-1, Sheet 1, Detail B). Remove spacer and boot assembly (Figure 11-4-57-1, Sheet 2, Detail C).

h. Remove bolts, washers, and nuts securing lower support and shims to airframe. Remove lower support and spacer tube from airframe (Figure 11-4-57-1, Sheet 2, Detail D).

i. Remove sealing compound from parts as follows:

(1) Remove sealing compound from lower support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

j. Remove bolts and washers securing upper support, shims, spacers, and rigging bracket to airframe. Remove spacers.

k. Remove sealing compound from parts as follows:

(1) Remove sealing compound from upper support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

11-4-57.1.2 Inspect.

a. Inspect tapered pin shank. Tapered surface must be smooth and free of damage, nicks, dents, scratches, or scuff marks. **NO DAMAGE ALLOWED.** If damaged, replace pin.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

b. Using dial indicator, inspect bearing in lower lever. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0025-INCH.**

c. Inspect bearing in upper and lower supports as follows:

(1) Using dial indicator, inspect radial play. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

(2) Check bearings for smooth operation. **IF BEARING FEELS ROUGH OR CANNOT BE ROTATED, REPLACE BEARING.**

d. Using dial indicator, inspect bearing in lateral lever and torque shaft. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

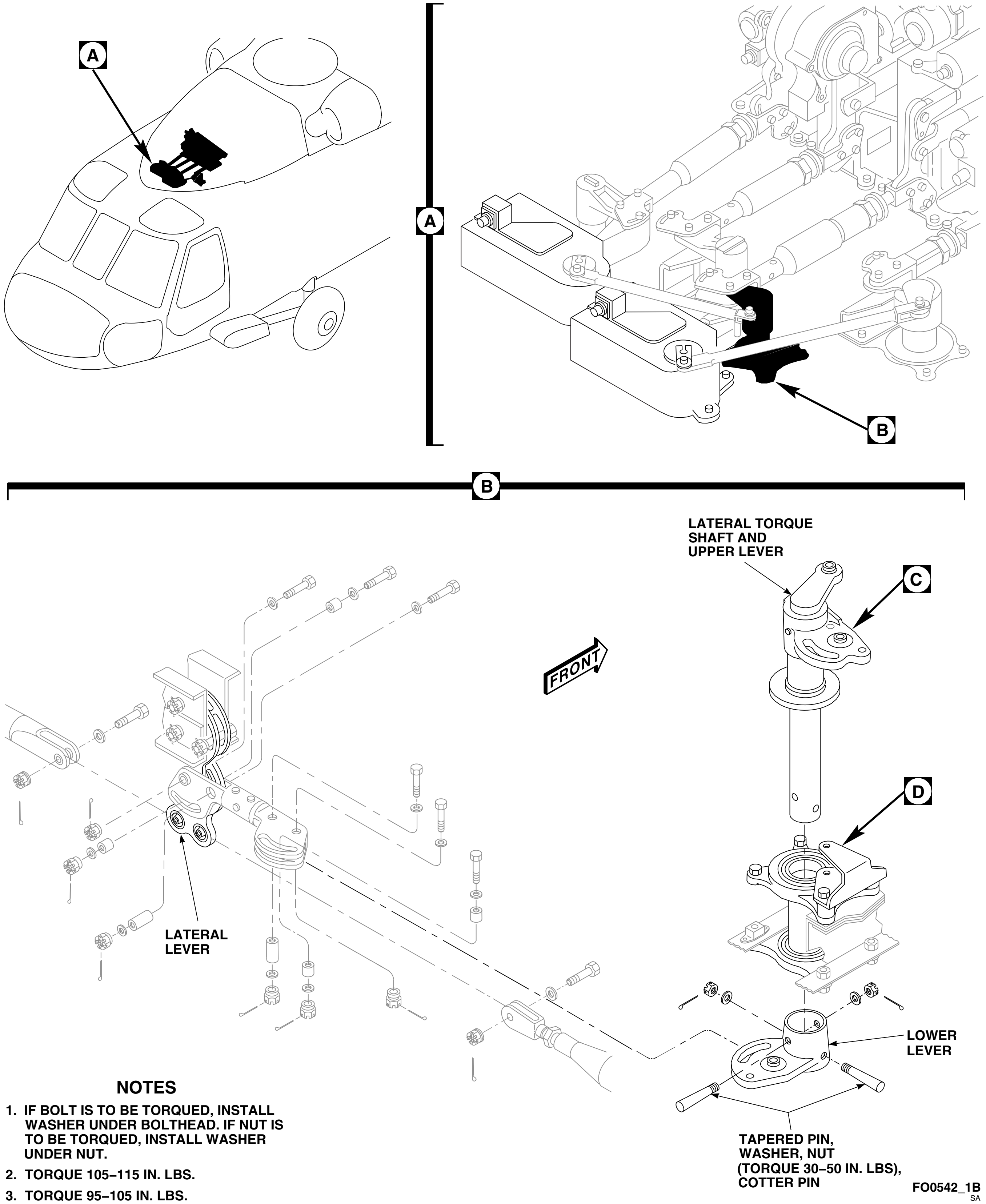


FIGURE 11-4-57-1. REPLACE LATERAL TORQUE SHAFT AND LEVERS (SHEET 1 OF 2)

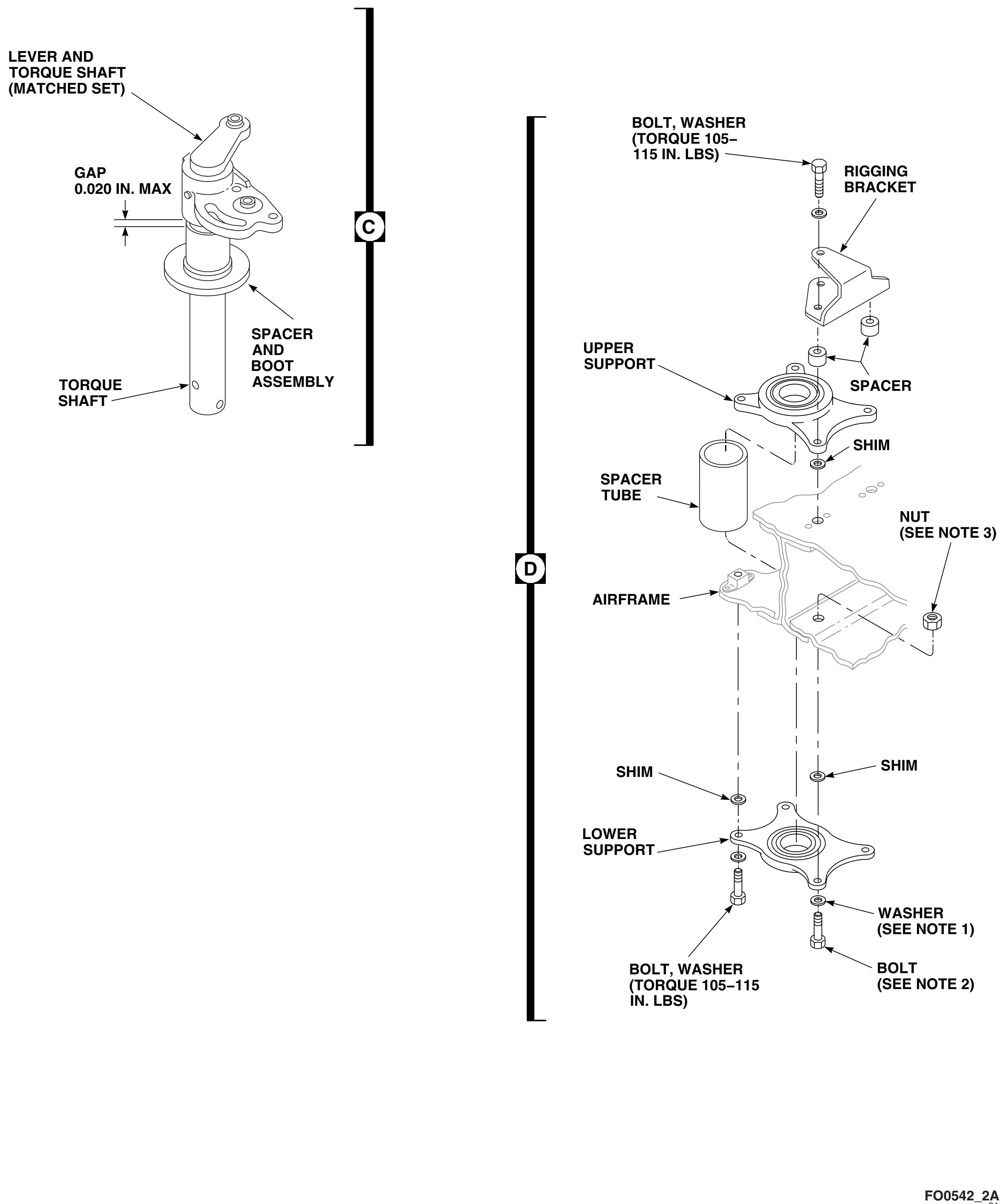


FIGURE 11-4-57-1. REPLACE LATERAL TORQUE SHAFT AND LEVERS (SHEET 2 OF 2)

e. Inspect rigging bracket for deformation. **NONE ALLOWED.** If bracket shows evidence of deformation, replace bracket.

11-4-57.1.3 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Bond replacement boot to spacer as follows:

(1) If old bond is on spacer, clean with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and wipe dry with clean cloth.

(2) Apply a coat of adhesive, Item 37, Appendix D, on one end of spacer, about as wide as boot collar. Apply a coat of adhesive on inside of boot collar.

(3) Allow adhesive to dry for at least 10 minutes.

(4) Apply a second coat of adhesive to mating surfaces and allow at least 10 minutes for it to dry.

(5) Slide boot on end of spacer and keep lower face of boot and end face of spacer flush. Press collar on boot against spacer all around to make sure of complete contact.

(6) Allow to dry for at least 8 hours.

c. Install upper support, lower support, and torque shaft as follows:

(1) Install two bolts and washers in right mounting holes of upper support (Figure 11-4-57-1, Sheet 2, Detail D).

(2) Place spacers and rigging bracket on right side of upper support. Face rigging hole in bracket toward center of support.

(3) Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

(4) Install spacer tube through opening in airframe.

(5) Install replacement lower support (flat side up), against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

(6) Install bolts (bolthead down), washers, and nuts on right side of support using a maximum of three washers in any combination. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(7) Install bolts (bolthead down) and washers on left side of support using a maximum of three washers in any combination. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(8) Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

(9) On replacement torque shaft, tap out tapered pins using hammer and brass drift. Remove lower lever from torque shaft.

(10) Install spacer and boot assembly on torque shaft (Figure 11-4-57-1, Sheet 2, Detail C), and slide shaft in upper and lower supports (Figure 11-4-57-1, Sheet 1, Detail B).

(11) Install lower lever on torque shaft, line up tapered pin holes and install one tapered pin. Tap pin in to make sure it is fully seated. Do not install nut and washer now.

d. On upper deck, pull up on torque shaft and use feeler gage to measure gap between upper lever and spacer (Figure 11-4-57-1, Sheet 2, Detail C). Record measurement.

e. Remove tapered pin and remove lower lever from torque shaft. Remove torque shaft from supports (Figure 11-4-57-1, Sheet 1, Detail B).

f. Remove bolts, washers, and nuts securing upper support, lower support, and spacer tube to airframe (Figure 11-4-57-1, Sheet 2, Detail D).

g. Shim torque shaft and levers as follows:

NOTE

- If measurement taken in step d. is 0.020-inch or less, shimming is not required.
- Shims may be installed at the upper support location, lower support location, or both locations.
- Total thickness of shims at the upper support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims at the lower support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims installed cannot exceed 0.162-inch.

(1) If measurement taken was 0.021-inch or more, add shims to obtain a maximum gap of 0.020-inch.

(2) Apply a light coat of adhesive, Item 32, Appendix D, on shims and around mounting holes on helicopter airframe. Place shims on helicopter airframe.

(3) Clean extra adhesive from holes in shims and lower support.

(4) Allow sealing compound to dry for 24 hours or use heat gun at temperature of 110° - 130°F (43° - 54°C) for two hours.

h. Install two bolts and washers in left mounting holes of upper support.

i. Place spacers and rigging bracket on right side of upper support. Face rigging hole in bracket toward center of support.

j. Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

k. Install spacer tube through opening in airframe.

l. Install replacement lower support (flat side up), against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

m. Install bolts (bolthead down), washers, and nuts securing lower support, using a maximum of three washers in any combination. Do not torque at this time.

n. Install spacer and boot assembly on torque shaft and slide shaft in upper and lower supports (Figure 11-4-57-1, Sheet 2, Detail C).

NOTE

Record hole and tapered pin positions in shaft and lever before cleaning and priming.

o. Install lower lever on torque shaft and line up tapered pin holes (Figure 11-4-57-1, Sheet 1, Detail B).

p. Wipe holes and surrounding surfaces using dry machinery towel, Item 211, Appendix D.

NOTE

Do not allow technical trichloroethylene to dry on parts.

q. Clean holes, shaft, and lever using clean machinery towel, Item 211, Appendix D, wet with technical trichloroethylene, Item 343, Appendix D. Wipe dry.

r. Using epoxy primer coating kit, Item 135, Appendix D, and acid brush, Item 84, Appendix D, apply one coat of epoxy primer to tapered pin holes in shaft and lever. Allow to dry.

s. Using acid brush, Item 84, Appendix D, apply a thin coat (0.002 - 0.003-inch) of sealing compound, Item 292 or Item 301A, Appendix D, covering tapered pins.

t. Install tapered pins, wet with sealing compound, into proper holes in shaft and lever. **SMALL DIAMETER END OF TAPERED PIN SHANK SHALL NOT EXTEND MORE THAN 1/16-INCH ABOVE LEVER SURFACE.** If end of tapered pin extends more than 1/16-inch, replace pin to obtain correct dimension.

u. Install washers and nuts making sure that sealing compound is visible between pin surfaces, washers, and both sides of lever. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins. If sealing compound, Item 301A, Appendix D, was used, make sure to wipe off excess sealing compound with clean machinery towel, Item 211, Appendix D.

v. **TORQUE LOWER SUPPORT NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-57-1, Sheet 2, Detail D).

w. Rotate torque shaft 360°. **TORQUE SHAFT MUST TURN FREELY AND FRICTION MUST NOT EXCEED 20 INCH-OUNCES.** If shaft does not turn freely or friction exceeds 20 inch-ounces, torque shaft is preloaded (less than zero gap). Disassemble and repeat step g. though step w.

x. Install pilot's and copilot's lateral stick rigging pins.

y. Install lateral torque shaft rigging pin.

z. Install lateral link (PARA 11-4-51).

aa. Install trim servo control rod (PARA 11-4-54).

ab. Install lateral input control rod (PARA 11-4-65).

ac. Connect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ad. Install lateral torque shaft rigging pin. **VERIFY LATERAL PILOT ASSIST ACTUATOR RIGGING PIN CAN BE INSTALLED.** If required, adjust lateral input control rod for lateral pilot assist actuator rigging pin to be installed.

ae. Remove all rigging pins.

af. Move collective sticks to full down position.

ag. Move cyclic stick to full left position. **VERIFY LEFT LIMITER IN MIXER IS CONTACTED BY ROLLER. VERIFY STICK STOPS ARE 0.010 - 0.030-INCH CLEAR.** If required, adjust stick stops.

ah. Move cyclic stick to full right position. **VERIFY RIGHT LIMITER IN MIXER IS**

CONTACTED BY ROLLER. VERIFY STICK STOPS ARE 0.010 - 0.030-INCH CLEAR. If required, adjust stick stops.

ai. Move collective sticks to full up position.

aj. Move cyclic stick to full left position. **VERIFY LEFT LIMITER IN MIXER IS CONTACTED BY ROLLER.**

ak. Move cyclic stick to full right position. **VERIFY RIGHT LIMITER IN MIXER IS CONTACTED BY ROLLER.**

al. Disconnect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

am. Make sure area is clean and free of foreign material.

an. Install soundproofing panels in cabin overhead (PARA 2-4-69).

ao. Close and latch main pylon rotor pylon sliding cover.

ap. Perform maintenance test flight (Appropriate MTF).

11-4-58 YAW TORQUE SHAFT AND LEVERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Brass Drift
- Dial Indicator
- Feeler Gage
- Hammer
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. oz
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 37, Appendix D
- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-46
- PARA 11-4-54
- PARA 11-4-66

11-4-58.1 Replace Yaw Torque Shaft and Levers.

11-4-58.1.1 Remove.

a. Remove trim servo control rod (PARA 11-4-54).

b. Remove yaw boost servo input control rod (PARA 11-4-66).

c. Disconnect overhead control rods from lower lever on yaw torque shaft (PARA 11-4-46).

d. Remove nuts and washers from tapered pins (Figure 11-4-58-1, Sheet 1, Detail B).

e. Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

f. Remove tapered pins using hammer and brass drift. Remove lower lever from torque shaft.

g. Remove torque shaft and upper lever from supports (Figure 11-4-58-1, Sheet 1, Detail B). Remove spacer and boot assembly (Figure 11-4-58-1, Sheet 2, Detail C).

h. Remove bolts and washers securing upper support, shims, spacers, and rigging bracket to airframe (Figure 11-4-58-1, Sheet 2, Detail D). Remove spacers.

i. Remove sealing compound from parts as follows:

(1) Remove sealing compound from upper support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

j. Remove bolts, washers, and nuts securing lower support and shims to airframe.

k. Remove sealing compound from parts as follows:

(1) Remove sealing compound from lower support, shims, and airframe using nonmetallic scraper.

(2) Clean remaining sealing compound from mating surfaces with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

11-4-58.1.2 Inspect.

a. Inspect tapered pin shank. Tapered surface must be smooth and free of damage, nicks, dents, scratches, or scuff marks. **NO DAMAGE ALLOWED.** If damaged, replace pin.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

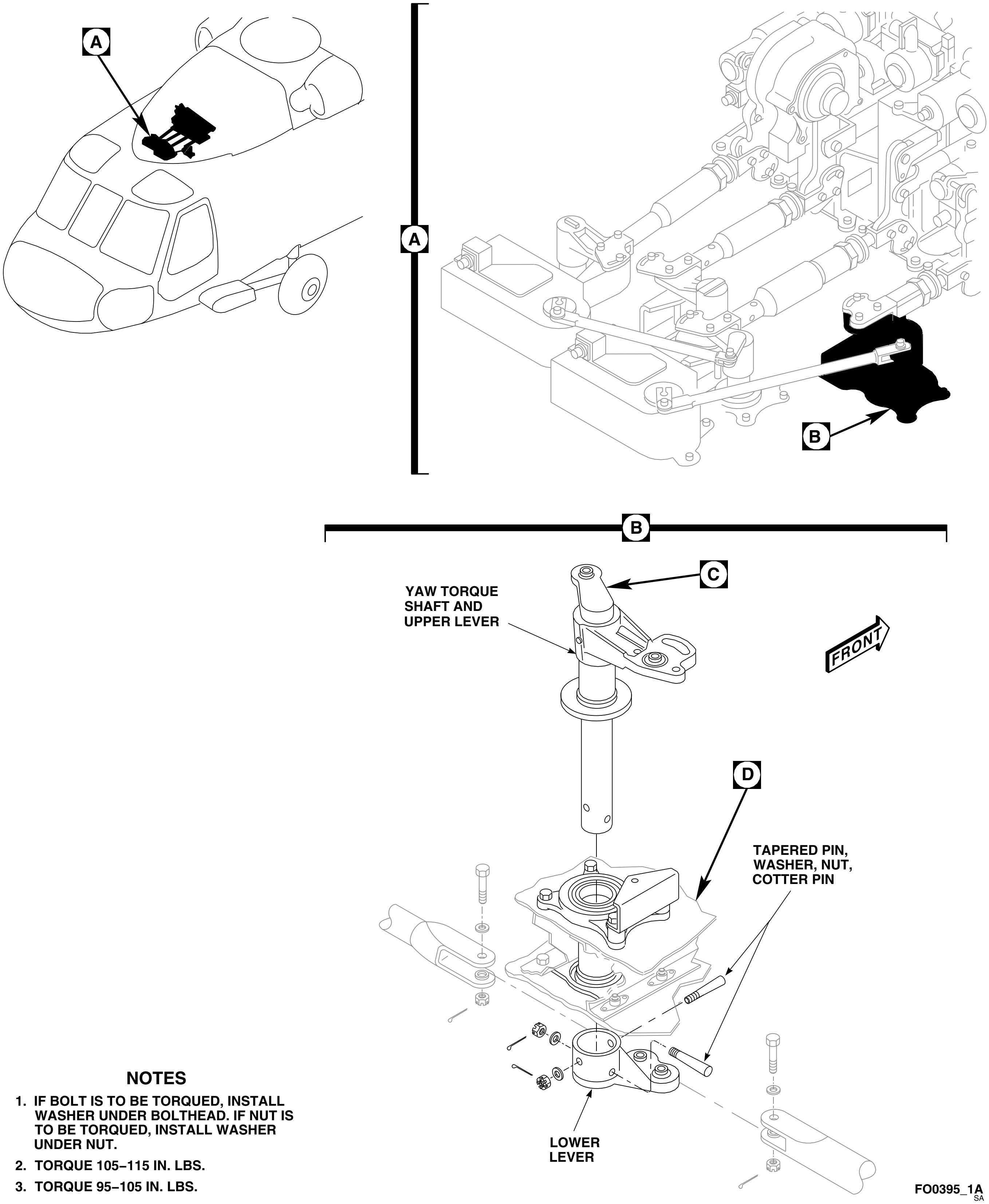
b. Using dial indicator, inspect bearing in lower lever. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0025-INCH.**

c. Inspect bearing in upper and lower supports as follows:

(1) Using dial indicator, inspect radial play. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

(2) Check bearings for smooth operation. **IF BEARING FEELS ROUGH OR CANNOT BE ROTATED, REPLACE BEARING.**

d. Using dial indicator, inspect bearing in yaw lever and torque shaft. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**



NOTES

1. IF BOLT IS TO BE TORQUED, INSTALL WASHER UNDER BOLTHEAD. IF NUT IS TO BE TORQUED, INSTALL WASHER UNDER NUT.
2. TORQUE 105–115 IN. LBS.
3. TORQUE 95–105 IN. LBS.

FIGURE 11-4-58-1. REPLACE YAW TORQUE SHAFT AND LEVERS (SHEET 1 OF 2)

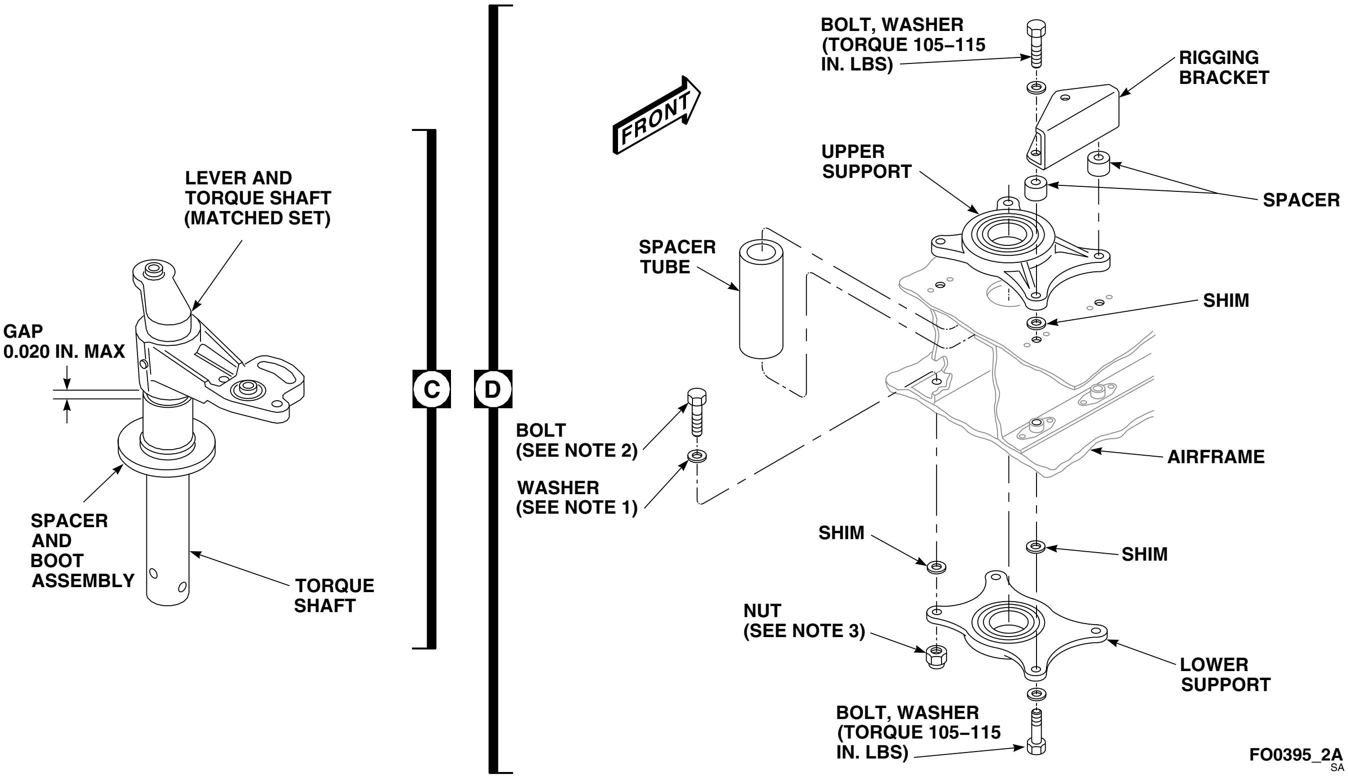


FIGURE 11-4-58-1. REPLACE YAW TORQUE SHAFT AND LEVERS (SHEET 2 OF 2)

e. Inspect rigging bracket for deformation. **IF BRACKET SHOWS EVIDENCE OF DEFORMATION, REPLACE BRACKET.**

11-4-58.1.3 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Bond replacement boot to spacer as follows:

(1) If old bond is on spacer, clean with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and wipe dry with clean cloth.

(2) Apply a coat of adhesive, Item 37, Appendix D, on one end of spacer, about as wide as boot collar. Apply a coat of adhesive on inside of boot collar.

(3) Allow adhesive to dry for at least 10 minutes.

(4) Apply a second coat of adhesive to mating surfaces and allow at least 10 minutes for it to dry.

(5) Slide boot on end of spacer and keep lower face of boot and end face of spacer flush. Press collar on boot against spacer all around to make sure of complete contact.

(6) Allow to dry for at least 8 hours.

c. Install upper support, lower support, and torque shaft as follows:

(1) Install two bolts and washers in left mounting holes of upper support (Figure 11-4-58-1, Sheet 2, Detail D).

(2) Place spacers and rigging bracket on right side of upper support. Face rigging hole in bracket toward center of support.

(3) Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

(4) Install spacer tube through opening in airframe.

(5) Install replacement lower support (flat side up) against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.
- When securing right side of lower support, install boltheads facing down. When securing left side of lower support, install boltheads facing up.

(6) Install bolts, washers, and nuts, if required, using a maximum of three washers in any combination. **TORQUE LOWER SUPPORT NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

(7) Before removing tapered pins, mark one tapered pin and its hole to make sure pin is installed in same hole from which it was removed.

CAUTION

Damage to equipment will occur if only metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift with hammer when removing tapered pins.

(8) On replacement torque shaft, tap out tapered pins using hammer and brass drift. Remove lower lever from torque shaft.

(9) Install spacer and boot assembly on torque shaft (Figure 11-4-58-1, Sheet 2, Detail C), and slide shaft in upper and lower supports (Figure 11-4-58-1, Sheet 1, Detail B).

(10) Install lower lever on torque shaft, line up tapered pin holes and install one tapered pin. Tap pin in to make sure it is fully seated. Do not install nut and washer now.

d. On upper deck, pull up on torque shaft and use feeler gage to measure gap between upper lever and spacer (Figure 11-4-58-1, Sheet 2, Detail C). Record measurement.

e. Remove tapered pin and remove lower lever from torque shaft. Remove torque shaft from supports (Figure 11-4-58-1, Sheet 1, Detail B).

f. Remove bolts, washers, and nuts securing upper support, lower support, and spacer tube to airframe (Figure 11-4-58-1, Sheet 2, Detail D).

g. Shim torque shaft and levers as follows:

NOTE

- If measurement taken in step d. is 0.020-inch or less, shimming is not required.
- Shims may be installed at the upper support location, lower support location, or both locations.
- Total thickness of shims at the upper support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims at the lower support location shall be within 0.032-inch thickness of each other.
- Total thickness of shims installed cannot exceed 0.162-inch.

(1) If measurement taken was 0.021-inch or more, add shims to obtain a maximum gap of 0.020-inch.

(2) Apply a light coat of adhesive, Item 32, Appendix D, on shims and around mounting holes on airframe. Place shims on airframe.

(3) Clean extra adhesive from holes in shims and lower support.

(4) Allow sealing compound to dry for 24 hours or use heat gun at temperature of 110° - 130°F (43° - 54°C) for two hours.

h. Install two bolts and washers in left mounting holes of upper support.

i. Place spacers and rigging bracket on right side of upper support. Face rigging hole in bracket toward center of support.

j. Install bolts and washers in rigging bracket and upper support. **TORQUE ALL BOLTS TO 105 - 115 INCH-POUNDS.**

k. Install spacer tube through opening in airframe.

l. Install replacement lower support (flat side up) against airframe.

NOTE

- If bolt is to be torqued, a minimum of one thin washer must be used under bolthead. If nut is to be torqued, a minimum of one thin washer must be used under nut.
- A maximum of three washers may be used under bolthead or nut to make sure of proper bolt length.

m. Install bolts (bolthead up), washers, and nuts on left side of lower support, using a maximum of three washers in any combination. Do not torque at this time.

NOTE

- A minimum of one thin washer must be used under bolthead.
- A maximum of three washers may be used under bolthead to make sure of proper bolt length.

n. Install bolts (bolthead down) and washers on right side of lower support. Do not torque bolts at this time.

o. Install spacer and boot assembly on torque shaft (Figure 11-4-58-1, Sheet 2, Detail C), and slide shaft in upper and lower supports (Figure 11-4-58-1, Sheet 1, Detail B).

NOTE

Record hole and tapered pin positions in shaft and lever before cleaning and priming.

p. Install lower lever on torque shaft and line up tapered pin holes.

q. Wipe holes and surrounding surfaces using dry machinery towel, Item 211, Appendix D.

NOTE

Do not allow technical trichloroethylene to dry on parts.

r. Clean holes, shaft, and lever using clean machinery towel, Item 211, Appendix D, wet with technical trichloroethylene, Item 343, Appendix D. Wipe dry.

s. Using epoxy primer coating kit, Item 135, Appendix D, and acid brush, Item 84, Appendix D, apply one coat of epoxy primer to tapered pin holes in shaft and lever. Allow to dry.

t. Using acid brush, Item 84, Appendix D, apply a thin coat (0.002 - 0.003-inch) of sealing compound, Item 292 or Item 301A, Appendix D, covering tapered pins.

u. Install tapered pins, wet with sealing compound, into proper holes in shaft and lever. **SMALL DIAMETER END OF TAPERED PIN SHANK SHALL NOT EXTEND MORE THAN 1/16-INCH ABOVE LEVER SURFACE.** If end of tapered pin extends more than 1/16-inch, replace pin to obtain correct dimension.

v. Install washers and nuts making sure that sealing compound is visible between pin surfaces, washers, and both sides of lever. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins. If sealing compound, Item 301A, Appendix D, was used, make sure to wipe off excess sealing compound with clean machinery towel, Item 211, Appendix D.

w. **TORQUE LOWER SUPPORT NUTS TO 95 - 105 INCH-POUNDS.** If torquing bolts, **TORQUE BOLTS TO 105 - 115 INCH-POUNDS** (Figure 11-4-58-1, Sheet 2, Detail D).

x. Rotate torque shaft 360°. **TORQUE SHAFT MUST TURN FREELY AND FRICTION MUST NOT EXCEED 20 INCH-OUNCES.** If shaft does not turn freely or friction exceeds 20 inch-ounces, torque shaft is preloaded (less than zero gap). Disassemble and repeat step g. through step x.

y. Install pilot's and copilot's yaw pedal adjustor rigging pins.

z. Install yaw torque shaft rigging pin.

aa. Connect overhead control rods to lower lever on yaw torque shaft (PARA 11-4-46).

ab. Install yaw boost servo input control rod (PARA 11-4-66).

ac. Install trim servo control rod (PARA 11-4-54).

ad. Connect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ae. Install yaw torque shaft rigging pin. **VERIFY YAW BOOST ACTUATOR RIGGING PIN CAN BE INSTALLED.** If required, adjust yaw boost servo input control rod.

af. Remove all rigging pins.

ag. Move collective sticks to full down position and lock in place.

ah. Move pedals full left. **VERIFY YAW BOOST SERVO BOTTOMS.**

ai. Move pedals full right. **VERIFY RIGHT YAW STOP IN CABIN OVERHEAD IS CONTACTED.**

aj. Move collective sticks to full up position and lock in place.

ak. Move pedals full right. **VERIFY YAW BOOST SERVO BOTTOMS.**

al. Move pedals full left. **VERIFY LEFT YAW STOP IN CABIN OVERHEAD IS CONTACTED.**

am. Disconnect external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

an. Make sure area is clean and free of foreign material.

ao. Install soundproofing panels in cabin overhead (PARA 2-4-69).

ap. Close and latch main rotor pylon sliding cover.

aq. Perform maintenance test flight (Appropriate MTF).

11-4-59 TORQUE SHAFT UPPER TAPERED PINS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Brass Drift
- Hammer
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- PARA 11-4-55
- PARA 11-4-56
- PARA 11-4-57
- PARA 11-4-58

11-4-59.1 Replace Torque Shaft Upper Tapered Pins.

NOTE

- This procedure is only required if tapered pin securing upper torque shaft lever to torque shaft is found to be loose. Under normal conditions, there would be no reason to remove upper tapered pins as upper lever and torque shaft are a matched set and cannot be replaced individually.
- Replacement of all upper tapered pins is the same.

11-4-59.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove nut and washer from torque shaft upper tapered pin (Figure 11-4-59-1, Detail A).

CAUTION

Damage to equipment will occur if metal hammer is used to remove tapered pins. Threads will be damaged. Use brass drift and hammer when removing tapered pins.

c. Using hammer and brass drift, remove tapered pin.

11-4-59.1.2 Install.

a. Wipe holes and surrounding surfaces using dry machinery towel, Item 211, Appendix D.

NOTE

Do not allow technical trichloroethylene to dry on parts.

b. Clean holes, shaft, and lever using clean machinery towel, Item 211, Appendix D, wet with technical trichloroethylene, Item 343, Appendix D. Wipe dry.

c. Using epoxy primer coating kit, Item 135, Appendix D, and acid brush, Item 84, Appendix D, apply two coats of epoxy primer to tapered pin holes in shaft and lever. Allow each coat to dry.

d. Using acid brush, Item 84, Appendix D, apply a thin coat (0.002 - 0.003-inch) of sealing compound, Item 292 or Item 301A, Appendix D, covering tapered pins.

e. Install tapered pins, wet with sealing compound, into proper holes in torque shaft and lever (Figure 11-4-59-1, Detail A). **SMALL DIAMETER END OF TAPERED PIN SHANK SHALL NOT EXTEND MORE THAN 1/16-INCH ABOVE LEVER SURFACE.** If end of tapered pin extends more than 1/16-inch, replace torque shaft and lever assembly (PARA 11-4-55, PARA 11-4-56, PARA 11-4-57, or PARA 11-4-58).

f. Install washers and nuts making sure that sealing compound is visible between pin surfaces, washers, and both sides of lever.

g. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

h. If sealing compound, Item 301A, Appendix D, was used, make sure to wipe off excess sealing compound with clean machinery towel, Item 211, Appendix D.

i. Check tapered pin for tightness. If tapered pin is still loose, replace torque shaft and lever assembly (PARA 11-4-55, PARA 11-4-56, PARA 11-4-57, or PARA 11-4-58).

j. Close and latch main rotor pylon sliding cover.

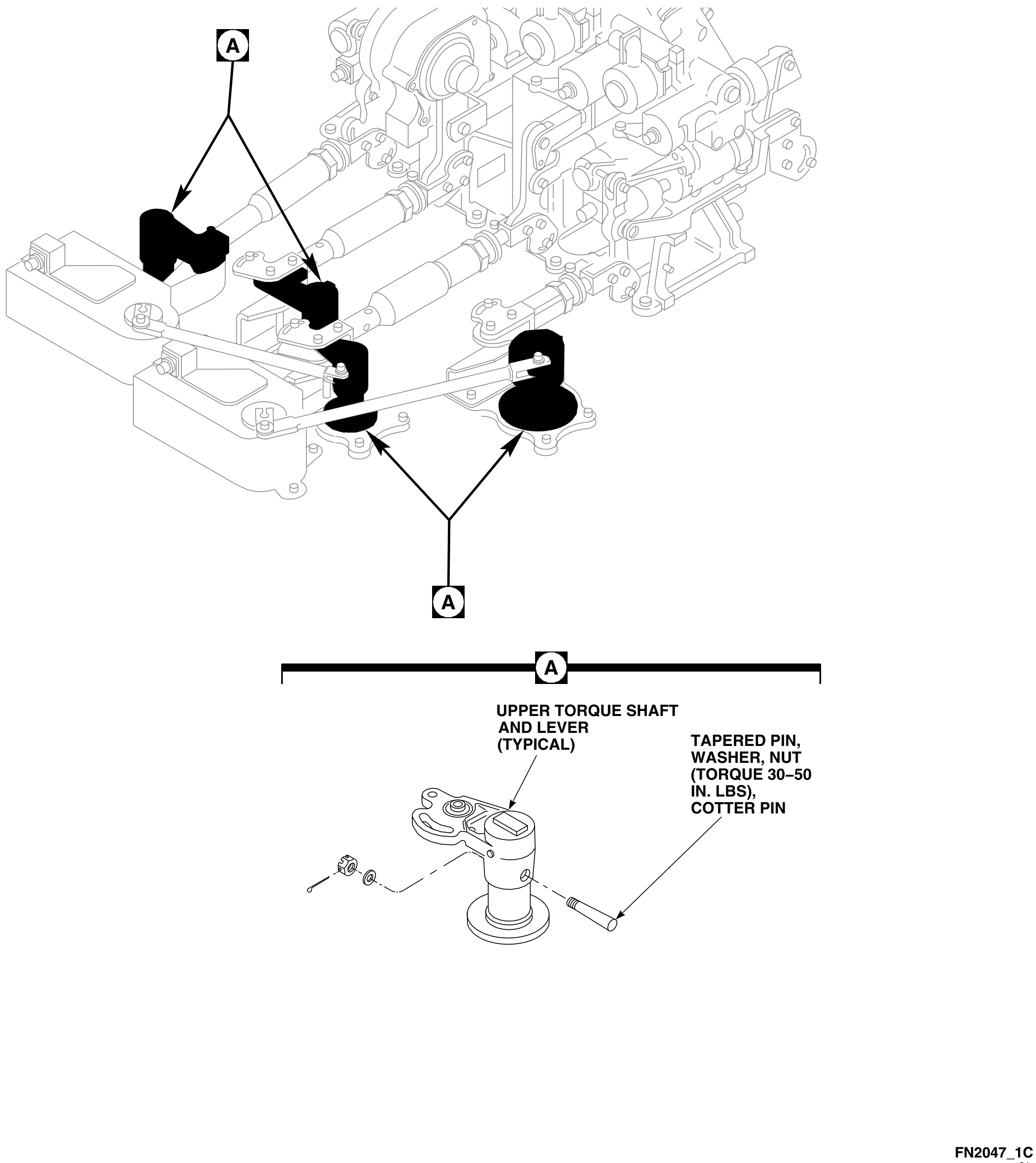


FIGURE 11-4-59-1. REPLACE TORQUE SHAFT UPPER TAPERED PINS

11-4-60 BALANCE SPRINGS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-61
- PARA 11-4-62
- PARA 13-4-11

11-4-60.1 Replace Balance Springs.

NOTE

Replacement of longitudinal (pitch) balance spring and collective balance spring is the same, except as noted.

11-4-60.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Move yaw pedals to midposition.

c. Remove soundproofing panels from cabin overhead, STA 247.0 to 265.0 (PARA 2-4-69).

d. Remove heater duct (PARA 13-4-11).

e. Loosen outer jamnut on adjusting screw until balance spring tension is released (Figure 11-4-60-1, Detail A).

f. Remove clevis pin and washer from each end of balance spring and remove spring.

g. If necessary, remove cotter pin, outer jamnut, adjusting screw, and inner jamnut from spring bracket.

11-4-60.1.2 Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. If inner jamnut, adjusting screw, and outer jamnut were removed, loosely install on spring bracket. Install outer jamnut just enough to hold adjusting screw in place. Install cotter pin (Figure 11-4-60-1, Detail A).

c. For collective balance spring, manually compress spring and secure with lockwire, Item 197, Appendix D.

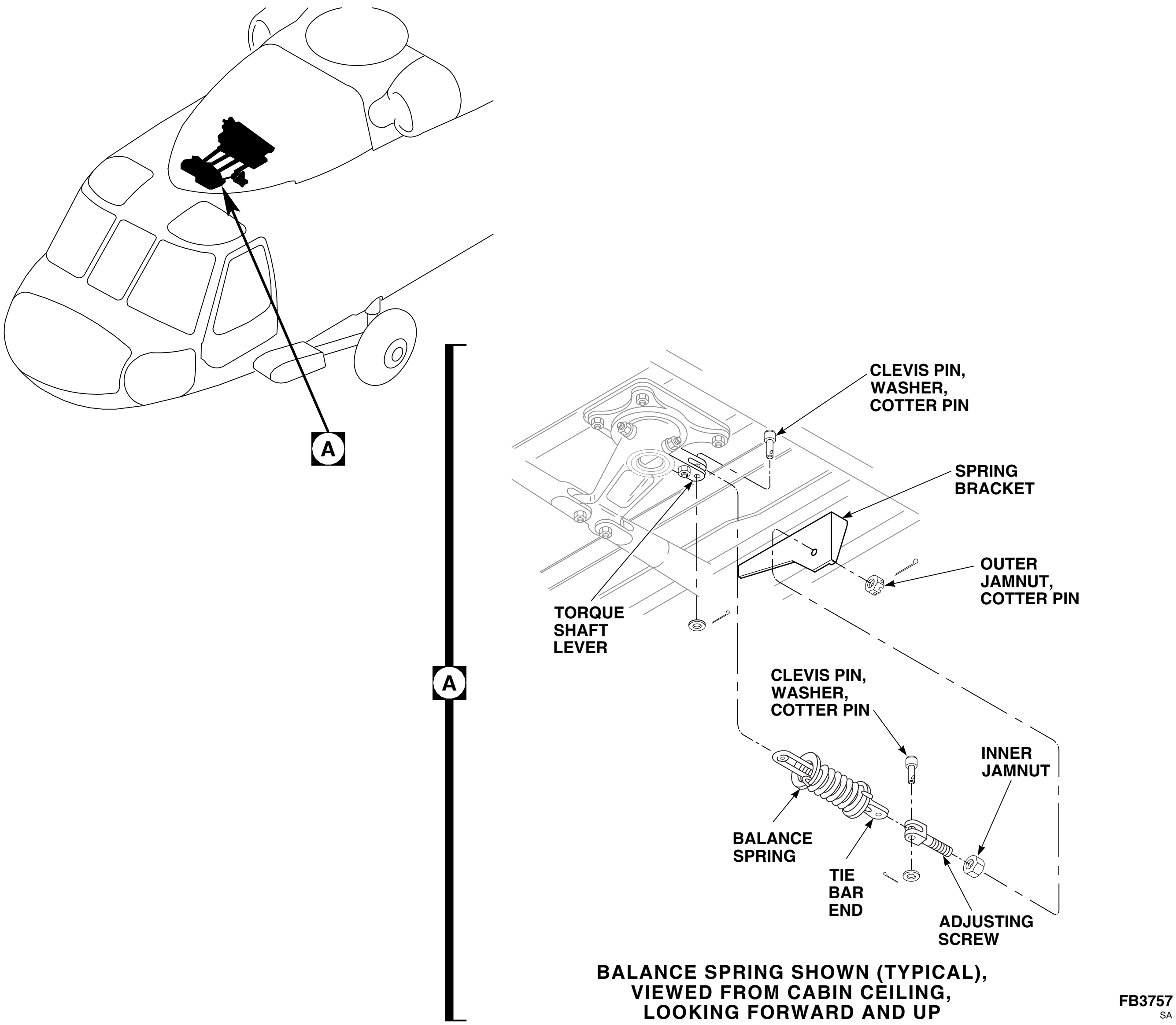


FIGURE 11-4-60-1. REPLACE BALANCE SPRINGS

NOTE

Make sure clevis pin is vertical when installed.

d. Position balance spring on adjusting screw and torque shaft lever, with tie bar end towards adjusting screw. Install clevis pin, washer, and cotter pin.

e. Connect balance spring to torque shaft lever using clevis pin, washer, and cotter pin.

f. For collective balance spring, remove lockwire to decompress spring.

g. Perform balance spring adjustment (PARA 11-4-61 or PARA 11-4-62).

h. Make sure area is clean and free of foreign material.

i. Install heater duct (PARA 13-4-11).

j. Install soundproofing panels in cabin overhead (PARA 2-4-69).

11-4-61 CYCLIC STICK BALANCE SPRING ADJUSTMENT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 2-4-69
- PARA 11-4-64
- PARA 13-4-11

11-4-61.1 Adjust Cyclic Stick Balance Spring.

NOTE

Balance spring adjustment is performed only after main rotor controls have been rigged to ensure proper relationship between cyclic sticks and cyclic stick balance springs.

a. Disconnect pitch/trim input control rod at pitch torque shaft lever (PARA 11-4-64).

b. Pull cyclic stick full aft against stop. Let go of stick. Measure gap between cyclic stick and aft stick stop. **GAP SHOULD BE 0.300-INCH MAXIMUM.** If gap is beyond limits, perform the following:

(1) Remove soundproofing panels in cabin overhead as required to access torque shaft (PARA 2-4-69).

(2) Remove heater duct (PARA 13-4-11).

NOTE

- Backing off inner jamnut and tightening outer jamnut increases spring tension, causing cyclic stick to move forward.
- Do not allow adjusting screw to turn while adjusting or tightening jamnuts. Tie bars could twist and disengage from slots in retainer tube.

(3) Adjust inner and outer jamnuts on spring bracket. **TIGHTEN JAMNUTS ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 11-4-61-1, Detail B).

(4) Remeasure gap per step b.

(5) Measure compressed length of balance spring. **COMPRESSED LENGTH TO BE NO LESS THAN 1.800-INCHES.**

(6) Move cyclic stick front and rear. Check force is same in both directions. At mid-position, cyclic stick should not fall forward.

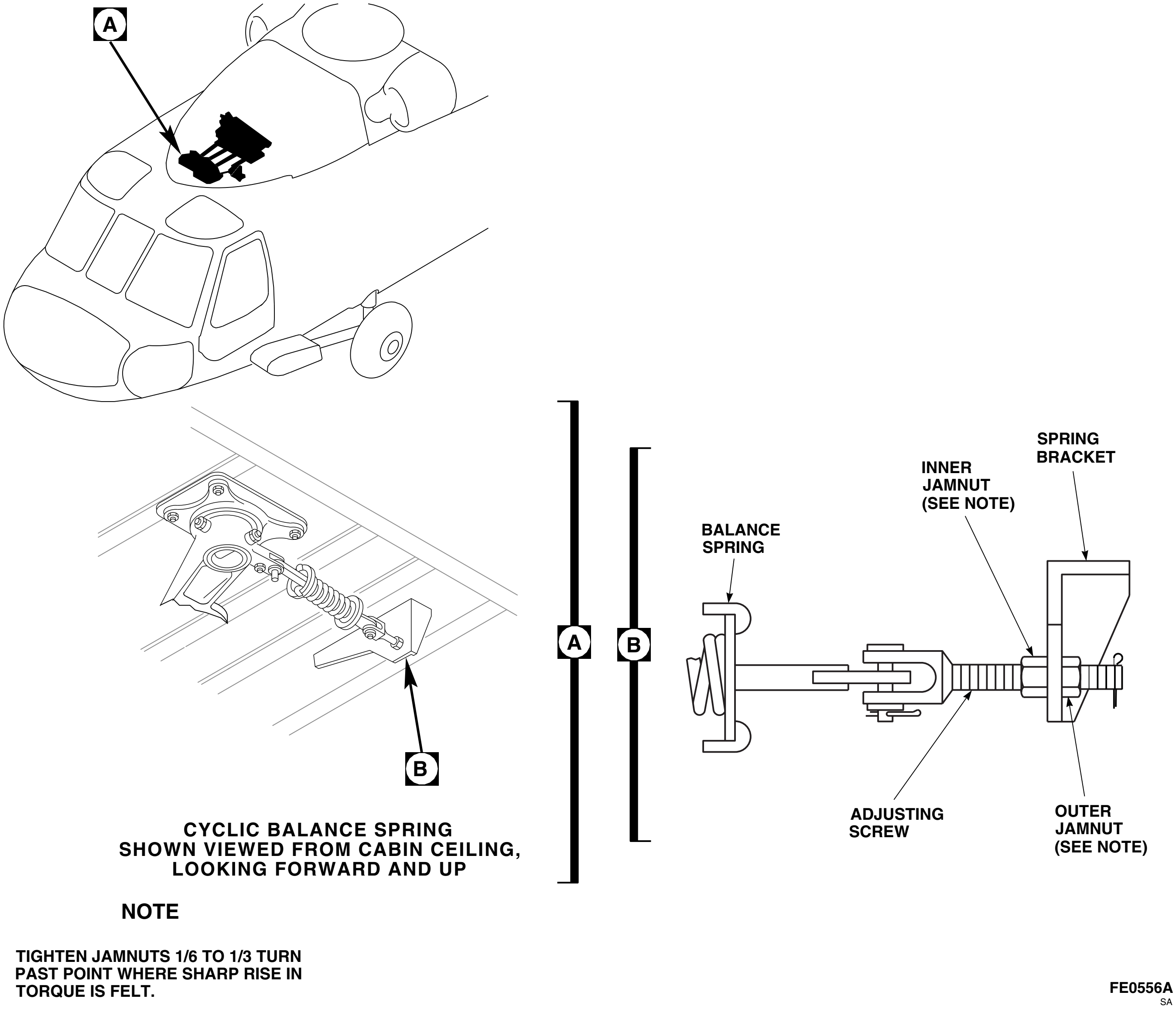


FIGURE 11-4-61-1. ADJUST CYCLIC STICK BALANCE SPRING

(7) Check that cotter pins are installed in clevis pins.

(8) Make sure area is clean and free of foreign material.

(9) Install heater duct (PARA 13-4-11).

(10) Install soundproofing panels (PARA 2-4-69).

c. Connect pitch/trim input control rod to pitch torque shaft lever (PARA 11-4-64).

11-4-62 COLLECTIVE STICK BALANCE SPRING ADJUSTMENT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Drill
- Drill Bit, No. 50, 0.070 inch-diameter

References

- PARA 2-4-69
- PARA 11-4-63
- PARA 13-4-11

11-4-62.1 Adjust Collective Stick Balance Spring.

a. Open main rotor pylon sliding cover.

b. Disconnect collective input control rod from collective torque shaft lever (PARA 11-4-63).

NOTE

Collective stick should not creep up or down when placed in any position.

c. Raise collective stick to midposition.

d. Remove soundproofing panels as required to access balance springs in cabin overhead (PARA 2-4-69).

e. Remove heater duct (PARA 13-4-11).

NOTE

- Backing off inner jamnut and tightening outer jamnut increases spring tension, causing collective stick to raise.
- Do not allow adjusting screw to turn while adjusting or tightening jamnuts. Tie bars could twist and disengage from slots in retainer tube.

f. If necessary, adjust inner and outer jamnuts on spring bracket to hold collective stick in any position (Figure 11-4-62-1, Detail B). If adjusting screw interferes with airframe structure preventing proper adjustment, perform the following:

(1) Remove between 0.250 and 0.300-inch from adjusting screw.

(2) Clean up threads and provide appropriate chamfer for run-on.

(3) Using drill with No. 50 drill bit, drill hole 0.110-inch from end of screw for cotter pin.

g. **TIGHTEN JAMNUTS ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

h. Raise collective to full up position.

i. Check that front edge of tie bar is at least 0.060-inch inside slot of retainer tube. If col-

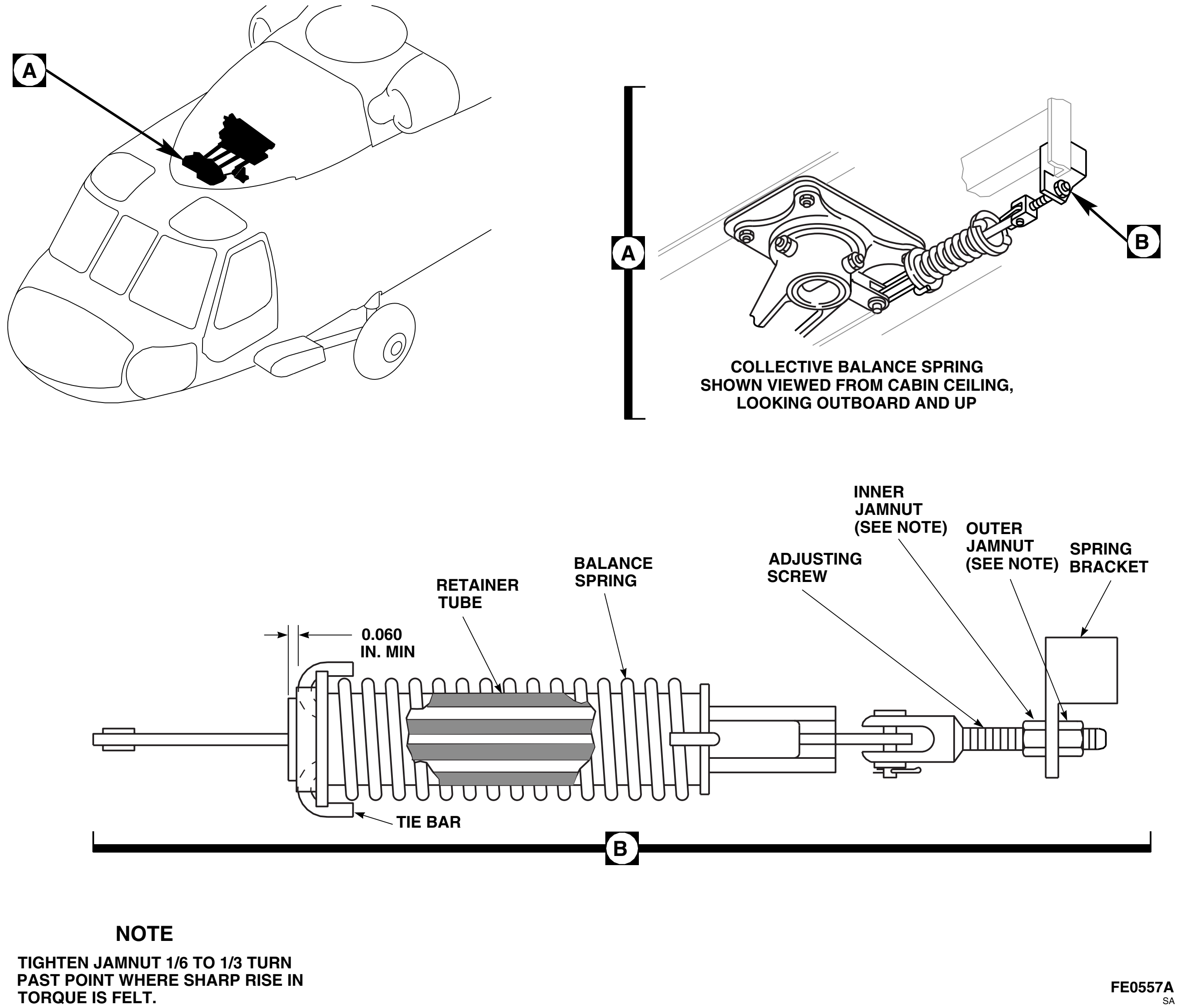


FIGURE 11-4-62-1. ADJUST COLLECTIVE STICK BALANCE SPRING

lective stick is adjusted and tie bar is too far towards front, replace spring, and readjust.

j. Check that cotter pins are installed in clevis pins.

k. Connect collective input control rod to collective torque shaft lever (PARA 11-4-63).

l. Make sure area is clean and free of foreign material.

m. Install heater duct (PARA 13-4-11).

n. Install soundproofing panels as required (PARA 2-4-69).

o. Close and latch main rotor pylon sliding cover.

11-4-63 COLLECTIVE INPUT CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-101
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-63.1 Replace Collective Input Control Rod.

11-4-63.1.1 Remove.

a. Remove main rotor pylon sliding cover (PARA 2-4-101).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Move collective stick to full down position.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

d. Install rigging pin in collective torque shaft (PARA 11-4-2).

e. Install long rigging pin in right side of mixer (PARA 11-4-2).

f. Remove bolts, washers, and nuts securing collective input control rod to collective torque shaft and collective servo arm (Figure 11-4-63-1, Sheet 1, Detail B).

11-4-63.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed control rod, as follows (Figure 11-4-63-1, Sheet 2, Detail C):

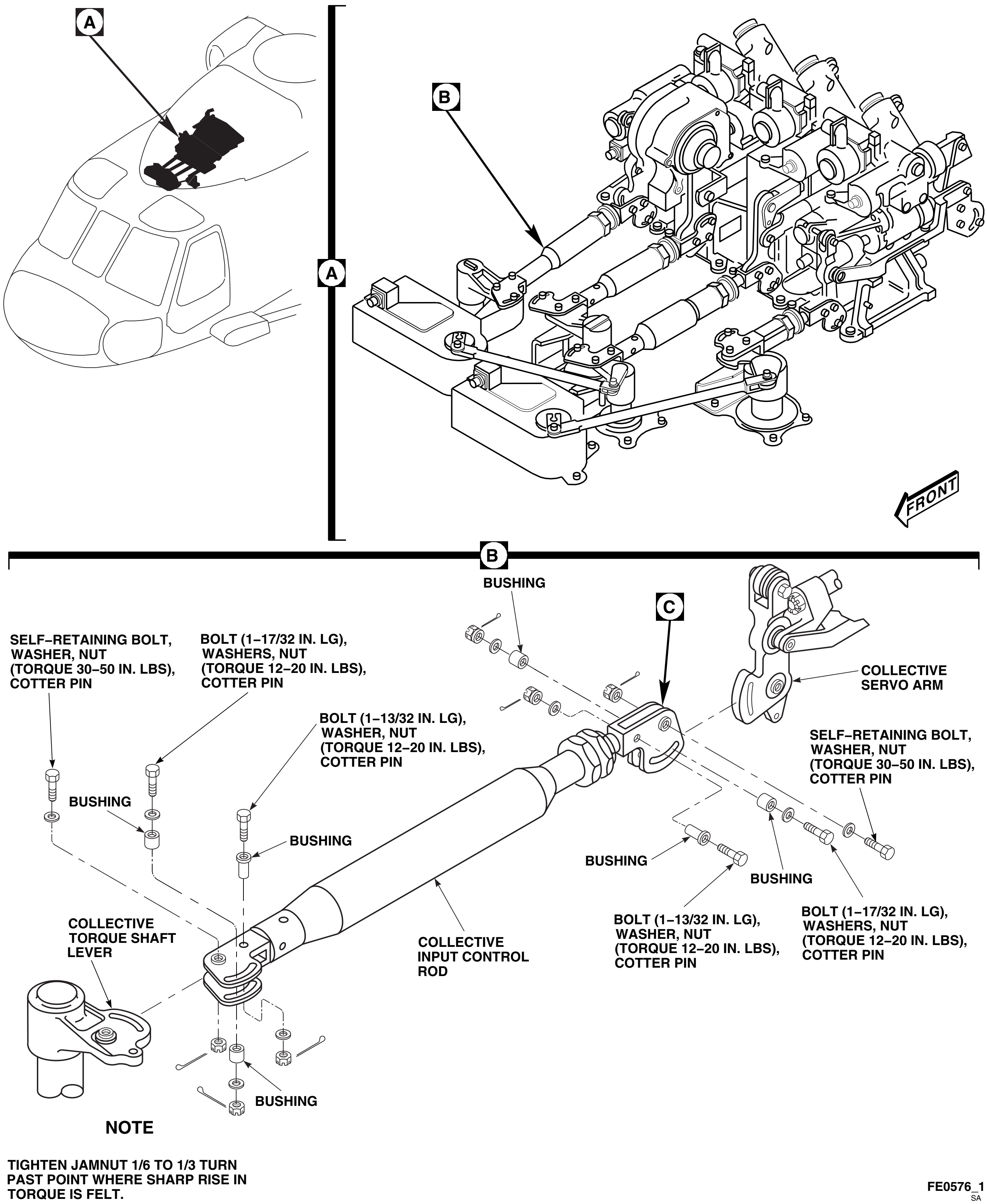


FIGURE 11-4-63-1. REPLACE COLLECTIVE INPUT CONTROL ROD (SHEET 1 OF 2)

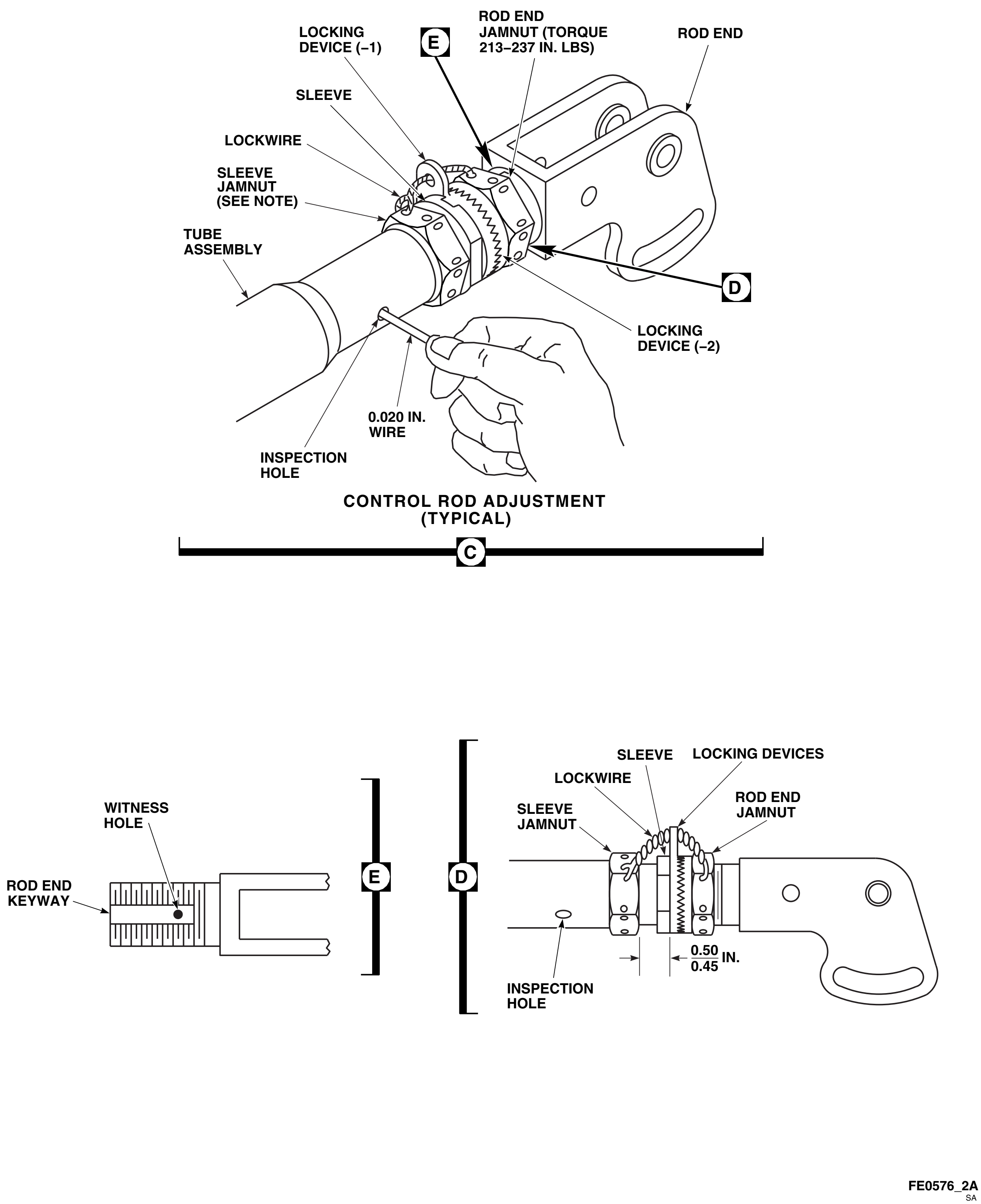


FIGURE 11-4-63-1. REPLACE COLLECTIVE INPUT CONTROL ROD (SHEET 2 OF 2)

(1) Slide locking device away from sleeve.

(2) Thread sleeve out of tube assembly to make control rod longer.

(3) Thread sleeve into tube assembly to make control rod shorter.

(4) Insert lockwire, Item 193, Appendix D, into inspection hole in tube assembly. Adjusted sleeve must stop wire from passing through.

(5) **TIGHTEN SLEEVE JAMNUT ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(6) Engage keys of locking device-1 with keyway in sleeve. Slide locking device-2 against locking device-1 to engage notches.

(7) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.**

(8) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

(9) Check that witness hole in rod end keyway cannot be seen outside rod end jamnut. Do not lockwire now (Figure 11-4-63-1, Sheet 2, Detail E).

b. Install control rod between torque shaft lever and servo arm, with adjustable end at servo arm (Figure 11-4-63-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

c. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

d. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

e. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Check that rigging pins slide in and out of their holes freely. Adjust control rod until this can be obtained.

g. When control rod length is correct, perform the following:

(1) Using lockwire, Item 197, Appendix D, lockwire jamnuts to tab of locking device (Figure 11-4-63-1, Sheet 2, Detail D).

(2) Perform quality check to ensure jamnuts have been properly lockwired.

h. Install bushings, bolts, washers, and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pin (Figure 11-4-63-1, Sheet 1, Detail B).

i. Remove rigging pins from torque shaft and mixer.

j. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

k. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

l. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

m. Make sure area is clean and free of foreign material.

n. Install main rotor pylon sliding cover (PARA 2-4-101).

11-4-64 PITCH/TRIM INPUT CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-101
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-64.1 Replace Pitch/Trim Input Control Rod.

11-4-64.1.1 Remove.

a. Remove main rotor pylon sliding cover (PARA 2-4-101).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Center cyclic stick.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

d. Install rigging pin in pitch torque shaft lever (PARA 11-4-2).

e. Install rigging pin in pitch trim assembly (PARA 11-4-2).

f. Remove bolts, washers, and nuts securing pitch/trim input control rod to torque shaft and pitch/trim servo lever (Figure 11-4-64-1, Sheet 1, Detail B)

11-4-64.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed control rod, as follows (Figure 11-4-64-1, Sheet 2, Detail C):

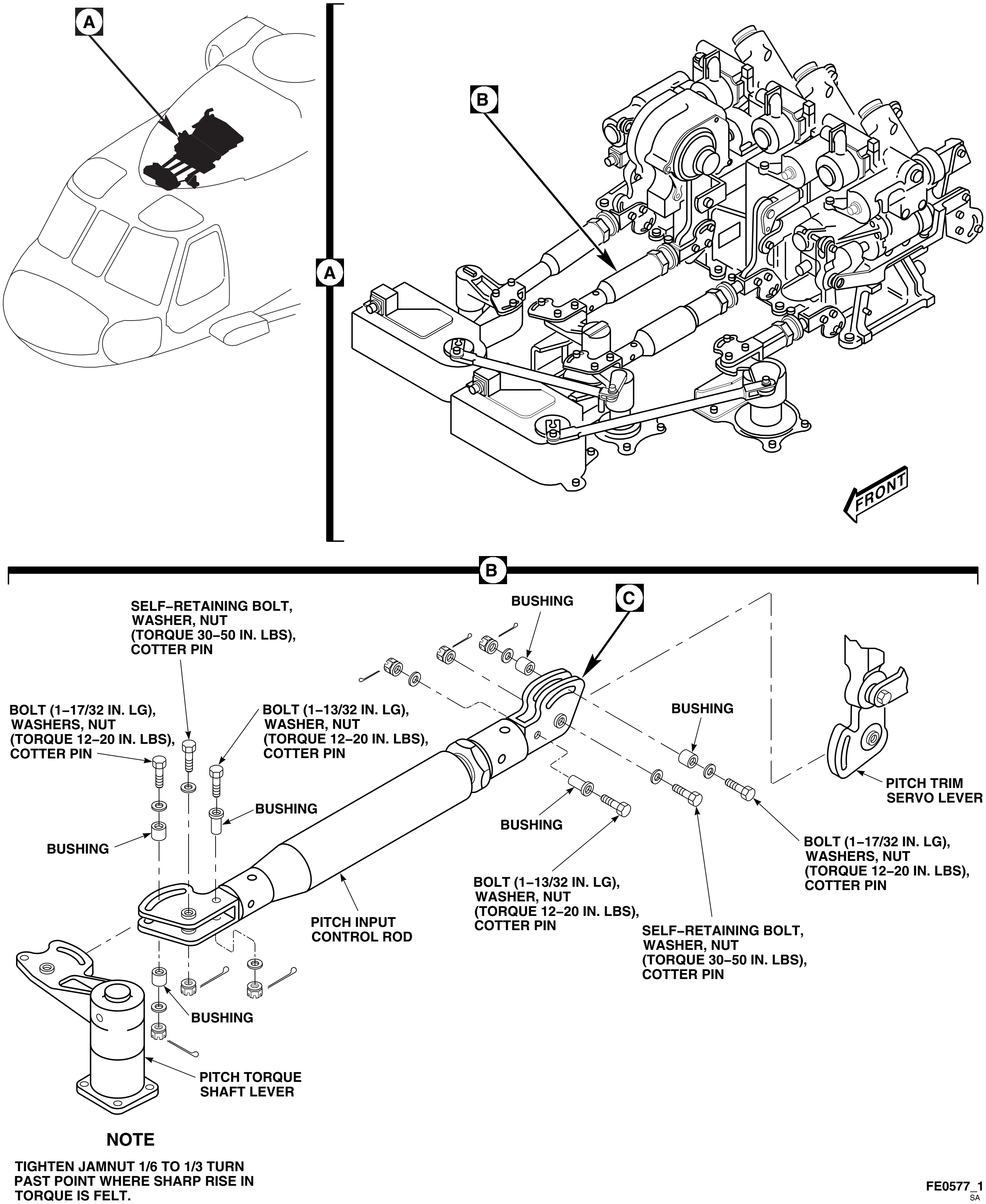
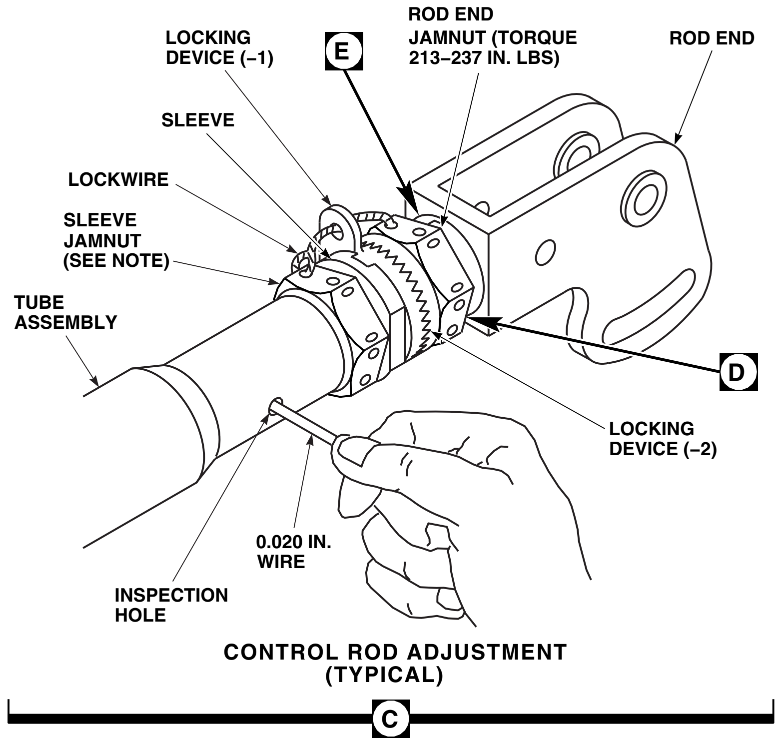


FIGURE 11-4-64-1. REPLACE PITCH/TRIM INPUT CONTROL ROD (SHEET 1 OF 2)



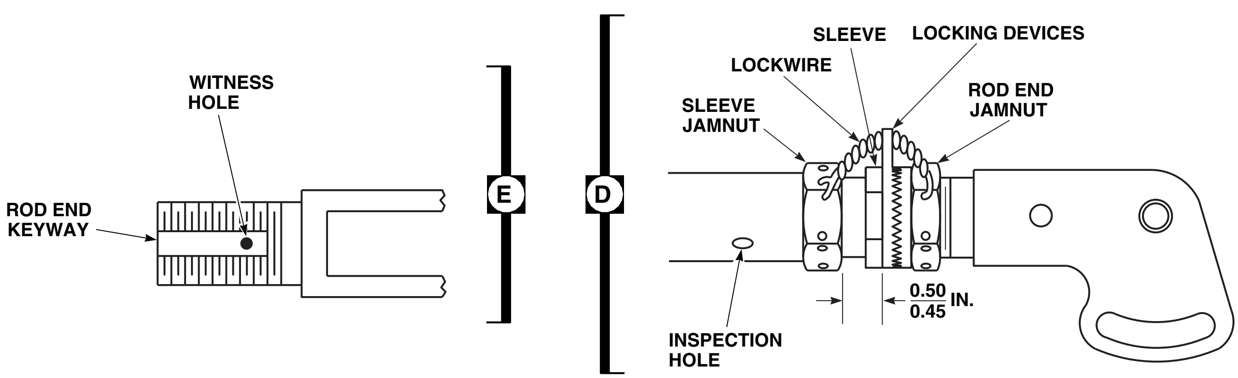


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FIGURE 11-4-64-1. REPLACE PITCH/TRIM INPUT CONTROL ROD (SHEET 2 OF 2)

(1) Slide locking device away from sleeve.

(2) Thread sleeve out of tube assembly to make control rod longer.

(3) Thread sleeve into tube assembly to make control rod shorter.

(4) Insert lockwire, Item 193, Appendix D, into inspection hole in tube assembly. Adjusted sleeve must stop wire from passing through.

(5) **TIGHTEN SLEEVE JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(6) Engage keys of locking device-1 with keyway in sleeve. Slide locking device-2 against locking device-1 to engage notches.

(7) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.**

(8) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

(9) Check that witness hole in rod end keyway cannot be seen outside rod end jamnut (Figure 11-4-64-1, Sheet 2, Detail E).

(10) Do not lockwire now.

b. Install control rod between torque shaft lever and servo arm with adjustable end at servo (Figure 11-4-64-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Boltheads can face either direction when installing bolts in pitch input control rod and pitch trim servo lever.

c. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

d. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

e. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Check that rigging pins slide in and out of their holes freely. Adjust control rod until this can be obtained. Remove rigging pins.

g. When control rod length is correct, using lockwire, Item 197, Appendix D, lockwire jamnuts to tab of locking device (Figure 11-4-64-1, Sheet 2, Detail D).

h. Perform quality check to ensure jamnuts have been properly lockwired.

i. Install bushings, bolts, washers, and nuts (Figure 11-4-64-1, Sheet 1, Detail B). **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pins.

j. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

k. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

l. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

m. Make sure area is clean and free of foreign material.

n. Install main rotor pylon sliding cover (PARA 2-4-101).

11-4-65 LATERAL INPUT CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-101
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-65.1 Replace Lateral Input Control Rod.

11-4-65.1.1 Remove.

a. Remove main rotor pylon sliding cover (PARA 2-4-101).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Center cyclic stick.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

d. Install rigging pin in lateral torque shaft lever (PARA 11-4-2).

e. Install rigging pin in roll actuator lever (PARA 11-4-2).

f. Remove bolts, washers, and nuts securing input control rod to torque shaft and roll actuator lever (Figure 11-4-65-1, Sheet 1, Detail B).

11-4-65.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed control rod, as follows (Figure 11-4-65-1, Sheet 2, Detail C):

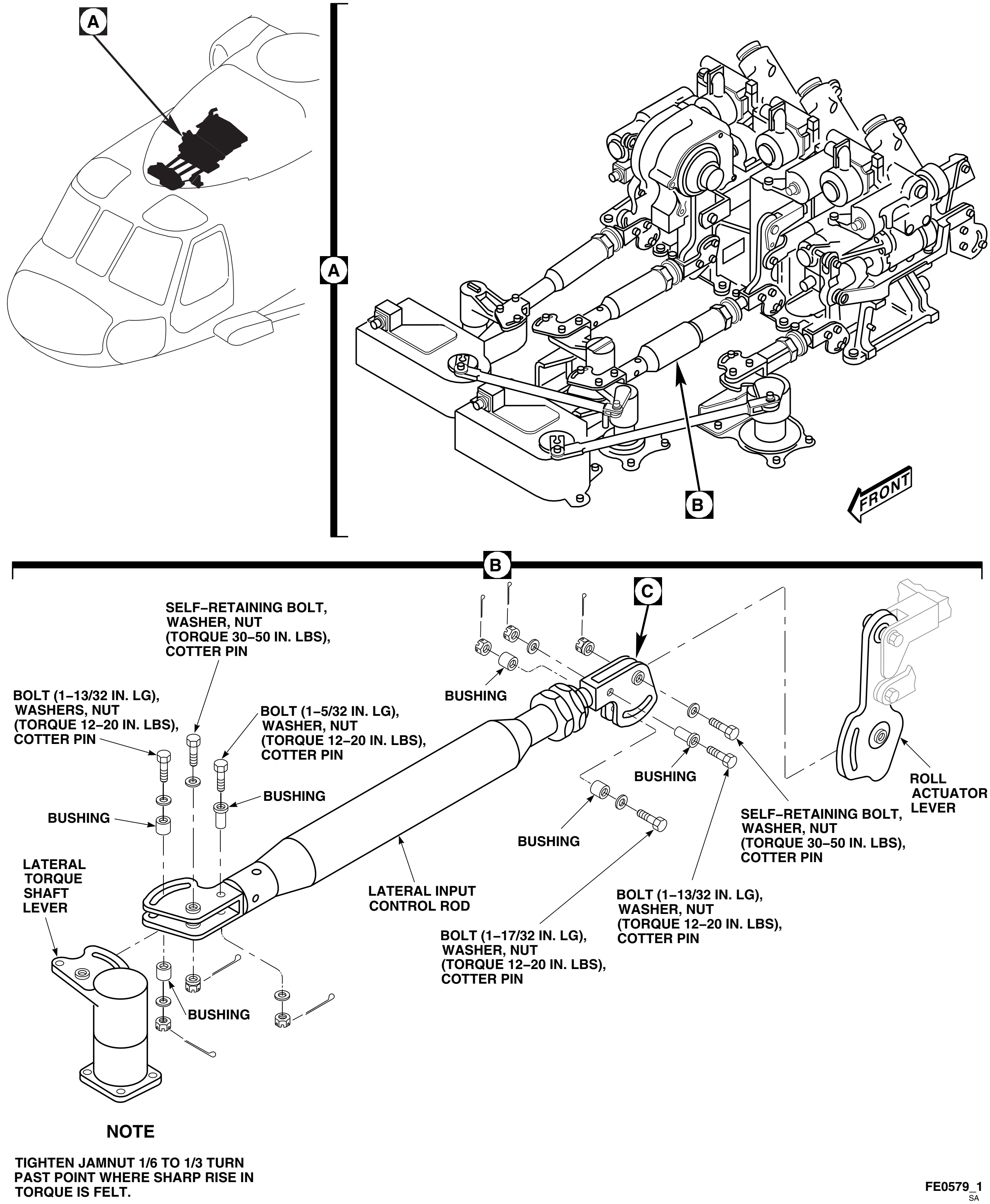
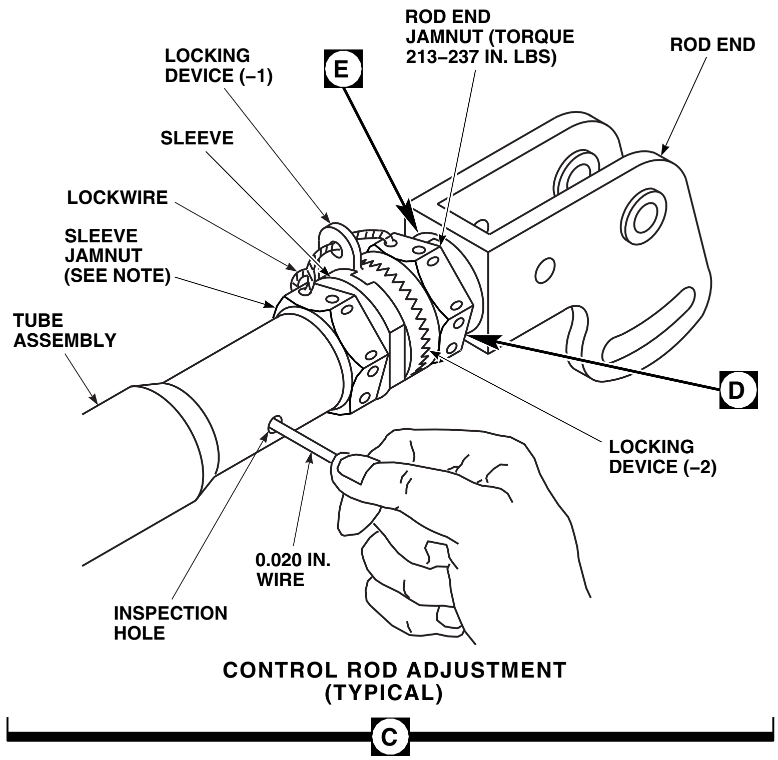


FIGURE 11-4-65-1. REPLACE LATERAL INPUT CONTROL ROD (SHEET 1 OF 2)



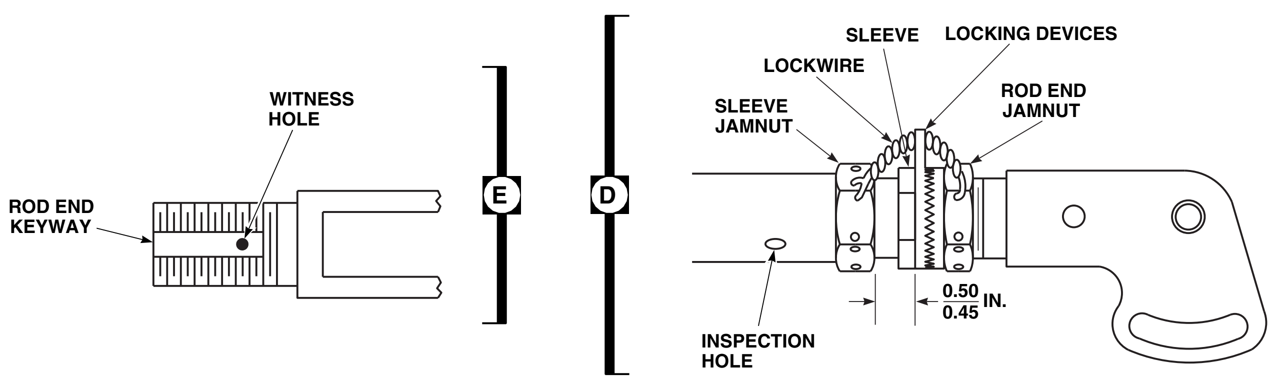


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FIGURE 11-4-65-1. REPLACE LATERAL INPUT CONTROL ROD (SHEET 2 OF 2)

(1) Slide locking device away from sleeve.

(2) Thread sleeve out of tube assembly to make control rod longer.

(3) Thread sleeve into tube assembly to make control rod shorter.

(4) Insert lockwire, Item 193, Appendix D, into inspection hole in tube assembly. Adjusted sleeve must stop wire from passing through.

(5) **TIGHTEN SLEEVE JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(6) Engage keys of locking device-1 with keyway in sleeve. Slide locking device-2 against locking device-1 to engage notches.

(7) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.**

(8) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

(9) Check that witness hole in rod end keyway cannot be seen outside rod end jamnut. Do not lockwire now (Figure 11-4-65-1, Sheet 2, Detail E).

b. Install control rod between torque shaft lever and actuator lever with adjustable end at actuator lever (Figure 11-4-65-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

c. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

d. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

e. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Check that rigging pins slide in and out of their holes freely. Adjust control rod until this can be obtained.

g. When control rod length is correct, using lockwire, Item 197, Appendix D, lockwire jamnuts to tab of locking device (Figure 11-4-65-1, Sheet 2, Detail D).

h. Perform quality check to ensure jamnuts have been properly lockwired.

i. Install bushings, bolts, washers, and nuts in rod ends. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pins (Figure 11-4-65-1, Sheet 1, Detail B).

j. Remove rigging pins from torque shaft and roll actuator.

k. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

l. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

m. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

n. Make sure area is clean and free of foreign material.

o. Install main rotor pylon sliding cover (PARA 2-4-101).

11-4-66 YAW BOOST SERVO INPUT CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-101
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-66.1 Replace Yaw Boost Servo Input Control Rod.

11-4-66.1.1 Remove.

a. Remove main rotor pylon sliding cover (PARA 2-4-101).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Center yaw pedals.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

d. Insert rigging pin in yaw torque shaft lever (PARA 11-4-2).

e. Insert rigging pin yaw boost servo lever (PARA 11-4-2).

f. Remove bolts, washers, and nuts securing yaw input control rod to torque shaft and yaw servo lever (Figure 11-4-66-1, Sheet 1, Detail B). Remove control rod.

11-4-66.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed control rod, as follows (Figure 11-4-66-1, Sheet 2, Detail C):

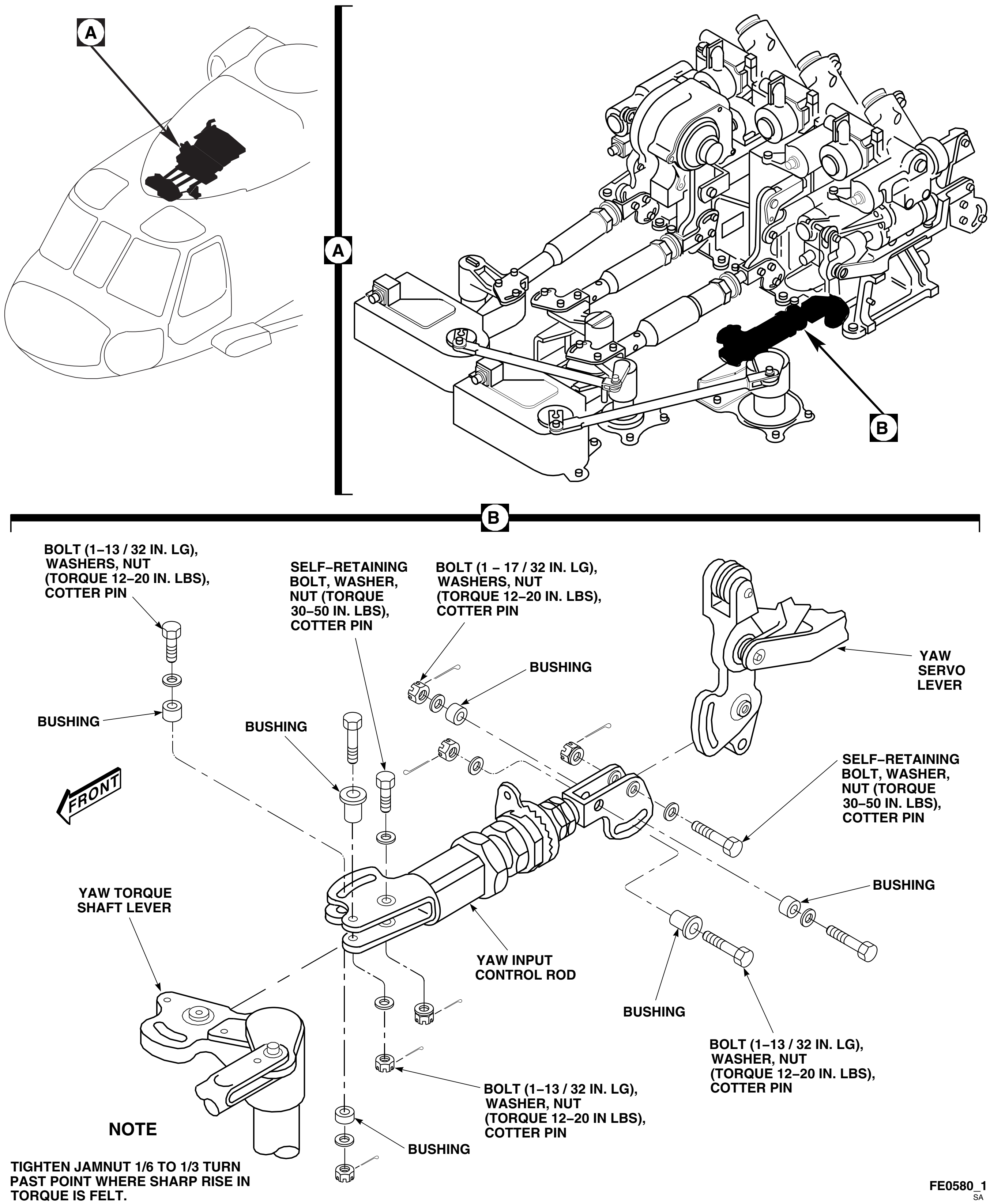
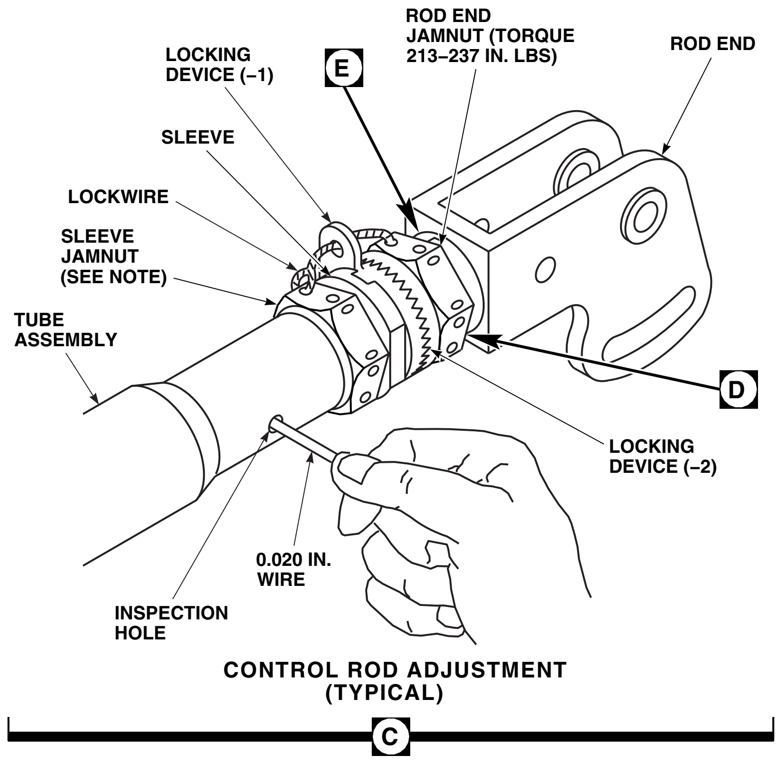


FIGURE 11-4-66-1. REPLACE YAW BOOST SERVO INPUT CONTROL ROD (SHEET 1 OF 2)



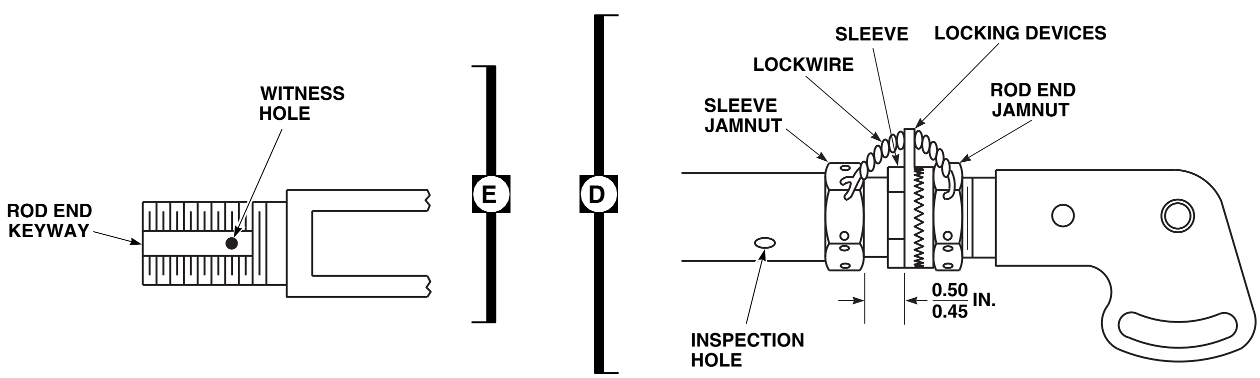


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FIGURE 11-4-66-1. REPLACE YAW BOOST SERVO INPUT CONTROL ROD (SHEET 2 OF 2)

(1) Slide locking device away from sleeve.

(2) Thread sleeve out of tube assembly to make control rod longer.

(3) Thread sleeve into tube assembly to make control rod shorter.

(4) Insert lockwire, Item 193, Appendix D, into inspection hole in tube assembly. Adjusted sleeve must stop wire from passing through.

(5) **TIGHTEN SLEEVE JAMNUT ⅙ TO ⅓ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(6) Engage keys of locking device-1 with keyway in sleeve. Slide locking device-2 against locking device-1 to engage notches.

(7) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.**

(8) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

(9) Check that witness hole in rod end keyway cannot be seen outside rod end jamnut. Do not lockwire now (Figure 11-4-66-1, Sheet 2, Detail E).

b. Install control rod between torque shaft lever and servo lever with adjustable end at servo (Figure 11-4-66-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

c. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

d. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

e. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Check that rigging pin and rigging tool slide in and out of their holes freely. Adjust control rod until this can be obtained.

g. When control rod length is correct, using lockwire, Item 197, Appendix D, lockwire jamnuts to tab of locking device (Figure 11-4-66-1, Sheet 2, Detail D).

h. Perform quality check to ensure jamnuts have been properly lockwired.

i. Install bushings, bolts, washers, and nuts (Figure 11-4-66-1, Sheet 1, Detail B). **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pins.

j. Remove rigging pins from torque shaft and yaw servo.

k. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

l. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

m. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

n. Make sure area is clean and free of foreign material.

o. Install main rotor pylon sliding cover (PARA 2-4-101).

11-4-67 COLLECTIVE BOOST ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 4-4-27
- PARA 7-2-1
- PARA 11-4-3
- PARA 11-4-63
- PARA 11-4-73

11-4-67.1 Replace Collective Boost Assembly.

11-4-67.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. If main rotor blades are installed, turn main rotor head to allow overhead clearance and make it easier to remove collective boost assembly.

d. Disconnect engine load-demand spindle (LDS) cables from mixer (PARA 4-4-27).

NOTE

Do not store self-retaining bolts in bearings.

e. Remove self-retaining bolt, bolts, washers, bushings, and nuts securing input control rod to input arm of collective boost assembly (Figure 11-4-67-1, Detail C). Remove input control rod.

f. Remove self-retaining bolt, bolts, washers, bushings, and nuts securing front end of output control rod to output link on collective boost assembly (Figure 11-4-67-1, Detail D).

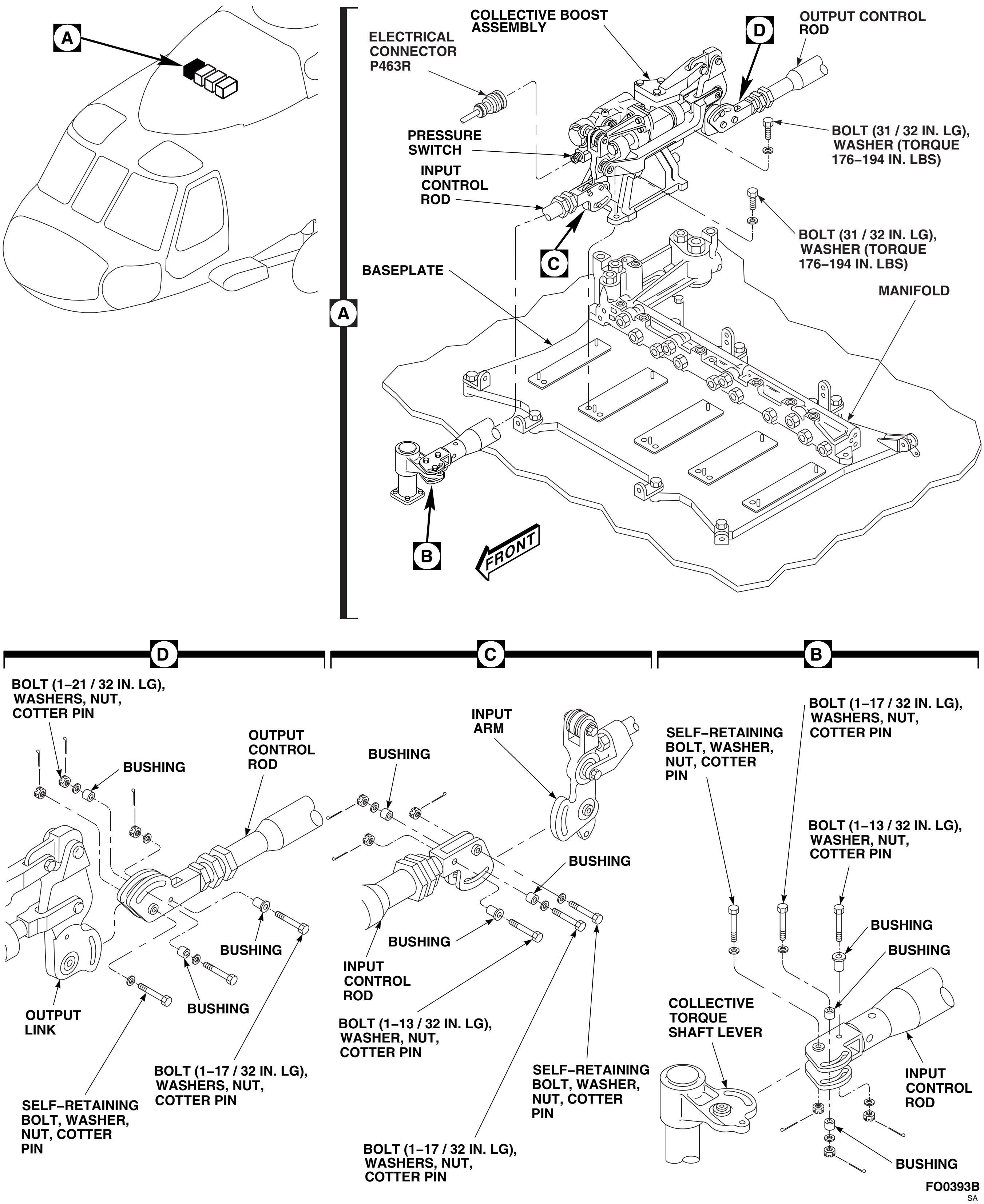


FIGURE 11-4-67-1. REPLACE COLLECTIVE BOOST ASSEMBLY

g. Disconnect electrical connector, P463R, from pressure switch on collective boost assembly (Figure 11-4-67-1, Detail A).

h. Install protective caps and plugs on electrical connectors.

i. Remove bolts and washers securing collective boost assembly to baseplate and manifold. Remove collective boost assembly.

j. Carefully pull collective boost assembly toward front to separate boost assembly self-sealing couplings from manifold.

k. Cover collective boost assembly and manifold to keep dirt from hydraulic system.

11-4-67.1.2 Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove cover from replacement collective boost assembly and from manifold.

c. Make sure manifold and collective boost assembly self-sealing couplings are clean. If needed, wipe clean with machinery towel, Item 211, Appendix D, and flush with ethyl alcohol, Item 71, Appendix D.

d. Thoroughly wipe deck mounting area and collective boost assembly feet with clean machinery towel, Item 211, Appendix D, and ethyl alcohol, Item 71, Appendix D. Final wiping must be in one direction only, and followed by a repeat wipe with clean machinery towel.

CAUTION

Damage to equipment may result if incorrect connecting link is installed. Incorrect link is prone to stress corrosion cracking. Make sure only link marked T73, is installed.

e. Inspect to make sure only connecting link, 70410-02334-045, marked T73, is installed.

f. Position collective boost assembly over mounting holes as close to manifold self-sealing couplings as possible (Figure 11-4-67-1, Detail A).

g. Carefully slide collective boost assembly toward manifold and connect self-sealing couplings between collective boost assembly and manifold.

h. Line up output control rod clevis with output link on collective boost assembly while sliding collective boost assembly toward manifold (Figure 11-4-67-1, Detail D).

WARNING

FSCAP

Verification of minimum torque and safety of bolts are critical characteristics.

i. Attach collective boost assembly to manifold and baseplate with bolts and washers. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.** Lockwire, Item 197, Appendix D, bolts (Figure 11-4-67-1, Detail A). Apply torque stripes (PARA 2-6-5).

j. Remove protective caps and plugs from electrical connectors.

k. Inspect and treat electrical connectors (PARA 1-7-20).

l. Connect electrical connector, P463R, to pressure switch on collective boost assembly.

m. Without applying hydraulic power, push collective boost assembly piston full aft.

n. Push input arm on collective boost assembly toward rear to bottom valve linkage.

o. Install rigging pin, 70700-20380-043, in collective torque shaft.

p. Adjust input control rod to connect to input arm on collective boost assembly while input arm is still in full rear position, then shorten input control rod by 1/3 turn to pull linkage off valve stop. Connect input control rod to input arm on collective boost assembly (PARA 11-4-63).

q. Connect input control rod to collective torque shaft lever (PARA 11-4-63).

r. Temporarily install output control rod to output link on collective boost assembly (Figure 11-4-67-1, Detail D).

s. Turn on all electrical power (PARA 1-6-16).

t. Place BACKUP HYD switch to ON.

u. Perform BOOST ON, SAS OFF procedure (PARA 11-4-3).

CAUTION

Damage to flight controls will result if hydraulic power is shut down with rigging pins installed. Make sure all rigging pins are removed before shutting down hydraulic power.

v. Install rigging pin, 70700-20380-044, in roll actuator.

w. Install rigging pin, 70700-20380-043, in pitch torque shaft.

x. Install rigging pin, 70700-20380-062, in yaw boost servo.

CAUTION

Damage to flight controls will result if control rod is adjusted with mixer rigging pin partially installed. If control rod is to be adjusted, first completely remove pin, adjust control rod, then attempt to install pin again.

y. Adjust collective boost assembly output control rod until long rigging pin, 70700-20380-046, can be inserted through right side of mixer. Connect output control rod to collective boost assembly (PARA 11-4-73).

z. Remove all rigging pins.

aa. Connect engine LDS cables to mixer (PARA 4-4-27).

ab. Place collective stick in full down position and verify that pilot's low collective stick stop is contacted. If collective stick stop is not contacted, do not adjust collective stick stops. Repeat step m. through step z. until collective boost assembly is bottomed and pilot's collective stick stop is contacted.

ac. Place collective stick in full up position and verify pilot's high collective stick stop is contacted.

ad. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ae. Perform operational check of No. 2 hydraulic system (PARA 7-2-1).

af. Make sure area is clean and free of foreign objects before closing cover.

ag. Close and latch main rotor pylon sliding cover.

ah. Perform maintenance test flight (Appropriate MTF).

11-4-68 PITCH/TRIM ASSEMBLY.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 7-2-1
- PARA 11-4-1
- PARA 11-4-3
- PARA 11-4-64
- PARA 11-4-74
- PARA 11-5-5
- PARA 11-5-6

11-4-68.1 Replace Pitch/Trim Assembly.

NOTE

If SAS servovalve or SAS actuator are to be removed, refer to PARA 11-5-5 or PARA 11-5-6.

11-4-68.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. If main rotor blades are installed, turn main rotor head to allow overhead clearance and make it easier to remove pitch/trim assembly.

d. Disconnect electrical connectors, P464R, and P460R, from pitch/trim assembly (Figure 11-4-68-1, Detail A).

e. Install protective caps and plugs on electrical connectors.

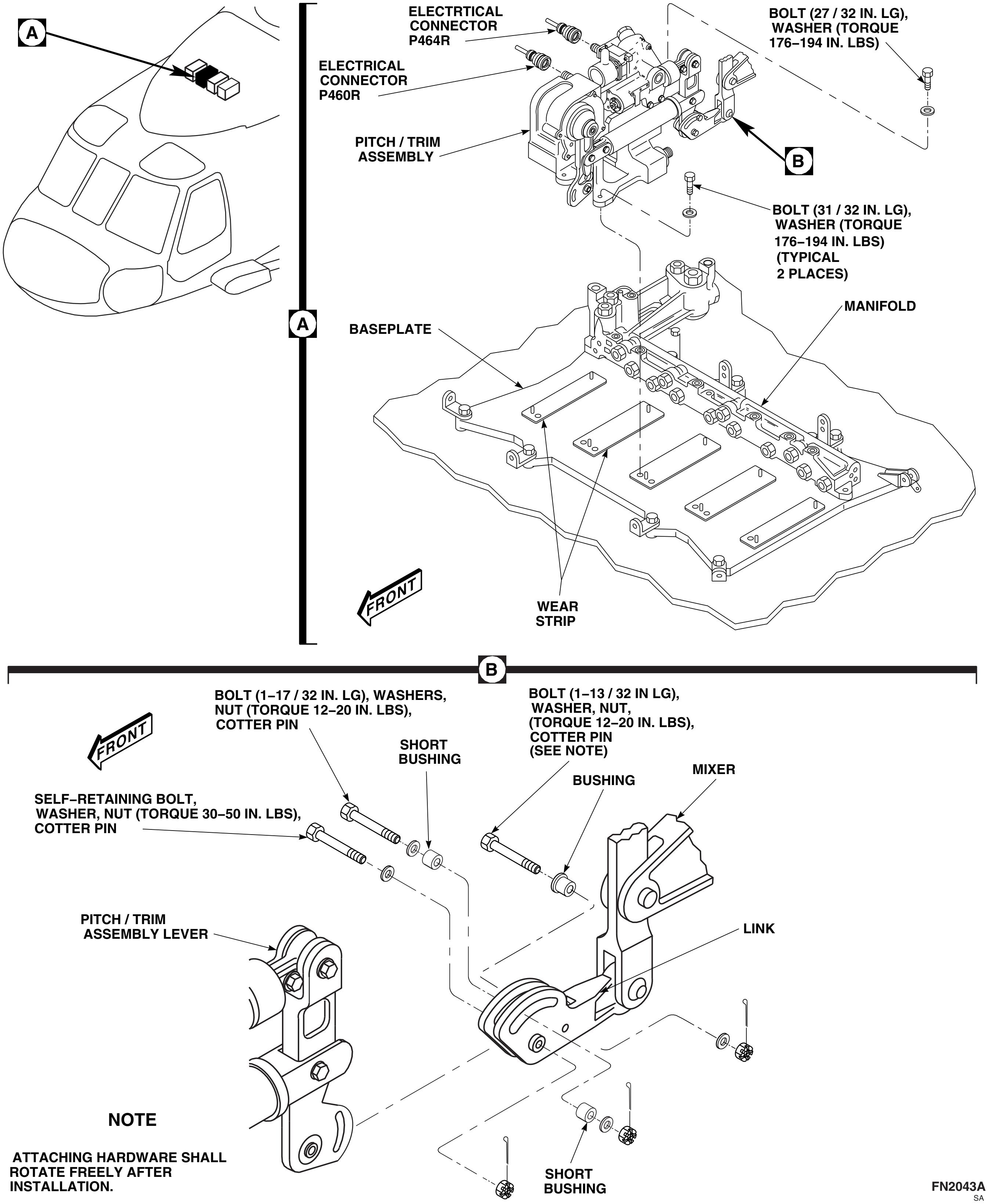


FIGURE 11-4-68-1. REPLACE PITCH/TRIM ASSEMBLY

f. Remove input control rod from pitch torque shaft and pitch/trim assembly (PARA 11-4-64).

g. Disconnect mixer link from pitch/trim assembly lever (PARA 11-4-74).

h. Remove bolts and washers securing pitch/trim assembly to baseplate (two bolts) and manifold (one bolt).

i. Carefully pull pitch/trim assembly forward to separate servo self-sealing couplings from manifold.

j. Cover assembly and manifold to prevent getting dirt in hydraulic system.

11-4-68.1.2 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove cover from replacement pitch/trim assembly and from manifold. Make sure manifold and servo self-sealing couplings are clean. Wipe with clean machinery towel, Item 211, Appendix D, and flush with ethyl alcohol, Item 71, Appendix D.

c. Thoroughly wipe deck mounting area and mounting feet with clean machinery towel, Item 211, Appendix D, and ethyl alcohol, Item 71, Appendix D. Final wiping must be in one direction only, and followed by a repeat wipe with clean, machinery towel, Item 211, Appendix D.

CAUTION

Damage to equipment may result if incorrect connecting link is installed. Incorrect link is prone to stress corrosion cracking. Make sure only link marked T73, is installed.

d. Inspect to make sure only connecting link, 70410-02334-045, marked T73, is installed.

e. Position pitch/trim assembly over mounting holes as close to manifold couplings as possible (Figure 11-4-68-1, Detail A).

WARNING

FSCAP

Verification of minimum torque and safety of bolts are critical characteristics.

f. Carefully slide pitch/trim assembly toward manifold and connect self-sealing couplings between assembly and manifold.

g. Attach pitch/trim assembly to manifold and to baseplate with bolts and washers (three bolts required, one 27/32-inch-long bolt, two 31/32-inch-long bolts). **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.** Lockwire, Item 197, Appendix D, bolts. Apply torque stripe (PARA 2-6-5).

h. Position pitch/trim assembly lever in clevis of mixer link (Figure 11-4-68-1, Detail B).

i. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

j. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

k. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

l. Install short bushings in slot on each side of link. Install 1-17/32-inch-long bolt, washers, and nut in slot of mixer link and pitch/trim assembly lever. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

m. Install bushing in rear hole of link and slot of pitch/trim assembly. Install 1-13/32-inch-long bolt, washer, and nut through spacer. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin. Attaching hardware shall rotate freely, with no more than finger torque required, after installation.

n. Install pitch/trim input control rod (PARA 11-4-64).

o. Remove protective caps and plugs from electrical connectors.

p. Inspect and treat electrical connectors (PARA 1-7-20).

q. Connect electrical connector, P464R, to pitch/trim actuator and electrical connector, P460R, to SAS module (Figure 11-4-68-1, Detail A).

r. Perform operational check of No. 2 hydraulic system (PARA 7-2-1).

s. Inspect for pitch system travel as follows:

CAUTION

Damage to flight control components will result if hydraulic power is shut down with rigging pins still installed. Make sure all rigging pins are removed prior to shutting down hydraulic power.

NOTE

Backup pump should be used only as an alternate source to power hydraulic system. External hydraulic power should be applied whenever available.

(1) Connect external hydraulic power to helicopter (PARA 1-6-15).

(2) Perform BOOST ON, SAS OFF procedure (PARA 11-4-3).

(3) Move pedals to approximately neutral position.

(4) Move collective stick to full up position and cyclic stick to full forward position. Verify forward limiter is contacted.

(5) Move collective stick to full down position with cyclic stick at full forward position. Verify forward limiter is contacted.

(6) With collective stick at full down position and move cyclic stick to full aft position. Verify aft limiter is contacted.

(7) Move collective stick to full up position with cyclic stick at full aft position. Verify aft stick stopbolt is contacted.

(8) Center controls and shut down all hydraulic power (PARA 1-6-15).

t. Make sure area is clean and free of foreign material before closing cover.

u. Close and latch main rotor pylon sliding cover.

v. Perform a maintenance test flight (Appropriate MTF).

11-4-69 ROLL ACTUATOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 4-4-27
- PARA 7-2-1
- PARA 11-4-3
- PARA 11-4-65
- PARA 11-4-75
- PARA 11-5-5
- PARA 11-5-6

11-4-69.1 Replace Roll Actuator.

NOTE

If SAS servo valve or SAS actuator are to be removed, refer to PARA 11-5-5 or PARA 11-5-6.

11-4-69.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. If main rotor blades are installed, turn main rotor head to allow overhead clearance and make it easier to remove roll actuator.

d. Disconnect engine LDS cables from mixer (PARA 4-4-27).

e. Disconnect electrical connector, P461R, from SAS servo (Figure 11-4-69-1, Detail A).

f. Install protective caps and plugs on electrical connectors.

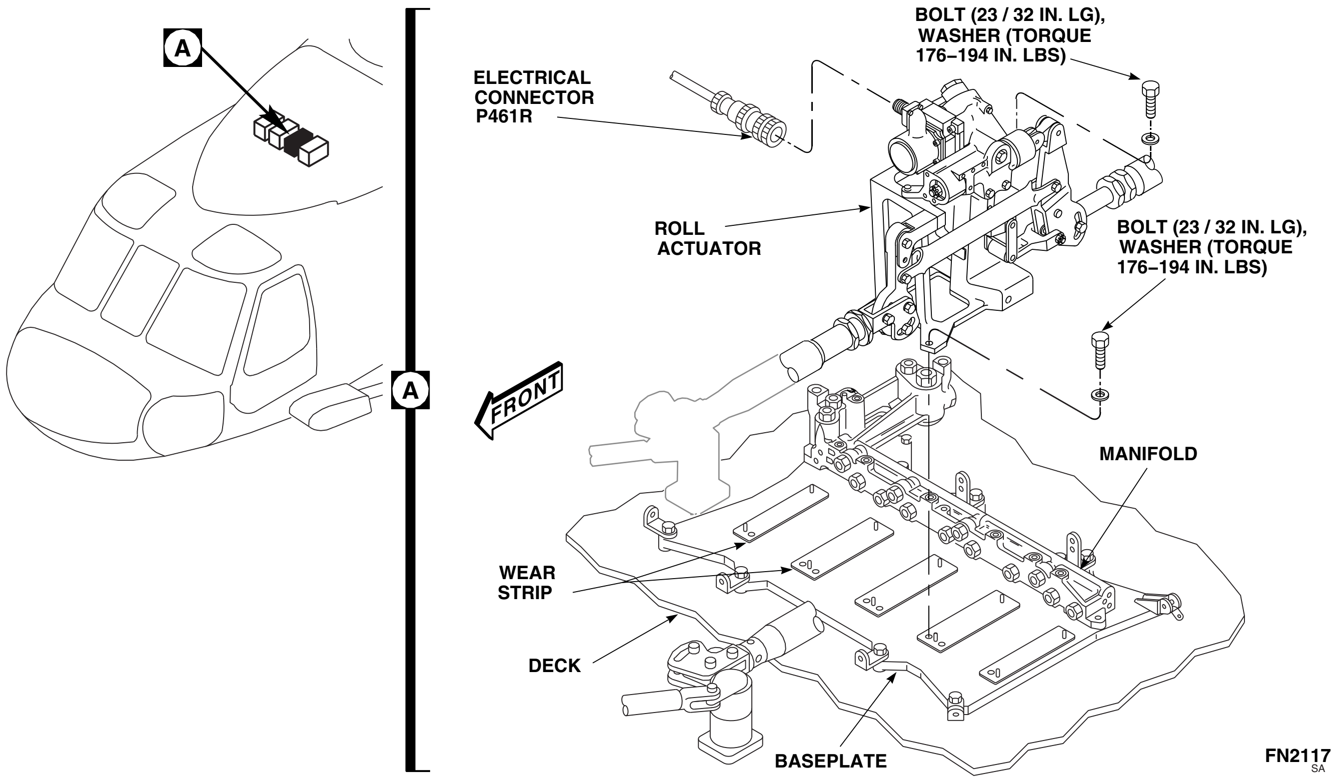


FIGURE 11-4-69-1. REPLACE ROLL ACTUATOR

CAUTION

Do not store self-retaining bolts in rod end bearings.

g. Remove lateral input control rod (PARA 11-4-65).

h. Disconnect lateral control rod from rear roll actuator lever (PARA 11-4-75).

i. Remove bolts and washers securing roll actuator to baseplate (two bolts) and manifold (one bolt).

j. Carefully pull roll actuator forward to separate self-sealing couplings from manifold.

k. Cover roll actuator and manifold to keep dirt from hydraulic system.

11-4-69.1.2 Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove cover from replacement roll actuator and from manifold.

c. Make sure manifold and roll actuator self-sealing couplings are clean. If needed, wipe with clean machinery towel, Item 211, Appendix D, and flush with ethyl alcohol, Item 71, Appendix D.

d. Thoroughly wipe deck mounting area and servo feet with clean machinery towel, Item 211, Appendix D, and ethyl alcohol, Item 71, Appendix D. Final wiping must be in one direction only, and followed by a repeat wipe with clean machinery towel, Item 211, Appendix D.

CAUTION

Damage to equipment may result if incorrect connecting link is installed. Incorrect link is prone to stress corrosion cracking. Make sure only link marked T73, is installed.

e. Inspect to make sure only connecting link, 70410-02334-045, marked T73, is installed.

f. Position servo over mounting holes as close to manifold couplings as possible.

g. Carefully slide servo toward manifold and connect self-sealing couplings between servo and manifold.

h. Attach roll actuator to manifold and to deck with bolts and washers (Figure 11-4-69-1, Detail A).

WARNING

FSCAP

Verification of minimum torque and safety of bolts are critical characteristics.

i. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.** Lockwire, Item 197, Appendix D, bolts. Apply torque stripe (PARA 2-6-5).

j. Remove protective caps and plugs from electrical connectors.

k. Inspect and treat electrical connectors (PARA 1-7-20).

l. Connect electrical connector, P461R, to SAS servo. Lockwire, Item 196, Appendix D, electrical connector.

m. Install rigging pin, 70700-20380-043, in roll torque shaft.

n. Install rigging pin, 70700-20380-044, in roll actuator.

o. Adjust and connect roll actuator input control rod to roll actuator (PARA 11-4-65).

p. Install roll actuator output control rod to actuator (PARA 11-4-75).

q. Turn on all electrical power (PARA 1-6-16).

CAUTION

Damage to flight controls will result if hydraulic power is shut down with rigging pins installed. Make sure all rigging pins are removed before shutting down hydraulic power.

r. Place BACKUP HYD switch to ON.

s. Perform BOOST ON, SAS OFF procedure (PARA 11-4-3).

t. Install rigging pin, 70700-20380-043, in collective torque shaft.

u. Install rigging pin, 70700-20380-043, in pitch torque shaft.

v. Install rigging pin, 70700-20380-062, in yaw boost servo.

CAUTION

Damage to flight controls will result if control rod is adjusted with rigging pin partially installed. If control rod is to be adjusted, first completely remove pin, adjust control rod, then attempt to install pin again.

w. Adjust roll actuator output control rod until long rigging pin, 70700-20380-046, can be installed through right side of mixer and short rigging pin, 70700-20380-068, can be installed through left side of mixer.

x. Remove all rigging pins.

y. Connect engine LDS cables to mixer (PARA 4-4-27).

z. Check for no movement of lateral (roll) limiters in mixer as follows:

(1) Lower collective stick to full down position and lock in place.

(2) Move cyclic stick to full left position. Check that roller in mixer is contacting lateral left limiter in mixer and cannot be rotated using finger pressure.

(3) With collective stick at full down position and cyclic stick at full left position, check gap at pilot's and copilot's left stick stops. Both stops should be 0.010 to 0.030-inch clear. If stops are outside limits, adjust as necessary.

(4) Move cyclic stick to full right position. Check that roller in mixer is contacting lateral right limiter in mixer and cannot be rotated using finger pressure.

(5) With collective stick at full down position and cyclic stick at full right position, check gap at pilot's and copilot's right stick stops. Both stops should be 0.010 to 0.030 inch clear. If stops are outside limits, adjust as necessary.

aa. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

ab. Perform operational check of the No. 2 hydraulic system (PARA 7-2-1).

ac. Make sure area is clean and free of foreign objects before closing cover.

ad. Close and latch main rotor pylon sliding cover.

ae. Perform a maintenance test flight (Appropriate MTF).

11-4-70 YAW BOOST SERVO.

PARAGRAPH BREAKDOWN

PAGE

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-5
- PARA 4-4-27
- PARA 7-2-1
- PARA 11-4-3
- PARA 11-4-5
- PARA 11-4-66
- PARA 11-4-76
- PARA 11-5-5
- PARA 11-5-6

11-4-70.1 Replace Yaw Boost Servo.

NOTE

If SAS servo valve or SAS actuator are to be removed, refer to PARA 11-5-5 or PARA 11-5-6.

11-4-70.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. If main rotor blades are installed, turn main rotor head to allow overhead clearance and make it easier to remove yaw boost servo assembly.

d. Disconnect engine load-demand spindle (LDS) cables at mixer (PARA 4-4-27).

e. Remove input control rod (PARA 11-4-66).

f. Remove bolts, washers, nuts securing front end of output control rod at servo (PARA 11-4-76).

g. Disconnect electrical connector, P462R, from SAS servo valve and electrical connector, P470R, from pressure switch (Figure 11-4-70-1, Detail A).

h. Install protective caps and plugs on electrical connectors.

i. Remove bolts and washers securing boost servo to baseplate and manifold.

j. Carefully pull boost servo forward to separate servo self-sealing couplings from manifold.

k. Cover boost servo and manifold to keep dirt from entering hydraulic system.

11-4-70.1.2 Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove cover from replacement yaw boost servo and from manifold.

c. Make sure manifold and servo self-sealing couplings are clean. If needed, wipe with clean machinery towel, Item 211, Appendix D, and flush with ethyl alcohol, Item 71, Appendix D.

d. Thoroughly wipe deck mounting area and servo feet with clean machinery towel, Item 211, Appendix D, and ethyl alcohol, Item 71, Appendix D. Final wiping must be in one direction only, and followed by a repeat wipe with clean machinery towel, Item 211, Appendix D.

CAUTION

Damage to equipment may result if incorrect connecting link is installed. Incorrect link is prone to stress corrosion cracking. Make sure only link marked T73, is installed.

e. Inspect to make sure only connecting link, 70410-02334-045, marked T73, is installed.

f. Position servo over mounting holes as close to manifold couplings as possible.

g. Carefully slide servo toward manifold and connect self-sealing couplings between servo and manifold (Figure 11-4-70-1, Detail A).

h. Line up output control rod clevis with boost servo arm while sliding servo toward manifold.

WARNING

FSCAP

Verification of minimum torque and safety of bolts are critical characteristics.

i. Attach boost servo to manifold and to baseplate with bolts and washers. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.** Lockwire, Item 197, Appendix D, bolts. Apply torque stripe (PARA 2-6-5).

j. Install yaw boost servo output control rod (PARA 11-4-76).

k. Install yaw boost servo input control rod (PARA 11-4-66).

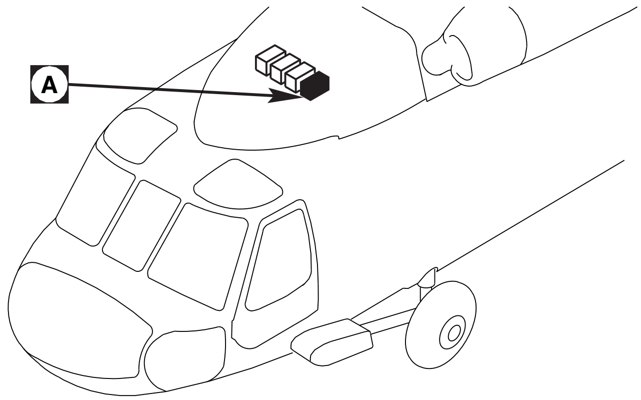
l. Remove protective caps and plugs from electrical connectors.

m. Inspect and treat electrical connectors (PARA 1-7-20).

n. Connect electrical connector, P462R, to SAS servo valve and electrical connector, P470R, to pressure switch.

o. Perform operational check of No. 2 hydraulic system (PARA 7-2-1).

p. Check for proper rigging as follows:



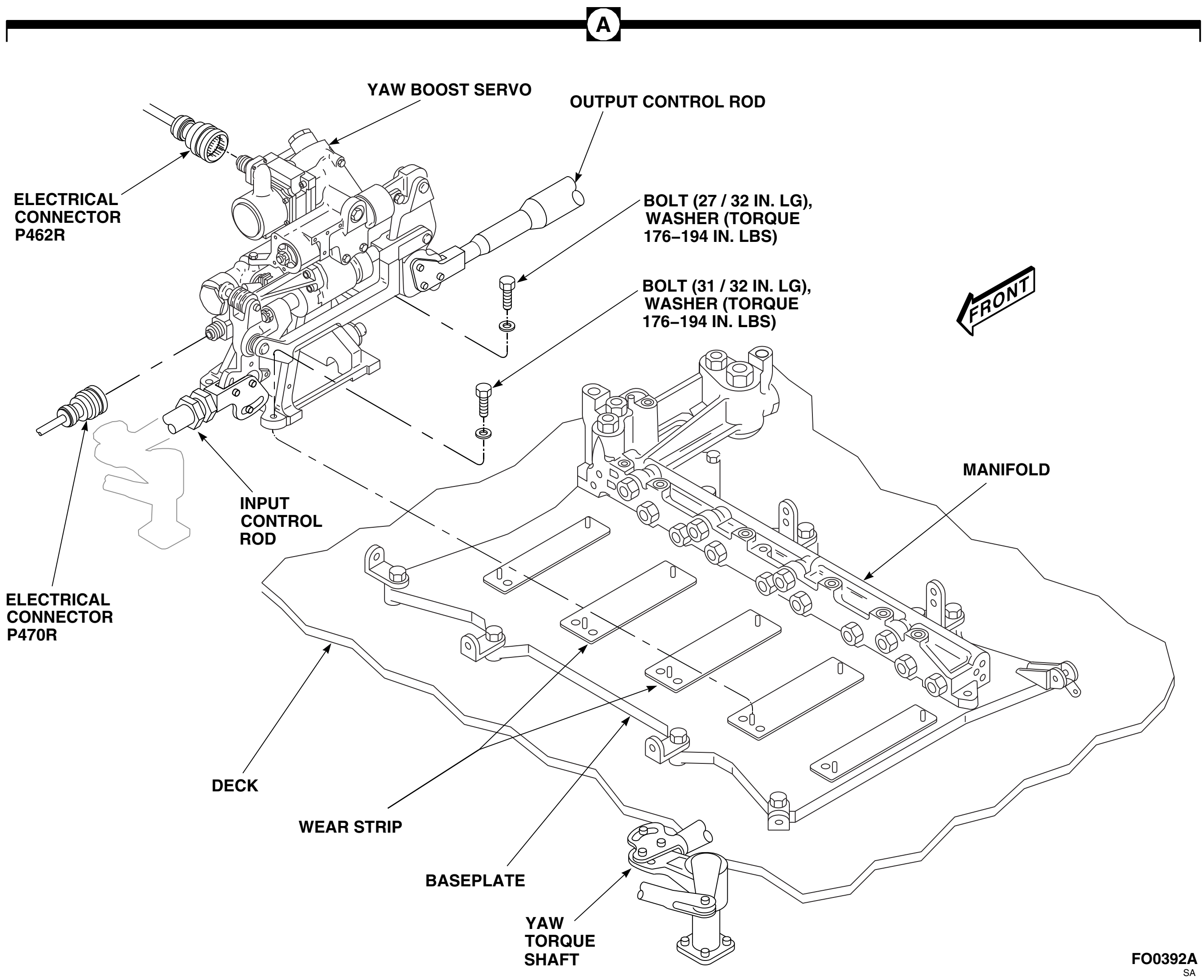


FIGURE 11-4-70-1. REPLACE YAW BOOST SERVO

CAUTION

Damage to flight control components will result if hydraulic power is shut down with rigging pins still installed. Make sure all rigging pins are removed prior to shutting down hydraulic power.

NOTE

Backup pump should be used only as an alternate source to power hydraulic system. External hydraulic power should be applied whenever available.

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Perform BOOST ON, SAS OFF procedure (PARA 11-4-3).

(3) Install rigging pin, 70700-20380-043, in yaw torque shaft on top deck.

(4) Attempt to install F-pin, 70700-20380-062, in yaw boost servo. If F-pin cannot be installed, adjust yaw boost servo input control rod until pin can be inserted.

(5) With collective stick in full down position, install rigging pin, 70700-20380-043, in collective torque shaft on top deck.

(6) Attempt to install rigging pin, 70700-20380-068, in left side of mixer and through yaw crank. If pin will not go through yaw crank in mixer, adjust yaw boost servo output control rod until pin will slide through yaw crank.

(7) Remove yaw torque shaft, collective torque shaft, yaw boost servo, and L/H mixer rigging pins.

(8) Connect engine LDS cables at mixer (PARA 4-4-27).

(9) Hinge down ceiling soundproofing panels at STA 311.0 to gain access to tail rotor pedal stopbolts located below mixer in cabin ceiling.

(10) Move collective stick to full up position and push full left pedal. Check that left pedal stopbolt in cabin ceiling is contacted. If stop is not contacted, perform tail rotor system complete rig (PARA 11-4-5).

(11) Move collective stick to full down position and push full right pedal. Check that right pedal stopbolt in cabin ceiling is contacted. If stop is not contacted, perform tail rotor system complete rig (PARA 11-4-5).

(12) Move collective stick to full up position and push full right pedal. Yaw boost servo should bottom in full extend direction.

(13) Move collective stick to full low position and push full left pedal. Yaw boost servo should bottom in full retract direction.

(14) Close ceiling soundproofing panels and secure.

(15) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

q. Make sure area is clean and free of foreign material before closing cover.

r. Close and latch main rotor pylon sliding cover.

s. Perform maintenance test flight (Appropriate MTF).

11-4-71 COLLECTIVE SAS ASSEMBLY/YAW BOOST SERVO PRESSURE SWITCH AND THERMAL RELIEF VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-4-65

11-4-71.1 Replace Collective SAS Assembly/Yaw Boost Servo Pressure Switch and Thermal Relief Valve.

CAUTION

Damage to collective SAS assembly/ yaw boost servo will result if dirt is allowed to enter servo during disassembly. To avoid getting dirt in components, make sure collective SAS assembly/yaw boost servo is clean before disassembly. Have replacement switch or valve ready for installation before old switch or valve is removed.

11-4-71.1.1. Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Disconnect electrical connector, P470R, from pressure switch on collective SAS assembly/ yaw boost servo (Figure 11-4-71-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

e. Remove pressure switch from collective SAS assembly/yaw boost servo.

f. Remove packings from pressure switch. Discard packings.

g. Unscrew thermal relief valve from servo.

h. Remove packing from valve. Discard packing.

11-4-71.1.2. Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Coat packing with hydraulic fluid, Item 154 or Item 155, Appendix D. Install packing on thermal relief valve (Figure 11-4-71-1, Detail A).

c. Screw relief valve in collective SAS assembly/yaw boost servo. **TORQUE VALVE TO 25 - 30 INCH-POUNDS.** Lockwire, Item 197, Appendix D, valve to servo.

d. Coat packings with hydraulic fluid, Item 154 or Item 155, Appendix D. Install packings on pressure switch.

e. Using wrench flats, screw switch into collective SAS assembly/yaw boost servo. **TORQUE PRESSURE SWITCH TO 142 - 157 INCH-POUNDS.** Lockwire, Item 197, Appendix D, switch to servo.

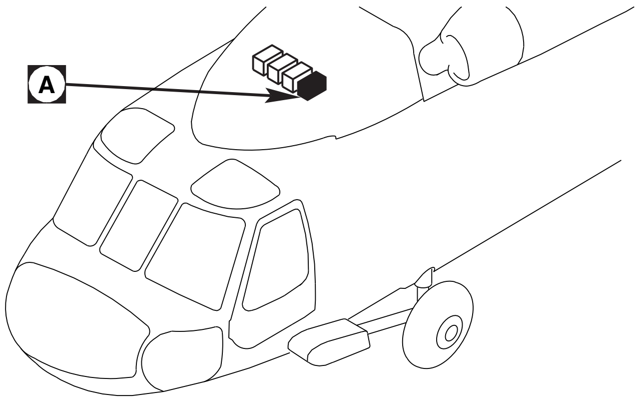
f. Remove protective caps and plugs from electrical connectors.

g. Inspect and treat electrical connectors (PARA 1-7-20).

h. Connect electrical connector, P470R, to pressure switch.

i. Bleed No. 1, No. 2, and backup hydraulic systems (PARA 7-4-65).

j. If required, service hydraulic system (PARA 1-3-8).



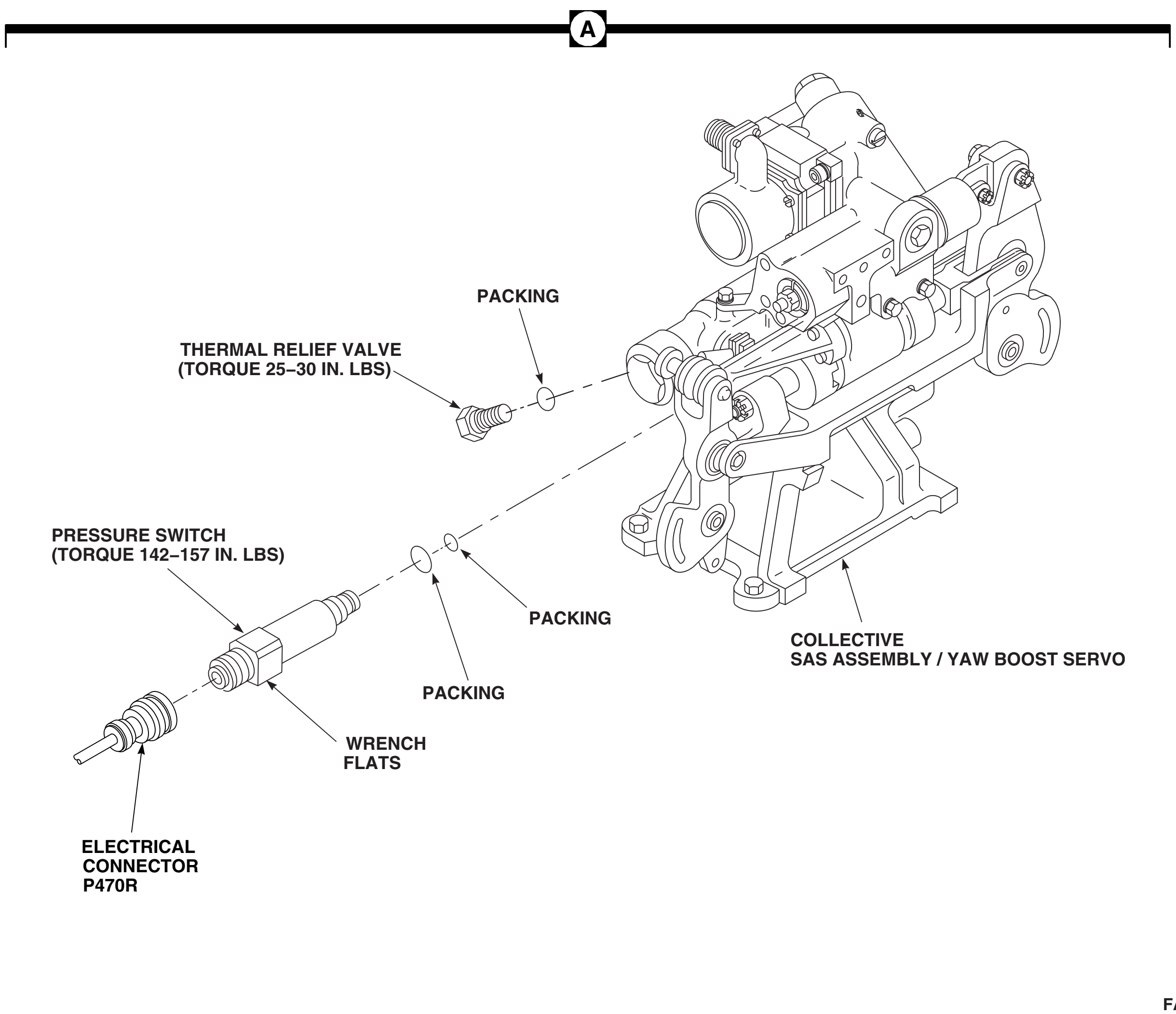


FIGURE 11-4-71-1. REPLACE COLLECTIVE SAS ASSEMBLY/YAW BOOST SERVO PRESSURE SWITCH AND THERMAL RELIEF VALVE

11-4-72 SAS ACTUATOR BLEED SCREW AND PACKING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Protection, Eye
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 7-3-1
- PARA 7-4-65

11-4-72.1 Replace SAS Actuator Bleed Screw and Packing.

NOTE

Replacement of pitch/trim assembly, roll trim assembly, and yaw boost servo SAS actuator bleed screw and packing is the same.

11-4-72.1.1 Remove.

a. Remove SAS actuator bleed screw (Figure 11-4-72-1). Remove and discard packing.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

b. Clean screw with ethyl alcohol, Item 71, Appendix D. Blow out screw bleed port with dry, filtered, compressed air.

11-4-72.1.2 Install.

a. Coat new packing and bleed screw threads with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install packing on bleed screw. Install screw in SAS actuator. **TORQUE SCREW TO 19 - 21 INCH-POUNDS.** Using lockwire, Item 196,

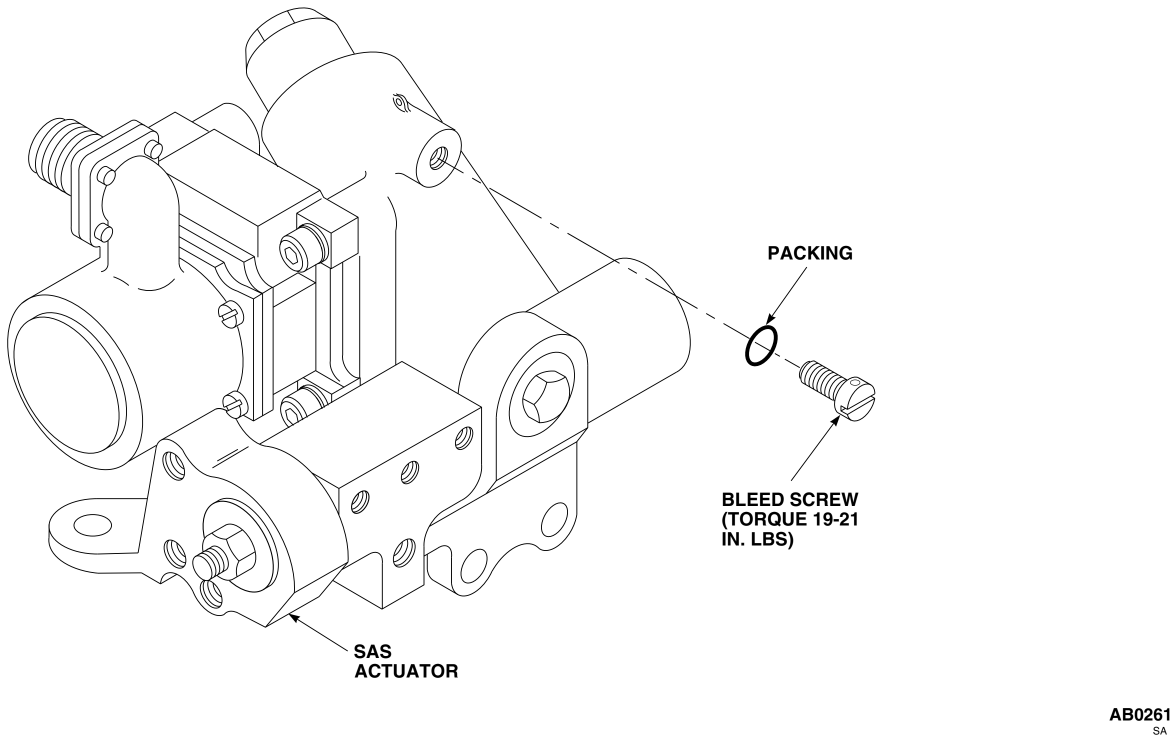


FIGURE 11-4-72-1. REPLACE SAS ACTUATOR BLEED SCREW AND PACKING

Appendix D, lockwire bleed screw to SAS actuator (Figure 11-4-72-1).

c. Bleed No. 1, No. 2, and backup hydraulic systems (PARA 7-4-65).

d. If required, service hydraulic system (PARA 1-3-8).

e. Check bleed screw installation for leakage (PARA 7-3-1).

11-4-73 COLLECTIVE SERVO TO MIXER CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-73.1 Replace Collective Servo to Mixer Control Rod.

11-4-73.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove bolt, washer, flanged bushing, and nut from mixer arm (Figure 11-4-73-1, Sheet 1, Detail B).

c. Remove bolt, washer, and nut securing collective control rod to mixer arm.

d. Remove bolts, washers, bushings, and nuts securing control rod to collective boost servo output arm. Remove control rod.

11-4-73.1.2 Install.

a. Adjust replacement control rod to exact length, bolthole-center to bolthole-center, as on removed rod as follows (Figure 11-4-73-1, Sheet 2, Detail C):

(1) Loosen jamnuts.

(2) Turn adjuster to lengthen or shorten control rod.

(3) Insert lockwire, Item 193, Appendix D, into inspection holes in tube and rod end.

(4) Adjuster threads should stop wire from passing through inspection holes.

(5) **TIGHTEN TUBE ASSEMBLY JAMNUT 1/6 TO 1/3 TURN PAST WHERE SHARP RISE IN TORQUE IS FELT.**

(6) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.** Do not lockwire now.

(7) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

b. Install control rod between mixer arm and output arm, with adjustable end at collective boost servo output arm (Figure 11-4-73-1, Sheet 1, Detail B).

NOTE

Boltheads can face either direction when installing bolts in servo arm, mixer arm, and control rod.

c. Temporarily install self-retaining bolts at each end.

d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

e. If replacement control rod has not been adjusted to exact length of replaced control rod or length of replaced control rod is not known, position collective servo at low collective (full rear).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

f. Install 5/16-inch by 36-inch rigging pin in right side of mixer. If necessary, adjust control rod so rigging pin can be installed freely (PARA 11-4-2).

g. If control rod length is correct. Using lockwire, Item 197, Appendix D, lockwire adjuster to tube jamnut and rod end jamnut to tab on locking device (Figure 11-4-73-1, Sheet 2, Detail D).

h. Perform quality check to ensure jamnuts have been properly lockwired.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

i. Install self-retaining bolt and washers (under bolthead) (Figure 11-4-73-1, Sheet 1, Detail B) (PARA 11-4-1).

j. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

k. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

NOTE

When installing link, boltheads can face either direction.

l. At mixer end, install flanged bushing through mixer arm clevis and install bolt, washer (under bolthead), and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

m. At collective boost servo, install long bushing through rod end and output arm. Install 1-17/32-inch long bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

n. Install short bushings in slots on each side of rod end. Install 1-21/32-inch long bolt, washers, and nut through bushings, rod end, and output arm. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

o. Remove rigging pin from mixer.

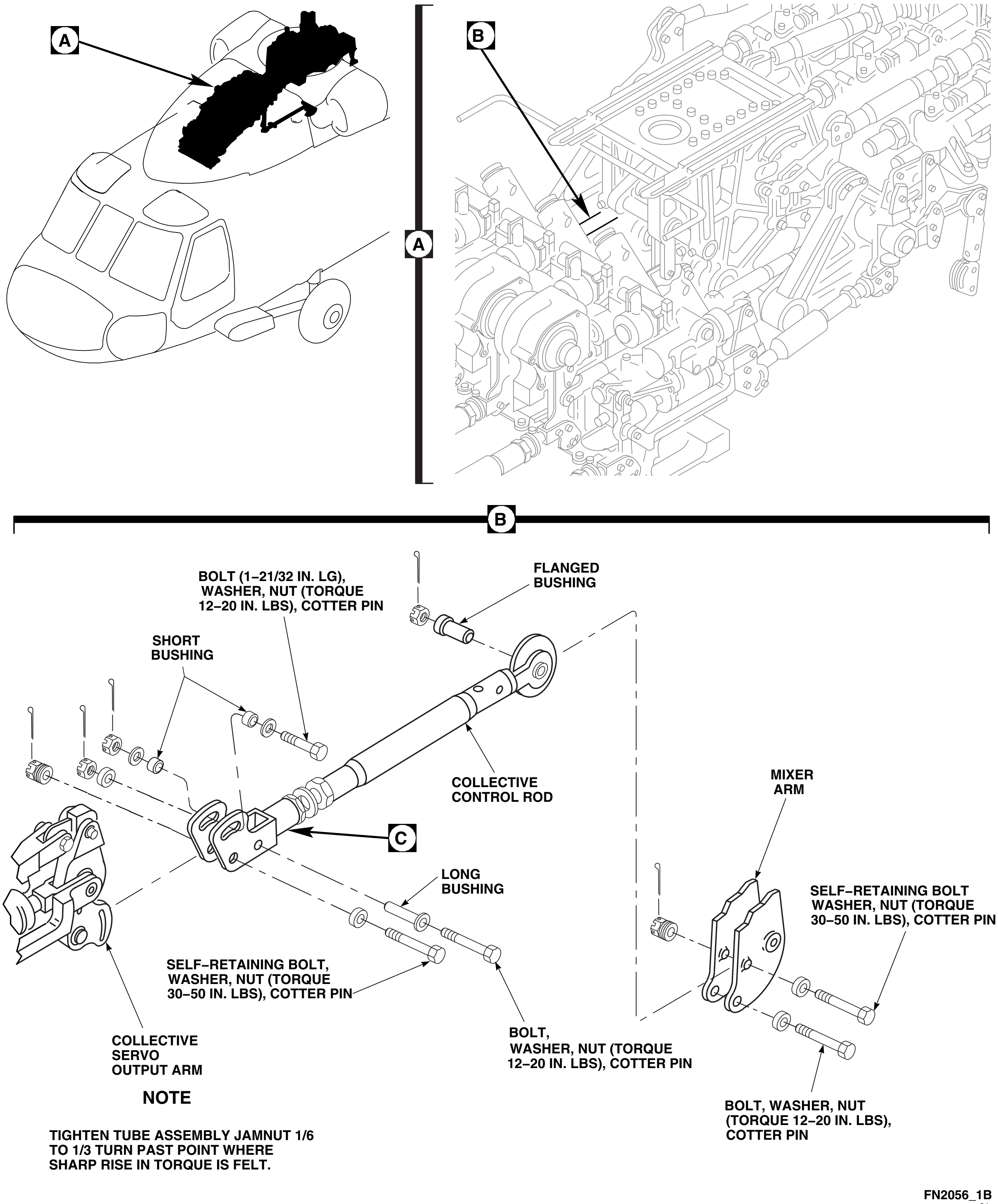
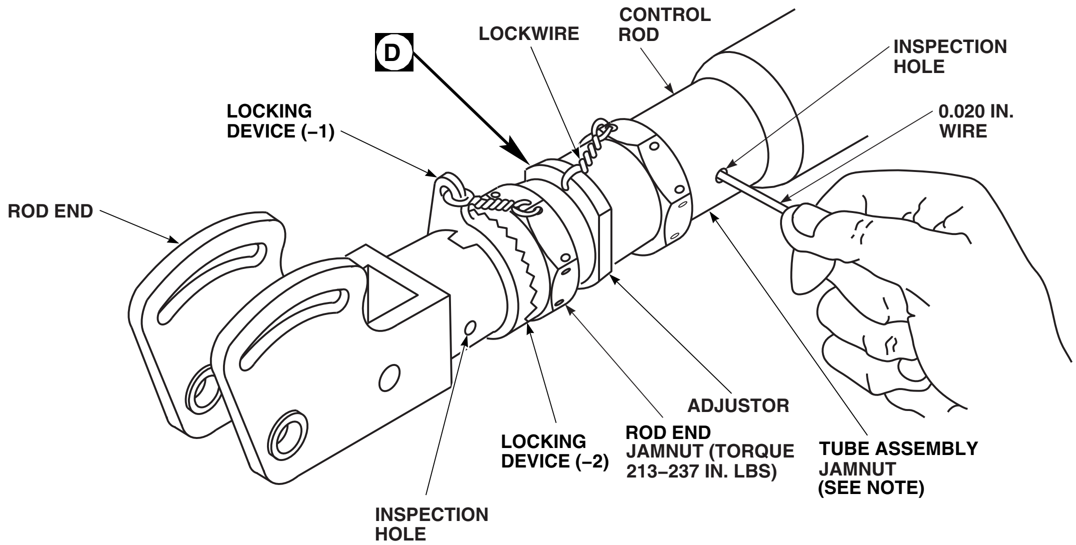
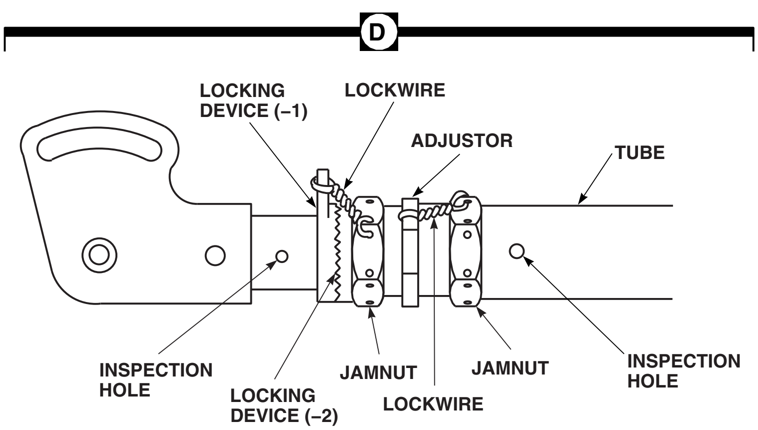


FIGURE 11-4-73-1. REPLACE COLLECTIVE SERVO TO MIXER CONTROL ROD (SHEET 1 OF 2)



CONTROL ROD ADJUSTMENT

C



COLLECTIVE

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FIGURE 11-4-73-1. REPLACE COLLECTIVE SERVO TO MIXER CONTROL ROD (SHEET 2 OF 2)

p. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

q. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

r. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

s. Make sure area is clean and free of foreign material.

t. Close main rotor pylon sliding cover.

11-4-74 PITCH/TRIM ASSEMBLY TO MIXER CONTROL LINK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-6-5
- PARA 11-4-1

11-4-74.1 Replace Pitch/Trim Assembly to Mixer Control Link.

11-4-74.1.1. Remove.

a. Open main rotor pylon sliding cover.

b. Remove bolts, washers, bushings, and nuts securing link to mixer and pitch/trim assembly lever (Figure 11-4-74-1, Detail B).

11-4-74.1.2. Install.

a. Install replacement link in mixer clevis and on pitch/trim assembly lever (Figure 11-4-74-1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

c. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

d. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. At pitch/trim assembly lever, install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

f. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

g. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

h. Install long bushing through link and slot in lever. Install 1-9/32-inch long bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

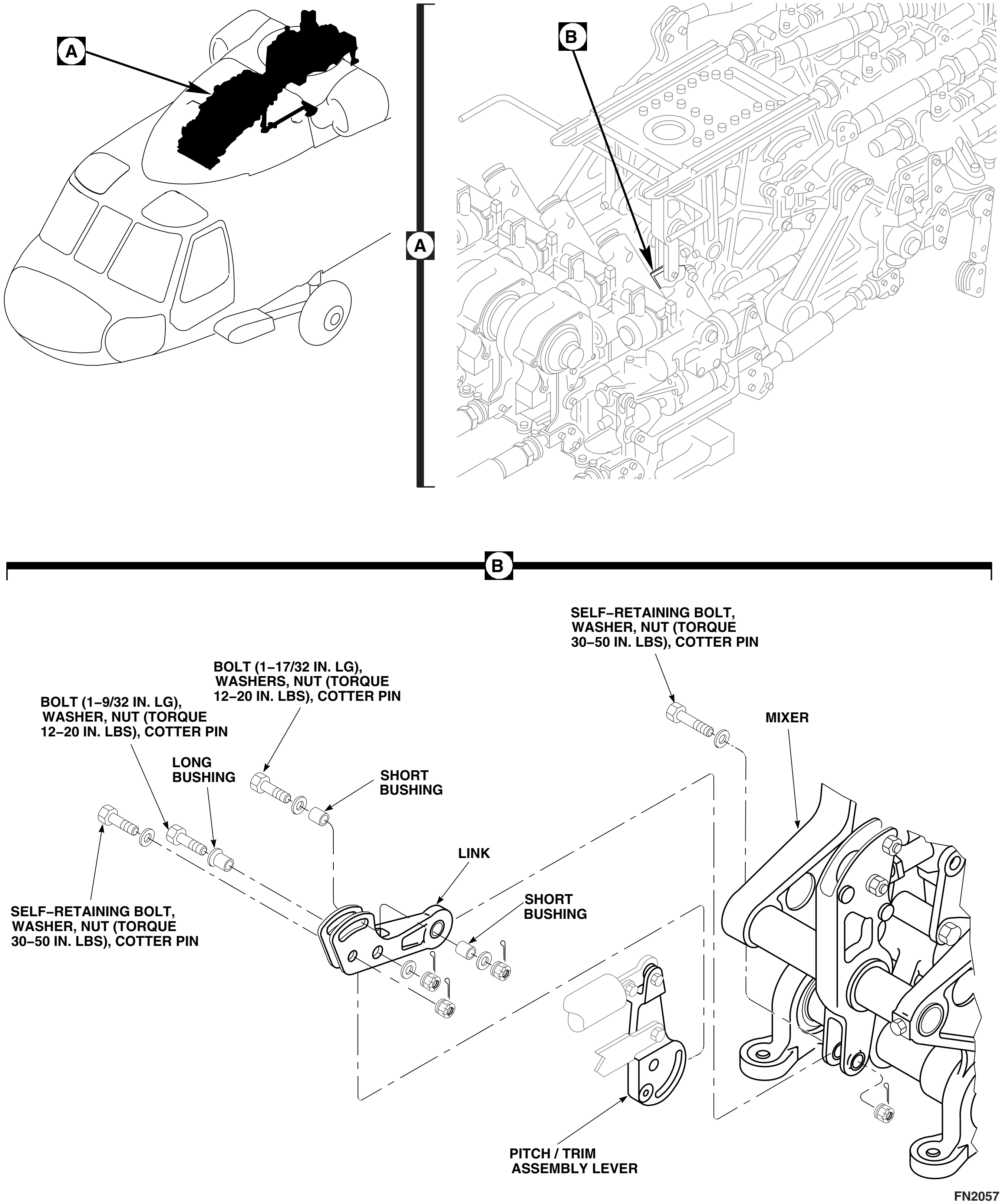


FIGURE 11-4-74-1. REPLACE PITCH/TRIM ASSEMBLY TO MIXER CONTROL LINK

NOTE

Bolt cannot be installed through link slot on helicopters with pitch/trim assembly, 70410-02760-044.

i. On helicopters with pitch/trim assemblies, 70410-02760-045 or 70410-02760-047, install short bushing in each side of link. Install 1-17/32-inch long bolt, washers, and nut through bushings and lever. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin (Figure 11-4-74-1, Detail B).

j. Make sure area is clean and free of foreign material.

k. Inspect installed control link for proper tri-pivot bolt installation (PARA 11-4-1).

l. Close and latch main rotor pylon sliding cover.

11-4-75 ROLL TRIM ASSEMBLY TO MIXER CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-75.1 Replace Roll Trim Assembly to Mixer Control Rod.

11-4-75.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove bolt, washer, bushing and nut from mixer lever (Figure 11-4-75-1, Sheet 1, Detail B).

c. Remove bolts, washers, and nuts securing lateral control rod to mixer lever.

d. Remove bolts, washers, and nuts securing control rod to roll trim assembly lever. Remove control rod.

11-4-75.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed rod as follows (Figure 11-4-75-1, Sheet 2, Detail C).

(1) Loosen jamnuts on tube and rod end.

(2) Turn sleeve to lengthen or shorten control rod.

(3) Insert lockwire, Item 193, Appendix D, in inspection hole in tube assembly. Adjusted sleeve must stop wire from passing through.

(4) Check that witness hole in rod and keyway cannot be seen outside rod end

jamnut (Figure 11-4-75-1, Sheet 2, Detail E).

(5) **TIGHTEN TUBE ASSEMBLY JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(6) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.** Do not lockwire now.

(7) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

b. Position control rod between mixer lever and roll actuator lever, with adjustable end at actuator (Figure 11-4-75-1, Sheet 1, Detail B).

NOTE

Boltheads can face either direction when installing control rod to actuator lever and mixer lever.

c. Temporarily install self-retaining bolts at each end of control rod.

d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

e. Install rigging pin in roll trim assembly and 5⁄16-inch by 36-inch rigging pin in right side of mixer. If necessary, adjust control rod so rigging pins can be installed (PARA 11-4-2).

f. If control rod length is correct, using lockwire, Item 197, Appendix D, lockwire jamnuts to tab of locking device (Figure 11-4-75-1, Sheet 2, Detail D).

g. Perform quality check to ensure jamnuts have been properly lockwired.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

h. Install self-retaining bolts and washers (under bolthead) at each end of control rod (PARA 11-4-1).

i. Install nuts on bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

j. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

k. At mixer lever, install long bushing through mixer lever clevis and install bolt, washer (under bolthead), and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin. Attaching hardware shall rotate freely, with no more than finger torque required, after installation.

l. At roll actuator lever, install long bushing through lever and rod end. Install 1-9/32-inch-long bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin. Attaching hardware shall rotate freely, with no more than finger torque required, after installation.

m. Install short bushings in slot on each side of actuator lever. Install 1-17/32-inch-long bolt, washers, and nut through bushing, rod end, and roll actuator lever. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

n. Remove rigging pins from mixer and roll actuator.

o. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

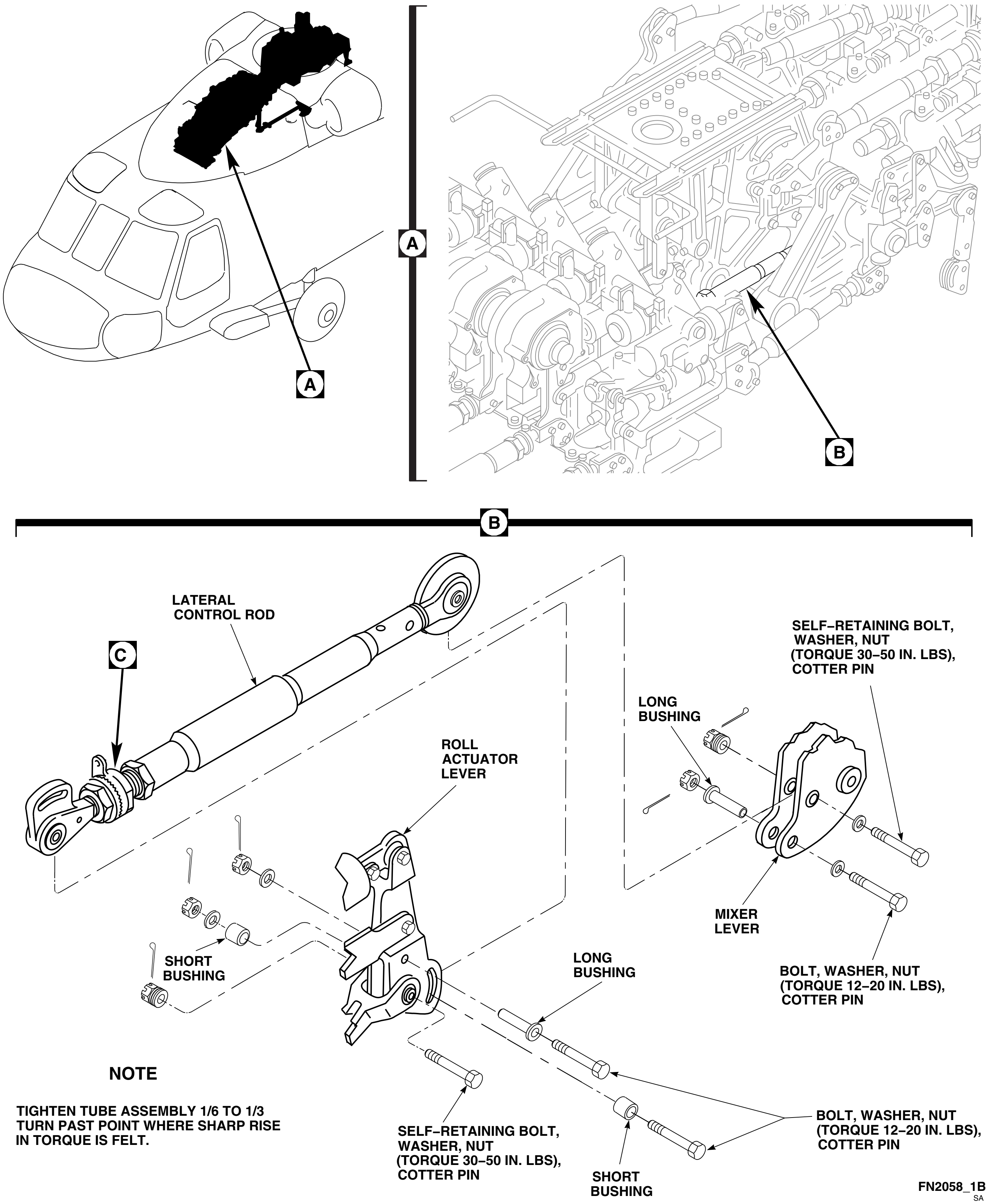


FIGURE 11-4-75-1. REPLACE ROLL TRIM ASSEMBLY TO MIXER CONTROL ROD (SHEET 1 OF 2)

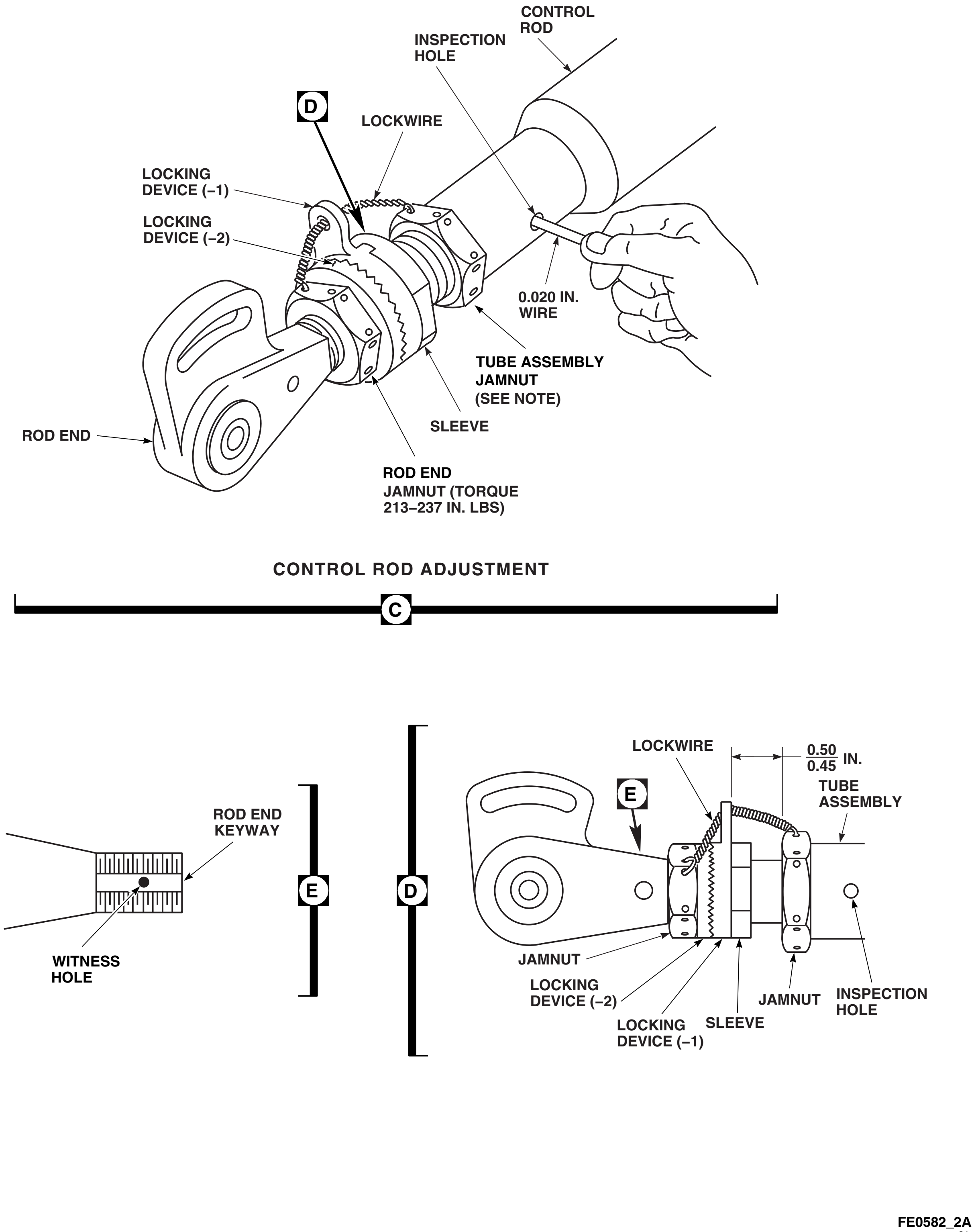


FIGURE 11-4-75-1. REPLACE ROLL TRIM ASSEMBLY TO MIXER CONTROL ROD (SHEET 2 OF 2)

p. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

q. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

r. Make sure area is clean and free of foreign material.

s. Close main rotor pylon sliding cover.

11-4-76 YAW BOOST SERVO TO MIXER CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 194, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-76.1 Replace Yaw Boost Servo to Mixer Control Rod.

11-4-76.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove bolts, washers, bushings, and nuts securing control rod to mixer lever and yaw boost servo arm (Figure 11-4-76-1, Sheet 1, Detail B). Remove control rod.

11-4-76.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed rod as follows (Figure 11-4-76-1, Sheet 2, Detail C):

(1) Loosen jamnuts on tube and rod end.

(2) Turn sleeve to lengthen or shorten control rod.

(3) Insert lockwire, Item 193, Appendix D, in inspection hole in rod assembly. Adjusted sleeve must stop wire from passing through.

(4) **TIGHTEN TUBE ASSEMBLY JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

(5) **TORQUE ROD END JAMNUT TO 213 - 237 INCH-POUNDS.** Do not lockwire now.

(6) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

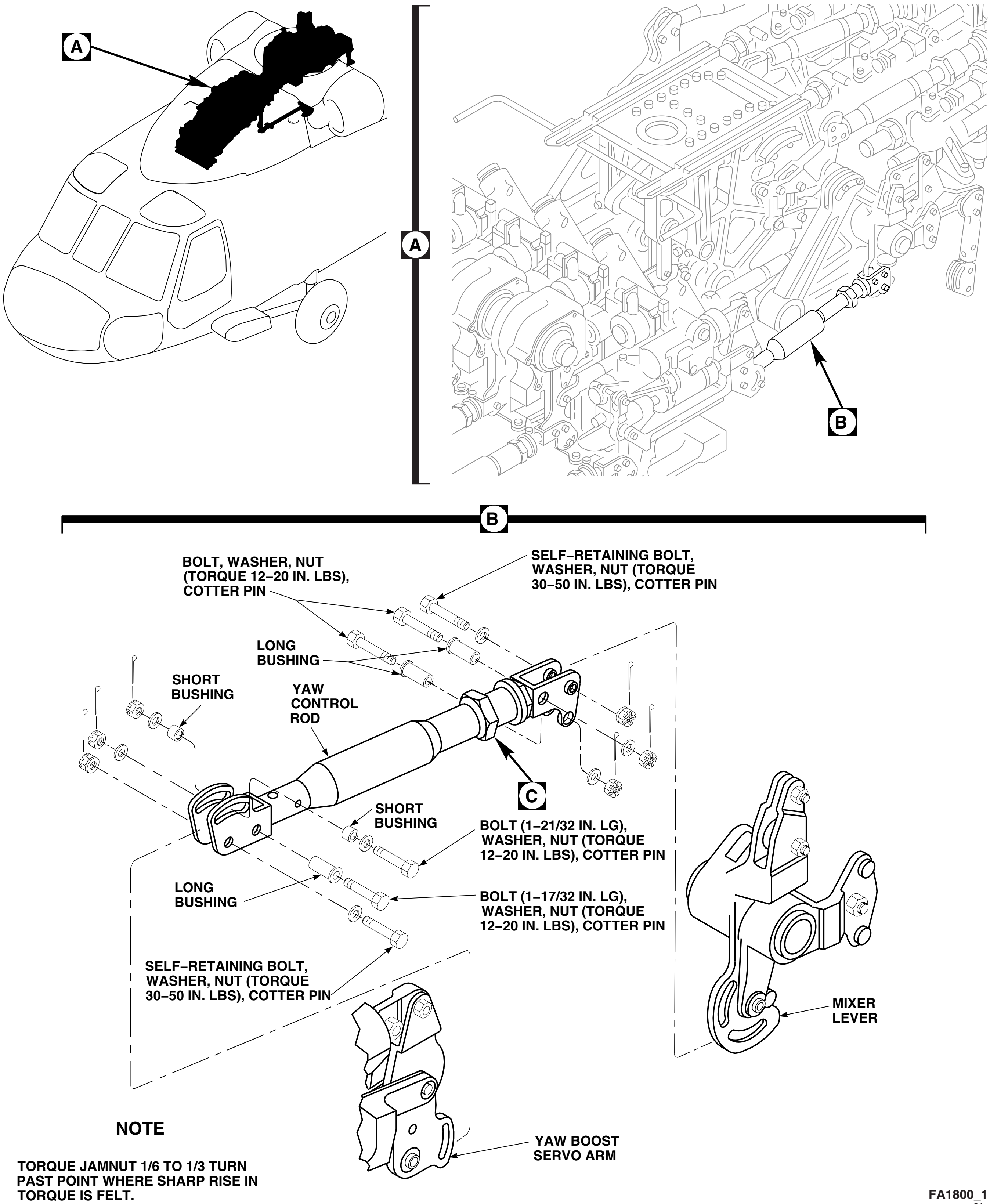
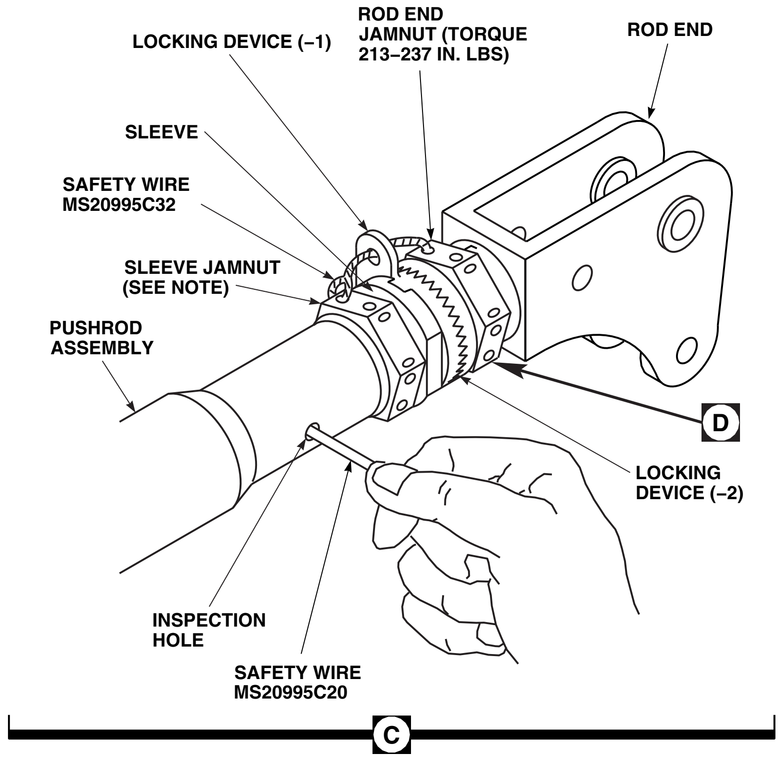


FIGURE 11-4-76-1. REPLACE YAW BOOST SERVO TO MIXER CONTROL ROD (SHEET 1 OF 2)



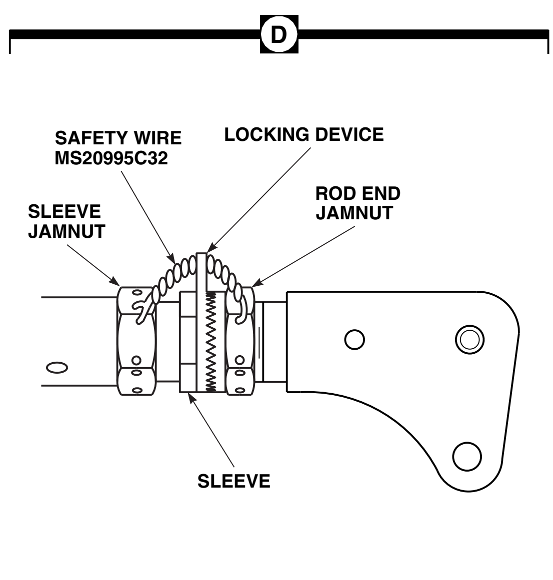


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FIGURE 11-4-76-1. REPLACE YAW BOOST SERVO TO MIXER CONTROL ROD (SHEET 2 OF 2)

b. Install control rod between mixer lever and servo arm with adjustable end at mixer (Figure 11-4-76-1, Sheet 1, Detail B).

NOTE

When installing control rod, boltheads can face either direction.

c. Temporarily install self-retaining bolts at each end. Place washers under each bolthead.

d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

e. Install rigging pin in yaw boost servo and 5/16-inch x 12-inch rigging pin in left side of mixer. If necessary, adjust control rods so rigging pins can be installed (PARA 11-4-2).

f. When control rod length is correct, using lockwire, Item 194, Appendix D, lockwire jamnuts to locking device tab (Figure 11-4-76-1, Sheet 2, Detail D).

g. Perform quality check to ensure jamnuts have been properly lockwired.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

h. Install self-retaining bolt and washers (under bolthead) (Figure 11-4-76-1, Sheet 1, Detail B) (PARA 11-4-1).

i. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

j. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

k. At mixer end, install long bushings through rod end and slots in mixer lever. Install bolts, washers, and nuts. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

l. At servo end, install long bushing through rod end and servo arm. Install 1-17/32-inch long bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

m. Install short bushings in slots on each side of rod end. Install 1-21/32-inch bolt, washers, and nut through bushings, rod end, and servo arm. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

n. Remove rigging pins from mixer and from yaw boost servo.

o. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

p. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

q. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

r. Make sure area is clean and free of foreign material.

s. Close main rotor pylon sliding cover.

11-4-77 MIXER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-69
- PARA 2-4-105
- PARA 2-6-5
- PARA 4-4-27
- PARA 7-4-26
- PARA 11-4-2
- PARA 11-4-4
- PARA 11-4-73
- PARA 11-4-74
- PARA 11-4-75
- PARA 11-4-76
- PARA 11-4-80
- PARA 11-4-81
- PARA 11-4-82
- PARA 11-4-83

11-4-77.1 Replace Mixer.

11-4-77.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove soundproofing panel from cabin ceiling between STA 281.0 and STA 295.0 (PARA 2-4-69).

c. Open main rotor pylon sliding cover.

d. If main rotor blades are installed, turn main rotor head to make removal of mixer easier.

e. Remove main rotor pylon work platform (PARA 2-4-105).

f. Disconnect electrical connectors, P317, P318, and P351, if installed, from J317, J318, and J351, if installed (Figure 11-4-77-1, Sheet 1, Detail A).

g. Install protective caps and plugs on electrical connectors.

h. Remove dust cap from electrical connector, J319.

i. Remove electrical connectors from bracket as follows:

(1) Remove screws, washers, and nuts securing electrical connectors, J317, J319, and J351, if installed, to bracket.

(2) Remove jamnut securing electrical connector, J318, to bracket.

j. Remove clamps, screws, washers, and nuts securing wiring harnesses to bracket. Remove harnesses.

k. Install rigging pins in collective, pitch, roll, and yaw torque shafts, pilot-assist and primary servos (PARA 11-4-2).

l. Remove and disconnect the following flight controls:

(1) Remove mixer to forward primary servo control rod (PARA 11-4-80).

(2) Remove mixer to aft primary servo control rod (PARA 11-4-81).

(3) Remove mixer to lateral primary servo control rod (PARA 11-4-82).

(4) Remove yaw boost servo to mixer control rod (PARA 11-4-76).

(5) Remove roll trim assembly to mixer control rod (PARA 11-4-75).

(6) Remove collective servo to mixer control rod (PARA 11-4-73).

(7) Disconnect mixer to yaw lever pushrod from mixer (PARA 11-4-83).

(8) Disconnect pitch/trim assembly to mixer control link from mixer (PARA 11-4-74).

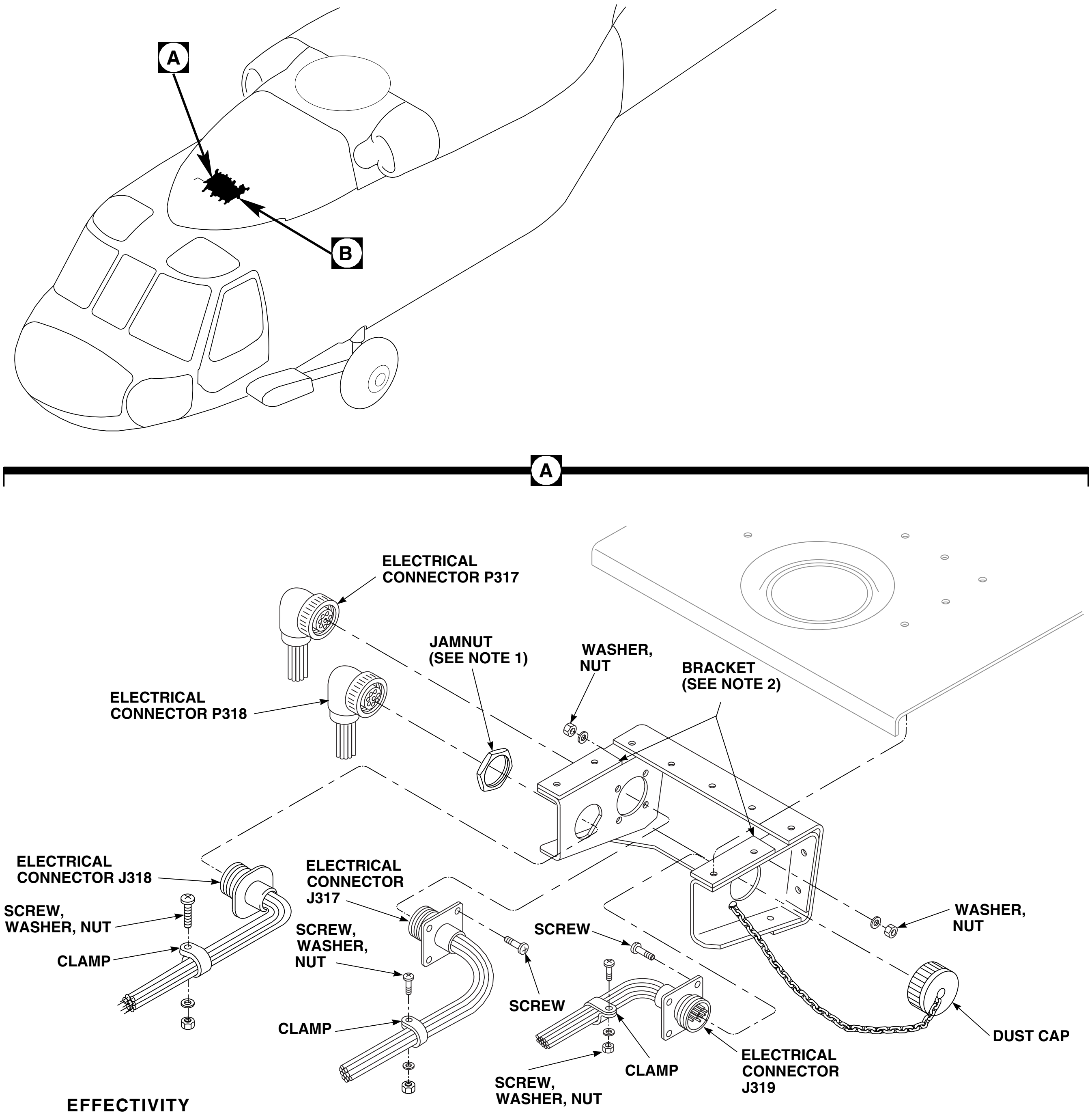
m. Disconnect both engine load-demand push/pull cables from mixer (PARA 4-4-27).

NOTE

There may be more than one countersunk washer under bolthead. Tag each bolt for which leg of mixer it was removed from. Record number of countersunk washers under each bolthead.

n. Remove bolts and countersunk washers securing mixer to deck. Keep countersunk washers with bolts (Figure 11-4-77-1, Sheet 3, Detail B).

o. Remove mixer from helicopter.



EFFECTIVITY

HELICOPTERS WITHOUT ELECTRICAL CONNECTOR, P351, INSTALLED.

NOTES

1. TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.
2. BRACKET SHOWN REMOVED FOR CLARITY.

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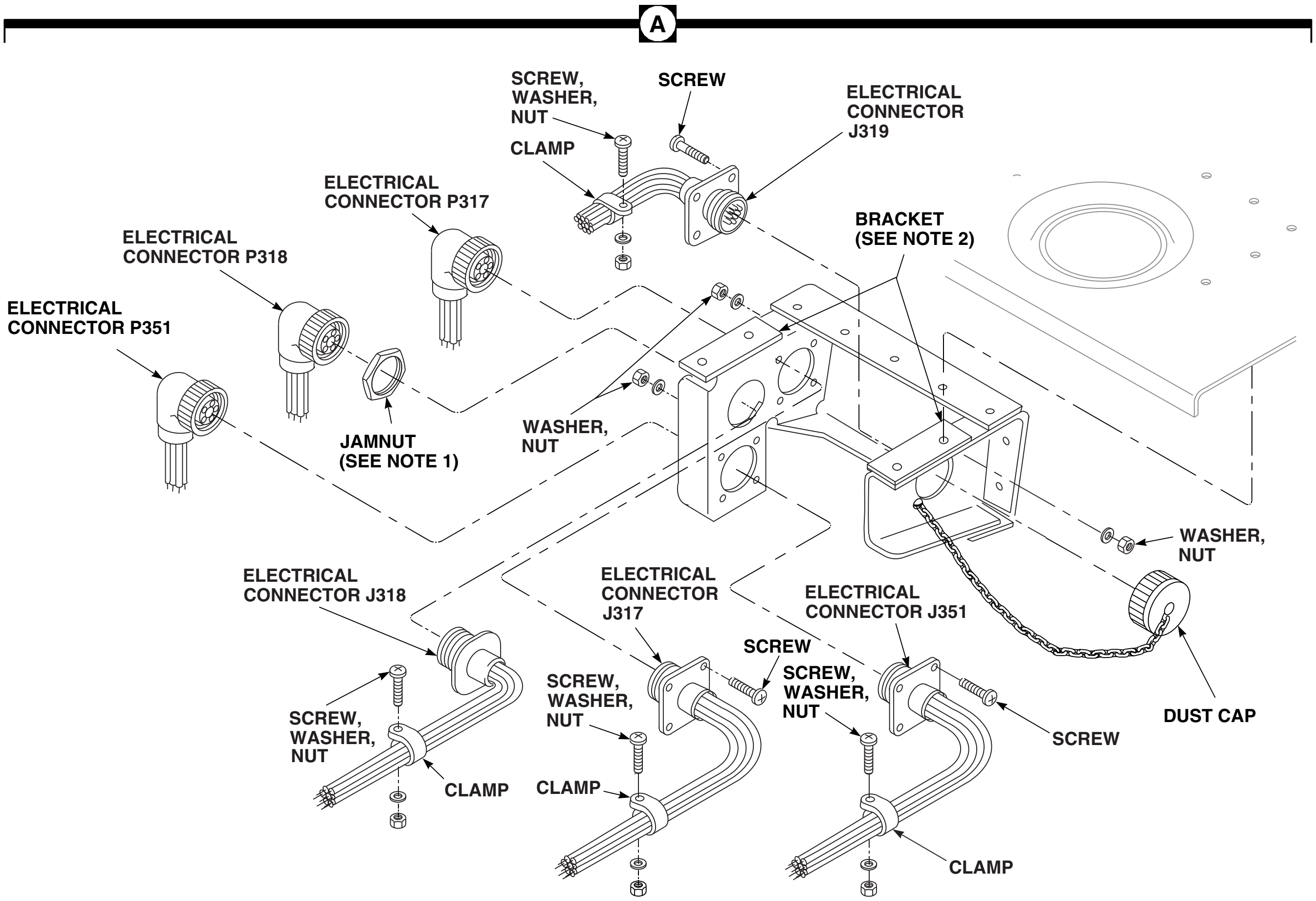
FIGURE 11-4-77-1. REPLACE MIXER (SHEET 1 OF 3)



EFFECTIVITY

HELICOPTERS WITH ELECTRICAL CONNECTOR, P351, INSTALLED.

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FIGURE 11-4-77-1. REPLACE MIXER (SHEET 2 OF 3)

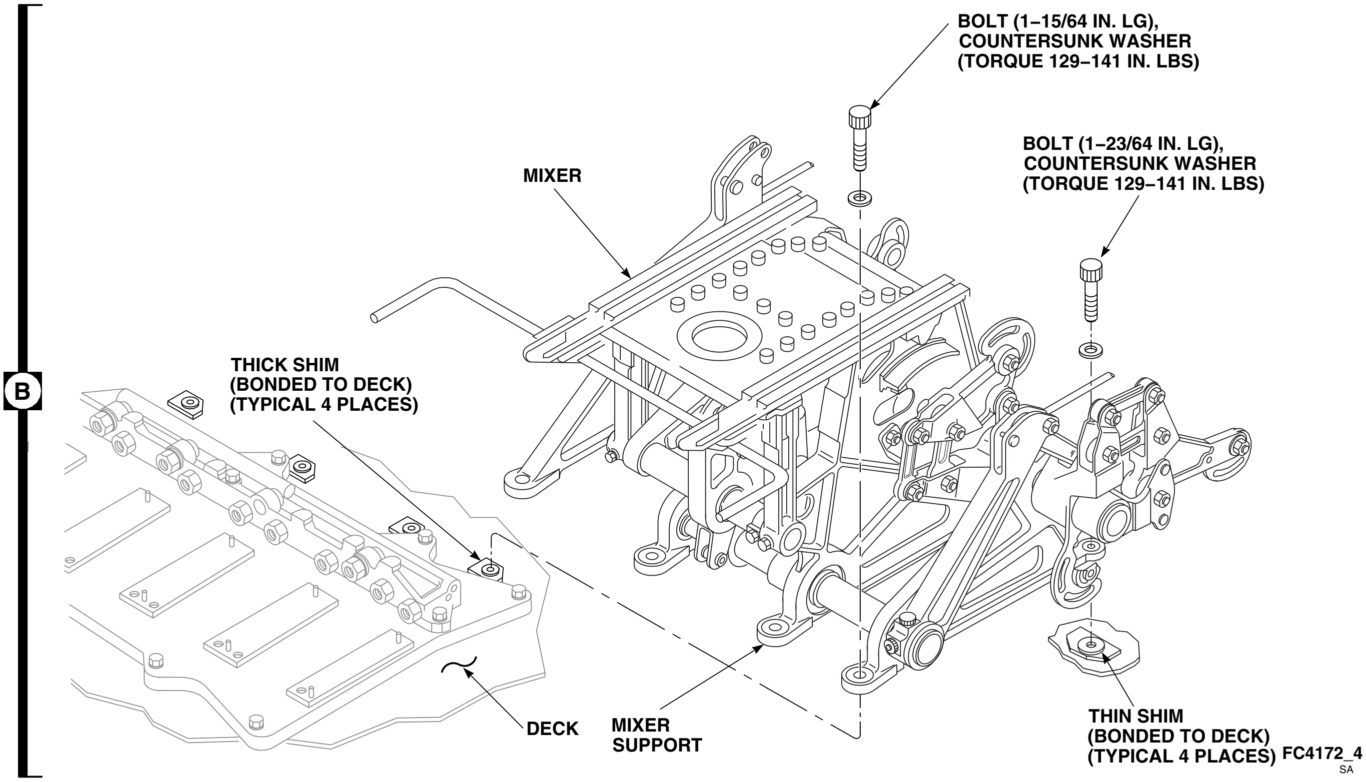


FIGURE 11-4-77-1. REPLACE MIXER (SHEET 3 OF 3)

11-4-77.1.2 Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. If shims are disbonded from deck, perform the following (Figure 11-4-77-1, Sheet 3, Detail B):

(1) Clean area of disbonded shim(s) using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Remove any remaining adhesive using nonmetallic scraper.

(2) Mix adhesive, Item 32, Appendix D, in ratio of 100 parts by weight of part A to 22 parts by weight of part B.

(3) Apply a light coat of adhesive to deck around mounting hole(s). Install shim(s). Wipe extra adhesive from inside diameter of hole in shim(s) using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(4) Allow adhesive to cure at 120° ± 10°F (49° ± 5°C) for at least 2 hours.

b. If shim(s) are lost or misplaced, perform the following (Figure 11-4-77-1, Sheet 3, Detail B):

NOTE

Helicopter may not have had shims or may not require shims if gap between mixer supports and deck is less than 0.010-inch.

(1) Position mixer over mounting holes. **INSTALL BOLTS FINGERTIGHT THROUGH MIXER SUPPORTS.**

(2) Using feeler gage, measure gap between mixer supports and deck. **MAXIMUM ALLOWABLE GAP BEFORE TORQUING BOLTS IS 0.010-INCH.**

(3) If any gap exceeds 0.010-inch, peel shims as required to allow mixer supports to seat.

(4) Bond replacement shim(s) per step a.

NOTE

- Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Install same number of countersunk washers under boltheads as recorded when removed.

c. Put replacement mixer on deck and/or deck shims, if installed, and install eight bolts and washers in same place they were removed from according to tags. Install four long bolts in rear of mixer, and four short bolts in front. Remove tags (Figure 11-4-77-1, Sheet 3, Detail B).

WARNING

FSCAP

Verification of minimum torque of bolts are critical characteristics.

d. **TORQUE BOLTS TO 129 - 141 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. In cabin, remove soundproofing panels in ceiling, as necessary, to get to nutplates in mixer supports (PARA 2-4-69).

f. Check that bolt is fully threaded into each nutplate and that two threads are showing past nutplate. Make sure bolt does not contact wiring cables routed under nutplates.

NOTE

There must be at least one washer but no more than three washers under bolthead.

g. If more than four threads are showing past nutplate, add washers under bolt as necessary or remove washers if bolt does not fully engage nutplate.

h. Seal areas around boltheads and mating surface of mixer and helicopter with sealing compound, Item 292, Appendix D.

i. Install and connect the following flight controls:

(1) Connect pitch/trim assembly to mixer control link from mixer (PARA 11-4-74).

(2) Connect mixer to yaw lever pushrod from mixer (PARA 11-4-83).

(3) Install collective servo to mixer control rod (PARA 11-4-73).

(4) Install roll trim assembly to mixer control rod (PARA 11-4-75).

(5) Install yaw boost servo to mixer control rod (PARA 11-4-76).

(6) Install mixer to lateral primary servo control rod (PARA 11-4-82).

(7) Install mixer to aft primary servo control rod (PARA 11-4-81).

(8) Install mixer to forward primary servo control rod (PARA 11-4-80).

j. Connect both engine load-demand push/pull cables to mixer (PARA 4-4-27).

k. Install electrical connectors to bracket as follows (Figure 11-4-77-1, Sheet 1, Detail A):

(1) Install electrical connector, J317, in rear mounting hole on inside of right bracket on mixer. Secure electrical connector to bracket with screws, washers, and nuts.

(2) If required, install electrical connector, J351, in lower front mounting hole on inside of right bracket on mixer. Secure electrical connector to bracket with screws, washers, and nuts.

(3) Install electrical connector, J319, in left mounting hole on inside of bracket. Secure electrical connector to bracket with screws, washers, and nuts.

(4) Install electrical connector, J318, in forward mounting hole on inside of right bracket. Secure electrical connector to bracket with jamnut. **TIGHTEN JAMNUT 1/6 TO 1/3 PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

l. Install dust cap on electrical connector, J319.

m. Install wiring harnesses on bracket using clamps, screws, washers, and nuts.

n. Remove protective caps and plugs from electrical connectors.

o. Inspect and treat electrical connectors (PARA 1-7-20).

p. Connect electrical connectors, P317, P318, and P351, if installed, to, J317, J318, and J351, if installed.

q. Check for minimum clearance between mixer lever and No. 1 transfer module (PARA 7-4-26).

r. Perform main rotor system rig check (PARA 11-4-4).

s. Install main rotor pylon work platform (PARA 2-4-105).

t. Make sure area is clean and free of foreign material.

u. Install soundproofing panels (PARA 2-4-69).

v. Close and latch main rotor pylon sliding cover.

w. Perform maintenance test flight (Appropriate MTF).

11-4-78 MIXER REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Slide Caliper
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Grease, Item 149, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-105
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-80
- PARA 11-4-82

11-4-78.1 Repair Mixer.

11-4-78.1.1. Prepare.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Several components of mixer assembly (left and right lateral limiter, limiter rollers, yaw/longitudinal coupling shaft, collective coupling shaft, and yaw/pitch coupling link) may become discolored (reddish-brown) while in service. Surface discoloration on these components is not cause for replacement.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Remove main rotor pylon work platform (PARA 2-4-105).

d. Remove screws, bolts, washers, and nuts securing work platform track assembly to mixer

(Figure 11-4-78-1, Sheet 1, Detail A). Remove work platform track assembly.

11-4-78.1.2. Replace Forward Input-Output Links.

a. Inspect small and large link bearings with dial indicator. **REPLACE LINK IF RADIAL PLAY IS 0.005 INCH OR GREATER.**

b. Check small and large link bearings for roughness, sticking, and seal damage. **REPLACE LINK WITH BEARINGS THAT HAVE ROUGHNESS, STICKING, AND SEAL DAMAGE.**

c. Inspect small and large link for loose bearings, corrosion, and damage. Blend out any corrosion with abrasive cloth, Item 10, Appendix D. **REPLACE LINKS IF CORROSION IS DEEPER THAN 0.010 INCH AFTER BLENDING.**

d. Inspect small and large link for nicks and scratches. Blend out any damage with abrasive cloth, Item 10, Appendix D. **REPLACE LINKS IF NICKS AND SCRATCHES ARE GREATER THAN 0.030 INCH AFTER BLENDING.**

e. Remove bolts, washers, and nuts securing small and large links to forward output and forward input bellcranks (Figure 11-4-78-1, Sheet 2, Detail B).

NOTE

- On helicopters with mixer, 70400-02300-041 or 70400-02300-043, install small link, 70400-02232-042, and large link, 70400-02154-042.
- On helicopters with mixer, 70400-02300-044, 70400-02300-045, 70400-02300-046, 70400-02300-048, or 70400-02300-050, install small link, 70400-02232-046.
- On helicopters with mixer, 70400-02300-044 or 70400-02300-045, install large link, 70400-02154-047.
- On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, install large link, 70400-02154-053.

f. Install replacement large link between bellcranks.

g. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

h. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

i. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. On helicopters with mixer, 70400-02300-041, 70400-02300-043, 70400-02300-044, or 70400-02300-045, install replacement small link between bellcranks. Install bolts, washers, and nuts. **TORQUE NUTS TO 20 - 45 INCH-POUNDS.** Install cotter pins.

k. On helicopters with mixer, 70400-02300-046, place replacement small link between bellcranks.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

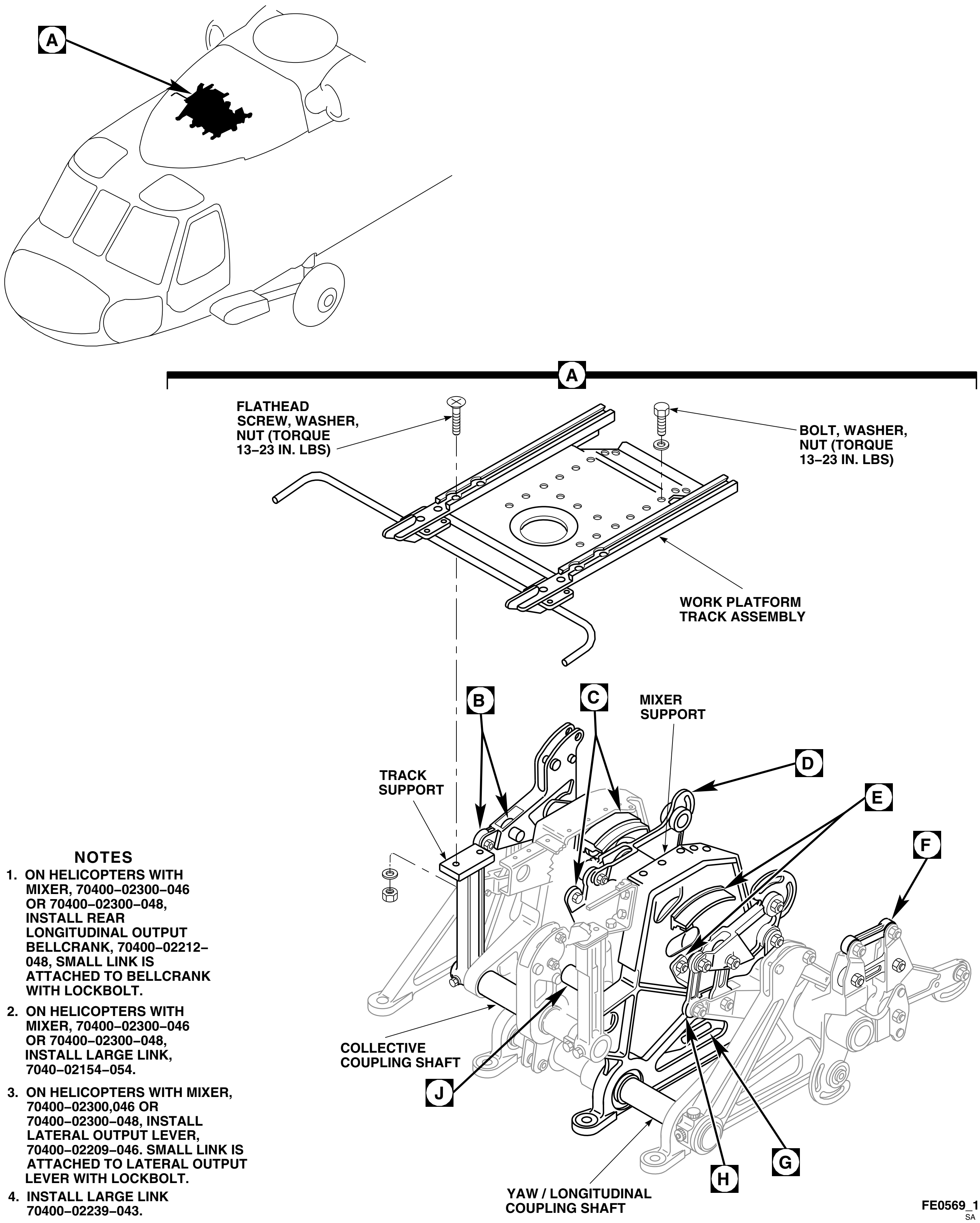
l. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

m. Install nut on bolt. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

n. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

11-4-78.1.3. Replace Longitudinal Front/ Longitudinal Rear Limiters.

a. Remove bolts, washers, and nuts securing longitudinal front limiter and longitudinal rear



NOTES

1. ON HELICOPTERS WITH MIXER, 70400–02300–046 OR 70400–02300–048, INSTALL REAR LONGITUDINAL OUTPUT BELLCRANK, 70400–02212–048, SMALL LINK IS ATTACHED TO BELLCRANK WITH LOCKBOLT.
2. ON HELICOPTERS WITH MIXER, 70400–02300–046 OR 70400–02300–048, INSTALL LARGE LINK, 7040–02154–054.
3. ON HELICOPTERS WITH MIXER, 70400–02300,046 OR 70400–02300–048, INSTALL LATERAL OUTPUT LEVER, 70400–02209–046. SMALL LINK IS ATTACHED TO LATERAL OUTPUT LEVER WITH LOCKBOLT.
4. INSTALL LARGE LINK 70400–02239–043.

FIGURE 11-4-78-1. REPAIR MIXER (SHEET 1 OF 7)

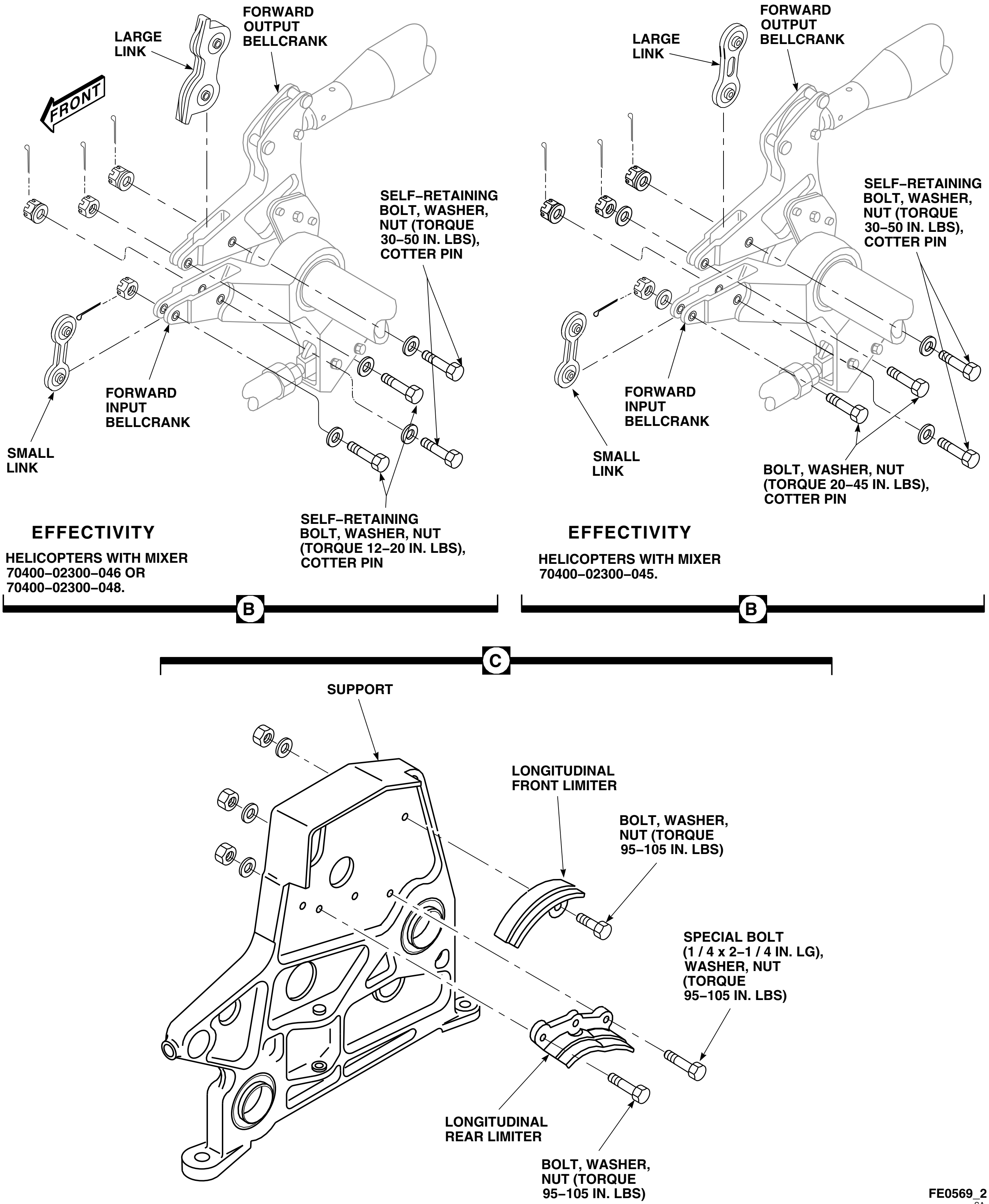
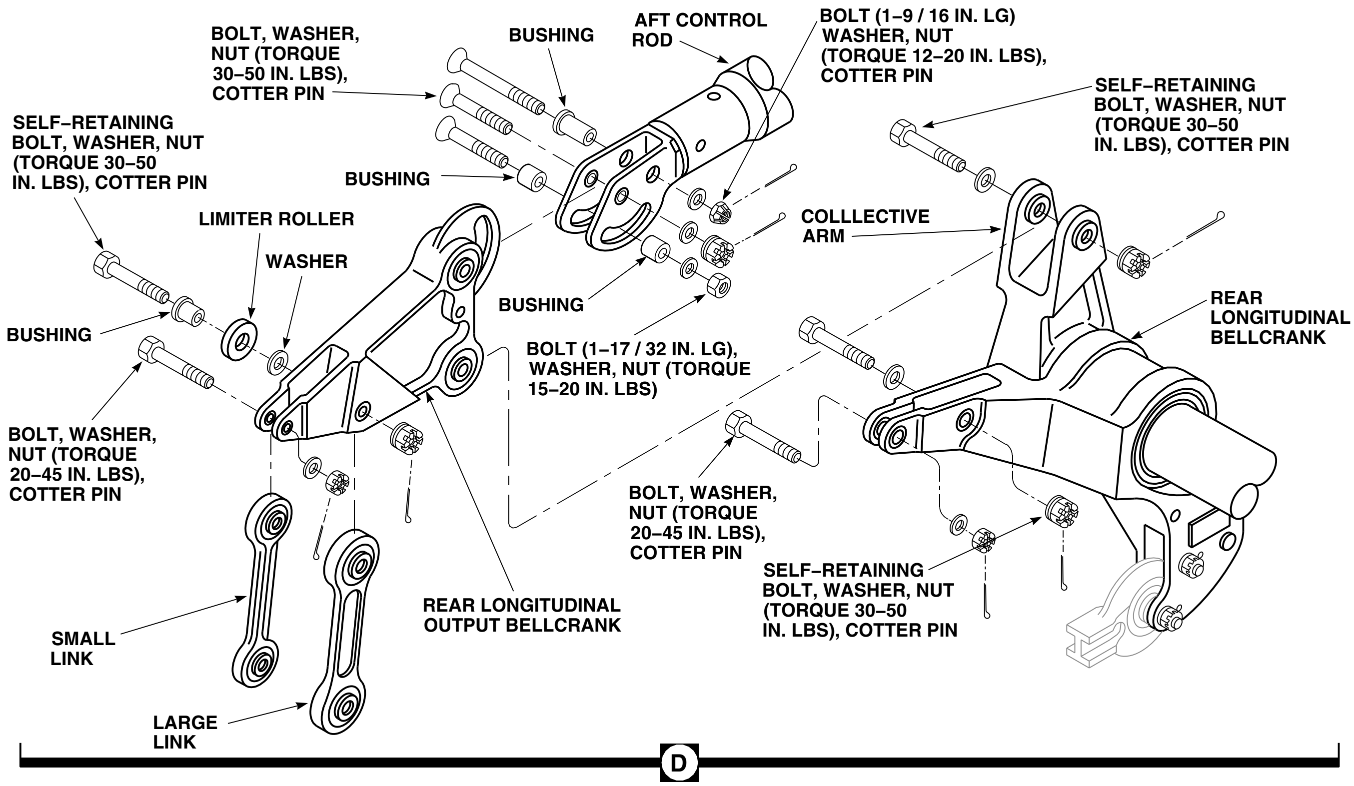


FIGURE 11-4-78-1. REPAIR MIXER (SHEET 2 OF 7)



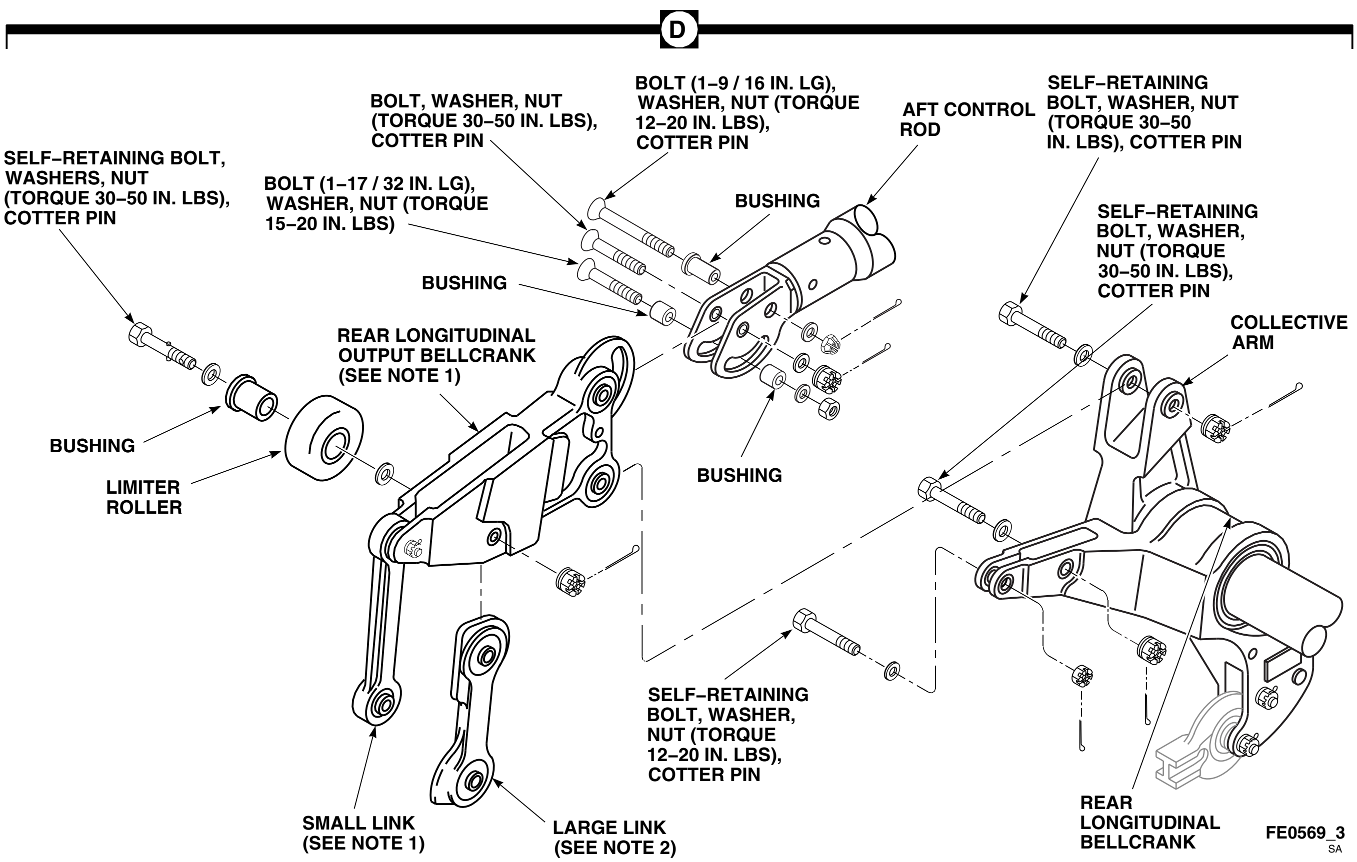


FIGURE 11-4-78-1. REPAIR MIXER (SHEET 3 OF 7)

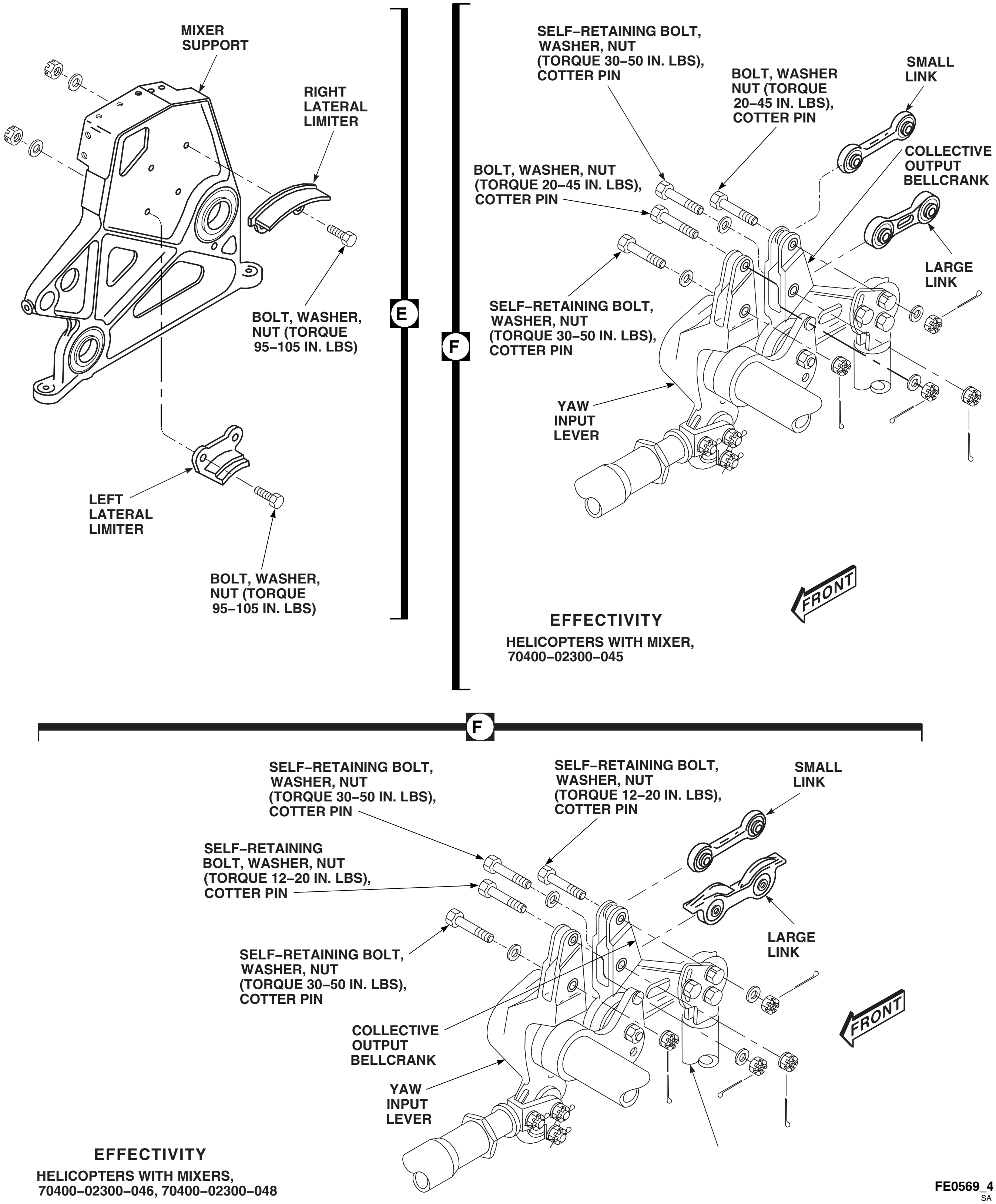


FIGURE 11-4-78-1. REPAIR MIXER (SHEET 4 OF 7)

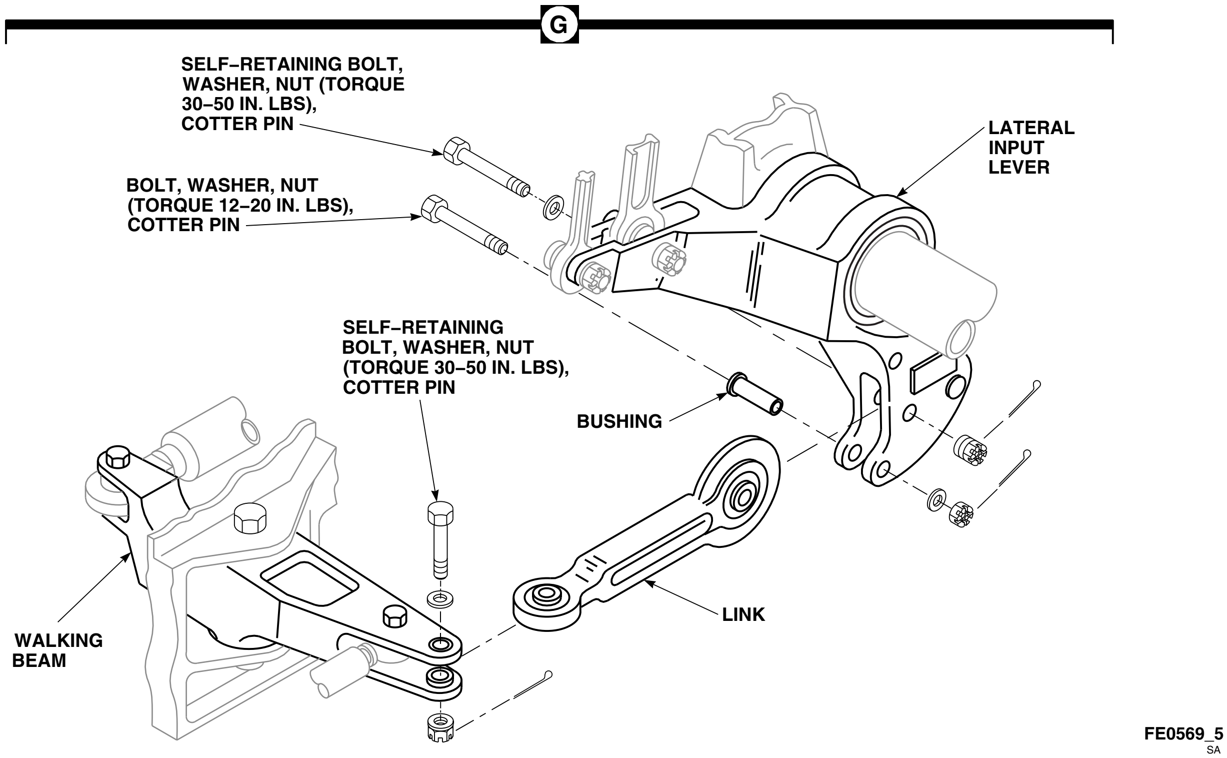


FIGURE 11-4-78-1. REPAIR MIXER (SHEET 5 OF 7)

limiter to mixer support (Figure 11-4-78-1, Sheet 2, Detail C). Remove limiters.

NOTE

The special bolt, 1/4 x 2-1/4 inches long, is installed in aft hole of longitudinal rear limiter. Torque on nut is same as on other bolts.

b. Secure replacement longitudinal front limiter and longitudinal rear limiter to mixer support with bolts, special bolt, washers, nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

11-4-78.1.4. Replace Rear Longitudinal Output Bellcrank and Links.

a. Remove bolts, washers, bushings, and nuts securing aft control rod to rear longitudinal output bellcrank (Figure 11-4-78-1, Sheet 3, Detail D).

b. Remove bolts, washers, bushing, limiter roller, and nuts securing small and large link to rear longitudinal and rear longitudinal output bellcranks.

c. Remove bolts, washers, and nuts securing rear longitudinal output bellcrank to collective arm.

d. Inspect small link, large link, and bellcrank bearings with dial indicator. **REPLACE LINKS AND BELLCRANKS IF RADIAL PLAY IS 0.005 INCH OR GREATER.**

e. Inspect small link, large link, and bellcrank bearings for roughness, sticking, and seal damage. **REPLACE LINKS AND BELLCRANKS THAT HAVE ROUGHNESS, STICKING, AND SEAL DAMAGE.**

f. Inspect small link, large link, and bellcrank for loose bearings, corrosion, and damage. **REPLACE LINKS AND BELLCRANKS IF CORROSION EXCEEDS 0.010 INCH.**

g. Inspect small link, large link, and bellcrank for nicks and scratches. **REPLACE LINKS AND BELLCRANKS IF NICKS AND SCRATCHES EXCEED 0.030 INCH.**

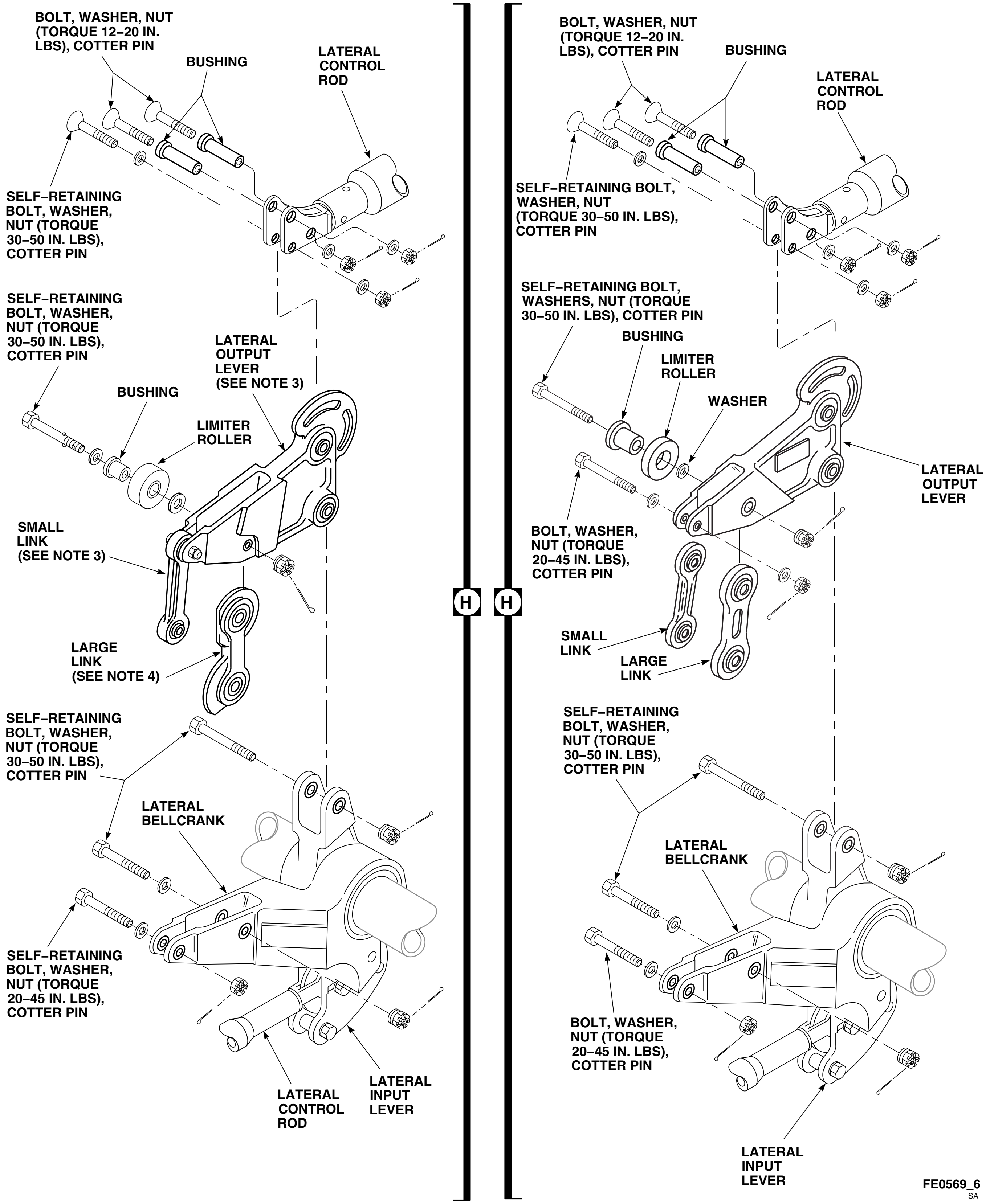


FIGURE 11-4-78-1. REPAIR MIXER (SHEET 6 OF 7)

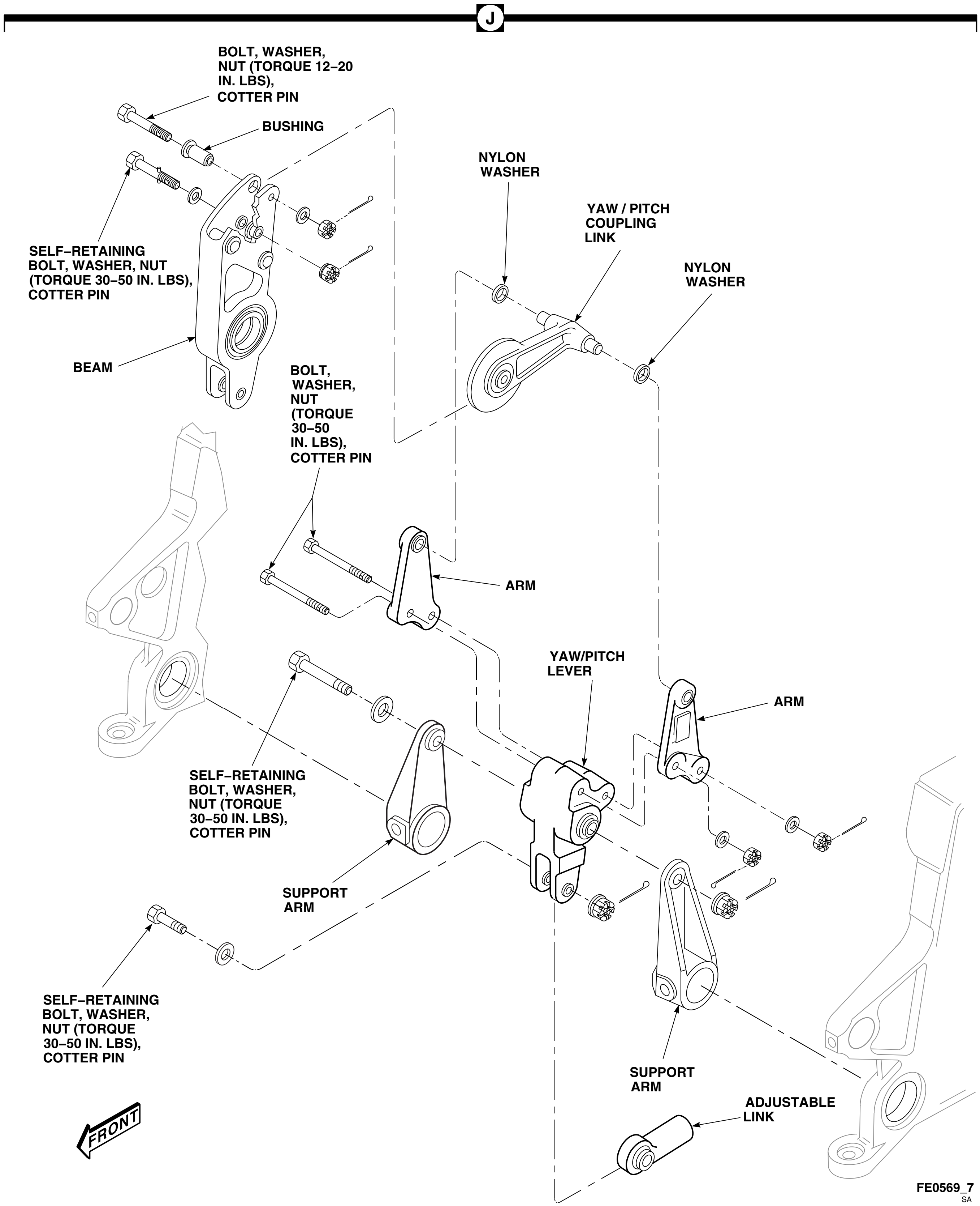


FIGURE 11-4-78-1. REPAIR MIXER (SHEET 7 OF 7)

h. Inspect bellcrank bushings, NAS537-4P86 and NAS538-4P11, using slide caliper. **INSIDE DIAMETER OF BUSHINGS SHALL NOT BE OVER 0.2505 INCH.** Inspect bellcrank bushings, NAS538-3P9 and SS4045-3P19, using slide caliper. **INSIDE DIAMETER OF BUSHINGS SHALL NOT BE OVER 0.2005 INCH.**

i. Inspect longitudinal limiter roller for gouges, cracks, sticking, and corrosion.

j. **REPLACE ROLLERS GOUGED DEEPER THAN 0.050 INCH.**

k. **NO CRACKS ALLOWED.** Repair or replace corroded parts.

l. Install replacement rear longitudinal output bellcrank in clevis of collective arm.

m. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

n. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

o. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

p. To install limiter roller on self-retaining bolt, do this:

   (1) Coat bushing with grease, Item 149, Appendix D.

   (2) Install washer(s) on self-retaining bolt (PARA 11-4-1).

   (3) Install bushing on self-retaining bolt.

   (4) Slip limiter roller over bushing.

NOTE

- On helicopters with mixer, 70400-02300-041, or 70400-02300-043, install large link, 70400-02154-043.
- On helicopters with mixer, 70400-02300-044, or 70400-02300-045, install large link, 70400-02154-048.
- On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, install large link, 70400-02154-054.

q. Install upper end of large link on rear longitudinal bellcranks.

r. Install self-retaining bolt with limiter roller and washers in upper end of large link (PARA 11-4-1).

s. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

t. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

u. Install lower end of large link on longitudinal bellcrank.

v. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

w. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

x. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

WARNING

To prevent damage to equipment and injury to personnel due to restriction of flight control travel, look through work platform track assembly to make sure of correct bolt installation. Physically hold large link and apply moderate force to confirm link is secured in place by bolt.

y. On helicopters with mixer, 70400-02300-041 or 70400-02300-043, install small link, 70400-02232-044, between upper and lower rear longitudinal bellcranks with bolts, washers, and nuts. **TORQUE NUTS TO 20 - 45 INCH-POUNDS.** Install cotter pin (Figure 11-4-78-1, Sheet 3, Detail D).

WARNING

To prevent damage to equipment and injury to personnel due to restriction of flight control travel, look through work platform track assembly to make sure of correct bolt installation. Physically hold large link and apply moderate force to confirm link is secured in place by bolt.

z. On helicopters with mixer, 70400-02300-045, install small link, 70400-02232-048, between upper and lower rear longitudinal bellcranks with bolts, washers, and nuts. **TORQUE NUTS TO 20 - 45 INCH-POUNDS.** Install cotter pins (Figure 11-4-78-1, Sheet 3, Detail D).

aa. On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, place small link attached to rear longitudinal output bellcrank assembly, 70400-02212-048, on lower rear longitudinal bellcrank.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

ab. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

ac. Install nut on bolt. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

ad. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

NOTE

All boltheads must face to right side.

ae. Connect aft control rod to rear longitudinal output bellcrank and primary servo (PARA 11-4-80).

11-4-78.1.5. Replace Left and Right Lateral Limiters.

a. Remove bolts, washers, and nuts securing right and left lateral limiter to mixer support (Figure 11-4-78-1, Sheet 4, Detail E). Remove limiters.

b. Install right lateral limiter on top holes of mixer support and install bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

c. Install left lateral limiter on bottom holes of mixer support and install bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

11-4-78.1.6. Replace Collective Output Links.

a. Remove bolts, washers, and nuts securing small and large links to yaw output lever and collective output bellcrank (Figure 11-4-78-1, Sheet 4, Detail F). Remove links.

b. Inspect small and large link bearings with dial indicator. **REPLACE LINK IF RADIAL PLAY IS 0.005 INCH OR GREATER.**

c. Inspect small and large link bearings for roughness, sticking, and seal damage. **REPLACE LINKS WITH BEARINGS THAT HAVE ROUGHNESS, STICKING, AND SEAL DAMAGE.**

d. Inspect small and large link for loose bearings, corrosion, and damage. Blend out any corrosion with abrasive cloth, Item 10, Appendix D. **REPLACE LINKS IF CORROSION EXCEEDS 0.010 INCH AFTER BLENDING.**

e. Inspect small and large link for nicks and scratches. Blend out any damage with abrasive cloth, Item 10, Appendix D. **REPLACE LINKS IF NICKS AND SCRATCHES EXCEED 0.030 INCH AFTER BLENDING.**

NOTE

- On helicopters with mixer, 70400-02300-041 or 70400-02300-043, install small link, 70400-02232-043.
- On helicopters with mixer, 70400-02300-044, 70400-02300-045, 70400-02300-046, 70400-02300-048, or 70400-02300-050, install small link, 70400-02232-047.
- On helicopters with mixer, 70400-02300-041 or 70400-02300-043, install large link, 70400-02154-046.
- On helicopters with mixer, 70400-02300-044, or 70400-02300-045, install large link, 70400-02154-049.
- On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, install large link, 70400-02154-050.

f. Install large link between yaw input lever and collective output bellcrank.

g. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

h. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

i. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. On helicopters with mixer, 70400-02300-041, 70400-02300-043, 70400-02300-044, or 70400-02300-045, install small link between yaw input lever and collective output bellcrank. Install bolts, washers, and nuts. **TORQUE NUTS TO 20 - 45 INCH-POUNDS.** Install cotter pin.

k. On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, place small link between yaw input lever and collective output bellcrank.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

l. Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

m. Install nuts on bolts. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

n. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

11-4-78.1.7. Replace Rear and Front Servo Link.

a. Remove bolt, washer, and nut securing bushing to clevis of lateral input lever (Figure 11-4-78-1, Sheet 5, Detail G).

b. Remove bolts, washers, and nuts securing link to lateral input lever and walking beam. Remove link.

c. Install replacement link between walking beam and lateral input lever.

d. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

e. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

f. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. Install bushing in clevis and install bolt, washer, nut. **TORQUE NUT 12 - 20 INCH-POUNDS.** Install cotter pin.

11-4-78.1.8. Replace Lateral Output Lever and Links.

a. Disconnect lateral control rod from lateral output lever (PARA 11-4-82).

b. Remove bolts, washers, and nuts securing small and large links to lateral output lever and lateral input lever. Remove limiter roller, bushing, and washer from self-retaining bolt (Figure 11-4-78-1, Sheet 6, Detail H).

c. Remove bolt, washer, and nut securing lateral output lever to lateral bellcrank.

d. Inspect small link, large link, and lever bearings using dial indicator. **REPLACE LINKS AND LEVERS IF RADIAL PLAY IS 0.005 INCH OR GREATER.**

e. Inspect small link, large link, and lever bearings for roughness, sticking, and seal damage. **REPLACE LINKS AND LEVERS THAT HAVE ROUGHNESS, STICKING, AND SEAL DAMAGE.**

f. Inspect small link, large link, and lever bearings for looseness, corrosion, and damage. **REPLACE LINKS AND LEVERS IF CORROSION EXCEEDS 0.010 INCH.**

g. Inspect small link, large link, and lever bearings for nicks and scratches. **REPLACE LINKS AND LEVERS IF NICKS AND SCRATCHES EXCEED 0.030 INCH.**

h. Inspect lever bushings, NAS537-4P86 and NAS538-4P9 using slide caliper. **INSIDE DIAMETER OF BUSHINGS SHALL NOT BE OVER 0.2505 INCH.** Inspect lever bushings, NAS538-3P9 and SS4045-3P18 using slide caliper. **INSIDE DIAMETER OF BUSHINGS SHALL NOT BE OVER 0.1900 INCH.**

i. Inspect lateral limiter roller for gouges, cracks, roughness, sticking, and corrosion.

j. **REPLACE ROLLERS SHOWING DAMAGE DEEPER THAN 0.050 INCH.**

k. **NO CRACKS ALLOWED.** Repair or replace corroded parts.

l. Install replacement lateral output lever in lateral bellcrank.

m. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

n. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

o. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

p. To install limiter roller on self-retaining bolt, do this:

(1) Coat bushing with grease, Item 149, Appendix D.

(2) Install washer(s) on self-retaining bolt (PARA 11-4-1).

(3) Install bushing on self-retaining bolt.

(4) Slip limiter roller over bushing.

WARNING

To prevent damage to equipment and injury to personnel due to restriction of flight control travel, check to make sure bolt is attaching rear lateral output bellcrank and upper end of large link. Physically hold large link and apply moderate force to confirm that link is secured.

NOTE

- On helicopters with mixer, 70400-02300-041 or 70400-02300-043, install small link, 70400-02232-041, and large link, 70400-02239-041 (Figure 11-4-78-1, Sheet 6, Detail H).
- On helicopters with mixer, 70400-02300-044 or 70400-02300-045, install small link, 70400-02232-045, and large link, 70400-02239-042.
- On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, install large link, 70400-02239-043. Small link should be attached to lateral output lever. Do not remove lockbolt (Figure 11-4-78-1, Sheet 6, Detail H).

q. Install upper end of large link on lateral output levers.

r. Install self-retaining bolt with limiter roller and washer on upper end of large link (PARA 11-4-1).

s. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

t. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

u. Install lower end of large link on lateral bellcrank.

v. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

w. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

x. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

y. On helicopters with mixer, 70400-02300-041, 70400-02300-043, 70400-02300-044, or 70400-02300-045, install small link between lateral output and input levers. Install bolts, washers, and nuts. **TORQUE NUTS TO 20 - 45 INCH-POUNDS.** Install cotter pins.

z. On helicopters with mixer, 70400-02300-046, 70400-02300-048, or 70400-02300-050, place small link attached to lateral output lever assembly, 70400-02209-046, on lateral bellcrank.

aa. Connect lateral control rod to lateral output lever and lateral primary servo (PARA 11-4-82).

11-4-78.1.9. Replace Yaw Pitch Coupling.

NOTE

Yaw/pitch lever and arms are a matched set. Do not try to replace separate parts.

a. Remove bolts, washers, and nuts securing arms and adjustable link from yaw/pitch lever (Figure 11-4-78-1, Sheet 7, Detail J).

b. Remove bolts, nuts and washers from arm assembly. Remove arms. Remove nylon washers from coupling link trunnion pin. Remove coupling link front end from beam. Remove coupling link.

c. Remove bolt, washer, and nut securing yaw/pitch lever to support arms.

d. Inspect yaw/pitch coupling link as follows:

NOTE

Yaw/pitch coupling link may become discolored (reddish-brown) while in service. Surface discoloration to this component is normal and is not cause for replacement and/or repair.

(1) Inspect for cracks, nicks, gouges, corrosion, or distortion. Check pivot ball bearing for condition and wear. Repair or replace link as required.

(2) Inspect large end outside diameter and trunnion pins dry film lubricant coatings for wear or damage, which shows base metal. Repair or replace link as required.

(3) Inspect trunnion pins for wear. **EACH TRUNNION PIN OUTSIDE DIAMETER SHALL MEASURE 0.2490 - 0.2495 INCH.** Repair or replace link as required.

e. Inspect self-retaining bolt (PARA 11-4-1). **REPLACE BOLT AS REQUIRED.**

NOTE

Do not reinstall pitch bias actuator.

f. Install yaw/pitch lever between support arms.

g. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

h. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

i. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. Connect adjustable link to yaw/pitch lever clevis.

k. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

l. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

m. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

n. Loosely install one arm, with bolts, through yaw pitch lever.

o. Install nylon washers (countersinks facing link) on link trunnion pins.

p. Position yaw/pitch coupling link in beam and on yaw pitch lever arm. Install second arm on lever over bolts. Put link trunnion pin through arm pivot hole. Install washers and nuts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

q. Install self-retaining bolt and washer through beam/link pivot hole (PARA 11-4-1).

r. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

s. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

t. Install bushing, bolt, washer, and nut in beam assembly. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

u. Install main rotor pylon work platform (PARA 2-4-105).

v. Make sure area is clean and free of foreign material.

w. Close and latch main rotor pylon sliding cover.

x. Do a maintenance test flight (Appropriate MTF).

11-4-79 NO. 3 COLLECTIVE STICK POSITION SENSOR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Multimeter
- Electrical Repairer's Toolkit
- Pencil
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-105
- PARA 4-2-1
- PARA 11-4-2
- PARA 11-4-78

11-4-79.1 Replace No. 3 Collective Stick Position Sensor.

11-4-79.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Remove main rotor pylon work platform (PARA 2-4-105) and track assembly (PARA 11-4-78).

d. Turn on all electrical power (PARA 1-6-16).

e. Make sure the following circuit breakers are pushed in:

(1) On upper console, DC ESNTL BUS circuit breaker panel - BACKUP HYD CONTR.

(2) On copilot's circuit breaker panel, NO. 1 DC PRI BUS - BACKUP PUMP PWR.

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use the following chart for limits when running backup pump.

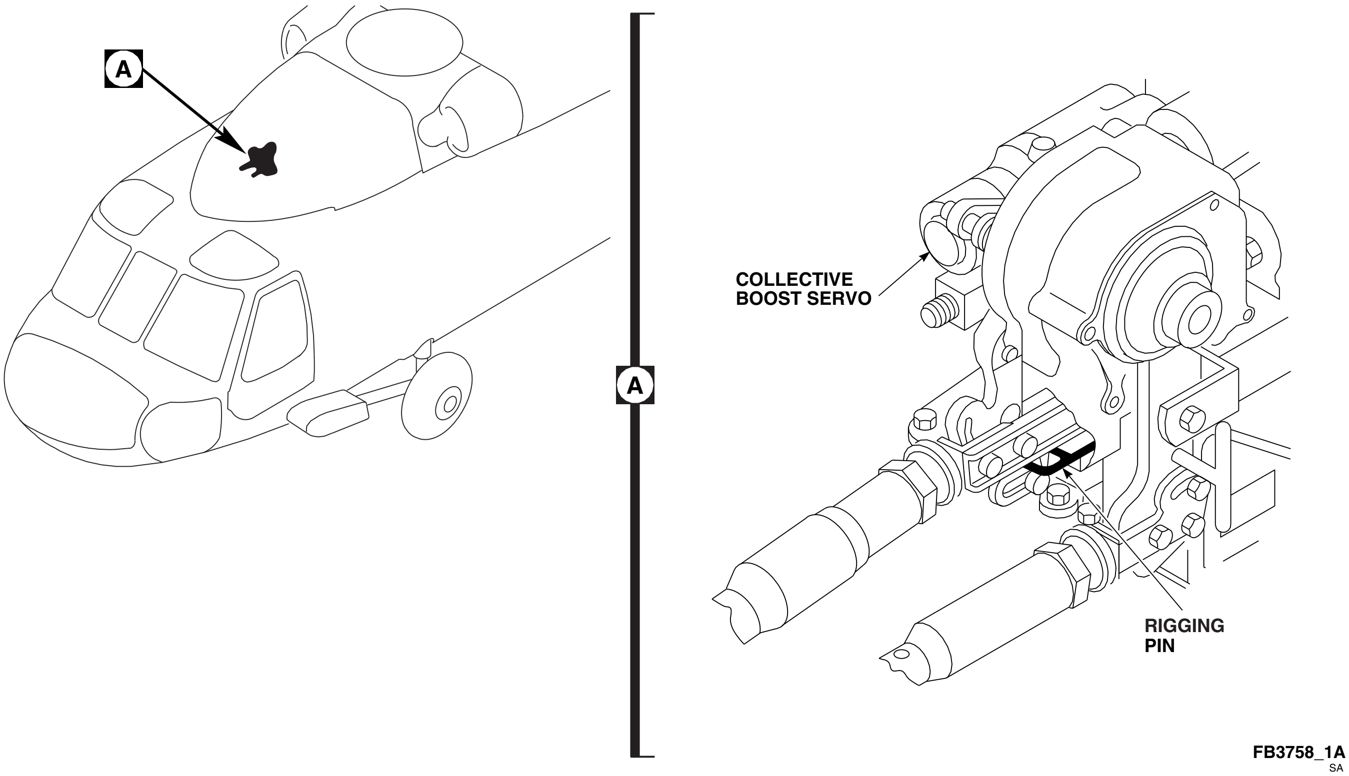


FIGURE 11-4-79-1. RIGGING PIN INSTALLATION

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33°- 38°)	24	72
100° - 125° (39° - 52°)	16	48

f. On upper console, place BACKUP HYD PUMP switch to AUTO.

g. Place pilot's or copilot's collective stick at mid-collective position.

h. Install rigging pin at collective boost servo (Figure 11-4-79-1, Detail A) (PARA 11-4-2).

i. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

j. Disconnect electrical connectors, P317, P318, and P351, if installed, from J317, J318, and J351, if installed, on mixer (Figure 11-4-79-2, Detail A).

k. Install protective caps and plugs on electrical connectors.

l. Remove screws, washers, and nuts securing longitudinal links to lever assembly on mixer.

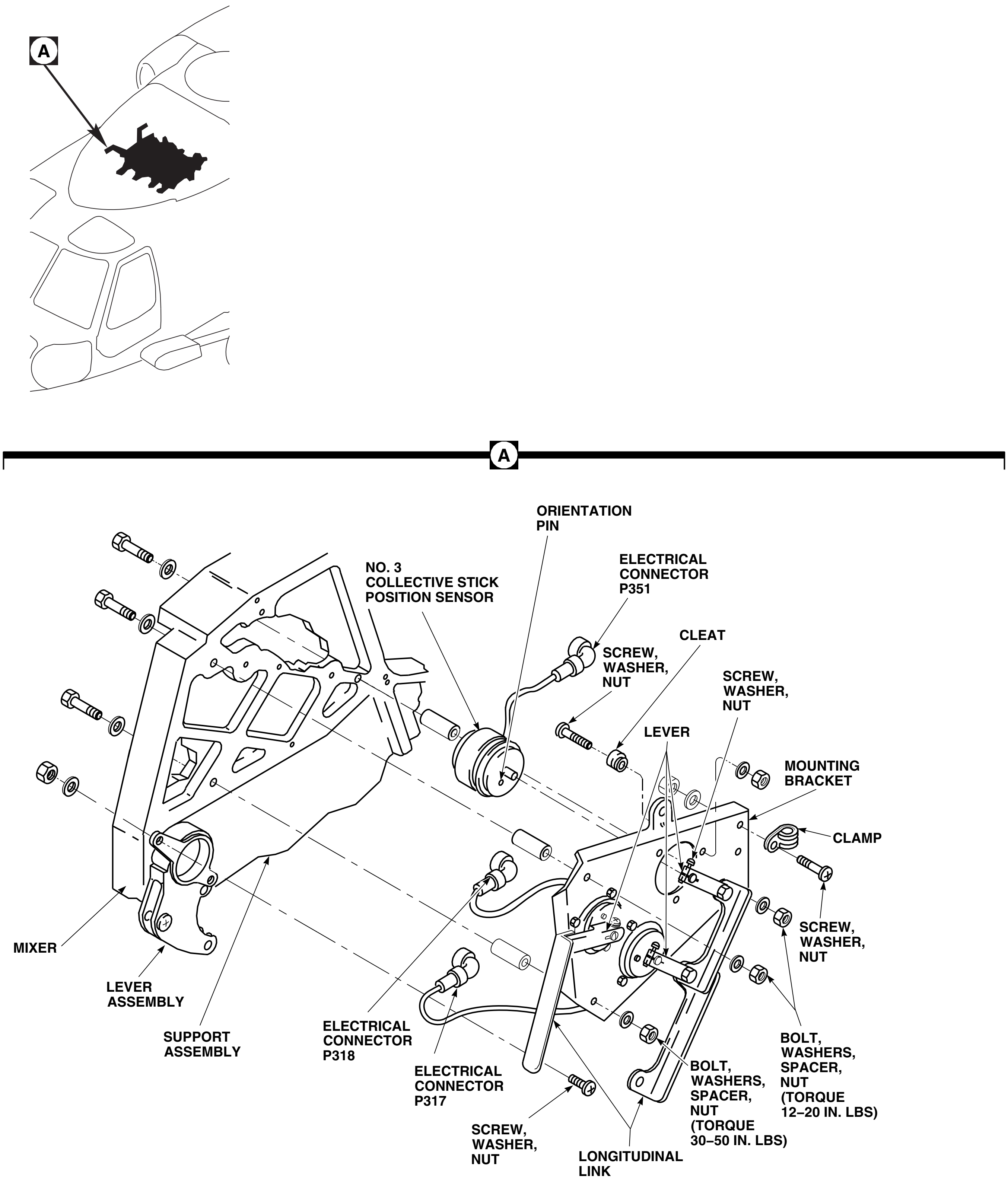
m. Remove bolts, washers, spacers, nuts, and mounting bracket from support assembly.

n. Loosen screw, washer, and nut securing lever to No. 3 collective stick position sensor shaft.

o. Make a pencil line on mounting bracket to mark location of orientation pin.

p. Remove screws, washers, nuts, and clamps securing position sensor cable to mounting bracket.

q. Remove screws, washers, nuts, and cleats securing No. 3 collective stick position sensor and remove sensor.



EFFECTIVITY
IF INSTALLED

FB1503A
SA

FIGURE 11-4-79-2. REPLACE NO. 3 COLLECTIVE STICK POSITION SENSOR

11-4-79.1.2 Install.

a. Position No. 3 collective stick position sensor shaft in mounting bracket and connect lever. Do not tighten screw and nut on lever. Line up position sensor orientation pin with pencil line on mounting bracket and secure sensor with screws, cleats, washers, and nuts (Figure 11-4-79-2, Detail A).

b. Install position sensor cable to mounting bracket with clamps, screws, washers, and nuts.

c. Install mounting bracket to support assembly with bolts, spacers, washers, and nuts. **TORQUE FRONT NUT AND UPPER REAR NUT TO 12 - 20 INCH-POUNDS. TORQUE LOWER REAR NUT TO 30 - 50 INCH-POUNDS.**

d. Install longitudinal links to lever assembly with screws, washers, and nuts.

e. Remove protective caps and plugs from electrical connectors.

f. Inspect and treat electrical connectors (PARA 1-7-20).

g. Connect electrical connectors, P317, P318, and P351, if installed, to J317, J318, and J351, if installed, on mixer (Figure 11-4-79-2, Detail A).

h. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

i. Remove rigging pin from collective boost servo (Figure 11-4-79-1, Detail A).

j. Adjust No. 3 collective stick position sensor (PARA 11-4-79.1.3).

11-4-79.1.3 Adjust.

a. Disconnect electrical connector, P801 from J801, on No. 1 engine.

b. Install protective caps and plugs on electrical connectors.

c. On pilot's and copilot's collective stick grips, place SERVO OFF switch to center position.

d. Turn on all electrical power (PARA 1-6-16).

NOTE

Make sure NO. 1 DC PRI BUS-BACKUP PUMP PWR circuit breaker is pulled out if an external source of hydraulic power is used.

e. Turn on all hydraulic power (PARA 1-6-15).

f. Connect digital multimeter to pin 29 and pin 30 on electrical connector, P801.

WARNING

Injury to personnel and damage to equipment will result if personnel or equipment interfere with movement of stabilator. Stabilator moves up and down during the following procedures. Do not allow personnel or equipment to interfere with stabilator.

g. On STABILATOR CONTROL/AUTO FLT CONT panel, engage BOOST switch.

h. Place and hold collective stick in full down position.

i. Check resistance on digital multimeter. **MULTIMETER SHALL READ 1348 - 1388 OHMS.** If resistance is not 1348 - 1388 ohms, adjust No. 3 collective stick position sensor as follows:

(1) Loosen screw and nut securing lever to No. 3 collective stick position sensor shaft, if required.

(2) Rotate No. 3 collective stick position sensor shaft until digital multimeter reads 1348 - 1388 ohms.

(3) **TIGHTEN SCREW AND NUT SECURING LEVER TO NO. 3 COLLECTIVE**

STICK POSITION SENSOR SHAFT. Make sure multimeter reads 1348 - 1388 ohms after tightening screw and nut.

j. Perform an operational check of engine and engine interface (PARA 4-2-1).

k. Disconnect multimeter.

l. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

m. Remove protective caps and plugs from electrical connectors.

n. Inspect and treat electrical connectors (PARA 1-7-20).

o. Connect electrical connector, P801 to J801, on No. 1 engine.

p. Install main rotor pylon work platform (PARA 2-4-105) and track assembly (PARA 11-4-78).

q. Make sure area is clean and free of foreign material.

r. Close main rotor pylon sliding cover.

11-4-80 MIXER TO FORWARD PRIMARY SERVO CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Protractor
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Universal Propeller Protractor

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-80.1 Replace Mixer to Forward Primary Servo Control Rod.

11-4-80.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

NOTE

Damper rigging block not required if blades are not installed.

c. Install damper rigging block at red blade and tie to damper piston with tiedown strap (PARA 11-4-2).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

d. Install rigging pins at four upper deck torque shafts, two rigging pins at pilot's cyclic stick and one rigging pin at yaw pedals (PARA 11-4-2).

e. Put red blade (or spindle) at 90° position. Make sure blade index marks on swashplate are lined up. Using digital or universal propeller protractor, measure blade angle. Record measurement.

f. Remove rigging pins.

CAUTION

Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to helicopter. Do not remove primary servo input control rods with hydraulic power on.

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. At mixer, remove bolts, washers, nuts, and bushing securing forward control rod to mixer forward output lever (Figure 11-4-80-1, Sheet 1, Detail B).

i. Remove bolts, washers, bearing retention plates, if installed, and nuts securing forward control rod to forward primary servo. Remove control rod.

11-4-80.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed rod, as follows (Figure 11-4-80-1, Sheet 2, Detail C):

(1) Loosen jamnuts at each end of barrel. Slide locking devices away from barrel.

NOTE

Jamnut at tube assembly end has left-hand thread.

(2) Turn barrel to make control rod longer or shorter.

(3) Insert lockwire, Item 193, Appendix D, into two inspection holes in barrel. Wire must not pass through either hole. Do not lockwire now.

(4) **TORQUE BOTH JAMNUTS TO 95 - 105 INCH-POUNDS.**

(5) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

b. Install control rod between primary servo and mixer arm (Figure 11-4-80-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of minimum torque and safety of bolts are critical characteristics.

c. At primary servo end, secure control rod to servo as follows:

(1) On helicopters with primary servos, 70410-02820-044 through 70410-02820-053, install bearing retention plates, bolts, washers, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).

(2) On helicopters with primary servos 70410-02820-054 and subsequent, install bolts, washers, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).

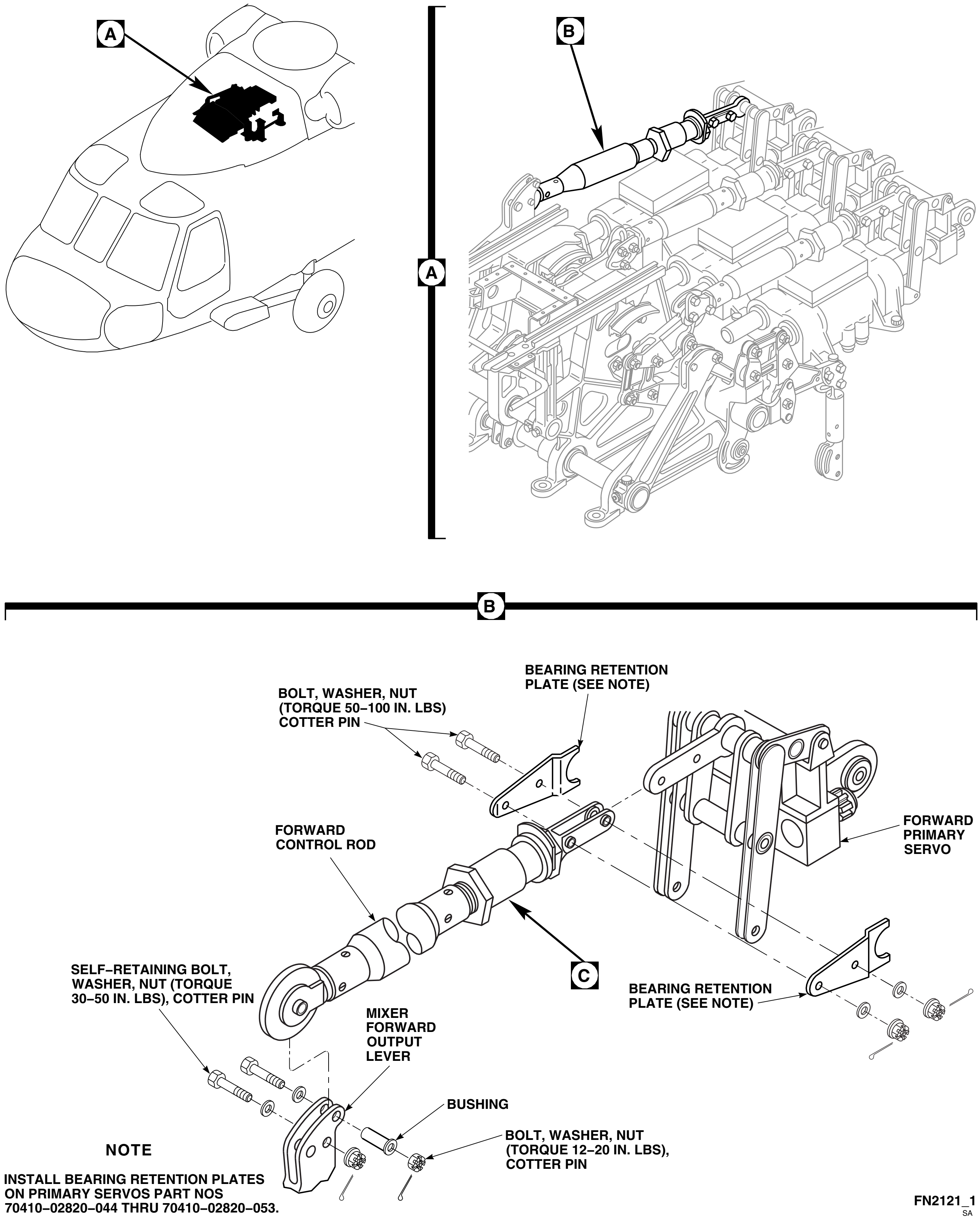


FIGURE 11-4-80-1. REPLACE MIXER TO FORWARD PRIMARY SERVO CONTROL ROD (SHEET 1 OF 2)

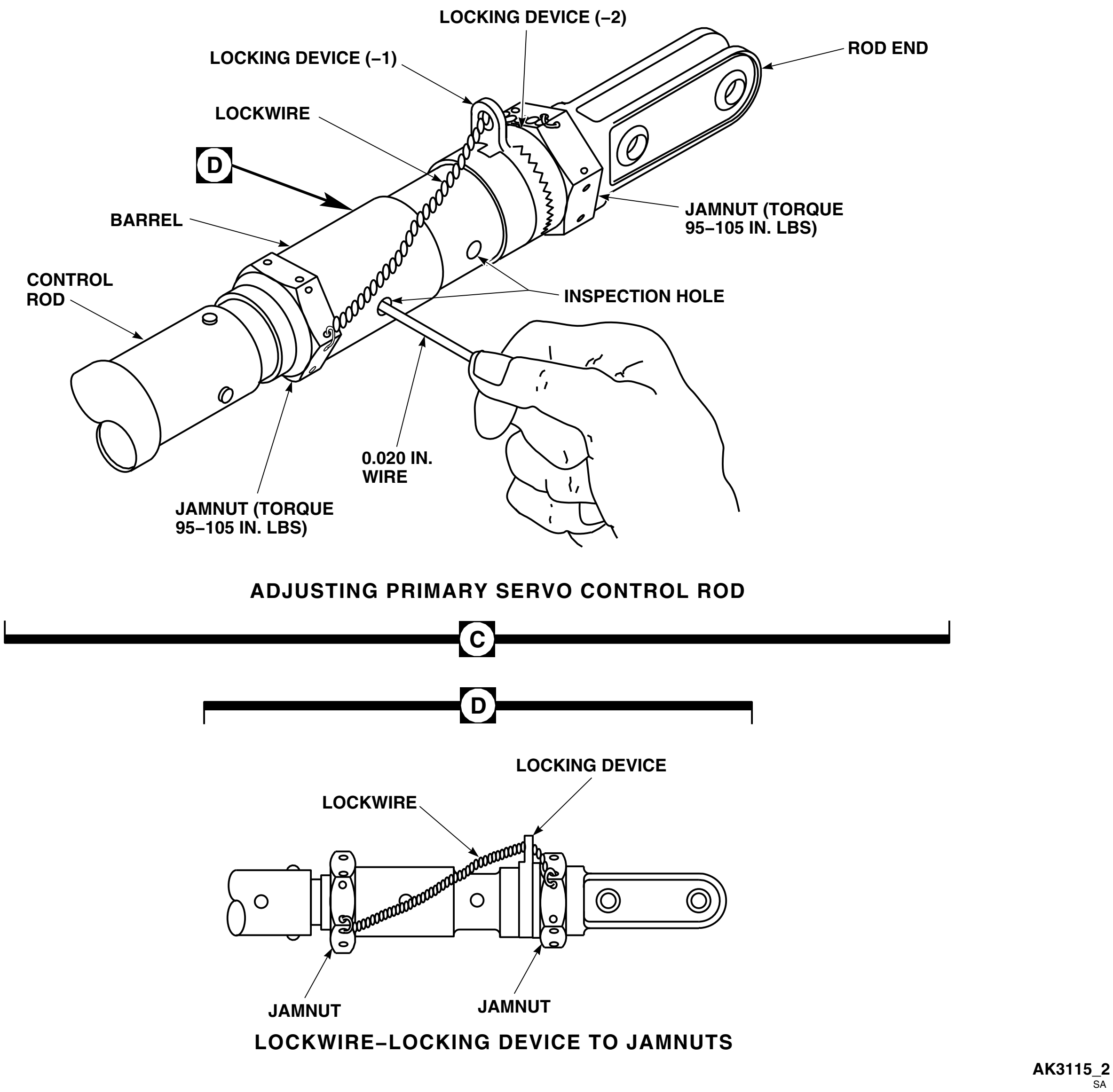


FIGURE 11-4-80-1. REPLACE MIXER TO FORWARD PRIMARY SERVO CONTROL ROD (SHEET 2 OF 2)

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Boltheads can face either direction when installing bolts in forward control rod and mixer forward output lever.

d. At mixer end, install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

e. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

f. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

g. Install bushing in mixer forward output lever, flanged end toward left. Install bolt, washer (under bolthead), and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

h. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

i. Install rigging pins at four upper deck torque shafts, two rigging pins at pilot's cyclic stick and one rigging pin at yaw pedals (PARA 11-4-2).

j. Using digital or universal propeller protractor, make sure that blade angle readings taken on red blade have not changed after servo rod replacement. Adjust main rotor servo input rod as required to reproduce the angle previously measured within ± 0.25°.

k. When control rod length is correct, using lockwire, Item 197, Appendix D, lockwire jamnuts to locking device tab (Figure 11-4-80-1, Sheet 2, Detail D).

l. Perform quality check to ensure jamnuts have been properly lockwired.

m. Remove rigging pins and damper block.

n. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

o. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

p. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

q. Make sure area is clean and free of foreign material.

r. Close main rotor pylon sliding cover.

11-4-81 MIXER TO AFT PRIMARY SERVO CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Protractor
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Universal Propeller Protractor

Materials/Parts

- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D
- Tiedown Strap, Item 338, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-81.1 Replace Mixer to Aft Primary Servo Control Rod.

11-4-81.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

NOTE

Damper rigging blocks not required if blades are not installed.

c. Install damper rigging block at red blade and tie to damper piston with tiedown strap, Item 338, Appendix D.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

d. Install rigging pins at four upper deck torque shafts, two rigging pins at pilot's cyclic stick and one rigging pin at yaw pedals (PARA 11-4-2).

e. Put red blade at the 90° position. Make sure blade index marks on swashplate are lined up. Using digital or universal propeller protractor, measure blade angle. Record measurement.

f. Remove all rigging pins.

CAUTION

Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to helicopter. Do not remove primary servo input control rods with hydraulic power on.

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Remove bolts, washers, countersunk bushings, and nuts securing aft control rod to rear longitudinal output bellcrank and aft primary servo (Figure 11-4-81-1, Sheet 1, Detail B). Remove control rod and bushings.

11-4-81.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed rod as follows (Figure 11-4-81-1, Sheet 2, Detail C):

(1) Loosen jamnuts at each end of barrel. Slide locking devices away from barrel.

NOTE

Jamnut at tube assembly end has left-hand thread.

(2) Turn barrel to make control rod longer or shorter.

(3) Insert lockwire, Item 193, Appendix D, into two inspection holes in barrel. Wire must not pass through either hole. Do not lockwire now.

(4) **TORQUE BOTH JAMNUTS TO 95 - 105 INCH-POUNDS.**

(5) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

b. Install control rod between primary servo and rear longitudinal output bellcrank (Figure 11-4-81-1, Sheet 1, Detail B).

WARNING

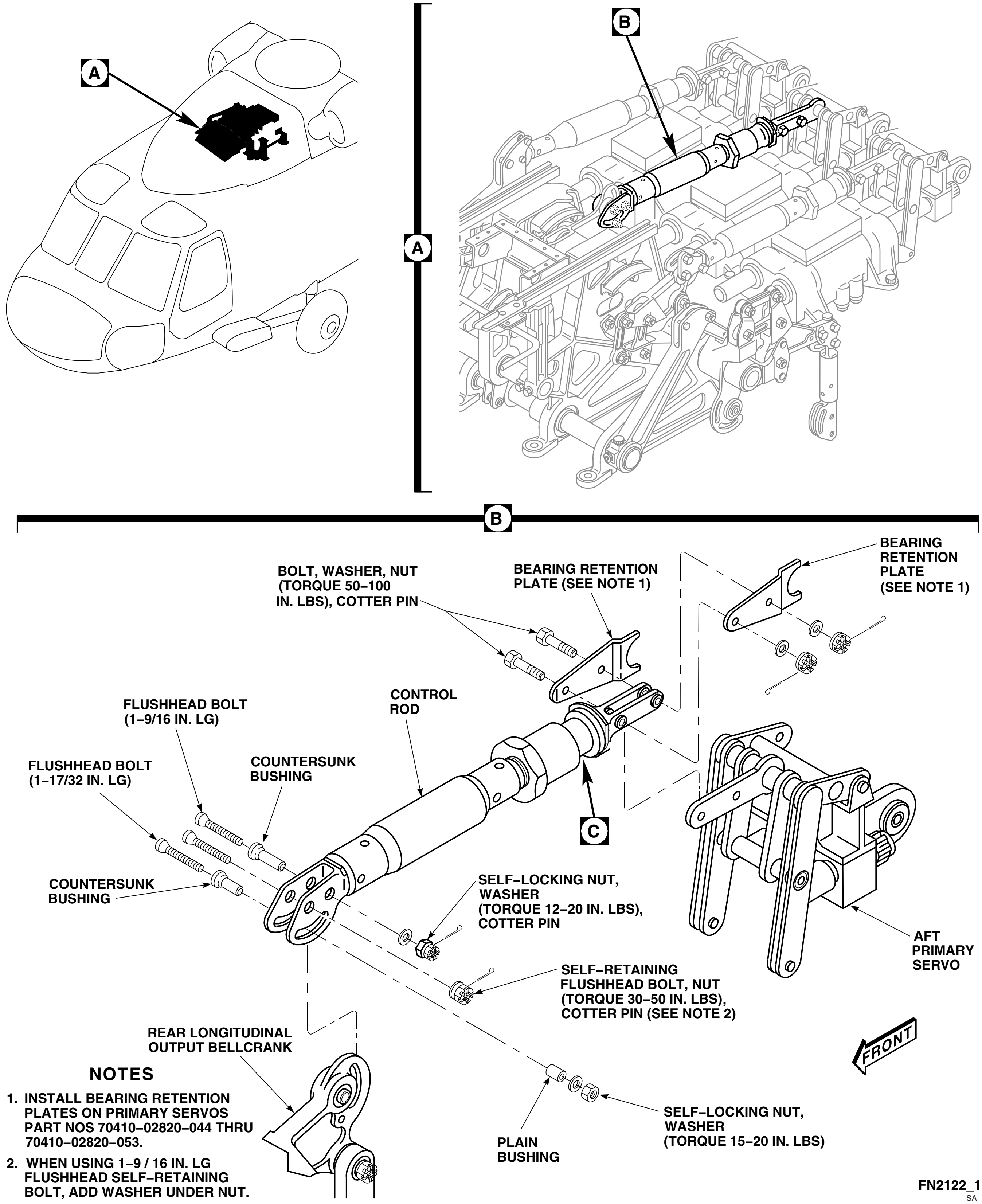
FSCAP

Verification of minimum torque and safety of nut are critical characteristics.

c. At primary servo end, secure control rod to servo as follows:

(1) On helicopters with primary servos, 70410-02820-044 through 70410-02820-053, install bearing retention plates, bolts, washers, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).

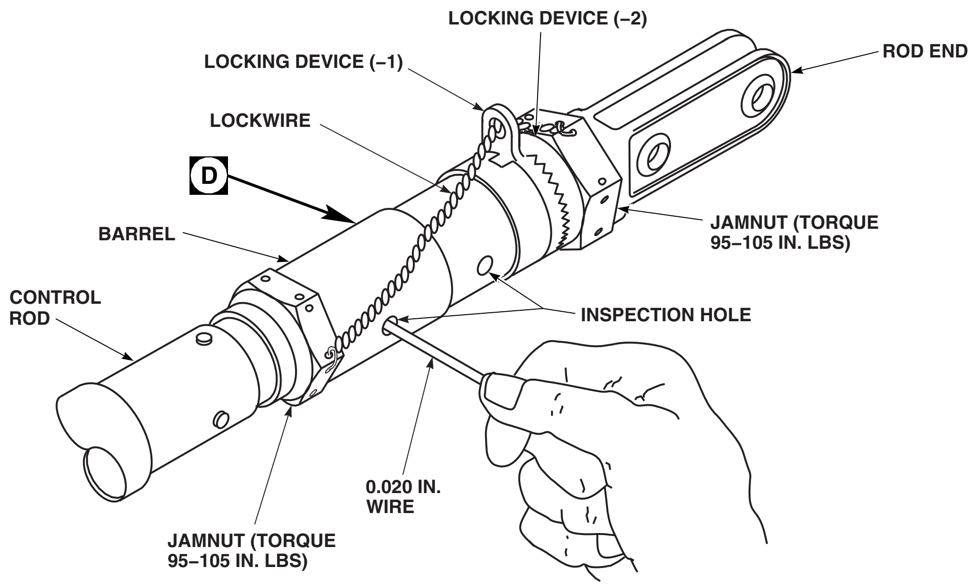
(2) On helicopters with primary servos 70410-02820-054 and subsequent, install bolts, washers, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).



NOTES

1. INSTALL BEARING RETENTION PLATES ON PRIMARY SERVOS PART NOS 70410–02820–044 THRU 70410–02820–053.
2. WHEN USING 1–9 / 16 IN. LG FLUSHHEAD SELF–RETAINING BOLT, ADD WASHER UNDER NUT.

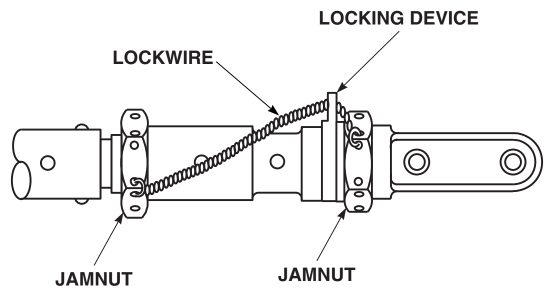
FIGURE 11-4-81-1. REPLACE MIXER TO AFT PRIMARY SERVO CONTROL ROD (SHEET 1 OF 2)



ADJUSTING PRIMARY SERVO CONTROL ROD

C

D



LOCKWIRE–LOCKING DEVICE TO JAMNUTS

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FIGURE 11-4-81-1. REPLACE MIXER TO AFT PRIMARY SERVO CONTROL ROD (SHEET 2 OF 2)

NOTE

All boltheads are toward right side.

d. At output bellcrank end, install flushhead self-retaining bolt and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

e. Install countersunk bushing through rod end right side and slot in rear longitudinal output bellcrank. Install 1 9/16-inch-long flushhead bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

NOTE

When torquing self-locking nuts, check and record amount of torque necessary to thread nut onto bolt before it touches washer. Add this torque to the given torque for the nut.

f. Install plain bushing in slot, left side of rod end. Install countersunk bushing (countersunk side facing out) in slot, right side of rod end. Install 1-17/32-inch-long flushhead bolt, washers, and nut.

g. **TORQUE NUT TO 15 - 20 INCH-POUNDS.**

h. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

i. Install rigging pins at four upper deck torque shafts, two rigging pins at pilot's cyclic stick and one rigging pin at yaw pedals (PARA 11-4-2).

j. Using digital or universal propeller protractor, make sure blade angle readings taken on red blade have not changed after servo rod replacement. Adjust main rotor servo input rod as necessary to reproduce angles previously measured within ± 0.25°.

k. When control rod length is correct, lockwire, Item 197, Appendix D, jamnuts to locking device tab (Figure 11-4-81-1, Sheet 2, Detail D).

l. Perform quality check to ensure jamnuts have been properly lockwired.

m. Remove rigging pins and damper block.

n. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

o. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

p. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

q. Make sure area is clean and free of foreign material.

r. Close main rotor pylon sliding cover.

11-4-82 MIXER TO LATERAL PRIMARY SERVO CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Digital Protractor
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Universal Propeller Protractor

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 180, Appendix D
- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D
- Tiedown Strap, Item 338, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2

11-4-82.1 Replace Mixer to Lateral Primary Servo Control Rod.

11-4-82.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

c. Install rigging pins at four upper deck torque shafts, two rigging pins at pilot's cyclic stick and one rigging pin at yaw pedals (PARA 11-4-2).

NOTE

Damper rigging block not required if blades are not installed.

d. Install damper rigging block on red blade and secure with tiedown strap, Item 338, Appendix D.

e. Align red blade at 0° position. Make sure blade index marks on swashplate are lined up. Using digital or universal propeller protractor, measure blade angle. Record measurement.

f. Remove all rigging pins.

CAUTION

Damage to swashplate will result if primary servo input control rod is removed while hydraulic power is applied to helicopter. Do not remove primary servo input control rods with hydraulic power on.

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Remove bolts, washers, bushings, and nuts securing lateral control rod to mixer lateral lever and lateral primary servo (Figure 11-4-82-1, Sheet 1, Detail B). Remove control rod.

11-4-82.1.2 Install.

a. Adjust replacement control rod to exact length, hole-center to hole-center, as removed rod, as follows (Figure 11-4-82-1, Sheet 2, Detail C):

(1) Loosen jamnuts at each end of barrel. Slide locking devices away from barrel.

NOTE

Jamnut at tube assembly end has left-hand thread.

(2) Turn barrel to make control rod longer or shorter.

(3) Insert lockwire, Item 193, Appendix D, into two inspection holes in barrel. Wire must not pass through either hole. Do not lockwire now.

(4) **TORQUE BOTH JAMNUTS TO 95 - 105 INCH-POUNDS.**

(5) Perform quality check to ensure proper jamnut torque and rod end thread engagement.

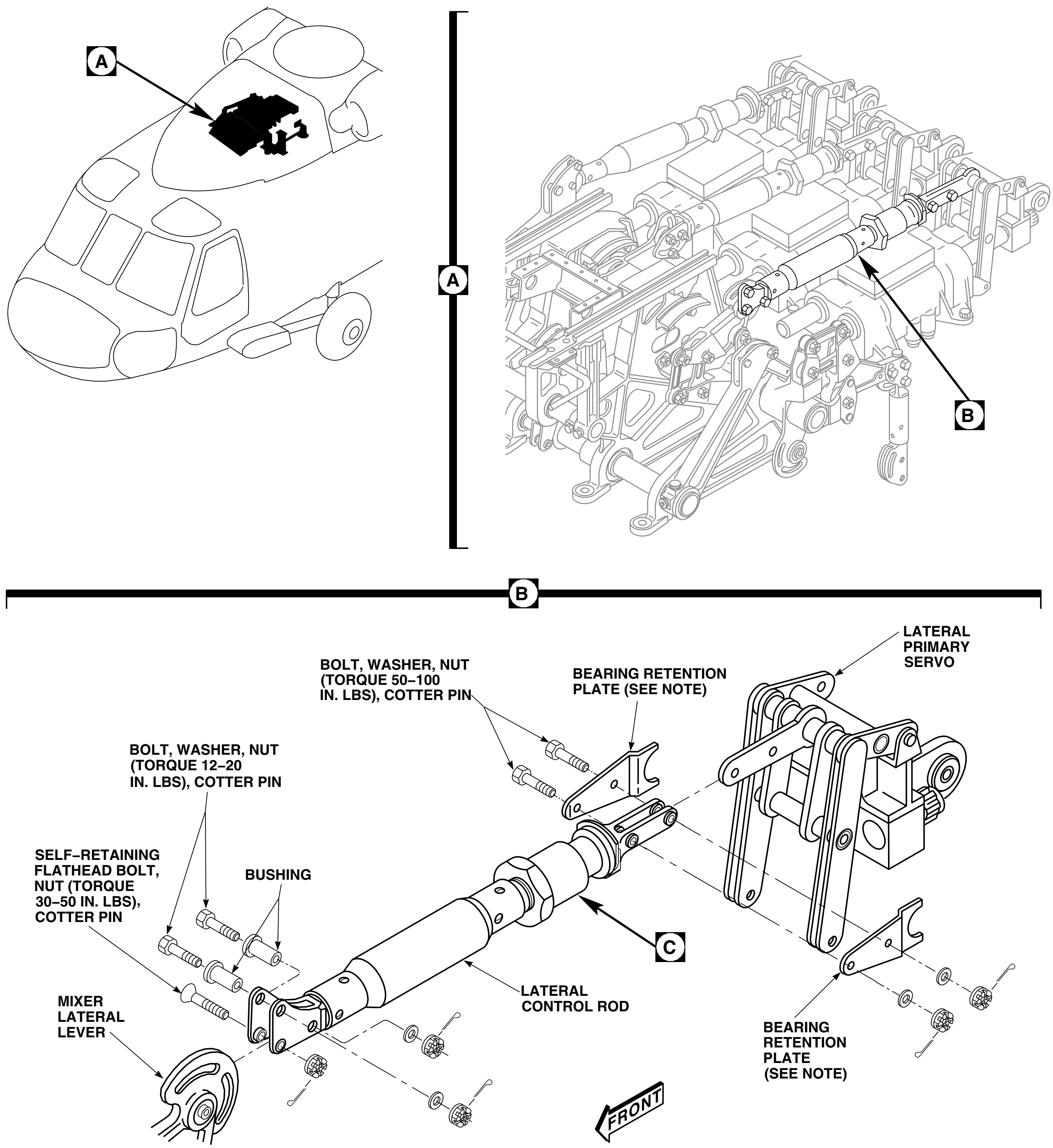
b. Install control rod between primary servo and mixer arm (Figure 11-4-82-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of minimum torque and safety of nut are critical characteristics.

c. At primary servo end, secure control rod to servo as follows:



NOTE

INSTALL BEARING RETENTION PLATES ON PRIMARY SERVOS PART NOS 70410–02820–044 THRU 70410–02820–053.

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FIGURE 11-4-82-1. REPLACE MIXER TO LATERAL PRIMARY SERVO CONTROL ROD (SHEET 1 OF 2)

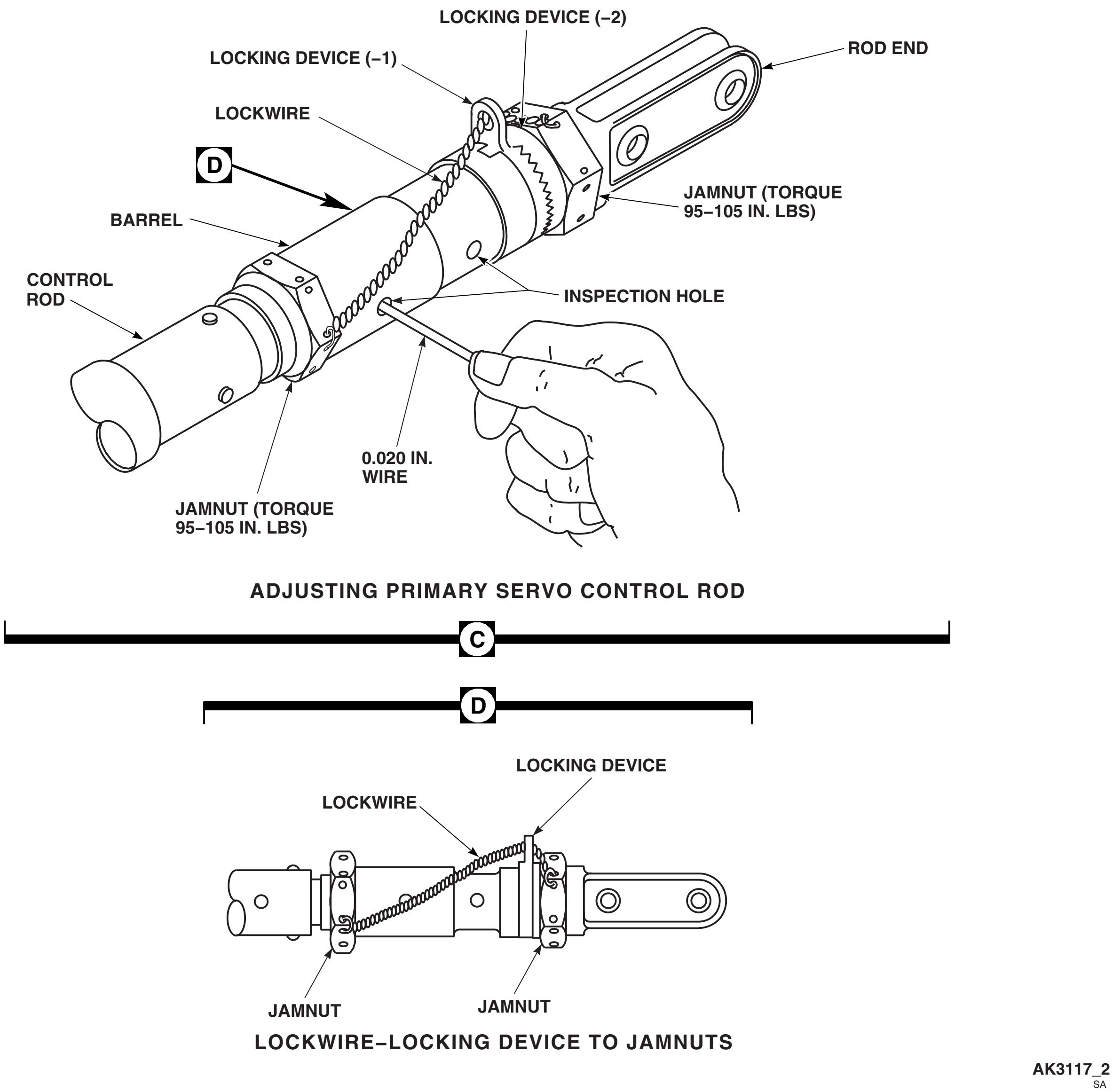


FIGURE 11-4-82-1. REPLACE MIXER TO LATERAL PRIMARY SERVO CONTROL ROD (SHEET 2 OF 2)

(1) On helicopters with primary servos, 70410-02820-044 through 70410-02820-053, install bearing retention plates, bolts, washers, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).

(2) On helicopters with primary servos 70410-02820-054 and subsequent, install bolts, washers, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).

NOTE

All boltheads must face towards right side.

d. At mixer end, install flathead self-retaining bolt and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

e. Install long bushings through rod end and slots in mixer arm. Install bolts, washers, and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pin.

f. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

g. Install rigging pins at four upper deck torque shafts, two rigging pins at pilot's cyclic stick and one rigging pin at yaw pedals (PARA 11-4-2).

h. Using digital or universal propeller protractor, make sure blade angle readings taken on red blade have not changed after servo rod replacement. Adjust main rotor servo input control rod as required to reproduce angles previously measured within ± 0.25°.

i. When control rod length is correct, lockwire, Item 197, Appendix D, jamnuts to locking device tab. Cover lockwire with heat-shrinkable insulation sleeving, Item 180, Appendix D (Figure 11-4-82-1, Sheet 2, Detail C).

j. Perform quality check to ensure jamnuts have been properly lockwired.

k. Remove rigging pins and damper block.

l. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

m. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

n. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

o. Make sure area is clean and free of foreign material.

p. Close main rotor pylon sliding cover.

11-4-83 MIXER TO YAW LEVER PUSHROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-3
- PARA 11-4-6

11-4-83.1 Replace Mixer to Yaw Lever Pushrod.

11-4-83.1.1 Remove.

a. Remove soundproofing panels at STA 311.0 in cabin ceiling (PARA 2-4-69).

b. Open main rotor pylon sliding cover.

c. Remove bolts, washers, bushings, and nuts securing pushrod to mixer lever and yaw lever (Figure 11-4-83-1, Detail B). Remove control rod.

11-4-83.1.2 Install.

a. Install pushrod between mixer lever and yaw lever (Figure 11-4-83-1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

c. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

d. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

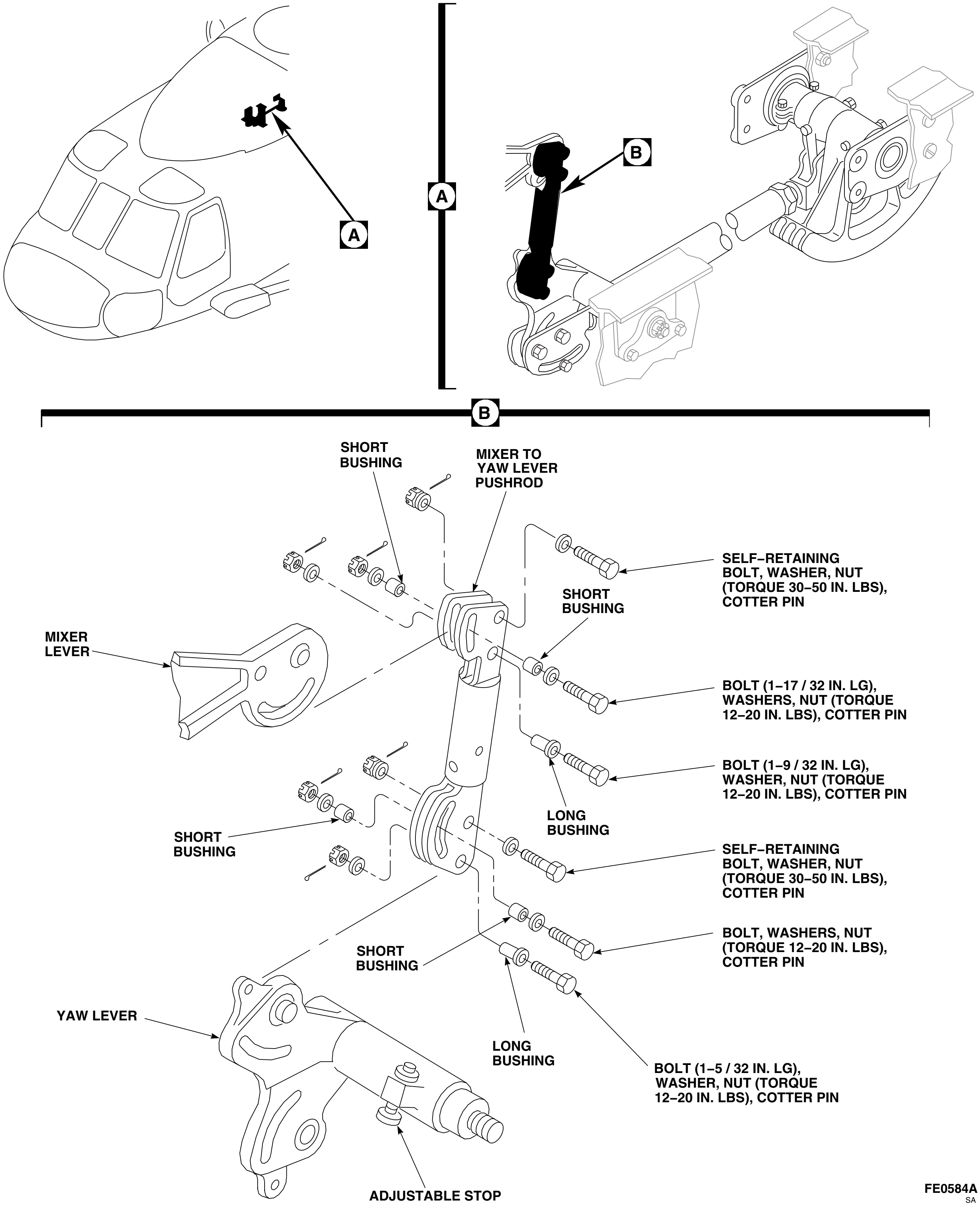


FIGURE 11-4-83-1. REPLACE MIXER TO YAW LEVER PUSHROD

e. At each end, install long bushing through rod ends and slots in levers. Install bolts through bushings and lever. Secure bolts with washers and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pins.

f. Install short bushings in slot on each side of rod end. Install bolts, washers and nuts. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.** Install cotter pins.

g. Check tail rotor servo piston for proper stroke as follows:

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Perform BOOST ON, SAS OFF procedure (PARA 11-4-3).

(3) Lower collective to full down position and push full right pedal. Measure and record gap between servo housing and piston rod yoke.

(4) Raise and hold collective in full up position and push full left pedal. Measure and record gap between servo housing and piston rod yoke.

(5) **DIFFERENCE BETWEEN TWO MEASUREMENTS MUST BE A MINIMUM OF 3.2-INCHES.**

(6) If difference is below limits, check installed control rod for proper tri-pivot bolt installation (PARA 11-3-3). Repeat step g. (2) through step g. (5).

(7) If difference is still below limits, perform tail rotor system rig check (PARA 11-4-6).

(8) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Make sure area is clean and free of foreign material.

i. Install soundproofing panels (PARA 2-4-69).

j. Close and latch main rotor pylon sliding cover.

11-4-84 YAW LEVER TO FORWARD QUADRANT CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 109, Appendix D
- Lockwire, Item 193, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-3
- PARA 11-4-6
- PARA 11-4-105
- PARA 11-4-118

11-4-84.1 Replace Yaw Lever to Forward Quadrant Control Rod.

11-4-84.1.1 Remove.

a. Remove soundproofing panels at STA 311.0 in cabin ceiling (PARA 2-4-69).

b. Open main rotor pylon sliding cover.

c. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

d. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

e. Perform BOOST ON, SAS OFF procedure (PARA 11-4-3).

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

f. Raise collective stick to midposition and install collective boost servo rigging pin (PARA 11-4-2). Lock pilot's collective friction lock.

g. Disconnect tail cone flight control cables from quick-disconnects (PARA 11-4-105).

h. Loosen jamnut on control rod (Figure 11-4-84-1, Sheet 1, Detail B).

NOTE

Keep bolts and bushings together as an assembly. Bushings are of different dimensions.

i. Remove bolts, washers, bushings, and nuts securing control rod to forward quadrant and yaw lever. Remove control rod.

j. Loosen jamnut on right pedal stop on yaw lever. Turn stop fully in until bolthead bottoms on yaw lever (Figure 11-4-84-1, Sheet 1, Detail C).

k. Move tail rotor pedals full right, to move yaw lever aside.

NOTE

Control rod must be taken out in two pieces.

l. Move forward quadrant aft as required and unscrew control rod from rod end. Count and record number of turns it takes to remove control rod end.

m. Remove control rod.

11-4-84.1.2 Install.

a. Move tail rotor pedals to full right. Install replacement control rod, rod end to forward quadrant (Figure 11-4-84-1, Sheet 1, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

b. Install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

c. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

d. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

e. Install long bushing through rod end and slots in lever. Install bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

f. Install short bushing in slot of rod end. Install bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

g. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

h. Apply corrosion preventive compound, Item 109, Appendix D, to threads of control rod.

i. Screw control rod onto rod end same number of turns as was required to remove control rod. Do not tighten control rod jamnut now.

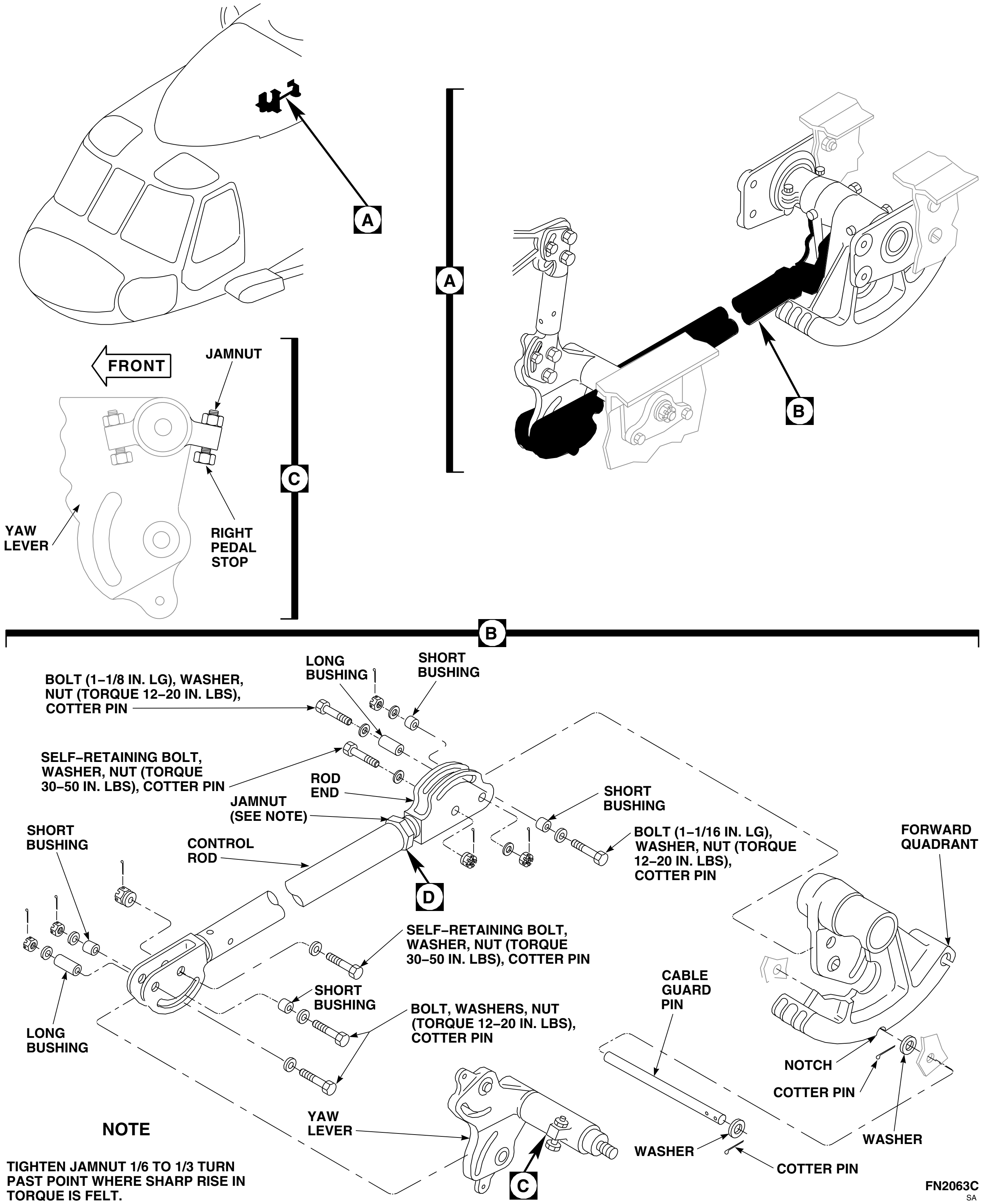


FIGURE 11-4-84-1. REPLACE YAW LEVER TO FORWARD QUADRANT CONTROL ROD (SHEET 1 OF 2)

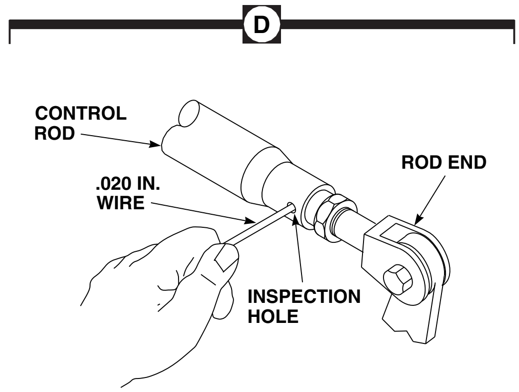


FIGURE 11-4-84-1. REPLACE YAW LEVER TO FORWARD QUADRANT CONTROL ROD (SHEET 2 OF 2)

j. Temporarily connect front part of control rod to yaw lever with self-retaining bolt, washer, and nut. Do not torque now.

CAUTION

Damage to flight control components will result if hydraulic power is shut down while rigging pins are installed. Make sure all rigging pins are removed before shutting down hydraulic power.

k. Install yaw boost servo rigging pin (PARA 11-4-2).

l. Check to see if alignment marks (rigging notch) on outside diameter of forward quadrant lines up with center of cable guard pin. If not, readjust rod until marks line up.

m. Insert lockwire, Item 193, Appendix D, into inspection hole in adjustable rod end. Adjusted thread end must stop wire from passing through inspection hole.

n. After marks line up, install self-retaining bolt and washers (under bolthead) (PARA 11-4-1).

o. Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

p. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

q. Install long bushing through rod end and slots in lever. Install bolt, washer, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

r. Install short bushing in slot of rod end. Install bolt, washer and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

s. **TIGHTEN JAMNUT ON CONTROL ROD 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

t. Adjustable end of control rods are to be inspected for minimum thread engagement. Check that adjustable rod end is threaded beyond inspection hole such that a 0.20-inch diameter wire cannot be inserted through inspection hole (Figure 11-4-84-1, Sheet 2 Detail D).

u. Connect tail cone flight control cables to quick-disconnects (PARA 11-4-105).

v. Remove collective and yaw boost servo rigging pins. Unlock pilot's friction device.

w. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

x. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

y. Inspect installed control rod for proper tri-pivot bolt installation (PARA 11-3-3).

z. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

aa. Perform tail rotor system rig check (PARA 11-4-6).

ab. Perform operational check of tail rotor flight control system (PARA 11-2-3).

ac. Make sure area is clean and free of foreign material.

ad. Install soundproofing panels at STA 311.0 in cabin ceiling (PARA 2-4-69).

ae. Close main rotor pylon sliding cover.

11-4-85 YAW LEVER AND SUPPORT (STA 301.0).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Feeler Gage
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Lockwire, Item 194, Appendix D

References

- Appendix D
- PARA 11-4-83
- PARA 11-4-84

11-4-85.1 Replace Yaw Lever and Support (STA 301.0).

11-4-85.1.1 Remove.

a. Remove mixer to yaw lever pushrod (PARA 11-4-83).

b. Remove yaw lever to forward quadrant control rod (PARA 11-4-84).

c. Remove bolts and washers securing retaining plate to fitting. Carefully remove pin from fitting and yaw lever, being careful not to lose washer(s) between fitting and yaw lever (Figure 11-4-85-1, Detail B).

d. Remove nut and large washer from yaw lever stud. Remove bolts, washers, and nuts securing support to airframe. Remove support.

e. Slide support from yaw lever and remove lever from between fitting and airframe.

11-4-85.1.2 Inspect.

Using dial indicator, inspect bearings in lever. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

11-4-85.1.3 Install.

NOTE

Install longest bolts in two bottom holes.

a. Slide replacement yaw lever up between beam fitting and airframe with long hub outboard. Slide support on lever hub and bolt support to airframe with bolts, washers, and nuts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS** (Figure 11-4-85-1, Detail B).

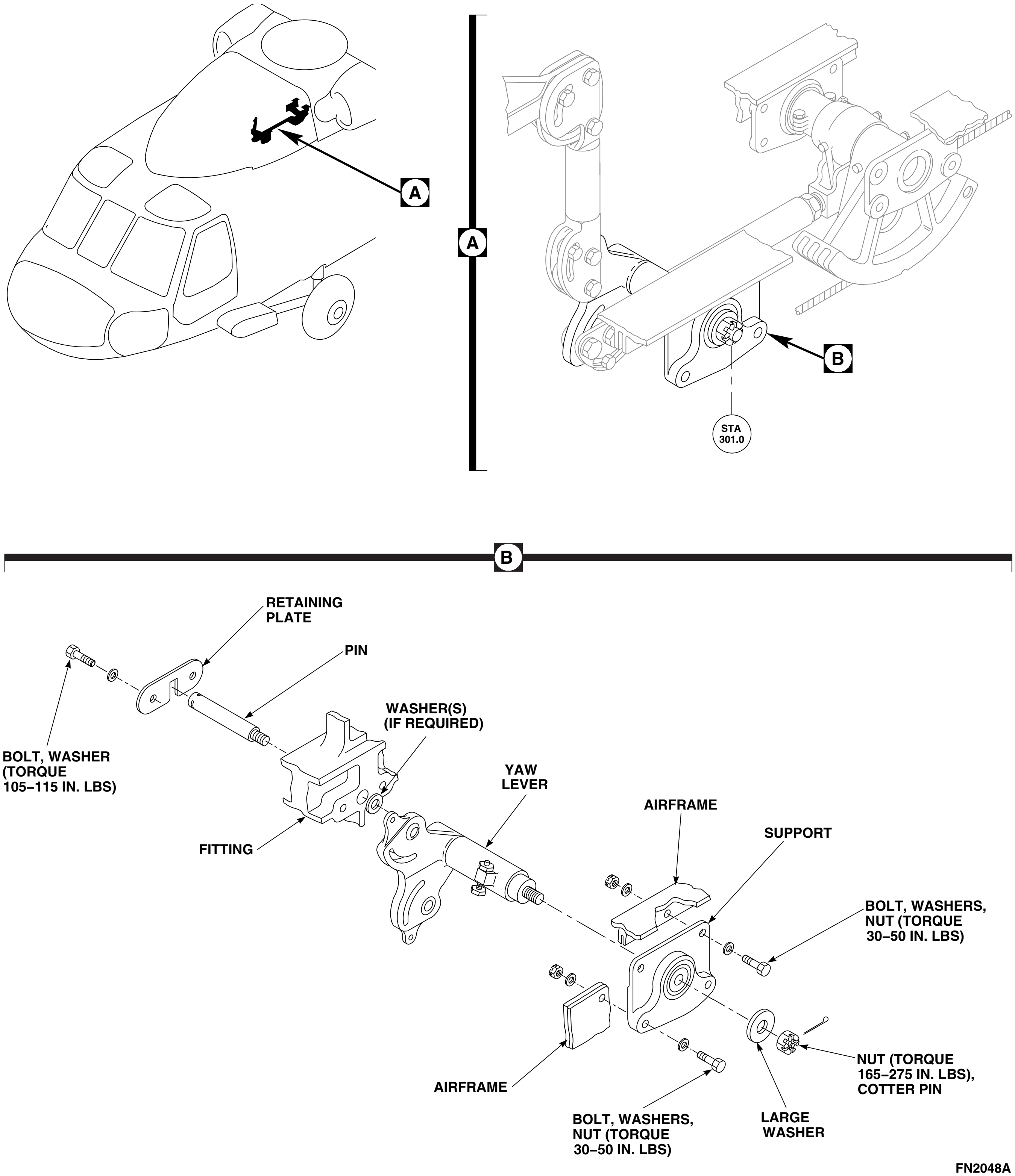


FIGURE 11-4-85-1. REPLACE YAW LEVER AND SUPPORT (STA 301.0)

WARNING

FSCAP

Verification of minimum gap between bearing and washer stackup is a critical characteristic.

b. Install pin through fitting and into bearing in short hub of yaw lever. Using feeler gage, measure gap between inner race of bearing and shoulder of pin. **MINIMUM GAP BETWEEN BEARING AND WASHER MUST BE 0.010-INCH.** If gap is more than 0.050-inch, perform step c. If gap is less than 0.050-inch, proceed to step d.

NOTE

A maximum of three washers may be used.

c. If shimming is required, slide pin far enough out of fitting and yaw lever to install washers.

WARNING

FSCAP

Verification of proper installation, torque, and safetying of retention plate and attaching bolts are critical characteristics.

d. Install retaining plate over pin head using bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Using lockwire, Item 194, Appendix D, lockwire bolts.

WARNING

FSCAP

Verification of torque and cotter pin installation are critical characteristics.

e. Install large washer and nut on lever stud. **TORQUE NUT TO 165 - 275 INCH-POUNDS.** Install cotter pin.

f. Install yaw lever to forward quadrant control rod (PARA 11-4-84).

g. Install mixer to yaw lever pushrod (PARA 11-4-83).

11-4-86 FORWARD QUADRANT AND SUPPORTS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Torque Wrench, 0 - 30 in. lbs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 11-4-6
- PARA 11-4-84
- PARA 11-4-105
- PARA 11-4-109
- PARA 11-4-118

11-4-86.1 Replace Forward Quadrant and Supports.

11-4-86.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove yaw lever to forward quadrant control rod (PARA 11-4-84).

c. Remove inboard washer. Remove cable guard pin from airframe beam (Figure 11-4-86-1, Sheet 1, Detail B). Slide cables and collars from forward quadrant. Remove collars from cable.

d. Remove bolts, washers, and nuts securing split spacers to quadrant shaft (Figure 11-4-86-1, Sheet 2, Detail C).

e. Remove screws, washers, and nuts securing inboard and outboard bearing supports to airframe beams. Remove bearing supports and forward quadrant from between airframe beams.

NOTE

Forward quadrant is a matched assembly. Lever, cable quadrant, and shaft is replaced as an assembled unit.

f. Remove laminated shim from quadrant shaft.

11-4-86.1.2 Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Install forward quadrant between airframe beams with quadrant outboard. Slide bearing supports on quadrant shaft with mounting bosses facing out (Figure 11-4-86-1, Sheet 2, Detail C).

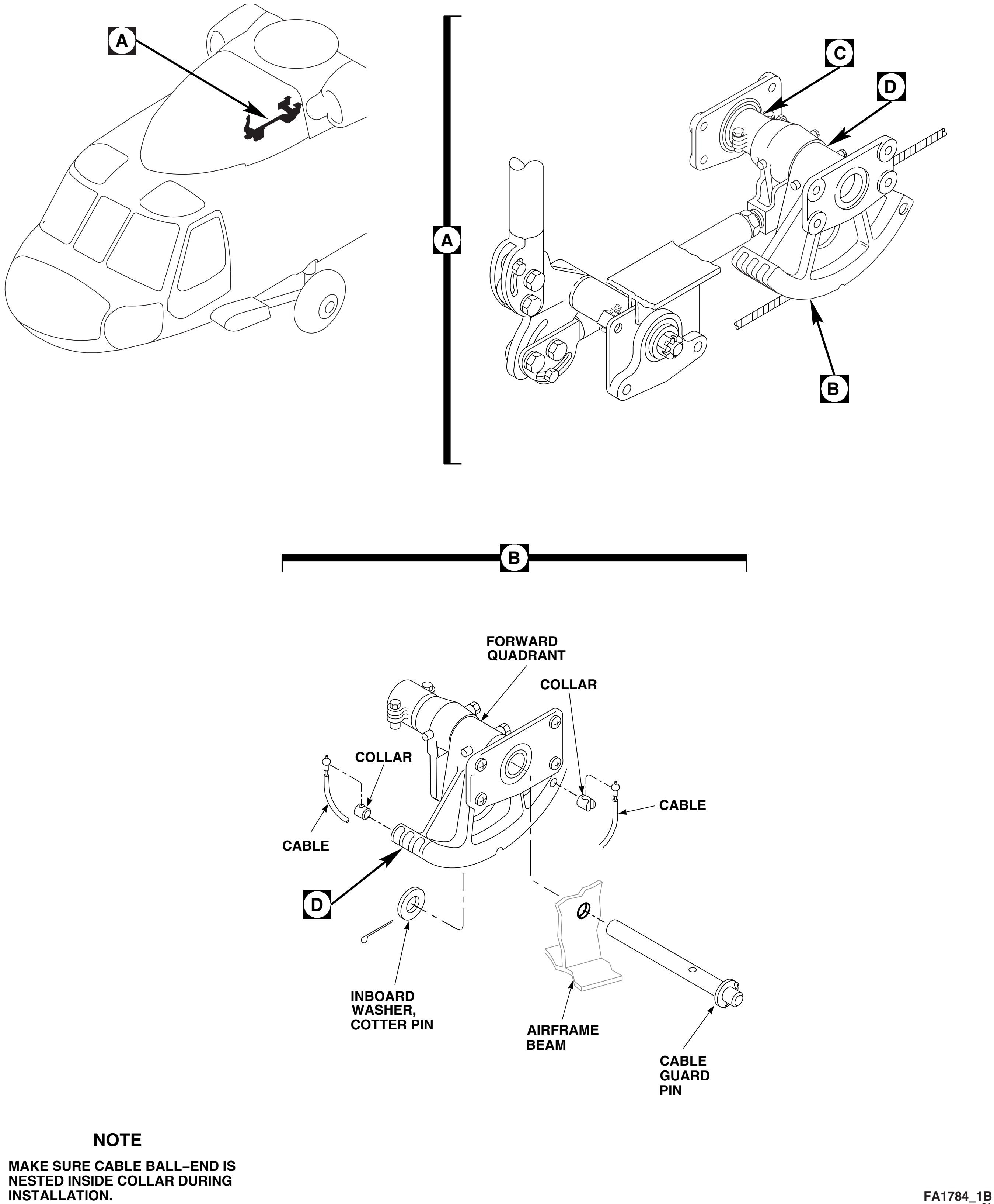


FIGURE 11-4-86-1. REPLACE FORWARD QUADRANT AND SUPPORTS (SHEET 1 OF 2)

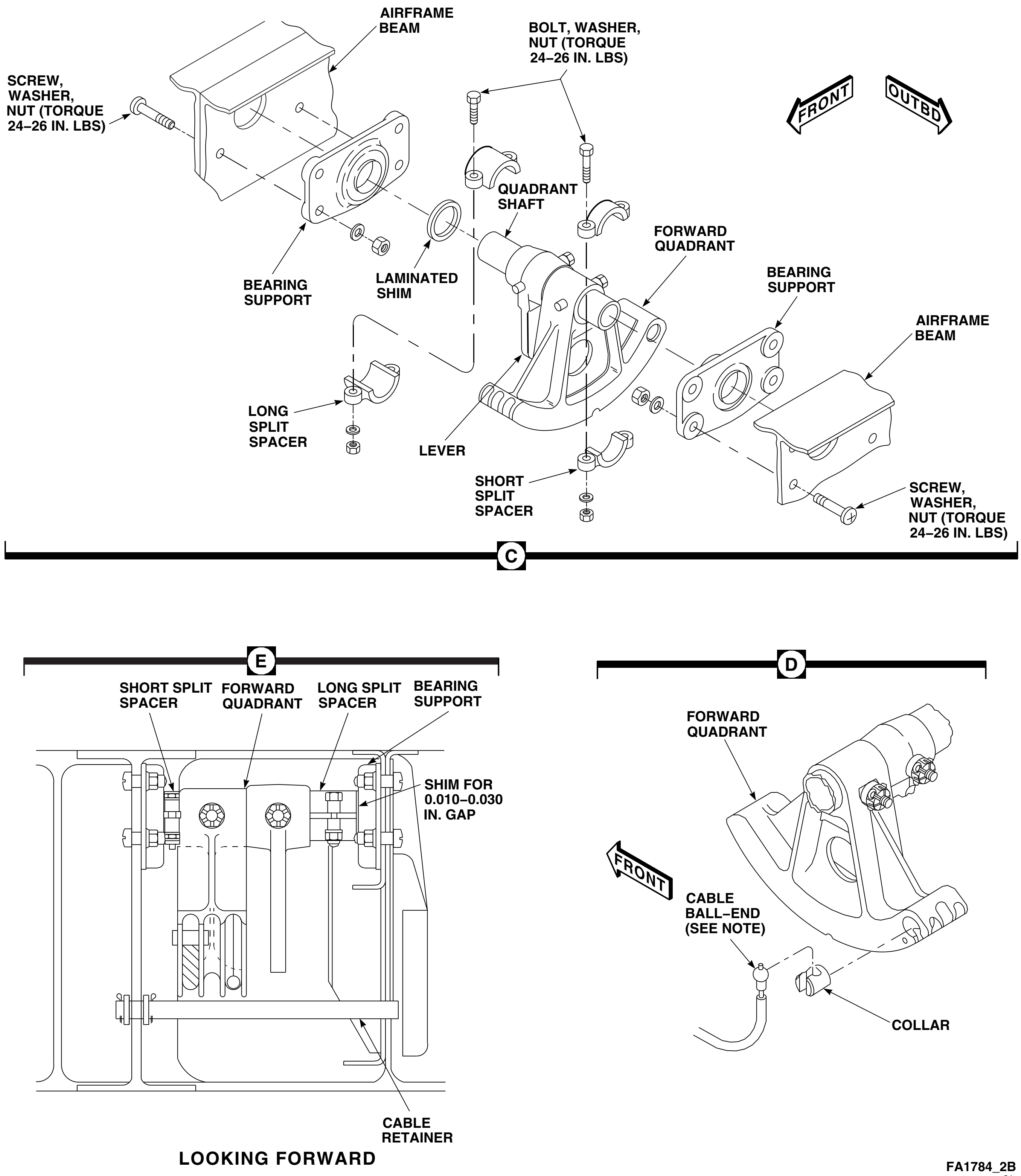


FIGURE 11-4-86-1. REPLACE FORWARD QUADRANT AND SUPPORTS (SHEET 2 OF 2)

c. Install bearing supports to airframe beams with screws, washers, and nuts (nuts on inside). **TORQUE NUTS TO 24 - 26 INCH-POUNDS.**

d. Install short split spacer on shaft between forward quadrant and outboard bearing support. Install long split spacer on shaft. Do not torque nuts at this time.

e. Push long split spacer, shaft, lever, and quadrant toward outboard bearing support. Using feeler gage, measure gap between inboard bearing support and long split spacer.

f. Remove split spacers, remove bolts, washers, and nuts securing bearing supports to airframe beams, and remove forward quadrant from between airframe beams. Remove inboard bearing support from shaft.

g. Peel laminated shim to a thickness of 0.010 - 0.030-inch less than measurement of gap between inboard bearing support and long split spacer. Allow 0.010 - 0.030-inch end play between split spacers and bearing supports.

h. Install laminated shim and inboard bearing support on quadrant shaft. Install forward quadrant and bearing supports between airframe beams using screws, washers, and nuts. **TORQUE NUTS TO 24 - 26 INCH-POUNDS.**

i. Install long split spacer between laminated shim and lever. Push spacer toward lever and outboard support. **TORQUE NUTS ON BOTH SPLIT SPACERS TO 24 - 26 INCH-POUNDS.** Using feeler gage, measure for 0.010 - 0.030-inch end play (Figure 11-4-86-1, Sheet 2, Detail E). Readjust laminated shim if required.

j. Install yaw lever to forward quadrant control rod (PARA 11-4-84).

NOTE

Make sure cable ball-end is nested inside collar during installation (Figure 11-4-86-1, Sheet 2, Detail D).

k. Install cables in collars. Install collars and cable in forward quadrant. Make sure cables are in quadrant grooves, then install cable guard pin. Put washer on cable guard pin inside of outboard beam as cable guard pin is being installed. Install cotter pin (Figure 11-4-86-1, Sheet 1, Detail B).

l. Connect tail cone flight control cables to quick-disconnects (PARA 11-4-105).

m. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118) or remove lockout blocks (PARA 11-4-109).

n. Perform tail rotor system rig check (PARA 11-4-6).

o. Make sure all rigging pins are removed and electrical and hydraulic power is off.

p. Make sure area is clean and free of foreign material.

q. Install soundproofing panels in cabin ceiling at STA 311.0 (PARA 2-4-69).

11-4-86.1.3 Inspect.

a. Using feeler gage, measure 0.010 - 0.030-inch end play between split spacers and bearing support (Figure 11-4-86-1, Sheet 2, Detail E).

b. Check that cables and collars have been properly installed.

c. Check for visual clearance between cable, cable guard pin, and forward quadrant.

11-4-87 PRIMARY SERVO.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Digital Protractor
- Feeler Gage
- Rigging Kit, 70700-20389-042
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Universal Propeller Protractor
- Vise, Soft-Jawed

Materials/Parts

- Alcohol, Ethyl, Item 71, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 180, Appendix D
- Lockwire, Item 193, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs
- Protective Covering
- Tiedown Strap, Item 338, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-101
- PARA 2-4-105
- PARA 2-6-5
- PARA 2-6-10
- PARA 5-4-22
- PARA 7-2-1
- PARA 7-3-1
- PARA 11-3-1
- PARA 11-3-24
- PARA 11-4-3
- PARA 11-4-4
- PARA 11-4-80
- PARA 11-4-81
- PARA 11-4-82
- PARA 11-4-90

• PARA 11-4-91

• PARA 11-4-92

11-4-87.1 Replace Primary Servo.

NOTE

- Replacement of primary servos is the same, except as noted.
- A main rotor rig check will be required if flight control rods are adjusted.
- If other maintenance was performed on flight control system in conjunction with primary servo replacement, a main rotor complete rig is required.

11-4-87.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. If main rotor blades are installed, turn main rotor head to allow overhead clearance and make it easier to remove primary servo.

d. Disconnect electrical connectors from primary servo being replaced as follows (Figure 11-4-87-1, Detail A):

(1) On forward primary servo, disconnect electrical connector, P426, from 1st stage pressure switch and electrical connector, P429, from 2nd stage pressure switch.

(2) On aft primary servo, disconnect electrical connector, P425, from 1st stage pressure switch and electrical connector, P428, from 2nd stage pressure switch.

(3) On lateral primary servo, disconnect electrical connector, P424, from 1st stage pressure switch and electrical connector, P427, from 2nd stage pressure switch.

(4) Install protective caps and plugs on electrical connectors.

e. Before removing primary servo control rods, inspect spherical bearing connections for radial and axial wear limits as follows:

(1) Move control rods up and down, measuring radial play on surface of connecting rods, using dial indicator. **MAXIMUM RADIAL PLAY MUST NOT BE MORE THAN 0.020-INCH.** If play is more, replace part.

(2) Measure combined total radial play from servos to swashplate links (PARA 11-3-24).

(3) Move control rods from side-to-side, measuring axial play on surface of connecting pushrods, using dial indicator. **MAXIMUM AXIAL PLAY MUST NOT BE MORE THAN 0.020-INCH.** If play is more, replace part.

f. Disconnect forward control rod (PARA 11-4-90), aft control rod (PARA 11-4-91), or lateral control rod (PARA 11-4-92) from applicable primary servo.

g. Remove mixer to forward primary servo control rod (PARA 11-4-80), mixer to aft primary servo control rod (PARA 11-4-81), or mixer to lateral primary servo control rod (PARA 11-4-82) from applicable primary servo.

h. Remove bolts, countersunk washers, and washers securing primary servo to primary servo manifold. Remove barrel-nuts and retainers from manifold.

i. Lift primary servo straight up and carefully remove it from primary servo manifold.

WARNING

FSCAP

The primary servo beam rail fitting upper surface areas are critical areas that must not be damaged. Make sure to install protective covering over fitting after removal of primary servo.

j. Install protective covering over servo beam rail fitting.

k. If replacement servo is not to be immediately installed, cover primary servo manifold quick-disconnect couplings (female end).

NOTE

Inspection of primary servo beam rail fittings is required to detect possible disbonds between wear strips and primary servo beam rails. Wear strip disbonds can lead to fretting of fitting attachment surfaces. Make sure inspection of primary servo beam rail fittings is performed every time primary servos are removed. Wear strips include both the strap on the mating surface with the primary servo and the liner in the barrel-nut hole.

l. Inspect primary servo beam rail fittings for wear strip disbond (PARA 2-6-10).

11-4-87.1.2 Inspect.

a. On servos, 70410-02820-044 through 70410-02820-053, check servo input link bearing spacers for grooves or gouges. **DAMAGE SHALL NOT BE MORE THAN 0.0626-INCH DEEP.** If damaged beyond limit, replace servo (Figure 11-4-87-1, Detail A).

b. Inspect for evidence of scraper ring displacement. If ring is displaced, check for seal leakage (PARA 7-3-1). If leakage criteria is not met, replace servo.

c. If scraper ring is completely detached from housing, remove scraper ring. Check for seal leakage (PARA 7-3-1). If leakage criteria is not met, replace servo.

NOTE

If all the criteria below is met and the collar is tight on the pin and the pin has no axial movement, the pin may or may not rotate.

d. Inspect pin-rivet connections as follows:

(1) Inspect pin rivets for looseness, using finger pressure. Collar may not rotate or have axial or radial play, relative to the pin. If loose pin-rivets are found, replace servo.

(2) Inspect pin-rivets for axial movement. **NONE ALLOWED.** If pin-rivets have axial play, replace servo.

(3) Inspect for gaps under head of pin-rivets with feeler gage.

(a) **GAPS WHICH PERMIT THE INSERTION OF A 0.004-INCH FEELER GAGE TO SHANK OF PIN ARE ACCEPTABLE, PROVIDED NO MORE THAN 40% OF CIRCUMFERENCE HAS A GAP.** If beyond limits, replace servo.

(b) **GAPS WHICH PERMIT INSERTION OF 0.005-INCH FEELER GAGE TO SHANK OF PIN ARE NOT ACCEPTABLE.** If beyond limits, replace servo.

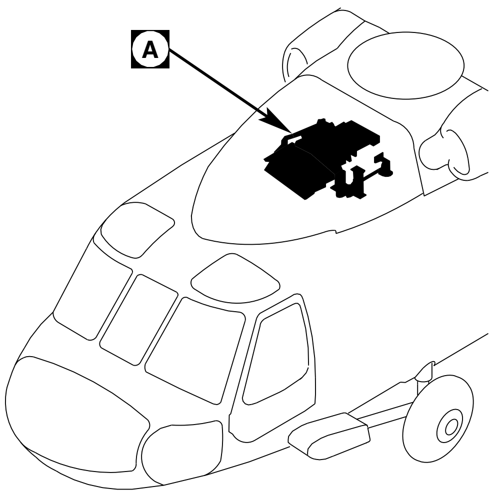
(4) Inspect pin-rivet collars for cracks. **NONE ALLOWED.** If cracked collar is found, replace servo.

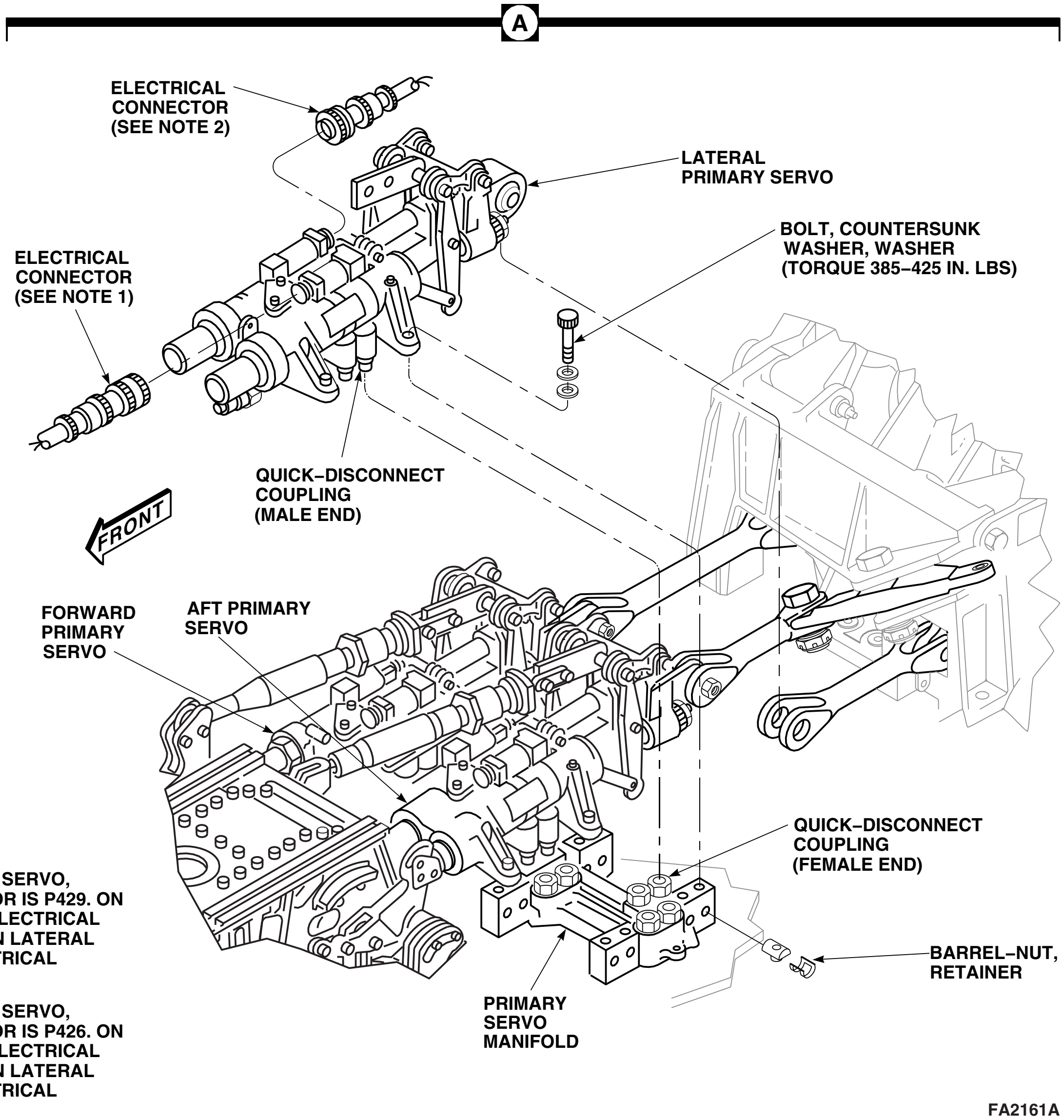
e. Check breakaway torque of bolts and barrel-nuts as follows:

NOTE

Make sure that only square barrel-nuts are used.

(1) Place barrel-nut in soft-jawed vise.





NOTES

1. ON FORWARD PRIMARY SERVO, ELECTRICAL CONNECTOR IS P429. ON AFT PRIMARY SERVO, ELECTRICAL CONNECTOR IS P428. ON LATERAL PRIMARY SERVO, ELECTRICAL CONNECTOR IS P427.
2. ON FORWARD PRIMARY SERVO, ELECTRICAL CONNECTOR IS P426. ON AFT PRIMARY SERVO, ELECTRICAL CONNECTOR IS P425. ON LATERAL PRIMARY SERVO, ELECTRICAL CONNECTOR IS P424.

FIGURE 11-4-87-1. REPLACE PRIMARY SERVO

(2) Thread bolt into barrel-nut until two threads are showing beyond nut.

(3) Using torque wrench, back out bolt. **MINIMUM BREAKAWAY TORQUE IS 9.5 INCH-POUNDS.**

(a) If breakaway torque is less than 9.5 inch-pounds, replace bolt. Repeat steps e. (1) through e. (3).

(b) If breakaway torque is still less than 9.5 inch-pounds, replace barrel-nut. Repeat steps e. (1) through e. (3).

11-4-87.1.3 Install.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Uncover primary servo manifold quick-disconnect couplings. Make sure they are clean. Wipe with clean machinery towel, Item 211, Appendix D, and flush with ethyl alcohol, Item 71, Appendix D.

WARNING

FSCAP

The primary servo beam rail fitting upper surface areas are critical areas that must not be damaged. Make sure to use care when installing primary servo.

c. Remove protective covering from servo beam rail fitting.

d. Line up primary servo quick-disconnect couplings, and push servo down until four feet contact servo beam fitting. Hold this position until bolts are torqued down.

CAUTION

Damage to barrel-nut fitting will occur if barrel-nut is incorrectly installed. To prevent damage make sure barrel-nut is installed with barrel (round side) towards bolthead.

NOTE

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters ″EN″ or ″SPS″ stamped on and end (additional marking may also include ″LH2577″).

e. Install barrel-nuts and retainers in servo beam fitting.

NOTE

Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

f. Install countersunk washers and flat washers on bolts.

WARNING

FSCAP

Verification of minimum torque and safety of bolts are critical characteristics.

g. Install mounting bolts. Torque four bolts down evenly, one turn at a time, to avoid cocking servo and possibly damaging quick-disconnect couplings. **TORQUE DIAGONALLY OPPOSITE BOLTS TO 385 - 425 INCH-POUNDS UNTIL ALL FOUR ARE FINALLY TORQUED.**

h. Lockwire, Item 197, Appendix D, boltheads as follows:

(1) Front and rear bolts right side of forward primary servo to each other. Cover lockwire with heat-shrinkable insulation sleeving, Item 180, Appendix D.

(2) Front and rear bolts left side of lateral primary servo to each other. Cover lockwire with heat-shrinkable insulation sleeving, Item 180, Appendix D.

(3) All other bolts - to primary servo mounting bolt next to it.

(4) Apply torque stripe to all boltheads (PARA 2-6-5).

NOTE

Do not adjust control rod.

i. Install mixer to forward primary servo control rod (PARA 11-4-80), mixer to aft primary servo control rod (PARA 11-4-81), or mixer to lateral primary servo control rod (PARA 11-4-82) to applicable primary servo.

j. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

k. Connect forward control rod (PARA 11-4-90), aft control rod (PARA 11-4-91), or lateral control rod (PARA 11-4-92) to applicable primary servo.

l. Connect electrical connectors to pressure switches as follows:

(1) Remove protective caps and plugs from electrical connectors.

(2) Inspect and treat electrical connectors (PARA 1-7-20).

(3) On forward primary servo, connect electrical connector, P426, to 1st stage pressure switch and electrical connector, P429, to 2nd stage pressure switch.

(4) On aft primary servo, connect electrical connector, P425, to 1st stage pressure switch and electrical connector, P428, to 2nd stage pressure switch.

(5) On lateral primary servo, connect electrical connector, P424, to 1st stage pressure switch and electrical connector, P427, to 2nd stage pressure switch.

m. Perform operational check of primary servo pressure switch (PARA 7-2-1).

n. Perform primary servo rig check (PARA 11-4-87.1.4).

11-4-87.1.4 Primary Servo Rig Check.

NOTE

- This procedure is used to check main rotor rigging after replacing a primary servo. Helicopter must be rigged with blade track/autorotation adjustments made. If helicopter is not rigged complete with blade track/autorotation adjustments, perform rigging (PARA 11-4-3).
- If a forward and/or aft servo was replaced, perform both the longitudinal and lateral blade angle checks. If only a lateral servo was replaced, perform only the lateral blade angle check.

a. Remove main rotor pylon sliding cover (PARA 2-4-101).

b. Remove main rotor pylon work platform (PARA 2-4-105).

c. Unzip pilot's and copilot's cyclic stick boot.

d. Perform BOOST ON, SAS OFF procedure as follows:

NOTE

If aircraft electrical power is not available for use, an alternate method to achieve BOOST ON, SAS OFF is to apply 28 volts DC to pin 1, and ground pin 2 on pilot-assist module side of connector, P468R.

(1) On STABILATOR CONTROL/AUTO FLT CONT panel, disengage BOOST, SAS 1, and SAS 2 switches (Figure 11-4-87-2, Detail A).

(2) Verify that BOOST SERVO OFF and SAS OFF capsules are on.

(3) Turn off all electrical power (PARA 1-6-16).

(4) Disconnect BOOST SHUTOFF valve electrical connector, P469R, on top of pilot-assist module.

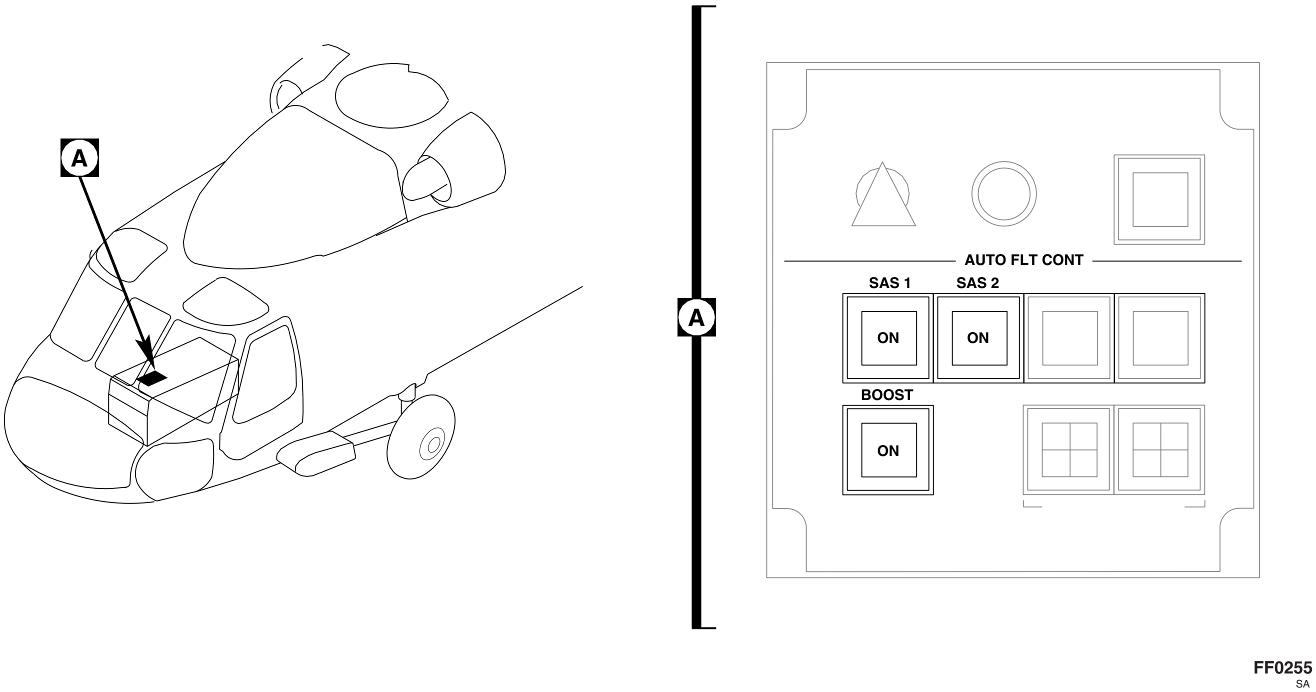


FIGURE 11-4-87-2. STABILATOR CONTROL/AUTO FLT CONT PANEL

(5) Install protective caps and plugs on electrical connectors.

(6) Turn on all electrical power (PARA 1-6-16).

(7) Verify that SAS OFF capsule is on and BOOST SERVO OFF capsule is off.

e. Apply external hydraulic power (PARA 1-6-15).

WARNING

Moving controls while installing rigging pins may cause personal injury or damage to equipment.

CAUTION

Damage to flight control components will result if hydraulic power is shut down during rigging with rigging pins still installed. Make sure all rigging pins are removed before shutting down hydraulic power.

f. Install 5⁄16 x 4 in. rigging pins, 70700-20380-043, in pitch, roll, and yaw upper torque shafts (Figure 11-4-87-4, Sheet 1, Detail A).

g. Apply main rotor gust lock to hold rotor head. Install damper rigging blocks on No. 1 (red) blade and hold with tiedown straps, Item 338, Appendix D. Release gust lock.

NOTE

If index marks on swashplate bearing retaining rings are missing, or cannot be matched, flight control rigging cannot be done. Align index marks (PARA 5-4-22).

h. Turn main rotor head until black index marks on swashplates line up and No. 1 pitch control rod is over front and rear input (90°) position (Figure 11-4-87-4).

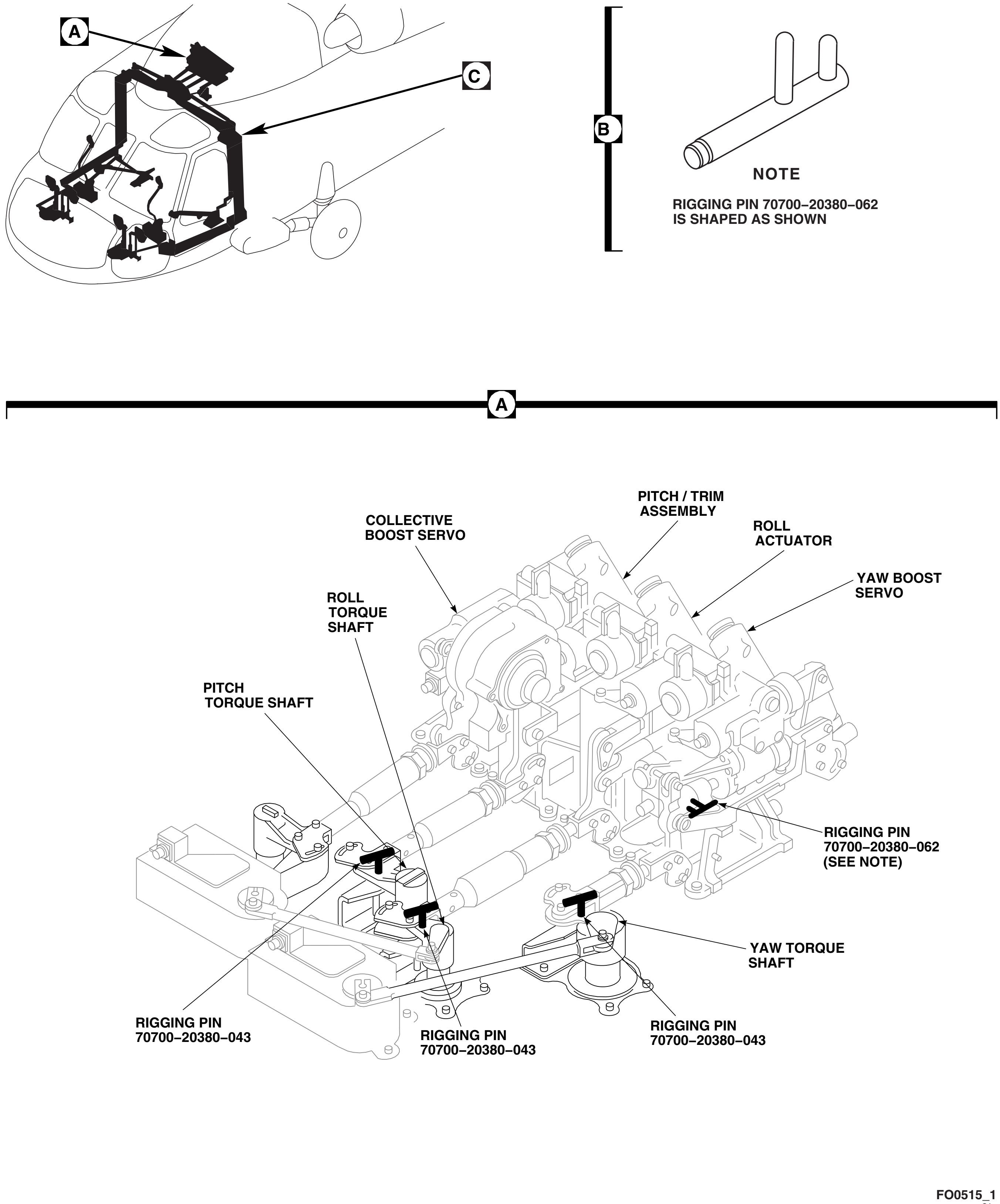


FIGURE 11-4-87-3. INSTALL TORQUE SHAFT RIGGING PINS (SHEET 1 OF 2)

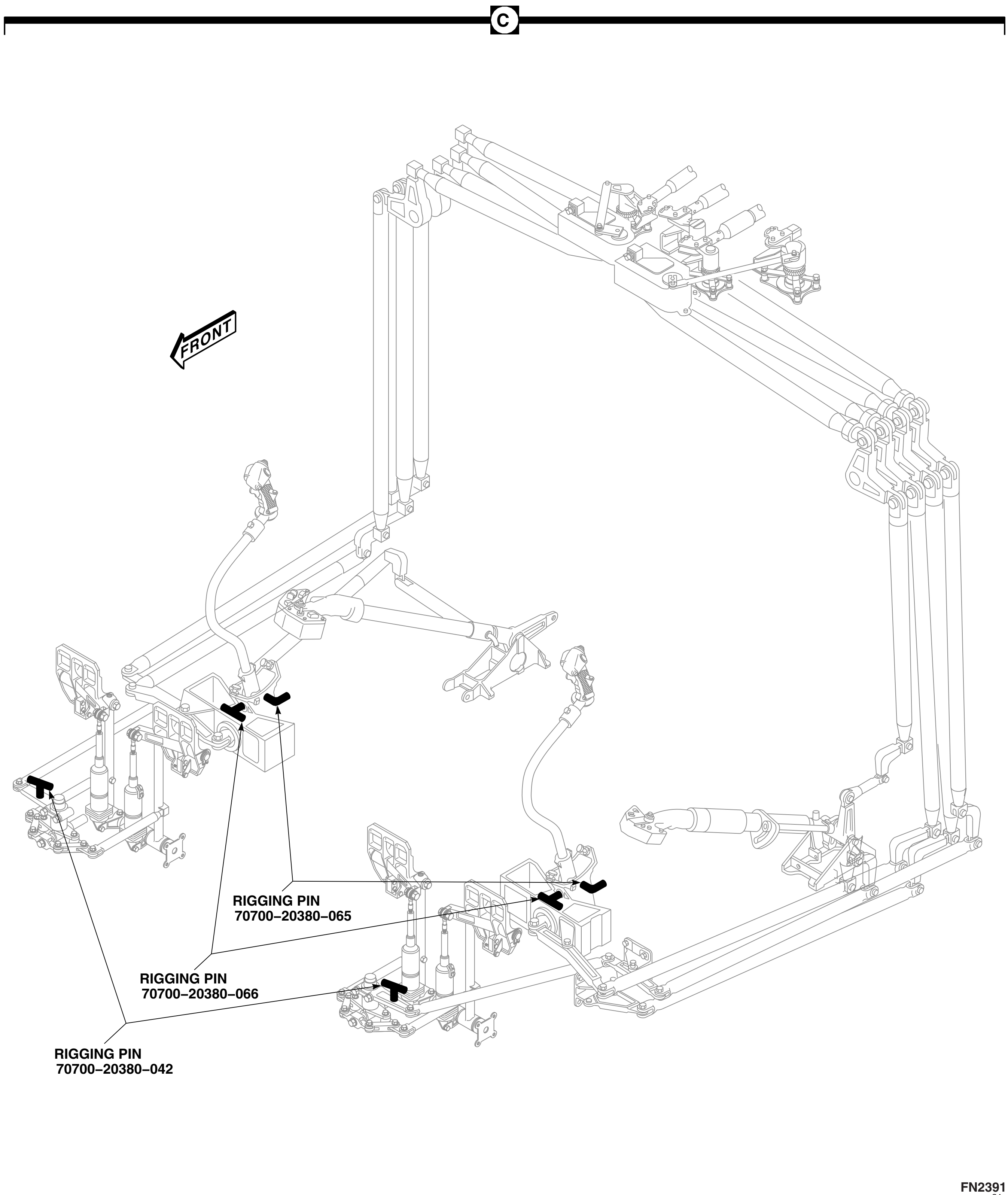


FIGURE 11-4-87-3. INSTALL TORQUE SHAFT RIGGING PINS (SHEET 2 OF 2)

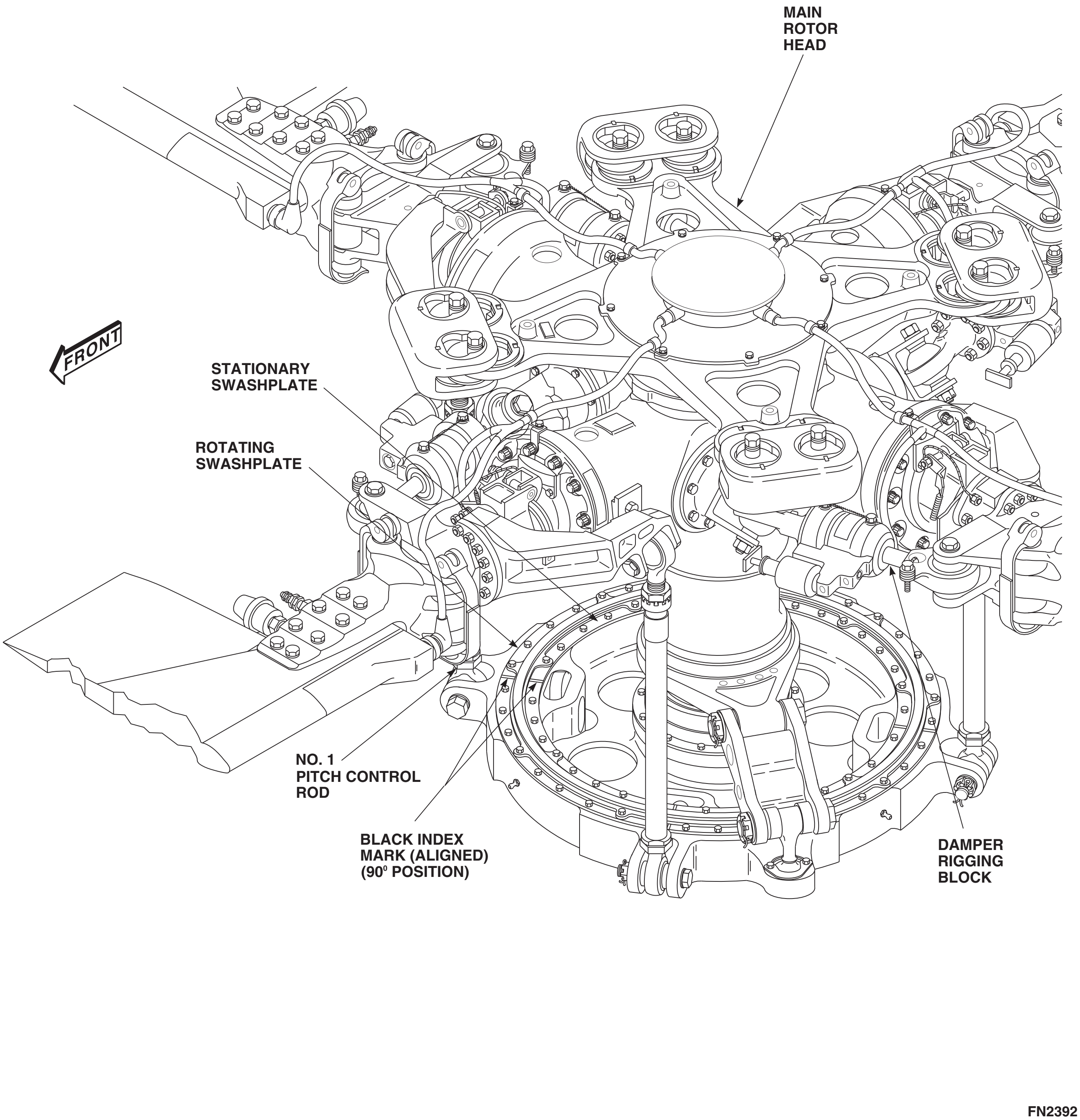


FIGURE 11-4-87-4. ALIGN SWASHPLATE INDEX MARKS

WORKSHEET

#1 (RED BLADE POSITION)

AIRCRAFT TAIL NO. ____________ DATE ____________

CONTROL POSITIONS		90° RT	180° FWD	270° LFT	0° AFT	TOTAL°	AVG	REQUIRED TOLERANCE
LOW COLL ANGLE LOW COLL NEUTRAL CYCLIC NEUTRAL YAW	#1							–.6°± .125° (–.475° TO –.725°)
BLADES	#2 #3 #4							BLADES #2, #3, #4 SAME AS #1 BLADE
HIGH COLL ANGLE HIGH COLL NEUTRAL CYCLIC NEUTRAL YAW								+15° TO +16° (SEE NOTE)
FWD CYCLIC ANGLE HIGH COLL NEUTRAL LAT FWD CYCLIC NEUTRAL YAW								DIFFERENCE BETWEEN 270° & 90° POSITIONS +33.0° MIN.
AFT CYCLIC ANGLE HIGH COLL NEUTRAL LAT AFT CYCLIC NEUTRAL YAW								DIFFERENCE BETWEEN 90° & 270° POSITIONS NOT LESS THAN +19°
LEFT CYCLIC ANGLE LOW COLL NEUTRAL LONG LEFT CYCLIC NEUTRAL YAW								SUM OF 180° & 0° POSITION TO BE 15° TO 17° AVG RANGE +7.5° TO 8.5°
RIGHT CYCLIC ANGLE LOW COLL NEUTRAL LONG RIGHT CYCLIC NEUTRAL YAW								SUM OF 180° TO 0° POSITION TO BE 14° TO 16° AVG RANGE +7.0° TO 8.0°
(FWD) 180° (LEFT) 270° 90° (RIGHT) 0° (AFT)		PRIMARY SERVO CONTROL ROD BIAS: FWD: AFT: LAT:						

NOTE

WHEN PERFORMING RIG CHECK, SEE TEXT FOR TOLERANCE WHEN PITCH CONTROL RODS HAVE BEEN ADJUSTED FOR AUTOROTATIONAL RPM AND/OR BLADE TRACK.

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SA

FIGURE 11-4-87-5. MAIN ROTOR RIGGING WORKSHEET (LOCALLY REPRODUCE)

i. Use Main Rotor Rigging Worksheet to record readings (Figure 11-4-87-5).

j. Install the following rigging pins (Figure 11-4-87-3, Sheet 1, Detail A and Sheet 2, Detail C):

(1) Pilot's and copilot's pedal adjustor rigging pins, 70700-20380-042.

(2) Yaw boost actuator ″F″ rigging pin, 70700-20380-062.

NOTE

This step is to be performed if a forward and/or aft servo was replaced.

k. Perform check of longitudinal angles as follows:

(1) Install pilot's and copilot's lateral cyclic stick rigging pins, 70700-20380-065 (Figure 11-4-87-3, Sheet 2, Detail C).

(2) Remove rigging pin from pitch torque shaft (Figure 11-4-87-3, Sheet 1, Detail A).

(3) While holding collective stick on full up stop, move cyclic stick forward. Measure blade angle at 90° and 270° location, using universal propeller or digital protrac-

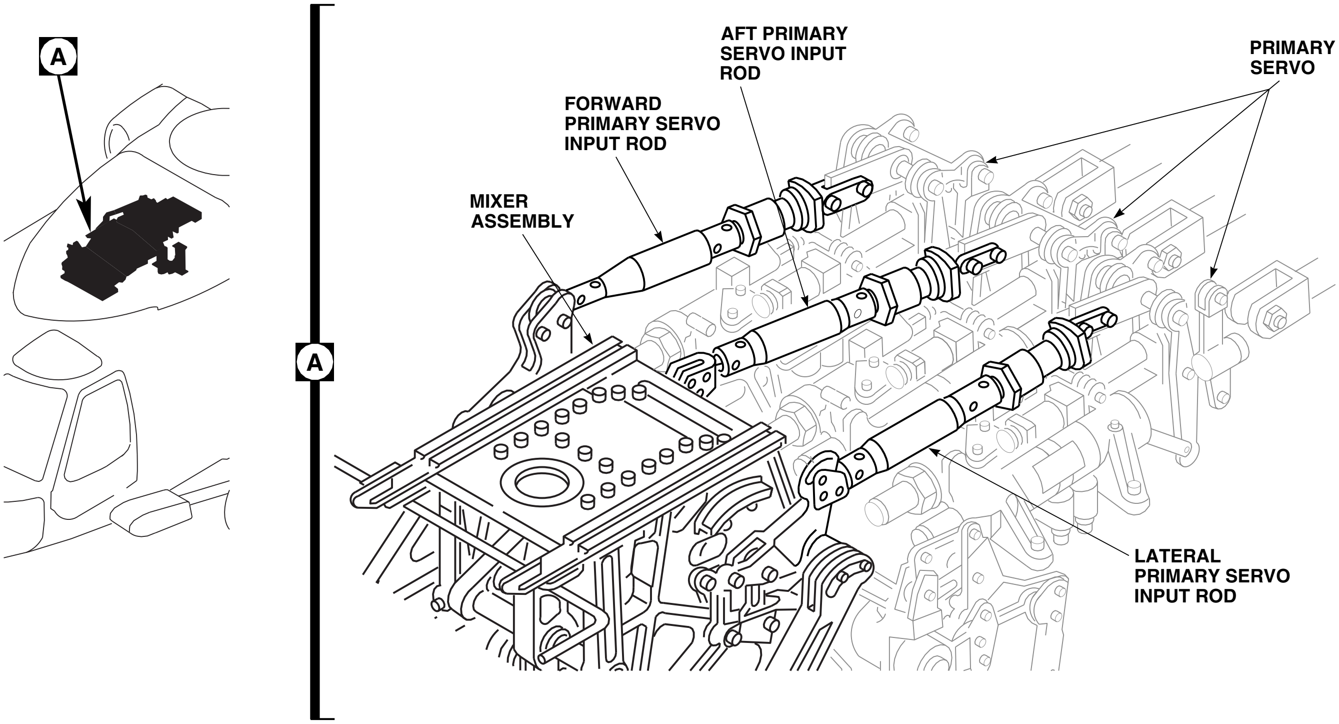


FIGURE 11-4-87-6. ADJUST PRIMARY SERVO CONTROL ROD

tor. **DIFFERENCE BETWEEN BOTH READINGS MUST BE 33° OR GREATER, AND PITCH FORWARD LIMITER ROLLER ON MIXER CANNOT BE FREELY TURNED.**

(4) If forward limiter is bottomed and roller cannot be turned, but blade angle difference is less than 33°, perform the following:

(a) Shorten mixer to forward primary servo control rod by turning barrel to right, while facing aft and lengthen mixer to aft primary servo control rod by turning barrel to left, same amount, while facing aft (Figure 11-4-87-6, Detail A).

(b) Perform this adjustment until difference between both readings is not less than 33° at full up collective/forward cyclic.

(c) Verify that aft servo is not bottomed.

(5) While holding full up collective, move cyclic stick full aft with No. 1 (red) blade at 90° location and at 270° location. Verify that the aft stick stops are being contacted. Read blade angles for both locations, using universal propeller or digital protractor. **DIFFERENCE BETWEEN 90° AND 270° READINGS MUST BE 19° OR GREATER.** If blade angle difference is not within limits, refer to main rotor complete rig (PARA 11-4-3).

(6) Move collective stick to low position, cyclic stick full forward. Verify mixer roller is contacting forward limiter (Figure 11-4-87-7, Detail A). Verify forward servo is not bottomed in the retract direction.

(7) Move collective stick to low position, cyclic stick full aft. Verify mixer roller is contacting aft limiter. Verify aft servo is not bottomed in the retract direction.

(8) Remove pilot's lateral cyclic stick rigging pin (Figure 11-4-87-3, Sheet 2, Detail C).

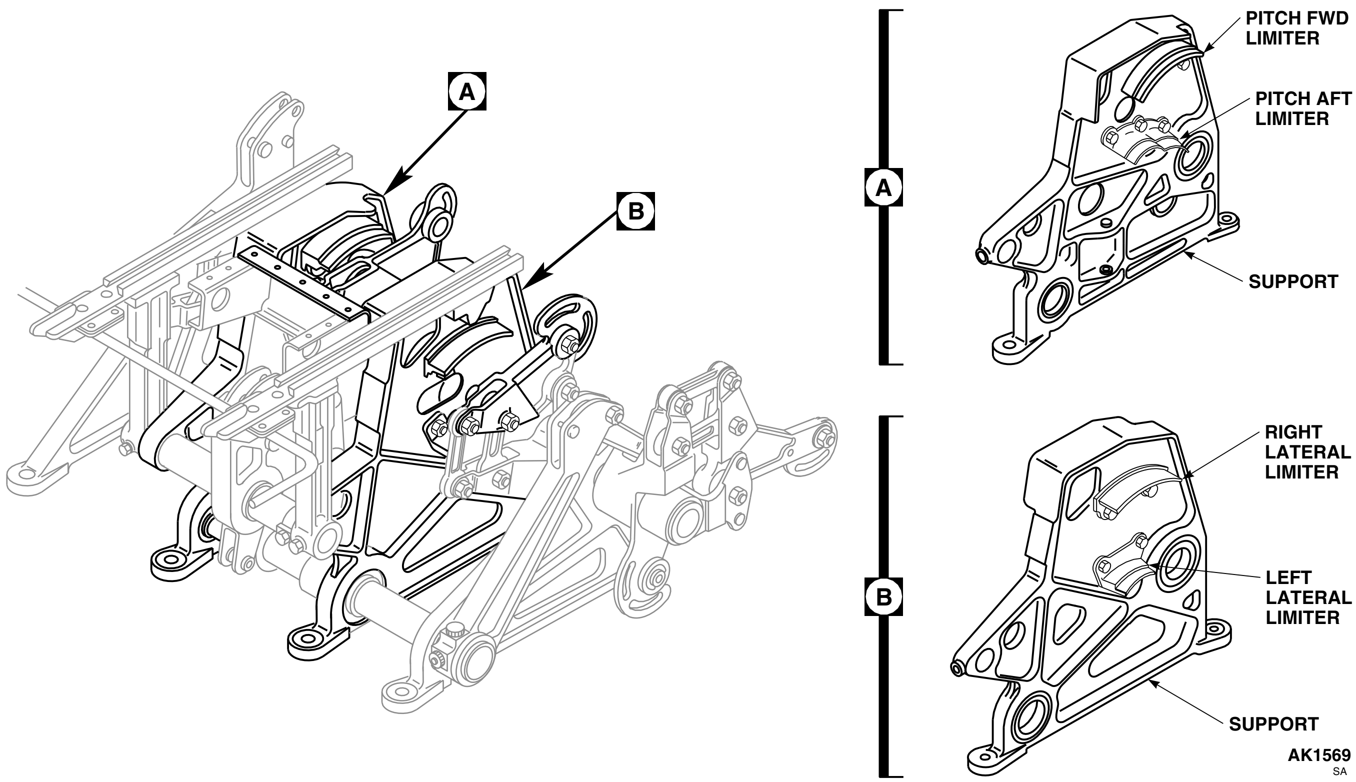


FIGURE 11-4-87-7. CHECK MIXER LIMITER

NOTE

This check is to be performed anytime a servo is replaced.

l. Perform check of lateral angles as follows:

(1) Remove rigging pin from roll torque shaft, and reinstall (5⁄16 x 4 in.) pin in pitch torque shaft (Figure 11-4-87-3, Sheet 1, Detail A).

(2) Install pilot's and copilot's longitudinal cyclic stick rigging pins, 70700-20380-066 (Figure 11-4-87-3, Sheet 2, Detail C).

(3) Move pilot's collective pitch stick to full low pitch position and hold in place.

NOTE

Stops may be adjusted in order to obtain proper clearance.

(4) Move cyclic stick full left until roller contacts left lateral limiter in mixer (Figure 11-4-87-7, Detail B). **CHECK FOR 0.010 TO 0.030-INCH CLEARANCE BETWEEN PILOT'S AND COPILOT'S CYCLIC STICK AND STOP.** Move red blade to 180° and 0° locations and measure blade angle at both locations, using universal propeller or digital protractor. Left lateral cyclic range is found by adding 180° reading to 0° reading, ignoring that this reading is negative. **SUM OF BOTH READINGS SHOULD BE 15° TO 17°.** If sum is less than 15°, adjust lateral primary servo input control rod by turning barrel to right while facing aft. Do this adjustment until total angle between 0° and 180° locations read 15° to 17°.

(5) Move cyclic stick full right until roller contacts right lateral limiter in mixer. **CHECK FOR 0.010 TO 0.030-INCH CLEARANCE BETWEEN PILOT'S AND COPILOT'S CYCLIC STICK AND STOP.** Move red blade to 180° and 0° locations and measure blade angle at

both locations, using universal propeller or digital protractor. Add 0° reading to 180° reading, ignoring that this reading is negative. **SUM OF THESE TWO READINGS SHOULD BE 14° TO 16°.** If reading cannot be obtained, adjust controls as follows:

(a) Lengthen lateral primary servo input control by turning barrel to left while facing aft. Perform this adjustment until low right cyclic reading is between 7.0° and 8.0°.

(b) Check full down collective/left cyclic range again, to make sure it is still within range. Adjust lateral primary servo control rod until both lateral range requirements are satisfied.

(6) Move collective stick to high position and move cyclic stick full left. Verify roller contacts left lateral limiter in mixer.

(7) With collective stick in high position, move cyclic stick full right. Verify right stick stops are contacted.

(8) Remove pilot's and copilot's longitudinal cyclic stick pins.

m. Perform the following quality assurance checks:

(1) No passage of lockwire, Item 193, Appendix D, through control rod inspection holes.

(2) Primary servo input rods for lockwire.

(3) Control rods for lockwire.

(4) Cotter pins installed in nuts where required.

(5) Cyclic and collective stick stopbolts for lockwire.

(6) All rigging pins and damper blocks removed.

n. Disconnect all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

o. Reset PLT VSI circuit breaker on No. 2 ac primary circuit breaker panel.

p. Zip up pilot's and copilot's stick boots.

q. Remove protective caps and plugs from electrical connectors.

r. Inspect and treat electrical connectors (PARA 1-7-20).

s. Connect electrical connector, P469R, to boost shutoff valve on top of pilot-assist module.

t. Make sure area is clean and free of foreign material.

u. Install main rotor pylon work platform (PARA 2-4-105).

v. Install main rotor pylon sliding cover (PARA 2-4-101).

w. Perform maintenance test flight of autorotational rpm and high collective stop (Appropriate MTF).

CAUTION

A main rotor complete rig is required, if other maintenance was performed on flight control system in conjunction with primary servo replacement.

x. If primary servo input control rods were adjusted during installation, perform main rotor rig check (PARA 11-4-4).

11-4-88 PRIMARY SERVO PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs
- Wrench

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 7-2-1

11-4-88.1 Replace Primary Servo Pressure Switch.

NOTE

Replacement of both pressure switches on forward, aft, and lateral primary servos is the same, except as noted.

11-4-88.1.1 Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Open main rotor pylon sliding cover.

c. Disconnect electrical connector from pressure switch (Figure 11-4-88-1, Detail A).

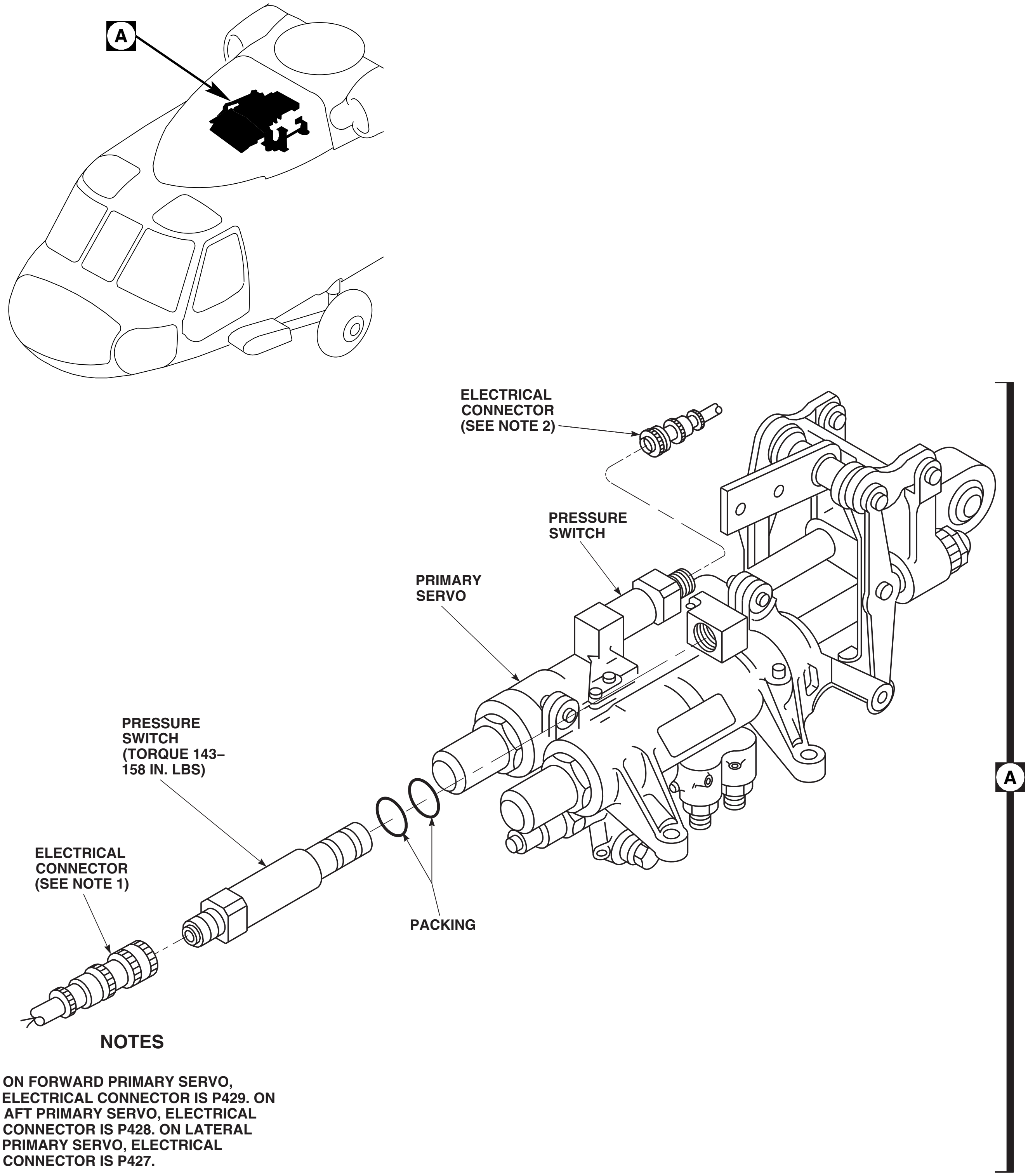
d. Install protective caps and plugs on electrical connectors.

CAUTION

- Damage to primary servo will result if dirt is allowed to enter components during disassembly. Make sure primary servo is clean before disassembly. Have replacement pressure switch ready for installation before old pressure switch is removed.
- To prevent equipment damage, use wrench to hold pressure switch and servo adapter while removing pressure switch.

e. Remove pressure switch from primary servo.

f. Remove packings from pressure switch. Discard packings.



NOTES

1. ON FORWARD PRIMARY SERVO, ELECTRICAL CONNECTOR IS P429. ON AFT PRIMARY SERVO, ELECTRICAL CONNECTOR IS P428. ON LATERAL PRIMARY SERVO, ELECTRICAL CONNECTOR IS P427.
2. ON FORWARD PRIMARY SERVO, ELECTRICAL CONNECTOR IS P426. ON AFT PRIMARY SERVO, ELECTRICAL CONNECTOR IS P425. ON LATERAL PRIMARY SERVO, ELECTRICAL CONNECTOR IS P424.

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FIGURE 11-4-88-1. REPLACE PRIMARY SERVO PRESSURE SWITCH

11-4-88.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

b. Coat packings and threads of pressure switch with hydraulic fluid, Item 154 or Item 155, Appendix D.

CAUTION

Damage to primary servo will result if pressure switch is installed incorrectly. Use wrench to hold pressure switch and servo adapter while torquing.

c. Install packings on pressure switch. Using wrench flats on pressure switch, screw switch in primary servo (Figure 11-4-88-1, Detail A). **TORQUE PRESSURE SWITCH TO 143 - 158 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire pressure switch.

d. Remove protective caps and plugs from electrical connectors.

e. Inspect and treat electrical connectors (PARA 1-7-20).

f. Connect electrical connector to primary servo.

g. Perform operational check of primary servo pressure switch (PARA 7-2-1).

h. Close main rotor pylon sliding cover.

11-4-89 PRIMARY SERVO BYPASS VALVE CAP PACKING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 53, Appendix D
- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D

References

- Appendix D

11-4-89.1 Replace Primary Servo Bypass Valve Cap Packing.

NOTE

- Replacement of both bypass valve cap packings on forward, aft, and lateral primary servos is the same.
- Bypass valve cap can be removed to replace packing, but cannot be replaced with a new bypass valve cap. Care should be taken to make sure no other part of bypass valve cap is removed as components are all part of a matched set.

11-4-89.1.1. Remove.

a. Open main rotor pylon sliding cover.

b. Remove bypass valve cap from primary servo (Figure 11-4-89-1, Detail A).

c. Remove packing from bypass valve cap. Discard packing.

11-4-89.1.2. Install.

a. Lubricate new packing with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install packing on bypass valve cap (Figure 11-4-89-1, Detail A).

c. Install bypass valve cap over spring and into primary servo.

d. **TORQUE BYPASS VALVE CAP TO 56 - 60 INCH-POUNDS.**

e. Seal bypass valve cap with adhesive, Item 53, Appendix D.

f. Lockwire, Item 197, Appendix D, bypass valve cap.

g. Close and latch main rotor pylon sliding cover.

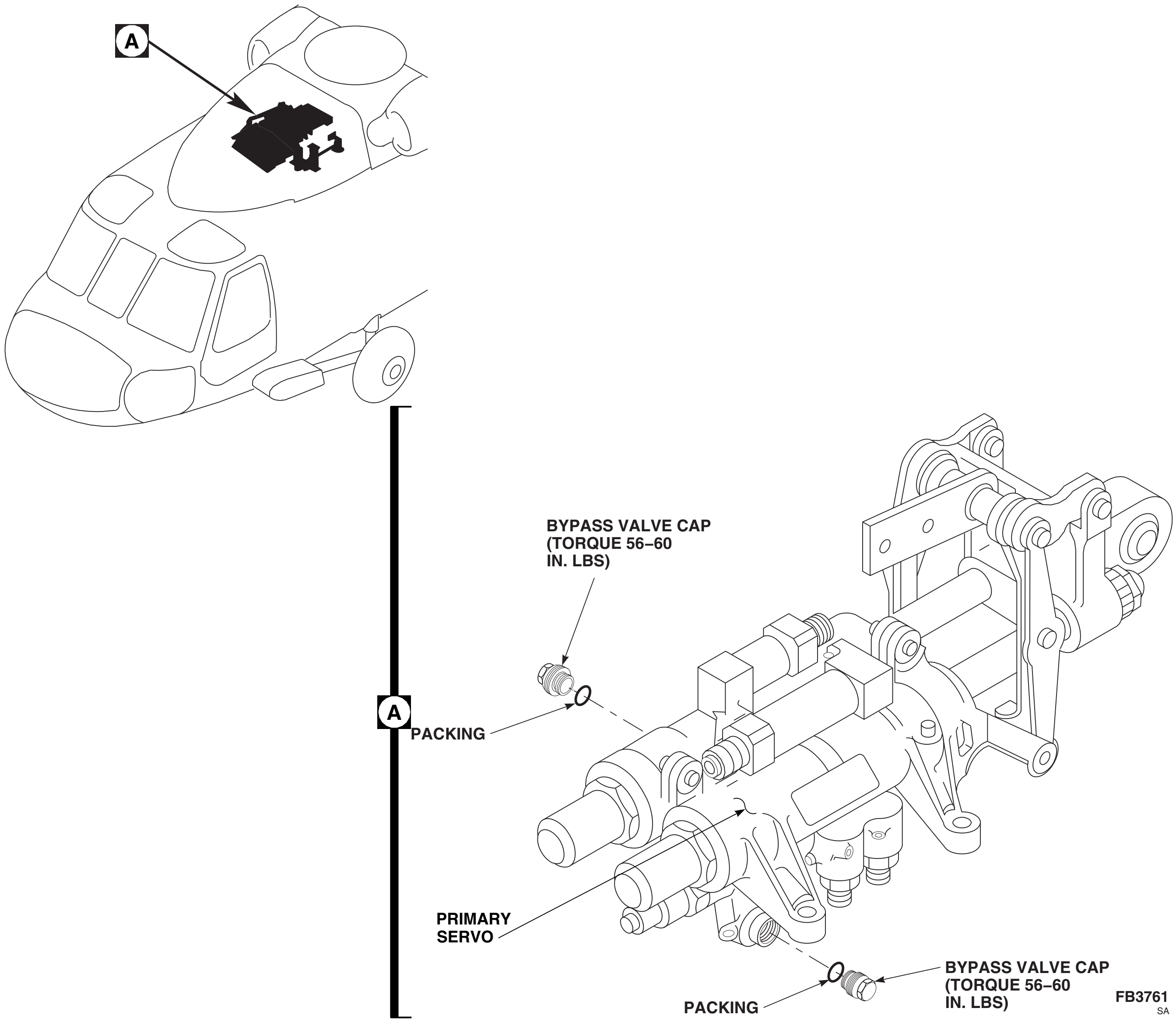


FIGURE 11-4-89-1. REPLACE PRIMARY SERVO BYPASS VALVE CAP PACKING

11-4-90 FORWARD CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 3/16-inch
- Crowfoot Attachment
- Protective Covering
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 1/2-inch thick

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-1
- PARA 11-3-17
- PARA 11-4-120
- PARA 11-5-31

11-4-90.1 Replace Forward Control Rod.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-90.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

d. Raise collective stick to raise swashplate.

e. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission. Lower swashplate onto wood blocks to support swashplate while removing forward control rod.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

f. Slowly lower collective stick until swashplate rests on wood blocks.

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 3⁄16-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

h. Remove special self-retaining bolts, countersunk/flat washers, additional washer(s), if installed, nuts, and cotter pins securing forward control rod to forward bellcrank and forward primary servo (Figure 11-4-90-1, Detail A). Remove control rod.

11-4-90.1.2 Inspect.

a. Perform main rotor control rod inspection (PARA 11-3-17).

b. Check bonded nylon washers for disbonding. **DISBONDING MUST NOT EXCEED 25% OF THE BONDED AREA.** If washer is disbonded, perform the following:

(1) If disbond is greater than 25% of the bonded surface area, replace nylon washer (PARA 11-4-120).

(2) If disbond is less than 25% of the bonded surface area, perform the following:

(a) Visually inspect control rod for evidence of corrosion in the disbonded area. **NONE ALLOWED.** If corrosion is found, replace control rod.

(b) Clean around bonding surfaces of nylon washer and control rod using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(c) Apply sealing compound, Item 292, Appendix D, to the disbonded edges of nylon washer (PARA 11-4-120).

(d) Annotate logbook that a repetitive 30 day visual inspection be performed for evidence of corrosion. **NONE ALLOWED.** If corrosion is found, replace control rod.

(e) Any control rod repaired while installed on the aircraft must be removed at the next 100 hour special inspection.

c. Inspect control rod bushings for nicks, scratches, corrosion, and wear. **DAMAGE MUST NOT EXCEED 0.010-INCH DEEP OR COVER MORE THAN 30% OF BORE.** If bushing is damaged beyond limits, replace control rod.

d. Inspect control rod bushings for looseness. **NONE ALLOWED.** If bushing is loose, replace control rod.

e. Inspect control rod for loose or missing identification plates. **NONE ALLOWED.** If

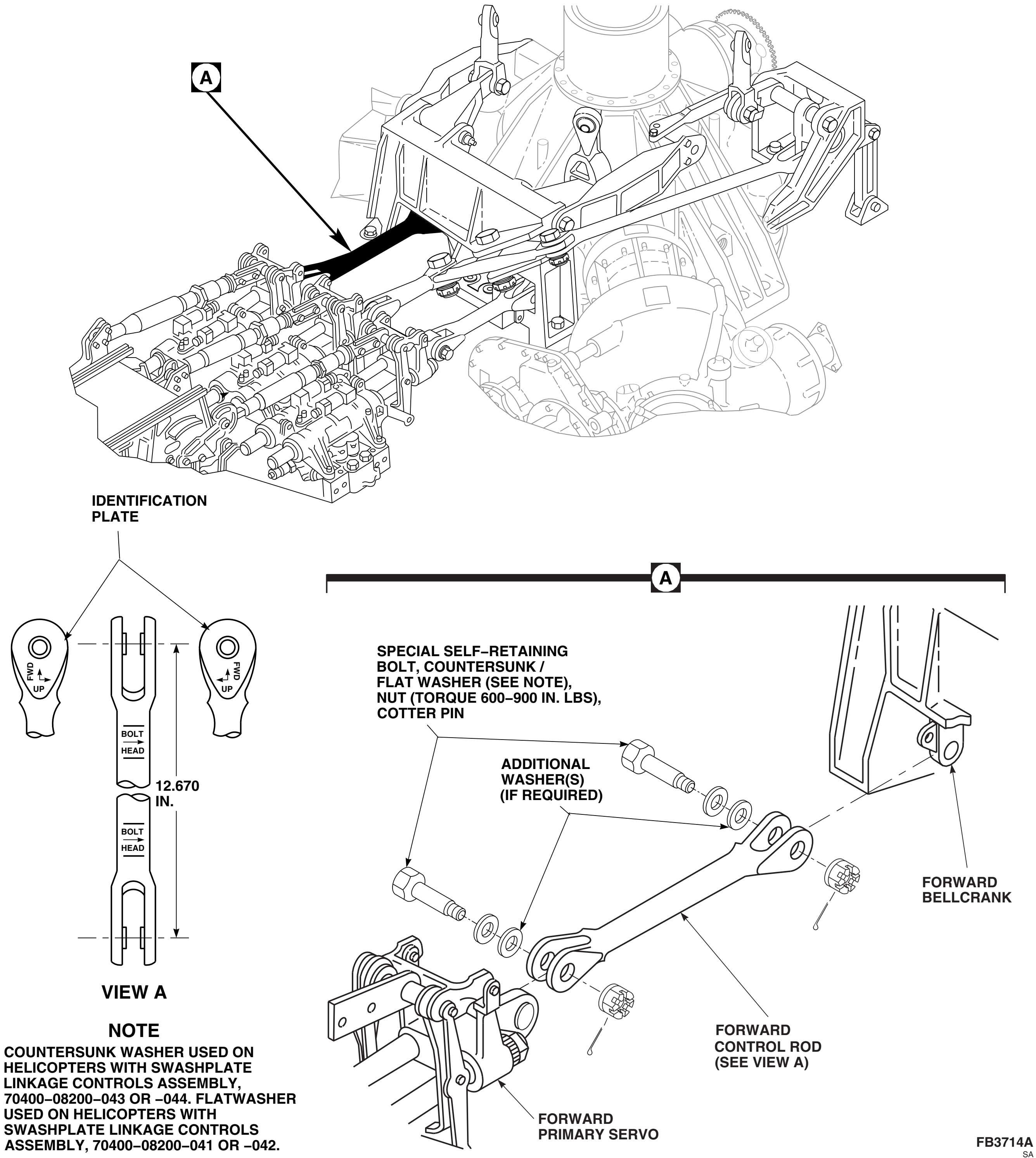


FIGURE 11-4-90-1. REPLACE FORWARD CONTROL ROD

loose or missing, replace identification plates (PARA 11-5-31).

f. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-90.1.3 Install.

WARNING

Injury to personnel and damage to equipment will result if incorrect control rod is installed. Forward and lateral primary servo output control rods are similar in length. Make sure forward control rod to be installed is 12.670-inches from bolthole-center to bolthole-center.

NOTE

- Verify nylon washers are installed and washer orientation is correct (PARA 11-4-120).
- Position control rod according to identification plates on rod (Figure 11-4-90-1, View A).

a. Place replacement forward control rod between forward primary servo and forward bellcrank (Figure 11-4-90-1, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is installed incorrectly. To install bolt, insert 3/16-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Add washers, as necessary, to make sure nut is bottomed against control rod and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolts and countersunk/flat washers (under bolthead). Add washers, as required (maximum of 2 additional washers under countersunk/flat washers) to provide correct grip length and cotter pin position (Figure 11-4-90-1, Detail A).

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nuts. **TORQUE NUTS TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET**

TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE. Install cotter pins.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Raise collective stick and remove wood blocks from between swashplate and main transmission.

i. Perform operational check of flight control system (PARA 11-2-3).

j. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

k. Make sure area is clean and free of foreign material.

l. Install main rotor pylon transmission cover (PARA 2-4-107).

m. Close and latch main rotor pylon sliding cover.

11-4-91 AFT CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 3⁄16-inch
- Crowfoot Attachment
- Protective Covering
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-1
- PARA 11-3-17
- PARA 11-4-120
- PARA 11-5-31

11-4-91.1 Replace Aft Control Rod.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-91.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

d. Raise collective stick to raise swashplate.

e. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing aft control rod.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

f. Slowly lower collective stick until swashplate rests on wood blocks.

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 3/16-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

h. Remove special self-retaining bolts, countersunk/flat washers, additional washer(s), if installed, nuts, and cotter pins securing aft control rod to aft primary servo and walking beam (Figure 11-4-91-1, Detail A). Remove control rod.

11-4-91.1.2 Inspect.

a. Perform main rotor control rod inspection (PARA 11-3-17).

b. Check bonded nylon washers for disbonding. **DISBONDING MUST NOT EXCEED 25% OF THE BONDED AREA.** If washer is disbonded, perform the following:

(1) If disbond is greater than 25% of the bonded surface area, replace nylon washer (PARA 11-4-120).

(2) If disbond is less than 25% of the bonded surface area, perform the following:

(a) Visually inspect control rod for evidence of corrosion in the disbonded area. **NONE ALLOWED.** If corrosion is found, replace control rod.

(b) Clean around bonding surfaces of nylon washer and control rod using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(c) Apply sealing compound, Item 292, Appendix D, to the disbonded edges of nylon washer (PARA 11-4-120).

(d) Annotate logbook that a repetitive 30 day visual inspection be performed for evidence of corrosion. **NONE ALLOWED.** If corrosion is found, replace control rod.

(e) Any control rod repaired while installed on the aircraft must be removed at the next 100 hour special inspection.

c. Inspect control rod bushings for nicks, scratches, corrosion, and wear. **DAMAGE MUST NOT EXCEED 0.010-INCH DEEP OR COVER MORE THAN 30% OF BORE.** If bushing is damaged beyond limits, replace control rod.

d. Inspect control rod bushings for looseness. **NONE ALLOWED.** If bushing is loose, replace control rod.

e. Inspect control rod for loose or missing identification plates. **NONE ALLOWED.** If loose or missing, replace identification plates (PARA 11-5-31).

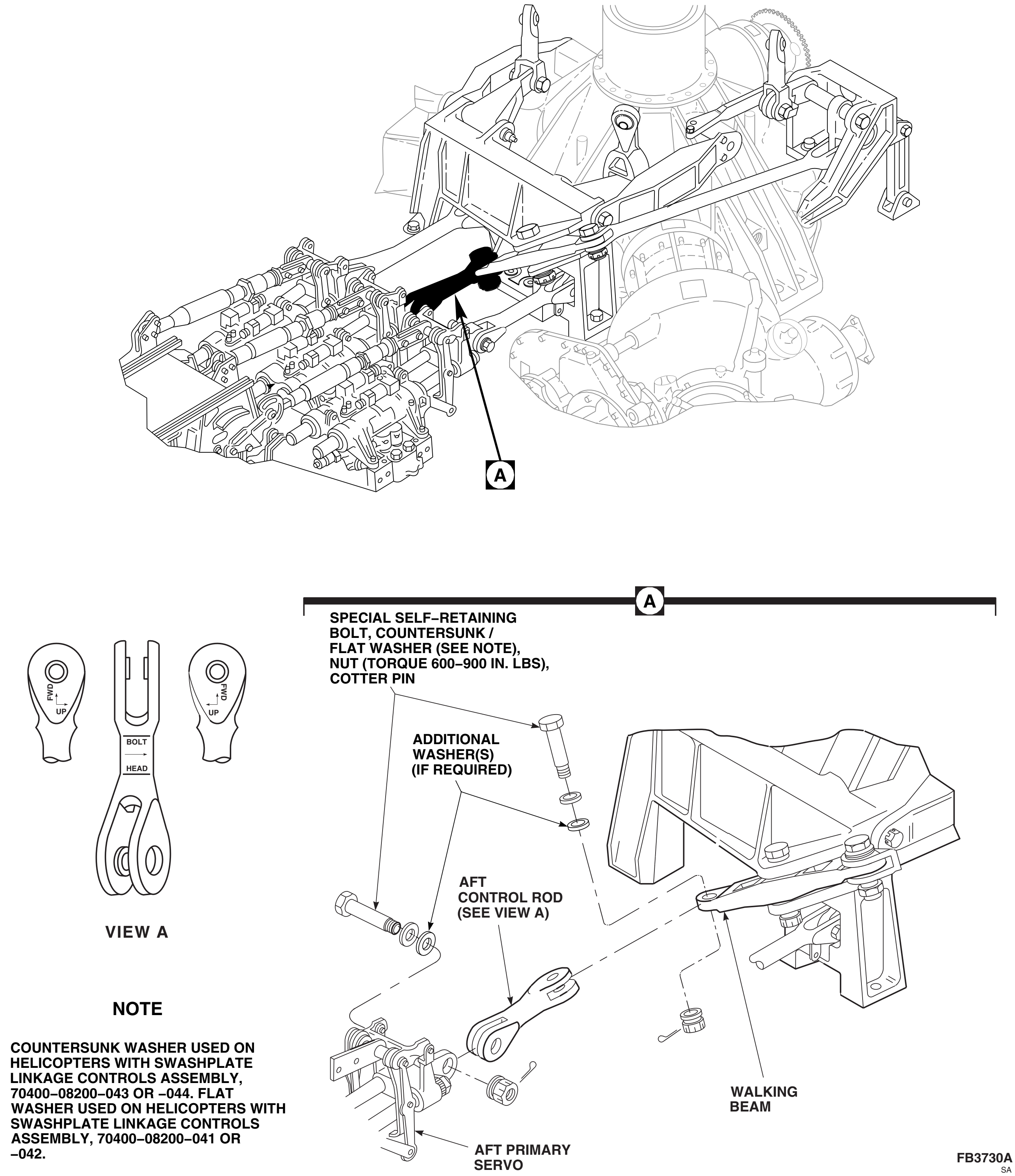


FIGURE 11-4-91-1. REPLACE AFT CONTROL ROD

f. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-91.1.3 Install.

NOTE

- Verify nylon washers are installed and washer orientation is correct (PARA 11-4-120).
- Position control rod according to identification plates on rod (Figure 11-4-91-1, View A).

a. Place replacement aft control rod between aft primary servo and walking beam (Figure 11-4-91-1, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is installed incorrectly. To install bolt, insert 3/16-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Add washers, as necessary, to make sure nut is bottomed against control rod and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolts and countersunk/flat washers (under bolthead). Add washers, as required (maximum of 2 additional washers under countersunk/flat washers) to provide correct grip length and cotter pin position (Figure 11-4-91-1, Detail A).

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nuts. **TORQUE NUTS TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pins.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Raise collective stick and remove wood blocks from between swashplate and main transmission.

i. Perform operational check of flight control system (PARA 11-2-3).

j. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

k. Make sure area is clean and free of foreign material.

l. Install main rotor pylon transmission cover (PARA 2-4-107).

m. Close and latch main rotor pylon sliding cover.

11-4-92 LATERAL CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 3⁄16-inch
- Crowfoot Attachment
- Protective Covering
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-1
- PARA 11-3-17
- PARA 11-4-120
- PARA 11-5-31

11-4-92.1 Replace Lateral Control Rod.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-92.1.1 Remove.

a. Open main rotor pylon sliding cover.

b. Remove main rotor pylon transmission cover (PARA 2-4-107).

c. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

d. Raise collective stick to raise swashplate.

e. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing lateral control rod.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

f. Slowly lower collective stick until swashplate rests on wood blocks.

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 3⁄16-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

h. Remove special self-retaining bolts, countersunk/flat washers, additional washer(s), if installed, nuts, and cotter pins securing lateral control rod to lateral primary servo and lateral bellcrank (Figure 11-4-92-1, Detail A). Remove control rod.

11-4-92.1.2 Inspect.

a. Perform main rotor control rod inspection (PARA 11-3-17).

b. Check bonded nylon washers for disbonding. **DISBONDING MUST NOT EXCEED 25% OF THE BONDED AREA.** If washer is disbonded, perform the following:

(1) If disbond is greater than 25% of the bonded surface area, replace nylon washer (PARA 11-4-120).

(2) If disbond is less than 25% of the bonded surface area, perform the following:

(a) Visually inspect control rod for evidence of corrosion in the disbonded area. **NONE ALLOWED.** If corrosion is found, replace control rod.

(b) Clean around bonding surfaces of nylon washer and control rod using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(c) Apply sealing compound, Item 292, Appendix D, to the disbonded edges of nylon washer (PARA 11-4-120).

(d) Annotate logbook that a repetitive 30 day visual inspection be performed for evidence of corrosion. **NONE ALLOWED.** If corrosion is found, replace control rod.

(e) Any control rod repaired while installed on the aircraft must be removed at the next 100 hour special inspection.

c. Inspect control rod bushings for nicks, scratches, corrosion, and wear. **DAMAGE MUST NOT EXCEED 0.010-INCH DEEP OR COVER MORE THAN 30% OF BORE.** If bushing is damaged beyond limits, replace control rod.

d. Inspect control rod bushings for looseness. **NONE ALLOWED.** If bushing is loose, replace control rod.

e. Inspect control rod for loose or missing identification plates. **NONE ALLOWED.** If

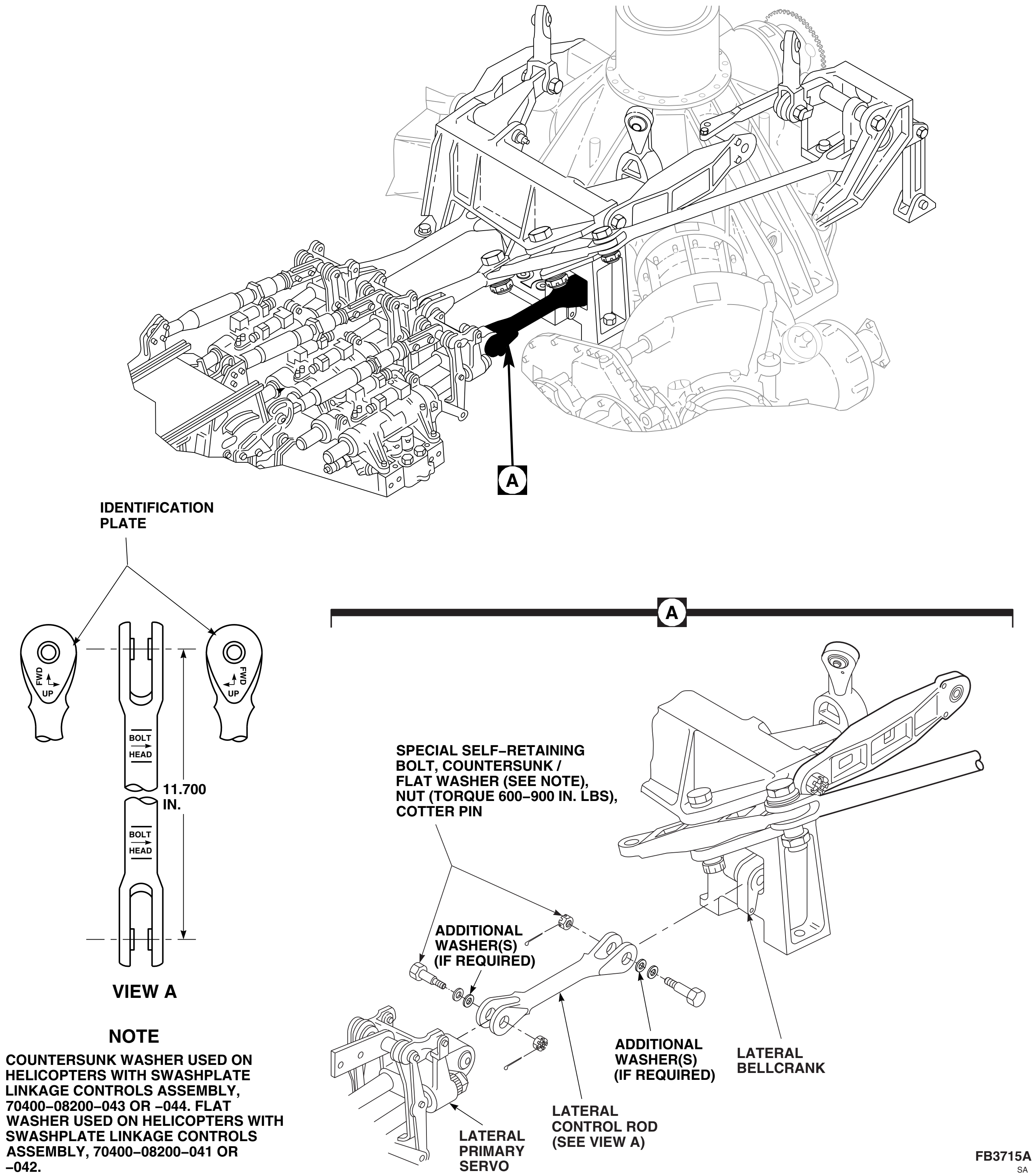


FIGURE 11-4-92-1. REPLACE LATERAL CONTROL ROD

loose or missing, replace identification plates (PARA 11-5-31).

f. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-92.1.3 Install.

WARNING

Injury to personnel and damage to equipment will result if incorrect control rod is installed. Forward and lateral primary servo output control rods are similar in length. Make sure lateral control rod to be installed is 11.700-inches from bolthole-center to bolthole-center.

NOTE

- Verify nylon washers are installed and washer orientation is correct (PARA 11-4-120).
- Position control rod according to identification plates on rod (Figure 11-4-92-1, View A).

a. Place replacement lateral control rod between lateral primary servo and lateral bellcrank (Figure 11-4-92-1, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is installed incorrectly. To install bolt, insert 3/16-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Add washers, as necessary, to make sure nut is bottomed against control rod and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolts and countersunk/flat washers (under bolthead). Add washers, as required (maximum of 2 additional washers under countersunk/flat washers) to provide correct grip length and cotter pin position (Figure 11-4-92-1, Detail A).

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nuts. **TORQUE NUTS TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET**

TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE. Install cotter pins.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Raise collective stick and remove wood blocks from between swashplate and main transmission.

i. Perform operational check of flight control system (PARA 11-2-3).

j. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

k. Make sure area is clean and free of foreign material.

l. Install main rotor pylon transmission cover (PARA 2-4-107).

m. Close and latch main rotor pylon sliding cover.

11-4-93 WALKING BEAM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ⅛-inch
- Crowfoot Attachment
- Micrometer
- Nonmetallic Drift
- Nonmetallic Punch
- Protective Covering
- Spacer Removal Tool, Locally-Made, Figure H-219, Appendix H
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

References

- Appendix H
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-1
- PARA 11-3-14
- PARA 11-4-91
- PARA 11-4-94
- PARA 11-4-100
- PARA 11-5-25

11-4-93.1 Replace Walking Beam.

11-4-93.1.1. Remove.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Disconnect aft control rod from walking beam (PARA 11-4-91).

c. Disconnect long control rod from walking beam (PARA 11-4-94).

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 1/8-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

d. Remove special self-retaining bolt, countersunk/flat washer, additional washer(s), if installed, nut, and cotter pin securing walking beam to forward bellcrank support (Figure 11-4-93-1, Detail A). Remove spacer and walking beam.

e. If spacer does not slide out with bolt, remove spacer as follows:

NOTE

If spacer removal tool is not available, it may be necessary to tap out spacers using nonmetallic punch.

(1) Locally make spacer removal tool (Figure H-219, Appendix H).

(2) Insert locally-made spacer removal tool into bolthead end of spacer.

(3) **TORQUE TOOL TO 200 - 250 INCH-POUNDS.**

(4) Tap out spacer using nonmetallic drift against underside of removal tool.

11-4-93.1.2. Inspect.

a. Perform walking beam inspection (PARA 11-3-14).

b. Visually inspect forward bellcrank support for damaged or worn pivot bushings (PARA 11-4-100).

c. Inspect walking beam flanged pivot bushings for cracks or evidence of corrosion. **NONE ALLOWED.** If cracked, or evidence of corrosion is found, replace bushing(s) (PARA 11-5-25).

d. Inspect face and journal of walking beam flanged pivot bushings for rips, tears, or disbonding of Teflon fabric (Figure 11-4-93-1, Detail B). **NONE ALLOWED.** If rips, tears, or disbonding is found, replace bushing(s) (PARA 11-5-25).

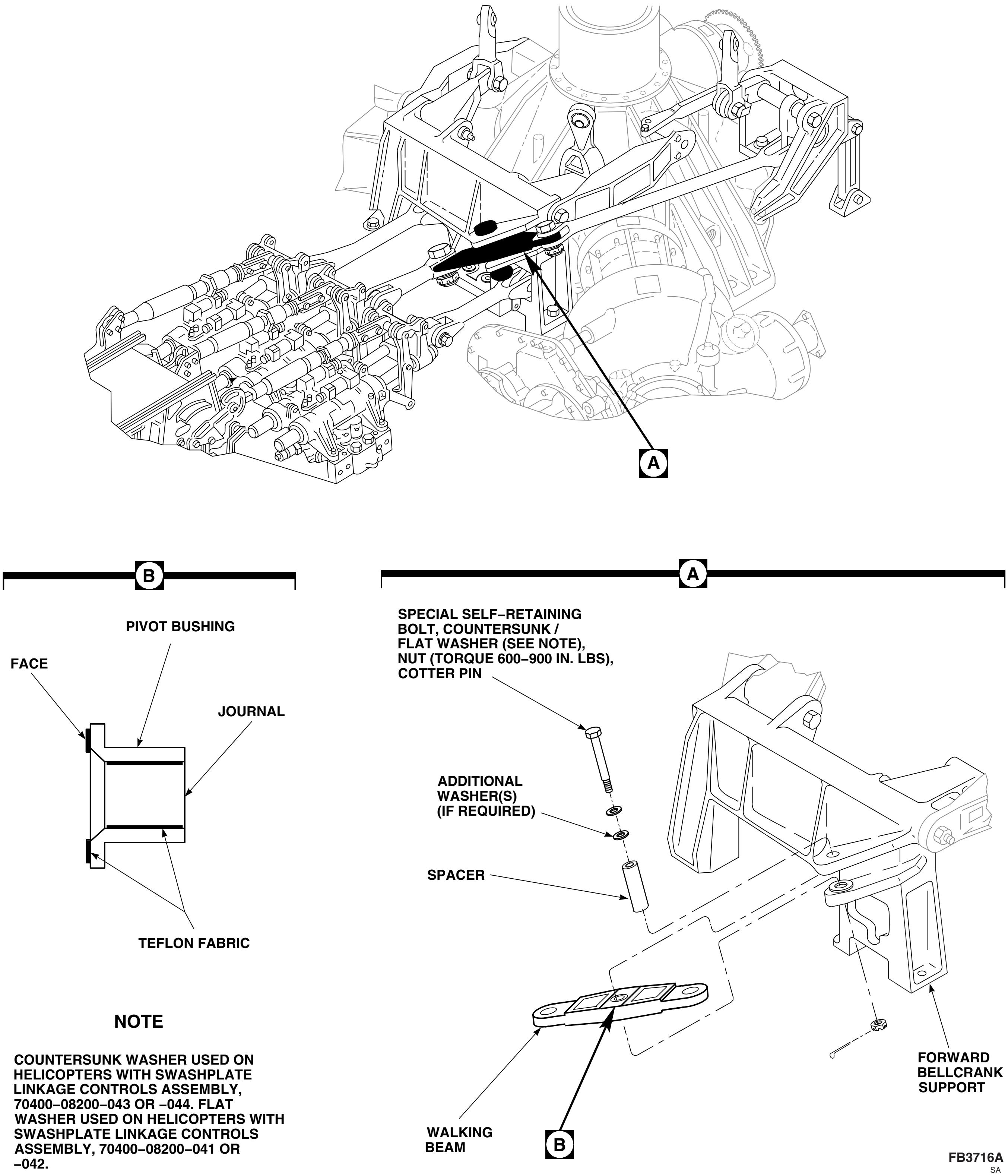
e. Inspect face and journal of walking beam flanged pivot bushings for fraying or Teflon fabric erosion. **NO FRAYING OR EROSION WHICH EXPOSES STRANDS OF TEFLON FABRIC ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

f. Inspect spacer along entire length for wear, using micrometer. **MAXIMUM ALLOWABLE WEAR ON BORE CIRCUMFERENCE SHALL NOT EXCEED 0.015-INCH.** If wear is beyond limit, replace spacer.

g. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-93.1.3. Install.

a. Install replacement walking beam in forward bellcrank support. Install spacer through support and walking beam (Figure 11-4-93-1, Detail A).



NOTE

COUNTERSUNK WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400−08200−043 OR −044. FLAT WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400−08200−041 OR −042.

FIGURE 11-4-93-1. REPLACE WALKING BEAM

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolt are critical characteristics.

CAUTION

Damage to special self-retaining bolt and walking beam will result if bolt is installed incorrectly. To install bolt, insert ⅛-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Add washers, as necessary, to make sure nut is bottomed against support and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolt and countersunk/flat washer (under bolthead). Add washers, as required (maximum of 2 additional washers under countersunk/flat washer) to provide correct grip length and cotter pin position.

c. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.

d. Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Connect aft control rod to walking beam (PARA 11-4-91).

h. Connect long control rod to walking beam (PARA 11-4-94).

i. Make sure area is clean and free of foreign material.

j. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-94 LONG CONTROL ROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 3⁄16-inch
- Crowfoot Attachment
- Protective Covering
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-1
- PARA 11-3-17
- PARA 11-4-120
- PARA 11-5-31

11-4-94.1 Replace Long Control Rod.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-94.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Raise collective stick to raise swashplate.

d. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing long control rod.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

e. Slowly lower collective stick until swashplate rests on wood blocks.

f. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to special self-retaining bolts, long control rod, walking beam, and aft bellcrank will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 3/16-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

g. Remove special self-retaining bolts, countersunk/flat washers, additional washer(s), if installed, nuts, and cotter pins securing long control rod to walking beam and aft bellcrank (Figure 11-4-94-1, Detail A). Remove control rod.

11-4-94.1.2 Inspect.

a. Perform main rotor control rod inspection (PARA 11-3-17).

b. Check bonded nylon washers for disbonding. **DISBONDING MUST NOT EXCEED 25% OF THE BONDED AREA.** If washer is disbonded, perform the following:

(1) If disbond is greater than 25% of the bonded surface area, replace nylon washer (PARA 11-4-120).

(2) If disbond is less than 25% of the bonded surface area, perform the following:

(a) Visually inspect control rod for evidence of corrosion in the disbonded area. **NONE ALLOWED.** If corrosion is found, replace control rod.

(b) Clean around bonding surfaces of nylon washer and control rod using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

(c) Apply sealing compound, Item 292, Appendix D, to the disbonded edges of the nylon washer (PARA 11-4-120).

(d) Annotate logbook that a repetitive 30 day visual inspection be performed for evidence of corrosion. **NONE ALLOWED.** If corrosion is found, replace control rod.

(e) Any control rod repaired while installed on the aircraft must be removed at the next 100 hour special inspection.

c. Inspect control rod bushings for nicks, scratches, corrosion, and wear. **DAMAGE MUST NOT EXCEED 0.010-INCH DEEP OR COVER MORE THAN 30% OF BORE.** If bushing is damaged beyond limits, replace control rod.

d. Inspect control rod bushings for looseness. **NONE ALLOWED.** If bushing is loose, replace control rod.

e. Inspect control rod for loose or missing identification plates. **NONE ALLOWED.** If

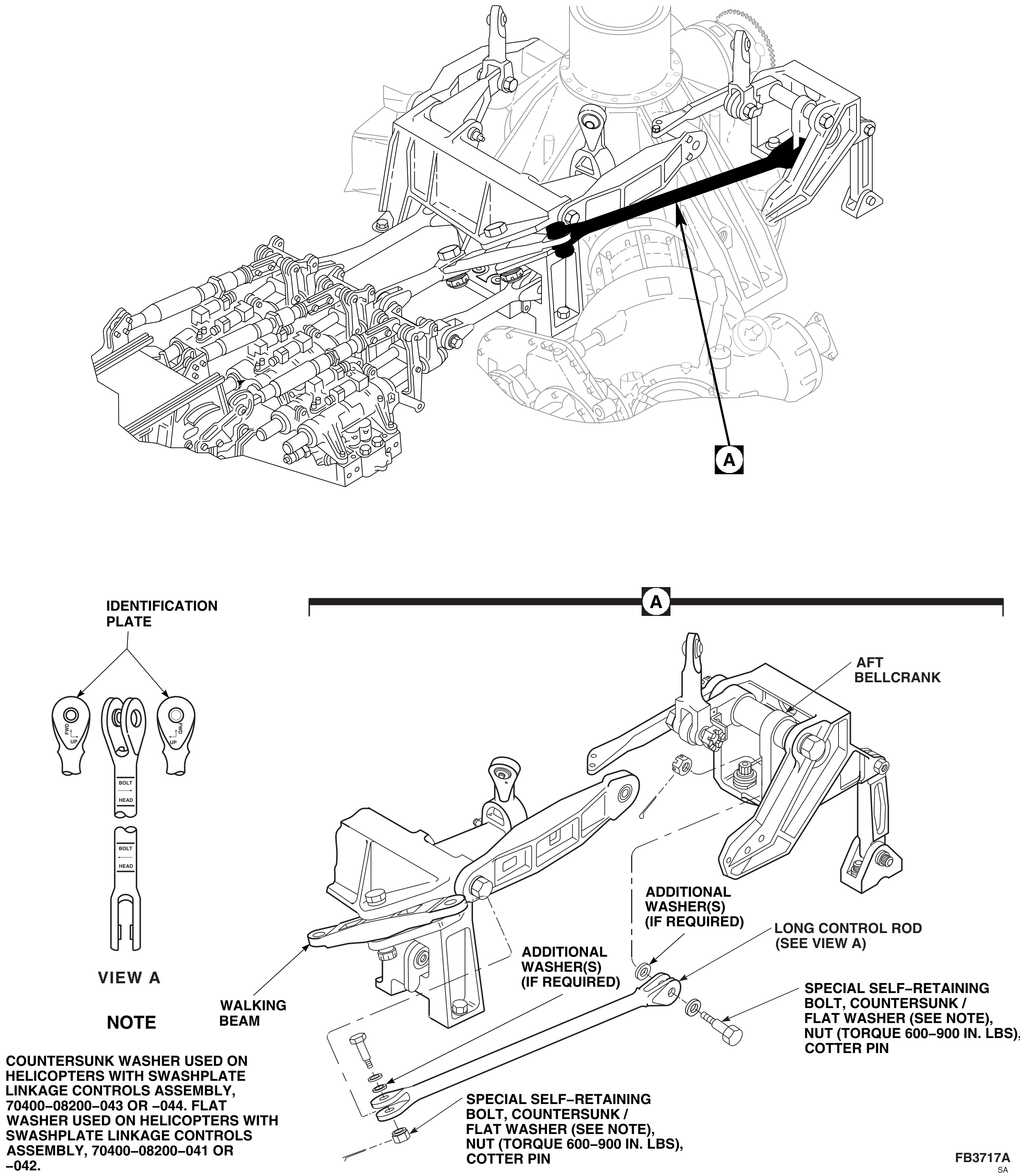


FIGURE 11-4-94-1. REPLACE LONG CONTROL ROD

loose or missing, replace identification plates (PARA 11-5-31).

f. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-94.1.3 Install.

NOTE

- Verify nylon washers are installed and washer orientation is correct (PARA 11-4-120).

- Position control rod according to identification plates on rod (Figure 11-4-94-1, View A).

a. Place replacement long control rod between walking beam and aft bellcrank (Figure 11-4-94-1, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is installed incorrectly. To install bolt, insert 3/16-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Add washers, as necessary, to make sure nut is bottomed against control rod and not on bolt.

- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolts and countersunk/flat washers (under bolthead). To provide correct grip length and cotter pin position, adjust as follows (Figure 11-4-94-1, Detail A):

(1) On forward end of long control rod, add a maximum of 2 washers under countersunk/flat washer, as required.

(2) On aft end of long control rod, add a maximum of 2 washers under nut, as required.

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nuts. **TORQUE NUTS TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pins.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., through step e.

f. Apply torque stripe (PARA 2-6-5).

g. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Raise collective stick and remove wood blocks from between swashplate and main gear box.

i. Perform operational check of flight control system (PARA 11-2-3).

j. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

k. Make sure area is clean and free of foreign material.

l. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-95 SWASHPLATE LINKS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ⅛-inch
- Allen Wrench, ¼-inch
- Crowfoot Attachment
- Protective Covering
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 ½-inch thick

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 5-4-24
- PARA 11-2-3
- PARA 11-3-1
- PARA 11-3-12

11-4-95.1 Replace Swashplate Links.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-95.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Raise collective stick to raise swashplate.

d. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing swashplate links.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

e. Slowly lower collective stick until swashplate rests on wood blocks.

f. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

g. Insert 1/4-inch Allen wrench into center of expandable pin securing swashplate link to stationary swashplate. While holding Allen wrench, remove nut. Do not allow pin to turn (Figure 11-4-95-1, Sheet 1, Detail A). Remove expandable pin.

CAUTION

Damage to special self-retaining bolts, lateral link, and lateral bellcrank will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 1/8-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

h. On lateral link, remove special self-retaining bolt, countersunk/flat washers, additional washer(s), if installed, spacer, nut, and cotter pin securing lateral link to lateral bellcrank (Figure 11-4-95-1, Sheet 2, Detail B). Remove lateral link.

CAUTION

Damage to special self-retaining bolts, forward link, forward bellcrank, aft link, and aft bellcrank will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 1/8-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

i. On forward and aft link, remove special self-retaining bolt, countersunk/flat washers, additional washer(s), if installed, nut, and cotter pin securing link to bellcrank (Figure 11-4-95-1, Sheet 2, Detail B). Remove forward or aft link.

11-4-95.1.2 Inspect.

a. Perform swashplate link inspection (PARA 11-3-12).

b. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

c. Inspect swashplate expandable pin(s) and nut(s) (PARA 5-4-24).

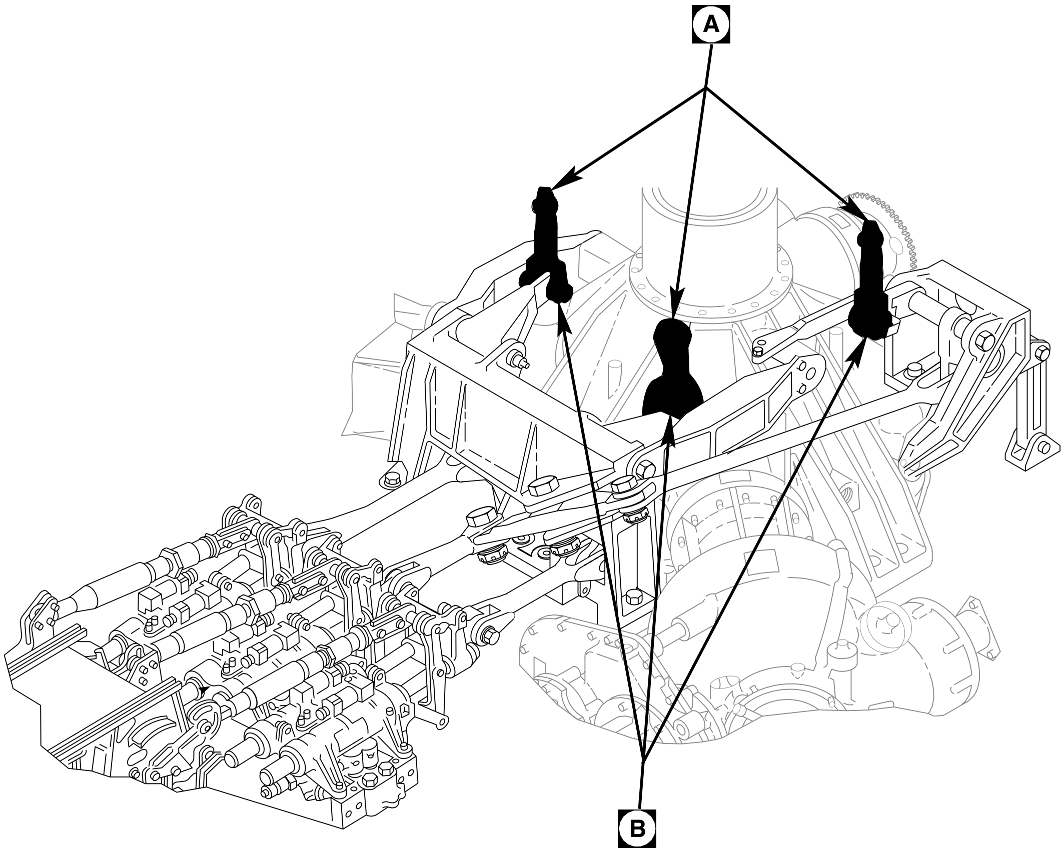
11-4-95.1.3 Install.

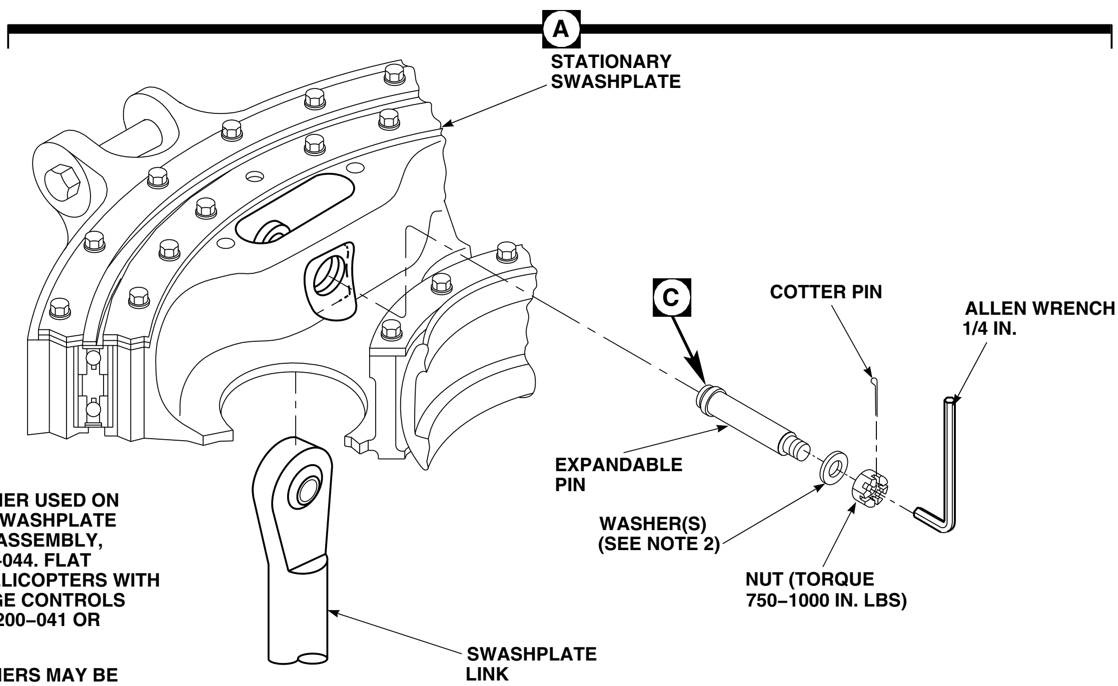
a. Install lateral link as follows (Figure 11-4-95-1, Sheet 2, Detail B):

NOTE

Install lateral link so spacer is installed from outboard side. Spacer should contact bolthead.

(1) Place replacement lateral link on lateral bellcrank. Install spacer in lateral link and bellcrank.





NOTES

1. COUNTERSUNK WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–043 OR –044. FLAT WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–041 OR –042.
2. IF NECESSARY, WASHERS MAY BE ADDED BETWEEN EXISTING WASHER AND NUT TO LINE UP COTTER PIN HOLE IN SWASHPLATE EXPANDABLE PIN WITH SLOT IN NUT.

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FIGURE 11-4-95-1. REPLACE SWASHPLATE LINKS (SHEET 1 OF 2)

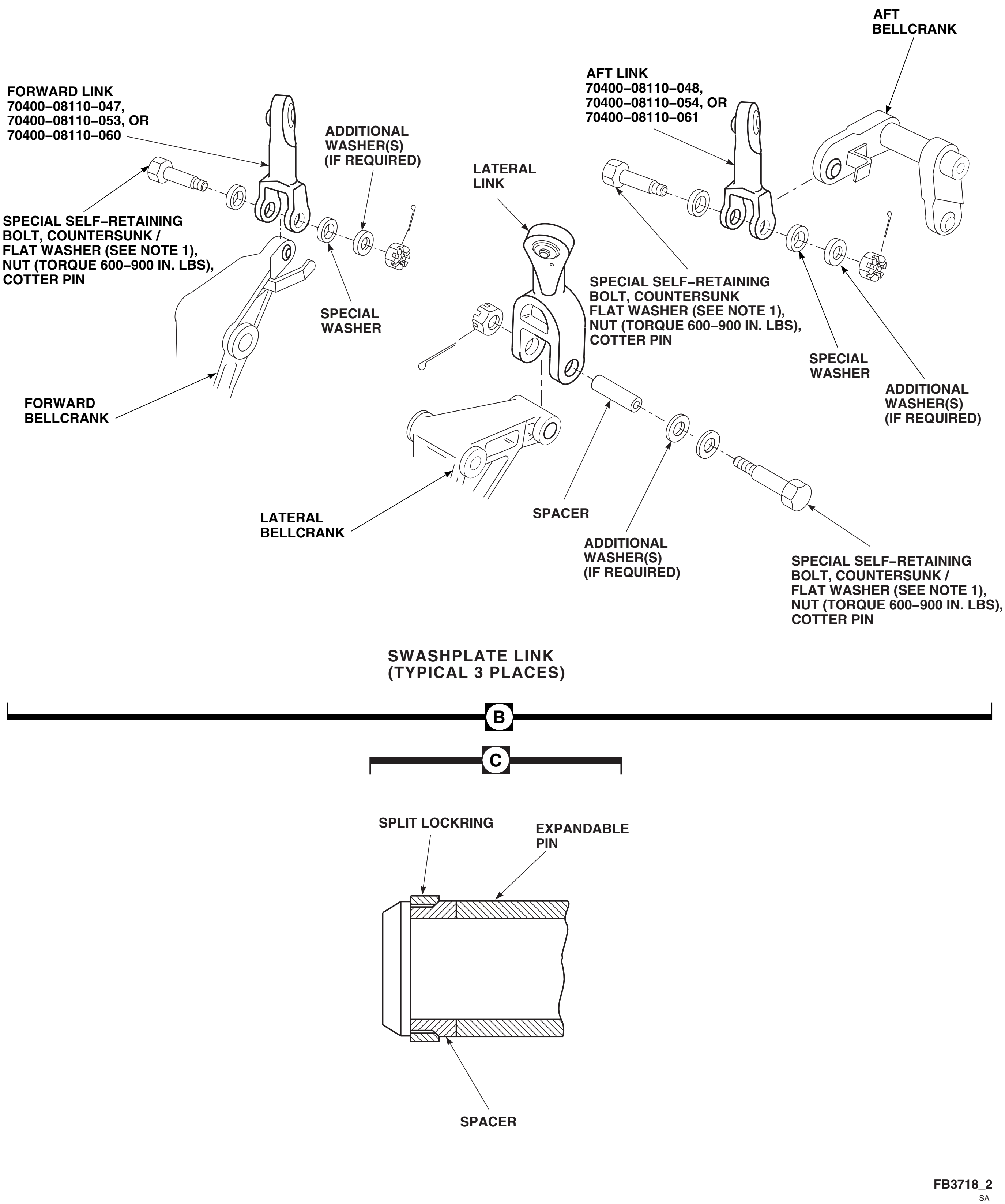


FIGURE 11-4-95-1. REPLACE SWASHPLATE LINKS (SHEET 2 OF 2)

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt, lateral link, and lateral bellcrank will result if bolt is installed incorrectly. To install bolt, insert ⅛-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

(2) Install special self-retaining bolt, countersunk/flat washer (under bolthead). Add washers, as required (maximum of 2 additional washers under countersunk/flat washer) to provide correct grip length and cotter pin position.

(3) Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.

(4) Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

(5) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(a) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(b) Repeat step a. (2), adjusting washer thicknesses or adding washers.

(c) Repeat step a. (3), step a. (4), and step a. (5).

(6) Apply torque stripe (PARA 2-6-5).

b. Install forward or aft link as follows (Figure 11-4-95-1, Sheet 2, Detail B):

WARNING

Injury to personnel and damage to equipment will result if forward and aft links are switched during installation. The forward and aft links look almost the same. To prevent damage to flight controls, make sure forward link is 70400-08110-047, 70400-08110-053, or 70400-08110-060, and aft link is 70400-08110-048, 70400-08110-054, or 70400-08110-061.

NOTE

To make sure link is oriented correctly during installation, the link should be positioned so that shouldered bushing in link is on same side as nut.

(1) Place replacement forward or aft link on forward or aft bellcrank.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

- Damage to equipment will result if more than one washer is added under bolthead when installing forward or aft link. If additional washers are required to seat nut correctly on link, they must be added under the nut.
- Damage to special self-retaining bolt, link, and bellcrank will result if bolt is installed incorrectly. To install bolt, insert 1/8-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

(2) Install special self-retaining bolt, countersunk/flat washer (under bolthead), and special washer. Add washers, as required (maximum of 2 additional washers under nut) to provide correct grip length and cotter pin position.

(3) Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

(4) Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

(5) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If either bolt rotates, perform the following:

(a) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(b) Repeat step b. (2), adjusting washer thicknesses or adding washers.

(c) Repeat step b. (3), step b. (4), and step b. (5).

(6) Apply torque stripe (PARA 2-6-5).

c. Install washer and nut on expandable pin, back off nut of expandable pin until split lockring segments are loose on pin (Figure 11-4-95-1, Sheet 1, Detail A and Sheet 2, Detail C).

NOTE

If necessary, add washers between existing washer and swashplate expandable pin nut, to line up cotter pin hole in swashplate expandable pin with slot on swashplate expandable pin nut.

d. Position swashplate link in swashplate and insert expandable pin through stationary swashplate and link. Make sure washer under nut is flush with inner side of swashplate.

e. **TIGHTEN NUT ON EXPANDABLE PIN HANDTIGHT.**

WARNING

FSCAP

- Verification of correct installation of swashplate expandable pin split lockring and male spacer is a critical characteristic.
- Verification of proper hardware installation and torque of swashplate expandable pin nut is a critical characteristic.

f. Insert ¼-inch Allen wrench into center of pin. **HOLD ALLEN WRENCH AND TORQUE NUT TO 750 - 1000 INCH-POUNDS** (Figure 11-4-95-1, Sheet 1, Detail A). Do not allow pin to turn.

g. **CHECK THAT CLEARANCE UNDER WASHER IS NOT MORE THAN 0.002-INCH.** This means that bolt is locked in place.

h. Install cotter pin.

i. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

j. Raise collective stick and remove wood blocks from between swashplate and main transmission.

k. Perform operational check of flight control system (PARA 11-2-3).

l. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

m. Make sure area is clean and free of foreign material.

n. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-96 FORWARD BELLCRANK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, 1/8-inch
- Crowfoot Attachment
- Micrometer
- Nonmetallic Drift
- Nonmetallic Punch
- Protective Covering
- Spacer Removal Tool, Locally-Made, Figure H-219, Appendix H
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

References

- Appendix H
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-1
- PARA 11-3-21
- PARA 11-4-90
- PARA 11-4-95
- PARA 11-4-100
- PARA 11-5-25

11-4-96.1 Replace Forward Bellcrank.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-96.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Disconnect forward control rod from forward bellcrank (PARA 11-4-90).

c. Disconnect forward swashplate link from forward bellcrank (PARA 11-4-95).

CAUTION

Damage to special self-retaining bolt, forward bellcrank, and forward bellcrank support will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 1/8-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

d. Remove special self-retaining bolt, countersunk/flat washer, additional washer(s), if installed, nut, and cotter pin securing forward bellcrank to forward bellcrank support. Remove spacer and remove bellcrank. If eccentric bushings come out with bolt, put them back in support (Figure 11-4-96-1, Detail A).

e. If spacer does not slide out with bolt, remove spacer as follows:

NOTE

If spacer removal tool is not available, it may be necessary to tap out spacer using nonmetallic punch.

(1) Locally make spacer removal tool (Figure H-219, Appendix H).

(2) Insert locally-made spacer removal tool into bolthead end of spacer.

(3) **TORQUE TOOL TO 200 - 250 INCH-POUNDS.**

(4) Tap out spacer using nonmetallic drift against underside of removal tool.

11-4-96.1.2 Inspect.

a. Perform forward bellcrank inspection (PARA 11-3-21).

b. Visually inspect forward bellcrank support for damaged or worn pivot bushings (PARA 11-4-100).

c. Inspect bellcrank flanged pivot bushings for cracks or evidence of corrosion. **NONE ALLOWED.** If cracked, or evidence of corrosion is found, replace bushing(s) (PARA 11-5-25).

d. Inspect face and journal of bellcrank flanged pivot bushing for rips, tears, or disbonding of Teflon fabric (Figure 11-4-96-1, Detail B). **NONE ALLOWED.** If rips, tears, or disbonding is found, replace bushing(s) (PARA 11-5-25).

e. Inspect face and journal of bellcrank flanged pivot bushing for fraying or Teflon fabric erosion. **NO FRAYING OR EROSION WHICH EXPOSES STRANDS OF TEFLON FABRIC ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

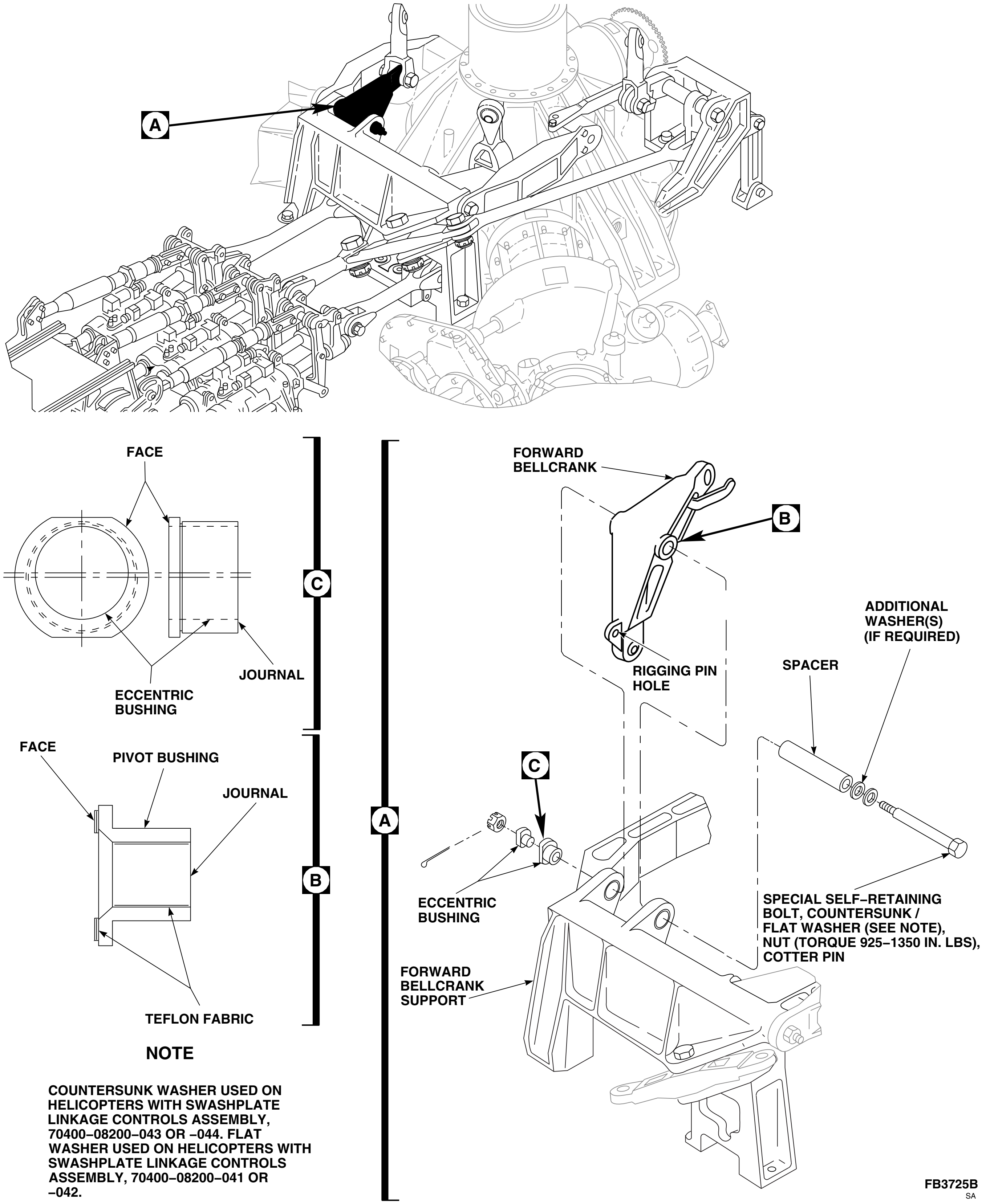
f. Inspect face and journal of eccentric bushings for damage (Figure 11-4-96-1, Detail C). **MAXIMUM ALLOWABLE DAMAGE TO THESE SURFACES SHALL NOT EXCEED 0.015-INCH.** Inspect inside and outside diameter of eccentric bushings for damage. **NONE ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

g. Inspect spacer along entire length, using micrometer. **MAXIMUM ALLOWABLE WEAR ON BORE CIRCUMFERENCE SHALL NOT EXCEED 0.015-INCH.**

h. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-96.1.3 Install.

a. Place replacement forward bellcrank in forward bellcrank support with long arm (with



NOTE

COUNTERSUNK WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–043 OR –044. FLAT WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–041 OR –042.

FIGURE 11-4-96-1. REPLACE FORWARD BELLCRANK rigging pin hole) down. Install spacer through inboard side of support (Figure 11-4-96-1, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt, forward bellcrank, and forward bellcrank support will result if bolt is installed incorrectly. To install bolt, insert 1/8-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Turn eccentric bushings as necessary to line up boltholes.
- Add washers, as necessary, to make sure nut is bottomed against support and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolt and countersunk/flat washer (under bolthead) through eccentric bushings and bellcrank. Add washers, as required (maximum of 2 additional washers under countersunk/flat washer) to provide correct grip length and cotter pin position.

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nut. **TORQUE NUT TO 925 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 1350 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 630 - 650 INCH-POUNDS** in tightening direction. If bolt rotates, perform the following:

   (1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

   (2) Repeat step b., adjusting washer thicknesses or adding washers.

   (3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Connect forward control rod to forward bellcrank (PARA 11-4-90).

h. Connect forward swashplate link to forward bellcrank (PARA 11-4-95).

i. Make sure area is clean and free of foreign material.

j. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-97 LATERAL BELLCRANK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ⅛-inch
- Crowfoot Attachment
- Micrometer
- Nonmetallic Drift
- Nonmetallic Punch
- Protective Covering
- Spacer Removal Tool, Locally-Made, Figure H-219, Appendix H
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

References

- Appendix H
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-1
- PARA 11-3-22
- PARA 11-4-92
- PARA 11-4-95
- PARA 11-4-100
- PARA 11-5-25

11-4-97.1 Replace Lateral Bellcrank.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-97.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Disconnect lateral control rod from lateral bellcrank (PARA 11-4-92).

c. Disconnect lateral swashplate link from lateral bellcrank (PARA 11-4-95).

CAUTION

Damage to special self-retaining bolt, lateral bellcrank, and forward bellcrank support will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 1/8-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

d. Remove special self-retaining bolt, countersunk/flat washer, additional washer(s), if installed, nut, and cotter pin securing lateral bellcrank to forward bellcrank support. Remove spacer and remove bellcrank. If eccentric bushings come out with bolt, put them back in support (Figure 11-4-97-1, Detail A).

e. If spacer does not slide out with bolt, remove spacer as follows:

NOTE

If spacer removal tool is not available, it may be necessary to tap out spacer using nonmetallic punch.

(1) Locally make spacer removal tool (Figure H-219, Appendix H).

(2) Insert locally-made spacer removal tool into bolthead end of spacer.

(3) **TORQUE TOOL TO 200 - 250 INCH-POUNDS.**

(4) Tap out spacer using nonmetallic drift against underside of removal tool.

11-4-97.1.2 Inspect.

a. Perform lateral bellcrank inspection (PARA 11-3-22).

b. Visually inspect forward bellcrank support for damaged or worn pivot bushings (PARA 11-4-100).

c. Inspect bellcrank flanged pivot bushings for cracks or evidence of corrosion. **NONE ALLOWED.** If cracked, or evidence of corrosion is found, replace bushing(s) (PARA 11-5-25).

d. Inspect face and journal of bellcrank flanged pivot bushing for rips, tears, or disbonding of Teflon fabric (Figure 11-4-97-1, Detail B). **NONE ALLOWED.** If rips, tears, or disbonding is found, replace bushing(s) (PARA 11-5-25).

e. Inspect face and journal of bellcrank flanged pivot bushing for fraying or Teflon fabric erosion. **NO FRAYING OR EROSION WHICH EXPOSES STRANDS OF TEFLON FABRIC ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

f. Inspect face and journal of eccentric bushings for damage (Figure 11-4-97-1, Detail C). **MAXIMUM ALLOWABLE DAMAGE TO THESE SURFACES SHALL NOT EXCEED 0.015-INCH.** Inspect inside and outside diameter of eccentric bushings for damage. **NONE ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

g. Inspect spacer along entire length, using micrometer. **MAXIMUM ALLOWABLE WEAR ON BORE CIRCUMFERENCE SHALL NOT EXCEED 0.015-INCH.**

h. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-97.1.3 Install.

WARNING

- **701657–SUBQ** Injury to personnel and damage to equipment will result if improved components are mixed with older parts. The improved components are as follows:

  - Lateral Bellcrank, 70400-08166-041
  - Right Tie Rod, 70400-08165-043
  - Aft Bellcrank Support, 70400-08163-043

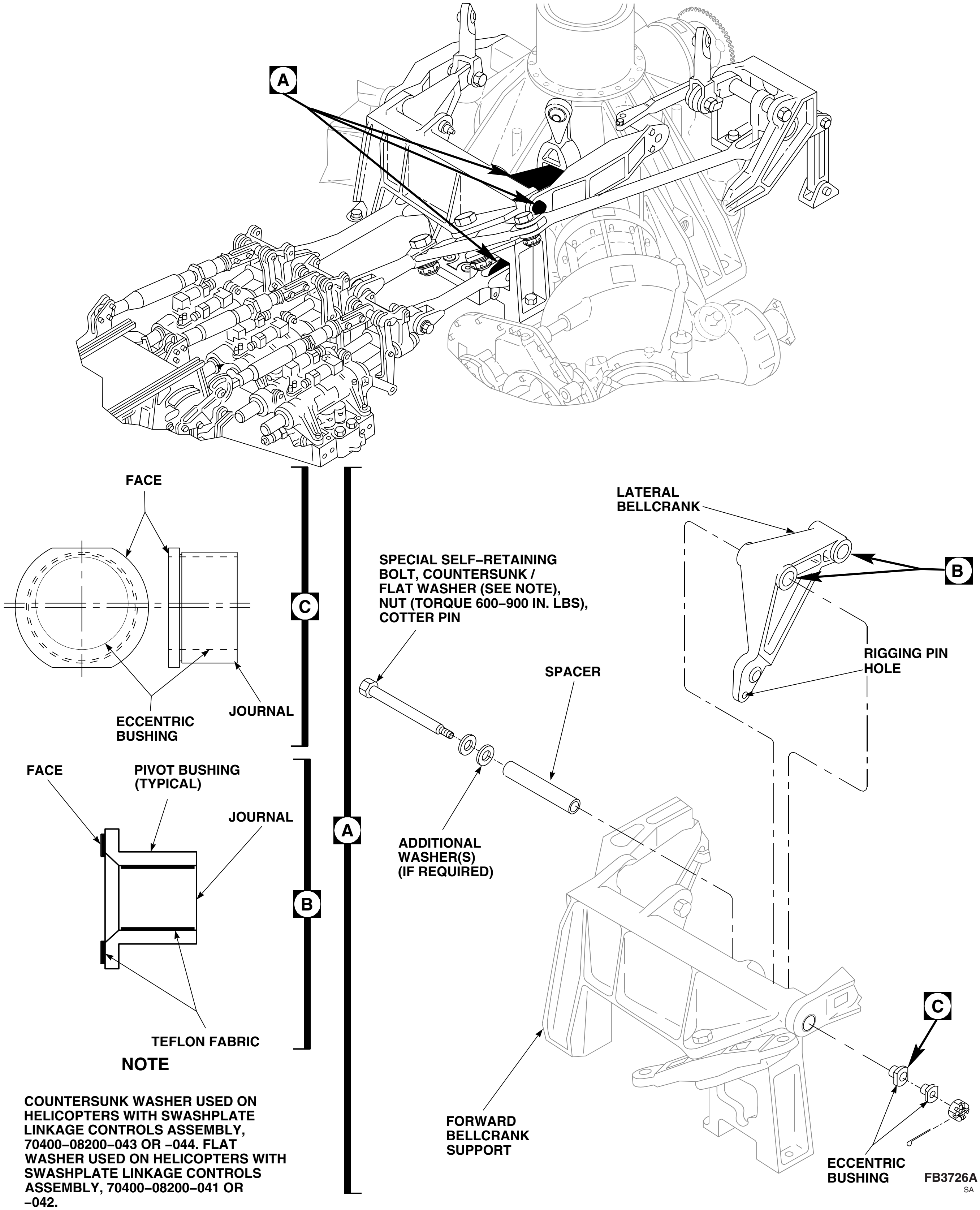


FIGURE 11-4-97-1. REPLACE LATERAL BELLCRANK

- Aft Tie Rod, 70400-08111-043
- Aft Tie Rod Support Fitting, 70400-08112-046

• Do not mix these parts with older parts.

a. Place replacement lateral bellcrank in forward bellcrank support with long arm (with rigging pin hole) down. Install spacer through inboard side of support (Figure 11-4-97-1, Detail A).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt, lateral bellcrank, and forward bellcrank support will result if bolt is installed incorrectly. To install bolt, insert ⅛-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

• Turn eccentric bushings as necessary to line up boltholes.

• Add washers, as necessary, to make sure nut is bottomed against support and not on bolt.

• Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

• Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolt and countersunk/flat washer (under bolthead) through eccentric bushings and bellcrank. Add washers, as required (maximum of 2 additional washers under countersunk/flat washer) to provide correct grip length and cotter pin position.

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Connect lateral control rod to lateral bellcrank (PARA 11-4-92).

h. Connect lateral swashplate link to lateral bellcrank (PARA 11-4-95).

i. Make sure area is clean and free of foreign material.

j. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-98 AFT BELLCRANK.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ⅛-inch
- Crowfoot Attachment
- Micrometer
- Nonmetallic Drift
- Nonmetallic Punch
- Protective Covering
- Spacer Removal Tool, Locally-Made, Figure H-219, Appendix H
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

References

- Appendix H
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-3-1
- PARA 11-3-18
- PARA 11-3-20
- PARA 11-4-94
- PARA 11-4-95
- PARA 11-5-25

11-4-98.1 Replace Aft Bellcrank.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-98.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Disconnect long control rod from aft bellcrank (PARA 11-4-94).

c. Disconnect aft swashplate link from aft bellcrank (PARA 11-4-95).

CAUTION

Damage to special self-retaining bolt, aft bellcrank, and aft bellcrank support will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use 1⁄8-inch Allen wrench to push pin in bolt-head. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

d. Remove special self-retaining bolt, countersunk/flat washer, additional washer(s), if installed, nut, and cotter pin securing aft bellcrank to aft bellcrank support. Remove spacer and remove bellcrank. If eccentric bushings come out with bolt, put them back in support (Figure 11-4-98-1, Detail A).

e. If spacer does not slide out with bolt, remove spacer as follows:

NOTE

If spacer removal tool is not available, it may be necessary to tap out spacer using nonmetallic punch.

(1) Locally make spacer removal tool (Figure H-219, Appendix H).

(2) Insert spacer removal tool into bolthead end of spacer.

(3) **TORQUE TOOL TO 200 - 250 INCH-POUNDS.**

(4) Tap out spacer using nonmetallic drift against underside of removal tool.

11-4-98.1.2 Inspect.

a. Perform aft bellcrank inspection (PARA 11-3-20).

b. Visually inspect aft bellcrank support for damaged or worn pivot bushings (PARA 11-3-18).

c. Inspect bellcrank flanged pivot bushings for cracks or evidence of corrosion. **NONE ALLOWED.** If cracked, or evidence of corrosion is found, replace bushing(s) (PARA 11-5-25).

d. Inspect face and journal of bellcrank flanged pivot bushing for rips, tears, or disbonding of Teflon fabric (Figure 11-4-98-1, Detail B). **NONE ALLOWED.** If rips, tears, or disbonding is found, replace bushing(s) (PARA 11-5-25).

e. Inspect face and journal of bellcrank flanged pivot bushing for fraying or Teflon fabric erosion. **NO FRAYING OR EROSION WHICH EXPOSES STRANDS OF TEFLON FABRIC ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

f. Inspect face and journal of eccentric bushings for damage (Figure 11-4-98-1, Detail C). **MAXIMUM ALLOWABLE DAMAGE TO THESE SURFACES SHALL NOT EXCEED 0.015-INCH.** Inspect inside and outside diameter of eccentric bushings for damage. **NONE ALLOWED.** If damage is beyond limit, replace bushing(s) (PARA 11-5-25).

g. Inspect spacer along entire length, using micrometer. **MAXIMUM ALLOWABLE WEAR ON BORE CIRCUMFERENCE SHALL NOT EXCEED 0.015-INCH.**

h. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-98.1.3 Install.

a. Place aft bellcrank in aft bellcrank support. Position so that arm (with retainer) is facing forward, with retainer facing outboard. Install spacer through outboard side of support (Figure 11-4-98-1, Detail A).

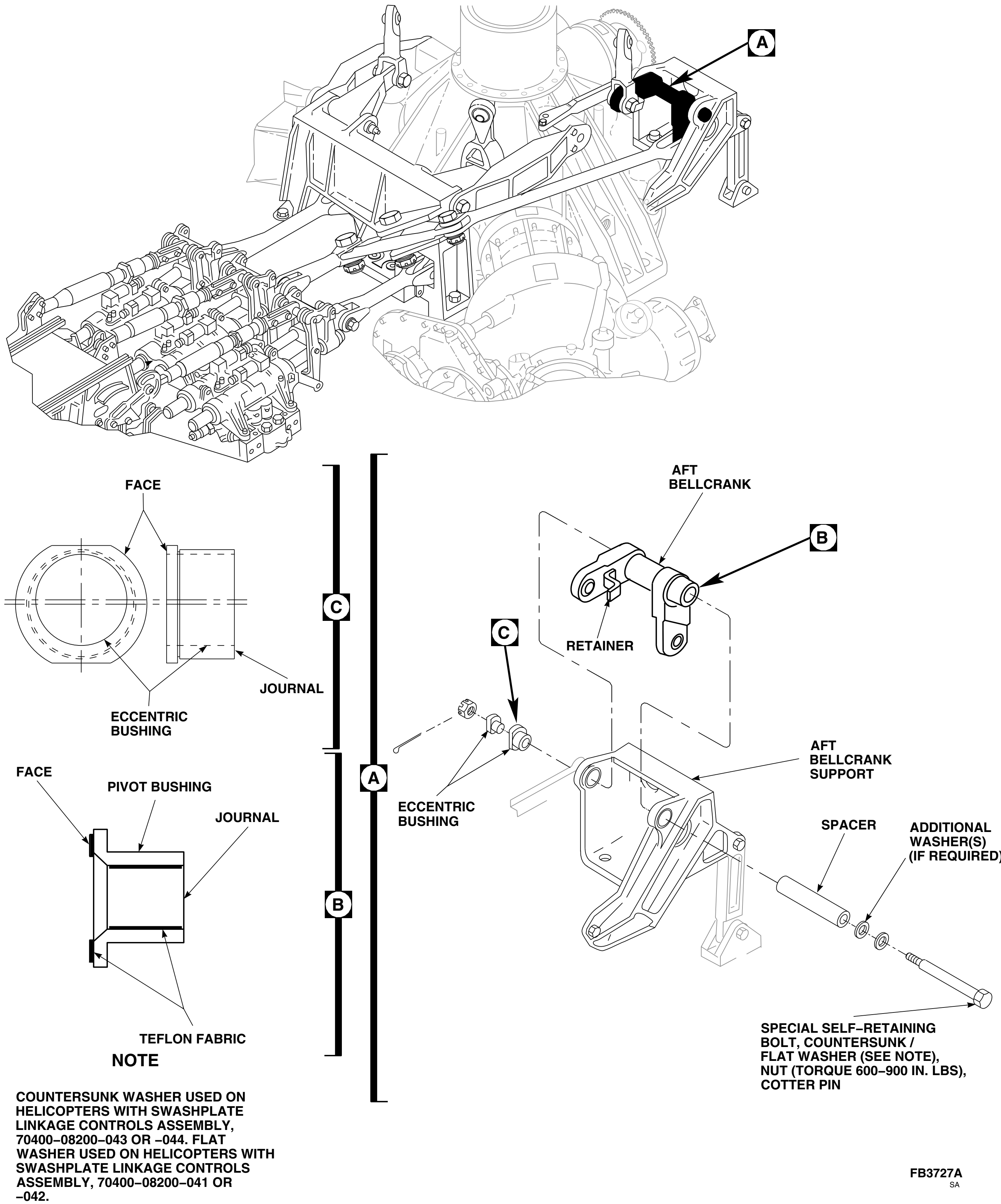


FIGURE 11-4-98-1. REPLACE AFT BELLCRANK

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt, aft bellcrank, and aft bellcrank support will result if bolt is installed incorrectly. To install bolt, insert ⅛-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Turn eccentric bushings as necessary to line up boltholes.
- Add washers, as necessary, to make sure nut is bottomed against support and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

b. Install special self-retaining bolt and countersunk/flat washer (under bolthead) through eccentric bushings and bellcrank. Add washers, as required (maximum of 2 additional washers under countersunk/flat washer) to provide correct grip length and cotter pin position.

c. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup.

d. Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

e. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step b., adjusting washer thicknesses or adding washers.

(3) Repeat step c., step d., and step e.

f. Apply torque stripe (PARA 2-6-5).

g. Connect long control rod to aft bellcrank (PARA 11-4-94).

h. Connect aft swashplate link to aft bellcrank (PARA 11-4-95).

i. Make sure area is clean and free of foreign material.

j. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-99 RIGHT AND LEFT TIE RODS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ⅛-inch
- Protective Covering
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-16
- PARA 11-4-1
- PARA 11-5-25

11-4-99.1 Replace Right and Left Tie Rods.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

NOTE

Replacement of left and right tie rods is the same, except as noted.

11-4-99.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Raise collective stick to raise swashplate.

d. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing tie rod.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

e. Slowly lower collective stick until swashplate rests on wood blocks.

f. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

g. Remove bolts, countersunk washers, and nylon washers securing aft end of tie rod to tie rod dowel pin on main transmission (Figure 11-4-99-1, Sheet 1, Detail A and Sheet 2, Detail C).

CAUTION

Damage to special self-retaining bolt, tie rod, and forward bellcrank support will result if bolt is withdrawn incorrectly. To remove bolt, first remove nut. Use ⅛-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while withdrawing bolt.

h. Remove nut securing forward end of tie rod to special self-retaining bolt on forward bellcrank support. Withdraw special self-retaining bolt enough to clear tie rod. Leave eccentric bushings in tie rod.

i. Pull tie rod straight outboard being careful not to damage tie rod dowel pin, tie rod, or main transmission. Remove insulator from dowel pin on main transmission.

11-4-99.1.2 Inspect.

a. Inspect tie rod lug and surface areas (PARA 11-3-16).

b. Inspect face and journal of eccentric bushings for damage (Figure 11-4-96-1, Sheet 2, Detail B). **MAXIMUM ALLOWABLE DAMAGE TO THESE SURFACES SHALL NOT EXCEED 0.015-INCH.** Inspect inside and outside diameter of eccentric bushings for damage. **NONE ALLOWED.** If damage is beyond limits, replace bushing(s) (PARA 11-5-25).

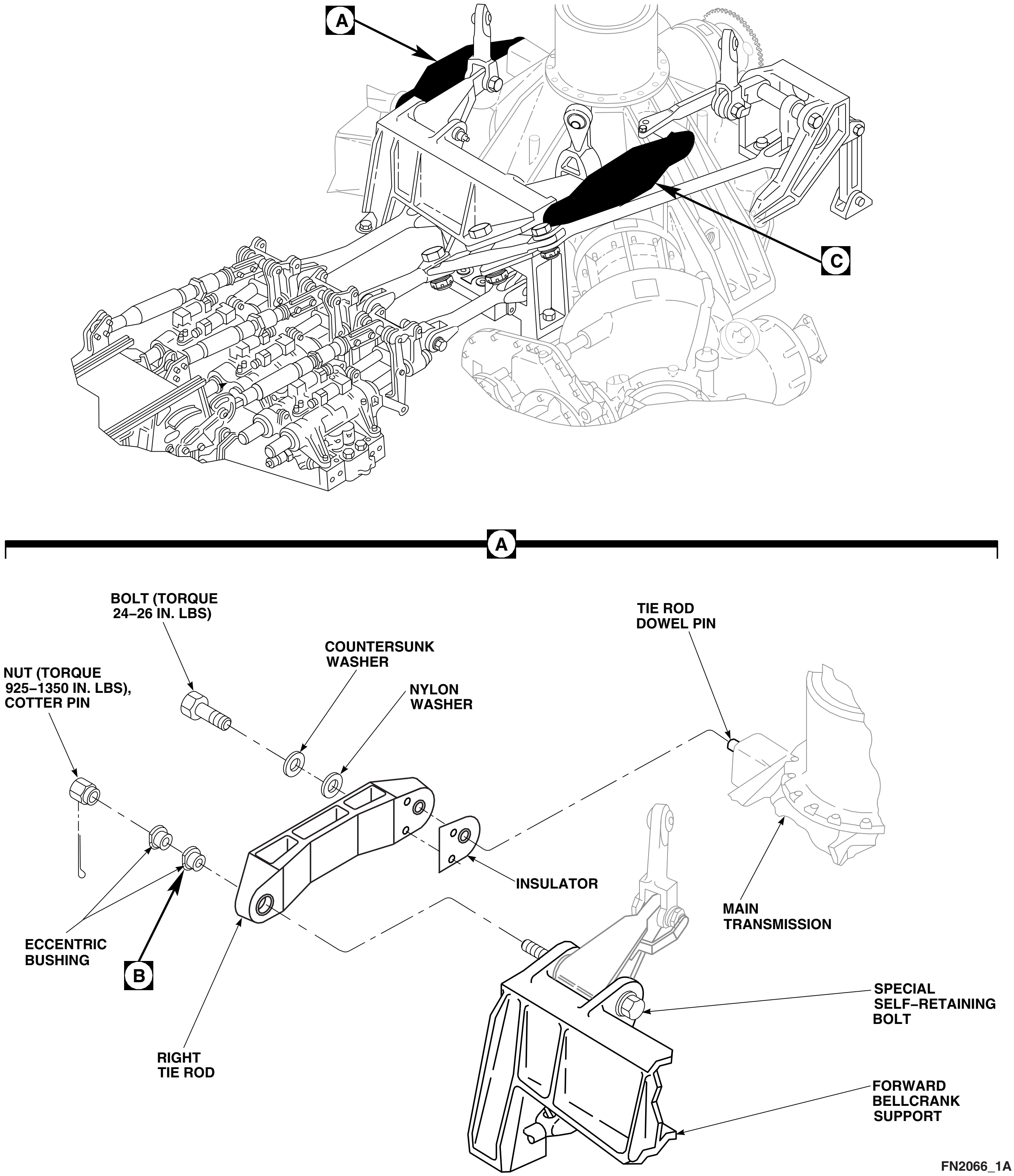
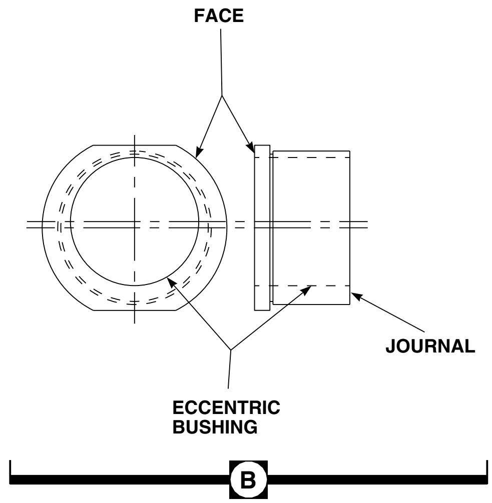


FIGURE 11-4-99-1. REPLACE RIGHT AND LEFT TIE RODS (SHEET 1 OF 2)



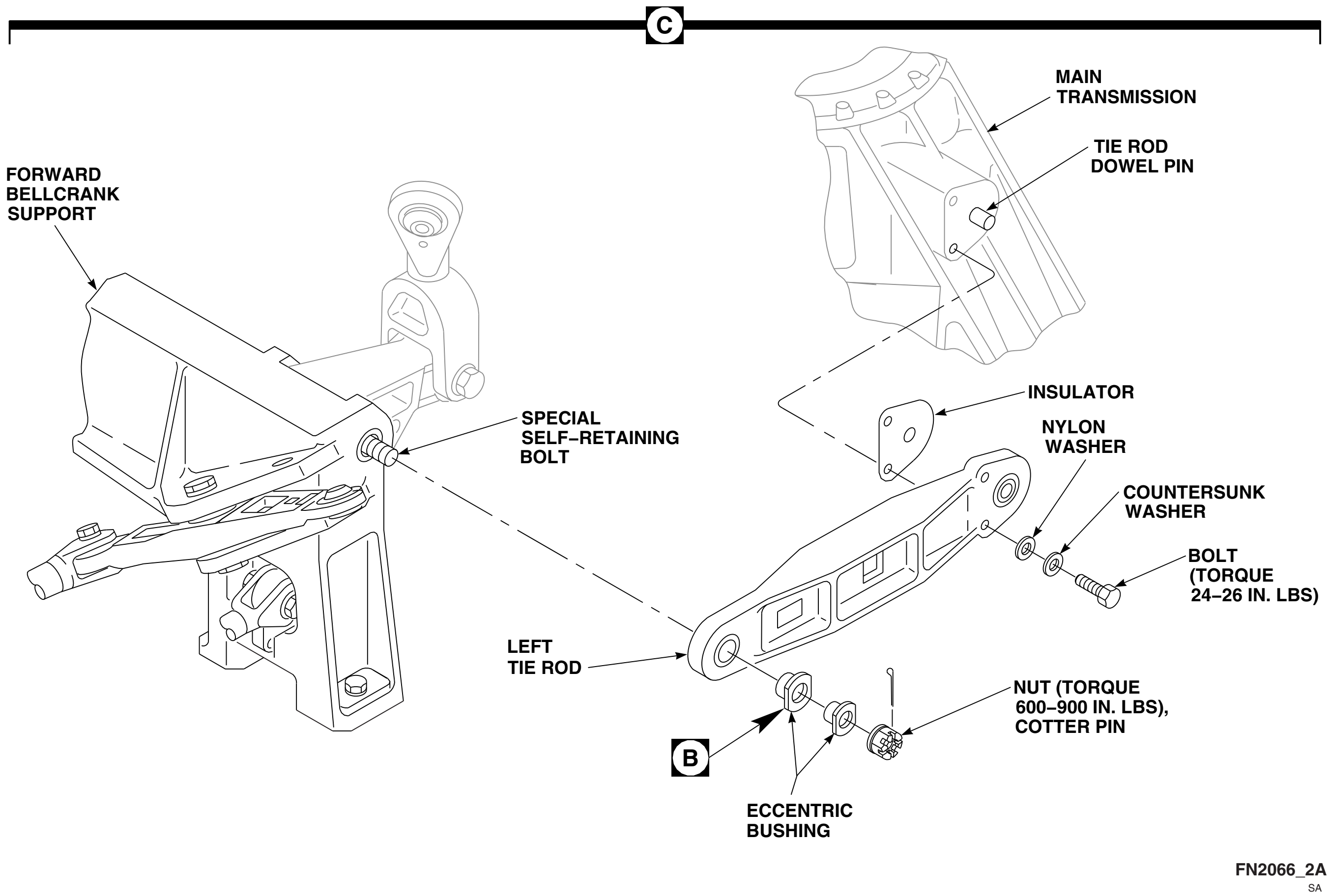


FIGURE 11-4-99-1. REPLACE RIGHT AND LEFT TIE RODS (SHEET 2 OF 2)

11-4-99.1.3 Install.

WARNING

- 701657–SUBQ Injury to personnel and damage to equipment will result if improved components are mixed with older parts. The improved components are as follows:
  - Lateral Bellcrank, 70400-08166-041
  - Right Tie Rod, 70400-08165-043
  - Aft Bellcrank Support, 70400-08163-043
  - Aft Tie Rod, 70400-08111-043
  - Aft Tie Rod Support Fitting, 70400-08112-046
- Do not mix these parts with older parts.

a. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to tie rod dowel pin. Install tie rod while primer is still wet.

CAUTION

Damage to special self-retaining bolt and tie rod will result if retaining balls are not released prior to installing tie rod. To release retaining balls, use ⅛-inch Allen wrench to push pin in bolthead. This will release retaining balls on threaded end of bolt so tie rod can be installed. Keep pressure on pin while installing tie rod.

NOTE

Turn eccentric bushings as necessary to line up boltholes.

b. Position aft end of tie rod and insulator on tie rod dowel pin on main transmission, and position forward end of tie rod over special self-retaining bolt (Figure 11-4-99-1, Sheet 1, Detail A and Sheet 2, Detail C).

c. Clean bolt using cleaning cloth, Item 102, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Inspect bolt as follows:

(1) Inspect bolt for evidence of corrosion in the area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt.

(2) Visually inspect bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack(s) is visible, replace bolt.

CAUTION

If resistance or binding of the retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free the mechanism. The inner core of the bolt is corroded and must be rejected.

d. Release retaining balls by holding retaining pin in. If resistance or binding of the retaining ball mechanism is observed when pin is actuated, discard bolt. If no resistance or binding of the retaining ball mechanism is observed when pin is actuated, apply corrosion preventive compound, Item 114, Appendix D, to bolt as follows:

(1) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.

(2) While working retaining pin, spray corrosion preventive compound, Item 114, Appendix D, into hole in bolthead and hole in end of bolt.

(3) Wipe excessive corrosion preventive compound from bolt using cleaning cloth,

Item 102, Appendix D, prior to installation.

(4) Apply corrosion preventive compound, Item 109, Appendix D, to the exterior retaining ball area and threads.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

e. Check bolts for free axial motion and that impedance feature is not restrained in part. If either bolt binds or impedance feature is depressed, change washer stackup under bolthead (maximum 3 washers under bolthead).

f. If installing right tie rod, perform the following (Figure 11-4-99-1, Sheet 1, Detail A):

(1) Install nut on right tie rod.

(2) **TORQUE NUT TO 925 - 1350 INCH-POUNDS.**

(3) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 630 - 650 INCH-POUNDS** in tightening direction. If bolt rotates, adjust by adding washers (PARA 11-4-1) and reinstall, repeating step f. Install cotter pin.

(4) Apply torque stripe (PARA 2-6-5).

g. If installing left tie rod, perform the following (Figure 11-4-99-1, Sheet 2, Detail C):

(1) Install nut on left tie rod.

(2) **TORQUE NUT TO 600 - 900 INCH-POUNDS.**

(3) **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If bolt rotates, adjust by adding washers (PARA 11-4-1) and reinstall, repeating step g. Install cotter pin.

(4) Apply torque stripe (PARA 2-6-5).

h. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer, to bolthead, shank, and area under bolthead of tie rod mounting bolts. Allow primer to dry.

i. Lubricate bolthead and shank with corrosion preventive compound, Item 109, Appendix D. Place countersunk washer and nylon washer on bolt. Make sure countersunk side of washer faces bolthead.

j. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer, to bolt threads. With threads still wet, install tie rod to main transmission with bolts, countersunk washers, and nylon washers. **TORQUE BOLTS TO 24 - 26 INCH-POUNDS.** Wipe off excess primer with machinery towel, Item 211, Appendix D. Using lockwire, Item 197, Appendix D, lockwire boltheads together.

k. Apply a bead of sealing compound, Item 292, Appendix D, around bolthead and washers.

l. Cover exposed end of tie rod dowel pin with epoxy coating, using epoxy coating kit, Item 126A, Appendix D.

m. Apply a thin coat of sealing compound, Item 292, Appendix D, around gap between transmission mounting surface and tie rod.

n. Make sure area is clean and free of foreign material.

o. Install main rotor pylon transmission cover (PARA 2-4-107).

p. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

q. Raise collective stick and remove wood blocks from between swashplate and main transmission.

r. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

s. Perform operational check of flight control system (PARA 11-2-3).

11-4-100 FORWARD BELLCRANK SUPPORT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Torque Wrench, 100 - 500 ft lbs

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cap Plug, NAS813-14
- Cap Plug, NAS813-20
- Corrosion Compound, Item 108, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Polyurethane Coating, Item 240, Appendix D
- Sealing Compound, Item 287, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-7-20
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-15
- PARA 11-4-4
- PARA 11-4-90
- PARA 11-4-91
- PARA 11-4-92
- PARA 11-4-93
- PARA 11-4-94
- PARA 11-4-95

- PARA 11-4-96
- PARA 11-4-97
- PARA 11-4-98
- PARA 11-4-99

11-4-100.1 Replace Forward Bellcrank Support.

11-4-100.1.1 Remove.

a. Remove forward, aft, lateral, and long control rods (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, and PARA 11-4-94).

b. Remove lateral, forward, and aft swashplate links (PARA 11-4-95).

c. Remove walking beam (PARA 11-4-93).

d. Remove lateral bellcrank (PARA 11-4-97).

e. Remove forward bellcrank (PARA 11-4-96).

f. Remove aft bellcrank (PARA 11-4-98).

g. Remove left and right tie rods (PARA 11-4-99).

NOTE

Do not remove insulator from main transmission unless it becomes disbonded when removing forward bellcrank support.

h. Remove bolts, countersunk washers, nylon washers, washers, and nuts securing 70583–701213 spring clip bushing retainer or 701214–SUBQ dowel pin retaining zee to forward bellcrank support tongue. Remove 70583–701213 spring clip bushing retainer or 701214–SUBQ dowel pin retaining zee and eccentric bushings from forward bellcrank support. If insulator is disbonded from main transmission, remove insulator using nonmetallic scraper (Figure 11-4-100-1, Sheet 1, Detail A).

i. Remove bolts and countersunk washers securing forward bellcrank support to forward bellcrank support beam (Figure 11-4-100-1, Sheet 2, Detail B and Detail C). Remove support by lifting straight up off forward dowel pin.

j. Clean primer from bolts, 70583–701213 spring clip bushing retainer or 701214–SUBQ dowel pin retaining zee, and areas on main transmission that mate with forward bellcrank support using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

k. Using nonmetallic scraper, clean any remaining sealing compound from main transmission.

11-4-100.1.2 Inspect/Repair.

a. Visually inspect pivot bushings for damaged or worn cadmium-plating. If cadmium-plating is damaged or worn, apply solid film lubricant, Item 201, Appendix D, for corrosion protection until part is replaced.

b. Visually inspect pivot bushings for damage or corrosion which penetrates into base metal. If damage or corrosion penetrates into base metal, repair bushings as follows:

(1) Remove damage or corrosion up to 0.003-inch deep using abrasive cloth, Item 7, Appendix D.

(2) Blend rework to a smooth transition with adjacent areas.

(3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace bushing.

(4) Apply corrosion compound, Item 108, Appendix D, to reworked areas using acid brush, Item 84, Appendix D. Allow compound to set on reworked area for 30 seconds then wipe off with clean machinery towel, Item 211, Appendix D.

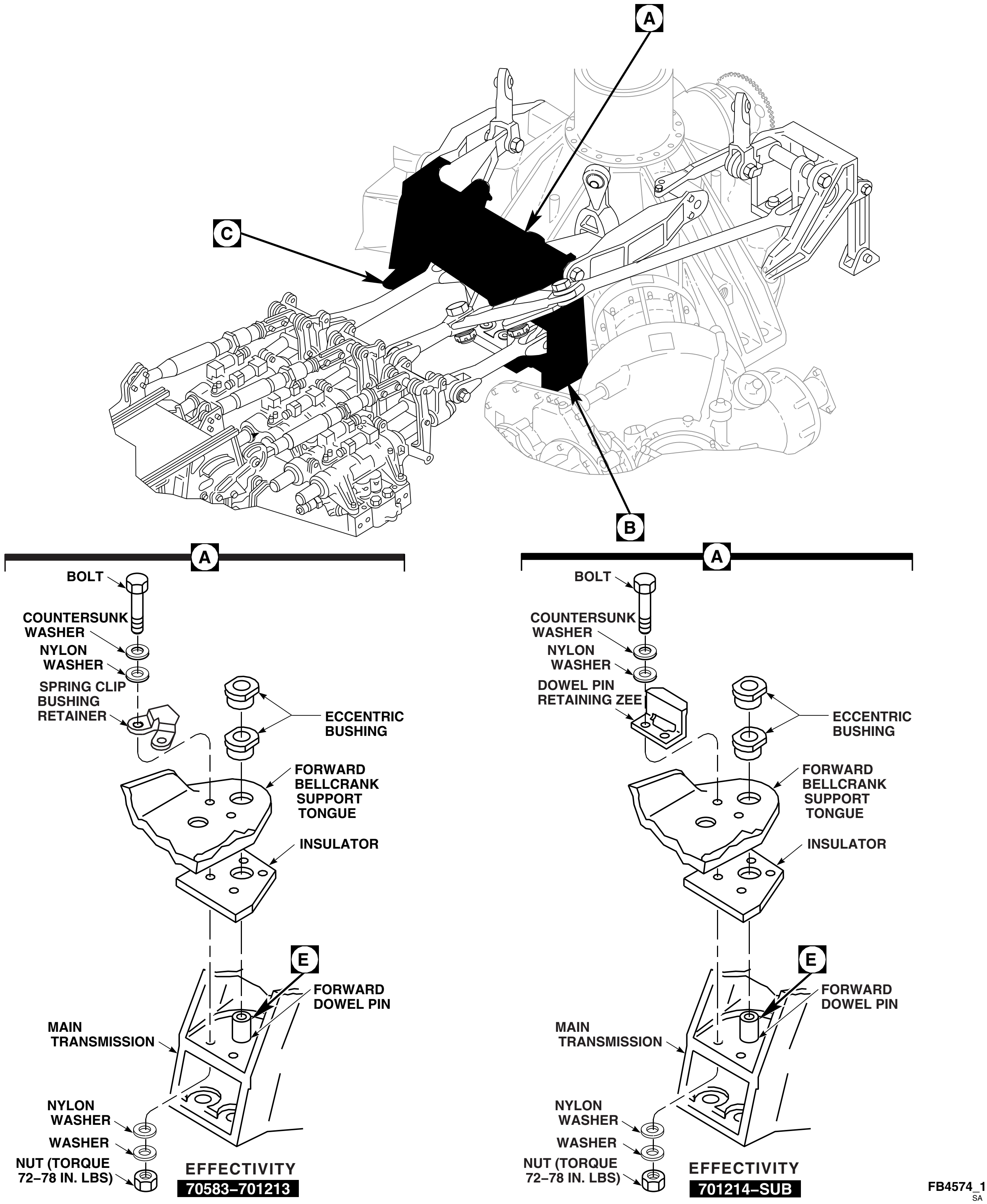


FIGURE 11-4-100-1. REPLACE FORWARD BELLCRANK SUPPORT (SHEET 1 OF 2)

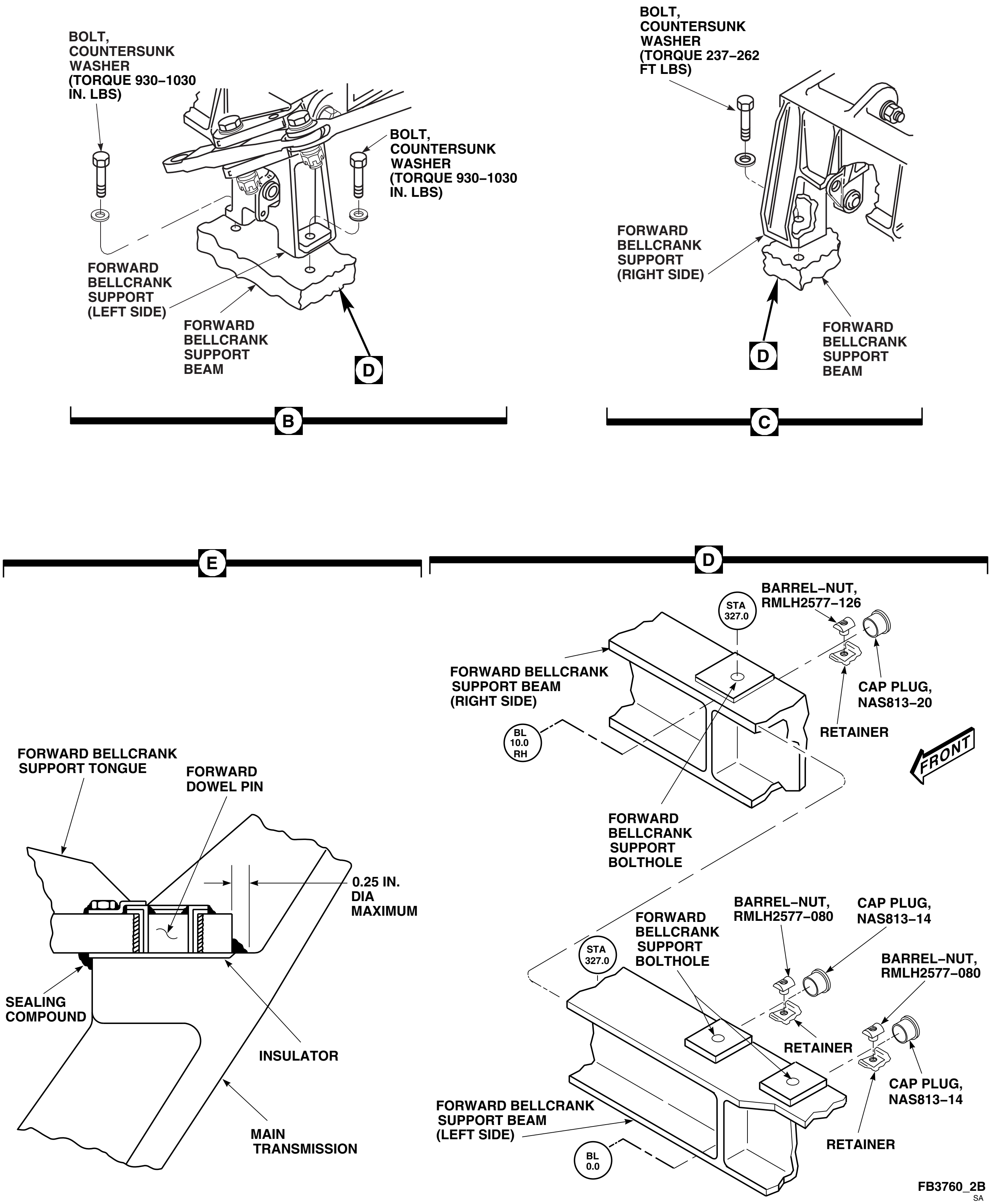


FIGURE 11-4-100-1. REPLACE FORWARD BELLCRANK SUPPORT (SHEET 2 OF 2)

(5) Apply solid film lubricant, Item 201, Appendix D, for corrosion protection until part is replaced.

c. Inspect main transmission forward bridge drain hole (PARA 1-7-20).

d. Perform forward bellcrank support inspection (PARA 11-3-15).

11-4-100.1.3 Install.

WARNING

- **701657–SUBQ** Injury to personnel and damage to equipment will result if improved components are mixed with older parts. The improved components are as follows:
  - Lateral Bellcrank, 70400-08166-041
  - Right Tie Rod, 70400-08165-043
  - Aft Bellcrank Support, 70400-08163-043
  - Aft Tie Rod, 70400-08111-043
  - Aft Tie Rod Support Fitting, 70400-08112-046
- Do not mix these parts with older parts.

a. If same insulator is to be used, clean insulator using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

b. Check breakaway torque of barrel-nuts and bolts as follows (Figure 11-4-100-1, Sheet 2, Detail D):

CAUTION

Damage to recess in airframe or forward bellcrank support beam will occur if barrel-nuts are incorrectly installed. To prevent damage to recess, make sure nuts are installed with barrel (round side) toward bolthead.

NOTE

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with the letters ″EN″ or ″SPS″ stamped on an end (additional marking may also include ″LH2577″).

(1) Install bolt into barrel-nut until two threads extend beyond nut.

(2) Using torque wrench, back bolt out of barrel-nut, noting torque required to start bolt turning out of nut.

(a) For right side barrel-nut, **MINIMUM BREAKAWAY TORQUE IS 50 INCH-POUNDS.** If breakaway torque is less than 50 inch-pounds, replace barrel-nut. Proceed to step b.(3).

(b) For left side barrel-nuts, **MINIMUM BREAKAWAY TORQUE IS 18 INCH-POUNDS.** If breakaway torque is less than 18 inch-pounds, replace barrel-nut(s). Proceed to step b.(3).

CAUTION

Damage to recess in frame or beam will occur if barrel-nuts are incorrectly installed. To prevent damage to recess, make sure nuts are installed with barrel (round side) toward bolthead.

NOTE

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with the letters ″EN″ or ″SPS″ stamped on an end (additional marking may also include ″LH2577″).

(3) If required, replace barrel-nut(s) as follows (Figure 11-4-100-1, Sheet 2, Detail D):

(a) From inside main transmission well, using nonmetallic scraper, remove sealing compound from around cap plug in airframe or forward bellcrank support beam. Remove cap plug.

(b) Remove barrel-nut and retainer from airframe or forward bellcrank support beam.

NOTE

Make sure that only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with the letters ″EN″ or ″SPS″ stamped on an end (additional marking may also include ″LH2577″).

(c) Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to barrel-nut and retainer.

CAUTION

Damage to recess in airframe or forward bellcrank support beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure barrel-nut is installed with barrel (round side) toward bolthead.

(d) Install barrel-nut and retainer in airframe or forward bellcrank support beam while primer is still wet.

NOTE

- Cap plug, NAS813-20, is installed in right side of forward bellcrank support beam.
- Cap plugs, NAS813-14, are installed in left side of forward bellcrank support beam.

(e) Apply a 0.25-inch diameter bead of sealing compound, Item 292, Appendix D, under lip of cap plug.

(f) Install cap plug in airframe or forward bellcrank support beam.

(g) Apply a 0.25-inch diameter bead of sealing compound, Item 292, Appendix D, around interface of cap plug and airframe or forward bellcrank support beam.

(h) Apply sealing compound, Item 292, Appendix D, to entire cap plug. Make sure cap plug is entirely sealed.

(4) If barrel-nut was replaced, recheck breakaway torque per step b. (2). **BREAKAWAY TORQUE SHALL NOT BE BELOW LIMIT.** If breakaway torque is below limit, replace bolt.

c. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean mount pad on forward bellcrank support tongue on main transmission. Wipe with clean machinery towel.

d. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to forward dowel pin on mount pad on main transmission.

e. If insulator was removed, install as follows:

(1) Apply a thin coat of sealing compound, Item 287, Appendix D, to entire surface of mount pad for forward bellcrank support tongue.

(2) Apply a thin coat of sealing compound, Item 287, Appendix D, to bottom of insulator.

(3) Install insulator over forward dowel pin and press onto mount pad for forward bellcrank support tongue. Align holes in insulator with holes in mount pad.

f. Check for minimum gap between forward bellcrank support tongue and main transmission as follows:

NOTE

Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

(1) Temporarily install forward bellcrank support on forward bellcrank support beam with bolts and countersunk washers. Tighten bolts until no relative motion is felt in forward bellcrank support.

(2) Using feeler gage, measure gap between forward bellcrank support tongue and main transmission. **MINIMUM GAP IS 0.001-INCH.**

(3) Remove bolts, countersunk washers, and forward bellcrank support from forward bellcrank support beam.

NOTE

Forward bellcrank support must be installed within two hours after application of sealing compound.

g. Apply a thin coat of sealing compound, Item 287, Appendix D, to entire lower surface of forward bellcrank support tongue.

h. Apply a thin coat of sealing compound, Item 287, Appendix D, to the entire upper surface of insulator.

i. Apply a thin coat of sealing compound, Item 292, Appendix D, to mounting surface of forward bellcrank support beam and bottom of forward bellcrank support.

j. Apply a fillet of sealing compound, Item 292, Appendix D, around forward dowel pin.

k. Position forward bellcrank support on forward bellcrank support beam and position forward bellcrank support tongue onto forward dowel pin on main transmission (Figure 11-4-100-1, Sheet 1, Detail A and Figure 11-4-100-1, Sheet 2, Detail B and Detail C).

WARNING

FSCAP

Make sure correct bolts are used when installing support fitting. Bolts used on controls installation, 70400-08200-011, are shorter than those used on controls installation, 70400-08200-012/-013 and cannot be interchanged.

NOTE

Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

l. Using epoxy primer coating kit, Item 135, Appendix D, coat shank, threads, and area under boltheads with epoxy primer. Install bolts and countersunk washers while primer is still wet.

WARNING

FSCAP

- Verification of minimum torque and safety of bolts are critical characteristics.

- To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

m. On left side of forward bellcrank support, **TORQUE BOLTS TO 930 - 1030 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolts to radius blocks (Figure 11-4-100-1, Sheet 2, Detail B). Apply torque stripes (PARA 2-6-5).

WARNING

FSCAP

- Verification of minimum torque of bolts is a critical characteristic.
- To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

n. On right side of forward bellcrank support, **TORQUE BOLT TO 237 - 262 FOOT-POUNDS** (Figure 11-4-100-1, Sheet 2, Detail C). On helicopters using bolt, SS5211-12H39, lockwire, Item 197, Appendix D.

o. Form a fillet with squeezed out sealing compound around joint between forward bellcrank support and forward bellcrank support beam. Make sure sealing compound covers entire joint line, but is no larger than 0.25-inch diameter.

p. Apply a bead of sealing compound, Item 292, Appendix D, around bolts and countersunk washers securing forward bellcrank support.

q. Install and adjust eccentric bushing in forward bellcrank support tongue to fit over forward dowel pin on main transmission (Figure 11-4-100-1, Sheet 1, Detail A).

r. **70583–701213** Position spring clip bushing retainer on forward bellcrank support tongue as follows (Figure 11-4-100-1, Sheet 1, Detail A):

(1) Apply a bead of sealing compound, Item 292, Appendix D, around each bolthole of spring clip bushing retainer on top of forward bellcrank support tongue.

NOTE

Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

(2) Position spring clip bushing retainer on forward bellcrank support tongue. Install bolts, countersunk washers, and nylon washers.

s. **701214–SUBQ** Position dowel pin retaining zee on forward bellcrank support tongue as follows (Figure 11-4-100-1, Sheet 1, Detail A):

WARNING

FSCAP

Verification of dowel pin retaining zee installation is a critical characteristic.

NOTE

Dowel pin retaining zee, 70400-08171-101, is used on main transmissions with tall forward dowel pin. Dowel pin retaining zee, 70400-08171-102, is used on main transmissions with short forward dowel pin.

(1) Apply a bead of sealing compound, Item 292, Appendix D, around each bolthole of dowel pin retaining zee on top of forward bellcrank support tongue.

NOTE

Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

(2) Position dowel pin retaining zee on forward bellcrank support tongue. Install bolts, countersunk washers, and nylon washers.

NOTE

Lower washers and nylon washers must be installed wet.

t. Using epoxy primer coating kit, Item 135, Appendix D, coat washers with epoxy primer.

u. Apply a bead of sealing compound, Item 287, Appendix D, to one side of nylon washers.

v. Install nylon washers (with sealing compound side towards main transmission), washers, and nuts. **TORQUE NUTS TO 72 - 78 INCH-POUNDS.**

NOTE

Do not use excess sealing compound to form fillet around forward bellcrank support tongue. Excessive sealing compound will hinder corrosion inspection.

w. Form a fillet with the squeezed out sealing compound around joint between main transmission and forward bellcrank support tongue. Make sure sealing compound covers entire joint line, but is no larger than 0.25-inch diameter (Figure 11-4-100-1, Sheet 2, Detail E).

x. Using polyurethane paint, Item 224, Appendix D, apply to mounting area of upper and lower surfaces of forward bellcrank support tongue (PARA 2-6-5).

y. Mask off adjacent area to prevent overspray, then apply 2 coats of polyurethane coating, Item 240, Appendix D, to mounting area of forward bellcrank support tongue and also to water entrapment area behind mounting area. Allow to cure between coats.

z. Install left and right tie rods (PARA 11-4-99).

aa. Install aft bellcrank (PARA 11-4-98).

ab. Install forward bellcrank (PARA 11-4-96).

ac. Install lateral bellcrank (PARA 11-4-97).

ad. Install walking beam (PARA 11-4-93).

ae. Install forward, aft, and lateral swashplate links (PARA 11-4-95).

af. Install forward, aft, lateral, and long control rods (PARA 11-4-90, PARA 11-4-91, PARA 11-4-92, and PARA 11-4-94).

ag. Make an entry in helicopter logbook indicating that an autorotational rpm check and high pitch stop check (V_h) must be performed during maintenance test flight.

ah. Perform operational check of flight control system (PARA 11-2-3).

ai. Perform main rotor system rig check (PARA 11-4-4).

11-4-101 AFT BELLCRANK SUPPORT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench, ⅛-inch
- Crowfoot Attachment
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Drift
- Nonmetallic Punch
- Nonmetallic Scraper, Locally-Made
- Protective Covering
- Spacer Removal Tool, Locally-Made, Figure H-219, Appendix H
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Torque Wrench, 100 - 500 ft lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 197, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 6-4-1
- PARA 11-2-3
- PARA 11-3-1

- PARA 11-3-18
- PARA 11-4-4
- PARA 11-4-94
- PARA 11-4-95

11-4-101.1 Replace Aft Bellcrank Support.

11-4-101.1.1 Remove.

a. Disconnect long control rod from walking beam (PARA 11-4-94).

NOTE

Reinstall expandable pin in swashplate after link is disconnected.

b. Disconnect aft swashplate link from stationary swashplate (PARA 11-4-95).

CAUTION

Damage to special self-retaining bolt, aft bellcrank, and aft bellcrank support will result if bolt is removed incorrectly. To remove bolt, first remove nut. Use ⅛-inch Allen wrench to push pin in bolt-head. This will release retaining balls on threaded end of bolt so bolt can be removed. Keep pressure on pin while removing bolt.

c. Remove special self-retaining bolt, countersunk/flat washer, additional washer(s), if installed, eccentric bushings, nut, and cotter pin securing aft bellcrank to aft bellcrank support. Remove spacer and bellcrank, with long control rod and aft swashplate link attached (Figure 11-4-101-1, Sheet 2, Detail A). If spacer does not slide out with bolt, remove spacer as follows:

NOTE

If spacer removal tool is not available, it may be necessary to tap out spacer using nonmetallic punch.

(1) Locally make spacer removal tool (Figure H-219, Appendix H).

(2) Insert spacer removal tool into bolthead end of spacer.

(3) **TORQUE TOOL TO 200 - 250 INCH-POUNDS.**

(4) Tap out spacer using nonmetallic drift against underside of removal tool.

d. Remove bolt, countersunk washer, washer, additional washer(s), if installed, and nut securing aft tie rod to aft bellcrank support. Position tie rod out of way.

e. Remove bolt, washer, and nylon washer securing leg of aft bellcrank support to main transmission.

f. Remove bolt, washer, and eccentric bushings securing aft bellcrank support to deck. Carefully pull support outboard from left aft dowel pin and remove from main transmission.

g. Remove insulators from main transmission and deck.

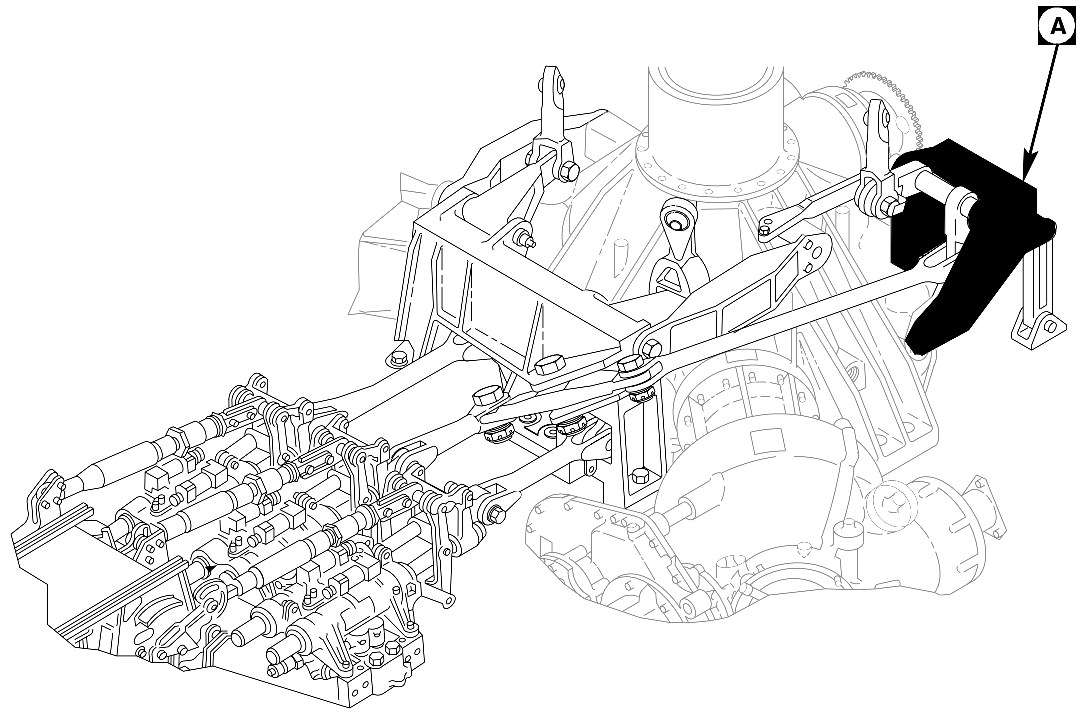
h. 701214–SUBQ Remove bolts, washers, and nuts securing retaining angle to aft bellcrank support.

i. Clean primer from bolts, insulators, retaining angle, and areas on main transmission that mate with aft support using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

j. Clean any remaining sealing compound from transmission using nonmetallic scraper.

11-4-101.1.2 Inspect/Repair.

a. Visually inspect pivot bushings for damaged or worn cadmium-plating. If cadmium-plating is damaged or worn, apply solid film lubricant, Item 201, Appendix D, for corrosion protection



NOTES

1. 701214–SUBQ
2. ON HELICOPTERS WITH AFT TIE ROD 70400–08111–041 OR –042, TORQUE NUT TO 532–588 INCH–POUNDS.
3. ON HELICOPTERS WITH AFT TIE ROD 70400–08111–043, TORQUE NUT TO 238–262 INCH–POUNDS.
4. ON HELICOPTERS WITH AFT TIE ROD, 70400–08111–041 OR 70400–08111–042, ADD WASHERS AS REQUIRED TO PREVENT NUT FROM BOTTOMING ON BOLT.
5. COUNTERSUNK WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–043 OR –044. FLAT WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–042.

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FIGURE 11-4-101-1. REPLACE AFT BELLCRANK SUPPORT (SHEET 1 OF 2)

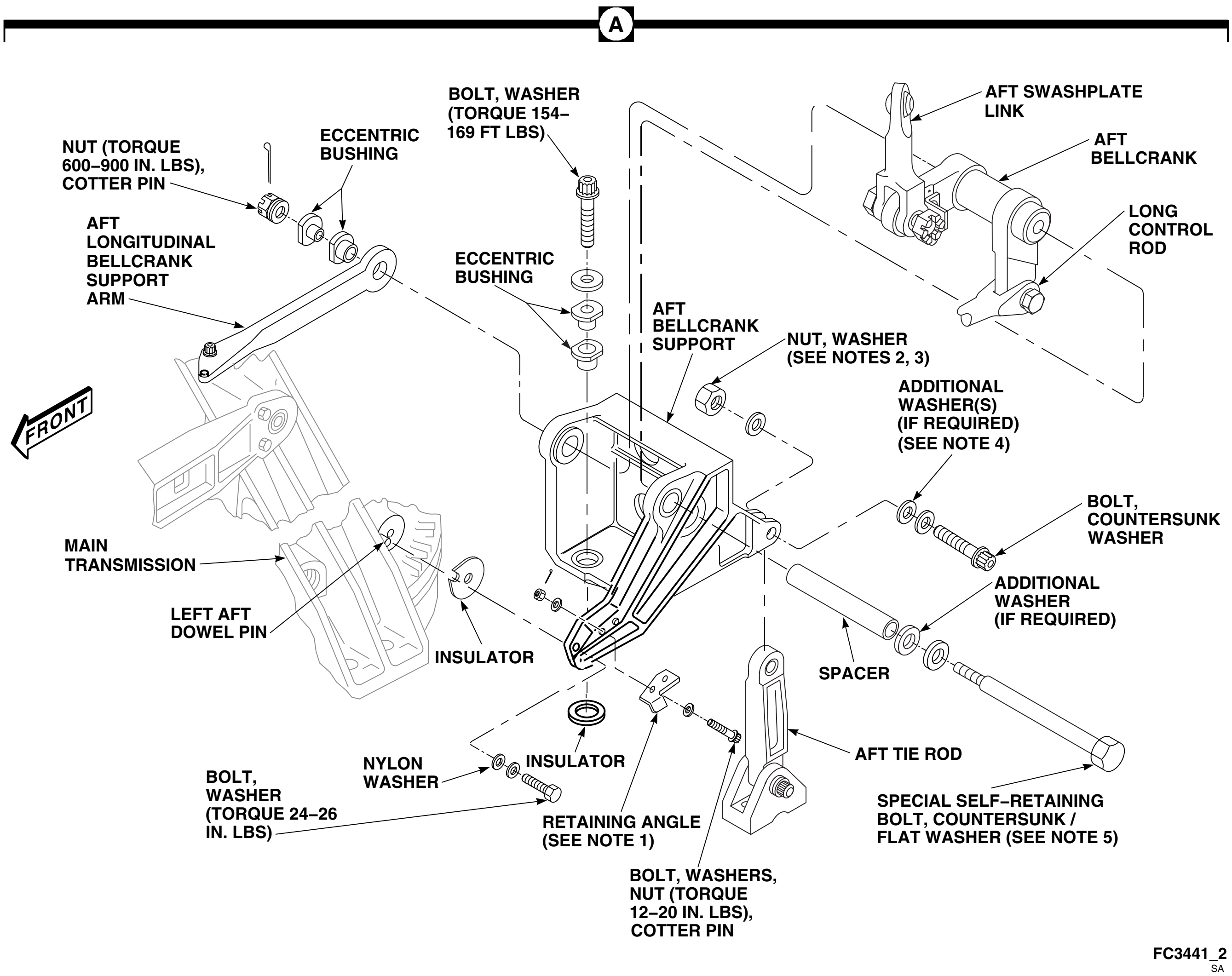


FIGURE 11-4-101-1. REPLACE AFT BELLCRANK SUPPORT (SHEET 2 OF 2)

until part is replaced.

b. Visually inspect pivot bushings for damage or corrosion which penetrates into base metal. **NO PENETRATION INTO BASE METAL ALLOWED.** If damage or corrosion penetrates into base metal, repair bushings as follows:

(1) Remove damage or corrosion up to 0.003-inch deep using abrasive cloth, Item 7, Appendix D.

(2) Blend rework to a smooth transition with adjacent areas.

(3) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect all reworked areas for cracks (ASTM E1417).

(4) Apply corrosion compound, Item 108, Appendix D, to reworked areas using acid brush, Item 84, Appendix D. Allow compound to set on reworked area for 30 seconds then wipe off with clean machinery towel, Item 211, Appendix D.

(5) Apply solid film lubricant, Item 201, Appendix D, for corrosion protection until part is replaced.

c. Perform aft bellcrank support inspection (PARA 11-3-18).

d. Perform special self-retaining bolt FSCAP inspection (PARA 11-3-1).

11-4-101.1.3 Install.

WARNING

- **701657–SUBQ** Injury to personnel and damage to equipment will result if improved components are mixed with older parts. The improved components are as follows:

  - Lateral Bellcrank, 70400-08166-041
  - Right Tie Rod, 70400-08165-043
  - Aft Bellcrank Support, 70400-08163-043
  - Aft Tie Rod, 70400-08111-043
  - Aft Tie Rod Support Fitting, 70400-08112-046

- Do not mix these parts with older parts.

a. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to area on main transmission that mates with aft bellcrank support.

b. Install insulator on aft support and carefully position support and insulator on main transmission while primer is still wet (Figure 11-4-101-1, Sheet 2, Detail A).

c. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to shank, threads, and area under bolthead.

WARNING

FSCAP

Verification of minimum torque and safety installation of bolts are critical characteristics.

d. Install and adjust eccentric bushings in aft bellcrank support. Install bolt and washer while primer is still wet. **TORQUE BOLT TO 154 - 169 FOOT-POUNDS.** Wipe excess primer from bolt using machinery towel, Item 211, Appendix D.

e. Apply a bead of sealing compound, Item 292, Appendix D, around bolthead and washer.

f. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to bolthead, shank, and area under bolthead. Allow primer to dry.

g. Lubricate bolthead and shank with corrosion preventive compound, Item 109, Appendix D. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to bolt threads. Install bolt while primer is still wet.

WARNING

To prevent damage to helicopter, do not use safety cable in this application. Use only lockwire that is specified in this step.

h. Secure leg of support to main transmission with bolt, washer, and nylon washer. Make sure countersunk side of washer faces bolthead. **TORQUE BOLT TO 24 - 26 INCH-POUNDS.** Wipe excess primer from bolt using machinery towel, Item 211, Appendix D. Using lockwire, Item 197, Appendix D, lockwire bolt to bolt installed in step d.

i. Apply a bead of sealing compound, Item 292, Appendix D, around bolt and washers.

j. 701214–SUBQ Install retaining angle as follows:

WARNING

FSCAP

Verification of retaining angle installation is a critical characteristic.

NOTE

Bolts securing retaining angle to leg of aft bellcrank support are installed with boltheads facing down.

(1) Install retaining angle to leg of aft bellcrank support and secure with bolts, washers, and nuts.

(2) Using feeler gage, measure clearance between left aft dowel pin and retaining angle. **MINIMUM CLEARANCE IS 0.010-INCH.** If clearance is less than 0.010-inch, reseat left aft dowel pin (PARA 6-4-1).

(3) **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**

(4) Install cotter pin.

k. Install eccentric bushings in aft longitudinal bellcrank support arm. Turn bushings to align boltholes.

l. Position aft bellcrank, with long control rod and aft swashplate link attached, in aft bellcrank support. Install spacer through outboard side of support.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to special self-retaining bolt and control rod will result if bolt is installed incorrectly. To install bolt, insert 1/8-inch Allen wrench to push pin in bolthead. Push on pin to release retaining balls at nut end of bolt. This will release retaining balls on threaded end of bolt so bolt can be installed. Keep pressure on pin while installing bolt.

NOTE

- Add washers, as necessary, to make sure nut is bottomed against support and not on bolt.
- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

m. Install special self-retaining bolt and countersunk/flat washer (under bolthead) through eccentric bushings and bellcrank. Add washers, as required (maximum of 2 additional washers under countersunk/flat washer) to provide correct grip length and cotter pin position.

n. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.

o. Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

p. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If bolt rotates, perform the following:

(1) Remove special self-retaining bolt, washer(s), nut, and cotter pin.

(2) Repeat step m., adjusting washer thicknesses or adding washers.

(3) Repeat step n. through step p.

q. Apply torque stripe (PARA 2-6-5).

r. Connect aft swashplate link to stationary swashplate (PARA 11-4-95).

NOTE

- Add washers (under countersunk washer), as necessary, to prevent nut from bottoming on bolt.
- Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

s. Install bolt, countersunk washer, washer, and nut securing aft tie rod to aft bellcrank support. On helicopters with aft tie rod, 70400-08111-041 or 70400-08111-042, add washers as required (under countersunk washer) to prevent nut from bottoming on bolt.

t. On helicopters with aft tie rod, 70400-08111-041 or 70400-08111-042, **TORQUE NUTS TO 532 - 588 INCH-POUNDS.** On helicopters with aft tie rod, 70400-08111-043, **TORQUE NUTS TO 238 - 262 INCH-POUNDS.**

u. Connect long control rod to walking beam (PARA 11-4-94). Do not lower swashplate now.

v. On helicopters with main transmission, 70351-08100-053 or 70351-08100-054, cover exposed end of left aft dowel pin with epoxy coating, using epoxy coating kit, Item 126A, Appendix D.

w. Apply a thin coat of sealing compound, Item 292, Appendix D, to exposed end of left aft dowel pin and around gap between main transmission mounting surface and support.

x. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

y. Raise collective stick and remove wood blocks from between swashplate and main transmission.

z. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

aa. Make sure area is clean and free of foreign material.

ab. Install main rotor pylon transmission cover (PARA 2-4-107).

ac. Perform an operational check of flight control system (PARA 11-2-3).

ad. Perform main rotor system rig check (PARA 11-4-4).

11-4-102 AFT LONGITUDINAL BELLCRANK SUPPORT ARM.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Crowfoot Attachment
- Protective Covering
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Preventive Compound, Item 109, Appendix D
- Corrosion Preventive Compound, Item 111, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lockwire, Item 194, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-107
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-3-23

11-4-102.1 Replace Aft Longitudinal Bellcrank Support Arm.

WARNING

FSCAP

All self-retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-102.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Raise collective stick to raise swashplate.

d. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing forward control rod.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

e. Slowly lower collective stick until swashplate rests on wood blocks.

f. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

g. Remove nut and eccentric bushings securing support arm to special self-retaining bolt on aft bellcrank support (Figure 11-4-102-1, Detail A).

h. Remove bolt, countersunk washer, washer, and retaining channel securing support arm to left vertical dowel pin on main transmission. Remove support arm.

11-4-102.1.2 Inspect.

a. Inspect aft longitudinal bellcrank support arm (PARA 11-3-23).

b. Inspect face and journal of eccentric bushings for damage (Figure 11-4-102-1, Detail B). **MAXIMUM ALLOWABLE DAMAGE TO THESE SURFACES SHALL NOT EXCEED 0.015-INCH.** Inspect inside and outside diameter of eccentric bushings for damage. **NONE ALLOWED.** If damage is beyond limit, replace bushing(s).

11-4-102.1.3 Install.

NOTE

Do not allow primer to dry.

a. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to left vertical dowel pin.

b. Install support arm on left vertical dowel pin on main transmission. Make sure support arm is fully seated and bolthole lines up with threaded hole in main transmission (Figure 11-4-102-1, Detail A).

c. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt shank and bolthead. Allow to dry.

d. Apply corrosion preventive compound, Item 111, Appendix D, to bolt shank and bolthead.

NOTE

Do not allow epoxy primer to dry.

e. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to bolt threads.

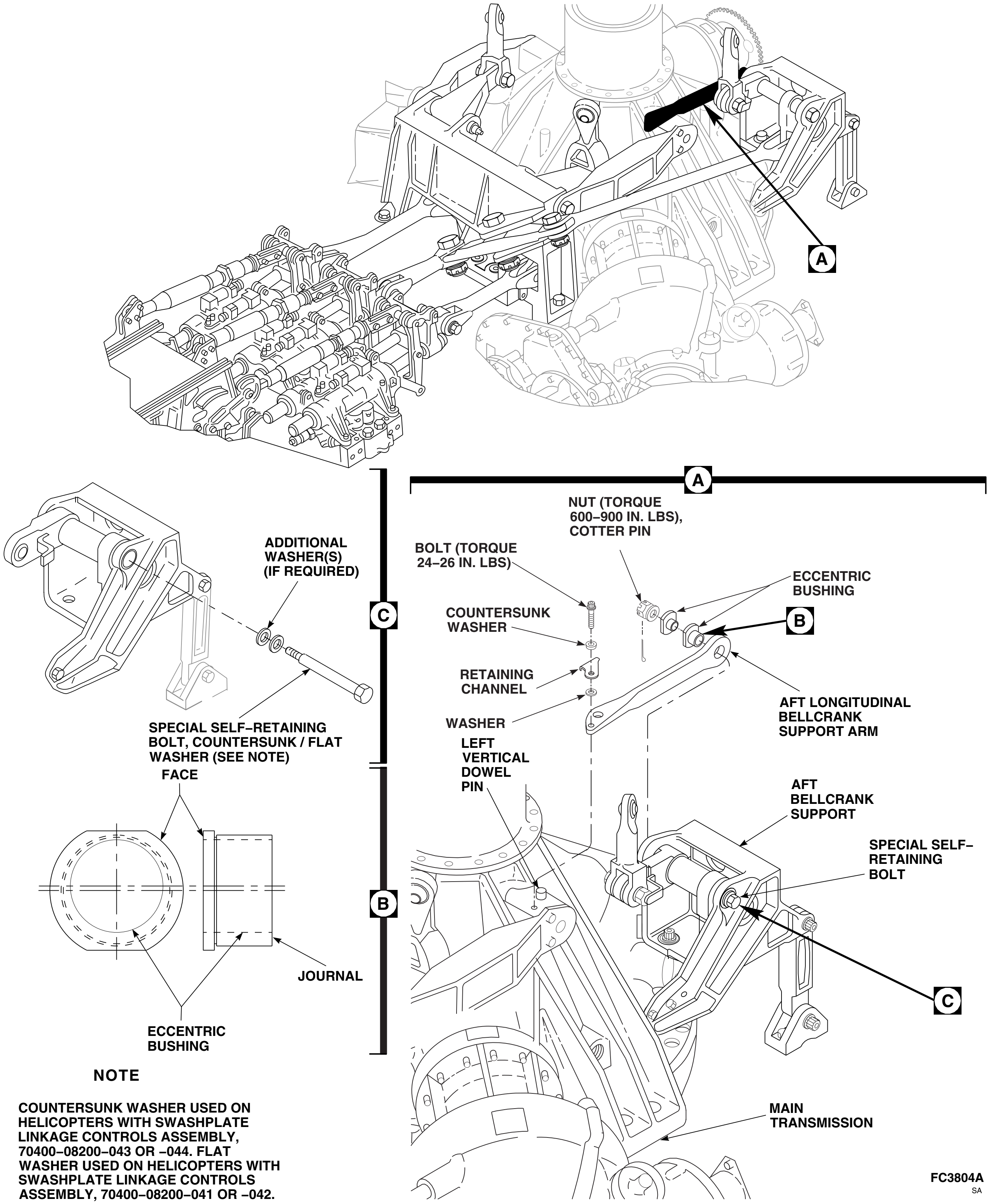
WARNING

FSCAP

Verification of presence of retaining channel is a critical characteristic.

NOTE

- Retaining channel is mounted between countersunk washer and washer.
- Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.



NOTE

COUNTERSUNK WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–043 OR –044. FLAT WASHER USED ON HELICOPTERS WITH SWASHPLATE LINKAGE CONTROLS ASSEMBLY, 70400–08200–041 OR –042.

FIGURE 11-4-102-1. REPLACE AFT LONGITUDINAL BELLCRANK SUPPORT ARM

f. Secure support arm to main transmission with bolt, countersunk washer, retaining channel, and washer. **TORQUE BOLT TO 24 - 26 INCH-POUNDS.**

g. Lockwire, Item 194, Appendix D, bolt to left tie rod upper attachment bolt.

h. Clean bolt using cleaning cloth, Item 102, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Inspect bolt as follows:

(1) Inspect bolt for evidence of corrosion in the area of retaining balls, threads, cotter pin hole, retaining pin hole in bolthead, and hole in threaded end of bolt. **NONE ALLOWED.** If corrosion is visible, replace bolt.

(2) Visually inspect bolt for cracks in area of retaining balls and threads. **NONE ALLOWED.** If crack(s) is visible, replace bolt.

CAUTION

If resistance or binding of the retaining ball mechanism is observed when pin is actuated, do not apply corrosion preventive compound to free the mechanism. Inner core of bolt is corroded and must be rejected.

i. Release retaining balls by holding retaining pin in. If resistance or binding of the retaining ball mechanism is observed when pin is actuated, discard bolt. If no resistance or binding of the retaining ball mechanism is observed when pin is actuated, apply corrosion preventive compound, Item 114, Appendix D, to bolt as follows:

(1) Work corrosion preventive compound through one retaining ball hole until excess is seen at opposite ball hole. Rotate bolt 180° and repeat procedure.

(2) While working retaining pin, spray corrosion preventive compound, Item 114, Appendix D, into hole in bolthead and hole in end of bolt.

(3) Wipe excessive corrosion preventive compound from bolt using cleaning cloth, Item 102, Appendix D, prior to installation.

(4) Apply corrosion preventive compound, Item 109, Appendix D, to the exterior retaining ball area and threads.

NOTE

Turn eccentric bushings as necessary to line up boltholes.

j. Install eccentric bushings in support arm and aft bellcrank support.

k. Install nut. **TORQUE NUT TO 600 INCH-POUNDS. IF SLOT IN NUT DOES NOT LINE UP WITH COTTER PIN HOLE, SET TORQUE WRENCH TO 900 INCH-POUNDS AND TIGHTEN NUT UNTIL SLOT IN NUT AND COTTER PIN HOLE LINE UP. DO NOT EXCEED MAXIMUM TORQUE.** Install cotter pin.

NOTE

If unable to torque bolthead, use crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

l. **APPLY TIGHTENING TORQUE (PROOF LOAD) OF 400 - 420 INCH-POUNDS** in tightening direction. If bolt rotates, perform the following:

(1) Remove special self-retaining bolt, countersunk/flat washer, additional washer(s), if installed, nut, and cotter pin (Figure 11-4-102-1, Detail A and Detail C).

NOTE

- Countersunk washer used on helicopters with swashplate linkage controls assembly, 70400-08200-043 or -044. Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.
- Flat washer used on helicopters with swashplate linkage controls assembly, 70400-08200-041 or -042.

(2) Install special self-retaining bolt and countersunk/flat washer (under bolthead) through eccentric bushings and aft bellcrank. Add washers, as required (maximum of 2 additional washers under countersunk/flat washers) to provide correct grip length and cotter pin position.

(3) Check bolts for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.

(4) Repeat step k. and step l.

m. Apply torque stripe (PARA 2-6-5).

n. Apply sealing compound, Item 292, Appendix D, to exposed end of left vertical dowel pin, around gap between main transmission mounting surface and support arm, and around bolt, washers, and retaining channel.

o. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

p. Raise collective stick and remove wood blocks from between swashplate and main transmission.

q. Perform an operational check of flight control system (PARA 11-2-3).

r. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

s. Make sure area is clean and free of foreign material.

t. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-103 AFT TIE ROD AND SUPPORT FITTING.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Protective Covering
- Torque Wrench, 150 - 750 in. lbs
- Wood Blocks, 1 ½-inch thick

Materials/Parts

- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-4-107
- PARA 11-2-3
- PARA 11-3-19

11-4-103.1 Replace Aft Tie Rod and Support Fitting.

WARNING

FSCAP

Aft bellcrank support, aft tie rod, and support fitting contain critical surfaces which must not be damaged during replacement of control rods and links. Provide protective covering, as required.

11-4-103.1.1 Remove.

a. Remove main rotor pylon transmission cover (PARA 2-4-107).

b. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Raise collective stick to raise swashplate.

d. Install three 1 ½-inch thick wood blocks equally spaced between swashplate and main transmission, lower swashplate onto wood blocks to support swashplate while removing aft tie rod and support fitting.

CAUTION

- Damage to swashplate will result if excessive pressure is applied when lowering collective stick to rest swashplate on wood blocks. Make sure observers are positioned to prevent further collective stick application after initial contact with wood blocks.
- Do not apply down collective with wood blocks installed.

e. Slowly lower collective stick until swashplate rests on wood blocks.

f. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

g. Remove bolts, countersunk washers, washers, additional washer(s), if installed, and nuts securing aft tie rod to aft bellcrank support and support fitting (Figure 11-4-103-1, Detail A). Remove tie rod.

h. Remove soundproofing panels, aft of drip pan, from cabin ceiling (PARA 2-4-69).

i. With aid of assistant inside cabin, remove bolts, countersunk washers, additional washer(s), if installed, and nuts securing support fitting to deck. Remove support fitting and shims.

11-4-103.1.2 Inspect.

Inspect aft tie rod, support fitting lug, and surface area (PARA 11-3-19).

11-4-103.1.3 Install.

WARNING

- 701657–SUBQ Injury to personnel and damage to equipment will result if improved components are mixed with older parts. The improved components are as follows:
  - Lateral Bellcrank, 70400-08166-041
  - Right Tie Rod, 70400-08165-043
  - Aft Bellcrank Support, 70400-08163-043
  - Aft Tie Rod, 70400-08111-043
  - Aft Tie Rod Support Fitting, 70400-08112-046
- Do not mix these parts with older parts.

a. Connect aft tie rod to aft bellcrank support with bolt. Do not install nut and washers now (Figure 11-4-103-1, Detail A).

b. Connect aft tie rod to support fitting with bolt. Do not install nut and washers now.

c. Using feeler gage, position support fitting on deck and measure gap between fitting and deck.

d. Remove bolts securing tie rod to aft bellcrank support and support fitting.

e. Peel shims to obtain a thickness of 0.000 - 0.010-inch thicker than measured gap. No more than three shims may be used.

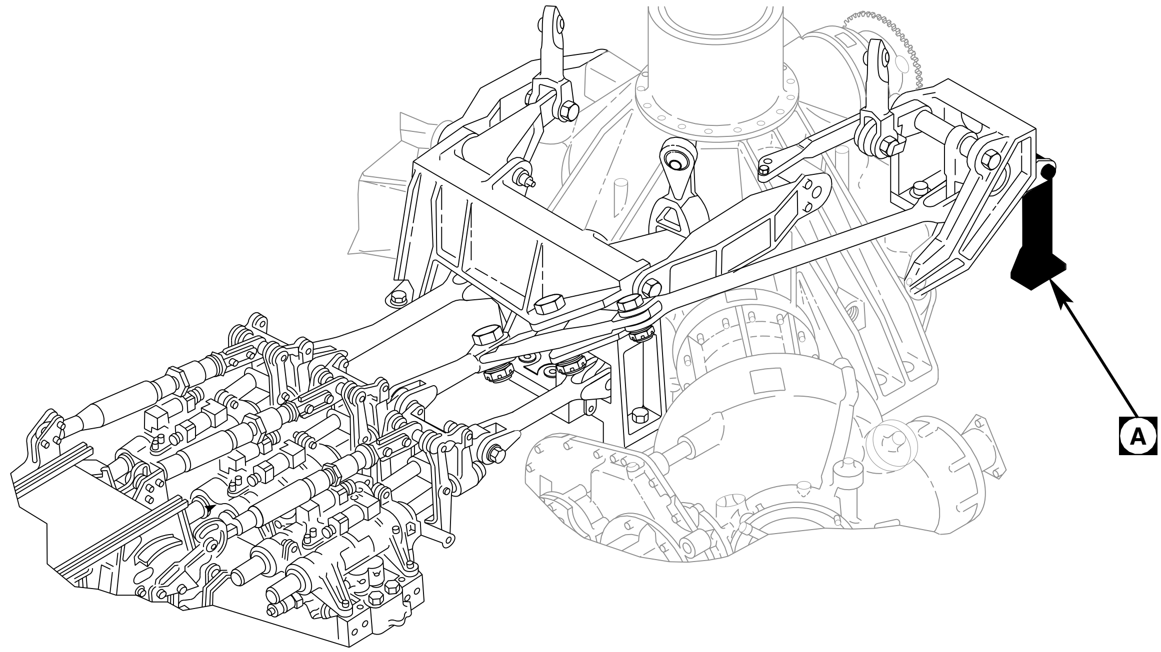
f. Treat shims with corrosion resistant coating, Item 118, Appendix D.

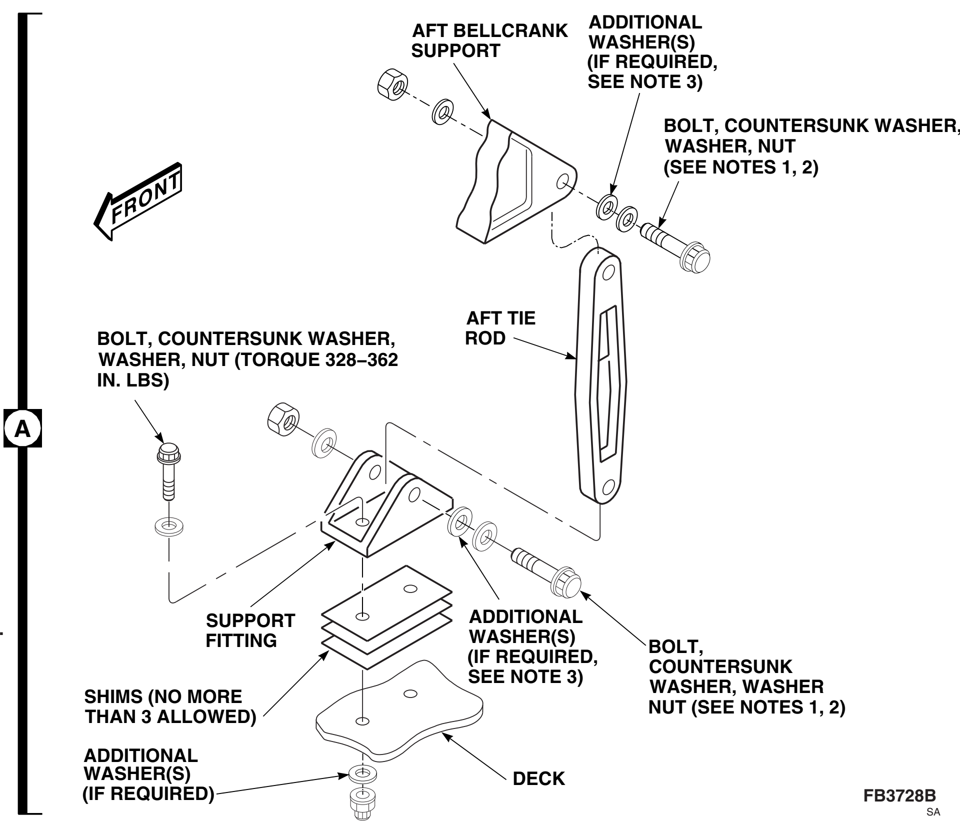
WARNING

FSCAP

Make sure correct bolts are used when installing support fitting. Bolts used on controls installation, 70400-08200-011, are shorter than those used on controls installation, 70400-08200-012/-013 and cannot be interchanged.

g. Using epoxy primer coating kit, Item 135, Appendix D, apply a light coat of epoxy primer to shank, threads, and area under bolthead of support fitting mounting bolts. Install bolts while primer is still wet.





NOTES

1. ON HELICOPTERS WITH AFT TIE ROD 70400–08111–041 OR 70400–08111–042, TORQUE NUT TO 532–588 INCH–POUNDS.
2. ON HELICOPTERS WITH AFT TIE ROD 70400–08111–043, TORQUE NUT TO 238–262 INCH–POUNDS.
3. ON HELICOPTERS WITH AFT TIE ROD, 70400–08111–041 OR 70400–08111–042, ADD WASHERS AS REQUIRED TO PREVENT NUT FROM BOTTOMING ON BOLT.

FIGURE 11-4-103-1. REPLACE AFT TIE ROD AND SUPPORT FITTING

NOTE

A maximum of three washers, in any combination, may be installed between deck and nut to make sure nut is bottomed against deck and not on bolt.

h. With aid of assistant inside cabin, secure shims and support fitting to deck with bolts, countersunk washers, additional washers (located between deck and nut, if required), and nuts. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.** Wipe off excess primer with machinery towel, Item 211, Appendix D.

NOTE

- Add washers (under countersunk washer), as necessary, to prevent nut from bottoming on bolt.
- Make sure countersunk washer is installed under bolthead and countersunk side faces bolthead.

i. Install bolts, countersunk washers, washers, and nuts securing aft tie rod to aft bellcrank support and support fitting. On helicopters with aft tie rod, 70400-08111-041 or 70400-08111-042, add washers as required (under countersunk washer) to prevent nut from bottoming on bolt.

j. On helicopters with aft tie rod, 70400-08111-041 or 70400-08111-042, **TORQUE NUTS TO 532 - 588 INCH-POUNDS.** On helicopters with aft tie rod, 70400-08111-043, **TORQUE NUTS TO 238 - 262 INCH-POUNDS.**

k. Apply a bead of sealing compound, Item 292, Appendix D, around support fitting mounting bolts, washers, and nuts.

l. Install soundproofing panels to cabin ceiling (PARA 2-4-69).

m. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

n. Raise collective stick and remove wood blocks from between swashplate and main transmission.

o. Perform operational check of flight control system (PARA 11-2-3).

p. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

q. Make sure area is clean and free of foreign material.

r. Install main rotor pylon transmission cover (PARA 2-4-107).

11-4-104 CABIN FLIGHT CONTROL CABLES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Flashlight
- Inspection Mirror
- Tensiometer
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-10
- PARA 2-4-69
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-4-1
- PARA 11-4-2
- PARA 11-4-6
- PARA 11-4-84
- PARA 11-4-109
- PARA 11-4-118
- TM 11-70-23

11-4-104.1 Replace Cabin Flight Control Cables.

11-4-104.1.1 Remove.

a. Remove soundproofing panels from cabin ceiling, STA 295.0 to rear of cabin (PARA 2-4-69).

b. Remove cargo netting from stowage compartments.

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring tension on tail rotor quadrant and cables has been removed before disconnecting cables.

c. Install lockout blocks (PARA 11-4-109) or disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

d. Remove locking clips from turnbuckles (Figure 11-4-104-1, Sheet 2, Detail D).

e. Turn body of turnbuckle to loosen cable. Continue turning and remove cable terminal end.

f. Remove cable guard pins and guard bracket from pulleys (-3), (-4), (-5), (-6), and quadrant. To reach pulley (-3) it may be necessary to remove stabilator amplifier (TM 11-70-23) (Figure 11-4-104-1, Sheet 1, Detail A and Detail B).

g. Remove yaw lever to forward quadrant control rod (PARA 11-4-84).

h. Remove self-retaining bolts, washers, and nuts securing pulleys (-1) and (-2) to bracket (Figure 11-4-104-1, Sheet 2, Detail F). Remove pulleys.

i. Remove cable guard pins from pulleys (-7), (-8), (-9), and (-10) (Figure 11-4-104-1, Sheet 1, Detail A and Detail B). Remove bolts and washers securing cable guard to airframe at pulley (-9).

j. Pull cable toward front of helicopter. Guide terminal end of cable through pulley brackets.

k. If installed, remove forward quadrant cable retaining cotter pins.

l. Slide cables and collars from forward quadrant. Remove collars from cable (Figure 11-4-104-1, Sheet 2, Detail G).

11-4-104.1.2 Install.

WARNING

FSCAP

Verification of proper cable routing between pulleys and retaining pins or guards is a critical characteristic.

a. Install right side cable. Start from the rear (-6) pulley and work forward. Lead with the cable ball-end as follows:

(1) Thread cable under (-6) pulley. Install cable guard pin (Figure 11-4-104-1, Sheet 1, Detail A and Detail B).

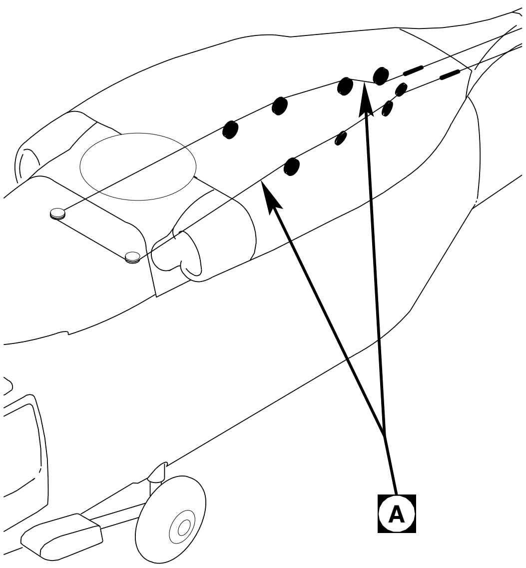
(2) Thread cable over top of (-5) pulley. Install cable guard pin.

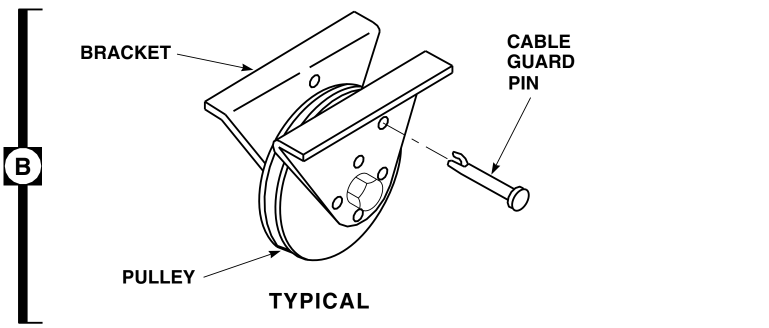
(3) Pass cable through grommets at STA 443.0 and STA 433.0, then over top of pulley (-4). Install cable guard to airframe with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 11-4-104-1, Sheet 1, Detail C).

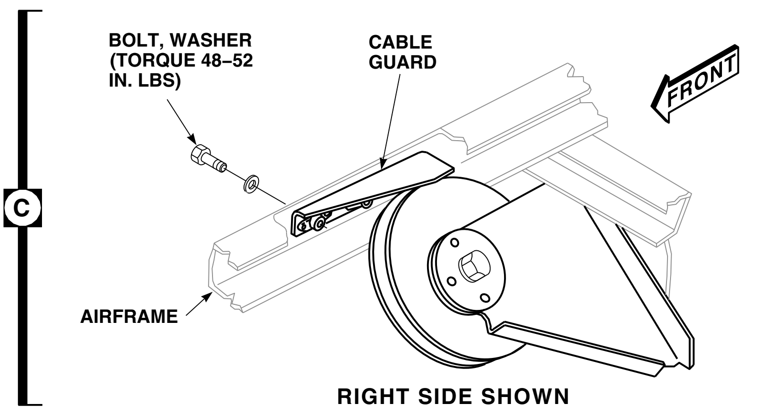
(4) Pass cable through grommet at STA 398.0, then over top of (-3) pulley. Install cable guard pin (Figure 11-4-104-1, Sheet 1, Detail A and Detail B).

(5) Pass cable through grommet at STA 379.0, then through grommets and cable conduit located between STA 360.0 and STA 327.0.

(6) Position cable through pulley bracket (-2), through grommets at STA 311.0, and







NOTES

1. THE DASH NUMBERS USED FOR PULLEYS ARE FOR REFERENCE ONLY.
2. MAKE SURE CABLE BALL END IS NESTED INSIDE COLLAR DURING INSTALLATION.

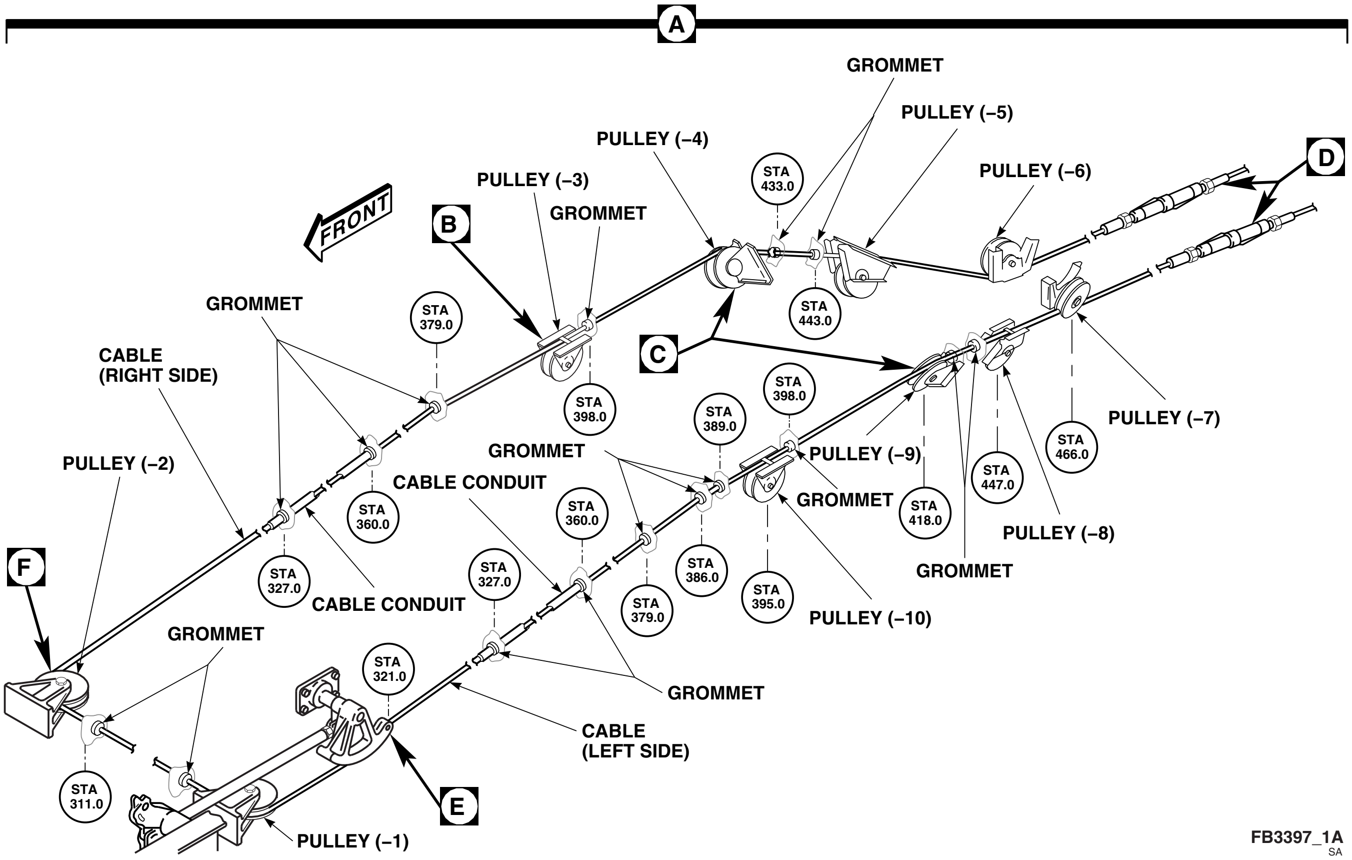


FIGURE 11-4-104-1. REPLACE CABIN FLIGHT CONTROL CABLES (SHEET 1 OF 2)

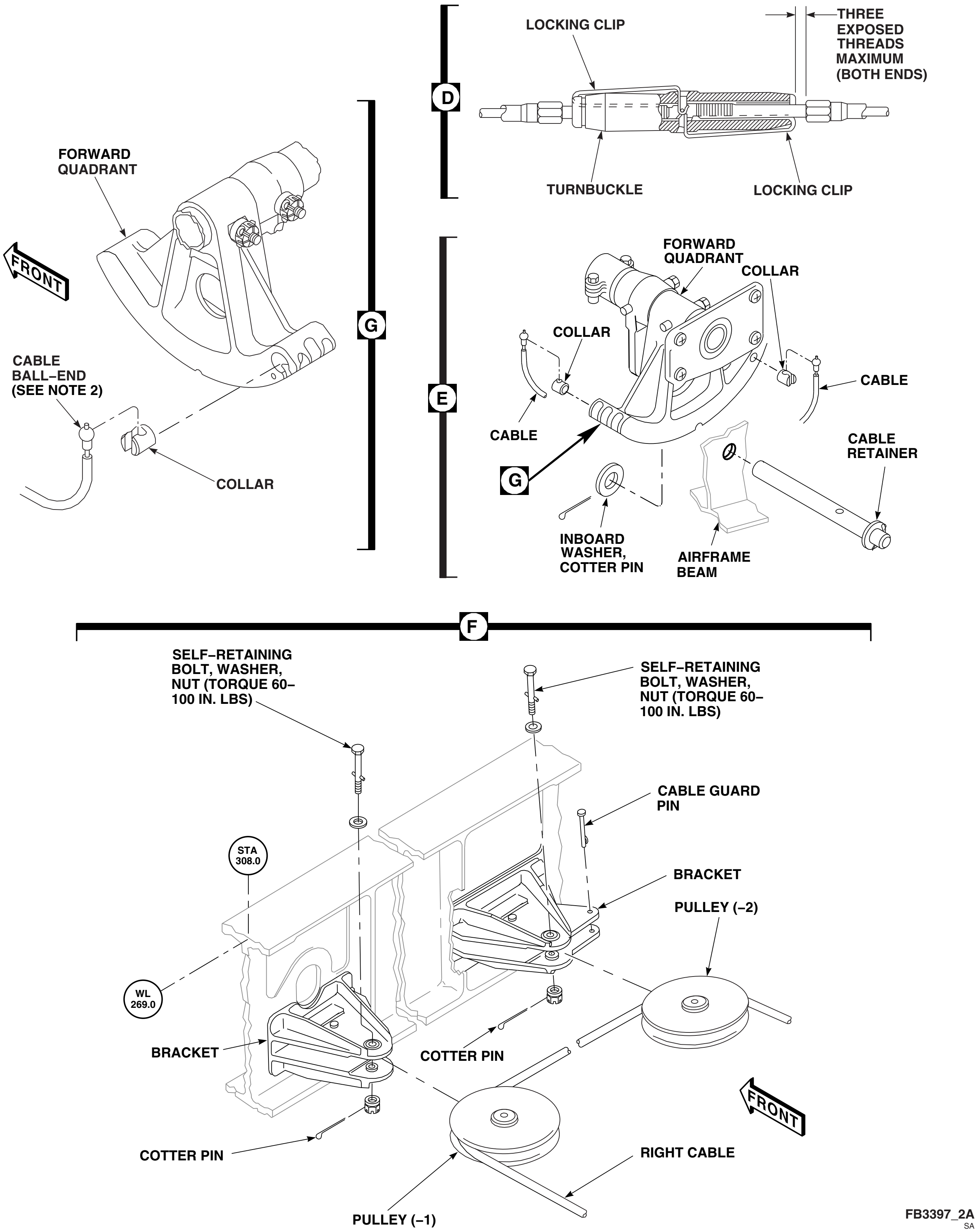


FIGURE 11-4-104-1. REPLACE CABIN FLIGHT CONTROL CABLES (SHEET 2 OF 2) through pulley bracket (-1). Install pulleys (-2) and (-1) with cable in pulley groove.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(7) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1) (Figure 11-4-104-1, Sheet 2, Detail F).

(8) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(9) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

NOTE

Make sure cable ball-end is nested inside collar during installation (Figure 11-4-104-1, Sheet 2, Detail G).

(10) Install cables in collars. Install collars and cable in forward quadrant outboard keyhole slot. Guide cable into outboard groove.

b. Install left side cable. Start from the rear (-7) pulley and work forward. Lead with the cable ball-end as follows:

(1) Thread cable under (-7) pulley. Install cable guard pin (Figure 11-4-104-1, Sheet 1, Detail A).

(2) Thread cable over top of (-8) pulley, through grommets at STA 443.0 and STA 433.0, then over top of (-9) pulley. Install cable guard pin at (-8) pulley and cable guard at (-9) pulley with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 11-4-104-1, Sheet 1, Detail C).

(3) Pass cable through grommet at STA 398.0 and over top of (-10) pulley. Install guard pin (Figure 11-4-104-1, Sheet 1, Detail A and Detail B).

(4) Guide cable forward through grommets at STA 389.0, STA 386.0, and STA 379.0, and cable conduit and grommets between STA 360.0 and STA 327.0 to front of quadrant.

NOTE

Make sure cable ball-end is nested inside collar during installation (Figure 11-4-104-1, Sheet 2, Detail G).

(5) Install cables in collars. Install collars and cable in forward quadrant inboard keyhole slot. Guide cable into inboard groove.

(6) Put washer on cable guard pin inside of outboard beam as cable guard pin is being installed. Install cotter pin (Figure 11-4-104-1, Sheet 2, Detail E).

c. Use inspection mirror and flashlight where required to see that cable is kept in pulley groove by the cable guards or pins at each pulley installation.

d. Install yaw lever to forward quadrant control rod (PARA 11-4-84).

e. Connect cabin cable to tail cone cable with turnbuckle barrels. Make sure terminal ends are threaded equally into turnbuckle.

f. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118) or remove lockout blocks (PARA 11-4-109).

g. Open main rotor pylon sliding cover.

h. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

i. On STABILATOR CONTROL/AUTO FLT CONT panel, disengage SAS 1, SAS 2, and TRIM switches. Engage BOOST switch.

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
100° - 125° (39° - 52°)	16	48

j. Install rigging pins in collective and yaw boost servos (PARA 11-4-2).

k. Install rigging pin in left side of mixer, then install rigging pin in tail rotor servo (PARA 11-4-2).

CAUTION

Damage to flight controls will result if rigging pins are allowed to bind up in controls when increasing tension in cables. To prevent damage while tensioning cables, constantly check rigging pins for free fit.

l. Using tensiometer, set cable tension by turning turnbuckle barrel. Set tension per Table 11-4-104-1. Check that tail rotor servo rigging pin slides freely in and out of rigging hole with correct tension set.

NOTE

Do not use quick-disconnects to adjust cable tension.

m. Check turnbuckle for safety. **NOT MORE THAN THREE THREADS OF TERMINAL MAY SHOW BEYOND END OF TURNBUCKLE.** If more than three threads are showing, check cable for proper installation.

n. Install turnbuckle locking clips (Figure 11-4-104-1, Sheet 2, Detail D).

Table 11-4-104-1. Tail Rotor Cable Temperature Correction Chart

Temperature Range Degrees (°F)	Temperature Range Degrees (°C)	Cable Tension (lbs)
+5 to +24	-15.0 to -5	61 to 85
+25 to +44	-4.9 to +7	85 to 109
+45 to +64	+7.2 to +18	109 to 133
+65 to +84	+18.3 to +29	133 to 168
+85 to +104	+29.4 to +40	168 to 191
+105 to +124	+41.6 to +51	191 to 215

Table 11-4-104-1. Tail Rotor Cable Temperature Correction Chart (Cont)

Temperature Range Degrees (°F)	Temperature Range Degrees (°C)	Cable Tension (lbs)
+125 to +144	+52.7 to +63	215 to 237
+145 to +160	+63.8 to +71	237 to 250

NOTES

1. Tail rotor cable tension must be stabilized on newly delivered helicopters, or on those having had recent cable replacement.

2. Check cable tension only after helicopter has been at an even temperature for 1 hour. Do not check tension with helicopter parked in hot sun. Tolerance on specified tension values is ± 2% difference between two cables.

3. If temperatures below +5°F (-15°C) are anticipated, set tension to 75 pounds. Reset tension to chart before helicopter is operated at temperatures above 40°F (4°C).

4. Flight control cables are 5/32-inch diameter, and with a plastic coating, total diameter is 7/32-inch. Measure cable tension with proper tensiometer setting.

   (a) When using adjustable risers, the No. 3 riser adapts to 7/32-inch total-diameter cable. Use scale for 7/32-inch diameter cables.

   (b) When tensiometer does not have adjustable risers, use the 5/32-inch diameter tension scale.

5. Record temperature and cable tension on helicopter logbook for use when performing future tension check.

6. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 to 11 flight hours (PARA 1-7-10).

---

o. Using inspection mirror and flashlight, check cable run over each pulley (-1) through (-10) (Figure 11-4-104-1, Sheet 1, Detail A). Make sure cable is in pulley groove and passes between pulley and cable guard pins; and make sure cable guard pins keep cable in pulley groove.

p. Perform tail rotor system rig check (PARA 11-4-6).

q. Remove rigging pins from tail rotor servo, then mixer.

r. Remove rigging pins from yaw and collective boost servos.

s. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

t. Make sure area is clean and free of foreign material.

u. Install cargo netting.

v. Install soundproofing panels (PARA 2-4-69).

w. Close main rotor pylon sliding cover.

x. Perform operational check of flight control system (PARA 11-2-3).

y. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 to 11 flight hours (PARA 1-7-10).

11-4-105 TAIL CONE FLIGHT CONTROL CABLES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Cord (Approximately 20 feet long)
- Flashlight
- Inspection Mirror
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-10
- PARA 2-4-69
- PARA 11-2-3
- PARA 11-4-2
- PARA 11-4-6
- PARA 11-4-104
- PARA 11-4-109
- PARA 11-4-118

11-4-105.1 Replace Tail Cone Flight Control Cables.

NOTE

Replacement of left or right tail cone flight control cable is the same.

11-4-105.1.1 Remove.

a. Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0 (Figure 11-4-105-1, Detail A). Remove access panel.

b. Remove soundproofing panels from rear bulkhead covering left or right stowage compartments (PARA 2-4-69).

c. Remove cargo netting from stowage compartments.

d. Install lockout blocks (PARA 11-4-109) or disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

e. Remove locking clips from turnbuckle on right or left side (Figure 11-4-105-1, Detail B).

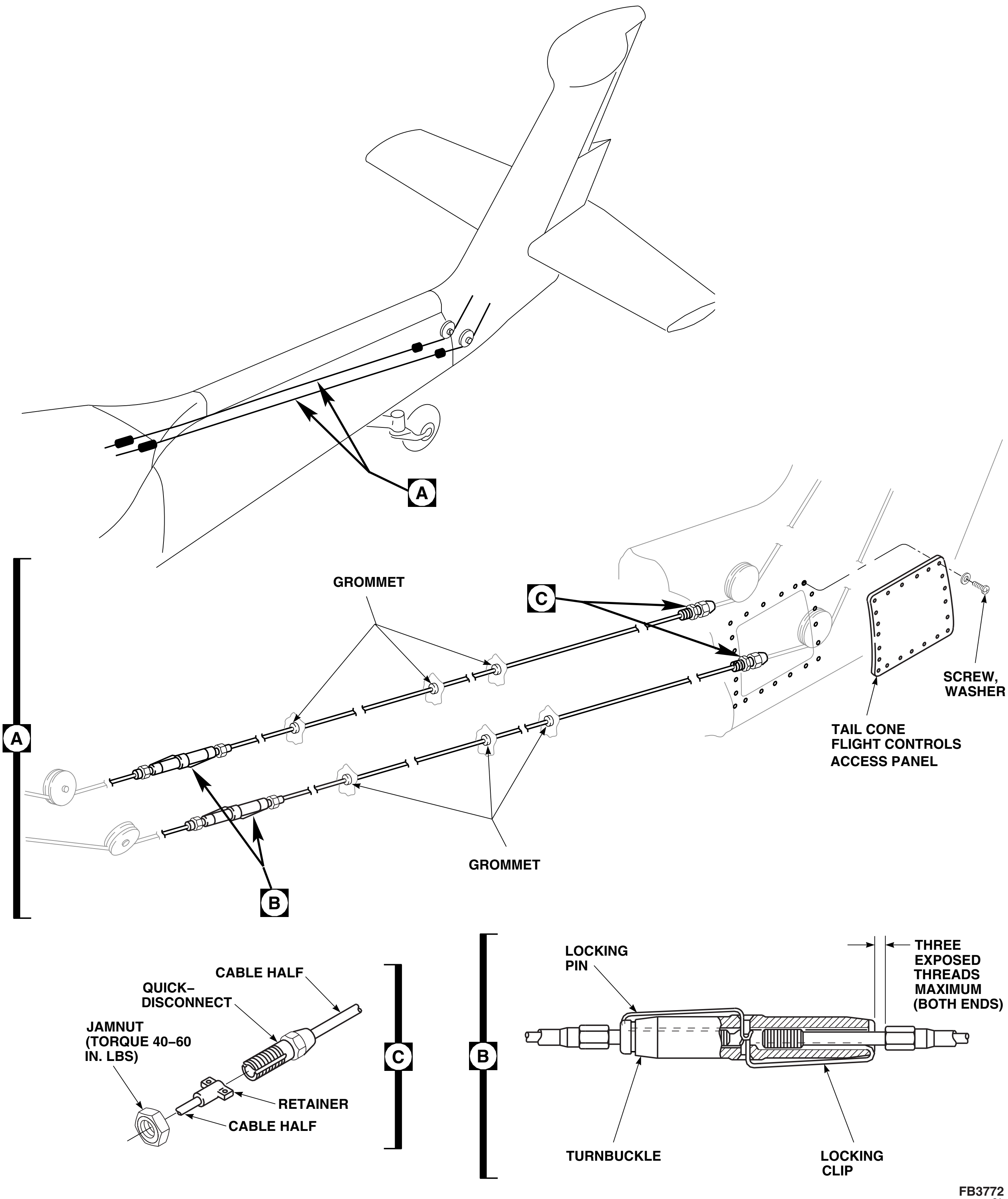


FIGURE 11-4-105-1. REPLACE TAIL CONE FLIGHT CONTROL CABLES

f. Turn body of turnbuckle to loosen cable. Continue turning and remove cable terminal end.

g. At left or right cable, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-105-1, Detail C).

h. Tie a strong cord or lockwire around cable terminal end.

i. From inside tail cone at cabin end, pull cable out through grommets until cord or lockwire appears.

j. Remove cord or lockwire from cable.

k. Remove quick-disconnect jamnut from cable by sliding jamnut along entire length of cable and over terminal end.

11-4-105.1.2 Install.

WARNING

FSCAP

Verification of proper cable routing between pulleys and retaining pins or guards is a critical characteristic.

a. Install jamnut by sliding nut over turnbuckle end and down entire length of replacement cable (Figure 11-4-105-1, Detail C).

b. Tie cord or lockwire around turnbuckle end of cable.

c. From inside tail cone, at tail end, pull on cable while assistant feeds cable through grommets.

NOTE

- Quick-disconnects have two configurations.
- Extension half must bottom in slot half (2.44-inch slot).
- Ball-end bottoms before extension half (2.63-inch slot).
- Both assemblies are acceptable when torqued to specifications.
- Do not use quick-disconnects to adjust cable tension.

d. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire jamnut to retainer.

e. Connect cabin cable to tail cone cable at turnbuckles (PARA 11-4-104).

f. Open main rotor pylon sliding cover.

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
100° - 125° (39° - 52°)	16	48

g. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. On STABILATOR CONTROL/AUTO FLT CONT panel, disengage SAS 1, SAS 2, and TRIM switches. Engage BOOST switch.

i. Remove lockout blocks (PARA 11-4-109) or connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

j. Install rigging pins in collective and yaw boost servos (PARA 11-4-2).

k. Install rigging pin in left side of mixer, then install rigging pin in tail rotor servo (PARA 11-4-2).

NOTE

Tail rotor cable tension must be stabilized on newly delivered helicopters, or on those that have had recent cable replacement. A tail rotor cable tension check must be performed every 9 to 11 flight hours until tension stabilizes.

l. Perform tail rotor cable tension check every 9 to 11 flight hours until tension stabilizes (PARA 1-7-10).

m. Adjust cable tension (PARA 11-4-104).

n. Install turnbuckle locking clips (Figure 11-4-105-1, Detail A).

o. Using inspection mirror and flashlight, check cable run over each pulley (Figure 11-4-105-1, Detail A). Make sure cable is in pulley groove and passes between pulley and cable guard pins; and make sure cable guard pins keep cable in pulley groove.

p. Perform tail rotor system rig check (PARA 11-4-6).

q. Remove rigging pins from tail rotor servo, then the mixer.

r. Remove rigging pins from yaw and collective boost servos.

s. Perform operational check of flight control system (PARA 11-2-3).

t. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

u. Make sure area is clean and free of foreign material.

v. Close main rotor pylon sliding cover.

w. Install tail cone flight controls access panel on tail cone and secure with screws and washers.

x. Install cargo netting.

y. Install soundproofing panels (PARA 2-4-69).

z. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 to 11 flight hours (PARA 1-7-10).

11-4-106 TAIL ROTOR PYLON FLIGHT CONTROL CABLES.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Cord (Approximately 8 feet long)
- Flashlight
- Inspection Mirror
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-9
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-10
- PARA 2-4-137
- PARA 11-2-3
- PARA 11-3-25
- PARA 11-4-2
- PARA 11-4-6
- PARA 11-4-104
- PARA 11-4-107
- PARA 11-4-108
- PARA 11-4-118

11-4-106.1 Replace Tail Rotor Pylon Flight Control Cables.

NOTE

Replacement of left or right tail rotor pylon flight control cable is the same, except as noted.

11-4-106.1.1. Remove.

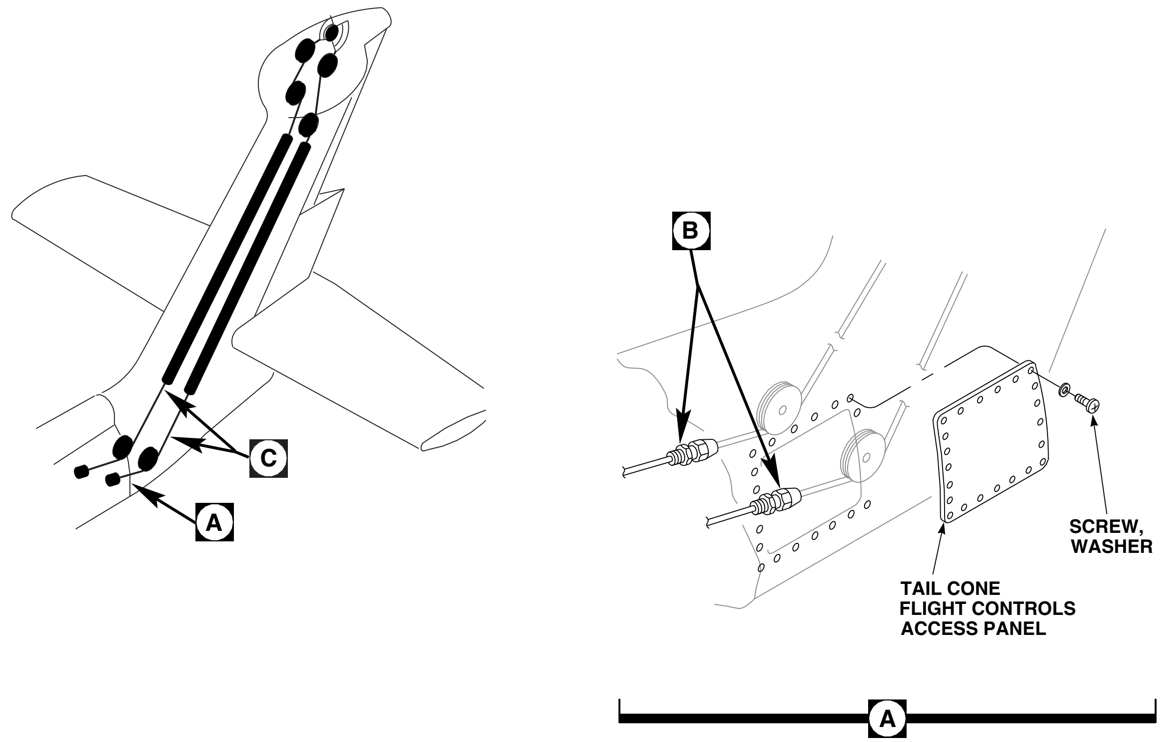
a. Open tail rotor pylon leading edge cover/antenna.

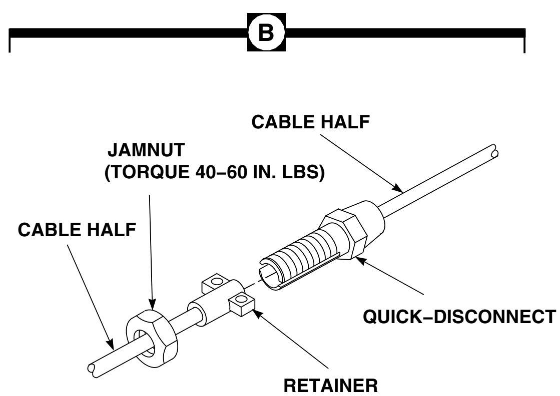
b. Remove tail gear box front fairing (PARA 2-4-137).

c. Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0. Remove access panel (Figure 11-4-106-1, Sheet 1, Detail A).

d. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

e. At left or right cable, back off jamnut securing quick-disconnect and retainer. Separate quick-





NOTE

LOWER HOLE NOT USED FOR
CABLE GUARD PIN INSTALLATION.

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FIGURE 11-4-106-1. REPLACE TAIL ROTOR PYLON FLIGHT CONTROL CABLES
(SHEET 1 OF 3)

disconnect and retainer (Figure 11-4-106-1, Sheet 1, Detail B).

f. Fold tail rotor pylon (PARA 1-6-9).

g. Remove pulleys and cable guard pins as follows:

(1) Pulley at PSTA 78.0 (PARA 11-4-107).

(2) Pulley at PSTA 185.0 for left cable or PSTA 193.0 for right cable (PARA 11-4-107).

(3) Cable guard pins at PSTA 199.0 (Figure 11-4-106-1, Sheet 2, Detail C).

h. Remove flathead screws, washers, and nuts securing cable guards to quadrant (Figure 11-4-106-1, Sheet 3, Detail E).

i. Turn cable 90° to tail rotor quadrant. Remove collar and cable from quadrant. Remove cable from collar (Figure 11-4-106-1, Sheet 3, Detail F).

j. Tie an 8-foot length of cord or wire to ball-end (upper end) of cable to be replaced.

k. Pull cable around pulleys, through cable conduit, and out lower end of tail rotor pylon. Remove cord or wire from cable.

l. Inspect cable conduit (PARA 11-3-25).

11-4-106.1.2. Install.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

a. Tie cord or wire to ball end of replacement cable and pull cable under pulleys and up through cable conduit and over pulleys in upper tail rotor pylon.

NOTE

- Make sure cable is not routed under sheet metal retaining device.
- Make sure cable is between pulley and pins when installing.

b. At PSTA 78.0, place cable in pulley groove. Install pulley (PARA 11-4-107).

WARNING

There are two holes in pulley bracket (PSTA 185.0) into which a cable guard pin will fit. Maintenance personnel are to make sure that no pin is inserted in the lower hole, as this may lead to premature wear, and eventual failure of the cable (Figure 11-4-106-1, Sheet 3, Detail G).

NOTE

- Make sure cable is not routed under sheet metal retaining device.
- Make sure cable is between pulley and pins when installing.

c. If installing left cable, at PSTA 185.0, place cable in pulley groove. Install pulley (PARA 11-4-107).

NOTE

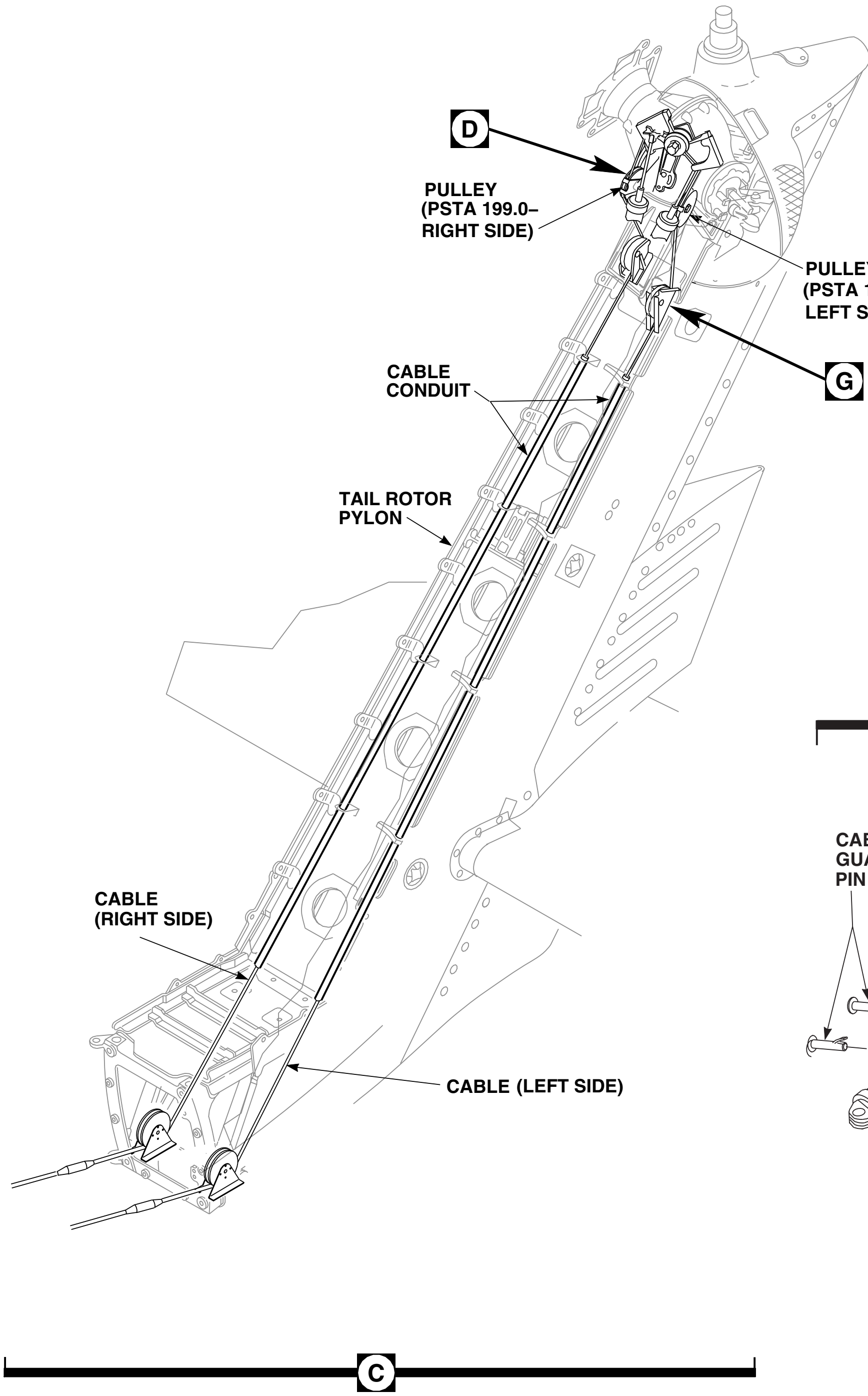
Make sure cable is between pulley and pins when installing.

d. If installing right cable, at PSTA 193.0, place cable in pulley groove. Install pulley (PARA 11-4-107).

NOTE

Make sure cable is between pulley and pins when installing.

e. At PSTA 199.0, route cable over pulley. Install cable guard pins.



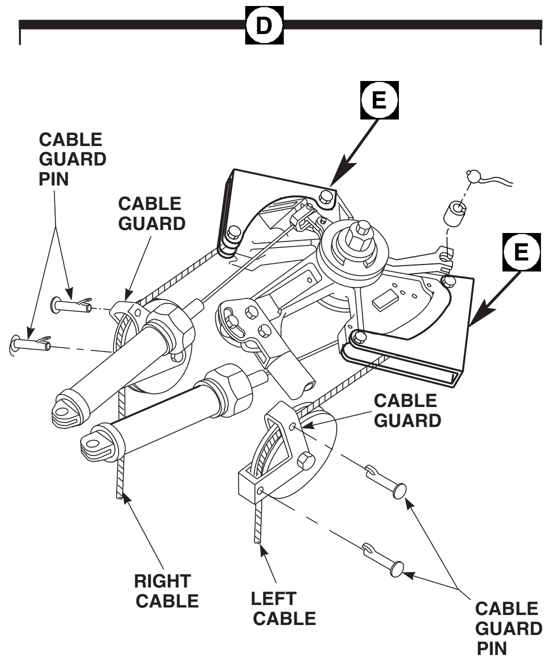


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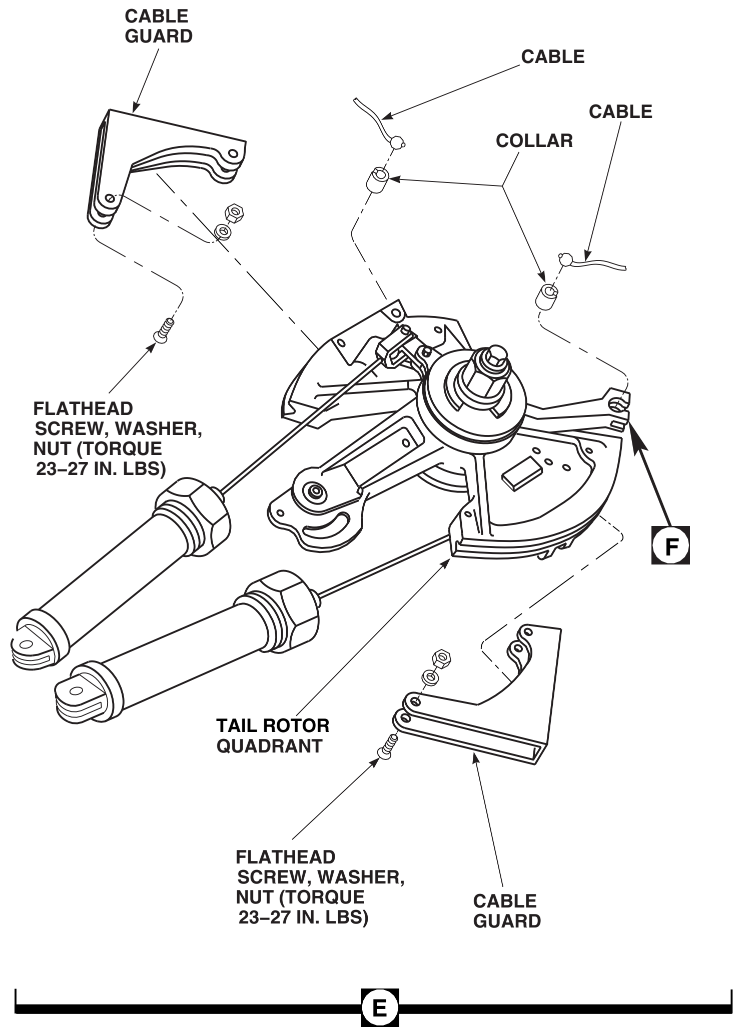
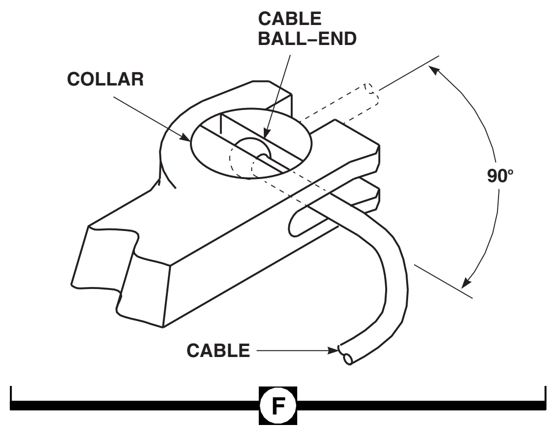
FIGURE 11-4-106-1. REPLACE TAIL ROTOR PYLON FLIGHT CONTROL CABLES (SHEET 2 OF 3)





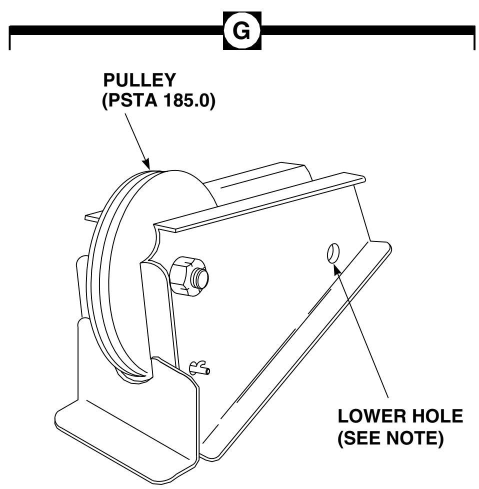


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FIGURE 11-4-106-1. REPLACE TAIL ROTOR PYLON FLIGHT CONTROL CABLES (SHEET 3 OF 3)

WARNING

FSCAP

Verification of proper cable routing over quadrant and pulleys, and under cable retainers are critical characteristics.

f. Verify cable has remained in pulley grooves. Insert cable guard pins at pulleys.

NOTE

Make sure collar is installed with slotted side up.

g. Insert cable into collar. Turn cable 90° to direction cable will be going. Slide collar and cable into quadrant. Make sure cable is in quadrant groove (Figure 11-4-106-1, Sheet 3, Detail F).

h. Install cable guard on quadrant and secure with flathead screws, washers, and nuts (nuts up). **TORQUE NUTS TO 23 - 27 INCH-POUNDS** (Figure 11-4-106-1, Sheet 3, Detail E).

i. Unfold tail rotor pylon (PARA 1-6-9).

NOTE

- Quick-disconnects have two configurations.
- Extension half must bottom in slot half (2.44-inch slot).
- Ball-end bottoms before extension half (2.63-inch slot).
- Both assemblies are acceptable when torqued to specifications.
- Do not use quick-disconnects to adjust cable tension.

j. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS.** Lockwire, Item 197, Appendix D, jamnut to retainer (Figure 11-4-106-1, Sheet 1, Detail B).

k. Turn on all external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

l. On AUTO FLIGHT CONTROL panel, disengage SAS 1, SAS 2, and TRIM switches. Engage BOOST switch.

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
100° - 125° (39° - 52°)	16	48

m. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

n. Install rigging pins in collective and yaw boost servos (PARA 11-4-2).

o. Install rigging pin in left side of mixer, then install rigging pin in tail rotor servo (PARA 11-4-2).

p. Adjust cable tension (PARA 11-4-104). Make sure turnbuckle is in proper safety when finished.

q. Perform tail rotor system rig check (PARA 11-4-6).

r. Remove rigging pins from tail rotor servo, then the mixer.

s. Remove rigging pins from yaw and collective boost servos.

t. Using inspection mirror and flashlight, check cable run over each pulley (Figure 11-4-106-1, Sheet 2, Detail C). Make sure cable is in pulley groove and passes between pulley and cable guard pins; and make sure cable guard pins keep cable in pulley groove.

u. Perform operational check of mechanical flight control system (PARA 11-2-3).

v. Turn off all external hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

w. Make sure area is clean and free of foreign material.

x. Install tail cone flight controls access panel on tail cone and secure with screws and washers (Figure 11-4-106-1, Sheet 1, Detail A).

y. Close tail rotor pylon leading edge cover/ antenna.

z. Install tail gear box front fairing (PARA 2-4-137).

aa. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 to 11 flight hours (PARA 1-7-10).

11-4-107 FLIGHT CONTROL CABLE PULLEYS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Flashlight
- Inspection Mirror
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 197, Appendix D

References

- Appendix D
- PARA 1-6-9
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-4-137
- PARA 2-6-5
- PARA 11-2-3
- PARA 11-4-1
- PARA 11-4-5
- PARA 11-4-84
- PARA 11-4-104
- PARA 11-4-118
- TM 11-70-23

11-4-107.1 Replace Flight Control Cable Pulleys.

11-4-107.1.1 Prepare to Remove.

NOTE

Cable pulley shall be replaced if any cracks, chips, nicks, or excessive wear is found on either pulley or pulley groove. Minute nicks, chips, or scratches not located in pulley groove are acceptable.

a. Remove soundproofing panels as required (PARA 2-4-69).

b. Remove screws and washers securing tail cone flight controls access panel to tail cone at STA 640.0 (Figure 11-4-107-1, Sheet 5, Detail L). Remove access panel.

c. If replacing pulleys at PSTA 185.0, PSTA 193.0, or PSTA 199.15, perform the following:

(1) Open tail rotor pylon leading edge cover/ antenna.

(2) Remove tail gear box front fairing (PARA 2-4-137).

11-4-107.1.2 Replace STA 311.75 Pulleys.

a. Replace left side pulley at STA 311.75 as follows (Figure 11-4-107-1, Sheet 2, Detail A):

(1) Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

(2) At left cable, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

(3) Remove yaw lever to forward quadrant control rod (PARA 11-4-84).

(4) Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 2, Detail A). Remove pulley.

NOTE

Do not remove cable guard pin.

(5) Verify cable guard pin is properly installed.

(6) Place cable in pulley and position pulley between brackets. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(a) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(7) Using feeler gage, check that 0.001-inch minimum clearance exists between face of pulley and bracket. Check that pulley rotates freely.

(8) Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

(9) Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pin

(10) Install yaw lever to forward quadrant control rod (PARA 11-4-84).

(11) Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

(12) Perform tail rotor system complete rig (PARA 11-4-5).

b. Replace right side pulley at STA 311.75 as follows (Figure 11-4-107-1, Sheet 2, Detail A):

(1) Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

(2) At right cable, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

(3) Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 2, Detail A). Remove pulley.

NOTE

Do not remove cable guard pins.

(4) Verify cable guard pins are properly installed.

(5) Place cable in pulley and position pulley between brackets. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(a) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(6) Using feeler gage, check that 0.001-inch minimum clearance exists between face of pulley and bracket. Check that pulley rotates freely.

(7) Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

(8) Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pins.

(9) Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

(10) Adjust cable tension (PARA 11-4-104).

11-4-107.1.3 Replace STA 395.5 Pulleys.

NOTE

- Replacement of left and right pulleys is the same, except as noted.
- Pulleys located at STA 395.5 can be replaced without disconnecting cables.
- When replacing right side pulley at STA 395.5, if bolt is installed with bolthead facing inboard, remove stabilator amplifier.

a. If required, remove stabilator amplifier (TM 11-70-23).

NOTE

Slight upward hand pressure may be required to remove bolt.

b. Remove self-retaining bolt, washer(s), bushings, and nut securing pulley to brackets (Figure 11-4-107-1, Sheet 2, Detail B). Remove pulley.

CAUTION

If more than one hole is present in pulley bracket, cable guard pin is to be installed through hole directly vertically above the bolt securing pulley.

NOTE

Do not remove cable guard pin.

c. Verify cable guard pin is properly installed.

NOTE

When positioning pulley, push pulley up between brackets while making sure cable is located in pulley groove.

d. Install flanged bushing on inboard bracket.

e. Position bushing and pulley between brackets. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

- Slight upward hand pressure may be required to install bolt.
- When installing right side pulley, bolthead may face either direction. Make sure washer, NAS1149D0563K, is installed between washer, MS25440-5, and bolthead.
- When installing left side pulley, bolthead must face outboard.

(1) Install self-retaining bolt and washer(s) (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

f. Verify cable is within pulley groove and that cable is located between the pulley and the cable guard pin.

g. If removed, install stabilator amplifier (TM 11-70-23).

h. Adjust cable tension (PARA 11-4-104).

11-4-107.1.4 Replace STA 418.2 Pulleys.

NOTE

Replacement of left and right pulleys is the same.

a. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

b. From same cable that pulley is being replaced, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

c. Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 3, Detail C). Remove pulley.

d. Verify cable guard is properly installed.

e. Place cable in pulley and position pulley on bracket. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Using feeler gage, check that 0.060-inch minimum clearance exists between cable and airframe.

g. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

h. Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard.

i. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

j. Adjust cable tension (PARA 11-4-104).

11-4-107.1.5 Replace STA 447.0 Pulleys.

NOTE

Replacement of left and right pulleys is the same.

a. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

b. From same cable that pulley is being replaced, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

c. Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 3, Detail D). Remove pulley.

NOTE

Do not remove cable guard pin.

d. Verify cable guard pin is properly installed.

e. Place cable in pulley and position pulley on bracket. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

g. Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pins.

h. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

i. Adjust cable tension (PARA 11-4-104).

11-4-107.1.6 Replace STA 466.7 Pulleys.

NOTE

Replacement of left and right pulleys is the same, except as noted.

a. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

b. From same cable that pulley is being replaced, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

c. Remove self-retaining bolt, washer, bushing, retainer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 3, Detail E). Remove pulley.

NOTE

Do not remove cable guard pins.

d. Verify cable guard pins are properly installed.

e. Place cable in pulley and position retainer, bushing, and pulley on bracket. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(1) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(2) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(3) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

g. Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pins.

h. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

i. Adjust cable tension (PARA 11-4-104).

11-4-107.1.7 Replace Tail Pylon Pulleys.

a. Remove pulleys at PSTA 78.0 as follows (Figure 11-4-107-1, Sheet 4, Detail F):

NOTE

Replacement of left and right pulleys is the same.

(1) Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

(2) From same cable that pulley is being replaced, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

(3) Fold tail pylon (PARA 1-6-9).

(4) Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 4, Detail F). Remove pulley.

NOTE

Do not remove cable guard pin.

(5) Verify cable guard pin is properly installed.

(6) Place cable in pulley and position pulley between brackets. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(a) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(7) Unfold tail pylon (PARA 1-6-9).

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

(8) Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pin.

(9) Using inspection mirror and flashlight, make sure cable is not routed under sheet metal retaining device.

(10) Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

(11) Adjust cable tension (PARA 11-4-104).

b. Remove pulley at PSTA 185.0 as follows (Figure 11-4-107-1, Sheet 5, Detail K):

(1) Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

(2) At left cable, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

(3) Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 5, Detail K). Remove pulley.

NOTE

Do not remove cable guard pin.

(4) Verify cable guard pin is properly installed.

(5) Place cable in pulley and position pulley in bracket. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(a) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(6) Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

(7) Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pin.

(8) Using inspection mirror and flashlight, make sure cable is not routed under sheet metal retaining device.

(9) Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

(10) Adjust cable tension (PARA 11-4-104).

c. Remove pulley at PSTA 193.0 as follows (Figure 11-4-107-1, Sheet 4, Detail G):

(1) Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

(2) At right cable, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

(3) Remove self-retaining bolt, washer, and nut securing pulley to bracket (Figure 11-4-107-1, Sheet 4, Detail G). Remove pulley.

NOTE

Do not remove cable guard pin.

(4) Verify cable guard pin is properly installed.

(5) Place cable in pulley and position pulley in bracket. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(a) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(6) Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

(7) Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pin.

(8) Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

(9) Adjust cable tension (PARA 11-4-104).

d. Remove pulleys at PSTA 199.15 as follows (Figure 11-4-107-1, Sheet 4, Detail H or Sheet 5, Detail J):

NOTE

Replacement of left and right pulleys is the same.

(1) Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

WARNING

Uncontrolled spring release may cause personal injury or damage to equipment. Make sure spring cylinders are disconnected.

(2) From same cable that pulley is being replaced, back off jamnut securing quick-disconnect and retainer. Separate quick-disconnect and retainer (Figure 11-4-107-1, Sheet 5, Detail M).

(3) Remove self-retaining bolt, washers, and nut securing pulley to guard (Figure 11-4-

107-1, Sheet 4, Detail H or Sheet 5, Detail J). Remove pulley.

NOTE

Do not remove cable guard pins.

(4) Verify cable guard pins are properly installed.

NOTE

0.032-inch washer to be located on outboard side of pulley.

(5) Place cable in pulley and position pulley and washer in guard. Secure pulley as follows:

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

(a) Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(6) Connect cable halves at retainer and quick-disconnect, and secure with jamnut. **TORQUE JAMNUT TO 40 - 60 INCH-POUNDS** (Figure 11-4-107-1, Sheet 5, Detail M). Lockwire, Item 197, Appendix D, jamnut to retainer.

WARNING

FSCAP

Verification of proper cable routing between cable retainers and quadrants or pulleys is a critical characteristic.

(7) Verify cable has remained in pulley groove and that cable is located between the pulley and the cable guard pins.

(8) Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

(9) Adjust cable tension (PARA 11-4-104).

11-4-107.1.8 Complete Installation.

a. Make sure area is clean and free of foreign material.

b. If removed, install soundproofing panels (PARA 2-4-69).

c. Install tail cone flight controls access panel on tail cone and secure with screws and washers (Figure 11-4-107-1, Sheet 5, Detail L).

d. If pulleys at PSTA 185.0, PSTA 193.0, or PSTA 199.15 were replaced, perform the following:

(1) Close tail rotor pylon leading edge cover/antenna.

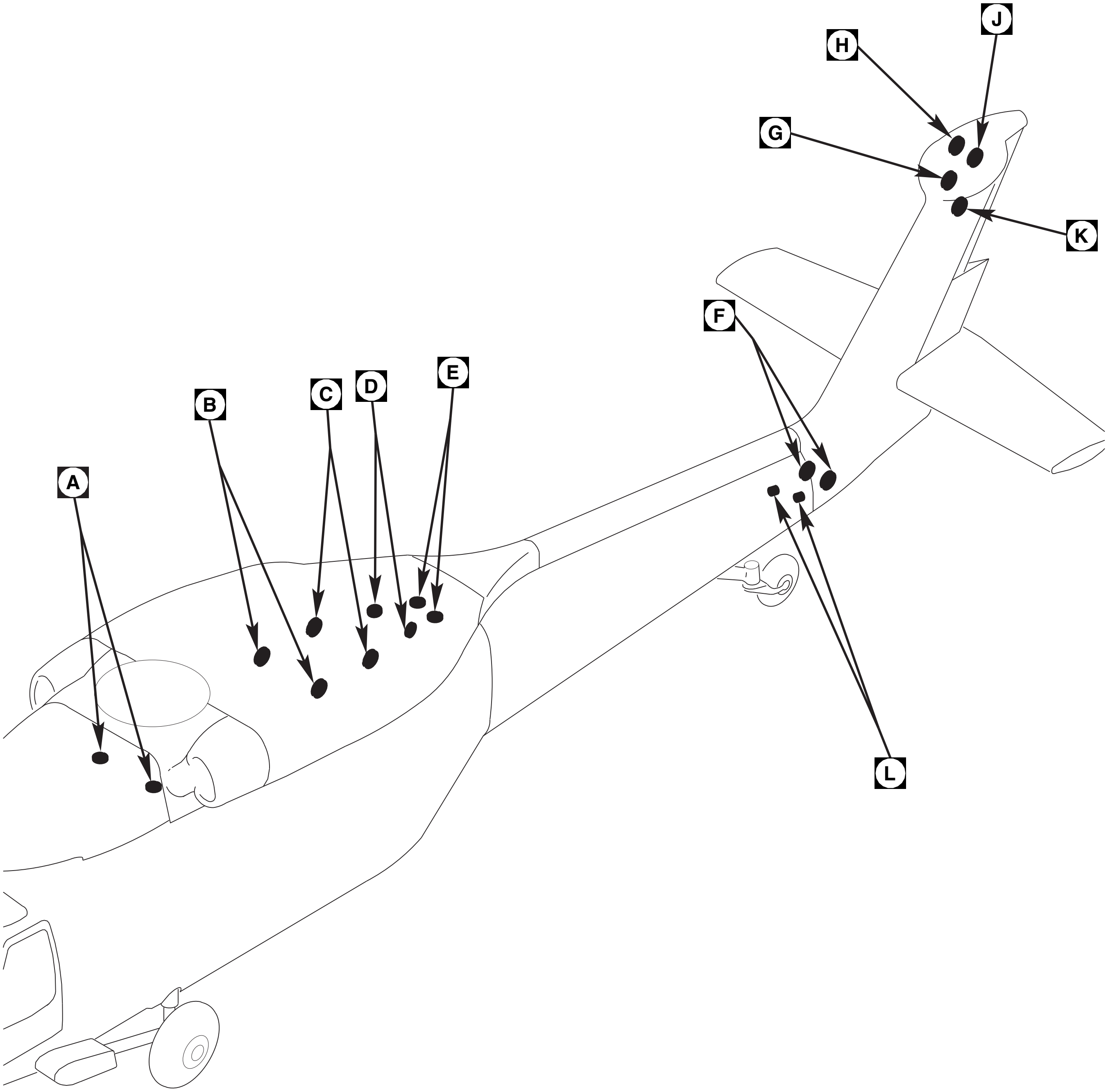
(2) Install tail gear box front fairing (PARA 2-4-137).

e. Turn on hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

f. Perform operational check of mechanical flight control system (PARA 11-2-3).

g. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

h. Check helicopter logbook to make sure tail rotor cable tension check is shown as due at 9 to 11 flight hours.



NOTE

RETAINER SHOWN REMOVED FOR CLARITY.

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FIGURE 11-4-107-1. REPLACE FLIGHT CONTROL CABLE PULLEYS (SHEET 1 OF 5)

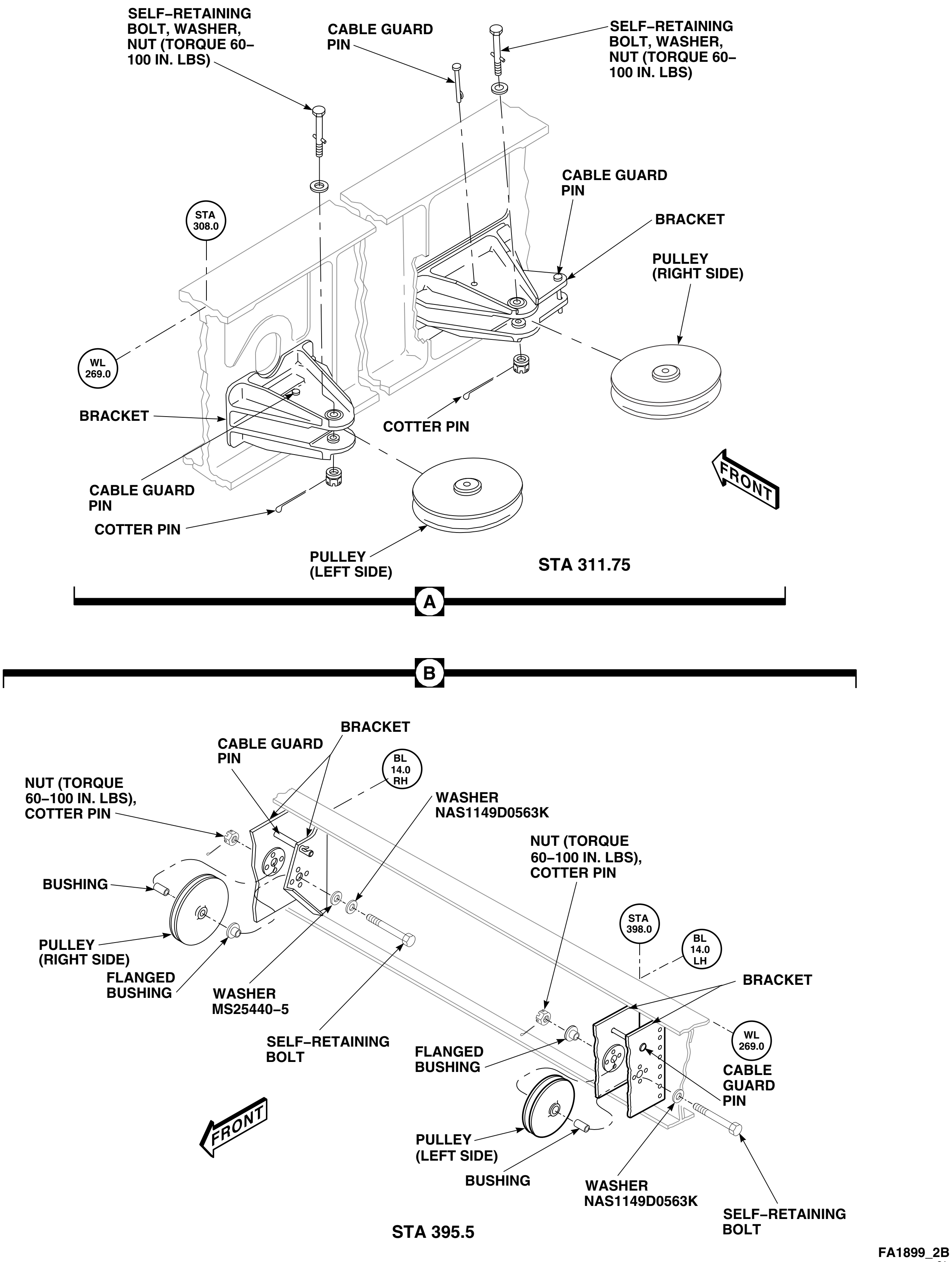


FIGURE 11-4-107-1. REPLACE FLIGHT CONTROL CABLE PULLEYS (SHEET 2 OF 5)

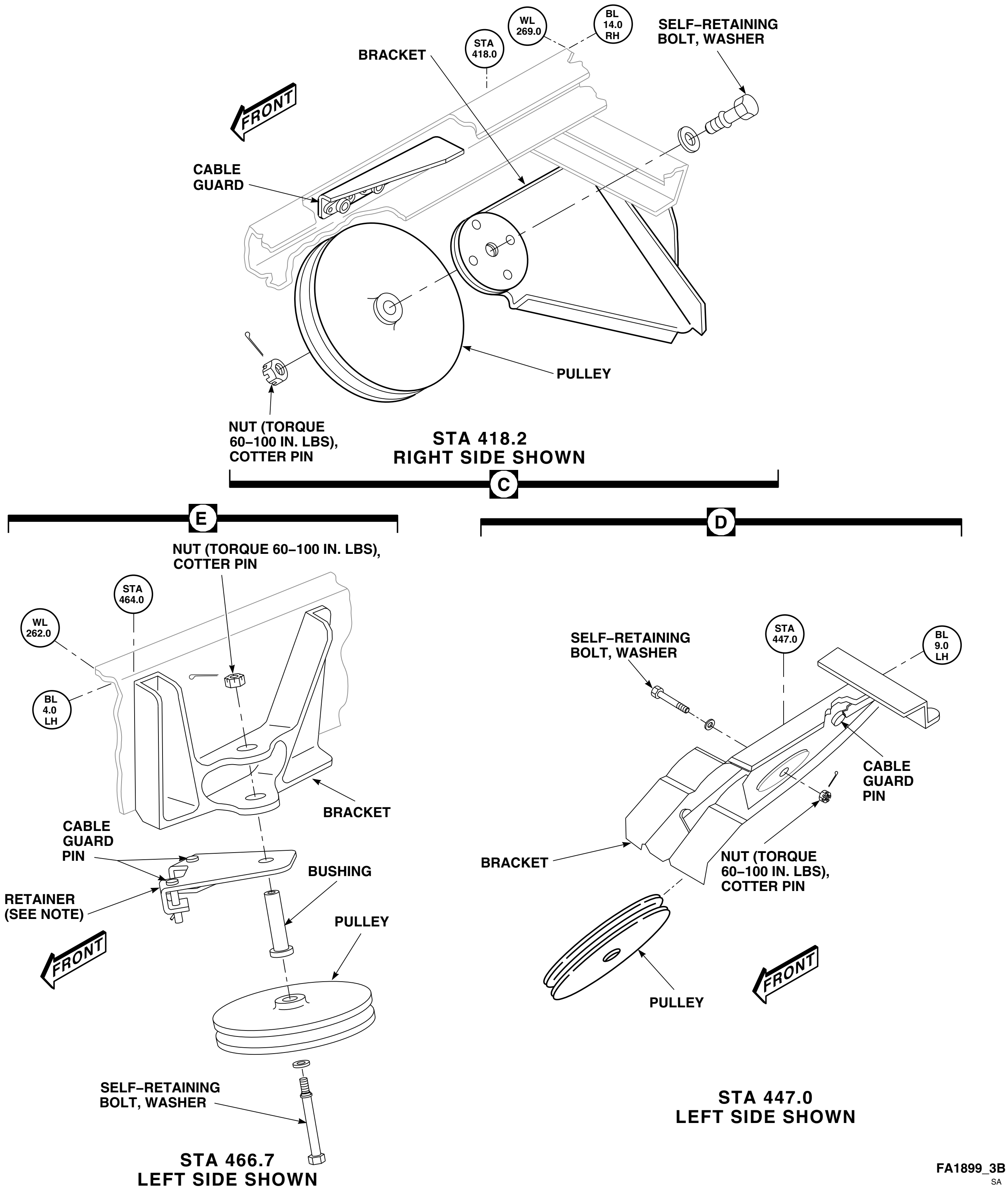


FIGURE 11-4-107-1. REPLACE FLIGHT CONTROL CABLE PULLEYS (SHEET 3 OF 5)

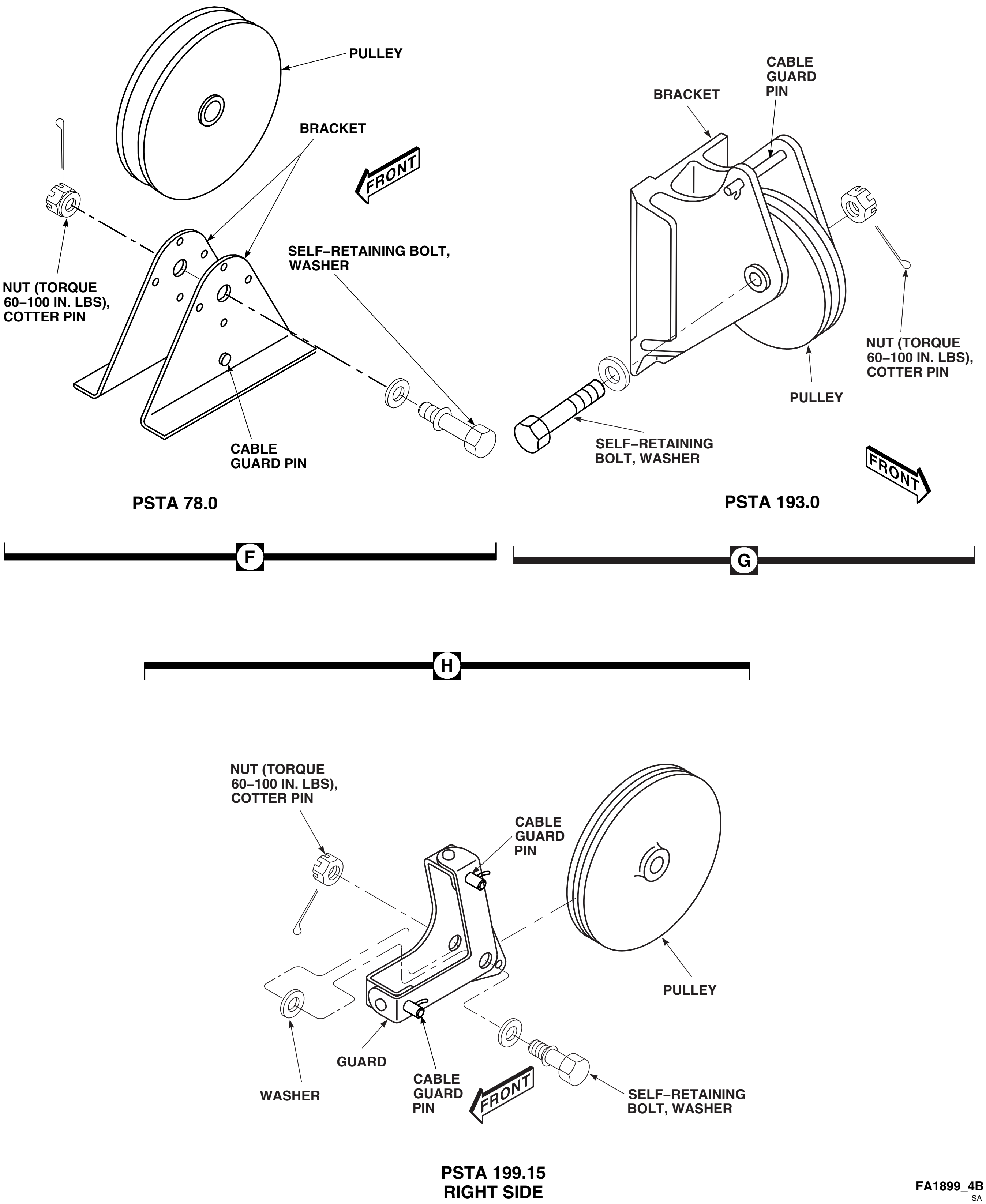


FIGURE 11-4-107-1. REPLACE FLIGHT CONTROL CABLE PULLEYS (SHEET 4 OF 5)

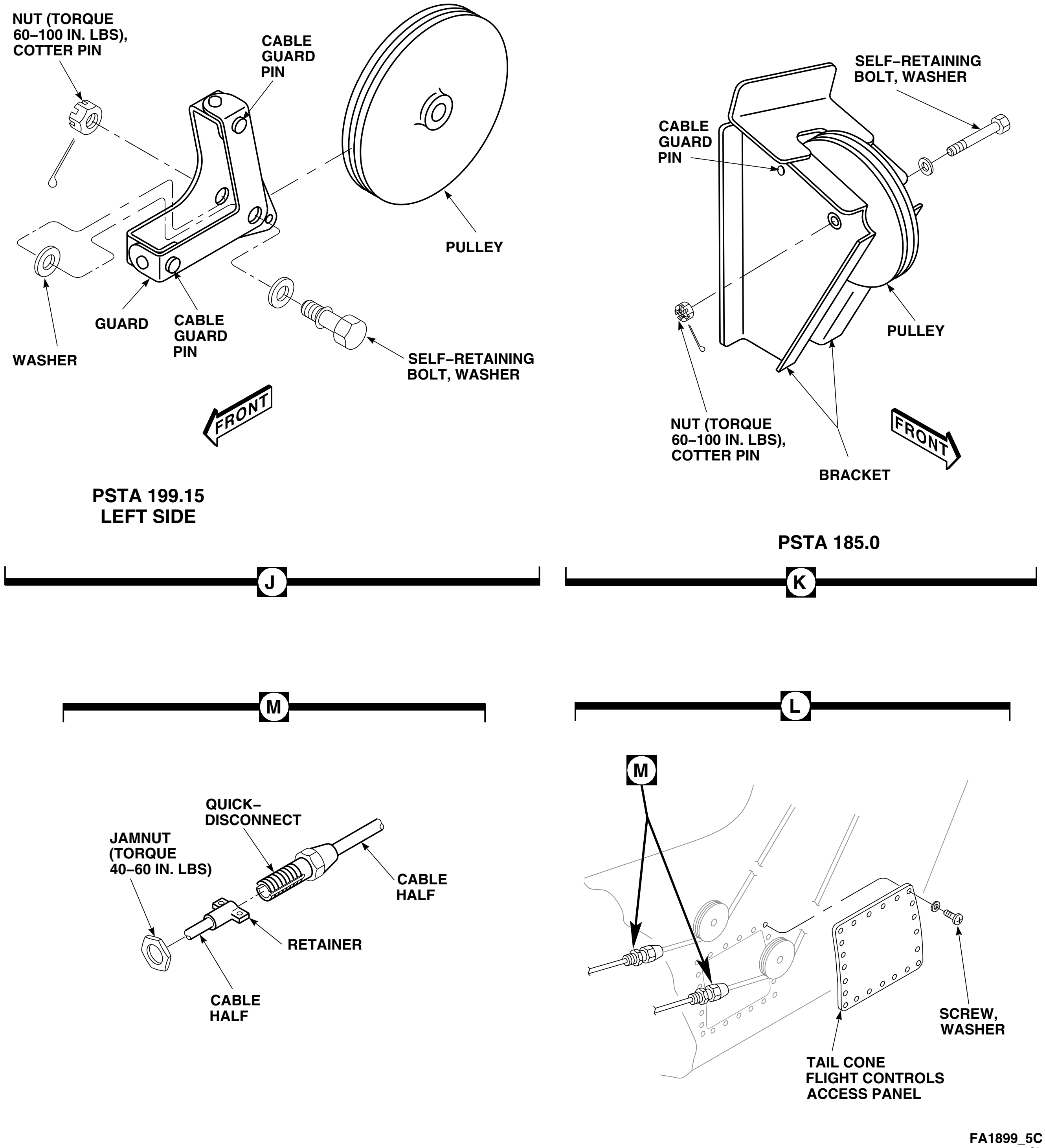


FIGURE 11-4-107-1. REPLACE FLIGHT CONTROL CABLE PULLEYS (SHEET 5 OF 5)

11-4-108 FLIGHT CONTROL CABLE CONDUIT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Adhesive, Item 38, Appendix D
- Adhesive Primer, Item 68, Appendix D
- Cheesecloth, Item 90, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-7-8
- PARA 1-7-10
- PARA 1-7-11
- PARA 6-4-41
- PARA 11-2-3
- PARA 11-4-106

11-4-108.1 Replace Flight Control Cable Conduit.

11-4-108.1.1. Remove.

a. Remove tail rotor drive shaft section IV (PARA 6-4-41).

b. Remove tail rotor pylon flight control cables (PARA 11-4-106).

c. Cut and remove conduit from tail rotor pylon (Figure 11-4-108-1, Detail A).

d. Remove all remaining nylon and adhesive debris from brackets with nonmetallic scraper.

11-4-108.1.2. Install.

a. Clean top and bottom of conduit brackets with methyl ethyl ketone, Item 217, Appendix D, and cheesecloth, Item 90, Appendix D. Wipe part with clean cheesecloth until no stain shows on cloth after last wipe. Do not touch cleaned brackets.

b. Using epoxy primer coating kit, Item 135, Appendix D, touch up bracket with epoxy primer. Allow primer to dry.

c. Trim replacement conduit to 93.31 - 93.56-inch length.

d. Measure and mark a line 0.25-inch up from one end of conduit. This is the length conduit is to

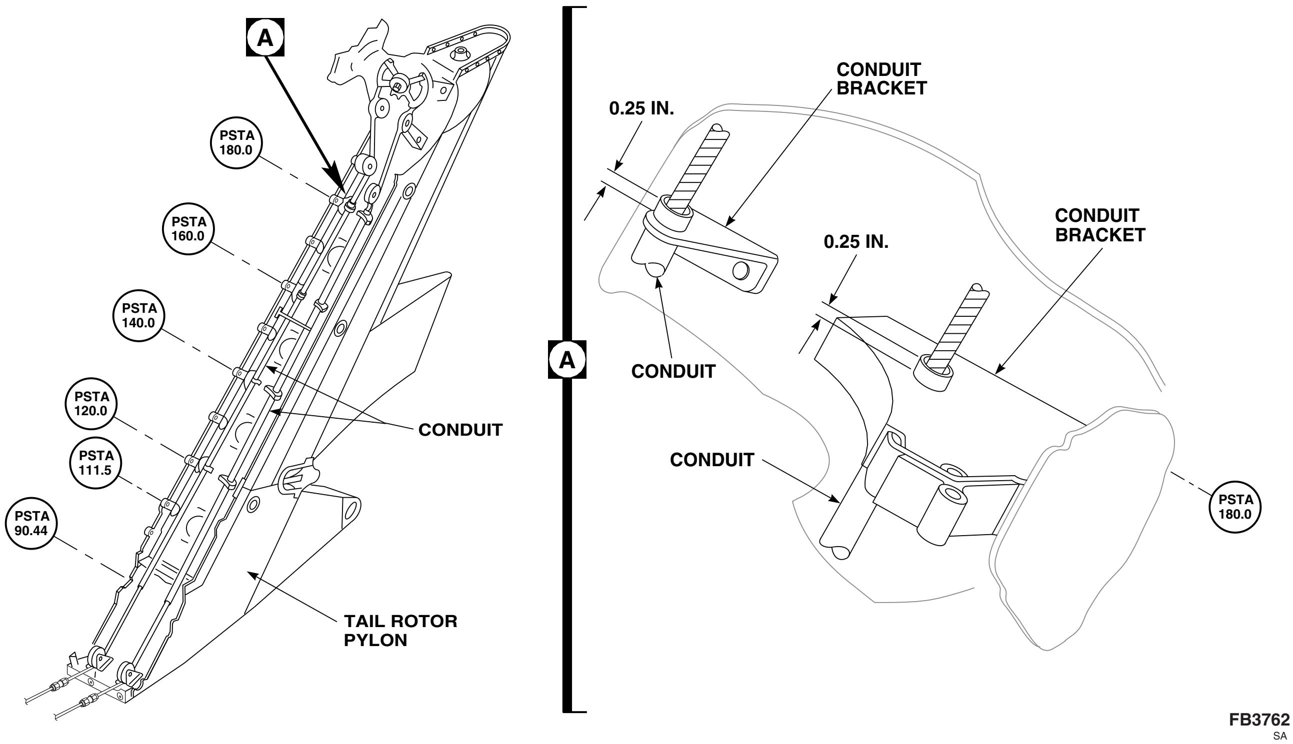


FIGURE 11-4-108-1. REPLACE FLIGHT CONTROL CABLE CONDUIT

be extended above pylon PSTA 180.0 brackets (Figure 11-4-108-1, Detail A).

e. Scuff areas of conduit that pass through brackets with abrasive cloth, Item 4, Appendix D. Wipe conduit clean with methyl ethyl ketone, Item 217, Appendix D, and clean cheesecloth, Item 90, Appendix D. Do not touch roughened areas after cleaning.

f. Prime conduit and bracket attach points with adhesive primer, Item 68, Appendix D. Allow primer to dry for at least 1 hour.

g. Position conduit in brackets with 0.25-inch-inch extending above pylon PSTA 180.0. Support conduit in this position.

h. Coat each contact point with adhesive, Item 38, Appendix D. Work adhesive into contact surfaces. Maintain support until adhesive is dry.

i. Install tail rotor pylon flight control cables (PARA 11-4-106).

j. Check for correctly installed lockwire, cotter pins, and clips. Make sure cable is in pulley grooves. Check cable retaining pins for correct installation.

k. Install tail rotor drive shaft section IV (PARA 6-4-41).

l. Perform operational check of mechanical flight controls (PARA 11-2-3).

m. Perform maintenance test flight (Appropriate MTF).

n. Check helicopter logbook to make sure tail rotor drive shaft torque check (PARA 1-7-8) and tail rotor cable tension check (PARA 1-7-10) are shown due 9 to 11 flight hours after installation, and tail rotor pylon-to-tail cone bolts torque check (PARA 1-7-11) is shown due 27 - 33 flight hours after installation.

11-4-109 FLIGHT CONTROL CABLE LOCKOUT BLOCKS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Lockout Blocks, 70700-20463-041

Materials/Parts

- Tiedown Strap, Item 333, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-137
- PARA 11-4-118

11-4-109.1 Install Flight Control Cable Lockout Blocks.

NOTE

Lockout blocks should be used to restrain spring cylinders for all cabin and tail cone flight control cable and pulley replacement. For all tail rotor pylon flight control cable and pulley replacement, the spring cylinders should be disconnected from tail rotor quadrant.

a. Remove tail gear box front fairing (PARA 2-4-137).

WARNING

FSCAP

- Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.
- Injury to personnel and damage to equipment will result if tail cone and tail rotor pylon flight control cable quick-disconnects are separated without cable tension being removed. Tail rotor quadrant spring cylinders maintain 51 - 223 pounds tension on cables. Make sure tail rotor quadrant spring cylinders have been restrained prior to disconnecting cables.
- To prevent injury to personnel, lockout blocks and tiedown straps must be installed on tail rotor quadrant, or spring cylinders must be disconnected, before disconnecting tail rotor cables.
- To prevent injury to personnel make sure there is communication between person in cockpit and person at quadrant.

b. If replacing tail rotor pylon flight control cables or pulleys, disconnect spring cylinders from support (PARA 11-4-118).

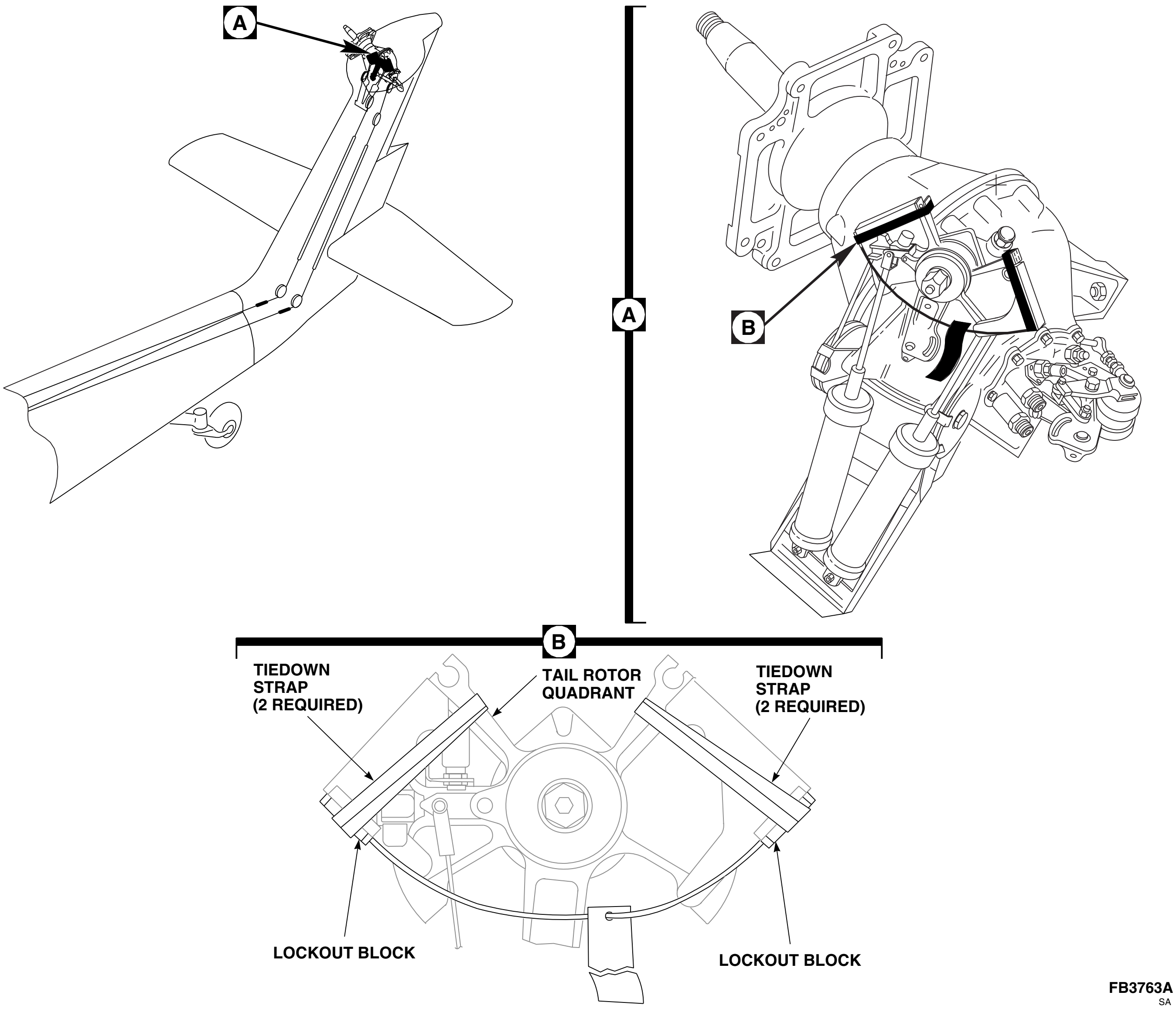


FIGURE 11-4-109-1. INSTALL/REMOVE FLIGHT CONTROL CABLE LOCKOUT BLOCKS

c. For all other flight control cable and/or pulley replacement, install lockout blocks on tail rotor quadrant. Secure lockout blocks with tiedown straps, Item 333, Appendix D (Figure 11-4-109-1, Detail B).

d. Flight control cable tension can now be released.

11-4-109.2 Remove Flight Control Cable Lockout Blocks.

a. After replacing pylon cables and adjusting tension, cut tiedown straps and remove lockout blocks (Figure 11-4-109-1, Detail B).

b. On caution/advisory panel, check that TAIL ROTOR QUADRANT light is off.

c. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

d. Install tail gear box front fairing (PARA 2-4-137).

11-4-110 TAIL ROTOR QUADRANT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Feeler Gage
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 114, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 198, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-137
- PARA 11-2-3
- PARA 11-3-26
- PARA 11-4-5
- PARA 11-4-105
- PARA 11-4-112
- PARA 11-4-117
- PARA 11-4-118

11-4-110.1 Replace Tail Rotor Quadrant.

11-4-110.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove tail gear box front fairing (PARA 2-4-137).

c. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

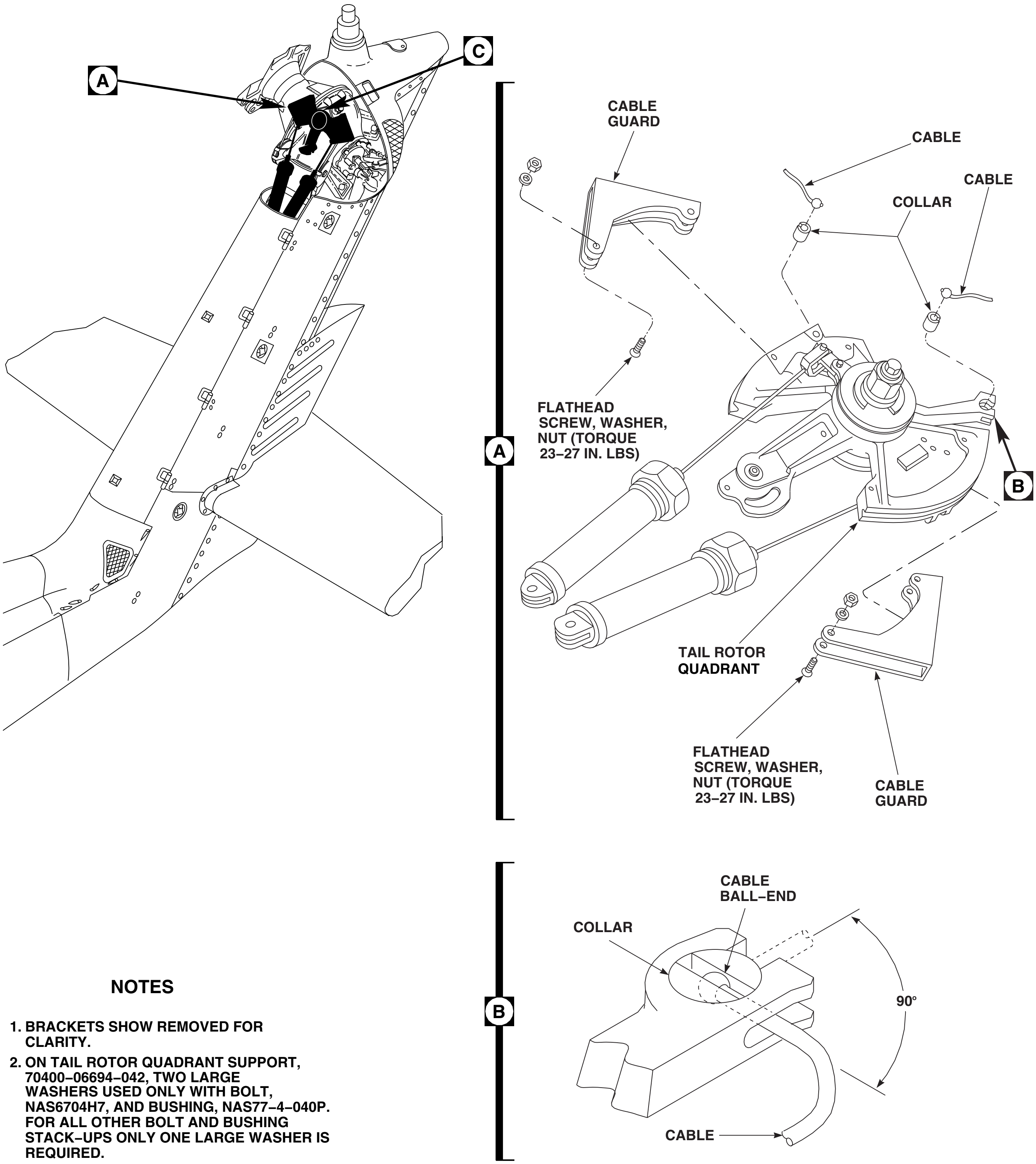
d. Disconnect tail cone flight control cables from quick-disconnects (PARA 11-4-105).

e. Remove tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

f. Remove flathead screws, washers, and nuts securing cable guards to quadrant (Figure 11-4-110-1, Sheet 1, Detail A).

g. Turn cable 90° to quadrant (Figure 11-4-110-1, Sheet 1, Detail B). Remove collar and cable from quadrant. Remove cable from collar (Figure 11-4-110-1, Sheet 1, Detail A).

h. Remove adjusting nuts securing right and left microswitches, S45 and S46, and keywashers to brackets on quadrant. Place microswitches



NOTES

1. BRACKETS SHOW REMOVED FOR CLARITY.
2. ON TAIL ROTOR QUADRANT SUPPORT, 70400–06694–042, TWO LARGE WASHERS USED ONLY WITH BOLT, NAS6704H7, AND BUSHING, NAS77–4–040P. FOR ALL OTHER BOLT AND BUSHING STACK–UPS ONLY ONE LARGE WASHER IS REQUIRED.
3. ON TAIL ROTOR QUADRANT SUPPORT, 70400–06694–043, ONLY ONE WASHER, 70400–06694–105, IS USED. VERIFY "THIS SIDE UP" CAN BE READ ON WASHER WHEN INSTALLED ON QUADRANT SUPPORT STUD.

FIGURE 11-4-110-1. REPLACE TAIL ROTOR QUADRANT (SHEET 1 OF 2)

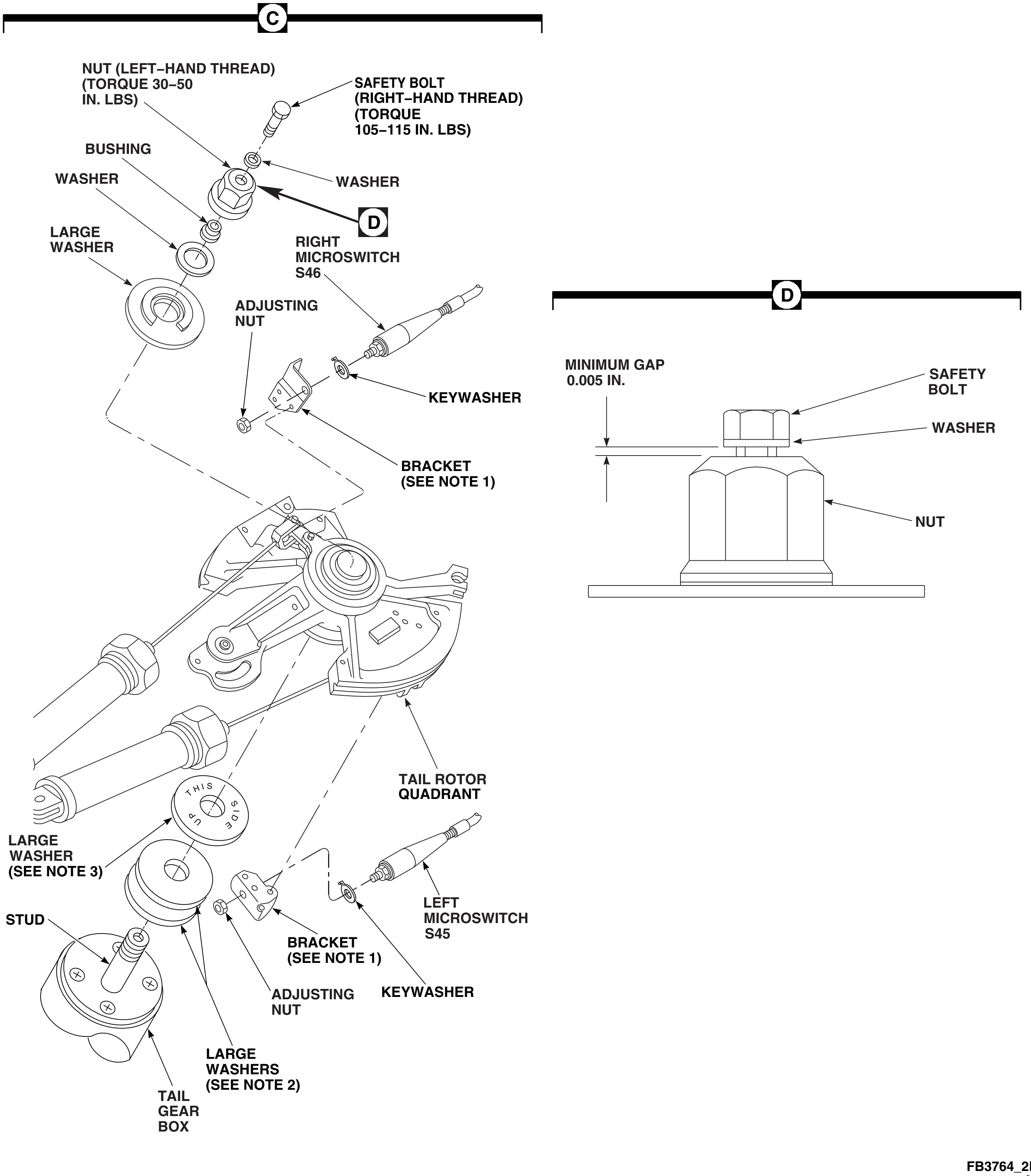


FIGURE 11-4-110-1. REPLACE TAIL ROTOR QUADRANT (SHEET 2 OF 2) out of way (Figure 11-4-110-1, Sheet 2, Detail C).

i. Remove safety bolt (right-hand thread) and washer from nut at top of quadrant. Remove nut (left-hand thread), bushing, washer, and large washer (with decals). Remove quadrant, with spring cylinders, from stud on tail gear box. Remove bottom large washer(s) from stud.

j. If necessary, repair tail rotor quadrant (PARA 11-4-117).

11-4-110.1.2 Install.

a. Turn off all electrical power (PARA 1-6-16).

CAUTION

Damage to equipment will result if incorrect hardware stackup is used when installing tail rotor quadrant. Two large washers must be used under quadrant if bolt, NAS6704H7, and bushing, NAS77-4-040P, are to be installed. If bolt, NAS6704H7, and bushing, NAS77-4-040P, are not available, only one large washer should be installed under quadrant.

NOTE

Prior to installing tail rotor quadrant, make sure toggle is correctly positioned in mating recess in quadrant.

b. Install tail rotor quadrant support as follows:

(1) On helicopters with tail rotor quadrant support, 70400-06694-042, perform the following:

(a) Install large washer(s) on stud of tail rotor quadrant support.

(b) Install replacement tail rotor quadrant on stud.

(c) Install large washer (with decals) on stud (Figure 11-4-110-1, Sheet 2, Detail C).

(2) On helicopters with tail rotor quadrant support, 70400-06694-043, perform the following:

WARNING

Prior to installing tail rotor quadrant, verify ″THIS SIDE UP’ can be read on bottom large washer when installed on quadrant support stud.

(a) Install large washer, 70400-06694-105, on stud of tail rotor quadrant support.

(b) Install replacement tail rotor quadrant on stud.

(c) Install large washer (with decals) on stud (Figure 11-4-110-1, Sheet 2, Detail C).

c. Install washer and bushing.

WARNING

FSCAP

Verification of proper hardware installation and safety installation are critical characteristics.

d. Install nut (left-hand thread) and thread on stud. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install safety bolt (right-hand thread) and washer through nut into stud. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.**

e. Check gap as follows:

(1) On helicopters with tail rotor quadrant support, 70400-06694-042, perform the following:

(a) Using feeler gage, check for 0.005-inch gap between washer and top of nut (Figure 11-4-110-1, Sheet 2, Detail D). If necessary, replace bolt and retorque.

(b) If gap cannot be achieved, make sure hardware stackup is as follows: (bolt, NAS6704H7, bushing, NAS77-4-040P, and 2 washers, SS4409-058). Repeat step c. and step d.

(2) On helicopters with tail rotor quadrant support, 70400-06694-043, perform the following:

(a) Using feeler gage, check for 0.005-inch gap between washer and top of nut (Figure 11-4-110-1, Sheet 2, Detail D). If necessary, replace bolt and retorque.

(b) If gap cannot be achieved, make sure hardware stackup is as follows: (bolt, NAS6704H7, bushing, NAS77-4-040P, and washer, 70400-06694-105). Repeat step c. and step d.

f. Using lockwire, Item 198, Appendix D, lockwire safety bolt to nut.

NOTE

Microswitches will be adjusted at end of procedure (PARA 11-4-110.1.3).

g. Install right microswitch, S46, and keywasher in bracket on quadrant. Install adjusting nut. Install left microswitch, S45, and keywasher in bracket on quadrant. Install adjusting nut.

h. Spray collars with corrosion preventive compound, Item 114, Appendix D.

WARNING

FSCAP

Verification of proper cable routing over quadrant and pulleys, and under cable retainers are critical characteristics.

NOTE

Make sure collars are installed with slotted side up.

i. Insert cables into collars (Figure 11-4-110-1, Sheet 1, Detail A). Turn cables 90° to direction cables will be going (Figure 11-4-110-1, Sheet 1, Detail B). Slide collars and cables into quadrant. Make sure cables are in quadrant grooves.

j. Install cable guards on quadrant and secure with flathead screws, washers, and nuts (nuts up). **TORQUE NUTS TO 23 - 27 INCH-POUNDS** (Figure 11-4-110-1, Sheet 1, Detail A).

k. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

l. Install tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

m. Connect tail cone flight control cables to quick-disconnects (PARA 11-4-105).

n. Perform tail rotor system complete rig (PARA 11-4-5).

o. Perform tail rotor quadrant microswitch adjustment (PARA 11-4-110.1.3).

p. Perform tail rotor quadrant function check (PARA 11-3-26).

11-4-110.1.3 Tail Rotor Quadrant Microswitch Adjustment.

a. Turn on all electrical power (PARA 1-6-16).

b. At tail rotor quadrant, adjust adjusting nuts on each microswitch to move switch away from quadrant arm (Figure 11-4-110-1, Sheet 2, Detail C).

c. On caution/advisory panel, check that TAIL ROTOR QUADRANT light is on.

d. Adjust nuts so microswitch will move toward quadrant arm until TAIL ROTOR QUADRANT light on caution/advisory goes off. **TIGHTEN NUTS.** Using lockwire, Item 196, Appendix D, lockwire nuts.

e. Turn off all electrical power (PARA 1-6-16).

f. Install tail gear box front fairing (PARA 2-4-137).

g. Perform operational check of tail rotor quadrant (PARA 11-2-3).

11-4-111 TAIL ROTOR QUADRANT REPAIR.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Drill
- Drill Bit, No. 32, 0.116-inch diameter
- Pin Punch

References

- PARA 11-4-110
- PARA 11-4-118

11-4-111.1 Repair Tail Rotor Quadrant.

11-4-111.1.1 Remove.

a. Remove tail rotor quadrant (PARA 11-4-110).

b. Remove upper and lower arms from quadrant with toggles and spring cylinders attached (Figure 11-4-111-1, Detail A).

c. Inspect spring cylinder clevises for wear. **MAXIMUM WEAR OR DAMAGE TO CLEVISES DUE TO MOUNTING BRACKETS CANNOT EXCEED 0.030-INCH METAL REMOVAL.** If clevis(es) exceeds limits, replace tail rotor quadrant spring cylinder (PARA 11-4-118).

d. Using drill with No. 32 drill bit, drill heads off rivets and remove rivets from toggle arms and spring cylinder clevises using pin punch. Remove spacers.

e. Remove spring cylinder clevis from toggle.

f. Remove toggle from arm, remove rings.

11-4-111.1.2 Install.

WARNING

FSCAP

Verification of toggle installation is a critical characteristic.

a. Install replacement toggle in arm. Install ring on each side of toggle. Install spacer and rivet (Figure 11-4-111-1, Detail A).

b. Install spring cylinder clevis on toggle. Install spacer. Install rivet.

c. Install upper and lower arms on quadrant. At same time, position toggle in mating recess in quadrant.

d. Install tail rotor quadrant (PARA 11-4-110).

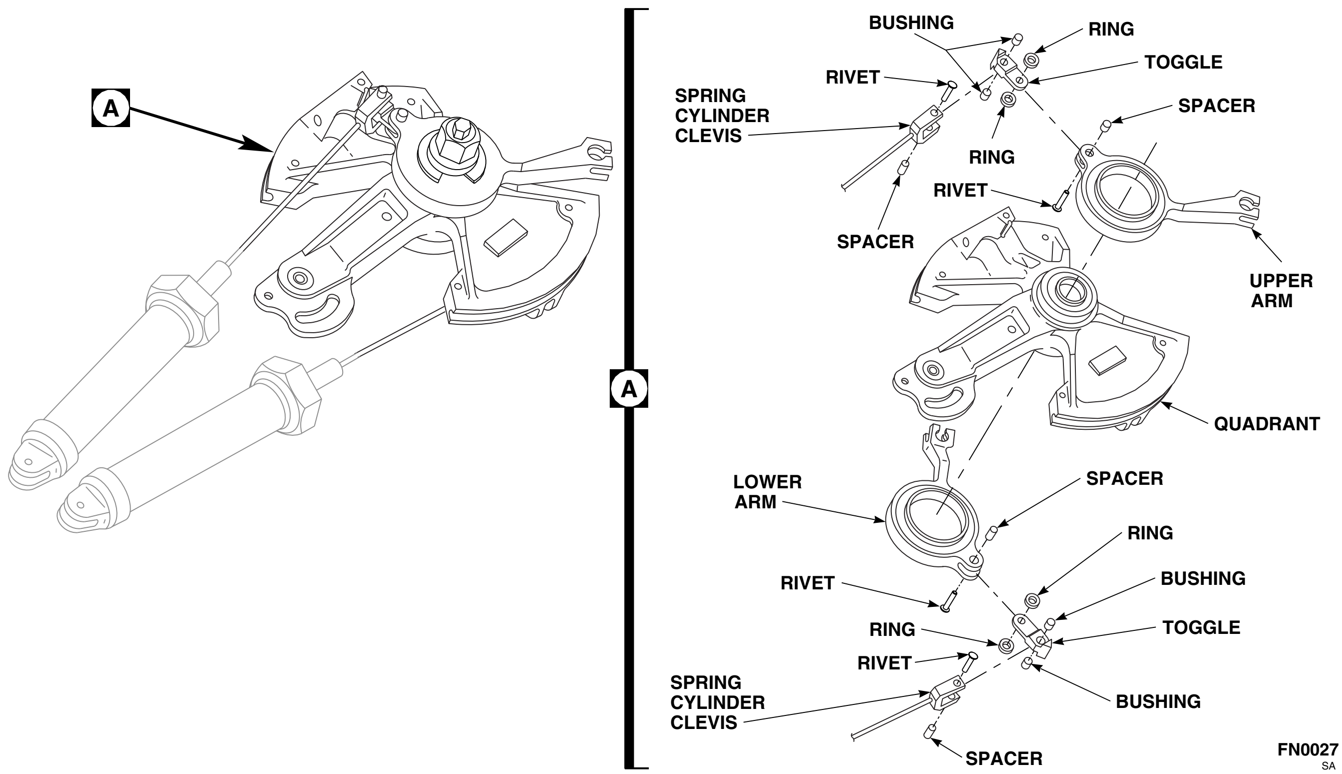


FIGURE 11-4-111-1. REPAIR TAIL ROTOR QUADRANT

11-4-112 TAIL ROTOR SERVO TO TAIL ROTOR QUADRANT PUSHROD.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-137
- PARA 2-6-5
- PARA 11-3-3
- PARA 11-4-1
- PARA 11-4-6

11-4-112.1 Replace Tail Rotor Servo to Tail Rotor Quadrant Pushrod.

11-4-112.1.1 Remove.

a. Remove tail gear box fairing (PARA 2-4-137).

b. Remove bolts, washers, long bushings, short bushings, flanged countersunk bushing, and nuts securing pushrod to tail rotor servo and tail rotor quadrant. Remove pushrod (Figure 11-4-112-1, Sheet 1, Detail A).

11-4-112.1.2 Install.

a. Adjust replacement pushrod to exact length, hole-center to hole-center, as removed rod, or adjust replacement rod so rod length between self-retaining bolthole centers is 12.56 - 12.68-inches. **TIGHTEN JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

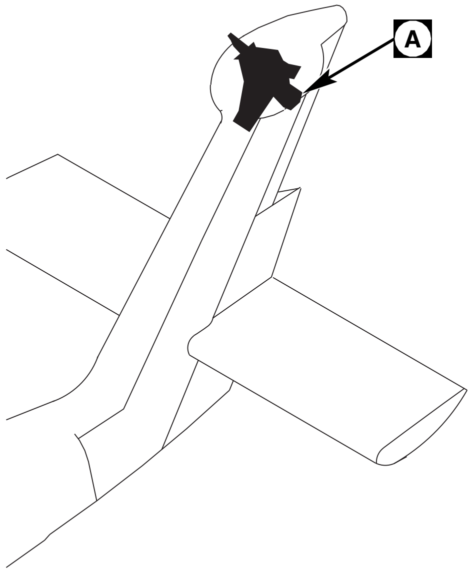
b. Adjustable end of control rods are to be inspected for minimum thread engagement. Check that adjustable rod end is threaded beyond inspection hole such that a 0.20-inch diameter wire cannot be inserted through inspection hole (Figure 11-4-112-1, Sheet 3, Detail D).

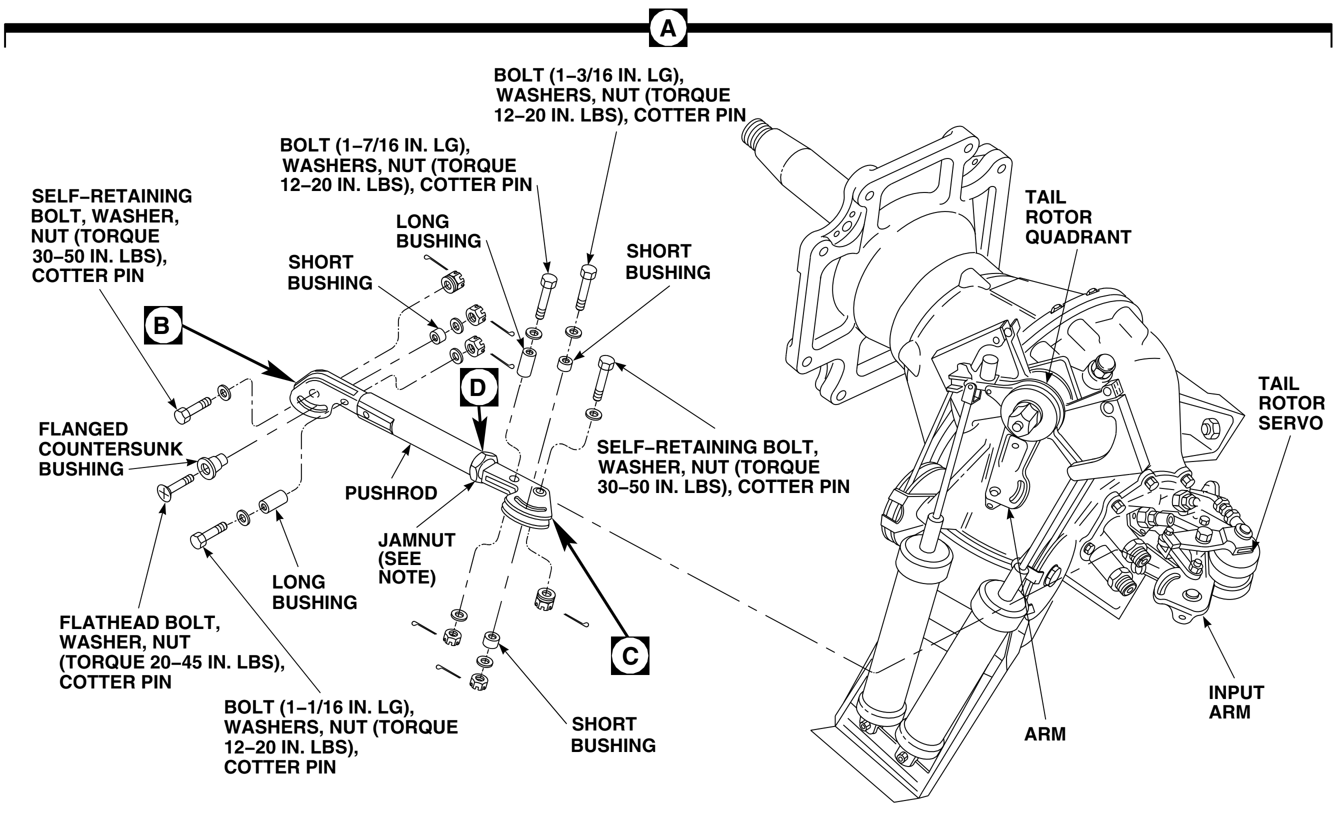
WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

c. Position replacement rod between input arm on tail rotor servo and arm on tail rotor quadrant, with adjustable rod end at servo.





NOTE

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 11-4-112-1. REPLACE TAIL ROTOR SERVO TO TAIL ROTOR QUADRANT PUSHROD (SHEET 1 OF 3)

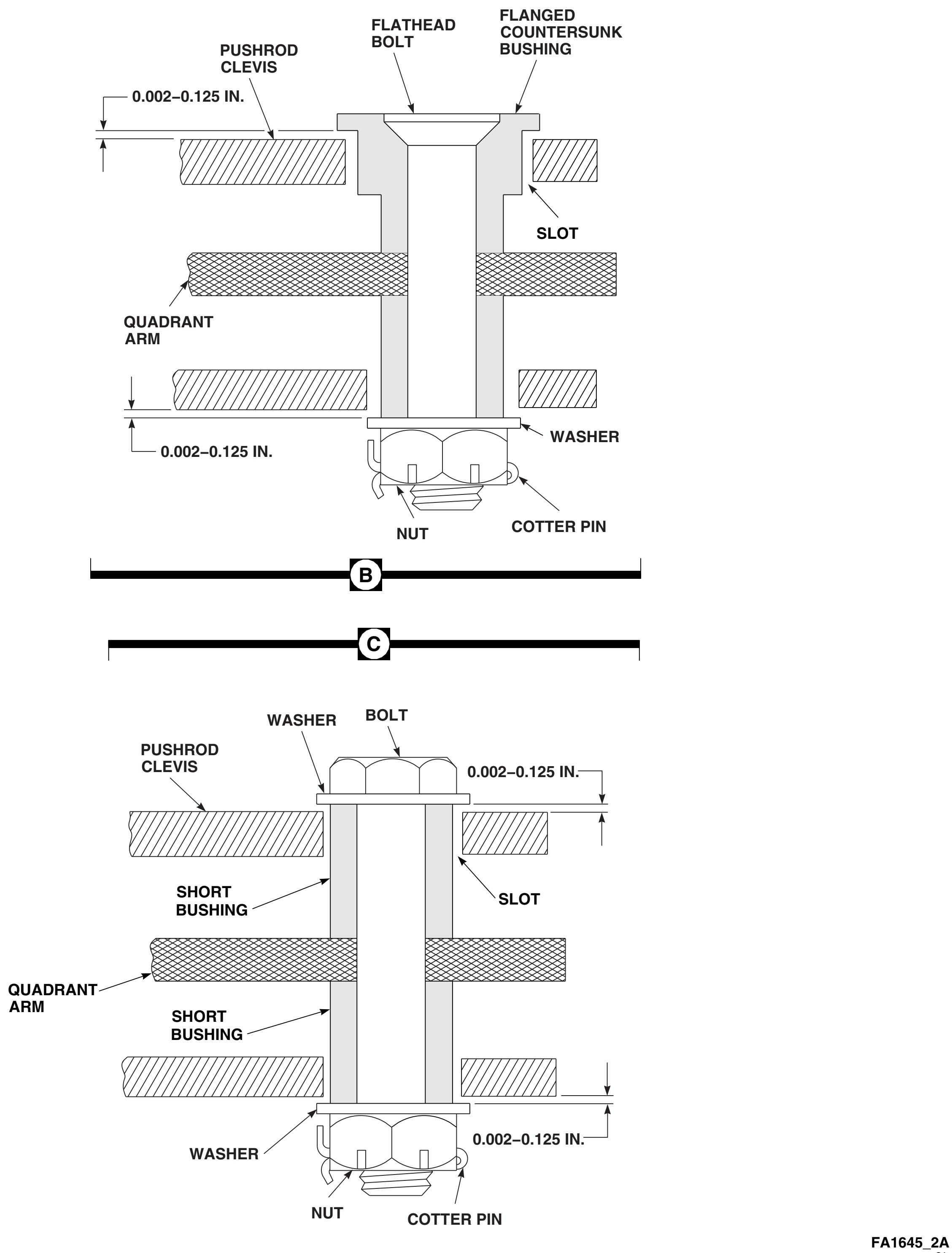


FIGURE 11-4-112-1. REPLACE TAIL ROTOR SERVO TO TAIL ROTOR QUADRANT PUSHROD (SHEET 2 OF 3)

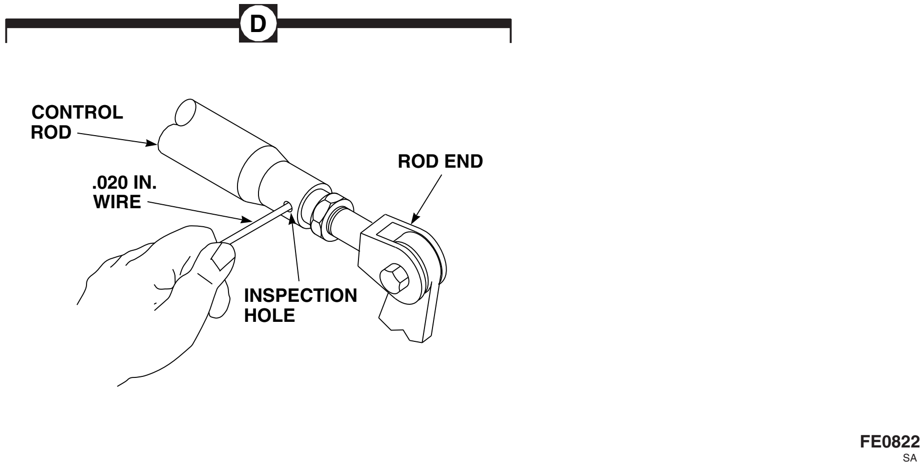


FIGURE 11-4-112-1. REPLACE TAIL ROTOR SERVO TO TAIL ROTOR QUADRANT PUSHROD (SHEET 3 OF 3)

d. Install self-retaining bolts and washers (under bolthead) (PARA 11-4-1).

e. Install nuts on self-retaining bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

f. **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

g. At input arm on tail rotor servo, install long bushing in rod end and slot of servo arm. Install 1 7/16-inch long bolt, washers, and nut. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

h. At tail rotor servo end, install short bushing in slot, each side of rod end. Install 1 3/16-inch long bolt, washers, and nut (Figure 11-4-112-1, Sheet 1, Detail A and Sheet 2, Detail B and Detail C). **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

i. At arm on tail rotor quadrant, install long bushing in rod end and slot of quadrant arm. Install 1 1/16-inch long bolt, washers, and nut (Figure 11-4-112-1, Sheet 1, Detail A and Sheet 2, Detail B and Detail C). **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

j. At arm on quadrant, install flanged countersunk bushing in top slot of rod end (Figure 11-4-112-1, Sheet 1, Detail A). Install short bushing in bottom slot of rod end. Install flathead bolt in bushings and quadrant arm. Install washer and nut. **TORQUE NUT TO 20 - 45 INCH-POUNDS.** Install cotter pin.

k. Check assembly for correct tri-pivot bolt installation (Figure 11-4-112-1, Sheet 2, Detail B and Detail C) (PARA 11-3-3).

l. Perform tail rotor system rig check (PARA 11-4-6).

m. Perform control rod corrosion protection coating touch up (PARA 11-4-1).

n. Make sure area is clean and free of foreign material.

o. Install tail gear box fairing (PARA 2-4-137).

11-4-113 TAIL ROTOR SERVO AND PITCH CONTROL SHAFT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicating Depth Gage
- Dial Indicator
- Micrometer
- Nonmetallic Scraper, Locally-Made
- Steel Rod, 5/64-inch diameter (0.078-inch diameter), Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Adhesive, Item 38, Appendix D
- Adhesive, Item 60, Appendix D
- Brush, Acid, Item 84, Appendix D
- Coating Compound, Black Oxide, Item 106A, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 149, Appendix D
- Lockwire, Item 193, Appendix D
- Lubricating Oil, Item 205, Appendix D
- Lubricating Oil, Item 206, Appendix D
- Lubricating Oil, Item 208, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 224, Appendix D
- Primer, Item 254, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Trichloroethane, 1,1,1, Item 341, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-3-8
- PARA 1-3-14
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 1-7-60

- PARA 2-4-137
- PARA 2-6-5
- PARA 5-4-40
- PARA 7-3-1
- PARA 7-4-50
- PARA 7-4-65
- PARA 11-2-3
- PARA 11-4-1
- PARA 11-4-6
- PARA 11-4-112
- PARA 11-4-116
- PARA 11-5-26

11-4-113.1 Replace Tail Rotor Servo and Pitch Control Shaft.

11-4-113.1.1. Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove tail gear box fairing (PARA 2-4-137).

c. Engage gust lock.

d. Remove tail rotor pitch beam (PARA 5-4-40).

e. Remove tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

f. Disconnect electrical connectors, P609, and P610, from 1st and 2nd stage pressure switches on tail rotor servo (Figure 11-4-113-1, Detail A).

g. Install protective caps and plugs on electrical connectors.

h. Remove tail rotor servo hydraulic connectors (PARA 11-4-116).

i. Cover connectors to keep dirt out of hydraulic systems while replacing tail rotor servo.

NOTE

Depending upon level of ground and position of helicopter, it may be necessary to drain some oil from tail gear box to prevent spilling when tail rotor servo is removed.

j. If required, drain some oil from tail gear box (PARA 1-3-14).

k. Remove nuts and washers securing tail rotor servo to tail gear box.

CAUTION

Damage to equipment will result if foreign objects enter tail gear box. Foreign objects may enter with tail rotor servo removed. Protect inside of tail gear box using cover.

l. With aid of assistant, carefully pull tail rotor servo off studs and out of tail gear box.

m. Remove self-retaining bolt, washer, and nut securing tail rotor servo to pitch control shaft.

n. Remove packing from tail rotor servo. Discard packing.

o. Perform tail gear box output shaft ultrasonic inspection (PARA 1-7-60).

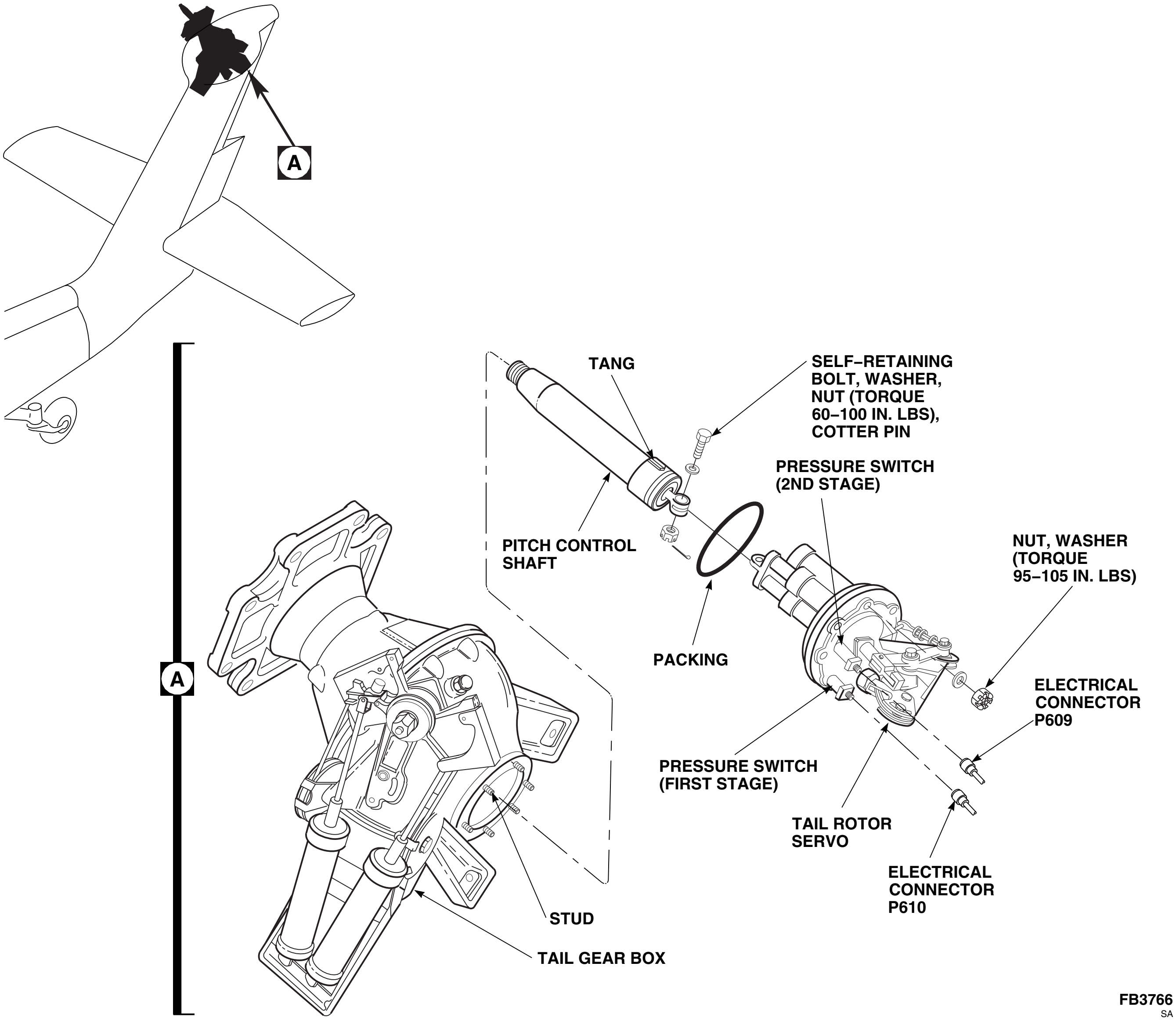


FIGURE 11-4-113-1. REPLACE TAIL ROTOR SERVO AND PITCH CONTROL SHAFT

11-4-113.1.2. Inspect/Repair.

NOTE

Several components of tail rotor servo assembly (input pivot shaft, summing lever, feedback drag link, adjustable feedback link and ball drive cover) may become discolored while in service. Surface discoloration on these components is not cause for replacement. Refer to Figure 11-4-113-3 for location.

a. Inspect self-retaining bolt connecting tail rotor servo to pitch control shaft for radial wear using micrometer. **RADIAL WEAR NOT TO EXCEED 0.002-INCH.**

NOTE

Bearing is located inside pitch control shaft at end of link assembly.

b. Check condition of bearing by holding pitch control shaft by link and spinning shaft. **BEARING MUST ROTATE FREELY.** If bearing sounds rough or if bearing chattering exists, replace pitch control shaft.

c. Inspect pitch control shaft link assembly for axial play by moving link in and out of shaft using hand force and measuring movement with dial indicator. **AXIAL PLAY NOT TO EXCEED 0.020-INCH.** If play is beyond limits, replace pitch control shaft.

d. Inspect washers bonded to clevis end of link assembly for wear. If either washer is worn 0.020-inch, perform the following:

(1) Loosen old adhesive from around washer using acid brush, Item 84, Appendix D, moistened with trichloroethane, 1,1,1, Item 341, Appendix D. Using nonmetallic scraper, remove washer from link.

(2) Remove old adhesive from washers and link using machinery towel, Item 211, Appendix D, dampened with trichloroethane, 1,1,1, Item 341, Appendix D.

(3) Apply a coat of primer, Item 254, Appendix D, to surface of link and washers. Allow to dry for a minimum of 3 minutes.

(4) Apply a bead of adhesive, Item 60, Appendix D, to surface of link. Position and rebond washers 90° to 270° from original position. Allow to air dry for 15 minutes, undisturbed.

e. Rebond loose washer(s) using above steps.

f. Inspect servo rod end, adjacent to bearing for wear on each side of lug area, using dial indicating depth gage. **MAXIMUM WEAR NOT TO EXCEED 0.003-INCH DEEP.**

g. Inspect servo rod end bearing for proper staking. **NO LOOSENESS ALLOWED.**

h. Inspect servo rod end bearing for axial and radial play as follows:

(1) **MAXIMUM RADIAL PLAY IS 0.005-INCH.**

(2) **MAXIMUM AXIAL PLAY IS 0.008-INCH.**

i. Inspect pressed bushing on pitch control shaft as follows:

(1) Check pressed bushing for security and that it cannot be rotated by hand. **IF BUSHING IS LOOSE, COCKED, OR CAN BE ROTATED BY HAND, REPLACE PRESSED BUSHING** (PARA 11-5-26).

(2) Check pressed bushing for nicks, dents, and scratches.

(a) Nicks, dents, and scratches up to 0.030-inch deep are acceptable provided that all sharp and raised edges are removed using abrasive cloth, Item 10, Appendix D, and that total damaged area is not more than 10% of surface area.

(b) If damage or repair is beyond limits, replace pressed bushing (PARA 11-5-26).

j. Inspect outside diameter of pressed bushing using micrometer. **MINIMUM OUTSIDE DIAMETER OF BUSHING IS 3.1135-INCHES.** If beyond limits, replace pressed bushing (PARA 11-5-26). Inspect bearing area for chips of debris.

NOTE

Spanner nut is located inside pitch control shaft at link assembly end.

k. Check for presence and condition of cotter pins securing spanner nut to pitch control shaft. If cotter pins are damaged or missing, replace pitch control shaft.

l. Inspect tang using micrometer. **MINIMUM WIDTH OF TANG IS 0.299-INCH. MINIMUM OUTSIDE DIAMETER OF TANG IS 3.275-INCHES.**

m. Inspect tang for wear. **MAXIMUM DEPTH OF WEAR IS 0.005-INCH. MINIMUM WIDTH OF TANG ACROSS WEAR SURFACE IS 0.289-INCH.**

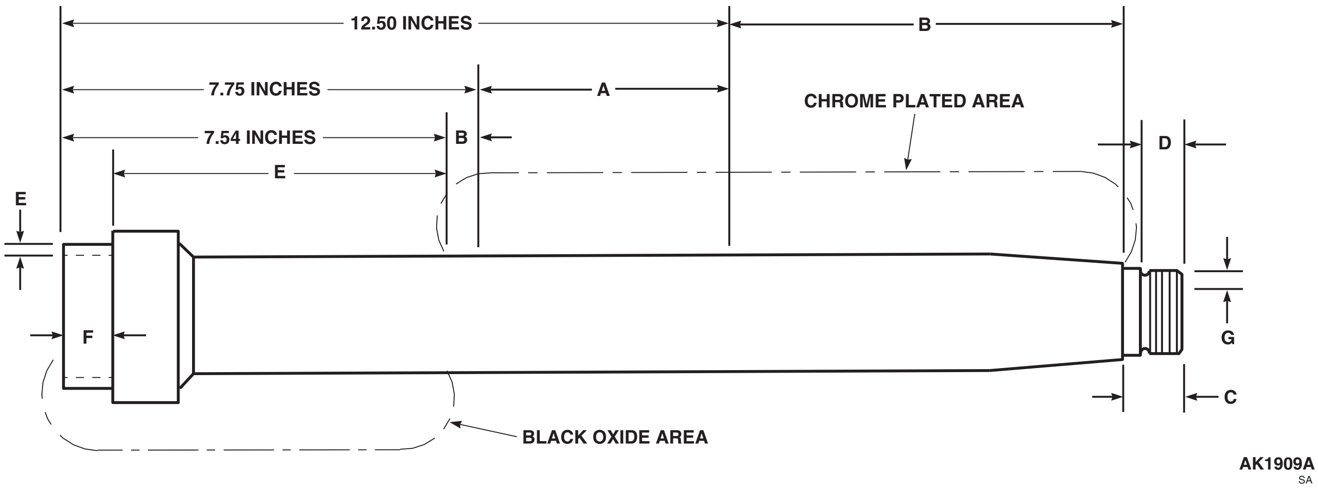


FIGURE 11-4-113-2. REPAIR PITCH CONTROL SHAFT

n. Check pitch control shaft chrome-plated sealing surface, AREA A, from 7.75 to 12.50-inches from end of shaft for damage (Figure 11-4-113-2).**AXIAL SCRATCHES WITH NO MEASURABLE DEPTH, AND OTHER NORMAL WEAR WHICH WILL NOT AFFECT SEAL WEAR IS ACCEPTABLE.** If damaged beyond limits, replace pitch control shaft.

o. Check pitch control shaft chrome-plated sealing surface, AREA A, from 7.75 to 12.50-inches from end of shaft for corrosion. Corrosion not permitted through chrome-plating into base metal may be blended out using abrasive cloth, Item 10, Appendix D. **REPAIRED AREA SHALL NOT EXCEED 10% OF THE SURFACE AREA.** If damaged or rework penetrates chrome-plated surface, replace pitch control shaft.

p. Check entire pitch control shaft chrome-plated area for blistering, cracking, or flaking. **NONE ALLOWED.** If blistering, cracking, or flaking is found, replace pitch control shaft.

q. Check pitch control shaft chrome-plated surface, AREA B, for scratches, nicks, and gouges. Damage not penetrating through chrome-plating into base metal may be blended out using abrasive cloth, Item 10, Appendix D. **REPAIRED AREA SHALL NOT EXCEED 10% OF SURFACE AREA.** If damage or rework penetrates chrome-plated surface, replace pitch control shaft.

r. Check pitch control shaft cadmium-plated area, AREA C, for damage. **NONE ALLOWED.** Corrosion not penetrating through cadmium-plating into base metal may be removed using abrasive cloth, Item 10, Appendix D. **REPAIRED AREA SHALL NOT EXCEED 10% OF SURFACE AREA.** If damage or rework penetrates into base metal, replace pitch control shaft.

s. Check pitch control shaft black oxide area, AREA E, for nicks and gouges. Blend out damage using abrasive cloth, Item 10, Appendix D. **BLEND AREA NOT TO EXCEED 0.100-INCH. IF DAMAGE EXCEEDS A MAXIMUM DEPTH OF 0.030-INCH, REPLACE PITCH CONTROL SHAFT.**

t. Corrosion may be blended out in surface, AREA E, using abrasive cloth, Item 10, Appendix D. **REPAIRED AREA SHALL NOT EXCEED 10% OF THE SURFACE AREA.** Touch up black oxide area after repair using black oxide coating compound, Item 106A, Appendix D.

u. Check pitch control shaft threads, AREA F, for stripping, crossing, and flattening. **NONE ALLOWED.** If threads are stripped, crossed, or flattened, replace pitch control shaft. Remove superficial corrosion using abrasive cloth, Item 10, Appendix D. Touch up black oxide area after repair using black oxide coating compound, Item 106A, Appendix D.

v. Check end of pitch control shaft, AREA G, for nicks or gouges. Blend out using abrasive cloth, Item 10, Appendix D. **BLEND AREA NOT TO EXCEED 0.100-INCH OR 10% OF THE SURFACE AREA. IF DAMAGE EXCEEDS A MAXIMUM DEPTH OF 0.030-INCH, REPLACE PITCH CONTROL SHAFT.**

w. Check pitch control shaft threads, AREA D, for stripping, crossing, and flattening. **NONE ALLOWED.** If threads are stripped, crossed, or flattened, replace pitch control shaft.

x. Visually inspect pitch control shaft for installation of dust plug. If dust plug is not installed, perform the following:

(1) Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean dust plug opening.

(2) Using acid brush, Item 84, Appendix D, apply a coat of adhesive, Item 38, Appendix D, around dust plug.

(3) Install dust plug in opening on pitch control shaft. Make sure dust plug is seated firmly against pitch control shaft.

(4) Apply a bead of sealing compound, Item 292, Appendix D, around edges of dust plug. Allow to dry.

y. Check piston rods for wear.

z. Linear marks on piston rods are normal and acceptable as long as they do not cause leakage greater than what is specified in PARA 7-3-1.

aa. Check tail rotor servo input arm bearing for play as follows:

(1) Apply finger pressure to the top of tail rotor servo input arm and move input arm all the way to one side (Figure 11-4-113-3, Detail B). Mount dial indicator to pick up one of two locations on servo input arm. Move input arm all the way to opposite side and record total side to side deflection. If deflection is 0.200-inch or less, the bearings are acceptable. **IF DEFLECTION IS GREATER THAN 0.200-INCH, REPLACE TAIL ROTOR SERVO.**

(2) Connect tail rotor servo to tail rotor quadrant pushrod to tail rotor servo input arm (PARA 11-4-112).

ab. Clamp dial indicator to tail rotor servo input arm (Figure 11-4-113-3).

NOTE

If pushrod has been removed from tail rotor servo input arm, install bolt, washer, and nut in bearing to measure play.

ac. Place tip of dial indicator on end of self-retaining bolt to measure axial play, and on side of nut to measure radial play.

ad. **AXIAL PLAY ON BEARING MUST NOT BE MORE THAN 0.008-INCH.** If bearing play is beyond limits, replace tail rotor servo.

ae. **RADIAL PLAY ON BEARING MUST NOT BE MORE THAN 0.005-INCH.** If bearing play is beyond limits, replace tail rotor servo.

af. Remove spring clips from adjustable feedback link spherical bearings (Figure 11-4-113-3, Detail A). Check spring clips for damage. **NONE ALLOWED.** If damaged, replace spring clip(s).

ag. Clamp dial indicator, to left adjustable feedback link.

ah. Place tip of dial indicator on end of attaching bolt to measure axial play, and on side of nut to measure radial play. Measure bearing at each end of link.

ai. Measure bearings on right adjustable feedback link in same way.

aj. **AXIAL PLAY ON BEARINGS MUST NOT BE MORE THAN 0.008-INCH.** If bearing play is beyond limits, replace tail rotor servo.

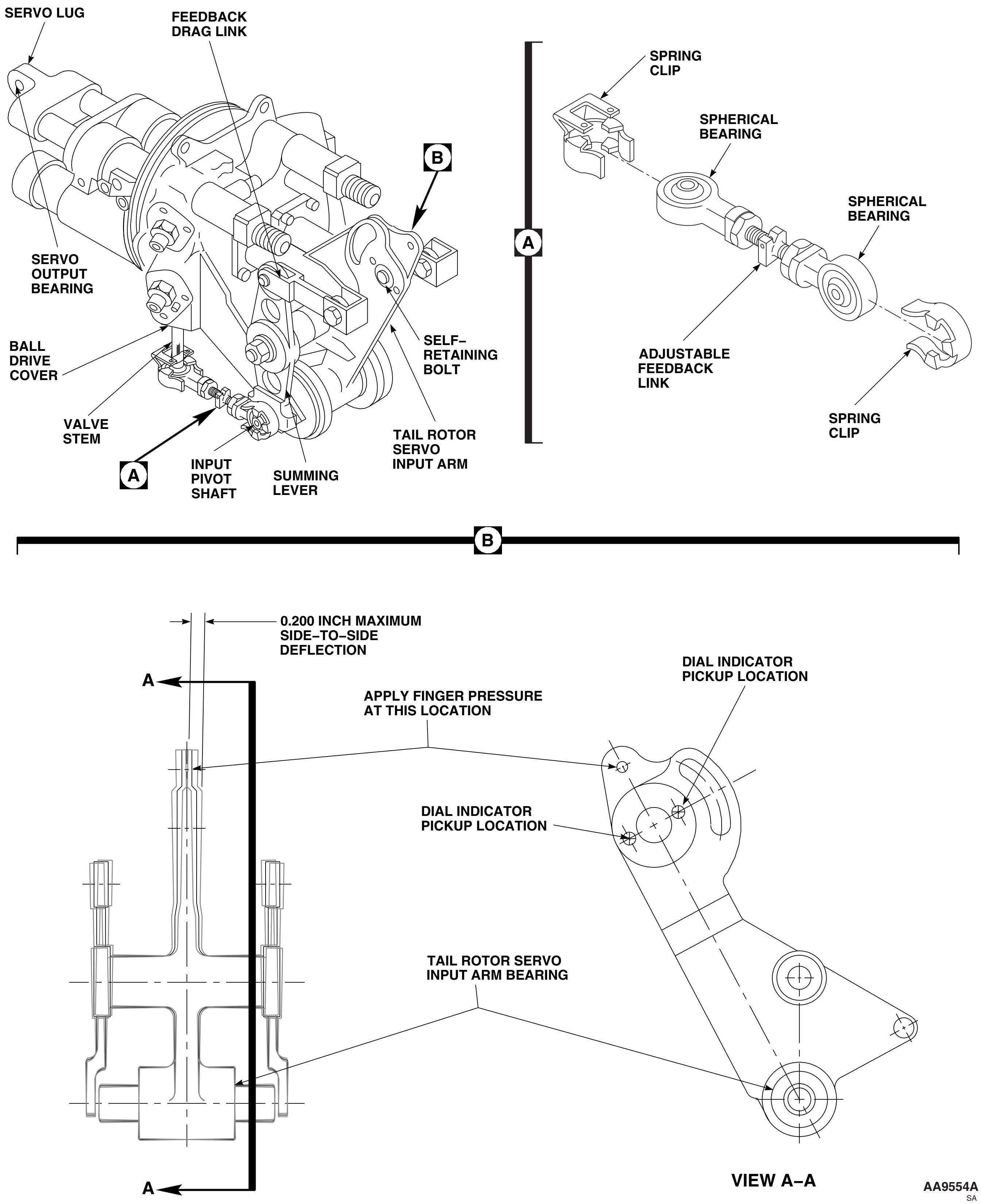


FIGURE 11-4-113-3. REPLACE SPRING CLIPS

ak. **RADIAL PLAY ON BEARINGS MUST NOT BE MORE THAN 0.005-INCH.** If bearing play is beyond limits, replace tail rotor servo.

al. Clamp dial indicator to left feedback drag link.

am. Install spring clips to adjustable feedback link spherical bearings. Lockwire, Item 193, Appendix D, spring clips.

an. Place tip of dial indicator on side of summing lever to measure summing lever bearing axial play, and against end of summing lever to measure radial play (Figure 11-4-113-3).

ao. Place tip of dial indicator on end of attaching bolt to measure feedback drag link bearing axial play and on side of nut to measure radial play.

ap. **AXIAL PLAY ON BEARINGS MUST NOT BE MORE THAN 0.008-INCH.** If bearing play is beyond limits, replace tail rotor servo.

aq. **RADIAL PLAY ON BEARINGS MUST NOT BE MORE THAN 0.005-INCH.** If bearing play is beyond limits, replace tail rotor servo.

11-4-113.1.3. Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

CAUTION

Damage to pitch control shaft bearing will result if exposed to cleaning solvents. Solvent can penetrate bearing seal and dissolve grease. Do not allow solvent to come in contact with pitch control shaft.

b. Thoroughly clean servo packing groove, tail gear box/servo interface flanges, studs, stud holes, washers, and nuts with acid brush, Item 84, Appendix D, and machinery towels, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry with clean machinery towel. Do not touch cleaned surfaces.

c. Coat packing with lubricating oil, Item 205, Item 206, or Item 208, Appendix D, before installing.

d. Install packing on tail rotor servo. Wipe off excess oil.

e. Install pitch control shaft to tail rotor servo (Figure 11-4-113-1, Detail A).

f. Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

g. Install nut on bolt. **TORQUE NUT TO 60 - 100 INCH-POUNDS.** Install cotter pin.

h. **PROOF TORQUE BOLTHEAD TO 30 - 32 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

CAUTION

Damage to tail rotor servo will result if valve stem is damaged during installation. Bending of valve stem will result in a sticking or stiff servo. Take care when handling servo to avoid bending or damaging servo linkages.

i. Using epoxy primer coating kit, Item 135, Appendix D, apply a thin, even coat of epoxy primer to tail gear box/servo flanges, studs, and stud holes. Install tail rotor servo in tail gear box while primer is wet.

j. Apply grease, Item 149, Appendix D, to tang on pitch control shaft and output gear slots.

k. With aid of assistant, line up shaft keys with key ways in output gear inside tail gear box. Slide pitch control shaft and tail rotor servo into tail gear box. Install servo on tail gear box studs. Using epoxy primer coating kit, Item 135, Appendix D, paint washer and nut contact surfaces with epoxy primer. Install washers and nuts securing tail rotor servo to tail gear box. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

CAUTION

Damage to tail gear box will result if seam between tail rotor servo and tail gear box housing is not sealed completely. Failure to properly seal servo/housing interface may lead to corrosion and result in premature removal of tail gear box. Make sure that all seams between servo and gear box housing are filled with sealant.

l. Cover nuts and washers, and fill seam between tail rotor servo and tail gear box with sealing compound, Item 292, Appendix D.

m. Make a bead around seam using excess sealing compound. Allow sealing compound to dry.

n. Using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer over sealant after sealing compound has cured.

o. Coat epoxy primer with polyurethane paint, Item 224, Appendix D.

p. Install tail rotor servo to tail rotor quadrant pushrod (PARA 11-4-112).

CAUTION

Damage to hydraulic system will result if dirt is allowed to enter system. To prevent getting dirt in hydraulic systems, make sure mounting faces of servo and connectors are clean before installing connectors.

q. Install tail rotor servo hydraulic connectors (PARA 11-4-116).

r. Remove protective caps and plugs from electrical connectors.

s. Inspect and treat electrical connectors (PARA 1-7-20).

t. Connect electrical connectors, P610 and P609, to 1st stage and 2nd stage pressure switches on tail rotor servo.

u. Install tail rotor pitch beam (PARA 5-4-40).

CAUTION

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation without oil (or oil below level specified in PARA 1-3-14) can cause loss of tail rotor drive. Lightly shake tail pylon to observe oil sloshing when checking oil level.

v. If required, service tail gear box (PARA 1-3-14).

w. Perform function check (PARA 11-4-113.1.4).

11-4-113.1.4. Function Check.

a. Turn on all electrical power (PARA 1-6-16).

CAUTION

Damage to backup hydraulic system will result if backup pump is run for an extended period of time when surrounding temperature is above 90°F (32°C). To prevent overheating backup hydraulic system, use this chart for limits when running backup pump.

Surrounding Temperature F° (C°)	Operating Time (Minutes)	Cooldown Time, Pump Off (Minutes)
90° (32°) max	Unlimited	-
91° - 100° (33° - 38°)	24	72
100° - 125° (39° - 52°)	16	48

b. On overhead console, place BACKUP HYD PUMP switch to ON.

c. On caution/advisory panel, check that #1 TAIL RTR SERVO and #2 TAIL RTR SERVO ON lights are off.

d. On lower console, place TAIL SERVO switch to BACKUP.

e. On caution/advisory panel, check that #1 TAIL RTR SERVO and #2 TAIL RTR SERVO ON lights are on.

f. On lower console, place TAIL SERVO switch to NORMAL and check that #1 TAIL RTR SERVO and #2 TAIL RTR SERVO ON lights are off on caution/advisory panel.

g. Check tail rotor servo for leaks.

h. Turn off all electrical power (PARA 1-6-16).

i. Service and bleed hydraulic reservoirs as necessary (PARA 1-3-8 and PARA 7-4-65).

j. If tail rotor servo hoses or tubes were disconnected, removed, or installed, test servo for correct plumbing connections (PARA 7-4-50).

k. Perform tail rotor system rig check and check tail rotor servo for bottoming (PARA 11-4-6).

CAUTION

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation without oil (or oil below level specified in PARA 1-3-14) can cause loss of tail rotor drive. Lightly shake tail pylon to observe oil sloshing when checking oil level.

l. If required, service tail gear box (PARA 1-3-14).

m. Inspect for bent input pilot valve stem as follows:

(1) Turn on all electrical power (PARA 1-6-16).

(2) On overhead control panel, place BACKUP HYD PUMP switch ON.

(3) Have assistant position and hold controls as follows:

CONTROL	POSITION
Collective	Full Down
Cyclic	Centered
Pedals	Full Right

(4) Check for a minimum clearance (0.078-inch) between pivot/stop pin and summing lever tangs on 1st and 2nd stage servos (four places). **IF CLEARANCE IS NOT AS SPECIFIED, REPLACE TAIL ROTOR SERVO** (Figure 11-4-113-4, Detail A).

NOTE

- Step m. (5) through step m. (8) apply only to tail rotor servos which do not have spring clips installed over input link spherical bearings.
- 1st stage valve stem is forward stem.

(5) Turn 1st stage valve stem 45° by hand. Recheck 1st stage summing lever tang

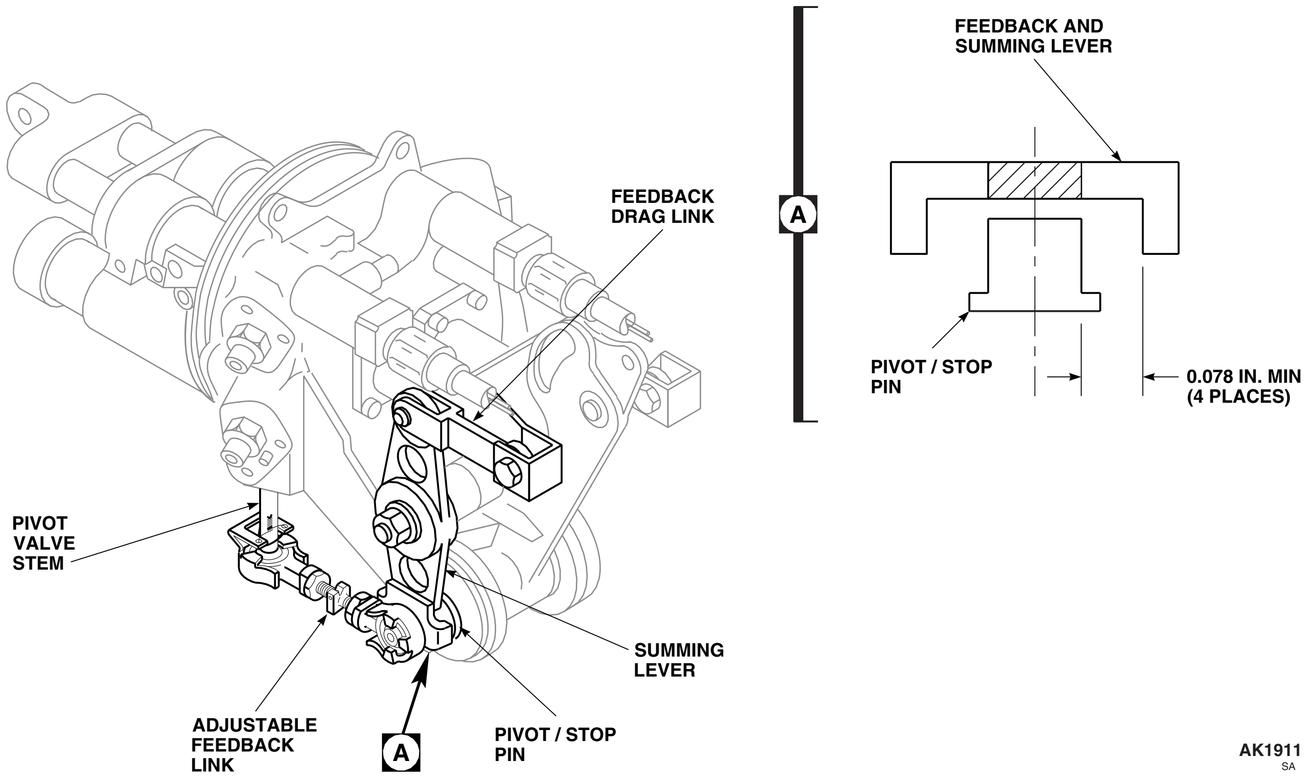


FIGURE 11-4-113-4. INSPECT FEEDBACK AND SUMMING LEVER

and pivot/stop pin clearance. If 5/64-inch diameter (0.078-inch diameter) steel rod can be inserted, continue turning valve stem by 45° steps until pilot valve stem has been turned 360° without binding or clamping measuring tool.

(6) On lower console, place TAIL SERVO switch to BACKUP.

(7) Turn backup stage valve stem 45° by hand. Recheck backup stage summing lever tang and pivot/stop pin clearance. If 5/64-inch diameter (0.078-inch diameter) steel rod can be inserted, continue turning valve stem by 45° steps until valve stem has been turned 360° without binding or clamping measuring tool.

(8) **IF 1ST STAGE OR BACKUP STAGE PILOT VALVE STEM FAILS CLEARANCE CHECK WHILE TURNING VALVE STEMS, REPLACE TAIL ROTOR SERVO.**

(9) On overhead control panel, place BACKUP HYD PUMP switch OFF.

(10) Turn off all electrical power (PARA 1-6-16).

n. Make sure area is clean and free of foreign material.

o. Perform tail rotor system rig check (PARA 11-4-6).

p. Perform operational check of tail rotor flight control system (PARA 11-2-3).

q. Install tail gear box fairing (PARA 2-4-137).

NOTE

If tail rotor outboard retention plate bolts were loosened or removed, a tail rotor balance check is to be completed prior to flight.

r. Perform maintenance test flight (Appropriate MTF).

11-4-114 TAIL ROTOR SERVO PRESSURE SWITCH.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-137
- PARA 7-4-65

11-4-114.1 Replace Tail Rotor Servo Pressure Switch.

NOTE

Replacement of 1st and 2nd stage tail rotor servo pressure switches is the same, except as noted.

11-4-114.1.1. Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove tail gear box front fairing (PARA 2-4-137).

NOTE

Have replacement pressure switch and packings ready to install before removing installed pressure switch. 1st and 2nd stage pressure switch part numbers are different. Make sure you have correct part number switch ready.

c. Disconnect electrical connector, P609 or P610, from pressure switch (Figure 11-4-114-1, Detail A).

d. Install protective caps and plugs on electrical connectors.

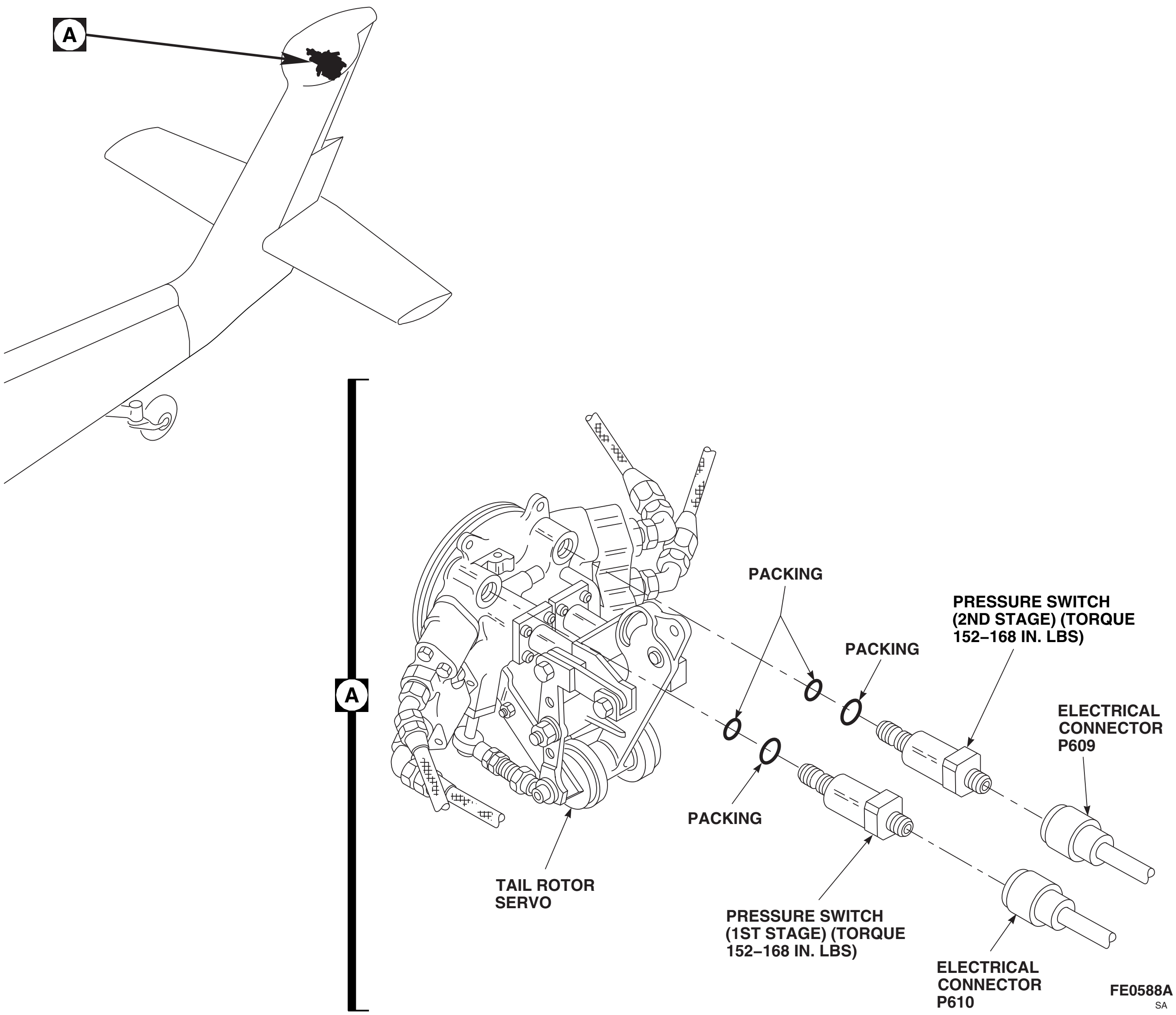


FIGURE 11-4-114-1. REPLACE TAIL ROTOR SERVO PRESSURE SWITCH

e. Use machinery towel, Item 211, Appendix D, to catch any spilled hydraulic fluid. Unscrew and remove installed pressure switch from tail rotor servo. Remove and discard packings.

11-4-114.1.2. Install.

a. Coat packings with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on replacement pressure switch. Coat threads of pressure switch with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install replacement pressure switch on tail rotor servo. **TORQUE TO 152 - 168 INCH-POUNDS.** Lockwire, Item 197, Appendix D (Figure 11-4-114-1, Detail A).

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P610, for 1st stage, P609, for 2nd stage, to pressure switch. Lockwire, Item 196, Appendix D.

f. Perform function check (PARA 11-4-114.1.3).

11-4-114.1.3. Function Check.

a. Apply external electrical power (PARA 1-6-16).

b. On overhead console, place BACKUP HYD PUMP switch to ON.

c. On caution/advisory panel, check that #1 TAIL RTR SERVO and #2 TAIL RTR SERVO ON capsules are off.

d. On lower console, place TAIL SERVO switch to BACKUP.

e. On caution/advisory panel, check that #1 TAIL RTR SERVO and #2 TAIL RTR SERVO ON capsules are on.

f. On lower console, place TAIL SERVO switch to NORMAL and check that #1 TAIL RTR SERVO and #2 TAIL RTR SERVO ON capsules are off.

g. Check tail rotor servo for leaks.

h. Bleed No. 1, No. 2, and backup hydraulic systems (PARA 7-4-65).

i. If necessary, service hydraulic pump modules (PARA 1-3-8).

j. On overhead console, place BACKUP HYD PUMP switch to OFF.

k. Shut off external electrical power (PARA 1-6-16).

l. Make sure area is clean and free of foreign material.

m. Install tail gear box front fairing (PARA 2-4-137).

11-4-115 TAIL ROTOR SERVO THERMAL RELIEF VALVE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-137
- PARA 7-4-65

11-4-115.1 Replace Tail Rotor Servo Thermal Relief Valve.

NOTE

Replacement of both tail rotor servo thermal relief valves is the same.

11-4-115.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove tail gear box front fairing (PARA 2-4-137).

NOTE

Have replacement thermal relief valve ready to install before removing installed valve.

c. Use machinery towel, Item 211, Appendix D, to catch any spilled oil. Remove thermal relief valve from tail rotor servo and discard packing (Figure 11-4-115-1, Detail B).

11-4-115.1.2 Install.

a. Coat replacement thermal relief valve and packing with hydraulic fluid, Item 154 or Item 155, Appendix D.

b. Install new packing on replacement valve (Figure 11-4-115-1, Detail B).

c. Install replacement thermal relief valve on tail rotor servo.

d. **TIGHTEN VALVE ¼ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using lockwire, Item 197, Appendix D, lockwire valves to each other.

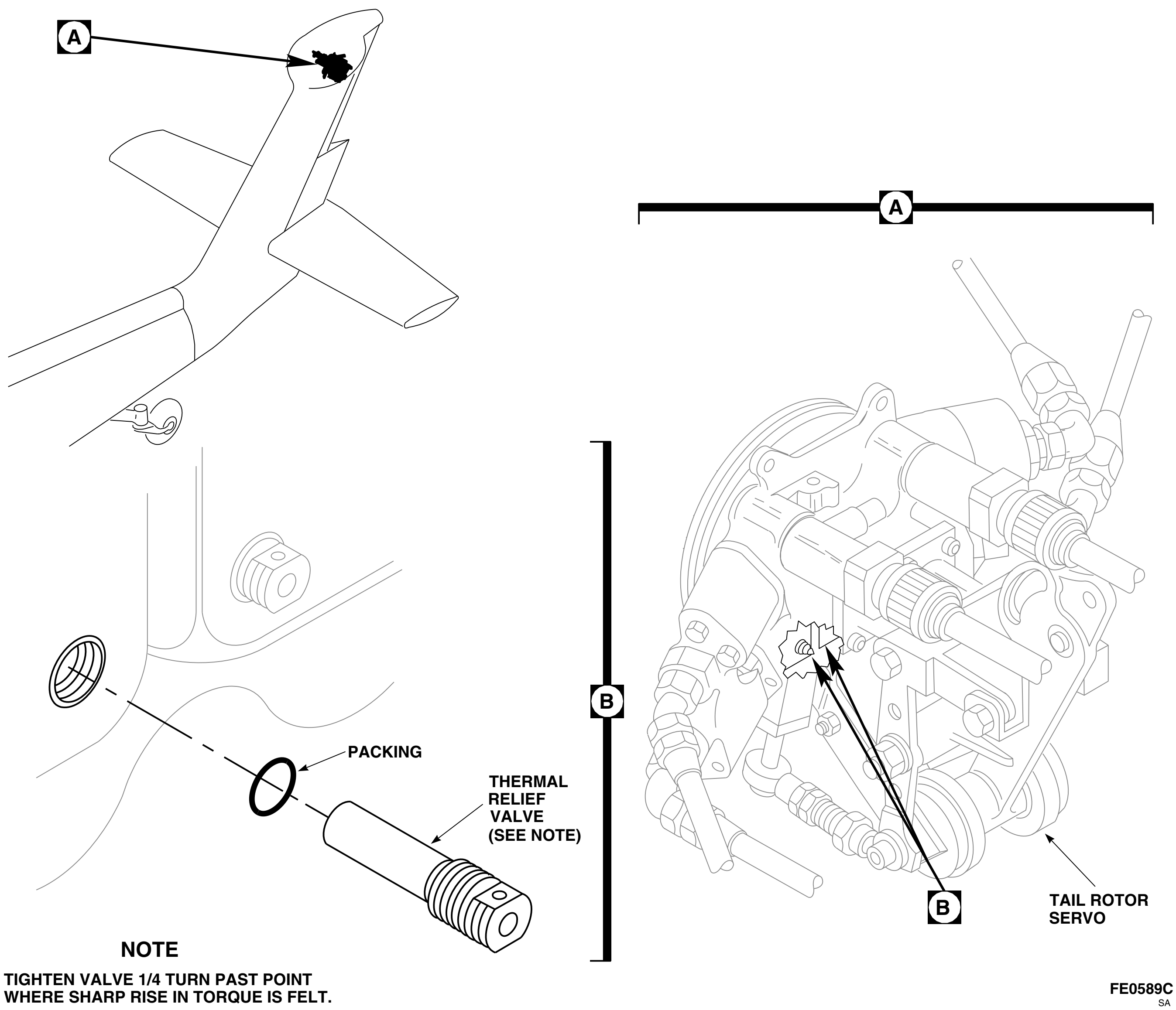


FIGURE 11-4-115-1. REPLACE TAIL ROTOR SERVO THERMAL RELIEF VALVE

e. Perform function check (PARA 11-4-115.1.3).

11-4-115.1.3 Function Check.

a. Apply external electrical power (PARA 1-6-16).

b. On overhead console, place BACKUP HYD PUMP switch to ON. Check for leaks.

NOTE

If second stage thermal relief valve is being replaced on lower console, place TAIL SERVO switch to BACKUP.

c. On lower console, place TAIL SERVO switch to NORMAL. On overhead console, place BACKUP HYD PUMP switch to OFF.

d. Bleed No. 1, No. 2, and backup hydraulic systems (PARA 7-4-65).

e. If necessary, service hydraulic pump modules (PARA 1-3-8).

f. Shut off external electrical power (PARA 1-6-16).

g. Make sure area is clean and free of foreign material.

h. Install tail gear box front fairing (PARA 2-4-137).

11-4-116 TAIL ROTOR SERVO HYDRAULIC CONNECTORS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-3-8
- PARA 1-6-15
- PARA 1-6-16
- PARA 7-4-50
- PARA 7-4-65
- PARA 11-4-112

11-4-116.1 Replace Tail Rotor Servo Hydraulic Connectors.

NOTE

Replacement of both tail rotor servo hydraulic connectors is the same, except as noted.

11-4-116.1.1. Remove.

a. Disconnect pressure and return hoses from hydraulic connector (PARA 7-4-50).

NOTE

- If hydraulic connector is not to be replaced now, plug hoses to keep dirt out.
- Bracket angles installed on 2nd stage connector.

b. Remove bolts and washers securing 1st or 2nd stage connector and bracket angles to tail rotor servo. Remove connector (Figure 11-4-116-1, Detail A).

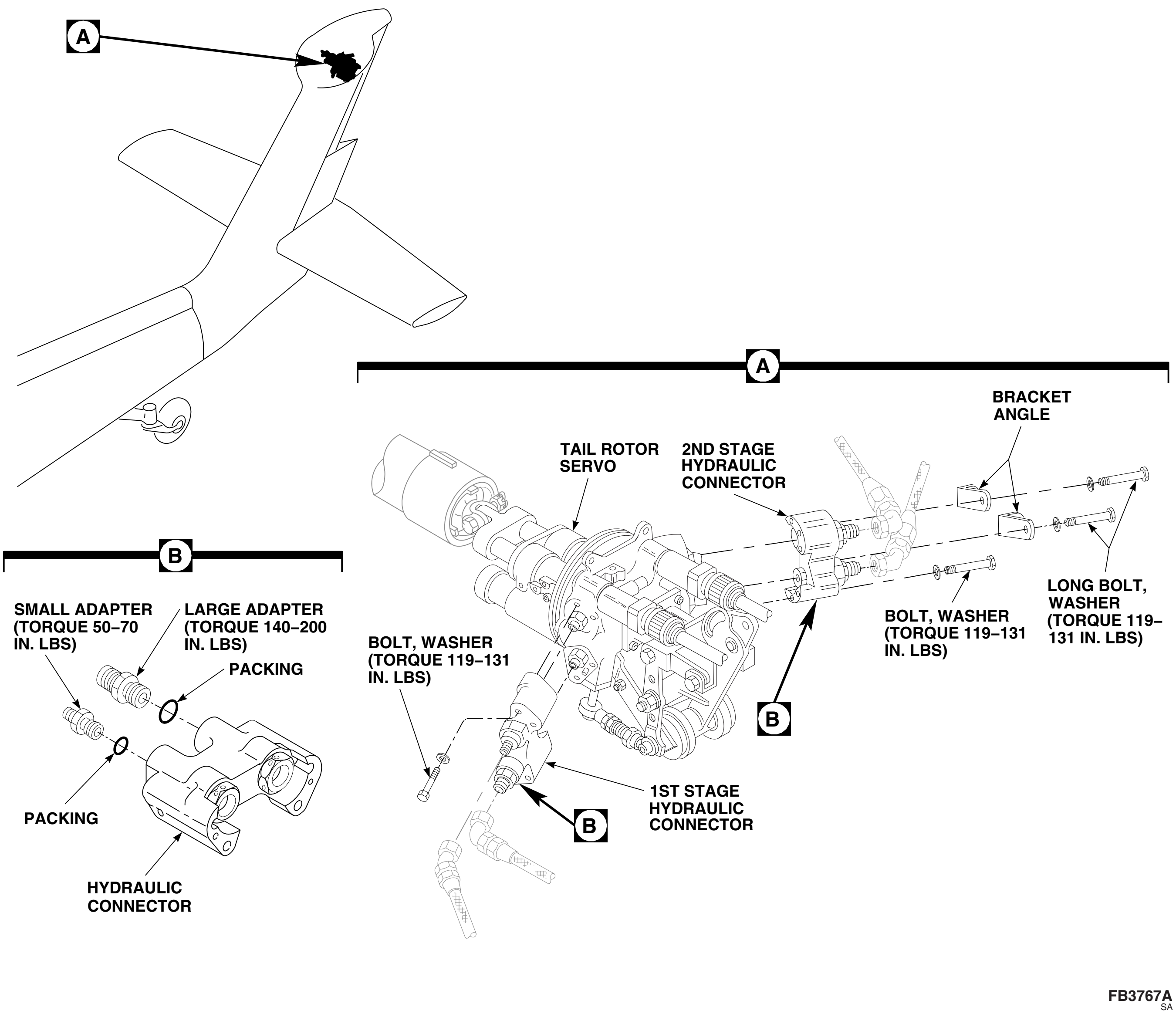


FIGURE 11-4-116-1. REPLACE TAIL ROTOR SERVO HYDRAULIC CONNECTORS

11-4-116.1.2. Disassemble.

CAUTION

Damage to hydraulic system will result if dirt or foreign material is allowed to enter system. Make sure area is clean and have replacement adapter ready for immediate installation before old adapter is removed.

Remove large or small adapter from hydraulic connector and remove packing from adapter (Figure 11-4-116-1, Detail B). Discard packing.

11-4-116.1.3. Assemble.

a. Lightly coat packing with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on small adapter (Figure 11-4-116-1, Detail B).

b. Lightly coat threads of small adapter with hydraulic fluid, Item 154 or Item 155, Appendix D, and install adapter in hydraulic connector. **TORQUE SMALL ADAPTER TO 50 - 70 INCH-POUNDS.**

c. Lightly coat packing with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on large adapter.

d. Lightly coat threads of large adapter with hydraulic fluid, Item 154 or Item 155, Appendix D, and install adapter in connector. **TORQUE LARGE ADAPTER TO 140 - 200 INCH-POUNDS.**

11-4-116.1.4. Install.

a. Line up 1st stage hydraulic connector with guide pins and tail rotor servo fittings and install connector. Secure 1st stage connector to tail rotor servo using bolts and washers. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.** Lockwire, Item 197, Appendix D, bolts (Figure 11-4-116-1, Detail A).

b. Install 2nd stage hydraulic connector as follows:

(1) Line up connector with guide pins and tail rotor servo fittings and install connector.

(2) Secure connector to tail rotor servo using bolt and washer.

(3) Position bracket angles on connector. Secure angles and connector to tail rotor servo using long bolts and washers.

(4) **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**

(5) Lockwire, Item 197, Appendix D, bolts.

c. Connect pressure and return hoses to connector (PARA 7-4-50).

d. Perform function check (PARA 11-4-116.1.5).

11-4-116.1.5. Function Check.

a. Open main rotor pylon sliding cover.

b. Disconnect tail rotor servo input pushrod from tail rotor servo input lever (PARA 11-4-112).

c. Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

d. If using backup pump, on overhead console, place BACKUP HYD PUMP switch to ON.

e. On lower console, place TAIL SERVO switch to NORMAL. Check servo for leaks

f. Hold right pedal to move servo input pushrod out of way as required.

g. Move tail rotor servo input lever to full extended position. Release lever. **SERVO SHALL MOVE BACK TO MIDSTROKE POSITION.**

h. If servo does not center, check for crossed 1st stage pressure and return lines. Correct crossed line condition. Recheck servo for correct hydraulic line installation.

i. On lower console, place TAIL SERVO switch to BACKUP. Check servo for leaks.

j. Move servo input lever to full extended position. Release lever. **SERVO SHALL MOVE BACK TO MIDSTROKE POSITION.**

k. If servo does not center, check for crossed 2nd stage pressure and return lines. Correct crossed line condition. Recheck servo for correct hydraulic line installation.

l. On lower console, place TAIL SERVO switch to NORMAL. On overhead console, place BACKUP HYD PUMP switch to OFF.

m. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

n. Connect tail rotor servo input pushrod to tail rotor servo input lever (PARA 11-4-112).

o. Bleed No. 1, No. 2, and backup hydraulic systems (PARA 7-4-65).

p. If necessary, service hydraulic pump modules (PARA 1-3-8).

q. Make sure area is clean and free of foreign material.

r. Close and latch main rotor pylon sliding cover.

11-4-117 TAIL ROTOR QUADRANT SUPPORT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-6-5
- PARA 11-4-110

11-4-117.1 Replace Tail Rotor Quadrant Support.

11-4-117.1.1. Remove.

a. Remove tail rotor quadrant (PARA 11-4-110).

b. Remove screws and quadrant support from tail gear box (Figure 11-4-117-1, Detail A).

11-4-117.1.2. Install.

WARNING

FSCAP

Verification of screw installation, torque and application of torque stripe are critical characteristics.

a. Install quadrant support to tail gear box with new self-locking screws. **TORQUE SCREWS TO 105 - 115 INCH-POUNDS** (Figure 11-4-117-1, Detail A). Apply torque stripe (PARA 2-6-5).

b. Install tail rotor quadrant (PARA 11-4-110).

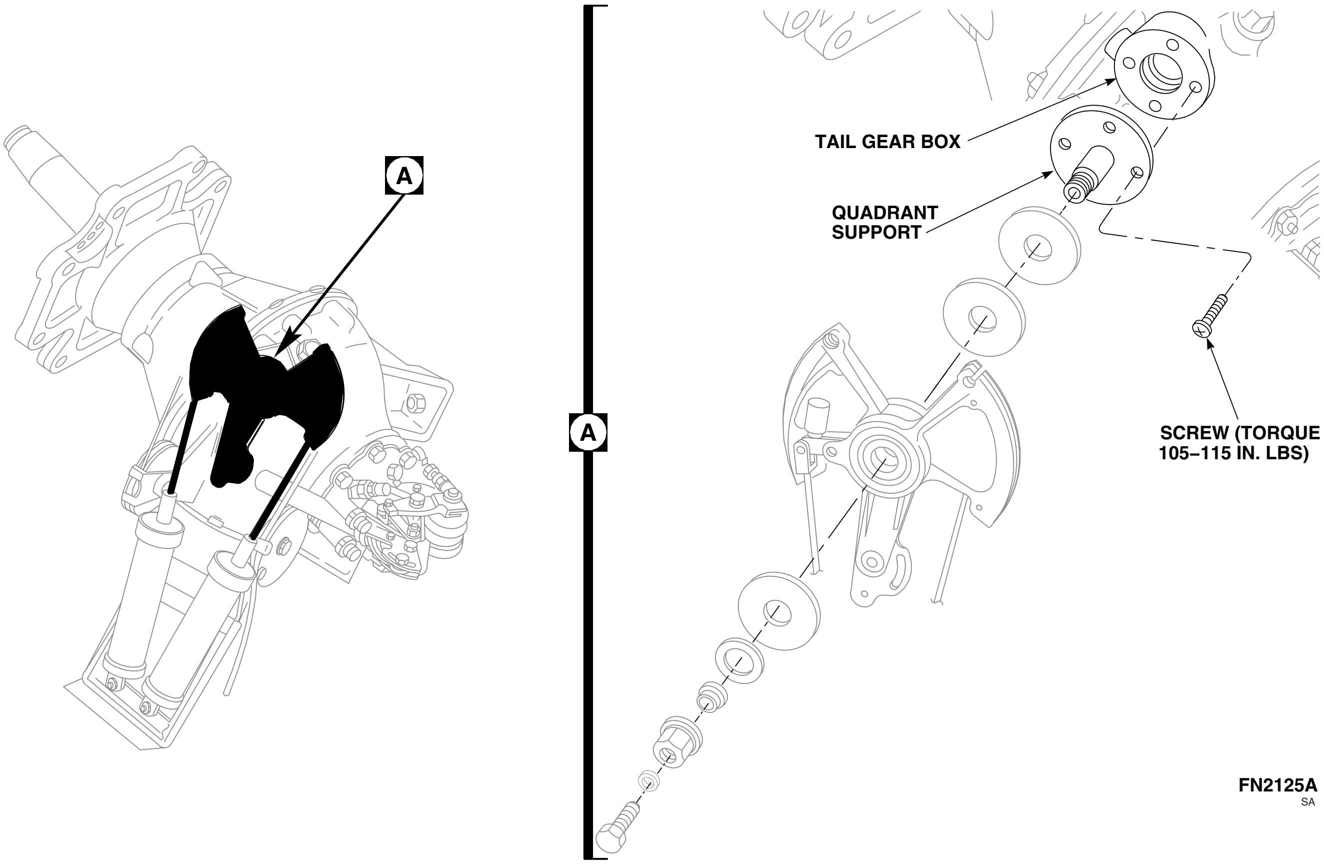


FIGURE 11-4-117-1. REPLACE TAIL ROTOR QUADRANT SUPPORT

11-4-118 TAIL ROTOR QUADRANT SPRING CYLINDERS.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Drill
- Drill Bit, No. 32, 0.116-inch diameter
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Adhesive, Item 59, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-6-5
- PARA 11-4-1
- PARA 11-4-110

11-4-118.1 Replace Tail Rotor Quadrant Spring Cylinders.

NOTE

- Spring cylinder shall be replaced every fifteen years. Refer to aircraft logbook for last record of spring cylinder replacement. After replacement, make sure helicopter logbook entry is made to reflect spring cylinder replacement.
- Replacement of both spring cylinders is the same, except as noted.
- If spring cylinders need only be disconnected for other maintenance, perform step b.

11-4-118.1.1 Remove.

a. Remove tail rotor quadrant (PARA 11-4-110).

b. Disconnect spring cylinders from support as follows (Figure 11-4-118-1, Detail B):

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Place collective stick in midposition.

(3) For right spring cylinder, apply full left pedal to reduce spring tension. Remove cotter pin and nut. Pull down on cylinder while removing bolt and washer. Disconnect spring cylinder from support.

(4) For left spring cylinder, apply full right pedal to reduce spring tension. Remove cotter pin and nut. Pull down on cylinder while removing bolt and washer. Disconnect spring cylinder from support.

(5) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Using drill with No. 32 drill bit, drill heads off rivet and remove rivet from arm and spring cylinder clevis using pin punch. Remove spacers (Figure 11-4-118-1, Detail C).

11-4-118.1.2 Inspect.

a. Inspect clevis of spring cylinder for security of washers. **NO LOOSE OR DISBONDED WASHER(S) ALLOWED.** If washer(s) is loose or disbonded, perform the following:

(1) If washer is loose or disbonded, remove washer from clevis using nonmetallic scraper (Figure 11-4-118-1, Detail D).

(2) Use machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area on clevis where washer is to be bonded.

(3) Using abrasive cloth, Item 1A, Appendix D, lightly scuff side of washer.

(4) Apply a light coat of adhesive, Item 59, Appendix D, to scuffed side of washer and on clevis.

(5) Install washer on clevis.

(6) Allow adhesive to air-dry for 2 hours before handling.

11-4-118.1.3 Install.

NOTE

If spring cylinders need only be connected for other maintenance, perform step b.

a. Install spring cylinder clevis on toggle. Install spacer. Install rivet (Figure 11-4-118-1, Detail C).

b. Connect spring cylinders to support as follows (Figure 11-4-118-1, Detail B):

(1) Turn on all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

(2) Place collective stick in midposition.

WARNING

FSCAP

Verification of toggle installation is a critical characteristic.

(3) Make sure toggle is in notch while connecting cylinders to support, causing microswitch to be depressed.

(4) To connect right spring cylinder, perform the following:

(a) Apply full left pedal. Pull on cylinder and install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

(c) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(5) To connect left spring cylinder, perform the following:

(a) Apply full right pedal. Pull on cylinder and install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

(b) Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

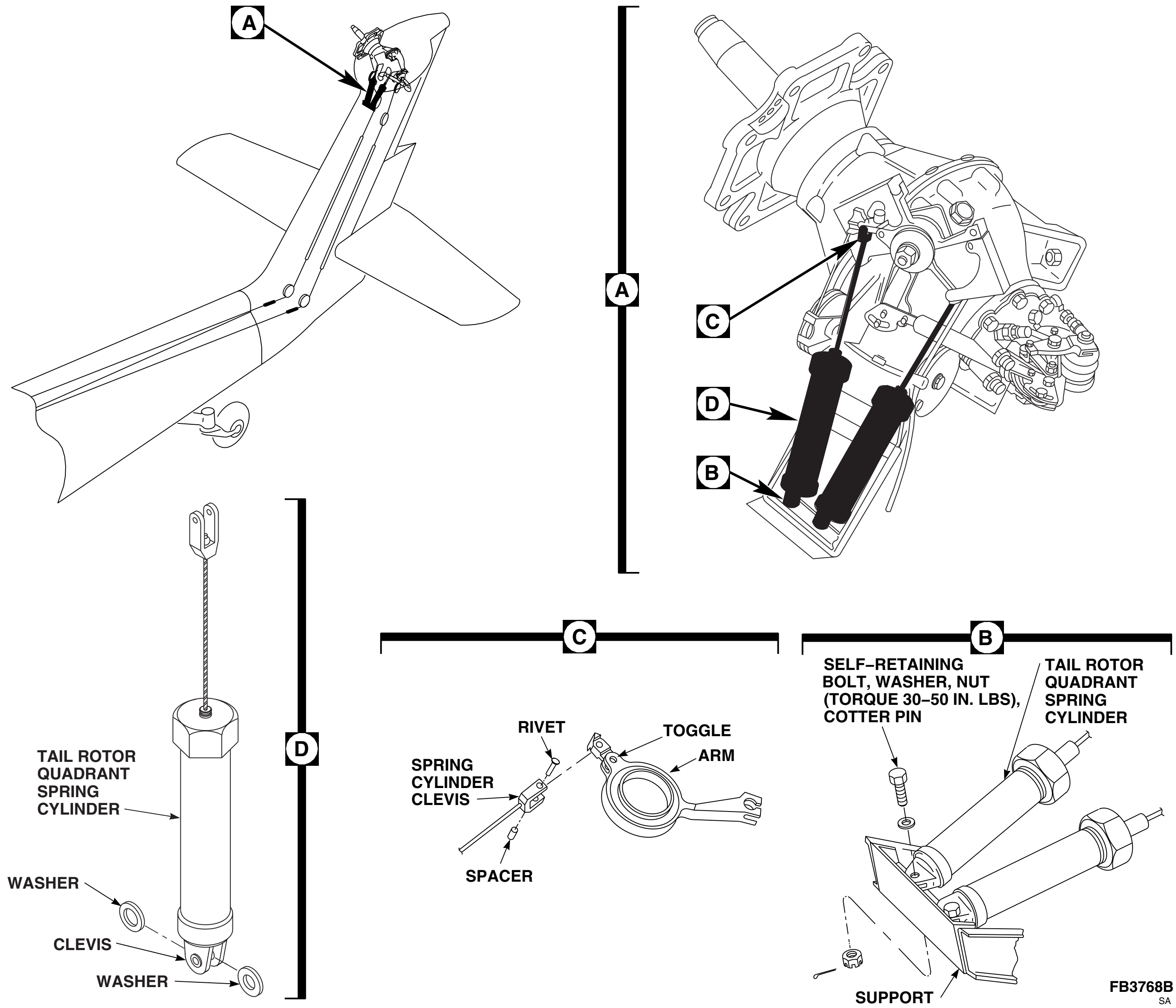


FIGURE 11-4-118-1. REPLACE TAIL ROTOR QUADRANT SPRING CYLINDERS

(c) **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

(6) Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

c. Install tail rotor quadrant (PARA 11-4-110).

NOTE

Make sure aircraft logbook entry is made to reflect spring cylinder replacement.

d. Annotate in aircraft logbook replacement date of spring cylinder.

11-4-119 TAIL ROTOR QUADRANT SPRING CYLINDER SUPPORT.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Drill
- Drill Bit, No. 32, 0.116-inch diameter
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Pin Punch
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Brush, Acid, Item 84, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 231, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-15
- PARA 1-6-16
- PARA 2-4-137
- PARA 2-6-5
- PARA 11-4-118

11-4-119.1 Replace Tail Rotor Quadrant Spring Cylinder Support.

11-4-119.1.1 Remove.

a. Turn off all hydraulic and electrical power (PARA 1-6-15 and PARA 1-6-16).

b. Remove tail gear box fairing (PARA 2-4-137).

c. Disconnect tail rotor quadrant spring cylinders from support (PARA 11-4-118).

d. Remove nuts and washers securing support to studs on tail gear box (Figure 11-4-119-1, Detail A).

e. Using nonmetallic scraper, remove sealing compound from between tail gear box and support.

f. Remove spring cylinder support from tail gear box.

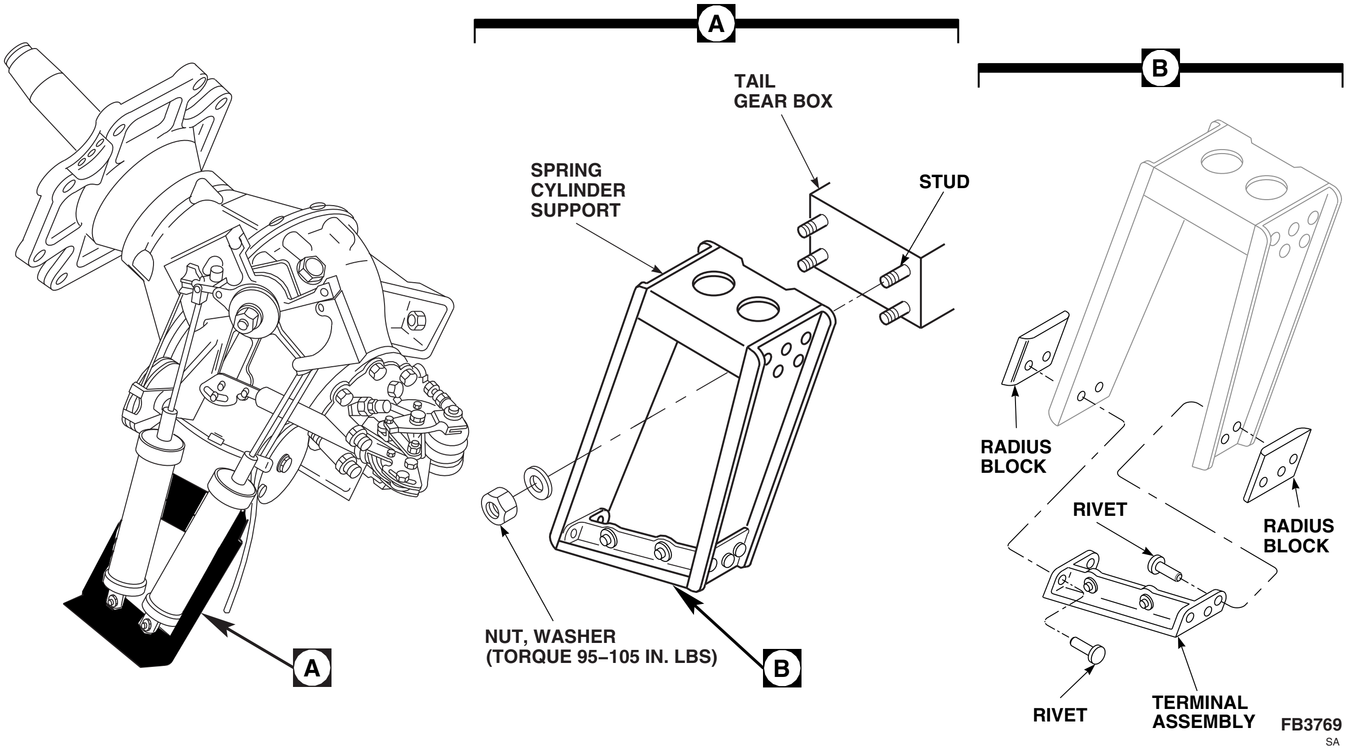


FIGURE 11-4-119-1. REPLACE TAIL ROTOR QUADRANT SPRING CYLINDER SUPPORT

g. Clean sealing compound from mating surfaces of tail gear box and support, using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

NOTE

Keep terminal assembly, spring cylinder support, and radius blocks, together for reassembly. If spring cylinder support or terminal assembly needs replacing, replace all components.

h. If required, using drill with No. 32 drill bit, drill heads off rivets and remove rivets from terminal assembly and spring cylinder support using pin punch. Remove terminal assembly from spring cylinder support (Figure 11-4-119-1, Detail B).

i. Using paint remover, Item 231, Appendix D, strip paint from surface of part around rivet holes.

j. Using fluorescent penetrant inspection kit, inspect spring cylinder support for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace spring cylinder support and terminal assembly.

k. Using epoxy primer coating kit, Item 135, Appendix D, coat paint-stripped areas with epoxy primer.

l. Touch up primer areas with paint (PARA 2-6-5).

11-4-119.1.2 Install.

a. If removed, position terminal assembly and radius blocks on spring cylinder support and secure with rivets (Figure 11-4-119-1, Detail B).

b. Apply sealing compound, Item 292, Appendix D, to mating surface of tail gear box using acid brush, Item 84, Appendix D. Avoid getting sealing compound on gear box mounting studs.

c. Install support on tail gear box studs with washers and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS** (Figure 11-4-119-1, Detail A).

d. Apply a bead of sealing compound, Item 292, Appendix D, around support and tail gear box.

e. Connect tail rotor quadrant spring cylinders to support (PARA 11-4-118).

f. Install tail gear box fairing (PARA 2-4-137).

11-4-120 SERVO TO SWASHPLATE CONTROL ROD NYLON WASHER.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Heat Gun
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 9A, Appendix D
- Abrasive Paper, Item 18, Appendix D
- Cheesecloth, Item 90, Appendix D
- Dry Ice, Item 125, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D

11-4-120.1 Replace Servo to Swashplate Control Rod Nylon Washer.

11-4-120.1.1. Remove.

NOTE

Dry ice and paint remover may be required to loosen bond on washer and remove adhesive if Hysol adhesive was used to secure washer to control rod. If sealing compound was used to secure washer to control rod, washer and any remaining sealing compound may be removed with nonmetallic scraper.

a. If required, pack dry ice, Item 125, Appendix D, on control rod end for 30 minutes to weaken bond on nylon washer.

b. Remove nylon washer using nonmetallic scraper.

c. Remove remaining adhesive on control rod using paint remover, Item 230, Appendix D, and nonmetallic scraper.

d. Clean bonding surface of control rod using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

e. Scuff bonding surface of nylon washer using abrasive paper, Item 18, Appendix D.

f. Clean bonding surface of nylon washer using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

11-4-120.1.2. Install.

NOTE

Washers, 70400-08155-120, and 70400-08155-121, are required to have a radius on outside corner of lipped area.

a. If required, blend corner of lipped washer, using abrasive cloth, Item 9A, Appendix D. Corner radius to be 0.070 - 0.130 inch maximum.

NOTE

Washers, 70400-08155-115, 70400-08155-120, and 70400-08155-121, have raised lips on one or two ends. During installation, these washers must be oriented so they are in an upright position (12 o'clock position) when washer has one lip and oriented at 12 and 6 o'clock position when washer has two lips (always viewed as if control rod is in its installed position). Refer to Figure 11-4-120-1, Detail A for typical view of washer orientation.

b. Apply sealing compound, Item 292, Appendix D, to bonding surfaces of control rod and nylon washer. Install nylon washer on control rod (Figure 11-4-120-1, Detail A and Detail B).

c. Gently smooth washer to remove any air bubbles in sealing compound. Remove any excess sealing compound with machinery towel, Item 211, Appendix D.

d. Cure sealing compound at 70° - 80°F (22° - 24°C) for 24 hours or 140° - 160°F (60° - 71°C) for 1 hour using heat gun.

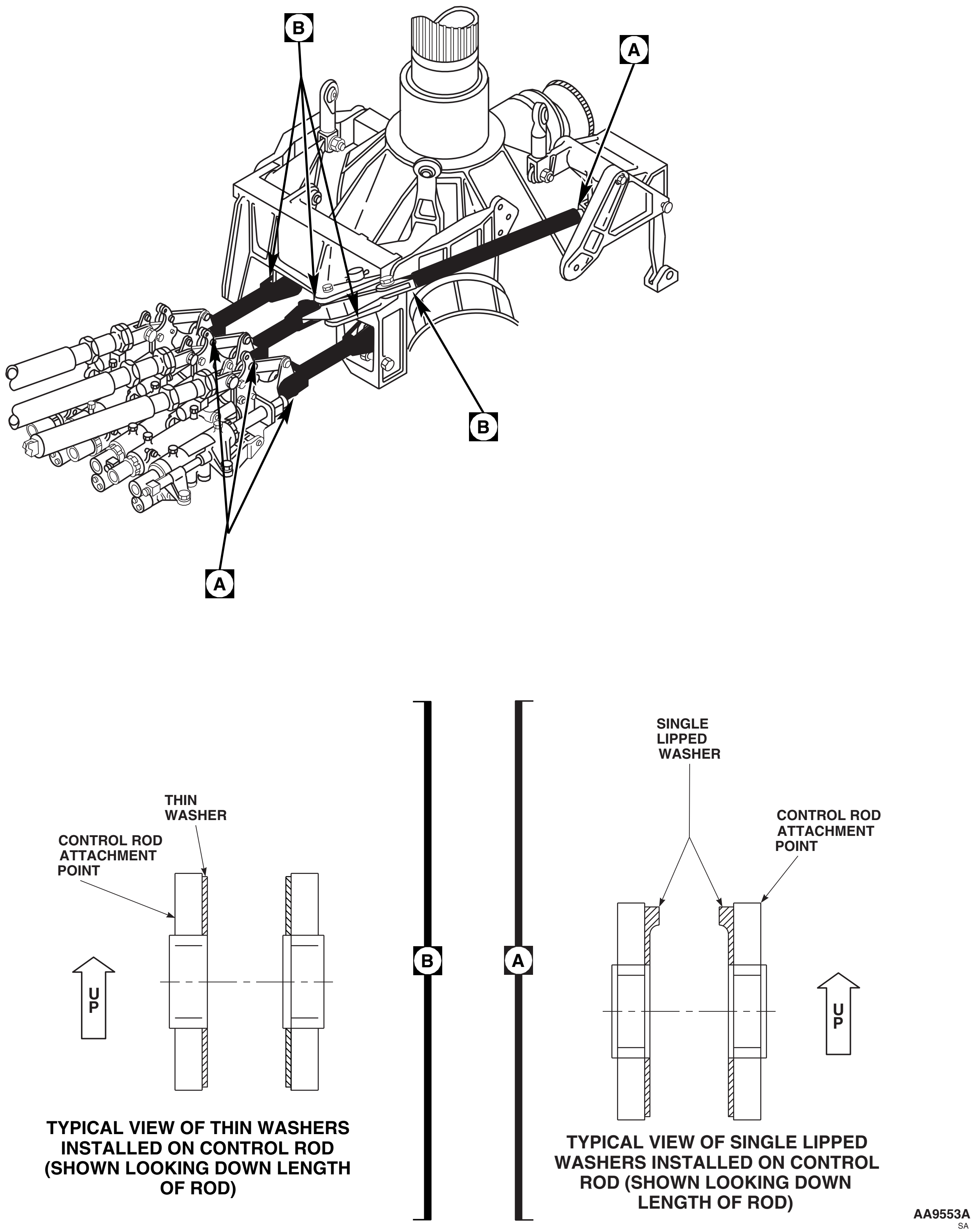


FIGURE 11-4-120-1. NYLON WASHER ORIENTATION (TYPICAL)

11-4-121 TAIL PYLON CABLE.

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Tape, Item 317C, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-137
- PARA 11-4-110

11-4-121.1 Replace Tail Pylon Cable.

11-4-121.1.1. Remove.

a. Turn off all electrical power (PARA 1-6-16).

b. Remove tail gear box fairing (PARA 2-4-137).

NOTE

Note location of tape for installation.

c. Remove tape securing left and right cables.

d. Remove adjusting nuts securing right and left microswitches, S45 and S46, and keywashers to brackets on quadrant (Figure 11-4-121-1, Detail A).

e. Disconnect electrical connector, P611 from J611.

f. Install protective caps and plugs on electrical connectors.

g. Remove bolt, washer, and nut securing clamp to bracket. Remove tail pylon cable from helicopter.

h. Remove clamp from tail pylon cable.

11-4-121.1.2. Install.

a. Install clamp on tail pylon cable (Figure 11-4-121-1, Detail A).

b. Install tail pylon cable and secure clamp to bracket with bolt, washer, and nut.

c. Remove protective caps and plugs from electrical connectors.

d. Inspect and treat electrical connectors (PARA 1-7-20).

e. Connect electrical connector, P611 to J611.

f. Install right microswitch, S46, and keywasher in bracket on quadrant. Install adjusting nut. Install left microswitch, S45, and keywasher in bracket on quadrant. Install adjusting nut.

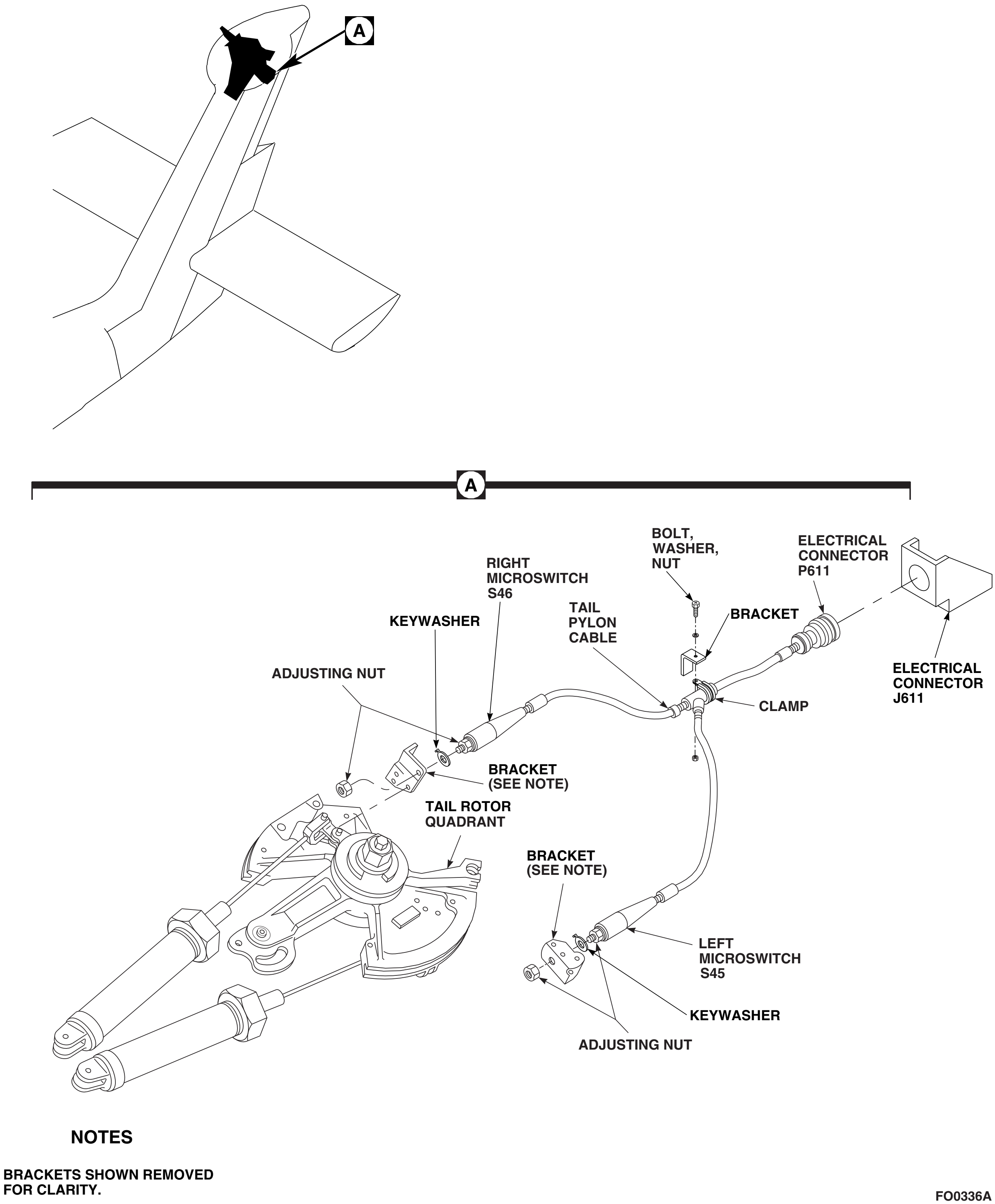


FIGURE 11-4-121-1. REPLACE TAIL PYLON CABLE

g. Install tape, Item 317C, Appendix D, in locations noted during removal securing left and right cables together.

h. Perform tail rotor quadrant microswitch adjustment (PARA 11-4-110).

SECTION 11-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
11-5-21	Pilot’s Collective Stick Bellcrank Support (BENCH)
11-5-22	Copilot’s Collective Stick Support (BENCH)
11-5-23	Copilot’s Collective Stick Bellcrank Bearings (BENCH)
11-5-24	Longitudinal Bellcrank Bearings (BENCH)
11-5-25	Forward, Lateral, and Aft Bellcrank/Walking Beam Flanged Bearings (BENCH)
11-5-26	Tail Rotor Pitch Control Shaft Pressed Bushing (BENCH)
11-5-27	Tail Rotor Quadrant Bearings (BENCH)
11-5-28	Tail Rotor Quadrant Upper and Lower Arm Bearings (BENCH)
11-5-29	Tail Rotor Quadrant Spring Cylinder Support Terminal Bearings (BENCH)
11-5-30	Yaw Control Rods (BENCH)
11-5-31	Control Rod Identification Plate (BENCH)

11-5-1 YAW CONTROL PEDAL (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Arbor Press
- Drill
- Drill Bit, No. 14
- Driver
- Fluorescent Penetrant Inspection Kit
- Pneumatic Rivet Setting Tool
- Protection, Eye
- Puller Set
- Reamer Set
- Size Block, 0.375 - 0.376-inch, Locally-Made
- Torque Wrench, 0 - 16 in. oz
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Acetone, Item 25, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-1.1 Repair Yaw Control Pedal (BENCH).

11-5-1.1.1 Disassemble.

a. Remove screw, nut, washer, spacer, and spring from pedal and actuator (Figure 11-5-1-1).

b. Remove long screw, washers, and nut, and remove actuator from pedal. Remove flanged bushings from actuator.

c. Using puller set, remove bearings from pedal.

d. Using arbor press and driver, remove flanged bushing and plain bushing from pedal.

e. Clean parts, except bearings, with acetone, Item 25, Appendix D.

11-5-1.1.2 Inspect/Repair.

a. Inspect screws for damage to threads.

b. Using paint remover, Item 230, Appendix D, remove paint from all painted parts.

c. Inspect actuator for bends and cracks. **NONE ALLOWED.** If bent or cracked, replace actuator.

d. Inspect parts for wear, cracks, elongated holes in bushings or attaching holes.

e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect pedal for cracks (ASTM E1417).

f. Inspect areas around bushings and bearings, and pedal arm clevis for cracks. **NONE ALLOWED.** If cracks are found, replace as necessary.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

g. Using acetone, Item 25, Appendix D, clean pedal. Blow dry with dry compressed air.

h. If necessary to replace angle, drill out two rivets using No. 14 drill bit. Be careful not to elongate holes. Secure new angle in place with rivets, using pneumatic rivet setting tool (Figure 11-5-1-1).

i. Using 1-inch masking tape, Item 212, Appendix D, mask pedal bearing bores and bushing holes in pedal.

j. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 223, Appendix D, paint pedal (PARA 2-6-5).

k. Using primer, Item 258, Appendix D, paint flanged and plain bushing outer surface. Using arbor press, and with primer still wet, install flanged bushing into pedal.

l. Insert a 0.375 - 0.376-inch size block over bushing flange, before installing plain bushing.

m. Using arbor press, install plain bushing. Measure space between ends of installed bushings. **CLEARANCE SHALL BE 0.375 - 0.376-INCH** (Figure 11-5-1-1, Detail A).

n. Using reamer set, line-ream bushing holes to 0.2495 - 0.2505-inch.

CAUTION

Damage to bearing will result if surface primer is allowed to contact bearings. Do not allow surface primer to come in contact with bearings.

o. Apply a light coat of sealing compound, Item 282, Appendix D, to outside diameter of bearing and inside diameter of pedal bores. Allow to air-dry for at least 30 seconds.

NOTE

Surface primer and sealing compound must be made by same manufacturer.

p. Apply about 1⁄16-inch bead of sealing compound, Item 280, Appendix D, to leading edge of bearing outside diameter and leading edge of pedal bore. Beads shall be all around edges.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

q. Within 10 minutes after applying sealing compound, press bearings into pedal using arbor press.

r. Using machinery towel, Item 211, Appendix D, damp with acetone, Item 25, Appendix D, wipe surfaces of bearings clean of any sealing compound.

s. Allow at least 10 minutes for sealing compound to dry before handling pedal.

11-5-1.1.3 Assemble.

a. Install flanged bushings in actuator with flanges on inside. Position actuator through pedal (Figure 11-5-1-1).

b. Install long screw, washers, and nut through actuator and pedal. **TORQUE NUT TO 13 - 23 INCH-POUNDS.**

c. Install spring on spacer and place spacer and spring between actuator and angle.

d. Install screw through actuator, spacer, and angle. Install washer and nut. **TORQUE NUT TO 13 - 23 INCH-POUNDS.**

e. Perform rotational drag check of bearings as follows:

(1) Install bolt and washers through bearings. Install nut on bolt.

(2) Using torque wrench and socket, turn bolthead and measure rotational drag of bearings. **ROTATIONAL DRAG SHALL NOT EXCEED 5 INCH-OUNCES.** If drag exceeds limit, replace bearings.

(3) Remove bolt, washer, and nut from bearings.

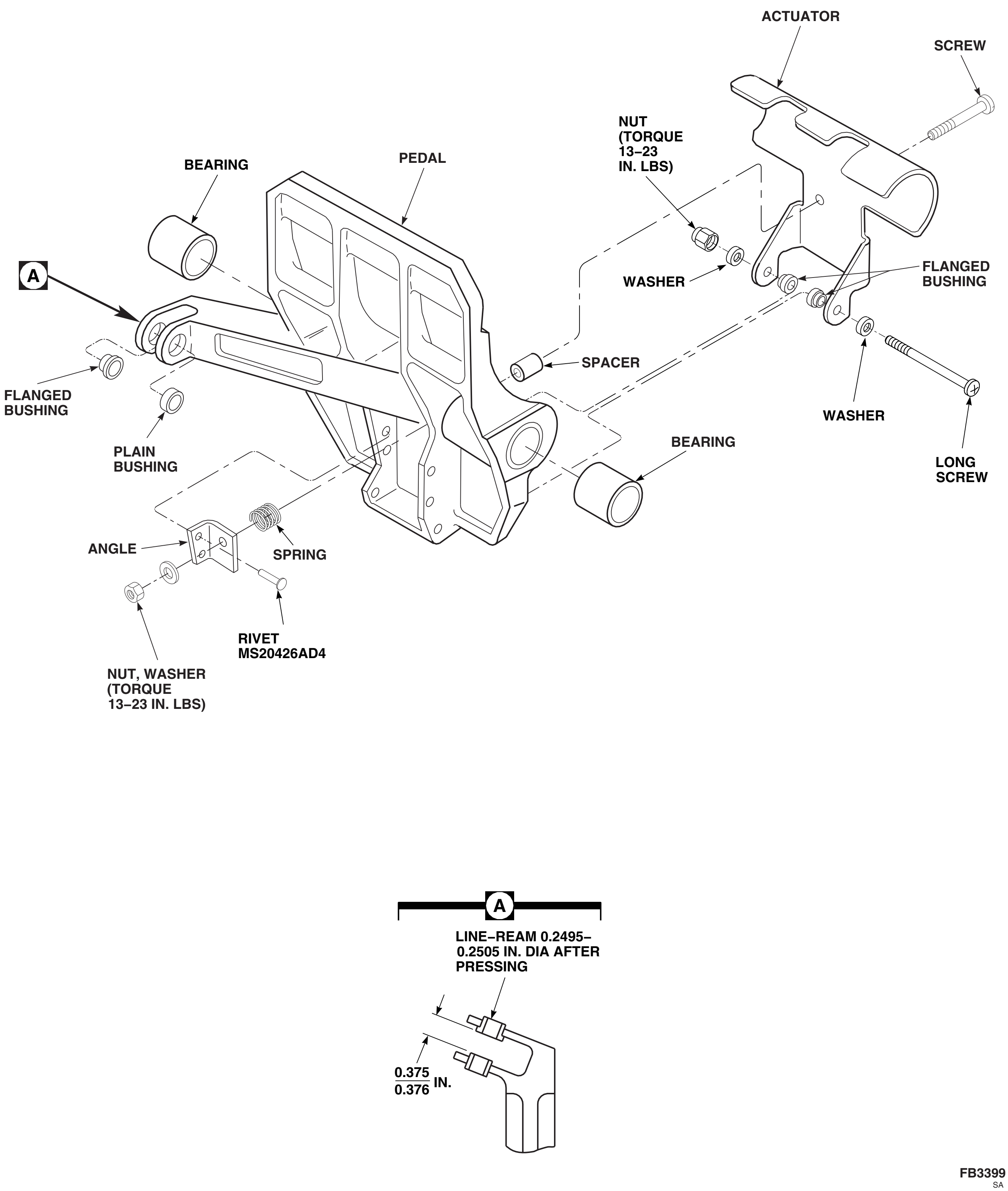


FIGURE 11-5-1-1. REPAIR YAW CONTROL PEDAL (BENCH)

11-5-2 YAW CONTROL PEDAL SUPPORT BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Dial Indicator
- Protection, Eye
- Puller Set
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Acetone, Item 25, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Grease, Item 149, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D
- Tags

References

- Appendix D
- ASTM E1444

11-5-2.1 Replace Yaw Control Pedal Support Bearings (BENCH).

11-5-2.1.1 Disassemble.

NOTE

Right shaft support is not secured to pedal pivot shaft.

a. Remove bolt, concave washers, and nut securing left shaft support to pedal pivot shaft. Remove shaft support (Figure 11-5-2-1 or Figure 11-5-2-2).

NOTE

Tag and identify location of each shim.

b. Remove retaining ring and shim. Slide pedal support tube and remaining shim from pedal pivot shaft.

c. Using puller set, remove bearings from pedal support tube.

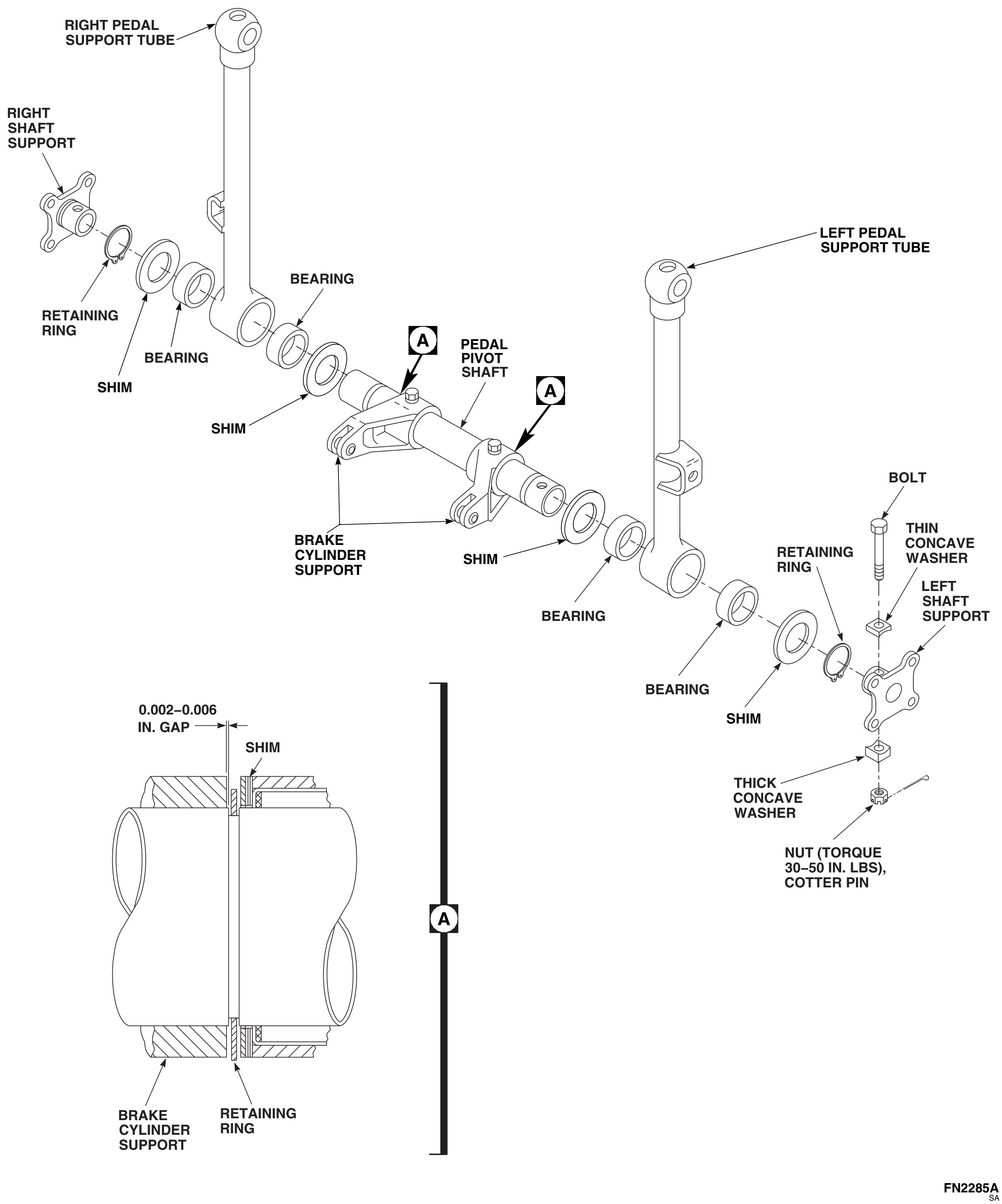


FIGURE 11-5-2-1. REPLACE YAW CONTROL PEDAL SUPPORT ASSEMBLY, 70400-01601-043, BEARINGS (BENCH)

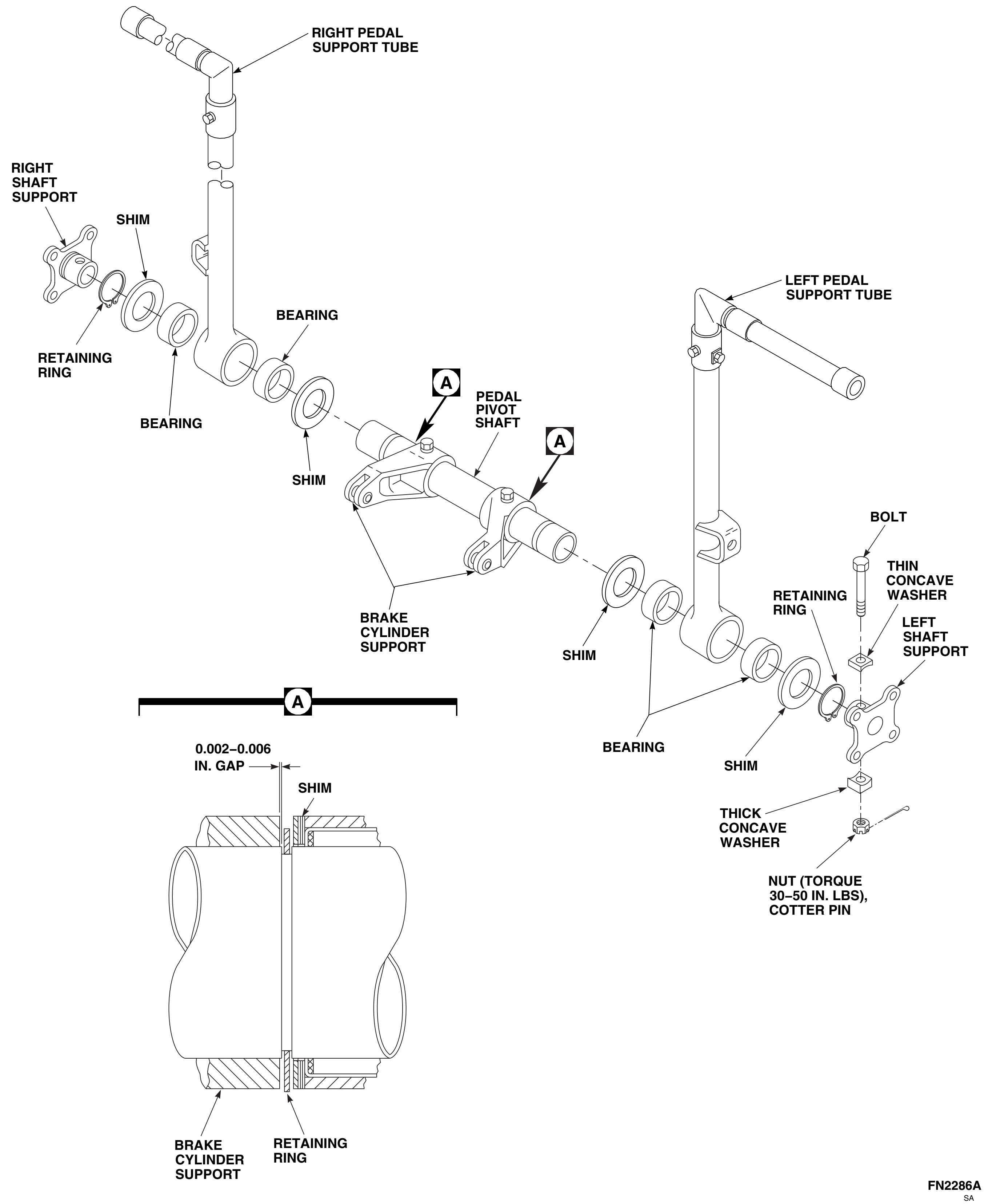


FIGURE 11-5-2-2. REPLACE YAW CONTROL PEDAL SUPPORT ASSEMBLY, 70400-01601-046, BEARINGS (BENCH)

11-5-2.1.2 Inspect/Repair.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

a. Clean pedal support tube, except bearings, with acetone, Item 25, Appendix D. Blow dry with dry compressed air.

b. Magnetic particle inspect pedal support tube and check for cracks in bearing area and welds at control rod attaching point (ASTM E1444). **NONE ALLOWED.** If crack is found, replace pedal support tube.

c. Inspect left and right pedal support tube bearings using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play is beyond limits, replace bearing.

11-5-2.1.3 Assemble.

CAUTION

Damage to bearing will result if surface primer is allowed to contact bearings. Do not allow surface primer to come in contact with bearings.

a. Apply a light coat of sealing compound, Item 282, Appendix D, to outside diameter of bearing and inside diameter of pedal support tube bore. Allow to air-dry for at least 30 seconds.

NOTE

Surface primer and sealing compound must be made by same manufacturer.

b. Before installing bearings, purge bearings and then hand pack with grease, Item 149, Appendix D. Clean outside diameter of bearings with cleaning cloth, Item 102, Appendix D, wet with acetone, Item 25, Appendix D.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

c. Apply about a 1⁄16-inch bead of sealing compound, Item 280, Appendix D, to leading edge of bearing outside diameter and leading edge of pedal support tube bore. Bead should be all around edges.

d. Within 10 minutes after applying sealing compound, press bearings in pedal support tube.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

e. Press bearings into pedal support tube, using arbor press. Each bearing must be seated flush to 0.010-inch below edge of support bore (Figure 11-5-2-1 or Figure 11-5-2-2).

f. Using cleaning cloth, Item 102, Appendix D, dampened with acetone, Item 25, Appendix D, wipe surfaces of bearing clean of any sealing compound.

NOTE

Sealing compound exposed to air (external to mating surfaces) will not dry.

g. Allow at least 10 minutes for sealing compound to dry before handling supports.

NOTE

If original shims are not available, proceed to step i.

h. Assemble pedal supports, with original shims, as follows:

(1) Remove tag from shim. Install shim on pedal pivot shaft.

(2) Install pedal support tube on pedal pivot shaft.

(3) Remove tag from shim. Install shim and retaining ring on pedal pivot shaft.

(4) Measure gap between brake cylinder support and retaining ring (Figure 11-5-2-1, Detail A or Figure 11-5-2-2, Detail A). **GAP SHALL BE 0.002 - 0.006-INCH.**

(5) If gap is less than 0.002-inch, peel shims evenly on both sides of pedal support tube to obtain a gap of 0.002 - 0.006-inch.

(6) If gap is greater than 0.006-inch, replace shims and perform step i.

(7) Install shims, pedal support tube, and retaining ring (Figure 11-5-2-1 or Figure 11-5-2-2).

i. Assemble pedal supports, with new shims, as follows:

(1) Temporarily, install pedal support tube and shaft support. Measure gap between brake cylinder support and retaining ring (Figure 11-5-2-1, Detail A or Figure 11-5-2-2, Detail A). **GAP SHALL BE 0.002 - 0.006-INCH.**

(2) Peel shims evenly on both sides of pedal support tube to obtain a gap of 0.002 - 0.006-inch.

(3) Install shims, pedal support tube, and retaining ring (Figure 11-5-2-1 or Figure 11-5-2-2).

j. Install left shaft support in pedal pivot shaft. Install thin concave washer on bolt with flat side toward bolthead. Install bolt through support and shaft. Install thick concave washer on bolt, curved side on shaft. Install nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

k. Install right shaft support in pedal pivot shaft. Tie to pedal support so it will not fall out.

11-5-3 YAW CONTROL PEDAL SUPPORT ASSEMBLY (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Airframe Repairer's Toolkit
- Arbor Press
- Drift
- Pneumatic Drill
- Reamer Set
- Size Block, 0.344 - 0.346-inch, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Acetone, Item 25, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Tags

References

- Appendix D
- ASTM E1444
- PARA 2-6-5

11-5-3.1 Repair Yaw Control Pedal Support Assembly (BENCH).

11-5-3.1.1. Disassemble.

NOTE

- Pedal pivot shaft may become discolored (reddish-brown) while in service. Surface discoloration to this component is not cause for replacement.
- Right shaft support is not secured to pedal pivot shaft.

a. Remove bolt, concave washers, and nut securing left shaft support to pedal pivot shaft. Remove shaft support (Figure 11-5-3-1 or Figure 11-5-3-2).

NOTE

Tag and identify location of each shim.

b. Remove retaining rings and shims. Slide pedal support tubes and remaining shims and retaining rings from pedal pivot shaft.

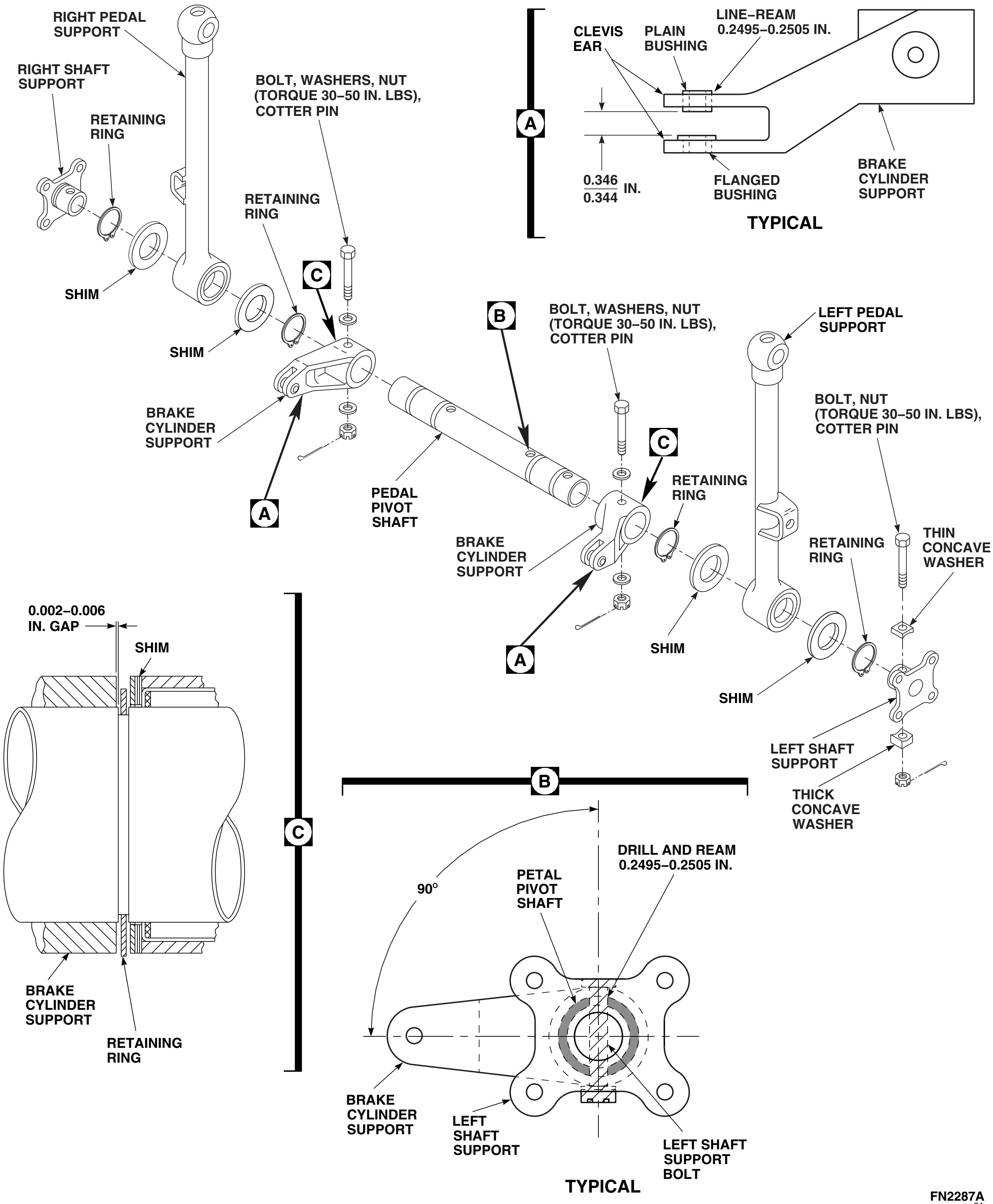


FIGURE 11-5-3-1. REPAIR YAW CONTROL PEDAL SUPPORT ASSEMBLY, 70400-01601-043 (BENCH)

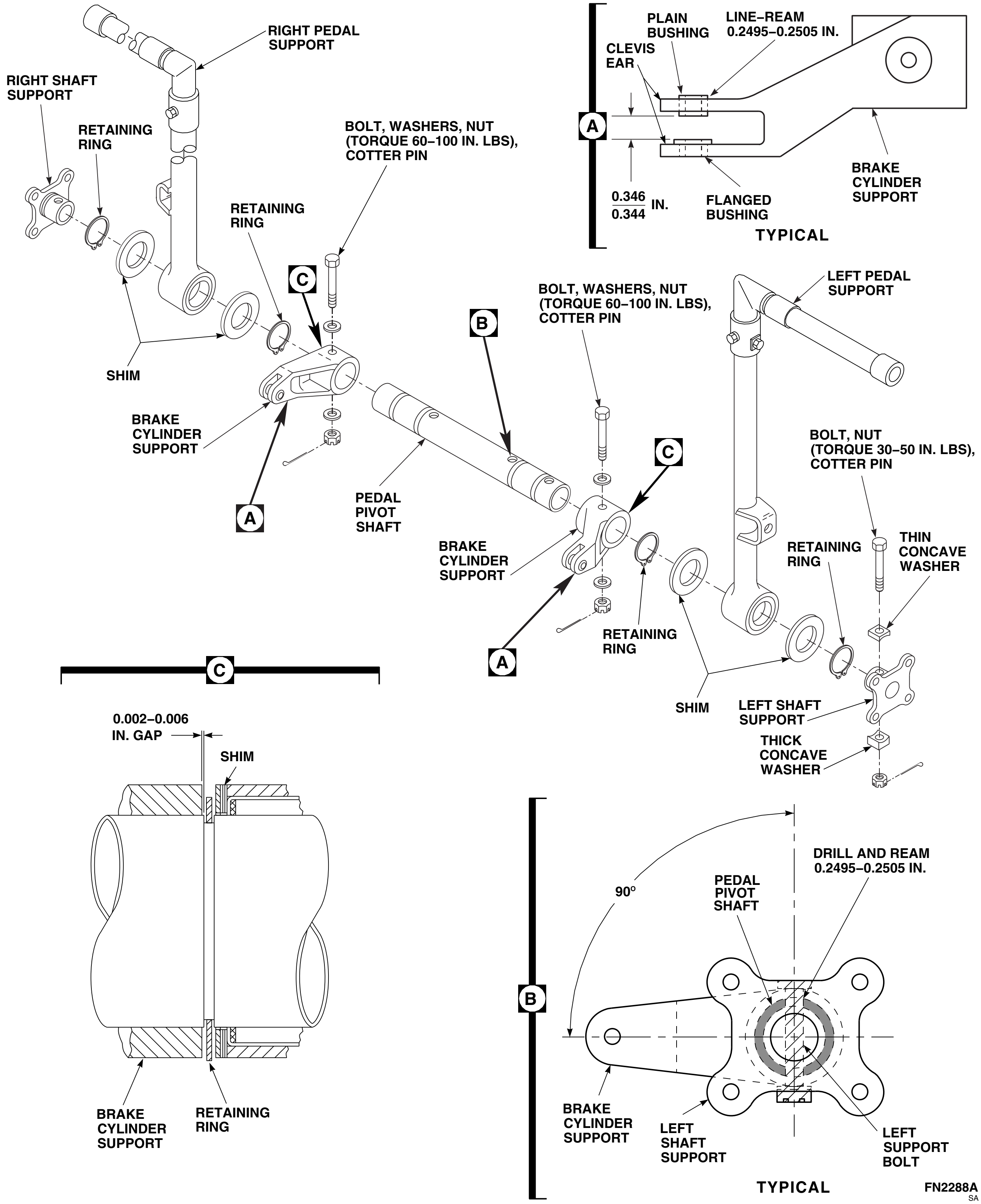


FIGURE 11-5-3-2. REPAIR YAW CONTROL PEDAL SUPPORT ASSEMBLY, 70400-01601-046 (BENCH)

c. Remove bolts, washers, and nuts securing brake cylinder supports to pedal support shaft. Slide supports from shaft.

d. If necessary, using arbor press and suitable drift, remove bushings from support clevis ears.

11-5-3.1.2. Inspect.

a. Using paint remover, Item 230, Appendix D, strip paint from pedal pivot shaft and brake cylinder supports.

b. Visually inspect pedal pivot shaft scoring for nicks on bearing area. Check for elongated holes.

c. Visually inspect brake cylinder supports for elongated bushing holes, and elongated bolt-holes.

d. Magnetic particle inspect shaft and brake cylinder supports (ASTM E1444).

e. Check pedal pivot shaft for cracks in retaining ring grooves and around boltholes. If crack is found, replace shaft.

f. Check brake cylinder supports for cracks around bushing holes, boltholes, where brake cylinder support attaches to shaft, and check clevis ears. If crack is found, replace support.

11-5-3.1.3. Repair.

a. Coat flanged bushing with primer, Item 258, Appendix D. Using an arbor press, with bushing still wet with primer, press bushing into clevis ear with flange on inside of clevis (Figure 11-5-3-1, Detail A).

b. Place a 0.344 - 0.346-inch size block on flange of flanged bushing before installing plain bushing.

c. Coat plain bushing with primer, Item 258, Appendix D. While still wet with primer, press plain bushing in clevis ear against size block. Remove size block. It should be just a snug fit, not tight.

d. Using reamer set, line-ream bushings to 0.2495 - 0.2505-inch diameter.

e. Clean brake cylinder support and pedal pivot shaft with acetone, Item 25, Appendix D.

f. Using 1-inch masking tape, Item 212, Appendix D, mask bushings and bore of brake cylinder support. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 223, Appendix D, paint support (PARA 2-6-5).

g. Using 1-inch masking tape, Item 212, Appendix D, mask bearing area between two snapring grooves at each end of pedal pivot shaft.

h. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 223, Appendix D, paint pedal pivot shaft (PARA 2-6-5).

11-5-3.1.4. Assemble.

NOTE

Replacement brake cylinder supports and left shaft support have pilot holes. Use same procedure for replacing them.

a. If new pedal pivot shaft is being installed, predrill left shaft support and brake cylinder support mounting holes as follows (Figure 11-5-3-1 or Figure 11-5-3-2):

(1) Temporarily install left shaft support to shaft.

(2) Line up hole in support with pilot hole in one end of shaft.

(3) Butt shaft against support.

(4) Using pneumatic drill and reamer set, drill and ream hole through shaft and support to 0.2495 - 0.2505-inch diameter (Figure 11-5-3-1, Detail B or Figure 11-5-3-2, Detail B). Remove support from shaft.

(5) Position brake cylinder supports on shaft so arm of support is at right angle to left shaft support mounting hole.

(6) Install retaining rings on two inside grooves (Figure 11-5-3-1 or Figure 11-5-3-2).

(7) Push supports against retaining rings. Using pneumatic drill and reamer set, drill and ream 0.2495 - 0.2505-inch hole through shaft and supports (Figure 11-5-3-1, Detail B or Figure 11-5-3-2, Detail B).

b. Install left and right brake cylinder support on shaft with arms bending outward. Install bolts, washers, and nuts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins (Figure 11-5-3-1 or Figure 11-5-3-2).

NOTE

If original shims are not available, go to step d.

c. Assemble pedal supports, with original shims, as follows:

(1) Install retaining rings in groove next to brake cylinder supports.

(2) Remove tags from shims. Install shims on pedal pivot shaft.

(3) Install pedal support tubes on shaft.

(4) Remove tags from shims. Install shims and retaining rings on shaft.

(5) Measure gap between brake cylinder support and retaining ring (Figure 11-5-3-1, Detail C or Figure 11-5-3-2, Detail C). **GAP TO BE 0.002 - 0.006-INCH.**

(6) If gap is less than 0.002-inch, peel shims evenly on both sides of pedal support tube to obtain a gap of 0.002 - 0.006-inch.

(7) If gap is greater than 0.006-inch, replace shims and perform step d.

(8) Install shims, pedal support tubes, and retaining rings (Figure 11-5-3-1 or Figure 11-5-3-2).

d. Assemble pedal supports, with new shims, as follows:

(1) Temporarily install pedal support tubes and shaft supports. Measure gap between brake cylinder support and retaining ring (Figure 11-5-3-1, Detail C or Figure 11-5-3-2, Detail C). **GAP TO BE 0.002 - 0.006-INCH.**

(2) Peel shims evenly on both sides of pedal support tube to obtain a gap of 0.002 - 0.006-inch.

(3) Install shims, pedal support tubes, and retaining rings (Figure 11-5-3-1 or Figure 11-5-3-2).

e. Install left shaft support in pedal pivot shaft. Install thin concave washer on bolt with flat side toward bolthead. Install bolt through support and shaft. Install thick concave washer on bolt, curved side on shaft. Install nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

f. Install right shaft support in pedal pivot shaft. Tie to pedal support so it will not fall out.

11-5-4 YAW CONTROL PEDAL ADJUSTER BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Anvil
- Arbor Press
- Bearing Remover, Step-Type
- Dial Indicator
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 280, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-4.1 Replace Yaw Control Pedal Adjuster Bearings (BENCH).

11-5-4.1.1 Inspect Bearings.

a. Inspect bearings in small arm and links using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play is beyond limit, replace bearing.

b. Inspect bearing in long arm using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.025-INCH.** If radial play is beyond limit, replace bearing.

11-5-4.1.2 Remove Bearings.

Using arbor press, a step-type bearing remover, and correct size anvil, press bearings from arm or link (Figure 11-5-4-1).

11-5-4.1.3 Inspect Arms and Links.

a. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect arms and links for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace arm or link.

b. Touch up paint arms and links as required (PARA 2-6-5).

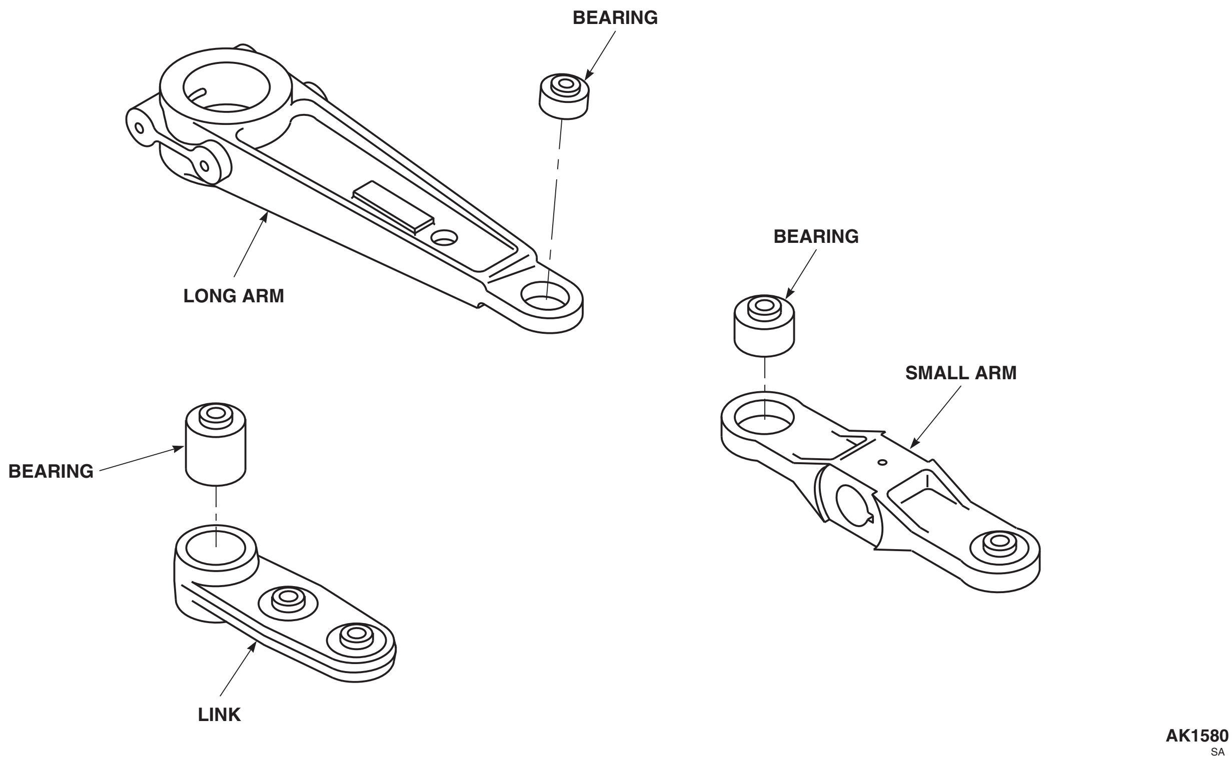


FIGURE 11-5-4-1. REPLACE YAW CONTROL PEDAL ADJUSTER BEARINGS (BENCH)

11-5-4.1.4 Install Bearings.

a. Clean inside bore of arm or link with acetone, Item 25, Appendix D.

b. Clean outside surface of replacement bearing with machinery towel, Item 211, Appendix D, wet with acetone, Item 25, Appendix D.

c. Apply about a 1/16-inch bead of sealing compound, Item 280, Appendix D, to leading edge of outside diameter and entrance edge of bearing bore. Bead shall be all around edge.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

d. Within 10 minutes after applying sealing compound, press bearing in arm or link (Figure 11-5-4-1). Make sure bearing is fully seated and there is no sealing compound in bearing.

e. Allow 10 minutes for sealing compound to dry before handling arm or link.

f. Perform rotational drag check of bearings as follows:

(1) Position washers on lip of bearing.

(2) Install bolt through bearing. Install nut on bolt. **TIGHTEN NUT ON BOLT ENOUGH SO THAT BEARING ROTATES.**

(3) Using torque wrench, turn bolthead and measure rotational drag of bearing. **IF DRAG IS MORE THAN 2 INCH-OUNCES FOR BEARINGS IN ARMS, REPLACE BEARING.**

(4) Using torque wrench, turn bolthead and measure rotational drag of bearing. **IF DRAG IS MORE THAN 8 INCH-OUNCES FOR BEARINGS IN LINK, REPLACE BEARING.**

(5) Remove bolt, washers, and nut from bearing.

11-5-5 SAS SERVOVALVE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 11-2-4
- PARA 11-2-5
- PARA 11-2-6

11-5-5.1 Replace SAS Servovalve (BENCH).

NOTE

Replacement of roll trim actuator SAS servovalve, yaw boost servo SAS servovalve, and pitch/trim assembly SAS servovalve is the same, except as noted.

11-5-5.1.1. Remove.

CAUTION

- Damage to servovalve will result if servovalve is mishandled during removal. Handle valve with extreme care. If valve is bumped or jarred, valve calibration will be disturbed.
- Damage to servovalve will result if dirt is allowed to enter components during disassembly. Make sure yaw boost servo is clean before removing servovalve from SAS actuator. Replacement servovalve should be ready for installation prior to removing old servo. Shipping cover, when available, should be transferred to old servo.
- Damage to SAS servovalve will result if follow-up wire is allowed to contact any surface. When removing replacement SAS servovalve from shipping cover, do not allow follow-up wire to contact any surface (Figure 11-5-5-1, Detail A). If follow-up wire contacts any surface, calibration will be disturbed and SAS servovalve must be replaced.

a. Remove screws and washers securing SAS servovalve to SAS actuator. Remove SAS servovalve carefully, so as not to damage follow-up wire (Figure 11-5-5-1).

b. Remove and discard packings.

c. Remove shipping cover, if installed, from replacement servovalve, and install on old servovalve.

11-5-5.1.2. Install.

CAUTION

- Damage to servovalve will result if servovalve is mishandled during installation. Handle valve with extreme care. If valve is bumped or jarred, valve calibration will be disturbed.
- Damage to SAS servovalve will result if follow-up wire is allowed to contact any surface. When installing servovalve, use care to properly position follow-up wire into SAS actuator. Follow-up wire sticks out of servovalve. It must be carefully put in hole of SAS actuator. Do not allow follow-up wire to contact any surface (Figure 11-5-5-1, Detail A). If follow-up wire contacts any surface, SAS servovalve calibration will be disturbed.

NOTE

SAS servovalve must be nulled prior to installation (PARA 11-2-4, roll trim actuator, PARA 11-2-5, yaw boost servo, or PARA 11-2-6, pitch/trim assembly).

a. Coat packings with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on SAS servovalve.

b. Position servovalve on SAS actuator so follow-up wire lines up with hole in SAS actuator, and carefully install servovalve on SAS actuator (Figure 11-5-5-1).

c. Install screws and washers in servovalve. **CAREFULLY TIGHTEN SCREWS IN A CRISSCROSS PATTERN UNTIL SERVOVALVE IS FULLY SEATED AGAINST SAS ACTUATOR. TORQUE SCREWS TO 24 - 26 INCH-POUNDS.**

d. Lockwire, Item 196, Appendix D, screws.

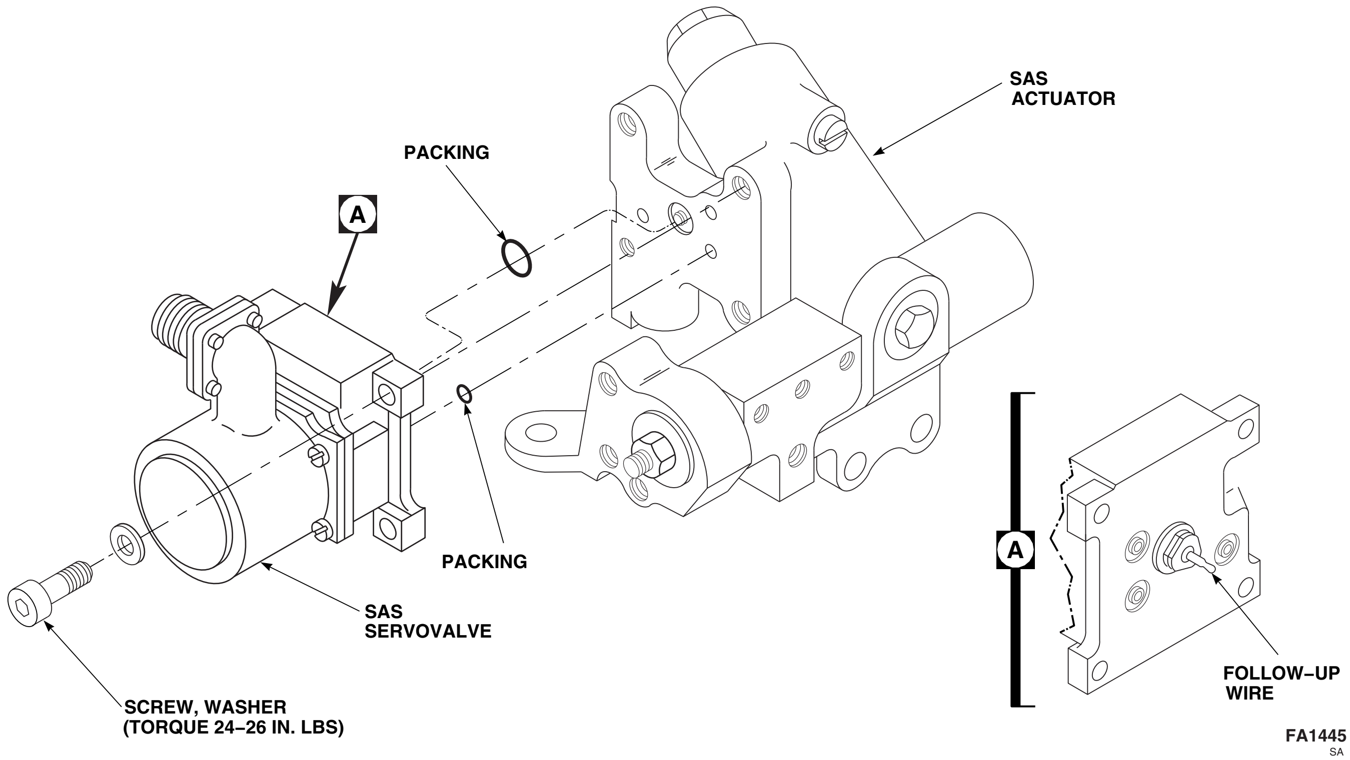


FIGURE 11-5-5-1. REPLACE SAS SERVOVALVE (BENCH)

11-5-6 SAS ACTUATOR (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 2-6-5
- PARA 11-2-4
- PARA 11-2-5
- PARA 11-2-6
- PARA 11-4-1

11-5-6.1 Replace SAS Actuator (BENCH).

CAUTION

Damage to yaw boost servo will result if dirt is allowed to enter components during disassembly. To avoid getting dirt in components, make sure yaw boost servo is clean before disassembly. Replace SAS actuator only in a clean place. Have replacement actuator ready for installation before old actuator is removed.

NOTE

Replacement of roll actuator SAS actuator, yaw boost servo SAS actuator, and pitch/trim assembly SAS actuator is the same, except where noted.

11-5-6.1.1 Remove.

a. Remove self-retaining bolt, washer, and nut securing output clevis to link (Figure 11-5-6-1, Sheet 1, Sheet 2, or Sheet 3).

b. Remove bolts and washers securing SAS actuator to roll actuator, yaw boost servo, or pitch/trim assembly.

c. Carefully lift actuator straight up and remove transfer tubes.

d. Remove seals from transfer tubes. Discard seals.

e. Remove self-retaining bolts, washers, and nuts securing link to roll actuator, yaw boost servo, or pitch/trim assembly. Remove link (Figure 11-5-6-1, Sheet 1, Sheet 2, or Sheet 3).

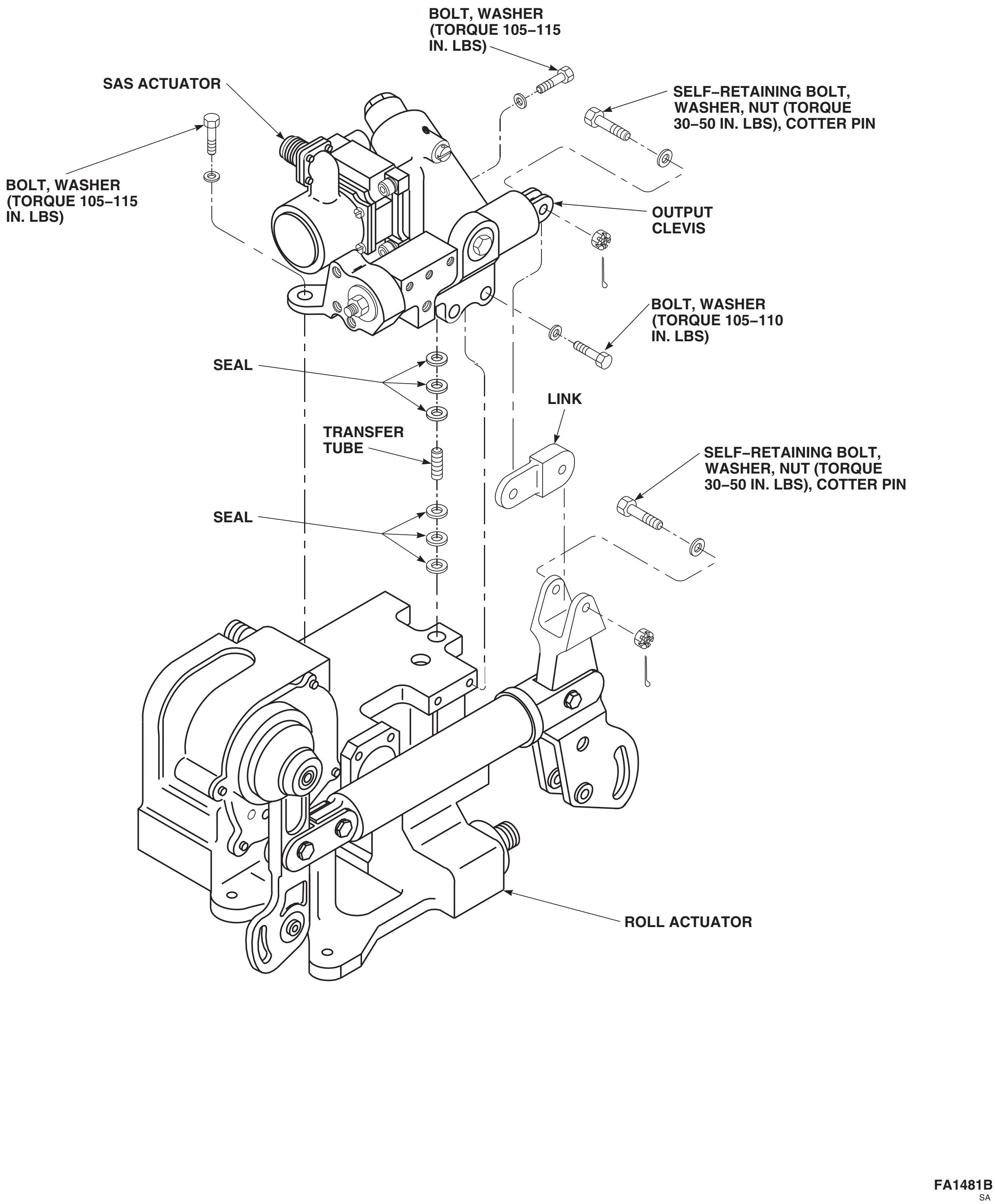


FIGURE 11-5-6-1. REPLACE ROLL ACTUATOR SAS ACTUATOR (BENCH) (SHEET 1 OF 3)

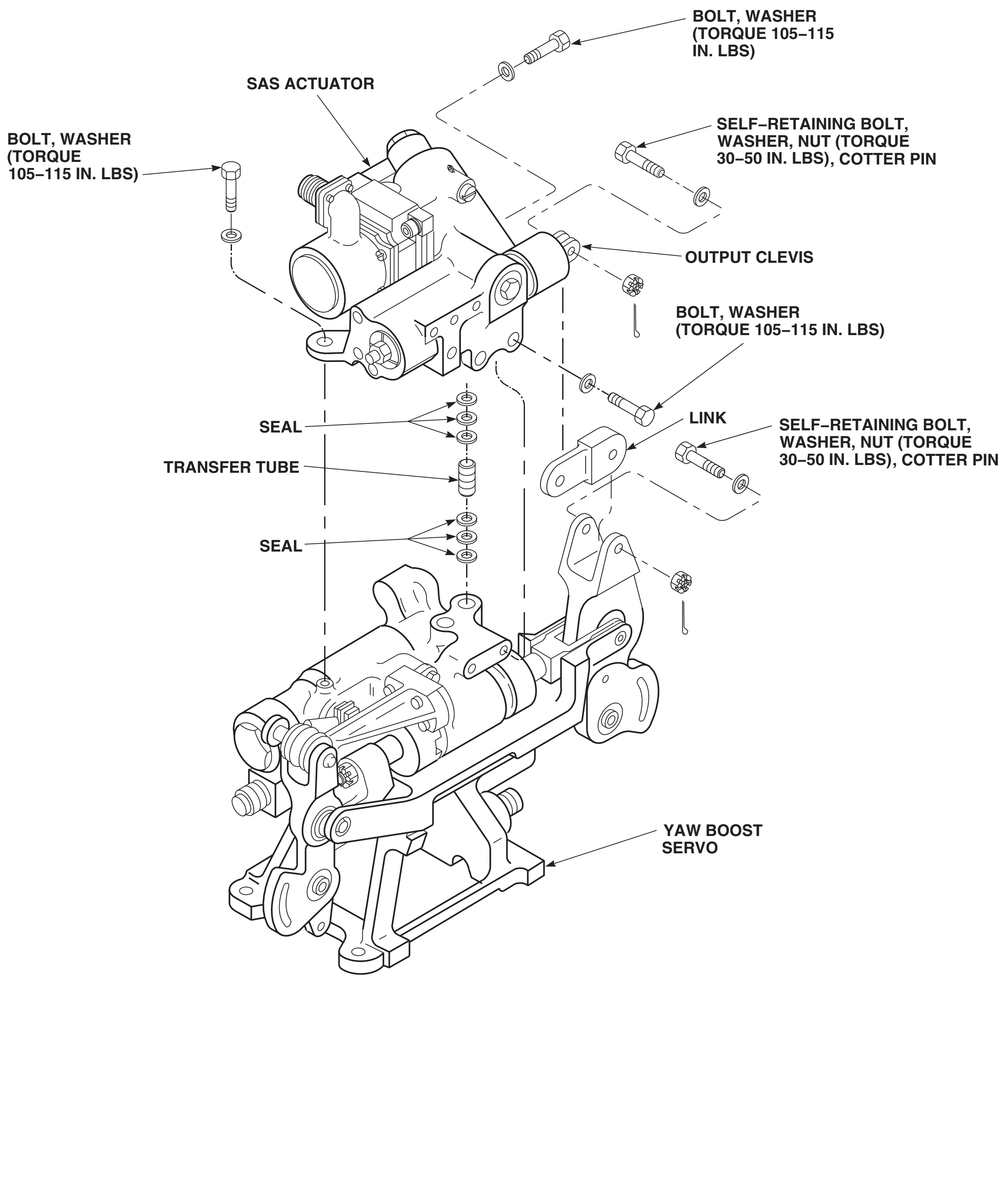


FIGURE 11-5-6-1. REPLACE YAW BOOST SERVO SAS ACTUATOR (BENCH) (SHEET 2 OF 3)

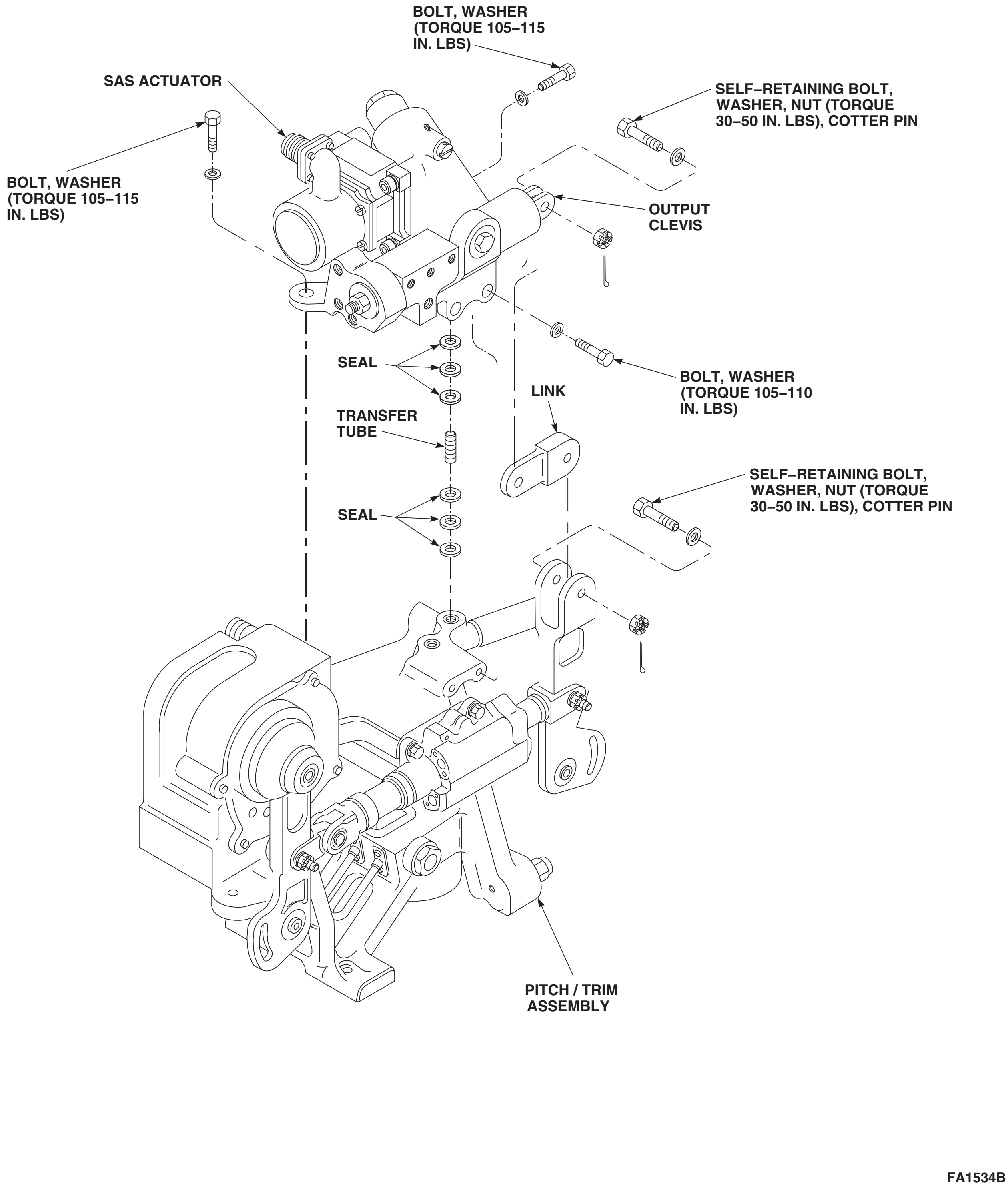


FIGURE 11-5-6-1. REPLACE PITCH/TRIM ASSEMBLY SAS ACTUATOR (BENCH) (SHEET 3 OF 3)

11-5-6.1.2 Install.

a. Coat seals with hydraulic fluid, Item 154 or Item 155, Appendix D, and install on transfer tubes.

b. Install transfer tubes in roll actuator, yaw boost servo assembly, or pitch/trim assembly (Figure 11-5-6-1, Sheet 1, Sheet 2, or Sheet 3).

c. Carefully install replacement SAS actuator on roll actuator, yaw boost servo assembly, or pitch/trim assembly, making sure transfer tubes line up with holes in actuator. Install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, bolts.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Damage to equipment may result if incorrect connecting link is installed. Incorrect link is prone to stress corrosion cracking. Make sure only link marked T73, is installed.

d. Install link, 70410-02334-045, marked T73, on roll actuator, yaw boost servo, or pitch/trim assembly.

e. Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

f. Install nut on self-retaining bolt. **TORQUE BOLTS TO 30 - 50 INCH-POUNDS.** Install cotter pin.

g. Install output clevis to link.

h. Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1).

i. Install nut on self-retaining bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

j. **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripes (PARA 2-6-5).

k. Perform operational check (PARA 11-2-4, roll actuator, PARA 11-2-5, yaw boost servo, or PARA 11-2-6, pitch/trim assembly).

11-5-7 ROLL ACTUATOR LINKS AND LEVERS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Dial Indicator
- Hydraulic Repairer's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

References

- PARA 2-6-5
- PARA 11-2-4
- PARA 11-4-1

11-5-7.1 Replace Roll Actuator Links and Levers (BENCH).

11-5-7.1.1. Remove.

a. Remove self-retaining bolt, washer, and nut securing rod end to output lever. Remove bushing and rod end (Figure 11-5-7-1 and Detail A).

b. Remove self-retaining bolts, washers, and nuts securing output lever to pushrod and scissor link (Figure 11-5-7-1). Remove output lever.

c. Remove self-retaining bolt, washer, and nut securing scissor link to vertical link. Remove bushing and scissor link.

d. Remove self-retaining bolt, washer, and nut securing pin to vertical link. Remove pin and vertical link.

e. Remove self-retaining bolts, washers, and nuts securing input lever to pushrod and support. Remove input lever and pushrod.

f. Remove bolt and washer securing support to housing. Remove support.

11-5-7.1.2. Inspect.

a. Inspect bearing in rod end using dial indicator. **RADIAL PLAY MUST BE LESS THAN 0.005-INCH.** If radial play exceeds limit, replace rod end. Remove rod end by removing self-retaining bolt, washer, and nut securing rod end to roll actuator (Figure 11-5-7-1).

b. Inspect bearing in output lever using dial indicator. **RADIAL PLAY MUST BE LESS THAN 0.005-INCH.** If radial play exceeds limit, replace output lever (Figure 11-5-7-1, Detail A).

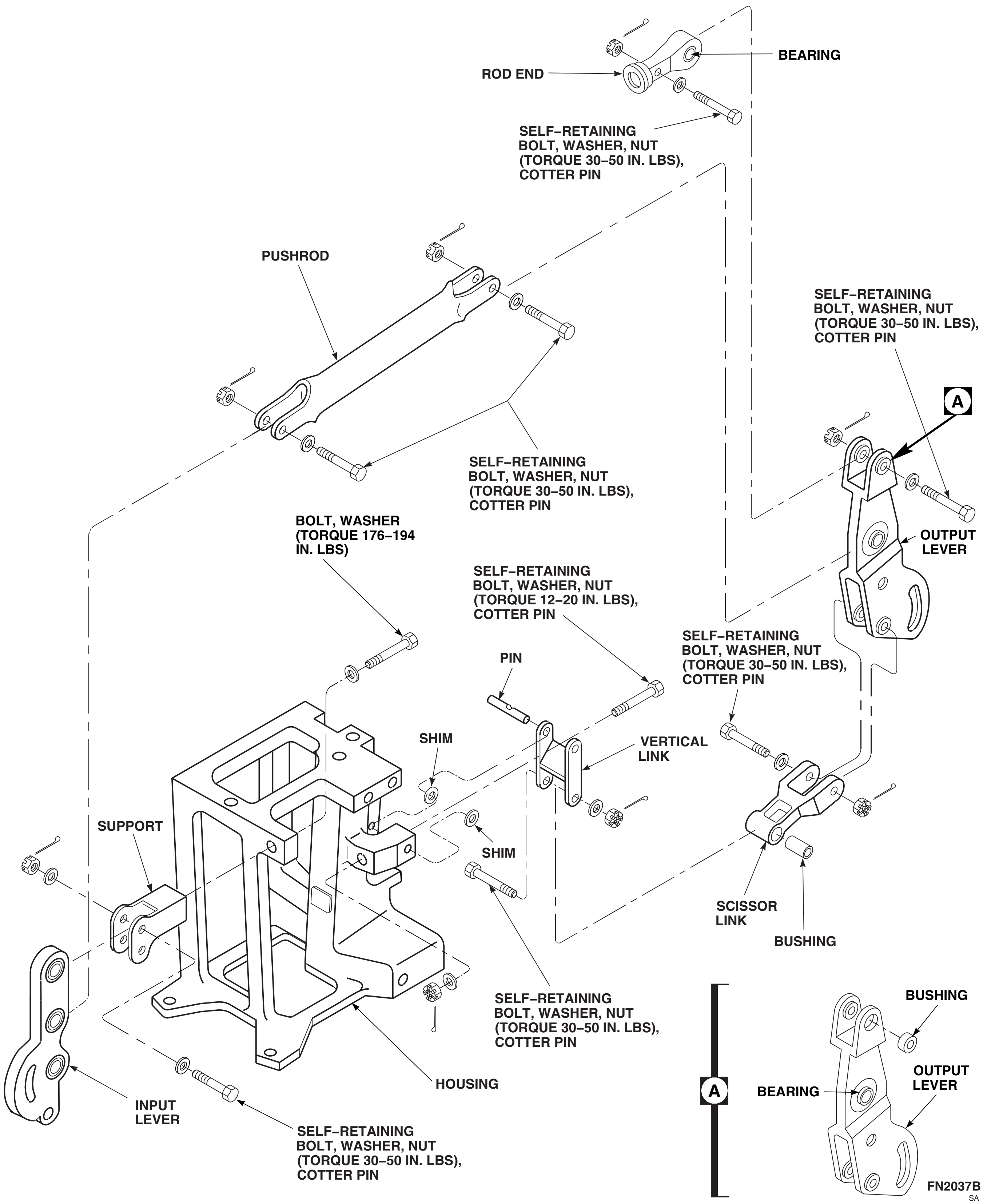


FIGURE 11-5-7-1. REPLACE ROLL ACTUATOR LINKS AND LEVERS (BENCH)

11-5-7.1.3. Install.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

a. Install rod end between actuator output clevis (if actuator is installed) and output lever (Figure 11-5-7-1).

b. Install bushing in output lever (Figure 11-5-7-1, Detail A).

c. Install self-retaining bolt and washer (under bolthead) (PARA 11-4-1) (Figure 11-5-7-1).

d. Install nuts on self-retaining bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

e. **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

f. Install replacement scissors link and pushrod on output lever.

g. Install self-retaining bolts and washers (under head) (PARA 11-4-1).

h. Install nuts on self-retaining bolts. **TORQUE NUTS TO 30 - 50 INCH-POUNDS.** Install cotter pins.

i. **PROOF TORQUE BOLTHEADS TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

j. Install lower end of vertical link on scissor link. Line up holes and install bushing.

k. Install self-retaining bolt and washer (under head) (PARA 11-4-1).

l. Install nut on self-retaining bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

m. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

NOTE

Install shims between vertical link and housing as required to maintain 0.002 - 0.016-inch end play.

n. Install vertical link on actuator housing. Line up holes and install shims, pin, self-retaining bolt, washer, and nut (washer under nut). **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Install cotter pin.

o. Install support on actuator housing with bolt and washer (under head). **TORQUE BOLT TO 176 - 194 INCH-POUNDS.**

p. Install input lever on support. Line upper holes of input lever with upper holes of support.

q. Install self-retaining bolt and washers (under head) (PARA 11-4-1).

r. Install nut on self-retaining bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

s. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

t. Install pushrod on center hole of input lever. Line up holes.

u. Install self-retaining bolt and washer (under head) (PARA 11-4-1).

v. Install nut on self-retaining bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

w. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

x. Perform operational check of roll actuator (PARA 11-2-4).

11-5-8 PILOT-ASSIST SERVO MODULE THREE-WAY VALVES (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D
- PARA 11-2-7

11-5-8.1 Replace Pilot-Assist Servo Module Three-Way Valves (BENCH).

CAUTION

Damage to pilot-assist servo module will result if dirt is allowed to enter components during disassembly. Make sure pilot-assist servo module is clean before disassembly. Replace three-way valves only in a clean place. Have replacement valve ready for installation before old valve is removed.

11-5-8.1.1. Remove.

a. Remove screws and three-way valve (boost shutoff, trim shutoff, or SAS shutoff, as necessary) from pilot-assist servo module (Figure 11-5-8-1).

b. Remove packings from trim shutoff valve, or packings and seals from SAS or boost shutoff valves. Discard packings and seals.

11-5-8.1.2. Install.

a. Coat packings and seals with hydraulic fluid, Item 154 or Item 155, Appendix D, before installing.

b. Install replacement valve as follows (Figure 11-5-8-1):

(1) If trim shutoff valve is being replaced, do this:

(a) Install new packings on valve.

(b) Install valve in module with screws. **TORQUE SCREWS TO 48 - 52 INCH-POUNDS.** Lockwire, Item 196, Appendix D, screws.

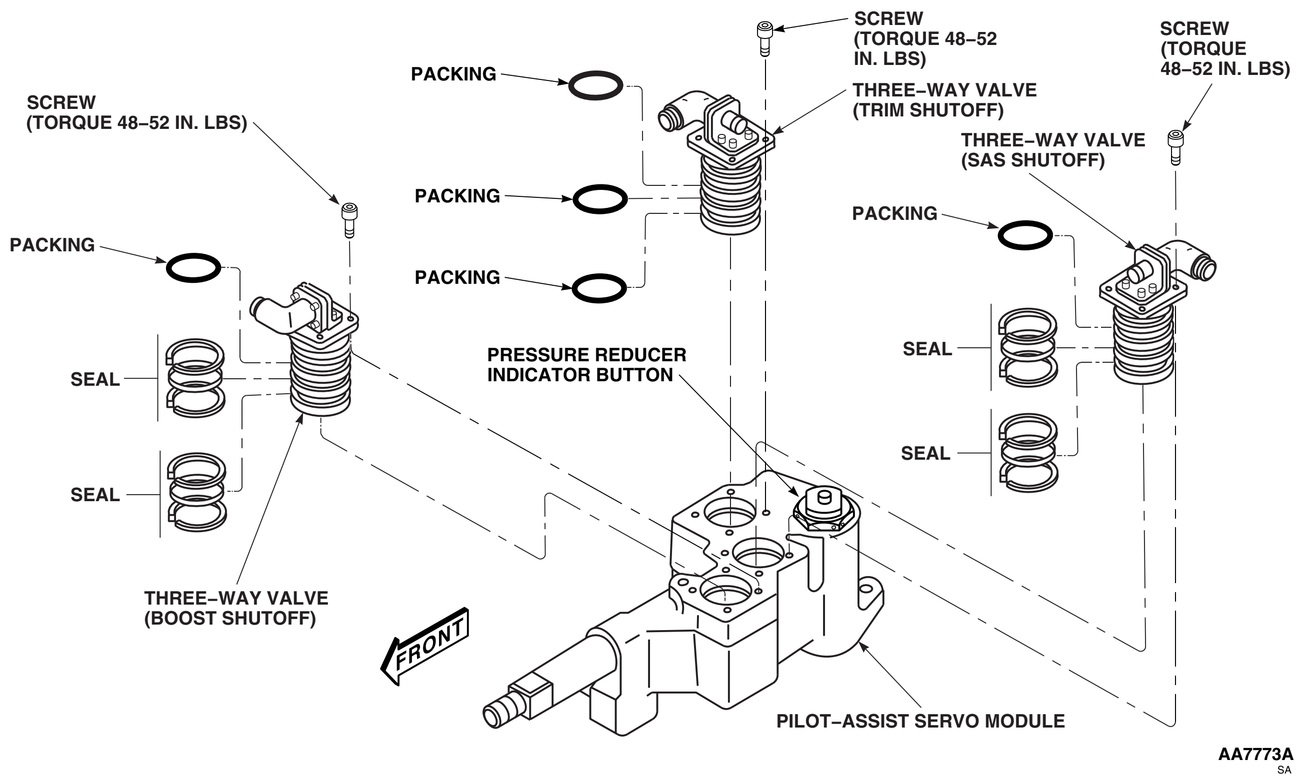


FIGURE 11-5-8-1. REPLACE PILOT-ASSIST SERVO MODULE THREE-WAY VALVES (BENCH)

(2) If boost or SAS shutoff valves are being replaced, do this:

(a) Install new packings and new seals on valve.

(b) Install valve in module with screws. **TORQUE SCREWS TO 48 - 52 INCH-POUNDS.** Lockwire, Item 196, Appendix D, screws.

c. Perform operational check of pilot-assist servo module (PARA 11-2-7).

11-5-9 PILOT-ASSIST SERVO MODULE PRESSURE REDUCER (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Hydraulic Repairer's Toolkit
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Hydraulic Fluid, Item 154, Appendix D
- Hydraulic Fluid, Item 155, Appendix D
- Lockwire, Item 196, Appendix D

References

- Appendix D

11-5-9.1 Replace Pilot-Assist Servo Module Pressure Reducer (BENCH).

CAUTION

Damage to pilot-assist servo module will result if dirt is allowed to enter components during reducer replacement. Make sure pilot-assist servo module is clean before replacing reducer. Replace pressure reducer only in a clean place. Have replacement reducer ready for installation before old reducer is removed.

11-5-9.1.1. Remove.

Unscrew pressure reducer from pilot-assist servo module. Discard packings and seal (Figure 11-5-9-1).

11-5-9.1.2. Install.

a. Coat packings, seal and threads of replacement pressure reducer with hydraulic fluid, Item 154 or Item 155, Appendix D, before installing.

b. Install new packings and new seal on replacement pressure reducer (Figure 11-5-9-1).

c. Screw pressure reducer into pilot-assist module. **TORQUE TO 238 - 262 INCH-POUNDS.** Lockwire, Item 196, Appendix D.

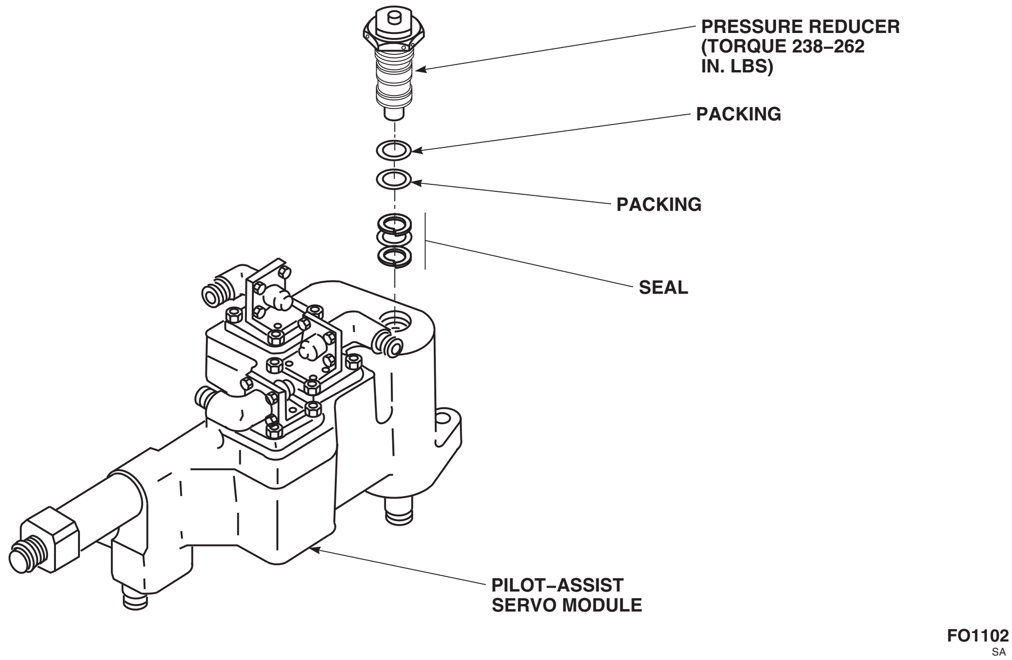


FIGURE 11-5-9-1. REPLACE PILOT-ASSIST SERVO MODULE PRESSURE REDUCER (BENCH)

11-5-10 MIXER SMALL LINK ON BELLCRANK/LEVER (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Dial Indicator
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Protection, Eye
- Vernier Caliper

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Acetone, Item 25, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Paint Remover, Item 230, Appendix D
- Polyurethane Coating, Item 240, Appendix D

References

- Appendix D
- ASTM E1417

11-5-10.1 Replace Mixer Small Link on Bellcrank/Lever (BENCH).

11-5-10.1.1 Remove.

a. Remove pin, washers, and collar to separate small link from bellcrank or lever (Figure 11-5-10-1, and Detail A).

b. Inspect bearings in small link. If bearings are beyond limits, replace small link.

11-5-10.1.2 Inspect.

a. Inspect bellcrank or lever bearings with dial indicator (Figure 11-5-10-1). **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.**

b. Inspect lever bushings, NAS538-3P9 and SS4045-3P18 (superseded by, NAS538-3P009 and NAS537-3P018), using vernier caliper. **INSIDE RADIAL DIAMETER OF BUSHINGS SHALL NOT BE OVER 0.1900-INCH.**

c. Inspect bellcrank bushings, NAS538-3P9 and SS4045-3P19 (superseded by, NAS538-3P009 and NAS537-3P019), using vernier caliper. **INSIDE RADIAL DIAMETER OF BUSHINGS SHALL NOT BE OVER 0.1900-INCH.**

d. Inspect bellcrank an lever bushings, NAS537-4P86 and NAS538-4P11 (superseded by, NAS537-4P086 and NAS538-4P011), using vernier caliper. **INSIDE RADIAL DIAME-**

TER OF BUSHINGS SHALL NOT BE OVER 0.2505-INCH.

e. Check all bearings for roughness, sticking, and seal damage. **NONE ALLOWED.** If roughness, sticking, or damage is found, replace bellcrank or lever.

f. Inspect lever for loose bushings, corrosion, and damage. **CORROSION SHALL NOT EXCEED 0.010-INCH.** If corrosion exceeds limit, replace lever.

g. Inspect lever for scratches, nicks, and dents. **SCRATCHES AND NICKS SHALL NOT EXCEED 0.030-INCH.** If scratches and nicks exceed limit, replace lever.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

h. Clean parts, except bearings, with acetone, Item 25, Appendix D. Blow dry with dry compressed air.

i. Using paint remover, Item 230, Appendix D, remove paint from lever.

j. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect lever for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace lever.

11-5-10.1.3 Repair.

a. Using abrasive cloth, Item 8, Appendix D, blend out any scratches, nicks, or corrosion within damage limits.

b. Mask bearings and bushings with 1-inch masking tape, Item 212, Appendix D.

c. Using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer to lever.

d. Apply two coats of polyurethane coating, Item 240, Appendix D, to lever.

11-5-10.1.4 Install.

NOTE

Make sure collar is on opposite side of plain bushing.

Install small link to bellcrank or lever and secure with pin, washers, and collar (Figure 11-5-10-1, and Detail A).

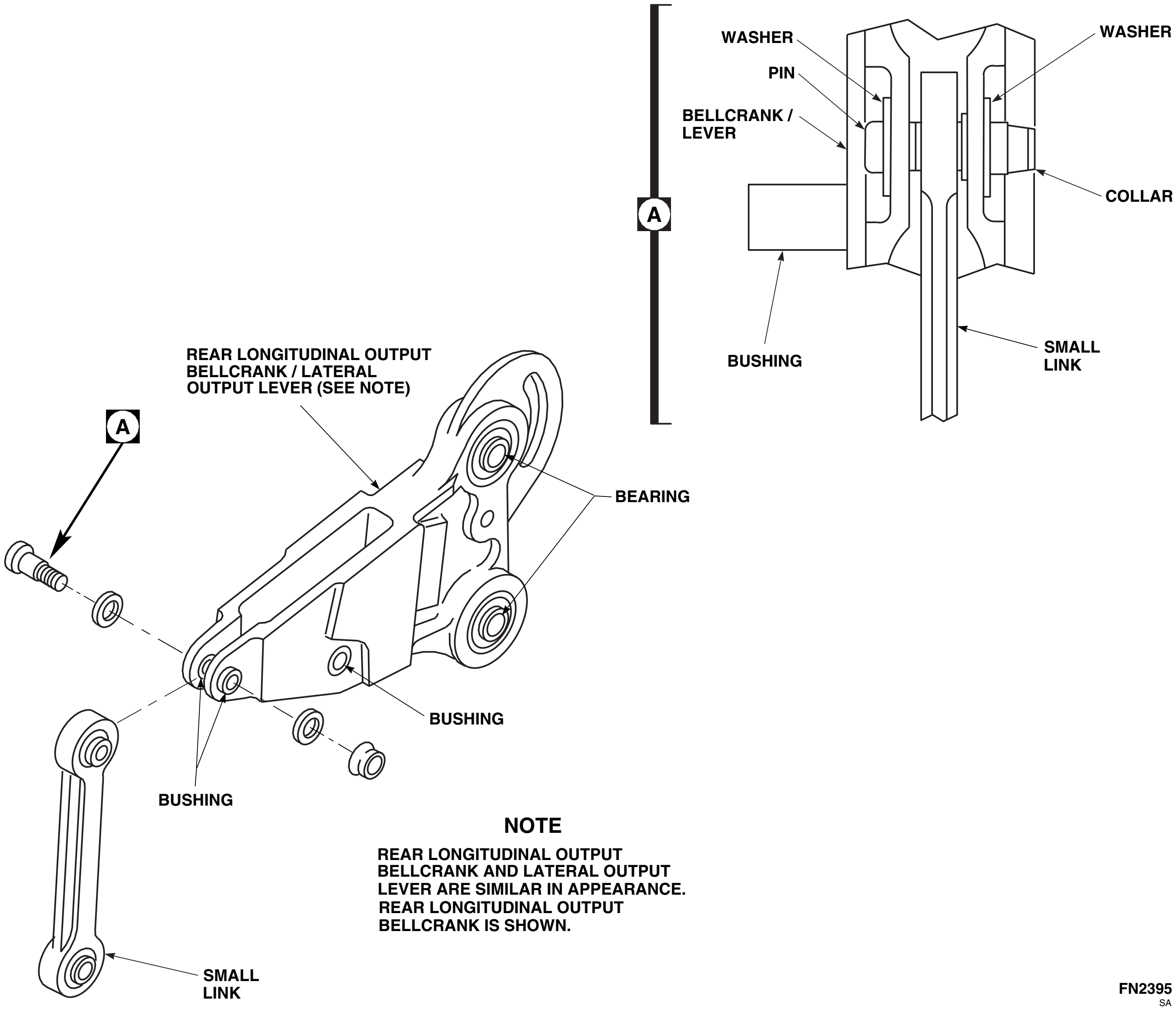


FIGURE 11-5-10-1. REPLACE MIXER SMALL LINK ON BELLCRANK/LEVER (BENCH)

11-5-11 CONTROL RODS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
11-5-11.1	Repair Control Rods (BENCH)	11-5-11-2

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Engine Borescope Kit
- Fluorescent Penetrant Inspection Kit
- Protection, Eye
- Small Arms Cleaning Kit

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Acetone, Item 25, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 108A, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Removing Compound, Item 116, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 11-5-12
- PARA 11-5-13
- PARA 11-5-14
- PARA 11-5-15
- PARA 11-5-30

11-5-11.1 Repair Control Rods (BENCH).

11-5-11.1.1 Disassemble.

Disassemble control rods (PARA 11-5-13, PARA 11-5-14, PARA 11-5-15, or PARA 11-5-30).

11-5-11.1.2 Inspect.

a. Visually inspect control rods. Inspect interior areas of adjustable control rods by removing adjustable ends and using engine borescope kit.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in a safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

b. Clean parts, except bearings, with acetone, Item 25, Appendix D. Blow dry with dry compressed air.

c. If evidence of cracks or corrosion is found, using fluorescent penetrant inspection kit, fluorescent penetrant inspect control rod for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace control rod.

d. Inspect for loose bushings, and rivets, damage to threads, worn or rounded wrench flats, corrosion, and damage to tube. Inspect and replace, if necessary, as follows:

(1) If internal corrosion cannot be cleaned with small arms cleaning kit, or if it extends in a band more than 30% around inside diameter, or if there are external pits greater than 0.010-inch deep, replace aluminum control rods.

(2) If there are corrosion pits greater than can be readily blended out with emery cloth, replace steel control rods. If there is any evidence of looseness between either insert and tube, or if there is any evidence of rust bleeding out from between insert(s) and tube, replace steel control rods. If corrosion pits are greater than 0.010-inch deep, replace rod ends.

(3) Replace control rods and fittings which have bushings, SS4045 and/or 70400-08107-110, installed in ends if pits on bushings are deeper than 0.030-inch, or if pitting is greater than 50% of reamed bore area.

e. Inspect for scratches, nicks, and dents no deeper than 0.020-inch, except on tube. Allow 0.010-inch on tube.

NOTE

Gouged or badly worn solid film lubricant coatings may be caused by too much pivot bearing wear. Check bearings as necessary (PARA 11-5-12).

f. Check control rods, 70400-02264-063 (roll trim assembly to mixer), 70400-02264-064 (collective servo to mixer), and 70400-02258-048 (mixer to forward primary servo), for condition of solid film lubricant coating on fixed end outside diameter. If coating is flaked or scratched through to base metal, repair (PARA 11-5-11.1.3), or replace control rod.

11-5-11.1.3 Repair.

NOTE

Repair of control rods is limited, except as noted, to replacing parts as necessary after inspection.

a. Perform internal repair of control rods as follows:

(1) Remove corrosion from internal surfaces using small arms cleaning kit.

(2) Flush control rod with acetone, Item 25, Appendix D. Rinse thoroughly with water and allow to dry.

(3) Apply corrosion removing compound, Item 116, Appendix D, to reworked areas of control rod. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow to dry completely.

(4) Apply corrosion preventive compound, Item 114, Appendix D, to all reworked areas of steel control rods. Remove excess corrosion preventive compound with machinery towel, Item 211, Appendix D. Allow to dry completely.

(5) Mask threads of control rod with 1-inch masking tape, Item 212, Appendix D.

(6) Using epoxy primer coating kit, Item 135, Appendix D, fill control rod with epoxy primer. Empty excess primer and allow to dry completely.

b. Perform external repair of control rods as follows:

(1) Damage to external surfaces of control rod may be blended out up to 0.010-inch deep using abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D.

(2) Clean repaired areas of control rod with acetone, Item 25, Appendix D. Rinse thoroughly with water and allow to dry.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of aluminum control rods.

(4) Using epoxy primer coating kit, Item 135, Appendix D, apply a thin coat of epoxy primer to reworked areas of control rod. Allow primer to dry.

(5) Touch up finish all reworked areas (PARA 2-6-5).

c. Perform thread repair of control rod as follows:

NOTE

Repair of control rod threads is limited to removal of minor surface corrosion only.

(1) Remove minor corrosion from threads with abrasive mat, Item 14, Appendix D.

(2) Clean threads with acetone, Item 25, Appendix D. Rinse thoroughly with water, and allow to dry.

(3) Apply corrosion removing compound, Item 116, Appendix D, to threads. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow to dry completely.

(4) Apply corrosion preventive compound, Item 108A, Appendix D, to threads. Remove excess corrosion preventive compound with machinery towel, Item 211, Appendix D.

d. Perform external repair of fixed and adjustable rod ends as follows:

NOTE

Minimum lug and clevis thickness shall not exceed limits as indicated in Figure 11-5-11-1, Detail A and Detail B.

(1) Damage to external surfaces of rod ends may be blended out up to 0.030-inch deep with abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D.

(2) Clean repaired areas of rod end with acetone, Item 25, Appendix D. Rinse rod end thoroughly with water and allow to dry.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of rod end.

(4) Using epoxy primer coating kit, Item 135, Appendix D, apply a thin coat of epoxy primer to reworked areas of rod end. Allow primer to dry.

(5) Touch up finish all reworked areas (PARA 2-6-5).

e. Perform thread repair of fixed rod ends, adjustable rod ends, and fixed adapters as follows:

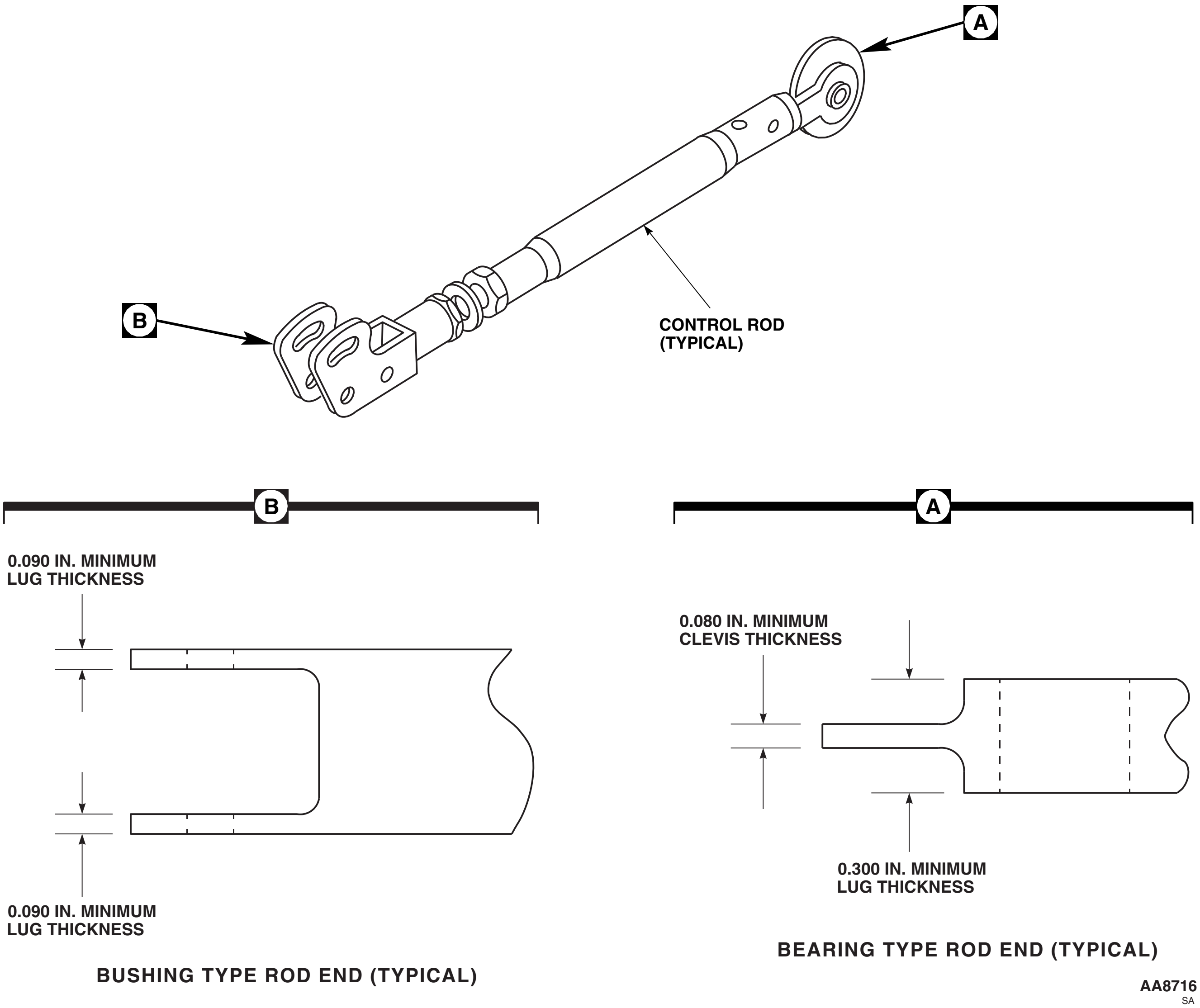


FIGURE 11-5-11-1. REPAIR CONTROL RODS (BENCH)

NOTE

Repair of threads is limited to removal of minor surface corrosion only.

(1) Remove minor corrosion from threads with abrasive mat, Item 14, Appendix D.

(2) Clean threads with acetone, Item 25, Appendix D. Rinse thoroughly with water, and allow to dry.

(3) Apply corrosion removing compound, Item 116, Appendix D, to threads. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow to dry completely.

(4) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of threads.

f. Perform internal repair of fixed adapters (riveted and threaded) as follows:

(1) Remove corrosion from internal surfaces using small arms cleaning kit.

(2) Flush adapter with acetone, Item 25, Appendix D. Rinse thoroughly with water and allow to dry.

(3) Apply corrosion removing compound, Item 116, Appendix D, to reworked areas of adapter. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow to dry completely.

(4) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of adapter.

(5) Using epoxy primer coating kit, Item 135, Appendix D, apply a thin coat of epoxy primer to reworked areas of adapter. Allow primer to dry.

(6) Touch up finish all reworked areas (PARA 2-6-5).

g. Perform external repair of fixed adapters (riveted and threaded) as follows:

(1) Damage to external surfaces of adapter may be blended out up to 0.030-inch deep with abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D.

(2) Clean repaired areas of adapter with acetone, Item 25, Appendix D. Rinse rod end thoroughly with water and allow to dry.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas.

(4) Using epoxy primer coating kit, Item 135, Appendix D, apply a thin coat of epoxy primer to reworked areas of adapter. Allow primer to dry.

(5) Touch up finish all reworked areas (PARA 2-6-5).

h. Perform internal repair of adjustable adapters as follows:

(1) Remove corrosion from internal surfaces using small arms cleaning kit.

(2) Flush adjustable adapters with acetone, Item 25, Appendix D. Rinse thoroughly with water and allow to dry.

(3) Apply corrosion removing compound, Item 116, Appendix D, to reworked areas of control rod. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow to dry completely.

(4) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of aluminum adapters.

(5) Using epoxy primer coating kit, Item 135, Appendix D, apply a thin coat of epoxy primer to reworked areas of adapter. Allow primer to dry.

(6) Touch up finish all reworked areas (PARA 2-6-5).

i. Perform external repair of adjustable adapters as follows:

(1) Damage to external surfaces of adjustable adapters may be blended out up to 0.030-inch deep using abrasive cloth, Item 8, Appendix D, or abrasive mat, Item 14, Appendix D.

(2) Clean repaired areas of adapter with acetone, Item 25, Appendix D. Rinse thoroughly with water and allow to dry.

(3) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of aluminum adapter.

(4) Using epoxy primer coating kit, Item 135, Appendix D, apply a thin coat of epoxy primer to reworked areas of adapter. Allow primer to dry.

(5) Touch up finish all reworked areas (PARA 2-6-5).

j. Perform thread repair of adjustable adapters as follows:

NOTE

Repair of threads is limited to removal of minor surface corrosion only.

(1) Remove minor corrosion from threads with abrasive mat, Item 14, Appendix D.

(2) Clean threads with acetone, Item 25, Appendix D. Rinse thoroughly with water, and allow to dry.

(3) Apply corrosion removing compound, Item 116, Appendix D, to threads. Allow to dry for 30 seconds then rinse area thoroughly with water. Allow to dry completely.

(4) Apply corrosion resistant coating, Item 118, Appendix D, to all reworked areas of aluminum adapter threads.

(5) Apply corrosion preventive compound, Item 108A, Appendix D, to all reworked areas of steel adapter threads. Remove excess corrosion preventive compound with machinery towel, Item 211, Appendix D.

k. Recondition solid film lubricant-coated surfaces on control rods, 70400-02264-063, 70400-02264-064, and 70400-02258-048, as follows:

CAUTION

Damage to bearings will result if abrasive dust is allowed to enter bearings. Do not allow abrasive dust to come in contact with bearings.

(1) Lightly roughen remaining film coating and/or bare metal with abrasive cloth, Item 8, Appendix D.

(2) Remove sanding dust and grime with clean cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. Turn cloth to a clean patch after each wipe until cloth remains clean after wiping part. Do not touch treated surfaces. Change cloth as often as needed until you get a clean wipe.

(3) Thoroughly mix solid film lubricant, Item 201, Appendix D, per manufacturer's instructions.

(4) Apply a thin even coat to rod end. Allow part to cure at room temperature for at least 12 hours.

NOTE

A serviceable solid film lubricant application will have a smooth flat black appearance. A high-gloss is caused by not enough mixing before application. A dry, easily-flaked condition is caused by a poor bond.

(5) Test solid film lubricant bond with base metal by sticking a 1-inch length of masking tape, Item 212, Appendix D, to cured film. Pull up masking tape. **COATING SHALL NOT SEPERATE FROM BASE METAL. A LIGHT POWDERY DUST LIFTED BY TAPE IS ALLOWED.** If coating separated from base metal, strip off unsatisfactory coatings and rework part as required.

11-5-11.1.4 Assemble.

Assemble control rod (PARA 11-5-13, PARA 11-5-14, PARA 11-5-15, or PARA 11-5-30).

11-5-12 CONTROL ROD PIVOT BEARING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Dial Indicator
- Mandrel
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Acetone, Item 25, Appendix D
- Applicator, Item 78, Appendix D
- Cheesecloth, Item 90, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Sealing Compound, Item 282, Appendix D
- Sealing Compound, Item 285, Appendix D

References

- Appendix D

11-5-12.1 Replace Control Rod Pivot Bearing (BENCH).

11-5-12.1.1. Inspect.

Check control rod pivot bearing for seal damage, lubricant loss and roughness. Check bearing wear at inner race with dial indicator. **IF DAMAGED OR WORN BEYOND LIMITS, REPLACE BEARING** (Table 11-5-12-1).

Table 11-5-12-1. Bearing Wear Limits

Bearing Part No.	Radial Wear Max. Inches *
SA4-14A1-504	0.0050

* Axial Wear Limits Not Required

11-5-12.1.2. Remove.

a. Remove bearings on control rods, 70400-02264-064, 70400-02264-063, and 70400-02258-048, as follows:

(1) Support rod end fitting. Press out old bearing with arbor press and suitable mandrel.

(2) Remove all old sealing compound from rod end bore with abrasive cloth, Item 8, Appendix D.

11-5-12.1.3. Install.

a. Install bearings on control rods, 70400-02264-064, 70400-02264-063, and 70400-02258-048, as follows:

(1) Remove all sanding dust and oil from replacement bearing outside diameter and rod end with cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. Turn or replace cloth after each wipe. Do not touch treated surface. Turn or replace cloth as needed until cloth remains clean after last wipe.

NOTE

Dipping a brush, swab or other tool into primer or sealing compound containers will contaminate all material in container. To avoid contamination, pour material onto brush or swab as needed.

(2) Apply a thin coat of sealing compound, Item 282, Appendix D, to rod end bore only with a clean applicator, Item 78, Appendix D. A break, or bird's-eye in wet primer indicates a contaminated base surface. Reclean and reprime contaminated work.

(3) Allow primer coat to dry undisturbed for 15 - 30 minutes. Do not touch primed surface.

(4) Use a clean applicator, Item 78, Appendix D, to cover bearing outside diameter and rod end bonding surfaces with sealing compound, Item 285, Appendix D. Assemble bearing into rod end within 4 minutes of sealing compound application. **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND APPLICATION AGAIN.**

(5) Dry bearing installation a minimum of 24 hours at room temperature. Sealing compound squeeze out will not dry. Wipe away compound squeeze out with cleaning cloth, Item 102, Appendix D, dampened with acetone, Item 25, Appendix D.

b. Perform bearing rotational drag check as follows:

(1) Install bolt and washers through rod end bearing. Install nut on bolt.

(2) Turn bearing inner race with a torque wrench and socket. Measure torque needed to get inner race to turn. If drag is greater than allowed by Table 11-5-12-2, replace bearing.

Table 11-5-12-2. Bearing Rotational Drag

Control Rod	Bearing	Bearing-Rod End Location	Maximum Rotational Drag (Inch-Ounces)
70400-02258-048	38950-00903-103 (Ball)	Fixed	2
70400-02264-063	38950-00903-103 (Ball)	Fixed	2
70400-02264-064	SA4-14A1-504 (Roller)	Adjustable	8
	38950-00903-103 (Ball)	Fixed	2

11-5-13 TORQUE SHAFT TO SERVO CONTROL RODS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 108A, Appendix D
- Corrosion Preventive Compound, Item 108B, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 11-5-11
- PARA 11-5-12

11-5-13.1 Repair Torque Shaft to Servo Control Rods (BENCH).

11-5-13.1.1. Disassemble.

a. Loosen rod end jamnut on adjustable rod end and back it away so locking device (-2) can be slid away from locking device (-1) (Figure 11-5-13-1).

b. Unscrew rod end from sleeve. Remove locking devices, and jamnut from adjustable rod end.

c. Loosen sleeve jamnut. Remove sleeve from tube assembly. Remove jamnut from sleeve.

11-5-13.1.2. Inspect/Repair.

Inspect and repair control rods (PARA 11-5-11 and PARA 11-5-12).

11-5-13.1.3. Assemble.

a. Coat all threaded, mating, and interior surfaces of control rod assembly with corrosion preventive compound, Item 108A, Appendix D.

b. Install sleeve jamnut on sleeve and thread sleeve into tube assembly.

c. Set 0.50 to 0.45-inch dimension from face of sleeve jamnut to face of sleeve (Figure 11-5-13-1, Detail A).

NOTE

Locking devices' serrated sides face each other.

d. Install jamnut, locking device (-2) and locking device (-1) on adjustable rod end. Thread rod end into sleeve (Figure 11-5-13-1).

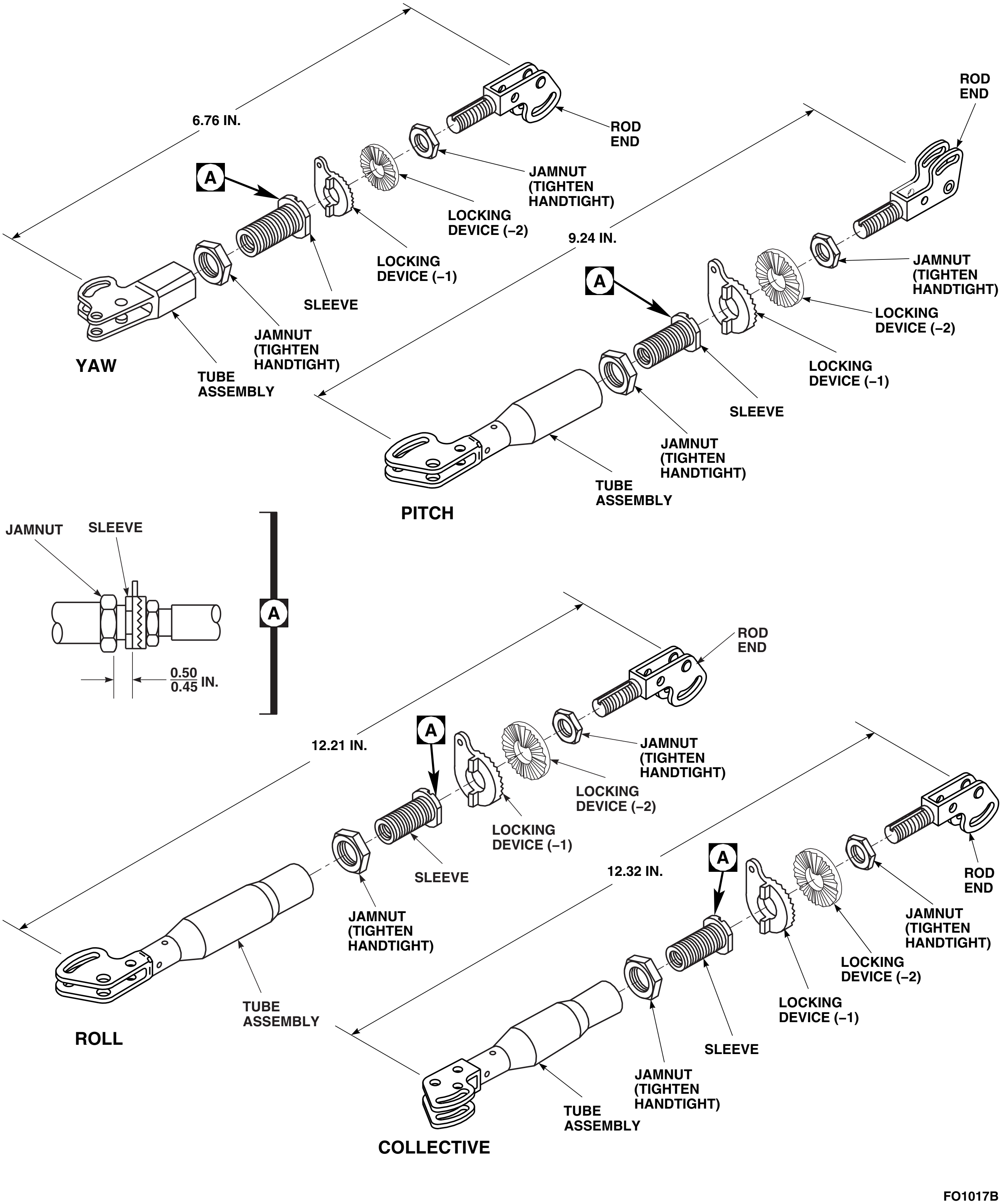


FIGURE 11-5-13-1. REPAIR TORQUE SHAFT TO SERVO CONTROL RODS (BENCH)

e. Set control rod length.

f. Set locking device (-1) key in rod end key slot.

g. **TIGHTEN JAMNUTS HANDTIGHT.**

h. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint as required, with epoxy primer.

i. After assembly and final adjustment of control rod, seal all mating and threaded surfaces, witness holes, slotted keyways, serrated locking devices, and exposed threads with corrosion preventive compound, Item 108B, Appendix D.

11-5-14 BOOST SERVO TO MIXER CONTROL RODS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 108A, Appendix D
- Corrosion Preventive Compound, Item 108B, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 11-5-11
- PARA 11-5-12

11-5-14.1 Repair Boost Servo to Mixer Control Rods (BENCH).

11-5-14.1.1. Disassemble.

a. Loosen rod end jamnut on adjustable rod end and back it away so locking device (-2) can be slid away from locking device (-1) (Figure 11-5-14-1).

b. On roll and yaw control rods, perform the following:

(1) Unscrew rod end from sleeve.

(2) Remove locking devices and jamnut from adjustable rod end.

(3) Loosen sleeve jamnut.

(4) Remove sleeve from tube assembly.

(5) Remove jamnut from sleeve.

c. On collective control rod, perform the following:

(1) Loosen jamnuts on adjustor.

(2) Unscrew rod end from adjustor.

(3) Remove locking devices and jamnut from adjustor.

(4) Unscrew adjustor from tube assembly.

(5) Remove jamnut from adjustor.

11-5-14.1.2. Inspect/Repair.

Inspect control rod (PARA 11-5-11 and PARA 11-5-12).

11-5-14.1.3. Assemble.

a. Coat all threaded, mating, and interior surfaces of control rod assembly with corrosion preventive compound, Item 108A, Appendix D.

b. Assemble collective control rod as follows (Figure 11-5-14-1):

NOTE

Adjustor has 5/8 - 11 thread on one end and 5/8 - 18 thread on the other end.

(1) On collective control rod, install jamnuts on adjustor.

(2) Install 5/8 - 18 thread end of adjustor in tube assembly.

NOTE

Locking devices' serrated sides face each other.

(3) Install locking device (-2) and locking device (-1) on adjustor (Figure 11-5-14-1).

(4) Thread rod end on adjustor and set length of control rod.

(5) Keep adjustor threaded about same distance into tube assembly and rod end.

(6) Keep rod ends parallel with each other.

(7) Set locking device (-1) key in rod end key slot.

NOTE

Control rod length will be adjusted when installed in helicopter.

(8) **HANDTIGHTEN JAMNUTS.**

c. Assemble roll and yaw control rods as follows:

(1) Install sleeve jamnut on sleeve and thread sleeve into tube assembly.

(2) Set 0.50 to 0.45-inch dimension from face of sleeve jamnut to face of sleeve (Figure 11-5-14-1, Detail A).

NOTE

Locking devices' serrated sides face each other.

(3) Install jamnut, locking device (-2) and locking device (-1) on adjustable rod end with serrations facing each other. Thread rod end into sleeve (Figure 11-5-14-1).

(4) Set control rod length.

(5) Set locking device (-1) key in rod end key slot.

(6) Keep rod ends parallel with each other.

NOTE

Control rod length will be adjusted when installed in helicopter.

(7) **TIGHTEN JAMNUTS HANDTIGHT.**

d. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint as required, with epoxy primer.

e. After assembly and final adjustment of control rod, seal all mating and threaded surfaces, witness holes, slotted keyways, serrated locking devices, and exposed threads with corrosion preventive compound, Item 108B, Appendix D.

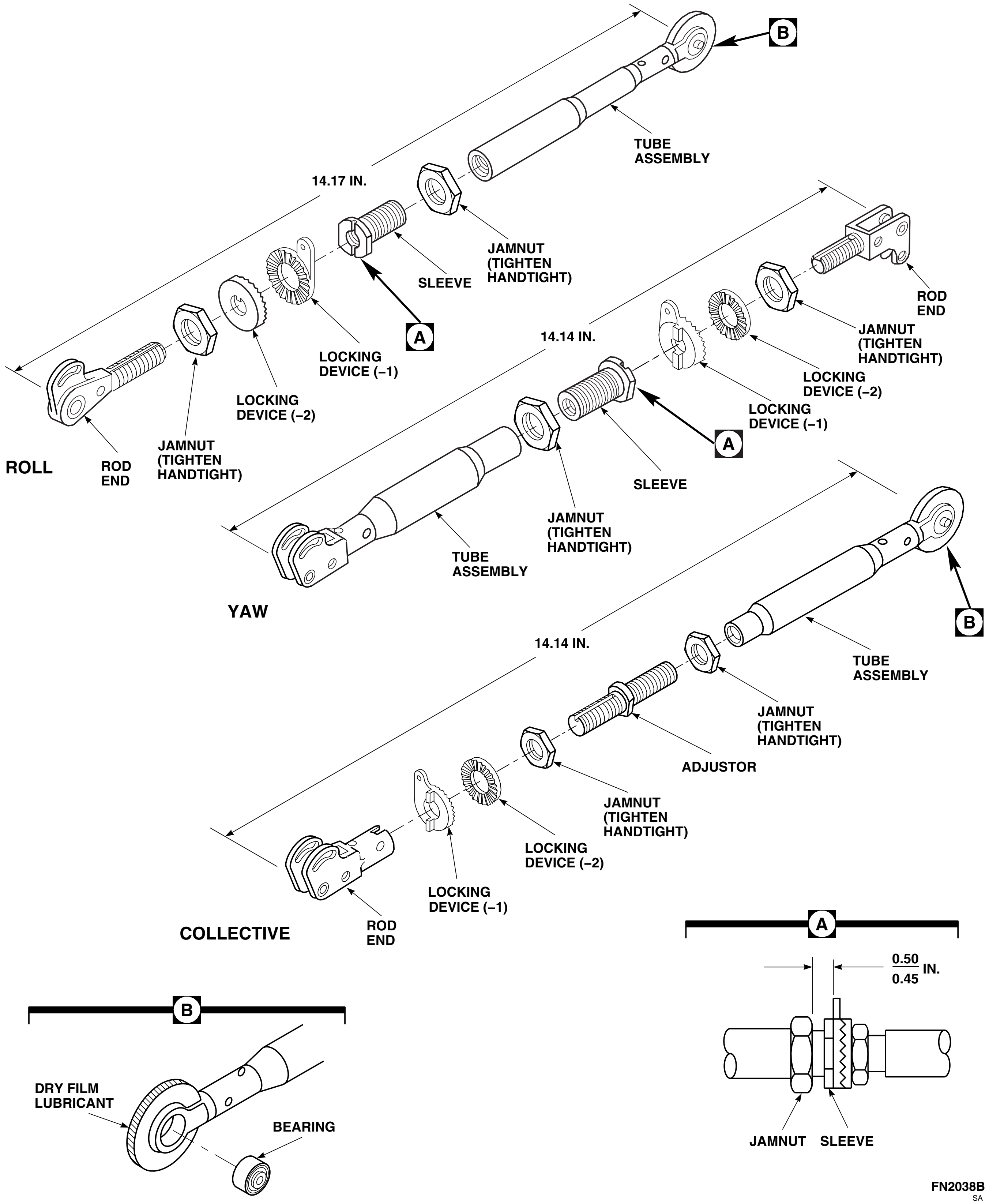


FIGURE 11-5-14-1. REPAIR BOOST SERVO TO MIXER CONTROL RODS (BENCH)

11-5-15 PRIMARY SERVO CONTROL RODS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 108A, Appendix D
- Corrosion Preventive Compound, Item 108B, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 11-5-11
- PARA 11-5-12

11-5-15.1 Repair Primary Servo Control Rods (BENCH).

11-5-15.1.1. Disassemble.

NOTE

Repair to primary servo control rod is limited to replacing parts. No repairs will be made to parts, except for blending out of minor scratches, nicks, or dents within limits.

a. Loosen two jamnuts (barrel jamnut has left-hand thread) (Figure 11-5-15-1).

b. Unscrew rod end clevis from barrel. Remove locking devices and jamnut from rod end clevis.

c. Unscrew barrel from tube assembly (left-hand thread). Remove jamnut from tube assembly.

11-5-15.1.2. Inspect/Repair.

Inspect and repair control rods (PARA 11-5-11 and PARA 11-5-12).

11-5-15.1.3. Assemble.

a. Coat all threaded, mating, and interior surfaces of control rod assembly with corrosion preventive compound, Item 108A, Appendix D.

b. Install barrel jamnut on tube assembly (left-hand thread). Thread barrel on tube assembly until tube assembly touches shoulder inside barrel. Back off about ¼ turn. **TIGHTEN BARREL JAMNUT HANDTIGHT** (Figure 11-5-15-1).

c. Install rod end jamnut and locking devices on rod end clevis. Install rod end clevis into barrel as far as possible.

NOTE

Locking devices' serrated faces face each other.

d. Adjust length of control rods by turning rod end clevis into barrel. Rod lengths for each control rod, measured between holes that are farthest apart, are as follows:

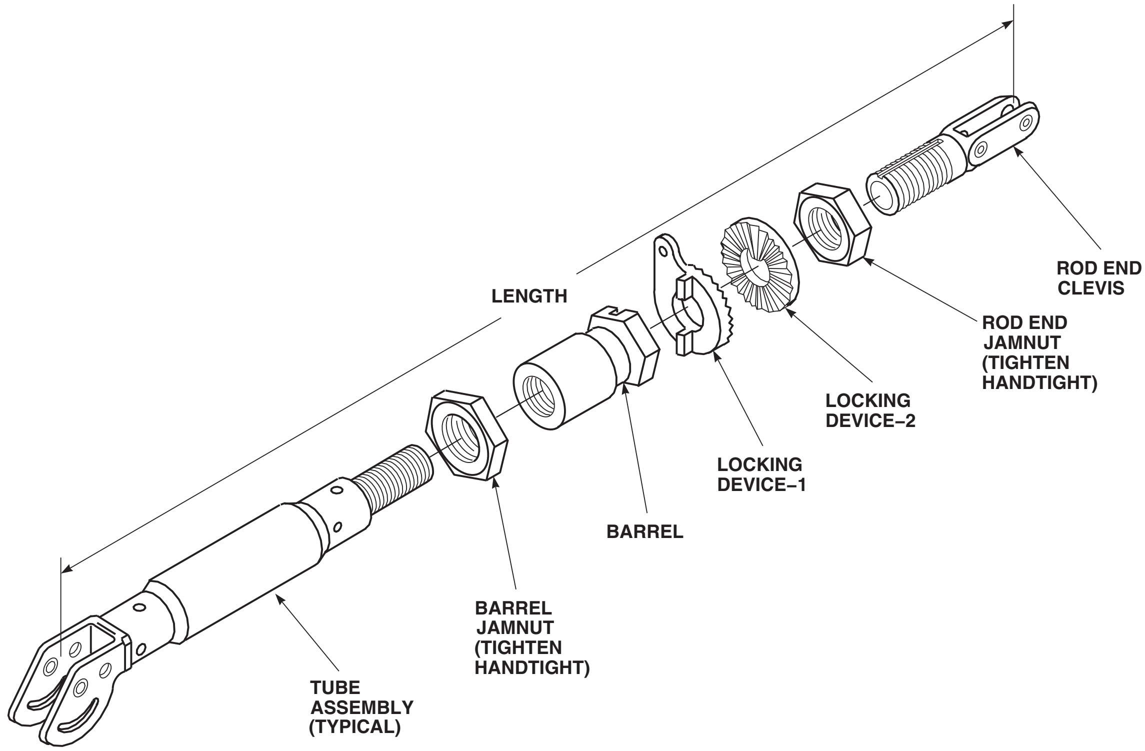


FIGURE 11-5-15-1. REPAIR PRIMARY SERVO CONTROL RODS (BENCH)

(1) Lateral 16 1⁄16-inches.

(2) Aft 15 3⁄4-inches.

(3) Forward 15 3⁄4-inches.

e. **TIGHTEN ROD END JAMNUT HANDTIGHT.** Final adjustment will be made when control rod is installed in helicopter.

f. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint as required, with epoxy primer.

g. After assembly and final adjustment of control rod, seal all mating and threaded surfaces, witness holes, slotted keyways, serrated locking devices, and exposed threads with corrosion preventive compound, Item 108B, Appendix D.

11-5-16 BELLCRANK BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Anvil
- Arbor Press
- Bearing Removal Tool, Step-Type
- Bolt, ¼-inch diameter
- Dial Indicator
- Drill Set
- Expansion Collet Knocker Puller
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Torque Wrench, 0 - 16 in. oz
- Washers, Standard or Locally-Made

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417

11-5-16.1 Replace Bellcrank Bearings (BENCH).

NOTE

- Replacement procedure is applicable for bellcrank assemblies listed in Table 11-5-16-1.
- Replacement for all bellcrank bearings is the same.

11-5-16.1.1 Remove.

a. For single bearings that can be pushed out, use arbor press, step-type bearing removal tool, and correct size anvil (Figure 11-5-16-1).

b. For bearings that have to be pulled, use an expansion collet knocker puller.

c. For double row bearings, with removable center spacer, remove rivets and press out bear-

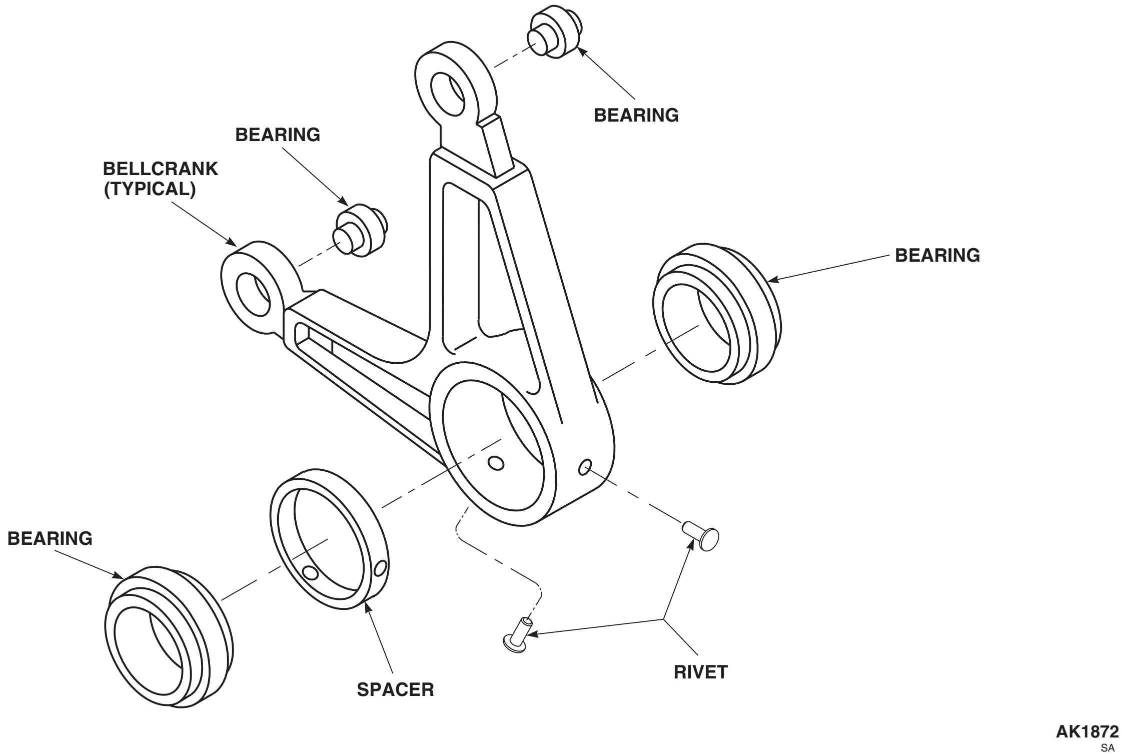


FIGURE 11-5-16-1. REPLACE BELLCRANK BEARINGS (BENCH)

ings and spacer using arbor press, step-type bearing removal tool, and correct size anvil.

11-5-16.1.2 Inspect.

a. Strip paint from surface of bellcranks around bearing holes using paint remover, Item 230, Appendix D.

b. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect bellcranks for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bellcrank.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

c. Inspect large bearings, MKP25B-A1173 (superseded by, 38950-00908-103), in directional, roll, pitch, and collective bellcranks using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play is beyond limits, replace bearing.

d. Inspect small bearing, SA4-14A1-504 (superseded by, SA4-14A1-512), in collective bellcrank using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play is beyond limits, replace bearing.

e. Inspect small bearing(s), MKSP4A-A2922 (superseded by, 38950-00903-104), in directional, roll, pitch, and collective bellcranks using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0025-INCH.** If radial play is beyond limits, replace bearing.

11-5-16.1.3 Install.

a. If bellcrank has two bearings with spacer between, install spacer and rivets (Figure 11-5-16-1).

b. Clean bearing surfaces of bellcrank and outside diameter of bearings with acetone, Item 25, Appendix D.

CAUTION

Damage to bearing will result if surface primer is allowed to contact bearings. Do not allow surface primer to come in contact with bearings.

c. Apply a light coat of sealing compound, Item 282, Appendix D, on outside diameter of bearings and on outside diameter of bellcrank.

d. Apply about a 1⁄16-inch bead of sealing compound, Item 280, Appendix D, to leading edge of bearing outside diameter and entrance edge of bellcrank bore. Bead should be all around edge.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

e. Within 10 minutes after applying sealing compound, press bearings into bellcrank.

f. Using machinery towel, Item 211, Appendix D, damp with acetone, Item 25, Appendix D, wipe surface of bearings clean of any sealing compound.

g. Allow at least 10 minutes for sealing compound to dry before handling bellcrank.

h. Perform rotational drag check of bearings as follows:

(1) For bearings with large inside diameters, use standard washers to fit on face of inside race, or make steel washers about 1⁄4-inch larger in diameter than inside diameter of inside race. Drill a 1⁄4-inch diameter hole in center of each washer.

(2) Install 1⁄4-inch diameter bolt through washers and bearing. Install nut on bolt.

(3) Using torque wrench and socket, turn bolthead and measure rotational drag of bearing. **IF DRAG IS MORE THAN ALLOWED IN TABLE 11-5-16-1, REPLACE BEARING.**

(4) Remove bolt, nut, and washers from bearing.

Table 11-5-16-1. Bearing Rotational Drag

Bellcrank Part Number	Bearing	Maximum Rotational Drag (Inch-Ounces)
70400-02170-044 Directional	MKSP4A-A2922 (superseded by, 38950-00903-104)	3
	MKP25B-A1173 (superseded by, 38950-00908-103)	5
70400-02171-049 Pitch	MKSP4A-A2922 (superseded by, 38950-00903-104)	3
	MKP25B-A1173 (superseded by, 38950-00908-103)	5
70400-02171-050 Roll	MKSP4A-A2922 (superseded by, 38950-00903-104)	3
	MKP25B-A1173 (superseded by, 38950-00908-103)	5
70400-02173-045 Collective (Left Side)	SA4-14A1-504 (superseded by, SA4-14A1-512)	3
	MKSP4A-A2922 (superseded by, 38950-00903-104)	3
	MKP25B-A1173 (superseded by, 38950-00908-103)	10
70400-02173-046 Collective (Right Side)	SA4-14A1-504 (superseded by, SA4-14A1-512)	3
	MKSP4A-A2922 (superseded by, 38950-00903-104)	3
	MKP25B-A1173 (superseded by, 38950-00908-103)	10

11-5-17 YAW LEVER AND SUPPORT BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Anvil
- Arbor Press
- Bearing Remover, Step-Type
- Dial Indicator
- Expansion Collet Knocker Puller
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Staking Tool, Three-Point Impression
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Cheesecloth, Item 90, Appendix D
- Cotton Swab, Item 119, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 274, Appendix D
- Sealing Compound, Item 282, Appendix D
- Trichloroethane, 1,1,1, Item 341, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- ASTM E1417

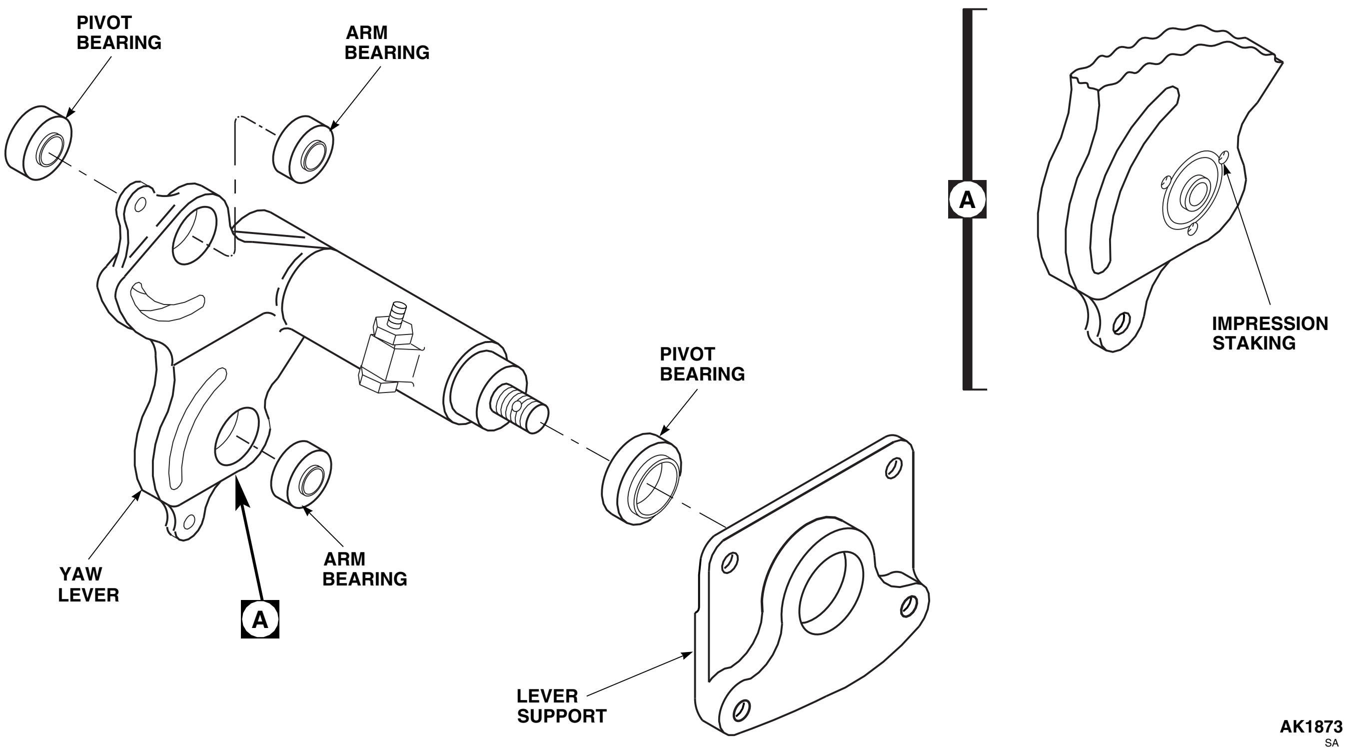


FIGURE 11-5-17-1. REPLACE YAW LEVER AND SUPPORT BEARINGS (BENCH)

11-5-17.1 Replace Yaw Lever and Support Bearings (BENCH).

11-5-17.1.1 Remove.

a. Using arbor press, a step-type bearing remover, and correct size anvil, press bearings from lever support and yaw lever (Figure 11-5-17-1).

b. Using expansion collet knocker puller, pull bearing from yaw lever.

11-5-17.1.2 Inspect.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

a. Inspect lever bearings with dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play exceeds 0.005-inch, replace lever bearing.

b. Check all bearings for roughness, sticking, and seal damage. **NONE ALLOWED.** If worn or damaged, replace bearing.

c. Strip paint from surface of parts around bearing holes using paint remover, Item 230, Appendix D.

d. Using fluorescent penetrant inspection kit, inspect support and lever for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support or lever.

11-5-17.1.3 Install.

a. Clean yaw lever, support, and outside diameter of bearings with machinery towel, Item 211, Appendix D, soaked in technical trichloroethylene, Item 343, Appendix D.

NOTE

Do not sand new parts. Parts should only be sanded if they are being reworked.

b. If reworking a used yaw lever or bearing, clean as follows:

(1) Remove sealing compound by lightly sanding lever inside diameter and bearing outside diameter with abrasive cloth, Item 8, Appendix D.

CAUTION

Damage to bearing will result if trichloroethane, 1,1,1 is allowed to contact bearings. Do not allow trichloroethane, 1,1,1 to come in contact with bearings.

(2) Remove any grime or grease with clean cheesecloth, Item 90, Appendix D, moistened with trichloroethane, 1,1,1, Item 341, Appendix D. Fold cloth until no stain shows on cloth after each use. Do not touch treated surfaces.

CAUTION

Damage to bearing will result if surface primer is allowed to contact bearings. Do not allow surface primer to come in contact with bearings.

c. Apply a light coat of sealing compound, Item 282, Appendix D, on outside diameter of bearings and inside diameter of lever or support. Allow primer to dry for 15 - 30 minutes. Do not touch primed surfaces.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

d. Apply an even coat of sealing compound, Item 274, Appendix D, to bearing outside diameter and lever bearing bore with cotton swab, Item 119, Appendix D.

e. Press bearings into lever or supports. Bearings must be installed within 4 minutes of sealing compound application (Figure 11-5-17-1). **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND APPLICATION PROCEDURE.**

f. Allow at least 15 minutes for sealing compound to dry before handling lever or support. Allow at least 24 hours drying time before staking bearings.

CAUTION

Damage to bearing will result if trichloroethane, 1,1,1 is allowed to contact bearings. Do not allow trichloroethane, 1,1,1 to come in contact with bearings.

g. Use machinery towel, Item 211, Appendix D, damp with trichloroethane, 1,1,1, Item 341, Appendix D, to wipe off sealing compound squeeze out.

h. Using three-point impression staking tool, stake arm bearings (Figure 11-5-17-1, Detail A). Stake on both sides. Do not stake over previous impressions.

i. Perform rotational drag check of bearings as follows:

(1) Install bolt and washer(s) through each yaw lever arm bearing. Install nut on each bolt.

(2) Using torque wrench and socket, turn bolthead and measure rotational drag of each lever arm bearing. **IF ROTATIONAL DRAG IS MORE THAN 5 INCH-OUNCES, REPLACE BEARING.**

(3) Using torque wrench and socket, turn bolthead and measure rotational drag of

pivot bearings. **IF ROTATIONAL DRAG IS MORE THAN 8 INCH-OUNCES, REPLACE BEARINGS.**

(4) Remove bolt, nut, and washers from bearing.

11-5-18 TORQUE SHAFT SUPPORT BEARING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Bolt and Nut, ¼-inch diameter
- Drill Set
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Torque Wrench, 0 - 16 in. oz
- Washers, Standard or Locally-Made

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Compound, Item 292, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D

11-5-18.1 Replace Torque Shaft Support Bearing (BENCH).

NOTE

Replacement of collective, pitch, lateral, and yaw upper and lower support bearings are the same, except as noted.

11-5-18.1.1. Remove.

a. Using nonmetallic scraper, remove gasket from upper support (Figure 11-5-18-1).

b. Remove remaining sealing compound from support using nonmetallic scraper and machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

c. Using arbor press, carefully remove bearing from support.

d. Clean inside bore of support with machinery towel, Item 211, Appendix D, dampened with technical trichloroethylene, Item 343, Appendix D.

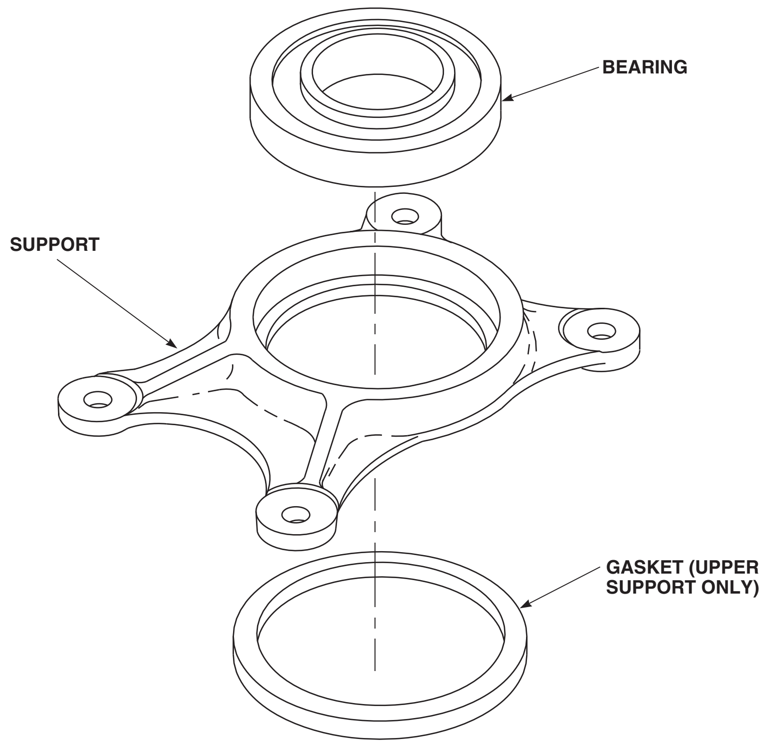


FIGURE 11-5-18-1. REPLACE TORQUE SHAFT SUPPORT BEARING (BENCH)

11-5-18.1.2. Install.

a. Clean outside surface of replacement bearing with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

b. Apply about a 1/16-inch bead of sealing compound, Item 273, Appendix D, to leading edge of bearing outside diameter and entrance edge of bearing bore. Bead shall be all around edge.

c. Within 10 minutes after applying sealing compound, press bearing into support using arbor press (Figure 11-5-18-1).

d. Allow 10 minutes for sealing compound to dry before handling support.

e. Perform rotational drag check of bearing as follows:

(1) For bearings with large inside diameters, use standard washers to fit on face of inside race, or make steel washers about 1/4-inch larger in diameter than inside diameter of inside race. Drill a 1/4-inch diameter hole in center of each washer.

(2) Install 1/4-inch diameter bolt through washers and bearing. Install nut on bolt.

(3) Using torque wrench and socket, turn bolthead and measure rotational drag of bearing. **IF DRAG IS MORE THAN 8 INCH-OUNCES, REPLACE BEARING.**

(4) Remove bolt, nut, and washers from bearing.

CAUTION

Damage to bearing will result if sealing compound is allowed to come in contact with bearing. Do not allow sealing compound to enter bearing.

f. Apply a light coat of sealing compound, Item 292, Appendix D, on gasket mounting area of upper support and mounting face of gasket.

g. Allow sealing compound to dry for 10 minutes, and place gasket on support.

h. Clean any extra sealing compound from upper support using nonmetallic scraper and methyl ethyl ketone, Item 217, Appendix D.

11-5-19 LEVER ASSEMBLY BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Anvil
- Arbor Press
- Bearing Remover, Step-Type
- Bolt and Nut, 3⁄16-inch diameter
- Drill Set
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Torque Wrench, 0 - 16 in. oz
- Washers, Standard or Locally-Made

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-19.1 Replace Lever Assembly Bearings (BENCH).

NOTE

Replacement of collective, pitch, lateral, and yaw lever assembly bearings are the same.

11-5-19.1.1 Remove.

Using arbor press, a step-type bearing remover, and correct size anvil, press bearings from lever (Figure 11-5-19-1).

11-5-19.1.2 Inspect.

a. Using paint remover, Item 230, Appendix D, strip paint from surface of parts around bearing holes.

b. Using fluorescent penetrant inspection kit, inspect lever for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace lever.

c. Coat paint-stripped area with primer, Item 258, Appendix D.

d. Touch up primed areas with paint (PARA 2-6-5).

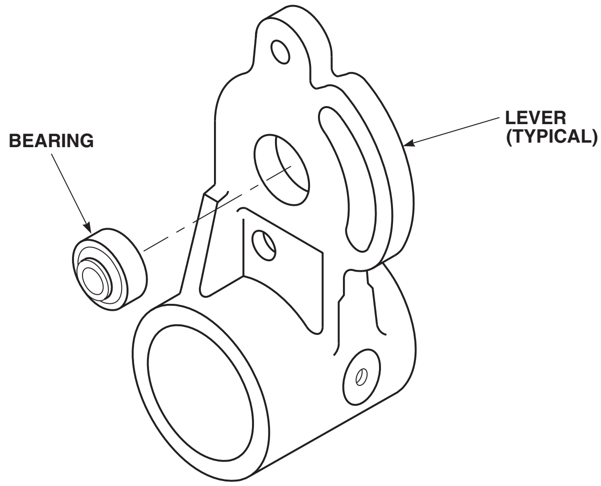


FIGURE 11-5-19-1. REPLACE LEVER ASSEMBLY BEARINGS (BENCH)

11-5-19.1.3 Install.

a. Clean bearing surfaces of lever and outside diameter of bearings with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D.

CAUTION

Damage to bearing will result if surface primer is allowed to contact bearings. Do not allow surface primer to come in contact with bearings.

b. Apply a light coat of sealing compound, Item 282, Appendix D, on outside diameter of bearings and inside diameter of lever.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

c. Apply about a 1/16-inch bead of sealing compound, Item 280, Appendix D, to leading edge of bearing outside diameter and entrance edge of bearing bore. Bead should be all around edge.

d. Within 10 minutes after applying sealing compound, press bearings into lever using arbor press (Figure 11-5-19-1).

e. Using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, wipe surface of bearings clean of any sealing compound.

f. Allow at least 10 minutes for sealing compound to dry before handling lever.

g. Perform rotational drag check of bearing as follows:

(1) For bearings with large inside diameters, use standard washers to fit on face of

inside race, or make steel washers about ¼-inch larger in diameter than inside diameter of inside race. Drill a ¼-inch diameter hole in center of each washer.

(2) Install 3⁄16-inch diameter bolt through washers and bearing. Install nut on bolt.

(3) Using torque wrench and socket, turn bolthead and measure rotational drag of bearing. **IF ROTATIONAL DRAG IS MORE THAN 2 INCH-OUNCES, REPLACE BEARING.**

(4) Remove bolt, nut, and washers from bearing.

11-5-20 PILOT'S COLLECTIVE STICK SUPPORT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Arbor Press
- Driver
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Reamer Set
- Sizing Block, 1.596 - 1.597-inches, Locally-Made
- Sizing Block, 1.998 - 2.000-inches, Locally-Made

Materials/Parts

- Acetone, Item 25, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 2-6-5

11-5-20.1 Repair Pilot's Collective Stick Support (BENCH).

11-5-20.1.1 Inspect.

a. Inspect upper support bushings, NAS537-5P-038 and NAS538-5P-023, and lower support bushings, NAS537-4P-038 and NAS538-4P-023 (Figure 11-5-20-1). **INSIDE RADIAL DIAMETER FOR UPPER SUPPORT BUSHINGS SHALL NOT BE OVER 0.4375-INCH. INSIDE RADIAL DIAMETER FOR LOWER SUPPORT BUSHINGS SHALL NOT BE OVER 0.3751-INCH.**

b. Using paint remover, Item 230, Appendix D, remove paint from support.

c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect support for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support.

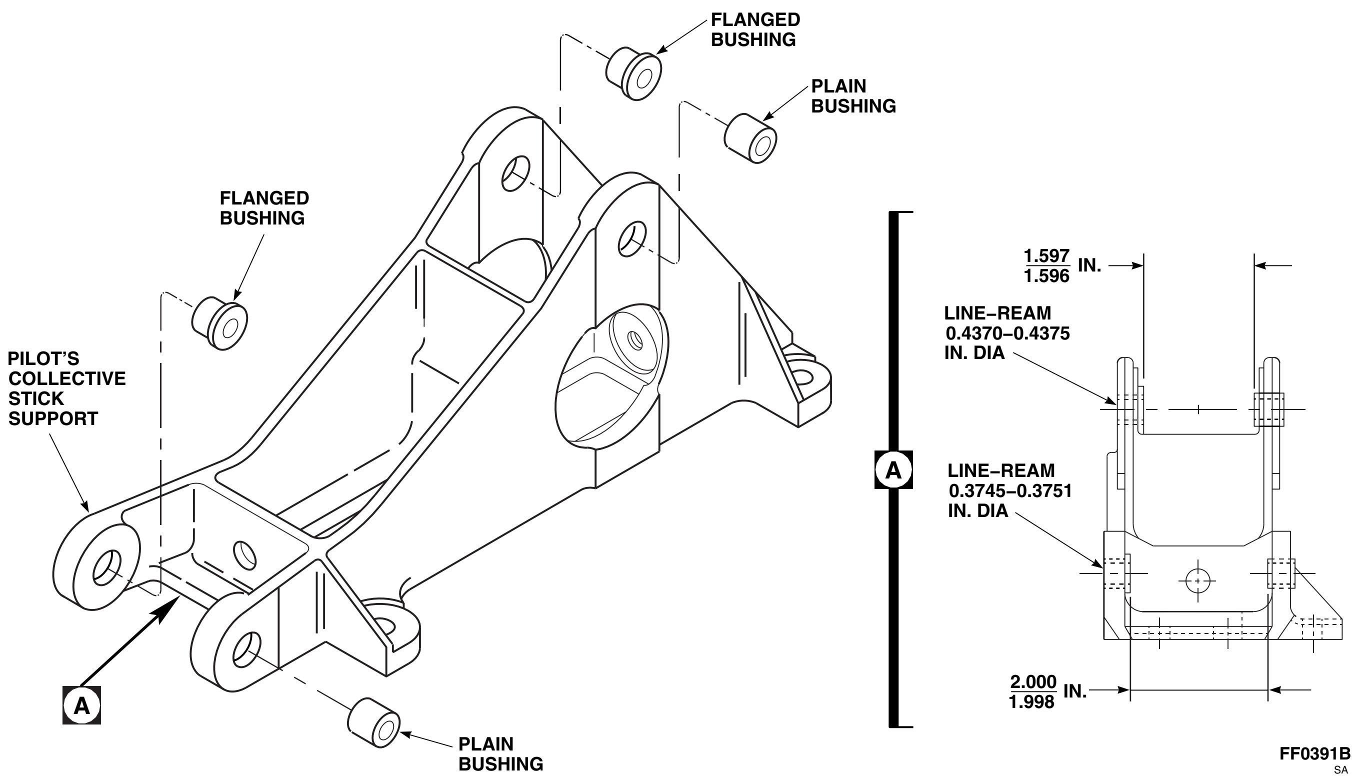


FIGURE 11-5-20-1. REPAIR PILOT'S COLLECTIVE STICK SUPPORT (BENCH)

d. Inspect support for wear and elongated bushing or attachment holes. **NONE ALLOWED.** If damaged, replace support.

11-5-20.1.2 Disassemble.

Using arbor press and driver, remove bushings from support (Figure 11-5-20-1).

11-5-20.1.3 Assemble.

a. Locally make 1.596 - 1.597-inch and 1.998 - 2.000-inch sizing blocks.

b. To replace bushings in support, perform the following:

(1) Freeze replacement bushings in dry ice, Item 125, Appendix D, for at least 30 minutes.

(2) Clean bushing holes in support with methyl ethyl ketone, Item 217, Appendix D. Coat flanged bushings with primer, Item 258, Appendix D.

(3) Using arbor press with bushings still wet with primer, press flanged bushings into upper and lower flanges on support. Bushing flanges to be on inside of support flanges (Figure 11-5-20-1).

c. On upper flange, place a 1.596 - 1.597-inch size block on flanged bushing before installing plain bushing (Figure 11-5-20-1, Detail A).

(1) Coat plain bushing with primer, Item 258, Appendix D. While still wet with primer, press plain bushing in upper support flange against sizing block.

(2) Remove sizing block. Measure space between ends of installed bushings. Clearance shall be 1.596 - 1.597-inches.

d. Using reamer set, line-ream upper flange bushing holes to 0.4370 - 0.4375-inch diameter.

e. On lower flange, place a 1.998 - 2.000-inch sizing block on flanged bushing before installing plain bushing.

(1) Coat plain bushing with primer, Item 258, Appendix D. While still wet with primer, press plain bushing in lower support flange against sizing block.

(2) Remove sizing block. Measure space between ends of installed bushings. Clearance shall be 1.998 - 2.000-inches.

f. Using reamer set, line-ream lower flange bushing holes to 0.3745 - 0.3751-inch diameter.

g. Clean support with acetone, Item 25, Appendix D.

h. Using 1-inch masking tape, Item 212, Appendix D, mask bushings.

i. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 223, Appendix D, paint support (PARA 2-6-5).

11-5-21 PILOT'S COLLECTIVE STICK BELLCRANK SUPPORT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Arbor Press
- Driver
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Reamer Set
- Sizing Block, 0.875 - 0.876-inch, Locally-Made

Materials/Parts

- Acetone, Item 25, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-21.1 Repair Pilot's Collective Stick Bellcrank Support (BENCH).

11-5-21.1.1 Inspect.

a. Inspect bellcrank support bushings inside diameter. **DIAMETER SHALL NOT EXCEED 0.2505-INCH.** If diameter is beyond limit, replace bushing.

b. Using paint remover, Item 230, Appendix D, remove paint from bellcrank support.

c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect bellcrank support for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace bellcrank support.

d. Inspect bellcrank support for wear and elongated bushing or attachment holes. **NONE ALLOWED.** If damaged, replace bellcrank support.

11-5-21.1.2 Disassemble.

Using arbor press and driver, remove bushings from bellcrank support (Figure 11-5-21-1).

11-5-21.1.3 Assemble.

a. Locally make 0.875 - 0.876-inch sizing block.

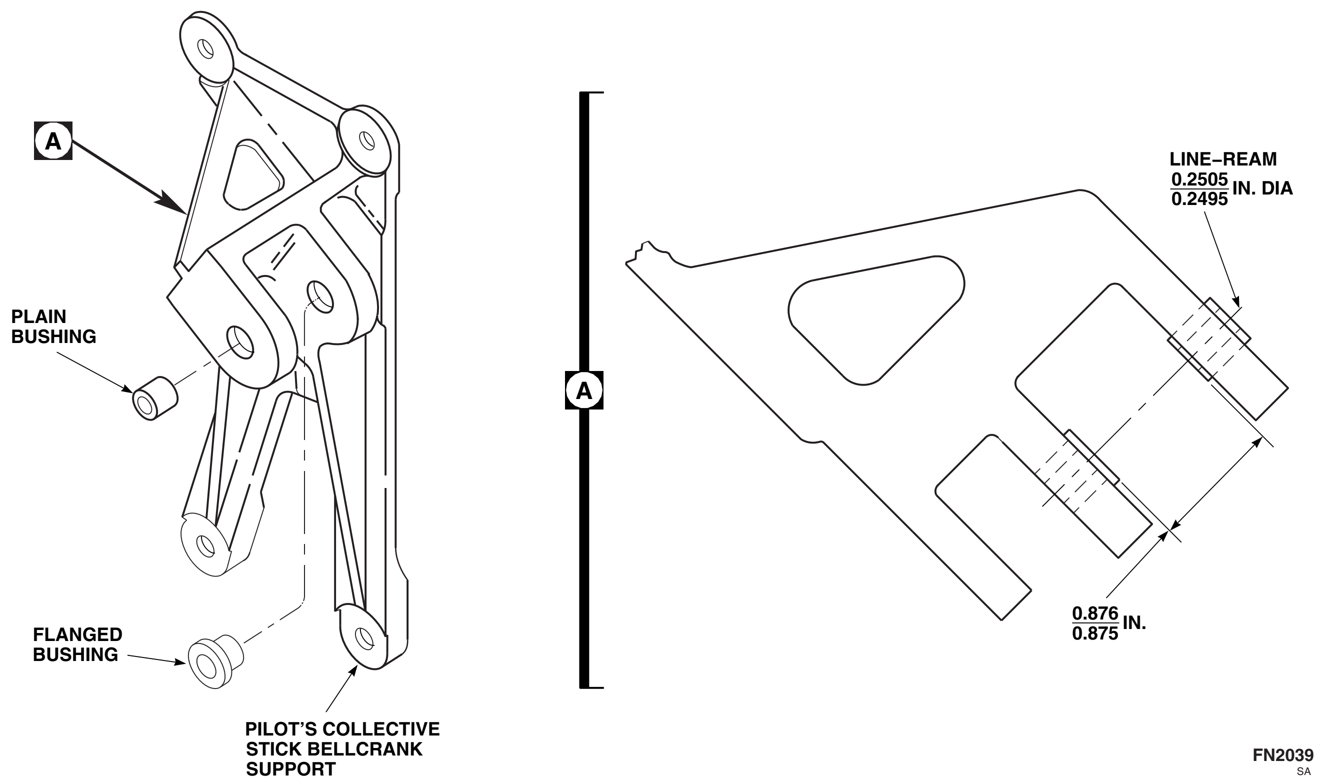


FIGURE 11-5-21-1. REPAIR PILOT'S COLLECTIVE STICK BELLCRANK SUPPORT (BENCH)

b. To replace bushings in bellcrank support, perform the following:

(1) Freeze replacement bushings in dry ice, Item 125, Appendix D, for at least 30 minutes.

(2) Clean bushing holes in support with methyl ethyl ketone, Item 217, Appendix D. Coat flanged bushings with primer, Item 258, Appendix D.

(3) With bushing still wet with primer, press bushing into clevis ear with arbor press. Flange to be on inside of clevis (Figure 11-5-21-1).

c. Place a 0.875 - 0.876-inch sizing block on flange of bushing before installing plain bushing (Figure 11-5-21-1, Detail A).

(1) Coat plain bushing with primer, Item 258, Appendix D. While still wet with primer, press plain bushing in clevis ear against sizing block.

(2) Remove sizing block. Measure space between ends of installed bushings. **CLEARANCE SHALL BE 0.875 - 0.876-INCH.**

d. Using reamer set, line-ream bushing holes to 0.2505 - 0.2495-inch diameter.

e. Clean support with acetone, Item 25, Appendix D.

f. Using 1-inch masking tape, Item 212, Appendix D, mask bushings.

g. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 223, Appendix D, paint support (PARA 2-6-5).

11-5-22 COPILOT'S COLLECTIVE STICK SUPPORT (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Arbor Press
- Driver
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Reamer Set
- Sizing Block, 3.816 - 3.817-inches, Locally-Made

Materials/Parts

- Acetone, Item 25, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 223, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-22.1 Repair Copilot's Collective Stick Support (BENCH).

11-5-22.1.1 Inspect.

a. Inspect support bushings inside diameter (Figure 11-5-22-1). **DIAMETER OF BUSHINGS SHALL NOT EXCEED 0.3130-INCH.** If diameter is beyond limit, replace bushing.

b. Using paint remover, Item 230, Appendix D, remove paint from support.

c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect support for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace support.

d. Inspect support for wear and elongated bushing or attachment holes. **NONE ALLOWED.** If damaged, replace support.

11-5-22.1.2 Disassemble.

Using arbor press and driver, remove bushings from support (Figure 11-5-22-1).

11-5-22.1.3 Assemble.

a. Locally make 3.816 - 3.817-inch sizing block.

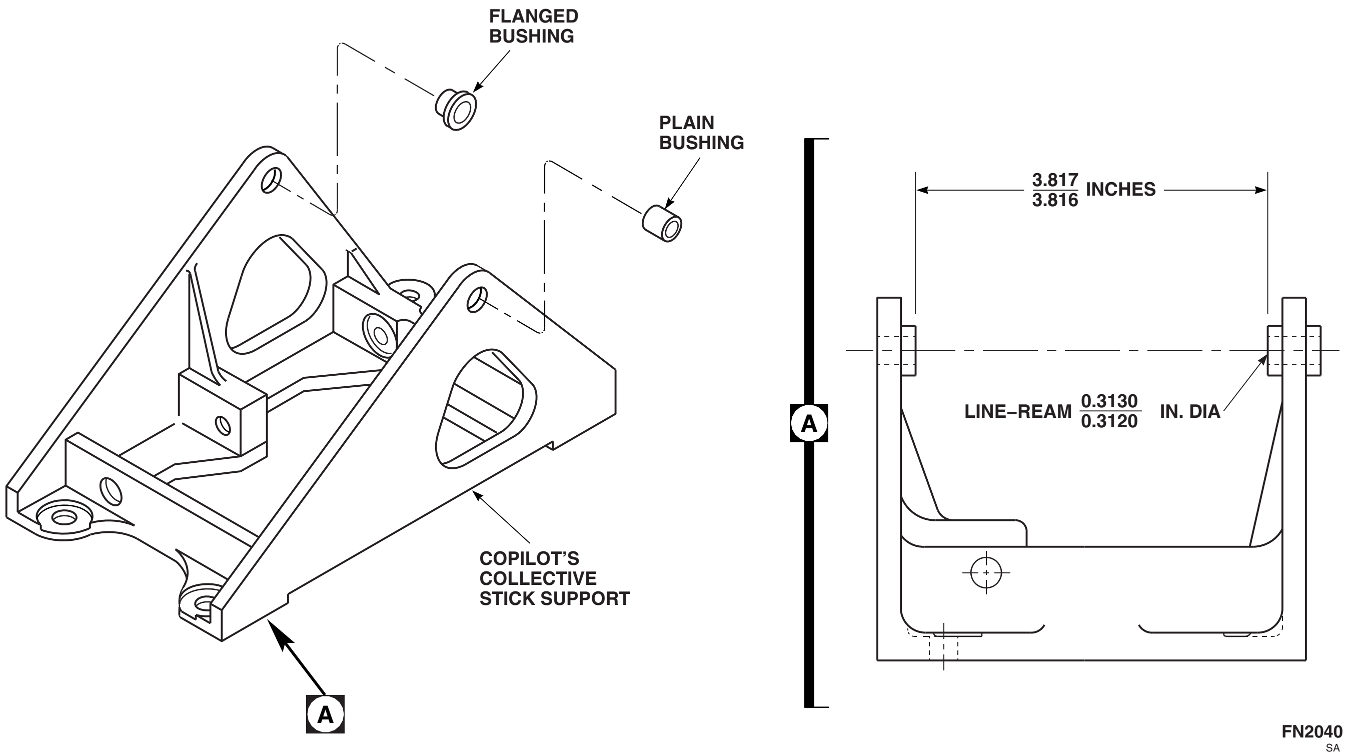


FIGURE 11-5-22-1. REPAIR COPILOT'S COLLECTIVE STICK SUPPORT (BENCH)

b. To replace flanged bushing in support, perform the following:

(1) Freeze replacement bushings in dry ice, Item 125, Appendix D, for at least 30 minutes.

(2) Clean bushing holes in support with methyl ethyl ketone, Item 217, Appendix D. Coat flanged bushings with primer, Item 258, Appendix D.

(3) Using arbor press with bushing still wet with primer, press bushing into support flange. Bushing flange to be on inside of support flange (Figure 11-5-22-1).

c. Place a 3.816 - 3.817-inch sizing block on flanged bushing before installing plain bushing (Figure 11-5-22-1, Detail A).

(1) Coat plain bushing with primer, Item 258, Appendix D. While still wet with primer, press plain bushing in support ear against sizing block.

(2) Remove sizing block. Measure space between bushings. **CLEARANCE SHALL BE 3.816 - 3.817-INCHES.**

d. Using reamer set, line-ream bushing holes to 0.3120 - 0.3130-inch diameter.

e. Clean support with acetone, Item 25, Appendix D.

f. Using 1-inch masking tape, Item 212, Appendix D, mask bushings.

g. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 223, Appendix D, paint support (PARA 2-6-5).

11-5-23 COPILOT'S COLLECTIVE STICK BELLCRANK BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Anvil
- Arbor Press
- Bearing Remover, Step-Type
- Dial Indicator
- Expansion Collet Knocker Puller
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Acetone, Item 25, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 274, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417

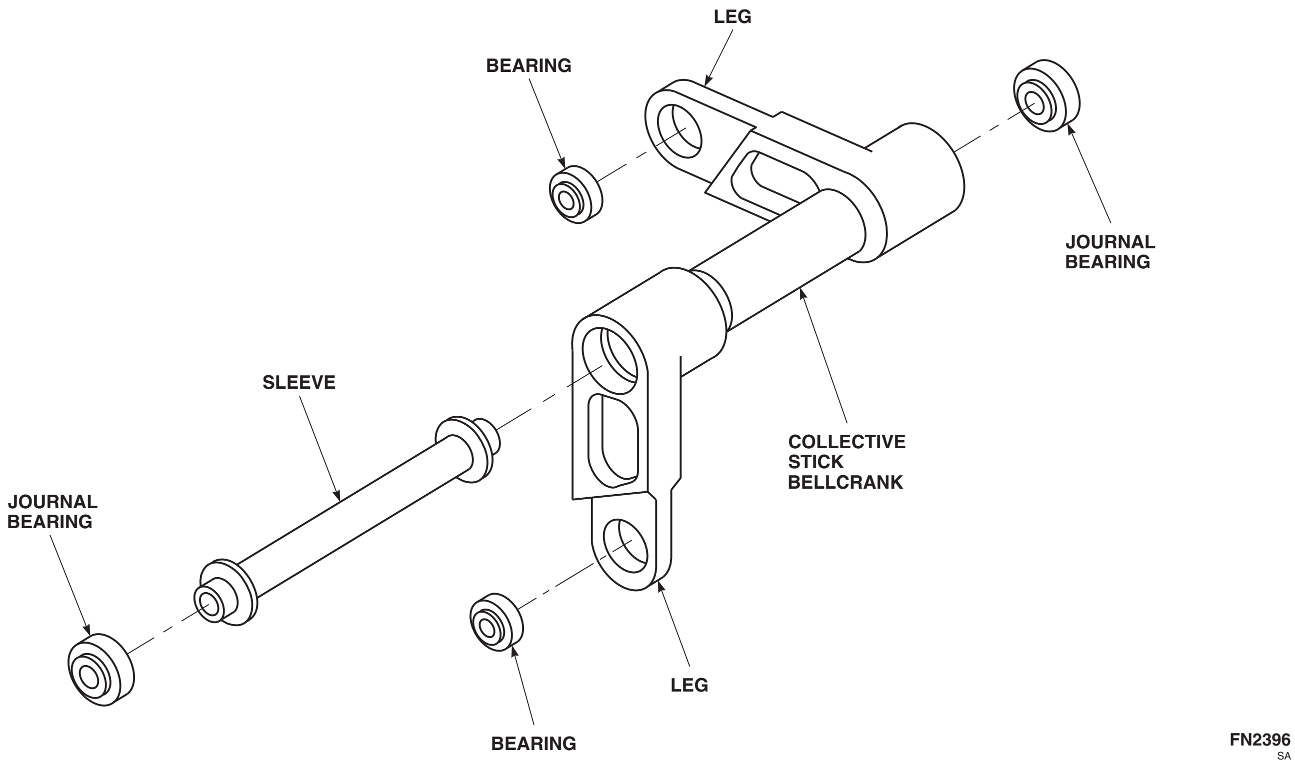


FIGURE 11-5-23-1. REPLACE COPILOT'S COLLECTIVE STICK BELLCRANK BEARINGS (BENCH)

11-5-23.1 Replace Copilot's Collective Stick Bellcrank Bearings (BENCH).

11-5-23.1.1 Inspect.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

a. Inspect bellcrank bearings with a dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0025-INCH.** If radial play is beyond limit, replace bearing.

b. Check all bearings for roughness, sticking, and damage. **NONE ALLOWED.** If worn or damaged, replace bearing.

c. Using paint remover, Item 230, Appendix D, remove paint from surface of parts around bearing holes.

d. Using fluorescent penetrant inspection kit, inspect bellcrank support for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace collective stick bellcrank.

11-5-23.1.2 Remove.

a. Using arbor press, a step-type bearing remover, and correct size anvil, press bearings from bellcrank legs (Figure 11-5-23-1).

b. Using expansion collet knocker puller, pull journal bearings from bellcrank.

c. Remove sleeve from collective stick bellcrank.

11-5-23.1.3 Install.

a. Clean bellcrank and outside diameter of bearings with machinery towel, Item 211, Appendix D, soaked with acetone, Item 25, Appendix D.

b. If reworking a used bellcrank or bearing, clean as follows:

NOTE

Do not sand new parts. Parts should only be sanded if they are being reworked.

(1) Remove sealing compound by lightly sanding bellcrank inside diameter and bearing outside diameter with abrasive cloth, Item 8, Appendix D.

(2) Remove any grime or grease with clean cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. Fold cloth until no stain shows on cloth after each use. Do not touch treated surfaces.

c. Apply a light coat of sealing compound, Item 282, Appendix D, on outside diameter of bearings and inside diameter of bellcrank. Allow sealing compound to dry for 15 - 30 minutes. Do not touch surfaces.

d. Apply an even coat of sealing compound, Item 274, Appendix D, to bearing outside diameter and bellcrank bearing bore with acid brush, Item 84, Appendix D.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

e. Press bearings into bellcrank legs. Bearings must be installed within 4 minutes of sealing compound application (Figure 11-5-23-1). **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND APPLICATION PROCEDURE.**

f. Install sleeve into bellcrank assembly.

g. Press journal bearings into bellcrank within 4 minutes of sealing compound application. **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND PROCEDURE.**

h. Allow at least 15 minutes for sealing compound to dry before handling bellcrank.

i. Using machinery towel, Item 211, Appendix D, damp with acetone, Item 25, Appendix D, wipe parts clean of sealing compound squeeze out.

j. Perform rotational drag check of bearings as follows:

(1) Position washers on lip of bearing.

(2) Install bolt through bearing. Install nut on bolt.

(3) Using torque wrench and socket, turn bolthead and measure rotational drag of bearing. **IF ROTATIONAL DRAG IS MORE THAN 2 INCH-OUNCES AT 70° - 90°F (21° - 32°C), REPLACE BEARING.**

(4) Remove bolt, washers, and nut from bearing.

11-5-24 LONGITUDINAL BELLCRANK BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Anvil
- Arbor Press
- Bearing Remover, Step-Type
- Dial Indicator
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer's Toolkit
- Puller Set
- Staking Tool, Three-Point Impression
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Acetone, Item 25, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 274, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417

11-5-24.1 Replace Longitudinal Bellcrank Bearings (BENCH).

11-5-24.1.1 Inspect.

NOTE

Check movement of bearings in radial direction only. Axial play check of bearings not required.

a. Inspect bearings using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.0005-INCH.** If radial play is beyond limit, replace bearing.

b. Check all bearings for roughness, sticking, and damage. If worn or damaged, replace bearing.

11-5-24.1.2 Remove.

a. Using arbor press, step-type bearing remover, and correct size anvil, press bearings from bellcrank legs (Figure 11-5-24-1).

b. Using puller set, pull pivot bearings from bellcrank.

c. Using puller set, remove bushing from bearing.

11-5-24.1.3 Inspect.

a. Inspect bushing for scoring and wear. **NONE ALLOWED.** If worn or damaged, replace bushing.

b. Using paint remover, Item 230, Appendix D, remove paint from surface of parts around bearing holes.

c. Using fluorescent penetrant inspection kit, inspect bellcrank support for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace longitudinal bellcrank.

11-5-24.1.4 Install.

a. Clean longitudinal bellcrank and outside diameter of bearings with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D.

b. If reworking a used bellcrank or bearing, clean as follows:

NOTE

Do not sand new parts. Parts should only be sanded if they are being reworked.

(1) Remove sealing compound by lightly sanding bellcrank inside diameter and bearing outside diameter with abrasive cloth, Item 8, Appendix D.

(2) Remove any grime or grease with clean cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. Fold cloth until no stain shows on cloth after each use. Do not touch treated surfaces.

c. Apply light coat of sealing compound, Item 282, Appendix D, on outside diameter of bellcrank leg bearings and inside diameter of bellcrank. Allow primer to dry for 15 - 30 minutes. Do not touch primed surfaces. If any break occurs in primer coat, reclean, and reprime.

d. Apply an even coat of sealing compound, Item 274, Appendix D, to leg bearings outside diameter and bellcrank bearing bore with a clean acid brush, Item 84, Appendix D.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

e. Press bellcrank leg bearings into bellcrank using arbor press. Bearings must be installed within 4 minutes of sealing compound application (Figure 11-5-24-1). **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND APPLICATION PROCEDURE.**

f. Press pivot bearing onto bushing using arbor press.

g. Apply light coat of sealing compound, Item 282, Appendix D, on outside diameter of pivot bearings and inside diameter of bellcrank. Allow primer to dry for 15 - 30 minutes. If any break occurs in primer coat, reclean, and reprime.

h. Apply an even coat of sealing compound, Item 274, Appendix D, to pivot bearings outside diameter and pivot bearing bore with a clean acid brush, Item 84, Appendix D.

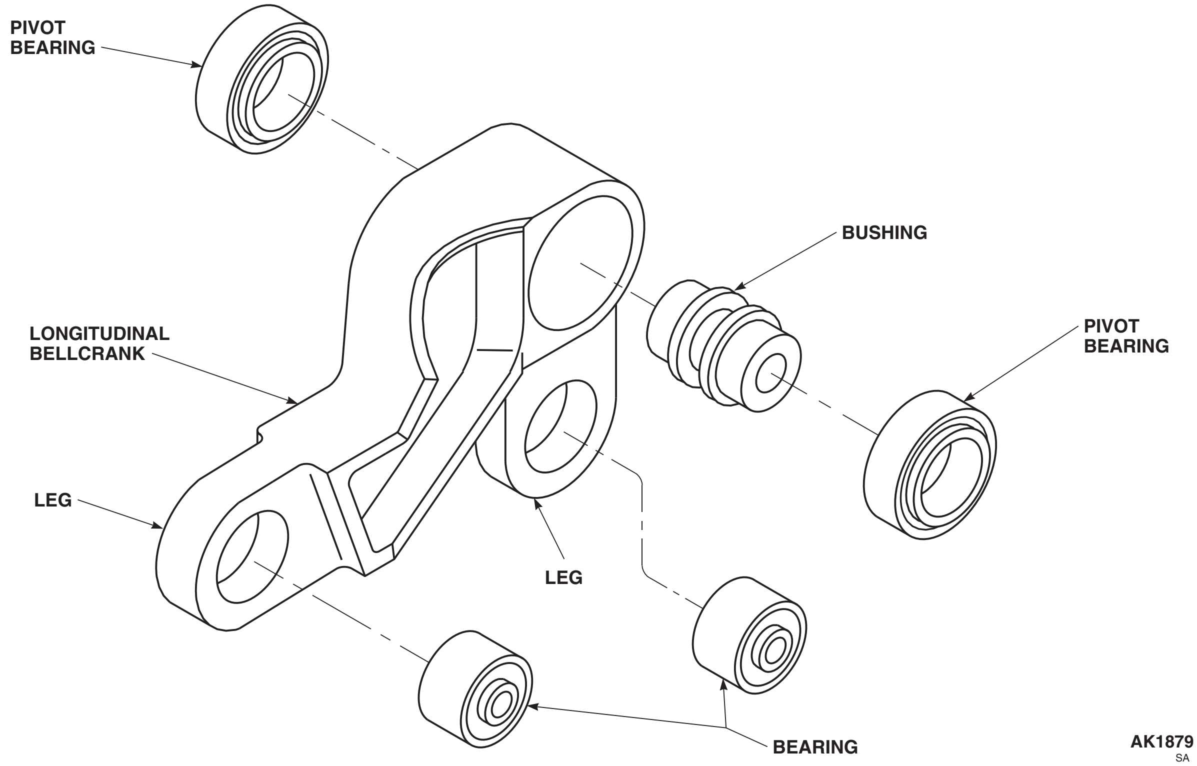


FIGURE 11-5-24-1. REPLACE LONGITUDINAL BELLCRANK BEARINGS (BENCH)

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

i. Press pivot bearing with bushing into bellcrank housing using arbor press. Bearing must be installed within 4 minutes of sealing compound application. **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND APPLICATION PROCEDURE.**

j. Allow at least 15 minutes for sealing compound to dry before handling bellcrank.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

k. Press pivot bearing into reverse side of bellcrank using arbor press. Bearing must be installed within 4 minutes of sealing compound application. **IF BEARING IS NOT INSTALLED WITHIN 4 MINUTES, REPEAT ENTIRE CLEANING, PRIMING, AND SEALING COMPOUND APPLICATION PROCEDURE.**

l. Allow at least 15 minutes for sealing compound to dry before handling bellcrank. Allow at least 24 hours drying time before staking bearings.

m. Using machinery towel, Item 211, Appendix D, damp with acetone, Item 25, Appendix D, wipe parts clean of sealing compound squeeze out. Be careful not to get solvent in bearings.

n. Using three-point impression staking tool, stake bearings.

o. Perform rotational drag check of bearings as follows:

(1) Position washers on lip of bearing.

(2) Install bolt through bearing. Install nut on bolt.

(3) Using torque wrench and socket, turn bolthead and measure rotational drag of bearing. **IF ROTATIONAL DRAG IS MORE THAN 2 INCH-OUNCES, REPLACE BEARING.**

(4) Remove bolt, washers, and nut from bearing.

11-5-25 FORWARD, LATERAL, AND AFT BELLCRANK/WALKING BEAM FLANGED BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Drift
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer's Toolkit
- Puller

Materials/Parts

- Adhesive, Item 32, Appendix D
- Brush, Acid, Item 84, Appendix D
- Dry Ice, Item 125, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Primer, Item 258, Appendix D

References

- Appendix D
- PARA 2-6-5

11-5-25.1 Forward, Lateral, and Aft Bellcrank/Walking Beam Flanged Bearings (BENCH).

NOTE

Replacement of forward, aft, and lateral bellcranks, and walking beam flanged bearings are the same, except as noted.

11-5-25.1.1 Remove.

a. On forward and aft bellcranks, remove retainer as follows (Figure 11-5-25-1):

(1) Remove rivet securing retainer to bellcrank.

(2) Loosen adhesive securing retainer using dry ice, Item 125, Appendix D, and nonmetallic scraper.

(3) Remove retainer from bellcrank.

b. Using arbor press and suitable puller, remove damaged flanged bearing.

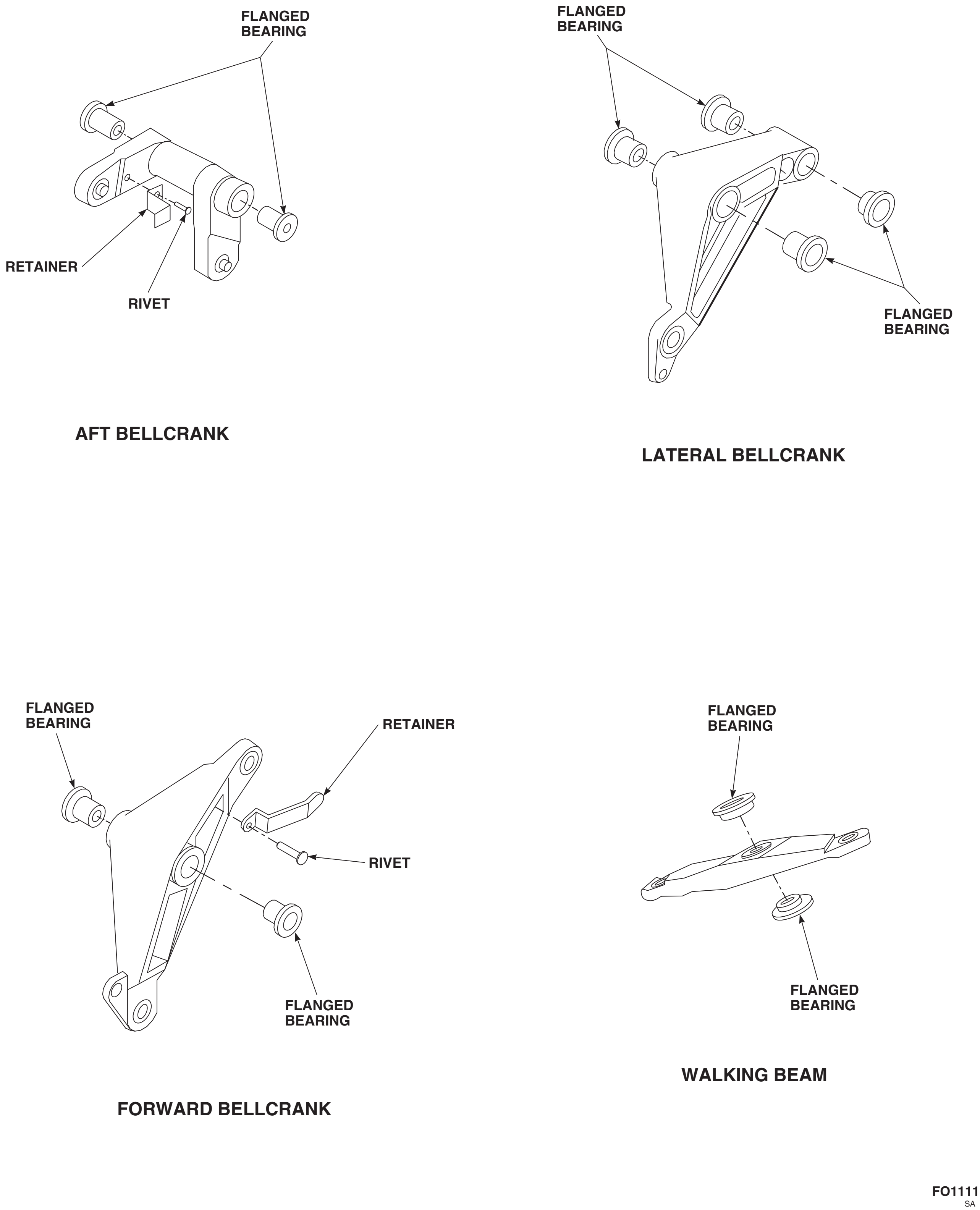


FIGURE 11-5-25-1. FORWARD, LATERAL, AND AFT BELLCRANK/WALKING BEAM FLANGED BEARINGS (BENCH)

11-5-20.1.2 Inspect.

Inspect liners for damage and disbonding. **NONE ALLOWED.** If damage or disbonding is found, replace bellcrank.

11-5-25.1.3 Install.

a. Apply light coat of primer, Item 258, Appendix D, to outside diameter of flanged bearing and inside diameter of liner using acid brush, Item 84, Appendix D.

b. Using arbor press and suitable drift, press in new flanged bearing while primer is still wet. Make sure shoulder of flanged bearing seats firmly on machined surface of bearing lug (Figure 11-5-25-1).

c. Clean excess primer from bearing lug area using machinery towel, Item 211, Appendix D.

d. On forward and aft bellcranks install retainer as follows:

NOTE

Retainer must be installed on same side as etched part number.

(1) Apply bead of adhesive, Item 32, Appendix D, to mating surface of retainer. Immediately position retainer over hole on bellcrank.

(2) Secure retainer to bellcrank with rivet.

(3) Clean any excess adhesive from bellcrank using machinery towel, Item 211, Appendix D.

(4) Allow to cure at room temperature for 24 hours, or 1 hour at 140° - 160°F (60° - 71°C), using heat gun.

e. Using 1-inch masking tape, Item 212, Appendix D, mask bearings and critical star stamp.

f. Touch up areas using epoxy primer coating kit, Item 135, Appendix D (PARA 2-6-5).

g. Touch up paint (PARA 2-6-5).

h. Remove all masking tape.

11-5-26 TAIL ROTOR PITCH CONTROL SHAFT PRESSED BUSHING (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Micrometer
- Nonmetallic Drift
- Nonmetallic Hammer

Materials/Parts

- Abrasive Cloth, Item 13, Appendix D
- Acetone, Item 25, Appendix D
- Barrier Material, Item 80, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D

References

- Appendix D

11-5-26.1 Replace Tail Rotor Pitch Control Shaft Pressed Bushing (BENCH).

NOTE

This procedure can only be used if bushing, 70358-06331-103, is to be installed as replacement bushing.

11-5-26.1.1. Remove.

a. Remove cotter pin securing spanner nut to pitch control shaft (Figure 11-5-26-1).

b. Cover linkage, hardware, and ball bearings with barrier material, Item 80, Appendix D. Secure barrier material in place with 1-inch masking tape, Item 212, Appendix D.

c. Cover chrome-plated surfaces, threads, and other exposed surfaces of pitch control shaft with barrier material, Item 80, Appendix D. Secure barrier material in place with 1-inch masking tape, Item 212, Appendix D.

d. Support shaft and remove pressed bushing from pitch control shaft with nonmetallic hammer and nonmetallic drift.

11-5-26.1.2. Install.

a. Smooth out any galling or score marks on pitch control shaft bushing journal using abrasive cloth, Item 13, Appendix D. Wipe clean with machinery towel, Item 211, Appendix D. **GALLING NO DEEPER THAN 0.010-INCH MAY BE BLENDED OUT, PROVIDED NOT MORE THAN 25% OF CONTACT AREA IS**

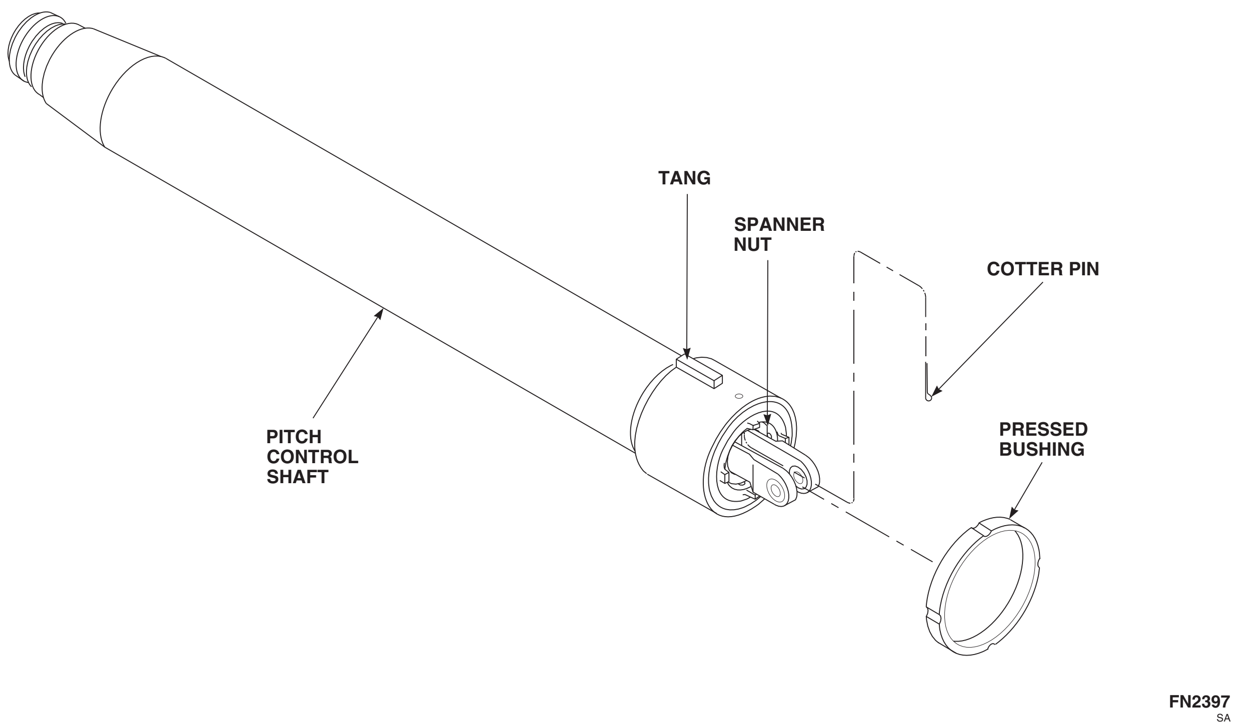


FIGURE 11-5-26-1. REPLACE TAIL ROTOR PITCH CONTROL SHAFT PRESSED BUSHING (BENCH)

AFFECTED. Thoroughly clean shaft bushing journal with machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D.

CAUTION

Damage to equipment will result if incorrect pressed bushing is installed on pitch control shaft. Bushing, 70358-06331-103, is the only acceptable replacement bushing as it is pre-machined and requires no further rework once it is pressed onto shaft.

NOTE

Slots on pressed bushing, 70358-06331-103, do not have to be aligned with pitch control shaft keys for installation.

b. Position pressed bushing, 70358-06331-103, on pitch control shaft and lightly tap pressed bushing evenly onto shaft using nonmetallic hammer and nonmetallic drift until bushing is fully seated against tangs on pitch control shaft (Figure 11-5-26-1).

c. Check that bushing is firmly pressed onto shaft and cannot be turned by hand.

d. Inspect outside diameter (OD) of pressed bushing using micrometer. **OD SHALL BE 3.1165 - 3.1185-INCHES.** If OD is beyond limits, replace bushing.

e. Remove barrier material from linkage, hardware, and ball bearings.

CAUTION

Damage to tail rotor gear box bore will result if cotter pin is installed incorrectly. Make sure that cotter pin is installed with head on inside of pitch control shaft and that ends of cotter pin are folded tightly against shaft outside diameter.

f. Install cotter pin through hole on inside of spanner nut (head on inside). Fold ends of cotter pin tightly against shaft outside diameter so that they are at least 0.030-inch below outside diameter of pressed bushing.

g. Remove barrier material from chrome-plated surfaces, threads, and other exposed surfaces of pitch control shaft.

11-5-27 TAIL ROTOR QUADRANT BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Bolt, ¼-inch diameter x ¾-inch long
- Dial Indicator
- Fluorescent Penetrant Inspection Kit
- Nut, ¼-inch diameter
- Powertrain Repairer's Toolkit
- Pusher, Step-Type
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-27.1 Replace Tail Rotor Quadrant Bearings (BENCH).

11-5-27.1.1 Inspect.

Using dial indicator, inspect bearings in tail rotor quadrant. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play exceeds limit, replace bearing(s).

11-5-27.1.2 Remove.

a. Using arbor press and step-type pusher, press all bearings and spacer, with rings, from tail rotor quadrant (Figure 11-5-27-1).

b. Using paint remover, Item 230, Appendix D, strip paint from surface of part around bearing holes and around hub.

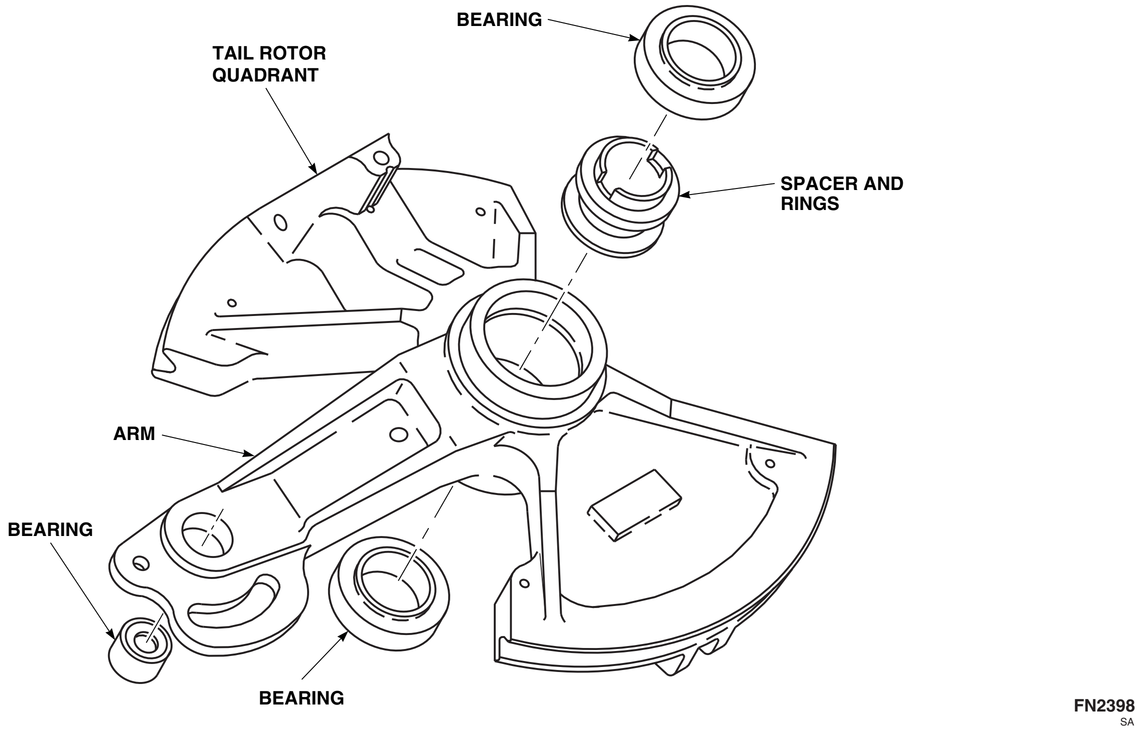


FIGURE 11-5-27-1. REPLACE TAIL ROTOR QUADRANT BEARINGS (BENCH)

c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect quadrant for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace quadrant.

d. Coat paint-stripped areas with primer, Item 258, Appendix D. Touch up primer areas with paint (PARA 2-6-5).

11-5-27.1.3 Install.

a. Clean bearing surfaces of tail rotor quadrant and outside diameter of bearings with acetone, Item 25, Appendix D.

CAUTION

Damage to bearing will result if surface primer is allowed to come in contact with bearing bore. Do not get surface primer on support bearing bore surface.

NOTE

Surface primer and sealing compound must be made by same manufacturer.

b. Apply a light coat of sealing compound, Item 282, Appendix D, to outside diameters of bearings. Allow to air-dry for at least 30 seconds.

c. Apply about a 1⁄16-inch bead of sealing compound, Item 273, Appendix D, to leading edge of bearings outside diameters and entrance edge of bearing bore in quadrant. Bead should be all around edges.

d. Within 10 minutes after applying sealing compound, press bearings into quadrant (Figure 11-5-27-1).

e. Press small bearing into arm of quadrant. Press one large bearing into quadrant. In other side, install spacer with rings. Press in large bearing. Be careful not to get sealing compound in bearings.

f. Using machinery towel, Item 211, Appendix D, damp with acetone, Item 25, Appendix D, wipe surfaces of bearings clean of any sealing compound. Be careful not to get solvent in bearings.

g. Allow at least 10 minutes for sealing compound to cure before handling quadrant.

h. Perform rotational drag check of all bearings as follows:

(1) Install ¼-inch diameter x ¾-inch long bolt through small bearings, and install nut on bolt.

(2) Using torque wrench and socket, turn bolthead and measure rotational drag of bearing. **IF ROTATIONAL DRAG IS MORE THAN 10 INCH-OUNCES, REPLACE BEARINGS.**

(3) Remove bolt and nut from bearings.

11-5-28 TAIL ROTOR QUADRANT UPPER AND LOWER ARM BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Arbor Press
- Dial Indicator
- Drill
- Drill Bit, ¼-inch
- Fluorescent Penetrant Inspection Kit
- Hexhead Bolt, ¼-inch diameter x 1-inch long
- Nut, ¼-inch diameter
- Powertrain Repairer's Toolkit
- Pusher, Step-Type
- Steel Washers, ⅛-inch thick x 2 ¼-inch diameter, Locally-Made
- Torque Wrench, 0 - 30 in. oz

Materials/Parts

- Acetone, Item 25, Appendix D
- Machinery Towel, Item 211, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Compound, Item 282, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

11-5-28.1 Replace Tail Rotor Quadrant Upper and Lower Arm Bearings (BENCH).

11-5-28.1.1 Inspect.

Inspect bearing, in arm using dial indicator. **MAXIMUM ALLOWABLE RADIAL PLAY IS 0.005-INCH.** If radial play exceeds 0.005-inch, replace bearing.

11-5-28.1.2 Remove.

a. Using arbor press and step-type pusher, press bearing from arm (Figure 11-5-28-1).

b. Using paint remover, Item 230, Appendix D, strip paint from surface of part around bearing hub and clevises.

c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect arm for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace arm.

d. Coat paint-stripped areas with primer, Item 258, Appendix D. Touch up primer areas with paint (PARA 2-6-5).

11-5-28.1.3 Install.

a. Clean bearing surfaces of arm and outside diameter of bearings with acetone, Item 25, Appendix D.

CAUTION

Damage to bearings will result if surface primer is allowed to enter bearings. Do not allow surface primer to enter bearings.

NOTE

Surface primer and sealing compound must be made by same manufacturer.

b. Apply a light coat of sealing compound, Item 282, Appendix D, to outside diameter of bearing and inside diameter of arm. Allow to air-dry for at least 30 minutes.

c. Apply about a 1/16-inch bead of sealing compound, Item 273, Appendix D, to leading edge of bearing outside diameter and entrance edge of bearing bore in arm. Bead should be all around edge.

CAUTION

Damage to bearing will result if sealing compound is allowed to contact bearings. Do not allow sealing compound to come in contact with bearings.

d. Within 10 minutes after applying sealing compound, press bearing into quadrant (Figure 11-5-28-1).

e. Using machinery towel, Item 211, Appendix D, damp with acetone, Item 25, Appendix D, wipe surfaces of bearing clean of any surface primer.

f. Allow at least 10 minutes for sealing compound to cure before handling arm.

g. Perform rotational drag check of bearing as follows:

(1) Locally make two 1/8-inch thick x 2 1/4-inch diameter steel washers. Using drill, drill 1/4-inch hole through center of each washer.

(2) Put washers on each side of bearing on center race.

(3) Install 1/4-inch diameter x 1-inch long hexhead bolt through bearing. Install nut on bolt.

(4) Using torque wrench and socket, turn bolthead and measure rotational drag of bearings. **IF ROTATIONAL DRAG IS MORE THAN 24 INCH-OUNCES, REPLACE BEARING.**

(5) Remove bolt, washers, and nut from bearing.

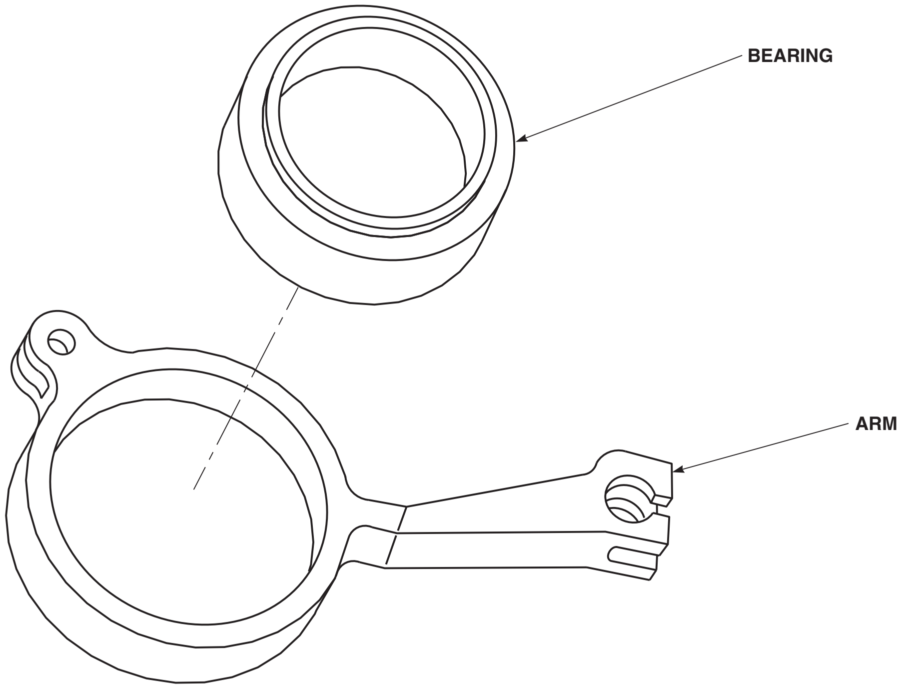


FIGURE 11-5-28-1. REPLACE TAIL ROTOR QUADRANT UPPER AND LOWER ARM BEARINGS (BENCH)

11-5-29 TAIL ROTOR QUADRANT SPRING CYLINDER SUPPORT TERMINAL BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Arbor Press
- Bolt, ¼-inch diameter x 1-inch long
- Bushing Installation Kit
- Fluorescent Penetrant Inspection Kit
- Nut, ¼-inch diameter
- Torque Wrench, 0 - 16 in. oz
- Torque Wrench, 30 - 150 in. lbs
- Washers

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Applicator, Item 78, Appendix D
- Cheesecloth, Item 90, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Primer, Item 258, Appendix D
- Sealing Compound, Item 302C, Appendix D
- Sealing Primer, Item 305, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

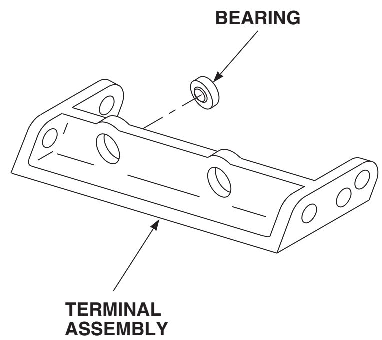


FIGURE 11-5-29-1. REPLACE TAIL ROTOR QUADRANT SPRING CYLINDER SUPPORT TERMINAL BEARINGS (BENCH)

11-5-29.1 Replace Tail Rotor Quadrant Spring Cylinder Support Terminal Bearings (BENCH).

11-5-29.1.1 Remove.

a. Using bushing installation kit and arbor press, remove bearing from terminal assembly (Figure 11-5-29-1).

b. Clean bearing bore in terminal assembly as follows:

(1) Using abrasive cloth, Item 8, Appendix D, remove retaining compound residue from fitting bores.

(2) Clean fitting bores, using applicator, Item 78, Appendix D, moistened with methyl ethyl ketone Item 217, Appendix D.

c. Inspect bearing bore in terminal assembly. **BORE TO BE NO GREATER THAN 0.6555-INCH.** If bore is beyond limit, replace terminal assembly.

d. Strip paint from surface of part around bearing hub and rivet holes, using paint remover, Item 230, Appendix D.

e. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect terminal assembly for cracks (ASTM E1417). **NONE ALLOWED.** If crack is found, replace terminal assembly.

f. Touch up stripped areas, with primer, Item 258, Appendix D. Touch up primer areas with paint (PARA 2-6-5).

11-5-29.1.2 Install.

a. Prepare bonding surfaces of bearing and bearing bore as follows:

(1) Mask bearing interior against contamination using 1-inch masking tape, Item 212, Appendix D.

NOTE

If a new bearing is being installed, sanding of bearing bonding surface is not required.

(2) Sand bonding surface of bearing and bearing bore lightly, using abrasive cloth, Item 8, Appendix D.

(3) Clean off abrasive residue using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone Item 217, Appendix D.

(4) Remove masking tape from bearing.

b. Clean bonding surfaces of fitting and bearing, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone Item 217, Appendix D.

c. Apply light coat of sealing primer, Item 305, Appendix D, to bearing outside diameter and inside diameter of terminal assembly bearing bore, using applicator, Item 78, Appendix D.

CAUTION

Protect primed surfaces from contamination during drying.

Do not touch primed surfaces with bare hands.

d. Let sealing primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step b. and step c.

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of sealing compound will make it unusable.

e. Apply even coat of sealing compound, Item 302C, Appendix D, to outside diameter of bearings and inside diameter of bearing bore in terminal assembly, using applicator, Item 78, Appendix D.

CAUTION

To avoid weak bond, bearing must be installed in terminal assembly within four minutes of retaining compound application. If this time is exceeded, step b. through step e. must be repeated.

f. Immediately install bearing in terminal support bearing bore with slight twisting motion to distribute retaining compound (Figure 11-5-29-1).

g. Allow fitting assembly to cure for 24 hours minimum.

h. Perform breakout torque check as follows:

(1) Install ¼-inch diameter x 1-inch long bolt through bearing with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**

(2) Rotate bearing inner races four complete turns.

(3) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360° TURN MUST NOT EXCEED 10 INCH-OUNCES.**

(4) Remove bolt, washers, and nut from bearing.

11-5-30 YAW CONTROL RODS (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 108A, Appendix D
- Corrosion Preventive Compound, Item 108B, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

References

- Appendix D
- PARA 11-5-11

11-5-30.1 Repair Yaw Control Rods (BENCH).

11-5-30.1.1. Disassemble.

a. Loosen rod end jamnut on adjustable rod end (Figure 11-5-30-1).

b. Unscrew rod end from tube assembly. Remove jamnut from rod end.

11-5-30.1.2. Inspect/Repair.

Inspect and repair control rods (PARA 11-5-11).

11-5-30.1.3. Assemble.

a. Coat all threaded, mating, and interior surfaces of control rod assembly with corrosion preventive compound, Item 108A, Appendix D.

b. Install jamnut on adjustable rod end and thread rod end into tube assembly (Figure 11-5-30-1).

c. Set control rod length.

d. **TIGHTEN ROD END JAMNUT HANDTIGHT.** Final adjustment will be made when control rod is installed in helicopter.

e. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint as required, with epoxy primer.

f. After assembly and final adjustment of control rod, seal all mating and threaded surfaces, witness holes, and exposed threads with corrosion preventive compound, Item 108B, Appendix D.

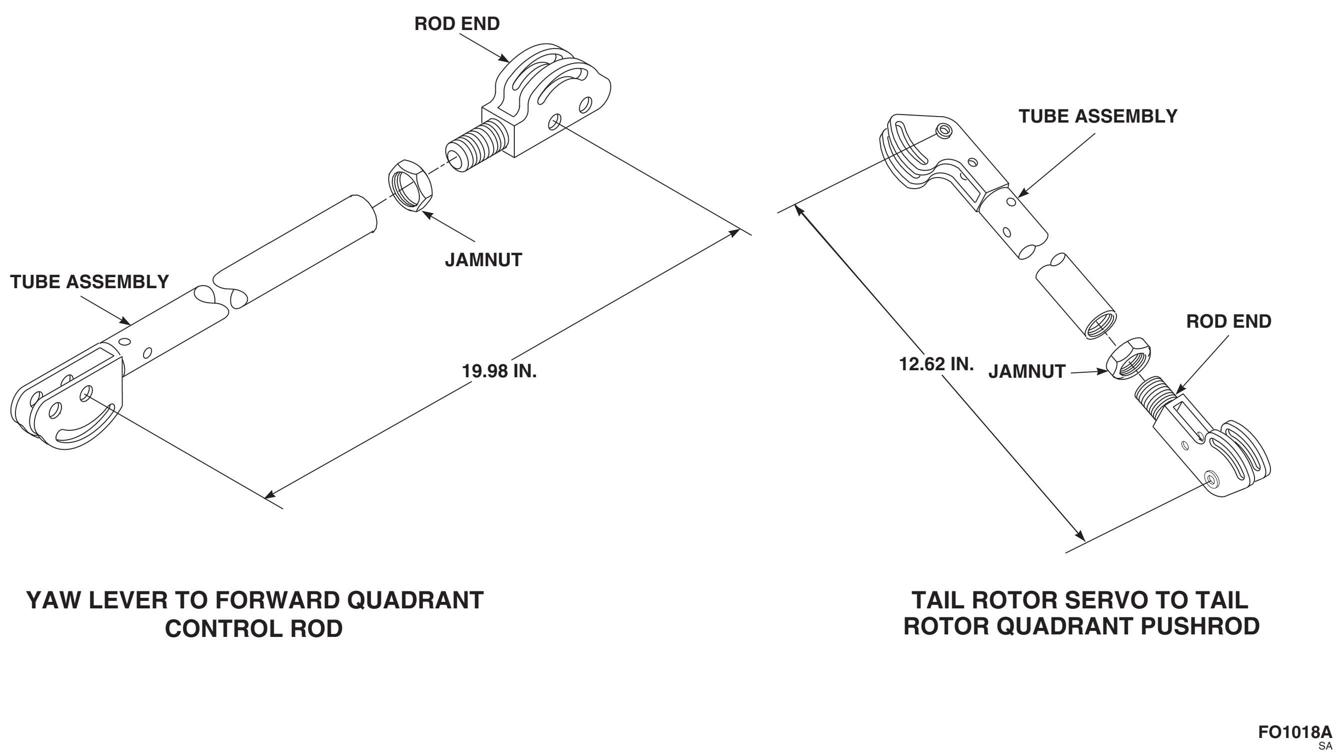


FIGURE 11-5-30-1. REPAIR YAW CONTROL RODS (BENCH)

11-5-31 CONTROL ROD IDENTIFICATION PLATE (BENCH).

PARAGRAPH BREAKDOWN

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Heat Gun
- Nonmetallic Scraper, Locally-Made
- Roller

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cheesecloth, Item 90, Appendix D
- Epoxy Coating Kit, Item 126B, Appendix D
- Polyurethane Coating, Item 241D, Appendix D

References

- Appendix D
- PARA 11-3-17

11-5-31.1 Replace Control Rod Identification Plate (BENCH).

11-5-31.1.1 Remove.

WARNING

FSCAP

Control rods contain critical surfaces which must not be damaged. Use care when removing identification plate.

a. Using nonmetallic scraper, remove identification plate(s) (Figure 11-5-31-1, Sheet 1 and Sheet 2).

b. Clean adhesive residue from control rod surface using cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D. Wipe surface dry using cheesecloth, Item 90, Appendix D.

c. Perform main rotor control rod inspection (PARA 11-3-17).

11-5-31.1.2 Install.

a. Clean control rod surface using cheesecloth, Item 90, Appendix D, dampened with isopropyl alcohol, Item 72, Appendix D. Wipe surface dry using cheesecloth, Item 90, Appendix D.

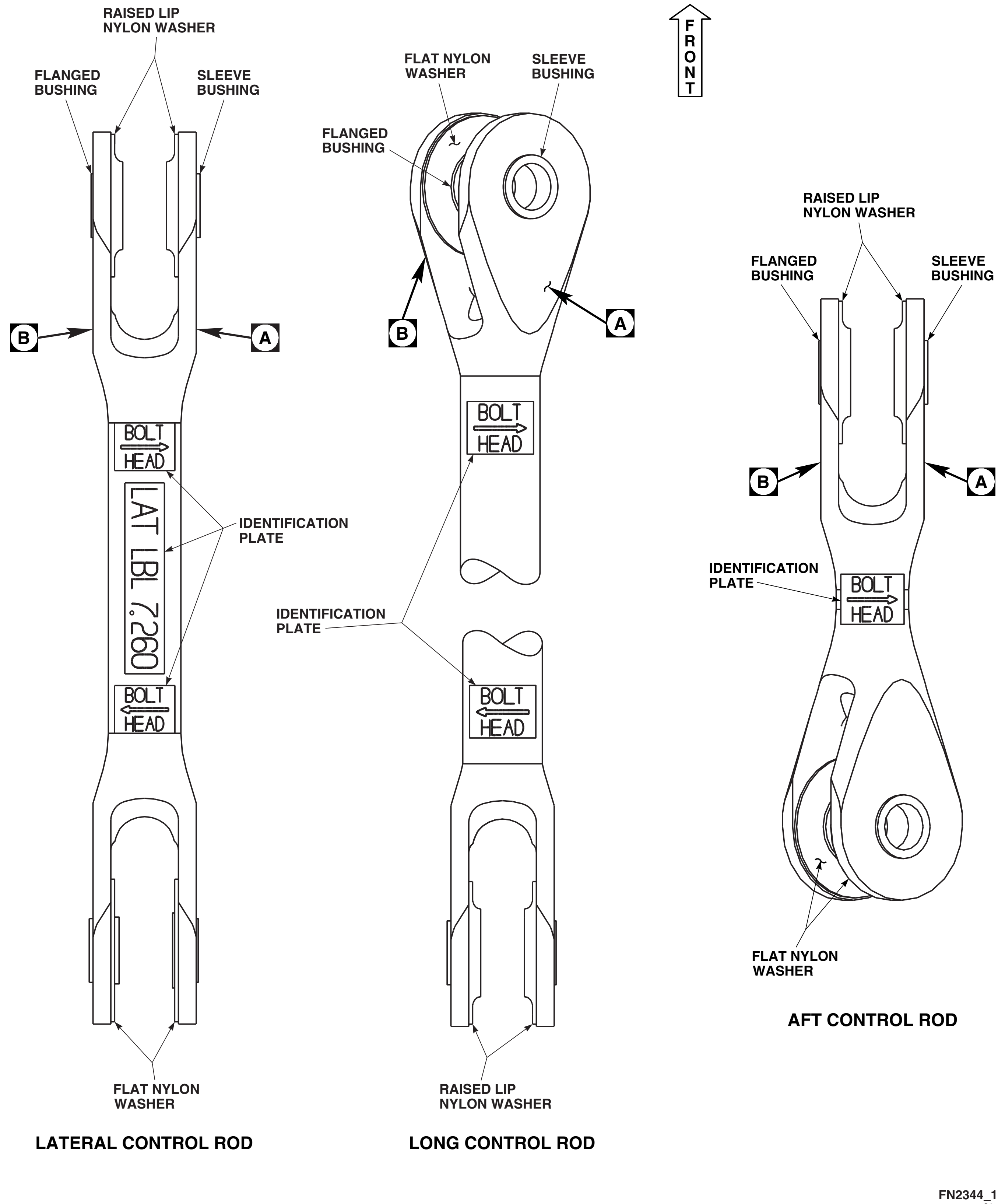
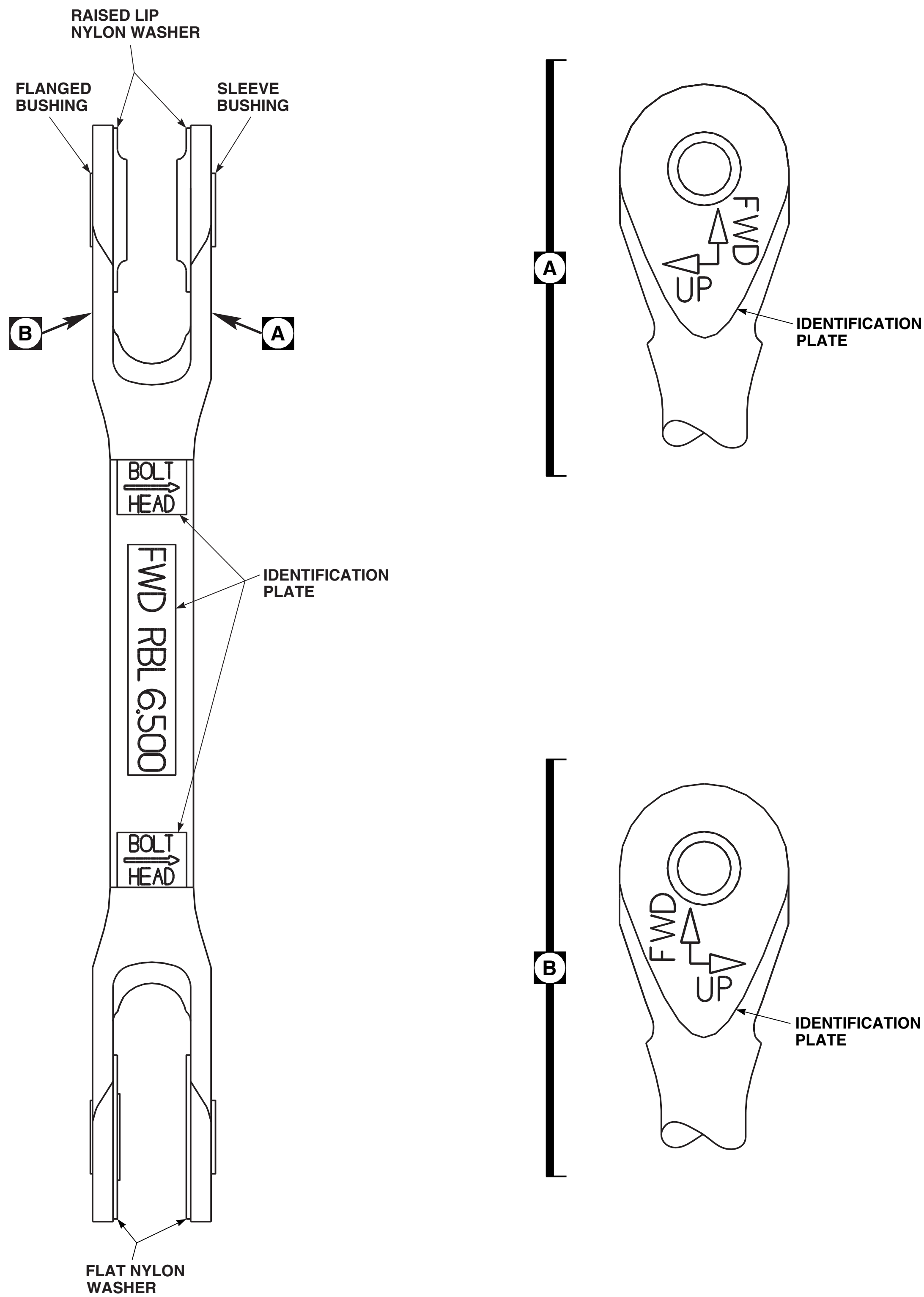


FIGURE 11-5-31-1. CONTROL ROD IDENTIFICATION PLATE (BENCH) (SHEET 1 OF 2)



FORWARD CONTROL ROD

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FIGURE 11-5-31-1. CONTROL ROD IDENTIFICATION PLATE (BENCH) (SHEET 2 OF 2)

NOTE

Make sure control rod is oriented correctly prior to installing identification plate.

b. Remove backing paper from identification plate and install identification plate on control rod (Figure 11-5-31-1, Sheet 1 and Sheet 2).

NOTE

Adhesive bond strength can be improved by applying heat to identification plate. Do not exceed 150°F (65°C).

c. Using roller, apply pressure to entire surface of identification plate. If required, using heat gun, apply heat to identification plate while applying pressure with roller.

d. Using epoxy coating kit, Item 126B, Appendix D, or polyurethane coating, Item 241D, Appendix D, seal edges of identification plate by applying a 1⁄8 to 3⁄8-inch wide coat around perimeter of identification plate. Make sure all edges are sealed.